Electrical Energy and Machines

Book 5 — The Electrical Energy & Machines Core

Magnetic circuits and energy conversion, transformers and electrical machines, motor drives, power generation and storage, power-system analysis, transmission, distribution and protection, power quality, smart grids and electrical safety.

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Part I — Magnetic Circuits and Energy Conversion

Almost every kilowatt-hour generated, transmitted or used in industry passes through a magnetic field at least twice: once in a generator, once or more in a transformer, and usually again in a motor. Part I builds the engineering tools for that field. It starts where *Engineering Physics: Top Down* stops, with quasi-static, lumped Maxwell equations and the motor as an ideal gyrator, and carries that physics into hardware: iron cores with air gaps that fringe and leak, steels that saturate and dissipate energy, windings whose flux linkage depends on current and on the position of a rotor, forces and torques computed from stored energy, a rotating field produced by stationary windings, and losses that set efficiency, temperature and rating. The examples, from data-centre filter reactors and refinery pump motors to lifting magnets and electric-vehicle traction motors, show the same principles at work across the energy field.

By the end of this part you should understand

- how to model an iron core with air gaps as a magnetic circuit of reluctances, compute its flux, inductance and ampere-turn requirement, and correct the result for fringing, leakage and the slotted gaps of machines;
- how saturation changes inductance, how to solve a nonlinear magnetic circuit from a $B{-}H$ curve, and where saturation sets limits on transformers, current transformers, machines and drives;
- how Faraday's law, written for a flux linkage that depends on current and position, separates transformer EMF from speed EMF, and why the applied voltage, not the current, sets the flux of a winding;
- how Lenz's law explains shading rings, eddy-current brakes, laminations and the trapped flux of disconnected motors;
- how self- and mutual inductances of real windings depend on saturation and position;
- how the energy balance of a lossless coupling field gives torque, and how co-energy gives force in saturating devices;
- how three-phase windings create a rotating magnetic field, and what sets synchronous speed, magnetising current, reversal and the effects of unbalance;
- where losses arise, how they set efficiency, and how insulation temperature limits turn losses into ratings.

Chapters

Chapter	Central question	What it delivers
Chapter 1 — Magnetic Circuit Fundamentals	How do you predict the flux, inductance and ampere-turn requirement of an iron core with air gaps, and where does the linear model fail?	Flux, field intensity, flux density and permeability as engineering data; MMF and reluctance; series and parallel paths in E–I and three-limb cores; air gaps and inductor design; fringing corrections and Carter's coefficient; saturation, the B–H curve, hysteresis and the permanent-magnet load line
Chapter 2 — Electromagnetic Induction	How does a changing flux linkage, from changing current or from motion, produce voltage, and how does the energy balance produce torque?	Faraday's law for $\lambda(i,\theta)$; transformer and speed EMF; Lenz's law in contactors, brakes and machines; motional EMF and its limits; volt-second control of flux; secant, incremental, magnetising, leakage and position-dependent inductances; mutual coupling in machines and with pipelines; the lossless coupling field; reluctance and alignment torque; the full electromechanical energy balance
Chapter 3 — Energy Conversion Principles	How do electromagnetic devices convert energy in both directions, and how do losses set efficiency, temperature and rating?	Air-gap shear stress and machine sizing; generator action and regenerative braking; co-energy and force in saturating actuators; magnetic pressure; energy-conversion loops; the rotating magnetic field and magnetising current; loss mechanisms and lamination choice; efficiency and IE classes; insulation classes, thermal time constants and short-time ratings

How this part connects to the rest of the book

Part I is the foundation for everything that converts energy through a magnetic field. In Part II the transformer of Chapter 4 is the magnetic circuit of Chapter 1 with the coupled windings of Chapter 2 and the losses of Chapter 3; the DC, induction, synchronous and special machines of Chapters 5–8 are the coupling field, rotating field and torque mechanisms of Chapters 2 and 3 in particular geometries; and Chapter 9 turns the loss and thermal principles into design and rating. The drives of Part III control flux and torque through the volt-second and co-energy relations developed here, and Parts IV–VII rely throughout on the saturation, inductance and loss behaviour introduced in this part.

Part I deliberately does not repeat its foundations. The physics (Maxwell's equations, constitutive relations, magnetostatics, the quasi-static limit, the effort–flow template, Poynting's theorem, thermodynamics and heat transfer) is taken from *Engineering Physics: Top Down*, chiefly Chapters 11, 14, 17 and 19. The circuit theory of inductance, coupled coils, AC power and three-phase circuits is taken from *Circuits, Signals and Communications*, chiefly Chapters 5 and 10. The microstructural origins of permeability, core loss and insulation ageing belong to *The Materials Core*.

Chapter 1 — Magnetic Circuit Fundamentals

Part I · Magnetic Circuits and Energy Conversion

A data-centre operator installs a 400 V, 50 kvar detuned power-factor-correction bank to keep its utility power factor above 0.95. Each phase has a capacitor in series with an iron-cored reactor whose nameplate reads 0.70 mH at 75 A. For a year nothing happens. Then a new row of servers with uncontrolled rectifier front ends is commissioned, the fifth-harmonic current in the bank rises, and the reactors begin to hum loudly and run hot. A technician measures the inductance at twice rated peak current and finds it 30 % below nameplate. The capacitor bank, designed to be tuned safely below the fifth harmonic, is now tuned closer to it, which draws still more harmonic current, which saturates the reactors further. Nothing in the reactor is broken. Its iron has simply left the straight part of its B–H curve, and every number on the nameplate was calculated on the assumption that it never would.

The same calculation, and the same assumption, sits behind every transformer, motor, generator, contactor, current transformer and lifting magnet in this book. This chapter builds the tool used to make it, the **magnetic circuit**, and then shows where it breaks.

> **How do you predict the flux, the inductance and the required ampere-turns of an iron core with air gaps, from its dimensions and material data, without solving a field problem? Where does the linear reluctance model stop working (leakage, fringing, saturation), and how large are the errors in real devices?**

Concept	Engineering destination
Magnetic flux	Transformer EMF and volts-per-turn, machine flux per pole, flux balance in three-limb cores
Magnetic field intensity	Ampere-turn budgets of cores and gaps, current-transformer burden limits
Flux density	Design limits of electrical steels, tooth and yoke sizing, magnetic loading of machines
Permeability	Material selection (steels, amorphous, ferrite, powder cores), inductance factor of cores
Reluctance	Inductor and reactor design, equivalent circuits of cores, reluctance motors
Magnetomotive force	Excitation of field windings, magnetising current of transformers and motors
Series and parallel paths	Shell-type and three-limb transformer cores, E-cores, flux sharing in machine yokes
Air gaps	Gapped reactors, contactor and relay magnets, machine air gaps, energy storage in inductors
Fringing	Effective gap area, Carter coefficient of slotted machines, stray loss near gaps
Saturation	Inductance collapse in reactors, transformer inrush, over-fluxing limits, open-circuit curves of generators

Roadmap. Flux, and where the physics book hands it over (1.1) → the field intensity and Ampère's law as an ampere-turn budget (1.2) → flux density as the design variable (1.3) → permeability as material data (1.4) →

magnetomotive force (1.5) → reluctance and the magnetic circuit (1.6) → series and parallel paths (1.7) → air gaps (1.8) → fringing and leakage (1.9) → saturation, hysteresis and permanent magnets (1.10).

1.1 Magnetic flux

Where *Engineering Physics* leaves off

Engineering Physics derives Maxwell's equations (Ch. 11 §11.4), adds the constitutive relation $\mathbf{B} = \mu\mathbf{H}$ for linear media (Ch. 11 §11.9), solves magnetostatics and defines inductance as $L = N\Phi/I$ with stored energy $\int B^2/2\mu_0 \, dV$ (Ch. 11 §11.11), and shows that at 50 or 60 Hz every power device is deep inside the quasi-static limit (Ch. 11 §11.12; Ch. 16 §16.10). It then lumps fields into effort–flow elements and lists a magnetic row whose effort is magnetomotive force and whose flow is the rate of change of flux (Ch. 17 §17.2). Those results are taken over here as given. What the physics book does not contain is the engineering model built on them: a core made of a nonlinear, lossy material, cut by air gaps, whose flux leaks and fringes, and whose ampere-turn requirement must be found from a datasheet curve rather than a constant μ. That model, the magnetic circuit, is the subject of this chapter.

Flux as a conserved quantity in a core

The magnetic flux through a surface S is $\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}$, measured in webers (Wb = V·s). Because $\nabla \cdot \mathbf{B} = 0$, the flux leaving any closed surface is zero. For an engineer this has one decisive consequence: **flux behaves like a current that cannot be stored at a junction.** Where a core divides, the fluxes in the branches add up to the flux arriving; where limbs meet, the fluxes sum to zero. This is the magnetic counterpart of Kirchhoff's current law, and unlike its electrical cousin it needs no lumping assumption at all, because no "magnetic charge" exists to accumulate.

What *is* approximate is the assumption that flux stays inside the iron. An electrical conductor is surrounded by insulation that is 10^{20} times less conductive; a magnetic core is surrounded by air that is only 10^3–10^5 times less permeable. Some flux therefore always takes paths through air, and the magnetic circuit has to account for it as leakage (Section 1.9).

Flux linkage

A coil of N turns that all encircle the same flux Φ has flux linkage $\lambda = N\Phi$ (weber-turns). In real windings different turns link different amounts of flux, and the flux linkage is the sum $\lambda = \sum_k \Phi_k$ over the turns. Flux linkage, not flux, is what Faraday's law turns into voltage (Chapter 2), and it is what a winding "sees" of the core.

Table 1.1 — Representative flux magnitudes in power equipment (typical).

Device	Quantity	Typical value
400 kVA, 11 kV/415 V distribution transformer	Flux per limb (peak)	40–50 mWb
100 MVA generator step-up transformer	Flux per limb (peak)	0.4–0.6 Wb
15 kW, 4-pole induction motor	Flux per pole (fundamental)	8–12 mWb
500 MW, 2-pole turbo-generator	Flux per pole	3–5 Wb
200 A/5 A protection current transformer	Core flux at rated burden	0.1–1 mWb
AC contactor magnet (230 V coil)	Flux in closed magnet	0.2–0.5 mWb

The range is six orders of magnitude, yet each value is fixed by the same relation that Chapter 2 develops: the volts per turn of the winding divided by $4.44 f$.

1.2 Magnetic field intensity

Ampère's law as an ampere-turn budget

The field intensity **H** (A/m) enters through the integral form of Ampère's law, which in the quasi-static limit (displacement current negligible at power frequency) reads

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

as derived in *Engineering Physics*, Ch. 11 §11.4 and §11.11. For a coil of N turns carrying i, a closed path through the core links the current N times, so the right-hand side is Ni.

The engineering reading of this equation is a **budget**. The ampere-turns Ni supplied by the winding are "spent" along the path: each segment k of length l_k in which H is roughly uniform consumes $H_k l_k$ ampere-turns, and the sum must balance:

$$Ni = \sum_{k} H_k \, l_k$$

A good core is one in which almost the whole budget is spent where it does something useful, usually in an air gap, and almost nothing is wasted in the iron.

Values and units

H is measured in amperes per metre (often "ampere-turns per metre" in older texts; the turn is dimensionless). In electrical steel at a working flux density of 1.0–1.3 T, H is only 100–400 A/m: a 1 m path costs a few hundred ampere-turns. In an air gap at the same flux density, $H = B/\mu_0 \approx 0.8\text{–}1.0 \times 10^6$ A/m, so a single millimetre of air costs 800–1000 ampere-turns. This contrast, roughly a factor of 3000, is the whole reason magnetic circuits are designed the way they are.

> **Status of the principle: Ampère's circuital law, $\oint \mathbf{H}\cdot d\mathbf{l} = Ni$**
> *Fundamental.* It is the Ampère–Maxwell equation in the quasi-static limit; at 50–60 Hz the displacement-current term is smaller than the conduction term by a factor of order $(L/\lambda)^2 \sim 10^{-12}$ for any power device, so the law is exact for engineering purposes. The approximation lies only in replacing $\oint$ by a sum $\sum H_k l_k$ with uniform H in each segment.

The segment-uniformity assumption

Replacing the line integral by $\sum H_k l_k$ assumes uniform H along each segment and a well-defined path length. In a rectangular core the flux crowds towards the inner corners, so the mean (centre-line) path is an approximation, good to a few percent when the window is large compared with the limb width. For machine teeth, where H varies strongly along the tooth, designers split the path into many short segments; IEC 60205 defines an effective path length l_e for standard ferrite core shapes so that the one-segment model gives the right answer.

1.3 Flux density

The design variable

Flux density $B = \Phi/A$ (tesla, T = Wb/m²) is the quantity designers actually limit, because the material properties (permeability, loss per kilogram, saturation) are all functions of B. Choosing B fixes the core cross-section for a given flux: a transformer limb carrying 45 mWb at 1.6 T needs $A = 0.045/1.6 = 0.028$ m² of net iron. Choose 1.8 T and the core is 11 % smaller, but its magnetising current and core loss rise steeply (Section 1.10 and Chapter 3).

Table 1.2 — Typical working flux densities (peak values unless stated).

Location	Typical B	What limits it
Earth's field	25–65 μT	—
Under a 400 kV line at ground level, full load	5–30 μT	ICNIRP public reference level 200 μT at 50 Hz
Ferrite core of a 100 kHz converter transformer	0.1–0.3 T	Core loss at high frequency; saturation near 0.4 T at 100 °C
Amorphous-core distribution transformer	1.3–1.4 T	Saturation of the alloy at about 1.56 T
Grain-oriented steel, distribution and power transformers	1.5–1.75 T	Magnetising current, no-load loss, noise, over-excitation margin
Induction-motor air gap (fundamental)	0.7–0.9 T	Magnetising current and tooth saturation
Induction-motor stator teeth	1.5–1.9 T	Saturation and tooth loss
Induction-motor stator yoke	1.2–1.6 T	Core loss
Surface-PM motor air gap	0.8–1.1 T	Magnet remanence and demagnetisation
Large turbo-generator air gap	0.8–1.1 T	Field-winding current and heating
Lifting-magnet pole face	1.2–1.5 T	Saturation of mild steel
MRI scanner (superconducting, air-cored)	1.5–3 T	Conductor critical field and cost

Iron-cored devices cluster between about 1 and 2 T because that is where the useful range of ferromagnetic materials ends (Section 1.10); a magnet that needs more must use superconductors. The air-gap values of machines are lower than their iron values because every tenth of a tesla in the gap costs magnetising current.

Uniform and non-uniform B

$B = \Phi/A$ is an average: a good one in a transformer limb, a poor one in machine teeth (where flux funnels into a narrow root), at corners, and at joints of stacked cores, where flux crosses between laminations and local densities exceed 2 T. Designers check the narrowest section of each path separately.

Stacking factor

Laminated cores are built from insulated sheets; the coating and the irregularity of the stack mean that the iron occupies only a fraction k_s of the gross cross-section. Typical stacking factors are 0.95–0.98 for 0.3–0.5 mm steels and 0.80–0.88 for amorphous ribbon about 25 μm thick. The flux density in the iron is $B = \Phi/(k_s A_{\text{gross}})$, while in an adjacent air gap the flux spreads over the gross area.

1.4 Permeability

Permeability as material data

In a linear medium $\mathbf{B} = \mu\mathbf{H}$ with $\mu = \mu_r\mu_0$ and $\mu_0 = 4\pi \times 10^{-7}$ H/m (*Engineering Physics*, Ch. 11 §11.9). The physics book treats μ as a material constant emerging from electron spin and exchange; the origin of ferromagnetism in iron's unpaired $3d$ electrons is outlined in *Engineering Physics*, Ch. 5 §5.7, and its

disappearance above the Curie temperature (770 °C for iron) as a Landau phase transition in Ch. 10 §10.9. For engineering purposes a ferromagnetic μ is not a constant. It depends on the flux density, on the history of magnetisation, on frequency, on temperature, on mechanical stress, and on the direction relative to the rolling direction of the sheet. It is read from data, and several different "permeabilities" are in use:

- **Amplitude (secant) permeability** $\mu_a = B/H$ at a point on the normal magnetisation curve. It is what the magnetic circuit uses to compute a reluctance at a given operating point.
- **Initial permeability** μ_i, the limit of μ_a at very low B. It matters for current transformers and sensors that work at low flux density.
- **Maximum permeability** $\mu_{\max}$, the peak of μ_a, usually reached between 0.5 and 1.2 T in electrical steels. Datasheets quote it as a figure of merit.
- **Incremental (differential) permeability** $\mu_\Delta = dB/dH$, the slope of the curve at the operating point. It sets the inductance seen by a small ripple current superimposed on a large DC or low-frequency current, for example in a DC-link choke or a smoothing reactor.

Table 1.3 — Magnetic materials for power equipment (typical values at room temperature).

Material	Relative permeability ($\mu_{r,\max}$ or range)	Saturation B_s (T)	Typical use
Low-carbon (mild) steel, solid	1000–2000	2.0–2.1	Lifting magnets, DC machine frames, solid rotor poles
Non-oriented silicon steel (2–3.5 % Si), 0.35–0.65 mm	4000–10 000	1.95–2.05	Motors, generators, small transformers
Grain-oriented silicon steel (3 % Si), 0.23–0.30 mm, rolling direction	30 000–80 000	2.03	Distribution and power transformers
Cobalt–iron (49 % Co)	5000–15 000	2.3–2.4	Aircraft generators, high-power-density motors
Amorphous iron-based ribbon, about 25 µm	50 000–600 000 (after annealing)	1.56	Low-loss distribution transformers
Nanocrystalline ribbon	20 000–200 000	1.2–1.25	Common-mode chokes, high-frequency transformers, CT cores
Manganese–zinc ferrite	2000–15 000 (initial)	0.4–0.5 at 25 °C, about 0.35–0.4 at 100 °C	Switch-mode power supplies, 20 kHz–1 MHz
Iron-powder and alloy-powder cores	10–150 (distributed gap)	1.0–1.5	PFC and DC-link chokes, output filters

Two lessons follow. Permeabilities span more than four orders of magnitude, but every one of these materials saturates between 0.4 and 2.4 T. And permeability is the variable that differs most between nominally identical cores: a ±25 % batch spread in μ_r is normal for ferrites and not unusual for steels. A design whose performance depends strongly on μ_r is therefore a fragile design, which is one of the reasons for deliberately adding air gaps (Section 1.8).

> **Status of the principle: Linear constitutive law $B = \mu H$ for ferromagnetic cores**
> *Approximate.* Valid for small excursions about an operating point or for a core whose reluctance is dominated by an air gap; it ignores saturation, hysteresis, frequency and temperature dependence, and anisotropy. For electrical steels the secant μ_r varies by more than a factor of ten between 0.5 T and 1.7 T, so a single value is meaningful only together with the flux density at which it was read.

1.5 Magnetomotive force

Definition and engineering meaning

The magnetomotive force (MMF) of a winding is $\mathcal{F} = Ni$, measured in amperes (often written ampere-turns, At). It is the left-hand side of the ampere-turn budget of Section 1.2: the "effort" that drives flux around the circuit. *Engineering Physics* places it in exactly this role, as the effort variable of the magnetic domain (Ch. 17 §17.2.2).

The engineering content of the definition lies in two practical facts.

Only the net enclosed current counts. A path that encloses the primary and the secondary windings of a loaded transformer sees $N_1 i_1 - N_2 i_2$, which is small, because the load ampere-turns of the secondary almost cancel those of the primary. The core flux of a transformer is set by the small difference, the **magnetising MMF**, not by the large load currents. This is why a transformer core can be sized for flux and not for load current (Chapter 4).

Distributed windings produce distributed MMF. In a machine the winding is spread over many slots, and the MMF is a function of position around the air gap. Its fundamental spatial component, produced by three phases together, is the rotating MMF wave of Chapter 3. The single-number $\mathcal{F} = Ni$ of this chapter is its one-dimensional ancestor.

MMF requirements in practice

Table 1.4 — Typical magnetising MMF and current (as a fraction of rated current).

Device	Where the MMF is spent	Magnetising current (% of rated)
Large power transformer, grain-oriented core	Joints and iron	0.1–0.5
Distribution transformer, 100–1000 kVA	Joints and iron	0.5–2
Induction motor, 4-pole, 15–200 kW	Air gap (about 70–85 % of total)	25–40
Induction motor, 8-pole or more, or small rating	Air gap	40–60
Salient-pole synchronous generator (field winding)	Air gap	Field current set by excitation system

The difference between the first two rows and the next two is the air gap. A transformer core has only the unavoidable gaps at its joints; a rotating machine must have a mechanical clearance between rotor and stator,

and that clearance dominates its magnetising requirement.

1.6 Reluctance

The magnetic Ohm's law

Combine the segment form of Ampère's law, $\mathcal{F} = \sum H_k l_k$, with $B_k = \mu_k H_k$ and flux continuity $\Phi = B_k A_k$ for a single path:

$$\mathcal{F} = \Phi \sum_k \frac{l_k}{\mu_k A_k} = \Phi\,\mathcal{R}, \qquad \mathcal{R}_k = \frac{l_k}{\mu_k A_k}$$

The **reluctance** $\mathcal{R}$ (A/Wb, or H^{-1}) plays the role of resistance, MMF that of voltage, and flux that of current. Its reciprocal is the **permeance** $\mathcal{P} = 1/\mathcal{R}$ (H). The relation between a winding's inductance and the reluctance of its path, $L = N^2/\mathcal{R} = N^2\mathcal{P}$, is stated and used in *Circuits, Signals and Communications*, Ch. 5 and is taken as known here. Manufacturers of cores quote the permeance per turn squared as the inductance factor A_L.

> **Status of the principle: The magnetic circuit $\mathcal{F} = \mathcal{R}\Phi$**
> *Approximate*. It follows from Ampère's law and flux continuity, both exact, plus three assumptions: flux confined to defined paths (no leakage), uniform B and H over each segment, and constant μ in each segment. It is accurate to a few percent for closed, gapped cores below saturation and fails progressively with leakage, fringing (gap length comparable to pole width) and saturation.

What the analogy does and does not carry

The electrical analogy is useful because every network method (series and parallel reduction, nodal analysis, Thévenin equivalents) carries over unchanged to linear magnetic circuits. Three differences must be kept in mind.

Electric circuit	Magnetic circuit	Where the analogy breaks
Current I (A)	Flux Φ (Wb)	Flux is not a flow of anything; nothing moves around a DC magnetic circuit
EMF V (V)	MMF $\mathcal{F}$ (A)	—
Resistance $R = l/\sigma A$	Reluctance $\mathcal{R} = l/\mu A$	Resistance dissipates; reluctance **stores** energy $\frac{1}{2}\mathcal{R}\Phi^2$
Conductivity ratio conductor/insulator about 10^{20}	Permeability ratio iron/air about 10^3–10^5	Flux leaks; there is no magnetic insulator
Ohm's law linear over many decades	μ depends on B	Reluctance changes with flux: saturation

The resistance–reluctance row is worth a second look, because it connects back to *Engineering Physics*. In the effort–flow template the magnetic effort is MMF and the magnetic flow is the *rate of change* of flux, $\dot{\Phi}$ (Ch. 17 §17.2.2). With that pairing a reluctance, $\mathcal{F} = \mathcal{R}\Phi = \mathcal{R}\int \dot{\Phi}\,dt$, relates effort to the integral of flow: it is a **capacitor-type** (storage) element, with permeance as its capacitance, not a resistor. A winding links the electrical and magnetic domains as $v = N\dot{\Phi}$ and $\mathcal{F} = Ni$, which is the gyrator law of Ch. 17 §17.5 with gyration

ratio N. A gyrator terminated in a capacitor looks like an inductor, and that is exactly how $L = N^2\mathcal{P}$ appears. This "gyrator–capacitor" picture is energy-consistent and is used in circuit simulators to model multi-winding magnetics; the familiar "flux as current" picture is simpler for static ampere-turn calculations and is the one used in the rest of this chapter. Both describe the same physics; they differ only in which variable is called the flow.

Worked example — Reluctance budget of a gapped filter reactor. The reactor of the opening problem has a laminated non-oriented steel core of gross section 60×100 mm ($A = 6.0 \times 10^{-3}$ m²), stacking factor 0.95 (net iron $A_{Fe} = 5.7 \times 10^{-3}$ m²), mean iron path $l_c = 0.50$ m, and two air gaps of 1.0 mm each in series ($g = 2.0$ mm in total). The winding has $N = 14$ turns. Take $\mu_r = 4000$ for the iron and neglect fringing for now.

Reluctances.

$$\mathcal{R}_c = \frac{l_c}{\mu_0 \mu_r A_{Fe}} = \frac{0.50}{4\pi \times 10^{-7} \times 4000 \times 5.7 \times 10^{-3}} = 1.745 \times 10^{4}\ \text{A/Wb}$$

$$\mathcal{R}_g = \frac{g}{\mu_0 A} = \frac{2.0 \times 10^{-3}}{4\pi \times 10^{-7} \times 6.0 \times 10^{-3}} = 2.653 \times 10^{5}\ \text{A/Wb}$$

The total is $\mathcal{R} = 2.827 \times 10^{5}$ A/Wb, of which the gaps contribute 93.8 %: two millimetres of air have fifteen times the reluctance of half a metre of iron.

Inductance. $L = N^2/\mathcal{R} = 196 / 2.827 \times 10^{5} = 0.693$ mH, which is the nameplate value.

Operating flux density. The rated current is 75 A rms; allowing a 20 % margin for harmonic content, the design peak is $75\sqrt{2} \times 1.2 = 127.3$ A. The flux is $\Phi = Ni/\mathcal{R} = 14 \times 127.3 / 2.827 \times 10^{5} = 6.30$ mWb, giving $B_{Fe} = 6.30 \times 10^{-3} / 5.7 \times 10^{-3} = 1.11$ T in the iron and 1.05 T in the gap. The ampere-turn budget is 1782 At, of which 1672 At are spent in the gaps and 110 At in the iron.

Where the energy is. The stored energy at peak current is $\tfrac{1}{2}Li^2 = 5.62$ J. Of this, $\tfrac{1}{2}\mathcal{R}_g\Phi^2 = 5.27$ J (94 %) sits in the 12 cm³ of air in the gaps, and only 0.35 J in the 2.85 litres of iron.

Without the gap. The same core with no gap would have $L = N^2/\mathcal{R}_c = 11.2$ mH, sixteen times more, but its iron would reach 1.2 T at only 8.5 A. An ungapped core is a good inductor for very small currents and a useless one for a 75 A filter. The gap trades inductance per turn for current capacity, and makes the inductance almost independent of the uncertain μ_r: a 25 % change in μ_r changes L by only about 2 %.

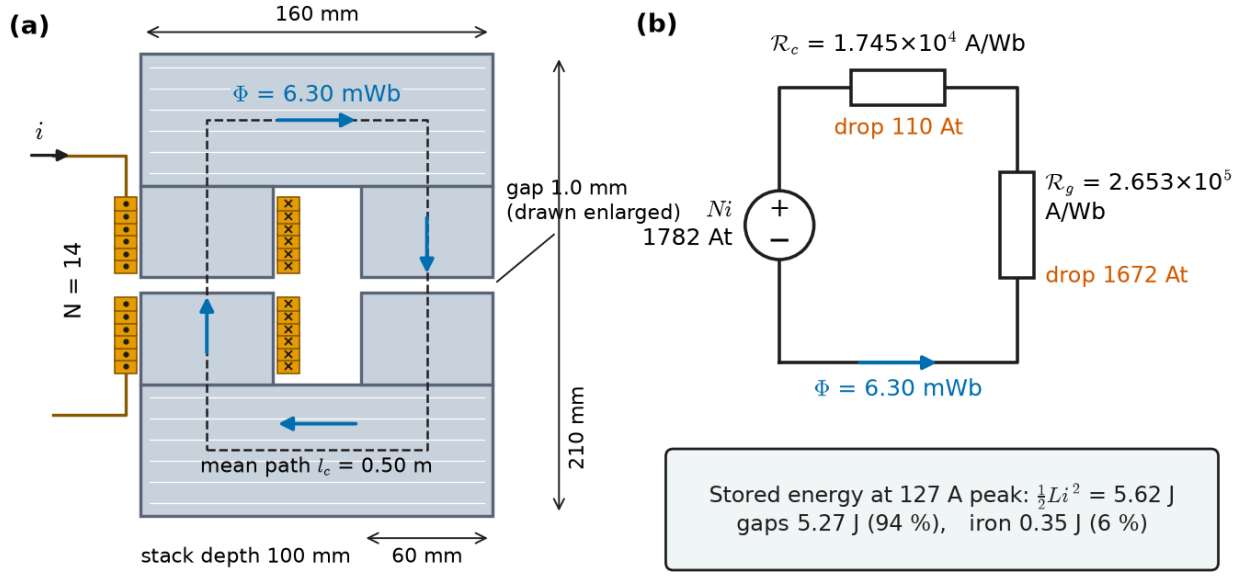


Figure 1.1 — A gapped filter reactor and its magnetic equivalent circuit: two 1 mm gaps carry 94 % of the reluctance and store 94 % of the energy.

Figure 1.1 shows the core and its equivalent circuit side by side; the circuit is a single loop because, so far, all flux follows one path.

1.7 Series and parallel magnetic paths

The two network laws

With reluctances as elements, a magnetic circuit is solved exactly like a resistive network:

- **Node law (flux continuity).** The fluxes entering a junction of limbs sum to zero, $\sum \Phi_k = 0$. This is exact, from $\nabla \cdot \mathbf{B} = 0$.
- **Loop law (Ampère).** Around any closed loop of the core, the sum of MMF drops $\mathcal{R}_k \Phi_k$ equals the sum of winding MMFs enclosed, $\sum \mathcal{R}_k \Phi_k = \sum N_j i_j$.

Elements carrying the same flux are in **series**, and their reluctances add. Elements sharing the same MMF drop between two junctions are in **parallel**, and their permeances add. For linear elements, the reduction rules, superposition of the MMFs of several windings, and nodal analysis all apply directly.

Shell-type and three-limb cores

Real cores are rarely a single loop. The shell-type (E–I) core of Figure 1.2, used for single-phase transformers, chokes and most small mains transformers, carries its winding on the centre limb; the flux returns through two outer limbs in parallel, each of half the centre limb's cross-section. The three-limb core of a three-phase transformer has three windings on three limbs joined by common yokes: each limb's flux returns through the other two, and because the three fluxes of a balanced supply sum to zero, no return limb is needed. A five-limb core adds two unwound outer limbs to provide a return path for zero-sequence flux (Chapter 4).

Worked example — Flux sharing in an E–I core with a faulty joint. An E–I core has a centre limb of 40×50 mm ($A_c = 2.0 \times 10^{-3}$ m²) and outer limbs of 20×50 mm ($A_o = 1.0 \times 10^{-3}$ m²). The centre path

(centre limb only) is 60 mm long and contains a deliberate gap of 0.5 mm; each outer path (yoke and outer limb, from the top to the bottom of the centre limb) is 180 mm long. The winding has 200 turns. Take $\mu_r = 3000$.

Balanced core. The reluctances are

$$\mathcal{R}_{cen} = \frac{0.060}{\mu_0\, 3000\, A_c} + \frac{0.5\times10^{-3}}{\mu_0 A_c} = 7.96\times10^{3} + 1.989\times10^{5} = 2.069\times10^{5}\ \text{A/Wb}$$

$$\mathcal{R}_{out} = \frac{0.180}{\mu_0\, 3000\, A_o} = 4.775\times10^{4}\ \text{A/Wb}$$

The two outer paths are in parallel (2.387×10^{4} A/Wb) and in series with the centre: $\mathcal{R}_{tot} = 2.308\times10^{5}$ A/Wb and $L = N^2/\mathcal{R}_{tot} = 173$ mH. For 1.0 T in the centre limb ($\Phi_c = 2.0$ mWb) the current is $\Phi_c\mathcal{R}_{tot}/N = 2.31$ A, and each outer limb carries 1.0 mWb at 1.0 T, exactly as intended by making the outer limbs half as wide.

One poor joint. Suppose a badly stacked lamination leaves an effective 0.1 mm butt gap in the left outer path. Its reluctance rises to $4.775\times10^{4} + 0.1\times10^{-3}/(\mu_0 A_o) = 1.273\times10^{5}$ A/Wb. At the same 2.31 A the parallel combination is 3.47×10^{4} A/Wb, the total centre flux falls to 1.91 mWb, and it divides in inverse proportion to the branch reluctances: 27 % (0.52 T) in the faulty limb and 73 % (1.39 T) in the good one. A gap one tenth of a millimetre wide has pushed one outer limb from 1.0 T to 1.39 T, closer to the knee of the curve, where its loss and magnetising current rise. The inductance falls only 4.5 %, to 166 mH, so a simple inductance test would hardly notice; a search-coil flux check on each limb would. Transformer manufacturers use step-lap joints and tight stacking tolerances precisely because joint gaps of a few hundredths of a millimetre already matter.

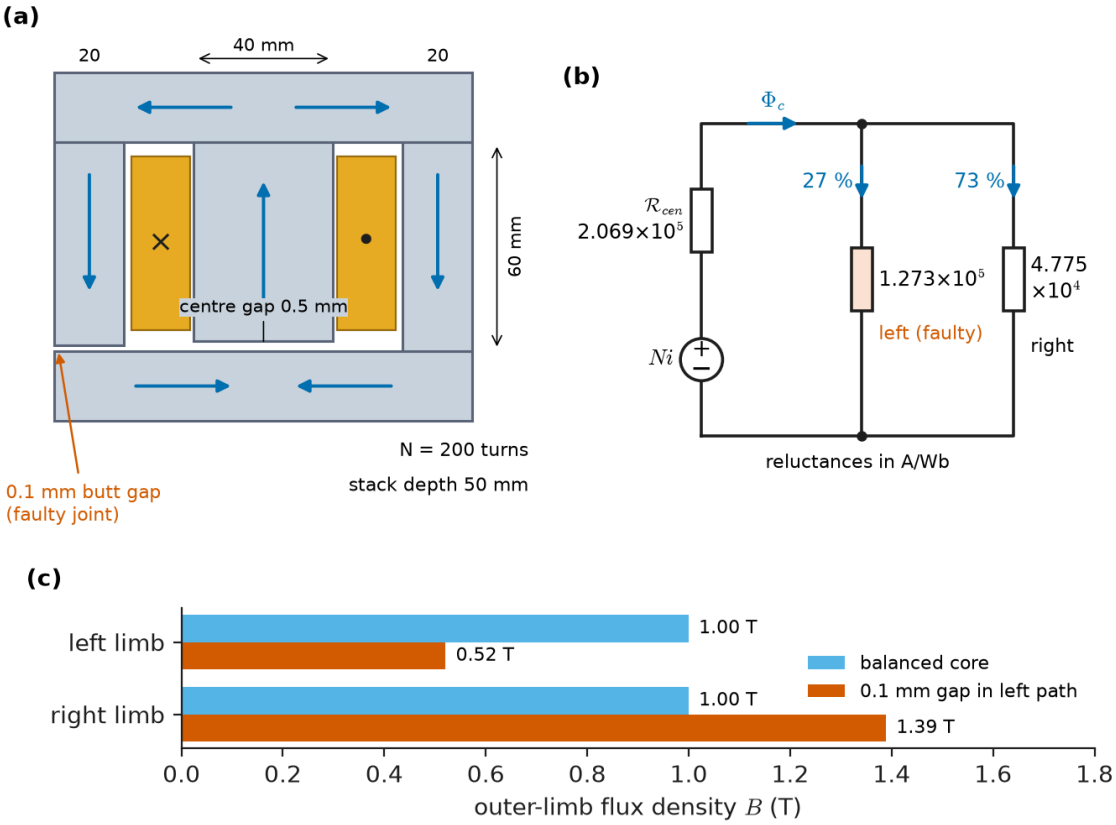


Figure 1.2 — Shell-type E-I core with its reluctance network: the centre-limb flux divides between two parallel outer paths, and a 0.1 mm joint gap in one path skews the division to 27/73.

Several windings and the superposition of MMF

When a core carries more than one winding, each contributes its MMF to every loop that encloses it, with a sign fixed by the winding direction (the physical basis of the dot convention of *Circuits, Signals and Communications*, Ch. 5). In a linear core the fluxes superpose; in a saturating core they do not. The same nodal method scales up: an induction motor can be modelled by a network of several hundred tooth, yoke and gap reluctances, iterated for saturation. Such magnetic-equivalent-circuit models run in seconds where finite elements take minutes, and remain the tool of choice for preliminary sizing.

1.8 Air gaps

Why gaps dominate

The reluctance ratio of a gap of length g to an iron path of length l_c and the same area is

$$\frac{\mathcal{R}_g}{\mathcal{R}_c} = \mu_r \frac{g}{l_c}$$

With $\mu_r = 4000$, a 1 mm gap is "worth" 4 m of iron. Equivalently, the iron path behaves like an extra gap of length l_c/μ_r, and the total reluctance of a gapped core is approximately $(g + l_c/\mu_r)/(\mu_0 A)$. As long as $g \gg l_c/\mu_r$, the inductance is set by geometry, $L \approx \mu_0 N^2 A/g$, and is insensitive to the material. That is the first reason designers add gaps.

Four reasons for a gap

1. **Stable inductance.** A reactor or choke must keep its inductance within a few percent despite batch spread, temperature and DC bias in μ_r. A gap that holds 90–98 % of the reluctance achieves that.
2. **Energy storage.** The energy stored in a linear magnetic circuit is $\tfrac{1}{2}\mathcal{R}\Phi^2$, distributed in proportion to the reluctances. Since the flux is limited by saturation, the energy a core can store is limited by how much reluctance it contains, which in practice means by its gap volume: $W \approx B^2\,(Ag)/(2\mu_0)$. Flyback transformers, boost inductors and DC-link chokes are therefore gapped.
3. **Mechanical clearance.** Any device with moving parts needs a gap: the rotor–stator clearance of a machine, the stroke of a solenoid or contactor, the clearance of a magnetic bearing. These gaps are set by mechanics (shaft deflection, bearing play, thermal growth, manufacturing tolerance), and the magnetic design must live with them.
4. **Less remanence.** A small gap lowers the flux left in a core after switch-off, which helps protection CTs recover from DC offsets (IEC 61869-2 classes PR and TPY).

Table 1.5 — Typical air gaps in power equipment.

Device	Gap (radial or total)	Function
Transformer core joint (step-lap)	0.01–0.05 mm effective per joint	Unavoidable; minimised
Protection CT, class TPY	0.1–0.5 mm total	Limit remanence
Gapped filter reactor, 50 Hz, tens of amperes	1–10 mm total, often split into several gaps	Linear inductance, energy storage
Induction motor, 1–20 kW	0.25–0.5 mm	Mechanical clearance, kept small to limit magnetising current
Induction motor, 1–5 MW	1.5–3 mm	Clearance for shaft deflection
PM synchronous motor (EV traction)	0.5–1 mm mechanical, plus 3–6 mm of magnet	Magnet thickness sets the effective gap
Salient-pole hydro generator	15–40 mm	Stability (low synchronous reactance)
Large two-pole turbo-generator	50–120 mm	Stability, cooling-gas flow
AC contactor, open	3–8 mm	Contact stroke

The bottom half of the table contains a lesson for later chapters. Large synchronous generators have enormous gaps not because they are mechanically necessary but because a large gap lowers the synchronous reactance and improves stability (Chapter 7). The price is a field winding that must supply tens of thousands of ampere-turns per pole.

Worked example — Choosing the gap of a reactor. Return to the filter reactor and design it rather than analyse it: required $L = 0.75$ mH at a design peak current of 127 A, with $B = 1.2$ T in the net iron ($A_{Fe} = 5.7 \times 10^{-3}$ m², gross $A = 6.0 \times 10^{-3}$ m²).

Turns from flux linkage. At peak, $Li = N\Phi$, so $N = Li/(BA_{Fe}) = 0.75 \times 10^{-3} \times 127/(1.2 \times 5.7 \times 10^{-3}) = 13.9$. Round to 14 turns.

Gap from reluctance. $\mathcal{R} = N^2/L = 196/0.75 \times 10^{-3} = 2.61 \times 10^5$ A/Wb. Subtracting the iron's 1.75×10^4 A/Wb leaves 2.44×10^5 A/Wb for the gaps, i.e. $g = \mathcal{R}_g \mu_0 A = 1.84$ mm. The manufacturer rounded up to two standard 1.0 mm spacers (2.0 mm total), trading about 8 % of inductance (0.75 to 0.693 mH, ignoring fringing) for about 9 % more current before the iron reaches 1.2 T (127 to 138 A). That gives the 0.693 mH of the previous example, or 0.711 mH with the fringing correction of the next section: the 0.70 mH on the nameplate.

The two steps, turns from the flux limit and gap from the inductance, are the core of every inductor design procedure. The losses and the temperature rise that then decide whether the core and winding are large enough are the subject of Chapter 3.

The gap as a source of loss and noise

A gap is not free. The flux fringing out of a gap (Section 1.9) enters the laminations and the winding conductors near it sideways, inducing eddy currents in directions the laminations do not interrupt; local heating near gaps is a common hot-spot in large reactors. The magnetic pressure across a gap, $B^2/2\mu_0$ (about 0.57 MPa at 1.2 T; Chapter 3), pulls the gap faces together at twice the supply frequency and is the main source of the 100 Hz hum of reactors. Large reactors therefore use many small gaps filled with ceramic or glass-fibre spacers, clamped hard, instead of one large gap.

1.9 Fringing

What fringing is

At an air gap the flux does not cross straight from face to face. It bulges outward, because nothing confines it, and some of it crosses through the air beside the gap (Figure 1.3a). The effect is to make the gap's effective area larger than the pole face, and its reluctance smaller, than the simple $g/\mu_0 A$ predicts. The larger the gap relative to the pole width, the larger the effect.

A widely used engineering correction adds the gap length to each side dimension of a rectangular pole face of size $a \times b$:

$$A_{\text{eff}} \approx (a + g)(b + g)$$

For a circular pole of diameter d the analogous rule is $A_{\text{eff}} \approx \pi(d + g)^2/4$. These rules are empirical fits to field solutions and are good to roughly ±10 % for g/a up to about 0.2; for larger ratios the bulging flux spreads further and dedicated fringing-permeance formulas or finite elements are used.

> **Status of the principle: Fringing correction** $A_{\text{eff}} \approx (a + g)(b + g)$
> *Empirical.* A fit to two-dimensional field solutions for gaps between flat, wide pole faces with no nearby iron; adequate for $g/a \lesssim 0.2$ and increasingly optimistic for longer gaps, for gaps close to the winding, or where adjacent iron captures the fringing flux.

Worked example — Fringing in the filter reactor. Each of the two gaps is 1.0 mm long between faces 60×100 mm. The effective area becomes $(60 + 1)(100 + 1) = 6161$ mm², 2.7 % larger than the face. The gap reluctance falls in proportion, the total reluctance to 2.759×10^5 A/Wb, and the inductance rises from 0.693 mH to 0.711 mH, a 2.5 % increase. For this short gap fringing is a small correction. Had the same 2 mm total been a single gap in a core of 20×20 mm section, the correction would have been $(22 \times 22)/(20 \times 20) = 1.21$: a 21 % increase in gap permeance. Fringing matters most in small, heavily gapped cores, which is exactly where ferrite inductor designers meet it.

The slotted gap: Carter's coefficient

In a rotating machine the gap is bounded on one or both sides by slotted iron (Figure 1.3b). Over a slot opening the gap is effectively much longer; the flux concentrates on the tooth tips. The average gap permeance is lower than that of a smooth gap, and the designer accounts for it by an **effective gap** $g' = k_C g$, where Carter's coefficient for one slotted surface is

$$k_C = \frac{\tau_s}{\tau_s - \gamma g}, \qquad \gamma = \frac{4}{\pi}\left[\frac{b_s}{2g}\arctan\frac{b_s}{2g} - \ln\sqrt{1 + \left(\frac{b_s}{2g}\right)^2}\right]$$

with τ_s the slot pitch and b_s the slot opening. The formula comes from a conformal-mapping solution of the two-dimensional field over an infinitely deep slot. When both surfaces are slotted, the product of the two coefficients is a good approximation.

> **Status of the principle: Carter's coefficient**
> *Derived.* An exact conformal-map result for a single, infinitely deep, rectangular open slot facing smooth iron of infinite permeability; the product rule for doubly slotted gaps, finite slot depth and saturated tooth tips are approximations with errors of a few percent in normal designs.

Worked example — Effective gap of an induction motor. A 15 kW, four-pole induction motor has a radial gap $g = 0.60$ mm. The stator slot pitch is $\tau_s = 15$ mm with slot opening $b_s = 3.0$ mm; the rotor has $\tau_r = 12$ mm and semi-closed slots with a 1.0 mm opening.

Stator. $b_s/2g = 2.5$, so $\gamma = (4/\pi)[2.5 \arctan 2.5 - \ln\sqrt{7.25}] = 2.528$ and $k_{C1} = 15/(15 - 2.528 \times 0.6) = 1.112$.

Rotor. $b_r/2g = 0.833$, giving $k_{C2} = 1.020$.

Combined. $k_C = 1.112 \times 1.020 = 1.135$, so the effective gap is 0.681 mm. To set up a fundamental gap flux density of 0.80 T across one gap crossing requires $Bg'/\mu_0 = 0.80 \times 0.681 \times 10^{-3}/4\pi \times 10^{-7} = 434$ At instead of the 382 At of a smooth gap. The slots alone cost 13.5 % more magnetising current. Open stator slots, used in medium-voltage motors so that preformed coils can be inserted, can raise k_C to 1.2–1.3, which is one reason those motors have higher magnetising currents; magnetic slot wedges are used to recover part of the loss.

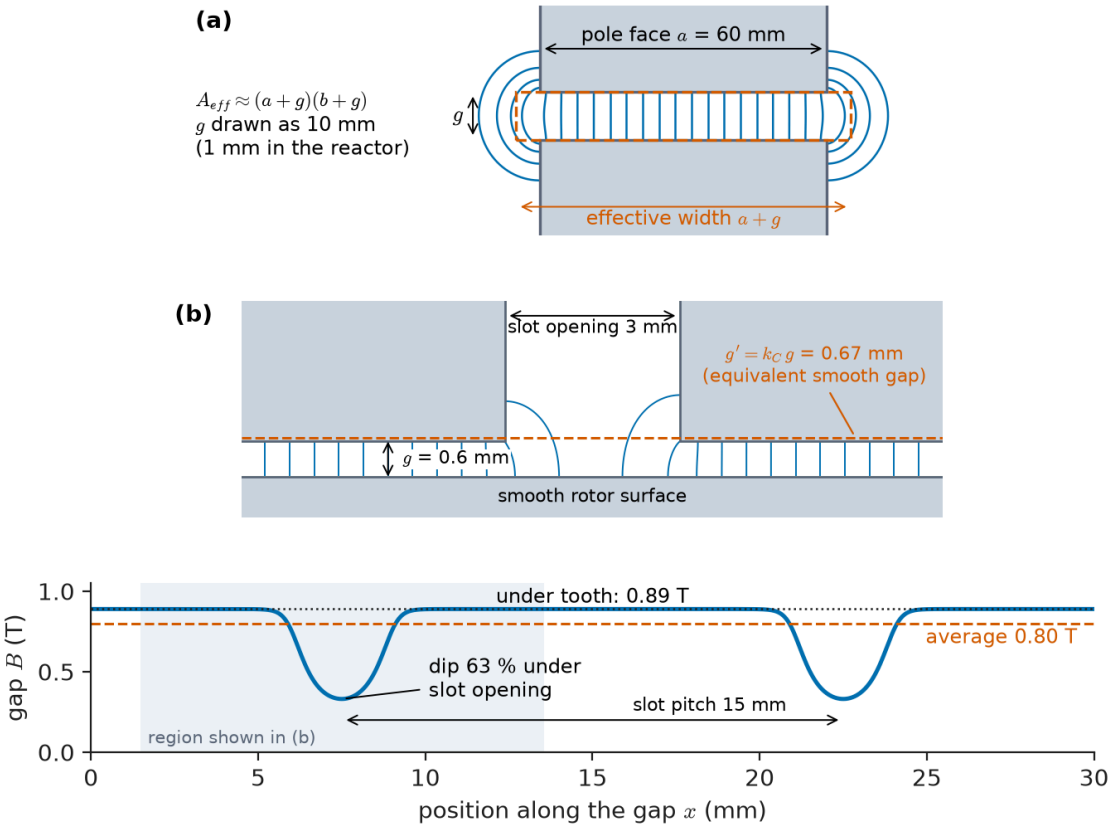


Figure 1.3 — Fringing at a plain gap and at a slotted machine gap.

Figure 1.3 shows both effects: the outward bulge at the edges of a plain gap, and the dip in gap flux density under each slot opening that Carter's coefficient averages out.

Leakage flux

Fringing is one form of a more general phenomenon: flux that does not follow the intended path. **Leakage flux** links one winding but not another (in a transformer), or crosses the slot from tooth to tooth without crossing the air gap (in a machine), or returns through air around a coil instead of through the gap of an actuator. In the reluctance network it is represented by **leakage permeances** in parallel with the main path.

Leakage is not a nuisance to be eliminated. It gives transformers the short-circuit impedance that limits fault currents (Chapters 4 and 19), gives induction motors the leakage reactance that shapes their torque–speed curve (Chapter 6), and is deliberately enlarged in welding transformers to make them current-limiting. Typical leakage inductances are 4–15 % of rated impedance in distribution and power transformers and 10–20 % (stator plus rotor) in induction motors. They are estimated from approximate formulas for standard slot and winding shapes, because the leakage field occupies air with ill-defined boundaries, and appear in later chapters as leakage reactances.

1.10 Magnetic saturation

The B–H curve as data

Above about 1.2 T in non-oriented silicon steel, and about 1.7 T in grain-oriented steel, the magnetic domains have almost all rotated into alignment with the field and further alignment becomes progressively harder. The slope dB/dH falls from several thousand times μ_0 towards μ_0 itself. The material's **normal magnetisation curve** (the locus of loop tips of symmetric hysteresis loops, measured to IEC 60404-2 on an Epstein frame or single-sheet tester) is the data the designer uses. Table 1.6 gives a typical curve for a 0.5 mm non-oriented grade of the M400-50A class.

Table 1.6 — Typical normal magnetisation curve, 0.5 mm non-oriented silicon steel (50 Hz, peak values).

B (T)	0.5	0.8	1.0	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9
H (A/m)	70	100	150	250	350	600	1500	3500	7000	13 000	22 000
Secant μ_r	5680	6370	5310	3820	2960	1860	800	360	190	110	70

Between 1.0 T and 1.6 T the secant permeability falls by a factor of 15 and the field intensity rises by a factor of 23 for a 60 % increase in flux density. The "knee" around 1.3–1.5 T is where most electrical machines are designed to work; it is the compromise between too much iron (low B) and too much magnetising current and loss (high B). Figure 1.4 plots this curve together with typical curves for grain-oriented steel, amorphous ribbon, ferrite and mild steel.

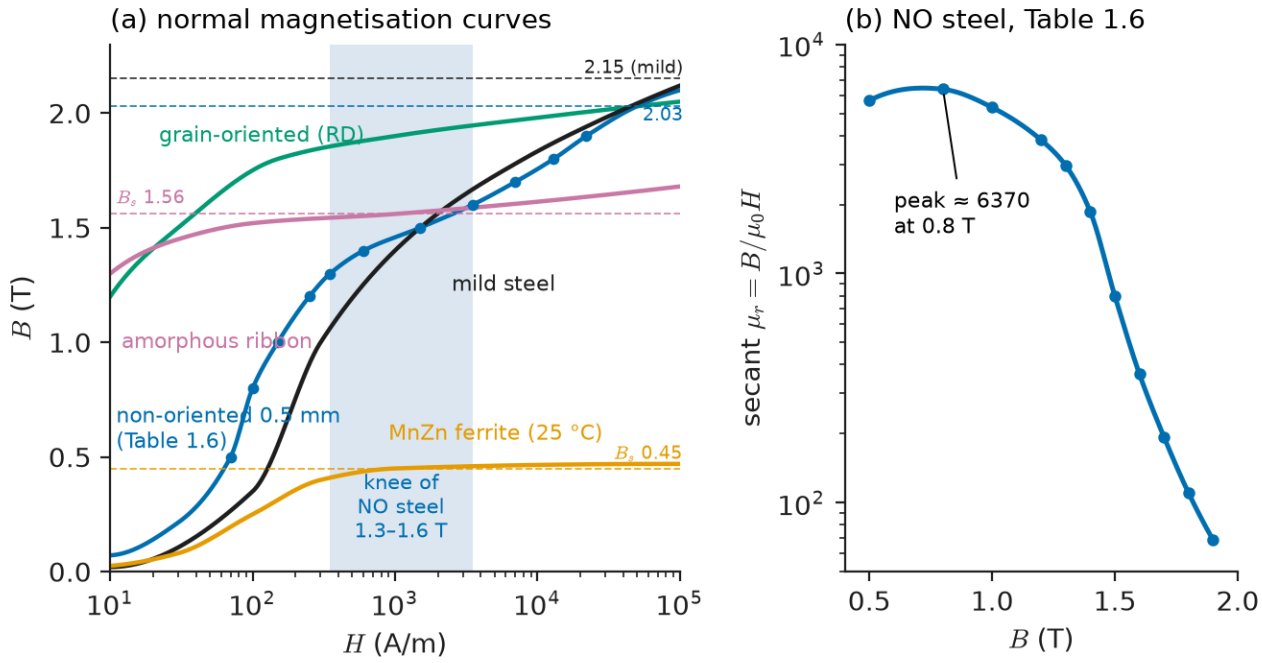


Figure 1.4 — Typical normal magnetisation curves of power core materials, with secant relative permeability.

> **Status of the principle: The normal magnetisation curve**
> *Empirical.* Measured data, reproducible to a few percent for a given grade under the standard test conditions (sinusoidal flux, 50 or 60 Hz, specified temperature, stress-free sample, rolling-direction or mixed-direction strips). It changes with mechanical stress from punching and shrink-fitting, with temperature, with frequency, and with any DC bias, so a core in a finished machine typically needs 10–30 % more MMF than the datasheet curve predicts.

Solving a saturating magnetic circuit

With a nonlinear iron segment, the magnetic circuit is still a single equation, but it can no longer be solved by dividing MMF by a fixed reluctance. For one iron path and one gap:

$$Ni = H(B)\,l_c + \frac{B A_{Fe}}{\mu_0 A_g}\,g$$

Two methods are standard. **Forward:** choose B, read $H(B)$ from the curve, compute the required Ni. This is how open-circuit characteristics of generators and magnetising curves of transformers are tabulated. **Inverse:** for a given current, find B by iterating, or graphically by drawing the gap's **load line** $B = \mu_0 (A_g/A_{Fe})(Ni - Hl_c)/g$ on the B–H plane and finding its intersection with the material curve (Figure 1.5a). A gap makes the load line steep and the operating point stable; without a gap the line is nearly horizontal and the flux becomes extremely sensitive to current near the knee.

Worked example — Why the reactor lost 30 % of its inductance. The filter reactor (14 turns, 2.0 mm total gap, $l_c = 0.50$ m, $A_{Fe} = 5.7 \times 10^{-3}$ m², $A = 6.0 \times 10^{-3}$ m²) is made of the steel of Table 1.6. Solving the saturating circuit by iteration for increasing peak current gives the following (fringing neglected):

Peak current (A)	B_{Fe} (T)	Iron MMF (At)	Gap MMF (At)	Secant $L = N\Phi/i$ (mH)
127 (design)	1.11	102	1676	0.697
150	1.28	165	1935	0.681
180	1.42	377	2143	0.628
200	1.46	587	2213	0.584
230	1.52	926	2294	0.527
254 (twice design)	1.55	1217	2339	0.486
300	1.60	1778	2422	0.426

Up to the design current, the gap takes 94 % of the MMF and the inductance is essentially the linear value. Once the iron passes 1.4 T its share of the budget grows fast; at twice design current the iron takes 34 % of the MMF and the secant inductance has fallen to 0.486 mH, 30 % below its low-current value, which is what the technician measured. For harmonic ripple riding on a peak, the relevant quantity is the **incremental** inductance $N\,d\Phi/di$, which is lower still: 0.19 mH at 200 A and 0.10 mH at 254 A, one seventh of nameplate. A capacitor bank detuned with a 0.70 mH reactor to a resonance at 189 Hz (3.8 times 50 Hz, 7 % detuning) would, for the ripple component, see its tuning frequency rise as $1/\sqrt{L}$ from 189 Hz to about 370 Hz at 200 A and 510 Hz at 254 A, sweeping straight through the fifth harmonic at 250 Hz that it was meant to stay below. The cure is a reactor specified for the real harmonic duty (IEC 60076-6 covers reactor ratings): a larger core or a longer gap so that B stays below the knee at the maximum expected current, not just at the rated fundamental.

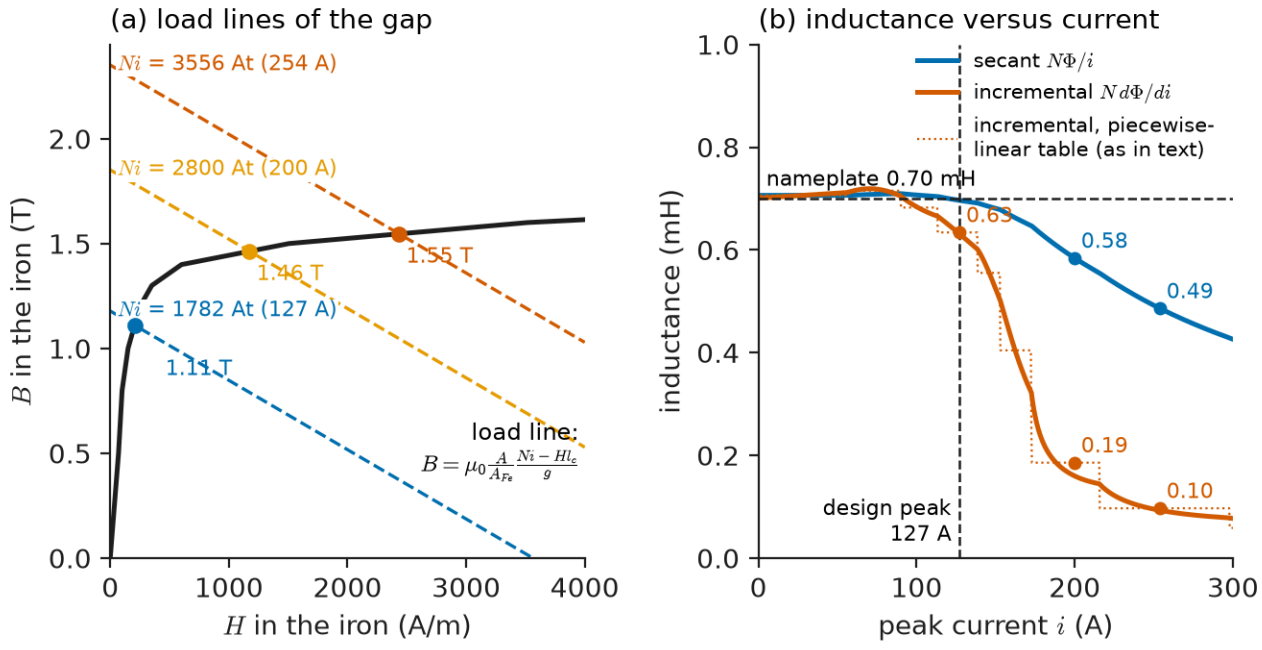


Figure 1.5 — Saturation of the gapped filter reactor: load-line solution and the collapse of secant and incremental inductance with current.

Figure 1.5 shows both views: on the left the load lines moving up the curve into the knee, on the right the secant and incremental inductances falling away from nameplate.

Saturation as a design limit across the power system

The same mechanism sets limits throughout this book:

- **Transformers** are designed at 1.6–1.75 T. A 10 % rise in voltage, or a 10 % fall in frequency, raises the peak flux by about 10 % (Chapter 2) and can double the magnetising current, which is why generators and step-up transformers carry volts-per-hertz protection (IEEE C37.102 describes settings typically between about 1.05 and 1.2 pu V/Hz). Energising a core that holds remanent flux drives it deep into saturation and draws an **inrush current** of 5–12 times rated (Chapter 4).
- **Current transformers** saturate when an offset fault current drives their flux past the knee, and protection relays then see less than the true fault current (Chapter 23).
- **Rotating machines** have nonlinear open-circuit characteristics and operating-point-dependent reactances (Chapters 6 and 7); **drives** hold the flux constant with V/f control and reduce it deliberately in field weakening (Chapter 12).

Hysteresis and remanence

If the field is reduced after magnetising, B does not retrace the normal curve. It follows a **hysteresis loop**: at $H = 0$ a **remanent flux density** B_r remains, and a reverse field equal to the **coercivity** H_c is needed to bring B to zero. For soft electrical steels H_c is only 30–80 A/m and B_r is 0.5–1.2 T, depending on the grade and the peak flux density; for grain-oriented steel H_c is below 10 A/m. Two engineering consequences follow. The area of the loop is energy dissipated per cycle per unit volume, the hysteresis loss treated with the other losses in Chapter 3. And the remanent flux left in a transformer core after de-energisation is what makes the next energisation's inrush current unpredictable.

Hard materials and the permanent-magnet load line

Materials with very high coercivity are **permanent magnets**. A magnet in a circuit with an air gap operates in the second quadrant of its loop, on the **recoil line** $B = B_r + \mu_0 \mu_{rec} H$ with recoil permeability μ_{rec} of about 1.03–1.10 for modern rare-earth and ferrite magnets. Ampère's law around a loop with no winding, $H_m l_m + H_g g' = 0$, together with flux continuity for a surface magnet of the same area as the gap, gives the gap flux density

$$B_g = \frac{B_r}{1 + \mu_{rec}\, g'/l_m}$$

A sintered NdFeB magnet ($B_r = 1.25$ T, $\mu_{rec} = 1.05$) 5 mm thick, facing an effective gap of 1.1 mm, sets up $B_g = 1.25/(1 + 1.05 \times 1.1/5) = 1.02$ T, consistent with the PM-motor row of Table 1.2. The magnet's own field is $H_m = -1.78 \times 10^5$ A/m, well inside its coercivity at room temperature. The danger lies at high temperature, where the coercivity of NdFeB falls by about 0.5–0.6 % per kelvin, and under stator currents that oppose the magnet: the operating point can then cross the knee of the demagnetisation curve and the magnet loses part of its flux permanently. This is a central design constraint of the PM machines of Chapter 8.

Where the magnetic circuit stops

With fringing corrections, leakage permeances and a measured B–H curve, the magnetic circuit predicts the inductance and magnetising requirement of a transformer or reactor to about 5 %, and of a rotating machine to about 10 %. It fails when the flux paths are not known in advance: saturated teeth pushing flux into slots, long gaps whose fringing reaches neighbouring iron, machine end regions, magnets near demagnetisation. Those cases need finite-element solutions of the magnetostatic equations of *Engineering Physics*, Ch. 11 §11.11 with the nonlinear B–H curve as input; the magnetic circuit remains the check on what such a run should give.

What remains open

Open question	Where it is answered
How does a changing flux produce the voltage of a winding, and how is the inductance of a real, saturable winding defined and measured?	Chapter 2 (electromagnetic induction)
How do the energy stored in gaps and the magnetic pressure across them turn into force and torque, and what does magnetisation cost in core loss and heat?	Chapter 3 (energy conversion principles)
How do the core flux, magnetising current, leakage and inrush of a transformer follow from its magnetic circuit?	Chapter 4 (transformers)
How do the air gap and saturation shape the magnetising reactance and torque of induction and synchronous machines?	Chapters 6 and 7
How are permanent magnets protected against demagnetisation in traction and servo motors?	Chapter 8 (special electrical machines)
How are core materials, flux densities and gaps chosen in a complete machine design?	Chapter 9 (machine design and performance)
How do saturating current transformers affect protection?	Chapter 23 (power-system protection)
How do harmonic currents and detuned capacitor banks interact on a real network?	Chapter 25 (power quality)
What in the microstructure of a steel sets its permeability, loss and saturation?	*The Materials Core* (magnetic materials)

Next: Chapter 2 lets the flux change in time and follows Faraday's law from the core into the windings, where it becomes induced voltage, inductance and, once something moves, electromagnetic torque.

Chapter 2 — Electromagnetic Induction

Part I · Magnetic Circuits and Energy Conversion

A refinery's 6.6 kV pump bus is fed from two transformers, with an automatic transfer scheme that moves the bus to the standby transformer if the normal supply fails. One night a fault upstream opens the normal breaker. The standby breaker should close within 50 ms; a sluggish mechanism makes it 100 ms. In those 50 extra milliseconds nothing seems to happen: the bus voltage, as seen by the transfer relay, has not collapsed at all. It is still above 75 % of nominal, but its frequency has fallen to 44 Hz and its phase has slipped more than 120° behind the incoming supply. When the standby breaker closes, the 1 MW pump motors are reconnected almost in phase opposition. The current surge trips the transformer protection, and the inspection afterwards finds a sheared coupling on one pump and a slipped rotor bar on another. Every one of those numbers follows from two facts: a rotor that carries current traps its flux for about a second after the stator is disconnected (Lenz's law), and a trapped flux carried round by a spinning rotor induces a voltage in the stator (Faraday's law). Industry standards recognise the hazard: ANSI/NEMA C50.41 limits the resultant volts per hertz between motor and incoming supply at reconnection to 1.33 pu.

This chapter turns Faraday's law into the working tool of machine engineering: flux linkage as a function of current and position, induced voltage in real windings and in conductors that were never meant to be part of the circuit, inductances that change with saturation and movement, and the energy balance that produces torque.

> **How does a changing flux linkage, caused by changing current or by motion, produce the voltages that every transformer, generator and motor depends on, and how large are they in real equipment?**
> **When a winding or a rotor moves, where does the electrical energy go, and how does the energy balance of a lossless coupling field yield electromagnetic torque?**

Concept	Engineering destination
Faraday's law	Transformer EMF, generator EMF, volts-per-hertz limits, search coils and flux measurement
Lenz's law	Shading rings, eddy-current brakes, damper windings, induction heating, trapped flux in disconnected motors
Motional EMF	Back-EMF of motors, generator voltage, DC-machine and PM-machine EMF constants
Induced voltage	Over-fluxing, inrush, induced voltages on pipelines and de-energised circuits
Self-inductance	Reactors, contactor coils, magnetising and leakage inductances, saturation-dependent inductance
Mutual inductance	Transformer coupling, stator–rotor coupling in machines, interference with parallel conductors
Energy conversion	The lossless coupling field as the core of every electromechanical model
Electromagnetic torque	Reluctance and mutual (alignment) torque, condition for average torque
Electromechanical energy balance	Motor and generator power accounting, coast-down, bus-transfer studies

Roadmap. Faraday's law for windings whose flux linkage depends on current and position (2.1) → Lenz's law and its engineering uses (2.2) → motional EMF and its limits (2.3) → induced voltage in windings and in neighbouring conductors (2.4) → self-inductance of real, saturable and moving windings (2.5) → mutual inductance in machines and power systems (2.6) → energy storage in the coupling field (2.7) → torque from the energy balance (2.8) → the full electromechanical energy balance (2.9).

2.1 Faraday's law

Where *Engineering Physics* leaves off

Engineering Physics derives Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ (Ch. 11 §11.4), integrates it around a loop to obtain $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$, and shows how, with flux confined to inductors, each inductor's share of the loop EMF is booked as a branch voltage $L\,di/dt$, giving KVL (Ch. 11 §11.12.2). The Lorentz force on a moving charge is recovered from the Lagrangian in Ch. 9 §9.5.2, and the motor appears in Ch. 17 §17.5 as an ideal gyrator with $V = K\omega$ and $\tau = KI$. *Circuits, Signals and Communications* adds the circuit forms $v = L\,di/dt$, mutual inductance and the dot convention (Ch. 5). All of these are used here without re-derivation. What they leave out is what a machine engineer needs: windings whose flux linkage depends on current *and* on rotor position, through iron that saturates; the separation of transformer and speed voltages; and the energy accounting that turns a gyrator constant K into a computable function of geometry.

Flux linkage as the state variable

For a winding of resistance R, Faraday's law with the motor (passive) sign convention gives the terminal equation

$$v = Ri + \frac{d\lambda}{dt}$$

where λ is the total flux linkage of the winding (Chapter 1). This is the one equation from which every winding model in this book starts. Its power is that λ can be any function of the variables that shape the field. In an electromechanical device those are the currents in all windings and the position θ (or x) of the moving part:

$$\lambda = \lambda(i_1, i_2, \ldots, \theta)$$

Differentiating with the chain rule separates two physically different contributions:

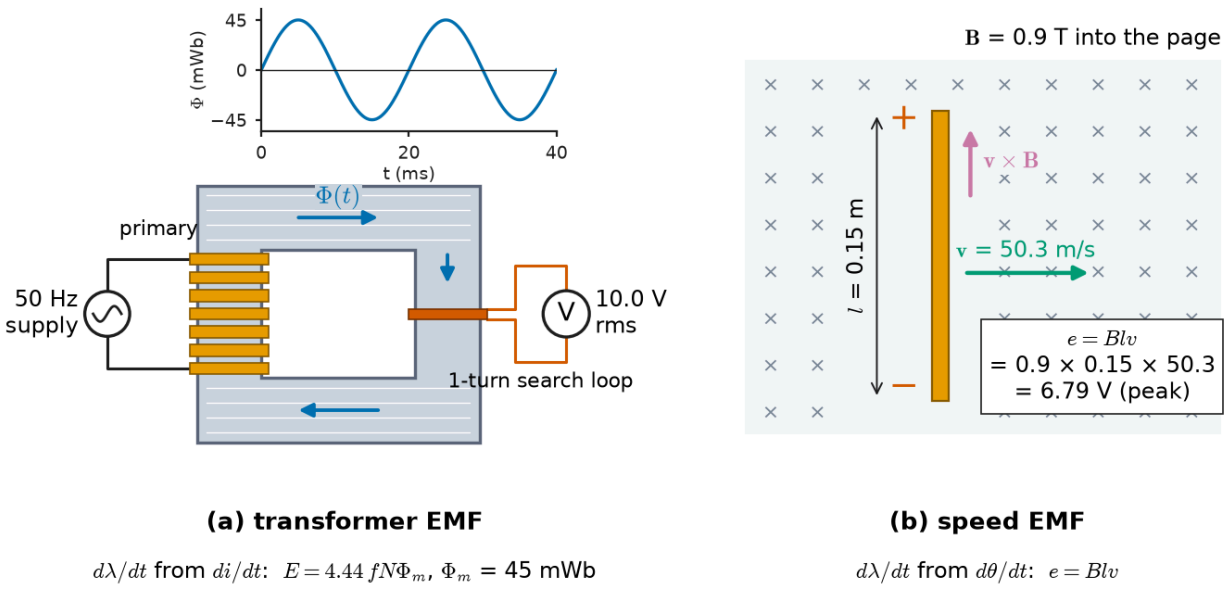
$$\frac{d\lambda}{dt} = \underbrace{\sum_k \frac{\partial \lambda}{\partial i_k}\frac{di_k}{dt}}_{\text{transformer EMF}} + \underbrace{\frac{\partial \lambda}{\partial \theta}\omega}_{\text{speed (motional) EMF}}$$

The **transformer EMF** is present whenever currents change, whether or not anything moves; it is the only term in a transformer or a stationary reactor. The **speed EMF** is present whenever the geometry changes, even with constant currents; it is the back-EMF of a DC motor and the generated voltage of an alternator. Every machine equation in Part II is this decomposition written out for a particular $\lambda(i, \theta)$.

> **Status of the principle: Faraday's law for a winding,** $v = Ri + d\lambda/dt$
> *Fundamental.* It is the integral form of the Maxwell–Faraday equation applied to the closed path formed by the winding conductor and its terminal leads, in the quasi-static limit. It holds for any dependence of λ on current, position, saturation and time; what is approximate is only the model chosen for $\lambda(i, \theta)$.

The two EMF mechanisms side by side

Figure 2.1 contrasts the two mechanisms. On the left, a stationary coil on a core sees its flux change because the current in another coil changes: pure transformer EMF. On the right, a conductor moves through a steady field: pure speed EMF. In a synchronous generator both appear together. The stator flux linkage changes because the rotor field moves (speed EMF) and because the stator currents change (transformer EMF of the armature reaction), and the machine's reactances are the bookkeeping of the transformer-EMF term.



(a) transformer EMF

$d\lambda/dt$ from di/dt: $E = 4.44\, fN\Phi_m$, Φ_m = 45 mWb

(b) speed EMF

$d\lambda/dt$ from $d\theta/dt$: $e = Blv$

Figure 2.1 — The two mechanisms of induced EMF: a stationary coil linking a changing flux (transformer EMF) and a conductor moving through a steady field (speed EMF).

Measuring flux by integrating voltage

Faraday's law also gives the most direct way to measure flux in a working device. A **search coil** of N_s turns wound around a limb, a tooth or a yoke carries no current, so its terminal voltage is $N_s\, d\Phi/dt$, and integrating it gives the flux. For sinusoidal flux $\Phi = \Phi_m \sin \omega t$ the RMS voltage is

$$E = \frac{2\pi}{\sqrt{2}} f N_s \Phi_m = 4.44\, f N_s \Phi_m$$

The same relation applied to a power winding is the EMF equation of the transformer (Chapter 4) and, with a winding factor, of the AC machine (Chapters 6 and 7).

Worked example — Search-loop check on a distribution transformer. A single turn of insulated wire is wrapped around one limb of a 400 kVA, 11 kV/415 V, 50 Hz transformer during a factory test. With rated voltage applied the loop reads 10.0 V rms.

Flux. $\Phi_m = E/(4.44 f N_s) = 10.0/(4.44 \times 50 \times 1) = 45.0$ mWb. The net limb area is 0.028 m², so $B_m = 1.61$ T, a normal design value for grain-oriented steel.

Volts per turn. The loop reads the transformer's volts per turn directly. The LV phase voltage is $415/\sqrt{3} = 240$ V, so the LV winding has $240/10.0 = 24$ turns, and the HV delta winding $11\,000/10.0 = 1100$ turns.

Wrong frequency. A 60 Hz transformer of identical design installed on a 50 Hz system at the same voltage would need $60/50 = 1.2$ times the flux, $B_m = 1.93$ T. Table 1.6 shows that at 1.93 T the steel needs about eight times the field intensity it needs at 1.6 T: the magnetising current would rise many-fold and the no-load loss sharply. A transformer is rated in volts per hertz, not in volts alone.

2.2 Lenz's law

The sign as an engineering rule

The minus sign in Faraday's law, read as a rule, is Lenz's law: **a current induced in a closed conductor flows in the direction that opposes the change of flux that produced it.** It is not a separate law. It is the sign of Faraday's law combined with energy conservation: if induced currents reinforced the change, a small disturbance would grow without an energy source.

> **Status of the principle: Lenz's law**
> *Fundamental.* It is the sign of the Maxwell–Faraday equation, enforced by energy conservation. Quantitatively it says nothing that Faraday's law with the correct sign does not already say; its value is as a quick rule for the direction of induced currents and forces.

The engineering consequence is that **flux linked by a closed, low-resistance circuit cannot change quickly.** A shorted winding of inductance L and resistance R holds its flux linkage with a time constant L/R; for a perfectly conducting loop the linkage would be frozen. Every application below exploits or fights this tendency.

Applications that use it

Shading rings in AC magnets. The force of an electromagnet is proportional to B^2 (Chapter 3), so on AC supply it falls to zero 100 times per second at 50 Hz. The return spring of a contactor then pulls the armature back slightly at every zero, and the contactor chatters. A copper ring embedded in part of each pole face carries an induced current that delays the flux in the shaded part by 60–80°. The two parts of the pole never reach zero force together, and the armature stays sealed. A cracked shading ring is a classic cause of contactor hum and contact wear.

Eddy-current brakes. A conductive disc or rail moving through a field carries induced currents that oppose the motion. The braking force is wear-free and independent of wheel–rail adhesion, which is why linear eddy-current rail brakes are fitted to high-speed trains such as the German ICE 3, and why disc-type eddy-current retarders are fitted to heavy trucks and coaches. Their force falls at low speed, so they supplement friction brakes rather than replace them.

Damper windings. Salient-pole synchronous machines carry copper or brass bars in their pole faces, short-circuited at the ends. In steady synchronous operation they carry no current; during a disturbance, any relative motion between rotor and stator field induces currents that oppose it, damping rotor oscillations (Chapters 7 and 20).

Induction heating and induction motors. Induction furnaces, hardening machines and cooktops use the induced currents for heat. The induction motor uses them for torque: the rotor currents are induced by the stator field sliding past the rotor, and their reaction with that field drags the rotor along (Chapter 6).

Shielding. A conductive plate in an AC field carries eddy currents whose field cancels much of the incident field on the far side. Aluminium enclosures around busbars and conductive screens near large reactors reduce stray field and the heating it would cause in nearby steel.

Applications that fight it

Laminations. Induced (eddy) currents in a solid iron core are a loss mechanism. Splitting the core into insulated sheets 0.23–0.65 mm thick interrupts the current paths; Chapter 3 quantifies the loss.

Trapped flux in disconnected machines. The rotor cage of an induction motor is a set of shorted, low-resistance loops. When the stator is disconnected, the rotor currents keep the air-gap flux alive and decaying with the rotor's open-circuit time constant, typically 0.3–3 s for motors from tens of kilowatts to several megawatts. The rotor keeps turning, carrying that flux with it, and the stator terminals show a voltage of falling magnitude and frequency. That is the voltage the refinery's transfer scheme reconnected against; Section 2.9 accounts for it.

2.3 Motional EMF

The flux-cutting form

For a conductor moving with velocity $\mathbf{v}$ through a field $\mathbf{B}$, each charge experiences the magnetic part of the Lorentz force, $q\mathbf{v} \times \mathbf{B}$ (*Engineering Physics*, Ch. 9 §9.5.2). Integrated along the conductor, this force per unit charge is an EMF:

$$e = \int (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} \qquad \Longrightarrow \qquad e = Blv$$

for a straight conductor of active length l, with $\mathbf{v}$, $\mathbf{B}$ and the conductor mutually perpendicular. This "Blv" form is the speed-EMF term of Section 2.1 viewed one conductor at a time. For a closed circuit whose shape changes in a steady field it gives exactly the same result as $d\lambda/dt$, and designers use whichever is easier.

> **Status of the principle: Motional EMF $e = Blv$**
> *Derived.* It follows from the Lorentz force on the conductor's charges; for a circuit whose shape changes in a field it agrees with $-d\Phi/dt$. The scalar form assumes uniform B over the active length and mutually perpendicular $\mathbf{v}$, $\mathbf{B}$ and conductor.

Worked example — Back-EMF of a traction PM motor. An eight-pole surface-PM motor has a rotor radius of 80 mm, an active length of 0.15 m and a sinusoidally distributed gap flux density of peak 0.9 T. At 6000 rpm:

Speed and frequency. $\omega_m = 2\pi \times 6000/60 = 628.3$ rad/s, so the surface speed is $v = r\omega_m = 50.27$ m/s. The electrical frequency is $f = pn/120 = 8 \times 6000/120 = 400$ Hz.

EMF per conductor. A conductor under the pole centre sees $e = Blv = 0.9 \times 0.15 \times 50.27 = 6.79$ V peak; as the sinusoidal field sweeps past, the conductor EMF is a 400 Hz sinusoid of that amplitude.

Phase EMF. With 48 conductors per phase in series and a winding factor of 0.95 (which accounts for conductors that are not all in phase; Chapter 6), the phase EMF is $48 \times 6.79 \times 0.95 = 309$ V peak, 219 V rms.

Check by flux linkage. The flux per pole is $\Phi_p = (2/\pi)B_m\tau l$, with pole pitch $\tau = 2\pi r/8 = 62.8$ mm, giving $\Phi_p = 5.40$ mWb. With $N_{ph} = 24$ series turns, $E = 4.44 \times 400 \times 24 \times 0.95 \times 5.40 \times 10^{-3} = 219$ V rms. The two routes agree, as they must. The EMF constant is $219/628.3 = 0.348$ V rms per rad/s per phase; Chapter 3 shows that, for current in phase with the EMF, the torque is $T = 3EI/\omega_m$, so the same constant times the number of phases is the torque per ampere.

Where the flux-cutting picture misleads

Conductors in slots. In a real machine the conductors lie in slots, and almost all the flux passes through the iron teeth on either side, not through the conductors. A conductor in a slot sits in a field of perhaps 0.05 T, not 0.9 T. Taken literally, Blv with the slot field would predict almost no EMF. The flux-linkage form is immune to the problem: the flux linked by the coil changes as the poles pass, wherever the flux lines physically run, and that is what the coil terminals see. The Blv calculation with the *gap* flux density, as in the example, gives the right total because it is equivalent to the flux-linkage calculation, not because the field at the conductor is 0.9 T. The same is true of force: the torque of a slotted machine acts mostly on the teeth, not on the conductors, yet "Bli" with the gap field gives the right total (Chapter 3).

Homopolar machines and sliding contacts. In a Faraday disc, or a homopolar generator in which current flows radially in a spinning disc with sliding brushes, the circuit's flux linkage does not obviously change, yet an EMF appears. Here the Lorentz-force form gives the right answer and naïve flux counting does not; the resolution is that the "circuit" includes a moving conductor whose material changes along the current path. Homopolar machines produce very large DC currents at low voltage and are used in pulsed-power and some ship-propulsion research.

The observer's frame. An airliner with a 64.8 m wingspan flying at 250 m/s through a 50 μT vertical component of the Earth's field has a motional EMF of 0.81 V across its wingtips. A voltmeter carried on board reads nothing, because its own leads move through the same field and develop the same EMF. Motional EMF

is real but appears only when part of the circuit moves relative to another part, which is exactly the situation inside every rotating machine.

2.4 Induced voltage

Volt-seconds set the flux

For a winding with negligible resistance, $v \approx d\lambda/dt$, so the flux linkage is the time integral of the applied voltage:

$$\lambda(t) = \lambda(0) + \int_0^t v\, dt$$

This reverses the usual cause-and-effect: **the voltage sets the flux, and the current adjusts to whatever the magnetic circuit needs to carry that flux.** Its consequences run through the whole book.

- **Volts per hertz.** For a sinusoidal voltage of RMS value V and frequency f, $\Phi_m = V/(4.44fN)$. Halving the frequency at constant voltage doubles the flux. Variable-frequency drives therefore reduce voltage with frequency (constant V/f) below base speed (Chapter 12), and the 60 Hz transformer of the previous example cannot run at 50 Hz on the same voltage.
- **Energisation and inrush.** If a transformer is switched on at a voltage zero, the integral of the first half-cycle of voltage is twice its steady-state amplitude: the flux swings from $\lambda(0)$ to $\lambda(0) + 2\lambda_m$. With remanent flux in the same direction, the peak can reach about $2.8\lambda_m$, far into saturation, and the core draws an inrush current of 5–12 times rated that decays over seconds (Chapter 4).
- **DC offset.** A small DC voltage across a winding integrates without limit. A transformer fed by an inverter with a 1 % DC error, or exposed to geomagnetically induced currents during a solar storm, walks into half-cycle saturation, draws large, distorted magnetising currents and heats its structural steel. Geomagnetically induced currents contributed to the 1989 Hydro-Québec blackout.
- **PWM drives.** An inverter applies rectangular voltage pulses, not sinusoids. The motor flux is still the integral of voltage, so the flux ripple is set by the volt-seconds of each pulse; that is why the switching frequency and DC-link voltage set the current ripple of a drive (Chapter 12).

> **Status of the principle: Flux equals integrated voltage, $\lambda = \int v\, dt$**
> *Derived*. Follows from Faraday's law when the winding resistance drop is negligible; at 50 Hz this holds to within the resistive drop, about 0.1–1 % of rated voltage in transformers and 1–5 % in small motors. It fails at low frequency, where Ri becomes comparable with v, which is why V/f drives add an "IR compensation" boost at low speed.

Induced voltage in conductors that are not part of the circuit

Faraday's law does not care whether the loop was designed as a winding. Any conductor forming a loop with a changing linked flux carries an induced EMF:

- **De-energised circuits beside live ones.** A circuit isolated for maintenance on a double-circuit tower links flux from the energised circuit. If earthed at both ends it forms a loop and carries induced current; if earthed at one end only, the far end floats at an induced voltage that can reach kilovolts. Work procedures (Chapter 29) require earthing at the worksite for this reason.

- **Pipelines and railways.** A buried pipeline or a fence running parallel to a power line links flux of the earth-return currents that flow during an earth fault. The **mutual inductance** of a conductor–earth loop is the subject of Section 2.6, which works the numbers.
- **Cable sheaths.** The metallic sheaths of single-core cables link the flux of the core currents. Sheaths bonded at both ends carry circulating currents and losses; cross-bonding and single-point bonding schemes trade those losses against standing sheath voltages (Chapter 22).

2.5 Self-inductance

What the circuit book provides and what it does not

Self-inductance $L = \lambda / i$, the formula $L = N^2 / \mathcal{R}$, the energy $\tfrac{1}{2} L i^2$ and the practical inductor model are covered in *Circuits, Signals and Communications*, Ch. 5. That treatment assumes a linear, stationary inductor. Power equipment breaks both assumptions. The inductance of an iron-cored winding depends on current through saturation, and the inductance of a winding on a machine or actuator depends on the position of the moving part. The single constant L has to be replaced by the function $\lambda(i, \theta)$, and several different "inductances" must be distinguished.

Secant and incremental inductance

At a given operating point, the **secant** (apparent) inductance is $L_s = \lambda / i$, and the **incremental** inductance is $L_\Delta = \partial \lambda / \partial i$. In a linear region they coincide. Beyond the knee, L_Δ falls much faster than L_s. The filter reactor of Chapter 1 had $L_s = 0.49$ mH but $L_\Delta = 0.10$ mH at twice design current. Which one matters depends on the question:

- the fundamental current drawn by a sinusoidally excited winding is set roughly by L_s at the peak;
- the response to a small ripple or a transient superimposed on an operating point is set by L_Δ;
- the energy stored is the area $\int i \, d\lambda$, which equals $\tfrac{1}{2} L_s i^2$ only in the linear case.

Current-dependent inductance is why the saturated magnetising reactance of a transformer is a poor constant in the equivalent circuit at overvoltage, why DC-link chokes specify inductance "at rated DC current", and why motor manufacturers quote magnetising inductance at rated flux.

Magnetising and leakage inductance

The flux linked by a winding can be divided into a part that also links other windings (the **magnetising** or mutual flux, mostly in the iron and across the air gap) and a part that links only the winding itself (the **leakage** flux, mostly in air around the coil and in slots). The self-inductance divides accordingly:

$$L_{11} = L_{\ell 1} + L_{m1}$$

The split matters because the two parts behave differently. The magnetising inductance passes through iron and saturates; the leakage inductance is mostly in air and stays nearly constant until very high currents saturate tooth tips. The magnetising inductance determines the no-load current; the leakage inductance determines the short-circuit current and the voltage drop under load. The equivalent circuits of transformers and induction machines in Chapters 4 and 6 are built on this division.

Table 2.1 — Typical magnetising and leakage reactances (per unit on the device rating).

Device	Magnetising reactance X_m (pu)	Total leakage reactance (pu)
Distribution transformer, 100–2500 kVA	50–500	0.04–0.07
Power transformer, 100–500 MVA	200–2000	0.10–0.20
Induction motor, 4-pole, 10–200 kW	2–4	0.15–0.25
Induction motor, 2–10 MW	3–5	0.15–0.25
Salient-pole synchronous machine (d-axis)	0.6–1.2	0.1–0.2
Turbo-generator (d-axis)	1.5–2.5	0.1–0.2

The contrast between the first two rows and the rest is the air gap again: a transformer's magnetising reactance is hundreds of per unit, a motor's only a few.

Inductance that depends on position

When the moving part of a device changes the reluctance of the flux path, L becomes a function of position. In a solenoid or contactor, L rises as the gap closes; in a salient-pole rotor, the inductance of a stator coil is maximum when the rotor's pole axis (d-axis) lines up with the coil and minimum when the interpolar axis (q-axis) does. To a good approximation for a salient rotor,

$$L(\theta) = L_0 + L_2 \cos 2\theta, \qquad L_0 = \frac{L_d + L_q}{2}, \quad L_2 = \frac{L_d - L_q}{2}$$

The period is 180° because the rotor looks the same to the coil after half a turn. The variation of L with position is what produces reluctance torque (Section 2.8) and is the operating principle of switched-reluctance and synchronous-reluctance motors (Chapter 8).

Worked example — Inrush and holding current of an AC contactor coil. A 230 V, 50 Hz contactor coil draws 120 VA at pick-up (armature open) and 12 VA once sealed (armature closed); its resistance is 150 Ω and it has 3000 turns.

Impedances. Open: $Z = 230^2/120 = 441\ \Omega$, $X = \sqrt{441^2 - 150^2} = 415\ \Omega$, $L_{open} = 1.32$ H. Closed: $Z = 4408\ \Omega$, $X = 4406\ \Omega$, $L_{closed} = 14.0$ H.

Currents. Pick-up current 0.52 A, holding current 0.052 A: a 10:1 ratio, which the control transformer and the PLC output driving the coil must be sized for.

What the magnetic circuit says. With a pole area of 3×10^{-4} m², the reluctances N^2/L correspond to equivalent air gaps of 0.24 mm closed and 2.6 mm open. The physical stroke is about 4 mm at each of two poles, 8 mm in total, but leakage and fringing paths in parallel with the open gap raise the open-armature inductance about threefold above the ideal gap value. A magnetic-circuit estimate with the bare 8 mm gap would give 0.42 H and an inrush of about 1.15 A; the real coil draws less than half of that.

Volt-seconds again. When closed, the coil voltage sets $\lambda_m = \sqrt{2} \times 230/(2\pi \times 50) = 1.04$ Wb-turn, so $\Phi_m = 0.345$ mWb and $B_m = 1.15$ T in the pole. When open, nearly the same flux linkage is needed (0.97 Wb-turn, because the 150 Ω resistance now takes part of the voltage and leaves $0.52 \times 415 = 216$ V across the inductance), but the reluctance is ten times larger, so ten times the current flows. The current adapts to the voltage-imposed flux, exactly as Section 2.4 says.

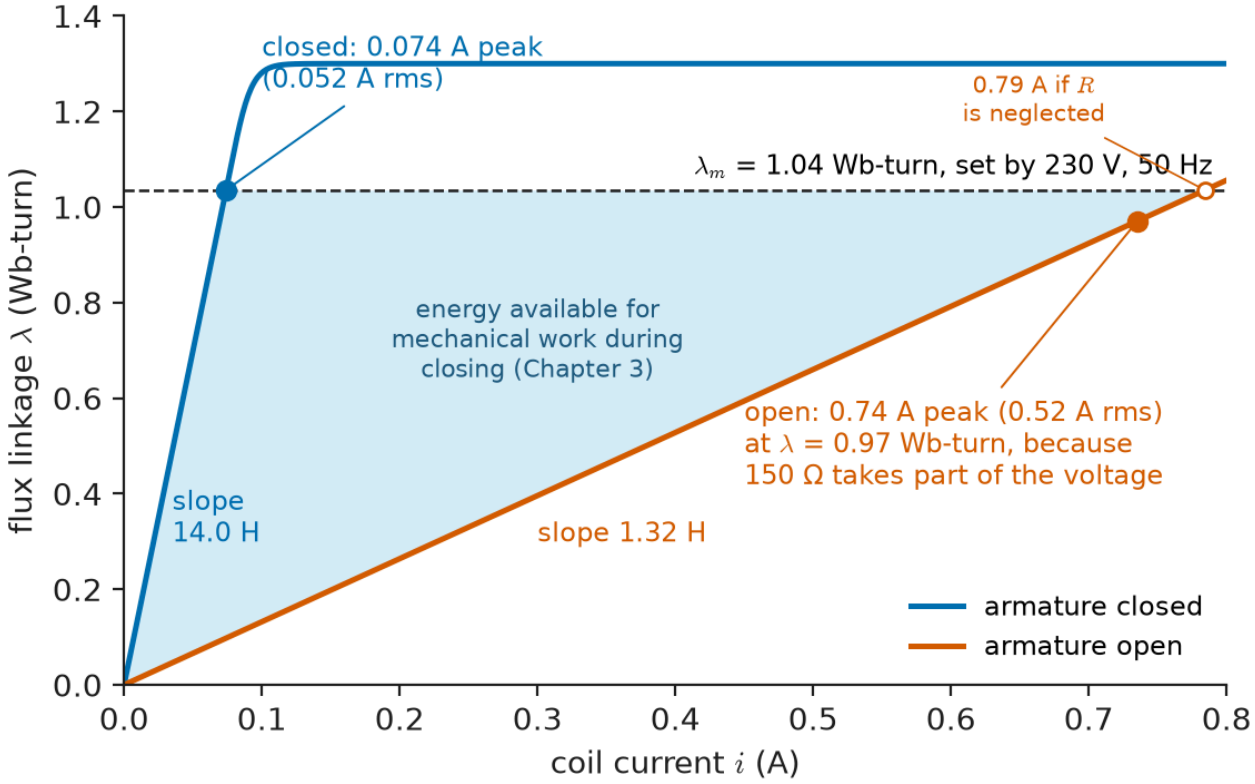


Figure 2.2 — Flux-linkage versus current of a contactor coil with armature open and closed: the operating flux linkage is fixed by the voltage, the current by the gap.

Figure 2.2 shows the two characteristics: the voltage fixes the operating flux linkage, and the armature position decides how much current that costs.

2.6 Mutual inductance

What carries over and what is new

The coupled-coil equations $\lambda_1 = L_{11} i_1 + M i_2$, $\lambda_2 = M i_1 + L_{22} i_2$, reciprocity $M_{12} = M_{21}$, the coupling coefficient $k = M/\sqrt{L_{11} L_{22}} \leq 1$ and the dot convention are taken from *Circuits, Signals and Communications*, Ch. 5. Three extensions matter for energy conversion.

Magnetising–leakage form and referral

Write each self-inductance as leakage plus magnetising, and note that both windings share the same magnetising flux path of reluctance $\mathcal{R}_m$. Then $L_{m1} = N_1^2/\mathcal{R}_m$, $L_{m2} = N_2^2/\mathcal{R}_m$ and $M = N_1 N_2/\mathcal{R}_m$, so that

$$M = \frac{N_2}{N_1} L_{m1}$$

Referring the second winding to the first by the turns ratio, $i_2' = (N_2/N_1) i_2$, turns the coupled pair into a T-network: two leakage inductances in series arms and one magnetising inductance in the shunt arm. That T-network is the equivalent circuit of the transformer (Chapter 4) and, with a speed-dependent rotor resistance, of the induction motor (Chapter 6). Its parameters are physical: the shunt branch is the core, the series branches are the leakage paths.

Mutual inductance that depends on position

Between a stator coil and a rotor coil the mutual inductance varies with rotor angle. For coils with sinusoidally distributed conductors,

$$M(\theta) = M_{\max}\cos\theta$$

where θ is the electrical angle between the coil axes. The coupling is maximum when the axes align and reverses sign after 180° electrical. This position-dependent coupling is the mechanism of every synchronous, induction and DC machine: Section 2.8 shows that its derivative $dM/d\theta$ is what produces torque.

Mutual inductance of earth-return circuits

Two parallel conductors that both return through the earth have a mutual inductance per unit length which, for earth-return circuits at power frequency, is given to engineering accuracy by Carson's simplified formula

$$M' = \frac{\mu_0}{2\pi}\ln\frac{D_e}{d}, \qquad D_e \approx 658\sqrt{\rho/f}\ \text{m}$$

where d is the separation of the conductors, ρ the earth resistivity in Ω·m and D_e the equivalent depth of the earth-return current.

> **Status of the principle: Carson's simplified earth-return mutual inductance**
> *Approximate.* The leading term of Carson's series solution for conductors over a homogeneous earth, valid for separations much smaller than D_e (hundreds of metres at power frequency); layered soil, nearby metallic structures and screening conductors modify it by tens of percent.

Worked example — Voltage induced on a pipeline during an earth fault. A steel gas pipeline runs parallel to a 132 kV line for 5 km at a separation of 30 m, in soil of resistivity 100 Ω·m. A single-phase-to-earth fault on the line drives 5 kA through the faulted phase with return through the earth.

Earth-return depth. $D_e = 658\sqrt{100/50} = 931$ m.

Mutual inductance and reactance. $M' = 2\times10^{-7}\ln(931/30) = 0.687$ μH/m, so $\omega M' = 2\pi\times50\times0.687\times10^{-6}\times1000 = 0.216$ Ω/km.

Induced EMF. $E = \omega M' l\, I = 0.216\times5\times5000 = 5.4$ kV along the pipeline for the duration of the fault.

Five kilovolts appears between the pipeline and remote earth, stressing the pipeline coating and creating touch-voltage hazards at valves and test posts. In practice the earth wires of the line carry part of the return current and screen the pipeline, and the pipeline is fitted with earthing mats and decouplers; standards such as EN 50443 set the permissible levels. The calculation is the mutual-inductance relation $e = M\,di/dt$ of the circuit book, applied to a loop nobody designed.

2.7 Energy conversion

The lossless coupling field

An electromechanical device has at least one electrical port, where a winding exchanges power vi with a source, and one mechanical port, where a shaft or plunger exchanges power $T\omega$ or Fv with a load.

Engineering Physics represents the ideal motor as a gyrator between these ports (Ch. 17 §17.5). To compute what the gyrator constant actually is, and to handle devices that store energy in their magnetic field, the device is split as in Figure 2.3:

- the winding resistance, taken out to the electrical side as an external R;
- core losses, taken out as an equivalent resistance (Chapter 3);
- friction and windage, taken out to the mechanical side;
- and what remains in the middle, a **lossless coupling field** that stores magnetic energy and exchanges energy with both ports.

All the electromechanical physics is in the coupling field. Because it is lossless and conservative, its stored energy is a function of state only, not of the path by which the state was reached.

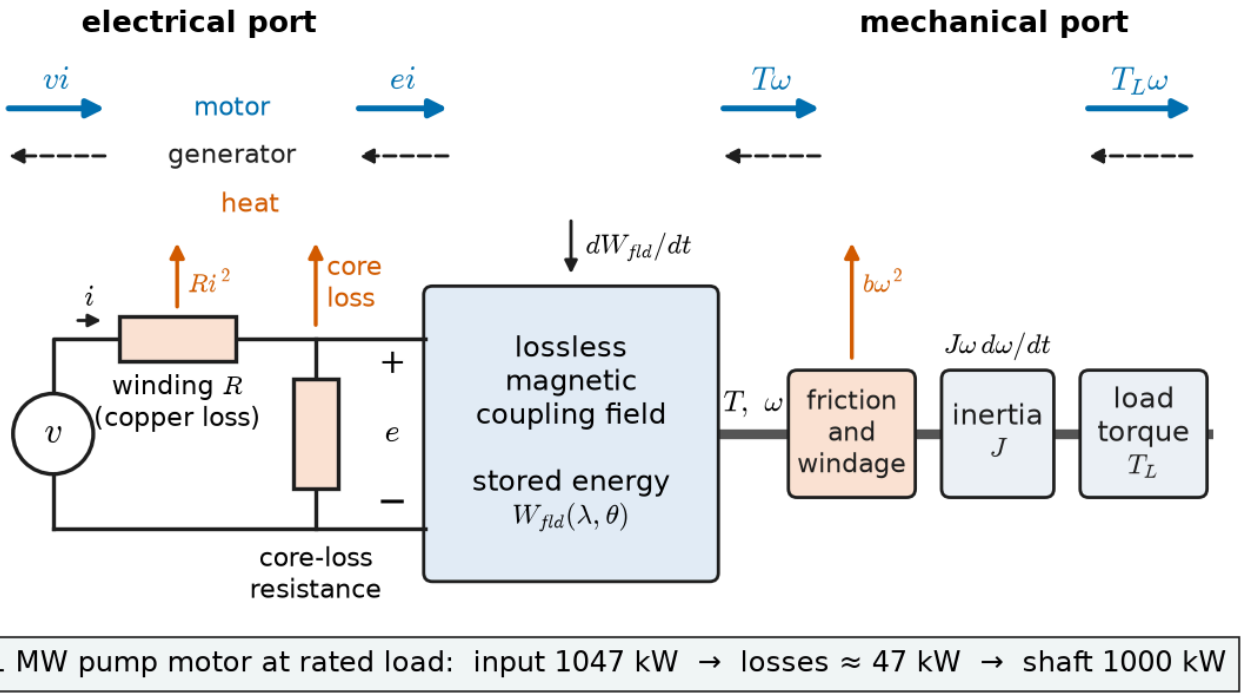


Figure 2.3 — The electromechanical device as a lossless coupling field between an electrical and a mechanical port, with the losses taken outside.

Field energy

At a fixed position, the coupling field behaves as an inductor. The electrical energy delivered into it while the flux linkage rises from 0 to λ is

$$W_{fld}(\lambda,\theta) = \int_0^{\lambda} i(\lambda',\theta)\, d\lambda'$$

which is the area between the λ–i curve and the λ axis (Figure 2.2). For a linear winding this becomes $\lambda^2/2L = \tfrac{1}{2}Li^2$, and for coupled linear windings $W_{fld} = \tfrac{1}{2}L_{11}i_1^2 + \tfrac{1}{2}L_{22}i_2^2 + Mi_1i_2$, as in the circuit book. For a saturating winding there is no such closed form; the integral is evaluated from the measured or computed curve.

Where in space the energy sits follows from Chapter 1: in a gapped device almost all of it is in the gaps, at density $B^2/2\mu_0$ (*Engineering Physics*, Ch. 11 §11.8). That is why force and torque are produced at gaps:

Chapter 3 shows that the force density on an iron surface is the magnetic pressure $B^2/2\mu_0$.

How energy enters and leaves

The electrical energy delivered to the coupling field in time dt is $e\,i\,dt = i\,d\lambda$. The mechanical energy delivered by the field to the shaft is $T\,d\theta$ (or $F\,dx$). Energy conservation for the lossless field is then

$$i\,d\lambda = dW_{fld} + T\,d\theta$$

Read in three ways, this equation contains all of electromechanical energy conversion. If nothing moves ($d\theta = 0$), all the electrical energy goes into storage, and the device is an inductor. If the flux linkage is held constant ($d\lambda = 0$), no electrical energy flows, and any mechanical work is drawn from the stored field energy. And if the device runs in a steady cycle, the field energy returns to its starting value each cycle, so the net electrical energy in equals the net mechanical energy out. Chapter 3 develops this cycle view into the energy-conversion loop of the λ–i plane.

2.8 Electromagnetic torque

Torque from the field energy

Hold the flux linkage constant in the energy balance. Then $i\,d\lambda = 0$ and $T\,d\theta = -dW_{fld}$, so

$$T = -\left.\frac{\partial W_{fld}(\lambda,\theta)}{\partial \theta}\right|_{\lambda}$$

This is the general result. The torque is the rate at which the stored field energy falls as the rotor moves, at constant flux linkage. It holds for saturating materials, provided W_{fld} is computed from the actual λ–i curves.

The linear case: torque from inductances

For linear windings it is more convenient to express torque in terms of currents, which are what an engineer controls. Consider two windings whose inductances depend on θ, with currents held constant during a small displacement $d\theta$. The electrical energy delivered is

$$i_1 d\lambda_1 + i_2 d\lambda_2 = i_1^2 dL_{11} + i_2^2 dL_{22} + 2 i_1 i_2\, dM$$

while the field energy changes by

$$dW_{fld} = \tfrac{1}{2} i_1^2 dL_{11} + \tfrac{1}{2} i_2^2 dL_{22} + i_1 i_2\, dM$$

exactly half as much. The other half goes to the shaft. Hence

$$T = \frac{1}{2} i_1^2 \frac{dL_{11}}{d\theta} + \frac{1}{2} i_2^2 \frac{dL_{22}}{d\theta} + i_1 i_2 \frac{dM}{d\theta}$$

> **Status of the principle: Torque from inductance derivatives**
> *Derived.* Follows from energy conservation in the lossless coupling field with linear flux–current relations; the "half to storage, half to work" split at constant current is exact for linear magnetics. With saturation the inductance form is invalid and the energy or co-energy form must be used (Chapter 3).

The three terms have names and engineering destinations:

- **Reluctance torque**, $\frac{1}{2} i^2 \, dL/d\theta$, arises because the rotor seeks the position of minimum reluctance (maximum inductance). It needs only one excited winding and a salient rotor, and it acts in the same direction whatever the sign of the current. It drives switched-reluctance and synchronous-reluctance motors and supplements the magnet torque of interior-PM traction motors (Chapter 8).
- **Mutual (alignment) torque**, $i_1 i_2 \, dM/d\theta$, arises because two excited windings seek alignment of their axes. With $M = M_{\max} \cos\theta$ it is $T = -M_{\max} i_1 i_2 \sin\theta$. It is the principal torque of DC, induction and synchronous machines.
- In a machine with a round rotor and a uniform air gap, the self-inductances do not depend on θ, and only the mutual term survives.

Worked example — Reluctance and alignment torque of an elementary machine. A small salient-pole synchronous machine has a stator coil with $L_d = 0.25$ H and $L_q = 0.10$ H, so $L_0 = 0.175$ H and $L_2 = 0.075$ H. The stator–rotor mutual inductance has peak $M_{\max} = 0.8$ H. The stator carries $i_s = 10$ A and the field $i_f = 20$ A, both DC (a "locked" test).

Reluctance torque alone (field open): $T = \frac{1}{2} i_s^2 \, dL/d\theta = -L_2 i_s^2 \sin 2\theta$, peak $0.075 \times 100 = 7.5$ N·m at $\theta = 45^\circ$.

Alignment torque: $T = -M_{\max} i_s i_f \sin\theta$, peak $0.8 \times 10 \times 20 = 160$ N·m at $\theta = 90^\circ$.

Both together: $T(\theta) = -160 \sin\theta - 7.5 \sin 2\theta$ N·m, with its largest magnitude, 160.7 N·m, near $\theta = 85^\circ$. The saliency adds less than 1 % to the peak but shifts the angle at which it occurs, and with a weak field (a lightly excited or under-excited machine) the reluctance term becomes significant. This is the origin of the two-term power–angle curve of salient-pole machines (Chapter 7), which Figure 2.4 previews.

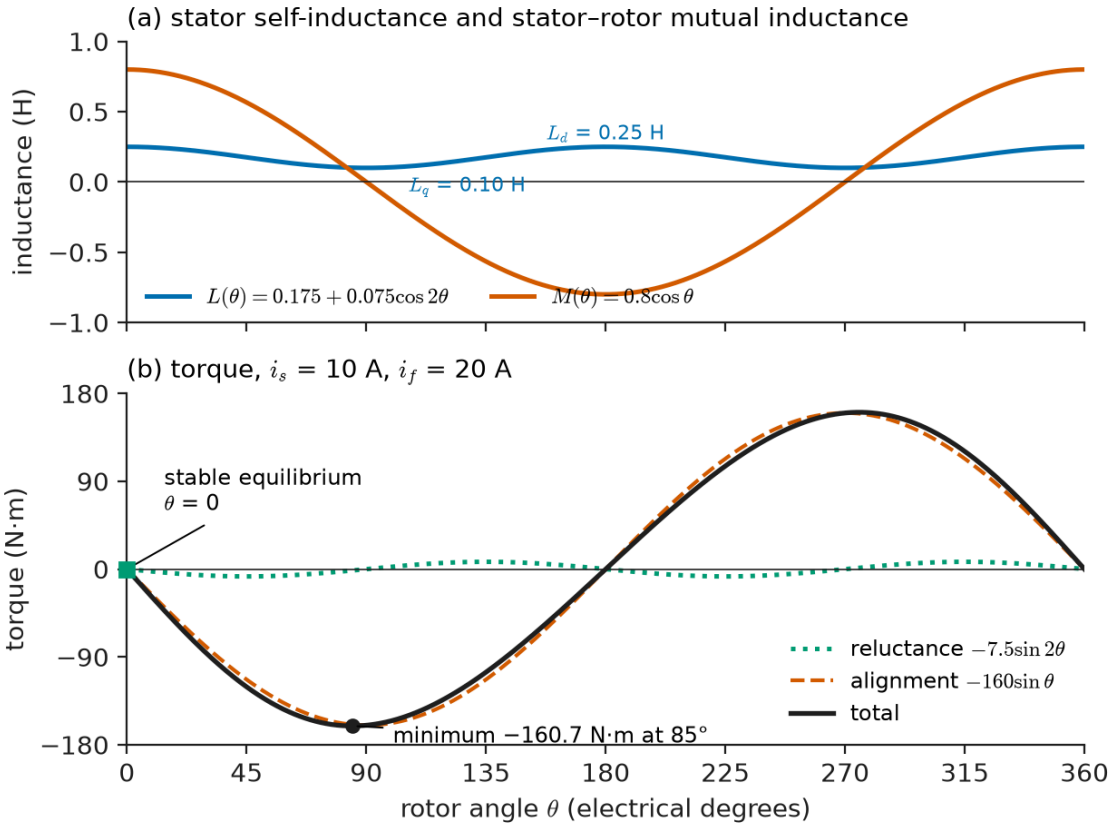


Figure 2.4 — Position-dependent inductances of an elementary salient-pole machine and the resulting reluctance and alignment torques.

Average torque needs synchronism

In a running machine the currents are AC and the rotor turns, so $\theta = \omega_m t + \delta$ and the currents vary at their own frequencies. The instantaneous torque $i_1 i_2 \, dM/d\theta$ is then a product of sinusoids at different frequencies, and its time average is zero unless the frequencies match. A machine produces steady average torque only when the speed of the rotor, the frequency of the stator currents and the frequency of the rotor currents satisfy a **synchronism condition**. In a synchronous machine the rotor turns exactly with the stator field; in an induction machine the rotor currents are induced at slip frequency, which makes their field turn in synchronism with the stator field whatever the rotor speed; in a DC machine the commutator does the same mechanically. Chapter 3 shows how a three-phase winding produces the rotating field with which this synchronism is achieved.

Two pictures of the same torque

The energy method computes the total torque on the rotor. The force on individual current-carrying conductors, $\mathbf{F} = i\mathbf{l} \times \mathbf{B}$, gives the same total for conductors in a smooth air gap, and with the gap field and the full rotor current it also gives the right total for slotted rotors, for the same reason that Blv gave the right EMF in Section 2.3. Where the two pictures differ is in *where* the force acts: on the teeth in a slotted machine, on the conductors in a slotless one. That matters for mechanical design (tooth vibration, coil bracing) but not for the torque itself.

2.9 Electromechanical energy balance

The complete balance

Put back the losses that were taken outside the coupling field. For a device with one winding and one rotating shaft, multiplying the terminal equation by i and adding the mechanical equation gives, at every instant,

$$v i = \underbrace{R i^2}_{\text{copper loss}} + \underbrace{\frac{dW_{fld}}{dt}}_{\text{field storage}} + \underbrace{J\omega \frac{d\omega}{dt}}_{\text{kinetic storage}} + \underbrace{b\omega^2}_{\text{friction}} + \underbrace{T_L \omega}_{\text{load}}$$

with core loss added as a further term when it matters. This is the lumped power balance of *Engineering Physics*, Ch. 17 §17.9 with an inductor whose inductance depends on position and a rotational mechanical system of the kind treated in Ch. 19 §19.2. For several windings, the first two terms become sums.

> **Status of the principle: Electromechanical energy balance**
> *Fundamental.* It is energy conservation applied to the electrical, magnetic and mechanical subsystems, valid instant by instant for any device in the quasi-static regime; the only modelling choices are how losses are represented and whether the field energy is computed from linear or saturating characteristics.

In steady operation the two storage terms average to zero and the balance reduces to the familiar statement that input power equals losses plus output. During transients, the storage terms dominate. The pump-motor transfer of the opening is a case where the storage terms are *all* that is left.

Worked example — Where the energy is when a pump motor is disconnected. The refinery pump motors are rated 1 MW, 6.6 kV, 1490 rpm (four-pole), efficiency 95.5 %, power factor 0.88. Each drives a

pump; the combined inertia is 30 kg·m². The magnetising current is 25 % of rated current, and the rotor open-circuit time constant is 1.0 s.

Rated and magnetising current. $I = 10^6/(\sqrt{3} \times 6600 \times 0.955 \times 0.88) = 104$ A, so $I_m = 26.0$ A. The per-phase magnetising reactance is $X_m = 3811/26.0 = 146\ \Omega$, and $L_m = 0.466$ H.

Field energy. $W_{fld} = 3 \times \tfrac{1}{2} L_m (\sqrt{2} I_m)^2 = 947$ J, about 1 kJ.

Kinetic energy. $\omega_m = 2\pi \times 1490/60 = 156.0$ rad/s, so $W_{kin} = \tfrac{1}{2} J \omega_m^2 = 365$ kJ, 386 times the field energy.

After disconnection. With the stator open, no electrical power flows ($vi = 0$ at the terminals). The magnetic energy, trapped by the rotor cage (Lenz), decays into rotor I^2R loss with the 1.0 s time constant. The kinetic energy drains into the pump load: with load torque proportional to speed squared and equal to the rated 6.41 kN·m at full speed, the motor decelerates at 214 rad/s² initially. The terminal voltage is the speed EMF of the trapped flux, proportional to flux times speed. Integrating the two decays:

Time after disconnection	Speed (pu)	Flux (pu)	Residual voltage (pu)	Frequency (Hz)	Phase slip behind supply
50 ms	0.94	0.95	0.89	46.5	35°
100 ms	0.88	0.90	0.80	43.7	124°
200 ms	0.79	0.82	0.64	39.0	439° (79° after a full slip cycle)
500 ms	0.59	0.61	0.36	29.5	many cycles

Consequence. At 50 ms the phasor difference between the 1.0 pu supply and the 0.89 pu residual voltage 35° out of phase is 0.58 pu, well inside the 1.33 pu limit of ANSI/NEMA C50.41. At 100 ms the residual voltage is still 0.80 pu but 124° out of phase, and the difference is about 1.6 pu: reconnection drives a current and a transient torque larger than a direct-on-line start, which is what broke the coupling. Fast transfer schemes therefore close within 3–6 cycles, or wait for the residual voltage to fall below about 0.25–0.3 pu before a "residual-voltage" transfer, or use in-phase monitoring to close at the next phase coincidence. The energy bookkeeping explains the time scales: the 1 kJ of field energy fixes how long the voltage persists, and the 365 kJ of kinetic energy, drained by the load, fixes how fast its phase slips.

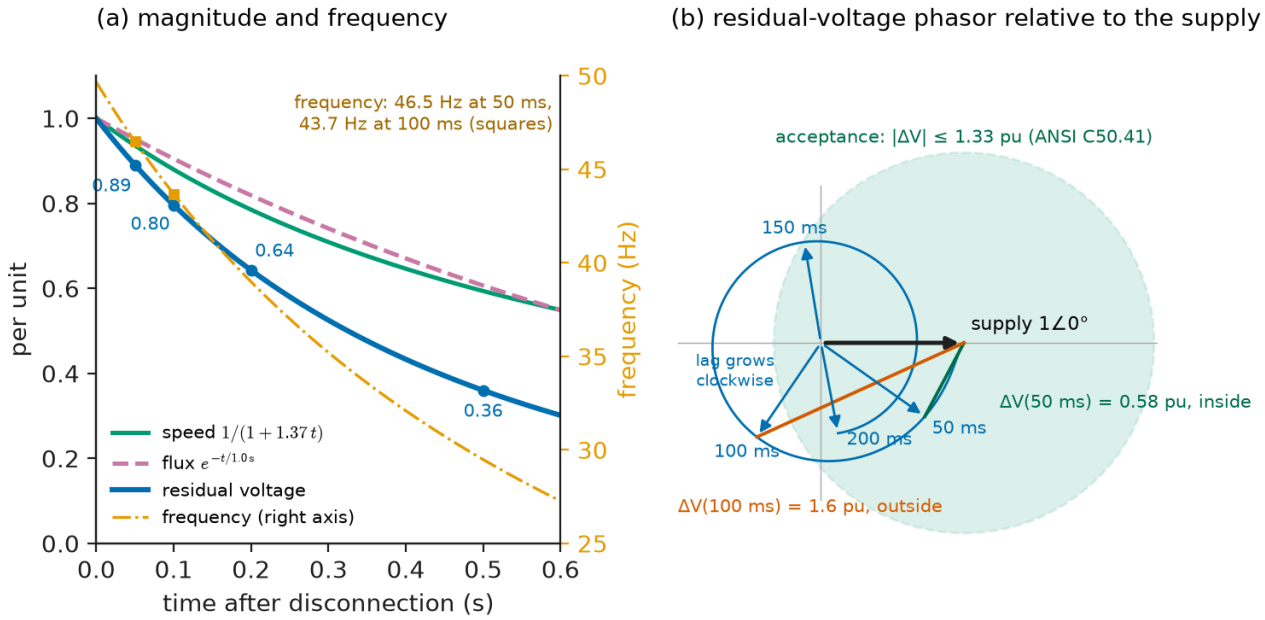


Figure 2.5 — Residual voltage of a disconnected 1 MW, 6.6 kV pump motor: magnitude, frequency and phase relative to the lost supply.

Figure 2.5 plots the decay. Every curve in it is one term of the energy balance: flux, governed by the rotor's trapped field energy; speed, governed by the stored kinetic energy and the load; and voltage, their product through Faraday's law.

Generator and motor in one equation

The energy balance does not distinguish motors from generators. In a generator, T_L is negative (the prime mover drives the shaft), $T\omega$ is negative, and vi is negative: electrical power leaves the terminals. Regenerative braking of a train, a lift or an EV, and the descent of a loaded mine hoist, are the same equation with the signs of the mechanical terms reversed. Chapter 3 uses that observation to treat electrical-to-mechanical and mechanical-to-electrical conversion as one process, and then asks what the losses in the balance cost in efficiency and temperature.

What remains open

Open question	Where it is answered
How is force and torque computed when the iron saturates, and what is co-energy?	Chapter 3 (energy conversion principles)
How does a three-phase winding create the rotating field needed for average torque?	Chapter 3; Chapter 6
How do the EMF equation, magnetising and leakage inductances and inrush appear in the transformer equivalent circuit?	Chapter 4 (transformers)
How do the EMF and torque constants of DC and PM machines follow from their windings?	Chapters 5 and 8
How do salient-pole reluctance torque and excitation shape the synchronous machine's power–angle curve?	Chapter 7 (synchronous machines)
How are motor bus transfers and motor contributions to faults handled in system studies?	Chapters 19 and 23
How are induced voltages on de-energised circuits and parallel infrastructure controlled during work?	Chapters 24 and 29
How do drives keep volts per hertz and flux under control?	Chapter 12 (AC motor drives)

Next: Chapter 3 generalises the torque of this chapter to saturating devices through co-energy, builds the rotating magnetic field, and follows the losses in the energy balance through to efficiency and temperature.

Chapter 3 — Energy Conversion Principles

Part I · Magnetic Circuits and Energy Conversion

An electric-vehicle traction motor is specified with two power ratings: 200 kW "peak" for 30 seconds and 100 kW "continuous". Nothing in the motor changes between the two: the same magnets, the same windings, the same iron, the same inverter. A battery-electric bus motor of similar construction is rated 160 kW continuous because its duty, hours of stop-start city driving, never lets it cool down. A 100 kW industrial pump motor that would do the same job in a factory weighs four times as much. The torque a given volume of machine can produce is set by magnetic and electric loading; how long it can produce that torque is set by how fast its losses can be carried away before the winding insulation reaches its temperature limit. Every machine rating in this book is, in the end, a thermal rating.

This chapter assembles the principles that connect those facts: how force and torque are produced (and computed when the iron saturates), how a three-phase winding creates the rotating field that makes continuous rotation possible, where the losses come from, and how losses become efficiency and temperature.

> **How do electromagnetic devices convert energy in either direction, and how are force and torque computed from co-energy in real, saturating geometries?**
> **Where do the losses come from, and how do they set efficiency, temperature rise and the rating of a machine?**

Concept	Engineering destination
Electrical-to-mechanical conversion	Motor sizing from air-gap shear stress; torque per rotor volume of industrial, traction and marine motors
Mechanical-to-electrical conversion	Generators, regenerative braking in EVs, trains, lifts and mine hoists
Force and torque production	Lifting magnets, contactors, magnetic bearings, reluctance and alignment torque
Magnetic co-energy	Force in saturating actuators, finite-element force calculation
Energy balance	Energy-conversion loops of switched-reluctance and stepper motors, plant energy accounting
Rotating magnetic fields	Induction, synchronous and PM machines; synchronous speed; reversal; space harmonics
Loss mechanisms	Copper, hysteresis, eddy-current, excess, mechanical and stray losses; lamination choice
Efficiency	IE efficiency classes, maximum-efficiency loading, life-cycle energy cost
Thermal constraints	Insulation classes, temperature-rise limits, short-time ratings, insulation life

Roadmap. Motor action and what sets torque per volume (3.1) → generator action and regeneration (3.2) → co-energy (3.3) → force and torque in linear and saturating devices (3.4) → the energy balance over a conversion cycle (3.5) → the rotating magnetic field (3.6) → loss mechanisms (3.7) → efficiency (3.8) → thermal constraints and ratings (3.9).

3.1 Electrical-to-mechanical conversion

Where *Engineering Physics* leaves off

Engineering Physics gives the electromagnetic energy density and the Poynting vector (Ch. 11 §11.8), the lumped power balance (Ch. 17 §17.9), and the motor as an ideal gyrator, $V = K\omega$, $\tau = KI$, with equal EMF and torque constants (Ch. 17 §17.5); thermodynamics and the thermal-resistance network come from Ch. 10 §10.5–10.6, Ch. 14 §14.7 and Ch. 19 §19.6. Chapter 2 of this book opened the gyrator into a lossless coupling field and derived torque from inductance derivatives for linear devices. This chapter adds what an engineer needs beyond that: force from co-energy when the iron saturates, the rotating field that turns alignment torque into continuous rotation, the loss mechanisms that the ideal gyrator omits, and the thermal limit that makes those losses decisive. The ideal gyrator has no size, no losses and no temperature; real machines are defined by all three.

Power crosses the air gap

In a motor, electrical power enters the stator and mechanical power leaves the rotor, with nothing but air between them. The power crosses the gap as electromagnetic energy flow: the axial electric field of the stator conductors and the circumferential gap field give the Poynting vector of *Engineering Physics*, Ch. 11 §11.8 a radial component whose integral over the gap surface is the **air-gap power**, the same quantity as the speed-EMF power $e\,i$ of Chapter 2. The field picture explains why all the torque passes through a gap a millimetre thick: the radial and circumferential components of $\mathbf{B}$ there exert a tangential stress on the rotor surface.

Air-gap shear stress and the sizing equation

That tangential stress, averaged over the rotor surface, is the **air-gap shear stress** σ (N/m² or Pa). For a rotor of diameter D and active length L, the tangential force is $\sigma \pi D L$, acting at radius $D/2$:

$$T = \sigma\,\pi D L\,\frac{D}{2} = 2\sigma\,V_r, \qquad V_r = \frac{\pi D^2 L}{4}$$

Torque is proportional to rotor volume, with the shear stress as the constant of proportionality. Power is torque times speed, so for a given power a faster machine is smaller, which is why traction motors run at 10 000–20 000 rpm through a reduction gear and why aircraft generators run at 12 000–24 000 rpm.

The shear stress is itself the product of the two loadings a designer chooses: the **magnetic loading** (the peak gap flux density $\hat{B}$, limited by saturation of the teeth to about 0.7–1.1 T) and the **electric loading** A (RMS ampere-conductors per metre of gap circumference, limited by cooling). For sinusoidal distributions, $\sigma \approx A\hat{B}\cos\varphi/\sqrt{2}$, where φ is the angle between the current and EMF distributions. With $A = 50$ kA/m and $\hat{B} = 0.9$ T in phase, $\sigma = 31.8$ kPa.

Table 3.1 — Typical air-gap shear stress, continuous rating.

Machine type	Cooling	Shear stress (kPa)
Small industrial induction motor, 1–20 kW	Totally enclosed, fan-cooled	8–20
Industrial induction motor, 100 kW–5 MW	Air, open or enclosed	15–35
Traction PM motor (EV, rail)	Liquid jacket, oil spray	30–60 continuous, up to about 100 for seconds
Direct-drive wind generator, 3–15 MW	Air or liquid	40–60
Large turbo-generator	Hydrogen and direct water cooling	80–130

The table is a table of cooling methods. The magnetic loading barely changes from row to row, because every machine uses steel that saturates near 2 T; the electric loading changes fivefold, because it is set by how much copper loss per square metre of gap surface the cooling can remove.

> **Status of the principle: Torque proportional to rotor volume, $T = 2\sigma V_r$**
> *Derived.* An exact consequence of defining σ as the average tangential stress on the rotor surface; the engineering content lies in the empirical ranges of σ in Table 3.1, which encode saturation and cooling limits and hold only for the stated cooling method and duty.

Worked example — Sizing an EV traction motor. A traction motor must deliver 310 N·m peak. With a liquid-cooled design capable of $\sigma = 45$ kPa at peak, the rotor volume is $V_r = T/2\sigma = 310/(2 \times 45 \times 10^3) = 3.44 \times 10^{-3}$ m³. With $L = D$, $D = (4V_r/\pi)^{1/3} = 164$ mm. Compare a 15 kW, four-pole, fan-cooled industrial motor with a 160 mm × 160 mm rotor: it delivers $15\,000/(2\pi \times 1460/60) = 98.1$ N·m, a shear stress of only 15.2 kPa. The traction motor's rotor is about the same size and gives three times the torque; it is the cooling that differs, and the price is that the traction motor can sustain its peak only for seconds (Section 3.9).

3.2 Mechanical-to-electrical conversion

One set of equations, two directions

Nothing in the coupling-field model of Chapter 2 prefers motoring. If the shaft is driven so that the mechanical power $T\omega$ flows *into* the field, the electrical power $e\,i$ flows out of the terminals: the device is a generator. The same machine changes role from instant to instant in a drive, according to the sign of $T\omega$; Chapter 10 maps the four quadrants of drive operation.

The reciprocity that makes this possible is the equality of EMF and torque constants. For the lossless coupling field, the electrical power absorbed by the speed EMF equals the mechanical power delivered, $e\,i = T\omega$. With $e = K\omega$ this gives $T = Ki$: the constant in V·s/rad is numerically equal to the constant in N·m/A. For a three-phase machine with current in phase with the EMF, $T = 3EI/\omega_m$; for the PM motor of Chapter 2, with $E/\omega_m = 0.348$ V rms per rad/s per phase, the torque is 1.045 N·m per ampere of RMS phase current.

> **Status of the principle: Equality of EMF and torque constants**
> *Derived.* Follows from energy conservation in the lossless coupling field; it holds exactly for the air-gap (internal) EMF and electromagnetic torque at any operating point. Measured terminal constants differ by the resistive and leakage drops, by core loss and friction, and, under heavy load, by saturation and armature reaction, which reduce the torque per ampere by 5–20 % in compact PM machines.

Generators in the energy system

Mechanical-to-electrical conversion is the source of nearly all grid electricity (Chapters 7, 14 and 15). Large generators convert at 98–99 %, so plant efficiency is decided by the thermodynamic cycle upstream, bounded by the Carnot limit of *Engineering Physics*, Ch. 10 §10.6, not by the electrical machine.

Regeneration

Motors become generators every time a vehicle or a load is braked electrically: trains, trams, lifts, cranes, mine hoists, EVs and, increasingly, industrial centrifuges and test rigs. The energy is returned to a battery, a DC link with a braking resistor, or the grid.

Worked example — Regenerative braking of an EV. An 1800 kg car decelerates from 90 km/h (25 m/s) at 3 m/s² using only its traction motor (road load neglected). The wheel radius is 0.33 m and the reduction gear ratio 9.0.

Force and power at the wheels. $F = ma = 5400$ N; at 25 m/s the braking power is $Fv = 135$ kW.

Motor operating point. Wheel speed $25/0.33 = 75.8$ rad/s; motor speed $9.0 \times 75.8 = 682$ rad/s (6510 rpm); motor torque $5400 \times 0.33/9.0 = 198$ N·m, negative (generating). The motor is working at a high-power generating point well inside its peak rating.

Energy. The kinetic energy is $\tfrac{1}{2}mv^2 = 562.5$ kJ (0.156 kWh). With a motor efficiency of 94 %, inverter 97 % and battery charging 96 %, the energy stored is $562.5 \times 0.875 = 492$ kJ, or 0.137 kWh per stop. The stop takes 8.3 s.

In stop-start urban driving with dozens of such stops per hour, regeneration recovers a substantial fraction of the traction energy. The losses that keep it from 100 %, about 70 kJ per stop here, appear as heat in the motor, inverter and battery, and are one reason city-bus motors need the high continuous rating of the opening example.

3.3 Magnetic co-energy

Definition

Chapter 2 computed torque from the field energy $W_{fld}(\lambda, \theta) = \int_0^{\lambda} i \, d\lambda'$ at constant flux linkage. In practice it is the current, not the flux linkage, that a designer specifies and a drive controls. The natural function of current is the **co-energy**

$$W'_{fld}(i, \theta) = \int_0^{i} \lambda(i', \theta) \, di'$$

which is the area between the λ–i curve and the i axis. Energy and co-energy together fill the rectangle λi:

$$W_{fld} + W'_{fld} = \lambda i$$

(Figure 3.1a). Co-energy is the Legendre transform of energy with respect to the pair (λ, i), the same operation that turns a Lagrangian into a Hamiltonian; it has no direct physical meaning as "stored energy", but its derivatives give force directly in terms of current.

For a linear device $\lambda = Li$, the curve is a straight line and the two triangles are equal: $W_{fld} = W'_{fld} = \frac{1}{2}Li^2$. For a saturating device the curve bends over, the co-energy exceeds the energy, and the distinction matters.

> **Status of the principle: Co-energy $W'_{fld} = \lambda i - W_{fld}$**
> *Derived.* A definition by Legendre transform; its usefulness rests on the energy balance of the lossless coupling field. Like the field energy it is a state function only for a lossless (non-hysteretic) characteristic; with hysteresis, energy and co-energy are path-dependent and the force must be computed from the field directly.

Force from co-energy

Insert $W_{fld} = \lambda i - W'_{fld}$ into the energy balance $i\,d\lambda = dW_{fld} + T\,d\theta$ of Chapter 2. The $i\,d\lambda$ terms cancel and

$$dW'_{fld} = \lambda\,di + T\,d\theta \qquad \Longrightarrow \qquad T = \left.\frac{\partial W'_{fld}(i,\theta)}{\partial \theta}\right|_{i}$$

and for linear motion $F = \partial W'_{fld}(i,x)/\partial x$ at constant current. The two formulas, from energy at constant λ and from co-energy at constant i, give the same instantaneous force; they differ only in which variable is held fixed in the derivative. The co-energy form is the one used in practice, including inside finite-element programs, which compute co-energy at two nearby positions with the same currents and difference them.

For several windings, $W'_{fld} = \int \sum_k \lambda_k\,di_k$ along any path in current space (all paths give the same answer for a lossless field), and for linear windings it reduces to the inductance form of Chapter 2.

3.4 Force and torque production

Force on an iron surface: the magnetic pressure

Apply co-energy to the simplest actuator, a magnet of pole area A pulling an armature across a gap x, with the iron reluctance neglected: $L = \mu_0 N^2 A/x$ for one gap, and

$$F = \frac{\partial}{\partial x}\left(\tfrac{1}{2}L(x)i^2\right) = -\frac{\mu_0 N^2 A i^2}{2x^2} = -\frac{B^2 A}{2\mu_0}$$

using $B = \mu_0 N i/x$. The negative sign means the force acts to close the gap. The magnitude per unit area, $B^2/2\mu_0$, is the **magnetic pressure**, the normal component of the Maxwell stress of *Engineering Physics*, Ch. 11 §11.8.1 at an iron surface. Field lines behave like stretched elastic bands that pull iron surfaces together.

> **Status of the principle: Magnetic pressure $F/A = B^2/2\mu_0$**
> *Derived*. From the co-energy of a gap (or equivalently the Maxwell stress tensor) for flux crossing normally between iron surfaces of very high permeability; it is accurate when the gap is short compared with the pole width and fringing is small, and it becomes an overestimate where fringing spreads the flux beyond the pole face.

Table 3.2 — Magnetic pressure at representative flux densities.

B (T)	0.4	0.8	1.2	1.6	2.0
$B^2/2\mu_0$ (kPa)	64	255	573	1019	1592
Equivalent load (tonnes per m²)	6.5	26	58	104	162

Worked example — An industrial lifting magnet. A scrap-yard lifting magnet has two pole faces of 0.05 m² each, both in the flux path. At 1.2 T in the gap the holding force is

$$F = \frac{B^2 (2A)}{2\mu_0} = \frac{1.2^2 \times 0.10}{2 \times 4\pi \times 10^{-7}} = 57.3 \text{ kN}$$

about 5.8 tonnes. With 1 mm effective gaps at each pole (paint, scale and surface roughness), the MMF needed to drive 1.2 T across both gaps is $B(2g)/\mu_0 = 1910$ At, plus the iron. If the load is rusty, uneven scrap and the gaps grow to 3 mm at the same MMF, the gap flux density falls to 0.40 T and the force to 6.4 kN, one ninth. The force falls as the inverse square of the gap: lifting-magnet ratings always assume clean, flat, thick steel, and are derated heavily for scrap, thin sheet and hot material (whose permeability falls towards the Curie point).

Force in a saturating actuator

In a saturating device, the inductance form $\frac{1}{2} i^2 \, dL/dx$ is wrong, and the error is large exactly where actuators are designed to work. Figure 3.1b illustrates why: the co-energy at a given current no longer grows as i^2 once the iron saturates.

Worked example — Saturation and the force of a plunger magnet. A plunger magnet has $N = 500$ turns, pole area 4×10^{-4} m², and a gap $x = 2$ mm, giving an unsaturated inductance $L(x) = \mu_0 N^2 A / x = 62.8$ mH. Its iron saturates near 1.6 T, so the flux linkage cannot exceed about $\lambda_s = N B_s A = 0.32$ Wb-turn. A simple model that reproduces both limits is

$$\lambda(i, x) = \lambda_s \tanh \frac{L(x)\, i}{\lambda_s}$$

whose co-energy is $W' = (\lambda_s^2 / L(x)) \ln\cosh(L(x) i / \lambda_s)$. Differentiating numerically with respect to x at constant current gives:

Current (A)	Pole B (T)	Force, co-energy (N)	Force, linear $\frac{1}{2} i^2 \, dL/dx$ (N)	Ratio
1	0.31	15.4	15.7	0.98
2	0.60	58.3	62.8	0.93
5	1.21	261	393	0.66
10	1.54	487	1571	0.31

Below about 0.6 T the linear formula is adequate. At 1.2 T it overestimates the force by half, and at 1.5 T by a factor of three: doubling the current from 5 A to 10 A gains less than a factor of two in force, not a factor of four. At 5 A the field energy is 0.52 J and the co-energy 0.68 J; the difference is the saturation. Contactor, solenoid-valve and relay designers therefore size coils from co-energy calculations (or finite-element runs) on the real B–H curve, never from $\tfrac{1}{2} i^2\, dL/dx$ with a constant L.

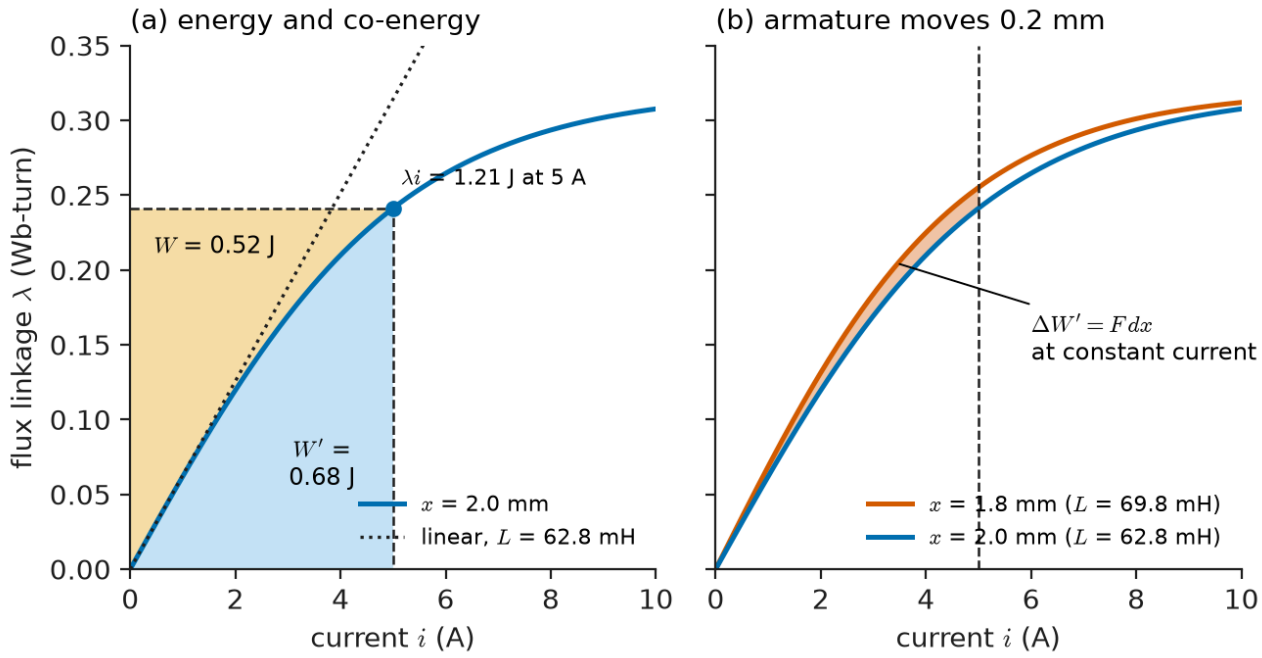


Figure 3.1 — Energy and co-energy in the lambda-i plane, and the work done when a saturating plunger magnet moves.

Torque in rotating machines

The same two terms found in Chapter 2 appear in every machine: **alignment torque** between two excited windings (or a winding and a magnet), and **reluctance torque** between an excited winding and a salient iron rotor. Table 3.3 places the machines of Part II on that map.

Table 3.3 — Torque mechanisms in electrical machines.

Machine	Main torque	Secondary torque	Chapter
DC machine	Alignment (armature–field), held at 90° by the commutator	Small reluctance	5
Induction machine	Alignment between stator field and induced rotor currents	—	6
Round-rotor synchronous	Alignment (stator–field winding)	—	7
Salient-pole synchronous	Alignment	Reluctance, 10–25 % of rated	7
Surface PM synchronous	Alignment (stator–magnet)	Cogging (zero average)	8
Interior PM (EV traction)	Alignment	Reluctance, often 30–50 % at peak torque	8
Synchronous and switched reluctance	Reluctance only	—	8

3.5 Energy balance

The conversion cycle in the λ–i plane

A machine in steady operation goes through a repeating cycle. Over one cycle the coupling field returns to its initial state, so its stored energy and co-energy return to their initial values, and the energy balance of Chapter 2, $\oint i\,d\lambda = \oint dW_{fld} + \oint T\,d\theta$, reduces to

$$W_{mech} = \oint i\,d\lambda$$

The mechanical work per cycle is the **area enclosed** by the trajectory of the operating point in the λ–i plane. A trajectory traversed anticlockwise (with λ on the vertical axis and i on the horizontal) converts electrical to mechanical energy; traversed clockwise it generates.

> **Status of the principle: Work per cycle equals the enclosed λ–i area**
> *Fundamental.* A direct consequence of energy conservation for a lossless coupling field over a closed cycle; it holds for any degree of saturation. When core hysteresis is present, the enclosed area of a stationary (non-moving) cycle is the hysteresis loss instead of work, and the two must be separated.

Switched-reluctance and stepper motors

The cycle view is the natural description of the switched-reluctance motor (SRM), which has no magnets and no rotor winding. Each phase is switched on as a rotor pole approaches alignment and off before it gets there. The flux linkage builds along the unaligned (low-inductance) curve, the rotor moves towards alignment at roughly constant current, and when the phase is switched off the flux linkage falls back to zero along a path inside the aligned (saturated) curve, because the rotor has not yet reached full alignment (Figure 3.2). The enclosed area is the work per stroke. Because the aligned curve saturates, the loop is "fatter" than the linear case would allow, and SRMs are deliberately designed to saturate.

Worked example — Power of a switched-reluctance motor from its loop area. A three-phase 12/8 SRM (12 stator poles, 8 rotor poles) has an energy-conversion loop of area 5.0 J per stroke at its rated current. Each phase strokes once for each rotor pole passing, so there are $3 \times 8 = 24$ strokes per revolution.

Average torque. $T_{av} = W \times 24/(2\pi) = 5.0 \times 24/2\pi = 19.1$ N·m.

Power at 3000 rpm. $P = T_{av}\omega = 19.1 \times 314.2 = 6.0$ kW, with $24 \times 50 = 1200$ strokes per second.

The torque is fixed by loop area and pole count; speed multiplies it into power. The torque ripple, from the discrete strokes, is the SRM's main drawback and the reason it is used where robustness and high temperature matter more than smoothness (mining, aerospace starter-generators, some EVs and household appliances).

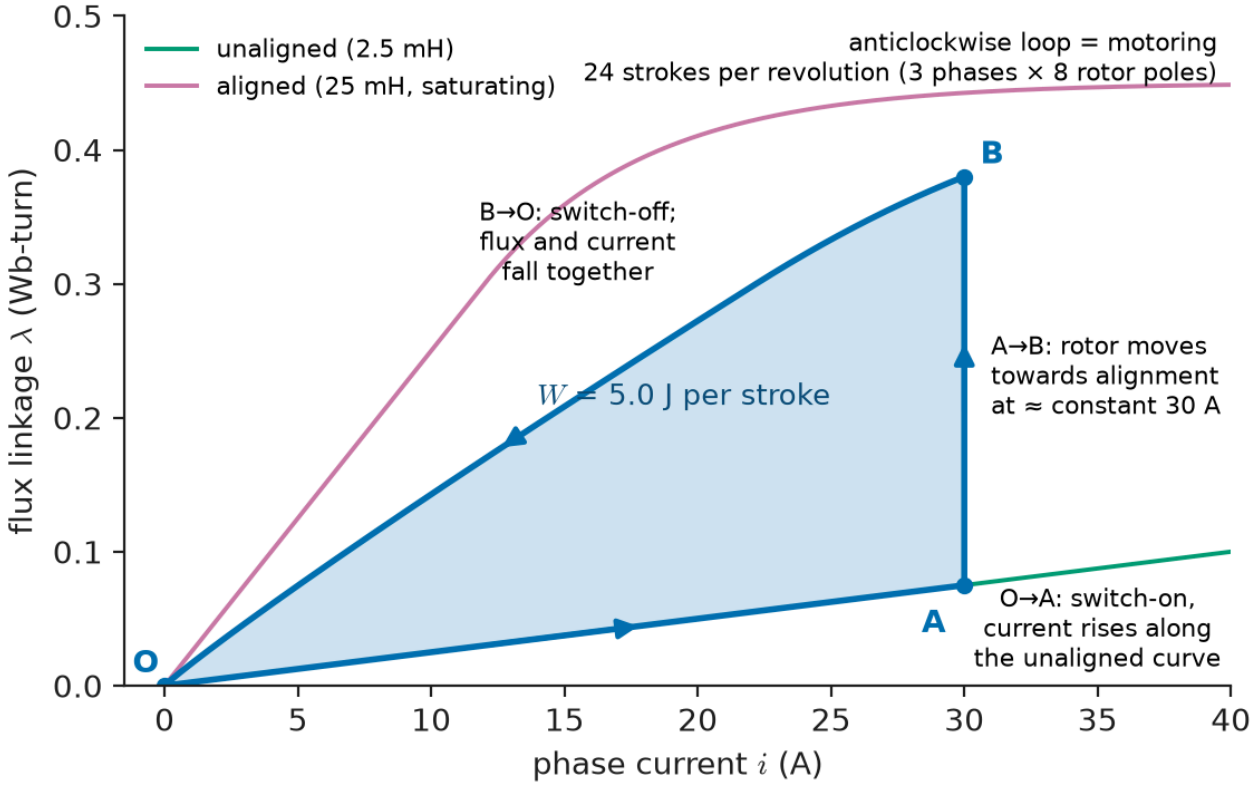


Figure 3.2 — Energy-conversion loop of one phase of a 12/8 switched-reluctance motor between its unaligned and aligned magnetisation curves.

Energy balance with losses

Adding the losses taken outside the coupling field gives the steady-state power balance used for every motor, generator and transformer test:

$$P_{in} = P_{out} + P_{Cu} + P_{core} + P_{fw} + P_{stray}$$

The losses are named by where they occur: copper (winding) loss, core loss, friction and windage, and stray load loss. In a generator, input and output swap roles. Section 3.7 treats each term; Section 3.8 turns the balance into efficiency.

3.6 Rotating magnetic fields

The problem the rotating field solves

Chapter 2 showed that a machine produces average torque only when its fields are in synchronism. A DC machine maintains that with a mechanical commutator. AC machines do it electromagnetically: a set of stationary windings fed with phase-displaced currents produces a magnetic field that **rotates**, and the rotor (magnetised, excited or induced) is dragged along with it. The rotating field is the reason three-phase power dominates industry, as *Circuits, Signals and Communications*, Ch. 10 notes from the circuit side. Here it is built from the windings.

From three pulsating fields to one rotating field

A distributed winding of one phase produces an MMF around the gap whose fundamental spatial component is $\mathcal{F}_a(\theta,t) = \hat{\mathcal{F}}_{ph}\,(i_a/\hat{I})\cos\theta$, where θ is the electrical angle around the gap measured from the axis of phase a. This is a standing wave: fixed in space, pulsating in time. Phases b and c are displaced by 120° and 240° electrical in space. Feed them with a balanced set $i_a = \hat{I}\cos\omega t$, $i_b = \hat{I}\cos(\omega t - 120^\circ)$, $i_c = \hat{I}\cos(\omega t - 240^\circ)$. Each phase contributes

$$\hat{\mathcal{F}}_{ph}\cos(\omega t - k\,120^\circ)\cos(\theta - k\,120^\circ) = \frac{\hat{\mathcal{F}}_{ph}}{2}\Big[\cos(\theta-\omega t) + \cos(\theta+\omega t - k\,240^\circ)\Big], \qquad k = 0,1,2$$

The first terms are identical for all three phases and add. The second terms form a balanced three-phase set in k and sum to zero. The result is

$$\mathcal{F}(\theta,t) = \frac{3}{2}\hat{\mathcal{F}}_{ph}\cos(\theta-\omega t)$$

a sinusoidal MMF wave of **constant amplitude**, 1.5 times the peak of one phase, whose crest travels around the gap at $d\theta/dt = \omega$ electrical radians per second (Figure 3.3).

> **Status of the principle: Rotating MMF of a balanced three-phase winding**
> *Derived.* An exact trigonometric consequence of three sinusoidally distributed windings displaced by 120° electrical carrying balanced sinusoidal currents. Real windings in slots add space harmonics, unbalanced currents add a backward-rotating component, and inverter supplies add time harmonics; these superpose on the fundamental forward wave rather than destroying it.

Synchronous speed and poles

A winding with p poles repeats its pattern $p/2$ times around the gap, so one electrical cycle moves the field by $2/p$ of a revolution. The field rotates at the **synchronous speed**

$$n_s = \frac{120 f}{p}\ \text{rpm}$$

1500 rpm for a four-pole machine at 50 Hz, 1800 rpm at 60 Hz, 3000 rpm for two poles at 50 Hz. Synchronous machines run exactly at n_s; induction machines run slightly below it, the difference being the slip (Chapter 6).

Amplitude and magnetising current

For a phase with N_{ph} series turns and winding factor k_w (Chapter 6), the fundamental MMF per pole of one phase is $\hat{\mathcal{F}}_{ph} = (4/\pi)(k_w N_{ph}/p)\,\hat{I}$, so the rotating wave has amplitude

$$\hat{\mathcal{F}} = \frac{3}{2}\cdot\frac{4}{\pi}\cdot\frac{k_w N_{ph}}{p}\,\hat{I}$$

Combining this with the gap reluctance of Chapter 1 gives the magnetising current of an AC machine from its dimensions.

Worked example — Magnetising current of a 15 kW induction motor. The 15 kW, 400 V, star-connected, four-pole motor of Chapter 1 has $D = L = 160$ mm, an air gap of 0.60 mm, Carter coefficient 1.135, $N_{ph} = 96$ series turns per phase and $k_w = 0.925$. The design gap flux density is 0.80 T peak. Allow 25 % extra MMF for the iron (saturation factor 1.25).

Required MMF per pole. $\hat{\mathcal{F}} = \hat{B} g k_C k_{sat} / \mu_0 = 0.80 \times 0.60 \times 10^{-3} \times 1.135 \times 1.25 / (4\pi \times 10^{-7}) = 542$ At.

Magnetising current. $\hat{I} = 542 / [1.5 \times (4/\pi) \times 0.925 \times 96/4] = 12.8$ A peak, 9.0 A rms.

Check against voltage. The flux per pole is $(2/\pi)\hat{B}\tau L$ with $\tau = \pi D/4 = 125.7$ mm, i.e. 10.2 mWb, and the phase EMF is $4.44 \times 50 \times 96 \times 0.925 \times 0.0102 = 202$ V, about 0.87 of the 231 V phase voltage, a typical allowance for the stator leakage and resistance drops. The rated current at 91 % efficiency and 0.85 power factor is 28.0 A, so the magnetising current is 32 % of rated, inside the range of Table 1.4. Half of the MMF, and so half of that current, would disappear if the gap were halved, which is why induction-motor gaps are as small as manufacturing and bearing tolerances allow.

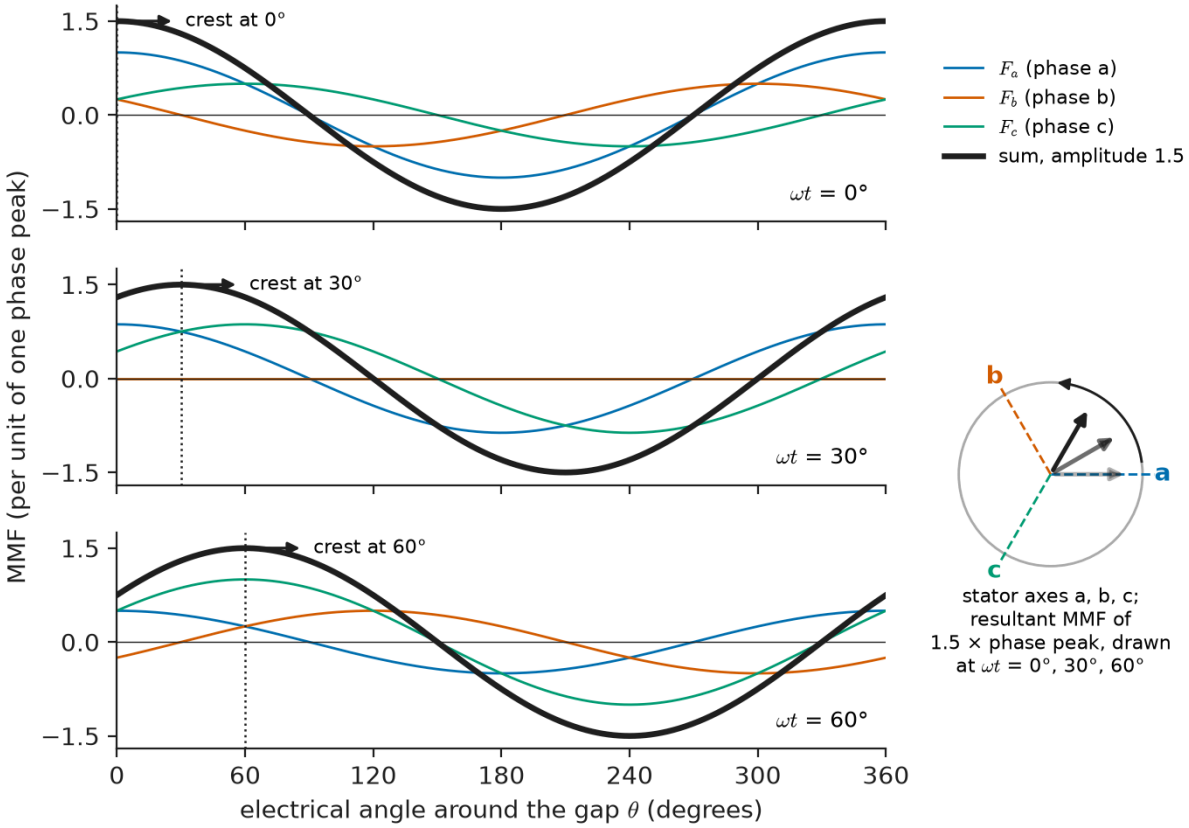


Figure 3.3 — The three phase MMFs and their sum at three instants: a constant-amplitude wave travelling around the gap.

Reversal, unbalance and single-phase supply

Three practical results follow from the same algebra.

- **Reversal.** Swapping any two supply leads reverses the phase sequence and the wave rotates at $-\omega$; this is how every three-phase motor is reversed.
- **Unbalance.** Unequal currents can be resolved into positive- and negative-sequence sets (Chapter 19). The negative-sequence currents produce a field rotating backwards at synchronous speed, which induces double-frequency currents in the rotor and heats it. As a rule of thumb from NEMA MG 1, the percentage increase in winding temperature rise is about twice the square of the percentage voltage unbalance, 8 % for a 2 % unbalance, which is why IEC 60034-26 and NEMA MG 1 derate motors on unbalanced supplies (Chapter 25).
- **Single phase.** A single winding gives a pulsating field, which by the same identity is two half-amplitude fields rotating in opposite directions, so a single-phase motor needs an auxiliary, phase-

shifted winding to start (Chapter 6).

Space harmonics

Windings in a finite number of slots also produce space harmonics of order $h = 6k \pm 1$ (triplens cancel): the 5th rotates backwards at one fifth of synchronous speed, the 7th forwards at one seventh. They cause parasitic torques, noise and loss, and can make an induction motor "crawl" near one seventh of synchronous speed; distributed, short-pitched windings suppress them at a small cost in fundamental (Chapter 6).

3.7 Loss mechanisms

Every conversion loses a few percent of the power that passes through it, and those few percent are what heat the machine. Table 3.4 lists the mechanisms, Figure 3.4 shows the core-loss data, and the rest of this section explains each.

Table 3.4 — Loss mechanisms in electrical machines and transformers.

Loss	Physical mechanism	Depends on	Typical share of total loss, 75 kW induction motor
Stator copper (I^2R)	Joule heating in windings	Current squared, winding temperature, frequency (skin and proximity)	30–40 %
Rotor copper or aluminium	Joule heating in cage or rotor winding	Rotor current squared (slip)	15–25 %
Core: hysteresis	Domain-wall motion, irreversible magnetisation	f, $B^{1.6–2}$, material	10–20 %
Core: eddy current (classical and excess)	Induced currents in laminations	f^2B^2, lamination thickness, resistivity	5–15 %
Friction and windage	Bearing friction, air drag, cooling fan	Speed (roughly cubic for fans)	5–15 %
Stray load loss	Leakage-flux eddy currents, harmonic fields, surface losses	Load current squared	10–20 %

Copper loss

The Joule loss in a winding is I^2R, with three corrections that matter in practice.

Temperature. Copper resistivity rises by 0.393 %/K; a winding at 155 °C has 1.53 times its 20 °C resistance. Efficiencies and losses are therefore always quoted at a reference temperature (IEC 60034-2-1 corrects test results to a reference coolant temperature of 25 °C plus the measured rise).

Skin and proximity effect. At 50 Hz the skin depth in copper is 9.3 mm, larger than most conductors, so the AC resistance is close to the DC value. But large bars in slots (the deep-bar rotors of induction motors, the

stator bars of large generators) carry current crowded towards the slot opening by the slot-leakage field, and the AC resistance of a deep bar can be several times its DC value at 50 Hz. In EV motors with rectangular "hairpin" conductors running at 400–1000 Hz (skin depth 3.3–2.1 mm), the AC loss factor can reach 1.5–3 at top speed.

Current harmonics. Inverter ripple currents add to the RMS current and, at high frequency, see a higher resistance.

Hysteresis loss

Each cycle of magnetisation traverses the hysteresis loop of Chapter 1, and the enclosed area is energy dissipated per cycle per unit volume. The loss per unit mass at frequency f and peak flux density $\hat{B}$ is described empirically by the Steinmetz law

$$p_h = k_h f \hat{B}^{\alpha}, \qquad \alpha \approx 1.6\text{–}2.0$$

with k_h and α fitted to material data. Hysteresis loss is proportional to frequency; it dominates the core loss of electrical steels at 50–60 Hz.

> **Status of the principle: Steinmetz hysteresis-loss law**
> *Empirical.* A power-law fit to measured loop areas, valid for sinusoidal flux within the fitted ranges of $\hat{B}$ (typically 0.5–1.7 T) and frequency; it fails for minor loops and DC-biased flux, for which corrected forms (the improved generalised Steinmetz equation) or loss maps are used.

Eddy-current loss

The changing flux induces a voltage around every closed path in the lamination (Chapter 2); the resulting currents dissipate power in the steel's resistivity. For a thin sheet of thickness d, resistivity ρ and density δ, with sinusoidal flux uniform across the thickness (sheet thinner than the skin depth), Faraday's and Ohm's laws give the **classical eddy-current loss**

$$p_c = \frac{\pi^2 d^2 f^2 \hat{B}^2}{6 \rho \, \delta} \ \text{W/kg}$$

> **Status of the principle: Classical eddy-current loss in a lamination**
> *Derived.* From Faraday's law and Ohm's law in a thin sheet with uniform flux; valid while the sheet is thinner than the skin depth (about 0.5–1.5 mm in electrical steel at 50 Hz, falling as $1/\sqrt{f}$). Measured eddy losses are typically 1.5–3 times larger because of domain-wall motion (the "excess" or "anomalous" loss), which the formula omits.

The $d^2 f^2$ dependence explains lamination practice. Thin, high-resistivity laminations cost money (more sheets, more coating, more punching), so each application uses the thickest sheet it can afford.

Worked example — Choosing lamination thickness for 50 Hz and 400 Hz. Silicon steel with $\rho = 0.48$ μΩ·m and $\delta = 7650$ kg/m³ is operated at $\hat{B} = 1.5$ T. The classical eddy loss is:

Thickness (mm)	50 Hz (W/kg)	400 Hz (W/kg)
0.50	0.63	40.3
0.35	0.31	19.8
0.20	0.10	6.5
0.10	0.025	1.6

At 50 Hz a 0.5 mm sheet loses only 0.63 W/kg to classical eddies, a small part of the roughly 4 W/kg total of an M400-50A grade: hysteresis dominates, and thinner sheet buys little. At 400 Hz, the frequency of aircraft electrical systems, the same sheet would lose 40 W/kg to classical eddies alone. Aircraft generators and transformers therefore use 0.1–0.2 mm laminations, often of cobalt–iron for its higher saturation, and EV traction motors, whose fundamental reaches 600–1000 Hz at top speed, use 0.20–0.27 mm "thin-gauge" non-oriented steels.

Worked example — Separating hysteresis and eddy loss by a two-frequency test. An Epstein test of a non-oriented steel at 1.5 T gives 4.0 W/kg at 50 Hz and 5.1 W/kg at 60 Hz. Write the loss per cycle as $p/f = k_h' + k_e' f$ (hysteresis constant per cycle, eddy proportional to frequency). The two measurements give $p/f = 0.0800$ and 0.0850 J/kg, so $k_e' = 5.0 \times 10^{-4}$ J·s/kg and $k_h' = 0.055$ J/kg. At 50 Hz, hysteresis is $0.055 \times 50 = 2.75$ W/kg and total eddy loss $5 \times 10^{-4} \times 2500 = 1.25$ W/kg. The classical formula gives only 0.63 W/kg for a 0.5 mm sheet; the other 0.62 W/kg is excess loss. The method is approximate, because it lumps the excess loss (which grows as $f^{1.5}$) into the f^2 term, but it shows clearly what thinner laminations can and cannot do: they reduce the 1.25 W/kg, not the 2.75 W/kg.

Table 3.5 — Typical specific core loss of electrical steels and alloys.

Material and grade	Thickness (mm)	Loss at 1.5 T, 50 Hz (W/kg)	Loss at 1.0 T, 400 Hz (W/kg)	Application
Non-oriented M800-65A	0.65	8.0 (max)	—	Small, low-cost motors
Non-oriented M400-50A	0.50	4.0 (max)	35–45	General-purpose motors
Non-oriented M235-35A	0.35	2.35 (max)	18–24	High-efficiency motors, generators
Thin non-oriented (e.g. NO20)	0.20	2.2–2.6	12–16	EV traction, high-speed machines
Grain-oriented M111-30P (rolling direction)	0.30	about 0.75 (1.11 at 1.7 T)	—	Distribution and power transformers
Amorphous iron-based ribbon	0.025	about 0.15 at 1.3 T	2–5	Low-loss distribution transformers

The grade designation of non-oriented steels (EN 10106 and IEC 60404-8-4) encodes the guaranteed maximum loss: M400-50A means at most 4.00 W/kg at 1.5 T and 50 Hz, in 0.50 mm sheet. Punching, welding, shrink-fitting into a frame and non-sinusoidal (PWM) flux all raise the loss in a finished machine above the Epstein value; a "building factor" of 1.3–2.0 is usual for motors and 1.1–1.3 for transformer cores.

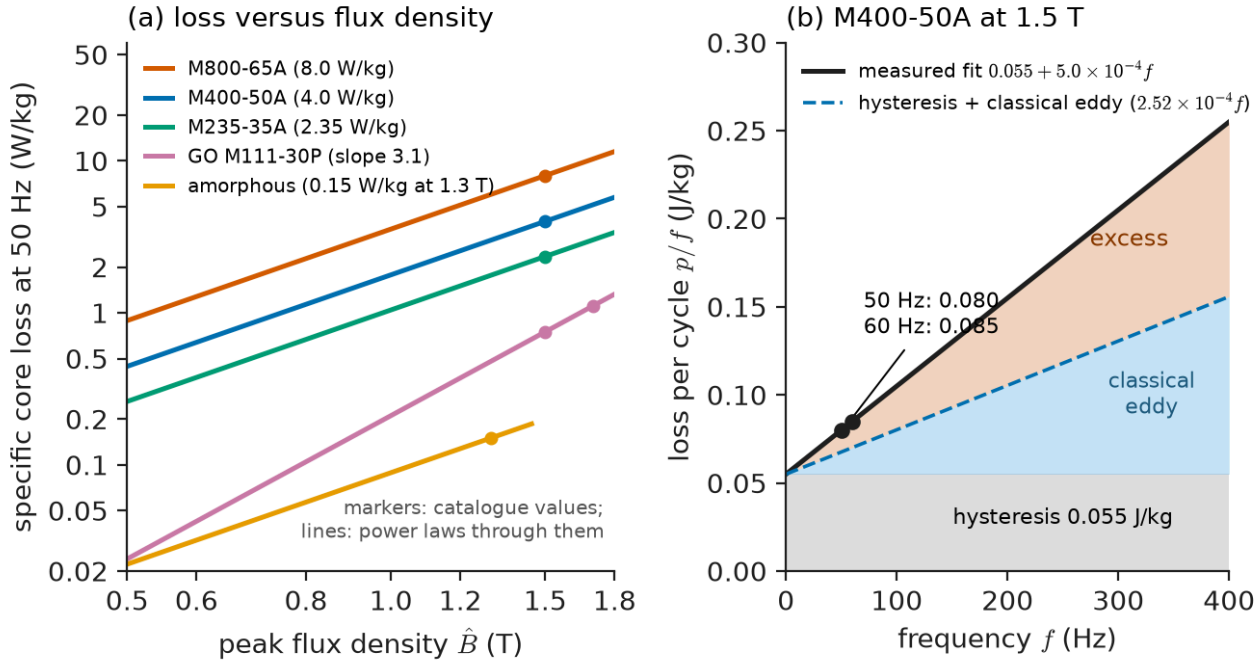


Figure 3.4 — Specific core loss of electrical steels: loss versus flux density at 50 Hz, and the split of loss per cycle into hysteresis and eddy components.

Mechanical and stray losses

Friction and windage come from bearings, air drag on the rotor and the cooling fan, roughly 0.5–1 % of rating in two-pole machines and less in slower ones; they are measured from a no-load run at different voltages (IEC 60034-2-1). **Stray load losses** are the catch-all for losses caused by load current outside the windings' I^2R: eddy currents in end-region structures and frame driven by leakage flux, high-frequency rotor surface losses from slot harmonics, and circulating currents in parallel strands. They are difficult to calculate, are determined in tests by the residual-loss method or assigned a standard allowance of 0.5–2.5 % of input depending on rating, and they are the reason that a motor's calculated and measured efficiencies differ.

A transformer has the same core and copper losses but no mechanical loss, and its load loss includes eddy losses in windings and tank driven by leakage flux (Chapter 4). Dielectric loss, $\omega C V^2 \tan\delta$, matters only in high-voltage insulation, where it is a diagnostic quantity (Chapter 24).

3.8 Efficiency

Definition and the loss-summation method

Efficiency is $\eta = P_{out}/P_{in} = P_{out}/(P_{out} + P_{loss})$. Above a few kilowatts it is not measured by subtracting two nearly equal power readings (0.5 % error in each would let a 95 % motor read anywhere from 94 % to 96 %); IEC 60034-2-1 and IEEE 112 determine each loss separately and sum them, and transformers are treated the same way from their open- and short-circuit tests (IEC 60076-1, Chapter 4).

Constant and variable losses

The losses split into those that do not depend on load (core loss, friction and windage at constant speed and voltage) and those that rise with the square of load current (copper and stray load losses). With constant loss

P_0, full-load variable loss P_v, rated output P_r, and load fraction x (approximately the current in per unit at constant voltage and power factor):

$$\eta(x) = \frac{x P_r}{x P_r + P_0 + x^2 P_v}$$

Setting $d\eta/dx = 0$ gives the maximum at

$$x_{\max} = \sqrt{P_0 / P_v}$$

the load at which variable loss equals constant loss.

> **Status of the principle: Maximum efficiency where variable loss equals constant loss**
> *Derived*. Exact for the two-component loss model above; approximate in a real machine because core loss and power factor change slightly with load and friction loss with speed. It is used to set the design point of transformers and motors to their expected load profile.

Worked example — Efficiency curve of a 75 kW pump motor. A 75 kW, four-pole premium-efficiency motor has constant losses of 1.2 kW (core 0.9 kW, friction and windage 0.3 kW) and full-load variable losses of 2.0 kW.

Full-load efficiency. $75/(75 + 1.2 + 2.0) = 95.9$ %, just under the IE4 minimum of 96.0 %; a slightly better rotor would be needed for certification.

Maximum efficiency. $x_{\max} = \sqrt{1.2/2.0} = 0.775$, where the losses are $1.2 + 2.0 \times 0.6 = 2.4$ kW and $\eta = 96.0$ %.

Part load. At 50 % load, $\eta = 95.7$ %; at 25 %, 93.4 %. Figure 3.5 plots the curve: flat between about 50 % and 100 % load, falling steeply below 25 %. Motors are deliberately designed with their maximum efficiency at 75–80 % load, because that is where most of them run.

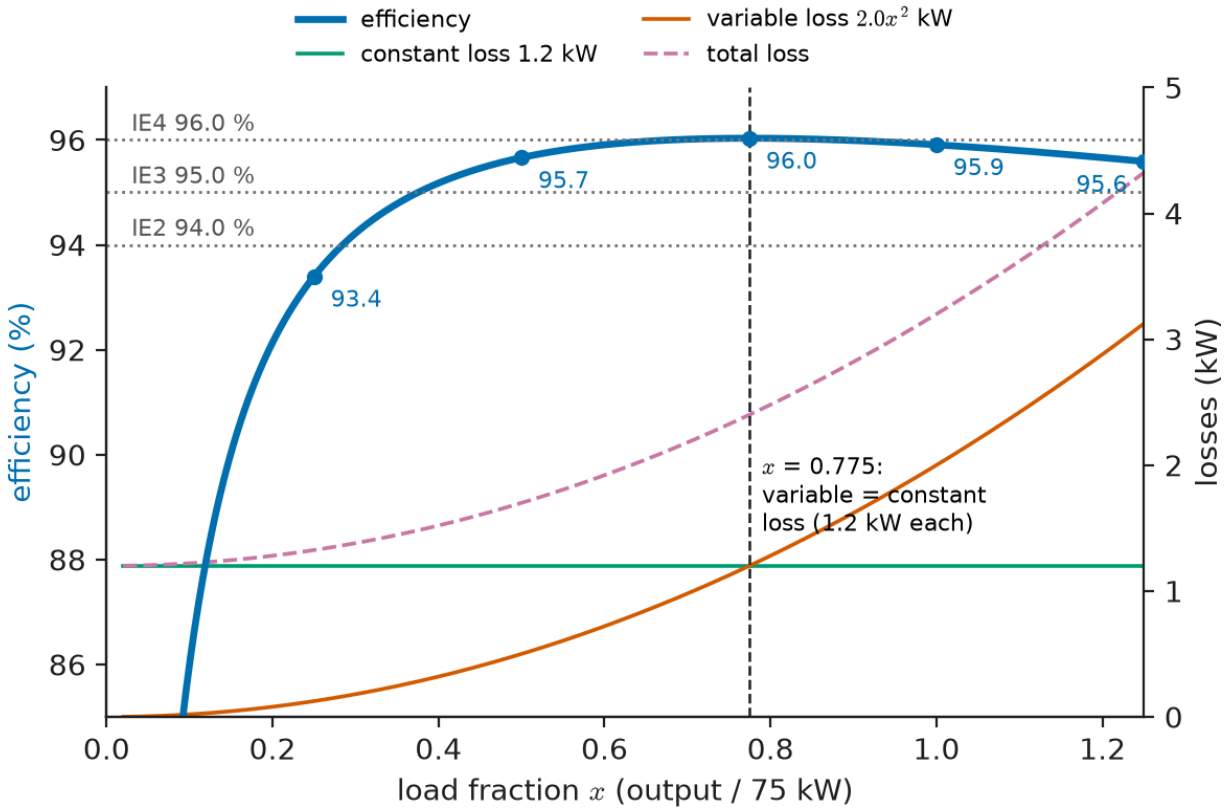


Figure 3.5 — Efficiency and losses of a 75 kW premium-efficiency motor versus load, with maximum efficiency where variable and constant losses are equal.

Efficiency classes

IEC 60034-30-1 defines efficiency classes IE1 (standard) to IE4 (super-premium) for line-operated motors from 0.12 kW to 1000 kW, and IEC TS 60034-30-2 adds IE5 for variable-speed motors. Minimum regulations follow them: in the European Union, Regulation (EU) 2019/1781 requires IE3 for most motors from 0.75 kW to 1000 kW and IE4 for those from 75 kW to 200 kW; the US Department of Energy and many other jurisdictions impose comparable levels (NEMA Premium is roughly IE3).

Table 3.6 — Minimum full-load efficiency, four-pole 50 Hz motors (IEC 60034-30-1).

Rated output	IE1	IE2	IE3	IE4
1.1 kW	75.0 %	81.4 %	84.1 %	87.2 %
11 kW	87.6 %	89.8 %	91.4 %	93.3 %
75 kW	92.7 %	94.0 %	95.0 %	96.0 %

Table 3.7 — Typical full-load efficiencies across the energy chain.

Converter	Typical efficiency
Large generator, 100–1000 MW	98.5–99.0 %
Power transformer, 100–1000 MVA	99.5–99.8 %
Distribution transformer, 400 kVA (EU Tier 2)	98.8–99.2 %
Induction motor, 75 kW, IE3	about 95 %
Induction motor, 1 kW, IE3	about 84 %
PM traction motor, best operating region	95–97 %

Small machines are the inefficient ones. At equal current density and flux density, doubling every linear dimension doubles the electric loading (deeper slots carry more current per metre of gap), so the torque rises sixteenfold while the losses, proportional to the volume of copper and iron, rise only eightfold: output scales roughly as the fourth power of size, losses as the third. Large machines are efficient; small ones are where efficiency regulation earns its money.

Worked example — What two efficiency classes are worth. A 75 kW pump in a water-treatment plant runs 6000 h/yr at close to full load. Replacing an IE2 motor (94.0 %) with an IE4 motor (96.0 %) reduces the input from $75/0.940 = 79.79$ kW to $75/0.960 = 78.13$ kW, a saving of 1.66 kW, or 9970 kWh/yr. At an industrial tariff of USD 0.12 per kWh, that is about USD 1200 per year, comparable to the price difference between the two motors, which is therefore recovered in a year or two. Over a 15-year life the motor's electricity bill is more than a hundred times its purchase price, which is why life-cycle cost (Chapter 28), not purchase price, is the basis of motor selection. Where the pump's flow can be reduced by speed control instead of throttling, a variable-speed drive saves far more than any efficiency class (Chapters 12 and 28).

3.9 Thermal constraints

From losses to temperature

Every watt of loss becomes heat that must flow from where it is produced, through insulation and iron, to a coolant: the surrounding air, a water jacket, oil or hydrogen. The steady-state temperature rise of each part is the heat flow times the thermal resistance of its path, and the transient response is governed by the thermal capacitances along the way. The network of conduction, convection and radiation resistances is the one built in *Engineering Physics*, Ch. 14 §14.7 and Ch. 19 §19.6; its laws are taken from there. What this book adds is the limit: how hot the winding may get, and why.

Insulation classes and temperature-rise limits

The weakest material in a machine is the organic winding insulation (enamel, impregnating resin, slot liners, mica tapes with resin binders). IEC 60085 classifies insulation systems by the temperature at which they give an acceptable life; IEC 60034-1 then sets the permitted winding temperature rise above a 40 °C coolant, measured by the resistance method, leaving a margin for the hot spot inside the winding.

Table 3.8 — Thermal classes and permitted winding temperature rise (typical values for medium machines, resistance method, 40 °C coolant).

Thermal class (IEC 60085)	Letter	Maximum hot-spot temperature	Permitted rise (IEC 60034-1)
130	B	130 °C	80 K
155	F	155 °C	105 K
180	H	180 °C	125 K

A common industrial practice is to build motors with Class F insulation but design them for a Class B rise of 80 K, leaving 25 K of margin that is spent on longer life, on operation in hotter climates, or on service factor. Ratings assume altitude up to 1000 m and coolant up to 40 °C; above these, IEC 60034-1 requires derating, because thinner air cools less.

Oil-immersed transformers are governed by a different rule set (IEC 60076-2 and IEC 60076-7): a top-oil rise of 60 K, average winding rise of 65 K, and a reference hot-spot temperature of 98 °C for thermally upgraded paper, at which the insulation ages at its normal rate.

Insulation life

The chemical degradation of organic insulation is a thermally activated process. Its rate follows the Arrhenius law, and over the working range this reduces to the engineering rule that **life halves for every 8–10 K of additional temperature** for machine insulation (Montsinger's rule), and for every 6 K for transformer paper above its reference temperature (IEC 60076-7).

> **Status of the principle: Insulation life halves per 8–10 K**
> *Empirical.* A linearisation of Arrhenius-type ageing fitted to accelerated-life tests of insulation systems (IEC 60216, IEC 60034-18); the halving interval depends on the material, and the rule ignores other ageing factors (partial discharge, vibration, moisture, thermal cycling) that often limit life in practice.

A winding that runs 10 K hotter than its design temperature for its whole life lasts about half as long. One that runs 10 K cooler lasts about twice as long. This is why voltage unbalance, poor ventilation, high ambient and supply harmonics, each worth only a few kelvin, matter so much to plant maintenance.

Thermal time constants and short-time ratings

The thermal capacitance of a machine lets it carry more than its continuous loss for a limited time. With a single-node model of thermal resistance R_{th} and capacitance C_{th}, the temperature rise under constant loss P approaches PR_{th} exponentially with time constant $\tau = R_{th}C_{th}$:

$$\Delta T(t) = PR_{th} + (\Delta T_0 - PR_{th})\, e^{-t/\tau}$$

The time to reach a limit $\Delta T_{\max}$ from an initial rise ΔT_0 is $t = \tau \ln[(PR_{th} - \Delta T_0)/(PR_{th} - \Delta T_{\max})]$.

Worked example — Why the EV motor's peak rating lasts tens of seconds. The traction motor of Section 3.1 is liquid-cooled with coolant at 65 °C and Class H insulation (180 °C limit), so the allowable winding rise is 115 K. At its 100 kW continuous rating it dissipates 4.0 kW.

Thermal resistance. $R_{th} = 115/4000 = 0.0288$ K/W, the value that makes 4.0 kW the continuous loss limit.

Thermal capacitance. The windings and stator core that heat together have an effective mass of about 25 kg at 500 J/(kg·K): $C_{th} = 12.5$ kJ/K, and $\tau = 0.0288 \times 12\,500 = 359$ s.

Peak operation. At 200 kW peak, the losses rise to about 12 kW (copper loss rises with the square of the higher torque current). The final rise would be $12\,000 \times 0.0288 = 345$ K, far beyond the limit. Starting from a rise of 70 K after normal driving, the time to reach 115 K is

$$t = 359 \ln \frac{345 - 70}{345 - 115} = 64 \text{ s}$$

and from a cold start, 146 s.

Where the single node fails. The copper alone, about 8 kg with a resistance of a few thousandths of a kelvin per watt to the stator iron, has its own time constant of roughly 20 s. Under a heavy overload the copper hot spot runs ahead of the stator iron, and the single-node model is optimistic. Manufacturers therefore use lumped-parameter thermal networks with 10–100 nodes, calibrated by test with embedded thermocouples, and quote peak ratings of 10–30 s, not 64 s. The 30 s peak rating of the opening example reflects exactly this.

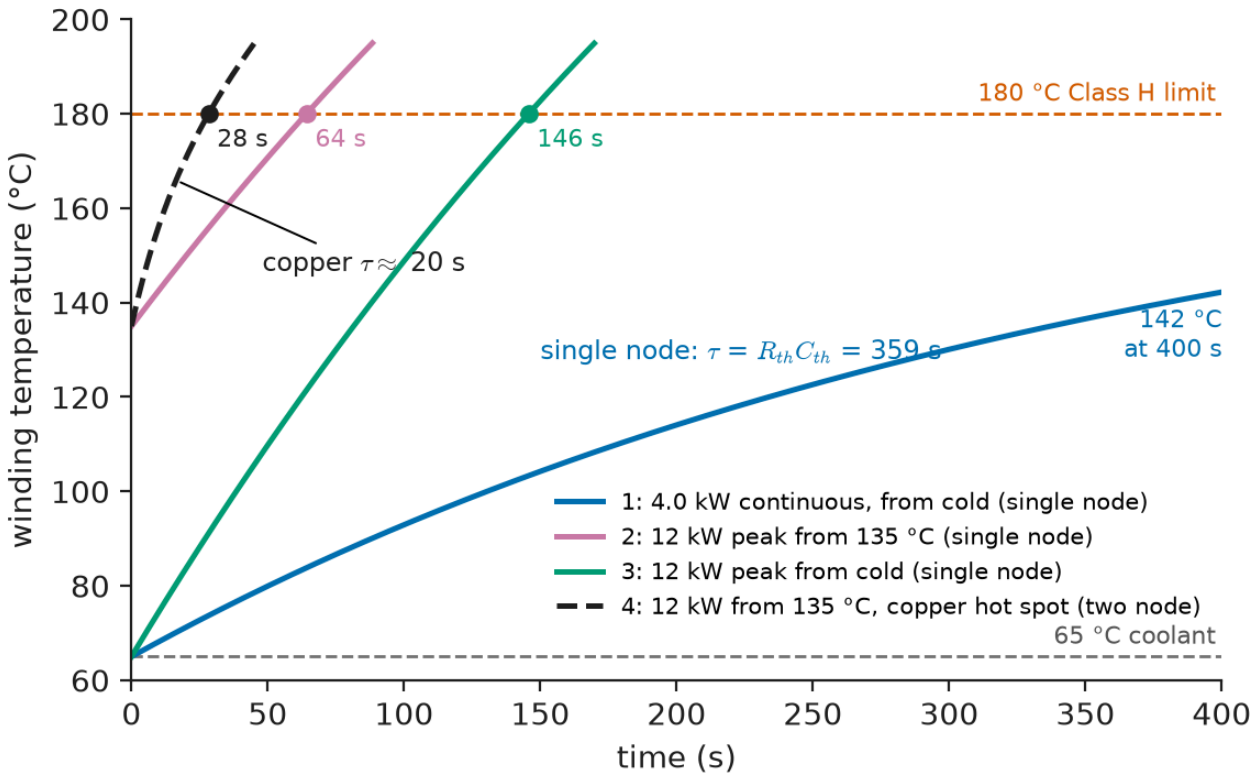


Figure 3.6 — Winding temperature of an EV traction motor under continuous and peak loss: single-node model and copper hot spot.

Figure 3.6 compares the curves. The continuous rating is a steady-state thermal limit; the peak rating is a transient one, set by the smallest thermal capacitance in the path; and both depend on the coolant, not on the electromagnetic design. Duty cycles that alternate between the two (the S1–S10 duty types of IEC 60034-1) are rated by the temperature they produce, as Chapter 9 explains.

Thermal constraints across the system

The same logic rates everything else in this book: cables by the I^2R that heats XLPE to 90 °C (Chapter 30), transformers by their oil and winding time constants (Chapter 4), overhead lines by wind and sun, which is why dynamic line rating raises their capacity on cool, windy days (Chapter 21). The electromagnetic design decides the losses; the thermal design decides how many of them can be tolerated.

What remains open

Open question	Where it is answered
How do the coupling-field principles appear in the transformer equivalent circuit, its tests and its efficiency?	Chapter 4 (transformers)
How does the rotating field produce slip, torque–speed curves and starting torque in induction motors?	Chapter 6 (induction machines)
How do alignment and reluctance torque combine in salient-pole synchronous and interior-PM machines?	Chapters 7 and 8
How are losses, cooling, insulation and duty cycles combined into a machine rating?	Chapter 9 (machine design and performance)
How is the operating point of a drive kept in the efficient, thermally safe region across four quadrants?	Chapters 10 and 12
How do generator and plant efficiencies combine into the heat rate of a power station?	Chapter 14 (conventional generation)
How do unbalance and harmonics add losses in motors and transformers on real networks?	Chapter 25 (power quality)
How are motor efficiency, load profiles and life-cycle cost used in plant energy management?	Chapter 28 (energy efficiency and management)
What sets the losses and saturation of steels and the thermal life of insulating materials at the microstructural level?	*The Materials Core* (magnetic and insulating materials)

Next: Part II begins with Chapter 4, where the magnetic circuit, Faraday's law and the loss mechanisms of Part I combine in the simplest and most efficient energy converter of all, the transformer.

Part II — Transformers and Electrical Machines

Part I established the physics of conversion in its engineering form: magnetic circuits with air gaps and saturation, induced EMF and inductance in real windings, force and torque from co-energy, the rotating magnetic field, and the losses and heat that set every rating. Part II applies that physics to the devices that carry almost all electrical energy from generation to use. Every kilowatt-hour from a power station passes through a synchronous generator and four or five transformers before it reaches a load, and most of it ends up in a motor. This part builds the model engineers use for each machine: an equivalent circuit whose parameters come from standard tests, a set of characteristics that show how the machine behaves under load, and the numbers (impedances, slips, efficiencies, temperature rises) that decide how it is specified, connected, started, controlled and protected. Each device is described by a model with stated assumptions, solved for its running performance, and followed to where the model fails: inrush, commutation, deep-bar effects, stability limits, demagnetisation, and heat in all of them.

By the end of this part you should understand

- How a real transformer departs from the ideal one, how its equivalent circuit, losses and regulation are measured, and how units are paralleled, connected in three-phase groups, or built as autotransformers and instrument transformers.
- How the commutator makes a DC machine's EMF and torque follow $E = k\Phi\omega$ and $T = k\Phi I_a$, and how the field connection shapes its characteristics, starting, speed control and braking.
- How slip governs the induction motor: its per-phase equivalent circuit, power flow, torque–speed curve, starting methods, speed control, efficiency classes and selection for a load.
- How a synchronous machine generates, how its power-angle relationship and excitation set active and reactive power, and how it is synchronised and kept stable on a grid.
- How permanent-magnet, brushless DC, switched-reluctance, stepper, universal, linear and servo motors work, and when each is the right choice.
- How temperature, insulation, duty and cooling set a machine's rating, and how machines are tested and maintained.

Chapters

Chapter	Central question	What it delivers
Chapter 4 — Transformers	How does a real transformer differ from the ideal one, and how are its losses, impedance and connections chosen and checked?	The equivalent circuit from open- and short-circuit tests; efficiency, all-day efficiency and regulation; polarity, parallel operation, vector groups, autotransformers and instrument transformers
Chapter 5 — DC Machines	How does a commutated machine convert energy, and how do its field connections fix its behaviour?	EMF and torque equations with winding constants; separately excited, shunt, series and compound characteristics; speed control, starter design, braking, losses and efficiency
Chapter 6 — Induction Machines	How does a short-circuited rotor produce torque, and how is the motor started, controlled and selected?	Slip, the per-phase equivalent circuit and power flow; torque–speed curve and design classes; starting methods and voltage dip; speed control; IE efficiency classes; single-phase motors; selection
Chapter 7 — Synchronous Machines	How does a generator turning at exactly synchronous speed deliver active and reactive power to a grid?	Phasor diagrams and regulation; the power-angle equation; synchronisation and parallel operation; synchronous motors; excitation and reactive power; stability fundamentals
Chapter 8 — Special Electrical Machines	Which machines replace the classical three when efficiency, power density or precise motion matter?	Principles and characteristics of PMSM, BLDC, switched-reluctance, stepper, universal, linear and servo motors and specialised generators; selection trade-offs
Chapter 9 — Electrical Machine Design and Performance	What limits a machine's rating, and how are performance and reliability verified?	Duty cycles, loss calculation, thermal limits, insulation systems, cooling, winding and material choices, performance testing, reliability and maintenance

How this part connects to the rest of the book

Part II rests on Part I and on two sibling books. The electromagnetic laws, the quasi-static limit and the transformer and gyrator elements of the bond graph come from *Engineering Physics: Top Down* (Ch. 11 and Ch. 17). Coupled coils, the ideal transformer, phasors, AC power and three-phase circuits come from *Circuits, Signals and Communications* (Chapters 5, 8, 9 and 10).

The machine models built here are the plants of Part III, where DC drives (Chapter 11) and AC drives (Chapter 12) control them with power electronics. The synchronous generator and the transformer are the building blocks of generation in Part IV, of the per-unit network models, load flow, fault and stability studies of Part V, and of the distribution and protection systems of Part VI, where instrument transformers and transformer differential protection reappear. Motor efficiency returns in Chapter 28 and equipment selection in Chapter 30.

Chapter 4 — Transformers

Part II · Transformers and Electrical Machines

A municipal utility replaces a forty-year-old 1000 kVA distribution transformer that feeds a shopping centre. The old unit never failed, never overheated and still tests healthy, yet the business case for replacing it is easy: its core dissipates about 1.7 kW around the clock, whether the shops are open or not, and its replacement's core dissipates 0.7 kW. That difference alone is nearly 9 MWh a year, and with the lower load losses of the new unit the saving reaches about 12.5 MWh a year, enough to pay for a sizeable part of the new unit over its life. At the other end of the scale, a 400/132 kV substation needs a second 240 MVA transformer to carry a growing city load, and the engineer must check that the new unit, bought from a different manufacturer, will share load with the old one instead of circulating current between them. Between these extremes sit the transformers in wind turbines, data-centre substations, traction supplies and ships, and the instrument transformers that let a relay see a 40 kA fault through a few amperes. The transformer is the most numerous, most efficient and least glamorous machine in the power system; this chapter turns the coupled-coil model into the engineering model that is used to specify, test, load and protect it.

> **How does a real transformer differ from the ideal one, and how are its losses, impedance and regulation measured and used?**
> **What decides whether transformers can share load, how three-phase windings are connected, and when an autotransformer or an instrument transformer is the right device?**

Concept	Engineering destination
Operating principle and construction	Specifying core material, cooling and winding arrangement; reading a nameplate
Ideal transformer	Referring impedances across a transformer; per-unit models (Chapter 17)
EMF equation	Choosing turns and core area; overfluxing (V/f) protection
Equivalent circuit	Load-flow and fault models (Chapters 17–19); efficiency and regulation calculations
Phasor diagrams	Understanding why secondary voltage falls with lagging load and can rise with leading load
Losses and efficiency	Loss capitalisation in procurement, Ecodesign limits, all-day efficiency
Voltage regulation	Tap-changer range, voltage drop at the end of a feeder (Chapter 22)
Open- and short-circuit tests	Factory acceptance tests to IEC 60076-1; parameter data for studies
Polarity	Correct connection of CTs, parallel units and three-phase banks
Parallel operation	Substation expansion, N−1 security, load sharing and circulating current
Three-phase transformers	Vector groups (Dyn11, YNd1), earthing, triplen-harmonic paths
Autotransformers	400/275/132 kV interbus units, motor starters, railway autotransformer feeding
Instrument transformers	Metering and protection (Chapter 23), CT saturation, VT ferroresonance

Roadmap. Why a transformer works and how it is built (4.1) → the ideal transformer as the starting point (4.2) → turns, flux and frequency (4.3) → the equivalent circuit (4.4) and how its parameters are measured (4.5) → phasor diagrams (4.6) → losses and efficiency (4.7) → voltage regulation (4.8) → polarity (4.9) → parallel operation (4.10) → three-phase transformers (4.11) → autotransformers (4.12) → instrument transformers (4.13).

4.1 Transformer operating principle

Two windings, one flux

A transformer is two or more windings linked by a common magnetic core. An alternating voltage applied to the primary drives an alternating flux in the core; the same flux links the secondary and induces in it a voltage in proportion to its turns. Power crosses from one winding to the other through the magnetic field, with no electrical connection and no moving part. Three results from earlier work make this possible, and none of them is re-derived here.

> **Boundary.**
> Faraday's law in integral form is in *Engineering Physics*, Ch. 11 §11.4; the transformer as a power-conserving two-port (the TF element of the bond graph) is in *Engineering Physics*, Ch. 17 §17.5. Mutual inductance, the dot convention, the coupling coefficient and the limit $k \to 1$, $L \to \infty$ that yields the ideal transformer are in *Circuits, Signals and Communications*, Ch. 5 §5.7–5.8. Magnetic circuits, reluctance and saturation are in Chapter 1, self- and mutual inductance of real windings in Chapter 2, and loss mechanisms in Chapter 3. This chapter starts from those results and builds the engineering model.

The engineering content lies in how a real transformer differs from perfectly coupled coils: the core needs magnetising current and dissipates power, some flux leaks between the windings, the windings have resistance, the core saturates at high volts per hertz, and the heat must be removed without ageing the insulation.

Construction

Two core geometries dominate (Figure 4.1). In a **core-type** (core-form) transformer the windings surround the limbs of a rectangular core; a three-phase unit has three limbs joined by top and bottom yokes, with the low-voltage winding nearest the limb and the high-voltage winding concentric outside it. In a **shell-type** (shell-form) transformer the core surrounds the windings, which are arranged as interleaved pancake coils; the geometry gives good short-circuit strength and is used for some large generator step-up and furnace transformers. Distribution transformers are almost all core type; wound-core (rolled ribbon) designs are common for amorphous-metal cores and small pole-mounted units.

The core is built from thin, insulated laminations of grain-oriented silicon steel, stacked with step-lap joints and with the rolling direction along the flux path, so that eddy currents are confined to thin sheets and hysteresis loss is minimised. The windings are copper or aluminium, as round wire, rectangular strip, continuously transposed cable (large units) or foil (LV windings of distribution transformers). The insulation is either mineral or ester oil with paper (liquid-immersed) or epoxy resin and air (dry-type, used indoors in buildings, ships and data centres where fire risk rules out oil).

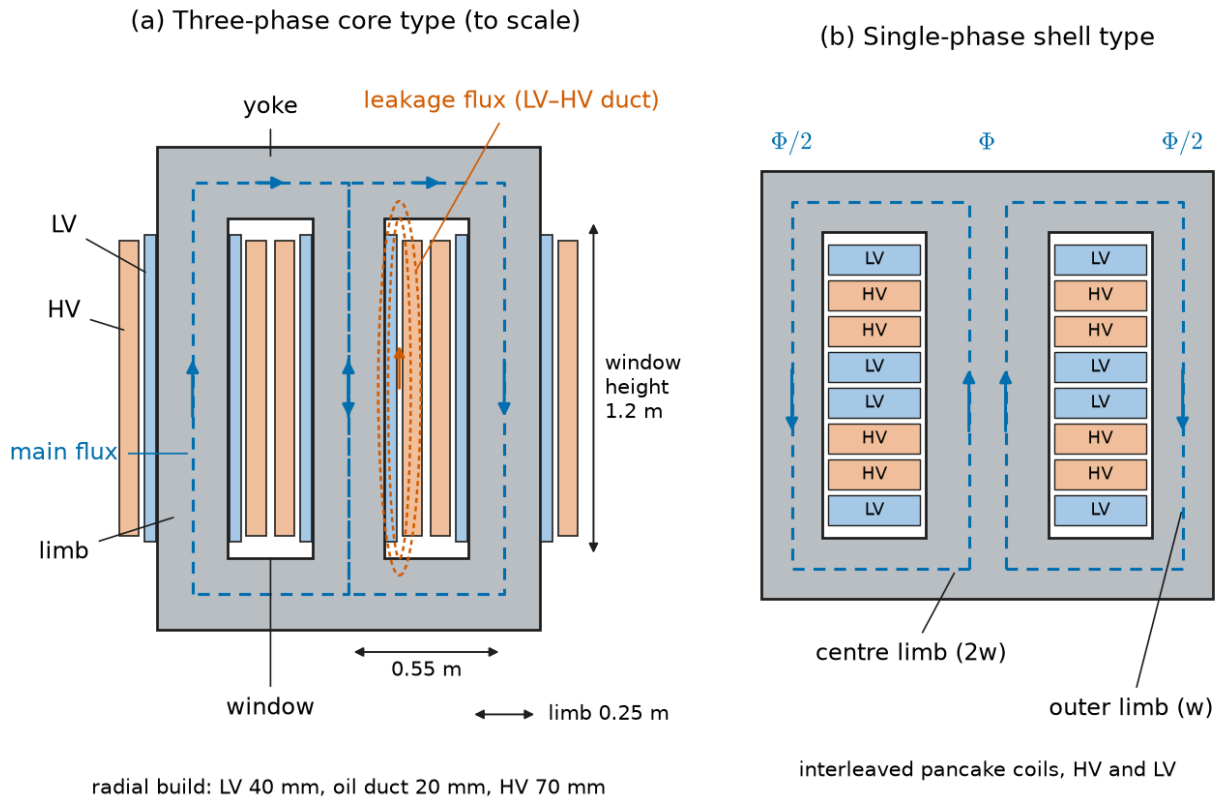


Figure 4.1 — Core-type and shell-type transformer construction, showing limbs, yokes, concentric LV and HV windings, main flux and leakage flux.

Table 4.1 — Core materials for power transformers (typical values at 50 Hz).

Material	Thickness	Saturation flux density	Typical design peak flux density	Specific core loss
Conventional grain-oriented steel (CGO)	0.27–0.35 mm	about 2.0 T	1.55–1.7 T	1.0–1.3 W/kg at 1.7 T
High-permeability GO, domain-refined (HiB, laser-scribed)	0.18–0.27 mm	about 2.0 T	1.6–1.75 T	0.65–0.90 W/kg at 1.7 T
Amorphous Fe-based ribbon	about 0.025 mm	1.56 T	1.25–1.4 T	0.1–0.2 W/kg at 1.3 T
Non-oriented steel (for comparison; machine laminations)	0.35–0.65 mm	about 2.0 T	1.3–1.5 T	2.5–5 W/kg at 1.5 T

Amorphous cores cut no-load loss by roughly 60–70 % compared with CGO, at the price of a lower working flux density (a larger, heavier core), higher cost and more audible noise, which is why they are attractive for distribution transformers that spend most of their life lightly loaded.

Table 4.2 — Representative transformer ratings across the energy field.

Application	Rating	Voltages	Notes
Pole-mounted distribution (rural)	25–100 kVA	11 kV / 240 V single-phase, or 11 kV / 415 V	Self-protected units common
Ground-mounted distribution / unit substation	315–2500 kVA	11 kV or 22 kV / 400–433 V, Dyn11	Buildings, data centres, commercial loads
Wind-turbine step-up	2.5–8 MVA	0.69–3.3 kV / 33–66 kV	Often in nacelle or tower base, dry-type or ester-filled
Solar-plant inverter station	2.5–6 MVA	0.6–0.8 kV / 33 kV	Often three-winding with two LV windings
Traction (on-board, AC locomotive)	5–10 MVA	25 kV / several secondaries	Weight-critical, high flux density, forced oil
Generator step-up (GSU)	100–1300 MVA	15–27 kV / 220–765 kV, YNd1 or YNd11	Efficiency about 99.7 %
Interbus autotransformer	240–1000 MVA	400/132 kV, 400/275 kV, 500/230 kV	With tertiary delta winding

4.2 Ideal transformer

The model and its two rules

The ideal transformer is the limit of perfectly coupled coils on a core of infinite permeability with lossless windings. With turns N_1 and N_2 and the turns ratio $a = N_1/N_2$, it obeys two rules, taken from *Circuits, Signals and Communications*, Ch. 5 §5.8:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a, \qquad \frac{I_1}{I_2} = \frac{1}{a}$$

with the currents taken as entering the primary dot and leaving the secondary dot. Power in equals power out, $V_1 I_1^* = V_2 I_2^*$, and no energy is stored. Book 3 states this limit and its status as an approximation; the engineering use made of it here is to move impedances, voltages and currents from one side of a transformer to the other.

Referring quantities across the transformer

A load Z_L on the secondary appears at the primary as

$$Z_L' = a^2 Z_L$$

and, conversely, any impedance on the primary side appears on the secondary side divided by a^2. This **referral** lets a network that contains transformers be redrawn as a single circuit at one voltage level, which is exactly what the equivalent circuit of Section 4.4 and the per-unit system of Chapter 17 do.

Worked example — Referring a cable impedance. A 1000 kVA, 11 kV/433 V transformer feeds a 400 V switchboard through a cable whose impedance is $0.010 + j0.006\ \Omega$ per phase. Seen from the 11 kV side the ratio of phase voltages is the same as the ratio of line voltages for a star–star equivalent, $a = 11000/433 = 25.40$, so the cable referred to 11 kV is $a^2(0.010 + j0.006) = 645.4 \times (0.010 + j0.006) = 6.45 + j3.87\ \Omega$. A few

milliohms on the LV side are several ohms at 11 kV; impedances on the low-voltage side of a distribution transformer dominate the fault level at the LV switchboard (Chapter 19) even though they look tiny.

> **Status of the principle: Ideal-transformer rules**
> *Approximate*. Exact for perfect coupling, infinite core permeability and lossless windings; for power transformers the voltage ratio holds to within the regulation (1–6 %) and the current ratio to within the magnetising current (0.1–2 % of rated), and both fail at DC, in saturation and at frequencies where winding capacitance matters.

Rating

Because the transformer conserves power and its losses depend on current (windings) and voltage (core), its rating is an apparent power, in kVA or MVA, not a real power. The same 1000 kVA transformer can supply 1000 kW at unity power factor or 800 kW at 0.8 lagging with the same winding temperature rise. The rating is fixed by the temperature rise the insulation can tolerate (IEC 60076-2: 65 K average winding rise for mineral-oil units with normal cooling), and the cooling class (ONAN, ONAF, OFAF, ODAF) often gives two or three ratings for one unit, for example 60/80/100 MVA ONAN/ONAF/OFAF.

4.3 EMF equation

Turns, flux and frequency

When the core flux is sinusoidal, $\phi = \Phi_m \sin \omega t$, Faraday's law gives a winding EMF $e = N\omega\Phi_m \cos \omega t$, whose RMS value is

$$E = \frac{2\pi}{\sqrt{2}} f N \Phi_m = 4.44\, f N \Phi_m = 4.44\, f N B_m A_c$$

where A_c is the net (iron) cross-section of the core and B_m the peak flux density. The coefficient 4.44 is $\sqrt{2}\pi$; it is the form factor 1.11 of a sine wave times 4. Read in the other direction, the equation says that the applied voltage fixes the flux: a transformer connected to a stiff supply runs at the flux density its designer chose, whatever the load, because the primary current adjusts itself to keep the back EMF equal to the supply voltage less a small internal drop.

> **Status of the principle: Transformer EMF equation**
> *Derived*. Follows from Faraday's law for a sinusoidal flux linking all N turns; it holds for the fundamental component even when the magnetising current is distorted, and it fails only if the flux is not sinusoidal (a non-sinusoidal supply voltage, or DC bias from geomagnetic currents or rectifier loads).

Volts per turn and the design trade-off

The **volts per turn** $E/N = 4.44 f B_m A_c$ is the designer's central choice. A larger core cross-section means fewer turns of copper for the same voltage (less load loss) but more iron (more no-load loss and weight). Large transformers run at 100–250 V per turn, distribution transformers at 5–20 V per turn. The peak flux density is set by the steel: a working B_m of 1.6–1.7 T leaves a margin below the knee of the B–H curve (Chapter 1) for the overvoltage the system may impose.

Worked example — Turns for a 1000 kVA distribution transformer. A three-phase 1000 kVA, 11 kV/433 V, 50 Hz, Dyn11 transformer has a net core limb area of $A_c = 0.040\ \text{m}^2$, and the designer starts from $B_m = 1.7$ T. The LV winding is star-connected, so its phase voltage is $433/\sqrt{3} = 250.0$ V; the HV winding is delta-connected, so its phase voltage is 11 000 V.

At 1.7 T the volts per turn would be $4.44 \times 50 \times 1.7 \times 0.040 = 15.10$ V, giving $250.0/15.10 = 16.56$ LV turns. Turns must be whole numbers, so the designer takes $N_2 = 17$, which gives 14.71 V per turn and an actual peak flux density of $14.71/(4.44 \times 50 \times 0.040) = 1.656$ T. The HV winding then needs $11000/14.71 = 748$ turns. The no-load ratio of line voltages is $11000/(748/17) \times \sqrt{3} = 433.0$ V, as required. A ±2×2.5 % off-circuit tap range is obtained by adding or removing $0.025 \times 748 \approx 19$ turns per step on the HV winding, which is where taps are placed because the HV current is small.

Overfluxing

Because $B_m \propto V/f$, a transformer is overfluxed either by overvoltage or by underfrequency. A generator step-up transformer on a unit that is excited while still running up to speed (low f) can be driven well into saturation; the magnetising current rises steeply, stray flux heats structural steel and tank walls, and in minutes local hot spots can damage insulation. Generator protection therefore includes a volts-per-hertz (ANSI 24) element, typically alarming at 1.05–1.1 pu V/Hz and tripping on an inverse-time curve up to about 1.2 pu (IEEE C37.106 and C57.12.00 allow continuous operation at 1.05 pu voltage at rated load and 1.1 pu at no load). Geomagnetically induced currents, quasi-DC currents driven through earthed neutrals during geomagnetic storms, cause similar half-cycle saturation.

Magnetising inrush

When a transformer is energised at a voltage zero, the flux must start from its residual value and rise for half a cycle; the steady-state flux would have been at its negative peak at that instant, so the transient flux reaches up to $2\Phi_m + \Phi_r$, far into saturation. The resulting **inrush current** can reach 8–12 times rated current for distribution transformers and 4–6 times for large units, rich in second harmonic and decaying over 0.1 s to several seconds. Differential protection (Chapter 23) uses the second-harmonic content to tell inrush from an internal fault; controlled (point-on-wave) switching is used for large units and for wind-farm transformers whose sequential energisation would otherwise cause voltage dips.

4.4 Equivalent circuit

From ideal to real, element by element

Each departure of the real transformer from the ideal one adds one element to the model (Figure 4.2):

- **Winding resistance** R_1, R_2: I^2R loss in the conductors, increased at power frequency by eddy currents in the conductors themselves.
- **Leakage reactance** X_{l1}, X_{l2}: the flux that links one winding but not the other, mainly in the duct between windings. It is nearly linear because its path is mostly through oil or air.
- **Magnetising reactance** X_m: the finite permeability of the core means a current is needed to set up the main flux. It is non-linear, rising sharply above about 1.7 T in grain-oriented steel.
- **Core-loss resistance** R_c: hysteresis and eddy-current loss in the core, modelled as a resistance across the applied voltage because the loss depends on flux density and hence on voltage.

The exact (T) equivalent circuit places $R_1 + jX_{l1}$ in series on the primary, the shunt branch $R_c \parallel jX_m$ across the induced EMF E_1, and $R_2 + jX_{l2}$ in series on the secondary, connected through an ideal transformer of ratio a. Referring the secondary to the primary removes the ideal transformer: $R_2' = a^2 R_2$, $X_{l2}' = a^2 X_{l2}$, $V_2' = aV_2$, $I_2' = I_2/a$.

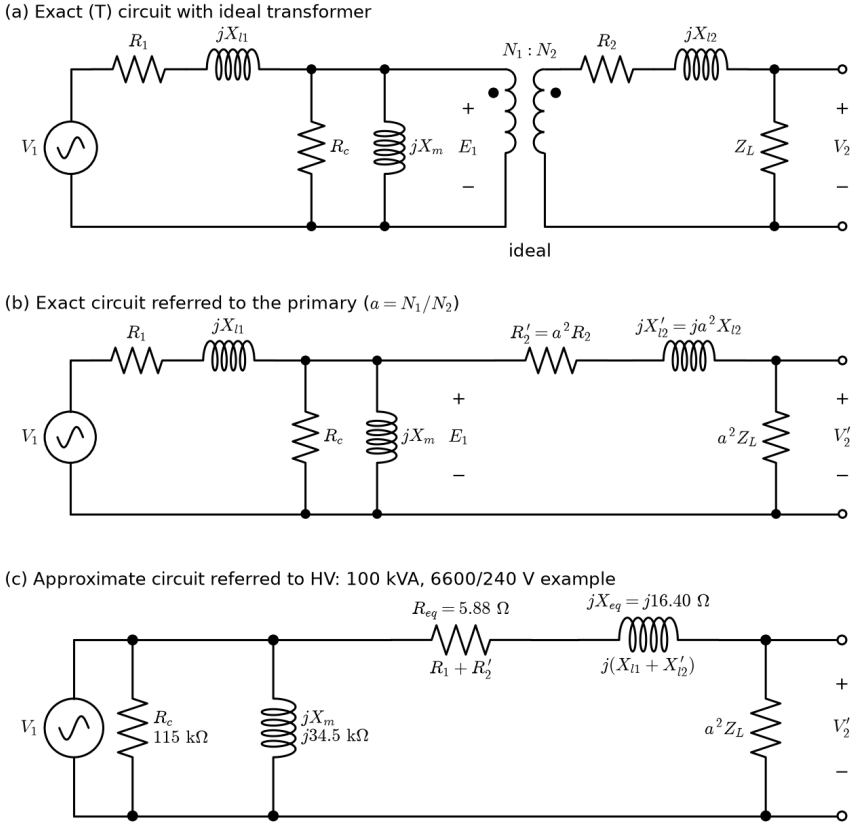


Figure 4.2 — Transformer equivalent circuits: exact T circuit with ideal transformer, circuit referred to the primary, and the approximate circuit with the shunt branch moved to the terminals.

The approximate circuit

In a power transformer the shunt branch impedance is 300–3000 times the series impedance, so moving it to the primary terminals changes the series voltage drop by a negligible amount. The result is the **approximate equivalent circuit**: a shunt branch across the supply, and a single series impedance

$$Z_{eq} = R_{eq} + jX_{eq} = (R_1 + a^2 R_2) + j(X_{l1} + a^2 X_{l2})$$

feeding the referred load. For load-flow and fault studies even the shunt branch is usually dropped, and the transformer becomes a series impedance (with an off-nominal ratio for tap position) in the network model of Chapter 17.

> **Status of the principle: Approximate transformer equivalent circuit**
> *Approximate*. Assumes linear magnetising and core-loss elements at constant frequency and a shunt branch large compared with the series impedance; it is accurate to a fraction of a per cent for regulation and efficiency at rated voltage and frequency, and it fails for inrush and overfluxing (non-linear X_m), for geomagnetic DC bias, for harmonics above a few kHz (winding capacitance), and for lightning surges, where the windings behave as distributed networks (Chapter 24).

Per-unit values

Expressed in per unit of the transformer's own rating (Chapter 17), the parameters of transformers of very different sizes fall into narrow bands, which is why a missing value can be estimated with confidence.

Table 4.3 — Typical transformer parameters in per unit on own rating.

Rating	Short-circuit impedance z	X/R ratio	No-load current	No-load loss	Load loss at rated current
25–100 kVA distribution	0.035–0.04	1–2	1.5–2.5 %	0.15–0.3 %	1.2–2.0 %
315–2500 kVA distribution	0.04–0.06	3–6	0.5–1.5 %	0.07–0.15 %	0.8–1.2 %
10–60 MVA primary substation	0.08–0.12	15–30	0.2–0.5 %	0.04–0.08 %	0.4–0.6 %
100–1000 MVA GSU and interbus	0.12–0.18	40–80	0.05–0.3 %	0.02–0.05 %	0.2–0.35 %

IEC 60076-5 sets minimum short-circuit impedances so that the transformer itself limits fault current: 4 % up to 630 kVA, 5 % from 631 to 1250 kVA, 6 % from 1251 to 2500 kVA, 7 % to 6300 kVA, 8 % to 25 MVA, rising to 12.5 % above 63 MVA. Impedance is a deliberate design choice: higher impedance limits fault current but raises voltage regulation and reactive-power consumption.

4.5 Open-circuit and short-circuit tests

Separating the shunt and series branches

The equivalent-circuit parameters are measured, not calculated, and two simple tests do it because the shunt and series branches can be separated by choosing the operating condition (IEC 60076-1 calls them the no-load and load-loss measurements).

Open-circuit (no-load) test. Rated voltage at rated frequency is applied to one winding, usually the LV, with the other winding open. The current is the no-load current I_0, a few per cent of rated or less, so the series drop is negligible and the wattmeter reads the core loss P_0. From V_{oc}, I_{oc} and P_{oc}:

$$\cos\phi_0 = \frac{P_{oc}}{V_{oc} I_{oc}}, \qquad R_c = \frac{V_{oc}}{I_{oc}\cos\phi_0}, \qquad X_m = \frac{V_{oc}}{I_{oc}\sin\phi_0}$$

Short-circuit (load-loss) test. One winding, usually the LV, is short-circuited, and a reduced voltage is applied to the other until rated current flows. That voltage is only the impedance voltage, 4–15 % of rated, so the core flux is a few per cent of normal and the core loss is negligible: the wattmeter reads the copper loss at rated current. From V_{sc}, I_{sc} and P_{sc}:

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}, \qquad R_{eq} = \frac{P_{sc}}{I_{sc}^2}, \qquad X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

referred to the side on which the measurement is made. The load loss is corrected to the reference temperature (75 °C for oil-immersed units under IEC 60076-1), since copper resistance rises by about 0.39 % per kelvin.

The open-circuit test is a Z-parameter measurement in the two-port language of *Circuits, Signals and Communications*, Ch. 12.

Worked example — Parameters from test data. A single-phase 100 kVA, 6600/240 V, 50 Hz pole-mounted transformer gives:

- Open-circuit test on the LV side: 240 V, 5.5 A, 380 W.
- Short-circuit test on the HV side: 264 V, 15.15 A, 1350 W.

Rated currents are $100000/6600 = 15.15$ A (HV) and $100000/240 = 416.7$ A (LV), so the short-circuit test was made at rated current, and the no-load current is $5.5/416.7 = 1.3$ % of rated.

Open circuit. $\cos\phi_0 = 380/(240 \times 5.5) = 0.288$. The core-loss component of current is $I_c = 5.5 \times 0.288 = 1.583$ A and the magnetising component $I_m = 5.5\sqrt{1 - 0.288^2} = 5.267$ A. Hence $R_c = 240/1.583 = 151.6\ \Omega$ and $X_m = 240/5.267 = 45.57\ \Omega$ on the LV side. With $a = 6600/240 = 27.5$, referred to the HV side these are $R_c = 114.6$ kΩ and $X_m = 34.46$ kΩ.

Short circuit. $Z_{eq} = 264/15.15 = 17.42\ \Omega$, $R_{eq} = 1350/15.15^2 = 5.88\ \Omega$ and $X_{eq} = \sqrt{17.42^2 - 5.88^2} = 16.40\ \Omega$, all referred to the HV side. The HV base impedance is $6600^2/100000 = 435.6\ \Omega$, so $z = 0.040$, $r = 0.0135$ and $x = 0.0377$ per unit: a 4 % impedance transformer with an X/R ratio of 2.8, typical for this size (Table 4.3).

The tests separate the two branches but cannot split R_{eq} into R_1 and R_2'; where the split matters (the exact circuit), a DC resistance measurement of each winding is used, or the two are assumed equal when referred to the same side.

Accuracy at low power factor

Both tests on large units run at very low power factor (0.02–0.1), where a small instrument-transformer phase error becomes a large power error (*Circuits, Signals and Communications*, Ch. 9 §9.9 quantifies this). Loss measurements on large transformers, which are guaranteed values with financial penalties, use class 0.05–0.1 instrument transformers and phase-corrected digital power analysers.

4.6 Phasor diagrams

No load

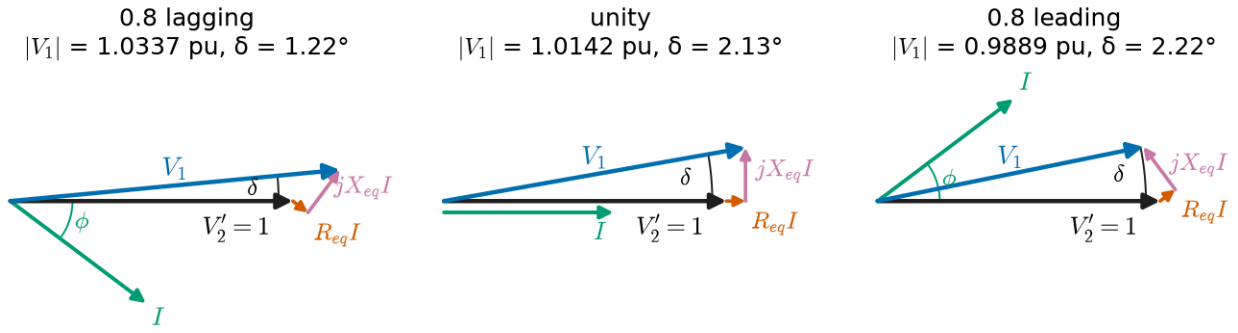
At no load the primary draws only I_0, made up of a core-loss component I_c in phase with the applied voltage and a magnetising component I_m lagging it by 90°, in phase with the flux. Since I_0 is small, $V_1 \approx E_1$, and the no-load phasor diagram is a voltage phasor with a small current lagging it by about 70–85°. The no-load current is distorted: because the B–H curve is non-linear, a sinusoidal flux needs a peaky magnetising current with a large third harmonic (typically 30–50 % of the fundamental magnetising component), which matters for the three-phase connections of Section 4.11.

On load

On load, the secondary current I_2 draws a balancing primary current $I_2' = I_2/a$, and the primary current is $I_1 = I_0 + I_2'$. Referred to the primary, the voltage equation of the approximate circuit is

$$V_1 = V_2' + I_2'(R_{eq} + jX_{eq})$$

Figure 4.3 draws this equation for three load power factors at the same current magnitude. With a lagging load, the $jX_{eq}I$ drop has a large component in phase with V_2' (at 0.8 power factor it leads V_2' by only 53°, and it swings into phase as the power factor falls towards zero), so V_1 must exceed V_2' by a large margin, and with constant V_1 the secondary voltage falls. At unity power factor the reactive drop is in quadrature with V_2' and contributes only a second-order rise in magnitude. With a sufficiently leading load, the reactive drop points backwards, and the secondary voltage can exceed the open-circuit value: this is the source of the voltage rise at the end of a lightly loaded cable network with a large capacitor bank or a PV-heavy feeder exporting reactive power.



Drops $R_{eq}I$ (0.0135 pu) and $jX_{eq}I$ (0.0377 pu) drawn 5× enlarged; I = 1.0 pu drawn at reduced length.

Figure 4.3 — Phasor diagrams of the approximate equivalent circuit referred to the primary for rated current at 0.8 lagging, unity and 0.8 leading power factor.

4.7 Losses and efficiency

Where the losses are

Transformer losses separate into two groups that behave differently with load:

- **No-load loss** P_0 (iron or core loss): hysteresis plus eddy-current loss in the core, a function of B_m and f only. It is constant whenever the transformer is energised, which for a distribution transformer means 8760 hours a year.
- **Load loss** P_k (copper loss): I^2R loss in the windings plus stray loss from leakage flux in winding conductors, clamps and tank walls. It varies as the square of the load fraction $x = S/S_{rated}$: $P_{cu} = x^2 P_k$.

The separation of core loss into its hysteresis and eddy components, and their dependence on frequency and flux density, are the subject of Chapter 3; for transformer work at fixed voltage and frequency P_0 is simply the measured no-load loss. Harmonic load currents raise the stray and winding eddy loss roughly as h^2 for each harmonic order h, which is why transformers feeding data-centre IT loads or variable-speed drives are derated or specified with a K-factor (IEEE C57.110).

Efficiency and its maximum

At load fraction x and power factor $\cos\phi$,

$$\eta = \frac{x S_r \cos\phi}{x S_r \cos\phi + P_0 + x^2 P_k}$$

Differentiating with respect to x shows the efficiency is greatest when the variable loss equals the fixed loss, $x^2 P_k = P_0$, that is at

$$x_{\eta\,\max} = \sqrt{\frac{P_0}{P_k}}$$

> **Status of the principle: Maximum efficiency when copper loss equals core loss**
> *Derived.* Follows from minimising total loss per unit output with a constant loss and a loss proportional to x^2; it assumes constant voltage, frequency and power factor, and shifts when load loss is corrected for winding temperature or harmonic content.

A power transformer that runs near full load around the clock (a GSU) is designed with $x_{\eta\,\max}$ near 0.7–1; a distribution transformer whose average load is 20–40 % of rating is designed with low no-load loss and $x_{\eta\,\max}$ near 0.3–0.5.

Worked example — Efficiency of the 100 kVA transformer. Using $P_0 = 380$ W and $P_k = 1350$ W from Section 4.5:

- Full load, 0.8 power factor: output 80 kW, losses $380 + 1350 = 1730$ W, $\eta = 80/81.73 = 97.88$ %.
- Half load, 0.8 power factor: output 40 kW, copper loss $0.25 \times 1350 = 337.5$ W, $\eta = 40/40.718 = 98.24$ %.
- Full load, unity power factor: $\eta = 100/101.73 = 98.30$ %.

Maximum efficiency occurs at $x = \sqrt{380/1350} = 0.531$, i.e. 53 kVA, where at unity power factor $\eta = 53.05/(53.05 + 0.76) = 98.59$ %.

Figure 4.4 shows the efficiency curve of this unit and of the 1000 kVA unit of the next example.

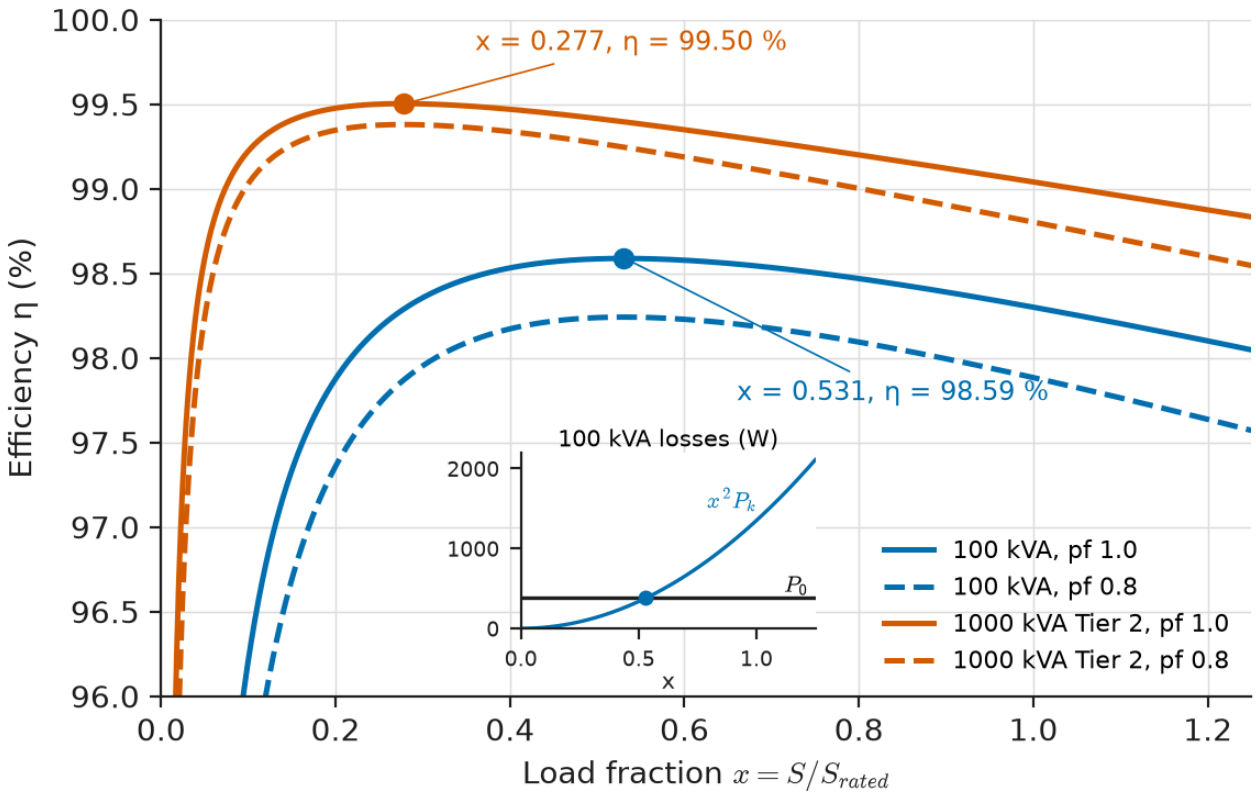


Figure 4.4 — Efficiency against load fraction for the 100 kVA (P0 = 380 W, Pk = 1350 W) and 1000 kVA Tier 2 (P0 = 693 W, Pk = 9000 W) transformers at unity and 0.8 power factor, with the maximum-efficiency points marked.

All-day efficiency and loss capitalisation

For a transformer with a varying load, the energy efficiency over a day or a year is the useful figure:

$$\eta_{\text{all-day}} = \frac{\sum S_i x_i \cos\phi_i \, t_i}{\sum S_i x_i \cos\phi_i \, t_i + 24 P_0 + \sum x_i^2 P_k \, t_i}$$

Worked example — Replacing an old distribution transformer. A 1000 kVA unit feeding a shopping centre carries 80 % load for 6 h, 50 % for 10 h and 20 % for 8 h each day, at 0.9 power factor. Compare an old unit with $P_0 = 1.7$ kW, $P_k = 10.5$ kW and a new unit meeting the EU Ecodesign Tier 2 limits (Regulation (EU) 548/2014) for 1000 kVA liquid-immersed transformers, $P_0 = 693$ W and $P_k = 9.0$ kW.

Energy delivered per day: $1000 \times 0.9 \times (0.8 \times 6 + 0.5 \times 10 + 0.2 \times 8) = 900 \times 11.4 = 10260$ kWh.

The sum $\sum x_i^2 t_i = 0.64 \times 6 + 0.25 \times 10 + 0.04 \times 8 = 6.66$ h. New unit: copper-loss energy $9.0 \times 6.66 = 59.94$ kWh, core-loss energy $0.693 \times 24 = 16.63$ kWh, total 76.57 kWh, $\eta_{\text{all-day}} = 99.26$ %. Old unit: $10.5 \times 6.66 = 69.93$ kWh plus $1.7 \times 24 = 40.8$ kWh, total 110.73 kWh, $\eta_{\text{all-day}} = 98.93$ %.

The difference is 34.2 kWh per day, or 12.5 MWh per year; at USD 0.12 per kWh that is about USD 1500 a year, and the loss reduction also frees capacity upstream. Note where the new unit's best efficiency sits: $\sqrt{693/9000} = 0.28$, close to the light-load periods that dominate the day. Utilities formalise this trade-off by **loss capitalisation**: each tender is evaluated on purchase price $+ A \cdot P_0 + B \cdot P_k$, where A and B (typically several thousand USD per kW, with A larger than B for distribution units) are the present value of a kilowatt of loss over the transformer's 30–40 year life.

Table 4.4 — Representative efficiencies at rated load, unity power factor.

Transformer	Typical efficiency
25–100 kVA distribution	98.0–98.8 %
1000 kVA distribution, Ecodesign Tier 2	99.0–99.2 %
1000 kVA dry-type cast resin (building, data centre)	98.8–99.0 %
40 MVA primary substation	99.4–99.6 %
500 MVA GSU	99.7–99.8 %

4.8 Voltage regulation

Definition and approximate formula

Voltage regulation is the change in secondary voltage from no load to a specified load, with the primary voltage held constant, expressed as a fraction of the full-load secondary voltage:

$$\text{regulation} = \frac{V_{2,nl} - V_{2,fl}}{V_{2,fl}}$$

(IEC 60076-1 references it to the rated no-load voltage instead; the difference is second order.) From the phasor equation of Section 4.6, with per-unit resistance r and reactance x and a load at power factor angle ϕ (positive for lagging),

$$\varepsilon \approx r\cos\phi + x\sin\phi + \frac{(x\cos\phi - r\sin\phi)^2}{2}$$

The first two terms are usually enough; the quadratic term matters only for high-impedance units at unity or leading power factor.

> **Status of the principle: Approximate regulation formula**
> *Approximate.* A Taylor expansion of the phasor equation of the series equivalent circuit; accurate to better than 0.1 % of rated voltage for impedances up to about 15 %, and it ignores the magnetising branch and any change in the load with voltage.

Worked example — Regulation of the 100 kVA transformer. With $r = 0.0135$ and $x = 0.0377$ pu:

- At 0.8 lagging ($\sin\phi = 0.6$): first-order $0.0135 \times 0.8 + 0.0377 \times 0.6 = 3.34$ %; with the quadratic term 3.37 %; an exact phasor solution gives $|V_1| = 1.0337$ pu for $V_2' = 1$, i.e. 3.37 %.
- At unity power factor: first-order 1.35 %, exact 1.42 %.
- At 0.8 leading: first-order -1.18 %, exact -1.11 %, a voltage *rise* on load.

A 240 V supply therefore falls by about 8 V from no load to full load at a typical domestic power factor, which is one reason distribution transformers are set with a no-load voltage a few per cent above nominal (433 V for a 400 V system).

Tap changers

Regulation is compensated by changing the turns ratio. **Off-circuit tap changers** (de-energised tap changers) give a fixed setting, typically ±2×2.5 %, chosen at commissioning or seasonally. **On-load tap changers** (OLTC) switch taps under load through a diverter switch with transition resistors; primary-substation and transmission transformers have ranges of ±10 % to ±15 % in 17–33 steps of 1.25–1.67 %, controlled by an automatic voltage control relay with line-drop compensation (Chapter 22). High-PV distribution networks increasingly fit OLTCs to 11 kV/400 V units too, because reverse power flow raises the voltage at noon.

4.9 Transformer polarity

Additive and subtractive polarity

The dot convention of *Circuits, Signals and Communications*, Ch. 5 §5.7 marks terminals whose voltages rise together. On a power transformer the same information is given by terminal markings: H1/X1 in IEEE practice, 1U/2U (or A/a) in IEC practice, where terminals with the same subscript are instantaneously of the same polarity. Polarity is called **subtractive** when H1 and X1 are physically adjacent (the standard for large transformers, and for distribution transformers above 200 kVA or 8.66 kV in IEEE C57.12.00) and **additive** when H1 is diagonally opposite X1 (the standard for small single-phase distribution transformers in North America). The names come from the test below.

Polarity test

Connect H1 to the adjacent LV terminal, apply a low AC voltage V across the HV winding and measure the voltage between H2 and the other LV terminal. If the reading is $V - V/a$, the polarity is subtractive; if it is $V + V/a$, additive. A DC kick test (a battery momentarily applied to the HV winding, observing the deflection of a DC voltmeter on the LV winding) gives the same answer and is used on CTs, where a polarity error reverses a differential or directional relay.

Polarity matters wherever transformer voltages or currents are added or compared: paralleling (wrong polarity is a dead short), three-phase banks (Section 4.11), differential and directional relays (Chapter 23), and revenue metering, where one reversed CT makes a three-element meter read a third of the true energy.

4.10 Parallel operation

Substations parallel transformers to carry a load larger than one unit, to provide N−1 security and to grow capacity in steps. (Data-centre and hospital substations often keep a normally open bus-coupler instead, to limit fault level.)

Conditions for satisfactory parallel operation

1. **Same polarity** (single-phase) or **same vector group** (three-phase, Section 4.11). This is mandatory: a phase displacement of 30° between secondaries drives a circulating current of about $2\sin 15^\circ/(z_A + z_B)$ pu, several times rated.

2. **Same voltage ratio** (within about ±0.5 %, the IEC 60076-1 tolerance on ratio). A difference Δv drives a circulating current $\Delta v/(z_A + z_B)$ at no load.
3. **Same per-unit impedance** (within about ±10 %), so that load divides in proportion to rating.
4. **Similar X/R ratio**, so that the currents are in phase and the sum of the two currents equals the load current.
5. **Same phase sequence** (three-phase).

For two transformers with the same ratio, the load divides inversely as the impedances expressed in ohms on a common base, i.e.

$$\frac{S_A}{S_B} = \frac{S_{rA}/z_A}{S_{rB}/z_B}$$

where z_A, z_B are per unit on their own ratings. A transformer with a lower per-unit impedance takes more than its share.

Worked example — Sharing between unequal transformers. A 1000 kVA unit with $z_A = 5$ % and a 1600 kVA unit with $z_B = 6$ % (same ratio, same X/R) supply 2400 kVA. The "admittance weights" are $1000/0.05 = 20000$ and $1600/0.06 = 26667$. Unit A takes $2400 \times 20000/46667 = 1029$ kVA, 2.9 % overload; unit B takes 1371 kVA, 86 % of its rating. The largest load the pair can carry without overloading either is $1000 \times 46667/20000 = 2333$ kVA, not the 2600 kVA nameplate sum. The fix, if the units cannot be matched, is to accept the derating or to specify the new unit's impedance to match the old one; manufacturers can design to an impedance within a tolerance of ±7.5–10 %.

Worked example — Circulating current from a ratio mismatch. Two identical 1000 kVA, 5 % transformers are paralleled with one on a tap 1 % higher than the other. At no load, the circulating current is $0.01/(0.05 + 0.05) = 0.10$ pu, i.e. 10 % of rated current, flowing round the loop and adding about 1 % of rated load loss in each unit (0.1^2 of P_k) with no useful output. On load it adds vectorially to the load current, overloading one unit while the other runs light. Paralleled units with on-load tap changers are therefore controlled by master–follower or circulating-current-minimisation schemes that keep the taps in step.

4.11 Three-phase transformers

Banks and three-phase cores

A three-phase transformation can be made with a **bank** of three single-phase transformers or with a **three-phase unit** on a common core. The common core is lighter, cheaper and more efficient, because in a three-limb core the return flux of each limb flows through the other two (the three fluxes sum to zero), and it occupies less space. Banks are used where a single three-phase unit would exceed transport limits (very large GSUs, 765 kV units) and where a spare single-phase unit gives cheaper security. A **five-limb** core provides return limbs for zero-sequence and third-harmonic flux and reduces the height of the yokes; it is common for large units.

Connections and vector groups

Each set of windings is connected in star (Y, with N for an accessible neutral), delta (D) or zigzag (Z). The **vector group** of IEC 60076-1 names the HV connection in capitals, the LV connection in lower case and the phase displacement of the LV voltages behind the HV voltages in multiples of 30°, read as a clock number: Dyn11 means delta HV, star LV with neutral brought out, LV leading HV by 30° (or lagging by 330°). The star–

delta voltage and current relations themselves come from *Circuits, Signals and Communications*, Ch. 10 §10.3–10.4; what a transformer adds is the phase shift between windings of different connection and the paths it offers to zero-sequence and triplen-harmonic currents.

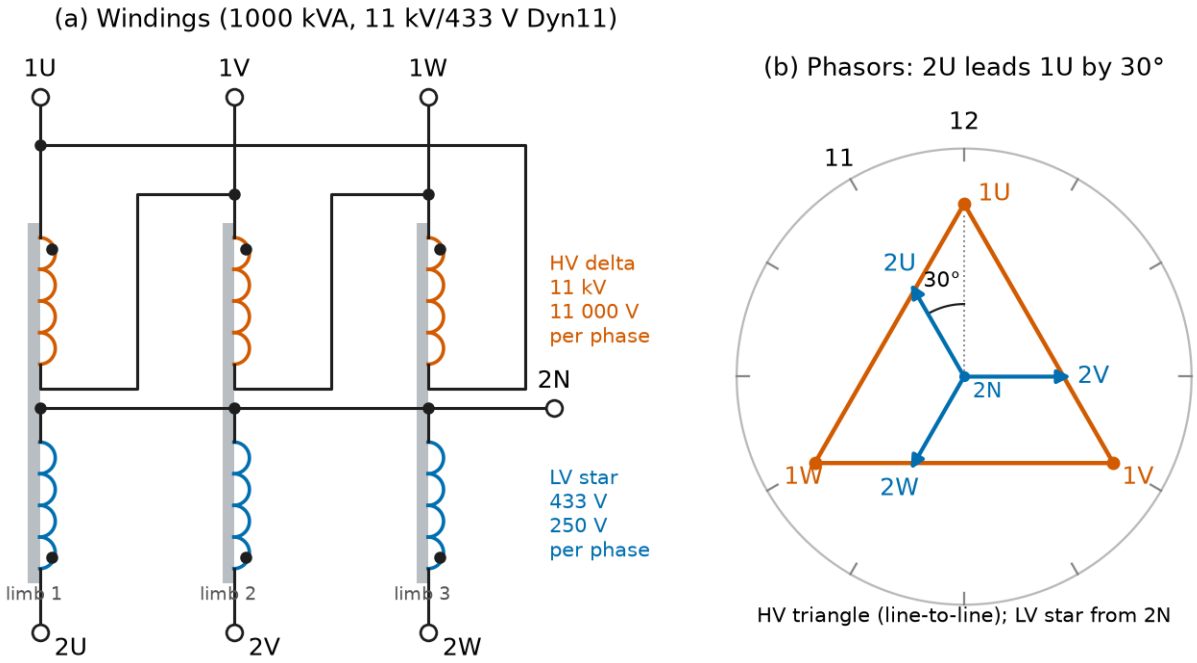


Figure 4.5 — Dyn11 three-phase transformer: winding connections and the HV and LV voltage phasors showing the 30 degree phase displacement.

Table 4.5 — Common three-phase connections and where they are used.

Vector group	Typical use	Why
Dyn11 (Dyn1 in some countries)	Distribution: 11–33 kV / 400 V	Delta HV traps triplen currents; earthed LV neutral supplies single-phase loads; LV earth faults draw positive-, negative- and zero-sequence current balanced in the delta
YNd1 or YNd11	Generator step-up; wind and solar collector step-up	Earthed HV neutral for the transmission system; delta on the generator side isolates generator from HV earth faults
YNyn0 with delta tertiary	Transmission interbus (also as autotransformer)	No phase shift between networks; tertiary provides zero-sequence path, stabilises neutral, can feed reactors or station supply
Ddo	Industrial plants, some rectifier supplies	No neutral needed; can run "open delta" (V–V) at 57.7 % rating with one unit out
Yd/Dd pair (30° apart)	12-pulse rectifiers for drives, HVDC, electrolysers	Cancels 5th and 7th harmonics in the supply
ZNyn or zigzag earthing transformer	Creating a neutral on a delta or unearthed system	Low zero-sequence impedance, high positive-sequence impedance

Why star–star needs a delta somewhere

A star–star transformer without a neutral connection or delta winding cannot carry the third-harmonic magnetising current that a sinusoidal flux requires, because triplen currents are in phase in all three lines and need a return path. The flux then contains a third harmonic, and in a bank or a five-limb core the phase voltages acquire a third-harmonic component that can exceed 50 % of the fundamental; in a three-limb core the triplen flux is forced out through the tank and heats it. Unbalanced single-phase loads on an unearthed-primary star–star unit also cause the neutral to shift. The remedy is a delta winding somewhere: as the main HV or LV winding, or as a **tertiary** winding (stabilising winding) of about one-third of the unit rating. This is why distribution transformers are Dyn and why star–star interbus units carry a delta tertiary. The zero-sequence equivalent circuits of each connection are developed in Chapter 19.

Phase shift and the per-phase model

For balanced studies a three-phase transformer is modelled per phase as a star–star equivalent with the per-phase impedance, plus a phase shift of the vector group. The phase shift is irrelevant within one voltage level, but it matters wherever two paths through different transformers meet: paralleling (Section 4.10), closing a ring between two substations fed from differently connected units, and setting transformer differential protection (Chapter 23), where the relay must compensate the 30° shift and remove the zero-sequence current that the delta winding blocks.

Worked example — Line currents through a Dyn11 unit. The 1000 kVA, 11 kV/433 V Dyn11 transformer at full load has LV line (and phase) current $1000000/(\sqrt{3}\times 433) = 1333$ A. The HV line current is $1000000/(\sqrt{3}\times 11000) = 52.5$ A, and the HV delta winding carries $52.5/\sqrt{3} = 30.3$ A. The check on the turns: winding current ratio $1333/30.3 = 44.0$, equal to the turns ratio $748/17 = 44.0$. An LV earth-fault current I_f flows in one LV winding and is balanced by $I_f/44.0$ in the corresponding HV delta winding, which appears in two HV lines in opposite directions, so no zero-sequence current flows at 11 kV; this is why an HV earth-fault relay cannot see an LV earth fault through a Dyn transformer.

4.12 Autotransformers

One winding, two voltages

An autotransformer has a single winding per phase with a tap: the lower voltage is taken between the tap and the common end. Part of the winding (the **common** winding) is shared by both circuits; the remainder (the **series** winding) carries only the HV current. For a ratio $a = V_H/V_L > 1$, power passes from one side to the other partly by conduction and partly by transformer action. The power actually transformed magnetically, which fixes the size of the core and windings, is

$$S_{\text{winding}} = S\left(1 - \frac{V_L}{V_H}\right) = kS$$

where $k = 1 - V_L/V_H$ is the **co-ratio**. For ratios close to 1 the saving is large.

> **Status of the principle: Autotransformer winding rating**
> *Derived*. Follows from the ideal-transformer rules applied to the series and common windings; it neglects magnetising current and losses, and holds only while the two circuits are galvanically connected.

Worked example — A 400/275 kV interbus autotransformer. A 500 MVA three-phase interbus unit connects a 400 kV and a 275 kV system. The co-ratio is $k = 1 - 275/400 = 0.3125$, so the windings are equivalent to a two-winding transformer of $0.3125 \times 500 = 156$ MVA. The HV line current at full load is $500 \times 10^6/(\sqrt{3} \times 400 \times 10^3) = 722$ A and the LV line current 1050 A; the series winding carries 722 A and the common winding the difference, $1050 - 722 = 328$ A. The unit is roughly a third of the size, weight and loss of a 500 MVA two-winding transformer, which is decisive when the transport limit on a road or rail route is 250–400 t.

Limitations

The price of the saving is galvanic connection between the two systems: an earth fault or a surge on the HV side reaches the LV side directly, so both systems must have compatible insulation levels and earthing (both solidly earthed, in practice). The per-unit short-circuit impedance on the throughput rating is roughly k times that of the equivalent two-winding transformer, so fault currents are higher. Autotransformers are therefore used where the ratio is below about 3 and both networks are effectively earthed: 400/132, 400/275, 500/230 and 345/138 kV interbus units; variable-voltage laboratory supplies (the "Variac"); autotransformer motor starters (Chapter 6), where a 65 % or 80 % tap reduces starting current; and 2×25 kV autotransformer traction feeding, where autotransformers every 10–15 km along an electrified railway halve the current in the overhead line and reduce interference in lineside cables.

4.13 Instrument transformers

Purpose and boundary

Relays and meters work with standard secondary values: 1 A or 5 A from **current transformers** (CTs) and 100–120 V (line-to-line) from **voltage transformers** (VTs). Instrument transformers scale down the primary quantities, isolate the low-voltage secondary circuits from the high-voltage system, and allow the secondary circuits to be earthed for safety. Their accuracy classes and the effect of phase displacement on power measurement at low power factor are given in *Circuits, Signals and Communications*, Ch. 9 §9.9; their application to protection schemes is developed in Chapter 23. This section explains how they differ from power transformers and where they fail.

Current transformers

A CT has its primary in series with the circuit, often a single turn (the busbar or cable passing through a ring core), and a secondary of N_2 turns feeding a low-impedance burden. It is a transformer operated close to short circuit: the primary current is imposed by the power system, and the secondary current is I_1/N_2 less the magnetising current needed to produce the flux that drives the secondary current through its burden. Errors therefore rise with the burden, and with the primary current when the core approaches saturation. Two consequences follow.

- **Never open-circuit a CT secondary under load.** With no secondary current to oppose it, the whole primary current becomes magnetising current; the core saturates every half-cycle and the rapid flux reversals induce peaks of several kilovolts across the open terminals, a shock and insulation hazard. CT secondaries are shorted by test blocks before relays or meters are removed.
- **Metering and protection CTs are different devices.** A metering CT (class 0.2S, 0.5 under IEC 61869-2) is designed to saturate early, to protect the meter during faults. A protection CT (class 5P or

10P, or PX) must reproduce currents up to its **accuracy-limit factor** (ALF) times rated current within 5 % or 10 % composite error, e.g. 5P20: 5 % error at 20 times rated current.

Worked example — Checking a protection CT. A 400/1 A, 5P20, 15 VA CT has a secondary winding resistance of 3.0 Ω. Its rated burden is $15/1^2 = 15\ \Omega$, so at the accuracy limit (20 A secondary) it must develop an EMF of $20 \times (3.0 + 15) = 360$ V without saturation. The actual burden is a digital relay (0.2 Ω) and 50 m of 2.5 mm² copper lead each way, $2 \times 0.050 \times 7.41 = 0.74\ \Omega$. At 20 times rated current the required EMF is only $20 \times (3.0 + 0.74 + 0.2) = 78.8$ V, well inside capability; the CT will reproduce symmetrical currents far beyond 20 times rated with the lighter burden. With a 25 kA fault the secondary current is $25000/400 = 62.5$ A, demanding $62.5 \times 3.94 = 246$ V, still inside the 360 V capability for the symmetrical current. The DC offset of a real fault can multiply the flux demand by $1 + X/R$, however (a factor of 15 for $X/R = 14$), so for fast protection on high-X/R systems the transient performance classes TPX, TPY and TPZ of IEC 61869-2 are used.

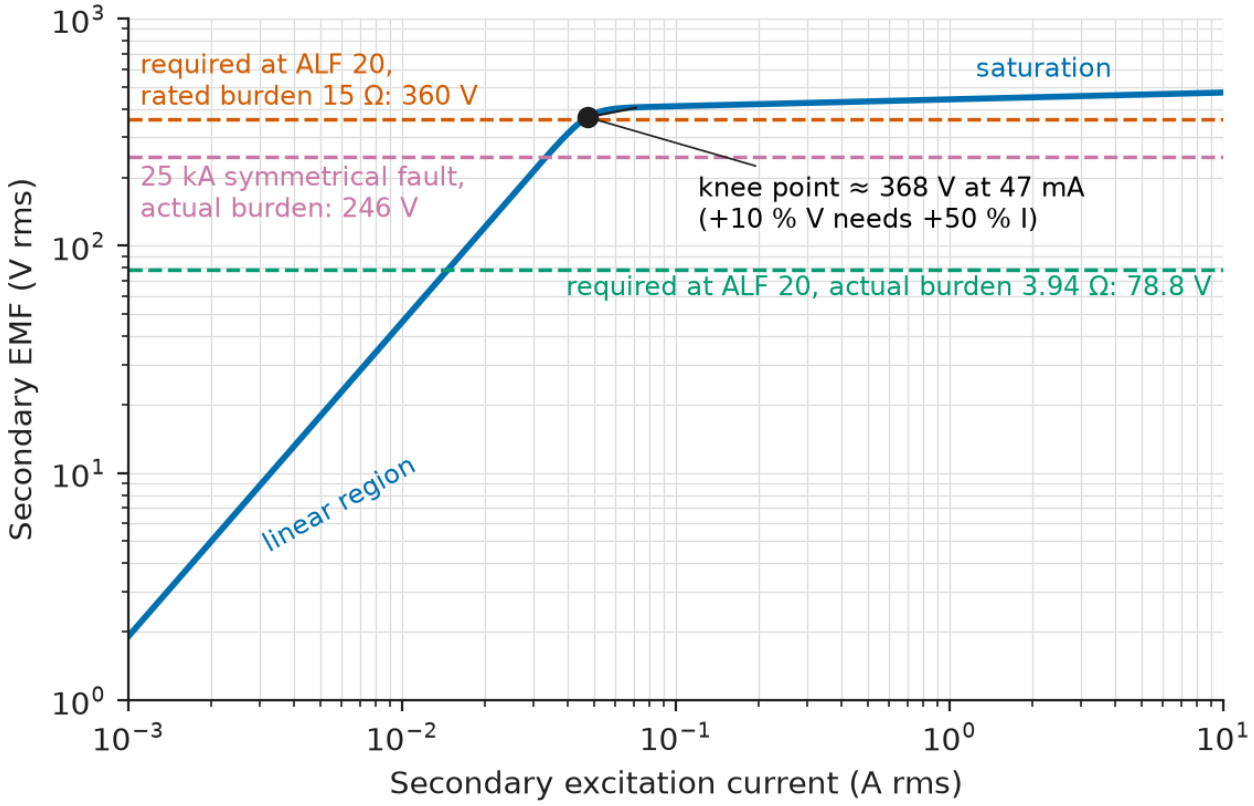


Figure 4.6 — Excitation characteristic of a 400/1 A 5P20 CT with the knee point and the EMF required for rated and actual burden at 20 times rated current.

Voltage transformers

An inductive VT is a small, accurate transformer operated near open circuit; its errors rise with burden. At transmission voltages above about 110 kV a **capacitor voltage transformer** (CVT) is cheaper: a capacitive divider steps the voltage down to 10–20 kV, and a small tuned inductive VT completes the transformation; the same capacitor stack doubles as a coupling capacitor for power-line carrier. CVTs have a transient response that can mislead fast distance relays during faults close to the relay (Chapter 23). Inductive VTs on unearthed or lightly loaded networks can fall into **ferroresonance** with system capacitance, a sustained non-linear oscillation that has destroyed VTs; damping resistors in the open-delta secondary are the usual remedy. In digital substations, low-power instrument transformers (Rogowski coils and dividers, IEC 61869-10 and -11) feed merging units on an IEC 61850 process bus.

Table 4.6 — Instrument transformer classes (IEC 61869).

Class	Device	Ratio error	Phase displacement	Typical use
0.2S	CT	±0.2 % (from 1 % to 120 % of rated current)	±10 minutes	Revenue metering at generator and grid connections
0.5	CT	±0.5 % at rated current	±30 minutes	Commercial and industrial metering
5P20	Protection CT	5 % composite error at 20 × rated	±60 minutes at rated	Overcurrent and earth-fault protection
PX (Class X)	Protection CT	Defined by knee-point voltage and winding resistance	—	Differential and distance protection
0.2	VT	±0.2 % (80–120 % voltage)	±10 minutes	Revenue metering
3P	VT	±3 % (5 % voltage to rated voltage factor)	±120 minutes	Protection

What remains open

Open question	Where it is answered
How is a transformer represented in per unit, with off-nominal taps, in a network model?	Chapter 17
How do tap changers and transformer reactive losses enter the load flow?	Chapter 18
What are the zero-sequence circuits of each vector group, and how do they shape earth-fault currents?	Chapter 19
How are distribution transformers sized and loaded on a feeder?	Chapter 22
How do differential protection, inrush restraint and CT saturation interact?	Chapter 23
How do windings respond to lightning and switching surges, and how is insulation tested?	Chapter 24
How do harmonic currents heat a transformer and how is it derated?	Chapter 25
How are cooling, insulation ageing and hot-spot temperature tied to the rating?	Chapter 9
How is a transformer selected for a building, plant or data centre?	Chapter 30

Next: Chapter 5 turns from the static transformer to the first rotating machine, the DC machine, in which the commutator makes a rotating winding behave like a stationary one and torque and speed follow from two simple linear relations.

Chapter 5 — DC Machines

Part II · Transformers and Electrical Machines

In an underground coal mine, a 1950s winder still hoists men and coal up a 600 m shaft with a DC motor of more than a megawatt, fed from a thyristor converter that replaced its original motor-generator set. The mine's engineers have been offered an AC drive many times, and they keep the DC machine for reasons a datasheet does not show: full torque from standstill, smooth control down to a crawl, regenerative braking of a descending cage with no extra hardware, and fifty years of maintenance records. At the other end of the size range, every car on the road still starts its engine with a 1–2 kW DC series motor, and millions of small permanent-magnet DC motors drive seats, wipers, pumps and fans. The DC machine has lost most of its industrial market to induction and permanent-magnet machines with power electronics, but it remains the clearest example of electromechanical conversion: two linear equations, $E = k\Phi\omega$ and $T = k\Phi I_a$, describe it completely, and every AC drive in Part III is designed to make an AC machine behave like it.

> **How does a commutator turn the alternating EMF of a rotating coil into DC, and what fixes the EMF and torque of a real machine?**
> **How do the field connections (separate, shunt, series, compound) shape a machine's characteristics, and how is it started, controlled and braked without destroying its commutator?**

Concept	Engineering destination
Construction	Inspecting commutators and brushes; replacing legacy machines in mills, mines and traction
Armature windings	Current and voltage ratings; commutation limits; armature reaction
EMF and torque equations	Motor constants K_e, K_t; drive design (Chapter 11); PMDC servo selection
Separately excited machines	Rolling-mill, hoist and paper-machine drives; test-bed dynamometers
Shunt and series machines	Self-excited generators; traction, cranes and engine starters
Compound machines	Legacy generators with flat voltage; elevator and press drives
Characteristics	Matching a motor to its load (Chapter 10)
Speed control	Armature-voltage and field-weakening regions; constant-torque and constant-power operation
Starting	Starting-resistor design; current limits for commutation
Braking	Dynamic, plugging and regenerative braking of hoists and vehicles
Losses and efficiency	Efficiency testing to IEC 60034-2-1; brush and commutator maintenance

Roadmap. How the machine is built (5.1) → how the armature winding and commutator rectify the EMF (5.2) → the EMF (5.3) and torque (5.4) equations → separately excited (5.5), shunt and series (5.6) and compound (5.7) machines → their characteristics side by side (5.8) → speed control (5.9) → starting (5.10) → braking (5.11) → losses and efficiency (5.12).

5.1 DC machine construction

Stator and rotor

A conventional DC machine (Figure 5.1) has a stationary field system and a rotating armature.

- The **yoke** is a steel frame that carries the flux between poles and supports the machine. In machines fed from thyristor or chopper converters, where the current ripples, the yoke is laminated to limit eddy currents.
- The **main poles**, bolted to the yoke, carry the **field windings** (or, in small machines, permanent magnets). Pole shoes spread the flux over the air gap, which is 1–5 mm in industrial machines.
- **Interpoles** (commutating poles), narrow poles between the main poles, carry a winding in series with the armature; they cancel the armature reaction in the commutating zone and induce the voltage that reverses the current in the coils being commutated (Section 5.2).
- **Compensating windings**, set in slots in the main pole faces and also in series with the armature, cancel armature reaction under the poles in large machines and in those that see rapid current changes (rolling mills, hoists).
- The **armature** is a laminated cylinder of non-oriented silicon steel with slots that carry the armature winding. It must be laminated because its iron sees a reversing flux at the frequency $f = pn/120$ as it turns under alternate poles.
- The **commutator** is a cylinder of hard-drawn copper segments insulated from each other by mica, with each segment connected to the junction of two armature coils. **Brushes** of carbon or electrographite, held in brush holders at a set pressure, slide on it and carry the armature current to the external circuit.

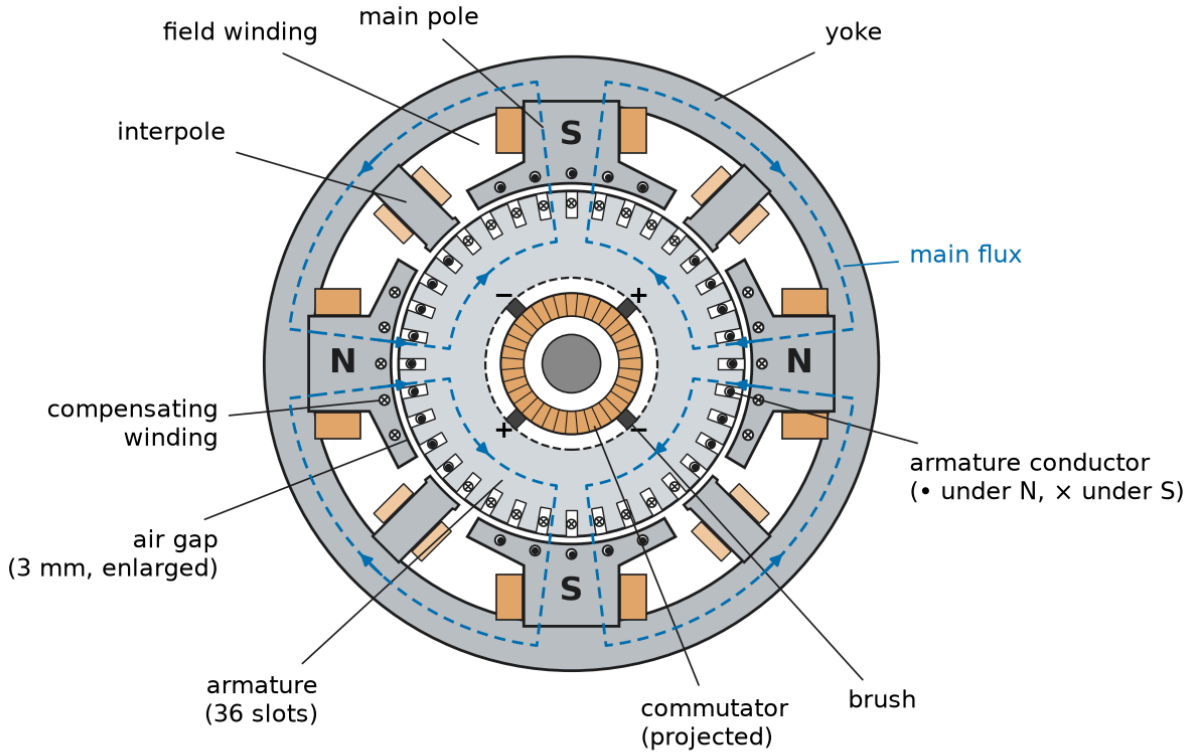


Figure 5.1 — Cross-section of a four-pole DC machine with interpoles and compensating windings, showing the main flux paths and the brush positions.

Where DC machines are still used

Table 5.1 — Representative DC machine applications and ratings.

Application	Type	Typical rating	Status
Automotive engine starter	Series (or PM) motor, 12/24 V	0.8–3 kW, short-time duty	Universal in internal-combustion vehicles
Seat, wiper, pump and fan actuators	Permanent-magnet DC (PMDC)	5–300 W	Very large volumes; moving to brushless
Mine winders, hoists	Separately excited, compensated	0.5–5 MW, 400–1000 V	Legacy, gradually replaced by AC drives
Rolling mills, paper machines	Separately excited, compensated	0.5–10 MW	Legacy, many still in service
Mining haul trucks, older locomotives	Series traction motors	300–600 kW per motor	Older fleets; new designs AC
Forklifts and golf carts	Series or separately excited, 24–80 V	2–20 kW	Moving to AC induction and PM
Electroplating, electrolysis supply (historic)	DC generators	up to several MW	Replaced by rectifiers
Laboratory dynamometers, exciters	Separately excited; DC exciters on old alternators	1 kW–1 MW	Legacy

The commutator limits the DC machine: it needs brush replacement and commutator maintenance, it cannot be used in flammable or dusty atmospheres without enclosure, it limits peripheral speed (about 40–50 m/s) and the voltage between adjacent segments (about 15–20 V average), and its sparking is a source of interference. These limits are why the induction machine (Chapter 6) and the permanent-magnet machine (Chapter 8) took over once power electronics could provide variable frequency.

5.2 Armature windings

Why a commutator is needed

A coil rotating in the field of the poles has an alternating EMF, as Chapter 2 showed for motional EMF. The commutator is a mechanical rectifier: as a coil passes from under a north pole to under a south pole, its ends move from one brush to the other, so the EMF it contributes to the external circuit always has the same sign. A single coil gives a rectified sine; a real armature has dozens to hundreds of coils, displaced around the periphery and connected in series between brushes, so the summed EMF has a ripple of well under 1 %.

Lap and wave windings

The armature coils are connected to the commutator in one of two patterns (Figure 5.2).

- In a **lap winding**, each coil's ends go to adjacent commutator segments, and the winding "laps back" on itself under each pair of poles. A simplex lap winding has as many parallel paths between the brushes as there are poles: $a = p$. It suits low-voltage, high-current machines, because each path carries only I_a/p. Paths under different poles can carry unequal currents if the air gaps are unequal, so large lap windings carry **equalising connections** joining points that should be at equal potential.

- In a **wave winding**, each coil's ends go to segments about two pole pitches apart, so the winding passes under every pole before returning near its starting segment. A simplex wave winding has only two parallel paths whatever the number of poles: $a = 2$. It suits higher-voltage, lower-current machines and needs only two brush sets (though all are usually fitted).

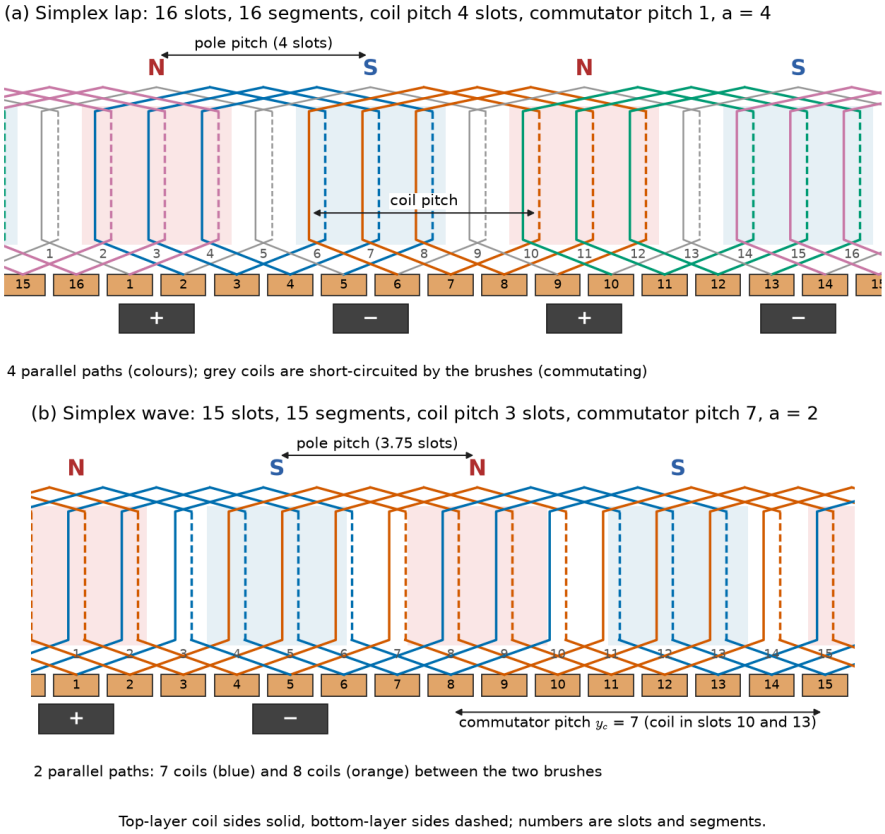


Figure 5.2 — Developed (unrolled) diagrams of a simplex lap and a simplex wave winding for a four-pole armature, with commutator segments and brushes.

Commutation

During the brief time a coil is short-circuited by a brush (about 1 ms in an industrial machine), its current must reverse from $+I_a/a$ to $-I_a/a$. The coil's self- and mutual inductance resists this change with a **reactance voltage** $L\,\Delta i/\Delta t$, and if the reversal is incomplete when the segment leaves the brush, the remaining current is broken as a spark. Sparking erodes the commutator and brushes; severe sparking can grow into a **flashover** around the commutator, a short circuit between brush arms that can wreck the machine. Three measures give good commutation: interpoles, whose field induces a voltage in the commutating coil that cancels the reactance voltage (proportional to I_a because the interpole winding carries armature current); carbon brushes, whose contact resistance helps force the current transfer (resistance commutation); and limits on the rate of change of armature current, which are why DC drives impose current-rise limits of a few times rated current per second.

Armature reaction

The armature current produces its own MMF, directed along the brush axis, at right angles to the main field. This **armature reaction** distorts the gap field, strengthening one half of each pole and weakening the other. Because the strengthened half saturates, the net flux per pole is reduced (a demagnetising effect of a few per cent at full load), and the magnetic neutral axis shifts, moving the zero-field point away from the brushes and

worsening commutation. In small machines without interpoles the brushes were once shifted to the new neutral, which works for only one load and one direction. Interpoles fix the commutating zone, and compensating windings cancel the cross field under the poles, which also keeps the segment-to-segment voltage even and prevents flashover during load transients.

5.3 EMF generation

The EMF equation

> **Boundary.**
> Motional EMF and the force on a current-carrying conductor follow from *Engineering Physics*, Ch. 11 §11.4; the motor/generator as a gyrator with $K_e = K_t$ is *Engineering Physics*, Ch. 17 §17.5.3; the DC-motor transfer function and its time constants are *Engineering Physics*, Ch. 21 §21.8.1. Rotational mechanics ($J\,d\omega/dt = T - T_L$) is *Engineering Physics*, Ch. 19 §19.2, and force from co-energy is Chapter 3. This chapter adds the winding constants, the field connections and the steady-state engineering of real machines.

What a real machine adds is the winding constant. Each of the Z armature conductors, moving at speed n rpm through a field that delivers flux Φ per pole, cuts $p\Phi$ webers per revolution, so its average EMF is $p\Phi n/60$. Between the brushes, Z/a conductors are in series. The armature EMF is therefore

$$E_a = \frac{p\,\Phi\,Z\,n}{60\,a} = k\,\Phi\,\omega, \qquad k = \frac{pZ}{2\pi a}$$

with $\omega = 2\pi n/60$ in rad/s. For a machine with constant field, $k\Phi$ is a single number, the **EMF constant** K_e in V·s/rad, which a datasheet for a permanent-magnet motor often gives in V per 1000 rpm.

> **Status of the principle: DC machine EMF equation**
> *Derived.* Follows from motional EMF summed over the conductors in series between brushes; exact for the average EMF, with Φ the net flux per pole including armature-reaction and saturation effects, which is where it becomes approximate in practice.

Worked example — EMF of a four-pole armature. A four-pole armature has 720 conductors and runs at 1200 rpm with a flux of 25 mWb per pole. Lap-wound ($a = 4$):

$$E_a = \frac{4 \times 0.025 \times 720 \times 1200}{60 \times 4} = 360\ \text{V}$$

Wave-wound ($a = 2$) the same armature gives 720 V, with half the current capacity, since each of the two paths must carry $I_a/2$ instead of $I_a/4$. The winding constant is $k = pZ/(2\pi a) = 114.6$ for the lap winding, so $k\Phi = 2.865$ V·s/rad.

The magnetisation curve

E_a is proportional to Φ, and Φ is a non-linear function of the field current through the magnetic circuit of yoke, poles, air gap, teeth and armature core (Chapter 1). The **open-circuit characteristic** (OCC, or magnetisation curve), measured by driving the machine at constant speed and recording E_a against field current I_f, captures this (Figure 5.3). It starts at a small **residual voltage** (2–5 % of rated) from the

remanence of the poles, rises linearly while the air gap dominates the reluctance, and bends over as the iron saturates. Machines are designed to operate just past the knee, where the field current needed is reasonable and the voltage is insensitive to small field changes.

5.4 Torque production

The torque equation

The electromagnetic torque follows from the power balance: the electrical power converted, $E_a I_a$, equals the mechanical power $T\omega$. Hence

$$T = k\,\Phi\,I_a = \frac{pZ}{2\pi a}\,\Phi\,I_a$$

The **torque constant** $K_t = k\Phi$ in N·m/A is numerically equal to the EMF constant K_e in V·s/rad: this is the gyrator identity of *Engineering Physics* Ch. 17 §17.5.3 with the winding constant made explicit.

> **Status of the principle: DC machine torque equation**
> *Derived.* Follows from the electromechanical energy balance with $E_a = k\Phi\omega$; it gives the electromagnetic (air-gap) torque, from which friction, windage and core-loss torques must be subtracted to get shaft torque.

Worked example — Torque of the four-pole armature. With the lap-wound armature of Section 5.3 carrying $I_a = 100$ A, $T = 2.865 \times 100 = 286.5$ N·m. Check through power: $E_a I_a = 360 \times 100 = 36.0$ kW, and at 1200 rpm ($\omega = 125.7$ rad/s), $36000/125.7 = 286.5$ N·m.

The two equations together

Combined with Kirchhoff's voltage law on the armature, $V_t = E_a \pm I_a R_a$ (plus for a motor, minus for a generator, with a brush drop of about 2 V for carbon brushes added to the $I_a R_a$ drop in more careful work), the two equations describe all steady-state DC machine behaviour. For a motor with constant flux, eliminating E_a and I_a gives the speed–torque relation

$$\omega = \frac{V_t}{k\Phi} - \frac{R_a}{(k\Phi)^2}\,T$$

a straight line drooping from the no-load speed $V_t/k\Phi$ with a slope set by R_a. Every speed-control method in Section 5.9 changes one of the three quantities in this equation: V_t, Φ or the armature-circuit resistance.

Table 5.2 — Typical per-unit parameters of industrial DC machines (on rated armature voltage and current).

Rating	Armature-circuit resistance R_a	Full-load speed drop (constant field)	Efficiency at rated load	Field power (% of rating)
1–10 kW	0.06–0.10 pu	5–10 %	80–87 %	3–5 %
10–100 kW	0.03–0.06 pu	3–6 %	87–92 %	1.5–3 %
100 kW–1 MW	0.015–0.03 pu	1.5–3 %	92–95 %	1–1.5 %
above 1 MW	0.008–0.015 pu	about 1 %	95–97 %	0.5–1 %

5.5 Separately excited machines

Connection and equations

In a **separately excited** machine the field winding is supplied from an independent source, today almost always a small controlled rectifier. The armature and field equations are uncoupled:

$$V_f = R_f I_f, \qquad \Phi = \Phi(I_f) \text{ from the OCC}, \qquad V_t = E_a + I_a R_a \ (\text{motor})$$

so the flux can be set independently of the load. This is the connection used for every large industrial DC drive: rolling mills, paper machines, mine winders, test-bed dynamometers. With a permanent-magnet field the machine is a PMDC motor, which behaves as a separately excited machine with fixed flux.

As a generator

Driven at constant speed with constant field, a separately excited generator's terminal voltage falls with load by the armature resistance drop and the armature-reaction loss of flux:

$$V_t = E_a(I_f, \omega) - I_a R_a - \Delta E_{ar}(I_a)$$

The fall from no load to full load is typically 5–10 % (Figure 5.4). The historical **Ward–Leonard** system used an AC motor driving a separately excited generator whose output fed the armature of a separately excited motor; controlling the generator field gave smooth, reversible, regenerative armature-voltage control. The thyristor converter replaced the generator from the 1960s onwards, and Chapter 11 develops the converter-fed drive.

As a motor

With constant field and supply, a separately excited motor has the drooping straight-line speed–torque characteristic of Section 5.4, with speed falling 1–10 % from no load to full load depending on size (Table 5.2). This stiff characteristic, independent of load, and the ability to control speed through V_t and Φ separately, are what made it the standard adjustable-speed drive before vector-controlled AC drives.

5.6 Shunt and series machines

Shunt connection

In a **shunt** machine the field winding is connected in parallel with the armature, across the terminals. The field has many turns of fine wire and draws 1–5 % of rated current. With a constant supply voltage, a shunt motor is indistinguishable from a separately excited one: $I_f = V_t / R_f$ is constant, and the speed falls slightly with load.

Worked example — Speed regulation of a shunt motor. A 220 V shunt motor has $R_a = 0.25\ \Omega$ and $R_f = 110\ \Omega$. At no load it draws 5 A and runs at 1000 rpm; at full load it draws 60 A. Neglecting armature reaction, find the full-load speed and torque.

The field current is $220/110 = 2.0$ A, so the armature current is 3 A at no load and 58 A at full load. The back EMFs are $E_0 = 220 - 3 \times 0.25 = 219.25$ V and $E_{fl} = 220 - 58 \times 0.25 = 205.5$ V. With constant flux, speed is proportional to back EMF:

$$n_{fl} = 1000 \times \frac{205.5}{219.25} = 937.3 \text{ rpm}$$

a speed regulation of $(1000 - 937.3)/937.3 = 6.7$ %. The developed power is $205.5 \times 58 = 11.92$ kW and the developed torque $11919/(937.3 \times 2\pi/60) = 121.4$ N·m. Armature reaction would weaken the flux slightly at full load and raise the speed a little, which in practice often makes the speed regulation smaller than this estimate. If it raises the speed above no-load speed, the characteristic rises with load and the drive can become unstable, which is one reason for a light series "stabilising" winding on shunt motors.

Self-excitation of a shunt generator

A shunt generator supplies its own field. Driven at speed, the residual voltage drives a small field current; if that current increases the flux (correct field polarity), the voltage rises, increasing the field current further, until the operating point settles where the OCC crosses the **field-resistance line** $V = R_{f,total} I_f$ (Figure 5.3). Three conditions must hold for the voltage to build up:

1. There must be residual magnetism. A machine that has lost it is "flashed" with a battery across the field.
2. The field connection must aid the residual flux; reversed, the voltage collapses to zero.
3. The field-circuit resistance must be below the **critical field resistance**, the slope of the linear part of the OCC. At a given resistance there is likewise a critical speed below which the machine will not build up.

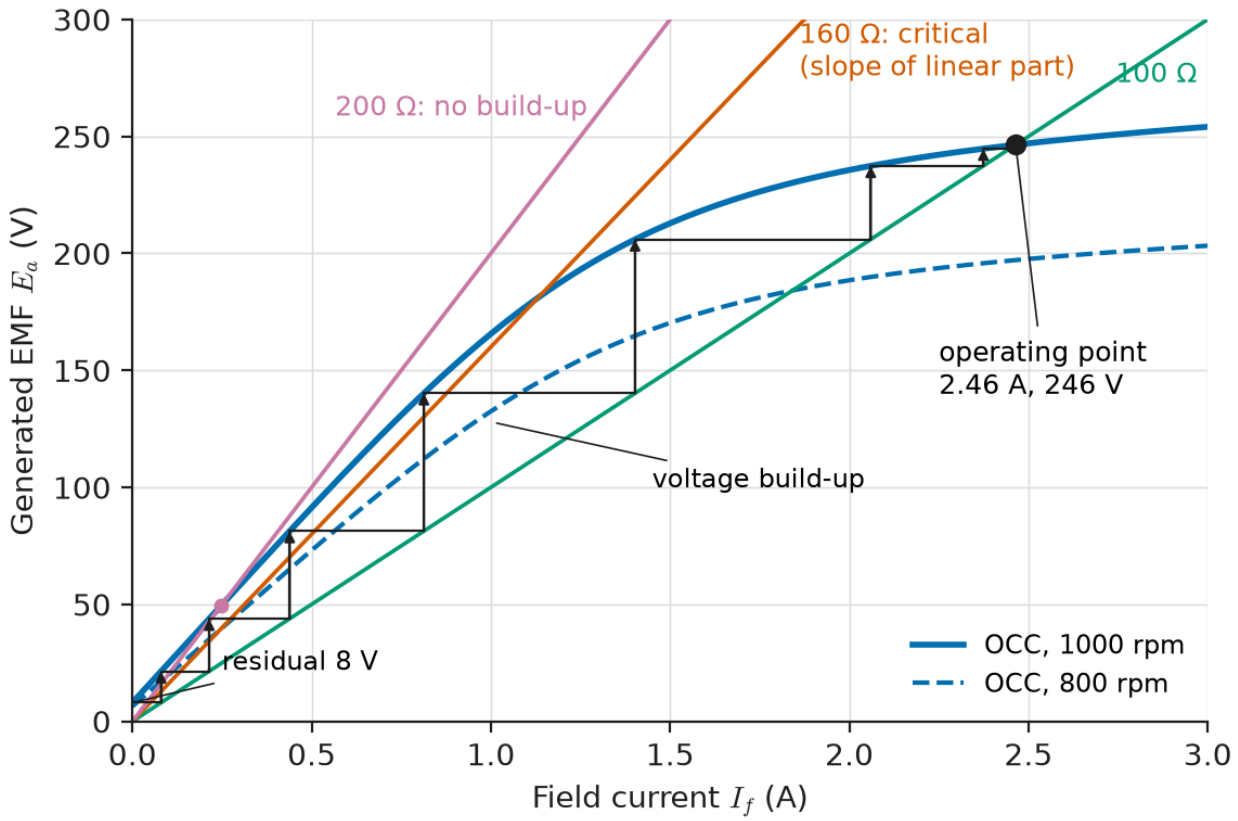


Figure 5.3 — Open-circuit characteristic of a 220 V shunt generator at 1000 rpm with field-resistance lines showing voltage build-up and the critical field resistance.

A shunt generator's terminal voltage falls more steeply with load than a separately excited one, because the falling terminal voltage also reduces the field current. Beyond a certain current, the characteristic turns back on itself: a short circuit collapses the voltage and the current settles at a value set by the residual voltage alone. This made the shunt generator self-protecting against short circuits but unsuitable as a stiff supply without a voltage regulator.

Series connection

In a **series** machine the field winding carries the armature current. It has a few turns of heavy conductor, with resistance comparable to that of the armature. Below saturation the flux is proportional to the current, $\Phi \approx c I_a$, so

$$T \approx k c\, I_a^2, \qquad \omega \approx \frac{V_t - I_a (R_a + R_{se})}{k c\, I_a}$$

The torque rises with the square of current, and the speed falls steeply as load increases and rises steeply as it decreases. The series motor is a natural traction motor: it gives very high starting torque from a limited current, and at high speed it automatically reduces its torque, delivering an approximately constant power over a wide range. Its danger is that at light load the speed rises without limit: a series motor must never be started or run uncoupled from its load, and series motors driving belts or ropes are fitted with overspeed protection.

Worked example — A series traction motor under changing load. A 600 V series motor of a mine locomotive has a total armature-plus-field resistance of 0.3 Ω and runs at 800 rpm drawing 100 A. The locomotive then meets a gradient that doubles the load torque. Assuming an unsaturated magnetic circuit ($\Phi \propto I_a$):

Torque doubles when the current rises by $\sqrt{2}$, to 141.4 A. The back EMFs are $600 - 100 \times 0.3 = 570$ V and $600 - 141.4 \times 0.3 = 557.6$ V. Since $n \propto E_a/\Phi \propto E_a/I_a$,

$$n_2 = 800 \times \frac{557.6}{570} \times \frac{100}{141.4} = 553 \text{ rpm}$$

The developed power rises only from 57.0 kW to 78.9 kW (38 %) while the torque doubles (680 N·m to 1361 N·m). A shunt motor in the same situation would hold speed and double its power and current. If instead the load falls until the motor draws 30 A, the same equations give 2765 rpm, an overspeed of nearly 3.5 times; in practice friction and windage limit it, but the example shows why a series motor is never run light.

5.7 Compound machines

Cumulative and differential compounding

A **compound** machine has both a shunt field and a series field on each main pole. In a **cumulative** compound machine the series field aids the shunt field; in a **differential** compound machine it opposes it. The shunt field may be connected across the armature alone (**short shunt**) or across the armature and series field (**long shunt**); the difference in performance is small.

Compound generators

In a cumulative compound generator, the series field adds flux in proportion to load current, compensating the armature resistance drop and armature reaction. By choosing the series turns (often with a diverter resistor across the series field to trim them), the designer can make the generator:

- **flat-compounded**: same terminal voltage at no load and full load;
- **over-compounded**: terminal voltage rising with load, to compensate the drop in the feeder to a distant load (historically used for DC tramway and mine feeders);
- **under-compounded**: voltage falling somewhat with load.

A differential compound generator has a steeply drooping characteristic, which gives an approximately constant current: it was used for arc-welding generators.

Compound motors

A cumulative compound motor combines a definite no-load speed (from the shunt field) with a strong torque rise under load (from the series field). Its speed falls more than a shunt motor's under load, which suits loads with heavy torque peaks and a flywheel, such as presses, shears and reciprocating compressors: the speed drop lets the flywheel deliver its stored energy and smooths the current drawn from the supply. Differential compound motors are unstable under heavy load (the series field can overpower the shunt field and reverse the torque) and are not used.

5.8 Characteristics

Generator external characteristics

The **external characteristic** of a generator is its terminal voltage against load current at constant speed and constant field-circuit setting. Figure 5.4 compares the connections for machines with the same no-load voltage.

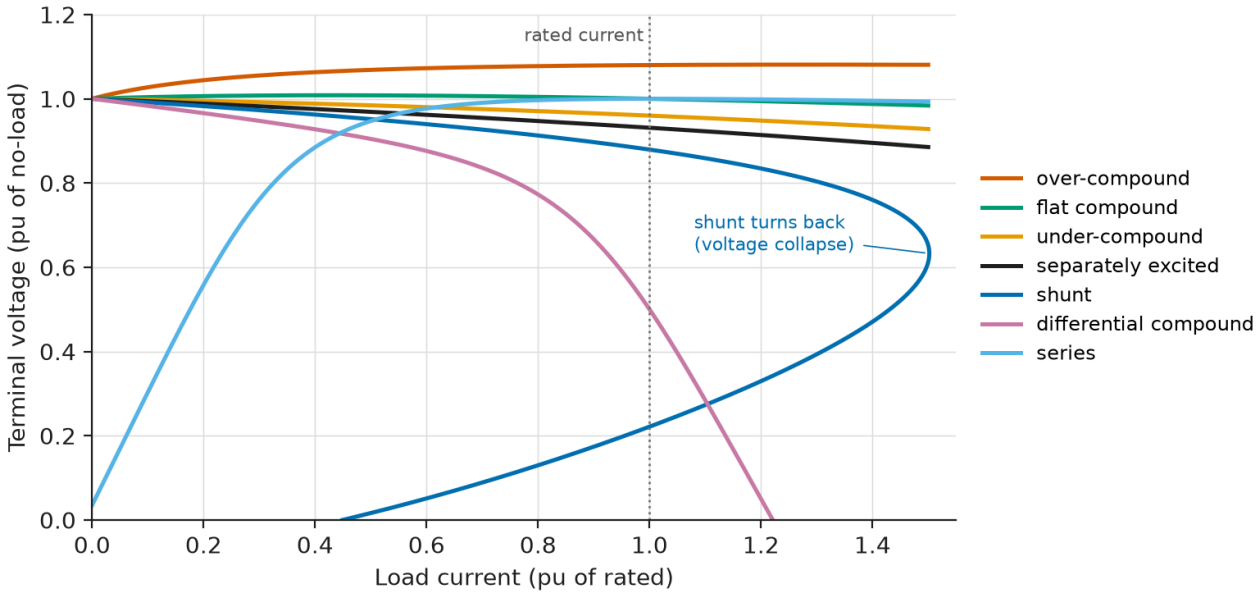


Figure 5.4 — External characteristics of DC generators at constant speed: separately excited, shunt, over-, flat- and under-compounded, differential compound and series.

Motor speed–torque characteristics

For motors the useful characteristic is speed against torque at rated supply voltage (Figure 5.5):

- **Separately excited and shunt**: nearly constant speed, falling 2–10 % to full load. Suits fans, machine tools, conveyors, and any load that needs a defined speed.
- **Series**: very high speed at light load, high torque at low speed; approximately $T \propto 1/\omega^2$ in the unsaturated range. Suits traction, cranes, hoists and engine starters, where high breakaway torque matters and the load is always connected.
- **Cumulative compound**: between the two, with a definite no-load speed.

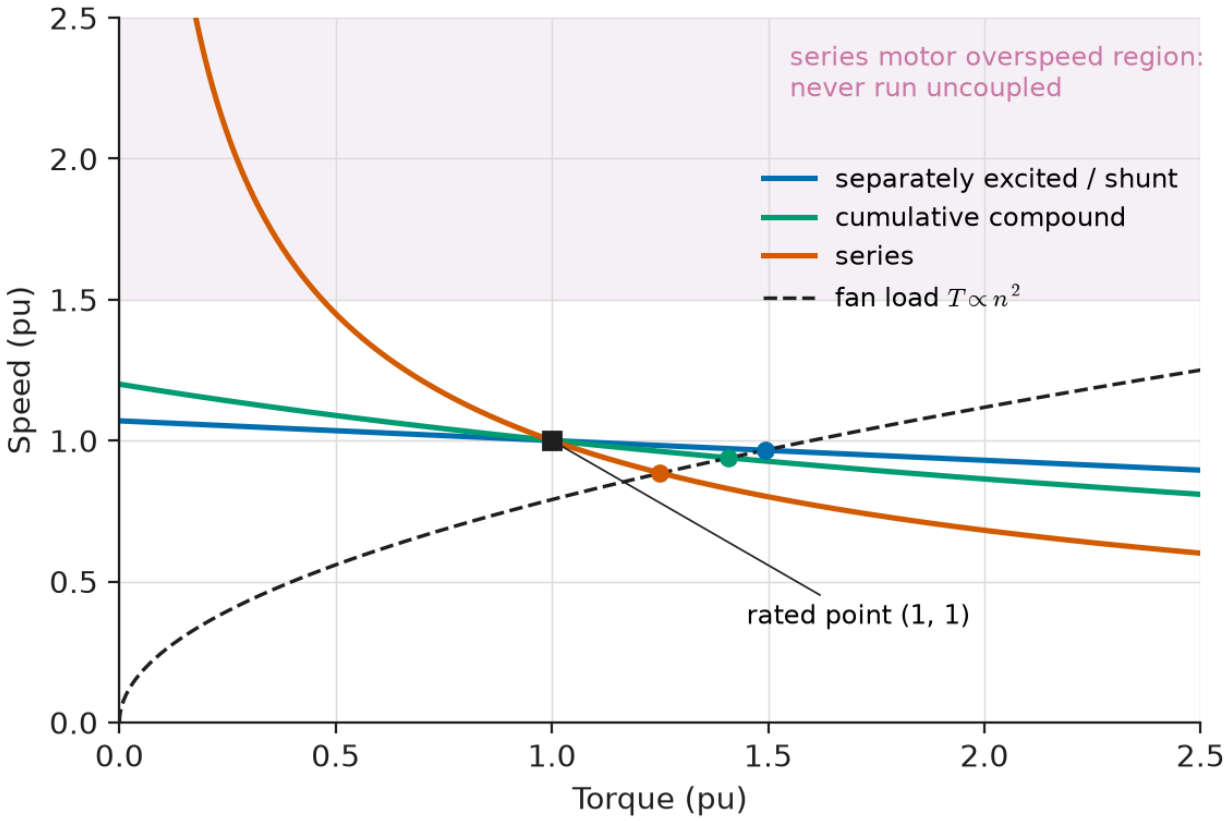


Figure 5.5 — Speed against torque at rated voltage for separately excited (or shunt), cumulative compound and series DC motors of equal rating.

The intersection of a motor's characteristic with the load's torque–speed curve sets the operating point; whether that point is stable depends on the relative slopes, a question developed with load characteristics in Chapter 10.

Table 5.3 — Comparison of DC motor connections.

Property	Separately excited / shunt	Series	Cumulative compound
Speed regulation (no load to full load)	2–10 %	50–100 %+ (very soft)	10–25 %
Starting torque at 2 × rated current	about 2 × rated	about 3–4 × rated (below saturation)	2.5–3 × rated
No-load speed	Defined	Dangerously high	Defined
Speed control	Excellent: armature voltage and field	Series resistance, field diverter, series–parallel	Armature voltage, shunt field
Typical loads	Mills, machine tools, winders, paper	Traction, cranes, starters	Presses, shears, compressors, elevators (historic)

5.9 Speed control

Three control quantities

From $\omega = (V_t - I_a R_a)/k\Phi$, speed can be changed by the armature voltage, the field flux or the armature-circuit resistance.

Armature-voltage control. Reducing V_t at full field lowers the no-load speed in proportion while keeping the slope of the characteristic: a family of parallel lines. The motor can deliver rated torque (rated I_a at full flux) at any speed from zero to base speed, so this is the **constant-torque region**. It is efficient because no power is dissipated in series resistors. The variable armature voltage came first from Ward–Leonard generators and now from thyristor converters or choppers (Chapter 11).

Field weakening. Reducing I_f above base speed (at full armature voltage) reduces Φ and raises speed. At rated armature current the torque falls in proportion to Φ while speed rises in inverse proportion, so the power stays roughly constant: the **constant-power region**. The range is limited by commutation (weaker main field means relatively stronger armature reaction) and mechanical speed limits to typically 2:1 to 4:1 above base speed. Field weakening has a long time constant (the field inductance is large, L_f/R_f of 0.5–3 s), so it is used for slow speed changes.

Armature-resistance control. Inserting resistance in the armature circuit steepens the speed–torque line and reduces speed at a given torque. It is simple but wasteful: at half speed with rated torque, about half the input power is burned in the resistor. It survives in small series-motor drives (older forklifts, cranes) and as a starting method (Section 5.10).

> **Status of the principle: Constant-torque and constant-power regions**
> *Approximate.* Follows from the EMF and torque equations with armature current held at its thermal limit; it assumes linear magnetics and ignores the reduced cooling of self-ventilated motors at low speed, which in practice requires a derating or a separately driven fan below about 50 % speed.

Worked example — Armature-voltage and field control of a mill motor. A 440 V, 150 kW separately excited mill motor has $R_a = 0.04\ \Omega$, rated armature current 370 A and base speed 600 rpm.

At base speed, $E_a = 440 - 370 \times 0.04 = 425.2$ V and $\omega = 62.83$ rad/s, so $k\Phi = 425.2/62.83 = 6.767$ V·s/rad and the rated electromagnetic torque is $6.767 \times 370 = 2504$ N·m (developed power 157.3 kW; the 150 kW rating is the shaft output after rotational losses).

Armature-voltage control. At $V_t = 220$ V and rated torque (still 370 A), $E_a = 220 - 14.8 = 205.2$ V and $\omega = 205.2/6.767 = 30.32$ rad/s, i.e. 290 rpm. Note that halving the voltage gives a little less than half the speed, because the $I_a R_a$ drop is a constant 14.8 V.

Field weakening. To run at 1000 rpm (104.7 rad/s) with 440 V and rated armature current, $k\Phi$ must fall to $425.2/104.7 = 4.060$ V·s/rad, i.e. 60 % of its full-field value. The torque available at rated current is $4.060 \times 370 = 1502$ N·m, and the developed power is still 157.3 kW: constant power. Figure 5.6 shows the resulting capability curve.

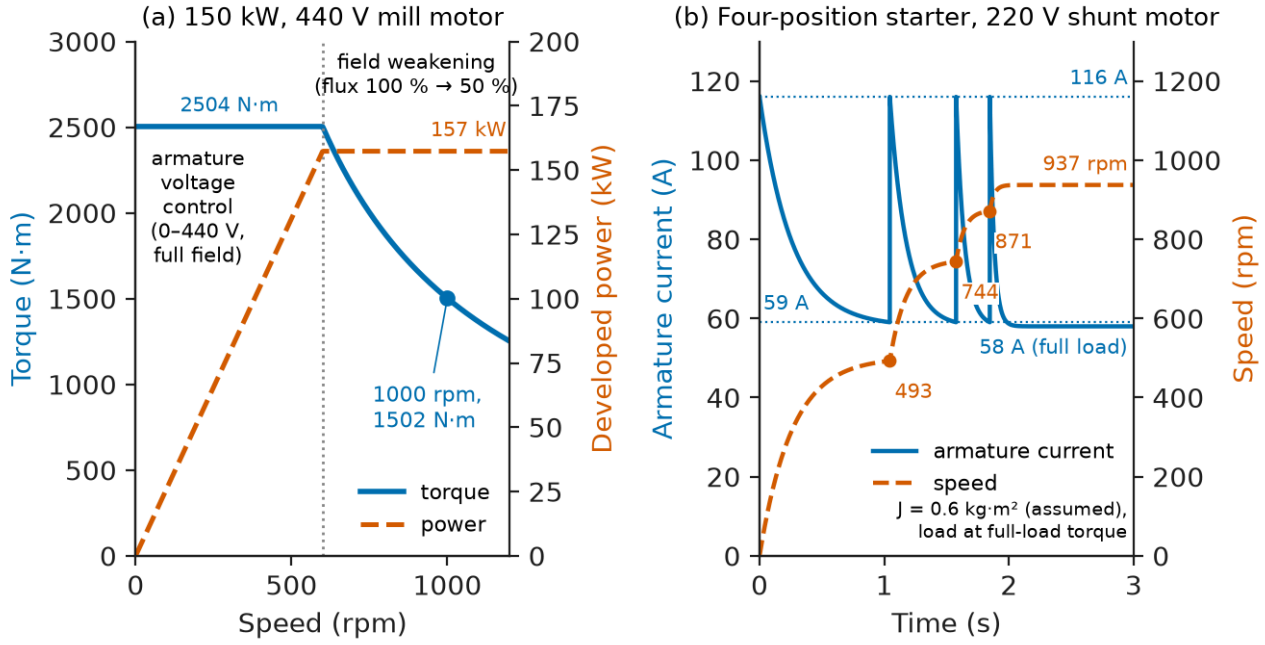


Figure 5.6 — Capability of the 150 kW, 440 V mill motor under armature-voltage and field control, and the armature current and speed during resistor starting of the 220 V shunt motor.

Series-motor control

Series motors are controlled by series resistance, by a **field diverter** (a resistor in parallel with the series field, which weakens the field and raises speed) and, in traction, by **series–parallel control**: two motors are started in series, each seeing half the line voltage, and switched to parallel at about half speed, halving the resistor losses during acceleration. Choppers replaced resistor control on DC rapid-transit vehicles from the 1970s, saving 20–30 % of traction energy on stop–start services.

5.10 Starting

Why a starter is needed

At standstill the back EMF is zero, so on full voltage the armature current is limited only by R_a: $I_{st} = V_t / R_a$. With R_a of 0.02–0.1 pu, this is 10–50 times rated current, which would flash over the commutator, trip the supply and shock the mechanical drive train. The commutator limits the safe current to 1.5–2.5 times rated during starting. A DC motor is therefore started either with reduced armature voltage (a converter that ramps the voltage, the method used by every modern drive) or with series resistance cut out in steps as the motor accelerates and the back EMF builds up.

Designing a resistor starter

With a step starter, the current is allowed to swing between an upper limit I_1 and a lower limit I_2. At each switching instant the speed, and hence the back EMF, is unchanged, so the resistances of successive steps satisfy $R_{k+1}/R_k = I_2/I_1$. Starting from $R_1 = V_t/I_1$ and ending at R_a, the number of steps m satisfies

$$\left(\frac{I_1}{I_2}\right)^{m} = \frac{R_1}{R_a}$$

Worked example — A four-position starter. For the 220 V shunt motor of Section 5.6 ($R_a = 0.25\ \Omega$, full-load armature current 58 A), the starting current is limited to 2 × full load, $I_1 = 116$ A. Then $R_1 = 220/116 = 1.897\ \Omega$ and $R_1/R_a = 7.59$. With I_2 at about full-load current, $I_1/I_2 = 2$ and $m = \ln 7.59 / \ln 2 = 2.92$, so three resistor sections are needed. Adjusting the ratio so that exactly three steps fit, $I_1/I_2 = 7.59^{1/3} = 1.965$, and the lower switching current is $I_2 = 116/1.965 = 59.0$ A. The total armature-circuit resistances on the successive positions are 1.897, 0.965, 0.491 and 0.25 Ω, so the three sections are 0.932, 0.474 and 0.241 Ω. The switching instants occur when the back EMF reaches $220 - 59.0 \times R_k$: 108.0 V, 163.0 V and 191.0 V, i.e. 493, 744 and 871 rpm on the motor's 219.25 V-per-1000-rpm characteristic (Figure 5.6b).

Manual faceplate starters include a **no-volt release** (a holding coil that returns the arm to "off" if the supply fails, so that the motor cannot restart on full voltage) and an **overload release**. A shunt motor must never lose its field while running: with Φ reduced to the residual value, the speed tries to rise without limit and the armature current becomes very large. Field-failure (field-loss) relays trip DC drives on loss of field current for this reason.

5.11 Braking

Three ways to brake electrically

Electrical braking makes the machine act as a generator, converting the kinetic or potential energy of the load into electrical energy. It is used to stop quickly, to hold speed on a descending load (hoists, haul trucks on downgrades, trains on gradients) and to recover energy.

Dynamic (rheostatic) braking. The armature is disconnected from the supply and connected to a braking resistor, with the field kept energised. The machine generates, driving current through the resistor and producing a braking torque $k\Phi I_a$. As the speed falls, so do the EMF, the current and the torque, so dynamic braking alone does not hold a load at standstill. Mining haul trucks of 200–400 t use dynamic braking (with large blower-cooled resistor grids rated at several megawatts) to descend ramps without wearing out friction brakes.

Plugging (reverse-current braking). The armature connections are reversed while the motor runs, so the supply voltage and back EMF add. A large resistor must be inserted to limit current. The braking torque remains even at zero speed, so the motor will reverse unless disconnected there; plugging is used where fast reversal is the goal (rolling-mill reversing tables, crane travel) and is energy-wasteful: both the kinetic energy and the energy drawn from the supply are burned in the resistor.

Regenerative braking. If the back EMF exceeds the supply voltage (the load drives the motor above its no-load speed, or the armature voltage is reduced quickly by a converter), the armature current reverses and power returns to the supply. No switching is required on a stiff DC supply. The receiving system must be able to absorb the power: a DC traction line needs another train, a stationary energy store or an inverting substation to accept it; a single-quadrant thyristor bridge cannot invert current and needs a second, antiparallel bridge (Chapter 11).

> **Status of the principle: Electrical braking requires generator operation**
> *Fundamental.* By energy conservation, a braking torque from the machine means mechanical power entering it, which must leave as electrical power to a resistor, a store or the supply; the method only decides where the energy goes.

Worked example — Braking the mill motor on a mine hoist. The 440 V, 150 kW motor of Section 5.9 ($k\Phi = 6.767$ V·s/rad at full field, $R_a = 0.04\ \Omega$, rated $I_a = 370$ A) drives a hoist that is lowering a load at 600 rpm.

Dynamic braking at rated current. At 600 rpm the EMF is 425.2 V. To hold 370 A, the total circuit resistance must be $425.2/370 = 1.149\ \Omega$, so the braking resistor is $1.109\ \Omega$. The braking torque is the rated 2504 N·m, the resistor dissipates $370^2 \times 1.109 = 151.8$ kW and the armature 5.5 kW. As the cage slows, the current and torque fall in proportion to speed unless resistance is cut out in steps.

Plugging at rated speed. With the armature reversed, the driving voltage is $440 + 425.2 = 865.2$ V. To limit the current to twice rated (740 A) the circuit needs $865.2/740 = 1.169\ \Omega$, an external resistor of $1.129\ \Omega$; the initial braking torque is 5008 N·m, and the supply as well as the load feed power into the resistor.

Regenerative braking. With the armature connected to the 440 V supply and full field, current reverses when the load drives the motor above its no-load speed. For rated regenerative current, $E_a = 440 + 370 \times 0.04 = 454.8$ V, i.e. 642 rpm; the machine then returns $440 \times 370 = 162.8$ kW to the supply. The hoist settles at a lowering speed only 3.4 % above its full-field no-load speed of $440/6.767 = 65.0$ rad/s (621 rpm), and 7 % above base speed, with no resistor at all, the reason DC winders were naturally regenerative long before AC drives could be.

5.12 Losses and efficiency

Loss components

The losses of a DC machine are grouped by how they vary with load (the physics of each mechanism is in Chapter 3):

- **Armature-circuit copper loss** $I_a^2 R_a$, including the interpole, compensating and series windings: varies as the square of load.
- **Brush contact loss**: approximately a constant voltage drop of about 2 V total for carbon or electrographite brushes (1 V per brush set), so $2I_a$ watts: varies linearly with load.
- **Field copper loss** $V_f I_f$: constant for a shunt or separately excited machine at a given field setting.
- **Core loss**: hysteresis and eddy loss in the armature teeth and core, where the flux reverses at $f = pn/120$; depends on flux and speed, so it is constant at a given speed and field.
- **Friction and windage loss**: bearing friction, brush friction on the commutator (a significant part in DC machines) and air resistance; depends on speed.
- **Stray-load loss**: extra loss from armature reaction distorting the flux and from eddy currents in conductors; conventionally taken as about 1 % of output when not measured.

Core loss and friction and windage together are the **rotational losses**, measured by running the machine as an unloaded motor at rated speed and flux and subtracting the small armature copper loss (Swinburne's test). Efficiency tests on DC machines are made to IEC 60034-2-1 by direct input–output measurement or by summation of separated losses.

Efficiency and its maximum

As for the transformer, the efficiency of a constant-flux, constant-speed machine is greatest when the load-dependent loss equals the constant loss, i.e. when $I_a^2 R_a \approx P_{field} + P_{rot}$.

Worked example — Efficiency of the 220 V shunt motor. The motor of Section 5.6 ($V_t = 220$ V, line current 60 A at full load, $R_a = 0.25\ \Omega$, field current 2 A, armature current 3 A at no load) has:

- input power: $220 \times 60 = 13.20$ kW;
- field copper loss: $220 \times 2 = 440$ W;
- armature copper loss: $58^2 \times 0.25 = 841$ W;
- brush loss: $2 \times 58 = 116$ W;
- rotational loss from the no-load test: $E_0 I_{a0} = 219.25 \times 3 = 658$ W (assumed constant, as the speed changes only 6 %);
- stray-load loss, 1 % of output: 111 W.

The output is $13200 - 440 - 841 - 116 - 658 - 111 = 11034$ W, an efficiency of 83.6 %. The constant losses total $440 + 658 = 1098$ W, so maximum efficiency (ignoring the small brush and stray terms) occurs at $I_a = \sqrt{1098/0.25} = 66$ A, slightly above full load: the machine was designed for its best efficiency near rated output.

Table 5.4 — Loss distribution at rated load (typical, % of input).

Loss	10 kW machine	500 kW machine
Armature-circuit copper	5–7 %	2–3 %
Brush contact	0.5–1 %	0.3–0.5 %
Field copper	3–4 %	1–1.5 %
Core	2–3 %	1–1.5 %
Friction and windage	1–2 %	0.5–1 %
Stray load	0.5–1 %	0.5–1 %
Efficiency	**83–88 %**	**92–95 %**

A DC machine of a given rating is typically 2–4 percentage points less efficient than an IE3 induction motor of the same rating (Chapter 6), mainly because of the field loss and the commutator. Permanent-magnet DC motors avoid the field loss, which is why they dominate below about 1 kW.

What remains open

Open question	Where it is answered
How does a converter-fed DC drive control armature current, speed and regeneration?	Chapter 11
How are load characteristics, stability of the operating point and four-quadrant operation analysed?	Chapter 10
How does the rotor-inertia dynamics of a DC motor appear in a speed loop?	*Engineering Physics*, Ch. 21 §21.8.1; *Circuits, Signals and Communications*, Ch. 33–35
How does an AC machine produce torque without a commutator?	Chapter 6 and Chapter 7
Which machines replaced the PMDC motor (brushless DC, universal motor)?	Chapter 8
How are machine temperature rise, insulation class and duty cycle linked to rating?	Chapter 9
How are motor efficiency and replacement decisions made in plants?	Chapter 28

Next: Chapter 6 replaces the commutator with a rotating magnetic field and the brushed armature with a short-circuited cage, giving the induction motor, the machine that drives most of the world's pumps, fans and compressors.

Chapter 6 — Induction Machines

Part II · Transformers and Electrical Machines

A water utility installs a new 37 kW borehole pump at a remote site fed by a 250 kVA pole-mounted transformer. On the first start the motor turns, the lights in the pump house dip, and the contactor drops out; on the second attempt the motor reaches speed but the transformer's LV fuse blows ten minutes later. The motor is healthy; the problem is that an induction motor started direct-on-line draws six to seven times its rated current at a power factor of 0.4 for several seconds, and the installation was not designed for it. In a chemical plant across the country, an energy audit finds that half the site's electricity goes to induction motors driving pumps and fans through throttling valves and dampers, and that variable-speed operation would save a third of it. Electric motors, most of them induction motors, use about two-thirds of industrial electricity and some 45 % of all electricity worldwide (IEA estimates); their starting, their speed control, their efficiency class and their selection are among the most consequential decisions an electrical engineer makes.

> **How does a short-circuited rotor produce torque from a rotating field, and what does the slip tell us about the machine's currents, losses and efficiency?**
> **How is an induction motor started without disturbing the supply, controlled in speed without wasting energy, and selected for its load?**

Concept	Engineering destination
Three-phase induction principle	Pumps, fans, compressors, conveyors, crushers, mills; induction generators
Rotating magnetic field	Synchronous speed, direction of rotation, pole-changing motors
Slip	Reading nameplates; rotor heating; slip measurement for load estimation
Equivalent circuit	Predicting current, power factor, efficiency; drive tuning (Chapter 12)
Torque–speed characteristics	Matching to load; design classes; stall and pull-out protection
Starting methods	Voltage dip limits, star–delta, soft starters, VFDs, wound-rotor resistance
Speed control	VFD energy savings on pumps and fans; DFIG wind turbines (Chapter 15)
Losses and efficiency	IE classes, IEC 60034-30-1, Ecodesign rules, life-cycle cost
Single-phase induction motors	Domestic appliances, small pumps, HVAC fans
Motor selection	Duty, enclosure, cooling, insulation, hazardous areas, efficiency, standards

Roadmap. How the machine works and is built (6.1) → the rotating field and synchronous speed (6.2) → slip (6.3) → the per-phase equivalent circuit and power flow (6.4) → the torque–speed curve (6.5) → starting (6.6) → speed control (6.7) → losses and efficiency (6.8) → single-phase motors (6.9) → selecting a motor (6.10).

6.1 Three-phase induction principle

Torque from induced current

Three-phase currents in a stator winding produce a magnetic field that rotates at synchronous speed (Section 6.2). A rotor that turns more slowly than the field is swept by it; Faraday's law (*Engineering Physics*, Ch. 11 §11.4) induces EMFs in the rotor conductors, the short-circuited rotor circuit carries currents, and the force on those currents in the field drags the rotor along in the direction of the field. The rotor can never quite reach synchronous speed while motoring, because at synchronism there would be no relative motion, no induced current and no torque. The difference between field speed and rotor speed, the **slip**, is what makes the machine work, and nearly every property of the induction motor is a function of it.

> **Boundary.**
> Faraday's and Ampère's laws are taken from *Engineering Physics*, Ch. 11 §11.4; balanced three-phase sources, phase sequence and star–delta relations from *Circuits, Signals and Communications*, Ch. 10; the rotating magnetic field from distributed windings, force and torque from co-energy and the loss mechanisms from Chapter 3; and the transformer equivalent circuit from Chapter 4. This chapter treats the induction machine as a transformer with a rotating, short-circuited secondary.

Construction

The **stator** is a laminated core of non-oriented silicon steel (0.35–0.65 mm) with slots carrying a distributed three-phase winding, connected in star or delta. The **rotor** is also laminated and takes one of two forms (Figure 6.1):

- A **squirrel-cage rotor** has bars of aluminium (die-cast, in motors up to a few hundred kilowatts) or copper (brazed, in large motors and in premium-efficiency designs) in the rotor slots, short-circuited at each end by end rings. It has no insulation, no slip rings and no brushes, so it is cheap, rugged and almost maintenance-free. More than 90 % of induction motors are cage motors.
- A **wound rotor** (slip-ring rotor) carries a three-phase winding like the stator's, brought out through slip rings and brushes to external resistors or a converter. It allows the rotor resistance to be changed for starting and speed control and is used for very high-inertia loads (large mills, crushers, cement kilns), hoists, and as the **doubly fed induction generator** (DFIG) of variable-speed wind turbines.

The air gap is as small as manufacturing allows (0.3–0.5 mm in small motors, 1–3 mm in large ones), because the magnetising current that sets up the field rises in proportion to the gap length (Chapter 1), and magnetising current is reactive current that lowers the power factor.

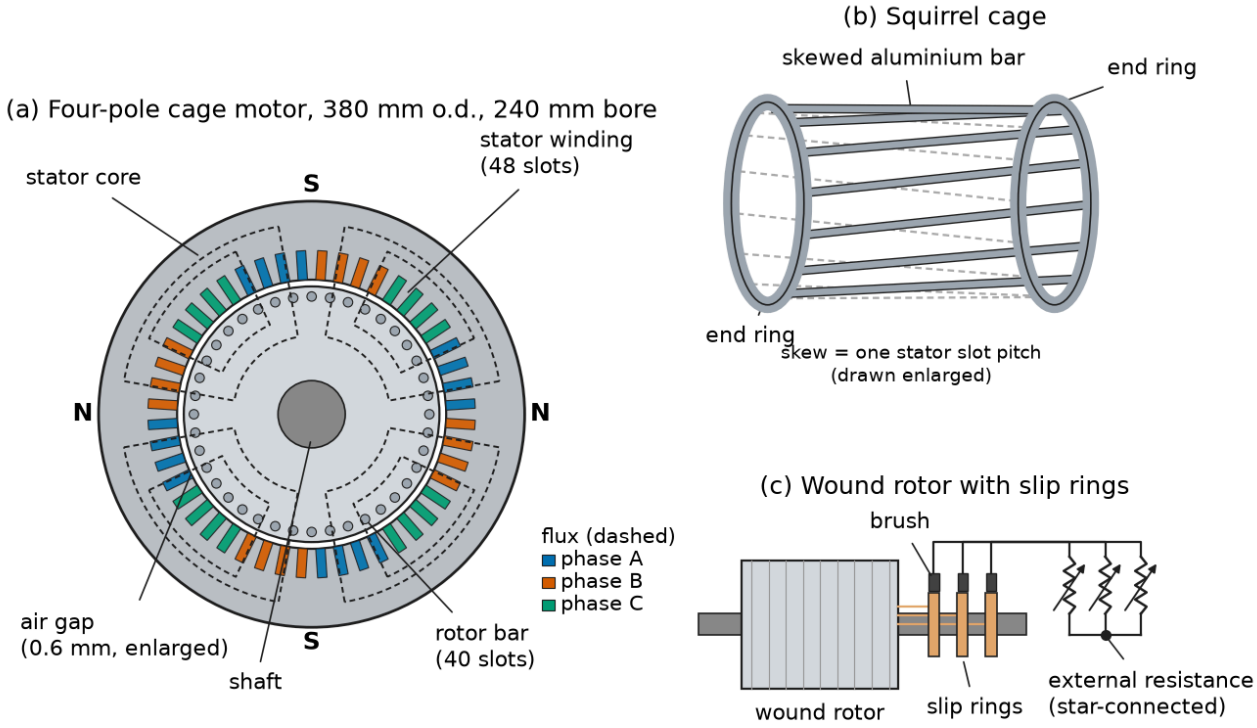


Figure 6.1 — Cross-section of a four-pole cage induction motor and the two rotor types: a squirrel cage with end rings and a wound rotor with slip rings.

Table 6.1 — Induction motors across the energy field (representative).

Application	Typical rating	Supply	Notes
Building HVAC fans and pumps	0.75–75 kW	400 V, 50 Hz (480 V, 60 Hz in North America)	Mostly VFD-fed; IE3/IE4
Data-centre chilled-water pumps, cooling-tower fans	15–200 kW	400–690 V	VFD-fed, often N+1 redundancy
Water and wastewater pumping	5 kW–5 MW	400 V to 11 kV	Large units at 3.3–11 kV
Mining conveyors, crushers, mills	100 kW–10 MW	690 V to 11 kV	Wound rotor or VFD for high-inertia starts
Oil and gas compressors	0.5–20 MW	6.6–13.8 kV	Ex-rated (IEC 60079) where gas may be present
Ship thrusters and pumps	50 kW–3 MW	440–690 V, 60 Hz; 6.6 kV	IEC 60092, marine classification rules
EV traction (e.g. some Tesla and Audi drivetrains)	100–300 kW	350–800 V DC via inverter	Copper rotor, 15–18 krpm, liquid-cooled
Small wind and micro-hydro induction generators	10 kW–2 MW	400–690 V	Grid-connected, slightly above synchronous speed

6.2 Rotating magnetic field

Synchronous speed

Chapter 3 showed that three windings displaced by 120 electrical degrees in space, carrying balanced three-phase currents displaced by 120° in time, produce an air-gap MMF wave of constant amplitude, $\frac{3}{2}$ times the peak MMF of one phase, rotating at one pole pair per cycle. For a winding with p poles fed at frequency f, the field turns at the **synchronous speed**

$$n_s = \frac{120 f}{p} \text{ rpm}, \qquad \omega_s = \frac{4\pi f}{p} \text{ rad/s}$$

> **Status of the principle: Rotating field from a balanced three-phase winding**
> *Derived.* Follows from superposition of three sinusoidally distributed MMFs carrying balanced currents; real windings in slots add space harmonics, and unbalanced supplies add a backward-rotating component, both treated below.

Table 6.2 — Synchronous speeds.

Poles	50 Hz	60 Hz	Typical full-load speed at 50 Hz	Typical use
2	3000 rpm	3600 rpm	2950–2980 rpm	Centrifugal pumps, compressors, blowers
4	1500 rpm	1800 rpm	1460–1490 rpm	The general-purpose standard
6	1000 rpm	1200 rpm	960–990 rpm	Fans, conveyors, mixers
8	750 rpm	900 rpm	720–740 rpm	Slow fans, mills, crushers
12	500 rpm	600 rpm	485–495 rpm	Large slow compressors (often synchronous instead)

The direction of rotation follows the phase sequence (*Circuits, Signals and Communications*, Ch. 10 §10.2): swapping any two supply leads reverses the field and the motor. Commissioning checklists include a "bump test" for this reason, because a pump or compressor run backwards can be damaged in seconds.

Real windings and their harmonics

A practical stator winding is distributed over several slots per pole per phase and short-pitched (coil span about 5/6 of a pole pitch) to make the MMF as sinusoidal as possible. The remaining space harmonics (5th, 7th, 11th, 13th, and slot harmonics) rotate at n_s/h forwards or backwards and produce parasitic torques. The 7th-harmonic asynchronous torque can produce a dip in the torque–speed curve near one-seventh of synchronous speed, where a heavily loaded motor may "crawl"; slot-harmonic synchronous torques can make a motor "cog" at standstill. Designers avoid these by choosing stator and rotor slot numbers that do not coincide and by **skewing** the rotor bars by about one stator slot pitch.

An unbalanced supply adds a negative-sequence field rotating backwards at n_s. The rotor sees it at a slip of about $2 - s$, i.e. at nearly twice supply frequency, where the rotor resistance is high because of skin effect, so a

small negative-sequence voltage drives a large negative-sequence current. A 1 % voltage unbalance causes about 6–10 % current unbalance; NEMA MG 1 requires derating above 1 % voltage unbalance (to about 0.75 at 5 %), and IEC 60034-1 limits continuous operation to 1 % negative-sequence voltage. Unbalance itself is treated in Chapter 25.

6.3 Slip

Definition and rotor frequency

The **slip** is the speed of the field relative to the rotor, as a fraction of synchronous speed:

$$s = \frac{n_s - n}{n_s}, \qquad n = n_s(1 - s)$$

The rotor conductors are cut by the field at the slip speed sn_s, so the frequency of rotor EMFs and currents is

$$f_r = sf$$

and the rotor EMF is proportional to slip: $E_r = sE_{r0}$, where E_{r0} is the rotor EMF at standstill. The rotor leakage reactance is likewise sX_{r0}. At standstill ($s = 1$) the motor is a short-circuited transformer; at full load (s of 1–5 %) the rotor currents alternate at only 0.5–2.5 Hz.

> **Status of the principle: Rotor frequency equals slip times supply frequency**
> *Fundamental.* A kinematic consequence of the field rotating at n_s relative to the stator and the rotor turning at n; it holds for any induction machine regardless of load, and it is how slip is measured with a search coil, a stroboscope or the sidebands in the stator current.

Worked example — Slip of a pump motor. A four-pole, 50 Hz motor runs at 1476 rpm. Synchronous speed is $120 \times 50/4 = 1500$ rpm, the slip is $(1500 - 1476)/1500 = 0.016$, i.e. 1.6 %, and the rotor currents alternate at $0.016 \times 50 = 0.8$ Hz. A six-pole 60 Hz motor nameplated 1164 rpm runs at $s = (1200 - 1164)/1200 = 3.0$ %, rotor frequency 1.8 Hz. Since slip is close to proportional to load in the normal operating range, a measured speed of 1488 rpm on the first motor ($s = 0.8$ %) indicates it is running at roughly half load, a quick estimate used in energy audits.

Table 6.3 — Typical full-load slip of cage induction motors (IE3, 50 Hz).

Rating	Full-load slip
0.75–4 kW	3–6 %
5.5–30 kW	1.5–3 %
37–200 kW	0.8–1.7 %
0.5–5 MW (medium voltage)	0.3–0.8 %

Slip and the power split

Slip divides the power crossing the air gap into two parts: a fraction s is lost as rotor copper loss, and the fraction $1 - s$ becomes mechanical power. This is derived from the equivalent circuit in Section 6.4, and it has

a direct consequence: an induction motor running at slip s can be no more efficient than $1 - s$, so a motor whose speed is reduced by running it at high slip (rotor-resistance or stator-voltage control) is inherently wasteful.

6.4 Equivalent circuit

The induction motor as a transformer

Per phase, the stator is a transformer primary with resistance R_1, leakage reactance X_1, magnetising reactance X_m and core-loss resistance R_c, exactly as in Chapter 4. The rotor is a short-circuited secondary whose EMF sE_{r0} and reactance sX_{r0} both scale with slip. Its current is

$$I_r = \frac{sE_{r0}}{R_r + jsX_{r0}} = \frac{E_{r0}}{R_r/s + jX_{r0}}$$

Dividing numerator and denominator by s converts the rotor circuit from slip frequency to supply frequency: the rotor now looks like a stationary circuit with an EMF E_{r0}, a reactance X_{r0} and a slip-dependent resistance R_r/s. Referring the rotor to the stator by the effective turns ratio (Chapter 4) gives the per-phase equivalent circuit of Figure 6.2, with referred rotor quantities R_2 and X_2:

$$Z_{in} = R_1 + jX_1 + \frac{jX_m\,(R_2/s + jX_2)}{R_2/s + j(X_2 + X_m)}$$

with the core-loss resistance either placed across the magnetising branch or, more simply, treated as a fixed loss.

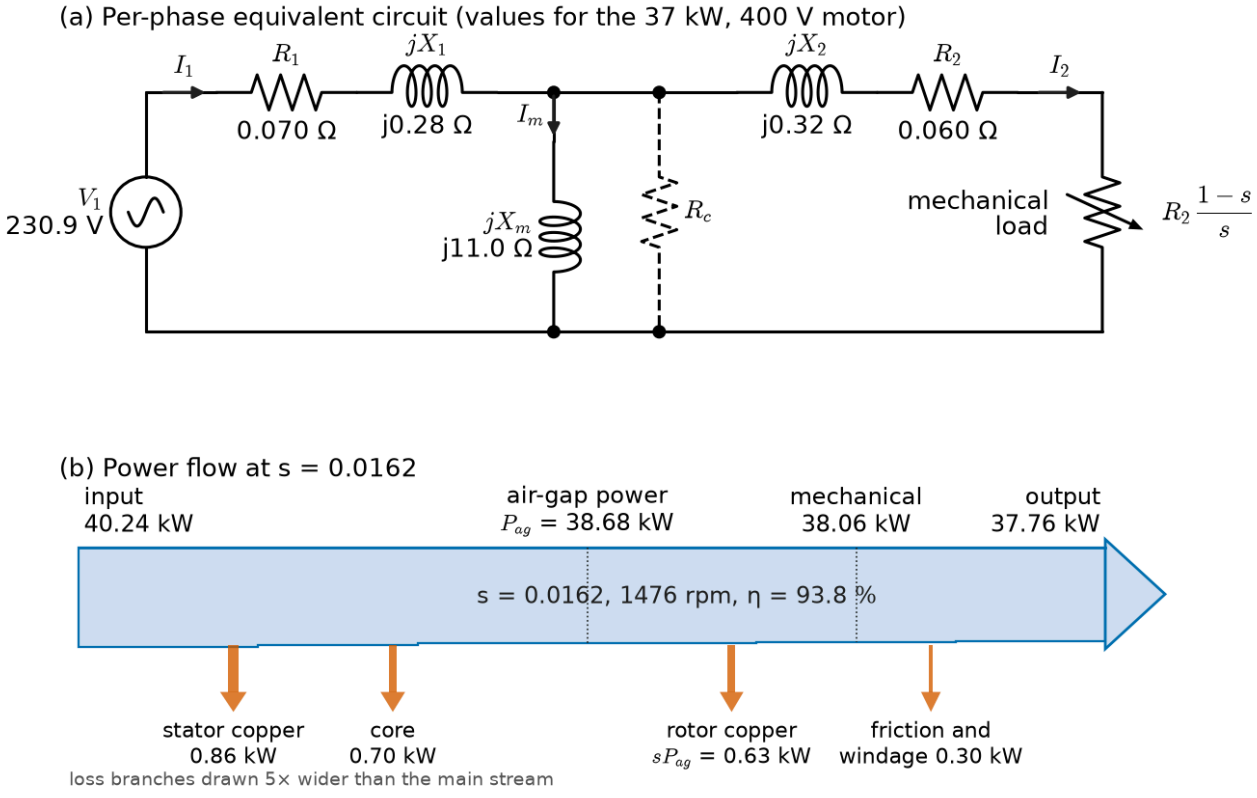


Figure 6.2 — Per-phase equivalent circuit of the induction motor with the rotor resistance split into copper-loss and mechanical-load parts, and the power-flow diagram for the 37 kW example motor.

Power flow

The resistance R_2/s is split into the rotor resistance R_2, which accounts for rotor copper loss, and the remainder $R_2(1-s)/s$, whose power is the power converted to mechanical form:

$$P_{ag} = 3I_2^2\frac{R_2}{s}, \qquad P_{cu2} = 3I_2^2 R_2 = sP_{ag}, \qquad P_{mech} = 3I_2^2 R_2\frac{1-s}{s} = (1-s)P_{ag}$$

The electromagnetic torque is the air-gap power divided by synchronous angular speed (equivalently mechanical power divided by rotor speed):

$$T = \frac{P_{ag}}{\omega_s} = \frac{P_{mech}}{\omega_s(1-s)}$$

> **Status of the principle: Per-phase induction-motor equivalent circuit**
> *Approximate.* Assumes a balanced sinusoidal supply, sinusoidally distributed windings, linear magnetics and constant parameters; it predicts running performance within a few per cent, but the constant-parameter rotor fails at high slip (deep-bar and double-cage rotors change R_2 and X_2 with rotor frequency), at saturation during starting (leakage paths saturate and X_1, X_2 fall), under unbalance (a negative-sequence circuit is needed) and with inverter supplies (time harmonics).

Finding the parameters

The parameters come from tests standardised in IEEE 112 and IEC 60034-28, analogous to the transformer tests of Chapter 4:

- **DC resistance test**: gives R_1 (corrected to operating temperature).
- **No-load test**: rated voltage, uncoupled shaft. Slip is nearly zero, so the rotor branch is open; the input gives $X_1 + X_m$ and the sum of core and friction–windage losses, which are separated by repeating the test at reduced voltages and extrapolating to zero voltage.
- **Locked-rotor test**: rotor blocked, reduced voltage to give rated current. The magnetising branch is bypassed by the low rotor impedance, so the input gives $R_1 + R_2$ and $X_1 + X_2$. To capture the running value of R_2 (not the starting value), the test is made at a reduced frequency, about 25 % of rated, because a deep-bar rotor's resistance at 50 Hz is much higher than at the 1 Hz of normal running.

Worked example — Running performance of a 37 kW motor. A 400 V, 50 Hz, four-pole, star-connected 37 kW cage motor has per-phase parameters $R_1 = 0.070\ \Omega$, $X_1 = 0.28\ \Omega$, $R_2 = 0.060\ \Omega$, $X_2 = 0.32\ \Omega$, $X_m = 11.0\ \Omega$, core loss 700 W and friction and windage loss 300 W. Find its performance at a slip of 1.62 %.

The phase voltage is $400/\sqrt{3} = 230.9$ V and synchronous speed is 1500 rpm ($\omega_s = 157.1$ rad/s). Solving the circuit at $s = 0.0162$ ($R_2/s = 3.70\ \Omega$):

- stator current $I_1 = 63.9$ A at power factor 0.893 lagging; referred rotor current $I_2 = 59.0$ A;
- power drawn by the circuit $3 \times 230.9 \times 63.9 \times 0.893 = 39.54$ kW; the circuit has no core-loss branch, so the 0.70 kW core loss is added to give a total input of 40.24 kW;
- stator copper loss $3 \times 63.9^2 \times 0.070 = 0.86$ kW, so the air-gap power is $39.54 - 0.86 = 38.68$ kW, equal to $3I_2^2R_2/s$;
- rotor copper loss $sP_{ag} = 0.0162 \times 38.68 = 0.63$ kW; mechanical power 38.06 kW;
- output after friction and windage: 37.76 kW, at $1500 \times (1 - 0.0162) = 1476$ rpm;

- efficiency $37.76/40.24 = 93.8$ %; electromagnetic torque $38680/157.1 = 246.2$ N·m, shaft torque $37760/154.5 = 244.4$ N·m.

The efficiency is close to the IE3 limit for a four-pole 37 kW motor (93.9 %, Table 6.6). The example has neglected stray-load loss, which IEC 60034-2-1 would add at about 1–1.5 % of input for this size; including it brings the efficiency to about 92.3–92.8 %. Figure 6.2b draws the power flow.

6.5 Torque–speed characteristics

The torque equation

Replacing the stator and magnetising branch by their Thévenin equivalent seen from the rotor (*Circuits, Signals and Communications*, Ch. 8 §8.6),

$$V_{th} = V_1 \frac{jX_m}{R_1 + j(X_1 + X_m)}, \qquad Z_{th} = R_{th} + jX_{th} = \frac{jX_m(R_1 + jX_1)}{R_1 + j(X_1 + X_m)}$$

the torque at any slip is

$$T = \frac{3V_{th}^2}{\omega_s} \cdot \frac{R_2/s}{(R_{th} + R_2/s)^2 + (X_{th} + X_2)^2}$$

Maximising with respect to R_2/s (maximum power transfer to R_2/s) gives the **breakdown** or **pull-out** torque and the slip at which it occurs:

$$s_{T\,\max} = \frac{R_2}{\sqrt{R_{th}^2 + (X_{th} + X_2)^2}}, \qquad T_{\max} = \frac{3V_{th}^2}{2\omega_s \left[R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2)^2}\right]}$$

Two consequences shape the whole of induction-motor engineering. First, $T_{\max}$ does not depend on R_2: adding rotor resistance moves the peak to a higher slip without changing its height, which is how wound-rotor starting works. Second, torque at a given slip is proportional to V^2: a 10 % voltage dip removes 19 % of the torque, which can stall a heavily loaded motor during a supply sag.

> **Status of the principle: Induction-motor torque–slip equation and breakdown torque**
> *Derived*. Follows from the equivalent circuit by Thévenin reduction and maximum power transfer; it inherits all of the circuit's approximations and is least accurate at high slip, where rotor parameters vary with frequency.

The shape of the curve

At small slip (R_2/s dominates), $T \approx 3V_{th}^2 s/(\omega_s R_2)$: torque is proportional to slip, the near-linear operating region. At large slip the reactance dominates and torque falls roughly as $1/s$. Figure 6.3 shows the whole curve from $s = 2$ to $s = -1$:

- **Motoring** ($0 < s < 1$): rotor slower than the field, in the same direction.
- **Generating** ($s < 0$): rotor driven above synchronous speed; the machine delivers power to the supply while still drawing its magnetising current. Small hydro and older fixed-speed wind turbines use cage induction generators at slips of -0.5 to -2 %; regenerative braking of a VFD-fed motor uses the same region.

- **Braking (plugging)** ($s > 1$): rotor turning against the field after two supply phases are reversed. Large currents flow and both electrical and mechanical power become rotor heat.

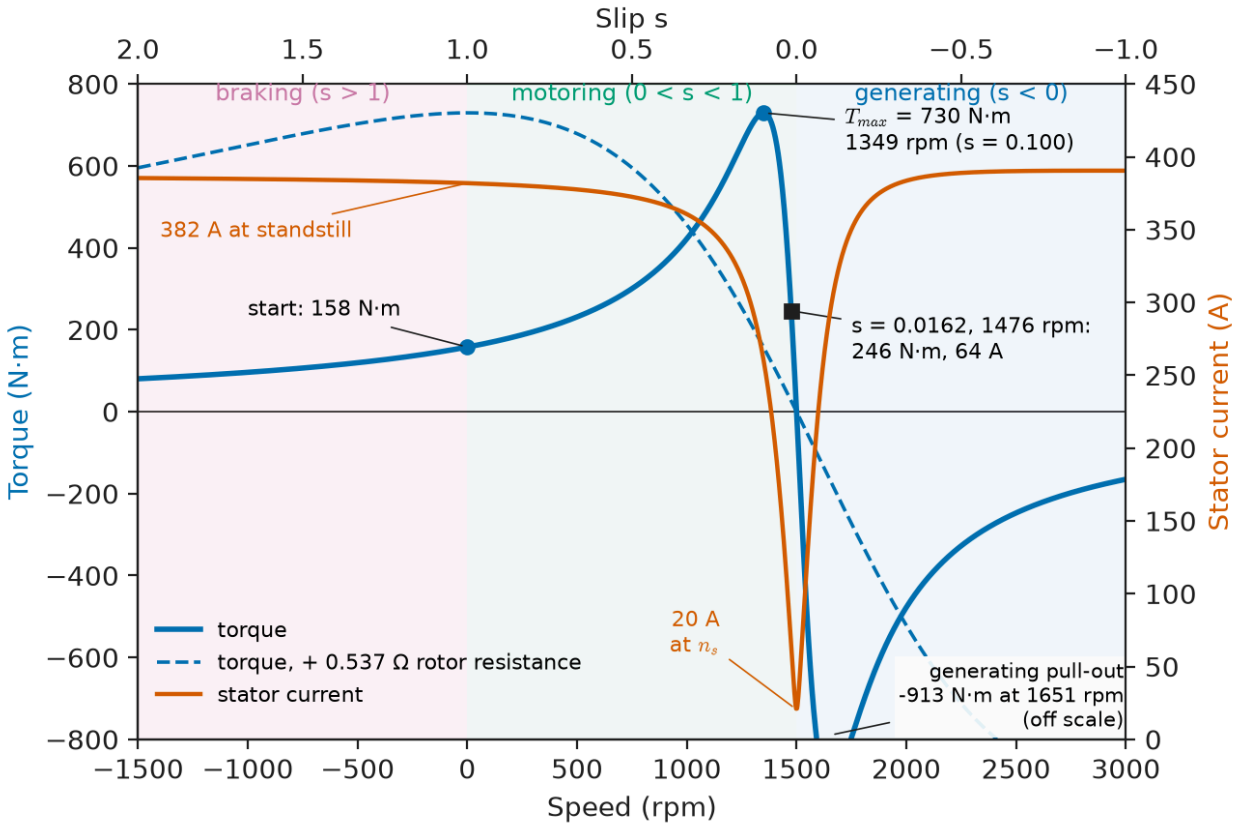


Figure 6.3 — Torque and stator current against speed for the 37 kW, four-pole motor, from plugging through motoring to generating, using the constant-parameter equivalent circuit.

Worked example — Breakdown and starting torque of the 37 kW motor. For the motor of Section 6.4, $V_{th} = 225.2$ V, $R_{th} = 0.0666\ \Omega$ and $X_{th} = 0.2735\ \Omega$. Then

$$s_{T\,\max} = \frac{0.060}{\sqrt{0.0666^2 + 0.5935^2}} = 0.100, \qquad T_{\max} = \frac{3 \times 225.2^2}{2 \times 157.1 \times (0.0666 + 0.5972)} = 730\ \text{N m}$$

i.e. 3.0 times rated torque, at 1349 rpm. With constant parameters the starting torque ($s = 1$) is 158 N·m, only 0.64 times rated, and the starting current 382 A, 6.0 times rated.

Real cage motors start much better than this, because the rotor is not a constant-parameter circuit. At standstill the rotor current is at 50 Hz, the current crowds into the top of each deep bar (skin effect, *Engineering Physics*, Ch. 16 §16.8.1), the effective rotor resistance rises and the leakage reactance falls. Taking the standstill values as $R_2 = 2.6 \times 0.060 = 0.156\ \Omega$ and $X_2 = 0.7 \times 0.32 = 0.224\ \Omega$, typical of a deep-bar rotor, gives a starting current of 422 A (6.6 times rated) at a power factor of 0.40 and a starting torque of 509 N·m (2.1 times rated), in line with the catalogue values of real 37 kW motors. The constant-parameter model fails here because the rotor frequency has changed by a factor of 60, and with it the physics of current distribution in the bars.

Wound-rotor starting. To get breakdown torque at standstill, $R_2 + R_{ext} = \sqrt{R_{th}^2 + (X_{th} + X_2)^2} = 0.597\ \Omega$, an external resistance of 0.537 Ω per phase referred to the stator (the dashed curve of Figure 6.3).

Design classes and rotor shape

Rotor-bar shape is the designer's tool for trading starting torque against running efficiency (Figure 6.4). A rotor with high resistance starts well and runs with high slip and high losses; one with low resistance runs efficiently but starts poorly. **Deep bars** and **double cages** (a high-resistance outer cage that carries most current at start, and a low-resistance inner cage that carries it when running) give both. IEC 60034-12 defines starting-performance classes N (normal), H (high starting torque) and their variants with suffix E for higher-efficiency designs; NEMA MG 1 uses designs A–D.

Table 6.4 — Cage-rotor design classes (typical values, 4-pole, 10–100 kW).

Class	Locked-rotor torque	Breakdown torque	Locked-rotor current	Full-load slip	Typical loads
IEC N / NEMA B	1.5–2.2 × rated	2.0–3.0 × rated	6–7 × rated	1–3 %	Pumps, fans, compressors, machine tools
IEC NE (high efficiency) / NEMA A	1.5–2.5 ×	2.5–3.5 ×	7–8.5 ×	0.8–2 %	As N; higher inrush must be checked
IEC H / NEMA C	2.0–2.5 ×	1.9–2.2 ×	6–7 ×	1–3 %	Loaded conveyors, crushers, reciprocating compressors
NEMA D	2.75 × or more	— (no distinct peak)	6–7 ×	5–13 %	Punch presses, hoists, high-inertia loads with flywheel

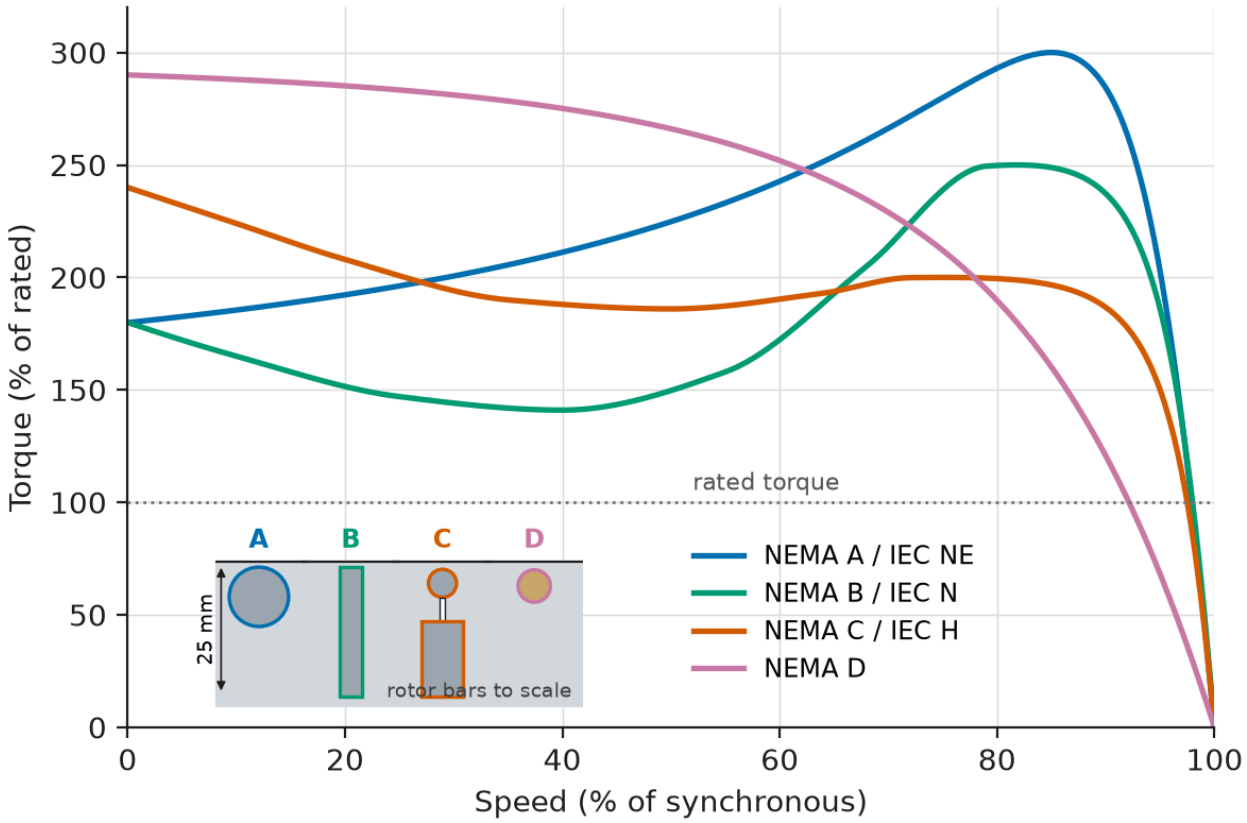


Figure 6.4 — Torque-speed curves of IEC/NEMA design classes and the rotor-bar shapes that produce them.

6.6 Starting methods

The starting problem

A cage motor started direct-on-line draws 6–8 times rated current at a power factor of 0.3–0.5 until it approaches full speed, which takes from a fraction of a second (a fan at no load) to 20–30 s (a large fan or mill with high inertia). This has three consequences: a **voltage dip** on the supply, which can disturb other loads and, if the dip is deep, stall the motor itself (torque scales as V^2); **mechanical shock** to couplings, gearboxes, belts and pipework (water hammer in pumping systems); and **rotor heating**, since the rotor dissipates energy equal to the kinetic energy given to the load during an unloaded start. Utility rules limit the dip at the point of common coupling, typically to 3–6 % for frequent starts on public LV networks (IEC 61000-3-3 and -3-11 and local distribution codes).

Worked example — Voltage dip from a direct-on-line start. The 37 kW motor of Section 6.5 (starting current 422 A at power factor 0.40) is fed from a 250 kVA, 400 V, 4 % impedance transformer with X/R = 2. The motor's locked-rotor impedance per phase is $230.9/422 = 0.548\ \Omega$ at 66.3°, i.e. $0.220 + j0.502\ \Omega$. The transformer impedance is $0.04 \times 400^2/250000 = 0.0256\ \Omega$ at 63.4°, i.e. $0.0114 + j0.0229\ \Omega$. As a voltage divider from a stiff 11 kV source, the motor-terminal voltage is

$$\frac{|Z_{mot}|}{|Z_{mot} + Z_t|} = 0.955 \text{ pu}$$

a dip of 4.5 % at the transformer LV terminals, before counting the cable and the 11 kV network impedance. Acceptable for an occasional start; for a borehole pump that starts several times an hour on a weak rural feeder

it is not, and it is exactly the case of this chapter's opening. With a star–delta starter the dip falls to about 1.5 %.

Methods compared

Figure 6.5 and Table 6.5 compare the methods for the 37 kW motor.

- **Direct-on-line (DOL)**: simplest and cheapest; full starting torque. Used up to about 7.5–11 kW on public LV supplies and much larger on stiff industrial supplies.
- **Star–delta**: a motor whose windings are rated for delta at the supply voltage is started in star, so each winding sees $1/\sqrt{3}$ of its rated voltage. Line current and torque both fall to one third of their DOL values. The motor must be able to accelerate the load on one third of its starting torque, so star–delta suits fans and unloaded compressors, not loaded conveyors. The open transition from star to delta causes a current and torque transient that can exceed the DOL value; closed-transition starters avoid it.
- **Autotransformer**: an autotransformer (Chapter 4) with 50/65/80 % taps; at tap k the motor current is k times DOL, the line current and torque k^2 times DOL. More flexible than star–delta and used for large LV and MV motors.
- **Soft starter**: antiparallel thyristors in each phase ramp the voltage; the current is limited (typically to 2.5–4 × rated) and torque again scales as V^2. Also provides soft stopping, which reduces water hammer in pumps. Bypassed by a contactor once at speed.
- **Variable-frequency drive (VFD)**: starts at low frequency with the motor running near rated slip, so full torque is available at about rated current. The best start for any load; justified when speed control is also needed (Section 6.7).
- **Rotor resistance (wound rotor)**: external resistors cut out in steps (or a liquid resistor) give high starting torque with low current; used for mills, crushers and kilns with very high inertia.

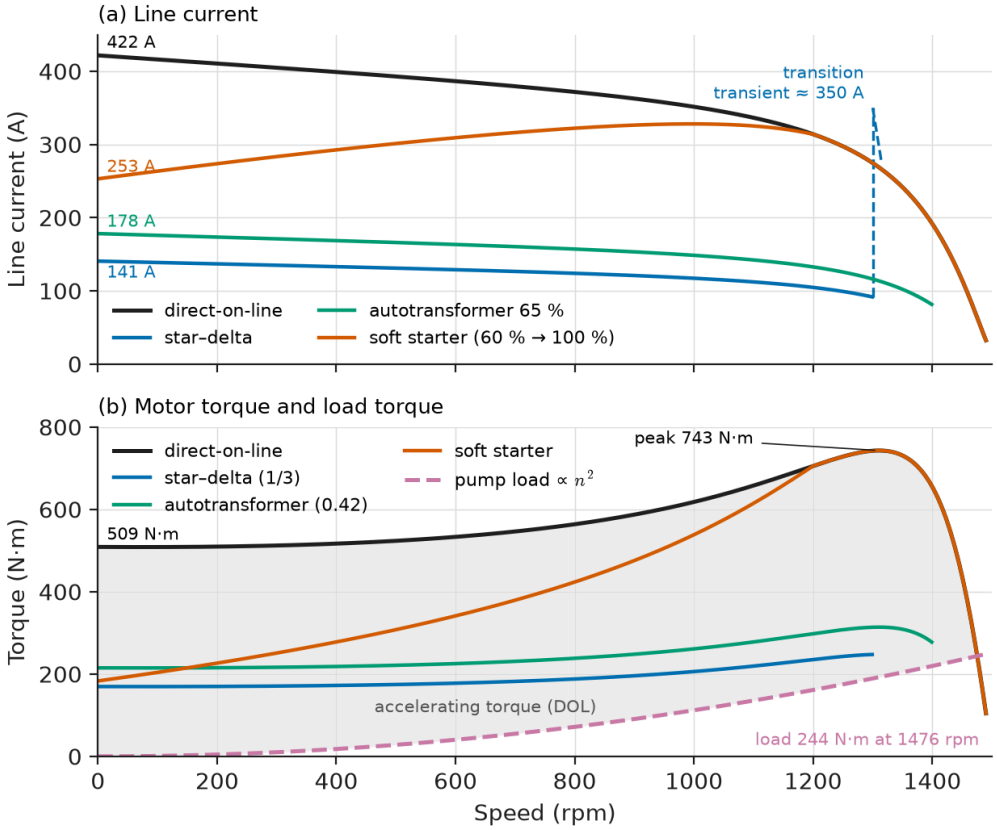


Figure 6.5 — Line current and motor torque against speed for direct-on-line, star-delta, 65 percent autotransformer and soft-starter starting of the 37 kW motor, with a centrifugal pump load torque.

Table 6.5 — Starting methods for the 37 kW motor (DOL: 422 A, 509 N·m at standstill).

Method	Line current at start	Starting torque	Torque / rated	Comments
Direct-on-line	422 A	509 N·m	2.1	Highest dip (4.5 % on 250 kVA)
Star–delta	141 A	170 N·m	0.69	Only for light starting loads; transition transient
Autotransformer, 65 % tap	178 A	215 N·m	0.87	Line current 0.65^2 of DOL
Soft starter, 60 % initial voltage	253 A	183 N·m	0.74	Ramp; soft stop for pumps
VFD	about 64–100 A	up to 1.5 × rated	1.5	Rated torque from standstill at rated current

Status of the principle: Starting torque scales with the square of applied voltage
Approximate. Follows from the torque equation at fixed slip with fixed parameters; it slightly overstates the reduction at low voltage, where leakage-path saturation is less than at full voltage and the effective reactances are higher.

6.7 Speed control

Why speed control matters

For a centrifugal pump or fan, flow is proportional to speed, head to speed squared and power to speed cubed (the affinity laws; the fluid physics is in *Engineering Physics*, Ch. 19 §19.7). Throttling the flow with a valve or damper wastes the excess head; reducing the speed does not. Since a fixed-frequency induction motor runs at nearly constant speed, the methods below are ways of changing $n = n_s(1 - s) = 120f(1 - s)/p$: change the poles, the slip or the frequency.

Methods

Pole changing. A **Dahlander** winding can be reconnected to give two speeds in a 2:1 ratio (e.g. 1500/750 rpm); separate windings give other ratios. Cheap and efficient, but only discrete speeds; used for two-speed fans, cooling towers and lifts.

Stator voltage control. Reducing voltage with a soft starter lowers the torque and so raises the slip. Because rotor loss is sP_{ag}, this is efficient only for loads whose torque falls steeply with speed (small fans) and then only over a narrow range. Its main use is starting.

Rotor-resistance control. Adding resistance to a wound rotor raises the slip for a given torque. Simple and robust, but the slip power sP_{ag} is burned in the resistors; used for cranes and hoists at intermittent duty.

Slip-power recovery. Instead of burning the slip power, a converter in the rotor circuit returns it to the supply (static Kramer and Scherbius drives). The modern form is the **doubly fed induction generator** of wind turbines: a back-to-back converter rated at about 25–30 % of the machine rating feeds the rotor, giving a speed range of about ±30 % around synchronous speed with a converter a third the size of a full-rated one. It is developed in Chapter 15 and Chapter 27.

Variable frequency. A VFD supplies variable frequency and voltage. Keeping V/f constant keeps the air-gap flux constant (Chapter 4's EMF equation applies equally to the stator winding), so the torque–speed curve shifts horizontally with nearly the same shape, and the motor operates at low slip, and hence efficiently, at every speed. Above base frequency, the voltage is held at its maximum and the flux falls, giving a constant-power region like the field weakening of the DC machine (Chapter 5). This is now the dominant method; its control (scalar V/f, field-oriented and direct torque control) is the subject of Chapter 12.

Worked example — Fan energy saving with a VFD. A 55 kW fan in a mine ventilation system delivers full flow at rated speed, absorbing 55 kW, and runs 6000 h a year at 80 % flow. By the affinity laws, a VFD at 80 % speed reduces the shaft power to $55 \times 0.8^3 = 28.2$ kW, or 29.0 kW from the supply allowing 3 % for drive losses. With an outlet damper, the same 80 % flow typically needs about 85–90 % of full power, say 48 kW. The saving is about 19 kW, i.e. some 114 MWh a year, worth about USD 14 000 at USD 0.12 per kWh, against a 55 kW VFD costing a fraction of that. This arithmetic, repeated across millions of pumps and fans, is why VFDs are the single largest electrical energy-efficiency measure in industry and buildings (Chapter 28).

6.8 Losses and efficiency

Loss components

The losses follow the power flow of Section 6.4. Their physics belongs to Chapter 3; here the engineering split matters:

- **Stator copper loss** $3I_1^2R_1$: 25–40 % of total loss at rated load.
- **Rotor copper loss** sP_{ag}: 15–25 %. Copper cages reduce it compared with aluminium by roughly 30–40 %.
- **Core loss**: 15–25 %; lower with thinner, lower-loss steel.
- **Friction and windage**: 5–15 %, largest in two-pole totally enclosed fan-cooled motors, where the cooling fan dominates.
- **Stray-load loss**: 10–20 %; harmonic fields, leakage-flux eddy currents and inter-bar currents. IEC 60034-2-1 determines it by test (residual-loss method) or assigns a value by formula.

Efficiency is nearly flat from about 50 % to 100 % load and falls sharply below about 25–30 % load, where the fixed core and friction losses dominate and, worse, the power factor drops because the magnetising current is unchanged. An oversized motor running at 30 % load is a common audit finding: its efficiency may be 5 points below nameplate and its power factor 0.5.

Efficiency classes

IEC 60034-30-1 defines efficiency classes IE1 (standard) to IE4 (super premium), with IE5 (ultra premium) introduced in IEC TS 60034-30-2, for line-operated motors from 0.12 kW to 1000 kW. Efficiency is measured according to IEC 60034-2-1. Since July 2021 the EU Ecodesign Regulation (EU) 2019/1781 requires IE3 for three-phase motors of 0.75–1000 kW (with 2-, 4-, 6- and 8-pole versions), and since July 2023 IE4 for 75–200 kW; the US (NEMA Premium, equivalent to IE3) and many other countries have similar rules.

Table 6.6 — Minimum efficiencies of IEC 60034-30-1 classes, four-pole, 50 Hz.

Rated output	IE1	IE2	IE3	IE4
0.75 kW	72.1 %	79.6 %	82.5 %	85.7 %
1.5 kW	77.2 %	82.8 %	85.3 %	88.2 %
4 kW	83.1 %	86.6 %	88.6 %	91.1 %
11 kW	87.6 %	89.8 %	91.4 %	93.3 %
37 kW	91.2 %	92.7 %	93.9 %	95.2 %
75 kW	92.7 %	94.0 %	95.0 %	96.0 %
200 kW and above	94.0 %	95.1 %	96.0 %	96.7 %

> **Status of the principle: IE efficiency-class limits**
> *Empirical.* The class boundaries are fitted to the efficiencies achievable by commercial designs of each size, defined by IEC 60034-30-1 and verified by the test methods of IEC 60034-2-1; they say nothing about efficiency at part load or on inverter supply (IEC 60034-2-3 covers the latter).

Worked example — Upgrading an 11 kW motor. A process-water pump motor of 11 kW runs 6000 h a year at 75 % load (8.25 kW shaft). Replacing an IE2 motor (89.8 %) by an IE4 motor (93.3 %), and assuming the efficiencies hold at 75 % load, reduces the input from $8.25/0.898 = 9.19$ kW to $8.25/0.933 = 8.84$ kW, a saving of 0.34 kW or 2.07 MWh a year, about USD 250 at USD 0.12/kWh. A motor's purchase price is typically only 2–3 % of its lifetime energy cost; the upgrade pays back in two to four years. One caution: higher-efficiency motors have lower slip and so run slightly faster; on a centrifugal pump, a 0.5 % speed increase raises power by 1.5 %, eroding part of the saving unless the impeller is trimmed or a VFD adjusts the speed.

6.9 Single-phase induction motors

No starting torque from a single phase

A single-phase winding produces a pulsating, not a rotating, field. Any pulsating field can be resolved into two fields of half the amplitude rotating in opposite directions at synchronous speed (the **double-revolving-field** picture, the same decomposition as the positive and negative sequence of *Circuits, Signals and Communications*, Ch. 10 §10.6). At standstill each produces an equal and opposite torque, so the net starting torque is zero. Once the rotor turns in either direction, the forward field (at slip s) produces more torque than the backward field (at slip $2 - s$), and the motor continues to run. The backward field causes a double-frequency torque pulsation (100 Hz on a 50 Hz supply) and extra rotor loss, so single-phase motors are noisier and less efficient than three-phase ones of the same rating.

> **Status of the principle: Double-revolving-field theory**
> *Derived*. A pulsating sinusoidal MMF is mathematically identical to two counter-rotating MMFs of half amplitude; the resulting two-part equivalent circuit is approximate in the same ways as the three-phase one.

Starting arrangements

To start, a second (**auxiliary**) winding displaced 90° in space carries a current shifted in time, giving an elliptical rotating field (Figure 6.6):

- **Split-phase**: auxiliary winding of high resistance-to-reactance ratio, disconnected by a centrifugal switch at about 75 % speed. Starting torque 1–1.5 × rated, high starting current. Small fans, washing machines, older appliances.
- **Capacitor-start**: a capacitor (electrolytic, 70–300 µF for a fractional to 1 kW motor) in series with the auxiliary winding gives nearly 90° phase shift at start; starting torque 2.5–4 × rated. Compressors, pumps, workshop machinery.
- **Permanent-split capacitor (PSC)**: a smaller run capacitor (film type, 2–40 µF) stays in circuit, improving power factor, efficiency and smoothness; low starting torque. HVAC fans, ceiling fans, circulating pumps.
- **Capacitor-start capacitor-run**: both; best performance for 1–3 kW single-phase loads such as farm and well pumps.
- **Shaded-pole**: a copper ring around part of each pole delays the flux in the shaded part, producing a weak sweeping field. Very cheap, efficiency 10–30 %; small fans below about 50 W.

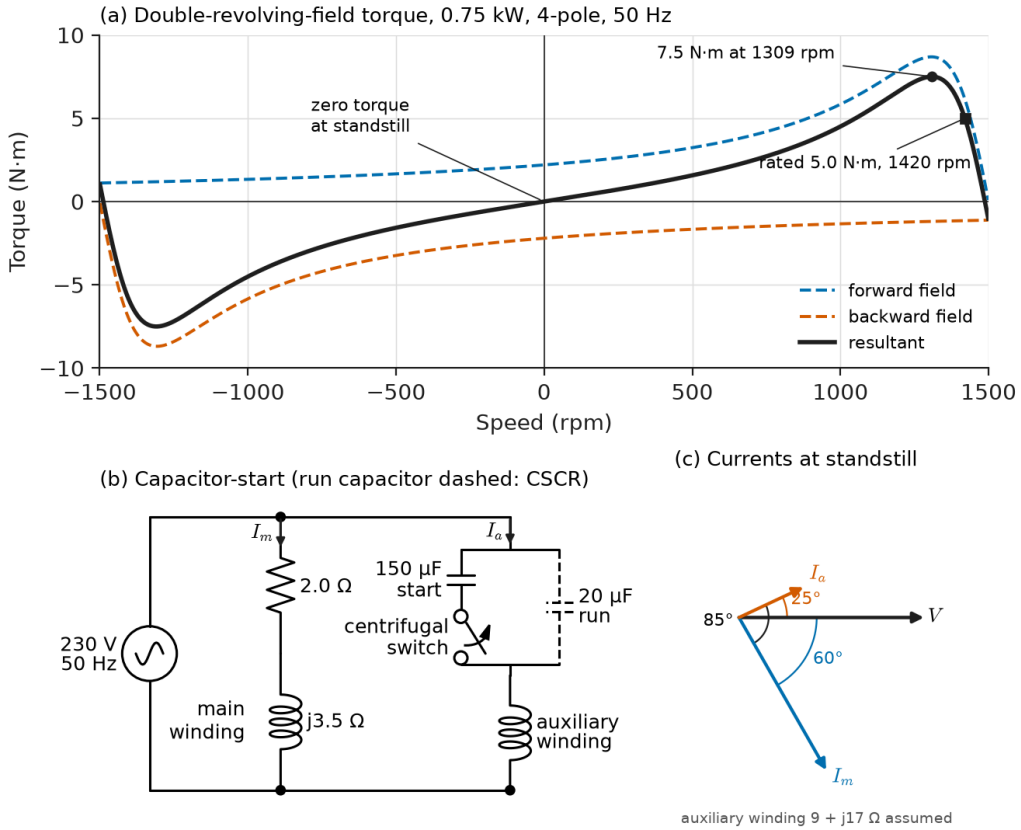


Figure 6.6 — Single-phase induction motor: forward and backward torque components and the capacitor-start connection.

Many applications that once used single-phase induction motors (refrigerator and air-conditioner compressors, washing machines, circulator pumps) have moved to brushless permanent-magnet motors with small inverters (Chapter 8), which are 20–40 % more efficient. Where a three-phase supply is unavailable for a larger motor, a single-phase-input VFD that produces three-phase output is now the usual solution up to a few kilowatts.

6.10 Motor selection

What the load demands

Selecting a motor starts from the load, not the motor catalogue. The engineer needs the load's torque–speed characteristic (Chapter 10), its inertia referred to the motor shaft, the starting duty (how often, how loaded), the speed range, the duty cycle and the environment. The motor must then satisfy:

1. **Power and speed**: rated output at least the maximum continuous load power, with a margin of 10–15 %, but not oversized (Section 6.8).
2. **Starting**: motor torque exceeds load torque at every speed during acceleration with margin (IEC 60034-12 recommends checking the minimum accelerating torque at the minimum expected voltage, often 80–90 %); acceleration time within the rotor's thermal limit (locked-rotor safe-stall time, typically 10–20 s hot).
3. **Duty type**: IEC 60034-1 defines S1 (continuous) to S10; a hoist or crane motor rated S3 40 % or S4 with a number of starts per hour has a different thermal rating from an S1 pump motor.

4. **Supply**: voltage, frequency and starting method; LV up to about 400 kW, MV (3.3–11 kV) above, because LV currents and cables become unwieldy.
5. **Enclosure and cooling**: IP55 (dust-protected, water-jet) is the industrial default, IP56/IP66 for wash-down and marine decks; cooling to IEC 60034-6, IC411 (totally enclosed fan-cooled) the most common, IC416 (separate fan) for VFD-fed motors running long periods at low speed.
6. **Insulation and temperature**: Class F insulation with Class B temperature rise is the standard combination, giving a thermal margin that roughly doubles insulation life (Chapter 9); derate above 40 °C ambient or 1000 m altitude.
7. **Hazardous areas**: motors in zones with flammable gas or dust require Ex certification (IEC 60079: Ex db flameproof, Ex eb increased safety, Ex ec non-sparking), with temperature classes T1–T6.
8. **Efficiency class and inverter suitability**: IE3/IE4 as required by regulation; for VFD use, reinforced winding insulation for voltage spikes and insulated bearings or shaft grounding against bearing currents above about 100 kW (IEC TS 60034-25).
9. **Mechanical**: frame size and mounting (IEC 60072, IM B3 foot, B5 flange), shaft and bearing loads, vibration grade, noise.

Table 6.7 — Typical motor choices by application.

Load	Torque characteristic	Typical choice
Centrifugal pump, fan	$T \propto n^2$, low starting torque	IEC N cage, IE3/IE4, VFD if flow varies; star–delta or soft start if fixed speed
Loaded belt conveyor	Constant torque, high breakaway (1.5–2 × rated)	IEC H cage with DOL or VFD; wound rotor or VFD above about 1 MW
Reciprocating compressor	Pulsating, needs flywheel	IEC N/H cage or synchronous motor (large); soft start with unloader
Crusher, ball mill	Very high inertia, constant torque	Wound rotor with liquid resistor, VFD, or synchronous motor above several MW
Hoist, crane	Constant torque, four-quadrant, S3–S5 duty	VFD with braking resistor or regenerative front end; wound rotor (legacy)
Marine thruster	$T \propto n^2$, frequent reversal	VFD-fed cage motor, IEC 60092 and class rules
EV traction	Constant torque then constant power, 10:1 speed range	PMSM or copper-rotor cage induction motor, inverter-fed

Worked example — Selecting a motor for a belt conveyor. A mining belt conveyor needs 30 kW at steady speed with a belt drive at 1480 rpm; breakaway torque when fully loaded is 1.8 times running torque, and the site supply is 400 V from a 500 kVA transformer. Running torque at 1480 rpm is $30000/(1480 \times 2\pi/60) = 194$ N·m, breakaway torque 349 N·m. A 37 kW IEC N motor (rated torque 244 N·m, locked-rotor torque about 2.1 × = 509 N·m) gives a 23 % power margin and 46 % starting-torque margin at full voltage. Star–delta starting is excluded: one third of 509 N·m is 170 N·m, below the 349 N·m breakaway. If the supply can stand a DOL start (on a 500 kVA, 4 % transformer the dip computed as in Section 6.6 is about 2.3 %), DOL with an IE3 37 kW, IP55, IC411, Class F motor is the simplest solution; at a 10 % undervoltage the locked-rotor torque falls to $0.81 \times 509 = 412$ N·m, still above breakaway. If belt tension control or frequent starts matter, a VFD sized for 150 % overload torque is the better engineering choice.

What remains open

Open question	Where it is answered
How does the synchronous machine, with a DC-excited rotor, differ from the induction machine?	Chapter 7
How do the permanent-magnet and reluctance machines that replace induction motors in EVs and appliances work?	Chapter 8
How are temperature rise, insulation class, duty and cooling turned into a rating, and how are motors tested?	Chapter 9
How are load characteristics, four-quadrant operation and drive protection handled?	Chapter 10
How do V/f, field-oriented and direct torque control work in a VFD?	Chapter 12
How does a DFIG wind turbine use the wound-rotor machine?	Chapter 15
How do large induction motors contribute to fault current and to voltage stability after faults?	Chapter 19 and Chapter 20
How do supply unbalance and harmonics heat motors, and how is motor starting assessed as a power-quality event?	Chapter 25
How are motor-replacement and VFD retrofit programmes evaluated over a life cycle?	Chapter 28

Next: Chapter 7 gives the rotor its own DC excitation, so that it turns exactly at synchronous speed, and develops the synchronous machine that generates almost all of the world's electricity.

Chapter 7 — Synchronous Machines

Part II · Transformers and Electrical Machines

A combined-heat-and-power plant at a paper mill runs a 20 MVA, 11 kV gas-engine generator in parallel with the utility. The plant engineer is asked to raise the generator's output from 12 MW to 16 MW and, at the same time, stop paying the utility's reactive-energy charge. The governor setpoint is raised, and the megawatts follow; the field current is raised, and the megavars follow. Nothing else changes: the terminal voltage stays at 11 kV and the frequency stays at 50 Hz, because the grid holds both. A month later a fault on the 33 kV line feeding the mill is cleared in 300 ms instead of the planned 120 ms, and the generator trips on pole slip. A second set at an offshore platform, running on its own with no grid behind it, behaves quite differently: there, raising the field current raises the voltage, and raising the fuel raises the frequency. All three behaviours come from one machine, the synchronous machine, and from where it is connected.

> **How does a synchronous machine produce a sinusoidal EMF at exactly grid frequency, and what sets its terminal voltage, its active power and its reactive power?**
> **How is it connected to a live grid, how do several machines share load, and how much power can it carry before it falls out of step?**

Concept	Engineering destination
Construction (cylindrical and salient rotors)	Turbo-generators, hydro units, gensets, wind and marine machines
Alternator operation and synchronous reactance	Per-phase models in power-flow and fault studies
EMF generation and winding factors	Waveform quality, harmonic voltage limits, winding design
Phasor diagrams	Field-current requirement, rotor angle, metering checks
Voltage regulation and OC/SC tests	Excitation sizing, test-certificate interpretation
Power-angle relationships	Transfer limits, rotor-angle monitoring, stability margins
Synchronization	Breaker-closing permissives, auto-synchronizer settings
Parallel operation	Governor droop, load sharing on islands and ships
Synchronous motors	Compressors, mills, LNG trains, power-factor correction
Excitation and reactive power	AVR, capability charts, grid-code reactive requirements, synchronous condensers
Stability fundamentals	Swing equation, inertia, critical clearing, RoCoF

Roadmap. What the machine is made of (7.1) → how it generates and how it is modelled (7.2) → the EMF and its waveform (7.3) → phasor diagrams for round and salient rotors (7.4) → voltage regulation and the standard tests (7.5) → the power-angle law (7.6) → connecting to a live bus (7.7) and sharing load (7.8) → the same machine as a motor (7.9) → what the field current does to reactive power (7.10) → how far the rotor angle can go before synchronism is lost (7.11).

7.1 Synchronous-machine construction

A synchronous machine has a stator carrying a distributed three-phase winding, the **armature**, and a rotor carrying a DC **field** winding (or permanent magnets, Chapter 8). The field produces a flux pattern fixed to the rotor; the rotor turns at exactly synchronous speed, so the armature sees a magnetic field moving at constant speed and an EMF at constant frequency. The frequency–speed–pole relation $f = pn/120$ is stated in *Circuits, Signals and Communications*, Ch. 10 §10.1; this chapter takes it as given.

Stator

The stator core is built from 0.35–0.65 mm laminations of non-oriented silicon steel (Chapter 9), stacked in segments with radial ventilation ducts every 40–60 mm in large machines. Slots on the bore hold the armature coils. Machines above a few MVA use **form-wound** coils or Roebel-transposed bars insulated with mica tape and resin (Chapter 9, §9.5), because terminal voltages of 11–27 kV demand a groundwall several millimetres thick. The armature is almost always star-connected, with the neutral earthed through a high resistance or a distribution transformer to limit earth-fault current to 5–15 A; a delta would let third-harmonic EMFs drive a circulating current.

Rotor: cylindrical and salient

Two rotor forms cover nearly every machine built (Figure 7.1):

- **Cylindrical (round, non-salient) rotors** are forged steel cylinders with axial slots milled along about two-thirds of the circumference, the field coils sitting in the slots and held by non-magnetic wedges and retaining rings. The air gap is uniform. They have two or four poles and run at 3000/3600 rpm or 1500/1800 rpm. Peripheral speeds reach 150–200 m/s, which limits rotor diameter to about 1.2–1.3 m; to reach high ratings the rotor is made long instead, up to 8 m of active length. Steam and gas turbo-generators use them.
- **Salient-pole rotors** carry separate pole bodies bolted or dovetailed to a hub or spider, each wound with a concentrated field coil. The air gap is shortest under the pole centre and much longer between poles. They have many poles, run slowly (60–750 rpm) and can be large in diameter: a 700 MW hydro generator at 107 rpm has a rotor of 16–20 m diameter. Hydro units, diesel and gas-engine gensets, and most synchronous motors use them.

The salient pole faces carry **damper (amortisseur) bars**, copper or brass bars short-circuited by end rings, like a partial induction-motor cage. They carry no current at steady synchronous speed; they damp rotor oscillations, carry negative-sequence currents under unbalance, and provide starting torque in synchronous motors (§7.9). Cylindrical rotors obtain the same effect from the solid forging and the slot wedges.

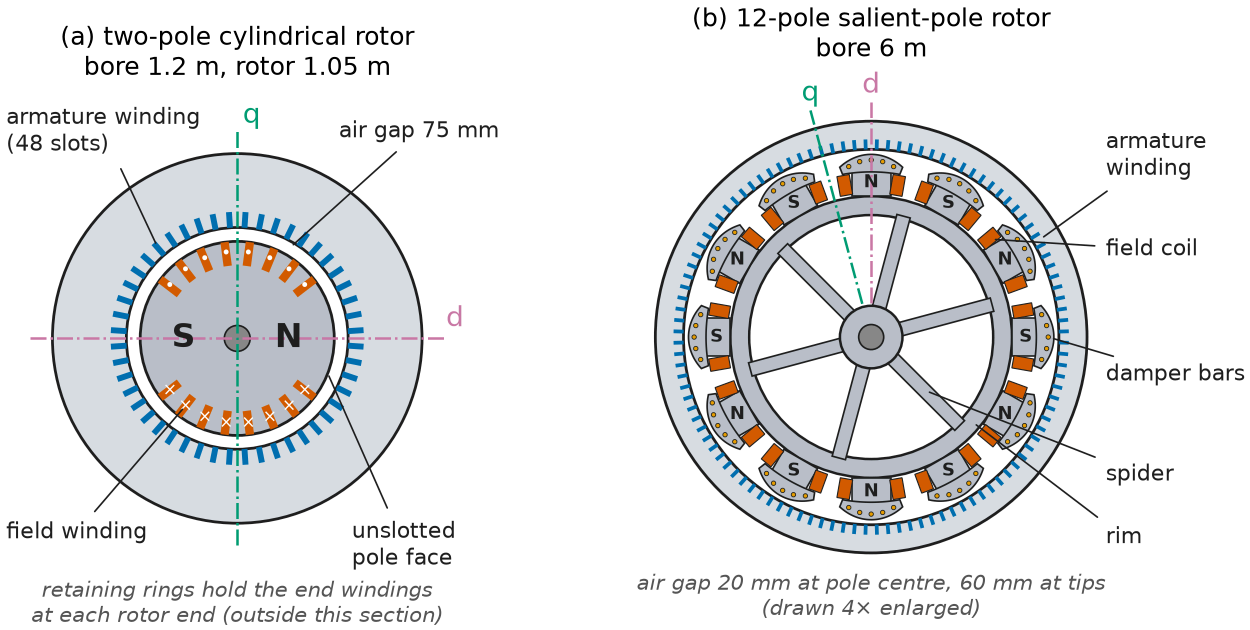


Figure 7.1 — Cross-sections of a two-pole cylindrical-rotor turbo-generator (a) and a 12-pole salient-pole hydro generator (b), drawn to show air-gap shape and field-winding placement.

Excitation systems

The field needs a DC supply of typically 0.5–1 % of the machine rating: a few hundred amperes at a few hundred volts for a 100 MVA unit, up to 6–7 kA in a 1000 MVA turbo-generator. Three arrangements are in use:

Excitation system	How the field is fed	Typical use	Response
DC exciter (legacy)	Shaft-driven DC generator through slip rings	Older units, small hydro	Slow (time constant 0.5–2 s)
Brushless (AC exciter + rotating diodes)	Inverted AC exciter on the shaft, rectified by a rotating diode wheel; no slip rings	Gensets, marine, most units up to ~300 MVA, synchronous motors	Medium (0.1–0.5 s)
Static (bus-fed thyristor)	Thyristor bridge supplied from the generator terminals through an excitation transformer, fed to the field via slip rings	Large turbo and hydro units, synchronous condensers	Fast (< 50 ms); ceiling voltage 2–3× rated

The automatic voltage regulator (AVR) commands the exciter; its dynamics belong to §7.10 and to Chapter 20. IEEE Std 421.5 gives the standard models used to represent all three in stability studies.

Ratings in the field

Table 7.1 — Representative synchronous machines.

Application	Rating	Voltage	Poles / speed (50 Hz)	Rotor	Cooling
Nuclear turbo-generator	1000–1750 MVA	22–27 kV	4 / 1500 rpm (some 2 / 3000)	Cylindrical	H_2 + water-cooled stator bars
Coal or CCGT steam/gas turbine	150–600 MVA	15–22 kV	2 / 3000 rpm	Cylindrical	Air or H_2
Large hydro	100–800 MVA	13.8–24 kV	28–96 / 125–214 rpm	Salient	Air, water-cooled stator in largest
Gas-engine CHP / data-centre genset	1–20 MVA	0.4–11 kV	4–6 / 1000–1500 rpm	Salient	Air (IC01, IC411)
Marine propulsion generator	3–15 MVA	6.6–11 kV	4–10 / 600–1500 rpm	Salient	Air–water (IC81W)
Synchronous motor (compressor, mill)	2–60 MW	6.6–13.8 kV	2–40	Salient or cylindrical	Air, air–water
Synchronous condenser	25–350 Mvar	11–20 kV	2–8	Salient or cylindrical	Air or H_2

7.2 Alternator operation

Armature reaction

When the alternator is loaded, the three armature currents produce their own rotating field, which, as shown in Chapter 3 for any balanced polyphase winding, travels at synchronous speed and is therefore stationary relative to the rotor. The air-gap field is the sum of the field-winding MMF and this **armature-reaction** MMF. Its effect depends on the phase of the current relative to the EMF:

- Current in phase with the EMF (resistive load) produces cross-magnetising MMF along the q-axis: it distorts the field and produces the electromagnetic torque.
- Current lagging the EMF by 90° (inductive load) produces MMF directly opposed to the field along the d-axis: it **demagnetises**, and the terminal voltage falls.
- Current leading by 90° (capacitive load, a long unloaded cable) **magnetises**: the terminal voltage rises, and an unregulated machine can self-excite.

Synchronous reactance and the per-phase model

Because the armature MMF is proportional to armature current and fixed in space relative to the rotor, its voltage effect in an unsaturated cylindrical machine is that of a reactance, the **armature-reaction reactance** X_{ad}. Adding the leakage reactance X_l of slot and end-winding flux gives the **synchronous reactance** $X_s = X_l + X_{ad}$, and the per-phase model of a cylindrical-rotor machine (generator convention) is

$$\mathbf{E}_f = \mathbf{V} + (R_a + jX_s)\,\mathbf{I}$$

where $\mathbf{E}_f$ is the open-circuit EMF set by the field current, $\mathbf{V}$ the terminal phase voltage and $\mathbf{I}$ the armature current. R_a is 0.002–0.01 pu and usually neglected beside X_s, which is 1–2.5 pu in turbo-generators. This is

the same Thévenin-source model that *Circuits, Signals and Communications*, Ch. 10 §10.5 uses for a three-phase source, now with a large internal reactance whose value comes from the machine's magnetic design.

> **Status of the principle: Synchronous-reactance model**
> *Approximate*. Valid for balanced steady state with a cylindrical rotor and an unsaturated (or linearised) magnetic circuit; it fails for salient rotors (use the two-reaction model of §7.4), under heavy saturation, under unbalance (negative-sequence reactance differs), and during transients, where the subtransient and transient reactances X''_d, X'_d govern (Chapter 19).

> **Boundary.**
> The machine as an energy converter is the gyrator of *Engineering Physics*, Ch. 17 §17.5: electrical effort and flow couple to torque and speed. Faraday's law (Ch. 11 §11.4) gives the induced EMF, and Chapters 2 and 3 of this book give the rotating field and the co-energy torque. Here we use those results in machine form.

Reactances and time constants

The synchronous reactance is only the steady-state member of a family. A sudden change in armature current is opposed by induced currents in the field and damper circuits, so the machine first behaves as a source behind the small **subtransient reactance** X''_d (decaying in tens of milliseconds), then behind the **transient reactance** X'_d (decaying over about a second), and finally behind X_d. The values matter in every chapter that follows: X''_d sets fault currents (Chapter 19), X'_d appears in transient stability (§7.11, Chapter 20), X_d and X_q set steady-state voltage and power behaviour.

Table 7.2 — Typical synchronous-machine parameters (per unit on machine rating; ranges collected from IEEE Std 1110 and manufacturers' data).

Parameter	Two-pole turbo	Four-pole turbo	Salient pole, with dampers	Synchronous condenser
X_d	1.6–2.3	1.0–2.0	0.6–1.5	1.5–2.5
X_q	1.5–2.2	1.0–1.9	0.4–1.0	0.9–1.5
X'_d	0.20–0.35	0.20–0.40	0.20–0.50	0.30–0.50
X''_d	0.12–0.25	0.15–0.25	0.15–0.35	0.18–0.30
X_l	0.10–0.20	0.10–0.20	0.10–0.25	0.10–0.20
T'_{d0} (s)	4–10	4–9	1.5–9	6–12
T''_{d0} (s)	0.02–0.05	0.02–0.05	0.02–0.05	0.02–0.05
Inertia constant H (s)	2.5–6 (incl. turbine)	4–8	2–4 (incl. turbine)	1–1.5 (more with flywheel)
Short-circuit ratio	0.4–0.6	0.5–0.7	0.9–1.5	0.4–0.6

7.3 EMF generation

The EMF equation

The rotor field sweeps each armature coil, and the induced phase EMF has the same form as the transformer EMF equation of Chapter 4, with two corrections for the fact that an armature phase is not one concentrated coil:

$$E_{ph} = 4.44\, f\, N_{ph}\, k_w\, \Phi$$

Here N_{ph} is the number of series turns per phase, Φ the fundamental flux per pole and $k_w = k_d\, k_p$ the **winding factor**.

> **Status of the principle: EMF equation of an AC winding**
> *Derived*. Follows from Faraday's law applied to a distributed winding in a sinusoidally distributed air-gap field; exact for the fundamental, and for each harmonic h with $f \to hf$, $\Phi \to \Phi_h$ and $k_w \to k_{wh}$.

Distribution and pitch factors

A phase belt spreads over q slots per pole per phase, adjacent slots being γ electrical degrees apart. The coil EMFs add as phasors with this angular spread, so their sum is less than the arithmetic sum by the **distribution factor**

$$k_d = \frac{\sin(q\gamma/2)}{q\sin(\gamma/2)}$$

A coil spanning less than a full pole pitch (a short-pitched or chorded coil, span ρ as a fraction of a pole pitch) links less than the full flux; the **pitch factor** is $k_p = \sin(\rho \cdot 90^\circ)$. For harmonic h the angles multiply by h:

$$k_{dh} = \frac{\sin(hq\gamma/2)}{q\sin(h\gamma/2)}, \qquad k_{ph} = \sin(h\rho \cdot 90^\circ)$$

The engineering point is that both factors are close to one for the fundamental and small for the harmonics. A 5/6 pitch reduces the fundamental by only 3.4 % and the 5th and 7th harmonics by about 74 %. Third and other triplen harmonics are in phase in all three phases and cancel in the line-to-line voltage of a star winding, so they never reach the grid; they do appear phase-to-neutral and must be considered when the neutral is earthed or when generators with different pitches are paralleled (circulating neutral current). IEC 60034-1 limits the telephone harmonic factor of generators above 300 kVA to 1.5 %, which in practice means a carefully pitched and distributed armature and skewed or shaped pole faces.

Worked example — EMF and flux of a 20 MVA CHP generator. A 20 MVA, 11 kV, 50 Hz, four-pole, star-connected generator has 48 stator slots and a two-layer winding with coil span 10 slots (full pitch 12 slots). Each phase has 48 series turns. Find the winding factor, the flux per pole for rated voltage, and the harmonic winding factors.

Slots per pole per phase: $q = 48/(4\times 3) = 4$. Slot angle: $\gamma = 180^\circ \times 4/48 = 15^\circ$ electrical.

$$k_d = \frac{\sin(4\times 15^\circ/2)}{4\sin(7.5^\circ)} = \frac{0.5}{4\times 0.1305} = 0.9577, \qquad k_p = \sin\!\left(\tfrac{10}{12}\times 90^\circ\right) = \sin 75^\circ = 0.9659$$

so $k_w = 0.9250$. The rated phase voltage is $11\,000/\sqrt{3} = 6351$ V, hence

$$\Phi = \frac{6351}{4.44 \times 50 \times 48 \times 0.9250} = 0.644 \text{ Wb}$$

With a bore diameter of 1.0 m and a core length of 1.3 m, one pole covers $\pi \times 1.0 \times 1.3/4 = 1.021$ m^2, so the average gap flux density is 0.63 T and the peak of a sinusoidal distribution is $0.63 \times \pi/2 = 0.99$ T, a typical design value. The harmonic winding factors are $k_{w5} = 0.205 \times 0.259 = 0.053$ and $k_{w7} = -0.158 \times 0.259 = -0.041$: the winding passes only 5.3 % and 4.1 % of any 5th and 7th harmonic present in the air-gap field, against 92.5 % of the fundamental. The 11th and 13th are reduced only to $|k_w| = 0.122$; they are slot harmonics ($h = 2q \cdot 3 \pm 1$), which is why the pole faces are skewed or the stator uses a fractional number of slots per pole per phase.

The open-circuit characteristic and saturation

With the rotor driven at synchronous speed and the armature open, the terminal voltage plotted against field current is the **open-circuit characteristic (OCC)**. At low field current it follows the straight **air-gap line**, because almost all the MMF is spent across the air gap; above about 0.8 pu voltage the iron begins to saturate and the OCC bends away. At rated voltage a typical machine needs 10–20 % more field current than the air-gap line predicts. This saturation is the magnetic-circuit saturation of Chapter 1 in a rotating machine, and it is the reason the linear model of §7.2 has to be used with a *saturated* synchronous reactance near rated voltage (§7.5).

7.4 Phasor diagrams

Cylindrical rotor

Take the terminal phase voltage $\mathbf{V}$ as reference and neglect R_a. The phasor $jX_s\mathbf{I}$ leads $\mathbf{I}$ by 90°, and $\mathbf{E}_f = \mathbf{V} + jX_s\mathbf{I}$. Figure 7.2 shows the three cases for the same current magnitude:

- **Lagging power factor** (the normal generator duty, supplying inductive load and reactive power): $jX_s\mathbf{I}$ points up and to the right, $\mathbf{E}_f$ is much larger than $\mathbf{V}$, and the machine is **over-excited**.
- **Unity power factor**: $jX_s\mathbf{I}$ is vertical, and $E_f = \sqrt{V^2 + (X_s I)^2}$.
- **Leading power factor** (absorbing reactive power): $jX_s\mathbf{I}$ points up and to the left; E_f may be less than V, and the machine is **under-excited**.

The angle δ between $\mathbf{E}_f$ and $\mathbf{V}$ is the **load angle** (power angle, torque angle). It is also, to within the pole-pair factor, the physical angle between the rotor's d-axis and the position it would occupy at no load, so it can be measured with a shaft-angle sensor, and modern generator protection relays compute it continuously.

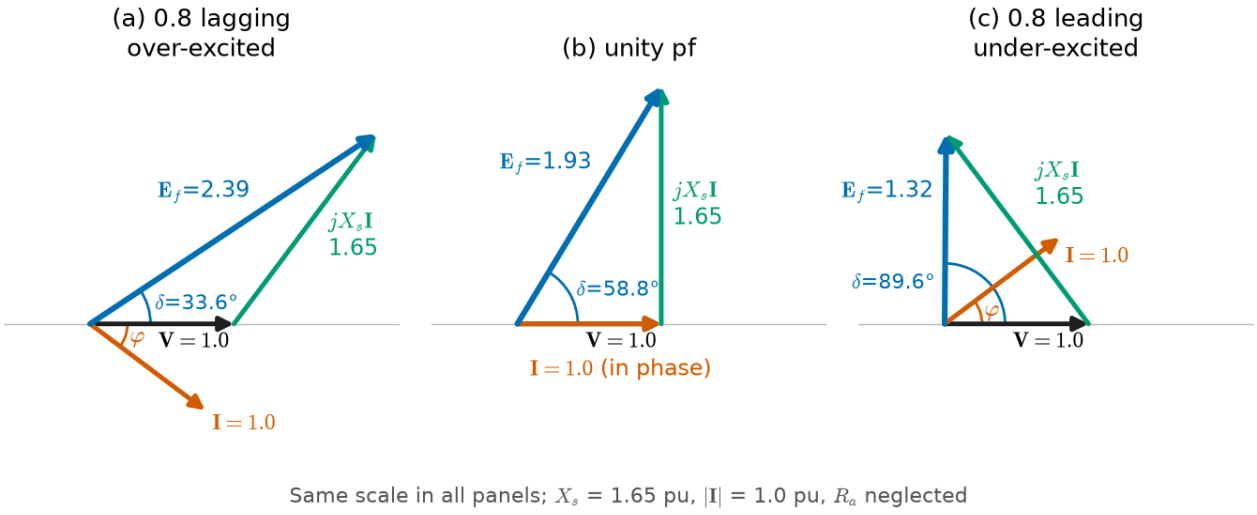


Figure 7.2 — Phasor diagrams of a cylindrical-rotor generator with Xs = 1.65 pu at rated current: (a) 0.8 lagging, (b) unity, (c) 0.8 leading power factor.

Salient rotor: the two-reaction model

In a salient-pole machine the air gap, and hence the reluctance, is very different along the pole axis (d-axis) and between poles (q-axis). An armature MMF along the d-axis produces more flux than the same MMF along the q-axis, so a single X_s cannot represent it. Blondel's **two-reaction model** resolves the armature current into components along the two axes and gives each its own reactance:

$$\mathbf{E}_f = \mathbf{V} + jX_d\mathbf{I}_d + jX_q\mathbf{I}_q, \qquad X_d > X_q$$

with $\mathbf{I}_d$ along the d-axis (in quadrature with $\mathbf{E}_f$) and $\mathbf{I}_q$ in phase with $\mathbf{E}_f$. Since the direction of $\mathbf{E}_f$ is unknown at the start, the construction uses a fictitious phasor $\mathbf{E}_Q = \mathbf{V} + jX_q\mathbf{I}$, which lies along $\mathbf{E}_f$ and so fixes δ; then $E_f = E_Q + (X_d - X_q)I_d$.

> **Status of the principle: Two-reaction (d–q) model**
> *Approximate.* Assumes sinusoidal winding distributions and neglects saturation and cross-coupling between axes; it is the basis of all dynamic machine models (Park's transformation) and is accurate to a few per cent in steady state for salient-pole and interior-PM machines (Chapter 8).

Worked example — load angle of a hydro generator. A salient-pole hydro generator has $X_d = 1.0$ pu and $X_q = 0.65$ pu (R_a neglected). It delivers rated current at rated voltage and 0.9 power factor lagging. Find δ and E_f.

With $\mathbf{V} = 1\angle 0^\circ$ and $\mathbf{I} = 1\angle -25.84^\circ = 0.9 - j0.4359$:

$$\mathbf{E}_Q = 1 + j0.65\,(0.9 - j0.4359) = 1.2833 + j0.585 = 1.410\angle 24.51^\circ$$

so $\delta = 24.51^\circ$. The angle between $\mathbf{I}$ and the q-axis is $\psi = 24.51^\circ + 25.84^\circ = 50.35^\circ$, giving $I_d = \sin 50.35^\circ = 0.770$ pu and $I_q = \cos 50.35^\circ = 0.638$ pu. Then

$$E_f = 1.410 + (1.0 - 0.65) \times 0.770 = 1.680 \text{ pu}$$

A cylindrical model with $X_s = X_d = 1.0$ would have given $|1 + j(0.9 - j0.4359)| = 1.695$ pu at $\delta = 32.1^\circ$: close in field current but 7.6° wrong in load angle, which matters for stability margins and for setting loss-of-

field and pole-slip protection.

7.5 Voltage regulation

Definition and what it is used for

The **voltage regulation** of an alternator is the rise in terminal voltage when rated load at a stated power factor is removed with field current and speed held constant:

$$\text{Regulation} = \frac{E_f - V_{\text{rated}}}{V_{\text{rated}}} \times 100\,\%$$

Unlike a transformer, whose regulation is a few per cent, an alternator with $X_s \approx 1.5$ pu has a regulation of 30–150 %. Nobody runs an alternator without an AVR, so regulation is not experienced as a voltage swing. It matters in three other ways: it fixes the **field current and exciter rating** needed at full load; it sets the overvoltage that appears on **sudden load rejection** before the AVR acts (the transient voltage rise of a genset that loses its load, limited by ISO 8528-5 for gensets); and it is the basis of the generator's test certificate.

Open-circuit and short-circuit tests

Two tests at synchronous speed characterise the machine (Figure 7.3):

1. **Open-circuit test**: armature open, field current raised in steps, terminal voltage recorded; this gives the OCC (§7.3).
2. **Short-circuit test**: the three terminals are shorted through a current transformer and the field raised until rated armature current flows. Because the current is almost purely reactive and demagnetising, the iron is unsaturated and the **short-circuit characteristic (SCC)** is a straight line.

From the two curves:

$$X_{s,\text{unsat}} = \frac{V_{\text{air-gap line}}(I_f)}{I_{sc}(I_f)}, \qquad \text{SCR} = \frac{I_f \text{ for rated voltage on OC}}{I_f \text{ for rated current on SC}} \approx \frac{1}{X_{s,\text{sat}}\ (\text{pu})}$$

The **short-circuit ratio** is the single number most quoted for a generator. A high SCR (hydro, 1.0–1.5) means a small X_s, a large air gap, a heavy and expensive machine, small load angles and good steady-state stability. A low SCR (modern turbo, 0.4–0.6) means a lighter, cheaper machine with larger load angles that relies on a fast AVR for stability.

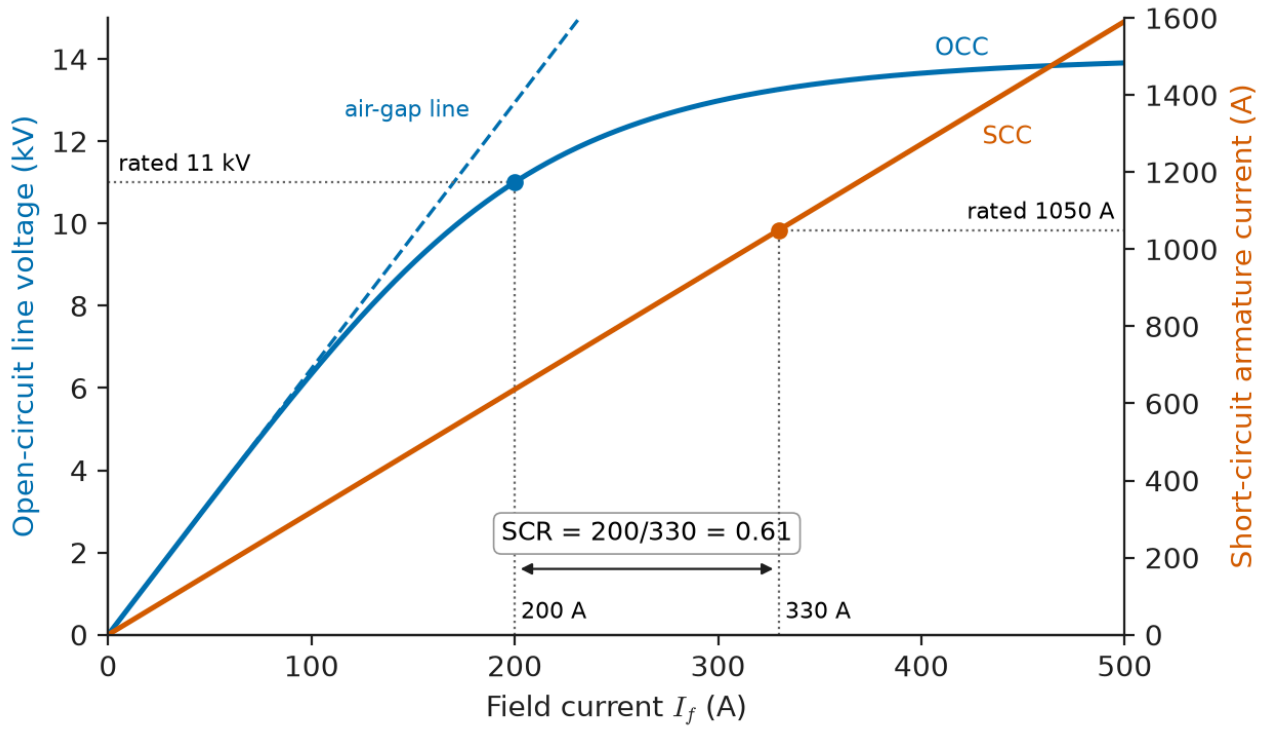


Figure 7.3 — Open-circuit and short-circuit characteristics of the 20 MVA, 11 kV generator, with the air-gap line and the field currents that define the short-circuit ratio.

Worked example — test data, synchronous reactance and regulation. The 20 MVA, 11 kV generator of §7.3 needs 200 A field current for rated voltage on open circuit and 330 A for rated armature current on short circuit. Find the SCR, the saturated synchronous reactance in per unit and ohms, and the regulation and field current at rated load for 0.8 lagging, unity and 0.8 leading power factor, using the linear (EMF) method.

Rated current $I_r = 20\times10^6/(\sqrt{3}\times 11\,000) = 1050$ A. Base impedance $Z_b = 11\,000^2/(20\times10^6) = 6.05\ \Omega$.

$$\text{SCR} = \frac{200}{330} = 0.606, \qquad X_s \approx \frac{1}{0.606} = 1.65\ \text{pu} = 1.65\times 6.05 = 9.98\ \Omega$$

With $\mathbf{V} = 1\angle 0^\circ$ pu and $|\mathbf{I}| = 1$ pu:

Power factor	$\mathbf{E}_f = \mathbf{V} + jX_s\mathbf{I}$	E_f (pu)	δ	Regulation	I_f (A, linear)
0.8 lagging	$1 + j1.65(0.8 - j0.6) = 1.99 + j1.32$	2.39	33.6°	139 %	478
Unity	$1 + j1.65$	1.93	58.8°	93 %	386
0.8 leading	$1 + j1.65(0.8 + j0.6) = 0.01 + j1.32$	1.32	89.6°	32 %	264

The field current is found by scaling 200 A by E_f, i.e. along the straight line through the rated OC point, which is how the saturated-reactance method treats the machine.

Two conclusions follow. First, at 0.8 leading the load angle is almost 90°: the machine is at its steady-state stability limit, which is why under-excited operation is restricted (§7.10). Second, the 139 % figure overstates what a real open-circuit voltage would be, because the OCC saturates: at 478 A the actual OC voltage is about 1.26 pu, not 2.39. The linear method is pessimistic about regulation and gives a conservative exciter size.

Better methods

Because the EMF (synchronous-impedance) method treats a saturating machine as linear, standards prescribe methods that separate the leakage drop from the saturation-dependent armature reaction. The **MMF method** adds field-current equivalents instead of voltages and is optimistic. The **zero-power-factor (Potier) method** uses a third test, rated current at zero power factor lagging, to extract the Potier reactance X_p (close to X_l) and reads the air-gap voltage behind X_p off the real OCC. The **ASA method** (IEEE Std 115) combines the two and is the basis of manufacturers' guaranteed field currents. For the example machine these methods give a rated-load field current of the order of 400 A rather than 478 A; read off the real OCC, that field gives about 13.6 kV on open circuit, a regulation of roughly 25 %, and the exciter can be sized correspondingly smaller.

7.6 Power-angle relationships

The cylindrical-rotor power-angle equation

With R_a neglected, the phasor diagram gives directly the active and reactive power delivered at the terminals, per phase:

$$P = \frac{E_f V}{X_s}\sin\delta, \qquad Q = \frac{E_f V\cos\delta - V^2}{X_s}$$

Three-phase totals are three times these with phase quantities, or the same expressions in per unit.

> **Status of the principle: Power-angle equation**
> *Derived*. Follows from complex power (*Circuits, Signals and Communications*, Ch. 9) applied to the source-behind-reactance model; it inherits that model's assumptions (balanced, steady state, $R_a \ll X_s$, constant X_s).

The same equation describes power transfer across any reactance between two voltage sources: a transmission line (Chapter 21), an interconnector, the coupling inductor of a grid-tied inverter (Chapter 27). Two facts dominate everything that follows:

1. **Active power is set by the load angle, reactive power mainly by the voltage magnitudes.** For small δ, $P \propto \delta$ and $Q \propto (E_f - V)$. Turbine torque moves P; field current moves Q.
2. **There is a maximum.** At $\delta = 90^\circ$, $P_{\max} = E_f V / X_s$. Beyond it, more mechanical input produces less electrical output, the surplus accelerates the rotor, and synchronism is lost.

Salient-pole power-angle equation

The two-reaction model adds a second term:

$$P = \frac{E_f V}{X_d}\sin\delta + \frac{V^2}{2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right)\sin 2\delta$$

The second term is **reluctance power**: torque produced by the rotor's tendency to align its low-reluctance axis with the stator field, with no field current at all. It is the principle of the reluctance and switched-reluctance motors of Chapter 8, and it is what lets a salient-pole synchronous motor keep running at light load after loss of excitation.

Worked example — power-angle curve of the hydro generator. For the machine of §7.4 ($E_f = 1.680$ pu, $X_d = 1.0$, $X_q = 0.65$, $V = 1$):

$$P(\delta) = 1.680 \sin\delta + 0.269 \sin 2\delta$$

At $\delta = 24.51^\circ$ this gives $P = 0.900$ pu, the load we started with. Searching for the maximum gives $P_{\max} = 1.758$ pu at $\delta = 74.2^\circ$, against 1.680 pu at 90° for the excitation term alone. The reluctance term raises the pull-out power by 4.6 % and moves it to a smaller angle. With zero excitation the machine can still carry 0.269 pu.

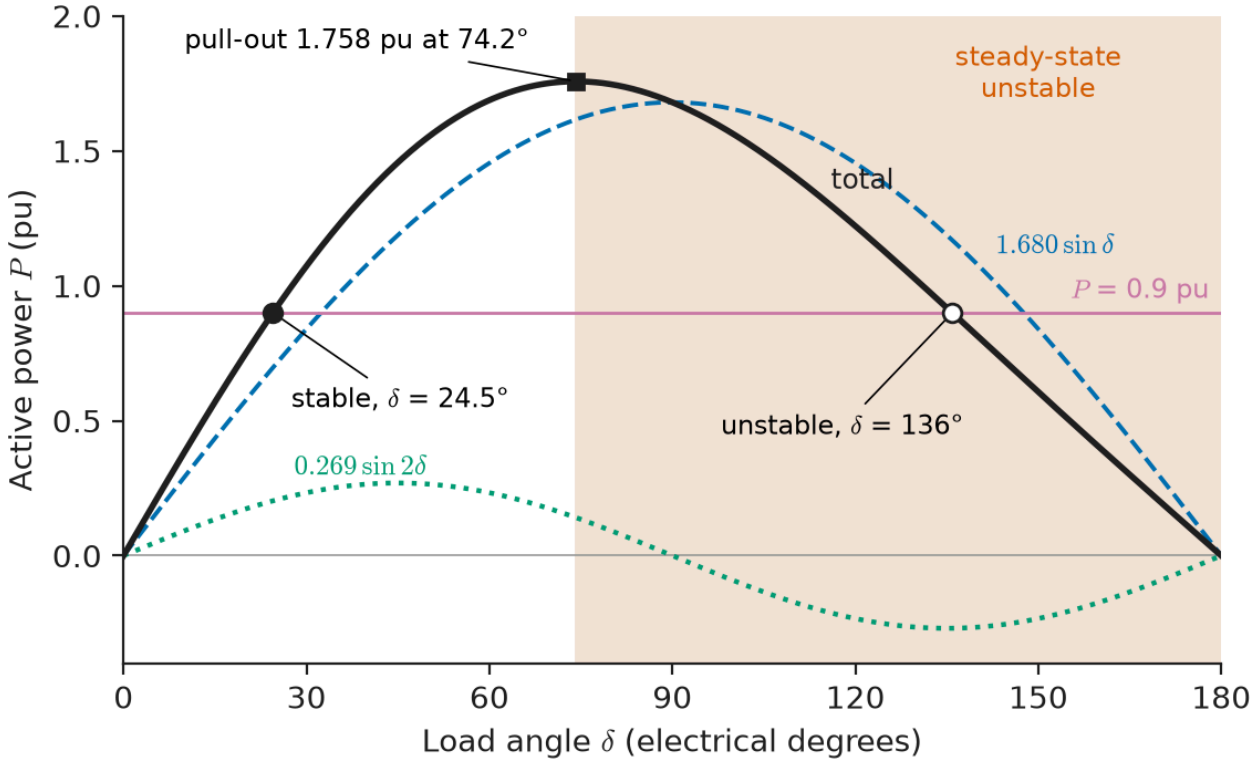


Figure 7.4 — Power-angle curves of the salient-pole hydro generator: excitation term, reluctance term and total, with the operating point and pull-out point.

The synchronizing power coefficient

The slope of the power-angle curve,

$$K_s = \frac{dP}{d\delta} = \frac{E_f V}{X_s}\cos\delta \quad \text{(cylindrical)}$$

is the **synchronizing power coefficient**: the restoring power per radian of rotor displacement. It is positive below the peak (stable) and zero at the peak. For the 20 MVA machine at rated 0.8 lagging load on an infinite bus, $P_{\max} = 2.388/1.65 = 1.447$ pu, $\delta = 33.6^\circ$ and $K_s = 1.447\cos 33.6^\circ = 1.21$ pu/rad. The steady-state stability margin $(P_{\max} - P)/P_{\max}$ is 45 % here, and much smaller in under-excited operation.

7.7 Synchronization

The four conditions

Closing a breaker between an incoming generator and a live bus is safe only if, at the instant the contacts touch, the two voltages are nearly identical. Four conditions must hold:

1. **Phase sequence** the same (checked once at commissioning with a sequence meter; a wrong sequence is a three-phase short circuit through two machines).
2. **Voltage magnitude** nearly equal.
3. **Frequency** nearly equal.
4. **Phase angle** nearly zero.

Any difference drives a current through the two machines' subtransient reactances. A phase error θ between two equal voltages V produces a voltage $2V\sin(\theta/2)$ across the closing breaker; with $X''_d = 0.2$ pu on the incoming machine and a stiff grid, a 10° error gives a current of $2\sin 5^\circ / 0.2 = 0.87$ pu, a tolerable transient, while an out-of-phase closure at 120° gives over 8 pu current and a shaft torque that can be several times the three-phase short-circuit torque, enough to crack couplings and loosen stator windings.

Acceptance limits

IEEE Std C50.12 (salient-pole) and C50.13 (cylindrical-rotor) give the closing window for large generators:

Quantity	Limit at breaker closing
Phase angle	within ±10°
Voltage	generator 0 to +5 % relative to bus
Frequency (slip)	within ±0.067 Hz

The slightly higher generator voltage and small positive slip ensure the machine picks up a little reactive and active power on closure rather than motoring. Smaller gensets use wider windows (±10–20°, ±0.1–0.3 Hz) set in the genset controller.

Synchronizing equipment

Older stations used **lamps** connected across the open breaker (dark-lamp method: lamps go dark when voltages are in phase) and a **synchroscope**, a rotating-pointer instrument whose speed and direction show the slip and whose position shows the phase angle. Closing was manual, at about five minutes to twelve on the dial with a slow clockwise rotation, so that the contacts touched at zero. Modern stations use an **auto-synchronizer** (ANSI device 25A) that adjusts the governor and AVR setpoints, predicts the zero-crossing of the phase angle from the slip, and issues the close command early by the breaker's closing time (40–100 ms). A separate **synch-check relay** (device 25) independently blocks closure outside the window. Data-centre switchgear that parallels four to sixteen standby gensets onto a common bus within 10–15 s of a utility failure relies on exactly this function, executed in parallel for every set.

7.8 Parallel operation

Connection to an infinite bus

An **infinite bus** is a supply whose voltage and frequency are unaffected by anything one machine does; a 20 MVA generator connected to a 10 GW grid is close to one. Once synchronized, the generator's speed is locked to the grid. Its two controls then act on different quantities:

- Raising the **governor setpoint** (more steam, gas or water) increases the mechanical torque, the rotor advances, δ increases, and the machine delivers more **active power**. Frequency does not change.
- Raising the **field current** increases E_f; with V fixed by the grid, the machine delivers more **reactive power** (lagging), and its current and power factor change. Active power does not change (to first order).

This decoupling is exactly what the paper-mill engineer observed in the opening.

Load sharing by governor droop

When several machines feed an isolated system (a ship, an oil platform, a mine site, a data centre on standby power, a microgrid) there is no infinite bus. Frequency is set by the balance between total mechanical input and total load, and voltage by the total reactive balance. For stable sharing each governor has a **droop** characteristic: its speed setpoint falls linearly with its output,

$$f = f_{nl} - R\, f_{\text{rated}}\, \frac{P}{P_{\text{rated}}}$$

where f_{nl} is the no-load frequency setpoint and R the droop, typically 3–5 %. All machines on the bus must run at the same frequency, so each settles where its own droop line crosses the common frequency.

> **Status of the principle: Droop load sharing**
> *Approximate.* Assumes linear governor characteristics and a common steady-state frequency; machines with equal per-unit droop share load in proportion to rating. It ignores dead bands and the dynamics of the prime mover, which dominate the first seconds (Chapter 20).

Worked example — two diesel generators on a ship. A 60 Hz ship's switchboard is fed by G1 (3 MW, droop 4 %, no-load setpoint 62.4 Hz) and G2 (2 MW, droop 5 %, no-load setpoint 62.0 Hz). The load is 3.5 MW. Find the sharing and the bus frequency, then the setpoints that restore 60 Hz with sharing in proportion to rating.

Slopes: G1 $0.04 \times 60/3 = 0.8$ Hz/MW; G2 $0.05 \times 60/2 = 1.5$ Hz/MW. Equal frequency and total load:

$$62.4 - 0.8P_1 = 62.0 - 1.5(3.5 - P_1) \;\Rightarrow\; P_1 = \frac{62.4 - 62.0 + 5.25}{2.3} = 2.457 \text{ MW}$$

so $P_2 = 1.043$ MW and $f = 62.4 - 0.8 \times 2.457 = 60.43$ Hz. G1 is at 82 % of rating, G2 at 52 %: unequal droops and setpoints produce unequal per-unit sharing. Proportional sharing at 60 Hz requires $P_1 = 2.1$ MW and $P_2 = 1.4$ MW, so the setpoints become $60 + 0.8 \times 2.1 = 61.68$ Hz and $60 + 1.5 \times 1.4 = 62.10$ Hz. A power-management system on board does this automatically (secondary control); grids do it with automatic generation control (Chapter 20).

Reactive sharing

Reactive power is shared the same way through **voltage droop** (reactive droop compensation in the AVR, typically 3–6 % voltage reduction at rated lagging reactive current) or through a cross-current compensation loop that equalises reactive currents. Without it, a machine whose AVR setpoint is slightly high takes on most of the reactive load, a circulating reactive current flows between machines, and one generator can end up over-excited while its neighbour is driven into under-excitation and trips on loss of field.

7.9 Synchronous motors

Motor operation

Reverse the power flow and the same machine is a motor. In motor convention $\mathbf{V} = \mathbf{E}_f + jX_s\mathbf{I}$, the load angle becomes negative (the rotor lags the stator field), and the motor runs at exactly synchronous speed at every load up to its **pull-out torque**,

$$T_{\max} = \frac{3E_f V}{\omega_s X_s}$$

(phase quantities, ω_s the mechanical synchronous speed), typically 1.5–2.5 times rated torque. There is no slip, no rotor copper loss in steady state and no magnetising current drawn from the supply, since the field supplies the flux. These three properties explain where synchronous motors are used: large, constant-speed, continuously running drives where efficiency and power factor are worth a more complex machine.

Application	Rating	Why synchronous
Centrifugal and reciprocating compressors (refineries, air separation, LNG)	5–60 MW	Efficiency 97–98.5 %, pf correction, constant speed
Ball and SAG mills (mining, cement)	5–28 MW (ring motors, gearless)	Low speed, very high torque, pf 1.0 or leading
Pumps for water transfer and pumped storage (as motor-generator)	10–500 MW	Bidirectional machine, efficiency
Ship propulsion with VFD	5–40 MW	Efficiency, high power at low speed
Wind-tunnel fans, rolling mills with VFD	10–100 MW	Precise speed, high torque

Starting

At standstill a synchronous motor connected to the grid produces only an oscillating torque at slip frequency, with zero average: the rotor cannot follow a field rotating at 1500 rpm from rest. Starting methods:

- **Induction (asynchronous) starting** on the damper cage, with the field winding short-circuited through a discharge resistor (5–10 times field resistance) to avoid dangerous induced field voltages. Near 95–97 % speed the DC field is applied at the right rotor angle and the motor **pulls into step**. Starting current is 4–7 pu, so large motors use reduced-voltage starting via reactor or autotransformer (Chapter 6).
- **Variable-frequency starting**: a load-commutated inverter or a medium-voltage VFD accelerates the motor synchronously from zero, then hands over to the line (soft-start of pumped-storage units and

large compressors), or runs it permanently (Chapter 12).
- **Pony motor**: a small auxiliary motor brings the unloaded machine to speed; used for synchronous condensers.

Synchronous-motor power-factor correction

Worked example — compressor motor improving plant power factor. An air-separation plant draws 5 MW at 0.75 lagging from its induction-motor loads. A new 2 MW synchronous compressor motor rated 2.5 MVA, 0.8 leading, $X_s = 1.2$ pu, is added. Find the excitation EMF at rated load and the plant power factor.

Motor convention, $\mathbf{V} = 1\angle 0^\circ$, $\mathbf{I} = 1\angle +36.87^\circ$:

$$\mathbf{E}_f = \mathbf{V} - jX_s\mathbf{I} = 1 - j1.2(0.8 + j0.6) = 1.72 - j0.96 = 1.970\angle -29.2^\circ \text{ pu}$$

The motor needs almost twice the no-load field. It supplies $2.5 \times 0.6 = 1.5$ Mvar to the plant. The plant's reactive demand is $5\tan(\cos^{-1}0.75) = 4.41$ Mvar; the net is $4.41 - 1.5 = 2.91$ Mvar, and the power factor at the utility meter becomes

$$\text{pf} = \frac{7}{\sqrt{7^2 + 2.91^2}} = 0.923 \text{ lagging}$$

compared with 0.75 before. The same motor at unity power factor would need only $E_f = |1 - j1.2 \times 0.8| = 1.386$ pu.

7.10 Excitation and reactive power

V-curves

Hold the shaft power of a synchronous machine constant and vary its field current. The active component of armature current stays fixed; the reactive component changes sign as the excitation passes through the value that gives unity power factor. Plotting armature current against field current gives the **V-curves** of Figure 7.5; the minimum of each V is unity power factor, and joining the minima gives the **unity-power-factor locus**. To the right of it the machine is over-excited (as a generator it supplies lagging vars; as a motor it draws leading current). To the left it is under-excited. The left end of each curve is cut off by the stability limit, $\delta = 90^\circ$, where $E_f = PX_s/V$.

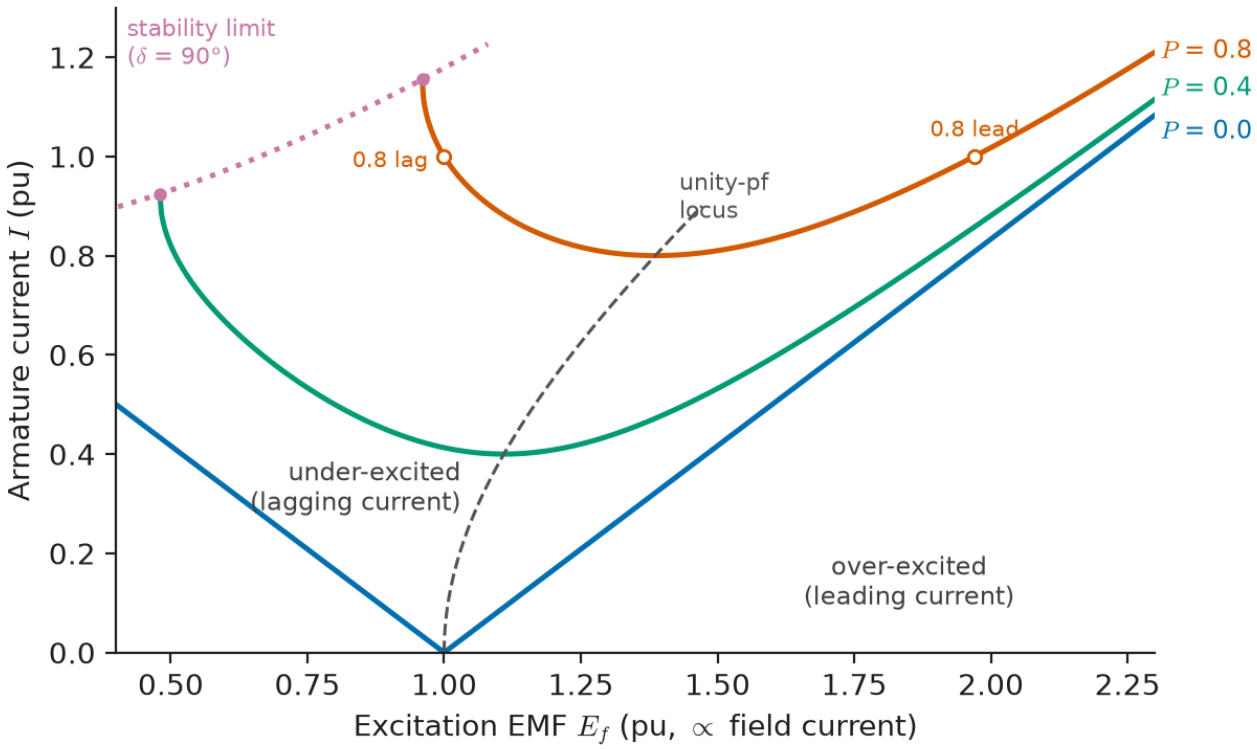


Figure 7.5 — V-curves of a synchronous motor with Xs = 1.2 pu at constant shaft power 0, 0.4 and 0.8 pu.

The synchronous condenser

A synchronous motor with no shaft load is a **synchronous condenser**: a continuously variable source or sink of reactive power controlled only by field current, from about −0.5 to +1.0 pu. After decades of replacement by static var compensators, condensers are being installed again (in South Australia, Great Britain, Denmark, Texas) because, unlike power-electronic var sources, they also supply **short-circuit current** (5–7 pu through X''_d), which protection and the grid-forming stability of inverter-dominated grids need, and **inertia**, especially when fitted with a flywheel (§7.11, Chapter 27). Retired coal-plant generators are often converted by decoupling the turbine.

The generator capability chart

A generator's allowable operating region in the P–Q plane is bounded by four limits (Figure 7.6):

1. **Armature (stator) heating**: $\sqrt{P^2+Q^2} \le S_{\text{rated}}$, a circle about the origin.
2. **Field heating**: $E_f \le E_{f,\max}$, set by the rated field current at rated MVA and power factor. In the per-unit P–Q plane this is a circle of radius $E_{f,\max}V/X_s$ centred at $(0, -V^2/X_s)$; it limits the over-excited region.
3. **Prime-mover limit**: a horizontal line at the turbine's maximum power.
4. **Under-excited limit**: steady-state stability (δ approaching 90°, with margin) and, for turbo-generators, heating of the stator core ends, which increases sharply under leading power factor because the end-region flux is then axial and induces eddy currents in the end laminations.

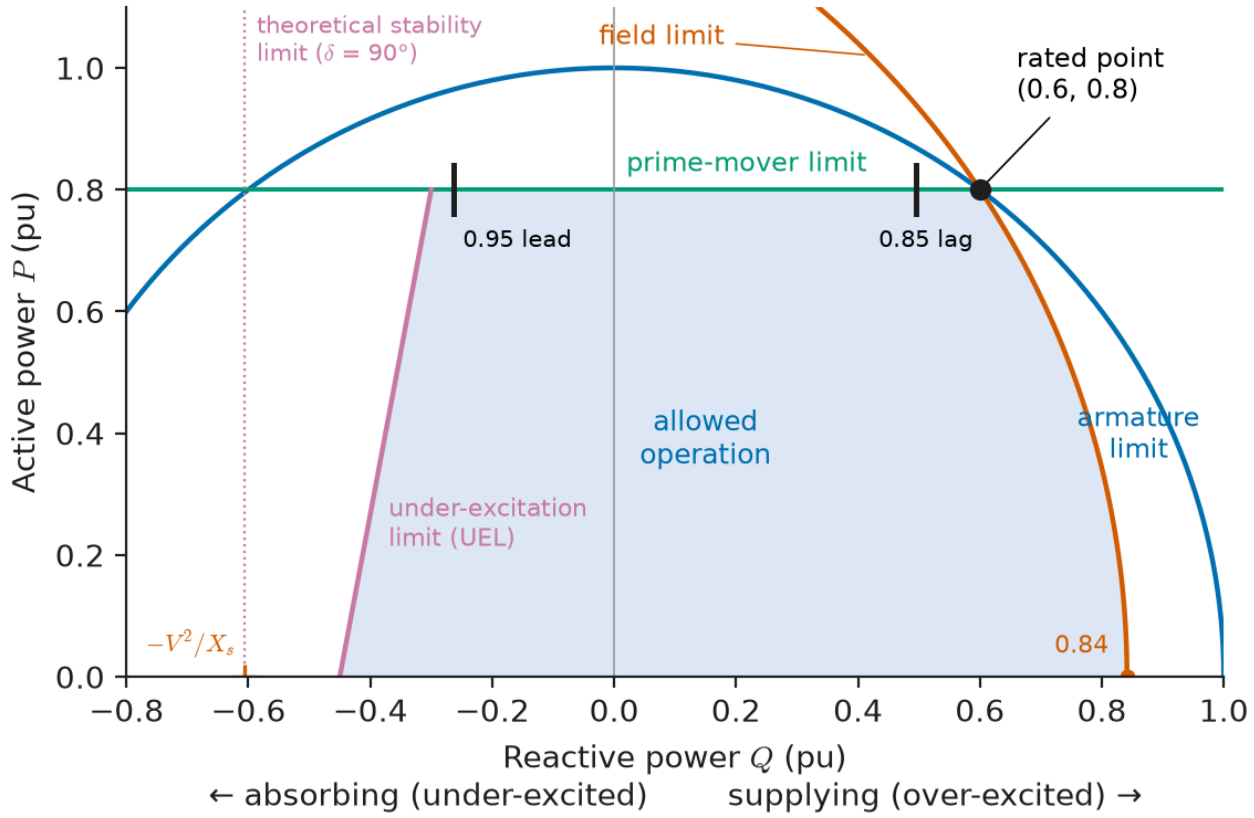


Figure 7.6 — Capability chart of the 20 MVA, 0.8 pf generator with Xs = 1.65 pu: armature, field, prime-mover and under-excitation limits.

Grid codes translate this chart into obligations. A typical transmission-connected synchronous unit must be able to operate between 0.85 power factor lagging and 0.95 leading at rated output (for example the GB Grid Code, CC.6.3.2), and within a voltage band of ±5 % (±10 % for limited times); the ENTSO-E Requirements for Generators (EU 2016/631) set comparable $Q/P_{\max}$ ranges by system and voltage. The capability chart shows whether a given machine can meet them.

The AVR and its limiters

The AVR measures terminal voltage and adjusts field voltage, typically holding voltage within ±0.5 %. Its gain, time constants and the ceiling voltage of the exciter decide how strongly the machine supports the grid during faults; the feedback design belongs to *Circuits, Signals and Communications*, Ch. 34. Around the AVR sit limiters that keep it inside the capability chart: the **over-excitation limiter** (OEL), which allows field forcing to 1.6–2 times rated for a few seconds during faults and then ramps back with an inverse-time characteristic matched to the field's thermal capability (IEEE C50.13); the **under-excitation limiter** (UEL), which stops the AVR from driving the machine below its stability and core-end limits; and the **V/Hz limiter**, which protects the generator and step-up transformer cores from overfluxing at low speed. A **power-system stabiliser** (PSS) adds a damping signal derived from speed or power to counter the electromechanical oscillations of §7.11, which a high-gain AVR otherwise tends to under-damp.

7.11 Stability fundamentals

Steady-state stability

From §7.6, a machine connected through total reactance X to an infinite bus can transfer at most $P_{\max} = EV/X$. Operating close to it leaves no margin for a voltage dip or a line outage that raises X. Planning criteria typically require the steady-state load angle across a transmission corridor to stay below 30–45° and the system to remain stable after the loss of any single element (the N−1 criterion, Chapter 13).

The swing equation

When the mechanical input P_m and electrical output P_e differ, the difference accelerates the rotor. Newton's law for a rotating body (*Engineering Physics*, Ch. 9 §9.7 and Ch. 19 §19.2) written in per unit on the machine rating, with the rotor angle measured in electrical radians against a synchronously rotating reference, gives the **swing equation**:

$$\frac{2H}{\omega_s}\frac{d^2\delta}{dt^2} = P_m - P_e(\delta) - D\frac{d\delta}{dt}$$

where $\omega_s = 2\pi f$ is the electrical synchronous speed, D a damping coefficient representing damper-winding and load effects, and H the **inertia constant**: kinetic energy at synchronous speed divided by rated MVA,

$$H = \frac{\tfrac{1}{2}J\omega_{sm}^2}{S_{\text{rated}}} \quad \text{(seconds)}$$

A 600 MVA steam unit with $H = 3.5$ s stores 2100 MJ in its rotating masses, enough to supply its full rating for 3.5 s.

> **Status of the principle: Swing equation**
> *Derived*. Newton's second law for rotation applied to the turbine–generator shaft, with power in place of torque (valid because speed stays within a few per cent of synchronous). It treats the shaft as a single rigid mass; torsional modes between turbine stages and the generator need multi-mass models (subsynchronous resonance, Chapter 20).

Small oscillations

Linearise about an operating point δ_0: $P_e \approx P_0 + K_s\Delta\delta$. With damping neglected the swing equation becomes a harmonic oscillator, and the rotor swings at the **natural frequency of electromechanical oscillation**

$$\omega_n = \sqrt{\frac{\omega_s K_s}{2H}}$$

> **Status of the principle: Linearised electromechanical oscillation**
> *Approximate*. Valid for small disturbances around a stable operating point with the classical model (constant EMF behind transient reactance); it is the starting point for small-signal stability analysis (Chapter 20).

Worked example — rotor oscillation and transfer margin of a 600 MVA unit. A 600 MVA, 50 Hz turbo-generator ($H = 3.5$ s) is connected through a step-up transformer ($X_t = 0.15$ pu) and two parallel lines (combined 0.35 pu) to a strong grid at $V = 1$ pu. In the classical transient model the machine is an EMF $E' =$

1.15 pu behind $X'_d = 0.30$ pu. It delivers 0.9 pu. Find the load angle, the synchronizing coefficient, the oscillation frequency, and the effect of losing one line.

Total reactance $X = 0.30 + 0.15 + 0.35 = 0.80$ pu, so $P_{\max} = 1.15/0.80 = 1.4375$ pu.

$$\delta_0 = \sin^{-1}\frac{0.9}{1.4375} = 38.8^\circ, \qquad K_s = 1.4375\cos 38.8^\circ = 1.121 \text{ pu/rad}$$

$$\omega_n = \sqrt{\frac{314.16 \times 1.121}{2 \times 3.5}} = 7.09 \text{ rad/s} \quad \Rightarrow \quad f_n = 1.13 \text{ Hz}$$

That is the typical local-mode frequency (0.8–2 Hz) seen on generator power traces after a disturbance. The transfer margin is $(1.4375 - 0.9)/1.4375 = 37\,\%$. If one line trips, the line reactance doubles to 0.70 pu, $X = 1.15$ pu and $P_{\max} = 1.0$ pu: the unit can still deliver 0.9 pu, but at $\delta = 64.2^\circ$ with only 10 % margin, and K_s falls to 0.44 pu/rad, lowering f_n to 0.70 Hz. Operators would reduce output; a protection scheme might do it automatically.

Transient stability and inertia

A large disturbance, a three-phase fault near the generator, collapses P_e almost to zero while P_m continues; the rotor accelerates for as long as the fault lasts. When the fault is cleared, P_e jumps above P_m and decelerates it. Whether the rotor swings back or passes the unstable equilibrium and **slips a pole** depends on the clearing time. The equal-area criterion and the critical clearing time are developed in Chapter 20; for the unit above, faults at the high-voltage terminal must be cleared in roughly 150–250 ms, which is why transmission protection is designed for 80–120 ms total clearing time (Chapter 23). The CHP generator in the opening, with a low H of the order of 1 s, lost synchronism because the 300 ms clearing time exceeded its critical value.

Inertia also governs the system frequency. When a generator of power ΔP trips, all remaining synchronous machines decelerate together, and the initial **rate of change of frequency** is

$$\text{RoCoF} = \frac{\Delta P \, f_0}{2 H_{\text{sys}} S_{\text{sys}}}$$

Losing 10 % of generation in a 50 Hz system with an aggregate H of 4 s gives $0.1 \times 50/(2 \times 4) = 0.625$ Hz/s. As inverter-based plant replaces synchronous machines, H_{sys} falls and RoCoF rises; several grid operators now limit it to 0.5–1 Hz/s and buy inertia from synchronous condensers. This connects the synchronous machine directly to the frequency-stability questions of Chapters 20 and 27.

What remains open

Open question	Where it is answered
How do the stator windings, losses and cooling of a synchronous machine set its rating, and how is it tested?	Chapter 9
How do permanent-magnet synchronous machines differ from wound-field ones, and why do they dominate EV traction?	Chapter 8
How are synchronous motors driven at variable speed by inverters?	Chapter 12
What turbines, boilers and dams drive these generators, and what are their auxiliary loads?	Chapter 14
How are generators represented in per-unit network models and power-flow studies?	Chapters 17 and 18
How large are generator fault currents, and how do X''_d and X'_d shape them?	Chapter 19
Equal-area criterion, critical clearing time, small-signal modes, PSS tuning and frequency control	Chapter 20
How are generators protected against faults, loss of field and pole slipping?	Chapter 23
How do grid-forming inverters emulate synchronous-machine inertia and voltage support?	Chapter 27
Feedback design of the AVR and governor loops	*Circuits, Signals and Communications*, Ch. 33–35

Next: Chapter 8 replaces the wound field with permanent magnets, the commutator with transistors and the rotating field with switched reluctance, and follows the synchronous principle into EVs, robots, drones and wind turbines.

Chapter 8 — Special Electrical Machines

Part II · Transformers and Electrical Machines

An electric-vehicle team has a 400 V battery, a gearbox ratio of 9:1 and a target of 0 to 100 km/h in seven seconds with a top speed of 180 km/h. An induction motor would do it, and so would a wound-field synchronous machine. The team picks an interior-permanent-magnet machine instead, because it fits in a 25 cm housing, reaches 95–97 % efficiency over most of the drive cycle, and adds range without adding battery. Down the corridor, a mining-equipment company has rejected the same machine for a battery shuttle car: magnets can be demagnetised by a hot short-circuit underground, and the duty is brutal, so it has chosen a switched-reluctance motor with no magnets and no rotor winding at all. Next door, a packaging-line integrator is choosing between a stepper motor that costs a fifth as much and a servo that will not lose position when a carton jams. These are the special electrical machines: built on the same physics as Chapters 5–7, but shaped by permanent magnets, power electronics and the application.

> **How do permanent magnets, electronic commutation and variable reluctance change what a motor can do, and where does each design reach its limits?**
> **How are these machines matched to traction, motion control, appliances, transport and wind generation?**

Concept	Engineering destination
Permanent-magnet synchronous motors	EV traction, industrial IE5 motors, compressors, elevators
Brushless DC motors	Fans, pumps, drones, e-bikes, disk drives, data-centre cooling
Switched-reluctance motors	Mining vehicles, aerospace starter-generators, harsh-duty drives
Stepper motors	3D printers, packaging, medical pumps, valve actuators
Universal motors	Power tools, vacuum cleaners, kitchen appliances
Linear motors	Maglev, metro LIM trains, aircraft-carrier launch, machine tools
Servo motors	Robots, CNC axes, pick-and-place, semiconductor equipment
Specialized generators	Wind turbines, aircraft, microturbines, vehicles, small hydro
Selection trade-offs	Choosing a machine against cost, efficiency, torque density and risk

Roadmap. Magnets replace the field winding (8.1) → electronic commutation of a trapezoidal machine (8.2) → torque from reluctance alone (8.3) → stepping instead of rotating (8.4) → the commutator machine that runs on AC (8.5) → unrolling the machine into a line (8.6) → closing the loop for precise motion (8.7) → generators shaped by their prime movers (8.8) → choosing among all of them (8.9).

Boundary.
Torque from co-energy and the rotating field are taken from Chapter 3; synchronous-machine phasors and the two-reaction model from Chapter 7; the electromechanical converter as a gyrator from *Engineering Physics*, Ch. 17 §17.5. Field-oriented and direct torque control are in Chapter 12; inverter and gate-driver circuits belong to *Electronics & Computing*; magnet metallurgy belongs to *The Materials Core*.

8.1 Permanent-magnet synchronous motors

Why replace the field winding

A wound field costs copper loss: 0.5–1 % of rating in a large synchronous machine, but 10–20 % of total losses in a 10 kW machine, where the field must still magnetise a gap of comparable size. A permanent magnet supplies that MMF with no loss and no slip rings. It also allows a rotor with no current-carrying parts, which is cooler (rotor heat is the hardest to remove) and smaller. The **permanent-magnet synchronous motor (PMSM)** is a synchronous machine whose field is fixed by its magnets; it runs at synchronous speed only when fed from an inverter at the corresponding frequency (or, in line-start designs, with a starting cage).

Magnet materials

Table 8.1 — Representative permanent-magnet materials (typical catalogue values at 20 °C).

Material	Remanence B_r (T)	Intrinsic coercivity H_{cJ} (kA/m)	$(BH)_{\max}$ (kJ/m³)	Max. operating temperature	B_r temperature coefficient
NdFeB, grade N42	1.28–1.32	≥ 960	320–340	80 °C	−0.12 %/K
NdFeB, grade 42SH (Dy/Tb grain-boundary diffused)	1.28–1.32	≥ 1590	320–340	150 °C	−0.11 %/K
NdFeB, grade 38UH	1.22–1.25	≥ 1990	290–310	180 °C	−0.11 %/K
SmCo (Sm_2Co_{17})	1.05–1.15	1400–2000	200–240	300–350 °C	−0.03 %/K
Sintered ferrite (Sr)	0.38–0.45	250–350	28–36	250 °C (risk of demagnetisation when cold)	−0.20 %/K
AlNiCo 5	1.25	50	40–44	500 °C	−0.02 %/K

Three engineering facts follow. Neodymium magnets give three times the flux density of ferrite and ten times the energy product, which is why NdFeB machines dominate where volume and mass are expensive (vehicles, aircraft, robots). Their coercivity falls with temperature, so high-temperature grades need dysprosium or terbium, which are scarce and price-volatile; this supply risk drives the development of magnet-free traction motors (§8.9). And every magnet has a **demagnetisation limit**: a stator current that drives the operating point below the knee of the B–H curve, most likely during a short circuit at high temperature, permanently

weakens the rotor. Machine design keeps a margin of 20–30 % between the worst-case demagnetising field and H_{cJ} at maximum temperature.

Surface and interior magnets

Two rotor geometries are used (Figure 8.1):

- **Surface-mounted PM (SPM)**: magnets glued to the rotor surface and held by a carbon-fibre or stainless sleeve. Since magnet permeability is close to that of air, the effective gap is large and uniform: $L_d \approx L_q$, low inductance, low armature reaction, smooth torque. Used in servo motors (§8.7), high-speed machines and BLDC motors.
- **Interior PM (IPM)**: magnets buried in slots inside the rotor laminations, often in a single or double V. The magnets are mechanically protected and shielded from stator harmonics, and the iron bridges make the rotor salient with $L_q > L_d$ (the opposite of the wound-field salient-pole machine, because the low-permeability magnets lie on the d-axis). The machine produces **reluctance torque** in addition to magnet torque. Used in nearly all EV traction motors and in variable-speed compressors.

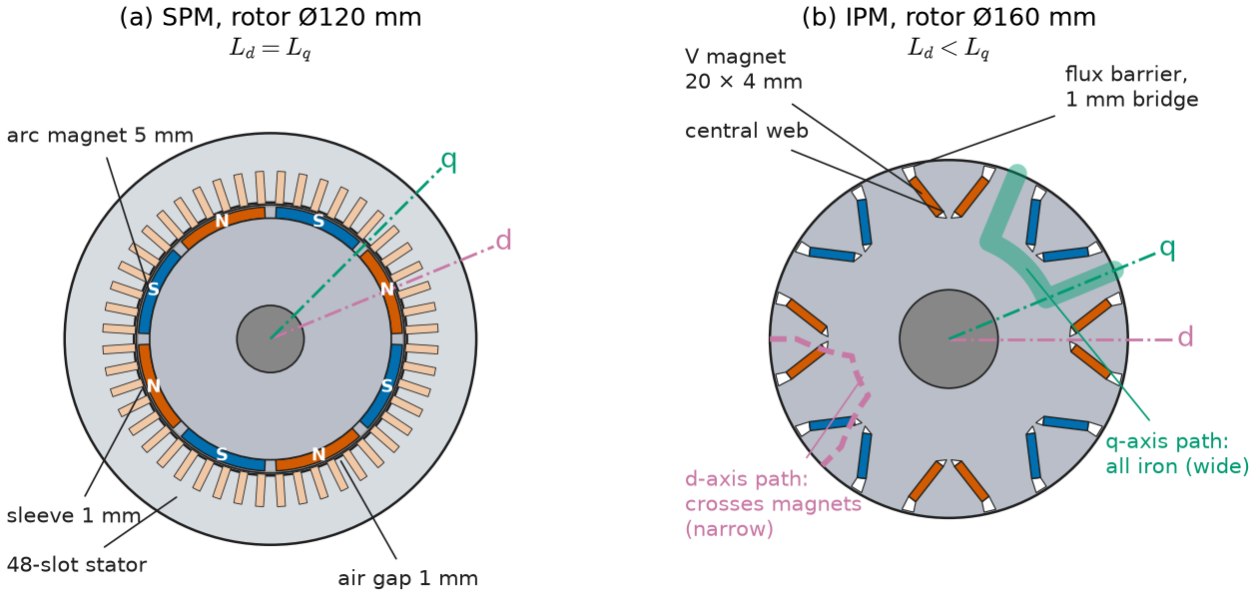


Figure 8.1 — Rotor cross-sections of an 8-pole surface-mounted PM motor (a) and an 8-pole interior PM traction motor with V-shaped magnets (b), with d- and q-axes marked.

Torque in the d–q frame

In the rotor (d–q) reference frame of Chapter 7, with p pole pairs, magnet flux linkage ψ_m and peak-value currents i_d, i_q, the electromagnetic torque is

$$T = \tfrac{3}{2}\, p\, [\psi_m i_q + (L_d - L_q)\, i_d i_q]$$

> **Status of the principle: PMSM torque equation**
> *Derived.* Follows from the co-energy of Chapter 3 expressed in the rotor frame (Park's transformation) for sinusoidally distributed windings; saturation makes L_d, L_q and even ψ_m functions of current, so production designs replace the constants with finite-element maps.

The first term is magnet torque; the second is reluctance torque, which in an IPM ($L_q > L_d$) is positive when i_d is negative. Driving current with a negative d-component therefore raises torque per ampere. The current angle that maximises torque for a given current magnitude defines **maximum torque per ampere (MTPA)** control (Chapter 12).

Field weakening and the speed range

The inverter can apply only so much voltage: for a DC link V_{dc} with space-vector modulation, the peak phase voltage is at most $V_{dc}/\sqrt{3}$. Neglecting resistance, the stator voltage magnitude is $\omega_e\sqrt{(\psi_m + L_d i_d)^2 + (L_q i_q)^2}$, which grows with speed until it reaches the limit at **base speed**. Above it the controller injects more negative i_d, cancelling part of the magnet flux (**field weakening**) and trading torque for speed at roughly constant power. The **characteristic current** ψ_m/L_d decides how far this can go: if it is less than or equal to the inverter current limit, the flux can in principle be cancelled entirely and the constant-power range is wide.

Worked example — torque, base speed and power of an EV IPM motor. An 8-pole ($p = 4$) IPM motor has $\psi_m = 0.08$ Wb, $L_d = 0.2$ mH and $L_q = 0.5$ mH. The inverter supplies 400 A peak from a 400 V battery. Compare the torque with $i_d = 0$ and with a 30° advance of the current vector, then find the base speed and power.

With $i_d = 0$, $i_q = 400$ A: $T = 1.5 \times 4 \times 0.08 \times 400 = 192$ N·m, magnet torque only.

With a 30° advance: $i_d = -400\sin 30^\circ = -200$ A, $i_q = 400\cos 30^\circ = 346.4$ A.

$$T = 1.5 \times 4\left[0.08 \times 346.4 + (0.2 - 0.5)\times 10^{-3} \times (-200) \times 346.4\right] = 166.3 + 124.7 = 291\ \text{N m}$$

The same current produces 52 % more torque. A search over current angle puts the MTPA point at about 34°, with 293 N·m: the 30° figure is almost optimal. The flux linkage at this operating point is $\sqrt{(0.08 - 0.2\times10^{-3}\times200)^2 + (0.5\times10^{-3}\times346.4)^2} = 0.178$ Wb. With $V_{\max} = 400/\sqrt{3} = 230.9$ V peak:

$$\omega_e = \frac{230.9}{0.178} = 1299\ \text{rad/s}, \qquad \omega_m = \frac{1299}{4} = 324.8\ \text{rad/s} = 3101\ \text{rpm}$$

and the power at base speed is $291 \times 324.8 = 94.5$ kW. With a 9:1 reduction and 0.33 m tyre radius, 3100 rpm corresponds to 43 km/h; the remaining speed range to 180 km/h and the motor's 13,000 rpm or so is covered by field weakening (Figure 8.2). At the true MTPA angle (34°, 293 N·m, flux linkage 0.169 Wb) the corner moves slightly up, to 3254 rpm and 100 kW. Note the characteristic current: $\psi_m/L_d = 0.08/0.2\times10^{-3} = 400$ A, exactly equal to the current limit, which is the classical design condition for an unlimited constant-power range in the ideal (lossless) machine. In this ideal model the current stays at 400 A above base speed while its angle advances, so power does not merely hold but keeps rising towards the asymptote $\tfrac{3}{2}V_{\max}I_{\max} = 1.5 \times 230.9 \times 400 = 139$ kW; at 13,000 rpm the envelope gives about 100 N·m and 137 kW. Real motors lose much of this rise to resistance, iron loss and saturation, and are usually rated for roughly constant power above base speed.

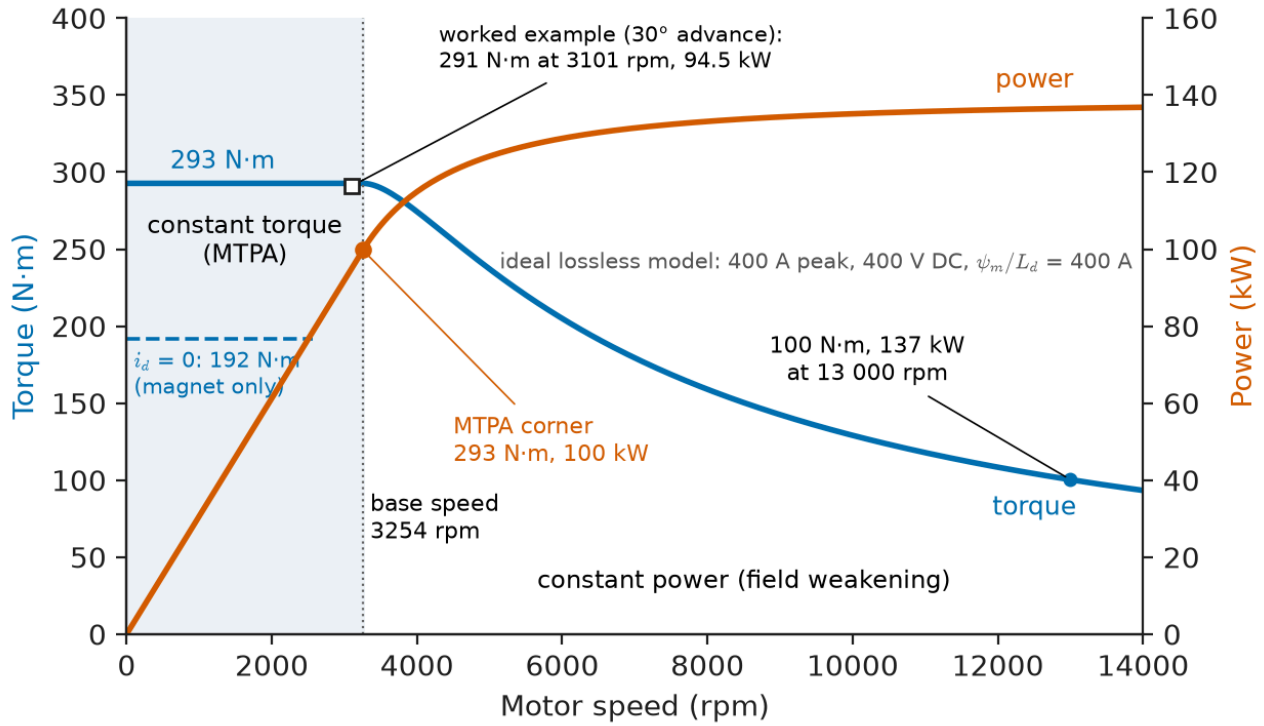


Figure 8.2 — Torque-speed and power-speed envelope of the 8-pole EV IPM motor at 400 A peak and 400 V DC, showing the constant-torque and field-weakening regions.

Efficiency, losses and uncontrolled generation

With no rotor copper loss, PMSMs reach IE4–IE5 efficiency classes (IEC 60034-30-2) in industrial sizes and 95–97 % peak efficiency in traction motors. Their losses are stator copper, stator iron at high frequency (a 12,000 rpm 8-pole motor runs at 800 Hz, so thin 0.20–0.27 mm steel is used, Chapter 9), and eddy loss in the magnets themselves, which are conductive and must be segmented. One hazard is specific to PM machines: when a vehicle coasts or a fan windmills with the inverter off, the magnets still generate a back-EMF proportional to speed. At top speed, an EV motor's open-circuit voltage can exceed the DC-link voltage, so an inverter fault can lead to **uncontrolled generator operation**, with current flowing back through the freewheel diodes. Designs limit the back-EMF ratio or provide active short-circuit modes, and maintenance staff must treat a rotating PM machine as energised.

8.2 Brushless DC motors

A DC motor turned inside out

The DC motor of Chapter 5 has a stationary field and a rotating armature commutated by brushes. The **brushless DC (BLDC) motor** moves the magnets to the rotor and the armature to the stator, and replaces the mechanical commutator with transistors switched according to rotor position. Electrically it behaves like a DC motor: speed proportional to voltage, torque proportional to current, and the gyrator relations $E = K_e\omega$, $T = K_t I$ with $K_e = K_t$ in SI units.

The difference from a PMSM lies in the back-EMF waveform and the drive. A BLDC motor is wound with concentrated or full-pitch windings and magnets that give a **trapezoidal** back-EMF with a 120° flat top. It is driven with **six-step (120° block) commutation**: at any instant two phases conduct a constant current and

the third floats, and every 60° electrical the switching pattern advances (Figure 8.3). When the flat top of the EMF coincides with the current block, the power $e_a i_a + e_b i_b + e_c i_c$, and therefore the torque, is ideally constant. A PMSM has a sinusoidal EMF and is driven with sinusoidal currents. In practice the boundary is blurred, and many small "BLDC" motors are driven sinusoidally for quietness.

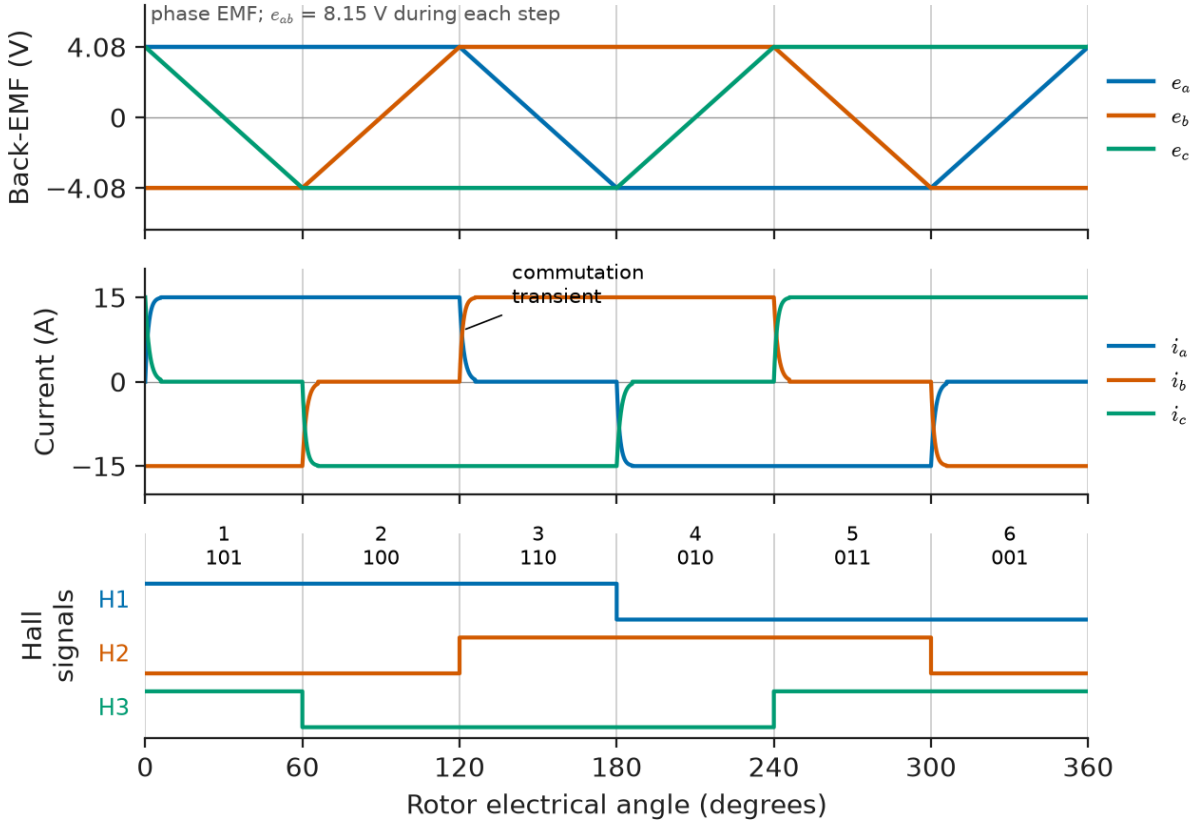


Figure 8.3 — Trapezoidal back-EMFs, six-step phase currents and Hall-sensor signals of a BLDC motor over one electrical cycle.

Position sensing and commutation

Six commutation instants per electrical cycle need only coarse position: three **Hall-effect sensors** spaced 120° electrical give six distinct codes. Sensorless drives detect the zero-crossing of the floating phase's back-EMF and commutate 30° later; they need a minimum speed (typically 5–10 % of rated) for the EMF to be detectable, so they start open-loop. The weaknesses of six-step drive are **commutation torque ripple** (10–20 % in practice, since the current cannot transfer instantly between phases through the winding inductance) and acoustic noise at the 6× electrical frequency.

Applications

BLDC motors cover a huge volume of small machines: 3–1000 W fans (including the EC fans in data-centre air handlers and server chassis), wet-rotor circulator pumps (which in the EU must meet an energy-efficiency index of 0.23 or better, achievable only with PM rotors), HDD and optical spindles, drone and model-aircraft propulsion, e-bike hub motors (250–750 W), cordless power tools and appliance compressors. Their speed control is a DC-link voltage or PWM duty cycle, which is why fan speed in a server follows a simple PWM signal from the motherboard.

Worked example — a drone propulsion motor. An outrunner BLDC motor rated $K_v = 920$ rpm/V drives a propeller from a four-cell lithium-polymer pack (14.8 V nominal). At hover it runs at 7500 rpm and draws 15 A; winding resistance (line-to-line) is 0.12 Ω. Find the torque, the required effective voltage, the PWM duty and the efficiency, neglecting iron and friction losses.

Converting K_v to SI: $920 \times 2\pi/60 = 96.34$ rad/s per volt, so $K_e = K_t = 1/96.34 = 0.01038$ V·s/rad = N·m/A.

- Torque: $T = 0.01038 \times 15 = 0.156$ N·m.
- Back-EMF at 7500 rpm: $7500/920 = 8.15$ V. Like the resistance, this is a line-to-line value: the EMF across the two conducting phases in series, each contributing a flat top of 4.08 V (Figure 8.3). Required voltage: $8.15 + 15 \times 0.12 = 9.95$ V, a duty cycle of $9.95/14.8 = 67\,\%$.
- Mechanical power: $0.156 \times (7500 \times 2\pi/60) = 0.156 \times 785.4 = 122$ W; electrical input $9.95 \times 15 = 149$ W; efficiency 82 %.

At full throttle the no-load speed would be $14.8 \times 920 = 13,600$ rpm, but propeller torque rises with the square of speed, so current and copper loss rise steeply. The K_v rating is the designer's choice of turns: a high-K_v motor gives speed on a small propeller, a low-K_v motor gives torque on a large one at the same voltage.

8.3 Switched-reluctance motors

Torque without magnets or rotor windings

A **switched-reluctance motor (SRM)** has salient poles on both stator and rotor, concentrated coils on the stator poles only, and a rotor of plain laminated steel. Common geometries are 6/4 (three phases) and 8/6 (four phases) stator/rotor poles. Exciting a phase pulls the nearest rotor pole pair into alignment, because that minimises reluctance; switching to the next phase before alignment pulls the rotor on. Torque comes from the variation of phase inductance with rotor angle (Figure 8.4). From the co-energy result of Chapter 3, with linear magnetics,

$$T = \tfrac{1}{2}\, i^2 \, \frac{dL}{d\theta}$$

> **Status of the principle: Reluctance torque $\tfrac{1}{2} i^2\, dL/d\theta$**
> *Approximate.* Exact for a magnetically linear machine; real SRMs are designed to run deep in saturation, where the torque must be computed from the co-energy of the measured flux-linkage–current–angle map. Saturation is beneficial: it raises the energy converted per stroke relative to the volt-amperes the converter must supply.

Torque is independent of current direction, so the converter can be unipolar: the **asymmetric half-bridge**, two switches and two diodes per phase, with no possibility of shoot-through. Each phase is electrically independent, so the machine continues to run with one phase failed: a valued property in aircraft fuel pumps and mining vehicles.

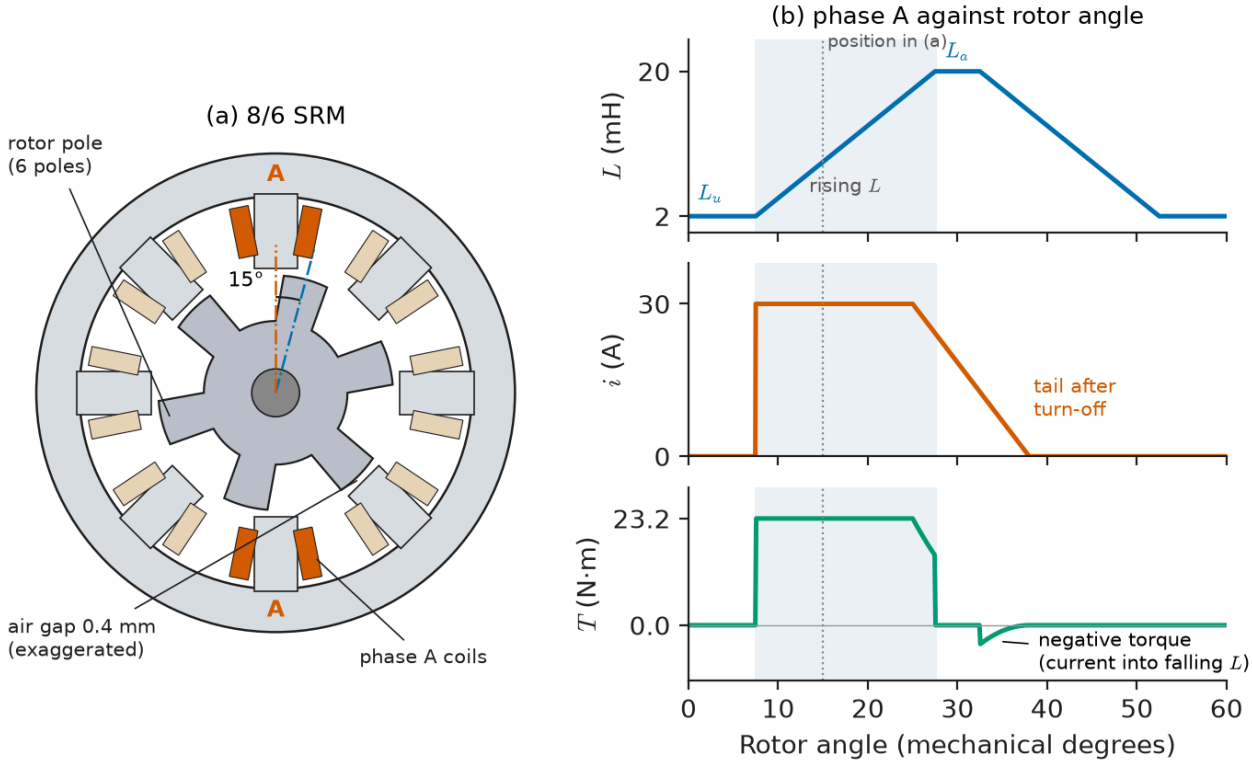


Figure 8.4 — An 8/6 switched-reluctance motor (a) and the idealised phase inductance, current and torque against rotor angle (b).

Worked example — torque of an 8/6 SRM. An 8/6 SRM phase has unaligned inductance 2 mH and aligned inductance 20 mH, with inductance rising linearly over 20° of rotation. The phase current is held at 30 A during the rising interval. Find the phase torque, the step angle and the average motor torque, assuming linear magnetics.

$$\frac{dL}{d\theta} = \frac{0.020 - 0.002}{20 \times \pi/180} = 0.0516\ \text{H/rad}, \qquad T = \tfrac{1}{2} \times 30^2 \times 0.0516 = 23.2\ \text{N m}$$

Strokes per revolution: $m \times N_r = 4 \times 6 = 24$, a stroke angle of 15°. Each stroke converts $\tfrac{1}{2} i^2 (L_a - L_u) = \tfrac{1}{2} \times 900 \times 0.018 = 8.1$ J, so the average torque is $8.1 \times 24/(2\pi) = 30.9$ N·m. The average exceeds the single-phase torque because the 20° rising region of each phase is longer than the 15° stroke, so phases overlap. In reality the current takes time to build against the back-EMF at speed, the converter must turn the phase off before alignment to avoid negative torque, and saturation reshapes everything.

Strengths and weaknesses

The SRM's rotor tolerates 200 °C and more and speeds above 50,000 rpm (it has nothing to fly apart but laminations), it contains no rare earths, and it is cheap to build. Against this, it has high **torque ripple** (30–100 % without current profiling), **acoustic noise** from radial forces that pull the stator into an oval at the stroke frequency, a small air gap (0.25–0.5 mm) that must be held tightly, and a need for position sensing and nonstandard converters. Applications are where robustness outweighs refinement: battery shuttle cars and wheel loaders in underground and surface mining, aircraft starter-generators and fuel pumps, some high-speed vacuum-cleaner and centrifuge drives, and textile machinery. Current research uses current-profiling control and multi-objective geometry to bring noise and ripple close to induction-motor levels, as a rare-earth-free traction option.

8.4 Stepper motors

Moving in steps

A **stepper motor** turns a fixed angle for each current pulse, so position is set by counting pulses, with no feedback sensor. Three types are built:

- **Permanent-magnet (tin-can)** steppers: a multipole magnet rotor between claw-pole stators; 7.5° or 15° steps; very cheap; used in HVAC damper actuators, automotive gauges and valve positioners.
- **Variable-reluctance** steppers: a toothed soft-iron rotor, the stepping version of the SRM; step angle $360^\circ/(mN_r)$, for example 15° for a three-phase motor with an 8-tooth rotor; now rare.
- **Hybrid** steppers: an axially magnetised magnet sandwiched between two toothed rotor cups offset by half a tooth pitch, combining PM and reluctance action. With $N_r = 50$ rotor teeth and two phases, there are $2mN_r = 200$ full steps per revolution, a **1.8° step**; 0.9° versions have 100 teeth. Holding torques range from 0.1–0.5 N·m in NEMA 17 frames (42 mm) to 1–3 N·m in NEMA 23 (57 mm) and 3–12 N·m in NEMA 34 (86 mm).

Drive modes

Full-step drive energises the phases in sequence; half-step alternates one and two phases on, doubling resolution. **Microstepping** feeds sine–cosine weighted currents to the two phases, dividing each full step into 4–256 microsteps. It smooths motion and reduces noise and resonance, but does not improve accuracy in proportion: microstep positions are only as accurate as the motor's torque–angle linearity, typically ±5 % of a full step.

Torque–speed behaviour, resonance and lost steps

Stepper torque falls with speed, because the winding inductance limits how fast phase current can rise within each step period; drivers therefore use chopper current control with supply voltages 10–20 times the winding rated voltage. Two curves describe the motor (Figure 8.5): the **pull-out torque** curve, the maximum load torque at which a running motor keeps step, and the lower **pull-in torque** curve, the maximum load at which it can start instantly at a given pulse rate without acceleration. Between them lies the **slew range**, reachable only by ramping the pulse rate. In a band of low-to-middle speeds (for the motor of Figure 8.5 about 200–300 full steps per second, 60–90 rpm) the rotor-load system behaves as an underdamped torsional oscillator, the **mid-band resonance**, and can lose torque or stall.

The fundamental limitation is that an open-loop stepper does not know when it has lost steps; a jam or overload silently corrupts position. For this reason **closed-loop steppers**, with an encoder and a drive that switches into servo-like control when the rotor lags, are increasingly used, and safety-related positioning uses servos.

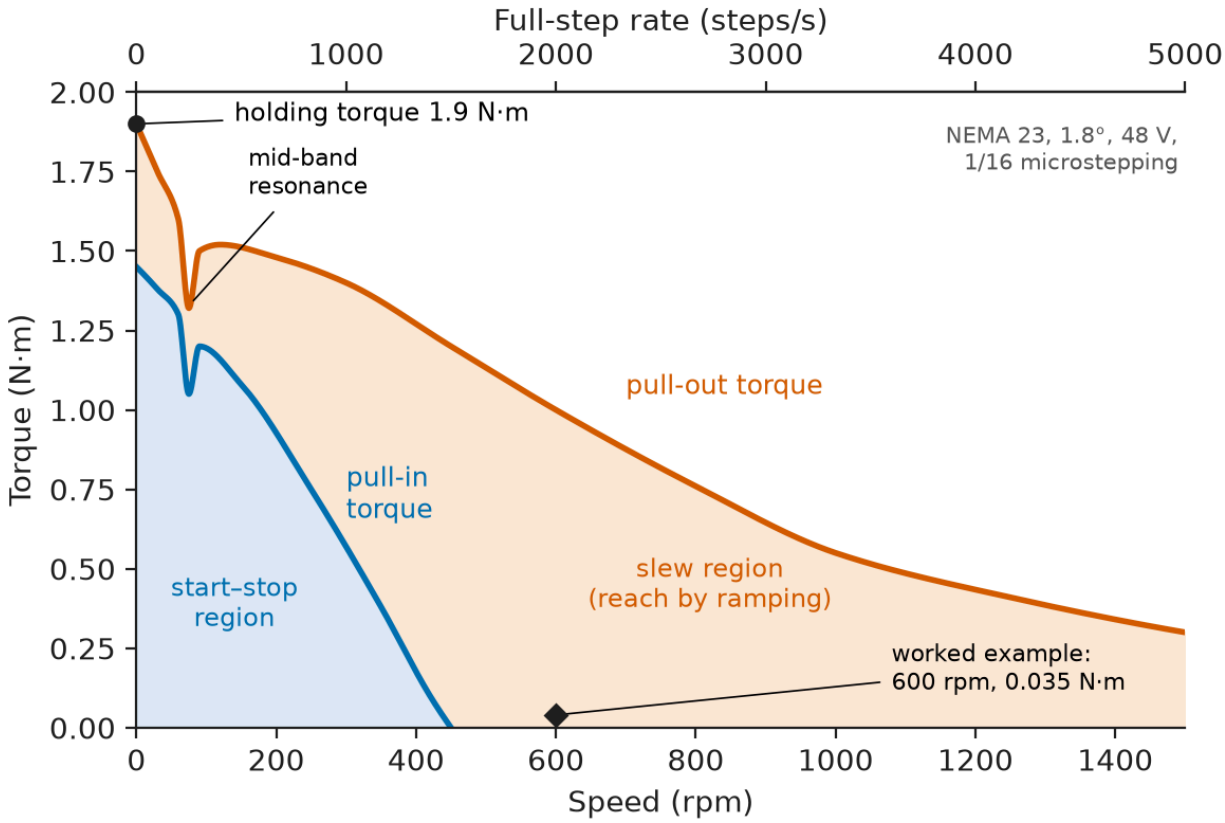


Figure 8.5 — Pull-out and pull-in torque curves of a NEMA 23 hybrid stepper (1.8 degree, 1.9 N m holding torque) driven at 48 V with 1/16 microstepping.

Worked example — a stepper-driven linear axis. A packaging machine moves a 20 kg carriage with a lead screw of 5 mm pitch driven by a 1.8° hybrid stepper with 1/16 microstepping. The carriage must travel at 50 mm/s and accelerate at 0.5 m/s² against 30 N friction; screw efficiency is 0.9. Find the motor speed, the full-step and microstep pulse rates, the positional resolution and the torque required.

- Screw speed $50/5 = 10$ rev/s = 600 rpm.
- Full-step rate $10 \times 200 = 2000$ steps/s; microstep rate $2000 \times 16 = 32,000$ pulses/s.
- Resolution $5\ \text{mm}/(200 \times 16) = 1.56\ \mu\text{m}$ per microstep (the achievable accuracy is several microsteps, since microsteps are not equally spaced).
- Force $20 \times 0.5 + 30 = 40$ N; torque $40 \times 0.005/(2\pi \times 0.9) = 0.035$ N·m (plus the motor's own rotor inertia).

On the curve of Figure 8.5 the pull-out torque at 600 rpm is about 1.0 N·m, a large margin. A smaller NEMA 17 motor could do it, which is why steppers are attractive at low cost; the margin is still needed for resonance and because an unnoticed lost step ruins every subsequent move.

8.5 Universal motors

A series DC motor on AC

A series-wound DC motor (Chapter 5) produces torque proportional to $\Phi I \propto I^2$, which does not change sign when the supply reverses. It therefore runs on AC as well as DC, hence the name **universal motor**. To run well on AC the stator must be laminated (the field flux alternates, so a solid yoke would have prohibitive eddy

loss), the field turns are kept low to limit reactance, and larger machines add a compensating winding to cancel armature reactance. Torque pulsates at twice supply frequency, which the rotor inertia smooths.

The universal motor's selling point is its speed. With no synchronous-speed limit, it runs at 10,000–35,000 rpm on 50/60 Hz, giving 1–2 kW from a motor the size of a fist and a series characteristic with high starting torque. It is the classic motor of drills, angle grinders, circular saws, vacuum cleaners, blenders and older washing machines. Speed is controlled by a triac phase-angle controller, the trigger in a power drill.

Limits

The brushes and commutator limit life to roughly 300–1000 operating hours in consumer products, generate carbon dust, and spark, causing radio-frequency interference that must be suppressed with capacitors and chokes to meet EN 55014 / CISPR 14. Efficiency is modest, typically 40–65 % in appliances, and acoustic noise high. Since about 2015 BLDC motors with integrated electronics have been replacing universal motors in cordless tools and premium vacuum cleaners (which now run at 100,000 rpm and more), driven by battery operation and by EU ecodesign rules that capped vacuum-cleaner input at 900 W.

A historical footnote explains a piece of grid engineering. Large series commutator motors were the first traction motors on AC railways, and their commutation improves at low frequency. That is why German, Austrian, Swiss, Swedish and Norwegian railways still run 15 kV at 16.7 Hz on separate traction grids fed by dedicated generators or frequency converters, while newer lines use 25 kV at 50 Hz with power-electronic drives.

8.6 Linear motors

Unrolling a rotating machine

Cut a rotating machine along a radius and unroll it: the stator becomes a flat **primary** carrying a travelling field, and the rotor becomes a **secondary** or **reaction plate**. The field travels at the **synchronous velocity**

$$v_s = 2\tau f$$

where τ is the pole pitch (metres) and f the supply frequency. There is no conversion from rotation to translation, no gearbox, no screw and no rack, and the force acts directly on the moving part. Two families exist, mirroring the rotating machines.

Linear induction motors

A **linear induction motor (LIM)** has a short three-phase primary on the vehicle and a long secondary in the track: an aluminium plate on a steel back-iron. The slip principle of Chapter 6 applies, with thrust $F = P_{ag}/v_s$, secondary loss sP_{ag} and mechanical power $(1-s)P_{ag}$. But LIMs have two handicaps a rotating motor does not: a large mechanical gap (10–15 mm for a vehicle on a track), which raises magnetising current and gives power factors of 0.5–0.65; and **end effects**, because fresh secondary enters the field at the leading edge and field leaves at the trailing edge, inducing eddy currents that brake the vehicle at high speed. Efficiencies are 60–80 %. In return, they give traction independent of wheel–rail adhesion, steep gradients (up to 6–8 %), tight curves and small tunnels. LIM metros include the Vancouver SkyTrain, Toronto Scarborough line (closed 2023), Kuala Lumpur's Kelana Jaya line and Tokyo's Toei Ōedo line; the US Navy's Electromagnetic Aircraft Launch System (EMALS) uses linear induction motors to accelerate 45 t aircraft to about 240 km/h in roughly 90 m.

Worked example — a LIM metro vehicle. A LIM has pole pitch 0.25 m and is fed at 50 Hz. At cruising, the air-gap power is 500 kW at a slip of 0.1. Find the synchronous and vehicle speeds, thrust, secondary loss and mechanical power.

$$v_s = 2 \times 0.25 \times 50 = 25 \text{ m/s} = 90 \text{ km/h}, \qquad v = 25(1 - 0.1) = 22.5 \text{ m/s} = 81 \text{ km/h}$$

Thrust $F = 500 \times 10^3 / 25 = 20$ kN; secondary (reaction-plate) loss $0.1 \times 500 = 50$ kW, deposited in the track; mechanical power 450 kW. The secondary loss heats the aluminium reaction plate, which must be designed to tolerate repeated acceleration near stations, and a variable-frequency inverter on the vehicle keeps slip small at all speeds.

Linear synchronous motors

A **linear synchronous motor (LSM)** carries magnets or a DC field on one member and a three-phase winding on the other. Long-stator maglev puts the winding in the guideway and the field on the vehicle, feeding only the guideway section under the train: the Shanghai Transrapid line (opened 2004, 431 km/h in service) and Japan's superconducting SCMaglev (603 km/h record, 2015) both work this way. In industry, **linear PM motors** (ironcore or ironless) drive machine-tool axes, laser cutters, semiconductor lithography stages and pick-and-place gantries with accelerations of 5–10 g, speeds up to 5 m/s, and nanometre-level positioning with linear encoders. Their drawbacks are cost (magnets along the whole stroke), strong attractive forces of ironcore types (several times the thrust, carried by the bearings), heat delivered directly into the machine structure, and the need for a full closed-loop servo drive.

8.7 Servo motors

What makes a motor a servo

A **servo motor** is not a distinct electromagnetic type but a motor designed to be part of a closed position or speed loop: motor, feedback device and servo drive tuned together. Modern AC servo motors are SPM synchronous motors (§8.1) with:

- **High-resolution feedback**: absolute encoders with 20–24 bits per revolution (about one million to 16.8 million counts), often multi-turn, sometimes resolvers in harsh environments.
- **Low rotor inertia** (long, slim rotors) for high acceleration, or medium-inertia versions for better inertia matching.
- **High peak torque**, typically three times continuous for a few hundred milliseconds, limited by the drive current and magnet demagnetisation margin.
- **Sinusoidal EMF and skewed or fractional-slot windings** for torque ripple below 1–2 %.
- Integrated brakes, thermistors and, increasingly, single-cable feedback.

Brushed DC servos remain in low-cost and battery-powered equipment; direct-drive torque motors (large-diameter, high-pole-count PM machines) replace gearboxes in rotary tables and robot joints.

Sizing a servo

Servo sizing follows the motion profile, not a steady load. Two checks dominate:

1. **Peak torque** must cover acceleration of the total inertia plus friction and process load at the highest speed in the profile.

2. **RMS torque** over the cycle must not exceed continuous torque, because copper loss is proportional to torque squared and the thermal time constant (10–60 min) is much longer than the cycle (Chapter 9, §9.1):

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cycle}}} \sum_{k} T_k^2 \, t_k}$$

> **Status of the principle: RMS-torque thermal equivalence**
> *Approximate.* Valid when the motion cycle is much shorter than the motor's thermal time constant and losses are dominated by copper loss proportional to T^2; it fails for long dwells at high torque and at high speeds where iron loss dominates.

A third rule concerns tuning: the **inertia ratio** $J_{\text{load}}/J_{\text{motor}}$ should usually be below 5–10 for responsive, stable tuning; above that, compliance in couplings and belts produces resonances the loop cannot suppress.

Worked example — servo for an indexing axis. A rotary indexer requires the motor to turn 10 revolutions in 0.2 s with a trapezoidal profile (one third accelerating, one third constant speed, one third decelerating), then dwell for 0.2 s. Motor rotor inertia is 1.2×10^{-4} kg·m², reflected load inertia 3.0×10^{-4} kg·m², friction 0.2 N·m. Find peak speed, peak torque and RMS torque.

Each segment is $t_a = 0.0667$ s. Distance $= \omega_{\max}(t_a/2 + t_a + t_a/2) = \omega_{\max} \times 0.1333$ s, so $\omega_{\max} = 20\pi/0.1333 = 471.2$ rad/s = 4500 rpm. Acceleration $\alpha = 471.2/0.0667 = 7069$ rad/s². Total inertia 4.2×10^{-4} kg·m², so the inertial torque is 2.97 N·m.

Segment	Duration (s)	Torque (N·m)
Accelerate	0.0667	$2.97 + 0.2 = 3.17$
Constant speed	0.0667	0.20
Decelerate	0.0667	$2.97 - 0.2 = 2.77$ (braking)
Dwell	0.2	0

$$T_{\text{rms}} = \sqrt{\frac{0.0667\,(3.17^2 + 0.20^2 + 2.77^2)}{0.4}} = 1.72 \text{ N m}$$

A 750 W-class servo (about 2.4 N·m continuous, 7.2 N·m peak, 6000 rpm maximum) fits: 72 % thermal utilisation, adequate peak margin at 4500 rpm, and an inertia ratio of 2.5. Peak mechanical power is $3.17 \times 471.2 = 1.49$ kW for a few tens of milliseconds, which is why servo drives are sized on peak current and why the braking energy (here up to about 1.3 kW) goes to a DC-bus brake resistor or a shared regenerative bus.

8.8 Specialized generators

Generators, too, are shaped by their prime movers. The wound-field synchronous generator of Chapter 7 serves constant-speed turbines; where the prime mover's speed varies, or where weight, cost or simplicity dominates, other machines take over.

Wind turbines: DFIG and full-converter PMSG

Wind turbines must run at variable speed to capture maximum power (Chapter 15). Two architectures dominate (Figure 8.6):

- **Doubly fed induction generator (DFIG)**: a wound-rotor induction machine whose stator is connected directly to the grid and whose rotor is fed through slip rings by a back-to-back converter. The converter injects rotor currents at slip frequency, so the machine can run from about 30 % below to 30 % above synchronous speed while delivering grid-frequency power. Since rotor power is $-s$ times stator power, the converter is rated at only 25–35 % of the turbine rating, which made DFIG the dominant onshore technology from about 2000 to 2015 at 1.5–6 MW.
- **Full-converter PMSG**: a permanent-magnet synchronous generator whose entire output passes through a back-to-back converter. Direct-drive versions, with 100–200 poles and diameters of 5–10 m, turn at 8–15 rpm and eliminate the gearbox; medium-speed versions use a one- or two-stage gearbox. The full converter decouples the generator from the grid, gives the best fault ride-through and grid support, and has become standard for offshore turbines of 8–15 MW and beyond.

Worked example — DFIG power flows. A 2 MW, four-pole, 50 Hz DFIG delivers 2.0 MW of mechanical power at 25 % above synchronous speed ($s = -0.25$). Neglecting losses, find stator and rotor power, rotor frequency and shaft torque.

From the induction-machine relations of Chapter 6, $P_{\text{mech}} = (1 - s)P_s$ and $P_r = -sP_s$:

$$P_s = \frac{2.0}{1.25} = 1.6\ \text{MW}, \qquad P_r = 0.25 \times 1.6 = 0.4\ \text{MW}$$

Both flow to the grid: 1.6 MW from the stator, 0.4 MW from the rotor via the converter. Rotor frequency is $|s|f = 12.5$ Hz. Speed is $1500 \times 1.25 = 1875$ rpm, so torque is $2 \times 10^6 / (1875 \times 2\pi/60) = 10.2$ kN·m. At $s = +0.2$ (sub-synchronous) the same 2 MW of shaft power would require $P_s = 2.5$ MW, with 0.5 MW flowing *into* the rotor from the grid. The converter must handle about $|s_{\max}|$ of the rating, here 0.4–0.5 MW: the economic reason for DFIG. The cost is the slip rings and brushes (maintenance every 6–12 months) and a weaker response to grid faults, because a voltage dip induces large rotor currents that the small converter cannot absorb, requiring a crowbar or chopper (Chapter 27).

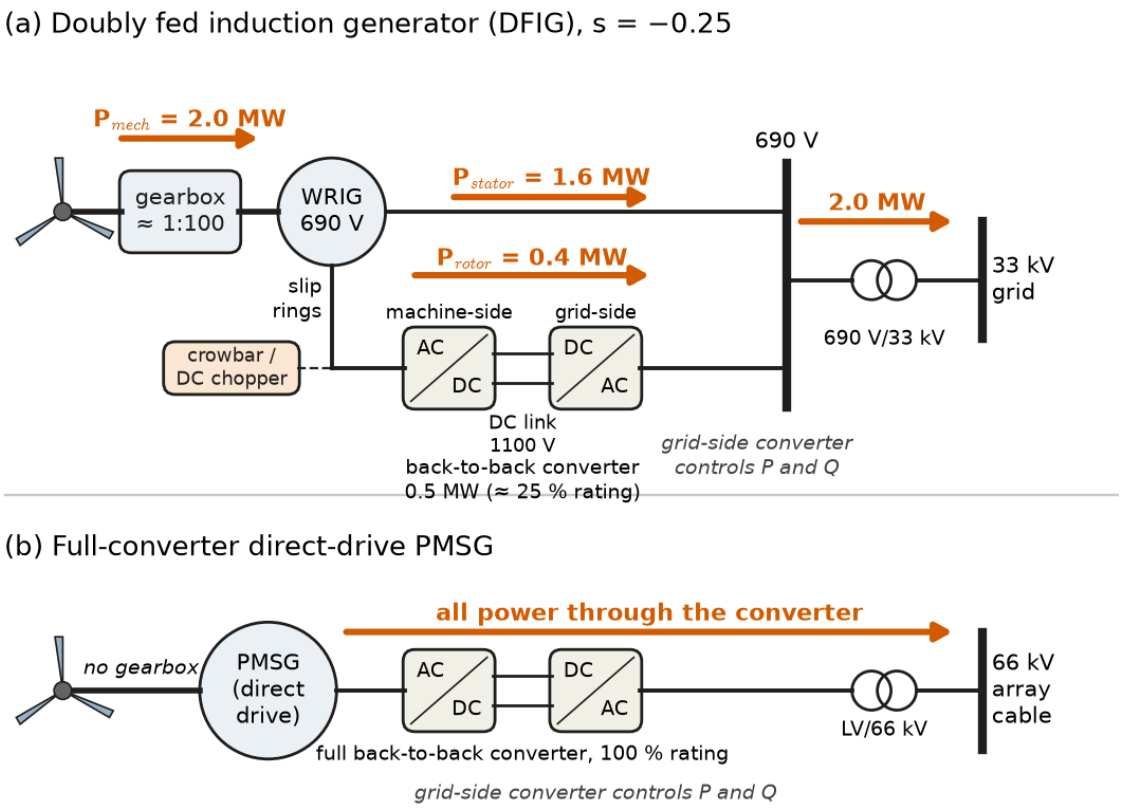


Figure 8.6 — Single-line diagrams of a doubly fed induction generator (a) and a full-converter direct-drive PMSG (b) wind turbine, with power flows from the worked example.

Induction generators

A squirrel-cage induction machine driven above synchronous speed generates (Chapter 6). Fixed-speed "Danish concept" wind turbines of the 1980s–90s used them with a gearbox, soft-starter and capacitor bank; many still run. Small hydro and micro-hydro plants (5 kW–5 MW) use grid-connected induction generators because they need no synchroniser and no excitation control. Off-grid, an induction machine can **self-excite** from capacitors across its terminals, but its voltage and frequency then vary with load, so it suits only simple stand-alone loads with an electronic load controller.

Aircraft and vehicle generators

Aircraft electrical systems run at 115 V (or 230 V) AC and 28 V DC. Traditional aircraft use an **integrated drive generator (IDG)**: a hydromechanical constant-speed drive turns a brushless synchronous generator at constant speed for 400 Hz output, typically 90–120 kVA per engine. Newer aircraft omit the constant-speed drive and accept a **variable-frequency** output tracking engine speed, 360–800 Hz: the Airbus A380 has four 150 kVA variable-frequency generators and the Boeing 787 four 250 kVA variable-frequency starter-generators at 235 V AC, which also start the engines as motors. These are brushless three-stage machines (PM pilot exciter, main exciter, main generator) on one shaft running at 7000–24,000 rpm, oil-cooled, with power densities of 2–3 kW/kg. Ships use **shaft generators** on the propeller line and, increasingly, variable-speed diesel gensets feeding a DC bus.

The car's **claw-pole (Lundell) alternator** is a wound-field synchronous generator with a diode rectifier, producing 14 V at 100–250 A. Its claw-pole rotor is cheap but leaky, and efficiency is only 50–65 %; each 100

W of electrical load costs of the order of 0.1–0.15 L of fuel per 100 km, so 48 V mild hybrids replace it with belt-driven starter-generators of 10–20 kW.

High-speed and unconventional generators

Microturbines (30–1000 kW) and organic-Rankine-cycle expanders couple directly to high-speed PM generators running at 45,000–100,000 rpm, with magnets retained by carbon-fibre or Inconel sleeves and an inverter to deliver grid frequency; the absence of a gearbox and the use of air bearings give long maintenance intervals. **Axial-flux** PM generators (disc-shaped) suit short, wide spaces such as small wind turbines and hub generators. **Superconducting generators**, using high-temperature or MgB_2 field windings (the physics is in *Engineering Physics*, Ch. 7 §7.7), have been demonstrated at multi-megawatt scale for wind and ship propulsion to cut the mass of direct-drive machines by a half, but cryogenic reliability has so far kept them out of series production.

8.9 Selection and application trade-offs

Comparing the machines

Table 8.2 — Comparison of motor types (typical, for machines of similar rating; IM = induction motor).

Machine	Typical power range	Peak efficiency	Torque density	Relative cost	Control	Main strengths	Main weaknesses
Induction motor (Ch. 6)	0.1 kW–20 MW	85–97 %	Medium	Low (1×)	Line or VFD	Rugged, cheap, line-start	Rotor loss, lower pf
Wound-field synchronous (Ch. 7)	1–1000 MW	96–99 %	Medium	High	Exciter + VFD/line	pf control, highest efficiency at scale	Exciter, brushes or diodes, complexity
PMSM (IPM/SPM)	10 W–10 MW	93–97.5 %	Very high	1.5–2×	Inverter essential	Efficiency, compactness, wide speed range	Magnet cost and supply, demagnetisation, back-EMF hazard
BLDC	1 W–10 kW	80–92 %	High	1.2–2× (with electronics)	Six-step or sinusoidal	Simple control, long life	Torque ripple, noise
Synchronous reluctance	1–500 kW	92–96 % (IE4–IE5)	Medium	≈ IM	Inverter essential	No magnets, cool rotor	Low pf, needs good control
Switched reluctance	100 W–500 kW	85–94 %	Medium–high	Low	Special converter	Robust, magnet-free, fault tolerant	Ripple, noise, special drive
Stepper	1–500 W	50–70 %	Medium at low speed	Low	Open-loop pulses	Simple positioning, holding torque	Lost steps, resonance, heats at standstill
Universal	50 W–2 kW	40–65 %	High (by speed)	Very low	Triac	High speed on mains	Brush life, noise, EMI
Linear (LIM/LSM)	100 W–50 MW	60–80 % (LIM), 85–95 % (LSM)	—	High	Inverter	Direct thrust, no adhesion limit	Large gap (LIM), cost per metre
Servo (SPM + encoder)	50 W–50 kW	90–95 %	Very high peak	High	Closed loop	Precision, dynamics	Cost, tuning

How choices are made

EV traction. The design target is efficiency over a drive cycle, not at one point, plus torque density and cost. IPM motors win on efficiency at low and medium speed and on volume. Induction motors win on cost and have no drag losses when idling (a reason Tesla used them on the secondary axle). Externally excited synchronous motors (BMW's fifth-generation drive, Renault) avoid rare earths at a small efficiency penalty. Rare-earth-free SRM and synchronous-reluctance designs are being developed for the same reason.

Industrial pumps, fans and compressors. For 0.75–375 kW at variable speed the choice is between an IE3/IE4 induction motor on a VFD and an IE5 PMSM or synchronous-reluctance motor on the same VFD. The PM or SynRM option saves 1–3 percentage points of efficiency; for a 55 kW fan running 8000 h/y that is 5–14 MWh/year, often a payback below two years (Chapter 28).

Motion control. Stepper for low-cost, low-speed, light-load positioning where a missed step is a nuisance rather than a hazard; closed-loop stepper for the middle ground; servo where dynamics, accuracy at speed or safety matter; linear motor where screw stiffness or speed limits the axis.

Harsh environments. SRM or induction motor where temperature, vibration and fault tolerance dominate (mining, downhole, aerospace fuel pumps). PM machines need care above about 150 °C rotor temperature and where short-circuit currents could demagnetise.

Generators. Wound-field synchronous for constant-speed turbines and grid strength; full-converter PMSG for modern wind and microturbines; DFIG where converter cost dominates; induction generators for small hydro and simple systems.

The final criterion is often not electromagnetic at all: supply-chain risk (rare earths), maintenance access (brushes on an offshore turbine), noise rules (appliances), or the existing skill of the maintenance team.

What remains open

Open question	Where it is answered
How are the losses, temperature rises and insulation of these machines calculated and tested?	Chapter 9
How is the load matched to the motor across four quadrants, and how are drives protected?	Chapter 10
How are PMSMs and BLDC motors controlled (FOC, MTPA, field weakening, sensorless)?	Chapter 12
How do wind and microturbine generators fit into renewable plants, and what is their capacity factor?	Chapter 15
How do DFIG and full-converter turbines meet grid codes for fault ride-through and reactive power?	Chapter 27
What efficiency gains justify replacing motors with PM or SynRM machines?	Chapter 28
Inverter, chopper and gate-driver circuits for these machines	*Electronics & Computing*
Magnet and electrical-steel materials science	*The Materials Core*
Position, speed and current control-loop design for servos	*Circuits, Signals and Communications*, Ch. 34–35

Next: Chapter 9 takes all of these machines into the design office and the test bay, where ratings, losses, temperature, insulation and cooling decide how much power a given frame can deliver and how long it will last.

Chapter 9 — Electrical Machine Design and Performance

Part II · Transformers and Electrical Machines

A cement plant replaces a failed 75 kW kiln-fan motor with a new one of the same nameplate rating from a different supplier. The new motor is 2 % more efficient on paper, yet eight months later its winding fails. The post-mortem finds nothing wrong with the motor: the fan had been retrofitted with a variable-frequency drive, the new motor's insulation was not rated for inverter voltage spikes, the plant sits at 1800 m where the air cools less, and the fan duty, which had crept up after a process change, kept the motor at 110 % load in a 45 °C motor room. Each of those facts changes how much power a motor can actually deliver. A nameplate kilowatt figure is not a property of the electromagnetics; it is the load at which losses, cooling and insulation reach an agreed balance under stated conditions. This chapter is about that balance.

> **What sets the rating of an electrical machine, and how do losses, temperature and insulation life connect to it?**
> **How are machines cooled, wound, built from which materials, tested, and kept running?**

Concept	Engineering destination
Rating and duty cycles	Motor selection for cranes, hoists, conveyors, traction; derating for altitude and ambient
Loss calculations	Efficiency guarantees, cooling design, energy bills
Thermal limits	Overload capability, protection settings, hot-spot management
Efficiency	IE classes, regulations, life-cycle cost
Insulation systems	Winding life, inverter-duty motors, HV generator bars
Cooling	Enclosure and cooling-method selection (IC/IP codes), large-generator hydrogen and water cooling
Winding design fundamentals	Sizing a machine, current density, slot fill
Material selection	Electrical steel grades, copper versus aluminium, magnets
Performance testing	Factory acceptance tests, commissioning, efficiency verification
Reliability and maintenance	Failure modes, condition monitoring, bearing life, maintenance planning

Roadmap. What a rating means (9.1) → where the losses come from (9.2) → how hot they make the machine (9.3) → efficiency and its regulation (9.4) → what the heat does to insulation (9.5) → how heat is removed (9.6) → how windings and machine size follow from loading limits (9.7) → which materials (9.8) → how all of it is verified (9.9) → how machines fail and how they are kept running (9.10).

> **Boundary.**
> Heat conduction, convection and the thermal-resistance network are owned by *Engineering Physics*, Ch. 14 §14.7 and Ch. 19 §19.6; resistivity and its temperature dependence come from Ch. 14 §14.1. Loss mechanisms and efficiency at the level of principle are in Chapter 3; the machines themselves are in Chapters 4–8. This chapter applies all of these to rating, design and testing.

9.1 Rating and duty cycles

What a rating promises

The **rated output** of a motor is the shaft power it can deliver continuously, at rated voltage and frequency, without exceeding the temperature limit of its insulation class, under the standard conditions of IEC 60034-1: ambient cooling air at no more than 40 °C, altitude no more than 1000 m, and a stated duty. A generator's rating is in kVA at a rated power factor, because its armature current, not its real power, sets its heating (Chapter 7). Change any of these conditions and the rating changes:

- **Ambient temperature.** Above 40 °C the permitted temperature rise must be reduced by the excess; a motor in a 50 °C ambient typically delivers about 90 % of its rating.
- **Altitude.** Air density falls with height, and with it the heat carried away by the fan. As a rule of thumb, rating falls by about 1 % per 100 m above 1000 m, unless the ambient is correspondingly cooler; IEC 60034-1 allows rise limits to be traded against lower ambient at altitude.
- **Supply quality.** Voltage unbalance raises losses sharply through negative-sequence currents (NEMA MG1 derates by about 10 % at 3 % voltage unbalance), and inverter supply adds harmonic losses (§9.2).

NEMA motors may carry a **service factor** (typically 1.15), allowing continuous operation at 115 % load with about 10 K more temperature rise and correspondingly shorter insulation life. IEC motors have no service factor; the margin, if any, comes from using class F insulation with class B rise.

Duty types

Many machines do not run at constant load. IEC 60034-1 defines ten standard **duty types**, which let a manufacturer rate a machine for a defined cycle rather than for continuous load:

Duty	Description	Typical application
S1	Continuous running at constant load, long enough for thermal equilibrium	Pumps, fans, compressors
S2	Short-time duty (e.g. S2 30 min), then rest until cold	Valve actuators, sluice gates, emergency pumps
S3	Intermittent periodic: identical cycles of constant load and rest, no significant starting heat; cyclic duration factor 15, 25, 40 or 60 % of a 10 min cycle	Cranes, hoists (without frequent starts)
S4	As S3 with significant starting heat	Cranes and hoists with frequent starts
S5	As S4 with electric braking	Lifts, mill auxiliaries
S6	Continuous-operation periodic duty: load and no-load, no rest	Machine tools, saws
S7	As S6 with starting and electric braking	Reversing drives
S8	Continuous operation with related load and speed changes	Multi-speed fans, pole-changing drives
S9	Non-periodic load and speed variations, including overloads	Most VFD-fed process drives, traction
S10	Discrete constant loads	Machines with a few defined operating states

A motor rated 30 kW S1 might be rated 37 kW S3 40 % and 45 kW S2 30 min: the shorter the time at load relative to the thermal time constant (§9.3), the more power it can deliver. Figure 9.1 shows the temperature histories behind these ratings.

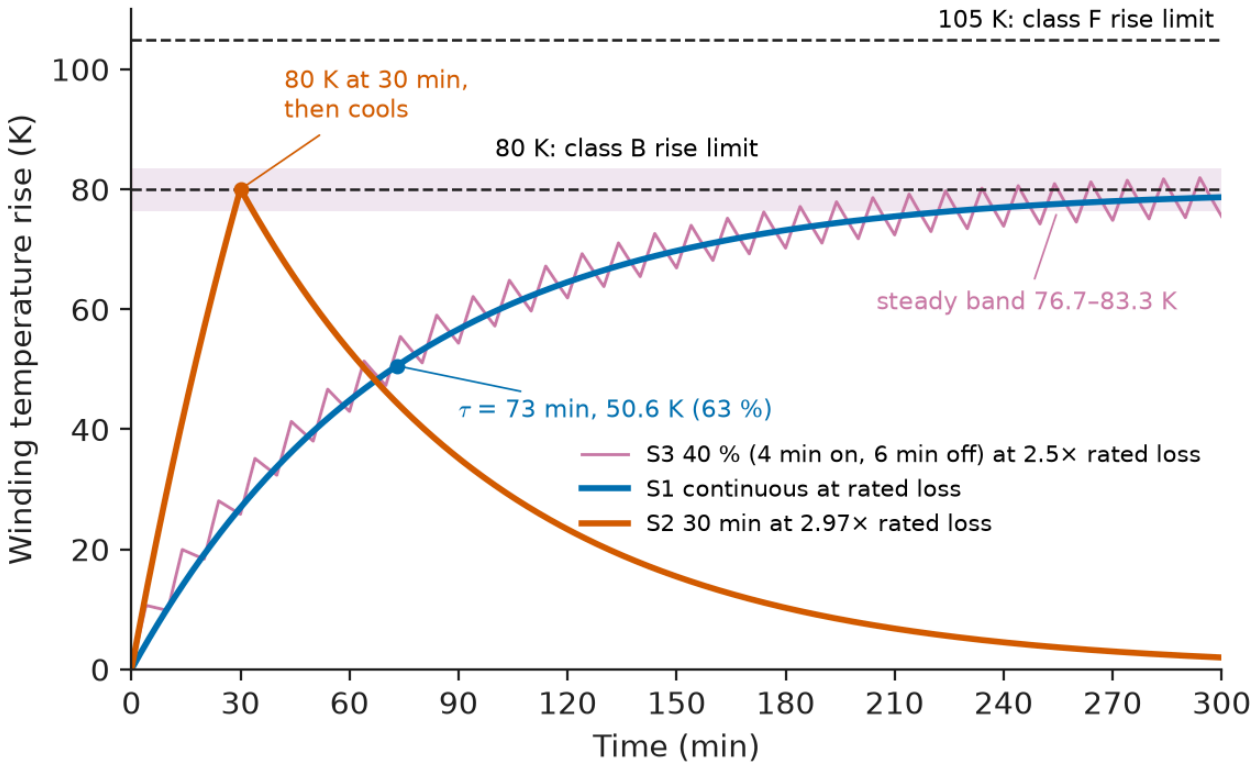


Figure 9.1 — Winding temperature rise under S1 continuous, S2 30 min and S3 40 percent duty for a motor with a 73 min thermal time constant and an 80 K permitted rise.

Equivalent continuous load

For a cyclic load with a period much shorter than the thermal time constant, the winding sees the time-average of its losses. With losses dominated by copper loss proportional to current squared, and current roughly proportional to torque in a machine at constant flux, the equivalent continuous load is the **RMS torque** (or RMS current) over the cycle, as already used for servo motors in Chapter 8. For a self-ventilated motor the cooling is worse at standstill, so rest periods count only partly: a common correction counts rest time at a fraction $k \approx 0.25$–0.5 of its duration.

Worked example — rating a mine-hoist motor. A forced-ventilated 6.6 kV, six-pole induction motor drives a skip hoist at 990 rpm. Each 80 s cycle comprises 8 s acceleration at 9.6 kN·m, 40 s full-speed hoisting at 6.0 kN·m, 8 s deceleration at 2.4 kN·m and 24 s at rest while the skip is loaded. Is a 630 kW motor adequate?

$$T_{\mathrm{rms}} = \sqrt{\frac{9.6^2 \times 8 + 6.0^2 \times 40 + 2.4^2 \times 8 + 0}{80}} = 5.27\ \mathrm{kN\,m}$$

Rated torque of a 630 kW motor at 990 rpm is $630/(990 \times 2\pi/60) = 6.08$ kN·m. Thermal utilisation is $5.27/6.08 = 87\,\%$, and the peak of 9.6 kN·m is 1.58 times rated, safely below a breakdown torque of typically 2.2–2.5 times. The equivalent RMS power is 547 kW and the average delivered power only 373 kW (speed rises and falls linearly during acceleration and deceleration, so those intervals deliver at half speed on average). If the motor were self-ventilated, with rest time counted at 40 %, the RMS torque would rise to 5.82 kN·m, 96 % of rated: still acceptable, but the margin for cycle-time creep, voltage dips and the extra heat of frequent starts (duty S4) would be gone. In practice a 710 kW or a forced-ventilated 630 kW machine would be chosen.

9.2 Loss calculations

Where the losses arise

The losses of every machine in Chapters 4–8 fall into five groups. Their separation matters because each scales differently with load, voltage, speed and temperature.

1. **Stator copper loss** $3I^2R$, with R corrected to operating temperature. For copper, $R_T = R_{20}[1 + 0.00393(T - 20)]$: the resistance at 115 °C is 37 % higher than at 20 °C, so efficiency tests always correct to a reference temperature (IEC 60034-2-1 uses the measured or class-specified winding temperature). At high frequency, **AC resistance** exceeds DC resistance because of skin and proximity effects in deep or wide conductors; this matters in large machines and in EV hairpin windings at several hundred hertz.
2. **Rotor copper loss**: sP_{ag} in an induction motor (Chapter 6), field $I_f^2 R_f$ in a synchronous machine, zero in PM machines apart from magnet eddy loss.
3. **Core (iron) loss**: hysteresis and eddy currents in the laminations, a function of flux density, frequency and lamination thickness (Chapter 1 and §9.8).
4. **Friction and windage**: bearings, seals, fan and rotor drag; nearly constant at fixed speed, roughly proportional to the cube of speed for the fan part.
5. **Stray (additional) load loss**: everything else that grows with load: leakage-flux eddy currents in end structures and frames, high-frequency losses from slot harmonics on the rotor surface, inter-bar currents in skewed cages.

Core-loss models

Core loss per unit mass is normally taken from the steel supplier's data at the operating flux density and frequency, and corrected with empirical factors (1.5–2 is typical) for punching damage, rotational fields and harmonics. To scale it, the classical separation is

$$p_{\text{core}} = k_h f B^{\alpha} + k_e (f B)^2 + k_{\text{ex}} (f B)^{1.5}$$

with $\alpha \approx 1.6$–2.2: hysteresis, classical eddy-current and excess (anomalous) loss, each with a fitted coefficient.

> **Status of the principle: Steinmetz/Bertotti core-loss separation**
> *Empirical*. The functional forms rest on physical arguments, but the coefficients are fitted to Epstein or single-sheet measurements for each steel grade; accuracy is ±10–20 % under sinusoidal flux and worse under PWM or with DC bias.

Stray load loss

Stray load loss is the hardest to calculate and was long the source of disagreement between test standards. IEC 60034-2-1 allows it to be measured (by the residual-loss method, §9.9) or **assigned** as a fraction of input power,

$$P_{LL} = P_1 \left[0.025 - 0.005 \log_{10}\!\left(\frac{P_2}{1\ \text{kW}} \right) \right] \qquad (1\ \text{kW} < P_2 < 10\ \text{MW})$$

For a 75 kW motor this gives 1.56 % of input, about 1.23 kW, which is roughly twice what a good modern design measures. The assigned value is deliberately conservative; a manufacturer that measures can claim a higher efficiency.

Inverter-supply losses

A PWM inverter adds current ripple at switching frequency and time harmonics in the flux. For induction motors these raise total losses by roughly 10–20 % at the same load (mostly iron loss and rotor-surface loss), which is why IEC TS 60034-25 and NEMA MG1 Part 30 recommend derating general-purpose motors on inverters, and why PM motors for high-speed drives need laminated or segmented magnets.

Worked example — loss budget and efficiency curve of a 75 kW IE3 motor. A 75 kW, four-pole, 400 V, 50 Hz induction motor has full-load efficiency 95.0 %. A typical breakdown of its losses is: stator copper 36 %, rotor copper 19 %, core 20 %, friction and windage 9 %, stray load 16 %. Find the losses at full load, the efficiency at 25–115 % load, and the load of maximum efficiency.

Input at full load: $75/0.95 = 78.95$ kW; total loss 3.947 kW. Split: stator copper 1.421 kW, rotor copper 0.750 kW, core 0.789 kW, friction and windage 0.355 kW, stray 0.632 kW.

Group into **fixed losses** (core plus friction and windage, nearly independent of load at fixed voltage and speed) $P_0 = 1.145$ kW, and **load losses** (copper plus stray, roughly proportional to the square of load fraction x) $P_L = 2.803x^2$ kW. Then $\eta(x) = 75x/(75x + 1.145 + 2.803x^2)$:

Load fraction x	Output (kW)	Losses (kW)	Efficiency
0.25	18.75	1.320	93.4 %
0.50	37.5	1.845	95.3 %
0.75	56.25	2.721	95.4 %
1.00	75.0	3.947	95.0 %
1.15	86.25	4.851	94.7 %

Figure 9.2 shows the components. Maximum efficiency occurs where load losses equal fixed losses:

$$x_{\text{opt}} = \sqrt{\frac{P_0}{P_{L,\text{rated}}}} = \sqrt{\frac{1.145}{2.803}} = 0.64, \qquad \eta_{\max} = 95.4\,\%$$

The curve is flat from about 50 % to full load, which is why a modestly oversized motor costs little efficiency; below 25 % load efficiency (and power factor) collapse. The simple x^2 scaling overstates the fall of stator copper loss at light load, because the magnetising current does not fall, so measured part-load efficiencies are slightly lower than this table.

> **Status of the principle: Maximum efficiency at equal fixed and variable losses**
> *Derived.* Setting $d\eta/dx = 0$ with losses $P_0 + P_L x^2$ gives $P_L x^2 = P_0$; it holds for any machine or transformer (Chapter 4) whose losses split cleanly into a constant and a square-law part.

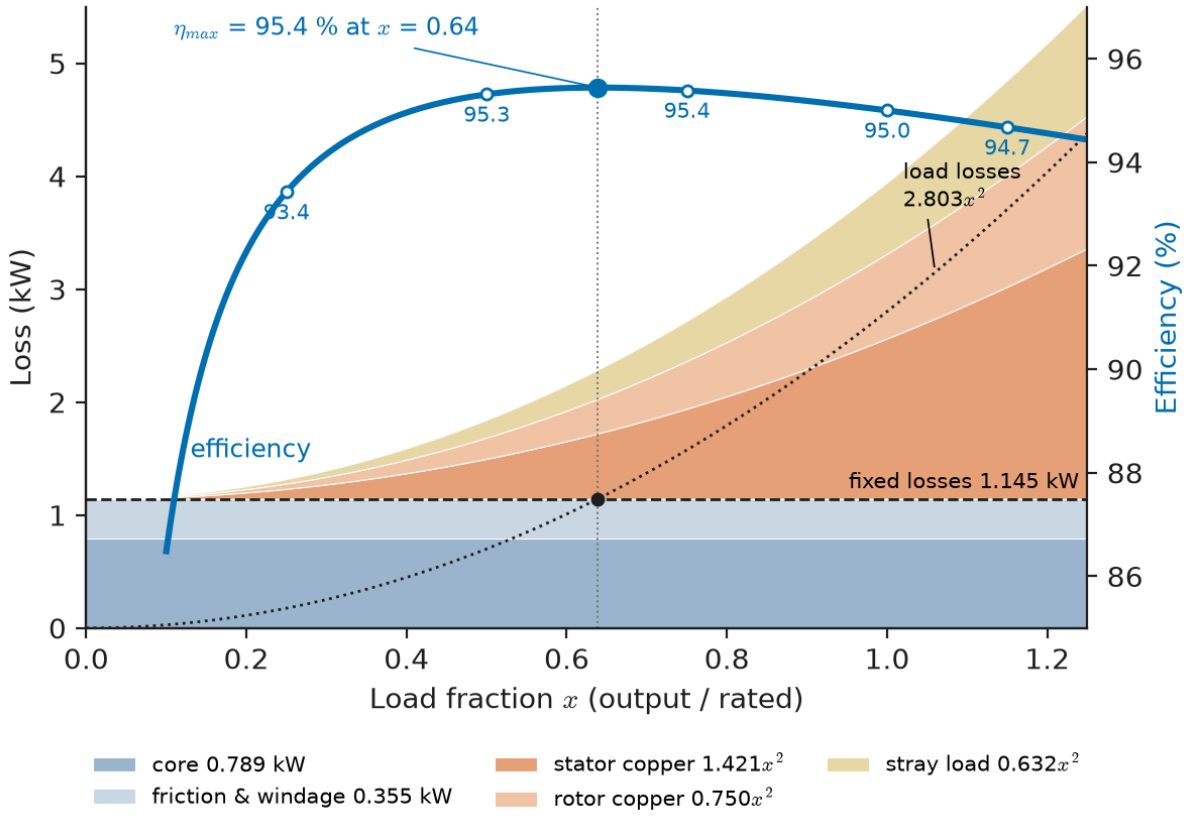
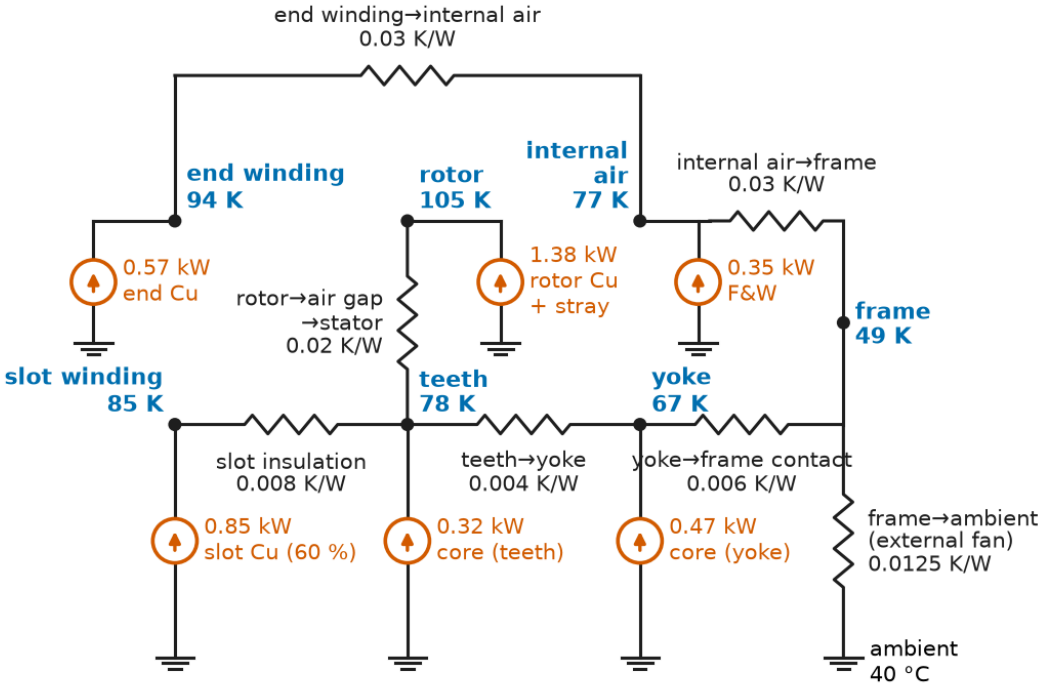


Figure 9.2 — Loss components and efficiency of the 75 kW IE3 induction motor against load, with the maximum-efficiency point.

9.3 Thermal limits

The thermal network

Losses heat the winding, core and rotor; the heat flows by conduction through insulation and iron, by convection to internal and external air, and by radiation from the frame. *Engineering Physics*, Ch. 14 §14.7, builds the thermal-resistance network that models this; machine designers use it with five to fifty nodes (Figure 9.3). The critical paths are almost always the same: **through the slot insulation** from copper to core (insulation has a thermal conductivity of 0.2–0.3 W/(m·K), against 400 for copper and about 25 for laminated steel across the stack), **across the frame-to-core interface** (contact pressure and fit tolerances matter), and **from the frame to the air** (fin area and air velocity). End windings, cooled only by internal air turbulence, are usually the hottest region in TEFC motors.



Solved rises above ambient (total loss 3.95 kW): winding average ≈ 88 K, end-winding hot spot ≈ 94 K, equivalent ≈ 0.022 K/W (single-body value of §9.3: 80 K / 3.95 kW = 0.020 K/W). Resistance values are illustrative.

Figure 9.3 — Lumped thermal network of a TEFC induction motor with losses as heat sources and the dominant resistances, for the 75 kW example.

Temperature limits by insulation class

The temperature a winding may reach is set by its insulation's **thermal class** (§9.5). IEC 60034-1 converts it into a permitted **temperature rise** above a 40 °C ambient, measured as the average winding temperature by the resistance method, leaving a margin for the hot spot:

Thermal class (IEC 60085)	Class temperature	Permitted average rise (resistance method, 40 °C ambient)	Hot-spot allowance
130 (B)	130 °C	80 K	10 K
155 (F)	155 °C	105 K	10 K
180 (H)	180 °C	125 K	15 K

Most modern industrial motors use class F insulation but are designed for class B rise ("F/B"), so that the hot spot runs about 25 K below the class limit. That reserve buys a long life (§9.5) and capacity for overload, high ambient or a poor supply.

Transient heating

Treated as one thermal mass C (J/K) with one resistance R_{th} to ambient (the thermal RC network of *Engineering Physics*, Ch. 14 §14.7.3), a machine's temperature rise after a step change in loss P follows

$$\theta(t) = \theta_\infty + (\theta_0 - \theta_\infty)\, e^{-t/\tau}, \qquad \theta_\infty = P R_{th}, \quad \tau = R_{th} C$$

> **Status of the principle: Single-body thermal model**
> *Approximate.* A good description of the frame and the average winding over tens of minutes; it underestimates the winding's fast rise during starting or stall, when the copper heats adiabatically in seconds with a time constant of its own (tens of seconds to a few minutes). Motor protection relays (IEC 60255-149) therefore use two-body thermal models.

Worked example — overload of the 75 kW motor. The motor of §9.2 has a full-load rise of 80 K, a mass of 450 kg and an average specific heat of 480 J/(kg·K). It has been running at full load and is then loaded to 120 % for 15 minutes. Find the thermal time constant, the ultimate rise at 120 % and the rise after 15 minutes.

$R_{th} = 80/3947 = 0.0203$ K/W; $C = 450 \times 480 = 216$ kJ/K; $\tau = 0.0203 \times 216\,000 = 4380$ s = 73 min.

At 120 % load, losses are $1.145 + 2.803 \times 1.2^2 = 5.18$ kW, so $\theta_\infty = 5180 \times 0.0203 = 105$ K, exactly the class F limit. After 15 minutes from 80 K:

$$\theta = 105 - (105 - 80)\,e^{-15/73} = 105 - 25 \times 0.814 = 84.6 \text{ K}$$

The short overload is harmless; even indefinitely, the 120 % load would sit at the class F limit. That is the F/B reserve at work. The real winding heats somewhat faster than this single-body estimate, so a protection relay set to this model should include the winding's shorter time constant.

Starting and stall

During starting an induction motor draws 6–8 times rated current, so copper loss is 36–64 times rated, and the rotor cage absorbs energy equal to the kinetic energy given to the load. IEC 60034-12 specifies how many consecutive starts a motor must tolerate (typically two from cold, one from hot), and data sheets give a **safe stall time**, typically 10–20 s hot, which the protection must respect. High-inertia loads (fans, centrifuges, crushers) are where motors most often fail thermally on starting.

9.4 Efficiency

Efficiency classes

Electric motors use about 45 % of the world's electricity. Since 2008 their efficiency has been classified by IEC 60034-30-1 into **IE classes**: IE1 (standard), IE2 (high), IE3 (premium), IE4 (super premium), with IE5 (ultra premium) defined as about 20 % lower losses than IE4. Table 9.1 gives the minimum full-load efficiencies for four-pole 50 Hz motors.

Table 9.1 — Minimum efficiencies, four-pole, 50 Hz (IEC 60034-30-1).

Rated output	IE1	IE2	IE3	IE4
0.75 kW	72.1 %	79.6 %	82.5 %	85.7 %
7.5 kW	86.0 %	88.7 %	90.4 %	92.6 %
15 kW	88.7 %	90.6 %	92.1 %	93.9 %
37 kW	91.2 %	92.7 %	93.9 %	95.2 %
75 kW	92.7 %	94.0 %	95.0 %	96.0 %
110 kW	93.3 %	94.5 %	95.4 %	96.3 %

Two patterns stand out: efficiency rises steeply with size (losses scale roughly with surface, output with volume), and each class step removes 15–25 % of the losses. Large machines go much further: a 10 MW synchronous motor reaches 97.5–98.5 %, a 500 MVA turbo-generator 98.7–99 %, and a large transformer 99.5–99.8 % (Chapter 4).

Regulation

Minimum efficiency is mandatory in most markets. In the EU, Regulation 2019/1781 has required IE3 for motors from 0.75 to 1000 kW since July 2021 and IE4 for 75–200 kW two-, four- and six-pole motors since July 2023; VFDs must meet IE2 under IEC 61800-9-2. The United States requires NEMA Premium (≈ IE3) under 10 CFR 431, and China, India, Brazil and others have comparable rules. IEC 61800-9-2 also defines **IES classes** for the motor-plus-drive system, recognising that what matters is the efficiency of the whole power drive system at its actual operating points.

Efficiency in operation

Rated efficiency is a single point. What the energy bill sees is efficiency weighted over the operating profile, which is why a VFD on a fan that spends most of its life at 60 % flow saves far more energy (the fan's power falls roughly with the cube of speed) than any motor upgrade (Chapter 28). Still, motor efficiency compounds over a life of 15–25 years: replacing the 75 kW IE3 motor (95.0 %) with an IE4 (96.0 %) at full load and 6000 h/y saves $(75/0.95 - 75/0.96) \times 6000 = 4.9$ MWh per year, about 590 per year at 0.12 per kWh, against a price premium of perhaps 15–25 % on a motor that costs a few thousand. The energy cost of a continuously running industrial motor over its life is typically 95–97 % of its life-cycle cost; the purchase price is 2–3 %.

9.5 Insulation systems

What the insulation has to do

A winding's insulation must withstand the working voltage between turns, between phases and to earth, for decades, at high temperature, under vibration, thermal cycling, contamination and moisture, and with steep-fronted surges from switching and inverters. It is the component that most often limits machine life (§9.10). A stator insulation system consists of:

- **Strand and turn insulation**: enamel (polyesterimide, polyamide-imide overcoat) on round wire; enamel plus glass or mica tape on rectangular strands of form-wound coils.

- **Groundwall (main) insulation**: slot liners of aramid paper or polyester film laminates (Nomex–Mylar–Nomex) in random-wound motors; mica-paper tape bonded with epoxy or polyester resin, 2–6 mm thick, in form-wound coils above about 1 kV, because mica alone resists partial discharge (see *Inverter-fed windings* below). Figure 9.4 shows a typical medium-voltage slot.
- **Phase separators**, slot wedges, end-winding bracing and lead insulation.
- **Impregnating resin**, applied by dipping or by **vacuum-pressure impregnation (VPI)**, which fills voids, bonds the winding into a rigid mass and improves heat transfer. Global VPI of a complete wound stator is standard for medium-voltage motors.
- In machines above about 6 kV, **semiconducting slot armour** to prevent discharges between coil surface and core, and **stress-grading** tape at the slot exit to control the surface field.

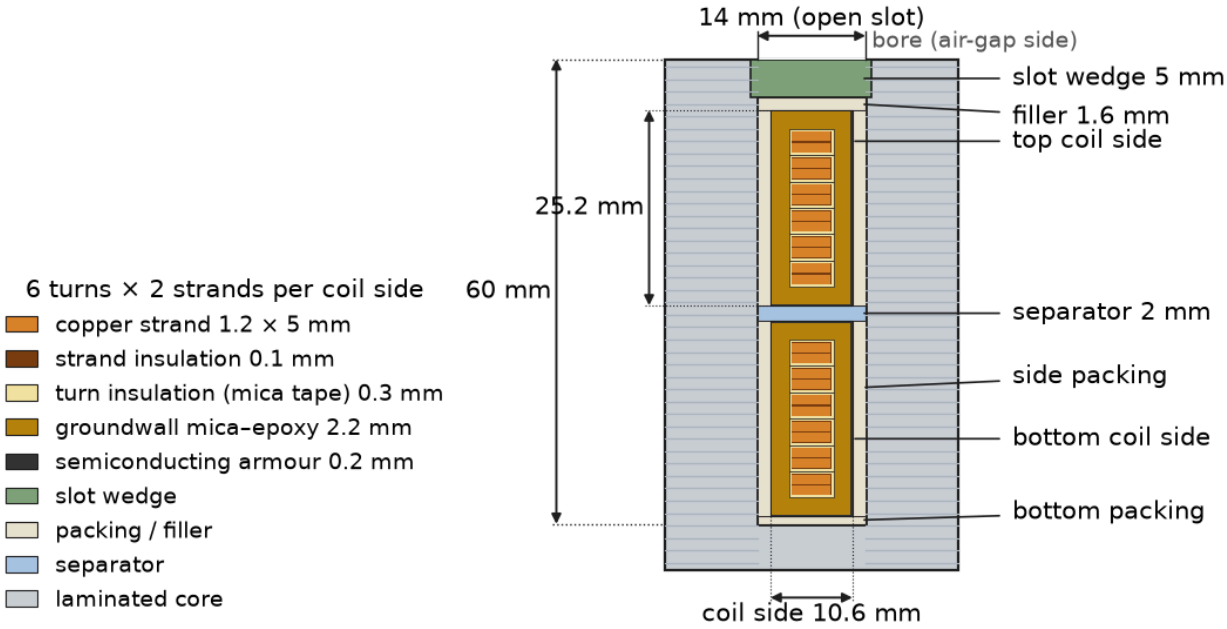


Figure 9.4 — Slot cross-section of a 6.6 kV form-wound two-layer stator winding showing the insulation system.

Thermal ageing

Insulation ages chemically: oxidation and depolymerisation of the resin binder make it brittle until vibration cracks it and moisture tracks across. The rate is a thermally activated (Arrhenius) process, as in *Engineering Physics*, Ch. 20 §20.7.1, so life falls exponentially with temperature. The thermal class is defined (IEC 60216, IEC 60085) as the temperature at which the material system reaches a nominal 20,000 h life in accelerated tests. In engineering practice this is expressed by the **Montsinger 10-degree rule**:

$$L(T) = L_0 \cdot 2^{(T_0 - T)/10\,\mathrm{K}}$$

> **Status of the principle: Ten-degree halving of insulation life**
> *Empirical.* An approximation to Arrhenius behaviour with an activation energy near 1.1 eV in the 120–180 °C range; real systems halve life every 8–12 K, and the rule ignores the electrical, mechanical and environmental stresses that often end a winding's life first (the multi-factor TEAM view: thermal, electrical, ambient, mechanical).

Worked example — insulation life against temperature. A class F (155 °C) winding has a nominal 20,000 h life at its class temperature. Estimate its thermal life for the 75 kW motor of §9.3 (hot spot = ambient

+ average rise + 10 K): at full load in a 40 °C ambient; at full load in a 30 °C ambient; and at the 120 % overload sustained continuously at 40 °C.

Case	Hot spot	Thermal life	Years at 8760 h/y
Full load, 40 °C	$40 + 80 + 10 = 130$ °C	$20\,000 \times 2^{2.5} = 113\,000$ h	13
Full load, 30 °C	120 °C	$20\,000 \times 2^{3.5} = 226\,000$ h	26
120 % load, 40 °C	$40 + 105 + 10 = 155$ °C	20,000 h	2.3

Figure 9.5 plots the rule for the three common classes. Ten degrees cooler doubles thermal life; a continuous 20 % overload cuts it by a factor of more than five. The cement-plant motor of the opening, at 110 % load in a 45 °C room at 1800 m, would have been consuming thermal life at several times the design rate even before its inverter problem.

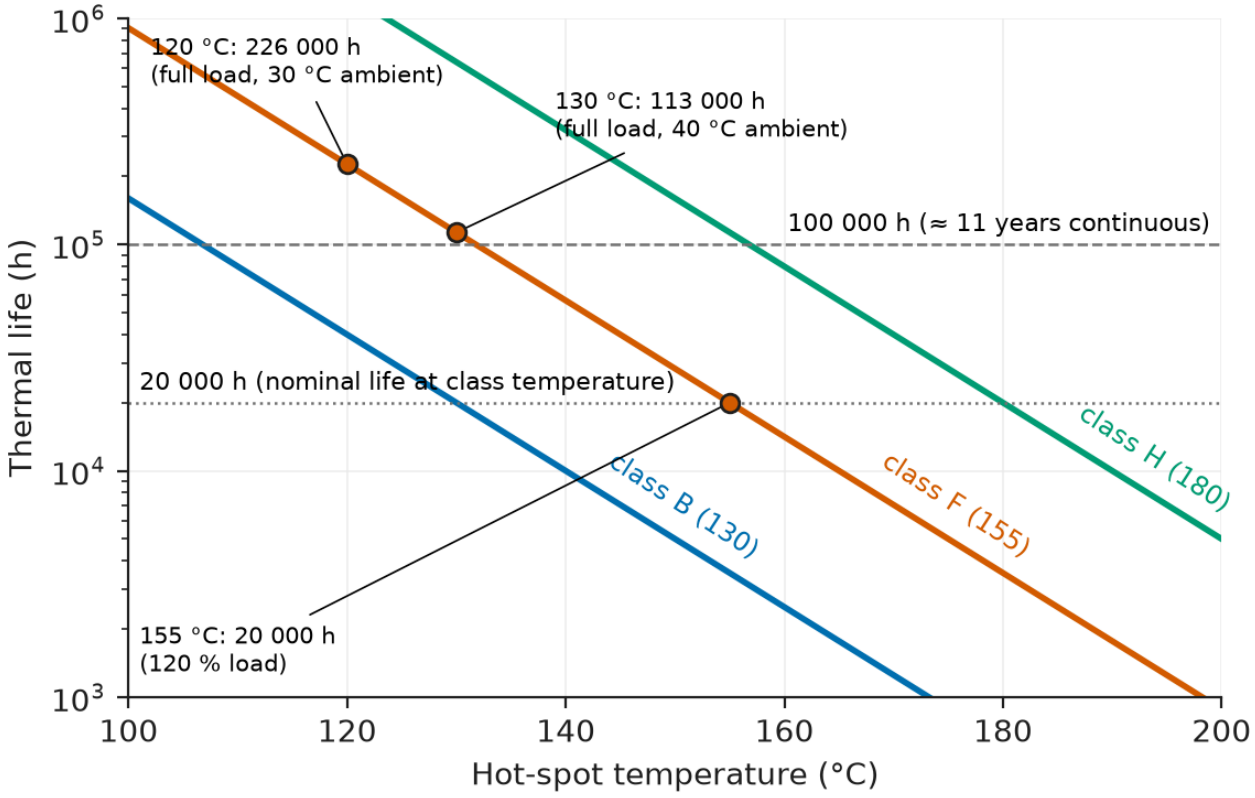


Figure 9.5 — Thermal life of class B, F and H insulation systems against hot-spot temperature according to the 10 K halving rule, with the worked-example points.

Inverter-fed windings

A PWM inverter delivers voltage steps with rise times of 50–200 ns. On a long cable these reflect at the motor terminals, where the high-frequency surge impedance of the motor is much larger than that of the cable, and the peak can approach twice the DC-link voltage (a 690 V drive with a 970 V DC link can produce 1.8–2 kV peaks) with most of the step falling across the first turns of the first coil. If the local field exceeds the inception voltage of air voids, **partial discharges** erode organic insulation within months. Inverter-duty design follows IEC 60034-18-41 (Type I, partial-discharge-free random-wound windings, qualified by impulse PD tests) and IEC 60034-18-42 (Type II, form-wound windings with PD-resistant mica); NEMA MG1 Part 31 requires inverter-duty motors up to 600 V to withstand peaks of 3.1 times rated voltage. Mitigations are corona-

resistant enamel, better phase separation, output dV/dt or sine filters, and shorter cables. This is the cause of the cement-plant failure in the opening.

9.6 Cooling

Cooling methods

Cooling determines how much loss, and therefore how much output, a given frame can carry. IEC 60034-6 codes the method as IC followed by the circuit arrangement and the method of moving the coolant; IEC 60034-5 codes the enclosure's protection against solids and water as IP.

Code	Arrangement	Typical use
IC01	Open, self-ventilated (drip-proof)	Clean indoor plant, gensets
IC411	Totally enclosed, fan-cooled (TEFC): shaft fan blows over a finned frame	The standard industrial LV motor, 0.1–1000 kW
IC416	TEFC with independent motor-driven fan	VFD motors running at low speed
IC511 / IC611	Totally enclosed with air-to-air heat exchanger (tube cooler)	MV motors 200 kW–5 MW
IC81W	Totally enclosed with air-to-water heat exchanger (TEWAC)	Large motors and generators, marine machines
IC71W / direct liquid	Water jacket or oil spray directly on windings	EV traction motors, high-speed machines

Table 9.2 — Representative enclosure codes (IEC 60034-5).

Code	Protection	Use
IP23	Fingers, solid objects > 12.5 mm; spraying water up to 60°	Indoor open machines
IP55	Dust-protected; water jets	Standard industrial TEFC
IP56 / IP66	Powerful water jets; dust-tight (66)	Marine deck, washdown, mining
IP67 / IP68	Temporary or continuous immersion	Submersible pumps, EV drive units

Large generators: hydrogen and water

As turbo-generators grew beyond a few hundred MVA, air could no longer carry the heat out of a rotor of limited diameter. **Hydrogen** at 2–6 bar gauge has about seven times the thermal conductivity of air and one-fourteenth of its density, so windage loss falls to roughly a tenth and heat transfer improves markedly. Hydrogen cooling is standard from roughly 300 MVA to over 1000 MVA, with seals to keep air out (the hydrogen purity is held above about 95 % to stay far from the flammable range) and CO_2 purging procedures. Beyond about 500–600 MVA the stator bars are hollow and cooled internally by **deionised water**, removing heat directly at the copper. Large hydro generators use air or, in the largest, water-cooled stators. These cooling advances, not new electromagnetics, are why unit ratings grew from 100 MVA in the 1950s to 1750 MVA today.

What cooling buys

The usable current density and electric loading (§9.7) rise with cooling intensity:

Cooling	Stator current density (A/mm²)	Electric loading A (kA/m)
TEFC, random-wound, small motors	4–7	15–40
Air-cooled, form-wound, MV motors and gensets	3–5	40–80
Indirect hydrogen-cooled turbo-generator	3–6	80–150
Direct water-cooled stator bars	6–12	150–250
EV traction, oil-spray or water jacket (peak, tens of seconds)	15–30	60–120

9.7 Winding design fundamentals

Windings

Two constructions dominate. **Random-wound** windings, used up to about 690 V and several hundred kW, are made of round enamelled wire inserted into semi-closed slots; turns lie in random positions, so adjacent turns may be from opposite ends of the coil and see the full coil voltage, which is acceptable at low voltage. Copper fill (copper area over slot area) is 0.35–0.45. **Form-wound** windings, used above about 1 kV and in large low-voltage machines, are made of pre-formed coils or bars of rectangular conductors in open slots, with turn and groundwall insulation applied before insertion; fill is 0.50–0.65 and every turn's position and voltage are controlled. **Hairpin** windings, rectangular U-shaped conductors inserted axially and welded at one end, give EV motors fill factors of 0.6–0.75 and automated manufacturing, at the cost of higher AC resistance at high frequency.

Within either construction the designer chooses the number of slots per pole per phase q, single- or double-layer, coil pitch (5/6 is common to suppress 5th and 7th harmonics, Chapter 7), the number of turns per coil, and the number of **parallel paths** a (used when the series turns needed are few and the current large). The constraint linking them is that the phase EMF $E = 4.44 f N_{ph} k_w \Phi$ (Chapter 7, §7.3) must match the supply voltage at the chosen flux density.

Electric and magnetic loading

Two quantities summarise how hard a machine is worked. The **magnetic loading** is the average air-gap flux density $\bar{B}$ over a pole pitch, typically 0.4–0.6 T in induction machines (limited by tooth saturation and magnetising current) and 0.5–0.7 T in synchronous and PM machines. The **electric loading** A is the total armature current per metre of bore circumference,

$$A = \frac{2 m N_{ph} I}{\pi D}$$

with m phases, N_{ph} series turns per phase, I RMS phase current and D the bore diameter. Magnetic loading is limited by the steel; electric loading by cooling.

The output equation

Substituting the EMF equation and the definition of A into the apparent power $S = mEI$ gives, for a three-phase machine with bore diameter D, core length L and synchronous speed n_s (rev/s),

$$S = \frac{\pi^3}{2\sqrt{2}} k_w \bar{B} A \, D^2 L n_s \approx 11 \, k_w \bar{B} A \, D^2 L n_s \quad \text{(VA)}$$

> **Status of the principle: Output equation**
> *Derived*. Follows from the EMF equation and the definitions of electric and magnetic loading, with sinusoidal flux distribution assumed; it is exact in form, but the loadings that may be inserted are empirical, set by cooling, saturation and experience.

The output equation says that output per unit speed scales with rotor volume $D^2 L$ at given loadings. Torque, equivalently, is proportional to rotor volume times the air-gap shear stress, so designers compare machines by **torque per rotor volume (TRV)**: about 2.5–7 kN·m/m³ for small totally enclosed motors, 7–30 for industrial motors, 15–50 for high-performance servos, 30–75 for aerospace machines and 100–250 kN·m/m³ for large liquid- or hydrogen-cooled machines.

Worked example — main dimensions of a 75 kW four-pole motor. Size the 75 kW, 400 V, 50 Hz, four-pole motor with $\eta = 0.95$, power factor 0.86, $k_w = 0.955$, $\bar{B} = 0.45$ T, $A = 30$ kA/m and a square pole face (core length equal to pole pitch).

$$S = \frac{75\,000}{0.95 \times 0.86} = 91.8 \text{ kVA}, \qquad n_s = 25 \text{ rev/s}$$

$$D^2 L = \frac{91\,800}{1.11 \pi^2 \times 0.955 \times 0.45 \times 30\,000 \times 25} = 0.0260 \text{ m}^3$$

With $L = \pi D/4$, $D^3 = 0.0260/0.785$, so $D = 0.321$ m and $L = 0.252$ m. These are close to a real frame-size 280 motor (bore about 0.30–0.32 m, core 0.25–0.30 m). Rated torque at 1480 rpm is 484 N·m, giving TRV $= 484/(\pi \times 0.321^2 \times 0.252/4) = 23.7$ kN·m/m³, in the industrial range. Line current is $91\,800/(\sqrt{3} \times 400) = 132.5$ A; in a delta-connected winding the phase current is 76.5 A, so at 5 A/mm² each phase needs 15.3 mm² of copper per turn, made of several parallel enamelled wires (for example 8 × 1.6 mm diameter), split if necessary between parallel paths.

9.8 Material selection

Electrical steel

Laminated non-oriented silicon steel carries the flux in rotating machines; grain-oriented steel, whose low loss exists only along the rolling direction, is used in transformers (Chapter 4). Grades are named by maximum loss at 1.5 T and 50 Hz and thickness (EN 10106): M270-35A means at most 2.70 W/kg, 0.35 mm.

Table 9.3 — Representative magnetic core materials.

Material	Thickness	Loss at 1.5 T, 50 Hz	Loss at 1.0 T, 400 Hz	Saturation (approx.)	Use
M800-65A (low Si)	0.65 mm	8.0 W/kg	—	2.0 T	Low-cost small motors
M400-50A	0.50 mm	4.0 W/kg	≈ 30–40 W/kg	2.0 T	General industrial motors
M270-35A	0.35 mm	2.7 W/kg	≈ 20–25 W/kg	1.95 T	IE3–IE4 motors, generators
NO20 (thin, high Si)	0.20 mm	≈ 2.2–2.5 W/kg	≈ 12–15 W/kg	1.9 T	EV traction, high-speed machines
Cobalt-iron (49 % Co)	0.1–0.35 mm	≈ 2–3 W/kg (at 1.5 T)	≈ 15–25 W/kg	2.35 T	Aerospace generators, high torque density
Amorphous ribbon	0.025 mm	≈ 0.2–0.3 W/kg (at 1.3 T)	≈ 3–5 W/kg	1.56 T	Distribution transformers, some high-speed motors
Soft magnetic composite	Bulk	High at 50 Hz	Competitive above ~1 kHz	1.5–1.8 T, μr ≈ 500	Axial-flux and claw-pole 3D flux paths
Grain-oriented (CRGO)	0.23–0.30 mm	0.8–1.1 W/kg (rolling direction, 1.7 T)	—	2.0 T	Transformers

The trade-off is plain: thinner and higher-silicon steels lose less, especially at high frequency, but cost more, have lower saturation and are harder to punch. Punching and welding damage the edges and raise local loss by 20–50 % in narrow teeth, which is why small motors' measured core losses exceed catalogue predictions.

Conductors

Copper (resistivity $1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$ at 20 °C, density 8.96 g/cm³) is standard for windings. Aluminium ($2.82 \times 10^{-8}\ \Omega\,\mathrm{m}$, 2.70 g/cm³) needs 64 % more cross-section for the same resistance but weighs half as much for that resistance, and costs less; it is used for die-cast squirrel-cage rotors in most induction motors below a few hundred kW. **Copper die-cast rotors** reduce rotor loss by 15–30 % and help motors reach IE3–IE4 at the cost of harder casting. Large induction motors use fabricated copper or copper-alloy bars brazed to end rings. Hollow copper conductors carry cooling water in the largest generators.

Magnets, insulation and structure

Permanent magnets are chosen from Table 8.1 against temperature, demagnetisation risk and cost. Insulation materials (§9.5) are chosen by thermal class and voltage: enamelled wire grades by thermal index 155–220 °C; mica for voltages above about 3 kV where partial discharge is unavoidable; aramid and polyimide films for slot liners. Structural parts matter too: non-magnetic steel retaining rings on turbo-generator rotors (18Mn18Cr, chosen for stress-corrosion resistance), carbon-fibre magnet sleeves for high-speed PM rotors, aluminium or cast-iron frames (cast iron for vibration damping and corrosion resistance in heavy industry), and shaft steels chosen for fatigue strength.

9.9 Performance testing

Type, routine and site tests

Machines are tested at three levels. **Type tests** are performed on the first machine of a design to verify every performance claim. **Routine tests** are performed on every machine before shipment. **Site (commissioning) tests** verify the installed machine and give the baseline for later condition monitoring. IEC 60034-1, IEC 60034-2-1 (efficiency), IEEE Std 112 (induction motors) and IEEE Std 115 (synchronous machines) define the procedures.

Test	Type / routine	What it shows
Winding resistance (with temperature)	Routine	Reference for copper loss and temperature measurement
No-load test (voltage sweep)	Routine	Core plus friction-windage loss, magnetising current
Locked-rotor test	Type (reduced voltage routine)	Starting current and torque, leakage impedance
Load test with dynamometer or back-to-back	Type	Efficiency, slip, power factor, stray load loss
Heat run (temperature-rise test) to equilibrium	Type	Temperature rise against class limit
Open- and short-circuit characteristics (synchronous)	Type and routine	OCC, SCC, SCR, X_d (Chapter 7)
Sudden short-circuit (synchronous)	Type	X''_d, X'_d, time constants
High-voltage withstand, 1 min	Routine	Insulation integrity
Insulation resistance and polarisation index	Routine and site	Moisture, contamination
Surge comparison (IEEE 522)	Routine for form-wound	Turn-to-turn insulation
Partial discharge (above ~6 kV)	Type and routine	Voids, delamination, slot discharge
Vibration and noise	Routine	Balance, alignment, bearing condition (IEC 60034-14, IEC 60034-9)

Efficiency determination

IEC 60034-2-1 distinguishes **direct** methods, measuring input power electrically and output power with a torque transducer (uncertainty limited by the torque sensor to a few tenths of a per cent in efficiency, which is large compared with the 1 % between IE classes), and **indirect** methods, which measure the losses and subtract them. The preferred method for induction motors, 2-1-1B (equivalent to IEEE 112 Method B), measures the input and output under load, separates the conventional losses using no-load and resistance data, and obtains the stray load loss as a **residual** smoothed by linear regression against torque squared. For large machines, where full-load testing is impossible, the **calorimetric** method measures the heat carried

away by the coolant, and the **summation-of-losses** method combines no-load, short-circuit and resistance tests.

Insulation tests

The **high-voltage test** applies 1000 V + $2U_N$ (minimum 1500 V) at power frequency for one minute between winding and earth: 1800 V for a 400 V motor, 14.2 kV for a 6.6 kV motor, 23 kV for an 11 kV generator. It is a pass–fail proof test, repeated later only at reduced voltage (typically 80 %) because each application consumes some life. **Insulation resistance** at 500–5000 V DC (IEEE Std 43) and the **polarisation index** (ratio of 10-minute to 1-minute resistance) reveal moisture and contamination; IEEE 43 recommends minimum one-minute values of 100 MΩ for form-wound windings and 5 MΩ for random-wound windings below 1 kV (corrected to 40 °C), and a PI of at least 2.0 for class B and above. **Surge tests** apply fast impulses to compare the ringing of phases and detect turn faults. **Partial-discharge** measurement, offline or online through coupling capacitors, trends the condition of medium- and high-voltage windings over years.

9.10 Reliability and maintenance

How machines fail

Surveys of large industrial motors by IEEE (1985) and EPRI give consistent pictures of failure causes (Figure 9.6):

Component	IEEE survey	EPRI survey
Bearings	44 %	41 %
Stator winding	26 %	36 %
Rotor (bars, end rings, shaft, core)	8 %	9 %
Other (accessories, external causes, unknown)	22 %	14 %

Bearing failures come from lubrication faults (too little, too much, wrong or contaminated grease), misalignment, excessive belt or coupling load, and, in VFD-fed motors, **bearing currents**: high-frequency common-mode voltage drives current through the bearing film, causing electrical discharge machining and fluting of the races. Mitigations are insulated bearings (non-drive end, or both ends in large motors), ceramic balls, shaft-grounding brushes and common-mode filters. Stator failures are the end-point of the ageing of §9.5, accelerated by contamination, moisture, thermal cycling, loose coils vibrating in their slots, and voltage surges. Rotor failures in induction motors are mostly broken bars from thermal and mechanical stress during frequent or high-inertia starts.

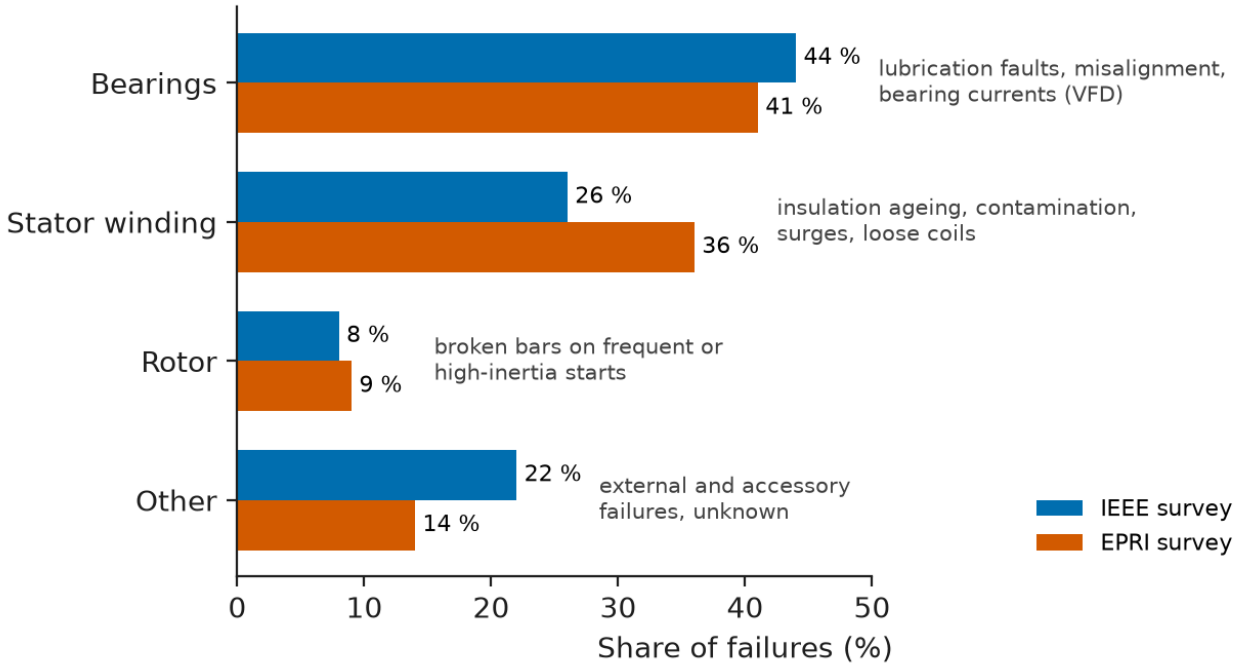


Figure 9.6 — Distribution of failure causes of large induction motors according to the IEEE and EPRI surveys.

Bearing life

Rolling-bearing fatigue life is rated by ISO 281 as the L_{10} life, the number of revolutions that 90 % of a large group of identical bearings will survive:

$$L_{10} = \left(\frac{C}{P}\right)^{p} \times 10^{6} \text{ revolutions}$$

with C the dynamic load rating, P the equivalent load, $p = 3$ for ball and $10/3$ for roller bearings.

> **Status of the principle: Bearing L_{10} life**
> *Empirical.* Fitted to large populations of fatigue tests; it predicts sub-surface fatigue under clean, well-lubricated conditions. Real bearings in motors usually fail earlier from lubrication, contamination or electrical damage, which ISO 281's life-modification factors only partly capture.

Worked example — effect of belt tension on bearing life. A four-pole motor at 1485 rpm has a drive-end deep-groove ball bearing with $C = 120$ kN. With a direct coupling the equivalent load is 6 kN; converting to a belt drive raises it to 8 kN. Compare the L_{10} lives.

$$L_{10} = 20^{3} \times 10^{6} = 8.0 \times 10^{9} \text{ rev} \;\Rightarrow\; \frac{8.0 \times 10^{9}}{1485 \times 60} = 89\,800 \text{ h}$$

$$L_{10} = 15^{3} \times 10^{6} = 3.375 \times 10^{9} \text{ rev} \;\Rightarrow\; 37\,900 \text{ h}$$

A 33 % increase in load cuts fatigue life by 58 %, from about ten years of continuous running to about four. Over-tensioned belts are among the commonest avoidable causes of motor bearing failure; a roller bearing at the drive end is chosen for heavy belt loads.

Maintenance strategies

Maintenance has moved from fixed intervals toward condition:

- **Reactive** (run to failure): acceptable for small, cheap, non-critical motors with spares on the shelf.
- **Preventive** (time-based): regreasing at intervals from the bearing manufacturer's formula (thousands of hours, shorter for hot or fast bearings), cleaning, insulation-resistance checks, brush replacement on DC and slip-ring machines.
- **Predictive / condition-based**: monitoring that detects deterioration before failure, so that repair is planned.
- **Reliability-centred maintenance**: choosing among these per machine according to the consequence of its failure.

Table 9.4 — Condition-monitoring techniques.

Technique	Detects	Typical use
Vibration analysis (velocity, envelope spectrum)	Unbalance, misalignment, looseness, bearing defects	All motors above ~30 kW in critical service; ISO 20816 severity zones
Thermography	Hot connections, blocked cooling, overloaded phases	Motors, switchgear, cable terminations
Motor-current signature analysis	Broken rotor bars, eccentricity, load faults	Induction motors, from the MCC without access to the machine
Insulation resistance and PI trending	Moisture, contamination	All windings, before restarting after long standstill
Online partial discharge	Insulation deterioration in MV and HV windings	Motors above 6 kV, generators
Lubricant and oil analysis	Bearing wear, contamination	Sleeve-bearing machines, gearboxes
Shaft-voltage measurement	Bearing-current risk	VFD-fed motors
Air-gap and flux monitoring	Rotor eccentricity, shorted field turns	Hydro and turbo-generators

NFPA 70B (a standard since 2023) requires an electrical maintenance programme for equipment in the United States, and IEC 60034-23 covers repair and overhaul of rotating machines. A large generator follows a major-inspection cycle of 6–10 years, with rotor removal, wedge-tightness checks, retaining-ring inspection and electrical tests; a fleet of small motors is managed by criticality ranking and spares strategy. For the plant manager the measures of success are mean time between failures (a well-maintained industrial motor fleet achieves failure rates of a few per cent per motor-year) and the cost of unplanned downtime, which for a kiln fan or a data-centre chiller motor can exceed the price of the motor many times over in a single day.

What remains open

Open question	Where it is answered
How are motors matched to their loads and protected within a drive system?	Chapter 10
How do inverter control strategies affect machine losses and heating?	Chapter 12
How do generator ratings and auxiliary loads appear in power-plant design?	Chapter 14
How do transformer and generator thermal ratings enter network and fault studies?	Chapters 17 and 19
How are motors protected against overload, stall, unbalance and earth faults?	Chapter 23
How are insulation systems tested at high voltage and coordinated against surges?	Chapter 24
How do harmonics and unbalance on the supply heat machines?	Chapter 25
How are motor efficiency upgrades, VFDs and life-cycle cost evaluated in an energy audit?	Chapter 28
How are motors selected, cabled and protected in an installation design?	Chapter 30
Heat-transfer correlations and thermal-network theory	*Engineering Physics*, Ch. 19 §19.6
Structure and properties of electrical steels, conductors, magnets and insulating polymers	*The Materials Core*

Next: Chapter 10 opens Part III by placing the machine in its drive system, where motor, load, converter and control must be matched across four quadrants of torque and speed.

Part III — Motor Drives

Part II treated electrical machines as they would behave on a fixed supply: a transformer on a constant-voltage bus, an induction motor started direct-on-line, a synchronous machine locked to the grid frequency. Almost no modern motor of any importance runs that way. Pumps and fans, conveyors and mills, hoists and lifts, machine tools and robots, ship propellers, trains and electric cars all run their motors through power converters that set voltage, current and frequency continuously, under a controller that closes loops on current, speed or position. Motor-driven systems use roughly 45 % of the world's electricity, and the drive decides most of their energy use, dynamics and failure modes. This part treats the motor, its converter, its controller and its mechanical load as one system: how to match them, how to control them, and what each type of drive can and cannot do.

By the end of this part you should understand

- How a load's torque–speed characteristic, inertia and duty cycle set the torque, speed and quadrants a drive must provide.
- Why motors and converters are sized from peak and RMS torque rather than running power, and how starting and braking energy is accounted for.
- How regeneration, brake resistors and active front ends deal with energy returned by overhauling loads, and what limits regenerative braking in traction.
- How the efficiency of the whole chain is evaluated, and why speed control of pumps and fans saves far more energy than any electrical improvement.
- How drives protect motors, converters and people, including thermal overload classes, safe torque off and the insulation and bearing stresses that PWM converters introduce.
- How a DC motor is controlled by armature voltage below base speed and by field weakening above it, using thyristor and chopper converters, cascaded current and speed loops and regenerative braking.
- How PWM inverters synthesise variable-frequency voltages and what limits their output.
- How V/f, field-oriented and direct torque control of induction motors and MTPA and flux-weakening control of PMSMs work, and how a drive is commissioned in the field.

Chapters

Chapter	Central question	What it delivers
Chapter 10 — Drive-System Fundamentals	How do we match a motor and converter to the torque, speed and energy flows its load really demands?	Shaft equation and referred quantities, operating-point stability, duty and RMS-torque sizing, load families, four-quadrant operation, starting and braking, chain efficiency and IEC 61800-9-2 classes, drive protection and safety functions, a selection procedure
Chapter 11 — DC Motor Drives	How are the speed and torque of a DC motor controlled, and what does a practical DC drive look like?	Armature and field control, thyristor-bridge and chopper supplies, power factor at low speed, speed regulation, cascaded current and speed control with tuning rules, regenerative braking and its limits, industrial architecture and DC-specific protection
Chapter 12 — AC Motor Drives	How does a variable-frequency drive make induction and permanent-magnet motors controllable, and how is it commissioned?	VFD structure, PWM and SVPWM voltage limits, V/f control with boost, induction-motor speed-control methods compared, space vectors and the d–q frame, field-oriented and direct torque control, PMSM MTPA and flux weakening, a commissioning procedure

How this part connects to the rest of the book

Part III stands on Part II. The DC machine equations of Chapter 5, the induction-motor equivalent circuit and torque–speed curve of Chapter 6, the synchronous and permanent-magnet machines of Chapters 7 and 8, and the ratings, losses and thermal limits of Chapter 9 are used here as known models; the rotating field and the co-energy torque expression of Part I underlie the d–q equations of vector control. Looking forward, the drive's input side reappears in the power-quality chapter (Chapter 25), where rectifier harmonics and the reactive demand of thyristor drives are measured and mitigated; the inverter control of Chapter 12, in its grid-connected form, is the basis of the renewable-energy interfaces of Chapters 15 and 27; regenerative braking connects to energy storage in Chapter 16; drive energy savings feed the efficiency audits of Chapter 28; and drive isolation and safe torque off return in the electrical safety of Chapter 29.

Two sibling books carry the theory this part uses. Rotational mechanics, gear trains as mechanical transformers and pump affinity laws are taken from *Engineering Physics: Top Down*, Ch. 19, and the averaged converter relations from its Ch. 18. Three-phase circuits, the DC-motor transfer function, feedback, PI tuning and digital control are taken from *Circuits, Signals and Communications*, Ch. 10 and Chs. 33–35. The circuit-level design of rectifiers, choppers and inverters, power semiconductors, gate drivers and the microcontrollers that run drive firmware belongs to *Electronics & Computing*, and robot and automation controllers to the robotics and automation books.

Chapter 10 — Drive-System Fundamentals

Part III · Motor Drives

A copper mine commissions a new overland belt conveyor: 1.8 km long, 3 m/s belt speed, driven by a 160 kW induction motor through a helical gearbox. On the first loaded start after a weekend shutdown the motor hums, draws six times its rated current, and the belt does not move; the overload relay trips after eight seconds. Nothing is broken. The motor was chosen from the steady running power, 125 kW, with a comfortable margin, but nobody counted the torque needed to break the cold, loaded belt free and to accelerate 60 tonnes of belt and ore, nor the fact that a direct-on-line induction motor gives its torque at the wrong speeds for this job. The same weekend, at a water utility across the valley, a 132 kW pump runs at full speed against a half-closed valve, burning 45 kW that does nothing but heat the water, and in an office tower a lift returns 10 kW of braking power to a resistor on the roof every time a full car comes down. Each of these is a *drive-system* problem: a motor, a power converter, a controller and a mechanical load that were not matched to each other.

> **How do we match a motor to the torque and speed its load actually demands, during starting, running and braking?**
> **How do we decide how many quadrants a drive needs, how efficient the whole chain is, how it is protected, and which drive to buy?**

Concept	Engineering destination
Motor-load matching	Gear ratios, referred inertia, operating-point stability of conveyors, pumps, hoists
Torque and speed requirements	Motor and converter sizing from duty cycles, peak and RMS torque
Load characteristics	Constant-torque, fan-law and constant-power loads; choosing the control method
Four-quadrant operation	Lifts, hoists, EVs, rolling mills, test benches, cranes
Starting and braking	Acceleration time, starting current, braking resistors, regenerative supply units
Drive efficiency	Pump and fan energy savings, IEC 61800-9-2 efficiency classes, life-cycle cost
Protection	Motor overload classes, drive trips, safe torque off, insulation and bearing stress
Drive selection	Specification sheets, normal and heavy duty ratings, supply and environment checks

Roadmap. What a drive system is and how motor and load meet on a common shaft (10.1) → the torque and speed a duty really needs (10.2) → the shapes of real load characteristics (10.3) → the four quadrants and which loads need them (10.4) → starting and braking, where most sizing errors live (10.5) → efficiency of the whole chain (10.6) → protection of motor, converter and people (10.7) → a selection procedure that ties it together (10.8).

10.1 Motor-load matching

The drive system as a chain

An electric drive is the chain of Figure 10.1: a **supply** (the grid, a battery, a DC traction line), a **power converter** that sets the voltage, current and frequency the motor sees, the **motor**, a **mechanical transmission** (coupling, gearbox, belt, screw, rope drum), the **load**, and a **controller** with its sensors that closes loops on current, speed or position. The machines themselves were the subject of Part II; this part is about the chain. A drive with an excellent motor can still fail its duty because the gearbox ratio is wrong, the converter cannot supply the starting current, or the controller has no way to absorb the energy that comes back from the load.

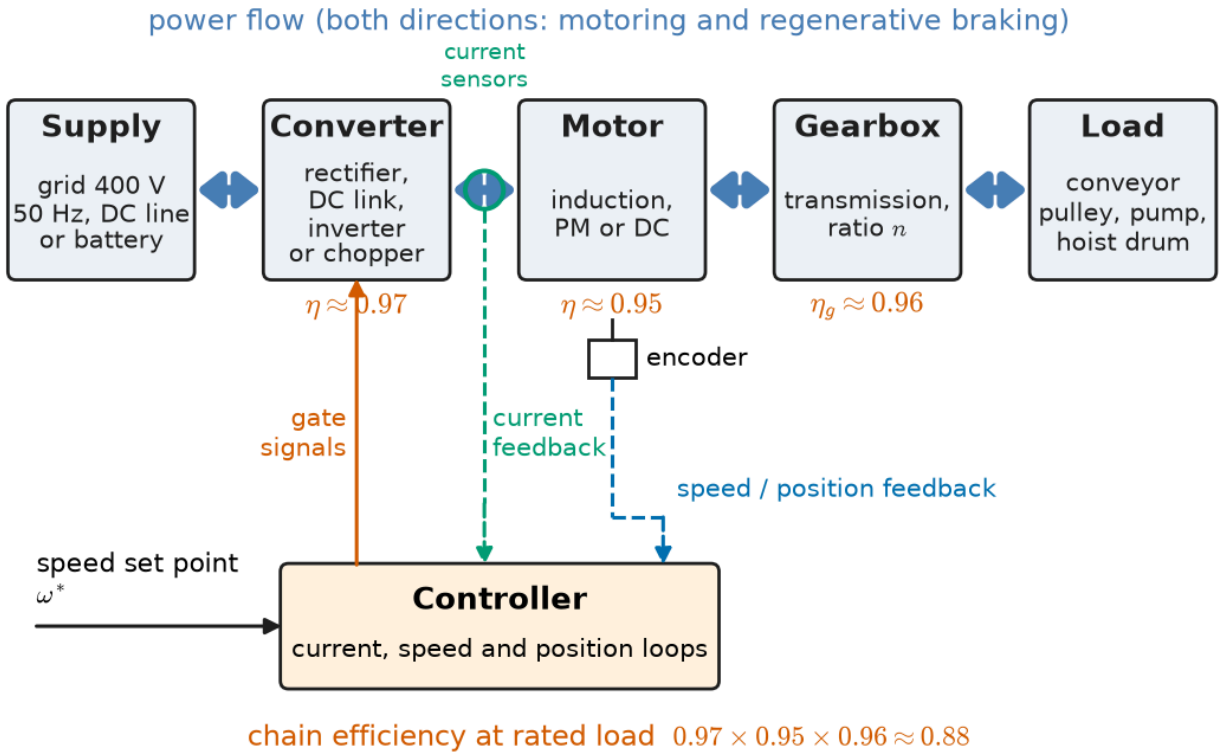


Figure 10.1 — The parts of an electric drive system and the power and signal flows between them.

Figure 10.1 shows power flowing in both directions. Whether the converter and the supply can actually accept power coming back from the load is one of the first questions a drive specification must answer (Section 10.4).

The equation of motion on the motor shaft

> **Boundary.**
> Rotational dynamics, the torque–angular-velocity effort–flow pair and the gear pair as a mechanical transformer are treated in *Engineering Physics*, Ch. 19 §19.2 (with rigid-body rotation in Ch. 9 §9.7); the motor as a gyrator between the electrical and mechanical domains is in Ch. 17 §17.5. Here those results are used, not derived.

Everything a drive does mechanically is governed by Newton's law for rotation applied to the motor shaft, with every inertia and torque referred to that shaft:

$$J_{eq}\frac{d\omega_m}{dt} = T_m - T'_L$$

where T_m is the electromagnetic torque of the motor (less its own friction), T'_L is the load torque referred to the motor shaft, and J_{eq} is the total inertia seen by the motor. When $T_m > T'_L$ the drive accelerates; when $T_m < T'_L$ it decelerates; in steady state the two torques are equal. All of starting, braking and speed control is the art of shaping T_m so that this equation produces the speed profile the process wants.

> **Status of the principle: Shaft equation of motion**
> *Fundamental.* Newton's second law for rotation of a rigid shaft; it fails only where shaft torsional compliance or backlash matters (long shafts, rope hoists, rolling-mill spindles), when the drivetrain must be modelled as several inertias joined by springs.

Referring torque and inertia through a transmission

A gearbox with ratio $n = \omega_m/\omega_L$ (motor speed over load speed) and efficiency η_g is a mechanical transformer (*Engineering Physics*, Ch. 19 §19.2.2). Referred to the motor shaft, a load torque and inertia become

$$T'_L = \frac{T_L}{n\,\eta_g}\ \text{(motoring)}, \qquad J'_L = \frac{J_L}{n^2}$$

When power flows backwards (the load drives the motor, as in a lowering hoist) the efficiency multiplies instead of divides: $T'_L = T_L\eta_g/n$. A translating mass m moving at speed v is referred by equating kinetic energies, $J'\omega_m^2 = mv^2$, so $J' = m(v/\omega_m)^2$. For a conveyor pulley or hoist drum of radius r driven through ratio n, $v/\omega_m = r/n$.

> **Status of the principle: Referred torque and inertia**
> *Derived.* Follows from power conservation across an ideal transmission and equality of kinetic energy; the efficiency correction is an approximation that treats transmission loss as proportional to transmitted torque, adequate for helical gears at more than about 25 % load.

The $1/n^2$ is why gearboxes are used. A conveyor pulley turning at 72 rpm with 11 600 kg·m² of referred belt inertia looks to a 1485 rpm motor like 27 kg·m², comparable with its own rotor. Without a gearbox a 72 rpm motor of the same power would need about 21 times the torque, and therefore roughly 21 times the active volume.

Matching ratio for dynamic loads

In servo applications (machine-tool axes, packaging, robots) the requirement is not steady torque but acceleration. For a load inertia J_L and motor inertia J_m with no load torque, the motor torque needed to give the load an angular acceleration α_L is

$$T_m = \left(J_m n + \frac{J_L}{n}\right)\alpha_L$$

which is smallest when $n_{opt} = \sqrt{J_L/J_m}$, i.e. when the referred load inertia equals the motor inertia. In practice servo designers accept inertia ratios J'_L/J_m of 1 to 10, because a gearbox ratio chosen for minimum torque may give too low a top speed, and because a stiff coupling tolerates a larger ratio than a belt. Ratios above about 10 make the speed loop hard to tune: the load resonance moves into the loop bandwidth.

Operating point and its stability

In steady state the motor runs where its torque–speed curve crosses the load curve, both referred to the same shaft (Figure 10.5 in Section 10.5). Not every crossing is a stable operating point. Suppose the speed rises slightly by $\Delta\omega$. Linearising the shaft equation,

$$J_{eq}\frac{d\,\Delta\omega}{dt} = \left(\frac{\partial T_m}{\partial \omega} - \frac{\partial T'_L}{\partial \omega}\right)\Delta\omega$$

and the disturbance dies away only if

$$\frac{\partial T'_L}{\partial \omega} > \frac{\partial T_m}{\partial \omega}$$

> **Status of the principle: Steady-state stability criterion**
> *Derived*. Linearisation of the shaft equation about the operating point (small-signal stability, as in *Circuits, Signals and Communications*, Ch. 33); it ignores the motor's electrical dynamics, which can destabilise an otherwise stable point (Chapter 12).

An induction motor on its normal low-slip region has a steeply falling torque curve, so it is stable against almost any load. On the high-slip side of its peak torque the motor torque *rises* with speed, and a constant-torque load crossing there is unstable: the drive either accelerates to the normal region or stalls. A DC series motor with a load that falls faster with speed than the motor torque does, or a fan driven by a motor whose torque rises with speed, are the classic textbook traps. Closed-loop drives make the question nearly academic, because the speed controller deliberately gives T_m a strong negative slope against speed error, but the criterion still governs drives in torque or open-loop scalar mode.

Worked example — matching a mine conveyor drive. The conveyor of the opening has belt speed $v = 3.0$ m/s on a drive pulley of diameter 0.80 m. The steady belt pull (friction of idlers, belt flexing and lift of the ore) is 40 kN; the total moving mass of belt and ore is 60 t, and the pulleys and idlers add an equivalent of 2000 kg·m² at the drive pulley. The motor is four-pole, 50 Hz, rated 160 kW at 1485 rpm; the helical gearbox has $\eta_g = 0.96$, and motor rotor plus coupling and gear input add 5.0 kg·m².

- Pulley speed: $\omega_L = v/r = 3.0/0.40 = 7.50$ rad/s (71.6 rpm). Motor speed $\omega_m = 1485 \times 2\pi/60 = 155.5$ rad/s, so $n = 155.5/7.50 = 20.7$.
- Pulley torque $T_L = 40\ \text{kN} \times 0.40\ \text{m} = 16.0$ kN·m; referred, $T'_L = 16\,000/(20.7 \times 0.96) = 804$ N·m. Belt power $Fv = 120$ kW; motor shaft power $120/0.96 = 125$ kW.
- Inertia at the pulley: $mr^2 + 2000 = 60\,000 \times 0.16 + 2000 = 11\,600$ kg·m²; referred, $11\,600/20.7^2 = 27.0$ kg·m². Total $J_{eq} = 27.0 + 5.0 = 32.0$ kg·m². The stored kinetic energy at full speed is $\tfrac{1}{2}J_{eq}\omega_m^2 = 387$ kJ.
- To reach full speed in 20 s (a typical limit for belt stresses) at constant acceleration needs an extra $J_{eq}\omega_m/t = 32.0 \times 155.5/20 = 249$ N·m.
- The motor's rated torque is $160\,000/155.5 = 1029$ N·m. Running load plus acceleration is $804 + 249 = 1053$ N·m, 1.02 times rated; if the cold belt needs 30 % more than its running pull to break away, the start demands $1.3 \times 804 + 249 = 1294$ N·m, 1.26 times rated, for the 20 s of the start.

The motor is correctly sized for running (78 % load) and thermally capable of the start, but a direct-on-line start gives the wrong torque in the wrong way. Its catalogue curve (Figure 10.5) offers about 2 times rated torque at standstill and a pull-up minimum of 1.6 near a third of speed, but it draws 6 to 7 times rated current, about 1.8 kA. On the mine's weak 400 V supply that current depresses the terminal voltage by 15 %, and induction-motor torque falls with the square of voltage: the pull-up torque drops to $1.6 \times 0.85^2 = 1.16$ times rated, below the 1.26 the loaded belt needs. The motor hangs at pull-up speed, drawing nearly locked-rotor

current, until the overload relay trips. A variable-frequency drive rated for 150 % torque for 60 s (a "heavy-duty" rating, Section 10.8) delivers the 126 % smoothly at about 130 % current, which is the solution adopted for most long conveyors. The example shows the rule that defeats most first-time drive specifications: size for the *torque profile*, not for the running power.

10.2 Torque and speed requirements

From process to torque–speed profile

The process sets a speed or position profile; the shaft equation turns it into a torque profile. A crane hoist must lift rated load at rated speed, accelerate it within a time fixed by productivity, and hold it at zero speed. A rolling mill must keep strip speed within 0.1 % while the bite of the next pass adds a torque step. An electric car must deliver high torque from standstill for acceleration, constant power over a wide speed range for overtaking, and a top speed four or five times the speed at which full torque ends. Table 10.1 collects representative requirements.

Table 10.1 — Representative torque and speed requirements (typical values).

Application	Speed range	Peak / rated torque	Accuracy or bandwidth	Quadrants
Centrifugal pump, fan	30–100 %	1.1	1–2 % speed	1
Belt conveyor	10–100 %	1.3–1.6 at start	1 % speed, load sharing	1 (2 on downhill)
Crane hoist	0–100 %, hold at zero	1.6–2.0	Smooth zero-speed holding	4
Passenger lift	0–100 % (1–6 m/s)	1.8–2.5	Levelling within 5 mm	4
Rolling mill stand	0–100 %, field weakening to 2–3×	2.5–3.0 for 10 s	0.01–0.1 % speed, 50–100 Hz current-loop bandwidth	4
Paper machine section	5–100 %	1.5	0.01 % speed (draw control)	1–2
CNC axis servo	0–6000 rpm	3–5 for 0.1 s	1–10 µm position, 100–300 Hz speed bandwidth	4
EV traction motor	0–12 000 rpm, constant power above 3000–4000 rpm	2–3 for 10–30 s	1–5 ms torque response	4
Ship propulsion (fixed-pitch propeller)	0–100 % both directions	1.2	1 % speed	4

Base speed, constant torque and constant power

A motor's continuous capability is limited by two different things in two speed regions (Figure 11.1 and Figure 12.6 show the DC and PMSM versions). Below **base speed** the limit is current, because current sets copper loss and therefore temperature; with full flux, full current means full torque, so the region is called **constant**

torque. Above base speed the converter has run out of voltage, because the motor's back-EMF $E \propto \Phi\omega$ has reached the supply limit; the only way to go faster is to reduce flux (**field weakening**), and with current still at its limit the torque falls as $1/\omega$ and power stays constant. At still higher speed the machine's own limits (commutation in DC machines, breakdown torque in induction machines, the voltage ellipse in PM machines) make torque fall faster, as $1/\omega^2$. A drive specification therefore states at least: rated torque up to base speed, the field-weakening range (2:1 for most industrial induction motors, 3:1 to 4:1 for traction), and the maximum speed.

Peak torque, RMS torque and duty

Two different motor limits apply to a cyclic duty. **Peak torque** is limited by the converter's current limit and, in induction motors, by breakdown torque (typically 2 to 3 times rated); in PM motors by demagnetisation of the magnets. **Continuous torque** is limited by temperature, which responds to the average of the losses over the thermal time constant of the motor (tens of minutes for a 100 kW motor). Because copper loss is proportional to current squared, and current to torque below base speed, the thermally equivalent constant torque of a cycle is its **RMS torque**:

$$T_{rms} = \sqrt{\frac{\sum_k T_k^2 \, t_k}{\sum_k t_k}}$$

A self-ventilated motor cools less well at standstill and low speed, so standstill periods are counted with a reduced weight $k_0 \approx 0.25$–0.5 in the denominator. The duty types S1 (continuous) to S10 of IEC 60034-1, and the thermal reasoning behind them, are covered in Chapter 9.

> **Status of the principle: RMS-torque sizing**
> *Approximate*. Valid when the cycle is much shorter than the motor's thermal time constant, when copper loss dominates and is proportional to torque squared (no field weakening), and when cooling is independent of speed or corrected by a standstill factor; it fails for long cycles, for operation in field weakening, and for motors whose iron loss is a large share of the total.

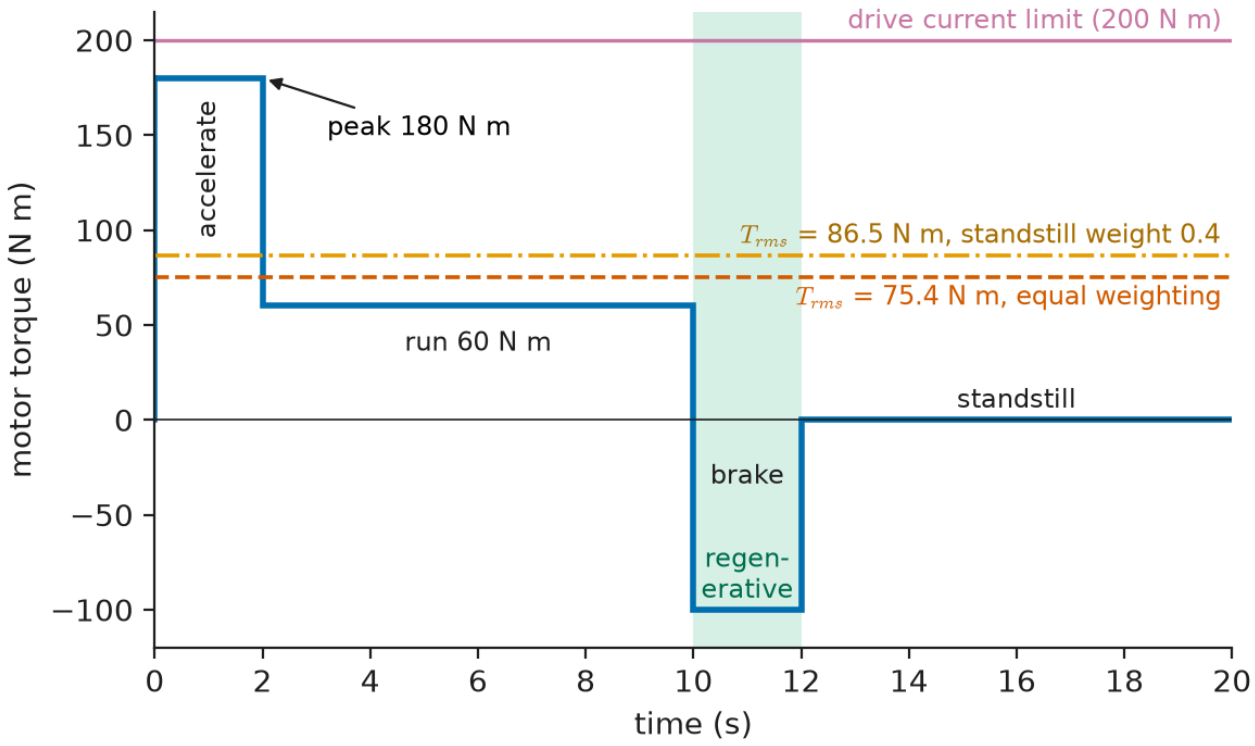


Figure 10.2 — Torque profile of a cyclic packaging-machine duty and its RMS equivalent.

Worked example — RMS torque of a cyclic duty. A packaging machine indexes a turret with the cycle of Figure 10.2: 180 N·m for 2 s to accelerate, 60 N·m for 8 s to run, −100 N·m for 2 s to brake, then 8 s at rest; the motor runs at 1470 rpm. With equal weighting

$$T_{rms} = \sqrt{\frac{180^2 \times 2 + 60^2 \times 8 + 100^2 \times 2}{20}} = \sqrt{\frac{64\,800 + 28\,800 + 20\,000}{20}} = 75.4\ \text{N m}$$

For a self-ventilated motor with standstill weight $k_0 = 0.4$ the denominator becomes $12 + 0.4 \times 8 = 15.2$ s, giving $T_{rms} = 86.5$ N·m. At 1470 rpm ($\omega = 153.9$ rad/s) the thermally equivalent power is $86.5 \times 153.9 = 13.3$ kW, while the peak demand is $180 \times 153.9 = 27.7$ kW. A 15 kW four-pole motor (rated torque about 97 N·m) is thermally adequate, its breakdown torque of 2.5 to 3 times rated covers the 180 N·m peak, and the converter must be sized for the peak current, about 1.9 times motor rated current for 2 s, not for 15 kW. The braking segment returns energy (Section 10.4) that must be absorbed somewhere.

10.3 Load characteristics

The main families

The torque a load demands as a function of speed is its **mechanical characteristic**. Most industrial loads fall into a few families (Figure 10.3), and the family largely decides the drive.

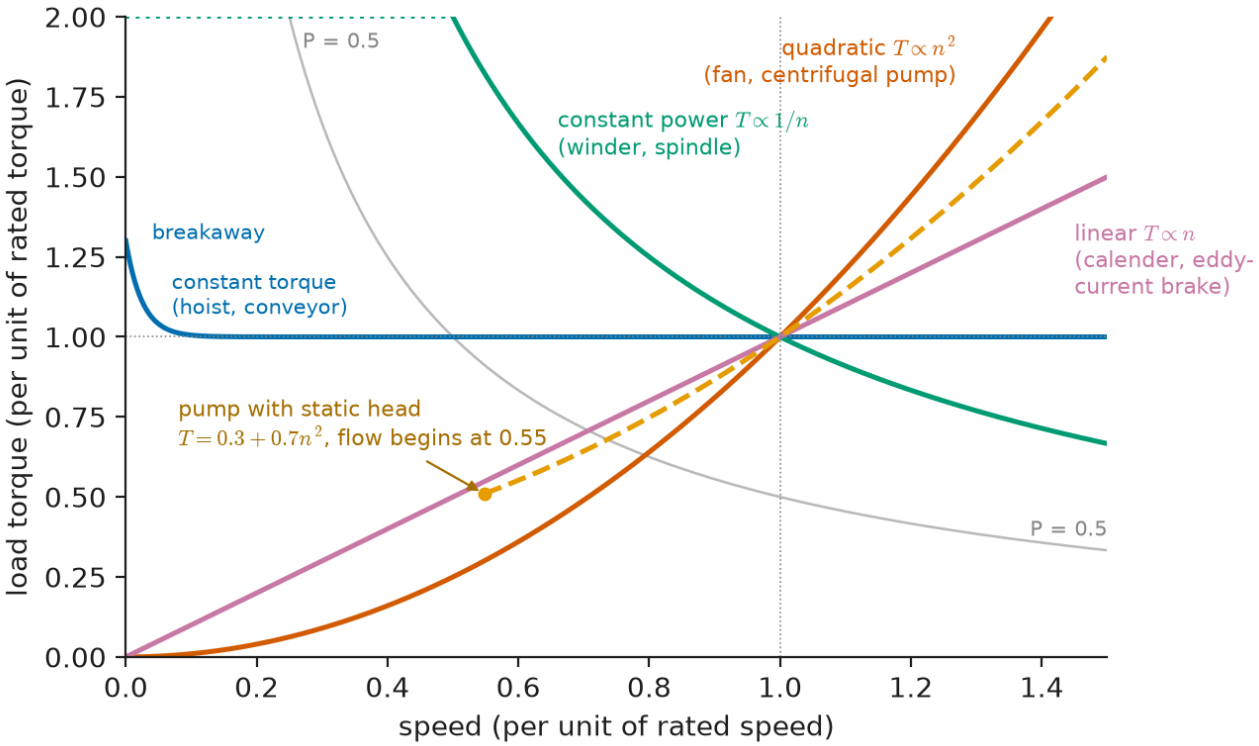


Figure 10.3 — Torque-speed characteristics of the main load families, normalised to rated torque and rated speed.

- **Constant torque**, $T_L \approx$ const. Hoists and cranes (gravity), conveyors (friction), positive-displacement pumps and compressors, extruders, mixers. Power is proportional to speed. Static friction usually adds a **breakaway torque** at standstill of 10 to 50 % above the running value, more after a cold standstill (grease, frozen belts, settled slurry).
- **Quadratic (fan law)**, $T_L \propto \omega^2$, $P \propto \omega^3$. Centrifugal fans, pumps and compressors. They follow the affinity laws, flow $\propto$ speed, head $\propto$ speed², power $\propto$ speed³ (*Engineering Physics*, Ch. 19 §19.7.3). With a static head (lifting water to a reservoir), flow begins only above the speed at which the pump's shut-off head equals the static head, and power then falls less steeply than the cube.
- **Constant power**, $T_L \propto 1/\omega$. Centre winders and unwinders (constant tension, so torque proportional to roll radius while speed is inversely proportional to it), lathe and milling spindles (constant cutting power across tool diameters), traction at speed. These loads want a motor with a wide field-weakening range.
- **Linear**, $T_L \propto \omega$. Calenders, viscous mixers at low Reynolds number, eddy-current brakes and generators feeding resistive loads.
- **Position-dependent and pulsating**. Reciprocating compressors and pumps, crank presses and beam pumps (oil-field "nodding donkeys") have torque that varies with shaft angle within each revolution; flywheels and inertia smooth it.

> **Status of the principle: Fan affinity laws**
> *Approximate.* Derived from dynamic similarity for a pump or fan running at different speeds against a purely frictional system; they fail when static head is a large share of the total, at very low speeds where Reynolds-number effects change efficiency, and for positive-displacement machines.

Active and passive loads

A second distinction matters more than the shape: is the load **passive** or **active**? A passive load (friction, fans, pumps, cutting) always opposes motion; remove the motor torque and it stops. An active load (gravity on a hoist or lift, a vehicle on a downhill grade, an unwinder, a wind or water turbine on a test bench) can drive the shaft by itself. An active load sends power back into the drive whenever it overhauls the motor, which is the reason four-quadrant drives exist.

Load characteristic decides control strategy

Table 10.2 links the load family to what the drive must do.

Table 10.2 — Load families and drive consequences.

Load family	Starting requirement	Typical control	Energy lever
Fan, centrifugal pump	Low (torque rises with speed)	Open-loop V/f, soft starter for fixed speed	Speed instead of dampers or valves: power $\propto$ speed^3
Conveyor, extruder, positive-displacement pump	High: breakaway plus acceleration	Sensorless vector, 150 % heavy-duty rating	Avoid oversizing; regenerate on downhill conveyors
Hoist, crane, lift	Full torque at zero speed, holding	Closed-loop vector with encoder, brake logic	Regeneration on lowering
Winder, spindle	Wide constant-power range	Vector or DC with field weakening	Correct base speed choice
Servo axis	High peak, fast response	PMSM servo, position loop	Low inertia, DC-bus sharing
Traction	High starting torque, constant power to 3–4× base	PMSM or induction vector, four-quadrant	Regenerative braking

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10.4 Four-quadrant operation

The torque–speed plane

Plot speed on the vertical axis and torque on the horizontal axis, both taken positive in the "forward" direction (Figure 10.4). The plane has four quadrants, and the product $P = T_m \omega_m$ tells which way power flows:

- **Quadrant I**, $T > 0, \omega > 0$: forward motoring. Power flows from supply to load.
- **Quadrant II**, $T < 0, \omega > 0$: forward braking. The motor runs forward but its torque opposes motion; power flows from load to supply (or to a resistor). The motor is a generator.
- **Quadrant III**, $T < 0, \omega < 0$: reverse motoring.
- **Quadrant IV**, $T > 0, \omega < 0$: reverse braking.

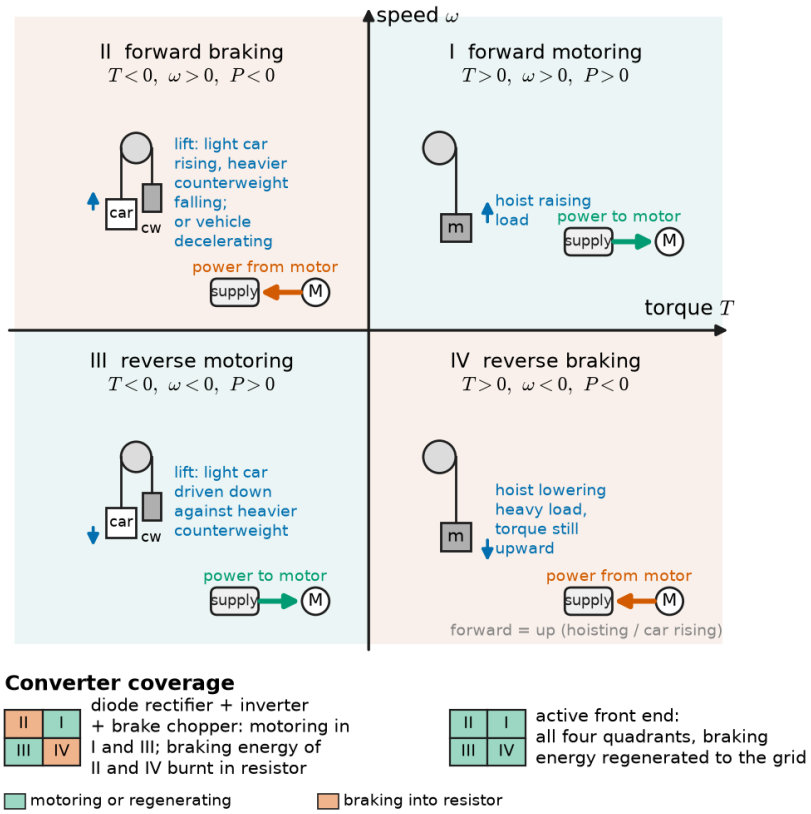


Figure 10.4 — The four quadrants of the torque-speed plane, with a hoist and a lift as examples and the direction of power flow in each.

Which quadrants a drive needs follows from the load. A pump or fan needs only Quadrant I. A conveyor needs I, plus II if it runs downhill or must stop faster than friction stops it. A crane hoist needs I (raising) and IV (lowering a load, with torque still upward to hold it against gravity). A lift, a rolling mill, a ship propeller and an electric vehicle use all four.

Two ways to brake: dissipate or regenerate

Braking energy that leaves the motor in Quadrants II and IV must go somewhere. A standard low-voltage VFD has a diode rectifier that cannot pass power back to the grid; the returned energy charges the DC-link capacitor, and its voltage rises. A 400 V drive (DC link about 540 V) typically trips on overvoltage near 800 V. Three remedies exist:

1. **Ramp-limited deceleration.** If the load's own friction or fan torque absorbs the energy, simply decelerate slowly enough. Drives include an overvoltage controller that automatically stretches the ramp.
2. **Brake chopper and resistor.** An IGBT switch connects a resistor across the DC link above a threshold (about 700 to 760 V for 400 V drives). The energy is burnt as heat. This is cheap and the usual choice for infrequent stops and for cranes up to a few hundred kW.
3. **Regeneration.** An active front end (an IGBT rectifier, essentially a grid-connected inverter, Chapter 27), a thyristor regenerative unit, or a **common DC bus** shared with other drives that are motoring at the same moment returns the energy. It costs more but saves energy where braking is frequent and heavy: lifts, cranes, test benches, centrifuges, downhill conveyors, mine winders.

In DC drives the same choice appears as rheostatic versus regenerative braking (Chapter 11); in traction and EVs regeneration is standard, and the limit is the receptivity of the battery or the traction line (Section 11.6).

Worked example — the regenerating lift. A traction lift has car mass 1000 kg, rated load 1000 kg and a counterweight of car mass plus half rated load, 1500 kg. Rated speed is 2.5 m/s, travel 30 m. With a full car descending, the car side is $2000 - 1500 = 500$ kg heavier than the counterweight. Gravity therefore drives the system with $F = 500 \times 9.81 = 4905$ N, and holding constant speed requires the machine to brake. Taking car-up as forward, the speed is negative and the torque still acts upward on the car side: Quadrant IV, like the hoist lowering its load. Mechanical power returned is $Fv = 4905 \times 2.5 = 12.3$ kW. With roping and sheave efficiency 0.85, motor 0.95 and inverter 0.97, the electrical power reaching the DC link is $12.3 \times 0.85 \times 0.95 \times 0.97 = 9.6$ kW for the 12 s run, 115 kJ per trip. An empty car going *up* is the mirror case (the counterweight is 500 kg heavier, so the machine must hold the car back: positive speed, negative torque, Quadrant II) and also regenerates; a full car up or an empty car down are motoring cases, drawing $12.3/(0.85 \times 0.95 \times 0.97) = 15.7$ kW.

With a brake resistor, the resistor must absorb up to the full 12.3 kW (sized conservatively on mechanical power) at the lower end of the chopper threshold band, about 700 V: $R \le V^2/P = 700^2/12\,260 = 40\ \Omega$, rated for the duty (a lift in an office tower may make 200 such trips a day). With a regenerative unit, about half of all trips return roughly 115 kJ each. At 200 trips a day, 100 regenerating, that is 100×115 kJ $= 11.5$ MJ ≈ 3.2 kWh a day returned instead of heating the machine room, a 15 to 30 % cut in lift energy, which is why modern lift standards for energy performance (ISO 25745) reward regeneration.

10.5 Starting and braking

Acceleration time

Integrating the shaft equation from standstill to the final speed ω_f gives the starting time:

$$t_{start} = J_{eq} \int_0^{\omega_f} \frac{d\omega}{T_m(\omega) - T'_L(\omega)}$$

The integrand is the reciprocal of the **accelerating torque** $T_m - T'_L$, the gap between the motor and load curves in Figure 10.5. Where the gap is small (near the induction motor's pull-up torque, or with a high-torque load) the drive spends most of its starting time. In practice the integral is evaluated graphically or numerically in steps of speed.

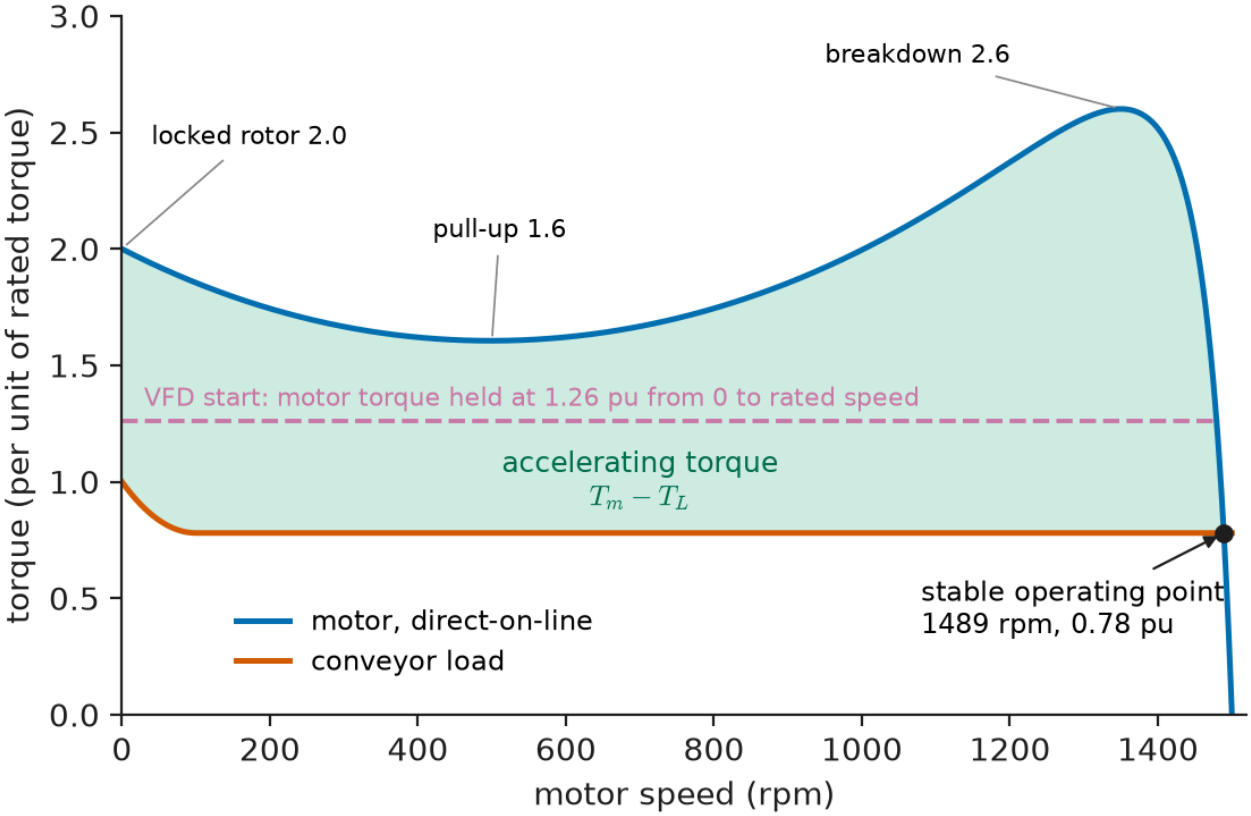


Figure 10.5 — Direct-on-line torque-speed curve of a four-pole induction motor against a conveyor load, showing the accelerating torque and the operating point.

Three practical consequences follow. First, in a direct-on-line start the motor's rotor losses during acceleration equal the kinetic energy finally stored in the load (for an unloaded induction motor), and more with load torque present, so frequent starts of high-inertia loads overheat rotors; motor data sheets give the permitted number of starts per hour and the maximum load inertia. Second, long starts with high current trip overload relays (Section 10.7). Third, the supply sees the starting current: a 160 kW motor started direct-on-line draws about 1.8 kA at 400 V, enough to depress the voltage of a weak site supply by 10 % or more (Chapter 25).

Starting methods at system level

The starting methods for induction motors (direct-on-line, star–delta, autotransformer, soft starter, VFD) and their current and torque ratios are described in Chapter 6; DC-motor starting in Chapter 5. From the drive-system point of view the choice compares three things the method controls: the starting current drawn from the supply, the starting torque available at the shaft, and whether the speed can be controlled during and after the start. Table 10.3 summarises.

Table 10.3 — Starting methods compared for a cage induction motor (typical values, per unit of rated).

Method	Supply current	Starting torque	Speed control	Typical use
Direct-on-line	6–8	1.5–2.5	No	Small motors, stiff supply
Star–delta	2–2.7	0.5–0.8	No	Light-start pumps, fans
Autotransformer (65 % tap)	2.5–3.4	0.6–1.0	No	Larger pumps, compressors
Soft starter (voltage ramp)	2.5–4.5 adjustable	Proportional to voltage²	Start and stop only	Pumps (water-hammer control), fans, conveyors with light start
Variable-frequency drive	1.0–1.5	1.5–2.0 at any speed	Full	Heavy start, speed control, frequent starts

Braking methods

Braking is the mirror of starting, with the important difference that the energy now flows back. The options:

- **Mechanical brakes** (disc, drum, band) dissipate in friction. They remain essential as holding and emergency brakes (a lift or hoist brake is a safety device, sized to stop and hold 125 % to 150 % of rated load), but they wear and cannot be modulated finely.
- **Dynamic (rheostatic) braking.** The motor runs as a generator into a resistor: across the armature of a DC machine, or via the brake chopper of a VFD. Braking torque falls as speed falls unless the resistance is stepped down.
- **DC injection braking** of an induction motor: DC in the stator produces a stationary field; the rotor's kinetic energy is dissipated in the rotor. Cheap and built into most drives, but all the energy heats the motor.
- **Plugging** (reverse-phase or reverse-polarity braking). Reversing two supply phases of an induction motor or the armature of a DC motor gives strong braking torque, but the motor dissipates about three times the kinetic energy of the load. Limited to small drives and emergency use.
- **Regenerative braking**, converting the kinetic or potential energy back to electrical energy in the supply or a battery. The most efficient; covered for DC drives in Section 11.6 and for AC drives in Chapter 12.

The time to stop under a constant braking torque T_b plus load torque is $t_{stop} = J_{eq}\omega_0/(T_b + T'_L)$, and the energy to be absorbed by the braking device is the stored kinetic energy minus the work done by the load. For the conveyor of Section 10.1, the 387 kJ stored at full speed would be dissipated by belt friction alone in about $J\omega/T'_L = 32.0 \times 155.5/804 = 6.2$ s, so a normal stop needs no electrical braking at all. Long *downhill* conveyors are the opposite: gravity on the loaded belt can exceed friction, and the drive must regenerate continuously, sometimes several megawatts on large overland conveyors.

10.6 Drive efficiency

The efficiency chain

The efficiency of a drive system is the product of the efficiencies of its stages (*Engineering Physics*, Ch. 19 §19.4.1):

$$\eta_{system} = \eta_{converter}\,\eta_{motor}\,\eta_{transmission}\,\eta_{process}$$

where the last factor, the efficiency with which the load turns shaft power into useful work (hydraulic power for a pump, air power for a fan), is often the smallest and the most variable. Table 10.4 gives representative values.

Table 10.4 — Representative efficiencies of drive-system stages at rated load.

Stage	Typical efficiency	Notes
LV VFD, 7.5–250 kW	96–98.5 %	Diode front end; active front end 1–1.5 points lower
MV drive, 1–20 MW	97–99 %	Including input transformer for multi-pulse designs: about 96–97.5 %
Induction motor 15 kW, IE3 / IE4 (IEC 60034-30-1)	92.1 % / 93.9 %	Four-pole, 50 Hz minimum values
Induction motor 160 kW, IE3 / IE4	95.8 % / 96.6 %	Four-pole, 50 Hz minimum values
PMSM, 15–200 kW	94–97 %	Higher at part load than induction
Helical gear stage	97–99 %	Per stage
Worm gear	40–90 %	Strongly ratio-dependent
V-belt / synchronous belt	93–97 % / 97–98 %	
Centrifugal pump at best-efficiency point	70–90 %	Falls rapidly away from it

The motor's efficiency classes and losses are dealt with in Chapter 9 and, as a plant energy measure, in Chapter 28. For the drive as a whole, IEC 61800-9-2 defines efficiency classes IE0 to IE2 for the **complete drive module** (the converter, CDM) and IES0 to IES2 for the **power drive system** (converter plus motor, PDS), measured at defined speed and torque points; IE2 for a CDM means losses at least 25 % below the standard's reference value, and IES2 for a PDS means losses at least 20 % below its reference.

Part-load efficiency and why it matters

Most drives spend most of their life at part load. A VFD has losses that are roughly a constant (control, fans, capacitor losses) plus a term proportional to current (switching and conduction), so its efficiency stays high to about 25 % load. Induction motors lose efficiency below about 50 % load because their magnetising losses do not fall, and an oversized motor running at 30 % load can be 3 to 5 points less efficient than its nameplate. The converter also adds losses *in the motor*: the PWM ripple current raises motor losses by roughly 5 to 15 % of their sinusoidal value at 4 to 8 kHz switching.

The biggest gains, though, are not in the electrical stages at all. They come from replacing throttling with speed control in fans and pumps, where the electrical chain's losses are a few percent and the throttle's are tens of percent.

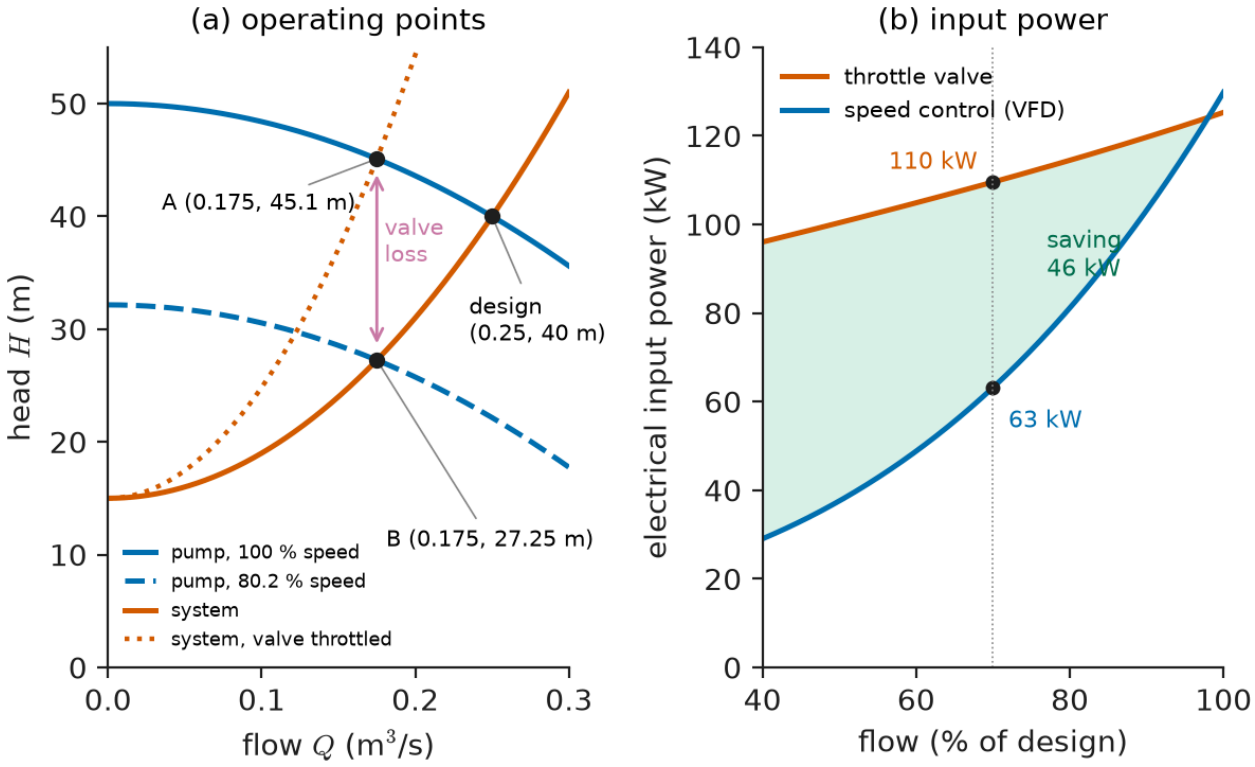


Figure 10.6 — Pump flow control by throttling versus by speed: head-flow operating points and the shaft power each method needs.

Worked example — throttle valve versus speed control. A water-utility booster pump has the curve $H = 50 - 160Q^2$ (head in m, flow in m³/s) at rated speed and works against a system $H = 15 + 400Q^2$, of which 15 m is static lift. The design point is 0.25 m³/s at 40 m, hydraulic power $\rho g Q H = 1000 \times 9.81 \times 0.25 \times 40 = 98.1$ kW, shaft power 120 kW at 82 % pump efficiency, on a 132 kW motor. For most of the year the required flow is 70 %, 0.175 m³/s (Figure 10.6).

Throttling. At full speed the pump delivers 0.175 m³/s at $H = 50 - 160 \times 0.175^2 = 45.1$ m; the valve burns the difference between 45.1 m and what the system needs. Hydraulic power is $9810 \times 0.175 \times 45.1 = 77.4$ kW; with pump efficiency 0.74 at this off-design point, shaft power is 104.6 kW and electrical input, at motor efficiency 0.955, is 109.6 kW.

Speed control. The system needs only $15 + 400 \times 0.175^2 = 27.25$ m. The pump curve at speed ratio k is $H = 50k^2 - 160Q^2$ (affinity laws), so $k^2 = (27.25 + 4.90)/50 = 0.643$, $k = 0.802$. Hydraulic power is $9810 \times 0.175 \times 27.25 = 46.8$ kW; with pump efficiency 0.80, shaft power 58.5 kW; with motor 0.95 and VFD 0.97, electrical input 63.5 kW.

The VFD saves $109.6 - 63.5 = 46.1$ kW. Over 6000 h/year that is 277 MWh, worth about USD 33 000 at USD 0.12 per kWh, against a 132 kW drive costing perhaps USD 15 000–25 000 installed: a payback of under a year. Note that the saving is far less than the cube law alone would suggest ($0.70^3 = 34$ % power at 70 % flow) because of the static head: speed fell only to 80 %, not 70 %. Pumps with mostly static head (borehole pumps, boiler feed against pressure) save much less, and a drive must not reduce speed below the point where the pump's shut-off head drops under the static head, or flow stops while the pump churns.

> **Status of the principle: Efficiency chain**
> *Derived*. Product of stage efficiencies from energy conservation through a series chain; it holds at each operating point, and the stage efficiencies themselves are load-dependent, so annual energy must be summed over the actual load profile, not computed from rated values.

10.7 Protection

A drive system must protect the motor, the converter, the mechanical load and the people around it, and must do so without nuisance trips. Protection of the supply network (feeder overcurrent and short-circuit protection upstream of the drive) is the subject of Chapter 23 and Chapter 29.

Motor thermal protection

The motor's limit is winding temperature (insulation class F: 155 °C hot-spot, usually used at class B rise, Chapter 9). Two approaches are used:

- **Thermal models.** An overload relay or the drive's software integrates current squared through a first-order thermal model with the motor's time constant. Electromechanical and electronic overload relays are characterised by **trip class** (IEC 60947-4-1): at 7.2 times setting current, class 10 trips in 4 to 10 s, class 20 in 6 to 20 s, class 30 in 9 to 30 s. A high-inertia load like a large fan or a centrifuge started direct-on-line needs class 20 or 30, or the relay trips before the motor is up to speed: one possible cause of the conveyor trip in the opening.
- **Direct temperature measurement.** PTC thermistors embedded in the windings (switching sharply at their rated temperature) or Pt100 sensors, wired to the drive or a relay. Essential when the motor runs at low speed on a VFD with a shaft-mounted fan, because cooling falls with speed while current does not, which no current-only model sees unless the drive includes the speed-dependent cooling curve.

> **Status of the principle: I^2t thermal overload model**
> *Approximate*. A single-time-constant model of winding temperature driven by copper loss; adequate for running overloads, but it underestimates rotor heating in long starts and stalls (rotor-limited motors) and ignores reduced cooling at low speed unless corrected.

Converter and system protection

Inside the drive, faster protection acts because semiconductors have thermal time constants of milliseconds:

- **Overcurrent and short-circuit**: hardware desaturation detection turns off IGBTs within about 10 µs of an output short; software current limit holds current at the overload rating (typically 150 % for 60 s in every 10 minutes for heavy duty, 110 % for normal duty).
- **DC-link overvoltage** (regeneration, supply surges; trip near 800 V on a 400 V drive) and **undervoltage** (supply dips; ride-through uses the load's kinetic energy to hold the DC link up for a few hundred milliseconds).
- **Earth fault** at the output, by summing the three output currents.
- **Input protection**: semiconductor (aR or gR) fuses or a circuit breaker coordinated with the converter's short-circuit rating, per IEC 61800-5-1.

- **Mechanical and process protection**: overspeed, stall detection, underload (a pump running dry or a broken belt shows as low torque at speed), torque limits that protect gearboxes and couplings.

Safety functions

For people, the important functions are standardised in IEC 61800-5-2. **Safe torque off (STO)** removes the gate signals by hardware so the motor cannot produce torque, without opening the power circuit; it replaces a motor contactor for emergency stops (stop category 0 per IEC 60204-1) and, with a controlled ramp first, gives safe stop 1 (SS1). Drive safety functions are rated by safety integrity level (SIL, per IEC 61508 and IEC 62061) or performance level (PL, per ISO 13849-1). STO is not isolation: the DC link remains charged and the motor terminals live, so electrical work still requires lockout and a wait for the DC-link capacitors to discharge, typically 5 to 15 minutes (Chapter 29).

Stresses the converter creates

A PWM converter protects the motor from overload but stresses it in two new ways. Fast voltage edges (rise times 50 to 200 ns) travel down the motor cable and reflect at the motor terminals, nearly doubling the voltage at cable lengths above roughly 20 to 50 m; inverter-duty insulation is specified for peak voltages up to 1.3–1.6 kV at 400–480 V (IEC TS 60034-25, NEMA MG 1 Part 31), or the drive is fitted with dv/dt or sine filters. The common-mode voltage of the inverter drives currents through the bearings, which pit races in months on large motors unless insulated bearings, shaft-grounding brushes or common-mode filters are used. These are converter phenomena owned by *Electronics & Computing*; their system consequences are why drive specifications state cable length and motor insulation class.

10.8 Drive selection

A selection procedure

Selecting a drive is a sequence of checks that moves from the load to the supply:

1. **Load profile.** Speed range, torque–speed characteristic, breakaway, inertia, duty cycle, required accuracy and dynamics, number of quadrants. Refer everything to the motor shaft through the chosen transmission ratio (Section 10.1).
2. **Motor.** Type (cage induction, PMSM, DC, special), rated torque from RMS torque, peak torque against breakdown or demagnetisation limits, base speed and field-weakening range, cooling at low speed (self-ventilated or forced), enclosure and hazardous-area certification, inverter-duty insulation.
3. **Converter.** Current rating from motor current and *peak* current, not from kW; normal-duty or heavy-duty rating; switching frequency (higher frequency reduces motor noise and ripple but requires derating of the converter); voltage class; braking arrangement (none, chopper and resistor, regenerative); control mode (V/f, sensorless vector, closed-loop vector, servo).
4. **Supply and installation.** Supply short-circuit level and harmonic limits (IEEE 519, IEC 61000-3-12; choose 6-pulse with DC choke, 12- or 18-pulse, or active front end, Chapter 25); EMC category per IEC 61800-3 (C1 residential to C4 large industrial); cable length and filters; ambient temperature and altitude derating (typically 1 % per 100 m above 1000 m and 1 to 2.5 % per °C above 40 °C); enclosure IP rating.
5. **Protection and safety.** Overload class, thermistor input, STO and other safety functions to the SIL or PL from the risk assessment, coordination of fuses or breakers.

6. **Life-cycle cost.** Purchase price is often under 10 % of the life-cycle cost of a continuously running drive; energy dominates (Chapter 28).

Drive types and their territory

Table 10.5 maps the main drive types to their territory. The DC drive's place (Chapter 11) has shrunk to legacy plants, retrofits and some very high-torque low-speed duties; the AC drive (Chapter 12) is now the default.

Table 10.5 — Drive types and typical applications.

Drive type	Power range	Strengths	Typical use
Soft starter with cage induction motor	5 kW–5 MW	Cheap, reduces starting current; no speed control	Fixed-speed pumps, fans, compressors
LV VFD, V/f or sensorless vector	0.2 kW–2 MW	Low cost, simple, energy saving	Pumps, fans, conveyors, mixers
LV VFD, closed-loop vector or DTC	1 kW–5 MW	Full torque at zero speed, fast torque response	Hoists, cranes, lifts, winders, extruders
PMSM servo drive	50 W–100 kW	Very high dynamics, high torque density	Machine tools, robots, packaging
PMSM or induction traction inverter	50–500 kW (road), up to 1.5 MW per bogie (rail)	Constant power over wide speed range, regenerative	EVs, buses, trams, locomotives
Thyristor DC drive	5 kW–10 MW	Simple four-quadrant, robust, retrofit	Legacy rolling mills, paper machines, excavators, cranes
MV drive (multilevel VSI, LCI)	0.5–100 MW	High power, high voltage	Pipeline compressors, mine mills, ship propulsion, pumped-hydro start-up

Normal duty and heavy duty

Converter manufacturers usually give two ratings for one frame. **Normal duty** (light overload, about 110 % for 60 s every 10 minutes) suits pumps and fans. **Heavy duty** (150 % for 60 s, often 200 % for 2–3 s) suits conveyors, hoists, extruders and anything with a high breakaway. The same hardware is rated one size lower in heavy duty, because the semiconductors' junction temperature, not the motor, sets the overload.

Worked example — selecting the conveyor drive. For the conveyor of Section 10.1 the specification reads: constant-torque load, 804 N·m running at 1485 rpm, 1294 N·m for 20 s at start (126 % of the 160 kW motor's rated torque), single quadrant on this level route, 400 V 50 Hz supply at a mine substation with 25 kA fault level, ambient up to 45 °C, altitude 1500 m.

- *Motor:* 160 kW four-pole IE3 cage motor (rated current about 280 A at 400 V), inverter-duty insulation, PTC thermistors. Running at 78 % load gives ample thermal margin at the 70 % to 100 % speeds the belt uses; no forced cooling needed.
- *Converter current:* the start needs about 126 % torque. In the constant-flux region of a vector-controlled induction motor, current rises less than proportionally with torque because the magnetising component is fixed (Section 12.6); about $1.3 \times 280 \approx 360$ A for 20 s.

- *Duty rating:* heavy duty. A frame with heavy-duty rating 300 A (normal duty about 360 A) gives 450 A for 60 s, covering the start.
- *Derating:* altitude 1500 m costs about 5 %, ambient 45 °C about 5 to 12 % depending on the manufacturer; the continuous 225 A running current (at 78 % load, roughly 0.8 × 280 A) is still well inside a derated 300 A × 0.85 ≈ 255 A.
- *Supply:* a 6-pulse front end with DC choke gives about 35–45 % current THD; whether that is acceptable depends on the point of common coupling (IEEE 519 limits relative to I_{SC}/I_L; here I_{SC}/I_L is roughly 25 000/230 ≈ 110, allowing 15 % TDD at the PCC), so a single drive is usually fine while a dozen would need a multi-pulse or active-front-end solution.
- *Braking:* none; friction stops the belt in about 6 s (Section 10.5).

The result is a 160 kW motor on a heavy-duty-rated drive of about 160 kW normal-duty size, with no brake resistor: the same motor that failed to start direct-on-line now starts every time in 20 s at 130 % current.

What remains open

Open question	Where it is answered
How is the speed of a DC motor controlled by armature voltage and field, and how does a chopper or thyristor converter implement it?	Chapter 11 (DC motor drives)
How do V/f, field-oriented and direct torque control make an induction or PM motor behave like a controllable torque source?	Chapter 12 (AC motor drives)
How are torque-speed curves, starting methods and losses of the motors themselves obtained?	Chapters 5, 6, 8 and 9
How are duty types, thermal ratings and insulation classes of motors defined?	Chapter 9 (electrical machine design and performance)
What do drive input harmonics do to the supply, and how are they mitigated?	Chapter 25 (power quality)
How much energy do drives save across a plant, and how is that audited?	Chapter 28 (energy efficiency and management)
How are drive and motor circuits isolated and worked on safely?	Chapter 29 (electrical safety)
How are the current and speed controllers of a drive tuned, and what limits their bandwidth?	Chapters 11 and 12; control theory in *Circuits, Signals and Communications*, Ch. 35
How do the IGBTs, gate drivers and DC-link circuits of a converter work internally?	*Electronics & Computing*

Next: Chapter 11 takes the simplest controllable machine, the DC motor, and shows how armature and field control, chopper and thyristor converters and cascaded current and speed loops turn it into a precise four-quadrant drive.

Chapter 11 — DC Motor Drives

Part III · Motor Drives

A hot-strip mill installed in the 1980s still runs its finishing stands on 3 MW DC motors fed by thyristor converters. The plant's power engineer has two complaints and one instruction. The complaints: the mill's 6.6 kV supply shows a power factor of 0.5 whenever the stands thread a new slab at low speed, and the utility charges for the reactive power; and stand 4 lost its field supply last year, after which its motor tried to run away before the protection caught it. The instruction from management: decide whether to keep the DC drives for another twenty years or replace them with AC. To answer, the engineer needs to know exactly what a DC drive does well (clean, fast, four-quadrant torque control with a converter of almost trivial structure), what it costs in power factor and maintenance, and how its control works, because the same structure, a fast inner current loop under a slower speed loop with the flux held separately, is the template that every AC vector drive in Chapter 12 imitates.

> **How are the speed and torque of a DC motor controlled through its armature voltage and its field, and what converter supplies each?**
> **How do chopper and thyristor drives regulate speed, control torque, recover braking energy, and what does a practical DC drive look like?**

Concept	Engineering destination
Armature control	Constant-torque speed control of mills, winders, excavators, cranes, legacy traction
Field control	Constant-power speed range above base speed; field-loss protection
Chopper-fed drives	Battery vehicles, forklifts, mining locomotives, DC trams and metros, small servos
Speed regulation	Draw control in paper and steel lines, process accuracy specifications
Torque control	Cascaded current and speed loops, tension control in winders, current limiting
Regenerative braking	Energy recovery in traction, hoists and mill reversals; line receptivity
Practical drive architecture	Specifying, maintaining or retrofitting DC drives; the template for AC vector control

Roadmap. The DC motor's two control inputs and the armature-voltage range below base speed (11.1) → field weakening above base speed (11.2) → the chopper as a DC-to-DC armature supply (11.3) → closing the speed loop and what regulation it buys (11.4) → the inner current loop and torque control (11.5) → braking with energy recovery (11.6) → how a complete industrial DC drive is built and protected (11.7).

11.1 Armature control

Two inputs, two time constants

> **Boundary.**
> The DC machine's construction, EMF and torque equations, characteristics and losses are in Chapter 5. The motor's transfer function, with its electrical and mechanical poles, is derived in *Circuits, Signals and Communications*, Ch. 33 §33.2. Converter circuit internals (thyristor commutation, chopper switching, gate drives) belong to *Electronics & Computing*; the averaged converter relations used here are summarised in *Engineering Physics*, Ch. 18 §18.8.

For a separately excited (or permanent-magnet) DC motor, Chapter 5 gives

$$E = k\Phi\,\omega_m, \qquad T_e = k\Phi\,I_a, \qquad V_a = E + R_a I_a + L_a \frac{dI_a}{dt}$$

so the steady-state speed is

$$\omega_m = \frac{V_a - R_a I_a}{k\Phi} = \frac{V_a}{k\Phi} - \frac{R_a}{(k\Phi)^2}\,T_e$$

Speed can be changed by the armature voltage V_a, by the flux Φ through the field current, or by adding resistance in the armature circuit. Resistance control wastes power in proportion to the speed drop and survives only in old cranes and traction starting rheostats. The other two are the basis of every modern DC drive, and they differ sharply in time scale: the armature time constant $\tau_a = L_a/R_a$ is 10 to 60 ms in industrial motors, while the field time constant $\tau_f = L_f/R_f$ is 0.5 to 5 s. Armature voltage is the fast input; field current is the slow one.

Constant torque below base speed

Armature voltage control keeps the field at its rated value and varies V_a from zero to rated. Equation above says the no-load speed is proportional to V_a, and the slope of the speed–torque line, $R_a/(k\Phi)^2$, does not change: armature control produces a family of parallel lines (Figure 11.2). At rated armature current the motor gives rated torque at every speed from zero to **base speed**, the speed at which V_a reaches rated with rated flux. Power rises linearly with speed. This is the constant-torque region of Section 10.2, and its ideal fit to hoists, conveyors, extruders and mills is the reason DC drives dominated variable-speed industry until the 1990s.

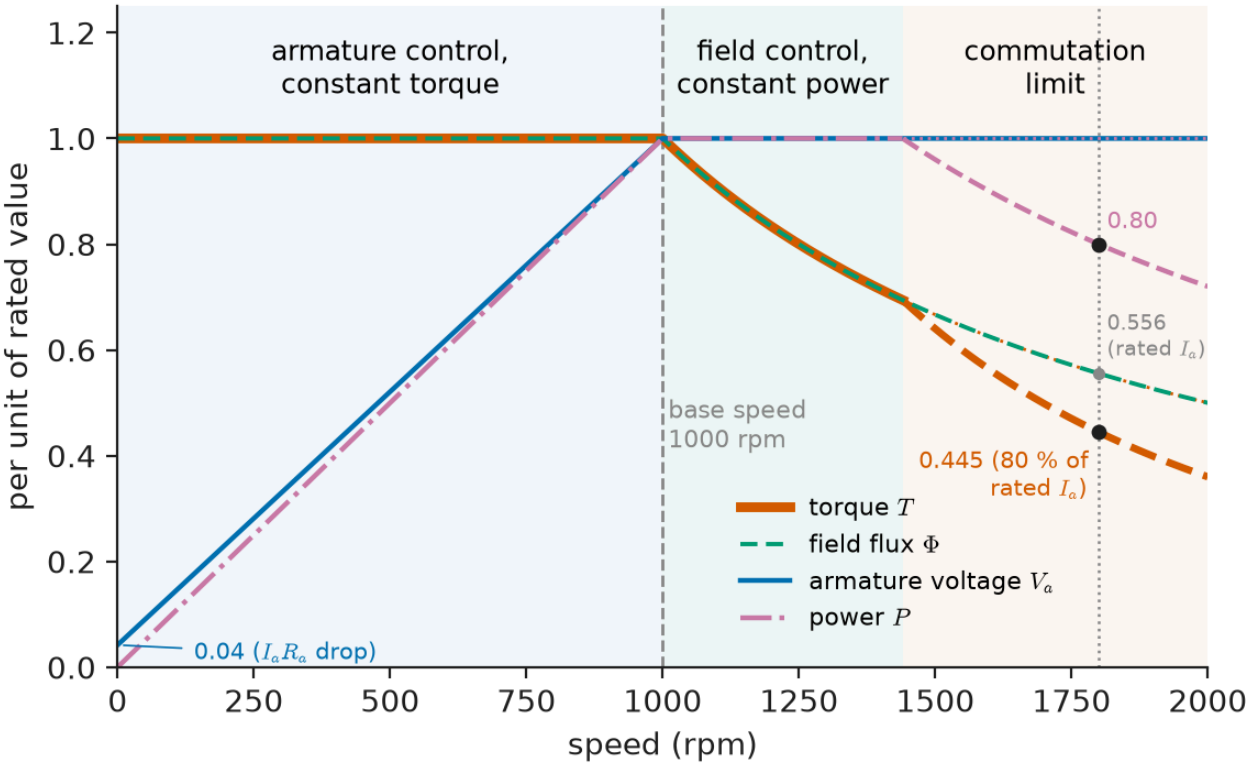


Figure 11.1 — Capability of a separately excited DC drive: armature voltage, field flux, torque and power against speed, with the armature-control, field-control and commutation-limited regions.

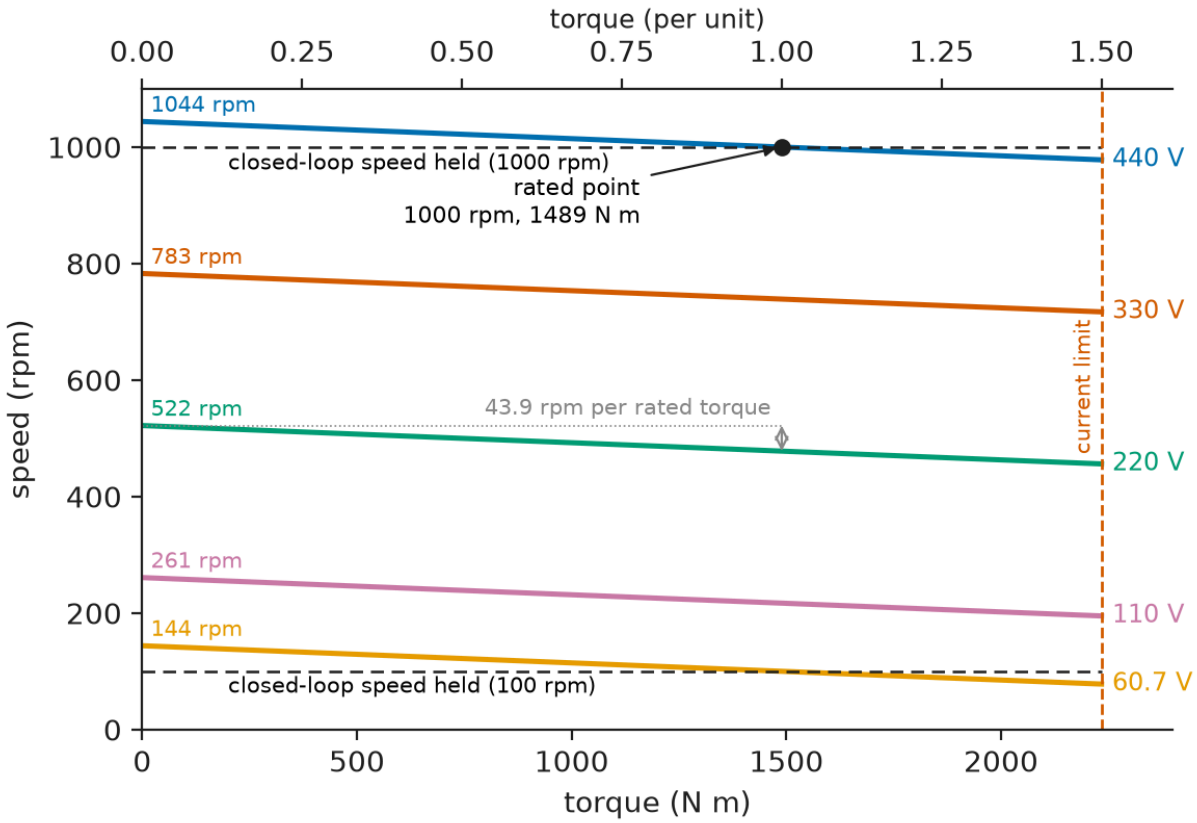


Figure 11.2 — Speed-torque lines of a 150 kW DC motor under armature voltage control at full field, open loop.

The thyristor converter as an armature supply

Most industrial DC drives above a few kilowatts take their armature voltage from a **three-phase fully controlled thyristor bridge**. Its averaged output, a result we take from converter theory, is

$$V_a = \frac{3\sqrt{2}}{\pi} V_{LL} \cos\alpha - \frac{3\omega L_c}{\pi} I_a = 1.35\, V_{LL} \cos\alpha - R_{ov} I_a$$

where α is the firing (delay) angle, V_{LL} the RMS line voltage of the supply and L_c the commutating (supply plus transformer) inductance per phase. The term $R_{ov} = 3\omega L_c/\pi$ is not a real resistance and dissipates nothing, but it lowers the output voltage like one: it represents the overlap time during which two thyristors share the current. The output has a six-pulse ripple at $6f$ (300 Hz at 50 Hz), smoothed by the armature inductance and often an added reactor so that the current stays continuous.

For $0 < \alpha < 90^\circ$ the bridge rectifies; for $90^\circ < \alpha < 150^\circ$ (the upper limit leaves time for commutation) its average voltage is negative while its current, which thyristors allow in only one direction, stays positive, so power flows *from* the DC side to the AC supply: the bridge **inverts**. A single bridge can therefore operate in two quadrants of the *voltage–current* plane, which for the motor means forward motoring and reverse-speed braking (Quadrants I and IV of Section 10.4). Section 11.6 shows how full four-quadrant operation is obtained.

> **Status of the principle: Averaged thyristor-bridge output**
> *Approximate.* Valid for continuous armature current, a stiff sinusoidal supply and an overlap angle under about 30°; it fails in discontinuous conduction at light load, where the average voltage rises and the drive's gain changes sharply, and on weak supplies whose voltage is notched by the commutations.

The power-factor price of low speed

The thyristor bridge draws line current in 120° blocks that lag the supply voltage by about $\alpha + \mu/2$, where μ is the overlap angle. Its displacement factor is therefore roughly $\cos(\alpha + \mu/2)$, and with the distortion of the quasi-square current (whose fundamental is $3/\pi = 0.955$ of its RMS value), the power factor is about $0.955\cos(\alpha + \mu/2)$. Low speed at full torque means low V_a, large α, and a reactive power demand that barely falls while real power does. This is the mill's complaint in the opening.

Worked example — firing angle and power factor of a mill-stand drive. A separately excited DC motor is rated 440 V, 370 A, 1000 rpm, with $R_a = 0.050\ \Omega$; it is fed by a fully controlled bridge from a 400 V, 50 Hz transformer secondary with commutating reactance $X_c = \omega L_c = 0.050\ \Omega$.

- Rated EMF: $E = 440 - 370 \times 0.050 = 421.5$ V. Motor constant: $k\Phi = 421.5/104.72 = 4.025$ V·s/rad. Rated torque: $k\Phi I_a = 4.025 \times 370 = 1489$ N·m; internal power $EI_a = 156$ kW, about 150 kW at the shaft after rotational losses.
- Converter: $1.35 \times 400 = 540.2$ V maximum; $R_{ov} = 3 \times 0.050/\pi = 0.0477\ \Omega$, an overlap drop of 17.7 V at 370 A.
- At 1000 rpm and rated current, the bridge must provide $440 + 17.7 = 457.7$ V before overlap, so $\cos\alpha = 457.7/540.2 = 0.847$, $\alpha = 32.1^\circ$. The overlap angle from $\cos\alpha - \cos(\alpha+\mu) = 2X_cI_a/(\sqrt{2}V_{LL})$ is $\mu = 6.5^\circ$, and the power factor is about $0.955\cos(35.3^\circ) = 0.78$.
- At 600 rpm and rated torque: $E = 4.025 \times 62.83 = 252.9$ V, $V_a = 271.4$ V, $\cos\alpha = (271.4 + 17.7)/540.2 = 0.535$, $\alpha = 57.6^\circ$, $\mu = 4.3^\circ$, power factor about 0.48. Shaft power has fallen to 60 % (94 kW of internal power) while the line current is unchanged at $\sqrt{2/3} \times 370 = 302$ A RMS.
- At 100 rpm (threading speed): $\alpha = 81.7^\circ$ and power factor about 0.11. The drive draws full current from the supply, about 210 kVA, to deliver 15.6 kW.

The supply sees the same 302 A at every speed while delivering power proportional to speed. That reactive demand, plus the 5th, 7th, 11th and 13th harmonic currents of a six-pulse bridge, is what forces mills to install static VAR compensators and harmonic filters (Chapter 25), and it is one of the strongest arguments for replacing thyristor DC drives with AC drives whose diode or active front end runs near unity displacement factor at all speeds.

11.2 Field control

Above base speed: weakening the flux

At base speed the armature voltage has reached its limit. Equation $\omega_m = (V_a - R_a I_a)/(k\Phi)$ shows that speed can still rise if Φ falls. In **field weakening** the drive holds V_a at rated and reduces the field current, so that $E = k\Phi\omega$ stays just below V_a. With I_a at its rated value, power EI_a is constant and torque $k\Phi I_a$ falls as $1/\omega$: the constant-power region of Figure 11.1.

> **Status of the principle: Constant-power field weakening**
> *Approximate.* Follows from $E = k\Phi\omega$ with armature voltage and current both held at rated values; it assumes $k\Phi$ is set by field current alone, but armature reaction reduces the effective flux at high current and low field, and commutation limits force armature current down at the top of the range.

How far field weakening can go

The field-weakening range of a DC motor is limited by the machine, not the converter:

- **Commutation.** The reactance voltage in the coils undergoing commutation rises with speed and current, while the interpole flux that cancels it stays fixed; above some speed sparking at the brushes forces armature current to be reduced. Typical field-weakening ranges are 2:1 to 3:1 for industrial motors and up to 4:1 for motors designed for winders and test benches.
- **Armature reaction.** At weak field the armature MMF distorts the main flux more, the neutral zone shifts and the risk of flashover rises; compensating windings in the pole faces of large motors counteract it.
- **Stability.** At very weak field a load increase raises I_a, armature reaction weakens the flux further, speed rises, and the drive may become unstable: the field-weakening controller must act on measured EMF, not on a fixed field-current schedule.
- **Mechanical speed limit** of the armature and commutator.

The flux is not proportional to field current: the magnetisation curve saturates (Chapter 1 and Chapter 5). Near rated flux, on the knee of the curve, a 10 % cut in flux may need a 20 % cut in field current; at weak field, on the nearly linear part, flux and field current fall almost in proportion. Practical drives therefore do not compute the field current from speed; they close an **EMF loop** that trims the field current until the measured armature EMF ($V_a - R_a I_a$) equals its set value, typically 95 % of rated armature voltage to leave headroom for current control.

Field supply and field loss

The field is supplied by a separate small converter, usually a single-phase half- or fully controlled bridge rated 1 to 3 % of the armature power. Because the field time constant is long, field weakening is slow: changing flux by

half takes roughly one field time constant, 1 to 3 s. Fully controlled field bridges can force the field current down faster by applying negative voltage.

The dangerous failure is **loss of field**. If the field current falls to a small residual value while the armature converter is still applying voltage, the motor tries to reach the speed at which $E = V_a$ with residual flux, which may be five to ten times rated: the motor overspeeds, or, if the load holds it, armature current rises to the current limit with almost no torque. DC drives therefore always include **field-failure protection** (a field-current relay that trips the armature converter when field current falls below about 50 to 70 % of its minimum expected value) and independent overspeed protection. The incident on the mill's stand 4 in the opening is the textbook case.

Worked example — field weakening for a coiler. The 150 kW motor of Section 11.1 drives a strip coiler that must run to 1800 rpm. Holding $V_a = 440$ V and $I_a = 370$ A:

- Required EMF is still 421.5 V, so $k\Phi = 421.5/(1800 \times 2\pi/60) = 421.5/188.5 = 2.236$ V·s/rad, 55.6 % of rated flux.
- Torque at rated current: $2.236 \times 370 = 827$ N·m, 55.6 % of rated; power $827 \times 188.5 = 156$ kW internal, unchanged.
- If the motor's commutation limit at 1.8 times base speed allows only 80 % of rated armature current, the available torque falls to $2.236 \times 0.8 \times 370 = 662$ N·m (0.445 per unit), and the power to 125 kW (0.80 per unit): the drive must be specified on the reduced curve. Because the reactance voltage grows roughly with the product $I_a n$, a limit of 80 % current at 1800 rpm means full current is available only up to about $0.8 \times 1800 = 1440$ rpm (Figure 11.1).

A coiler is a constant-power load (Section 10.3): tension times strip speed is constant while the coil diameter grows, so the full-field, full-current torque is needed at the largest coil diameter (lowest speed) and the weakest field at the empty mandrel (highest speed). A 1.8:1 diameter build-up matches the 1.8:1 field-weakening range almost exactly, which is how coiler motors are specified.

11.3 Chopper-fed drives

From a DC source to a variable armature voltage

Where the supply is DC (a battery, a traction line at 600, 750, 1500 or 3000 V, or the DC link of a larger converter), the armature voltage is controlled by a **chopper**: a switch that connects the source to the armature for a fraction D (the duty ratio) of each switching period $T_s = 1/f_s$. The averaged result, the buck-converter relation of *Engineering Physics* Ch. 18 §18.8.2, is

$$V_a = D\,V_{dc}, \qquad 0 \le D \le 1$$

The armature inductance plays the role of the converter's filter inductor, and the motor's back-EMF that of the output voltage. Choppers switch with IGBTs or MOSFETs at 1 to 20 kHz in modern drives (GTO thyristors at 200 to 500 Hz in the trams and metros of the 1980s and 1990s, many still running).

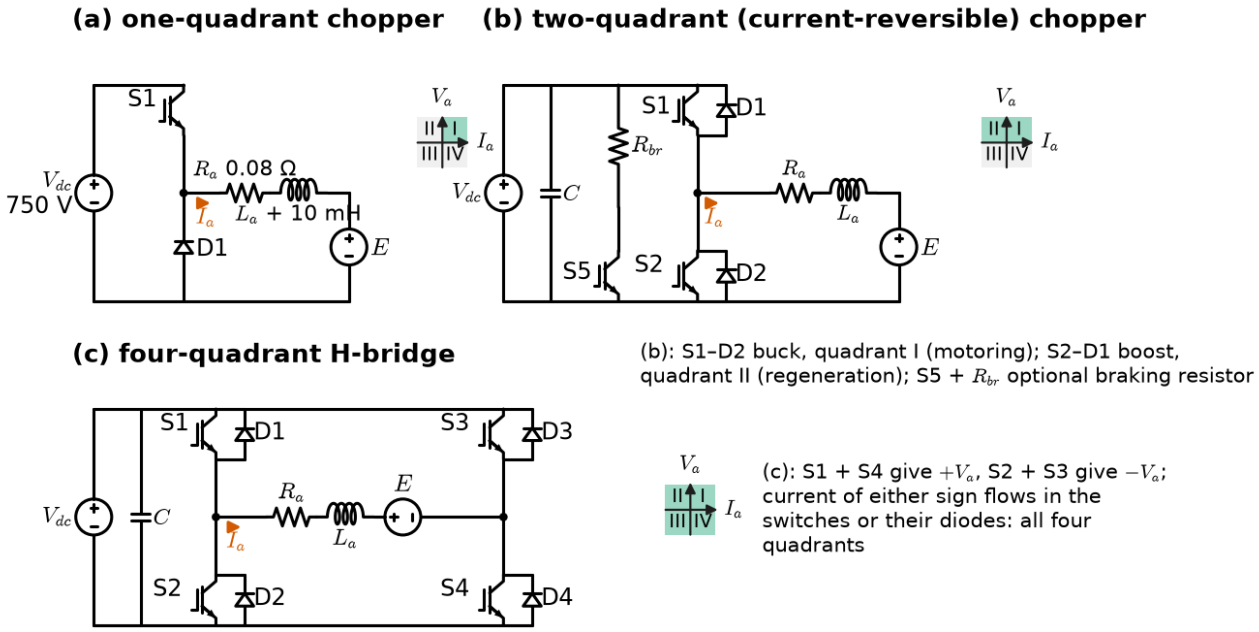


Figure 11.3 — One-, two- and four-quadrant chopper configurations feeding a DC motor armature, with the quadrants each can reach.

Configurations and quadrants

Figure 11.3 shows the three standard configurations:

- **One-quadrant (step-down) chopper**: a switch and a freewheeling diode. Positive voltage, positive current; forward motoring only. Simple traction and forklift drives use it with a mechanical or contactor-switched reversing arrangement.
- **Two-quadrant (current-reversible) chopper**: two switches with antiparallel diodes in a half-bridge. Positive voltage, current in both directions: forward motoring as a buck converter, forward braking as a boost converter that pumps current back into the source (Section 11.6). This is the standard DC traction chopper.
- **Four-quadrant H-bridge chopper**: four switches; voltage and current of both signs. The standard for DC servo amplifiers, automotive actuators (window lifts, electric power steering in older designs, throttle bodies) and battery-vehicle drives with electronic reversing.

Current ripple and discontinuous conduction

Between switching instants the armature current ramps up and down. If the armature circuit (motor plus any smoothing reactor) has inductance L, resistance small compared with L/T_s, and the switching is fast compared with τ_a, the peak-to-peak current ripple of a one- or two-quadrant chopper is

$$\Delta I = \frac{V_{dc}\, D(1-D)}{L\, f_s} \le \frac{V_{dc}}{4 L f_s}$$

with the maximum at $D = 0.5$. The ripple matters for three reasons. It adds copper and iron loss in the motor (the RMS current exceeds the average), it worsens commutation at the brushes (a "form factor" limit of about 1.02 to 1.05 on RMS-to-average current is typical for DC motors on choppers), and if the average current is less than $\Delta I/2$ the current falls to zero in each cycle: **discontinuous conduction**, in which the simple relation $V_a = D V_{dc}$ no longer holds and the speed rises under light load.

> **Status of the principle: Chopper averaged model and ripple formula**
> *Approximate.* Holds for continuous conduction, ideal switches and a switching period much shorter than the armature time constant; it fails at light load (discontinuous conduction), at very low duty ratio where minimum on-time limits resolution, and when the source has significant impedance, as on a long traction feeder.

Worked example — ripple in a tram chopper. A tram motor on a 750 V line has armature resistance 0.080 Ω and total armature-circuit inductance (motor plus smoothing reactor) 10 mH; its average current at full tractive effort is 300 A.

- At a GTO-era switching frequency of 400 Hz the worst-case ripple is $750/(4 \times 0.010 \times 400) = 46.9$ A peak to peak, 16 % of the average current.
- At 2 kHz with IGBTs: $750/(4 \times 0.010 \times 2000) = 9.4$ A, 3 % of the average current.
- At low speed with $E = 200$ V, the duty ratio is $D = (200 + 300 \times 0.080)/750 = 0.299$ and the ripple at 2 kHz is $750 \times 0.299 \times 0.701/(0.010 \times 2000) = 7.9$ A; at $E = 500$ V, $D = 0.699$ and the ripple is again 7.9 A, symmetrical about $D = 0.5$.
- Conduction becomes discontinuous below an average current of about 4.7 A at 2 kHz (half the worst-case ripple), but below 23 A at 400 Hz, which is why GTO choppers needed large smoothing reactors and why the reactor of a modernised tram can be several times smaller.

The fivefold frequency increase cut the ripple fivefold, or allowed the reactor to shrink fivefold for the same ripple; it also moved the switching noise from a strongly audible 400 Hz whine toward the edge of hearing, and moved the line-current harmonics away from the 50 Hz signalling frequencies of the track circuits, an EMC constraint that railway standards (EN 50121, EN 50238) police strictly.

Choppers in battery and traction systems

Chopper DC drives were the first efficient electronic traction drives: the Paris and London metro choppers of the 1970s–1990s, DC trams worldwide, and still the battery forklifts and mining locomotives of today at 24 to 80 V and 5 to 30 kW, where a series or separately excited DC motor and a MOSFET chopper remain cheap and robust. Their weaknesses are the commutator (brush wear, 1000 to 5000 hours between brush changes in traction, and a sparking hazard in gassy mines that demands flameproof enclosures) and the weight per kW. New traction designs use inverters and AC motors (Chapter 12), but the chopper principle survives in every DC–DC converter of an EV and in the boost stage of every PV inverter.

11.4 Speed regulation

Open-loop speed droop

Without feedback, a DC drive holding V_a constant slows as load increases, by the resistive drop over $k\Phi$:

$$\Delta\omega = \frac{(R_a + R_{ov} + R_{reactor})\, I_a}{k\Phi}$$

The industrial measure is **speed regulation**, the change in speed from no load to full load as a percentage of full-load speed (or of rated speed, which matters at low settings). Because the drop is a fixed number of rpm, regulation as a percentage of *set* speed worsens in proportion as speed falls.

Worked example — open- and closed-loop regulation of the mill motor. For the 150 kW motor of Section 11.1 fed by its thyristor bridge, the effective circuit resistance is $R_a + R_{ov} = 0.050 + 0.0477 = 0.0977\ \Omega$. At rated current the speed drop is

$$\Delta\omega = \frac{0.0977 \times 370}{4.025} = 8.99\ \text{rad/s} = 85.8\ \text{rpm}$$

At a 1000 rpm setting that is 8.6 % regulation; at a 100 rpm threading setting the same 86 rpm drop is 86 %, and a load step can nearly stall the stand. Figure 11.2 shows the parallel lines in terms of terminal voltage, where only R_a appears (43.9 rpm per rated torque); the converter's overlap drop doubles the slope seen from the firing-angle command.

With speed feedback from a tachogenerator or encoder and a *proportional* speed controller of loop gain K (the product of controller, converter, motor and sensor gains around the loop at steady state), the droop is divided by $1 + K$: with $K = 20$ the drop is $85.8/21 = 4.1$ rpm, with $K = 100$ it is 0.85 rpm. A controller with integral action (PI) removes the steady-state droop completely, leaving the error set by the speed measurement itself.

What limits regulation accuracy

Once the integral action has removed load droop, regulation is limited by the sensor and by how fast the loop recovers from load steps:

Table 11.1 — Speed feedback options and achievable steady-state accuracy (typical).

Feedback	Steady-state accuracy (of rated speed)	Notes
None (open loop)	5–10 % at base speed, worse at low speed	Plus temperature drift of R_a
Armature voltage with IR compensation	1–2 %	No tachometer; compensation tuned to hot R_a
DC tachogenerator	0.1–0.5 %	Ripple and brush wear; still common on old mills
AC tachogenerator, rectified	0.5–1 %	
Incremental encoder, 1024–4096 lines	0.01 %	Standard for paper and steel lines, cranes
High-resolution sin/cos encoder	0.001 %	Test benches, precision lines

Dynamic regulation, the speed dip and recovery time after a load step, depends on the loop bandwidth, and that depends on the inner current loop (Section 11.5). A rolling-mill stand might specify a dip of under 1 % for a 100 % impact load, recovered in 0.3 s. Paper machines specify the *draw* (speed difference between sections) to 0.01 %, because the sheet tears if a downstream section pulls too hard; they achieve it by digital speed references and encoder feedback with a common time base.

> **Status of the principle: Steady-state regulation with feedback**
> *Derived*. The droop of a proportional loop is the open-loop droop divided by $1 + K$, and integral action drives it to zero (internal-model principle, *Circuits, Signals and Communications*, Ch. 34 §34.7); it holds only while the converter is not saturated at its voltage or current limit.

11.5 Torque control

Why control current first

Torque is $k\Phi I_a$. With the field held, controlling armature current *is* controlling torque, and the armature current can be changed in milliseconds while speed changes in tenths of seconds or seconds. Industrial DC drives therefore use **cascade control** (Figure 11.4): an inner current loop that makes the converter and armature behave as a fast, current-limited torque source, and an outer speed loop whose output is the current reference. The structure has three advantages that made it universal, and that AC vector drives copy:

1. **Current limiting is built in.** Clamping the current reference at, say, 150 % of rated limits torque and protects converter, motor and gearbox, without any separate protective action: a stalled hoist simply produces limit torque.
2. **The fast electrical dynamics are hidden.** Seen from the speed loop, the current loop is a small first-order lag, so the speed controller sees a nearly pure integrator ($T_e \to \omega$ through $1/Js$).
3. **Each loop is tuned separately**, inner first, outer second.

Many applications use the inner loop alone: **torque control**. A winder or unwinder holds strip tension F at a coil radius r by commanding torque Fr (plus inertia and friction compensation) while the line's master drive sets speed; test benches and dynamometers set torque to emulate a road load; load-sharing between two motors on one conveyor belt sets one in speed control and the other as a torque follower.

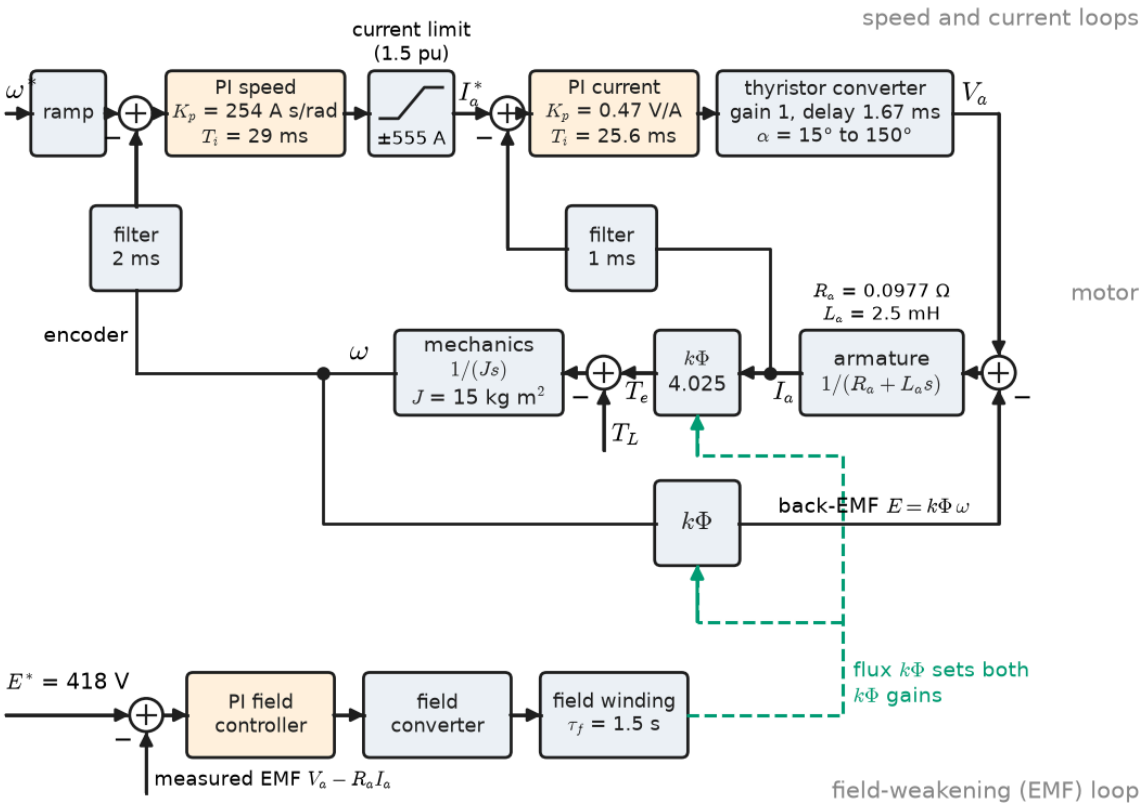


Figure 11.4 — Cascade speed and current control of a separately excited DC drive, with field-weakening (EMF) controller.

Tuning the current loop

The current loop plant is the converter (a gain with a small delay) and the armature circuit $1/(R + L_a s)$ with time constant τ_a. Its small delays are lumped into one time constant T_σ: for a six-pulse thyristor bridge at 50 Hz the average firing delay is about half a pulse interval, $1/(2 \times 300) = 1.67$ ms, plus the current-measurement filter. A standard rule, the **technical (modulus) optimum**, places the zero of a PI controller on the armature pole and sets the gain for a well-damped loop:

$$T_i = \tau_a = \frac{L_a}{R}, \qquad K_p = \frac{L_a}{2\,K_{conv}\,T_\sigma}$$

which gives a closed current loop behaving approximately as a first-order lag of time constant $2T_\sigma$ (step overshoot about 4 %). The back-EMF acts as a slow disturbance that the integral action removes. The general theory of PI tuning, anti-windup and margins is in *Circuits, Signals and Communications*, Ch. 35 §35.2–35.4; what the drive engineer adds is where T_σ comes from and how small it can be made.

Tuning the speed loop

With the current loop reduced to a lag $2T_\sigma$ and the speed-measurement filter added, the speed loop plant is an integrator $k\Phi/(Js)$ with a small equivalent lag T_{eq}. For an integrator plant, a PI controller tuned by the **symmetrical optimum** places the crossover frequency at $1/(2T_{eq})$, geometrically centred between the controller zero and the lag, maximising the phase margin (about 37° for the basic form, raised by a reference pre-filter or by widening the spacing):

$$K_{p\omega} = \frac{J}{2\,k\Phi\,T_{eq}}, \qquad T_{i\omega} = 4T_{eq}$$

The symmetrical optimum gives excellent rejection of load-torque steps; its set-point overshoot (about 40 %) is suppressed by a first-order filter on the speed reference or by a ramp generator.

> **Status of the principle: Technical and symmetrical optimum tuning**
> *Approximate.* Rules for loops whose small delays can be lumped into one equivalent time constant much shorter than the dominant plant time constant; they assume linear operation and fail when the converter saturates, when the mechanical drive train has torsional resonances inside the speed-loop bandwidth, or in discontinuous conduction.

Worked example — tuning the mill drive. The 150 kW drive has $R = 0.0977\ \Omega$, $L_a = 2.5$ mH, $k\Phi = 4.025$ V·s/rad and motor-plus-load inertia $J = 15$ kg·m².

- Armature time constant $\tau_a = 2.5 \times 10^{-3}/0.0977 = 25.6$ ms; mechanical time constant $\tau_m = JR/(k\Phi)^2 = 15 \times 0.0977/16.20 = 90.5$ ms.
- Current loop: $T_\sigma = 1.67 + 1.0 = 2.67$ ms. With converter gain normalised to 1 V/V, $T_i = 25.6$ ms and $K_p = 2.5 \times 10^{-3}/(2 \times 2.67 \times 10^{-3}) = 0.47$ V/A. The closed current loop has time constant $2T_\sigma = 5.3$ ms, a bandwidth of about $1/(2\pi \times 5.3\ \text{ms}) = 30$ Hz, and a step response that reaches the reference in roughly 12 ms.
- Speed loop: $T_{eq} = 5.3 + 2.0 = 7.3$ ms including a 2 ms speed filter. $K_{p\omega} = 15/(2 \times 4.025 \times 7.33 \times 10^{-3}) = 254$ A per rad/s, $T_{i\omega} = 29$ ms; crossover at $1/(2T_{eq}) = 68$ rad/s (11 Hz).
- Current limit at 1.5 times rated, 555 A, gives $4.025 \times 555 = 2234$ N·m, so an unloaded run-up to 1000 rpm takes $J\omega/T = 15 \times 104.7/2234 = 0.70$ s, during which the speed controller saturates and needs anti-windup.

With an IGBT chopper at 5 kHz instead of the thyristor bridge, T_σ falls to about 0.2 ms and the current-loop bandwidth rises to about 400 Hz: the converter's switching rate, not the motor, sets the dynamics. That is why thyristor drives give 20 to 40 Hz current loops and transistor drives 300 Hz to 2 kHz.

11.6 Regenerative braking

The principle

A DC motor whose EMF exceeds the applied armature voltage drives current backwards: $I_a = (V_a - E)/R_a < 0$, torque reverses and the machine generates. To brake *regeneratively*, the converter must accept that reversed current and pass the power on to the supply. The converter decides what is possible:

- A **one-quadrant chopper** or a **single thyristor bridge** cannot carry reversed armature current; it can brake only rheostatically (with a resistor switched across the armature) or, for the thyristor bridge, regenerate in Quadrant IV, with speed reversed by an overhauling load, as in a hoist lowering.
- A **two-quadrant chopper** regenerates in Quadrant II as a **boost converter**: switching the lower device short-circuits the armature for a fraction D of the period, building current from the EMF; releasing it forces the current through the upper diode into the source. The averaged relation is $E \approx (1 - D)V_{dc}$.
- A **dual converter** (two thyristor bridges in antiparallel) gives all four quadrants: one bridge carries positive current, the other negative, and the controller blocks one before firing the other with a dead time of a few milliseconds (circulating-current-free operation). The armature current reverses within 10 to 20 ms, fast enough for reversing mills that change direction several times per slab. A cheaper alternative for slow reversals reverses the *field* instead, at the cost of a reversal time of a second or more.

Limits on regeneration

Regeneration needs a voltage to push against and a receiver. At low speed the EMF falls, and a boost converter needs $D \to 1$ to keep current flowing; below roughly 5 to 10 % of base speed the available braking current fades, and mechanical or rheostatic braking takes over to stop and hold. At the receiving end, a **battery** must have charge acceptance (cold or full batteries refuse charge, Chapter 16); a **thyristor bridge** must keep α below about 150° or the commutation fails (an "inverter shoot-through" or commutation failure, a short circuit of the supply through the converter); and a **DC traction line** must have another train accelerating nearby to absorb the power. Railway supply standards fix how far the line voltage may rise: for a 750 V DC system, EN 50163 permits a highest permanent voltage of 900 V and a highest non-permanent voltage of 1000 V. If the line is not receptive, the chopper diverts the energy to an on-board resistor; reversible substations or wayside storage (supercapacitors or batteries) raise receptivity.

Worked example — regenerative braking of a tram. A 30 t tram on a 750 V DC network brakes from 60 km/h (16.7 m/s) at 1.0 m/s².

- Kinetic energy: $\tfrac{1}{2} \times 30\,000 \times 16.7^2 = 4.17$ MJ (1.16 kWh), ignoring rotating masses.
- Braking force $ma = 30$ kN; initial braking power $Fv = 500$ kW, falling linearly to zero over the 16.7 s stop.
- With motor efficiency 0.92 and chopper 0.97 in the generating direction, about $0.892 \times 4.17 = 3.72$ MJ (1.03 kWh) reaches the line, minus the share absorbed by running resistance during the stop.
- At 5 km/h the motor EMF is only $5/60 = 8$ % of its 60 km/h value; the chopper's duty ratio must approach 1 and armature resistance drop takes most of the EMF, so electric braking fades and the

friction brake completes the stop.

On a busy line with trams accelerating nearby, almost all of the 1 kWh is reused; at night, with one tram on a feeder section, the line voltage climbs toward 900 V and the energy goes to the roof resistor. Metro operators report that regenerative braking returns 15 to 35 % of traction energy, the lower figure without storage or reversible substations, the higher with them.

> **Status of the principle: Regenerative energy recovery**
> *Derived*. Energy conservation through the motor and converter efficiency chain; the recoverable share is further limited by the receptivity of the source, the fade of generated EMF at low speed and the current limit at high speed, which no efficiency figure captures.

11.7 Practical drive architecture

Building blocks

A complete industrial thyristor DC drive (Figure 11.5) contains:

- **Supply side**: a circuit breaker or switch-fuse, a line reactor or dedicated converter transformer (to limit commutation notching and fault current, and to match voltage), semiconductor fuses, and an RC snubber network.
- **Armature converter**: a six-pulse fully controlled bridge (two-quadrant) or a dual bridge (four-quadrant); for multi-megawatt mill drives, two bridges in series or parallel fed from transformer secondaries 30° apart, forming a 12-pulse converter with lower harmonics.
- **Field converter**: a single-phase half- or fully controlled bridge, 1 to 3 % of armature rating, with its own current loop and the EMF controller of Section 11.2.
- **DC side**: armature smoothing reactor where needed, a DC circuit breaker or contactor on large drives, armature current measurement (shunt, Hall-effect or DC current transformer).
- **Motor**: a separately excited DC machine with compensating winding above a few hundred kW, forced ventilation (blower motor with airflow switch), brush-wear and temperature monitoring.
- **Feedback**: encoder or tachogenerator; armature voltage measurement.
- **Control**: a digital controller running the ramp generator, speed and current loops, field control, firing-angle generation synchronised to the supply by a phase-locked loop, and the protection logic; communication to the plant control system over fieldbus.

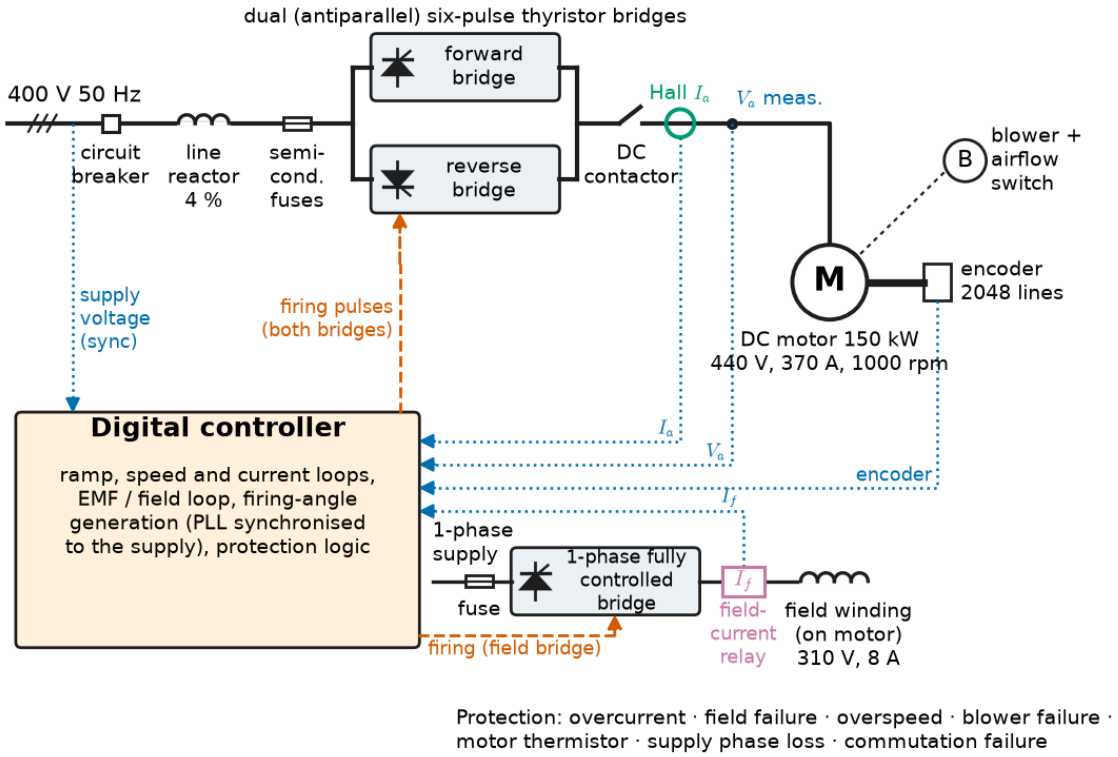


Figure 11.5 — Architecture of an industrial four-quadrant thyristor DC drive for a 150 kW mill motor.

Protection specific to DC drives

Besides the generic drive protection of Section 10.7, a DC drive protects against faults that AC drives do not have:

Table 11.2 — DC-drive-specific protection (typical settings).

Fault	Consequence	Protection
Field loss	Overspeed or uncontrolled armature current	Field-current relay at 50–70 % of minimum field; trips armature converter and applies brake
Overspeed	Mechanical failure of commutator	Independent overspeed trip at 110–120 % of maximum speed
Commutation failure in inverting mode	Short circuit of the supply through the bridge	Firing-angle limit (about 150°), fast DC breaker or fuses
Blower failure	Motor overheating, especially at low speed and full torque	Airflow switch, winding thermistors
Brush wear and flashover	Commutator damage, arcing	Brush-length sensors, compensating winding, current-rise limit di/dt
Supply phase loss or dip	Commutation failure, loss of control in regeneration	Supply monitoring, controlled pulse suppression

Where DC drives stand today

DC drives still offer: the simplest four-quadrant power circuit (a dual thyristor bridge has no DC-link capacitor and no PWM), very low converter cost per kW at high power, a decoupled flux and torque structure that needs no motor model, and easy retrofit when only the converter is replaced. They also bring: brush and commutator maintenance (brushes every few thousand hours, commutator resurfacing), larger and heavier motors than induction or PM motors of the same rating, a speed ceiling set by commutation, and the poor and variable power factor of Section 11.1. New installations have been overwhelmingly AC since about 2000; existing DC installations in steel, paper, mining excavators and draglines, cranes, lifts and some marine winches are either kept with new digital converters on the old motors (the cheapest modernisation, often a few years' payback on reliability alone) or replaced by AC drives when the motors themselves reach end of life.

For the engineer of the opening, the arithmetic is: new thyristor converters keep the 0.5 power factor at threading speed and need SVC or filter investment; AC drives with active front ends remove the reactive demand and the commutator maintenance but require new motors, foundations and cabling. The control architecture, though, transfers unchanged. Field-oriented control of an AC motor (Section 12.6) is nothing other than the construction, inside a controller, of a "field current" and an "armature current" from the three-phase currents, so that the cascade of Figure 11.4 can be applied.

What remains open

Open question	Where it is answered
How is an AC motor made to behave like a separately excited DC motor with independent flux and torque control?	Chapter 12 (vector and field-oriented control)
How do PWM inverters synthesise variable-frequency voltages, and how are induction and PM motors controlled with them?	Chapter 12 (AC motor drives)
Where do the DC machine's EMF, torque and commutation limits come from?	Chapter 5 (DC machines)
How are the six-pulse harmonics and poor power factor of thyristor drives measured and corrected at plant level?	Chapter 25 (power quality); Chapter 28 (energy efficiency and management)
How do batteries accept regenerated energy, and what limits charge acceptance?	Chapter 16 (energy storage)
How are traction substations and DC railway supplies organised?	Chapter 22 (distribution systems); Chapter 13 (power-system structure)
How do thyristor commutation, chopper switching and gate drives work at circuit level?	*Electronics & Computing*
How are PI controllers designed, and how is anti-windup implemented?	*Circuits, Signals and Communications*, Ch. 35

Next: Chapter 12 replaces the commutator with a PWM inverter and shows how V/f, field-oriented and direct torque control turn induction and permanent-magnet motors into drives that match, and now surpass, the DC drive.

Chapter 12 — AC Motor Drives

Part III · Motor Drives

A district-cooling plant replaces the inlet-vane control of its four 250 kW chiller-condenser fans with variable-frequency drives and saves 40 % of their energy in the first summer. Its engineers then try the same drives on a 37 kW cooling-tower hoist that lifts maintenance platforms, and the platform sags by a few centimetres every time it starts from rest: the drive, still in its factory V/f mode, cannot produce full torque at zero speed. In a car factory two floors away, a 300 N·m permanent-magnet motor in an electric-vehicle test cell reverses from full motoring to full regeneration in 2 ms, and an operator pressing the emergency stop at 12 000 rpm learns that the inverter cannot simply switch off, because the motor's magnets would then pump 560 V back into a 400 V battery. These are all AC drives, and they differ only in what the controller knows about the motor and what it does with that knowledge. This chapter builds the AC drive from the inverter outward: how a PWM inverter makes a variable-frequency, variable-voltage supply, how scalar control uses it, and how vector control, field orientation, direct torque control and PMSM control make an AC motor behave like the separately excited DC motor of Chapter 11, only faster and without a commutator.

> **How does a variable-frequency drive synthesise the voltages an AC motor needs, and when is simple V/f control enough?**
> **How do field-oriented and direct torque control decouple flux and torque in induction and permanent-magnet motors, and how is such a drive commissioned?**

Concept	Engineering destination
Variable-frequency drives	Pumps, fans, compressors, conveyors, HVAC, the default industrial drive
PWM inverters	Output voltage limits, switching-frequency choice, motor insulation and bearing stress
Scalar (V/f) control	Simple drives for fans, pumps and group drives; low-speed boost and slip compensation
Induction motor speed control	Choosing between pole changing, slip control, slip recovery and VFDs
Vector control fundamentals	Space vectors and the d–q frame used by every high-performance drive
Field-oriented control	Cranes, hoists, lifts, winders, traction; full torque at zero speed
Direct torque control	Fast torque response without a modulator; rolling mills, marine propulsion
PMSM control fundamentals	EV traction, servo drives, high-efficiency pumps and compressors
Drive commissioning	Parameter identification, loop tuning, ramps, braking and protection settings

Roadmap. The VFD and what it must deliver (12.1) → how the PWM inverter delivers it, and its limits (12.2) → the simplest control, constant V/f (12.3) → the induction motor speed-control options compared (12.4) → the space-vector and d–q tools (12.5) → field-oriented control of the induction motor (12.6) → direct torque control (12.7) → PMSM control, MTPA and field weakening (12.8) → commissioning a drive (12.9).

12.1 Variable-frequency drives

What the motor needs

> **Boundary.**
> The rotating magnetic field, slip and the induction-motor equivalent circuit are developed in Chapter 3 and Chapter 6; synchronous and permanent-magnet machines in Chapter 7 and Chapter 8. Three-phase phasors and the operator $a = 1\angle 120^\circ$ are from *Circuits, Signals and Communications*, Ch. 10. The internal circuits of rectifiers and inverters, IGBT physics and gate drives belong to *Electronics & Computing*.

An AC motor runs near the synchronous speed set by the supply frequency, $n_s = 120 f / p$. Changing speed efficiently means changing f. But the stator flux of an AC machine is set by the ratio of voltage to frequency: Faraday's law makes the air-gap flux linkage approximately $\psi \approx V / (2\pi f)$ when the stator resistance drop is small. To keep the motor at rated flux, and therefore at rated torque per ampere, the voltage must change in proportion to frequency. A **variable-frequency drive** (VFD) is a converter that supplies three-phase voltage of independently adjustable frequency and amplitude, typically 0 to 50 or 60 Hz at constant V/f, then up to 2 or 3 times base frequency at constant (maximum) voltage in field weakening.

The voltage-source drive structure

Nearly every low-voltage and most medium-voltage drives are **voltage-source inverters** (VSI) with a DC link (Figure 12.1):

1. A **rectifier** turns the AC supply into DC. A six-pulse diode bridge is the cheapest; a DC choke or AC line reactor (3 to 5 %) limits harmonic current; 12- and 18-pulse versions with phase-shifting transformers, or an **active front end** (an IGBT rectifier with an LCL filter), meet stricter harmonic limits and allow regeneration.
2. A **DC link** with electrolytic or film capacitors holds a stiff voltage, about $1.35 V_{LL}$ under load: 540 V for a 400 V supply, 650 V for 480 V, 930 V for 690 V.
3. A **PWM inverter** of six IGBTs (two per phase) switches each motor terminal between the positive and negative DC rails, at 2 to 16 kHz, so that the *average* terminal voltage over each switching period follows a sinusoidal reference (Section 12.2).
4. **Braking**: a brake chopper and resistor across the DC link, or regeneration through an active front end (Section 10.4).
5. **Control**: a digital controller that measures two or three phase currents and the DC-link voltage (and speed, when an encoder is fitted), runs the control algorithm of Sections 12.3 to 12.8 every 50 to 250 µs, and generates the switching pattern.

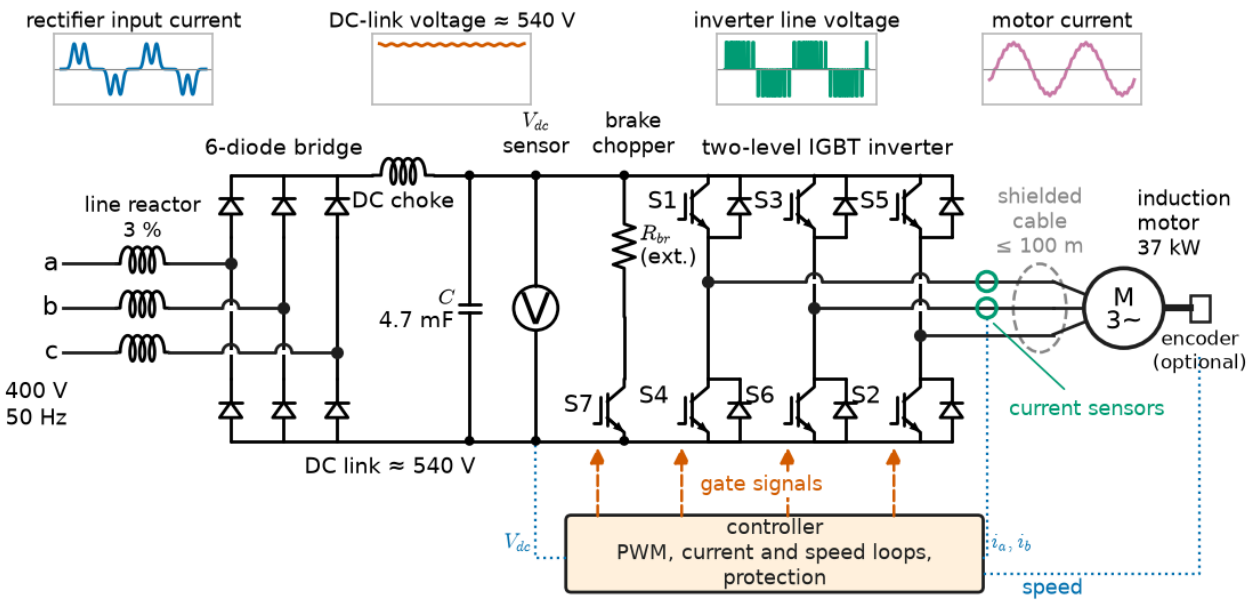


Figure 12.1 — Power circuit of a low-voltage voltage-source variable-frequency drive.

Current-source inverters and load-commutated inverters (LCI) with thyristors survive in multi-megawatt synchronous-motor drives (compressors, ship propulsion, pumped-storage start-up), and cycloconverters in a few very large low-speed mills and ship drives; medium-voltage VSIs use multilevel topologies (three-level neutral-point clamped, cascaded H-bridge, modular multilevel) to reach 3.3 to 13.8 kV with lower-voltage devices.

What a VFD delivers and costs

Table 12.1 — Representative VFD characteristics (typical values).

Quantity	Low voltage (0.4–0.69 kV)	Medium voltage (2.3–13.8 kV)
Power range	0.2 kW–5 MW	0.3–100 MW
Output frequency	0–300 Hz (up to 1–2 kHz for spindles)	0–75 Hz typical
Switching frequency	2–16 kHz	0.3–1.5 kHz per device
Efficiency at rated load	96–98.5 %	96.5–99 %
Input current THD, 6-pulse with choke	30–45 %	— (multi-pulse standard)
Input current THD, AFE or 18-pulse	3–8 %	3–5 %
Overload	110 % / 60 s (normal duty), 150 % / 60 s (heavy duty)	110–150 % / 60 s
Output voltage rise time	50–200 ns	1–5 µs (with multilevel steps)

12.2 PWM inverters

Making a sinusoid from two levels

Each leg of a two-level inverter connects its motor terminal either to $+V_{dc}/2$ or to $-V_{dc}/2$ (measured from the DC-link midpoint). In **sine-triangle PWM** (Figure 12.2), a sinusoidal reference for each phase, of amplitude m (the **modulation index**) relative to a triangular carrier of amplitude 1, is compared with the carrier: when the reference exceeds the carrier the upper switch is on. Averaged over a carrier period, the leg voltage then follows the reference:

$$\bar{v}_{a0}(t) = m\,\frac{V_{dc}}{2}\sin\omega t, \qquad 0 \le m \le 1$$

and the fundamental line-to-line RMS output is

$$V_{LL1} = \frac{\sqrt{3}}{2\sqrt{2}}\,m\,V_{dc} = 0.612\,m\,V_{dc}$$

The harmonics of the output lie in sidebands around multiples of the carrier frequency, far above the fundamental, where the motor's leakage inductance filters them into a small current ripple.

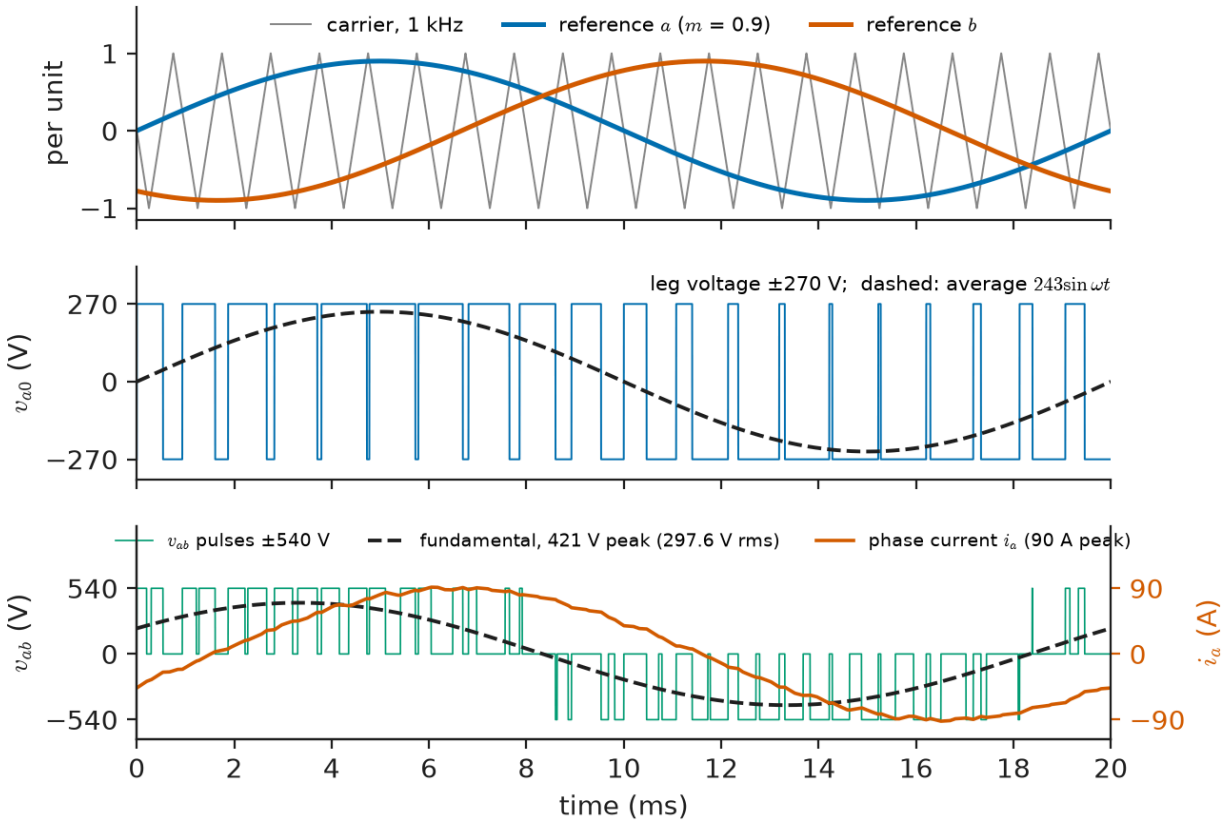


Figure 12.2 — Sine-triangle PWM of one inverter leg and the resulting line voltage and motor current.

Using the DC link fully

Sine-triangle PWM uses only 86.6 % of the available DC voltage, because the line voltage is the *difference* of two phase legs and the legs' common-mode part is wasted. Adding a third-harmonic (or, equivalently, a min-max offset) common-mode component to all three references, or using **space-vector PWM** (SVPWM), which

chooses the inverter's eight switching states directly (Section 12.5), extends the linear range to $m = 2/\sqrt{3} = 1.155$:

$$V_{LL1,max} = \frac{V_{dc}}{\sqrt{2}} = 0.707\,V_{dc}$$

Beyond this, **overmodulation** drops pulses and the output approaches **six-step** (square-wave) operation, the maximum a two-level inverter can give:

$$V_{LL1,six-step} = \frac{\sqrt{6}}{\pi}\,V_{dc} = 0.780\,V_{dc}$$

at the cost of large 5th and 7th harmonic currents and torque ripple at six times the fundamental frequency.

> **Status of the principle: Averaged PWM inverter model**
> *Approximate*. The inverter is replaced by its switching-period average, valid when the carrier frequency is at least about 10–20 times the output frequency and the dead time between complementary switches is negligible; dead time distorts the output at low voltage (low speed), and the model fails in overmodulation and six-step operation.

Worked example — can a 400 V drive give a 400 V motor its rated voltage? A 400 V, 50 Hz supply through a diode bridge gives about $\sqrt{2} \times 400 = 566$ V on the DC link at no load, and about $1.35 \times 400 = 540$ V under load.

- Sine-triangle PWM at $m = 1$: $0.612 \times 540 = 330$ V, only 83 % of rated motor voltage.
- SVPWM at the end of the linear range: $540/\sqrt{2} = 382$ V, 95.5 %.
- Six-step: $0.780 \times 540 = 421$ V.

A drive in the linear range therefore cannot quite deliver 400 V to the motor at full load; at rated frequency the motor runs with about 5 % less voltage and so about 5 % less flux, drawing a few percent more current for rated torque. Drives handle this by entering mild overmodulation near base speed, by setting the field-weakening point a few percent below the nameplate frequency, or by being specified at one voltage class above the motor (a 480 V drive for a 460 V motor). It also explains why supply dips of only 5 to 10 % immediately reduce available torque at top speed.

Switching frequency and its side effects

Raising the switching frequency reduces current ripple (and audible motor noise, which is why HVAC drives in occupied buildings run at 8 to 16 kHz) but raises IGBT switching losses roughly in proportion, forcing the drive to be derated. A typical 37 kW drive rated at 4 kHz loses 10 to 20 % of its current rating at 8 kHz and 30 to 40 % at 16 kHz.

Fast voltage edges have system consequences treated in Section 10.7: reflected-wave overvoltages on long cables, stressing turn-to-turn insulation (IEC TS 60034-25 and NEMA MG 1 Part 31 limits); **common-mode voltage**, which steps by $V_{dc}/3$ at each switching and drives capacitive currents through the motor bearings and into the earth; and radiated and conducted emissions that IEC 61800-3 limits by environment category. Output dv/dt filters, sine filters (an LC filter that makes the motor voltage nearly sinusoidal, at the cost of voltage drop and lost dynamic performance), shielded symmetrical cables and correct 360° shield bonding are the mitigation toolkit.

12.3 Scalar control

Constant volts per hertz

Scalar or **V/f control** commands only the amplitude and frequency of the stator voltage, with no knowledge of the motor's internal state. The speed reference sets f; a characteristic sets $V = V_{rated}\, f / f_{rated}$ up to base frequency and $V = V_{max}$ above it (Figure 12.3a). The motor then slips naturally: from the equivalent circuit of Chapter 6, if the air-gap flux is constant, torque near synchronism is proportional to the **slip frequency** $f_2 = sf$ and does not depend on f itself. The whole torque–speed curve shifts sideways with frequency, keeping its shape (Figure 12.3b), and the slip speed for a given torque stays roughly constant in rpm.

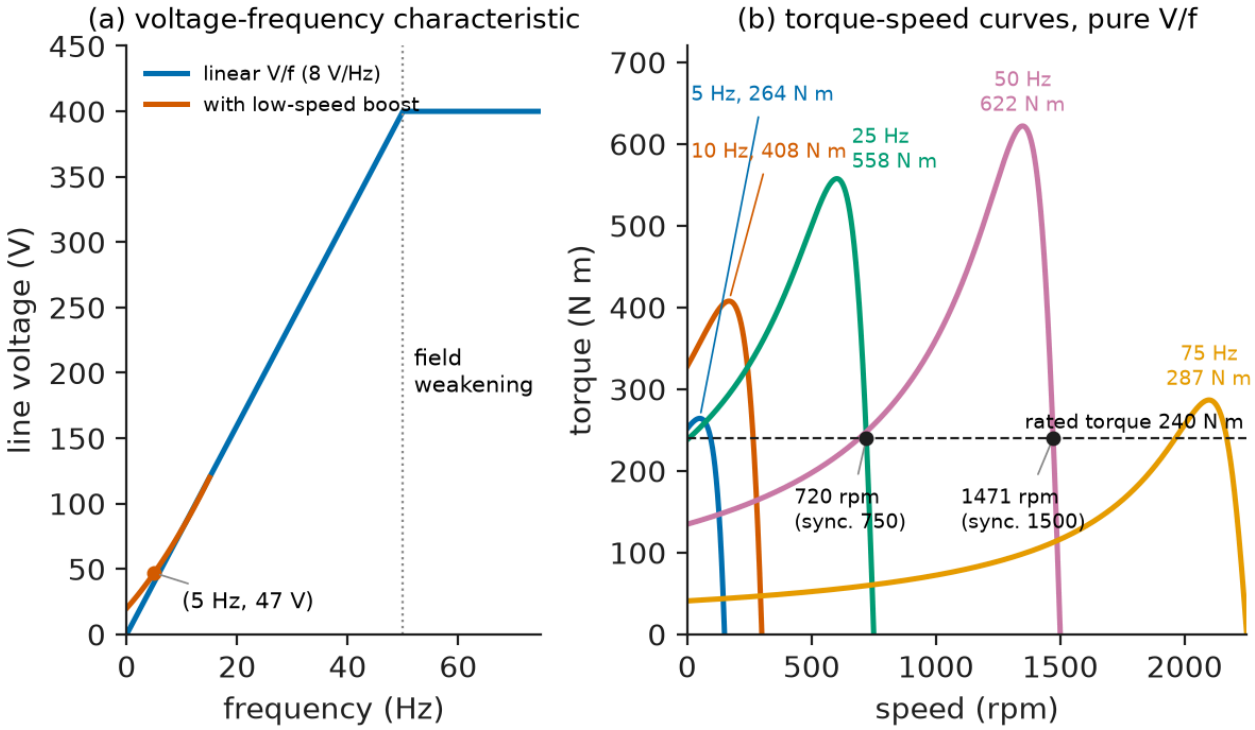


Figure 12.3 — Constant V/f control: (a) the voltage-frequency characteristic with low-speed boost; (b) the resulting torque-speed curves of a 37 kW four-pole motor.

Low-speed boost and slip compensation

Two corrections make V/f usable. At low frequency the stator resistance drop $R_1 I_1$ is no longer small compared with the shrinking applied voltage, so the air-gap flux falls and so does available torque. **Voltage boost** (or "IR compensation") adds voltage at low frequency, either as a fixed offset or in proportion to measured current. **Slip compensation** raises the frequency slightly in proportion to the measured active current, so that the shaft speed rather than the synchronous speed follows the reference, improving open-loop speed accuracy from about 2–3 % to about 0.5–1 %.

Worked example — V/f operation of a 37 kW motor. A 37 kW, 400 V, 50 Hz, four-pole induction motor has per-phase (star-equivalent) parameters $R_1 = 0.080\ \Omega$, $R_2' = 0.070\ \Omega$, $X_1 = X_2' = 0.35\ \Omega$ and $X_m = 11.0\ \Omega$ at 50 Hz (core loss neglected). Solving the equivalent circuit of Chapter 6 numerically:

- At 50 Hz, rated torque (241 N·m electromagnetic, 240 N·m at the shaft) occurs at slip 0.0190, 1471 rpm, with stator current 63.6 A and power factor 0.88. Breakdown torque is 622 N·m (2.6 times rated) at slip 0.10.

- At 25 Hz with exactly half the voltage (115.5 V per phase), rated torque occurs at 720.5 rpm: a slip of 29.5 rpm, almost the same as the 28.5 rpm at 50 Hz, as constant-flux reasoning predicts. But breakdown torque has fallen to 558 N·m, because R_1 now takes a larger share of the voltage.
- At 10 Hz and 5 Hz, pure V/f gives breakdown torques of only 408 and 264 N·m. At 5 Hz the motor can barely carry rated torque.
- To restore the 50 Hz air-gap EMF per hertz at 5 Hz and rated torque, the phase voltage must be 27.2 V instead of the proportional 23.1 V: a boost from 40 V to 47 V line-to-line, which is the point marked on Figure 12.3a.

This is the cooling-tower hoist of the opening: in V/f mode near zero frequency the drive has neither the flux nor any knowledge of the load, and a fixed boost large enough for full load over-fluxes and overheats the motor at light load. Hoisting needs vector control.

> **Status of the principle: Constant-flux V/f control**
> *Approximate.* Keeps air-gap flux close to rated when the stator resistance drop is small compared with applied voltage, roughly above 10 % of base frequency; below that, flux and torque depend on load unless the voltage is boosted, and V/f gives no control of transient torque at any speed.

Where scalar control is the right choice

V/f control needs no motor model beyond the nameplate, tolerates almost any motor (including several motors in parallel on one drive, as on roller tables and textile machines), and is stable without tuning. It is the correct choice for fans, centrifugal pumps and compressors whose torque falls at low speed, and for group drives. It is the wrong choice where full torque is needed at low speed, where torque must be controlled dynamically, or where the motor can be overhauled by the load near zero speed, as in hoists and lifts. Its other weakness is light-load instability: some motors, especially large and efficient ones, show low-frequency oscillations of speed and current at light load on open-loop V/f, damped in practice by a stabilising loop that modulates frequency with measured active current.

12.4 Induction motor speed control

The VFD has made most older speed-control methods obsolete, but each survives somewhere, and comparing them shows why frequency control won. The fundamental fact, from Chapter 6, is that the rotor copper loss equals slip times air-gap power:

$$P_{cu2} = s\,P_{ag}, \qquad P_{mech} = (1 - s)\,P_{ag}$$

Any method that reduces speed by increasing slip at fixed supply frequency dissipates the fraction s of the air-gap power in the rotor circuit, so its efficiency cannot exceed $1 - s$.

> **Status of the principle: Rotor loss proportional to slip**
> *Derived.* Follows from the induction-machine equivalent circuit (Chapter 6) for sinusoidal steady state; it is exact for the fundamental and bounds the efficiency of every slip-control method, though harmonic losses from converter supplies come on top.

Table 12.2 — Induction motor speed-control methods compared.

Method	Principle	Speed range	Efficiency at reduced speed	Where it survives
Pole changing (Dahlander, separate windings)	Changes p in $n_s = 120f/p$	Two or three fixed speeds (e.g. 1500/750 rpm)	High	Two-speed fans, lift motors of older design, machine tools
Stator voltage control (thyristor or triac)	Reduces torque, slip rises	Narrow, load-dependent	Poor: rotor loss sP_{ag}	Small fans, ceiling fans, soft starters
Rotor resistance (wound rotor)	External resistors raise slip	50–100 %	Poor: loss in resistors	Older cranes, hoists, mine winders
Slip-energy recovery (static Kramer, Scherbius)	Rotor power fed back to supply via converter	60–100 %	Good	Large pumps; and the doubly fed induction generator of wind turbines, Chapter 15
Variable frequency, V/f	Shifts the whole torque–speed curve	5–200 %	High (slip stays small)	Pumps, fans, conveyors
Variable frequency, vector control	Controls flux and torque separately	0–300 %, full torque at zero speed	High	Hoists, lifts, traction, process lines

Worked example — slowing a fan by voltage or by frequency. A fan absorbing 37 kW at 1475 rpm must run at 1050 rpm. By the fan law (Section 10.3) shaft power falls to $37 \times (1050/1475)^3 = 13.3$ kW.

- *Stator voltage control.* Synchronous speed stays 1500 rpm, so slip is $(1500 - 1050)/1500 = 0.30$. Air-gap power is $13.3/(1 - 0.30) = 19.0$ kW and rotor loss $0.30 \times 19.0 = 5.7$ kW, all of it inside the rotor of a motor whose rated rotor loss is about $0.019 \times 37.7 \approx 0.7$ kW. The motor overheats unless it is a special high-slip design, and the system efficiency cannot exceed 70 %.
- *VFD.* The load torque is $13\,300/110.0 = 121$ N·m, half the motor's rated torque, so the slip speed is about half the rated 28.5 rpm, roughly 14 rpm. The drive runs at about 35.5 Hz (synchronous 1064 rpm), slip is $14/1064 = 0.013$, and rotor loss is about $0.013 \times 13.3/0.987 = 0.18$ kW.

The VFD keeps the rotor loss about 30 times smaller. This is the physical reason voltage control is restricted to small fans with high-resistance rotors, and wound-rotor resistance control to duties where the heat goes into external resistors.

12.5 Vector control fundamentals

The idea

A DC motor's torque $k\Phi I_a$ is the product of two quantities controlled by separate windings whose fields are at right angles, so torque responds instantly to armature current while flux is held by the field. In an AC motor the flux and the torque-producing current are both carried by the same three stator currents, and both rotate. **Vector control** recovers the DC motor's structure mathematically: it describes the three phase currents as a

single rotating vector, expresses that vector in a coordinate frame rotating with the flux, and splits it there into a flux-producing component and a torque-producing component that the controller can regulate independently, like field and armature current.

Space vectors

For three phase quantities x_a, x_b, x_c (currents, voltages or flux linkages) the **space vector** is the complex number

$$\vec{x} = \frac{2}{3}\left(x_a + a\,x_b + a^2 x_c\right), \qquad a = e^{j2\pi/3}$$

It is a single complex quantity in the stationary plane of the stator whose real and imaginary parts are the α and β components (the **Clarke transform**). The factor $2/3$ makes it amplitude-invariant: for balanced sinusoidal currents of peak $\hat{I}$, $\vec{i} = \hat{I}e^{j\omega t}$, a vector of constant length $\hat{I}$ rotating at the electrical angular frequency. The space vector is not a phasor: a phasor represents one sinusoid at one frequency, while the space vector represents the instantaneous values of all three phases and remains defined during transients, with harmonics, and at any waveform. Physically, it points in the direction of the air-gap MMF wave produced by the three windings (Chapter 3).

The eight switching states of a two-level inverter map to eight voltage space vectors: six active vectors of length $2V_{dc}/3$ at 60° intervals and two zero vectors. SVPWM synthesises any reference inside the hexagon they span by time-averaging the two adjacent active vectors and the zero vectors in each switching period; the inscribed circle of radius $V_{dc}/\sqrt{3}$ is the linear limit of Section 12.2.

The rotating d–q frame

Viewing the space vector from axes that rotate at angle θ (the **Park transform**),

$$i_d + j\,i_q = \vec{i}\,e^{-j\theta}$$

turns sinusoidal currents of any frequency into constant values when θ follows the rotating field. If the d-axis is aligned with a chosen flux vector, i_d is the component along the flux and i_q the component at right angles to it, and the electromagnetic torque of any three-phase machine is

$$T_e = \frac{3}{2}\,p_p\,\left(\psi_d\,i_q - \psi_q\,i_d\right)$$

with p_p the number of pole pairs and the factor $3/2$ compensating for the amplitude-invariant scaling. With perfect alignment ($\psi_q = 0$) this is $T_e = \frac{3}{2}p_p\psi_d i_q$: flux times a perpendicular current, the DC motor's $k\Phi I_a$ in different clothes.

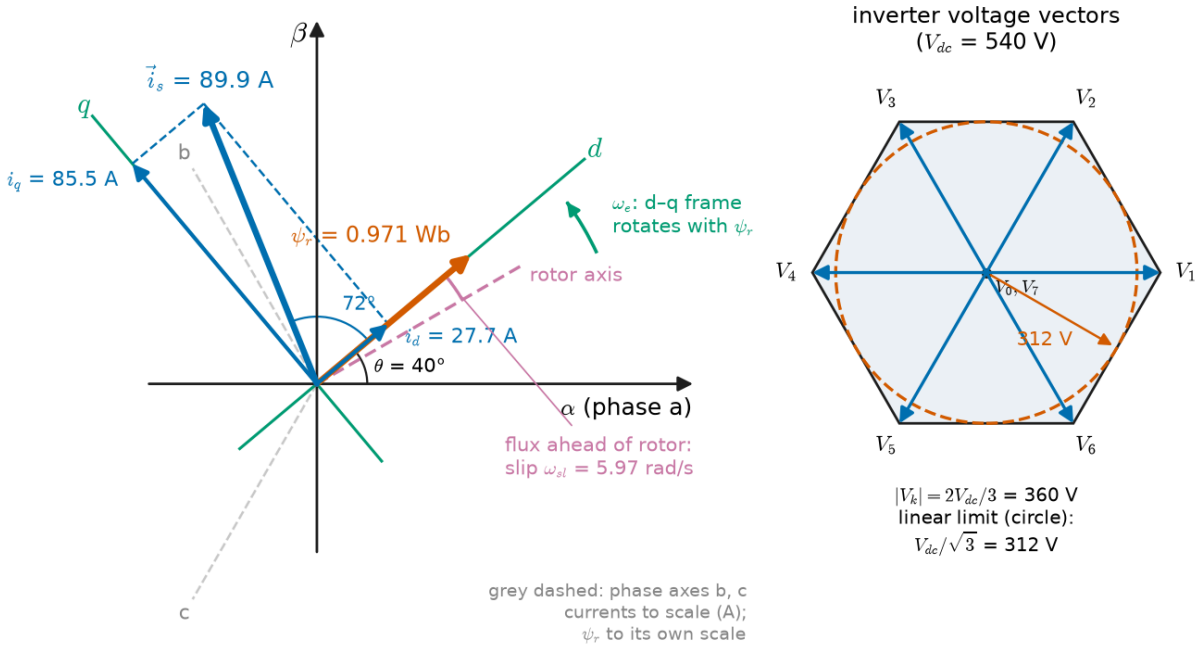


Figure 12.4 — The stator current space vector in the stationary alpha-beta frame and in a d-q frame aligned with the rotor flux.

> **Status of the principle: d–q torque equation**
> *Derived.* Follows from the co-energy of a machine with sinusoidally distributed windings (Chapter 3) under the Park transform; it assumes sinusoidal winding distribution, linear magnetics in each axis and no slotting effects, so saturation, cross-coupling between axes and space harmonics appear as parameter variations and torque ripple.

12.6 Field-oriented control

Rotor-flux orientation of the induction motor

In **field-oriented control** (FOC) of an induction motor the d-axis is aligned with the **rotor flux linkage** $\vec{\psi}_r$. In that frame the rotor equations reduce to two simple results. The rotor flux follows the d-axis current through a first-order lag with the **rotor time constant** $\tau_r = L_r/R_r$:

$$\tau_r \frac{d\psi_r}{dt} + \psi_r = L_m\, i_d$$

and the torque is set by the q-axis current:

$$T_e = \frac{3}{2}\, p_p\, \frac{L_m}{L_r}\, \psi_r\, i_q$$

Here L_m is the magnetising inductance and $L_r = L_m + L_{2l}$ the rotor self-inductance. In steady state $\psi_r = L_m i_d$. So i_d plays the role of field current (slow, with τ_r of 0.1 to 1 s, like a DC machine's field), and i_q that of armature current (instant torque, as fast as the current loop can change it). Holding i_d constant and controlling i_q gives exactly the DC drive of Chapter 11.

To stay aligned, the controller must know where the rotor flux is. The flux rotates relative to the rotor at the **slip angular frequency**

$$\omega_{sl} = \frac{L_m\, i_q}{\tau_r\, \psi_r}$$

and the flux angle is the integral of rotor electrical speed plus slip.

> **Status of the principle: Rotor-flux-oriented induction motor model**
> *Approximate*. Exact for the linear d–q model of the machine; correct decoupling of flux and torque requires the rotor time constant and magnetising inductance to be known, and both vary with rotor temperature (rotor resistance rises about 30 % from 20 °C to 100 °C) and saturation.

Indirect and direct FOC

- **Indirect FOC** computes the slip from the commanded currents and the estimated τ_r, adds it to the measured rotor speed from an encoder, and integrates to get the flux angle. It works down to and at zero speed and is the standard for cranes, lifts and process lines. Its weakness is sensitivity to τ_r: a detuned estimate misaligns the frame, so flux and torque deviate from their commands by several percent up to over 10 %. Drives adapt τ_r online from a thermal model or a model-reference estimator.
- **Direct FOC** estimates the flux vector itself from measured voltages and currents (a "voltage model", integrating $\vec{v}_s - R_s\vec{i}_s$) at medium and high speed, blended with a current model at low speed. **Sensorless** vector drives replace the encoder with a speed estimator (model-reference adaptive systems, observers); they give full torque down to about 1 to 2 % of base speed, but not reliably at sustained zero frequency under load, where the induction motor's voltage carries no speed information. Hoists and lifts therefore keep their encoders.

The control structure

The FOC drive of Figure 12.5 is the DC cascade of Figure 11.4 rebuilt in d–q coordinates. Two PI current controllers regulate i_d and i_q with a bandwidth of 0.5 to 2 kHz (limited by the PWM frequency and sampling, *Circuits, Signals and Communications*, Ch. 35 §35.8); decoupling terms cancel the cross-coupling voltages $\omega_e \sigma L_s i$ between the axes. A speed PI controller produces the torque reference, hence i_q^*; a flux controller, or a field-weakening schedule above base speed, produces i_d^*. The reference voltages are rotated back into the stationary frame and passed to the SVPWM modulator.

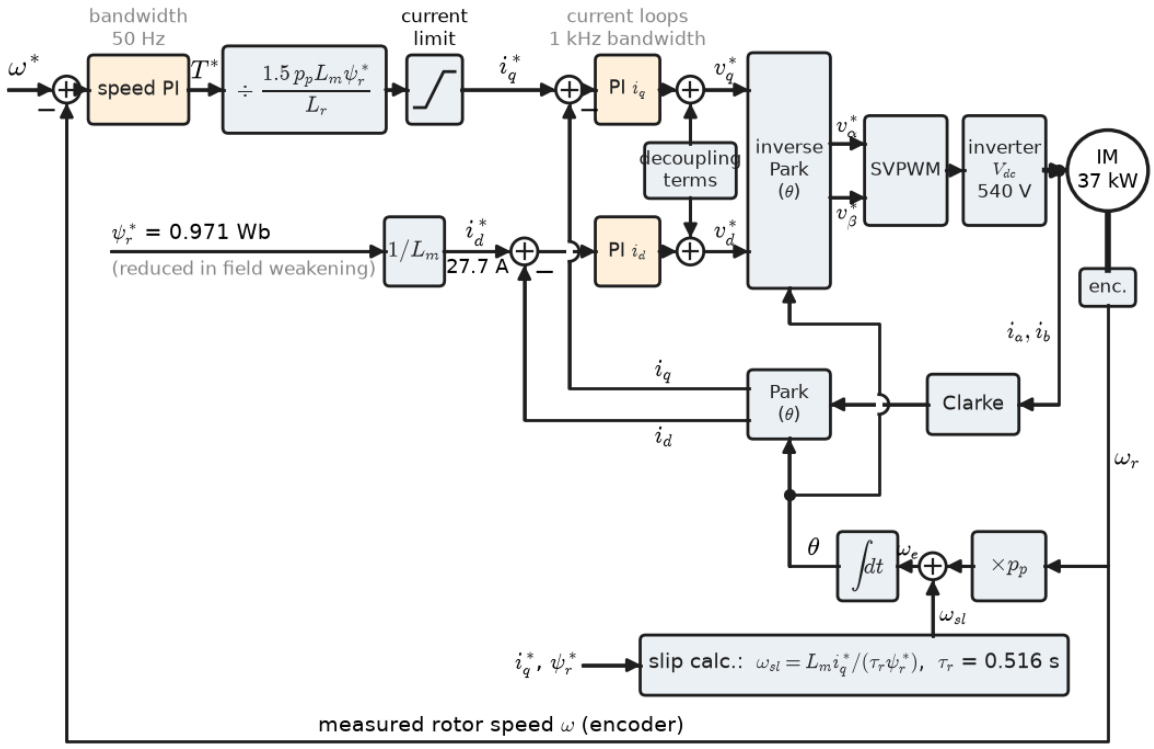


Figure 12.5 — Indirect rotor-flux-oriented control of an induction motor with encoder feedback.

Worked example — FOC operating point of the 37 kW motor. Use the motor of Section 12.3. With $\omega = 2\pi \times 50$: $L_m = 11.0/314.2 = 35.0$ mH, leakage inductances 1.11 mH each, $L_r = 36.1$ mH, $R_r = 0.070\ \Omega$, so $\tau_r = 0.516$ s. At rated load (slip 0.0190, 63.6 A RMS) the equivalent circuit gives, after the transformation to rotor-flux coordinates (peak, amplitude-invariant values):

- Rotor flux $\psi_r = 0.971$ Wb; $i_d = \psi_r / L_m = 27.7$ A (19.6 A RMS equivalent); $i_q = 85.5$ A (60.5 A RMS equivalent); $|\vec{i}_s| = \sqrt{27.7^2 + 85.5^2} = 89.9$ A, matching $\sqrt{2} \times 63.6$.
- Torque: $T_e = \frac{3}{2} \times 2 \times (35.0/36.1) \times 0.971 \times 85.5 = 241$ N·m, matching the equivalent-circuit result.
- Slip: $\omega_{sl} = L_m i_q / (\tau_r \psi_r) = 0.0350 \times 85.5 / (0.516 \times 0.971) = 5.97$ rad/s, i.e. 0.951 Hz, exactly $sf = 0.0190 \times 50$.

For 150 % torque at constant flux, i_q rises to 128 A while i_d stays 27.7 A; the current magnitude becomes 131 A peak (92.8 A RMS), 146 % of rated current. Torque per ampere improves at overload because the magnetising component is fixed, which is why vector drives deliver more starting torque per ampere than any other way of driving an induction motor (and the basis of the conveyor sizing in Section 10.8).

Field weakening of an induction motor

Above base speed, voltage is limited, so the flux reference is reduced roughly as $1/\omega$ and available torque falls. Induction motors weaken their field easily (there is no magnet to fight), but breakdown torque falls as $1/\omega^2$ at constant voltage, so above about 1.5 to 2 times base speed the drive is limited by breakdown torque rather than current and power falls. Traction motors designed for a 3:1 or 4:1 constant-power range have low leakage reactance to push that limit out.

12.7 Direct torque control

Controlling torque and flux by switching directly

Direct torque control (DTC) dispenses with current controllers and the PWM modulator. It estimates the stator flux vector $\vec{\psi}_s$ (by integrating $\vec{v}_s - R_s\vec{i}_s$) and the torque $T_e = \frac{3}{2}p_p(\psi_{s\alpha}i_{s\beta} - \psi_{s\beta}i_{s\alpha})$ every 25 to 50 µs, compares them with their references through hysteresis comparators, and chooses one of the inverter's eight voltage vectors from a switching table. The reasoning is simple: over one short interval, $\Delta\vec{\psi}_s \approx \vec{v}\,\Delta t$, so applying a voltage vector moves the flux vector's tip in the direction of that vector. A vector ahead of the flux and outward increases flux and torque; ahead and inward increases torque and reduces flux; a zero vector stops the flux and lets torque fall slowly. With the flux plane divided into six sectors, the table needs only three inputs: flux error sign, torque error sign (with a three-level comparator), and sector.

Table 12.3 — Basic DTC switching table (flux in sector k, voltage vectors numbered counter-clockwise).

Flux error	Torque error	Selected vector
Increase	Increase	V_{k+1}
Increase	Hold	Zero vector
Increase	Decrease	V_{k-1}
Decrease	Increase	V_{k+2}
Decrease	Hold	Zero vector
Decrease	Decrease	V_{k-2}

Strengths and weaknesses

DTC gives the fastest torque response of any industrial method, about 1 to 2 ms to full torque, because each decision applies the most effective voltage vector immediately; it needs no rotor parameters except R_s, no coordinate transformations and no current-loop tuning, and it is inherently robust to rotor-parameter changes. Its costs are a switching frequency that varies with speed and load (spreading acoustic noise and complicating filter design), higher torque and current ripple than FOC with PWM at the same average switching frequency, demanding sampling rates, and the stator-flux estimator's drift at very low speed, where the integral of a small voltage is dominated by errors in R_s. Industrial DTC drives (introduced commercially in 1995 for drives from a few kW to tens of MW, including marine propulsion and rolling mills) add flux observers, ripple reduction and speed-dependent hysteresis bands. Modern **model-predictive control** generalises the DTC idea: at each sampling instant it predicts the effect of every possible switching state over a short horizon and chooses the best against a cost function, which is now used in medium-voltage multilevel drives.

> **Status of the principle: Direct torque control**
> *Approximate.* Rests on the stator-flux estimate from voltage integration and on hysteresis control over very short intervals; it is accurate wherever $|\vec{v}_s| \gg R_s|\vec{i}_s|$, and fails without additional observers near zero frequency, where estimator drift and R_s errors dominate.

Table 12.4 — FOC and DTC compared (typical industrial values).

Property	FOC with SVPWM	DTC
Torque step response	2–5 ms	1–2 ms
Switching frequency	Fixed (2–16 kHz)	Variable (average 2–5 kHz)
Current ripple at same average frequency	Lower	Higher
Parameters needed	τ_r, L_m, σL_s, R_s	R_s (plus L_s for observers)
Coordinate transforms	Yes	No
Low-speed behaviour without encoder	Down to 1–2 % speed	Down to 1–2 % speed with observer
Typical use	General industry, servo, traction	Rolling mills, marine propulsion, large pumps and fans, general industry

12.8 PMSM control fundamentals

Rotor-oriented control of the permanent-magnet motor

A permanent-magnet synchronous motor (PMSM, Chapter 8) has its flux on the rotor, so the d-axis is simply the rotor's magnet axis, known from a position sensor (resolver, encoder) or a position estimator. In rotor coordinates, with magnet flux linkage ψ_{PM} and d- and q-axis inductances L_d and L_q:

$$\psi_d = L_d\, i_d + \psi_{PM}, \qquad \psi_q = L_q\, i_q$$

$$T_e = \frac{3}{2}\, p_p \left[\psi_{PM}\, i_q + (L_d - L_q)\, i_d\, i_q\right]$$

The first term is **magnet torque**. The second is **reluctance torque**, present only when the rotor is salient: in an **interior PM** (IPM) motor, with magnets buried in the rotor iron, $L_q > L_d$, so a negative i_d adds torque. In a **surface PM** (SPM) motor $L_d \approx L_q$, the reluctance term vanishes, and the best choice below base speed is $i_d = 0$, giving torque strictly proportional to i_q, like a DC motor with a permanent field. That is the standard servo drive.

The stator voltage in steady state, neglecting resistance, is

$$v_d = -\omega_e L_q\, i_q, \qquad v_q = \omega_e \left(L_d\, i_d + \psi_{PM}\right)$$

and the inverter limits its magnitude to about $V_{dc}/\sqrt{3}$ (peak phase, SVPWM limit) minus a margin for the current controllers. The current is limited to I_{max} by the inverter and by the motor's heating.

Maximum torque per ampere and field weakening

Below base speed only the current limit is active. For an IPM motor, the **maximum-torque-per-ampere** (MTPA) trajectory gives each torque with the least current, at a current angle β ahead of the q-axis that grows with load. Above base speed the back-EMF $\omega_e \psi_{PM}$ would exceed the voltage limit; a negative i_d reduces ψ_d and so the voltage, which is **flux weakening** by current rather than by a field winding. In the i_d–i_q plane (Figure 12.6), the current limit is a circle and the voltage limit an ellipse centred at $(-\psi_{PM}/L_d,\, 0)$ that shrinks as speed rises; the operating point must lie inside both.

The **characteristic current** ψ_{PM}/L_d decides the high-speed behaviour. If it is less than I_{max}, the ellipse centre lies inside the current circle and the motor can, in principle, produce power at any speed: this is the design rule for traction motors with wide constant-power ranges. If it is greater, power falls to zero at a finite speed.

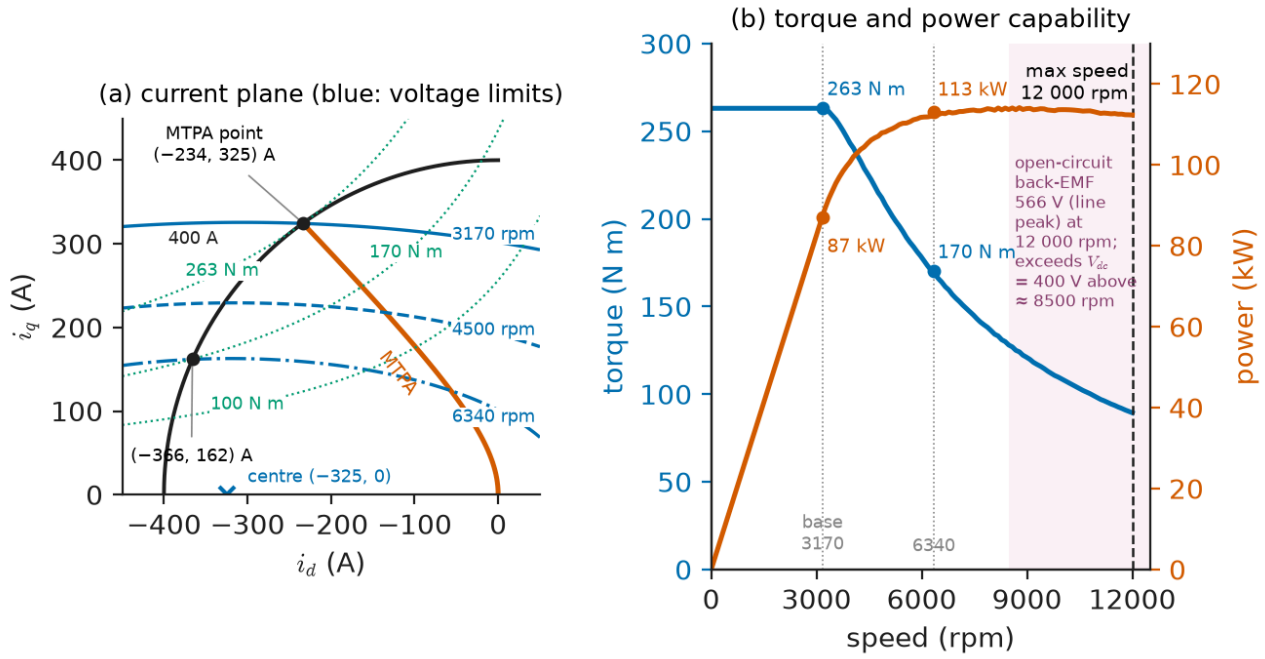


Figure 12.6 — IPM traction motor: (a) the current plane with current-limit circle, MTPA trajectory and voltage-limit ellipses; (b) torque and power capability against speed.

Worked example — an IPM motor for an electric car. An eight-pole ($p_p = 4$) IPM traction motor has $\psi_{PM} = 0.065$ Wb, $L_d = 0.20$ mH, $L_q = 0.50$ mH; the inverter runs from a 400 V battery and limits current to 400 A peak. The voltage limit, with a 5 % control margin, is $0.95 \times 400/\sqrt{3} = 219$ V peak per phase.

- *Pure q-axis current* ($i_d = 0$, as for an SPM): $T_e = \frac{3}{2} \times 4 \times 0.065 \times 400 = 156$ N·m.
- *MTPA*: searching the current angle gives the maximum at $\beta = 35.8°$: $i_d = -234$ A, $i_q = 325$ A. Magnet torque $\frac{3}{2} \times 4 \times 0.065 \times 325 = 127$ N·m; reluctance torque $\frac{3}{2} \times 4 \times (0.20 - 0.50) \times 10^{-3} \times (-234) \times 325 = 137$ N·m; total 263 N·m, 69 % more than with $i_d = 0$ for the same current.
- *Base speed*: the stator voltage $\omega_e \sqrt{(L_d i_d + \psi_{PM})^2 + (L_q i_q)^2}$ (resistance 10 mΩ included in the calculation) reaches 219 V at $\omega_e = 1327$ rad/s, i.e. 211 Hz or $1327/4 \times 60/2\pi = 3170$ rpm. Power there: $263 \times 331.7 = 87$ kW.
- *At twice base speed* (6340 rpm), the best point inside both limits is $i_d = -366$ A, $i_q = 162$ A at full current: 170 N·m and 113 kW. Power rises above base speed because the characteristic current $\psi_{PM}/L_d = 325$ A is below I_{max}, and the motor gains reluctance torque as i_d grows.
- *At 12 000 rpm* ($\omega_e = 5027$ rad/s) the magnets alone generate $5027 \times 0.065 = 327$ V peak per phase, 566 V peak line-to-line, well above the 400 V battery. Under control the drive still delivers about 89 N·m there (112 kW): the power stays nearly flat from twice base speed to maximum speed.

The last line is the emergency-stop problem of the opening. If the inverter simply blocked its gate signals at 12 000 rpm, its freewheeling diodes would form an uncontrolled rectifier and the motor would pump current into the battery and DC-link capacitors with no control over its magnitude, causing overvoltage and a large braking torque on the drivetrain. EV inverters therefore respond to faults at high speed by an **active short circuit** (turning on all three lower or upper switches, so the motor current circulates at a safe level set by ψ_{PM}/L_d) rather than by switching off, and their DC-link components are rated for the uncontrolled-generator voltage.

> **Status of the principle: PMSM d–q model with MTPA**
> *Approximate.* Uses constant ψ_{PM}, L_d and L_q; in real IPM motors L_q falls by 30 to 50 % from no load to peak current through saturation and cross-saturation, and ψ_{PM} falls by about 0.1 % per kelvin (NdFeB), so production drives use measured flux maps or lookup tables of optimal i_d, i_q against torque and speed instead of the closed-form trajectory.

Position sensing

PMSM control needs the rotor angle, to within 1 to 2 electrical degrees for good torque accuracy. Servo drives use high-resolution encoders; EV and aircraft drives use **resolvers**, robust to temperature and vibration; low-cost pump and compressor drives (and HVAC compressors in heat pumps) run **sensorless**, estimating the angle from back-EMF at medium and high speed and, for IPMs, from the saliency by injecting a high-frequency signal at low speed. Brushless DC motors (Chapter 8) with trapezoidal back-EMF use three Hall sensors and six-step commutation, the simplest form of the same idea.

12.9 Drive commissioning

A commissioning sequence

Commissioning turns a drive from a box of parameters into a working system. Its steps, in the order a field engineer follows them:

1. **Pre-power checks.** Supply voltage and earthing; cable types, shield bonding at both ends, cable length against the drive's limit and the need for output filters; motor insulation resistance tested with the drive *disconnected* (a megger on the drive's output would damage it); motor insulation suitable for inverter supply (Section 10.7); brake resistor value and rating; correct fuse types.
2. **Nameplate data.** Rated voltage, current, frequency, speed, power, power factor and connection (star or delta); encoder type and resolution; motor thermistor connection.
3. **Motor identification (autotune).** A **stationary** identification injects DC and pulsed voltages to measure R_s and the leakage inductance σL_s; a **rotating** identification, with the motor uncoupled or lightly loaded, measures the magnetising inductance and τ_r (induction) or ψ_{PM}, L_d, L_q and encoder offset (PMSM). The rotating test is preferred for vector control; when the load cannot be uncoupled (a pump, a lift already roped) the drive relies on estimates from nameplate data and the stationary test.
4. **Direction and feedback check.** Run at low speed in V/f to confirm rotation direction (and pump or fan direction from the process side) and that the encoder counts in the same direction; a reversed encoder makes a vector drive run away to current limit.
5. **Current-loop tuning.** Usually automatic from the identified R_s and σL_s, giving bandwidths of 0.5 to 2 kHz.
6. **Speed-loop tuning.** From the measured or estimated inertia (an acceleration test identifies J), set for 40 to 60° phase margin (*Circuits, Signals and Communications*, Ch. 35 §35.4); check for mechanical resonances with a step test, and add notch filters if the coupling or gearbox resonates.
7. **Ramps and limits.** Acceleration and deceleration ramps, minimum and maximum speed, current and torque limits, skip frequencies (to avoid running a fan or pump at a structural resonance), flying-start enable for fans that may be windmilling.
8. **Braking and DC-link control.** Brake chopper threshold and resistor thermal model, or regeneration limits; overvoltage controller.

9. **Protection.** Motor thermal model with low-speed cooling curve, thermistor trip, stall and underload detection, earth-fault, phase-loss and STO wiring tests (Section 10.7).
10. **Process tests and documentation.** Load tests across the speed range; harmonic measurement at the point of common coupling against IEEE 519 or IEC 61000-3-12; parameter backup; commissioning report with settings, measured values and safety-function validation.

Worked example — the deceleration ramp of a large fan. A 75 kW induction-motor fan runs at 1480 rpm ($\omega = 155.0$ rad/s) with a total inertia, fan plus motor, of 60 kg·m². Its full-speed shaft torque is $75\,000/155.0 = 484$ N·m and falls with speed squared. The drive has a diode front end and no brake resistor. The commissioning engineer must set a deceleration ramp to half speed.

- Stored energy at full speed: $\tfrac{1}{2} \times 60 \times 155.0^2 = 721$ kJ.
- Coasting, the fan torque alone decelerates the rotor as $J\,d\omega/dt = -T_0(\omega/\omega_0)^2$, whose solution is $\omega(t) = \omega_0/(1 + T_0 t/J\omega_0)$. It reaches half speed in $J\omega_0/T_0 = 60 \times 155.0/484 = 19.2$ s.
- A 10 s linear ramp to half speed demands a decelerating torque of $J\,\Delta\omega/\Delta t = 60 \times 77.5/10 = 465$ N·m. The fan supplies 484 N·m at full speed but only 121 N·m at half speed; the deficit must come from the motor as braking torque. Regeneration starts almost immediately (at 98 % speed, where $T_0(\omega/\omega_0)^2 = 465$ N·m) and reaches $(465 - 121) \times 77.5 = 26.7$ kW at half speed.
- With no brake resistor, the DC link charges up and the drive trips on overvoltage within a fraction of a second, unless its overvoltage controller stretches the ramp.

So the engineer either sets the ramp to about 20 s or longer (matching the fan's natural deceleration), enables the drive's overvoltage controller to do so automatically, or specifies a brake chopper if the process really needs the 10 s stop. The same reasoning, the shaft equation of Section 10.1 applied to the actual load curve, settles most commissioning disputes about ramps.

> **Status of the principle: Parameter identification**
> *Empirical.* Autotune results are fitted to measured responses under test conditions; they drift with temperature (stator and rotor resistance), saturation and load, so drives that need accurate torque, such as winders and test benches, combine identification with online adaptation and periodic re-identification.

Typical commissioning problems

Table 12.5 — Common commissioning faults and causes.

Symptom	Likely cause	Remedy
Overcurrent trip at start	Wrong motor data, flying fan not caught, output short or earth fault	Check nameplate entry, enable flying start, test cables
DC-link overvoltage on stopping	Ramp too short for the inertia, no braking resistor	Lengthen ramp, enable overvoltage control, add brake chopper
Vector drive runs to current limit, wrong direction	Encoder channels or motor phases swapped	Swap encoder A/B or two motor phases, re-identify
Speed oscillates at low frequency	Speed gain too high for the inertia, or mechanical resonance	Re-identify inertia, reduce gain, add notch filter
Hoist load sags on start	V/f mode, brake opened before torque established	Vector mode with encoder, torque-proving before brake release
Motor bearing failure after months	Common-mode bearing currents	Insulated bearing, shaft grounding, symmetrical shielded cable
Nuisance earth-fault trips with long cables	Capacitive leakage current	Output reactor, reduce switching frequency
Supply transformer overheating, capacitor bank failures	Drive harmonics resonating with PF capacitors	Detuned capacitor bank, harmonic filter (Chapter 25)

The hoist of the opening needs two of these fixes: vector control with an encoder, so that full torque exists at zero speed, and brake logic that proves the motor holds the load before the mechanical brake releases. After that the platform does not move a millimetre.

What remains open

Open question	Where it is answered
How do inverter harmonics and drive input currents affect the supply, and how are they limited?	Chapter 25 (power quality)
How do grid-connected inverters synchronise to the grid and control active and reactive power, using the same d–q tools?	Chapter 27 (grid integration of renewable energy)
How do doubly fed induction generators and full-converter PMSM generators control wind turbines?	Chapter 15 (renewable energy systems); Chapter 27
How much energy do VFDs save across a plant, and how are motor efficiency classes used?	Chapter 28 (energy efficiency and management)
How are battery packs that feed EV inverters rated and managed?	Chapter 16 (energy storage)
How are motors designed for inverter supply, and how are their thermal limits set?	Chapter 9 (electrical machine design and performance)
How are drive cabinets isolated and worked on safely after STO?	Chapter 29 (electrical safety)
How do IGBT inverters, gate drivers and SVPWM modulators work at circuit and firmware level?	*Electronics & Computing*
How are digital current and speed controllers discretised and their sampling rates chosen?	*Circuits, Signals and Communications*, Ch. 35

Next: Chapter 13 leaves the motor shaft for the network that feeds it, opening Part IV with the structure of the power system: how generation, transmission and distribution are arranged to deliver energy reliably to every drive, building and plant.

Part IV — Power Generation

Parts I–III built the converters one at a time: the magnetic circuit, the transformer, the induction, synchronous and special machines, and the drives that control them. Part IV puts them to work at the scale of a nation. It begins with the structure of the power system as a whole, then follows energy into it from every source: conventional plant through steam and gas turbines and hydro runners, renewable plant often through power electronics rather than a rotating generator, and storage that moves energy in time. In every case the questions are the same: how much energy is available, how efficiently it is converted, how fast and how predictably the plant can respond, what it costs and what it emits, and what the plant looks like to the network it connects to.

By the end of this part you should understand

- How a power system is organised into generation, transmission and distribution, why it uses the voltage levels, meshed and radial topologies and substation arrangements it does, and where DC wins over AC.
- How load curves, load factor, diversity, N-1 security, adequacy measures (LOLE) and distribution indices (SAIFI, SAIDI) are used to plan and operate the system, and how a screening curve gives a first estimate of the least-cost generation mix.
- How steam, gas-turbine, combined-cycle, hydro and nuclear plants convert their primary energy, what limits their efficiency and flexibility, and how heat rate, fuel price and carbon price set their place in the merit order.
- How the electrical side of a power station works: generators, excitation, governors and droop, capability curves and auxiliary power.
- How photovoltaic, wind, small hydro, biomass and geothermal plants are designed and rated, how a resource is assessed and turned into P50 and P90 energy estimates, and what capacity factor, capacity credit and forecast error mean for the system.
- How batteries, supercapacitors, flywheels, pumped hydro, thermal stores and hydrogen are rated, managed, compared on cost and kept safe.

Chapters

Chapter	Central question	What it delivers
Chapter 13 — Power-System Structure	How is a power system organised from generator to socket, and how are demand, reliability and growth quantified?	Voltage levels, meshed and radial architecture, components and substations, AC versus DC, power flow, load curves, N-1 and reliability indices, resilience, screening-curve planning
Chapter 14 — Conventional Generation	How do thermal, hydro and nuclear plants turn a primary resource into electricity, and what sets their efficiency and flexibility?	Steam conditions, real gas turbines and ambient derating, combined cycles, hydro turbines and water hammer, reactors and decay heat, generators and droop, heat rate and merit order, auxiliaries, emissions and capture
Chapter 15 — Renewable Energy Systems	How are sun, wind, water, biomass and geothermal heat converted, and how much energy do they deliver, how predictably?	PV string and inverter design, the Betz limit and wind-generator types, run-of-river design flow, biomass CHP, geothermal plants, P50 and P90, performance ratio, capacity factor and credit, variability and forecasting
Chapter 16 — Energy Storage	How is electrical energy moved in time, and which storage technology suits which job?	Battery chemistries, energy and power ratings, battery management, state of charge and state of health, supercapacitors, flywheels, pumped hydro, thermal storage, hydrogen, and the economics and safety of storage

How this part connects to the rest of the book

Part IV draws on everything before it. The synchronous generator of Chapter 7 is the heart of every conventional plant; the induction and permanent-magnet machines of Chapters 6 and 8 reappear as wind generators; the thermal ratings of Chapter 9 set generator limits; and the converter control of Chapter 12 underlies every inverter that connects a PV array, a wind turbine or a battery. Looking forward, Part V turns the structure of Chapter 13 into per-unit models, power flow, fault and stability calculations (Chapters 17–20) in which the generators and inverters described here are the active elements; Part VI carries the power through lines, HVDC, distribution and protection; Part VII returns to renewables and storage as grid-integration and smart-grid problems (Chapters 26–28); and Part VIII uses these ratings in electrical design (Chapter 30).

Two sibling books supply the theory this part uses without repeating it. Thermodynamic laws and the Carnot limit, the Rankine and Brayton cycles, heat exchangers, fluid systems as hydraulic circuits, nuclear binding energy and fission, band theory and the solar-cell equations are taken from *Engineering Physics: Top Down*, Chs. 6, 10, 14, 15 and 19. AC and complex power, three-phase circuits and transmission-line wave theory are taken from *Circuits, Signals and Communications*, Chs. 9, 10 and 14. Battery materials belong to *The Materials Core*, and inverter circuit internals to *Electronics & Computing*.

Chapter 13 — Power-System Structure

Part IV · Power Generation

A developer wants to build a 300 MW data-centre campus on the edge of a growing city and asks the local utility for a connection. The answer comes back in a letter: the nearest 132 kV substation has 40 MW of spare transformer capacity, the 400 kV substation that feeds it is already loaded close to its firm rating on summer evenings, a new 400/132 kV transformer and two new cable circuits will be needed, and the earliest date the transmission operator can offer is six years away because three new generation projects ahead in the queue need the same reinforcement. Nothing in the letter is about a machine or a circuit law. It is about structure: where power is produced, the voltage levels through which it travels, which elements are allowed to fail without loss of supply, how much demand there will be at the moment of system peak, and how long it takes to build the steel and copper that carry it. Those are the subjects of this chapter. Parts I–III built the converters (transformers, machines, drives); this chapter assembles them into the largest machine engineers have ever made, the interconnected power system, and sets out the vocabulary that the rest of the book uses to analyse it.

> **How is a power system organised from generator to socket, and why is it built at the voltages, in the topologies and with the components it has?**
> **How are demand, energy flow, reliability and future growth quantified, and how do those numbers drive what gets built?**

Concept	Engineering destination
Generation, transmission, distribution	Voltage-level choice, loss budgets, network ownership and regulation
Grid architecture	Meshed transmission, radial distribution, synchronous areas, interconnectors
Power-system components	Substation layouts, transformer and breaker specifications, compensation
AC versus DC	HVDC links, offshore wind export, DC data centres, EV and PV systems
Energy flow	Sankey diagrams, line loading, the power-angle transfer limit
Load demand	Load curves, load factor, diversity, peak forecasts, tariffs
Reliability and resilience	N-1 security, LOLE, SAIDI and SAIFI, black start, hardening
Power-system planning	Generation mix, screening curves, transmission reinforcement, connection queues

Roadmap. The supply chain from fuel to socket (13.1) → how the networks are wired together (13.2) → the hardware in them (13.3) → why most of it is AC and where DC wins (13.4) → how energy and power actually move (13.5) → what the customers demand and when (13.6) → how continuity of supply is measured and secured (13.7) → how the system is planned years ahead (13.8).

13.1 Generation, transmission, and distribution

> **Boundary.**
> AC power, complex power and power factor are taken from *Circuits, Signals and Communications*, Ch. 9; three-phase connections, line and phase quantities and three-phase power from Ch. 10; distributed line parameters and surge impedance from Ch. 14. This chapter uses them to describe the system, not to re-derive them.

The three functional layers

Every public power system performs three functions, usually by different organisations and always at different voltages.

- **Generation** converts a primary resource (fuel, falling water, wind, sunlight, nuclear fission) into three-phase electrical power, typically at 11–27 kV at the terminals of a synchronous generator or at 0.4–0.8 kV at the output of an inverter or wind-turbine converter. A generator step-up (GSU) transformer raises it to transmission voltage.
- **Transmission** moves bulk power over tens to thousands of kilometres at 220–765 kV AC (and ±320 kV to ±1,100 kV DC), between large plants, between regions and between countries. It is meshed, so that power has several paths, and it is where the system's frequency and stability are managed.
- **Distribution** takes power from bulk-supply substations down through subtransmission (33–132 kV) and medium voltage (6.6–33 kV) to low voltage (230/400 V or 120/240 V) and delivers it to customers, mostly through radial feeders.

The boundaries are not sharp. A 132 kV network is "transmission" in one country and "distribution" in another; a 50 MW solar plant connects at 33 kV inside the distribution system; a large industrial customer (an aluminium smelter, a steel mill with an electric arc furnace, a hyperscale data centre) may take supply directly at 132–400 kV. Figure 13.1 shows the chain with typical voltages.

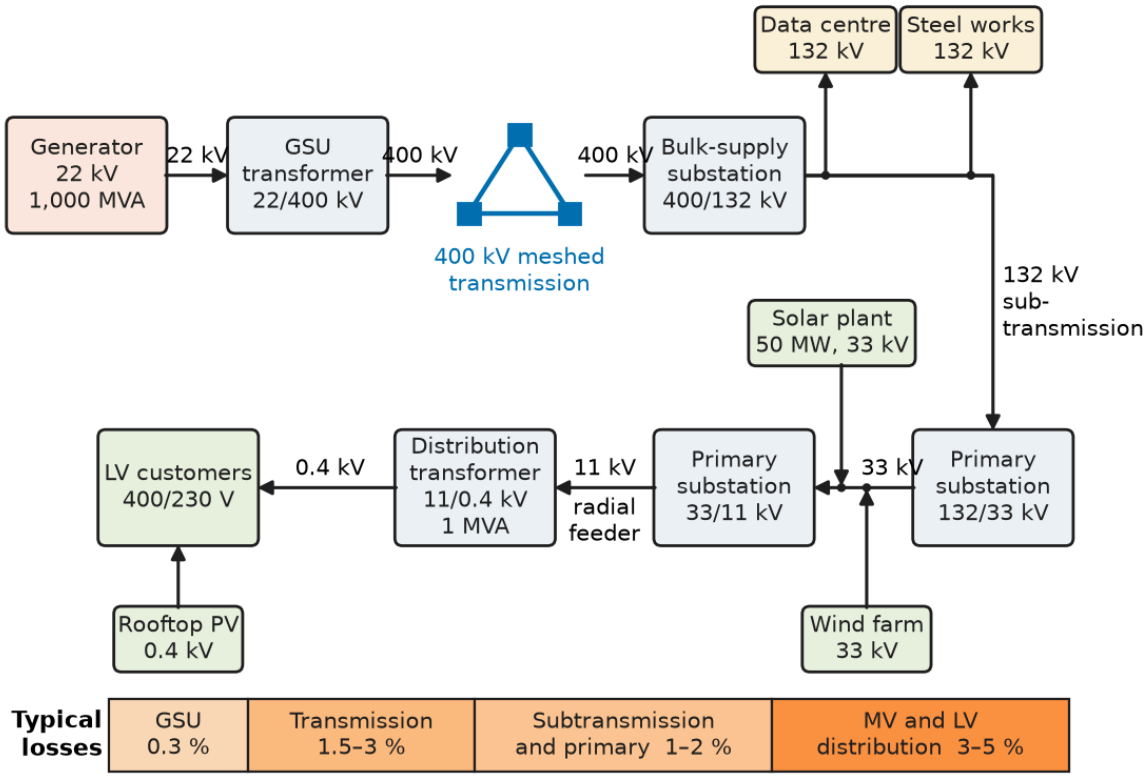


Figure 13.1 — The supply chain from generator to customer with typical voltage levels and the share of losses at each stage.

Why voltage rises with distance and power

The reason for the ladder of voltages is the conductor loss $3I^2R$. For a fixed power P at power factor $\cos\varphi$, the line current is $I = P/(\sqrt{3}\,V\cos\varphi)$, so the loss on a given conductor falls as $1/V^2$. The three-phase power relations and the copper saving of three-phase over single-phase transmission are in *Circuits, Signals and Communications*, Ch. 10 §10.1 and §10.7; here the question is the engineering one of choosing the level.

Worked example — choosing a transmission voltage. A 500 MW wind cluster must be carried 300 km to a load centre at power factor 0.95. A twin-conductor bundle has an AC resistance of 0.035 Ω/km per phase, so $R = 10.5\ \Omega$ per phase. At 220 kV the line current is

$$I = \frac{500 \times 10^6}{\sqrt{3} \times 220 \times 10^3 \times 0.95} = 1381\ \text{A}, \qquad P_{\text{loss}} = 3I^2R = 3 \times 1381^2 \times 10.5 = 60.1\ \text{MW}$$

which is 12 % of the transported power; if the line ran at full output half the hours of the year it would lose about 260 GWh, worth some USD 16 million a year at USD 60/MWh. At 400 kV the current falls to 760 A and the loss to 18.2 MW (3.6 %); at 765 kV it would be 397 A and 5.0 MW (1.0 %). The higher voltage costs more per kilometre (taller towers, longer insulator strings, wider right of way, more expensive substations), so the economic voltage rises with both power and distance. Rules of thumb that fall out of studies of this kind are summarised in Table 13.1.

Table 13.1 — Representative AC voltage levels and their roles (line-to-line, nominal).

Level	Typical voltages	Typical power per circuit	Typical distance	Examples
Low voltage (LV)	230/400 V, 120/240 V, 480 V, 690 V	up to ~1 MVA	< 0.5 km	Homes, offices, motor control centres, wind-turbine converters
Medium voltage (MV)	6.6, 11, 13.8, 20, 22, 33 kV	1–30 MVA	1–30 km	Urban and rural feeders, large motors, solar and wind collector systems
Subtransmission / high voltage (HV)	66, 110, 132, 138 kV	50–250 MVA	10–100 km	Regional rings, industrial supplies, distribution bulk-supply points
Transmission (EHV)	220, 275, 330, 345, 400, 500 kV	300–2,000 MVA	50–500 km	National backbones, large power stations
Ultra-high voltage (UHV)	765, 800, 1,000, 1,150 kV	2,000–6,000 MVA	300–1,500 km	India, China, South Africa, Russia long-distance corridors

The nominal values and the highest voltages for equipment are standardised in IEC 60038 and ANSI C84.1; equipment insulation levels follow from them (Chapter 24).

Who owns what

Until the 1990s most systems were vertically integrated: one utility owned generation, transmission and distribution and sold to captive customers. Many countries have since unbundled them. A typical liberalised structure has competing generators selling into a wholesale market; a **transmission system operator** (TSO; in North America an ISO or RTO with transmission owners) that runs the high-voltage network, balances supply and demand second by second and procures reserves; **distribution network operators** (DNOs or DSOs) that own and run the wires below about 132 kV; and retailers who bill the customer. The physics does not care about ownership, but planning, connection procedures and the incentives that decide which reinforcement is built are shaped by it. The data-centre letter of the opening is an artefact of that structure: the DSO owns the 132 kV substation, the TSO owns the 400 kV one, and both must agree before the connection is offered.

13.2 Grid architecture

Meshed transmission, radial distribution

Transmission networks are **meshed**: every major substation is connected to at least two others, so that the loss of any single line or transformer redistributes power over the remaining paths rather than interrupting it. Meshing buys security at the cost of control. In a mesh the operator cannot simply send power down a chosen line; power divides among parallel paths in inverse proportion to their impedances (Section 13.5), and a single tripped circuit can overload its neighbours.

Distribution networks are mostly **radial** in operation even where they are built as loops. An 11 kV urban feeder may form a ring between two primary substations, but a **normally open point** splits it, so each half is radial:

protection is simple (current flows one way), fault levels are lower, and voltage control is a matter of setting the transformer tap at the source. After a fault the faulted section is isolated and the open point is moved to restore the rest. Dense city centres (parts of London, New York, Paris, Singapore) use interconnected LV networks and meshed MV networks with network protectors to reach very high reliability, at higher cost. Distribution architectures are developed in Chapter 22.

Synchronous areas and interconnections

All AC generators connected through AC lines must turn in electrical synchronism, at one common frequency. The set of networks that share a frequency is a **synchronous area**. Examples:

- **Continental Europe** (ENTSO-E): one synchronous area from Portugal to Turkey and from Denmark to Greece, more than 400 million people, 50 Hz.
- **Nordic**, **Great Britain**, **Ireland** and **Baltic** areas are separate synchronous areas linked to Continental Europe only by HVDC.
- **North America** has four main synchronous areas at 60 Hz: the Eastern Interconnection, the Western Interconnection, ERCOT (most of Texas) and Québec, linked by a few back-to-back HVDC ties.
- **India** operates a single national synchronous grid at 50 Hz since 2013; **Japan** is split between 50 Hz (east) and 60 Hz (west) with frequency-converter stations of a few GW between them.

A large synchronous area pools inertia and reserves: when a 1,000 MW unit trips, every generator in the area contributes a share of the replacement power through its governor droop (Chapter 14 §14.6), so the frequency dip is small. The price is that disturbances also propagate: a cascading outage can spread across a whole area, as in the 2003 North American and Italian blackouts and the 2006 split of the European system into three islands. DC links between areas transfer scheduled power without coupling their frequencies and act as firewalls.

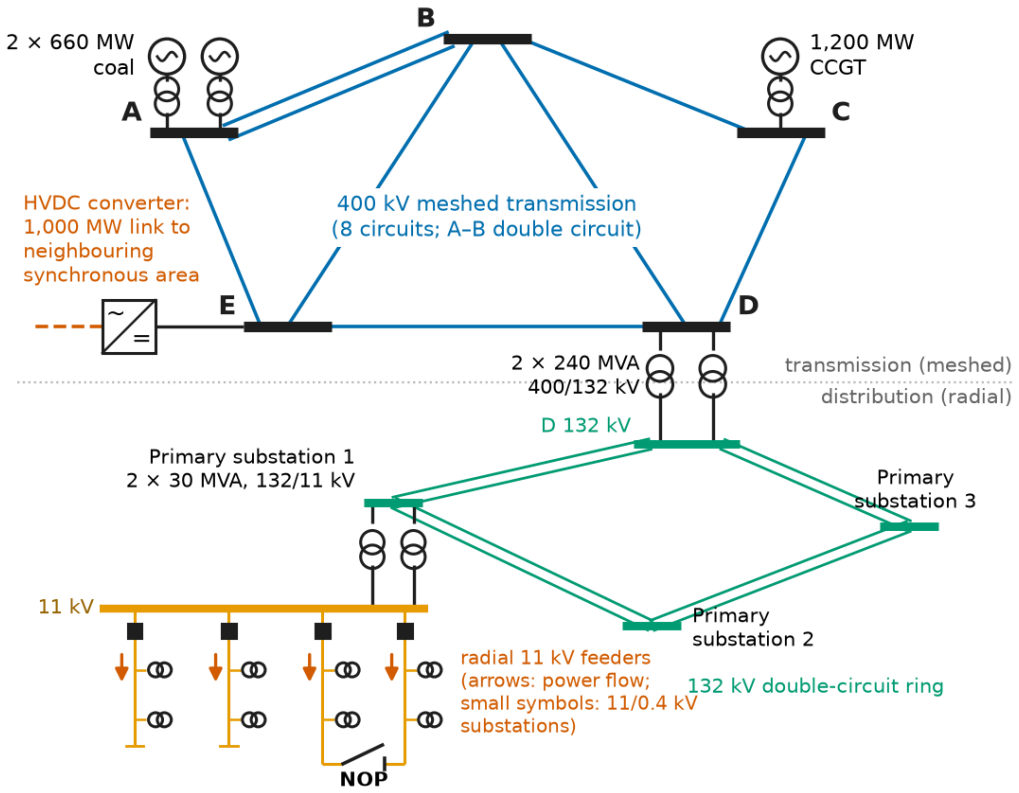


Figure 13.2 — A meshed 400 kV transmission ring feeding radial 132 kV and 11 kV distribution with a normally open point.

Figure 13.2 shows the two layers together. Within a synchronous area the grid is divided into **control areas** (balancing authorities), each responsible for keeping its own interchange with its neighbours at the scheduled value; the combined action of their automatic generation control keeps the frequency at nominal (Chapter 20).

Islands, microgrids and isolated systems

At the other extreme are isolated systems: islands (Hawaii, Cyprus, the Canary Islands), remote mines, offshore platforms, ships and aircraft. They have small inertia, a few large units and no neighbour to lean on, so a single unit trip can be 20–50 % of the load. Their design rules are harsher: spinning reserve at least equal to the largest unit, fast load shedding, often battery storage for frequency support. A large cruise ship is a 60–90 MW isolated diesel-electric power system at 6.6 or 11 kV; a mine in the Atacama desert can be a 200 MW islanded system with gas engines and solar. A **microgrid** is a portion of a distribution network that can disconnect from the main grid and run as an island (a university campus, a military base, a hospital, a village); its control is treated in Chapter 26.

13.3 Power-system components

The inventory

Every power system is assembled from a small number of component types. Their electrical models are introduced across the book (machines in Part II, network models in Chapter 17, lines in Chapter 21, protection in Chapter 23); here they are placed in the system with their typical ratings.

Table 13.2 — Principal power-system components and representative ratings.

Component	Function	Representative ratings and parameters
Synchronous generator	Converts shaft power to three-phase AC; supplies inertia, fault current and reactive power	50–1,750 MVA, 11–27 kV; $X''_d \approx 0.15$–0.25 pu
Inverter-based resource	Converts DC (PV, battery) or variable-frequency AC (wind) to grid AC	3–6 MVA per central inverter, 0.1–0.35 MVA per string inverter; fault current 1.1–1.5 pu
Generator step-up transformer	Raises generator voltage to transmission voltage	22/400 kV, 100–1,000 MVA, impedance 12–16 %
Interbus (auto)transformer	Couples two transmission voltages	400/220 or 400/132 kV, 240–1,000 MVA, impedance 10–15 %, on-load tap changer ±10 %
Distribution transformer	Final step to LV	11/0.4 kV, 0.1–2.5 MVA, impedance 4–6 %
Overhead line	Bulk transport	400 kV double circuit, 1.5–3 GVA per circuit (thermal); $x \approx 0.25$–0.33 Ω/km
Underground / submarine cable	Transport where lines cannot go	132–400 kV XLPE, 200–1,000 MVA; capacitance 0.15–0.3 μF/km
Circuit breaker	Interrupts load and fault currents	420 kV, 4 kA continuous, 50–63 kA interrupting (IEC 62271-100)
Disconnector, earthing switch	Visible isolation for maintenance	Off-load only
Shunt reactor	Absorbs charging reactive power of long lines and cables	50–300 Mvar at 400 kV
Shunt capacitor bank	Supplies reactive power, supports voltage	10–200 Mvar at 132–400 kV
SVC, STATCOM	Fast controllable reactive power	±100 to ±400 Mvar
Series capacitor	Cancels part of line reactance to raise transfer	30–70 % compensation of a long line
HVDC converter station	AC–DC–AC conversion	0.5–12 GW per bipole; losses 0.7–1 % per station
Instrument transformers, relays	Measurement and protection	CT 1 A or 5 A secondary, VT 110 V secondary
Load	Motors, lighting, electronics, heating, electrolysers	From watts to hundreds of MW

Substations and busbar arrangements

A **substation** is a node of the network: a set of busbars to which lines, transformers, generators and compensation devices connect through circuit breakers. The way the busbars and breakers are arranged decides what happens when a breaker or a busbar fails, and how much of the substation can be maintained

without an outage. Table 13.3 compares the common arrangements, and Figure 13.3 draws a breaker-and-a-half bay.

Table 13.3 — Busbar arrangements.

Arrangement	Breakers per circuit	Busbar fault consequence	Breaker maintenance	Typical use
Single busbar	1	Whole substation lost	Circuit out of service	Small MV substations, industrial switchboards
Single busbar with bus section	1 (+ section)	Half the substation lost	Circuit out	MV primary substations
Double busbar	1 (+ coupler)	Circuits on the faulted bar lost until transferred	Circuit out (unless bypass)	HV and EHV substations in Europe
Ring (mesh)	1	No circuit lost for a busbar section fault	No outage	Small EHV substations, 4–6 circuits
Breaker-and-a-half	1.5	No circuit lost	No outage	EHV substations, power-station switchyards (North America, India)
Double breaker, double bus	2	No circuit lost	No outage	Critical nuclear and large-plant switchyards

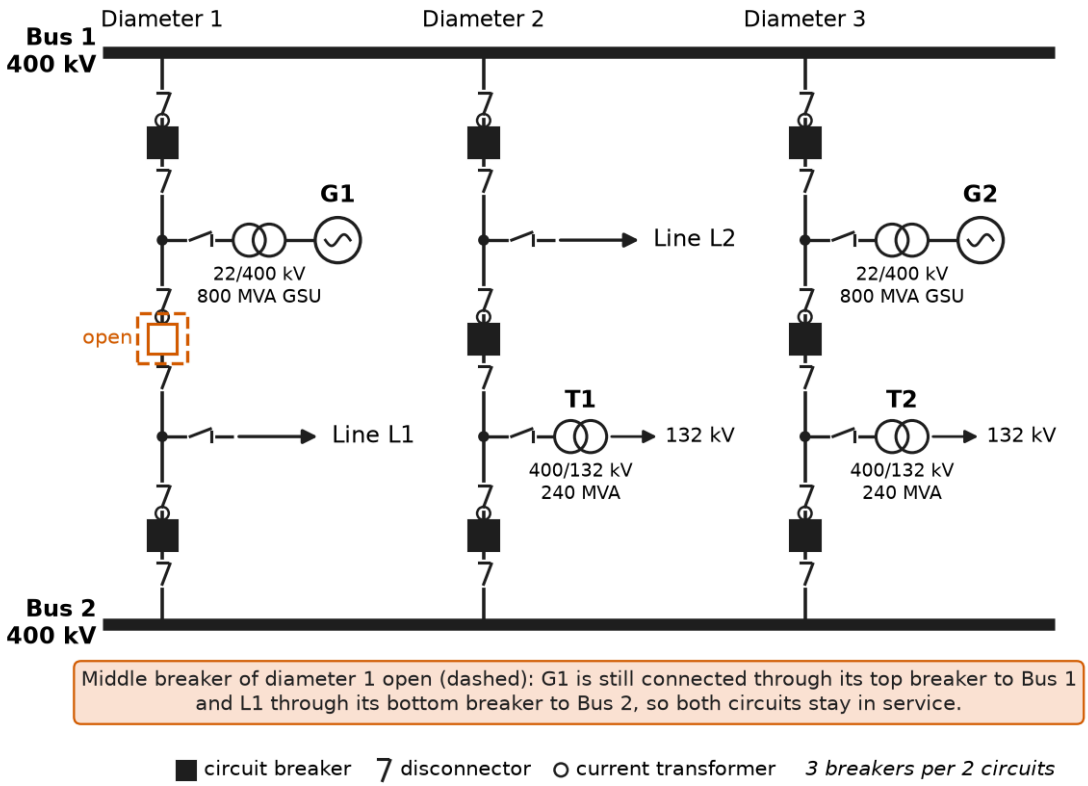


Figure 13.3 — A 400 kV breaker-and-a-half switchyard with three diameters serving two generators, two lines and two transformers.

Secondary systems

Behind the primary plant sit the secondary systems: protection relays and their current and voltage transformers (Chapter 23), the substation automation system with IEC 61850 communications, the DC auxiliary supply (110 V or 125 V batteries that must trip breakers when the AC supply is gone), revenue metering, and the supervisory control and data acquisition (SCADA) link to the control centre where the energy management system runs state estimation, power flow and contingency analysis (Chapter 26). A modern 400 kV substation costs perhaps 15–30 % of its price in secondary equipment, and a large share of major disturbances begin with a protection or control misoperation rather than a primary-plant failure.

13.4 AC versus DC systems

Why AC won the first round

The commercial systems of the 1880s were DC, at about 110 V, because DC lamps and motors existed and DC generators were well understood. They could serve customers only within a kilometre or two of the station, because the voltage could not be raised for transmission and lowered for use. The transformer (Chapter 4) made AC voltage conversion cheap, efficient (over 99 % at large ratings) and passive, and the induction motor (Chapter 6) gave AC a rugged, brushless load. Three-phase AC also gives constant instantaneous power and a rotating field from stationary windings (*Circuits, Signals and Communications*, Ch. 10 §10.7). By 1900 AC had won public supply. It still dominates because transformers, synchronous generators and induction motors are mature, cheap and robust, and because AC current passes through zero twice a cycle, which makes interrupting a fault current with a mechanical breaker far easier than in DC.

Where AC runs out

AC has three structural limits, all of which come from reactance.

1. **Transfer limit of long lines.** The power transferable over a line falls with its reactance (Section 13.5), and the reactance grows with length. Beyond a few hundred kilometres, uncompensated AC lines are stability-limited well below their thermal rating (Chapter 21).
2. **Cable charging current.** A cable is a long capacitor. Its charging current, which flows whether or not any power is delivered, uses up the conductor's thermal capacity.
3. **Synchronism.** Two AC systems can only be joined if they share a frequency; joining 50 Hz and 60 Hz, or two systems that are not to be coupled dynamically, needs conversion.

Worked example — the critical length of an AC cable. A 220 kV XLPE cable has a capacitance of 0.20 µF/km per phase and a thermal rating of 1,000 A. The charging current per kilometre at 50 Hz is

$$I_c = \frac{V}{\sqrt{3}}\,\omega C = \frac{220\times10^{3}}{\sqrt{3}} \times 2\pi \times 50 \times 0.20 \times 10^{-6} = 7.98\ \text{A/km}$$

and the cable generates reactive power at $V^2\omega C = 3.04$ Mvar/km. If the cable is fed from one end only, the charging current alone reaches the thermal rating at $1000/7.98 = 125$ km, leaving nothing for real power. In practice shunt reactors at both ends (and sometimes midway) are used to stretch AC cables to 100–150 km, which is why the export cables of offshore wind farms more than about 80–100 km from shore are almost always HVDC. A 400 kV cable with 0.18 µF/km has a charging current of 13 A/km and generates 9 Mvar/km, so the limit is similar despite the higher rating.

Where DC wins

HVDC transmission carries no reactive power, has no charging current in steady state, has no stability limit set by line reactance, and uses two conductors (or one plus earth or sea return) instead of three. Its costs are concentrated in the converter stations at each end. The result is the classic break-even picture of Figure 13.4: AC is cheaper for short distances, DC beyond a break-even distance of roughly 600–800 km for overhead lines and 50–100 km for submarine or underground cables.

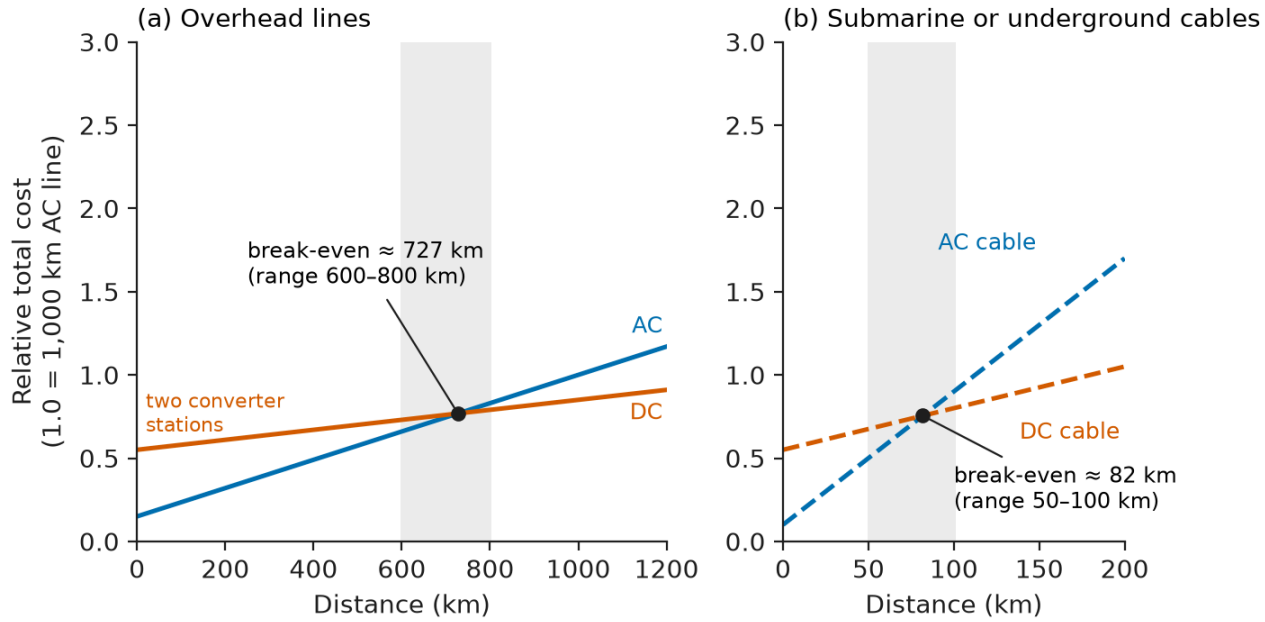


Figure 13.4 — Total cost of AC and DC transmission versus distance, showing the break-even distance for overhead lines and for cables.

Other cases where DC is chosen regardless of distance: asynchronous links between synchronous areas (back-to-back converters, as between Japan's 50 and 60 Hz grids or between ERCOT and the Eastern Interconnection); controllable power flow inside a congested AC mesh; and offshore wind connection. Two converter families are used: line-commutated converters (LCC) with thyristors, cheaper and with lower losses for very large bulk links (±800 kV, 8–12 GW in China and India), and voltage-source converters (VSC) with IGBTs, which control active and reactive power independently, can energise a dead network and suit offshore wind and multi-terminal grids. HVDC fundamentals are in Chapter 21; converter circuit internals belong to *Electronics & Computing*.

DC at the edges of the system

DC is also returning at low and medium voltage because the sources and loads are increasingly DC themselves. PV modules, batteries, fuel cells and electrolysers are DC devices; nearly every electronic load, LED driver, EV battery and variable-speed drive has a DC link inside. Examples:

- **Data centres** use 380 V DC or 48 V DC distribution in some designs to remove one or two conversion stages; a conversion stage at 97–98 % efficiency saved on a 30 MW IT load is 0.6–0.9 MW of losses and cooling.
- **Ships** with electric propulsion are moving from AC to DC switchboards with batteries on the DC bus; **aircraft** use 270 V DC in more-electric designs; **traction** systems use 750 V and 1.5 kV DC for metros and 3 kV DC for some main lines.

- **PV plants** collect at 1,000–1,500 V DC before inversion; MVDC collection for large solar and wind farms is being demonstrated.

The case against DC in distribution is protection: with no current zero, DC fault interruption needs solid-state or hybrid breakers or current-limiting converters. The structure of the future grid is therefore AC backbone with DC links and DC islands, not a wholesale return to DC.

13.5 Energy flow

From primary energy to useful work

Energy flows through a power system from a primary resource to an end use, with losses at every conversion. A Sankey diagram, with band widths proportional to energy, makes the chain visible. For a fossil-fuelled system, starting from 100 units of fuel energy:

- a thermal plant at 40 % net efficiency on the lower heating value delivers 40 units at its terminals and rejects 60 as heat (Chapter 14);
- the plant's own auxiliaries take about 5 % of gross output, so 38 units are sent out;
- transmission and distribution lose about 8 % of what enters them, leaving 35 units at customers' meters;
- a pump driven through a variable-speed drive and an IE3 motor at a combined 90 % delivers about 31.5 units of shaft work.

Less than a third of the fuel energy reaches the pump shaft. The chain explains why efficiency improvements at the end use have the largest leverage on fuel: a kilowatt-hour saved at the motor saves about three kilowatt-hours of fuel (Chapter 28). For wind, solar and hydro there is no fuel to account for and the relevant efficiency is that of turning a free but variable resource into deliverable energy, which is the subject of Chapter 15.

Active and reactive power flow in a network

Inside the network, energy flows as active power P at the system frequency, accompanied by reactive power Q that oscillates between fields without net transport of energy. The definitions and conservation of complex power are in *Circuits, Signals and Communications*, Ch. 9 §9.2–9.4. What the system engineer needs on top is how P and Q divide among the branches of a network. For a predominantly inductive link of reactance X between two buses with voltages $V_1\angle\delta$ and $V_2\angle 0$, neglecting resistance,

$$P = \frac{V_1 V_2}{X}\sin\delta, \qquad Q_2 = \frac{V_2\,(V_1\cos\delta - V_2)}{X}$$

where Q_2 is the reactive power arriving at bus 2. The equations are derived for a synchronous machine in Chapter 7 and generalised to any network in Chapter 18.

> **Status of the principle: Two-bus power transfer**
> *Approximate*. Exact for a lossless series reactance between two fixed voltages; good for EHV lines and transformers where $R \ll X$ (typically $X/R > 10$), poor for LV and MV cables where resistance is comparable to reactance and voltage drop is set mainly by P, not Q.

Two consequences shape the whole of power-system operation:

- **Active power flows toward the lagging voltage angle.** Angle differences across a transmission system are a direct measure of loading; phasor measurement units now record them in real time across continents.
- **Reactive power flows toward the lower voltage magnitude**, and it travels badly: its transport causes voltage drops that grow with X. Reactive power is therefore produced locally, by generators, capacitor banks, STATCOMs and cables, close to where it is consumed.

Worked example — the power-angle limit of a 400 kV line. A single 400 kV circuit 250 km long has a series reactance of 0.32 Ω/km, so $X = 80\ \Omega$. With both ends held at 400 kV,

$$P = \frac{(400 \times 10^3)^2}{80} \sin\delta = 2000 \sin\delta \ \text{MW}$$

At $\delta = 20^\circ$ the transfer is 684 MW; at 30° it is 1,000 MW; the theoretical steady-state maximum at 90° is 2,000 MW. Operators keep the angle across such a line well below about 30–40° so that the system can survive a fault and the loss of a parallel path (Chapter 20). A line twice as long has twice the reactance and half the transfer at the same angle, which is why long AC corridors need series compensation or are replaced by HVDC.

Loop flows

Because power in a mesh divides according to impedance, the path that contracts specify and the path the electrons take can differ. Power sold from northern to southern Germany flows partly through Poland and the Czech Republic; power scheduled within the US Midwest loops through Ontario. These **loop flows** load neighbouring networks without their consent, and they are controlled with phase-shifting transformers (which insert an adjustable angle in series with a line) and HVDC links. They are a direct consequence of the structure of a meshed AC network, and a reason power-flow analysis (Chapter 18) is the central computational tool of the industry.

13.6 Load demand

Load curves

The demand on a power system is the sum of millions of individual decisions to switch something on. Taken together they are remarkably predictable. A **daily load curve** plots demand against time of day; its shape depends on climate, season, day of the week and the mix of residential, commercial and industrial customers. A temperate winter weekday shows a morning rise around 07:00, a daytime plateau, and an evening peak around 17:00–19:00 when lighting, cooking and heating coincide with the end of the working day. Hot climates peak in the afternoon with air conditioning.

The spread of rooftop and utility solar has changed the curve that conventional generators see. **Net load** (demand minus variable renewable output) dips at midday and rises steeply in the evening as the sun sets and demand peaks: the "duck curve" first described for California. Figure 13.5 shows a representative summer weekday. The evening ramp is the hardest operating requirement on such systems; it can exceed 10 GW in three hours in California, and it is the main business case for four-hour battery storage (Chapter 16).

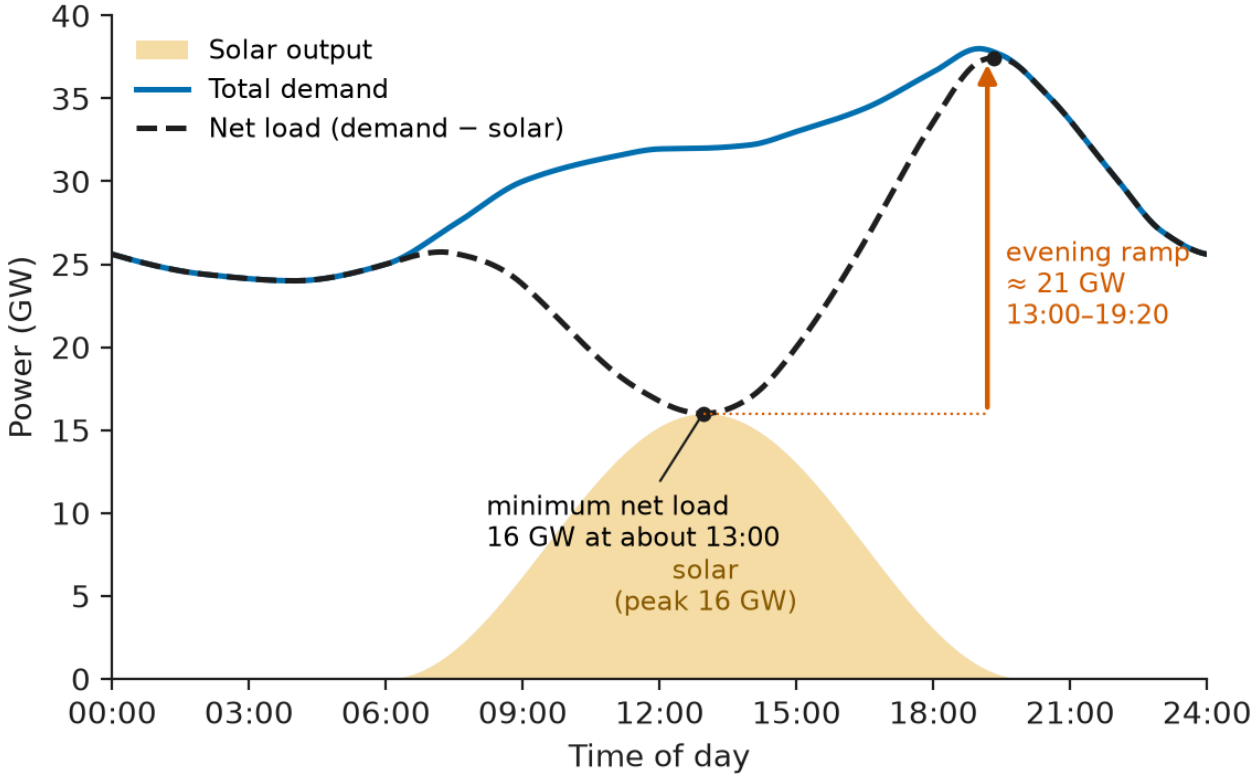


Figure 13.5 — Daily demand and net load on a summer weekday in a system with high solar penetration.

Measures of demand

A small set of numbers describes demand for planning and tariff purposes.

- **Maximum (peak) demand** $P_{\max}$: the highest demand averaged over a demand interval (15 or 30 min for billing, one hour for system planning).
- **Load factor** $LF = \bar{P}/P_{\max}$, where $\bar{P}$ is average demand over the period. A system or a customer with high load factor uses its connection and its generators intensively. Typical annual system load factors are 55–70 %; an aluminium smelter or a data centre may exceed 85–90 %; a residential feeder may be 30–40 %.
- **Demand factor**: maximum demand divided by connected load. A house with 25 kW of appliances may never draw more than 7 kW.
- **Diversity factor**: the sum of the individual maximum demands of a group of customers divided by the maximum demand of the group. Because individual peaks do not coincide, it exceeds 1; its reciprocal is the **coincidence factor**.

> **Status of the principle: Diversity and coincidence factors**
> *Empirical.* They are statistical summaries of measured demand for a class of customers, region and season; values transfer poorly between climates and change as new loads (heat pumps, EV chargers) alter behaviour.

Worked example — sizing a primary substation from customer data. A primary substation will supply four groups: an industrial estate with a measured maximum demand of 4.2 MW, a hospital campus with 3.1 MW, a residential district with 2.6 MW and a shopping centre with 1.8 MW. The individual peaks sum to 11.7 MW, but they occur at different times; the coincident maximum demand measured on a similar

substation is 9.0 MW, a diversity factor of $11.7/9.0 = 1.30$. Sizing the transformers on the arithmetic sum would oversize them by 30 %. With an annual load factor of 0.58, the substation delivers $9.0 \times 0.58 \times 8760 =$ 45.7 GWh a year. A firm-capacity design (Section 13.7) uses two 12 MVA transformers, so that either alone can carry the 9 MW peak at power factor 0.9 (10 MVA) with margin for growth.

Load composition and models

For network studies a load is represented by how its power changes with voltage. Resistive heating draws power proportional to V^2 (constant impedance); a motor driving a fixed mechanical load draws roughly constant power until it stalls; electronics behind a regulated power supply draw constant power. A composite load is often modelled as a weighted sum of constant-impedance, constant-current and constant-power parts (the ZIP model) plus a dynamic motor component. Constant-power loads are the dangerous ones for voltage stability: when voltage sags they draw more current, which deepens the sag (Chapter 20).

Demand growth and electrification

Demand is not static. After two decades of flat consumption in many developed countries, electrification of transport and heat and the growth of data centres are now increasing it. Some orders of magnitude: an electric car at 15 kWh/100 km over 15,000 km a year uses 2,250 kWh a year, so a million of them add 2.25 TWh, about the output of a 300 MW CCGT running at 85 %; their effect on peak depends almost entirely on when they charge, from a few hundred megawatts for managed overnight charging to several gigawatts for uncontrolled early-evening charging at 7 kW. A heat pump can double a house's winter peak. A single hyperscale data-centre campus of 300–1,000 MW is as large as a city. Planners must forecast all of this, as Section 13.8 discusses.

13.7 Reliability and resilience

Adequacy and security

Reliability has two parts. **Adequacy** is having enough generation, transmission and distribution capacity to meet demand, allowing for scheduled maintenance and random failures. **Security** is the ability to withstand sudden disturbances (a short circuit, the unexpected loss of a generator or line) without uncontrolled loss of load. A system can be adequate but insecure (enough plants, but one fault cascades) or secure but inadequate (stable, but short of capacity on a cold windless evening).

The N-1 criterion

The central security rule of transmission planning and operation is **N-1**: the system must remain within thermal, voltage and stability limits after the loss of any single element (a line, a transformer, a generator, a busbar section where applicable), without loss of load. Many operators extend it to N-1-1 (a second outage after re-dispatch) or to specified common-mode events such as the loss of a double-circuit tower line. N-1 is why a primary substation has two transformers, each able to carry the peak (firm capacity), and why the 400 kV substation of the opening letter was "loaded to its firm rating" long before its transformers reached their nameplate sum.

> **Status of the principle: N-1 security criterion**
> *Empirical.* It is a deterministic engineering rule calibrated by decades of operating experience, not a law; it ignores the probability of each outage and the cost of the load lost, which is why probabilistic methods are increasingly used alongside it.

Generation adequacy measures

The probability that available generation falls short of demand is quantified by the **loss-of-load expectation** (LOLE), the expected number of hours or days per year in which demand exceeds available supply, and the **expected energy not served** (EENS). Common standards are 0.1 days per year ("one day in ten years") in much of North America and 3 hours per year in Great Britain, France and Belgium. A traditional proxy is the **reserve margin**, installed firm capacity above forecast peak, typically 12–20 %. With large shares of wind and solar the reserve margin no longer works, because a megawatt of wind is not a megawatt of firm capacity; its contribution is measured by its effective load-carrying capability (Chapter 15 §15.8).

Distribution reliability indices

At the customer end, interruptions are counted with the indices of IEEE 1366:

$$\mathrm{SAIFI} = \frac{\sum N_i}{N_T}, \qquad \mathrm{SAIDI} = \frac{\sum r_i N_i}{N_T}, \qquad \mathrm{CAIDI} = \frac{\mathrm{SAIDI}}{\mathrm{SAIFI}}$$

where N_i is the number of customers interrupted in event i, r_i the duration of that interruption and N_T the total number of customers served. SAIFI is interruptions per customer per year, SAIDI minutes of interruption per customer per year, CAIDI the average restoration time. Only sustained interruptions (longer than 5 min in IEEE 1366, 3 min in European practice) are counted; momentary interruptions have their own index, MAIFI. Major event days are excluded by a statistical rule so that storms do not swamp the trend.

Table 13.4 — Typical annual distribution reliability (sustained interruptions, excluding major events).

System	SAIDI (min/yr)	SAIFI (interruptions/yr)
Dense urban underground networks (Germany, Netherlands, Singapore)	5–20	0.1–0.3
National averages, Western Europe	15–60	0.3–1.0
US investor-owned utilities, average	100–150	1.0–1.4
Rural overhead systems, developed countries	150–400	1.5–3
Many developing-country systems	1,000–10,000	10–100

Worked example — reliability indices for a distribution utility. A utility serving 48,000 customers records six sustained interruptions in a year (major events excluded): 12,000 customers for 95 min, 3,500 for 240 min, 800 for 30 min, 22,000 for 45 min, 1,500 for 600 min and 6,000 for 180 min. Then

$$\sum N_i = 45{,}800, \qquad \sum r_i N_i = 4{,}974{,}000 \text{ customer-minutes}$$

$$\mathrm{SAIFI} = \frac{45{,}800}{48{,}000} = 0.95, \qquad \mathrm{SAIDI} = \frac{4{,}974{,}000}{48{,}000} = 104 \text{ min}, \qquad \mathrm{CAIDI} = \frac{104}{0.95} = 109 \text{ min}$$

The average service availability is $1 - 103.6/525{,}600 = 0.99980$, "three nines and a bit". The 22,000-customer event contributes almost half the SAIFI; automatic sectionalising that restored three quarters of those customers within 3 min would remove 16,500 customer interruptions from the count and cut SAIFI to 0.61, which is why feeder automation (Chapter 26) is usually the cheapest reliability investment.

Component reliability and redundancy

The reliability of a system of components follows from their failure rates and repair times. With a constant failure rate λ and mean repair time r, a component's unavailability is approximately $U \approx \lambda r$ when $\lambda r \ll 1$. Two elements in series (both needed) have $U_s \approx U_1 + U_2$; two in parallel (either suffices) have $U_p \approx U_1 U_2$ if failures are independent. A 132/11 kV transformer with $\lambda = 0.02$ per year and a repair or replacement time of 1,000 h has $U = 0.02 \times 1000/8760 = 2.3 \times 10^{-3}$, about 20 h a year on average; two in parallel give 5.2×10^{-6}, under 3 min a year. The same arithmetic underlies data-centre "2N" and "N+1" designs and the Uptime Institute tiers.

> **Status of the principle: Constant-failure-rate reliability model**
> *Approximate*. Valid in the useful-life part of the bathtub curve, with independent failures; it fails for ageing equipment, for common-mode events (one storm, one tower carrying both circuits, one protection error) and for repair times that depend on spares.

Resilience

Reliability indices deliberately exclude major events. **Resilience** is about exactly those: high-impact, low-probability events such as hurricanes, ice storms, wildfires, floods, geomagnetic storms, extreme cold (Texas, February 2021, when more than 50 GW of generation was forced out and about 4.5 million customers lost supply), cyberattacks (Ukraine 2015) and cascading blackouts. Resilience has four phases: resist (hardening: undergrounding, stronger poles, flood walls, winterised plant), absorb (islanding, microgrids, load shedding that sacrifices part to save the whole), recover (spares, mutual-aid crews, black-start plans), and adapt (rebuilding to a higher standard).

Black start is the restoration of a system from total shutdown. It needs units that can start without external supply (hydro, small gas turbines, diesel generators, increasingly batteries with grid-forming inverters), energise a skeleton of lines, pick up load in steps small enough for their governors, and synchronise islands progressively. Restoration of a large system takes hours to a day; the 2025 Iberian blackout showed again that the planning, testing and documentation of black-start paths are an integral part of system structure, not an afterthought.

13.8 Power-system planning

Time horizons

Planning spans eight orders of magnitude of time, from the protection settings that act in tens of milliseconds to the transmission corridors that take a decade to permit and build.

Table 13.5 — Planning and operating horizons.

Horizon	Decisions	Tools
Milliseconds–seconds	Protection, inertia and primary frequency response, fast reserves	Relay settings, dynamic simulation (Chapters 20, 23)
Minutes–hours	Economic dispatch, secondary and tertiary reserves, redispatch	Energy management system, optimal power flow
Day ahead–week	Unit commitment, outage scheduling, market clearing	Forecasting, security-constrained unit commitment
1–5 years	Maintenance plans, operational adequacy, connection offers	Adequacy studies, short-circuit and power-flow studies
5–40 years	Generation mix, transmission reinforcement, new voltage levels, interconnectors	Capacity expansion models, cost-benefit analysis, scenarios

Load forecasting

Every plan starts with a forecast of energy (GWh) and peak (MW) by region and year. Short-term forecasts (hours to days) use weather forecasts and calendar effects and reach 1–3 % error for system demand a day ahead. Long-term forecasts combine population, economic growth, efficiency trends and electrification scenarios, and their errors are tens of percent over twenty years. Because the error is large, planners now work with several scenarios (high electrification, low growth, high distributed generation) and choose investments that are robust across them.

Generation expansion and the screening curve

Generation technologies differ in how their cost divides between fixed (capital, fixed operation and maintenance, per kilowatt per year) and variable (fuel, carbon, variable maintenance, per megawatt-hour). Which is cheapest depends on how many hours a year a unit runs. The **screening curve** plots annual cost per kilowatt against hours of operation; combined with the load-duration curve it gives a first estimate of the least-cost mix.

Worked example — screening curve for a thermal mix. Three options have the following annualised fixed and variable costs:

Option	Fixed cost (USD/kW-yr)	Variable cost (USD/MWh)
Open-cycle gas turbine (peaking)	60	120
Combined-cycle gas turbine (mid-merit)	110	65
Coal-fired steam (baseload)	230	35

The annual cost per kilowatt of a unit running h hours is $C(h) = F + V\,h/1000$. The OCGT and CCGT lines cross where $60 + 0.120h = 110 + 0.065h$, at $h = 909$ h; the CCGT and coal lines where $110 + 0.065h = 230 + 0.035h$, at $h = 4{,}000$ h. Reading these hours on the load-duration curve (Figure 13.6): the slice of load needed fewer than about 900 hours a year should be met by peaking turbines, the slice needed between 900 and 4,000 hours by combined cycles, and the slice needed more than 4,000 hours by baseload plant. If the system peak is 10 GW and the load-duration curve shows demand above 8.6 GW for 900 hours and above 6.4 GW for 4,000 hours, the screening answer is about 6.4 GW of baseload, 2.2 GW of CCGT and 1.4 GW of OCGT, plus a reserve margin.

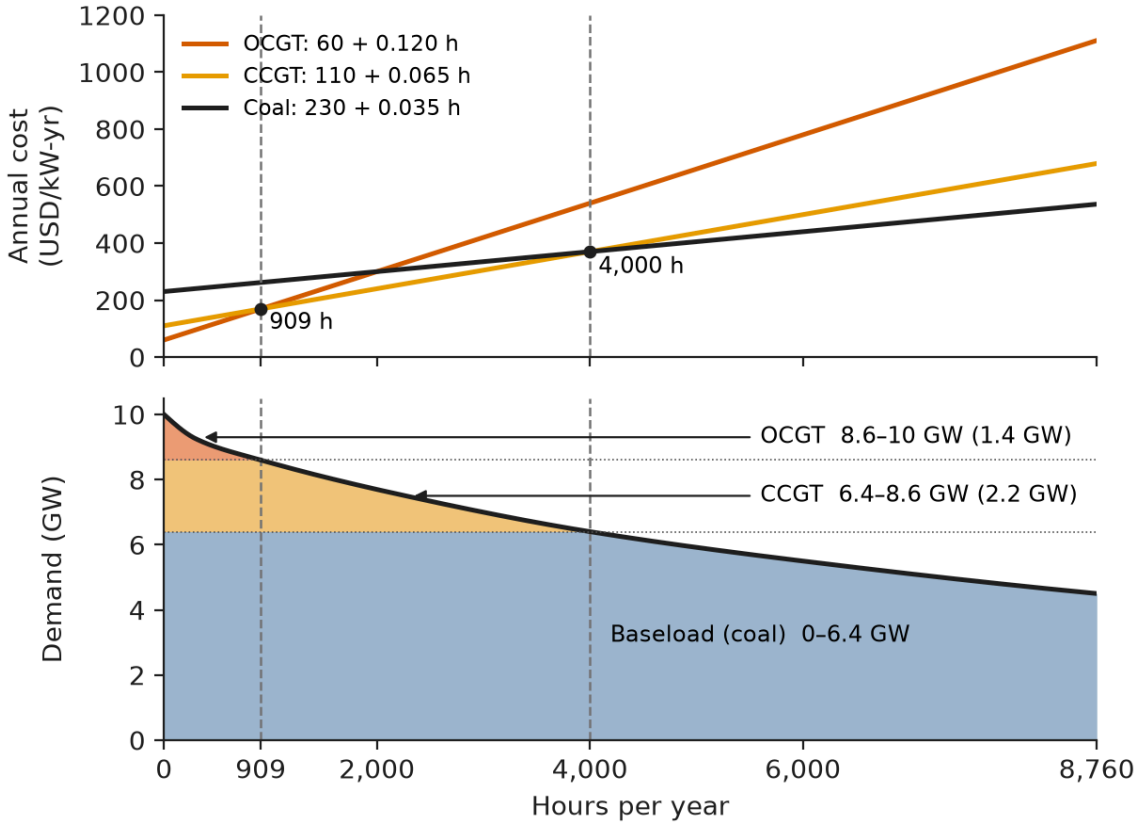


Figure 13.6 — Screening curves for three technologies above the annual load-duration curve; the crossings fix the least-cost mix.

> **Status of the principle: Screening-curve method**
> *Approximate.* It assumes linear costs, perfectly flexible units and a load-duration curve that is the only thing that matters; it ignores start-up costs, ramp limits, network constraints, chronology and storage, and it cannot handle variable renewables, whose output is not dispatchable. Production-cost and capacity-expansion models with hourly chronology replace it for real decisions.

With wind and solar the same idea is applied to **net load**: renewables have high fixed and almost zero variable cost, and they reshape the load-duration curve that the other plant sees. Their value falls as their share rises, because they produce at the same times (Chapter 15).

Transmission and distribution planning

Transmission plans are tested against N-1 (and specified N-2) contingencies across many demand and generation scenarios, using power-flow, short-circuit and stability studies (Part V). Reinforcement options include reconductoring with high-temperature low-sag conductors, dynamic line rating, series compensation, phase-shifting transformers, new circuits, new voltage levels and HVDC overlays. Distribution plans (Chapter 22) test feeder loading and voltage at peak and, with distributed generation, at minimum load and maximum generation, the condition under which rooftop PV raises feeder voltages at noon.

The connection queue

The opening letter describes a common contemporary bottleneck. In many countries the queue of generation, storage and large-load projects waiting for grid connection is several times the installed capacity of the system.

Projects that would each need the same reinforcement are increasingly studied as a cluster, connection offers are made "flexible" (accepting curtailment in return for an earlier date), and anticipatory investment ahead of firm demand is being allowed by regulators. The data-centre developer may well be offered 200 MW in three years on a non-firm basis, with an obligation to shed load or run on-site generation and batteries during the hours when the 400 kV transformers are at their firm limit. That offer is the planning process of this section, the N-1 criterion of Section 13.7 and the load curves of Section 13.6 acting together.

What remains open

Open question	Where it is answered
How do steam, gas, combined-cycle, hydro and nuclear plants convert fuel to electricity, and what limits their efficiency and flexibility?	Chapter 14
How are wind, solar, small hydro, biomass and geothermal resources converted, and how variable are they?	Chapter 15
How does storage shift energy in time and support the evening ramp?	Chapter 16
How are mixed voltage levels and components reduced to one network model?	Chapter 17
How are power flows and voltages calculated in a meshed network?	Chapter 18
How large are fault currents and what must breakers interrupt?	Chapter 19
How do rotor-angle, voltage and frequency stability limit transfers?	Chapter 20
How are long AC lines modelled and compensated, and how does HVDC work?	Chapter 21
How are radial and ring distribution systems designed and regulated?	Chapter 22
How does protection isolate faults selectively?	Chapter 23
How do automation, microgrids and demand response change the structure?	Chapter 26
Where do the three-phase power relations and the AC power definitions come from?	*Circuits, Signals and Communications*, Ch. 9 and Ch. 10

Next: Chapter 14 opens the first block of Figure 13.1, the conventional power station, and follows the energy from fuel through steam or gas turbine to the generator terminals.

Chapter 14 — Conventional Generation

Part IV · Power Generation

On the hottest afternoon of the year a grid operator calls on a 450 MW combined-cycle plant for its full output, and the plant delivers 405 MW. Nothing is broken. At 40 °C the air entering the gas turbines is 8 % less dense than at the 15 °C rating condition, so less mass flows through the machines; the compressors absorb more of the turbine work; the cooling towers cannot pull the condenser down to its design pressure, so the steam turbine loses output too. The same plant, cycled down to 40 % load at noon every day because rooftop solar has filled the midday demand, burns 15 % more gas per megawatt-hour than at full load and wears its hot-gas-path parts out faster with every start. Down the coast a 660 MW coal unit is losing money on the same days, and a nuclear unit is asked whether it can follow load at all. Every one of these questions is a question about how thermal and hydraulic energy is turned into three-phase electricity, and how the electrical side of the plant, from generator to auxiliaries, shapes what the system can ask of it. This chapter follows that chain through the five conventional technologies that still supply most of the world's electricity.

> **How do steam, gas-turbine, combined-cycle, hydro and nuclear plants turn a primary resource into electrical power, and what sets their efficiency, flexibility and rating?**
> **What does the electrical system of a power station consist of, how much of the output does the station consume itself, and what does conventional generation cost the environment?**

Concept	Engineering destination
Steam power plants	Coal, lignite, biomass, nuclear and solar-thermal stations; condenser and cooling design
Gas-turbine plants	Peaking plant, island and offshore power, data-centre and hospital backup
Combined-cycle plants	Mid-merit and baseload gas generation, industrial cogeneration
Hydroelectric generation	Large dams, frequency control, black start, water-hammer and surge design
Nuclear generation	Baseload supply, decay-heat removal, safety-class electrical systems
Generator systems	Generator ratings, cooling, excitation, governors, capability curves, unit connection
Plant efficiency	Heat rate, fuel cost, merit order, LHV and HHV accounting
Auxiliary loads	Station service transformers, large motor starting, drive retrofits, black start
Environmental considerations	Emission limits, carbon cost, cooling water, waste, capture

Roadmap. The steam plant (14.1) → the gas turbine (14.2) → the two combined (14.3) → water power (14.4) → nuclear heat (14.5) → the electrical machine they all drive and its controls (14.6) → efficiency compared and costed (14.7) → the plant's own consumption (14.8) → emissions, water and waste (14.9).

14.1 Steam power plants

> **Boundary.**
> The laws of thermodynamics and the Carnot limit $\eta_C = 1 - T_c/T_h$ are in *Engineering Physics*, Ch. 10 §10.5–10.6; the Rankine cycle with superheat, reheat and regeneration, and its efficiency $\eta = 1 - Q_{\text{cond}}/Q_{\text{boiler}}$, are in Ch. 19 §19.5.1–19.5.2; heat-exchanger LMTD and NTU methods are in Ch. 19 §19.6.3. This section takes the cycle as given and follows it into a real plant.

From cycle to plant

A utility steam unit realises the Rankine cycle with five major systems (Figure 14.1). The **boiler** (or steam generator) burns pulverised coal, lignite, biomass or oil in a furnace whose walls are lined with water tubes; superheaters and a reheater in the hot flue-gas path raise the steam to its final temperature. The **turbine** is split into a high-pressure (HP), an intermediate-pressure (IP) and two or three double-flow low-pressure (LP) cylinders on one shaft, turning a two-pole generator at 3,000 rpm (50 Hz) or 3,600 rpm (60 Hz). The **condenser** under the LP turbines condenses the exhaust steam at a pressure of 4–10 kPa. The **feedwater train** returns the condensate through six to nine feedwater heaters, heated by steam bled from the turbine, and a boiler feed pump raises it to boiler pressure. The **flue-gas train** removes particulates, sulphur and nitrogen oxides before the stack (Section 14.9).

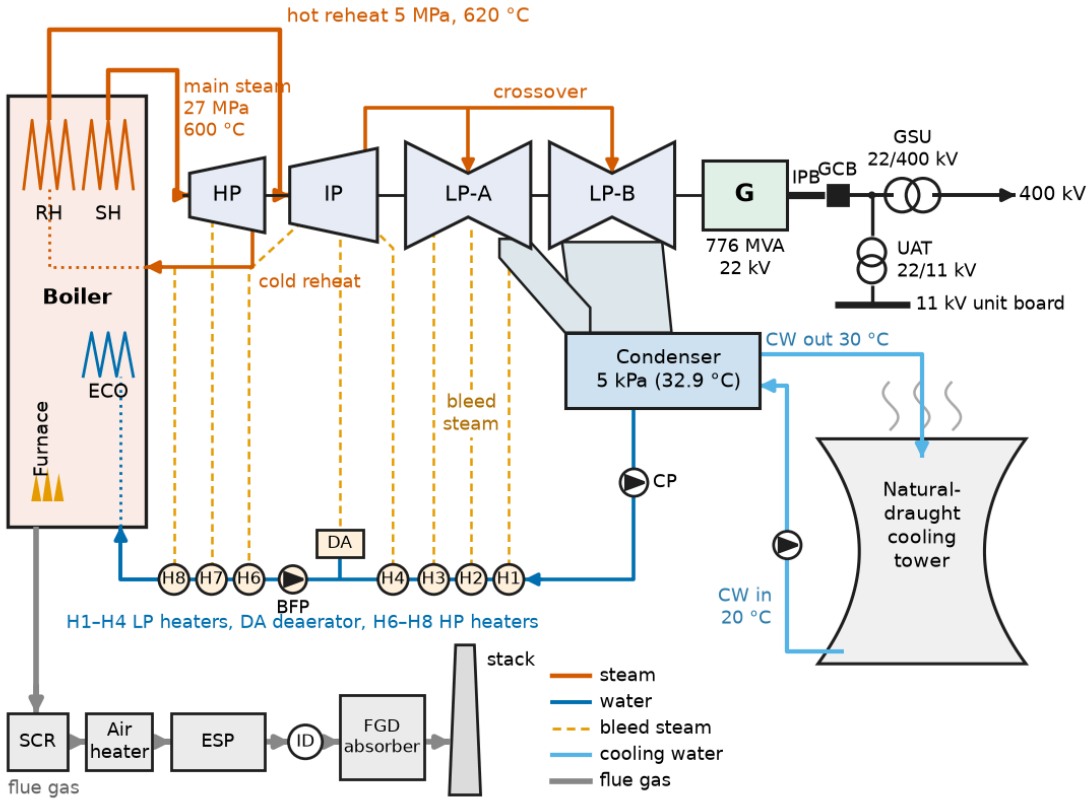


Figure 14.1 — A 660 MW ultra-supercritical coal unit with reheat, regenerative feed heating and the main electrical connections.

Steam conditions and efficiency

The efficiency of a steam unit is set by the mean temperature at which heat is added and the temperature at which it is rejected. Raising the main-steam temperature and pressure past the critical point of water (22.06 MPa, 374 °C) removes the boiling plateau and lifts the mean heat-addition temperature; reheating the steam once (twice in a few units) does the same for the second half of the expansion. Table 14.1 shows how plant efficiency has followed steam conditions, which in turn have followed the creep strength of boiler and turbine steels: 9–12 % chromium martensitic steels for 600–620 °C, nickel-base alloys for the 700 °C class.

Table 14.1 — Steam conditions and typical net efficiency of steam units (LHV basis).

Class	Main steam	Reheat	Net efficiency	Net heat rate (kJ/kWh)
Subcritical drum unit	16–17 MPa, 538 °C	538 °C	35–38 %	9,500–10,300
Supercritical	24 MPa, 565 °C	565 °C	39–42 %	8,600–9,200
Ultra-supercritical (USC)	25–30 MPa, 600 °C	610–620 °C	43–47 %	7,700–8,400
Advanced USC (demonstration)	35 MPa, 700 °C	720 °C	48–50 %	7,200–7,500
Light-water reactor secondary	6–7 MPa, ~285 °C saturated	moisture separator reheater	32–36 %	10,000–11,250
Small biomass or waste boiler	4–10 MPa, 450–540 °C	none	20–32 %	11,250–18,000

The cold end matters as much as the hot end. Condenser pressure is fixed by the cooling-water temperature: at 5 kPa the steam condenses at 32.9 °C, at 10 kPa at 45.8 °C. A unit cooled by cold sea water (Nordjylland in Denmark, Isogo in Japan) gains one to two percentage points of efficiency over the same unit with a dry air-cooled condenser in a desert, and on a hot day any unit loses output as its condenser pressure rises. This is the steam-side part of the opening problem.

Worked example — fuel, carbon and cooling water of a 660 MW unit. An ultra-supercritical unit delivers 660 MW net at 42 % net efficiency (LHV), burning bituminous coal of lower heating value 25 MJ/kg. Fuel power input is

$$\dot{Q}_f = \frac{660}{0.42} = 1571\ \text{MW}, \qquad \dot{m}_{\text{coal}} = \frac{1571}{25} = 62.9\ \text{kg/s} = 226\ \text{t/h}$$

about 5,400 tonnes a day, two trainloads. The net heat rate is $3600/0.42 = 8{,}571$ kJ/kWh. With the IPCC default emission factor for bituminous coal of 94.6 kg CO₂ per GJ, each net kilowatt-hour emits $8.571 \times 0.0946 = 0.811$ kg of CO₂. For the condenser: auxiliaries take 7 % of gross output, so gross output is $660/0.93 = 710$ MW; boiler and stack losses take about 8 % of fuel input (126 MW); the rest, $1571 - 710 - 126 = 736$ MW, is rejected in the condenser. With a cooling-water temperature rise of 10 K the circulating-water flow is

$$\dot{m}_{cw} = \frac{736 \times 10^6}{4186 \times 10} = 1.76 \times 10^4\ \text{kg/s} \approx 17.6\ \text{m}^3/\text{s}$$

a small river. In a closed loop with a cooling tower, about 1.5–2 % of that flow evaporates, roughly 0.3 m³/s, which is why siting a thermal plant is as much a water question as a fuel question (Section 14.9).

Part load and flexibility

Steam units were designed for continuous full-load operation. With wind and solar displacing them for part of the day they now cycle, and their limits become important: a minimum stable load of 30–50 % (set by furnace flame stability and boiler circulation), ramp rates of 1–4 % of rating per minute (set by thermal stress in thick-walled headers and turbine rotors), a hot start of 2–4 h and a cold start of 6–10 h. Each start consumes fatigue life of the thick components; operators assign an equivalent cost, often tens of thousands of dollars per cold start for a large unit, which enters the unit-commitment problem of Chapter 13 §13.8. Retrofits to lower the minimum load (indirect firing, better mill control, turbine bypass) are now common.

14.2 Gas-turbine plants

> **Boundary.**
> The ideal Brayton cycle and its efficiency $\eta = 1 - r_p^{-(\gamma-1)/\gamma}$, together with the materials that permit high turbine inlet temperatures, are in *Engineering Physics*, Ch. 19 §19.5.3. Here the cycle is made non-ideal and put on a generator.

Machines and ratings

A gas turbine compresses air, burns fuel in it at constant pressure and expands the hot gas through a turbine that drives both the compressor and the load. Two families serve power generation.

- **Heavy-duty (industrial, frame) machines** are built for power: single-shaft, 3,000 or 3,600 rpm direct-coupled to a two-pole generator (or geared for smaller sizes), pressure ratios 15–25, turbine inlet temperatures 1,400–1,600 °C in F-, H- and J-class machines, and ratings of 250–570 MW in simple cycle at 38–44 % efficiency.
- **Aeroderivative machines** are adapted jet engines: lighter, higher pressure ratio (25–40), two or three shafts, a free power turbine driving the generator, ratings of 30–120 MW at 40–44 % efficiency. They start in 5–10 min and tolerate many starts, which suits peaking duty, offshore platforms, LNG plants, ships and fast-reserve service.

The real cycle

The ideal Brayton efficiency predicts 57.5 % for a pressure ratio of 20; no simple-cycle machine achieves it. The main reason is that the compressor and turbine are not isentropic, and because the compressor absorbs a large fraction of the turbine work (the **back-work ratio**), small component inefficiencies are amplified.

Worked example — a non-ideal Brayton cycle. Take an air-standard cycle with $\gamma = 1.4$, $c_p = 1.005$ kJ/(kg·K), ambient $T_1 = 288.15$ K (ISO 15 °C), pressure ratio $r_p = 20$, turbine inlet $T_3 = 1600$ K, compressor isentropic efficiency 0.88 and turbine isentropic efficiency 0.90. The isentropic temperature ratio is $r_p^{0.2857} = 2.354$, so

$$T_2 = T_1 + \frac{T_1(2.354 - 1)}{0.88} = 731\ \text{K}, \qquad T_4 = T_3 - 0.90\,T_3\left(1 - \frac{1}{2.354}\right) = 772\ \text{K}$$

Compressor work is $c_p(T_2 - T_1) = 445$ kJ/kg, turbine work $c_p(T_3 - T_4) = 832$ kJ/kg, heat added $c_p(T_3 - T_2) = 873$ kJ/kg. The net work is 387 kJ/kg, the back-work ratio 0.54 and the efficiency $387/873 = 44.3$ % against the ideal 57.5 %. Real machines lose a few points more to the compressed air bled off to cool the first

turbine stages, to pressure losses in the inlet, combustor and exhaust, and to variable gas properties, which brings them to the 36–42 % of Table 14.4. A machine of 400 MW at 387 kJ/kg would swallow about 1,030 kg/s of air. Repeating the calculation for other pressure ratios shows why designs settle where they do: efficiency keeps rising slowly with r_p (46 % at 30, 47 % at 40), but specific work peaks near $r_p \approx 15$ and falls above it, so heavy-duty machines intended for combined cycles choose moderate pressure ratios and high exhaust temperature, while aeroderivatives choose high pressure ratios.

> **Status of the principle: Air-standard cycle with component efficiencies**
> *Approximate*. Treats the working fluid as air with constant specific heats and lumps all irreversibility into two isentropic efficiencies; it captures trends with pressure ratio, firing temperature and ambient temperature well but overestimates absolute efficiency by several points because it omits blade cooling, combustion products and duct losses.

The ambient-temperature penalty

A gas turbine is a constant-volume-flow machine: at fixed shaft speed its compressor swallows a nearly fixed volume of air per second, so its mass flow falls with air density. Hotter air also costs more compressor work per kilogram. Repeating the worked example with $T_1 = 308$ K (35 °C) gives net specific work 356 kJ/kg (8 % lower) and a mass flow lower by the density ratio $288/308 = 0.935$; output falls to $0.920 \times 0.935 = 0.86$ of its ISO value, about 0.7 % per kelvin. Manufacturers quote output and heat rate at ISO conditions (15 °C, 60 % relative humidity, 101.325 kPa; ISO 3977-2) with correction curves like Figure 14.2. Inlet chilling, evaporative coolers and fogging recover part of the loss on hot days, exactly when the power is most valuable.

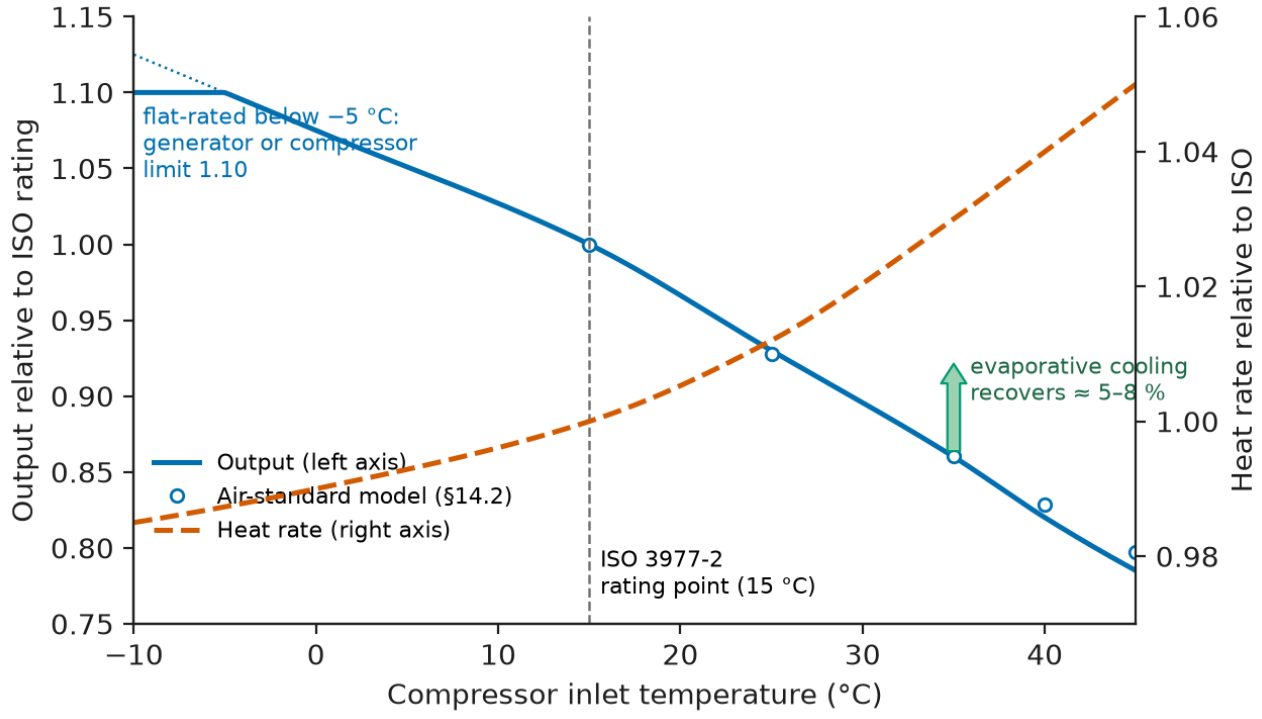


Figure 14.2 — Gas-turbine output and heat rate relative to ISO rating versus compressor inlet temperature.

Role in the system

Open-cycle gas turbines are the classic peaking plant: low capital cost, high fuel cost per megawatt-hour (Chapter 13 §13.8), fast starts. Reciprocating gas engines (4–20 MW per engine, 44–49 % efficiency, start in

2–5 min) compete for the same role and for island grids, and diesel gensets (1–3 MW each) provide the backup supply of hospitals, data centres and nuclear safety systems. Gas turbines burn natural gas, distillate oil and, increasingly, hydrogen blends; most current heavy-duty combustors accept 30–50 % hydrogen by volume, with full-hydrogen combustion under development, limited by flame speed and NOx formation.

14.3 Combined-cycle plants

The bottoming cycle

A gas turbine exhausts at 550–650 °C, which is hot enough to raise steam for a Rankine cycle. A **combined-cycle gas turbine** (CCGT) plant puts a **heat-recovery steam generator** (HRSG) in the exhaust duct and drives a steam turbine from it (Figure 14.3). If the gas turbine converts a fraction η_{GT} of fuel energy to work and rejects the rest in its exhaust, the HRSG recovers a fraction ε of that exhaust heat, and the steam cycle converts a fraction η_{ST} of the recovered heat, energy conservation gives

$$\eta_{CC} = \eta_{GT} + \varepsilon\,\eta_{ST}\,(1 - \eta_{GT})$$

> **Status of the principle: Combined-cycle efficiency**
> *Derived.* A direct first-law balance on two cascaded heat engines with no supplementary firing; it holds exactly for the stated definitions, and the physics lies in the values of ε and η_{ST}, which are limited by the HRSG pinch point and stack temperature.

With $\eta_{GT} = 0.40$, $\varepsilon = 0.90$ and $\eta_{ST} = 0.35$, $\eta_{CC} = 0.40 + 0.90 \times 0.35 \times 0.60 = 0.589$, nearly 59 %. The best H- and J-class plants have reached about 63–64 % net LHV in commercial operation; typical new plants are 58–62 %, older F-class plants 55–58 %. Roughly two thirds of the output comes from the gas turbine and one third from the steam turbine.

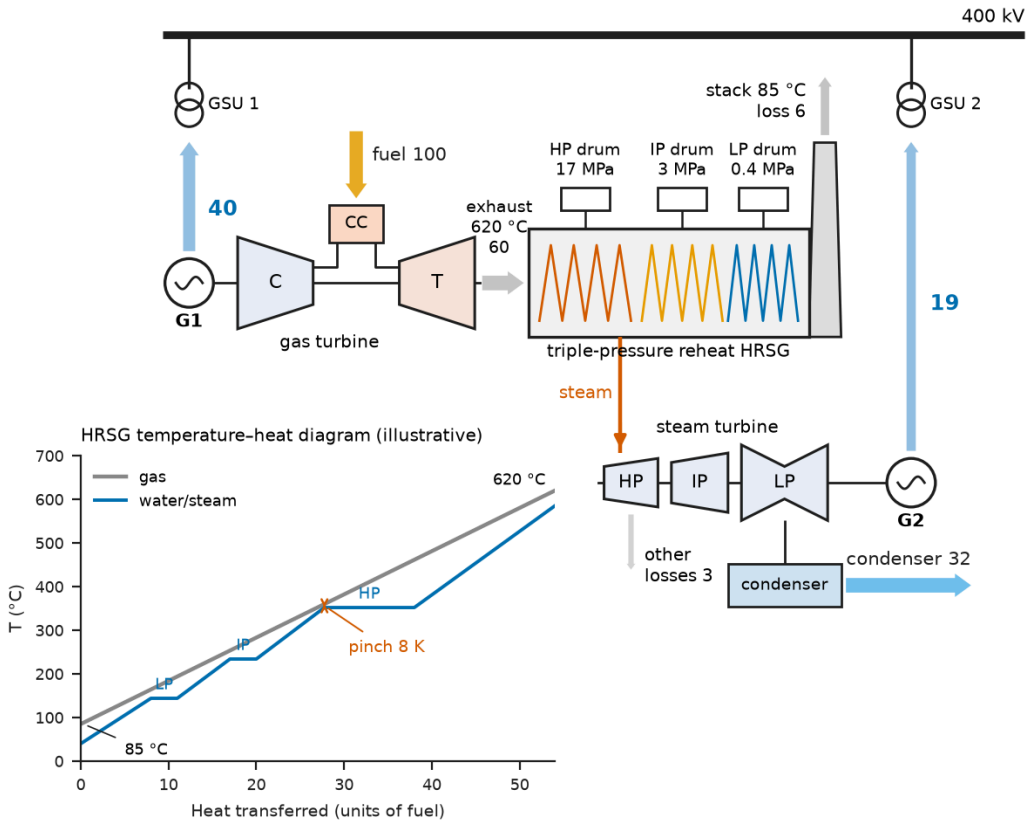


Figure 14.3 — A 1-on-1 combined-cycle unit with triple-pressure reheat HRSG and its energy flows per 100 units of fuel.

Configurations and operation

A CCGT is built as a single-shaft unit (gas turbine, generator and steam turbine on one shaft, with a synchro-self-shifting clutch for starting) or as a multi-shaft "2-on-1" block in which two gas turbines with their HRSGs feed one steam turbine. A 2-on-1 H-class block reaches 1,100–1,300 MW. The HRSG is the slow element: its thick drums limit the warm-up rate, and the steam turbine cannot be loaded until steam conditions match its metal temperatures. Modern "fast-start" designs with once-through HP sections and steam-turbine bypass reach full output in 30–40 min from a hot start; gas-turbine power is available in about 10 min, before the steam side contributes.

Part-load efficiency falls because the gas turbine must reduce firing temperature to cut output, which cools its exhaust and degrades the bottoming cycle: a 60 % plant at full load runs at roughly 52–54 % at 50 % load. A 2-on-1 block can shut one gas turbine down and run the other near full load, which keeps efficiency high at half output.

Cogeneration

Where there is a use for low-temperature heat (district heating, paper and chemical plants, refineries, desalination), the steam turbine can exhaust at 0.2–1 MPa instead of a vacuum and deliver its condensing heat to the process. Electrical efficiency falls, but the total useful output (electricity plus heat) reaches 75–90 % of fuel energy. The value of the heat depends on its temperature; an exergy view counts a megawatt of 120 °C district heat as worth roughly a quarter of a megawatt of electricity.

14.4 Hydroelectric generation

> **Boundary.**
> Bernoulli's equation and hydrostatics are in *Engineering Physics*, Ch. 15 §15.5; fluid systems as hydraulic circuits, pipe losses and the pump affinity laws are in Ch. 19 §19.7. This section covers large reservoir and pumped-storage plant; small and run-of-river hydro as a renewable resource are treated in Chapter 15 §15.3.

Power from head and flow

A hydro plant converts the potential energy of water stored at a height H above the turbine. The power delivered to the grid is

$$P = \eta_t \, \eta_g \, \rho g Q H_n$$

where Q is the turbine flow, H_n the net head (gross head minus losses in intake, tunnel and penstock), η_t the turbine efficiency (90–95 % at the best point for large units) and η_g the generator efficiency (98–99 %). Water-to-wire efficiencies of 85–92 % make hydro by far the most efficient large-scale conversion technology.

Worked example — a Francis unit. A plant has a gross head of 180 m and a design flow of 120 m³/s per unit. Penstock and intake losses at design flow are 4 % of gross head, so $H_n = 172.8$ m. The hydraulic power is $1000 \times 9.81 \times 120 \times 172.8 = 203.4$ MW, and with $\eta_t = 0.93$ and $\eta_g = 0.985$ the unit delivers $P = 186$ MW. Over a year with 4,000 equivalent full-load hours it generates 745 GWh.

Turbine types

The turbine is chosen by head and flow (Figure 14.4), conventionally through the specific speed:

- **Pelton** (impulse) turbines for high heads, about 200–1,800 m: water jets strike buckets on a wheel at atmospheric pressure. Flat efficiency curves down to 10–20 % of rated flow.
- **Francis** (reaction) turbines for about 20–700 m: the most common type, from a few megawatts to the 1,000 MW units at Baihetan. Efficiency falls off at part load, and a "rough zone" of vortex-rope pressure pulsations is typically avoided between about 40 % and 60 % load.
- **Kaplan** and **bulb** turbines for low heads, about 2–70 m for Kaplan and 2–25 m for bulb units, with adjustable runner blades that keep efficiency high over a wide flow range: run-of-river plants on large rivers and tidal barrages.
- **Reversible pump-turbines** for pumped storage, usually of Francis type, with motor-generators (Chapter 16).

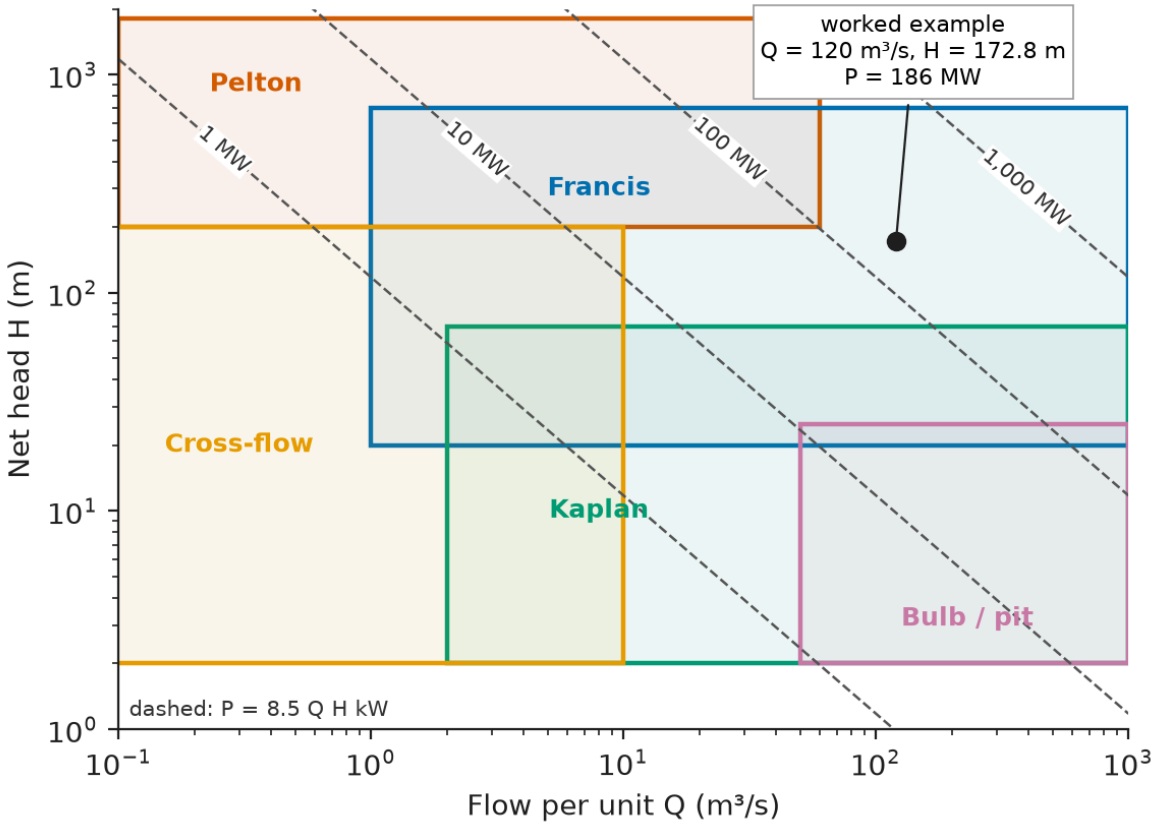


Figure 14.4 — Turbine-type application ranges by net head and flow, with lines of constant unit power.

Water hammer, surge and governing

A hydro unit is hydraulically coupled to a long column of water. Two consequences follow, both absent from thermal plants.

First, **water hammer**. If the wicket gates close faster than a pressure wave can travel up the penstock and back, $2L/a$ where $a \approx 1{,}000$ m/s is the wave speed in a steel penstock, the column is stopped abruptly and the pressure rise follows the Joukowsky relation $\Delta p = \rho a \, \Delta v$. In the worked-example plant a 650 m penstock of 5.5 m diameter carries $v = 120/23.8 = 5.05$ m/s; sudden closure would raise the pressure by $1000 \times 1000 \times 5.05 = 5.05$ MPa, a head of 515 m on top of the 180 m static head. Gates are therefore closed over 5–15 s, much longer than $2L/a = 1.3$ s, and long tunnels are protected by surge tanks. This is why a hydro unit that loses its grid connection (load rejection) cannot simply shut its gates: it overspeeds to 130–160 % before the governor brings it back, and the generator and its rotor are designed for that runaway.

Second, the **water starting time**

$$T_w = \frac{L\,v}{g\,H_n}$$

is the time needed to accelerate the water column from rest to rated velocity under the net head. For the worked example $T_w = 650 \times 5.05/(9.81 \times 172.8) = 1.9$ s; typical values are 0.5–4 s. When the governor opens the gates to increase output, the water column cannot accelerate instantly, so the pressure at the turbine first falls and power briefly *decreases* before it rises. In control terms the turbine has a right-half-plane zero; the governor must be detuned (transient droop) to stay stable, which limits how quickly a hydro unit can respond to a frequency drop despite its fast mechanical start.

Status of the principle: Rigid-water-column turbine model
Approximate. Treats the water as incompressible in a rigid conduit, which is adequate for governor and frequency studies when T_w is small compared with the wave travel time effects; it fails for long penstocks and fast gate movements, where the elastic travelling-wave model of water hammer is needed.

The system value of hydro

Large hydro with reservoirs is the most flexible conventional source: units start from standstill to full load in 1–5 min, ramp at tens of percent per minute, provide black-start capability (Chapter 13 §13.7) and can run as synchronous condensers with the runner dewatered. Its output is limited by water, not by capacity, so reservoir plants are dispatched to the hours of highest value and their capacity factors (35–60 %) reflect hydrology and strategy rather than technical availability.

14.5 Nuclear generation

Boundary.
Nuclear binding energy, the energy of about 200 MeV per fission of uranium-235, the fission barrier, the chain reaction and the four-factor formula for k_{eff} are in *Engineering Physics*, Ch. 6 §6.9–6.11; the radioactive-decay law is in §6.10.3. This section is about the reactor as a heat source for a steam plant and about its electrical consequences.

Reactor types

A power reactor is a controlled heat source for a Rankine cycle. Its coolant and moderator define its type (Table 14.2).

Table 14.2 — Principal power-reactor types.

Type	Coolant / moderator	Coolant outlet	Steam to turbine	Net efficiency	Examples
Pressurised water reactor (PWR, VVER)	Light water at 15.5 MPa	~325 °C	6–7.5 MPa saturated, via steam generators	33–37 %	Most of the world fleet; EPR 1,650 MW, AP1000
Boiling water reactor (BWR)	Light water boiling at 7 MPa	~286 °C	Direct cycle	33–35 %	US, Japan, Sweden, ABWR 1,350 MW
Pressurised heavy-water reactor (CANDU, PHWR)	Heavy water	~310 °C	~4.7 MPa saturated	30–33 %	Canada, India, Romania
Advanced gas-cooled reactor (AGR)	CO_2 / graphite	~640 °C	16 MPa, 540 °C superheated	~41 %	UK
High-temperature gas reactor (HTGR)	Helium / graphite	750–950 °C	13–17 MPa, 540–570 °C	40–45 %	China HTR-PM
Sodium-cooled fast reactor	Liquid sodium / none	~550 °C	14–17 MPa, ~500 °C	38–42 %	Russia BN-800
Small modular reactors (SMR)	Mostly light water	300–325 °C	Saturated	30–33 %	50–300 MW designs

The low steam conditions of light-water reactors, limited by the pressure a reactor vessel can safely hold, explain their modest efficiency: the steam leaves the steam generators saturated, so the turbine has very wet steam in its later stages and needs moisture-separator reheaters between HP and LP cylinders. To keep blade tip speeds of the enormous LP last stages within limits, large nuclear turbines run at half speed (1,500 or 1,800 rpm) and drive four-pole generators of up to about 2,000 MVA, the largest rotating electrical machines built.

Worked example — fuel use of a 1,000 MWe PWR. At 33 % net efficiency the reactor produces $1000/0.33 = 3{,}030$ MW of heat. At 200 MeV (3.20×10^{-11} J) per fission the fission rate is

$$\frac{3.03 \times 10^{9}}{200 \times 1.602 \times 10^{-13}} = 9.46 \times 10^{19}\ \text{s}^{-1}$$

which fissions $9.46 \times 10^{19} \times 86{,}400/6.022 \times 10^{23} \times 235\ \text{g} = 3.19$ kg of uranium-235 per day, a little over a tonne a year (plus some plutonium bred in the fuel, and more U-235 lost to non-fission capture). The reactor is refuelled by fuel assemblies, accounted for by burn-up: at 50 GWd per tonne of uranium and a capacity factor of 90 %, the core must be supplied with $3030 \times 365 \times 0.9/50{,}000 = 19.9$ tonnes of enriched uranium a year, against about 2.7 million tonnes of coal for a coal plant of the same output. The spent fuel is the same 20 tonnes, and it is the waste-management question of Section 14.9.

Decay heat and the electrical consequences

Shutting down the chain reaction does not shut down the heat. The fission products continue to decay, releasing about 6–7 % of full thermal power immediately after shutdown and falling slowly. A widely used approximation (Way–Wigner) for a reactor that has run long at power P_0 gives

$$\frac{P_d}{P_0} \approx 0.066\, t^{-0.2}$$

with t in seconds after shutdown: 1.3 % after 1 hour (39 MW for the 3,030 MW core), 0.7 % after a day (21 MW), 0.5 % after a week.

> **Status of the principle: Way–Wigner decay-heat approximation**
> *Empirical.* A power-law fit to the summed decay of fission products, accurate to roughly ±30 % between about 10 s and 100 days; licensing analyses use the tabulated standards (ANS-5.1, ISO 10645), which are slightly lower in the first hours.

Removing that heat needs pumps and therefore electricity, at the very moment the generator has stopped. This single fact drives the electrical design of a nuclear station: two independent offsite supplies from the grid, safety-class (Class 1E, IEEE 308) AC buses fed by redundant emergency diesel generators of 4–8 MW each that start and load within 10–15 s, large DC batteries for instrumentation and control, and, in newer designs, passive cooling that needs no AC power for 72 h. The 2011 Fukushima Daiichi accident was a station blackout: the earthquake removed offsite power and the tsunami flooded the diesels and switchgear, so decay heat could not be removed. Grid engineers also feel the consequence: a nuclear unit is usually the largest single infeed in its system, so its trip sets the size of the frequency reserve (Chapter 20), and grid voltage and frequency excursions that trip the reactor's coolant pumps are design cases for the grid code.

Flexibility

Nuclear plants have very low marginal cost and are normally run at full output. They can follow load: French PWRs routinely ramp between about 50 % and 100 % at 1–5 % per minute, using grey control rods to avoid distorting the power distribution, and German units provided load-following and primary frequency response. Limits arise from xenon-135 poisoning (after a power reduction xenon builds up for about 8–10 h and can prevent a return to power late in the fuel cycle) and from fuel-cladding stress.

14.6 Generator systems

> **Boundary.**
> Synchronous-machine theory (EMF generation, phasor diagrams, power-angle relations, synchronisation and excitation) is the subject of Chapter 7. This section is about the generator as a power-station component.

Turbo-generators and hydro generators

Every conventional plant ends in a synchronous generator, and the prime mover sets its shape. Steam and gas turbines run fast, so their **turbo-generators** have two (or, for nuclear, four) poles, cylindrical rotors of solid forged steel 1–1.3 m in diameter and long active lengths of up to 8 m. Hydro turbines run slowly, so **hydro generators** have many salient poles on a large-diameter rotor: the 700 MW units of Three Gorges turn at 75 rpm with 80 poles and a rotor about 19 m across. Table 14.3 compares them.

Table 14.3 — Representative large generator data.

Quantity	2-pole turbo-generator	4-pole nuclear generator	Salient-pole hydro generator
Rating	100–1,300 MVA	1,000–2,000 MVA	10–1,000 MVA
Terminal voltage	11–27 kV	20–27 kV	11–24 kV
Speed (50 Hz)	3,000 rpm	1,500 rpm	75–750 rpm
Synchronous reactance X_d	1.8–2.5 pu	1.8–2.2 pu	0.8–1.2 pu
Subtransient reactance X_d''	0.15–0.25 pu	0.25–0.30 pu	0.20–0.30 pu
Inertia constant H	2–5 s (with turbine)	4–7 s	2–4 s
Rated power factor	0.85–0.9 lagging	0.9	0.9–0.95
Full-load efficiency	98.7–99.0 %	~99 %	98–99 %
Cooling	Air to ~300–500 MVA; hydrogen; water in stator bars above ~600 MVA	Hydrogen and water	Air, water in large units

Cooling sets the rating. Losses of about 1 % of a 1,000 MVA machine, some 10 MW, must leave a rotor only a metre across. Hydrogen at 3–5 bar gauge has about seven times the thermal conductivity of air and about one-fourteenth of its density, cutting windage loss and raising heat removal; it needs shaft seals, purity monitoring and purging procedures because hydrogen–air mixtures are explosive between 4 % and 75 %. The largest machines also pump deionised water through hollow stator conductors. The thermal limits that follow are developed in Chapter 9.

Unit connection

A large generator is connected to the grid as a **generator–transformer unit** (Figure 14.1). The terminal current is large: a 776 MVA, 22 kV generator carries

$$I = \frac{776 \times 10^6}{\sqrt{3} \times 22 \times 10^3} = 20.4\ \text{kA}$$

per phase, carried by **isolated-phase bus**, each phase in its own earthed aluminium enclosure so that phase-to-phase faults are almost impossible and magnetic forces on adjacent structures are contained. A **generator circuit breaker** (IEEE C37.013 / IEC 62271-37-013) between the generator and the GSU lets the unit be synchronised and tripped on the low-voltage side while the unit auxiliary transformer stays energised from the grid. The GSU, typically delta on the generator side and earthed star on the HV side, blocks zero-sequence current from the generator and its earthing (Chapter 19).

Excitation, voltage regulation and the capability curve

The field winding is supplied either by a **static exciter** (a thyristor bridge fed from the generator terminals through an excitation transformer, with brushes and slip rings) or by a **brushless exciter** (an inverted AC generator on the same shaft feeding a rotating diode rectifier). Field currents of large machines are 3–6 kA at a few hundred volts. The automatic voltage regulator (AVR) holds terminal voltage; a power-system stabiliser (PSS) adds damping of electromechanical oscillations (Chapter 20); limiters keep the machine inside its **capability curve** (Figure 14.5), the region of the P–Q plane bounded by stator current heating, rotor (field)

current heating at lagging power factor, stator end-region heating and steady-state stability at leading power factor, and the prime mover's maximum and minimum power. Dynamic models of exciters for stability studies are standardised in IEEE 421.5.

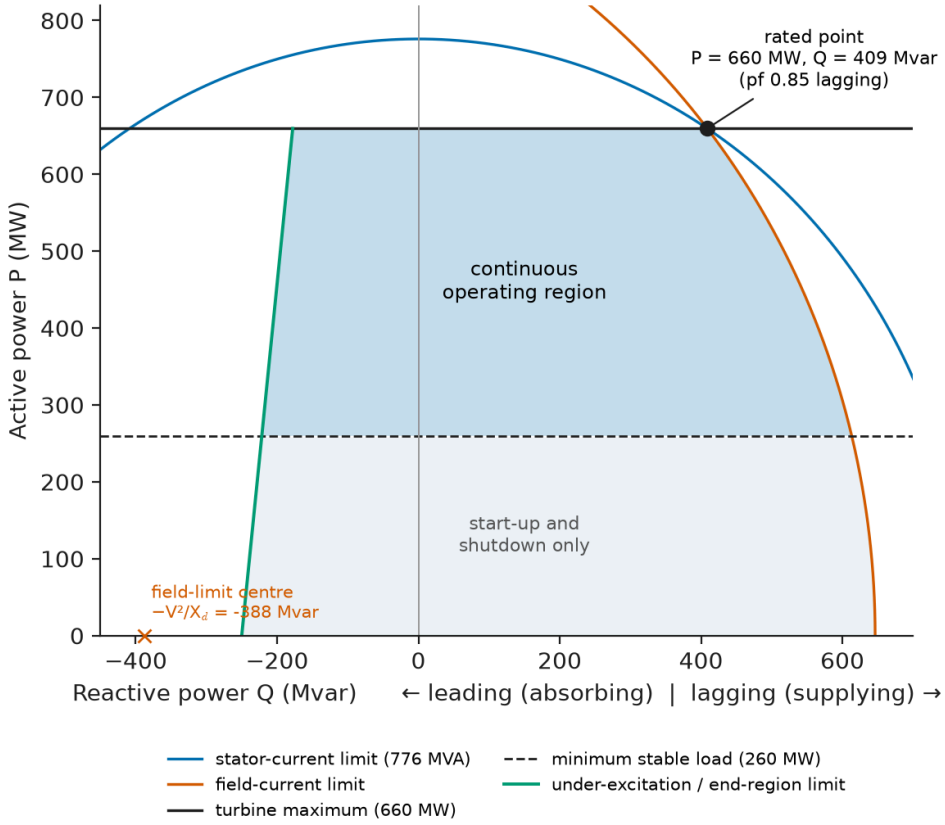


Figure 14.5 — Capability curve of a 776 MVA, 0.85 power-factor, 22 kV turbo-generator.

Governors and droop

The turbine governor adjusts steam valves, gas-turbine fuel or wicket gates. For frequency control each unit is given a **droop** R: the fractional change in frequency that drives its output from zero to rated, typically 4–5 %. A unit with droop R responds to a frequency deviation Δf with

$$\frac{\Delta P}{P_{\text{rated}}} = -\frac{1}{R}\,\frac{\Delta f}{f_0}$$

> **Status of the principle: Proportional (droop) governing**
> *Approximate.* Describes the steady-state response of a governor outside its dead band and within headroom limits; the actual response is delayed by valve, boiler, combustion or water-column dynamics and saturates at maximum output or minimum load.

Worked example — primary response of a 600 MW unit. After a large infeed is lost, a 50 Hz system's frequency settles 0.2 Hz low. A 600 MW unit with 4 % droop increases output by $(0.2/50)/0.04 = 0.10$ pu, 60 MW, provided it has 60 MW of headroom and its boiler can supply it. Units with no headroom (running flat out), units with governors in a wide dead band and inverter-based plant without frequency response contribute nothing, which is why grid codes now require all units to reserve and deliver this response (Chapter 27). Because all droop-controlled units share any change in load in proportion to their ratings, the system's frequency settles at a new value rather than returning to 50 Hz; secondary control restores it (Chapter 20).

14.7 Plant efficiency

Heat rate and heating value

Power-station efficiency is usually expressed as a **heat rate**, the fuel energy consumed per kilowatt-hour generated: $HR = 3600/\eta$ kJ/kWh (or $3412/\eta$ Btu/kWh in US practice). Two conventions coexist. The **higher heating value** (HHV) counts the latent heat of the water vapour formed in combustion; the **lower heating value** (LHV) does not, because no conventional plant condenses its flue gas. HHV/LHV is about 1.11 for natural gas and 1.03–1.05 for coal. European and manufacturer data are normally LHV; US utility data are normally HHV. A CCGT at 58 % LHV is 52.3 % HHV, a heat rate of about 6,530 Btu/kWh (HHV): the same machine, numbers that differ by almost six points. Equally important is **gross** (at the generator terminals) versus **net** (after the plant's own auxiliary consumption, Section 14.8) output.

Table 14.4 — Typical net efficiency and flexibility of conventional plant (LHV).

Technology	Net efficiency	Net heat rate (kJ/kWh)	Hot start	Ramp rate	Minimum stable load
Subcritical coal	33–38 %	9,500–10,900	2–4 h	1–2 %/min	40–50 %
Ultra-supercritical coal	43–47 %	7,700–8,400	2–4 h	2–4 %/min	30–40 %
Open-cycle gas turbine, heavy duty	36–42 %	8,600–10,000	10–20 min	8–15 %/min	20–50 %
Aeroderivative gas turbine	40–44 %	8,200–9,000	5–10 min	20–50 %/min	20–50 %
Reciprocating gas engine	44–49 %	7,350–8,200	2–5 min	20–100 %/min	10–30 % (per engine)
Combined cycle (F to J class)	55–63 %	5,700–6,550	30–60 min	3–8 %/min	30–50 %
Light-water nuclear	32–36 %	10,000–11,250 (thermal)	Hours (typically baseload)	1–5 %/min in load-follow	30–50 %
Large hydro (water to wire)	85–92 %	Not applicable	1–5 min	50–100 %/min	20–40 % (Francis)

Part-load, degradation and ambient effects

Efficiency figures are at full load and design conditions. A real unit spends much of its life elsewhere. Part load lowers efficiency in every thermal plant, most strongly in gas turbines; deposits, blade erosion, seal wear and compressor fouling degrade heat rate by 2–5 % between overhauls (compressor washing recovers part of it); and ambient temperature moves both gas-turbine output (Section 14.2) and condenser pressure (Section 14.1). Plant performance testing to ASME PTC 46 or ISO 2314 corrects measured performance to reference conditions so that guarantees can be checked.

Efficiency in money

The fuel and carbon cost of a megawatt-hour of output is $c = (p_f + p_c\, e_f)/\eta$, where p_f is the fuel price per megawatt-hour of fuel energy, e_f the fuel's emission factor in tonnes of CO2 per megawatt-hour of fuel and p_c the carbon price.

Worked example — merit order with a carbon price. Natural gas costs €30/MWh (LHV) and emits 56.1 kg CO2/GJ, which is 0.202 t/MWh of fuel; coal costs €12/MWh and emits 94.6 kg/GJ, 0.341 t/MWh. The carbon price is €70/t. A 58 % CCGT has a fuel cost of $30/0.58 = 51.7$ €/MWh, emits $0.202/0.58 = 0.348$ t/MWh and pays €24.4/MWh for carbon: €76.1/MWh. A 40 % coal unit has a fuel cost of $12/0.40 = 30.0$ €/MWh, emits 0.851 t/MWh and pays €59.6/MWh for carbon: €89.6/MWh. The gas plant is cheaper to run despite fuel two and a half times as expensive per unit of energy. Setting the two costs equal gives the **fuel-switching carbon price**, $(51.7 - 30.0)/(0.851 - 0.348) = 43$ €/t: below it coal runs first in the merit order, above it gas does. The coal plant could close the gap only by raising its efficiency; at 45 % its cost would fall to €79.6/MWh, still above the CCGT.

14.8 Auxiliary loads

What the station consumes

A power station is also a large electricity consumer. Its pumps, fans, mills, conveyors, cooling-tower fans, emission-control equipment, lighting, HVAC and controls are its **auxiliary load** or **station service**, and its net output is gross output minus this load. Table 14.5 gives typical values.

Table 14.5 — Auxiliary load as a share of gross output (full load, typical).

Plant	Auxiliary load	Largest consumers
Coal unit with FGD and SCR	6–10 %	Motor-driven boiler feed pumps 2.5–3.5 %, ID/FD/PA fans 1.5–2.5 %, CW pumps 0.8–1.2 %, FGD 1–1.5 %, coal mills 0.5–1 %
Lignite unit	8–12 %	As coal, plus fuel handling and drying
Combined cycle	1.5–3 %	Boiler feed pumps, CW pumps or air-cooled condenser fans, gas compressors
Open-cycle gas turbine	0.5–1.5 %	Lube oil, cooling, fuel-gas compressor if needed
PWR nuclear	4–7 %	Reactor coolant pumps (four of 6–9 MW), condensate and CW pumps (main feed pumps often turbine-driven)
Hydro	0.3–1 %	Cooling water, governors, drainage, lighting

The auxiliary electrical system

Figure 14.6 shows a typical arrangement. The **unit auxiliary transformer** (UAT), tapped off the generator bus between generator breaker and GSU, feeds the unit's medium-voltage switchboards at 6.6, 10 or 11 kV when the unit runs. A **station service transformer** (or start-up transformer) fed from the grid supplies the same boards for start-up and when the unit is off, with automatic transfer between sources. Large motors (boiler feed pumps of 8–20 MW, fans of 2–8 MW, CW pumps of 2–5 MW, reactor coolant pumps) connect at

MV; smaller drives at 400–690 V through motor control centres. Essential loads (turning gear, lube-oil pumps, control systems, emergency lighting) are backed by diesel generators and by 110–250 V DC batteries, so that a turbine tripped with loss of all AC can coast down without wiping its bearings.

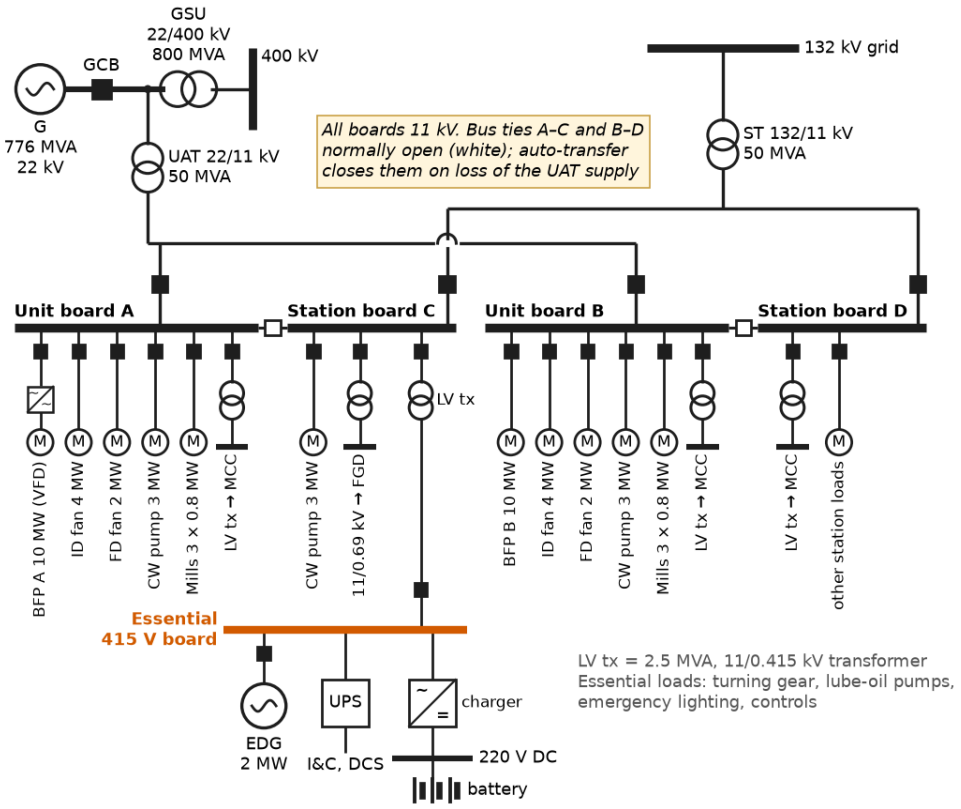


Figure 14.6 — Auxiliary power system of a 660 MW unit with unit and station transformers, MV boards and emergency supplies.

Starting large auxiliaries

Starting a 10 MW boiler feed pump direct-on-line draws about six times rated current at low power factor. With a motor rating of $10/(0.97 \times 0.9) = 11.5$ MVA, the starting demand is about 69 MVA. On an 11 kV board with a fault level of 25 kA (476 MVA), the voltage dip is roughly $69/(476 + 69) = 13$ %, enough to drop out contactors and disturb other drives on the board. The usual remedies are a variable-frequency drive, a soft starter, or a starting sequence that brings the biggest motors on first; motor starting is treated in Chapters 6 and 10.

Saving auxiliary power

Because auxiliary consumption is a direct deduction from saleable output, it is one of the most rewarding places for efficiency work. Fans and pumps controlled by dampers or throttle valves waste the pressure they destroy; speed control follows the affinity laws of *Engineering Physics*, Ch. 19 §19.7.3, in which shaft power scales with the cube of speed.

Worked example — a variable-speed drive on an induced-draught fan. A 4 MW ID fan runs at 70 % of design flow for 6,000 h a year as the unit cycles. With inlet-damper control it draws about 80 % of rated power, 3.2 MW. With a variable-frequency drive the ideal fan power is $4 \times 0.7^3 = 1.37$ MW, and with a drive efficiency of 96 % the input is 1.43 MW. The saving is $3.2 - 1.43 = 1.77$ MW, or about 10,600 MWh a year: at a

wholesale price of €60/MWh, €640,000 a year from one fan, against a drive costing perhaps €1–1.5 million. (The real saving is somewhat smaller because the system curve has a static component, so the fan speed must stay above 70 % at 70 % flow.)

14.9 Environmental considerations

Air emissions

Combustion produces carbon dioxide, which cannot be removed by end-of-pipe treatment short of capture, and several pollutants which can. Table 14.6 gives representative values and the main controls.

Table 14.6 — Direct emissions per net kWh and controls (typical).

Plant	CO_2 (kg/kWh)	Main pollutants	Controls and removal
Subcritical coal	0.90–1.0	SO_2, NOx, particulates, mercury	FGD (95–99 % SO_2), SCR (80–90 % NOx), ESP or fabric filter (> 99.5 % dust)
Ultra-supercritical coal	0.75–0.85	As above	As above
Lignite	1.0–1.2	As coal, more SO_2 and dust	As above
Combined cycle	0.33–0.40	NOx	Dry low-NOx combustors, SCR
Open-cycle gas turbine	0.48–0.60	NOx	Dry low-NOx combustors, water injection
Diesel genset	0.65–0.75	NOx, particulates	SCR, particulate filters
Nuclear, hydro	~0 direct	None in operation	Not applicable

Emission limits for large combustion plants are set by law: in the EU, Annex V of the Industrial Emissions Directive (2010/75/EU) sets limits of the order of 150 mg/Nm³ for SO_2 and NOx and 10 mg/Nm³ for dust for new coal units above 300 MW thermal, tightened by the best-available-technique conclusions for large combustion plants; in the US, New Source Performance Standards and the Mercury and Air Toxics Standards. Each control system also costs auxiliary power (Section 14.8) and, in the case of FGD, limestone and gypsum handling.

Carbon capture

Post-combustion capture scrubs CO_2 from flue gas with an amine solvent, which is regenerated with steam bled from the turbine. The energy penalty is large.

Worked example — the energy penalty of carbon capture. The 660 MW unit of Section 14.1 at 42 % efficiency emits 0.811 kg CO_2/kWh. Fitting 90 % capture with solvent regeneration, CO_2 compression and the associated auxiliaries reduces net efficiency by about 9 percentage points to 33 %. The fuel needed per net kilowatt-hour rises by $0.42/0.33 - 1 = 27$ %, so the CO_2 produced per kilowatt-hour rises to 1.03 kg, of which 10 % is released: 0.103 kg/kWh, an 87 % reduction rather than 90 %. For the same coal input the unit's net

output falls from 660 MW to about 519 MW. The penalty is lower on CCGTs (about 7 percentage points) because their flue gas is a smaller stream per kilowatt-hour, though more dilute in CO_2.

Water

Thermal plants are among the largest users of water. Once-through cooling withdraws 75,000–190,000 litres per megawatt-hour and returns it 8–12 K warmer, which is regulated to protect river and coastal ecosystems and forces plants to reduce output during heat waves when rivers are warm and low (as French nuclear plants on the Rhône and Garonne have done repeatedly). Closed-loop cooling with towers withdraws much less but consumes, by evaporation, roughly 2,000–2,700 L/MWh for coal and nuclear plants and 700–1,000 L/MWh for combined cycles (less steam per kilowatt-hour). Dry air-cooled condensers consume almost no water at a cost of 1–2 points of efficiency and higher auxiliary power.

Solid waste, nuclear waste and land

A coal plant produces fly ash, bottom ash and FGD gypsum, together about 10–15 % of the coal mass; much is used in cement and plasterboard, the rest landfilled. A nuclear plant produces about 20 t of spent fuel per gigawatt-year (the worked example of Section 14.5), which is highly radioactive and must be cooled in pools for years, then stored in dry casks and eventually emplaced in a deep geological repository; Finland's Onkalo is the first in the world to approach operation. Hydro reservoirs flood land, displace people, block fish migration and trap sediment, and in warm climates the decay of flooded vegetation releases methane, which can make the life-cycle emissions of some tropical reservoirs comparable with those of gas plants.

Life-cycle emissions

Direct stack emissions miss fuel extraction, transport, methane leakage, construction and decommissioning. Life-cycle assessments summarised by the IPCC (AR5, Annex III) give median values of roughly 820 g CO_2-eq/kWh for coal, 490 g for combined-cycle gas, 24 g for hydro and 12 g for nuclear, with wide ranges for each. These numbers, together with the costs and flexibility of Table 14.4, are the starting point for the comparison with renewable sources in Chapter 15.

What remains open

Open question	Where it is answered
How are the synchronous generator's EMF, reactances, power-angle curve and excitation analysed?	Chapter 7
How are generator losses, cooling and insulation designed to a thermal rating?	Chapter 9
How are wind, solar, small hydro, biomass and geothermal plant built and rated?	Chapter 15
How does pumped storage use reversible hydro units, and how does storage compete with peaking plant?	Chapter 16
How are generators represented in network models and in per unit?	Chapter 17
How much fault current does a generator deliver and what breaker duty results?	Chapter 19
How do inertia, governors, AVRs and PSSs determine frequency and rotor-angle stability?	Chapter 20
How is a generator and its transformer protected?	Chapter 23
What do grid codes require of generating units, and how do inverter-based plants provide the same services?	Chapter 27
Where do the thermodynamic cycles and fission physics come from?	*Engineering Physics*, Ch. 10, Ch. 19 and Ch. 6

Next: Chapter 15 turns from fuels to flows of energy that cannot be stored at the source, sunlight, wind, water, biomass and the Earth's heat, and asks how much of each can be converted, at what capacity factor and with what predictability.

Chapter 15 — Renewable Energy Systems

Part IV · Power Generation

A 50 MW solar plant in a semi-arid region finishes its first year of operation 6 % below the energy its lenders were promised. The modules are fine: flash tests on a sample match their datasheets. The shortfall is a stack of small things. The year was slightly less sunny than the long-term average; the satellite irradiance data used in the yield study were a few percent high for that site; dust storms soiled the modules more than the 2 % allowed for; the inverters clipped more noon energy than modelled because the DC array was oversized; and the grid operator curtailed output on 40 spring afternoons when the regional network was full. None of these is a failure of physics; all of them are failures to quantify a resource that varies from second to second and from year to year, and to follow it through every conversion step to the meter. Wind farms, small hydro schemes, biomass plants and geothermal stations each have their own version of the same problem. This chapter takes the five main renewable technologies from resource to grid connection, and then asks the three questions every investor and grid operator asks of them: how much energy, at what capacity factor, and how predictably.

> **How are sunlight, wind, flowing water, biomass and the Earth's heat converted into grid electricity, and where are the losses in each chain?**
> **How is a renewable resource measured and turned into a bankable energy estimate, and how do capacity factor, variability and forecast error shape the role of these plants in the power system?**

Concept	Engineering destination
Solar photovoltaic generation	Module and string design, inverter sizing, rooftop and utility PV plants
Wind generation	Turbine selection, generator topologies, wind-farm layout and grid interface
Hydroelectric systems	Small and run-of-river schemes, design-flow selection, mini-grids
Biomass generation	Sugar-mill and wood-fired CHP, biogas engines, co-firing
Geothermal energy	Flash and binary plants, brine management, baseload renewables
Renewable-resource assessment	Yield studies, P50 and P90 estimates, project finance
Energy conversion	Loss chains, performance ratio, inverter-based versus synchronous interfaces
Capacity factor	Plant comparison, capacity credit, replacement of thermal plant
Variability and forecasting	Reserves, ramp limits, storage sizing, market bidding

Roadmap. Photovoltaics (15.1) → wind (15.2) → small and run-of-river hydro (15.3) → biomass (15.4) → geothermal (15.5) → how the resource is measured and turned into an energy estimate (15.6) → the conversion chains compared (15.7) → capacity factor and capacity credit (15.8) → variability and its forecasting (15.9).

15.1 Solar photovoltaic generation

> **Boundary.**
> Band gaps of the common semiconductors are in *Engineering Physics*, Ch. 6 §6.6; the p-n junction, the Shockley diode equation, the illuminated cell $I = I_0(e^{eV/k_BT} - 1) - I_{SC}$, its open-circuit voltage and the Shockley–Queisser limit of about 33 % are in Ch. 14 §14.5.4–14.5.5; the blackbody spectrum behind that limit is in Ch. 10 §10.8.4. This section starts from the single cell and builds the plant.

From cell to module

A single crystalline-silicon cell gives about 0.6–0.7 V at open circuit and, in today's 182 mm or 210 mm wafer formats, 13–18 A at short circuit. A **module** connects 60, 72 or (as half-cut cells) 120, 132 or 144 cells in series, laminated between glass and a polymer or glass back sheet, with **bypass diodes** across groups of 20–24 cells. Ratings are given at **standard test conditions** (STC: irradiance 1,000 W/m², cell temperature 25 °C, AM1.5 spectrum; IEC 60904-3), and design and type approval follow IEC 61215 and IEC 61730. Table 15.1 gives representative commercial values.

Table 15.1 — Representative PV module technologies (commercial, typical).

Technology	Module efficiency	Power temperature coefficient	Notes
Mono-Si PERC	21–22 %	−0.34 to −0.37 %/K	Mainstream until about 2023
Mono-Si TOPCon	22–23.5 %	−0.29 to −0.32 %/K	Current mainstream, often bifacial
Silicon heterojunction (HJT)	22.5–24 %	−0.24 to −0.26 %/K	Best hot-climate performance
Back-contact silicon	23–24.5 %	−0.26 to −0.30 %/K	Premium rooftop and utility
Cadmium telluride thin film	19–20 %	−0.28 to −0.32 %/K	Utility scale, good in diffuse light and heat
Perovskite–silicon tandem	Lab cells above 33 %	Under evaluation	Pre-commercial; stability is the open issue

A typical 144-half-cell TOPCon module rated 550 W has, at STC, $V_{OC} = 49.8$ V, $I_{SC} = 14.0$ A, $V_{mp} = 41.9$ V and $I_{mp} = 13.1$ A, on a 2.58 m² area (21.3 %).

Irradiance and temperature

Two environmental variables move the operating point (Figure 15.1). Short-circuit current, and therefore power, is nearly proportional to the irradiance G reaching the cells. Voltage falls with cell temperature at roughly −0.27 %/K for V_{OC} and −0.3 to −0.35 %/K for maximum power, the consequence of the exponential temperature dependence of I_0. Cell temperature is estimated from ambient temperature T_a with the nominal operating cell temperature (NOCT, typically 42–46 °C at 800 W/m², 20 °C ambient, 1 m/s wind):

$$T_c \approx T_a + \frac{\mathrm{NOCT} - 20}{800}\,G, \qquad P \approx P_{\mathrm{STC}}\,\frac{G}{1000}\,\left[1 + \gamma\,(T_c - 25)\right]$$

where γ is the power temperature coefficient (negative).

> **Status of the principle: NOCT cell-temperature and linear power model**
> *Empirical.* Fitted to open-rack modules in moderate wind; it underestimates cell temperature for roof-integrated modules with poor rear ventilation, ignores wind speed and thermal lag, and the linear power model loses accuracy below about 200 W/m², where the module's low-irradiance efficiency falls. Yield-modelling software uses more detailed fits (Faiman, Sandia, PVsyst models).

Worked example — module output on a hot afternoon. The 550 W module ($\gamma = -0.35$ %/K, NOCT 45 °C) sees 800 W/m² at an ambient temperature of 35 °C. Its cells run at $T_c = 35 + (45 - 20)/800 \times 800 = 60$ °C, 35 K above STC, and

$$P \approx 550 \times 0.8 \times (1 - 0.0035 \times 35) = 386 \text{ W}$$

The temperature alone costs 12 % of the output. An HJT module with $\gamma = -0.25$ %/K would lose 8.75 %, which is why low temperature coefficients command a premium in hot climates.

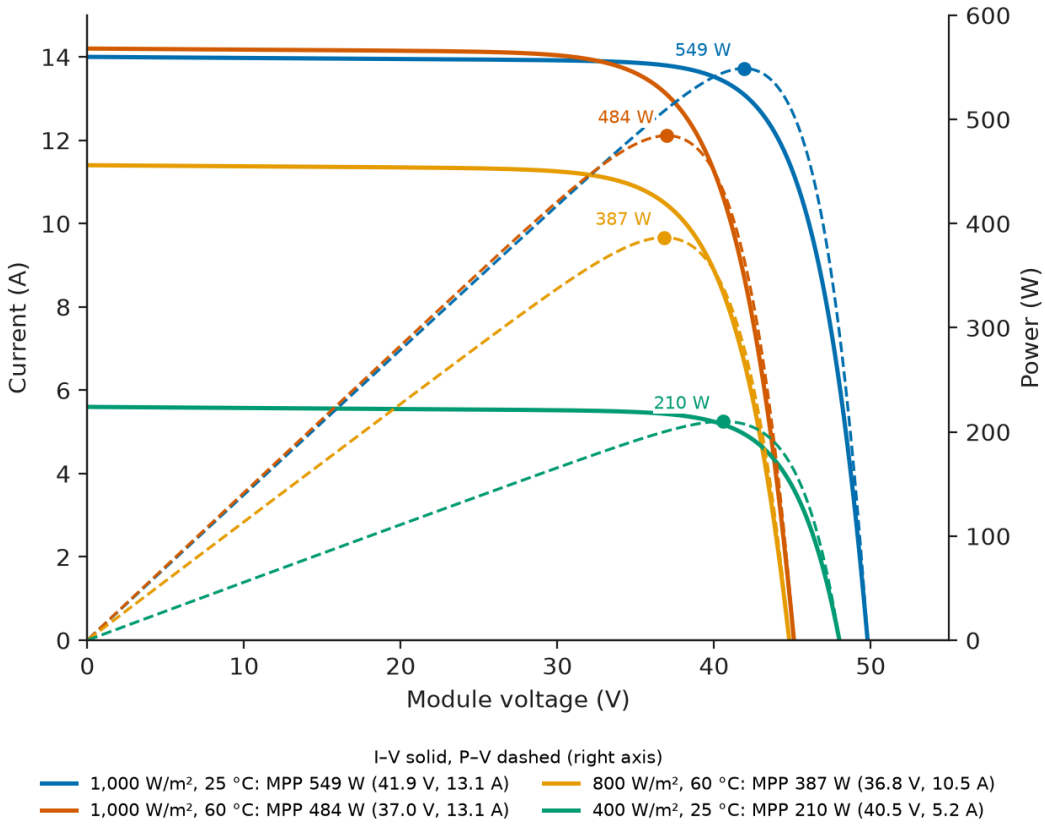


Figure 15.1 — Current-voltage and power-voltage curves of a 550 W module at three irradiances and two cell temperatures.

Strings, arrays and inverters

Modules are connected in series into **strings** to reach the inverter's input voltage, and strings in parallel to reach its power. The upper limit is the maximum system voltage (1,000 V or, for utility plants, 1,500 V DC under IEC 60364-7-712 and the module's own rating), which must not be exceeded even at the lowest expected cell temperature, when V_{OC} is highest.

Worked example — string sizing for a 1,500 V system. The site minimum ambient temperature is −10 °C; at dawn on such a day the cells are at ambient with irradiance enough to bring the string to open circuit. With a V_{OC} temperature coefficient of −0.27 %/K, at 35 K below STC

$$V_{OC,\max} = 49.8 \times (1 + 0.0027 \times 35) = 54.5\ \text{V}, \qquad N_{\max} = \left\lfloor \frac{1500}{54.5} \right\rfloor = 27$$

A 27-module string has a maximum-power voltage of 1,131 V at STC and, with cells at 70 °C on a summer afternoon (V_{mp} falling at −0.35 %/K), $27 \times 41.9 \times (1 - 0.0035 \times 45) = 953$ V, which must stay above the inverter's minimum MPPT voltage. The string power at STC is $27 \times 550 = 14.85$ kW. String fuses and cables are sized on 1.25 I_{SC}, often with a further 1.25 factor (IEC 62548), about 22 A here.

Three inverter architectures dominate: **central inverters** of 2–6 MVA with 1,000–1,500 V DC input for utility plants; **string inverters** of 3–350 kW, each with several MPPT inputs, for commercial roofs and increasingly for utility plants; and **microinverters or optimisers** at module level for shaded residential roofs. Every inverter runs a **maximum power point tracking** (MPPT) algorithm, typically perturb-and-observe or incremental conductance, which adjusts its DC input voltage continuously to hold the array at the knee of its P–V curve. Inverter circuits are the subject of *Electronics & Computing*; their grid behaviour is the subject of Chapter 27.

Partial shading and bypass diodes

Series connection makes a string only as strong as its weakest cell. If one cell is shaded, the string current forces it into reverse bias, where it dissipates power as heat and can form a **hot spot** that damages the module. Bypass diodes conduct around the shaded substring, limiting the loss to that substring, but they create several local maxima in the string's P–V curve (Figure 15.2). A simple MPPT can lock onto a local peak; modern inverters perform periodic global scans. Shading by a chimney or a neighbouring row of modules at low sun angles is the most common avoidable loss in rooftop systems, and inter-row spacing on ground-mounted plants is chosen to keep winter-noon shading acceptable.

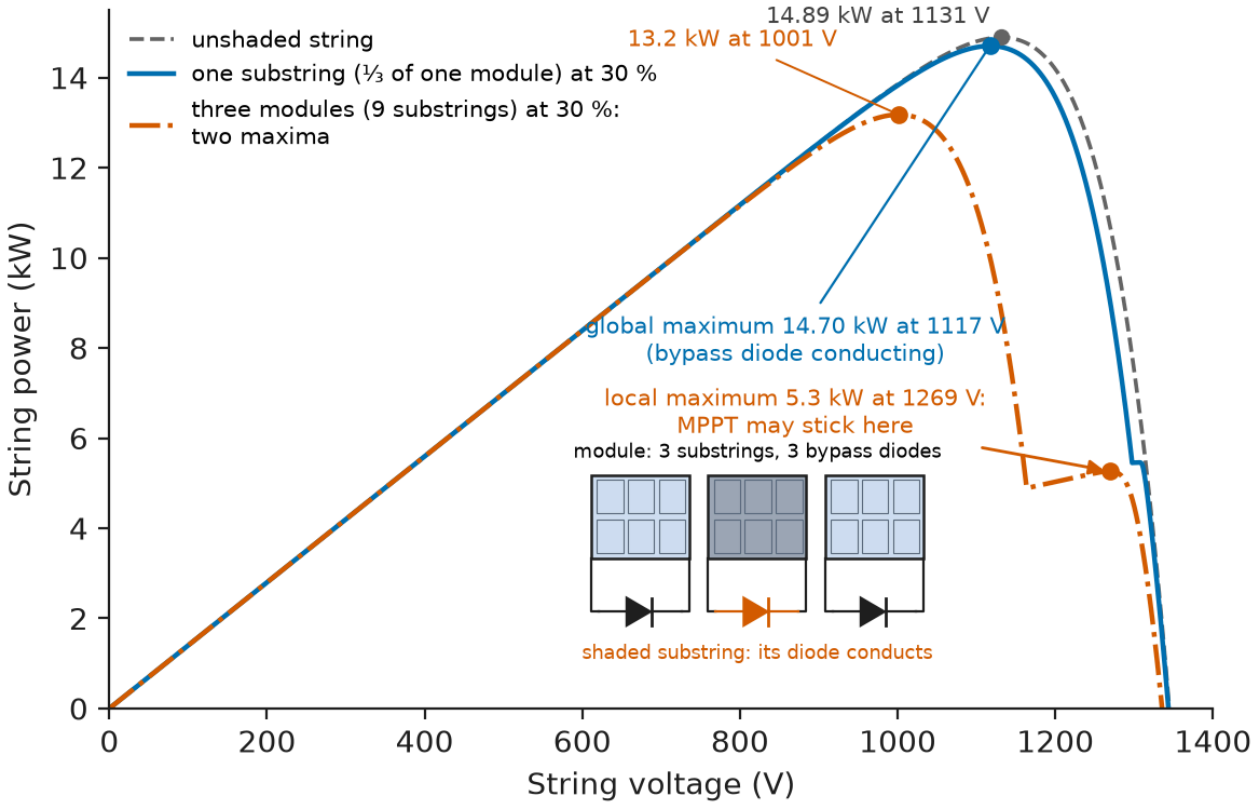


Figure 15.2 — Power-voltage curves of a 27-module string under partial shading, showing how bypass diodes create a local maximum that can trap a simple MPPT.

Plant design choices

Utility plants have converged on a few design decisions:

- **DC/AC ratio** (array STC power over inverter AC rating) of 1.2–1.4. Oversizing the array fills the inverter for more hours at the cost of clipping a few percent of noon energy; inverters are cheaper per watt than ever, but the grid connection is not.
- **Single-axis trackers** that turn rows east to west raise annual yield by 15–25 % at low latitudes for a 5–10 % cost increase, and are standard in sunny regions; **bifacial** modules collect 5–10 % extra from ground-reflected light, more over bright sand or snow.
- **Degradation** of 0.4–0.55 % a year is assumed over a 30-year life; light- and elevated-temperature-induced degradation in the first months adds 1–2 %.
- Grid-connected plants of tens of megawatts collect at 33 kV through inverter transformers; larger ones step up to 132–400 kV.

15.2 Wind generation

> **Boundary.**
> Mass and momentum conservation in a fluid and Bernoulli's equation are in *Engineering Physics*, Ch. 15 §15.4–15.5; turbulence as the regime that has no exact model is in §15.8.2. The rotor-aerodynamic limit below is not in that book and is given here in brief.

Power in the wind and the Betz limit

The kinetic energy carried per second through an area A perpendicular to a wind of speed v and density ρ is

$$P_w = \tfrac{1}{2}\rho A v^3$$

The cube makes wind power acutely sensitive to speed: 10 % more wind gives 33 % more power. A rotor cannot capture all of it, because the air must leave with some speed or it would pile up. Treat the rotor as a thin actuator disc that slows the wind from v far upstream to $v(1-a)$ at the disc and $v(1-2a)$ far downstream, where a is the axial induction factor. Applying mass conservation, the momentum balance and Bernoulli's equation on each side of the disc gives the power extracted as $P = 2\rho A v^3 a(1-a)^2$, a power coefficient $C_p = P/P_w = 4a(1-a)^2$, and a maximum where $dC_p/da = 0$:

$$a = \tfrac{1}{3}, \qquad C_{p,\max} = \frac{16}{27} = 0.593$$

> **Status of the principle: Betz limit**
> *Derived*. Follows from conservation of mass, momentum and energy for an ideal, frictionless, non-rotating actuator disc in an unbounded uniform stream; real rotors fall below it because of wake rotation, finite blade number (tip losses) and profile drag, and reach aerodynamic C_p of about 0.45–0.50.

Rotor control and the power curve

A rotor's power coefficient depends on the **tip-speed ratio** $\lambda = \omega R / v$ and on the blade pitch angle β. Three-bladed rotors reach their maximum C_p at $\lambda \approx 7$–9. A variable-speed turbine therefore runs in four regions, which together make the **power curve** of Figure 15.3:

1. Below the **cut-in** speed (typically 3 m/s) the rotor idles.
2. Between cut-in and **rated** speed (10–13 m/s), the controller varies rotor speed in proportion to wind speed to hold λ at its optimum and C_p at its maximum, so power rises roughly as v^3.
3. Between rated and **cut-out** speed (typically 25 m/s), the blades are pitched to shed excess power and hold rated output; generator torque and rotor speed are held near their limits.
4. Above cut-out the turbine feathers its blades and stops, to protect the structure; many modern turbines ramp down gradually between 25 and 30 m/s instead.

Rotor tip speed is limited to about 80–90 m/s onshore (noise) and around 100 m/s offshore (leading-edge erosion), so large rotors turn slowly. A 236 m rotor at $\lambda = 8$ in a 10 m/s wind turns at $\omega = 8 \times 10/118 = 0.68$ rad/s, 6.5 rpm, with an 80 m/s tip speed.

Worked example — a 15 MW offshore turbine. A turbine with a 236 m rotor ($A = 43{,}740$ m²) in air of density 1.225 kg/m³ sees, at 10 m/s, a wind power of $\tfrac{1}{2} \times 1.225 \times 43{,}740 \times 10^3 = 26.8$ MW, of which the Betz limit allows 15.9 MW. At its rated speed of 11 m/s the wind power is 35.7 MW; delivering 15 MW through a drivetrain and converter with a combined efficiency of 94 % needs an aerodynamic C_p of $15/(0.94 \times 35.7) = 0.45$, close to the practical maximum. Now place it at a site where the hub-height wind follows a Rayleigh distribution (a Weibull distribution with shape factor $k = 2$) of mean 9.5 m/s, and idealise its power curve as $P = 15\,(v^3 - 3^3)/(11^3 - 3^3)$ MW between 3 and 11 m/s, 15 MW from 11 to 25 m/s and zero otherwise. Integrating the power curve against the distribution numerically gives a mean output of 8.10 MW, a gross capacity factor of 54 % and 71.0 GWh a year. The wind is above rated speed 35 % of the time and below cut-in 7.5 %. After 10 % wake losses within the farm and 3 % for availability and electrical losses, the net figures are 47 % and 62 GWh. The same turbine at sites with mean speeds of 7.5 and 6.5 m/s would reach gross capacity factors of 39 % and 30 %.

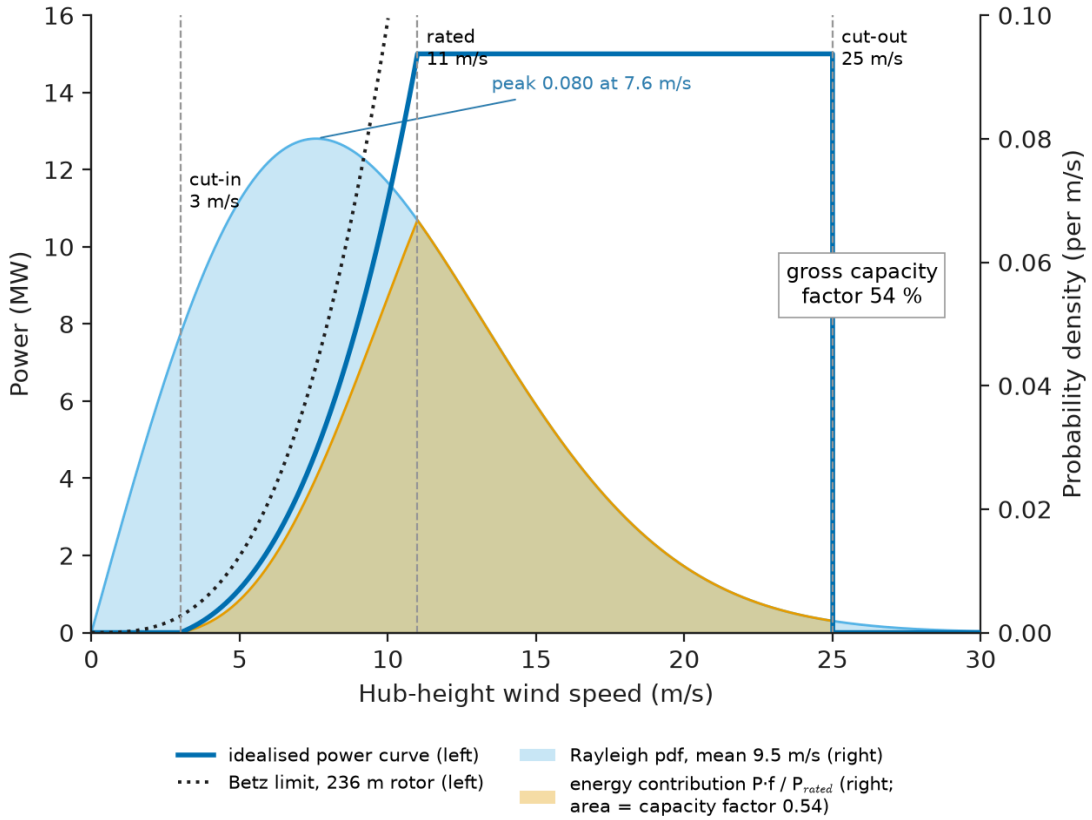


Figure 15.3 — Power curve of a 15 MW turbine with the Rayleigh wind-speed distribution of a 9.5 m/s site and the resulting energy distribution.

The wind resource near the ground

Wind speed increases with height above ground because of surface friction. Over open terrain the profile is often approximated by the power law $v(h) = v_{\text{ref}}(h/h_{\text{ref}})^{\alpha}$ with $\alpha \approx 0.14$ (the "one-seventh law") over flat open land, 0.10–0.12 offshore and 0.2–0.3 over forest or in stable night-time conditions. Raising a hub from 100 m to 150 m with $\alpha = 0.14$ gains $1.5^{0.14} = 1.058$ in speed and $1.5^{0.42} = 1.19$ in power, which is why onshore hub heights have risen from 80 m to 120–170 m. Turbines are designed for a wind class under IEC 61400-1: classes I, II and III correspond to reference 10-min extreme speeds of 50, 42.5 and 37.5 m/s (annual means up to 10, 8.5 and 7.5 m/s), with turbulence categories A, B and C. A low-wind class III turbine has a larger rotor for the same generator.

Turbines in a farm shade each other. A wake recovers over 5–10 rotor diameters; typical spacing is 5–9 diameters in the prevailing wind direction and 3–5 across it, and wake losses are 5–15 % of the energy of an isolated turbine, more in large offshore arrays.

Generator topologies

The electrical side of a wind turbine has gone through four generations, classified in IEEE and IEC grid-integration work as Types 1 to 4 (Figure 15.4).

Table 15.2 — Wind-turbine generator types.

Type	Generator and converter	Speed range	Grid behaviour	Status
1	Squirrel-cage induction generator, directly connected, capacitor bank	Fixed (about 1 % slip)	Absorbs reactive power; power fluctuates with gusts; large fault current	Legacy, small units
2	Wound-rotor induction generator with variable rotor resistance	About 10 % above synchronous	As Type 1, softer torque	Legacy
3	Doubly-fed induction generator (DFIG), rotor fed by back-to-back converter of 25–35 % rating	About ±30 %	Controllable P and Q; stator directly on grid; fault ride-through needs a crowbar or chopper	Common onshore up to ~6 MW
4	Synchronous (often permanent-magnet) or induction generator behind a full-rating converter, geared or direct-drive	Full range	Fully decoupled from grid; behaviour set by converter control; fault current limited to 1.1–1.5 pu	Standard offshore and for the largest onshore turbines

The machines themselves (induction machines, PMSMs, specialised generators) are in Chapters 6–8, and the drive-side control of their converters follows the methods of Chapter 12. A Type 4 turbine is electrically an inverter-based resource like a PV plant: it has no natural inertia and no large fault current, which is a central theme of Chapter 27.

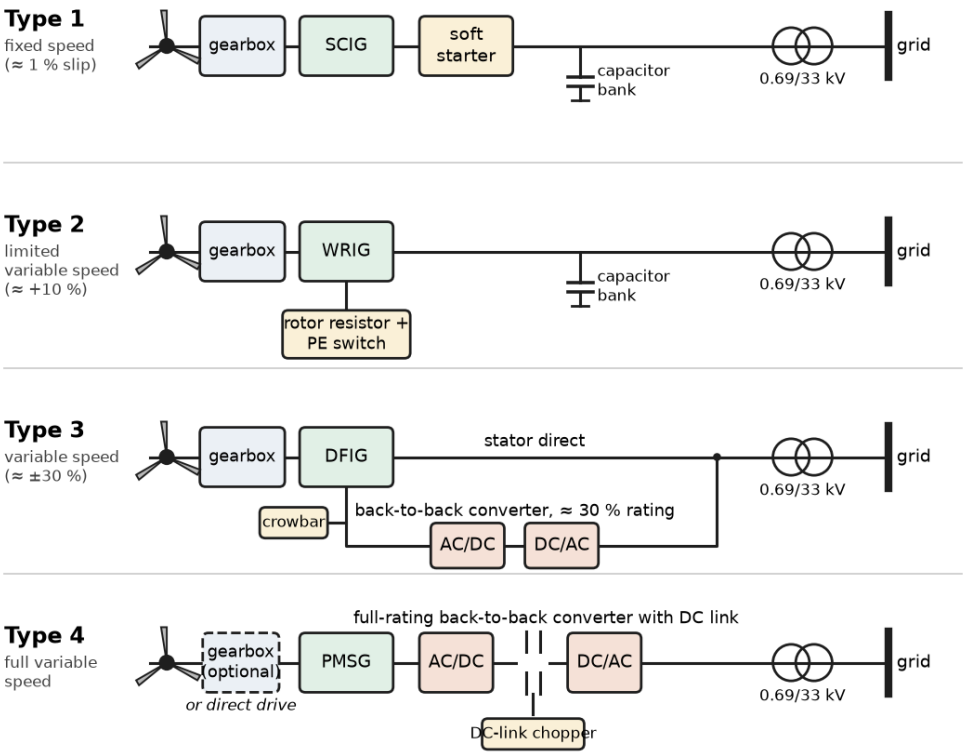


Figure 15.4 — The four wind-turbine generator types from fixed-speed induction to full-converter machines.

15.3 Hydroelectric systems

> **Boundary.**
> The hydro power equation $P = \eta_t \eta_g \rho g Q H_n$, turbine types, water hammer and governing are in Chapter 14 §14.4. This section treats hydro as a variable renewable resource: small and run-of-river schemes without large storage, whose output follows the river.

Small hydro and run-of-river

"Small hydro" is defined by capacity, with thresholds that vary by country from 10 MW to 50 MW; mini (below about 1 MW), micro (below 100 kW) and pico (below 5 kW) schemes serve villages, farms and remote telecommunications sites. Most are **run-of-river**: a low weir diverts part of the river through an intake, a headrace canal or pipe and a penstock to a powerhouse, and returns it downstream. With little or no storage, output follows the river's flow from day to day and season to season, and the key resource description is the **flow-duration curve** (FDC): the flow plotted against the percentage of time it is equalled or exceeded.

Choosing the design flow

The design (rated) flow of the turbine sets both its cost and its energy. A turbine sized for a high flow captures flood flows but runs at part load or stops for much of the year; one sized for a low flow runs full most of the time but spills the rest. Two further constraints enter: an **environmental (residual) flow** must be left in

the river reach between intake and outfall, and each turbine type has a minimum operating flow (about 10–20 % of rated for Pelton and Kaplan, 30–40 % for Francis).

Worked example — sizing a run-of-river scheme. A river at a proposed intake has the flow-duration curve of Figure 15.5: 30 m³/s exceeded 0 % of the time (peak flood), 13 m³/s exceeded 10 %, 6.8 m³/s exceeded 30 %, 4.4 m³/s exceeded 50 %, 2.5 m³/s exceeded 80 % and 1.2 m³/s exceeded 100 % of the time. The net head is 60 m, the overall water-to-wire efficiency 85 %, an environmental flow of 0.5 m³/s must pass the weir, and the turbine shuts down below 25 % of its rated flow. Integrating the power over the FDC numerically for several design flows gives:

Design flow (m³/s)	Rated power (MW)	Annual energy (GWh)	Capacity factor
4.4 (Q_{50})	2.20	14.5	75 %
5.4 (Q_{40})	2.70	16.0	68 %
6.8 (Q_{30})	3.40	17.5	59 %
9.0 (Q_{20})	4.50	18.8	48 %
13.0 (Q_{10})	6.50	19.2	34 %

Rated power is $9.81 \times 0.85 \times 60 \times Q_d$ kW. Raising the design flow from Q_{40} to Q_{20} adds 67 % to the turbine and generator size but only 17 % to the energy; going on to Q_{10} adds almost nothing. A design flow around Q_{25}–Q_{35} is a common optimum, and the choice of two unequal units rather than one extends the low-flow operating range.

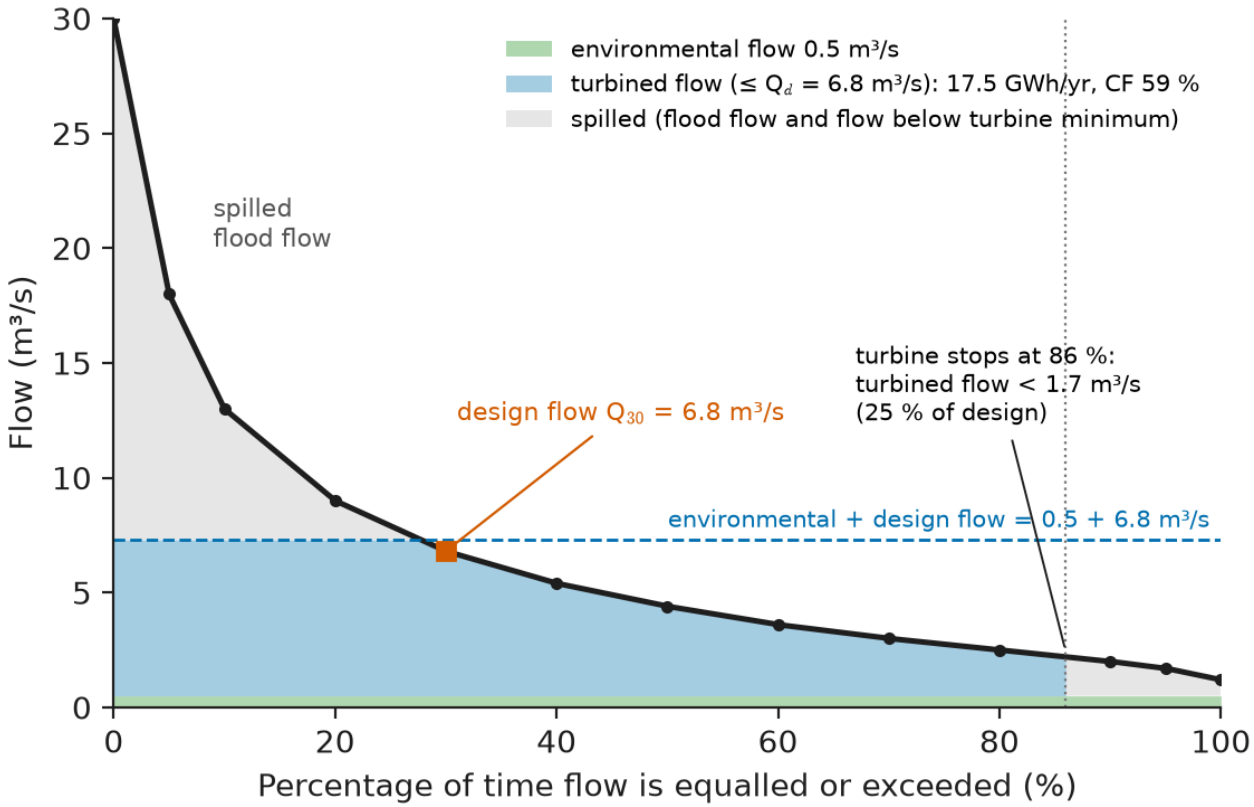


Figure 15.5 — Flow-duration curve of a run-of-river site with design flow, environmental flow and the energy captured.

Electrical side

Small hydro units of up to a few megawatts often use induction generators, which need no synchroniser or exciter and simply run slightly above synchronous speed when connected to the grid; synchronous generators are used where the unit must run islanded or support a weak grid. In isolated mini-grids, where a governor is too expensive, an **electronic load controller** keeps the turbine fully loaded and diverts surplus power to ballast heaters, holding frequency by dumping energy rather than adjusting water. Grid-connected small hydro is classed as distributed generation and connects under the distribution rules of Chapter 22.

15.4 Biomass generation

> **Boundary.**
> The Rankine cycle is in *Engineering Physics*, Ch. 19 §19.5.2; its realisation in a utility steam plant is in Chapter 14 §14.1. Biomass plants are small steam plants with an awkward fuel.

Fuels

Biomass for power is mainly wood (forestry residues, chips, pellets), agricultural residues (sugar-cane bagasse, rice husk, straw, palm-oil residues), energy crops and wet wastes converted to biogas. The decisive property is moisture, because every kilogram of water in the fuel must be evaporated in the furnace and leaves up the stack. A useful approximation for the lower heating value of wet fuel with moisture fraction w (wet basis) is

$$\mathrm{LHV}_{\mathrm{wet}} \approx \mathrm{LHV}_{\mathrm{dry}}\,(1 - w) - 2.44\,w \quad \mathrm{MJ/kg}$$

where 2.44 MJ/kg is the latent heat of water at 25 °C.

Table 15.3 — Representative biomass fuel properties.

Fuel	Moisture (wet basis)	LHV as received (MJ/kg)	Ash (dry)
Wood pellets	6–10 %	16.5–17.5	0.5–1 %
Fresh wood chips	40–55 %	7–10	1–3 %
Sugar-cane bagasse	48–52 %	7.5–8.5	2–4 %
Rice husk	8–12 %	13–15	15–20 %
Cereal straw (baled)	12–18 %	13–15	5–8 %
Biogas (60 % CH_4)	Not applicable	about 21.5 MJ/m³	Not applicable

Conversion routes and efficiencies

- **Dedicated steam plants** of 5–50 MW burn chips, straw or residues on grates or in fluidised beds. Their steam conditions are modest (4–10 MPa, 450–540 °C) because alkali chlorides in the ash corrode superheaters, so net electrical efficiencies are 20–32 % (Table 14.1). Large converted coal units burning pellets (Drax in the UK) reach 36–38 %.
- **Co-firing** a few to 20 % biomass in a coal unit uses the coal plant's efficiency and emission controls at little extra cost.

- **Combined heat and power** is the natural use of biomass in industry: sugar mills, paper mills and sawmills have both residues and a heat demand. Modern high-pressure sugar-mill cogeneration exports 100–130 kWh per tonne of cane crushed.
- **Biogas** from anaerobic digestion of manure, food waste or sewage sludge, and **landfill gas**, are burned in reciprocating gas engines of 0.5–4 MW at 38–42 % electrical efficiency, often with heat recovery.
- **Gasification** converts solid biomass to a combustible gas for an engine or turbine; it promises higher efficiency at small scale but has struggled with tar removal.

Worked example — bagasse for a sugar-mill power plant. A mill plans a cogeneration plant with 25 MW net electrical output at a net electrical efficiency of 24 % (the rest of the steam serves the mill's process). Bagasse at 50 % moisture with a dry LHV of 18.5 MJ/kg has

$$\mathrm{LHV}_{\mathrm{wet}} = 18.5 \times 0.5 - 2.44 \times 0.5 = 8.03\ \mathrm{MJ/kg}$$

The fuel power is $25/0.24 = 104$ MW, so the plant burns $104/8.03 = 13.0$ kg/s, 46.7 t/h. Over a 200-day crushing season at 24 h a day that is about 224,000 t of bagasse; at a bagasse yield of 30 % of cane mass, the mill must crush about 750,000 t of cane to fuel the plant. Outside the season the plant stops or burns stored or purchased fuel, so its annual capacity factor is often only 50–60 %, and a mill that wants year-round export must stockpile bagasse or buy wood chips.

Is biomass carbon-neutral?

The CO_2 released by burning biomass was absorbed from the atmosphere as the plant grew, so its stack emissions are conventionally counted as zero in national energy statistics. The real climate effect depends on what would have happened otherwise. Residues that would decay within a few years anyway (bagasse, straw, sawdust) come close to neutral; harvesting whole trees from slow-growing forests creates a carbon debt that may take decades to repay. Fuel collection, drying, pelletising and long-distance shipping add 5–30 % of the energy content. Combined with carbon capture (BECCS), biomass is one of the few routes to net removal of CO_2, at the energy penalty discussed in Chapter 14 §14.9.

15.5 Geothermal energy

> **Boundary.**
> The Carnot efficiency $\eta_C = 1 - T_c/T_h$ is in *Engineering Physics*, Ch. 10 §10.6. Geothermal plants are the clearest case in this book of a heat engine running close to that limit's floor, at low source temperature.

Resources and plant types

The Earth's crust warms by about 25–30 K per kilometre of depth on average, much more in volcanic regions and along plate boundaries, where water-filled reservoirs at 150–350 °C lie 1–3 km down. Power plants tap them through production wells and return the cooled fluid through reinjection wells. Four plant types match four resource types:

- **Dry steam** plants pass steam from the reservoir directly to a turbine (The Geysers, California; Larderello, Italy). Rare resources, simplest plant.
- **Flash** plants take a high-pressure hot-water or two-phase fluid, flash part of it to steam in a separator at reduced pressure, and send the steam to a condensing turbine; double-flash plants flash the remaining

brine again at lower pressure for 15–25 % more output. Most of the world's geothermal capacity (Indonesia, the Philippines, Kenya's Olkaria, Iceland, New Zealand) is flash.

- **Binary** plants pass moderate-temperature brine (100–180 °C) through a heat exchanger that boils a low-boiling organic working fluid (isopentane, n-pentane, R245fa) in an organic Rankine cycle (ORC). The brine never touches the turbine and is fully reinjected, so emissions of non-condensable gases are near zero.
- **Enhanced geothermal systems** (EGS) create permeability in hot dry rock by hydraulic stimulation and circulate water through it; recent projects using horizontal drilling from the oil-and-gas industry are moving them toward commercial scale.

Why the efficiency is low

The resource is hot water, not a fuel, and it gives up its heat over a falling temperature range. The maximum work obtainable from brine cooled from T_h to T_r with a heat sink at T_0 is its exergy, $\dot{m} c_p \left[(T_h - T_r) - T_0 \ln(T_h/T_r) \right]$, well below the Carnot fraction of the heat because only the first part of the heat is delivered at T_h.

Worked example — a binary geothermal plant. A well field delivers 100 kg/s of brine ($c_p = 4.25$ kJ/(kg·K)) at 160 °C; to avoid silica scaling in the reinjection wells the brine may not be cooled below 70 °C; the air-cooled condenser rejects heat at an effective 30 °C. The heat available is

$$\dot{Q} = 100 \times 4.25 \times (160 - 70) = 38.3 \ \text{MW}_{\text{th}}$$

The Carnot efficiency between 160 °C and 30 °C would be 30 %, but the exergy of the brine over this range is only

$$\dot{W}_{\max} = 100 \times 4.25 \left[90 - 303.15 \ln \frac{433.15}{343.15} \right] = 8.24 \ \text{MW}$$

or 21.5 % of the heat. A real ORC converts about 12 % of the heat to gross power, 4.6 MW. The feed pumps of the organic cycle, the condenser fans and the brine production and reinjection pumps take 20–30 % of gross output in binary plants; at 25 % the net output is 3.4 MW, a net thermal efficiency of 9 % but a second-law (exergy) efficiency of 42 %. The low thermal efficiency is not a sign of a poor machine; it is the thermodynamics of a low-temperature source.

Characteristics

Geothermal plants run around the clock with capacity factors of 75–95 %, which makes them one of the few renewable sources that can serve baseload. Their capital cost is dominated by drilling: wells cost several million dollars each, and exploration risk (dry or low-productivity wells) is the main barrier to development. Reservoirs decline if fluid is extracted faster than heat is replenished, so field management with make-up wells and reinjection is a continuous task. Flash plants release dissolved CO_2 and H_2S from the reservoir fluid; emissions vary from near zero to a few hundred grams of CO_2 per kilowatt-hour depending on the field, and H_2S abatement is mandatory in most jurisdictions. Reinjection can trigger small earthquakes, a particular concern for EGS. The generators are conventional synchronous machines, so geothermal plants contribute inertia and fault current like thermal plant.

15.6 Renewable-resource assessment

Why assessment matters

A renewable plant's costs are almost entirely fixed at financial close; its revenue depends on a resource that varies and can only be estimated. A 1 % error in mean wind speed is a 2–3 % error in energy at a typical site. Resource assessment is therefore a measurement and statistics discipline, and its products, the **P50** and **P90** energy estimates, decide how much debt a project can carry.

Solar resource

Solar resource is described by three irradiance components: **global horizontal irradiance** (GHI), **direct normal irradiance** (DNI) and **diffuse horizontal irradiance** (DHI). Long-term estimates come from satellite-derived datasets with 10–30 years of hourly or sub-hourly data, summarised as a typical meteorological year (TMY) for simulation. A transposition model converts GHI and DHI to the **plane-of-array** (POA) irradiance on tilted or tracking modules. Annual GHI ranges from about 1,000 kWh/m² in northern Europe through 1,600–1,900 in southern Europe, India and the southern US to 2,200–2,500 in the deserts of Chile, Namibia, the Middle East and Australia. Satellite data carry 3–6 % uncertainty in annual GHI; a year of on-site measurement with secondary-standard pyranometers (IEC 61724-1 Class A monitoring) reduces it through site adaptation. Interannual variability of annual GHI is 2–5 %.

Wind resource

Wind assessment needs on-site measurement because wind speed depends strongly on local terrain, roughness and height. A campaign of at least one year uses a met mast with calibrated cup anemometers and vanes at several heights, or a ground-based lidar, ideally both. The short record is related to a long-term reference (a nearby station or a reanalysis dataset) by **measure–correlate–predict** to estimate the long-term mean; the shear profile extrapolates it to hub height; a flow model (linear models for simple terrain, computational fluid dynamics for complex terrain) spreads it across the turbine positions; and the measured frequency distribution, often fitted with a Weibull distribution of shape factor $k = 1.8$–2.5, is combined with each turbine's power curve, corrected for site air density, and with a wake model. Interannual variability of annual wind energy is typically 4–8 %, larger than for solar.

Hydro, biomass and geothermal

Hydro assessment relies on long river-gauge records, synthesised for ungauged catchments from rainfall and catchment area; climate change makes the past an imperfect guide. Biomass assessment is a supply-chain study: residue availability, competing uses and transport distance. Geothermal assessment starts with surface geology and geochemistry and ends with test wells and flow tests; its uncertainty is the largest and is resolved only by drilling.

From P50 to P90

The **P50** is the annual energy that is expected to be exceeded with 50 % probability; the **P90** is exceeded with 90 % probability. Independent uncertainties (expressed as standard deviations in percent of energy) are combined in quadrature, and with a normal distribution

$$P_{90} = P_{50}\,(1 - 1.282\,\sigma)$$

where σ is the combined relative standard deviation. Interannual variability averages out over a debt term, so a ten-year P90 divides that component by $\sqrt{10}$.

> **Status of the principle: Normal uncertainty model for energy yield**
> *Approximate*. Assumes independent, normally distributed error components; it understates the tails when errors are correlated or skewed (curtailment, long outages, multi-year climate anomalies) and says nothing about systematic bias in the P50 itself, which industry back-casting studies have shown to be several percent optimistic for many portfolios.

Worked example — P90 for the 50 MW solar plant. The opening plant's yield study gave a P50 of 90 GWh/yr with uncertainties of 4.0 % (interannual variability of irradiance), 4.5 % (long-term irradiance dataset), 2.0 % (transposition model), 3.0 % (simulation and loss model) and 2.0 % (soiling and availability assumptions). Combined in quadrature, $\sigma_1 = \sqrt{4.0^2 + 4.5^2 + 2.0^2 + 3.0^2 + 2.0^2} = 7.3\ \%$, so the one-year P90 is $90 \times (1 - 1.282 \times 0.073) = 81.6$ GWh. Over ten years the interannual term falls to $4.0/\sqrt{10} = 1.26\ \%$, $\sigma_{10} = 6.2\ \%$ and the ten-year P90 is 82.8 GWh. A first-year yield 6 % below P50, 84.6 GWh, lies well inside the one-year distribution: unlucky, not a design failure, which is why lenders size debt on the P90 and not the P50.

15.7 Energy conversion

The loss chains compared

Each renewable technology converts its resource through a chain of steps, each with a loss. For PV the chain from plane-of-array irradiance to AC energy at the grid connection is summarised by the **performance ratio** (PR), the ratio of actual AC yield to the yield a lossless system would give at STC efficiency (IEC 61724-1). A typical loss budget for a well-designed utility plant is soiling 2 %, temperature 6 %, mismatch and light-induced degradation 2 %, DC cabling 1.5 %, low-irradiance and spectral effects 1.5 %, inverter 2 %, MV transformer 1 %, AC cabling 0.5 % and availability 1 %. Multiplying the remaining fractions,

$$PR = 0.98 \times 0.94 \times 0.98 \times 0.985 \times 0.985 \times 0.98 \times 0.99 \times 0.995 \times 0.99 = 0.837$$

Worked example — specific yield and capacity factor of a tracker plant. A single-axis-tracked plant in a sunny region receives 2,100 kWh/m² a year in the plane of array. With PR = 0.837 the specific yield is $2100 \times 0.837 = 1{,}758$ kWh per kWp of STC array power. The capacity factor on the DC rating is $1758/8760 = 20.1\ \%$; on the AC rating, with a DC/AC ratio of 1.25 and clipping already counted in the losses, it is about 25 %. The 50 MWp plant of the opening, with 40 MW AC, would deliver about 88 GWh a year at this site.

Table 15.4 — Conversion chains and typical overall efficiencies.

Technology	Conversion chain	Overall efficiency (resource to grid)	Grid interface
Crystalline-Si PV	Photons → electron–hole pairs → DC → inverter → AC	17–20 % of incident sunlight (module 21–24 % × PR 0.80–0.87)	Inverter (grid-following, increasingly grid-forming)
Wind (Type 4)	Kinetic energy → rotor (Cp 0.45–0.50) → gearbox or direct drive → generator → converter → AC	40–45 % of kinetic energy through the rotor disc at below-rated speeds	Full converter
Wind (Type 3)	As above, with partial converter on rotor circuit	Similar	Stator direct, rotor via converter
Small hydro	Potential energy → turbine → generator	75–88 % water to wire	Synchronous or induction generator
Biomass steam	Chemical energy → combustion → steam cycle → generator	20–32 % electrical (75–85 % with CHP)	Synchronous generator
Biogas engine	Digestion → gas engine → generator	38–42 % of biogas energy	Synchronous generator
Geothermal binary	Brine heat → ORC → generator	8–12 % net thermal (35–50 % exergy)	Synchronous generator
Geothermal flash	Steam → condensing turbine → generator	10–17 % net thermal	Synchronous generator

Synchronous versus inverter-based interfaces

The last column of Table 15.4 divides renewable plants into two electrically different classes. Hydro, biomass and geothermal plants drive synchronous generators: they contribute rotating inertia, deliver 5–7 times rated current into a nearby fault and hold their internal EMF through disturbances, exactly like the thermal plant of Chapter 14. PV, Type 4 wind and batteries are **inverter-based resources** (IBRs): they have no intrinsic inertia, their fault current is limited by semiconductor ratings to 1.1–1.5 pu, and their response to a disturbance is whatever their control software makes it. Most present IBRs are **grid-following**: they measure the grid voltage with a phase-locked loop and inject current synchronised to it, so they rely on a strong grid formed by others. **Grid-forming** inverters, which behave as controlled voltage sources and can provide synthetic inertia and black-start, are now required for new plants in some grid codes. This difference, and not the variability of the resource, is the deepest engineering consequence of the energy transition for system stability and protection (Chapters 20, 23 and 27).

15.8 Capacity factor

Definition and typical values

The **capacity factor** is the ratio of the energy a plant actually delivers in a period to the energy it would deliver at its rated power throughout:

$$CF = \frac{E}{P_{\text{rated}}\,T}$$

For renewable plants it is set mainly by the resource, for thermal plants mainly by economics and availability. It is always necessary to say which rating: PV capacity factors quoted on the DC (kWp) rating are 15–25 % lower than on the AC rating. Table 15.5 gives typical ranges.

Table 15.5 — Typical annual capacity factors (net).

Technology	Capacity factor	Main determinant
Rooftop PV, northern Europe (DC basis)	9–12 %	Latitude, cloud
Utility PV, fixed tilt (AC basis)	17–25 %	Irradiance, DC/AC ratio
Utility PV, single-axis tracking, desert (AC basis)	25–33 %	Irradiance, DC/AC ratio
Onshore wind	25–45 %	Mean wind speed, rotor-to-generator ratio
Offshore wind	40–55 %	Mean wind speed, availability (access)
Run-of-river hydro	35–65 %	Hydrology, design flow
Reservoir hydro	35–60 %	Water availability, dispatch strategy
Biomass steam and CHP	50–85 %	Fuel supply, seasonal operation
Geothermal	75–95 %	Reservoir, availability
Nuclear (for comparison)	80–93 %	Refuelling outages, availability
Combined-cycle gas (for comparison)	40–65 %	Merit order

A low capacity factor is not a defect but a cost driver. A wind turbine with a larger rotor on the same generator ("low specific rating") has a higher capacity factor and produces more energy in low winds, when power is scarcer and more valuable; a PV plant with a high DC/AC ratio does the same.

Worked example — replacing a nuclear unit's energy. A 1,000 MW nuclear unit at a 92 % capacity factor delivers $1 \times 0.92 \times 8760 = 8.06$ TWh a year. Delivering the same energy needs $8.06/(0.22 \times 8.76) = 4.2$ GW of utility PV at 22 %, or 2.6 GW of onshore wind at 35 %, or 1.9 GW of offshore wind at 48 %. The energy is the same; its timing is not. The PV produces nothing at night; the wind may produce little for a week. Replacing the nuclear unit's *firm capacity* is a separate question, answered with storage, other dispatchable plant or demand response.

Capacity credit

The contribution of a variable plant to meeting peak demand is measured by its **capacity credit**, defined most rigorously as its **effective load-carrying capability** (ELCC): the amount by which the system's demand could rise, with the plant added, while keeping the same loss-of-load expectation (Chapter 13 §13.7). Typical ELCCs are 10–30 % of nameplate for wind, depending on the correlation of wind with peak demand, and for solar from 50–70 % at low penetration in summer-peaking systems with afternoon peaks to below 10 % at high penetration once the net peak has moved into the evening. The ELCC of every additional unit of the same technology declines, because they produce at the same times. Storage can raise the capacity credit of a PV portfolio by moving its energy into the evening peak (Chapter 16).

> **Status of the principle: Declining marginal capacity credit of correlated resources**
> *Empirical*. It is observed in every probabilistic adequacy study and follows qualitatively from the correlation of output across plants of one type, but its magnitude is system-specific and must be computed with chronological demand and resource data for each case.

15.9 Variability and forecasting

Time scales of variability

Variability is not one phenomenon but several, each with its own engineering response:

Time scale	Cause	Engineering response
Seconds	Cloud edges over a PV plant, gusts on a turbine	Inverter ramp control, smoothing by aggregation, small battery
Minutes	Cloud fields, wind fronts	Automatic generation control, frequency reserves (Chapter 20)
Hours	Diurnal solar cycle, weather systems	Dispatchable plant, storage, demand response
Days–weeks	Anticyclones with low wind ("dark doldrums")	Long-duration storage, firm capacity, interconnection
Seasons–years	Winter versus summer, dry versus wet years	Seasonal hydro storage, hydrogen, capacity planning

A single PV plant under broken cloud can swing by 50–70 % of its output in under a minute. Island grids such as Puerto Rico and Hawaii have imposed ramp-rate limits of the order of 10 % of rated power per minute on new plants, usually met with a battery.

Smoothing by aggregation

Fast fluctuations at different sites are weakly correlated, so they partly cancel when plants are combined. For N equal plants whose fluctuations have the same standard deviation and a pairwise correlation ρ, the standard deviation of the aggregate, relative to that of N times a single plant, is

$$\frac{\sigma_{\text{agg}}}{N\sigma_1} = \sqrt{\frac{1 + (N-1)\rho}{N}}$$

For 20 uncorrelated plants it is 0.22; for 20 plants with $\rho = 0.3$ it is 0.58, and even 100 such plants only reach 0.55, approaching the floor $\sqrt{\rho}$. Fluctuations of seconds to minutes are almost uncorrelated over a few kilometres and smooth very effectively (Figure 15.6); hourly and daily variations driven by the sun's position or large weather systems are strongly correlated over hundreds of kilometres and do not. Geographic diversity and interconnection reduce short-term reserve needs a great deal and the multi-day problem very little.

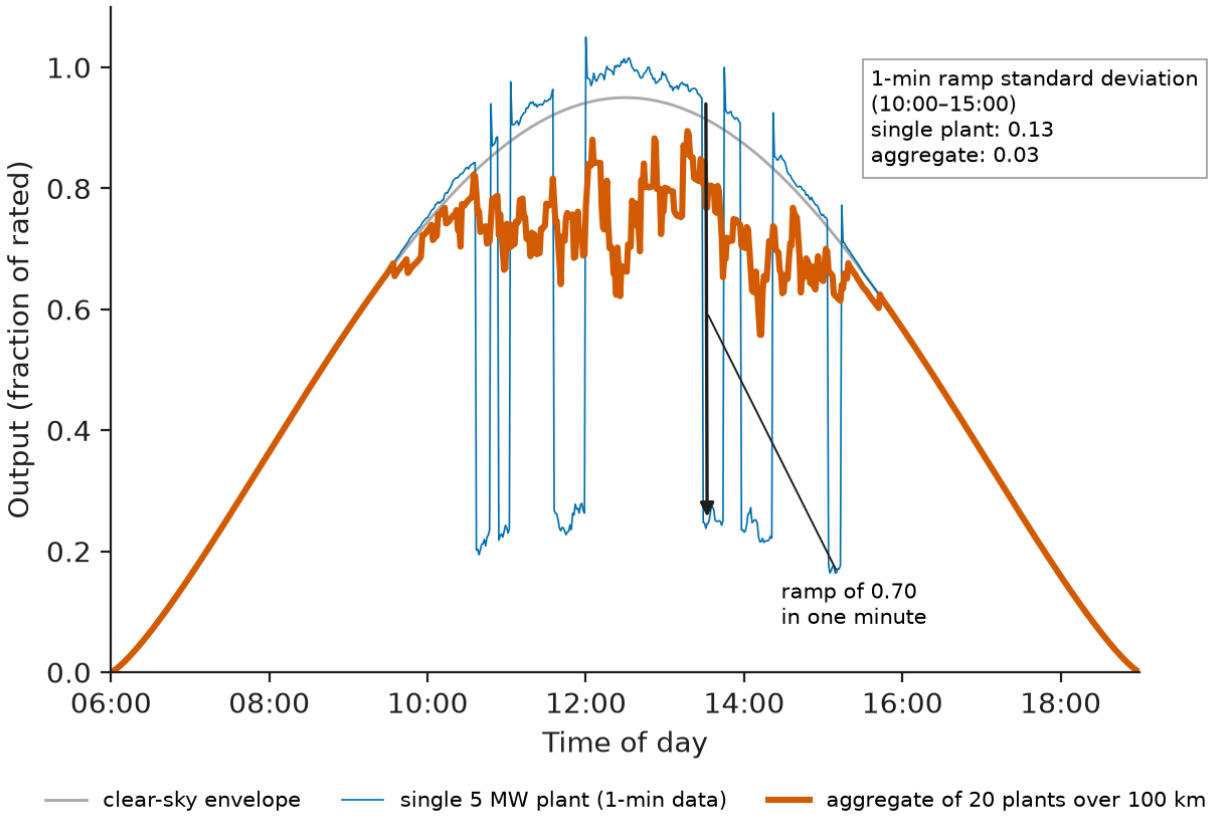


Figure 15.6 — Normalised output of one PV plant and of an aggregate of twenty plants on a partly cloudy day.

Forecasting

Forecasts turn variability, which is a physical fact, into uncertainty, which is what costs money. Different methods dominate at different horizons:

- **Seconds to minutes**: persistence (the next value equals the present one) is hard to beat; sky cameras and upstream sensors help PV plants anticipate cloud shadows.
- **Minutes to six hours**: satellite cloud-motion vectors for solar, statistical and machine-learning models trained on recent plant output and weather observations.
- **Hours to ten days**: numerical weather prediction (NWP) ensembles, post-processed with plant-specific statistical models.

Forecast errors grow with horizon and shrink with aggregation. Typical normalised mean absolute errors (as a percentage of installed capacity) for day-ahead forecasts are 8–15 % for a single wind farm but 3–6 % for a whole national fleet, and 5–10 % for a single PV plant but 2–5 % for a national PV fleet. Error distributions have heavy tails: the rare large miss, such as a storm front arriving hours early or fog that does not lift, drives reserve requirements.

Worked example — reserves for forecast error. A system has 20 GW of wind with a day-ahead forecast error of standard deviation 4 % of capacity for the fleet, 800 MW, and a day-ahead demand-forecast error of standard deviation 600 MW. If the two are independent, the combined standard deviation is $\sqrt{800^2 + 600^2} = 1{,}000$ MW, and an operator who holds reserve for three standard deviations must hold about 3 GW; without the wind it would hold 1.8 GW. Shortening the forecast horizon helps more than anything else: intraday markets and re-dispatch close to real time, where wind errors are half or less of their day-ahead size, reduce this reserve substantially, which is one reason market design has moved toward shorter gate-closure times.

What variability means for the system

Variable renewable output enters the system as a reduction in net load (Chapter 13 §13.6). At low shares it is absorbed by the flexibility that thermal and hydro fleets already have. At higher shares it deepens the midday dip, steepens the evening ramp, lowers the minimum net load below the minimum stable output of thermal units, and causes curtailment when there is nowhere for the energy to go, as the opening plant found on spring afternoons. The responses, storage (Chapter 16), demand response and smart-grid control (Chapter 26), grid codes that require renewable plants to provide frequency and voltage support (Chapter 27), and stronger interconnection (Chapter 21), are the subject of much of the rest of this book.

What remains open

Open question	Where it is answered
How do induction, permanent-magnet and doubly-fed machines behave as wind generators?	Chapters 6–8
How are the generator-side converters of wind turbines controlled?	Chapter 12
How is renewable energy stored and shifted to the hours of demand?	Chapter 16
How do inverter-based resources change fault levels and protection?	Chapters 19 and 23
How do low inertia and grid-following inverters affect frequency and rotor-angle stability?	Chapter 20
How are offshore wind farms connected by AC and HVDC cables?	Chapter 21
How does rooftop PV raise feeder voltages, and how are distribution networks adapted?	Chapter 22
What harmonics and flicker do inverters and wind turbines inject?	Chapter 25
How do microgrids and demand response absorb variability?	Chapter 26
What must renewable plants do under grid codes: ride-through, frequency and voltage support, anti-islanding?	Chapter 27
Where do the cell physics and the Carnot limit come from?	*Engineering Physics*, Ch. 14 §14.5 and Ch. 10 §10.6

Next: Chapter 16 takes up the missing element that variability makes essential, energy storage, from batteries and their management systems to pumped hydro, thermal storage and hydrogen.

Chapter 16 — Energy Storage

Part IV · Power Generation

At 4:30 on a winter afternoon a regional grid operator watches 3 GW of solar output fall away in ninety minutes while demand climbs to its evening peak. A growing share of the gap now comes from battery plants that charged at midday, when wholesale prices went negative; and when a 660 MW coal unit trips, the same batteries inject full power within a quarter of a second, long before any steam governor has moved. Elsewhere, a data centre rides through a 10 s utility outage on a spinning steel flywheel; a tram recovers its braking energy into a supercapacitor bank on its roof; and a steelworks studies hydrogen stored in a salt cavern as a way to carry summer wind energy into winter. Each is the same problem, moving energy in time, solved with different physics, costs and risks.

> **How are storage devices rated, and why do energy rating, power rating, efficiency and lifetime have to be specified separately?**
> **How does a battery management system know how full and how worn a battery is, and keep it safe? When is a battery the right answer, and when do supercapacitors, flywheels, pumped hydro, thermal stores or hydrogen win?**

Concept	Engineering destination
Battery technologies	Choosing a chemistry for a grid plant, an EV, a UPS or a ship
Energy and power ratings	MW/MWh specification of a storage plant; C-rate; usable versus nameplate energy
Battery management systems	Protection limits, balancing, thermal management, functional safety
State of charge	Dispatch of a battery plant, EV range display, UPS autonomy
State of health	Warranties, augmentation, end-of-life and second-life decisions
Supercapacitors	Regenerative braking in trams and cranes, wind-turbine pitch back-up, power smoothing
Flywheels	Data-centre ride-through, frequency regulation, kinetic-energy recovery
Pumped hydro	Bulk daily and weekly shifting, black start, system inertia
Thermal storage	Concentrating solar plants, building ice storage, Carnot batteries
Hydrogen storage	Seasonal storage, sector coupling, gas-turbine fuel
Economics and safety	Levelised cost of storage, revenue stacking, NFPA 855 and UL 9540A siting

Roadmap. What a battery is made of and how chemistries differ (16.1) → how storage is rated in energy, power and efficiency (16.2) → the management system that keeps a battery inside its limits (16.3) → estimating how full it is (16.4) and how worn it is (16.5) → the short-duration alternatives: supercapacitors (16.6) and flywheels (16.7) → the bulk alternatives: pumped hydro (16.8), thermal storage (16.9) and hydrogen (16.10) → what storage costs and how it fails (16.11).

16.1 Battery technologies

> **Boundary.**
> The electrochemistry of electrodes and electrolytes, and the crystal structures of cathode materials, belong to *The Materials Core*. The DC/DC converters and inverters that connect a battery to a bus belong to *Electronics & Computing*; *Engineering Physics*, Ch. 18 §18.8 gives the outline. This chapter treats a cell as an engineering component with a voltage, a capacity, an internal resistance, an efficiency, a lifetime and a failure mode.

The cell as a component

Every electrochemical cell has two electrodes, an electrolyte that conducts ions but not electrons, and a separator. The open-circuit voltage (OCV) is set by the chemistry and the state of charge; the terminal voltage under load differs from it by the internal voltage drops. For engineering purposes a cell is represented by the equivalent circuit of Figure 16.3 later in the chapter: an OCV source that depends on state of charge, a series ohmic resistance R_0, and one or two RC pairs that represent charge-transfer and diffusion polarisation with time constants of seconds to tens of minutes.

Cells come in cylindrical, prismatic (up to 300 Ah or more for stationary use) and pouch formats, and are connected in series for voltage and in parallel for capacity: a "96S2P" pack has 96 series groups of two parallel cells. The string is only as good as its weakest cell (Section 16.3).

Lead-acid

Lead-acid (2.0 V per cell) is the oldest rechargeable chemistry and still the cheapest per nameplate kWh. Flooded cells need watering and ventilation for the hydrogen released on overcharge; valve-regulated (VRLA) cells recombine most of it internally. Cycle life at 80 % depth of discharge is only a few hundred cycles, and sulphation sets in if a cell is left partly discharged. Its domains today are standby duty (substation DC systems, telecom, older UPS installations) and engine starting. Sizing follows IEEE 485.

Nickel–cadmium (1.2 V per cell) is robust from −40 °C to +50 °C and tolerates abuse, which keeps it in aircraft, rolling stock and offshore platforms despite cadmium's toxicity.

Lithium-ion

A lithium-ion cell shuttles lithium ions between a graphite negative and a metal-oxide or phosphate positive in an organic electrolyte; the cathode names the family:

- **LFP** (lithium iron phosphate, 3.2 V): moderate energy density, a very flat OCV curve, long cycle life and the highest thermal-runaway onset temperature of the common chemistries; it dominates stationary storage.
- **NMC** and **NCA** (nickel-rich oxides, 3.6–3.7 V): high energy density for long-range EVs and aircraft, with lower runaway onset temperatures and more energetic failures.
- **LTO** (lithium titanate negative, 2.3–2.4 V): low energy density, but tens of thousands of cycles and charging at 6C or more; used in opportunity-charged buses.

Sodium-ion, high-temperature and flow batteries

Sodium-ion cells need no lithium, cobalt or nickel; energy density is below LFP but cold performance is better. **Sodium–sulphur** (NaS) batteries run at 300–350 °C with molten electrodes and have served six-hour grid duty in Japan since the early 2000s.

In a **flow battery**, led commercially by the vanadium redox flow battery (VRFB), energy is held in liquid electrolytes in external tanks and power is set by the stack, so the two are sized independently; the electrolyte does not degrade and cannot run away thermally, at the price of low energy density, pumps and 65–75 % round-trip efficiency.

Table 16.1 — Representative properties of battery technologies (typical cell-level values; packs are lower by 20–40 % in energy density).

Technology	Nominal cell voltage	Specific energy (Wh/kg)	Energy density (Wh/L)	Cycle life to 80 % capacity	DC round-trip efficiency	Typical use
Lead-acid (VRLA)	2.0 V	30–45	60–110	300–1,500 (depth dependent)	75–85 %	Substation DC, telecom, standby UPS
Nickel–cadmium	1.2 V	40–60	50–150	1,500–3,000	65–80 %	Aircraft, rail, offshore
Li-ion LFP	3.2 V	140–205	300–450	4,000–10,000	92–96 %	Grid storage, EVs, buses, marine
Li-ion NMC / NCA	3.6–3.7 V	200–300	550–750	1,000–3,000	92–96 %	Long-range EVs, aviation, power tools
Li-ion LTO	2.3–2.4 V	60–110	120–250	10,000–30,000	95–98 %	Fast-charge buses, frequency regulation
Sodium-ion	3.0–3.1 V	100–160	250–350	2,000–5,000	90–94 %	Low-cost stationary, small EVs
Sodium–sulphur	2.1 V	100–150	150–250	4,000–5,000	75–85 % (incl. heating)	Six-hour grid storage
Vanadium redox flow	1.2–1.4 V	10–30 (system)	15–25 (electrolyte)	over 15,000	65–75 % (AC)	Long-duration stationary

Figure 16.1 places these technologies, and the non-battery stores of later sections, on a Ragone plot of specific power against specific energy. The plot is the quickest way to see why no single technology serves every application: moving up the energy axis costs power, and the diagonals of constant discharge time separate the seconds-scale devices from the hours-scale ones.

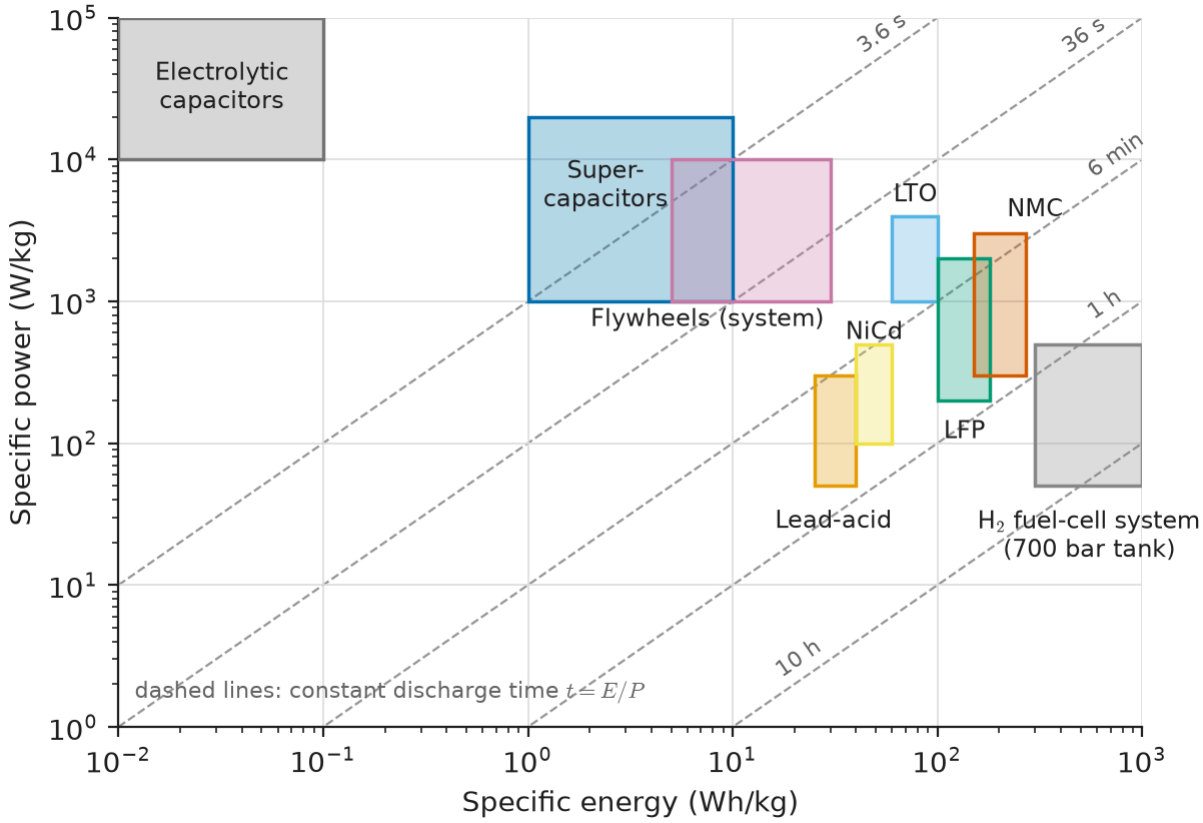


Figure 16.1 — Ragone plot of specific power against specific energy for storage technologies, with lines of constant discharge time.

16.2 Battery energy and power ratings

Capacity, energy and C-rate

A cell is rated by its **capacity** Q in ampere-hours, measured at a stated discharge rate and temperature down to a stated cut-off voltage. Its **energy** is the integral of terminal voltage times current over that discharge, approximately

$$E \approx Q\,V_{\mathrm{nom}},$$

so a 280 Ah LFP cell at 3.2 V nominal holds 0.896 kWh. The **C-rate** expresses current relative to capacity: 1C discharges the rated capacity in one hour, 0.25C in four hours, 4C in fifteen minutes. A battery plant rated "100 MW / 400 MWh" is a four-hour or 0.25C system; a frequency-regulation plant at "50 MW / 25 MWh" runs at 2C.

Capacity falls at high discharge rate, because internal drops bring the terminal voltage to cut-off earlier, and at low temperature. For lead-acid the effect is large and follows Peukert's law, $t = H\,(C/(IH))^{k}$ with $k \approx 1.1$–1.3; for lithium-ion it is a few per cent between 0.2C and 1C at 25 °C, but significant below 0 °C, where charging must also be restricted to avoid lithium plating.

> **Status of the principle: Peukert's law and rate-dependent capacity**
> *Empirical.* Fitted to constant-current discharge tests of lead-acid cells; it does not hold for variable currents, high temperatures or lithium-ion chemistries, for which manufacturers publish capacity-versus-rate curves instead.

Power rating and internal resistance

The power a battery can deliver is limited by voltage limits, current limits and heat. With OCV V_{oc} and internal resistance R_i, the terminal voltage at discharge current I is $V = V_{oc} - IR_i$, and the Joule loss is $I^2 R_i$. Maximum power transfer at $V = V_{oc}/2$ is never approached: the minimum cell voltage (2.5 V for LFP, about 3.0 V for NMC) and the temperature rise limit the current long before. Datasheets give a continuous current, a 10 s or 30 s pulse current and a power capability that depends on SoC and temperature, which the BMS (Section 16.3) turns into a live "state of power" for the inverter.

Internal resistance is small but decisive. A 280 Ah prismatic LFP cell has $R_i \approx 0.2$–0.3 mΩ when new; at 0.5C (140 A) its loss is about $140^2 \times 0.25 \times 10^{-3} \approx 4.9$ W against 448 W delivered, about 1 %. At 2C the loss fraction quadruples and the cooling system must remove it.

From nameplate to usable energy

Four layers separate the energy printed on a battery's cells from the energy delivered at the grid connection:

1. **State-of-charge window.** Operating between, say, 5 % and 95 % SoC protects the cells and leaves room for estimation error. Usable DC energy is 90 % of nameplate.
2. **Conversion losses.** The power conversion system (PCS: the inverter), the medium-voltage transformer and, for a transmission-connected plant, the main step-up transformer each lose energy in both directions.
3. **Auxiliary loads.** HVAC, fire detection, BMS and controls consume a few per cent of throughput; in hot climates the HVAC load can reach 3–5 % of discharged energy.
4. **Degradation.** Capacity falls with age (Section 16.5). The plant contract either over-builds at the start, or adds new cells ("augmentation") every few years to maintain the guaranteed energy at the point of interconnection (POI).

Round-trip efficiency (RTE) is the ratio of energy delivered to energy absorbed over a full cycle, measured at a stated point. A lithium-ion cell has a DC round-trip efficiency of about 95 %; a complete grid plant measured at the POI, including transformers and auxiliaries, achieves 82–90 %. IEC 62933-2-1 defines the test procedures.

Worked example — Sizing a 100 MW / 400 MWh battery plant. A transmission-connected battery must deliver 100 MW for four hours at the POI. LFP prismatic cells of 280 Ah, 3.2 V are used; the PCS efficiency is 98.5 %, the medium-voltage transformers 99.3 %, the main 33/132 kV transformer 99.5 %. The SoC window is 5–95 %.

One-way efficiency, DC to POI: $0.985 \times 0.993 \times 0.995 = 0.9732$.

Usable DC energy: $400 / 0.9732 = 411.0$ MWh.

Nameplate energy at beginning of life: with a 90 % window, $411.0 / 0.90 = 456.7$ MWh.

Cells: each holds $280 \times 3.2 = 0.896$ kWh, so $456.7 \times 10^3 / 0.896 \approx 5.10 \times 10^5$ cells. With 416 cells in series per string (nominal DC voltage $416 \times 3.2 = 1331$ V, inside the 1500 V limit of the PCS), one string holds 372.7 kWh, and 1,226 strings are needed (456.97 MWh). At twelve strings per container that is 102.2 containers, rounded up to 104 so that 26 four-container blocks, each with its own 4 MVA PCS, are identical.

C-rate: the DC power at full output is $100 / 0.9732 = 102.8$ MW, or $102.8 / 457.0 = 0.225$C; each string carries $102.8 \times 10^6 / (1226 \times 1331) = 63$ A, comfortably within the continuous rating of a 280 Ah cell.

Degradation: if the plant is guaranteed to deliver 400 MWh after ten years, when the cells may have only 80 % of their original capacity, the initial build would have to be $456.7 / 0.80 = 571$ MWh, a 25 % over-build. Most developers instead build close to 457 MWh and augment in years 4–7, when cells are cheaper.

Round-trip efficiency: with a cell-level DC efficiency of 95 %, $0.9732^2 \times 0.95 = 0.90$ before auxiliaries; with 2–3 % HVAC and control load, about 87–88 % at the POI.

Figure 16.2 shows the arrangement: battery containers in DC blocks, each block feeding a PCS and a medium-voltage transformer, collected on a 33 kV bus and stepped up to the transmission network. The structure is the same for a 2 MWh commercial system, with fewer blocks and a low-voltage connection.

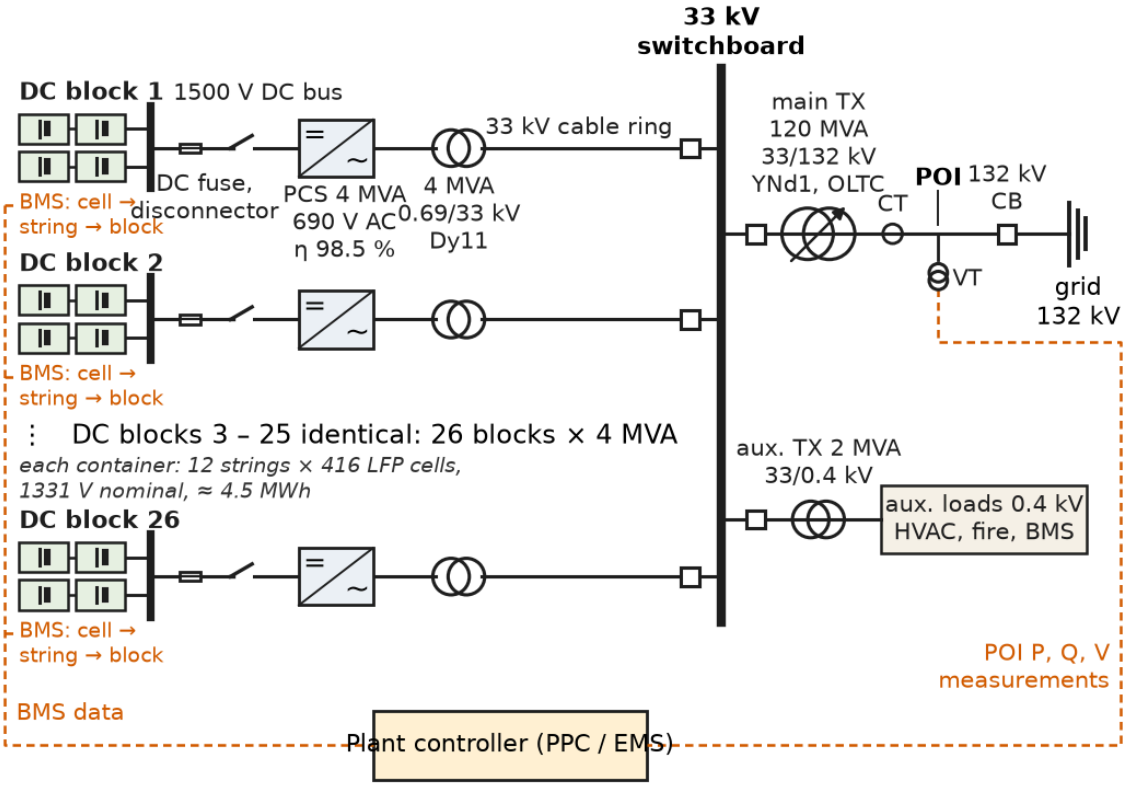


Figure 16.2 — Single-line diagram of a 100 MW / 400 MWh battery energy storage plant connected at 132 kV.

16.3 Battery management systems

What the BMS is for

A lithium-ion cell is safe and long-lived only inside a narrow operating window: voltage between about 2.5 and 3.65 V (LFP) or 3.0 and 4.2 V (NMC), temperature between about −20 and +55 °C on discharge and 0 to +45 °C on charge, and current within the datasheet limits. Outside it, the cell degrades quickly or, at the extreme, enters thermal runaway. The **battery management system** (BMS) enforces that window. It measures every cell voltage to a few millivolts, the pack current and enough temperatures to catch the hottest cell; it opens contactors on over-voltage, under-voltage, over-current, over-temperature or isolation fault, and derates current as limits approach; it estimates state of charge (Section 16.4), state of health (Section 16.5) and state of power (the current available for the next 2, 10 or 30 s); it balances cells, commands cooling and heating, and reports to the vehicle, inverter or plant controller over CAN, Modbus or IEC 61850.

Architecture

Large packs use a hierarchy: cell-monitoring circuits (one integrated circuit per 12–18 series cells) on each module, a string controller that owns the contactors and current measurement, and a system controller (three levels in Figure 16.2). The monitoring chain is galvanically isolated, because a 1331 V string puts its top module 1.3 kV above the bottom one.

Because the BMS is a safety function, its hardware and software are developed to functional-safety standards: ISO 26262 for road vehicles (traction batteries usually at ASIL C or D for over-voltage protection), IEC 61508 for industrial systems, and the product standards IEC 62619 and UL 1973 for stationary cells and batteries. A typical requirement is that an over-voltage condition be detected and the current interrupted within a few hundred milliseconds, with a probability of dangerous failure below 10^{-7} per hour.

Cell balancing

Cells differ in capacity by 1–2 % at manufacture, in self-discharge and in temperature across a pack. In a series string the first cell to reach its upper limit ends the charge and the first to reach its lower limit ends the discharge, so without correction usable capacity drifts below that of the weakest cell.

- **Passive balancing** bleeds charge from the highest cells through resistors (typically 50–200 mA) near the top of charge. It is cheap and universal but wastes energy as heat and is slow: at 100 mA, correcting a 1 % imbalance on a 280 Ah cell takes 28 h.
- **Active balancing** moves charge from high cells to low cells through inductive or capacitive DC/DC converters, at amperes rather than milliamperes and with 80–90 % transfer efficiency. It recovers capacity in packs of aged or second-life cells, at higher cost and complexity.

> **Status of the principle: Weakest-cell limit of a series string**
> *Derived.* Follows directly from the series connection (the same current flows in every cell) and the per-cell voltage limits; balancing changes the charge distribution but not the rule.

Thermal management

Cell temperature drives both ageing (Section 16.5) and safety. Liquid cooling through cold plates holds the cell-to-cell spread to 3–5 K; air-cooled systems may see 10 K or more. The heat-transfer models of *Engineering*

Physics, Ch. 14 §14.7 and Ch. 19 §19.6, apply directly: a thermal-resistance network from cell core to coolant, with the cooling plant sized for the worst-case duty and ambient.

16.4 State of charge

Definition

State of charge is the remaining charge as a fraction of the present full capacity:

$$\mathrm{SoC}(t) = \mathrm{SoC}(t_0) - \frac{1}{Q_{\mathrm{act}}} \int_{t_0}^{t} \eta_c \, i(\tau)\, d\tau,$$

with discharge current positive, Q_{act} the actual (aged) capacity and η_c the coulombic efficiency (above 99.9 % for lithium-ion, 85–95 % for lead-acid on charge). SoC cannot be measured, only estimated, and an error of a few per cent becomes stranded energy in a dispatched plant or an unexpected shutdown in a UPS.

Coulomb counting

Integrating the measured current is accurate over short periods but needs a correct starting value and Q_{act}, and accumulates every sensor error: an offset grows linearly with time, a gain error with charge throughput.

OCV-based estimation

After a rest of tens of minutes, the terminal voltage relaxes to the OCV, which is a known function of SoC for each chemistry (Figure 16.3). An OCV reading resets the coulomb counter. For NMC the OCV curve has a useful slope, about 6–10 mV per per cent of SoC through the mid-range. For LFP it is almost flat, about 0.5 mV per per cent between 30 % and 90 % SoC, with hysteresis between charge and discharge of a few tens of millivolts. That flatness is the main reason SoC estimation for LFP is hard, and why LFP plants periodically drive to full charge to recalibrate.

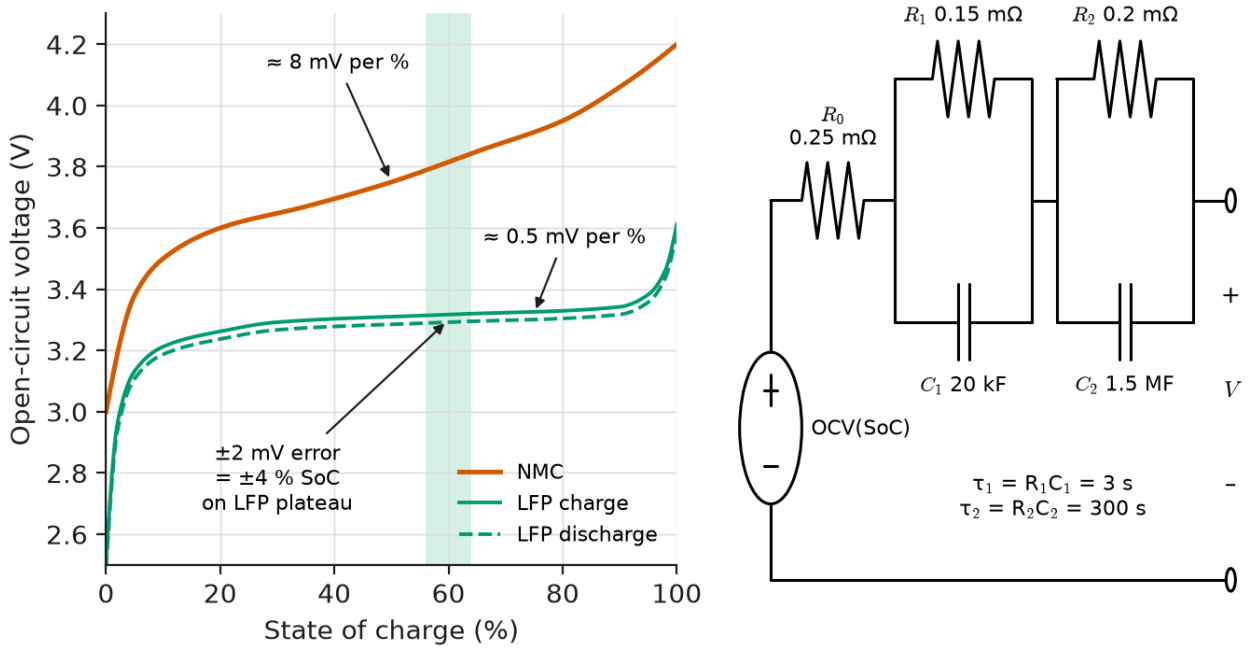


Figure 16.3 — Open-circuit voltage against state of charge for NMC and LFP cells, with the equivalent-circuit model used by battery management systems.

Model-based estimators

Modern BMSs combine both sources of information in a state observer, most often an extended Kalman filter. The model is the equivalent circuit of Figure 16.3 with SoC as a state; the filter predicts SoC by coulomb counting, compares the predicted terminal voltage with the measurement, and corrects the estimate in proportion to the mismatch, weighting each source by its uncertainty. When the OCV curve is steep the voltage correction is strong; on the LFP plateau it is weak and the filter rightly trusts the current integral. Well-implemented estimators keep SoC error within about 2–3 % for NMC and 3–5 % for LFP over the pack's life.

> **Status of the principle: Equivalent-circuit model of a cell**
> *Approximate.* A lumped fit to measured pulse responses; valid within the temperature, current and SoC range over which its parameters were identified, it does not capture lithium plating, ageing mechanisms or behaviour near the voltage limits, and its parameters drift as the cell ages.

Worked example — Why an LFP bus loses track of its SoC. An electric bus has a 200 Ah LFP pack and a Hall-effect current sensor with a 0.2 A residual offset after calibration. The bus runs two shifts a day and is opportunity-charged at the depot to about 85 % SoC, never to full.

Offset drift. Over 24 h the offset integrates to $0.2 \times 24 = 4.8$ Ah, which is $4.8/200 = 2.4$ % of capacity per day. After ten days without a reset the coulomb count can be wrong by 24 %.

Can OCV correct it? At the depot the bus rests for 30 min. With a cell-voltage measurement accuracy of ±2 mV and an LFP plateau slope of 0.5 mV per per cent, the OCV reading constrains SoC only to $\pm 2/0.5 = \pm 4$ %, and hysteresis adds more. For an NMC pack with a slope of 8 mV per per cent the same measurement would fix SoC to ± 0.25 %.

Consequence and remedy. The LFP pack needs a periodic full charge (the top of the OCV curve is steep, and the BMS resets SoC to 100 % when the charge current tapers) or a better current sensor, typically a shunt with

offset below 20 mA. Fleet operators schedule a full "balancing charge" weekly for exactly this reason.

16.5 State of health

Definitions

State of health (SoH) describes how much of the battery's original capability remains. Two definitions are used:

$$\mathrm{SoH}_Q = \frac{Q_{\mathrm{act}}}{Q_{\mathrm{rated}}}, \qquad \mathrm{SoH}_R = \frac{R_{\mathrm{EOL}} - R_{\mathrm{act}}}{R_{\mathrm{EOL}} - R_{\mathrm{new}}}.$$

Capacity fade limits energy; resistance growth limits power. Conventional **end of life** is 80 % of rated capacity for EV and stationary cells, or a doubling of resistance for power applications. A battery at end of life for an EV may still be useful in a less demanding stationary "second life".

Ageing mechanisms

Two kinds of ageing act together:

- **Calendar ageing** happens whether or not the battery is used. Its main mechanism is the slow growth of the solid-electrolyte interphase on the graphite, which consumes cyclable lithium. It accelerates with temperature and with high SoC, and is often fitted as proportional to the square root of time.
- **Cycle ageing** depends on charge throughput, depth of discharge, C-rate and temperature. Mechanical stress from repeated expansion cracks particles; at low temperature and high charge rate, lithium plates on the negative electrode as metal, which both consumes capacity and seeds internal short circuits.

Both are commonly modelled with an Arrhenius temperature factor (the Boltzmann-factor rate law of *Engineering Physics*, Ch. 20 §20.7.1). For calendar fade,

$$\Delta Q(t, T) = k_0 \, e^{-E_a/(RT)} \sqrt{t},$$

with an activation energy E_a fitted to test data, typically 30–60 kJ/mol for lithium-ion. Figure 16.4 shows representative fade curves.

> **Status of the principle: Battery ageing laws**
> *Empirical.* Square-root-of-time calendar fade, Arrhenius temperature acceleration and Wöhler-type cycle-life curves are fitted to accelerated tests for each cell design; they extrapolate poorly beyond the tested range and miss the "knee" at which fade suddenly accelerates late in life.

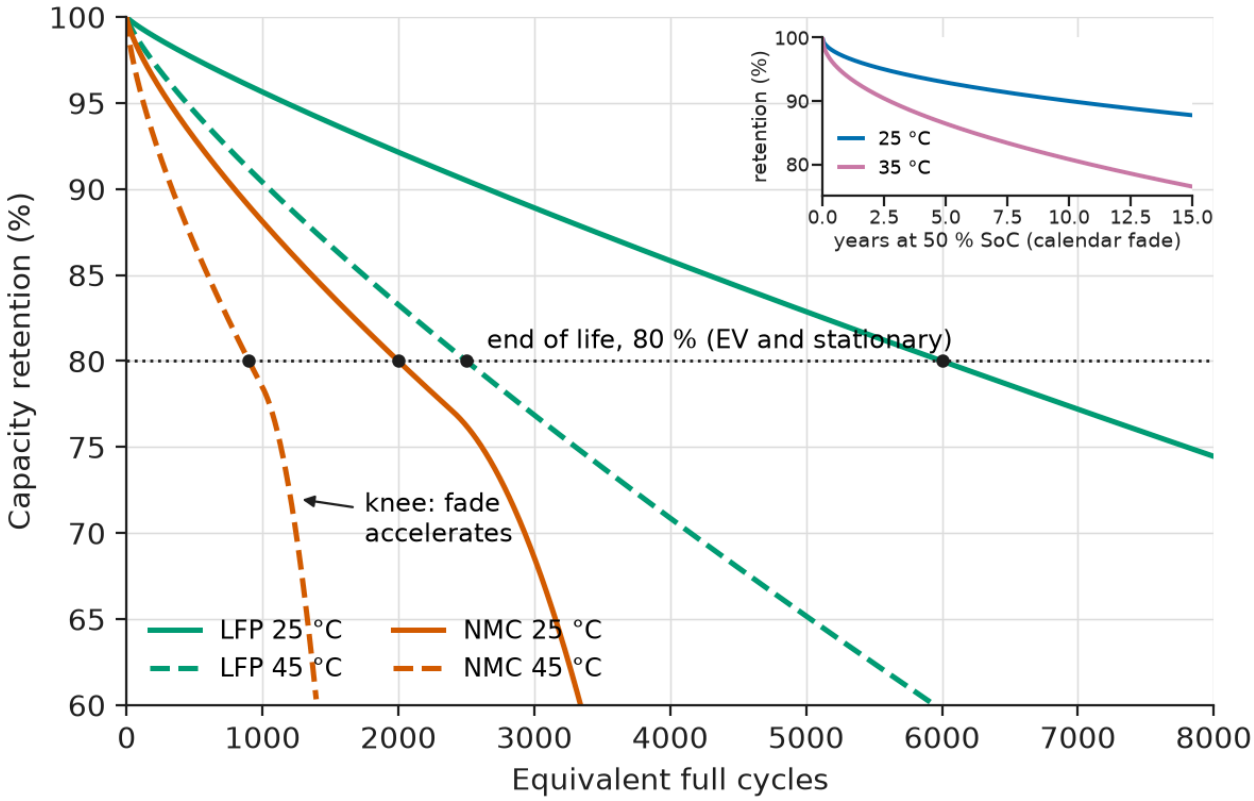


Figure 16.4 — Capacity retention of LFP and NMC cells against equivalent full cycles at different temperatures, showing the end-of-life threshold and the late-life knee.

Estimating SoH in the field

In service a full capacity test is rarely possible, so the BMS estimates SoH from partial data: charge counted between two well-defined OCV points, the voltage step at a current change, or incremental-capacity analysis (dQ/dV) of slow charges. Grid plants also run an annual capacity test at the POI, because the guaranteed MWh is contractual.

Worked example — A UPS battery in a hot electrical room. A data-centre UPS uses NMC modules whose calendar fade at 25 °C is 10 % after 10 years at 50 % SoC. The electrical room's cooling is reduced to save energy and the battery now sits at 35 °C. Take $E_a = 50$ kJ/mol.

Acceleration factor:

$$\frac{k(35^\circ\mathrm{C})}{k(25^\circ\mathrm{C})} = \exp\left[\frac{50\,000}{8.314}\left(\frac{1}{298.15} - \frac{1}{308.15}\right)\right] = 1.92.$$

Fade after ten years at 35 °C: $10 \times 1.92 = 19.2$ %, essentially at end of life, against 10 % at 25 °C.

Time to 20 % fade. With $\Delta Q = k\sqrt{t}$ and $k_{25} = 10/\sqrt{10} = 3.16$ % per $\sqrt{\text{year}}$, 20 % fade takes $(20/3.16)^2 = 40$ years at 25 °C (cycle ageing and other mechanisms will end life sooner) but $(20/(3.16 \times 1.92))^2 = 10.8$ years at 35 °C. A 10 K rise has roughly quartered the calendar life (the square-root law turns a 1.92-fold rate into a $1.92^2 \approx 3.7$-fold reduction in time). The money saved on cooling is small beside the cost of a premature battery replacement, which is why IEEE 1188 and manufacturers' warranties specify 20–25 °C for stationary batteries.

16.6 Supercapacitors

Principle and ratings

An **electric double-layer capacitor** (EDLC, supercapacitor or ultracapacitor) stores charge electrostatically in the double layer, about a nanometre thick, at the interface between a porous carbon electrode and an electrolyte. The huge internal area (over 1,000 m² per gram of carbon) and tiny separation give capacitances of thousands of farads per cell, but the electrolyte limits cell voltage to about 2.7–3.0 V. The stored energy is the capacitor energy of *Circuits, Signals and Communications*, Ch. 5 §5.4, $E = \tfrac{1}{2}CV^2$.

Because no chemical reaction is involved, supercapacitors charge and discharge in seconds, last 500,000 to over a million cycles, work down to −40 °C and have round-trip efficiencies of 90–98 % at moderate currents. Their weaknesses are low specific energy (3–8 Wh/kg for cells, 2–5 Wh/kg for modules, against 150+ Wh/kg for LFP cells), self-discharge of a few per cent per day, and a terminal voltage that falls in proportion to charge.

Usable energy and the voltage window

The falling voltage means that a converter must interface the bank to a fixed DC bus, and that energy cannot be fully extracted. Discharging from $V_{\max}$ to $V_{\min}$ releases

$$E_{\text{use}} = \tfrac{1}{2}C\left(V_{\max}^2 - V_{\min}^2\right) = E_{\max}\left[1 - \left(\frac{V_{\min}}{V_{\max}}\right)^2\right].$$

Stopping at half voltage recovers 75 % of the stored energy, while the converter current doubles. Going lower gains little energy at the cost of a much larger converter, so $V_{\min} = V_{\max}/2$ is the usual design point.

Applications

Supercapacitors serve where energy flows in and out in seconds, many times an hour: regenerative braking in trams, metros and container-port cranes, back-up for wind-turbine pitch systems, and smoothing of pulsed loads in hybrid ship and aircraft power systems. Hybrid **lithium-ion capacitors** reach 10–30 Wh/kg and fill part of the gap to batteries.

Worked example — Supercapacitor bank for a catenary-free tram. A 40 t tram brakes from 60 km/h at 1.2 m/s² and must store its braking energy on board. The drive chain from wheel to storage is 80 % efficient. The bank is built from 125 V, 63 F heavy-transport modules (each about 60 kg) connected in series and discharged to half voltage.

Kinetic energy: $v = 60/3.6 = 16.67$ m/s, so $E_k = \tfrac{1}{2} \times 40\,000 \times 16.67^2 = 5.56$ MJ $= 1.54$ kWh.

Energy to store: $0.80 \times 5.56 = 4.44$ MJ.

Per module: $\tfrac{1}{2} \times 63 \times 125^2 = 492$ kJ stored at full voltage, of which 75 % (369 kJ) is usable down to 62.5 V.

Number of modules: $4.44/0.369 = 12.04$, so 13 modules in series, rated $13 \times 125 = 1625$ V, total mass about 0.8 t.

Power and current: the braking force is $40\,000 \times 1.2 = 48$ kN, so at 16.67 m/s the wheel power is 800 kW and about 640 kW reaches the bank. If braking begins with the bank near half voltage (812 V), the current is $640\,000/812 \approx 790$ A; with the bank nearly full (1625 V) it is half that. A bidirectional DC/DC converter rated about 800 A links the bank to the 750 V traction bus. A lithium-ion pack of equal mass would hold forty times the energy, but it could not accept 0.8 MW per tonne every two minutes for a million stops.

16.7 Flywheels

Energy and the material limit

A flywheel stores kinetic energy in a rotating mass. With moment of inertia I and angular speed ω, $E = \tfrac{1}{2} I\omega^2$; rotational mechanics, moment of inertia and the steel-disk example are in *Engineering Physics*, Ch. 19 §19.2, and Ch. 9 §9.7. The engineering extension is the question of how much energy a rotor of given mass can hold before it bursts. The hoop stress in a rotating body scales with ρv^2, where v is the rim speed, while the stored energy per unit mass scales with v^2. Eliminating v gives the maximum specific energy

$$\left.\frac{E}{m}\right|_{\max} = K\,\frac{\sigma_{\text{allow}}}{\rho},$$

where K is a shape factor: about 0.5 for a thin rim, about 0.6 for a solid disc of steel ($\nu = 0.3$), and up to 1 for an ideal constant-stress profile.

> **Status of the principle: Specific-energy limit of a flywheel rotor**
> *Derived*. Follows from the energy of a rotating body and the stress in a rotating disc or rim; the shape factor assumes isotropic linear-elastic material, and real composite rotors are limited by radial (through-thickness) stress and fatigue, so design values are half or less of the ideal.

The ratio σ/ρ favours strong, light materials. High-strength steel ($\sigma \approx 800$ MPa usable, $\rho = 7850$ kg/m³) gives at most about 17 Wh/kg in a solid disc; a carbon-fibre composite rim ($\sigma \approx 1500$ MPa, $\rho \approx 1600$ kg/m³) gives about 130 Wh/kg in theory. Complete systems achieve 5–30 Wh/kg.

Construction and losses

A modern flywheel unit has the rotor in an evacuated housing (to eliminate windage), magnetic or hybrid bearings, and an integral motor-generator, usually a permanent-magnet synchronous machine (Chapter 8) driven by a bidirectional converter. The usable speed range is limited, as with the supercapacitor, by the converter: discharging from $\omega_{\max}$ to $\omega_{\max}/2$ releases 75 % of the energy. Self-discharge from bearing, residual windage and magnetic losses is significant, typically 1–5 % of stored energy per hour, so flywheels serve seconds-to-minutes duties, not hours. Burst containment is a design requirement: the housing must absorb the energy of a failed rotor, which is why rotors are often installed below ground level.

Applications

The main applications are UPS ride-through, bridging the 10–20 s between a utility outage and a diesel generator taking load, and frequency regulation, where tens of thousands of shallow cycles a year would wear out a battery; the 20 MW Stephentown plant in New York uses 200 composite flywheels of 100 kW each. Flywheels also smooth pulsed loads such as mine hoists.

Worked example — Flywheel ride-through for a data centre. A 1 MW UPS must ride through 15 s until the standby generators have started and taken load. The converter and machine efficiency is 95 %, and the flywheel operates between its maximum speed of 7,700 rpm and half speed. The rotor is a solid steel disc of radius 0.35 m.

Energy: usable $1 \times 10^6 \times 15/0.95 = 15.8$ MJ; total at full speed $15.8/0.75 = 21.1$ MJ.

Speed: $\omega = 7700 \times 2\pi/60 = 806$ rad/s; rim speed $806 \times 0.35 = 282$ m/s.

Mass: for a solid disc $E = \tfrac{1}{4} m r^2 \omega^2$, so $m = 4 \times 21.1 \times 10^6 / (0.35^2 \times 806^2) = 1057$ kg, a disc 0.35 m thick.

Stress check: the peak (central) stress of a solid disc is $\tfrac{3+\nu}{8}\rho v^2 = 0.4125 \times 7850 \times 282^2 = 258$ MPa, a safety factor of about 3 against an 800 MPa allowable.

Specific energy: $21.1 \times 10^6 / 1057 / 3600 = 5.5$ Wh/kg in the rotor alone. A battery would weigh less but need replacing every 5–8 years; the flywheel lasts twenty, and its state of charge is simply its speed.

16.8 Pumped hydro

Principle and scale

Pumped-storage hydroelectricity moves water between two reservoirs at different heights: pumping uphill when electricity is cheap or surplus, generating when it is valuable. It provides over 90 % of the world's installed storage energy, about 180 GW of power capacity, with individual plants from tens of megawatts to 3.6 GW. The stored energy is the gravitational potential energy of the water; the flow through penstock and turbine is the hydraulic circuit of *Engineering Physics*, Ch. 19 §19.7. The electrical energy delivered from a volume V falling through a net head h is

$$E = \rho\, g\, h\, V\, \eta_g,$$

with η_g the generating efficiency (turbine, generator and transformer, about 0.88–0.92). Pumping the same water back takes $\rho g h V / \eta_p$ with η_p about 0.86–0.90. Round-trip efficiency is 70–82 % for modern plants.

Figure 16.5 shows the layout: upper reservoir, intake, pressure tunnel and steel-lined penstock, an underground powerhouse with reversible pump-turbines, tailrace and lower reservoir. Head is the key design parameter: energy per cubic metre is proportional to it, so high-head sites (300–700 m) need much smaller reservoirs. One cubic metre falling 500 m yields only $1000 \times 9.81 \times 500 = 4.9$ MJ, or 1.36 kWh: pumped hydro has the lowest energy density of any technology in this chapter and makes up for it in volume.

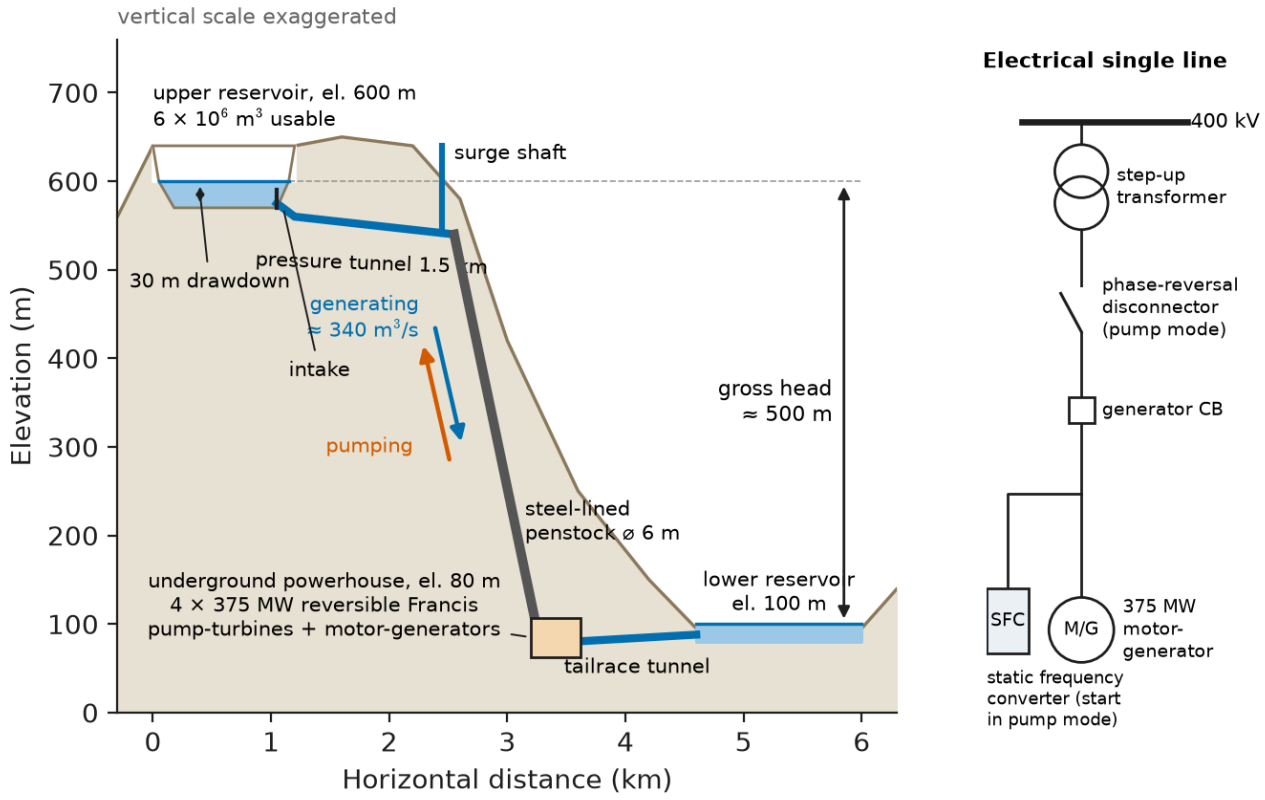


Figure 16.5 — Cross-section of a pumped-storage plant with reversible pump-turbines in an underground powerhouse.

Machines and control

Most pumped-storage plants use reversible Francis pump-turbines coupled to synchronous motor-generators (Chapter 7). The machine reverses rotation between modes, so a disconnector swaps the phase sequence, and pump-mode starting needs a static frequency converter. A fixed-speed pump draws fixed power; **variable-speed** units with doubly fed induction machines or full-power converters (Chapter 12) adjust pumping power by ±10–20 % and so can regulate frequency while pumping.

What pumped hydro adds to the grid

Pumped storage also contributes synchronous inertia and fault current, fast ramping (Dinorwig in Wales, 1,728 MW and about 9 GWh, reaches full load from standstill in under 2 minutes) and black-start capability. Its limits are geographic: two reservoirs with a large height difference, a decade of construction and high capital cost, eased in recent schemes by closed-loop "off-river" sites and old mines.

Worked example — Energy of an upper reservoir. A pumped-storage plant has a usable upper reservoir volume of 6×10^6 m³, a head of 500 m (the gross head of Figure 16.5, taken as constant over the 30 m drawdown), a generating efficiency of 90 % and a pumping efficiency of 88 %; the friction losses in tunnel and penstock, a few percent of the head, are counted in these two efficiencies rather than by reducing h. Four units give 1,500 MW.

Stored electrical energy (generating): $1000 \times 9.81 \times 500 \times 6 \times 10^6 \times 0.90 = 2.65 \times 10^{13}$ J $= 7.36$ GWh.

Duration at full output: $7360 / 1500 = 4.9$ h.

Energy to refill: $1000 \times 9.81 \times 500 \times 6 \times 10^6 / 0.88 = 3.34 \times 10^{13}$ J $= 9.29$ GWh.

Round-trip efficiency: $0.90 \times 0.88 = 79\ \%$.

Flow at full output: $Q = P/(\rho g h \eta_g) = 1500 \times 10^6/(1000 \times 9.81 \times 500 \times 0.90) = 340\ \text{m}^3/\text{s}$.

The same energy in LFP would take about 8.2 million 280 Ah cells; batteries win economically at 1–4 h, pumped hydro at 6 h and longer where a site exists.

16.9 Thermal storage

Forms of thermal storage

Thermal storage holds energy as heat (or cold) and comes in three forms:

- **Sensible heat**: raising the temperature of a material, $Q = mc_p\Delta T$. Molten nitrate salt, water, concrete, rocks and sand are used.
- **Latent heat**: melting or freezing a phase-change material, which stores energy at nearly constant temperature. Ice (334 kJ/kg) is the commonest; salts and paraffins cover other temperatures.
- **Thermochemical**: reversible chemical reactions, still largely at demonstration scale.

Thermal storage is cheap per kWh of heat. Its weakness appears when the energy must come back as electricity: the conversion from heat to work is bounded by the Carnot efficiency of *Engineering Physics*, Ch. 10 §10.6, and real Rankine cycles (*Engineering Physics*, Ch. 19 §19.5) reach only 35–45 %. Thermal storage therefore works best where heat is the input (concentrating solar power), the output (industrial process heat, district heating) or both (building cooling).

Concentrating solar power with molten salt

A concentrating solar power (CSP) plant collects sunlight as heat, so storing the heat before the turbine costs little efficiency. Most commercial plants use a two-tank system: "solar salt" (60 % $NaNO_3$, 40 % KNO_3 by mass, $c_p \approx 1.5$ kJ/(kg·K), melting near 220 °C) is pumped from a cold tank at about 290 °C through the receiver to a hot tank at about 565 °C, and from there through a steam generator that drives a conventional turbine (Chapter 14).

Worked example — Salt inventory for a 110 MW CSP plant with ten hours of storage. The turbine cycle efficiency is 41.5 %, the salt cycles between 290 °C and 565 °C, $c_p = 1.5$ kJ/(kg·K), and the hot salt density is about 1900 kg/m³.

Thermal energy: $110 \times 10 = 1100$ MWh of electricity requires $1100/0.415 = 2651\ \text{MWh}_{\text{th}} = 9.54 \times 10^{12}$ J.

Salt mass: $m = Q/(c_p\Delta T) = 9.54 \times 10^{12}/(1500 \times 275) = 2.31 \times 10^7$ kg, about 23,000 t.

Tank: $2.31 \times 10^7/1900 = 12\,200$ m³; a 40 m diameter tank needs a salt depth of $12\,200/(\pi \times 20^2) = 9.7$ m. Real plants add a heel of salt that is never drawn down and a margin for heat losses (tanks lose about 1 K per day), so built inventories are 20–40 % larger.

Power-to-power check: if the same tank were charged by electric heaters (a "Carnot battery"), round-trip efficiency would be about $1.0 \times 0.415 = 41.5$ % before heat losses, far below a battery, and the ideal limit with heat rejected at 40 °C would still be only $1 - 313/838 = 63$ %.

Thermal storage in buildings and industry

In buildings, ice made at night cools offices in the afternoon: a 10,000 ton-hour ice store (35 MWh of cooling) shifts about 7–9 MWh of chiller electricity off the peak. Hot-water cylinders and storage heaters are distributed thermal stores that utilities control with ripple control and smart meters (Chapter 26). In industry, high-temperature stores (refractory brick or sand at 600–1500 °C) charged by electric resistance heating replace fossil fuel in process heat, the "power-to-heat" route that avoids the Carnot penalty entirely.

> **Status of the principle: Carnot bound on power-to-heat-to-power storage**
> *Fundamental.* Follows from the second law of thermodynamics; it caps the electrical round-trip efficiency of any storage that passes through heat, whatever the storage medium. Heat-pump-charged Carnot batteries raise the attainable figure to about 50–70 % but cannot escape the bound.

16.10 Hydrogen energy storage

The chain

Hydrogen storage converts electricity to chemical energy by electrolysis, stores the hydrogen as compressed gas, liquid or a carrier compound, and reconverts it in a fuel cell or a combustion turbine. Figure 16.6 traces the chain and its losses. The chain is inefficient, but it is the only option here whose cost scales to weeks and months, because caverns hold energy at very low cost per kWh.

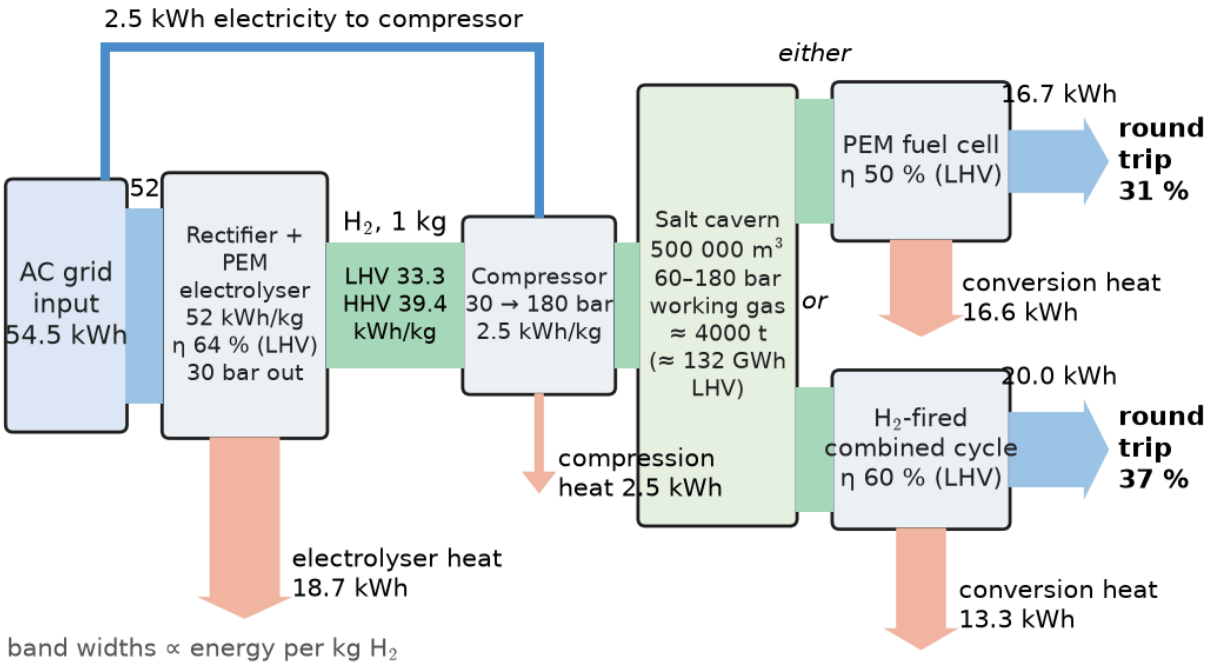


Figure 16.6 — Power-to-hydrogen-to-power chain with energy flows per kilogram of hydrogen.

Electrolysers

Three electrolyser types matter: **alkaline** (mature, cheapest per kW, best at steady load), **proton-exchange-membrane** (PEM: responds in seconds, suits variable renewable input, delivers gas at 30 bar or more) and **solid-oxide** (700–850 °C, highest electrical efficiency given waste heat, early commercial).

System consumption is about 50–55 kWh of electricity per kilogram of hydrogen for alkaline and PEM plants, against a lower heating value (LHV) of 33.3 kWh/kg and a higher heating value (HHV) of 39.4 kWh/kg: about 60–67 % efficiency on an LHV basis. Electrically, an electrolyser is a large DC load fed by a thyristor or IGBT rectifier; a 100 MW electrolyser plant is a significant source of harmonics and a candidate for demand response and frequency services (Chapters 25 and 27).

Storage and reconversion

Hydrogen's volumetric density is poor: about 0.09 kg/m³ at atmospheric pressure, 40 kg/m³ at 700 bar and 71 kg/m³ as a liquid at −253 °C. Compression to 200 bar costs 2–3 kWh/kg, liquefaction 10–12 kWh/kg. For grid-scale storage the attractive option is a solution-mined **salt cavern**, used for hydrogen at Teesside and in Texas for decades. Reconversion uses PEM or solid-oxide fuel cells (45–60 % LHV) or gas turbines adapted to burn hydrogen or hydrogen blends (combined-cycle efficiency 55–62 % LHV).

Worked example — Round-trip efficiency and cavern capacity. Electrolysis consumes 52 kWh/kg, compression to 180 bar 2.5 kWh/kg.

Round trip via a fuel cell at 50 % LHV: $33.3 \times 0.50/(52 + 2.5) = 16.7/54.5 = 31\ \%$.

Round trip via a hydrogen-fired combined cycle at 60 % LHV: $20.0/54.5 = 37\ \%$.

Cavern capacity. A 5×10^5 m³ salt cavern at about 320 K cycles between 60 and 180 bar. With compressibility factors of about 1.035 and 1.11, the gas density $\rho = pM/(ZRT)$ is $60 \times 10^5 \times 0.002016/(1.035 \times 8.314 \times 320) = 4.39$ kg/m³ at 60 bar and 12.29 kg/m³ at 180 bar. The working gas is $(12.29 - 4.39) \times 5 \times 10^5 \approx 3.95 \times 10^6$ kg, about 3,950 t, holding $3.95 \times 10^6 \times 33.3 = 132$ GWh (LHV), which returns about 66 GWh of electricity through fuel cells.

That single cavern returns about nine times the electricity of the pumped-hydro reservoir of Section 16.8 and over 160 times that of the 400 MWh battery plant of Section 16.2; measured as stored hydrogen energy (LHV) the ratios are about 18 and 330. The two-thirds of input energy lost on the way is the price. Hydrogen makes sense for storage only where duration, not efficiency, is the binding constraint: seasonal balancing in a high-renewable grid, or where the hydrogen is also needed as a feedstock (steel, ammonia, refining), so that only part of it is ever converted back to electricity.

16.11 Storage economics and safety

Rating storage by service

Storage earns money by providing services, and the service determines the rating. Table 16.2 maps the main services to the duration and response they need, and to the technologies that usually win.

Table 16.2 — Grid and customer storage services (typical values).

Service	Response time	Duration	Cycles per year	Technologies usually chosen
Fast frequency response, synthetic inertia	0.1–1 s	seconds to 15 min	many partial cycles	Li-ion (LFP, LTO), flywheel, supercapacitor
Frequency regulation (secondary)	seconds	15–60 min	thousands of shallow cycles	Li-ion, flywheel
UPS ride-through	under 10 ms	10 s to 15 min	few	Li-ion, VRLA, flywheel
Peak shaving, demand-charge reduction	minutes	1–4 h	50–300	Li-ion
Energy arbitrage, solar shifting	minutes	2–8 h	250–365	Li-ion, pumped hydro, flow
Capacity adequacy (peak reserve)	minutes	4–10 h	few to 100	Li-ion, pumped hydro
Black start	minutes	hours	0–1	Pumped hydro, Li-ion with grid-forming inverters
Seasonal balancing	hours	weeks to months	1–2	Hydrogen, large hydro reservoirs

Most battery plants "stack" several services, because one revenue stream rarely pays for the plant.

Levelised cost of storage

Storage is compared by the **levelised cost of storage** (LCOS): the constant price per MWh of discharged energy that recovers capital, operating and charging costs over the plant's life. In its simplest annualised form,

$$\mathrm{LCOS} = \frac{C_{\mathrm{cap}}\,\mathrm{CRF} + C_{\mathrm{O\&M}}}{E_{\mathrm{out}}} + \frac{c_{\mathrm{charge}}}{\eta_{\mathrm{RT}}}, \qquad \mathrm{CRF} = \frac{i(1+i)^{n}}{(1+i)^{n}-1},$$

where CRF is the capital recovery factor at discount rate i over n years, E_{out} is the annual discharged energy and c_{charge} the price paid for charging energy.

Worked example — Levelised cost of a four-hour battery. The 100 MW / 400 MWh plant of Section 16.2 costs \$280 per kWh of rated energy installed (\$112 M), with fixed O&M of 2 % of capital per year, a 20-year life, a 7 % discount rate and one full cycle a day at an average of 90 % of rated energy (allowing for degradation and availability). Charging energy costs \$30/MWh and the AC round-trip efficiency is 85 %.

Capital recovery factor: $0.07 \times 1.07^{20} / (1.07^{20} - 1) = 0.0944$; annualised capital $112 \times 0.0944 = 10.57$ \$M/yr; O&M 2.24 \$M/yr.

Annual discharge: $400 \times 365 \times 0.90 = 131\,400$ MWh.

Fixed cost per MWh: $(10.57 + 2.24) \times 10^{6} / 131\,400 = 97.5$ \$/MWh.

Charging cost per MWh delivered: $30 / 0.85 = 35.3$ \$/MWh.

LCOS: $97.5 + 35.3 = 133$ \$/MWh. The plant must on average sell at about \$100/MWh more than it buys to break even on arbitrage alone. Cycling twice a day halves the fixed component and gives an LCOS of about

$84/MWh, provided the cells' cycle life allows it, which is why frequency-response and capacity payments, not daily arbitrage alone, have underpinned most battery projects.

Rough mid-2020s ranges: lithium-ion $200–350 per kWh installed for four-hour plants; pumped hydro $1,500–3,000 per kW but cheap per kWh at long durations; hydrogen dominated by the per-kW cost of electrolysers and reconversion, with the cavern costing well under $1 per kWh.

Failure modes and thermal runaway

Flywheels can burst; hydrogen is flammable from 4 to 75 % in air with a minimum ignition energy about a tenth of methane's; lead-acid batteries release hydrogen. For lithium-ion the defining hazard is **thermal runaway**: once a cell's internal temperature exceeds a threshold (about 150–200 °C for NMC, 200–270 °C for LFP), exothermic decomposition reactions heat it faster than it can lose heat, and the temperature rises to 500–900 °C in seconds to minutes. The cell vents flammable and toxic gases (hydrogen, carbon monoxide, hydrocarbons, hydrogen fluoride). If the heat ignites neighbouring cells, runaway **propagates** through the module and rack. Triggers include internal short circuits from manufacturing defects or lithium plating, over-charge, external short circuits, mechanical damage and external fire.

At McMicken, Arizona, in 2019, a runaway in a 2 MWh container filled it with unignited gas, which exploded when firefighters opened the door and injured four of them. More than twenty fires in South Korea in 2017–2019, and the January 2025 fire in an indoor battery building at Moss Landing, California, pushed the industry toward outdoor containerised designs with spacing between units.

Safety standards and design defences

Safety is designed in layers, each covered by a standard:

- **Cell and battery**: IEC 62619 and UL 1973 (stationary), IEC 62660 and UN 38.3 (transport), with abuse tests for over-charge, short circuit, crush and thermal stability.
- **System**: UL 9540 lists complete energy storage systems; **UL 9540A** is the test method that measures whether runaway propagates from cell to module to unit to installation, and quantifies vent gas composition.
- **Installation**: **NFPA 855** (and the International Fire Code) sets limits on unit size (typically 50 kWh per unit in occupied buildings unless tests justify more), spacing, explosion control (deflagration venting per NFPA 68 or gas detection and ventilation per NFPA 69), fire suppression and emergency response plans. IEC 62933-5-2 covers the same ground internationally.
- **Electrical protection**: DC arc-fault and ground-fault detection, correctly rated DC fuses and breakers (Chapter 23), isolation monitoring, and the arc-flash assessment of Chapter 29. A 1500 V DC battery string can deliver tens of kiloamperes into a fault, with no current zero to help a breaker interrupt.

> **Status of the principle: Layered defence against thermal runaway**
> *Empirical.* The test-based approach of UL 9540A and NFPA 855 rests on observed propagation behaviour in full-scale fire tests; it bounds the consequences of a single-cell failure for the tested configuration, not for untested layouts, chemistries or ambient conditions.

The shift to LFP in stationary storage is partly a safety decision: its higher onset temperature and lower heat release make propagation slower and less violent than in NMC, though not impossible.

What remains open

Open question	Where it is answered
How storage fits into the structure, planning and adequacy of the whole power system	Chapter 13
How the synchronous machines of pumped-storage plants are built, excited and synchronised	Chapters 7 and 14
How the variability of wind and solar sets the storage duration a grid needs	Chapter 15 and Chapter 27
How storage is represented as a bus injection in a load-flow model	Chapters 17 and 18
How fast storage helps to arrest frequency decline after a generator trip	Chapter 20
How battery inverters provide grid-forming control, frequency and voltage support under grid codes	Chapter 27
How storage operates inside a microgrid and an energy management system	Chapter 26
How DC faults in a battery plant are detected and cleared	Chapter 23
How harmonics from electrolysers and battery inverters are limited	Chapter 25
How arc-flash and shock hazards of battery rooms are assessed	Chapter 29
Electrode materials, electrolytes and cell chemistry	*The Materials Core*
Bidirectional converters, battery chargers and balancing circuits	*Electronics & Computing*

Next: Chapter 17 opens Part V by building the per-unit system and the network models that let generation, storage, transformers and lines of every voltage level be analysed as a single network.

Part V — Power-System Analysis

Part IV described the sources of electrical energy one plant at a time. Part V treats the power system as what it really is: one large, interconnected circuit in which a generator in one region, a battery plant in another and a factory load hundreds of kilometres away all interact through the network. The engineer's questions now concern the whole: what voltages and flows result from a given pattern of generation and load, what currents flow when something short-circuits, and whether the system survives the disturbance and settles into a new steady state. Answering them requires a common language, the per-unit system and the bus-and-branch network model, and a set of calculations built on it: power flow for the normal state, fault analysis for the abnormal state, and stability analysis for the transition between them. These are the studies behind every connection agreement, switchgear rating, protection setting and operating limit in a modern grid.

By the end of this part you should understand

- Why power engineers work in per unit, how base values are chosen so that transformers disappear from the network, and how manufacturers' data are converted to a common system base.
- How generators, transformers (including off-nominal taps, phase shifts and three-winding units), lines, cables, loads, motors and inverter-based resources are modelled for steady-state, fault and stability studies.
- How a network is reduced to Thevenin and external equivalents, and how the bus admittance and impedance matrices are built, used and checked.
- How bus types and the power-balance equations define the load-flow problem, why it is nonlinear, and why a solution may not exist beyond the nose of the P–V curve.
- How the Gauss–Seidel, Newton–Raphson and fast-decoupled methods solve it, how fast each converges, and how voltage profiles and reactive-power controls are read from and enforced in the solution.
- How symmetrical and unsymmetrical fault currents are calculated with symmetrical components and sequence networks, and how they set the ratings of breakers, cables and switchgear.
- How rotor-angle, voltage and frequency stability are defined, assessed, simulated and improved.

Chapters

Chapter	Central question	What it delivers
Chapter 17 — Per-Unit Systems and Network Modeling	How is a system with many voltage levels and many kinds of equipment turned into one network that can be analysed?	Per-unit quantities and base values, change of base, single-line diagrams, equipment models, Thevenin and Kron reduction, the bus admittance and impedance matrices, data preparation and validation
Chapter 18 — Power-Flow Analysis	What voltages and flows result when loads and generators are specified as powers, and how are they kept within limits?	Bus types, power-balance equations, Gauss–Seidel, Newton–Raphson and fast-decoupled solutions, nose curves, voltage limits, reactive-power and tap-changer control, convergence and contingency analysis
Chapter 19 — Fault Analysis	How large are the currents when a network is short-circuited, and what ratings must equipment have to survive them?	Short-circuit transients, symmetrical faults, symmetrical components and sequence networks, unsymmetrical faults, fault-current calculation, equipment fault ratings, fault location
Chapter 20 — Power-System Stability	Will the system stay in synchronism, hold its voltage and arrest its frequency after a disturbance?	Rotor-angle, voltage and frequency stability, small-signal and transient stability, disturbance response, stability margins, improvement measures, simulation and its interpretation

How this part connects to the rest of the book

Part V turns the hardware of earlier parts into network models. The transformer equivalent circuits and vector groups of Chapter 4, the induction-motor model of Chapter 6 and the synchronous-machine reactances, capability and power-angle relations of Chapter 7 become per-unit equipment models; the grid structure of Chapter 13 and the generation and storage plants of Chapters 14–16 supply the buses and injections. Looking forward, Part VI uses the fault currents of Chapter 19 to set protection (Chapter 23) and the line parameters it develops (Chapter 21) to refine these models; distribution feeders, with their high R/X ratios and unbalance, extend the power-flow methods in Chapter 22. Part VII applies power flow and stability to renewable integration and smart grids (Chapters 26 and 27), and Part VIII uses fault levels and voltage drops in safety and design (Chapters 29 and 30).

Part V rests on two sibling books. Phasors, complex power, three-phase circuits, nodal analysis, two-port and transmission-line theory, and the Laplace-domain and feedback methods used in stability analysis come from *Circuits, Signals and Communications*, Chs. 3, 8–10, 12, 14 and 33–35. Rigid-body rotation, which underlies the swing equation, comes from *Engineering Physics: Top Down*, Chs. 9 and 19. The numerical methods themselves, Newton's method, sparse linear algebra and the integration of differential equations, belong to *The Mathematics Universe*.

Chapter 17 — Per-Unit Systems and Network Modeling

Part V · Power-System Analysis

A consulting engineer is asked whether a 5 MW induction motor can be added to the 6.6 kV board of a mineral-processing plant. The question sounds simple: will the board's voltage stay inside its limits, and can the switchgear still interrupt the fault current? But the data arrive in a dozen incompatible forms. The utility quotes a fault level of 3000 MVA at its 132 kV substation. The main transformer nameplate says 60 MVA, 132/33 kV, 12 % impedance. The 33 kV cable manufacturer gives ohms per kilometre. The plant transformer is 20 MVA, 33/6.6 kV, 8 %. The motor datasheet gives locked-rotor current as a multiple of full-load current. Adding these numbers directly is meaningless: an ohm on the 132 kV side is not the same as an ohm on the 6.6 kV side, and a percentage on a 60 MVA base is not the same as one on a 20 MVA base. Before any analysis can begin, the plant has to be turned into one network, on one set of bases, with one model for each piece of equipment. That translation is the subject of this chapter, and it underlies every load-flow, fault and stability study in Part V.

> **Why do power engineers work in per unit, and how are base values chosen so that transformers disappear from the network?**
> **How is each piece of equipment represented, and how is a whole system reduced to the bus-and-branch model a computer can solve?**

Concept	Engineering destination
Per-unit quantities	Nameplate impedances, relay settings, study reports, grid-code limits
Base values	Consistent analysis across 400 kV, 33 kV, 6.6 kV and 400 V in one network
Impedance conversion	Using manufacturers' data on a common system base
Single-line diagrams	Drawings that every study, protection scheme and switching order starts from
Equipment models	Generators, transformers, lines, cables, loads, motors and inverters in studies
Network reduction	Thevenin equivalents, fault levels, external-grid equivalents
Bus and branch representation	The admittance matrix solved by load-flow and fault programs
System-data preparation	Building and validating the study model that decisions rest on

Roadmap. What a per-unit quantity is and why it helps (17.1) → choosing bases so that ideal transformers vanish (17.2) → converting manufacturers' data to the system base (17.3) → the single-line diagram (17.4) → per-unit models of each kind of equipment (17.5) → reducing a network to equivalents (17.6) → the bus-and-branch model and the admittance matrix (17.7) → preparing and checking the data (17.8).

> **Boundary.**
> Phasors, complex power, the per-phase equivalent of a balanced three-phase system and nodal analysis come from *Circuits, Signals and Communications*: Ch. 8 (§8.7, AC nodal analysis), Ch. 9 (§9.4, complex power), Ch. 10 (§10.5, per-phase equivalent) and Ch. 3 (§3.8, matrix formulation and stamping). They are used here without re-derivation. Everything in this chapter assumes a balanced system solved per phase; unbalanced conditions need the sequence networks of Chapter 19.

17.1 Per-unit quantities

Definition

A **per-unit** (pu) quantity is the actual quantity divided by a chosen base quantity of the same dimension:

$$\text{quantity in pu} = \frac{\text{actual quantity}}{\text{base quantity}}.$$

A bus voltage of 10.45 kV on an 11 kV base is 0.95 pu; a current of 875 A on a 1000 A base is 0.875 pu; a load of 45 MW on a 100 MVA base is 0.45 pu. Percent is per unit times 100, and transformer and machine nameplates traditionally use it: "12 % impedance" means 0.12 pu on the transformer's own rating.

Per-unit values are complex where the quantity is complex. The base is a real number, so the angle is unchanged: $\bar{V} = 10.45\angle -5^\circ$ kV becomes $0.95\angle -5^\circ$ pu.

Why per unit

Per-unit normalisation is not a convenience of hand calculation that computers made obsolete; every load-flow, fault and stability program works internally in per unit, and every study report and grid code states its results that way. The reasons are practical:

1. **Ideal transformers disappear.** With bases chosen correctly (Section 17.2), a transformer's turns ratio vanishes from the per-unit network, and a system with five voltage levels becomes one circuit.
2. **Equipment parameters fall into narrow ranges.** A 2 MVA distribution transformer and a 1000 MVA generator step-up transformer have impedances of a few ohms (referred to 11 kV) and a few tens of ohms (referred to 400 kV) respectively, but 6 % and 15 % on their own ratings. An engineer who sees a transformer impedance of 0.6 % or 60 % knows immediately that the data are wrong.
3. **Normal operating values are near 1.** Bus voltages are 0.9–1.1 pu, so violations and errors stand out.
4. **The $\sqrt{3}$ factors vanish.** With the base choices of Section 17.2, three-phase and per-phase per-unit quantities are numerically the same.
5. **Numerical conditioning improves.** Quantities near unity, rather than spanning 10^{-3} Ω to 10^{5} V, keep the matrices of Chapters 18–20 well scaled.

Table 17.1 — Typical per-unit impedances of power-system equipment, on the equipment's own rating (typical ranges).

Equipment	Parameter	Typical value (pu, own base)
Turbo-generator (2-pole, 100–1500 MVA)	X_d / X'_d / X''_d	1.6–2.3 / 0.2–0.35 / 0.15–0.25
Salient-pole hydro generator	X_d / X'_d / X''_d	0.8–1.3 / 0.25–0.4 / 0.18–0.3
Distribution transformer (up to 2.5 MVA)	Z	0.04–0.06
MV/MV and HV/MV transformer (5–100 MVA)	Z	0.07–0.14
Generator step-up and EHV transformer	Z	0.12–0.18
Induction motor (locked rotor)	$Z_{LR} \approx X''$	0.15–0.20 (locked-rotor current 5–7 pu)
Current-limiting reactor	X	0.03–0.10
Wind or PV inverter (fault contribution)	current limit	1.0–1.5 pu of rated current

Typical X/R ratios rise with rating: 5–10 for distribution transformers, 20–50 for large transmission transformers and generators.

17.2 Base values

Choosing the bases

Four quantities are linked by two circuit laws, $S = VI^*$ and $V = ZI$, so only two bases can be chosen freely. Power-system practice chooses:

- **Base power** S_b: one value for the whole system, usually 100 MVA for transmission studies, 10 MVA or 1 MVA for industrial and distribution studies. It is a three-phase power.
- **Base voltage** V_b: one value per **voltage zone**, a zone being a part of the network bounded by transformers. It is a line-to-line voltage, normally the nominal system voltage of that zone.

The remaining bases follow:

$$I_b = \frac{S_b}{\sqrt{3}\,V_b}, \qquad Z_b = \frac{V_b^2}{S_b}, \qquad Y_b = \frac{1}{Z_b}.$$

With line-to-line V_b in kV and S_b in MVA, Z_b is in ohms and I_b in kA. A per-phase impedance in ohms is converted by $Z_{\text{pu}} = Z_\Omega / Z_b$.

Why the $\sqrt{3}$ vanishes

The per-phase base quantities are $S_b/3$ and $V_b/\sqrt{3}$, giving the same I_b and the same $Z_b = (V_b/\sqrt{3})^2/(S_b/3) = V_b^2/S_b$. A line-to-line voltage and the corresponding line-to-neutral voltage have the same per-unit value, and the three-phase power and the per-phase power have the same per-unit value. A per-unit per-phase calculation therefore gives three-phase answers directly: $S_{3\phi,\text{pu}} = V_{\text{pu}} I^*_{\text{pu}}$, with no factor of 3 or $\sqrt{3}$.

Why transformers disappear

Take an ideal transformer of ratio $V_1 : V_2$ (line-to-line, rated) and choose base voltages in the two zones in the same ratio, $V_{b1}/V_{b2} = V_1/V_2$, with the same S_b on both sides. Then a voltage on side 1 and the voltage it produces on side 2 have the same per-unit value, and so do the corresponding currents, because $I_{b1}/I_{b2} =$

V_{b2}/V_{b1} matches the current ratio. The ideal transformer becomes a 1:1 connection, and only its leakage impedance, the same per-unit value seen from either side, remains in the circuit. An impedance Z referred across the transformer changes by the square of the ratio in ohms, but its per-unit value does not change, because Z_b changes by the same square.

> **Status of the principle: Per-unit normalisation**
> *Derived*. A change of units applied to linear circuit equations: with one base power and zone base voltages in the ratio of the transformer turns ratios, ideal transformers become unity ratio and per-phase per-unit quantities equal three-phase ones. It is exact; any error comes from transformers whose actual ratio differs from the zone base ratio (Section 17.3) or from phase shifts in star–delta transformers (Section 17.5).

Table 17.2 — Base impedance and base current at $S_b = 100$ MVA.

Base voltage (kV, line-to-line)	Z_b (Ω)	I_b (A)
400	1600	144.3
132	174.24	437.4
33	10.89	1,749.5
11	1.21	5,248.6
6.6	0.4356	8,747.7
0.69	0.004761	83,674
0.4	0.0016	144,338

The table makes the scaling problem visible. One ohm is 0.000625 pu at 400 kV and 625 pu at 400 V; a network whose impedances were left in ohms would mix values six orders of magnitude apart.

17.3 Impedance conversion

Change of base

Manufacturers give impedances in per unit (or percent) on the equipment's own rating, S_{rated} and V_{rated}. The study needs them on the system bases S_b and V_b. Since the ohmic value is the same,

$$Z_{\text{pu,new}} = Z_{\text{pu,old}} \, \frac{S_{b,\text{new}}}{S_{b,\text{old}}} \, \left(\frac{V_{b,\text{old}}}{V_{b,\text{new}}} \right)^2 .$$

A larger base power makes per-unit impedances larger; a larger base voltage makes them smaller. The voltage factor equals one when the equipment rated voltage equals the zone base, which is why zone bases are usually set to nominal voltages.

> **Status of the principle: Change-of-base formula**
> *Derived*. Follows from $Z_{\text{pu}} = Z_\Omega S_b / V_b^2$ applied with both sets of bases to the same ohmic impedance; exact for any linear impedance.

Transformers whose ratio does not match the bases

Choose a 230 kV base for a transmission zone and the zone bases are fixed by the transformer ratios. If the generator step-up transformer is 14.4/230 kV, the generator zone has a 14.4 kV base, even though the generator is rated 13.8 kV. Its nameplate reactance must then be converted with the voltage factor $(13.8/14.4)^2$. Where two transformers of different ratio connect the same two zones, or a transformer is on a tap other than nominal, the bases cannot match every ratio. The mismatch is represented by an **off-nominal tap ratio** in the transformer model (Section 17.5), never by bending the bases.

Worked example — Generator reactance on the system base. A generator is rated 125 MVA, 13.8 kV, with subtransient reactance $X''_d = 0.18$ pu on its own rating. It connects through a 14.4/230 kV step-up transformer, and the study uses $S_b = 100$ MVA with a 230 kV base in the transmission zone.

Zone base at the generator: $230 \times 14.4/230 = 14.4$ kV.

Converted reactance:

$$X''_{d} = 0.18 \times \frac{100}{125} \times \left(\frac{13.8}{14.4}\right)^{2} = 0.18 \times 0.8 \times 0.9184 = 0.1323 \text{ pu}.$$

Check in ohms: on its own rating, $0.18 \times 13.8^2/125 = 0.274\ \Omega$; on the system base, $Z_b = 14.4^2/100 = 2.0736\ \Omega$ and $0.274/2.0736 = 0.1323$ pu. Omitting the voltage correction would overstate the reactance by 8.9 % and understate the generator's fault contribution by about 8 %.

Impedances given in ohms

Line and cable data come in ohms per kilometre. Multiply by length and divide by the zone Z_b. A 5 km, 33 kV cable with $0.13 + j0.11\ \Omega$/km has $0.65 + j0.55\ \Omega$, and on a 100 MVA base ($Z_b = 10.89\ \Omega$) $0.0597 + j0.0505$ pu. Shunt susceptance in siemens is multiplied by Z_b.

Fault levels and percent impedance

A network infeed is usually described by its three-phase **fault level** S_k (MVA) at nominal voltage, and an X/R ratio. Its Thevenin impedance is

$$Z_{\text{pu}} = \frac{S_b}{S_k},$$

split into R and X by the X/R ratio. Conversely, a transformer of impedance z pu on its own rating S_T, fed from an infinite bus, would pass a fault level of S_T/z: 60 MVA at 12 % limits a secondary fault to 500 MVA. This one-line arithmetic is how engineers first size switchgear, long before a full study.

17.4 Single-line diagrams

What the diagram shows

A **single-line diagram** (SLD, one-line diagram) represents a balanced three-phase system by one line per circuit, with standard symbols for each item of equipment (IEC 60617 or IEEE 315). It is the master document of an installation: studies, protection schemes, switching orders and arc-flash labels all refer to it. Figure 17.1 shows the plant of the opening problem.

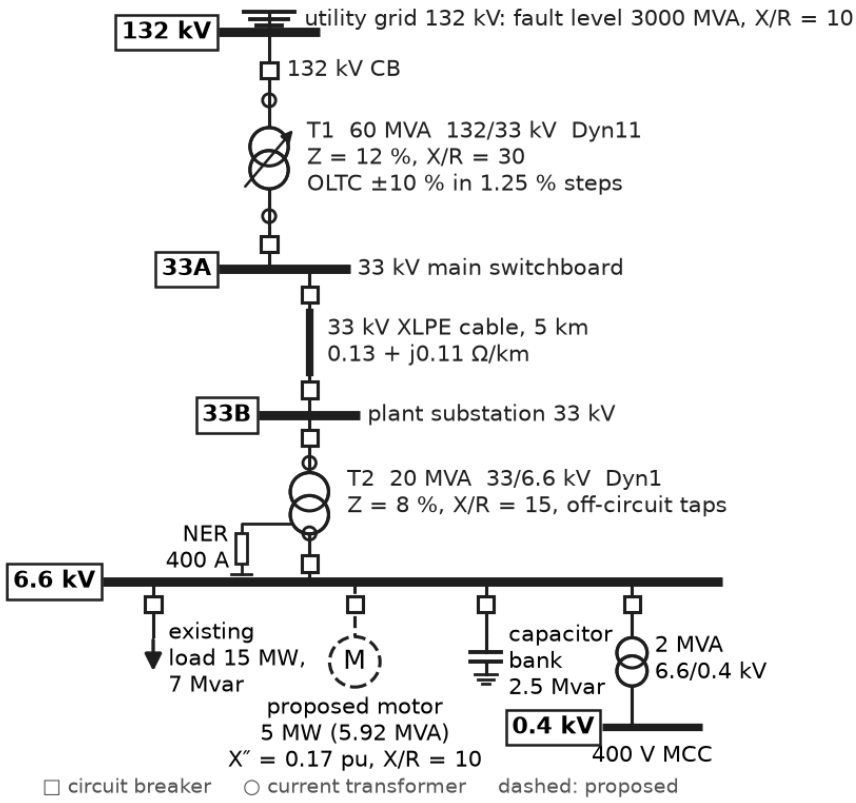


Figure 17.1 — Single-line diagram of a mineral-processing plant supplied at 132 kV, used throughout the chapter.

A complete SLD shows, for each item: rating (MVA, kV, A), impedance in percent or ohms, vector group and earthing of transformers, tap range, breaker and fuse ratings, current and voltage transformer ratios, cable sizes and lengths, and the normal open points. A study SLD may omit protection details but must show every impedance that the study uses.

Conventions worth knowing

- **Breakers and switches** are drawn in their normal state; normally open points (between two feeders in a ring, Chapter 22) are marked.
- **Vector groups** such as Dyn11 or YNd1 record winding connections and the 30° phase shifts they introduce (Section 17.5). A positive-sequence study can ignore the shift if it is the same around every loop; a negative-sequence or unbalanced study cannot.
- **Earthing** of transformer neutrals (solid, resistance, reactance or isolated) does not affect balanced studies at all, but it determines earth-fault currents (Chapter 19) and must be on the diagram.
- **Bus names and numbers** used in the SLD should match those in the study model, so that results can be traced back to the hardware.

17.5 Equipment models

Each item of equipment is represented in a balanced study by a per-phase per-unit model. The models are chosen for the question asked: steady-state load flow, short circuit or transient stability needs different detail from the same machine. Table 17.3 summarises.

Table 17.3 — Positive-sequence equipment models used in system studies.

Equipment	Load flow (Chapter 18)	Short circuit (Chapter 19)	Stability (Chapter 20)
Synchronous generator	P and V (or P and Q) specified at terminals, within capability limits	Voltage behind X''_d	Voltage behind X'_d or detailed d–q model with controls
Transformer	Series $R + jX$, off-nominal tap, phase shift	Series $R + jX$	Series $R + jX$, tap changer dynamics if slow
Overhead line, cable	Nominal or equivalent π	Series impedance (shunt often neglected)	π model
Load	Constant P and Q, or ZIP / exponential in V	Usually neglected (static), or constant impedance	Voltage- and frequency-dependent, motor models
Induction motor	Load (P, Q)	Voltage behind X'' for a few cycles	Dynamic motor model
Inverter-based resource	P and Q or V control	Current-limited source, 1.0–1.5 pu	Control-loop model per grid code
Shunt capacitor or reactor	Constant admittance	Constant admittance (capacitor discharge neglected)	Constant admittance

> **Status of the principle: Lumped per-phase equipment models**
> *Approximate.* Each model assumes balanced, fundamental-frequency, quasi-steady operation and linear magnetics; they fail under unbalance (use sequence models), during fast electromagnetic transients (use distributed and frequency-dependent models), and where saturation, harmonics or converter control limits dominate.

Synchronous generators

The machine models of Chapter 7 enter the network in three forms. For load flow the generator is a terminal constraint: it holds a scheduled active power and a voltage set-point while its reactive output stays within the capability curve of Chapter 18. For short circuit it is an internal voltage behind the subtransient reactance X''_d (first cycles) or transient reactance X'_d. For stability it is a dynamic model with excitation and governor controls. In each case the reactances come from the manufacturer on the machine's rating and are converted to the system base as in Section 17.3.

Two-winding transformers and tap changers

The approximate equivalent circuit of Chapter 4 reduces, in per unit, to a series leakage impedance $R + jX$; magnetising current (0.5–2 % of rated) is neglected in most system studies, or modelled as a small shunt admittance where no-load losses matter. Two refinements remain.

Off-nominal tap. Let the transformer have per-unit series admittance $y = 1/(R + jX)$ and an ideal transformer of ratio $a : 1$ on its tap side i, where a is the actual turns ratio divided by the ratio of the zone bases. The bus admittance entries are

$$Y_{ii} = \frac{y}{a^2}, \qquad Y_{jj} = y, \qquad Y_{ij} = Y_{ji} = -\frac{y}{a}.$$

These are reproduced exactly by the π circuit of Figure 17.2: a series branch y/a, a shunt $y(1-a)/a^2$ at the tap side and a shunt $y(a-1)/a$ at the other side. When $a = 1$ the shunts vanish. When $a \neq 1$ one shunt is capacitive and the other inductive; they have no physical existence and represent the voltage transformation of the tap.

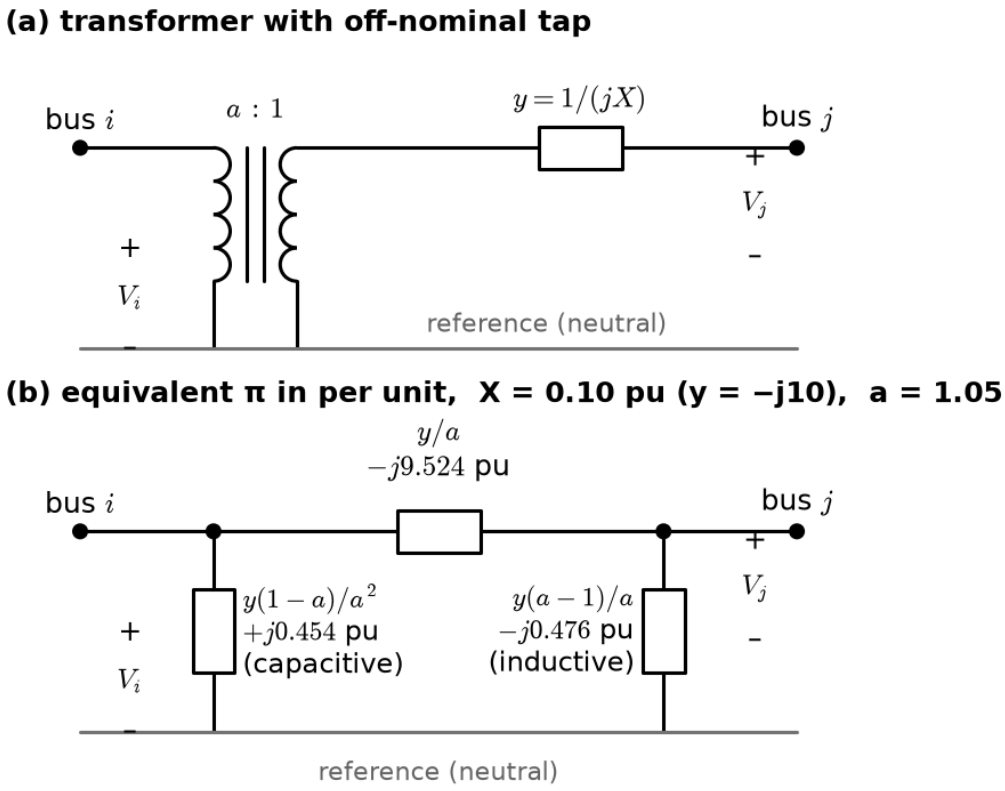


Figure 17.2 — Transformer with off-nominal tap ratio a:1 and its equivalent pi circuit in per unit.

For example, a transformer with $X = 0.10$ pu ($y = -j10$) on a tap of $a = 1.05$ gives $Y_{ii} = -j9.070$, $Y_{jj} = -j10$, $Y_{ij} = +j9.524$ pu; the π shunts are $+j0.454$ pu at the tap side and $-j0.476$ pu at the other. On-load tap changers (typically ±10 % in 16 or more steps of 1.25 %) make a a control variable that the load flow of Chapter 18 adjusts to hold a bus voltage.

Phase shift. A star–delta transformer shifts positive-sequence voltages by 30° (Dyn11: low side leads by 30°; YNd1: low side lags by 30°). In a radial network the shift does not change magnitudes or power flows and is usually dropped. In meshed networks, and for phase-shifting transformers built to control power flow, the ratio becomes complex, $a = |a|\angle\theta$, and $Y_{ij} \neq Y_{ji}$: the admittance matrix loses its symmetry.

Three-winding transformers

A three-winding transformer (for example 132/33/11 kV, with the 11 kV tertiary feeding a capacitor bank or station supply) is tested in three short-circuit tests, each between a pair of windings with the third open, giving Z_{ps}, Z_{pt} and Z_{st}. After conversion to a common base, the star equivalent of Figure 17.3 has

$$Z_p = \tfrac{1}{2}\left(Z_{ps} + Z_{pt} - Z_{st}\right), \quad Z_s = \tfrac{1}{2}\left(Z_{ps} + Z_{st} - Z_{pt}\right), \quad Z_t = \tfrac{1}{2}\left(Z_{pt} + Z_{st} - Z_{ps}\right).$$

A typical set, $Z_{ps} = 12$ % on 60 MVA, $Z_{pt} = 20$ % on 60 MVA and $Z_{st} = 8$ % on 20 MVA, converts on a 100 MVA base to 0.200, 0.333 and 0.400 pu, giving $Z_p = 0.067$, $Z_s = 0.133$ and $Z_t = 0.267$ pu. One branch of the star can come out small or even negative; this is normal, because the star point is fictitious.

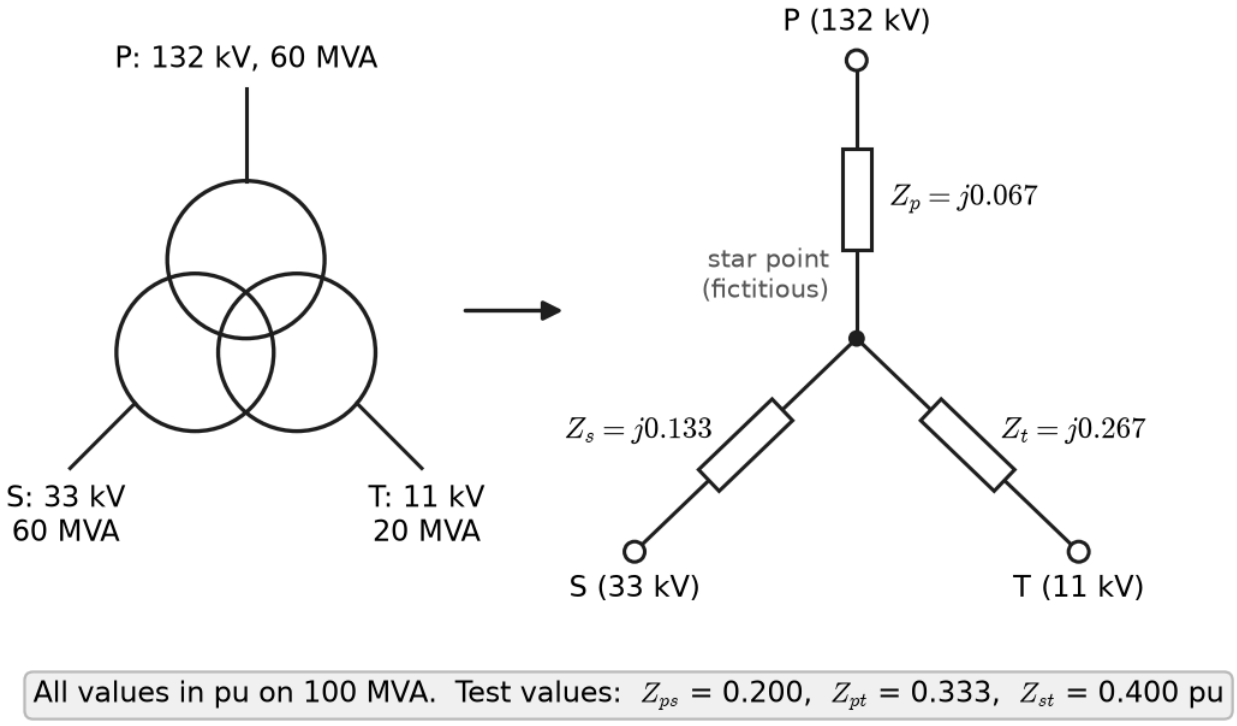


Figure 17.3 — Star equivalent circuit of a three-winding transformer in per unit.

Lines and cables

Overhead lines and cables are represented by π circuits: series impedance $R + jX$ and half the total shunt susceptance $B/2$ at each end. Lines shorter than about 80 km (overhead) may drop the shunt branch; lines longer than about 250 km need the hyperbolic correction to give an equivalent π. The wave theory behind these models is in *Circuits, Signals and Communications*, Ch. 14, the two-port forms in Ch. 12 (§12.3 and §12.5), and the calculation of line parameters from conductor geometry in Chapter 21. Cables have much larger capacitance than lines of the same voltage (a 132 kV cable generates 1–2 Mvar per km against about 0.05 Mvar per km for a line), which matters for voltage control and is never neglected.

Loads

Loads are the least certain part of any model. The aggregate load at a substation is a mixture of motors, lighting, electronics, heating and drives whose power depends on voltage (and, more weakly, frequency). Two families of static model are used:

$$P = P_0\left(a_Z\,\bar{V}^2 + a_I\,\bar{V} + a_P\right) \quad \text{(ZIP)}, \qquad P = P_0\,\bar{V}^{\,n_p}, \quad Q = Q_0\,\bar{V}^{\,n_q} \quad \text{(exponential)},$$

with $\bar{V}$ the voltage in per unit of its nominal value and $a_Z + a_I + a_P = 1$. Constant power ($n_p = 0$), constant current ($n_p = 1$) and constant impedance ($n_p = 2$) are special cases. Figure 17.4 compares them. Constant-power load is the most pessimistic for voltage stability, because its current rises as voltage falls; constant-impedance load is the most optimistic.

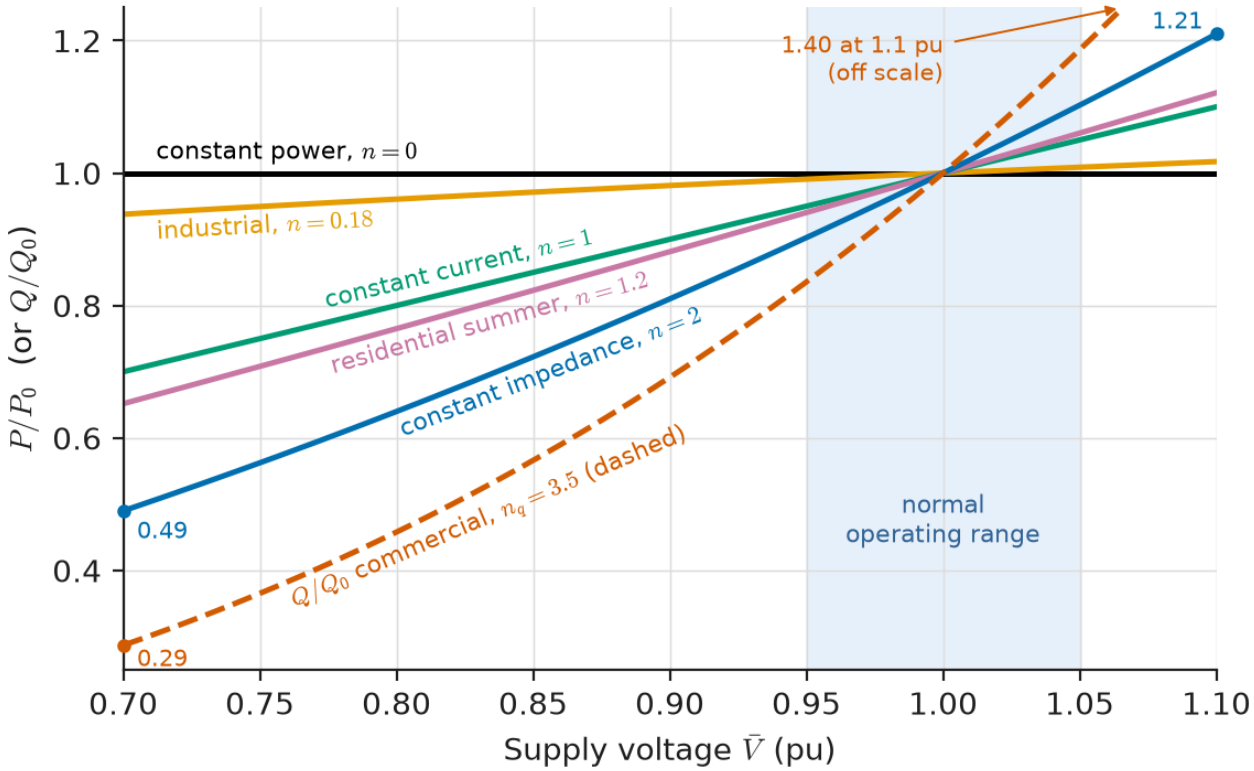


Figure 17.4 — Active power drawn by static load models as a function of supply voltage.

Table 17.4 — Typical voltage exponents of aggregate loads (after the IEEE Task Force on Load Representation; typical values).

Load class	n_p	n_q
Residential, summer (air-conditioning)	1.2	2.9
Residential, winter (heating)	1.5	3.2
Commercial, summer	0.99	3.5
Commercial, winter	1.3	3.1
Industrial	0.18	6.0
Power-plant auxiliaries	0.1	1.6

> **Status of the principle: Static load models**
> *Empirical.* ZIP and exponential coefficients are fitted to field measurements of voltage steps at substations; they vary with season, time of day and the growing share of power-electronic loads (which behave closer to constant power), and they say nothing about dynamic motor stalling.

Motors, inverters and shunt devices

An induction motor (Chapter 6) appears in a load flow as a load and in a short-circuit study as a source behind its locked-rotor reactance for the first few cycles, contributing 4–6 times its rated current. Inverter-based resources (PV, wind, batteries) appear in load flow as P–Q or P–V injections and in short-circuit studies as current-limited sources, because their converters will not supply more than 1.0–1.5 times rated current. Shunt

capacitors and reactors are constant admittances, so their reactive output varies with V^2: a 10 Mvar capacitor bank at 0.95 pu supplies only 9.0 Mvar.

17.6 Network reduction

Series, parallel and star–delta

Once every element is in per unit on a common base, the network is an ordinary linear circuit and its reduction uses the familiar tools: series and parallel combination, star–delta transformation (*Circuits, Signals and Communications*, Ch. 4 §4.8) and Thevenin's theorem (§4.2). The engineering use of the result is the **Thevenin impedance at a bus**, which gives both the fault level at the bus and the sensitivity of the bus voltage to load.

Worked example — Fault level and voltage drop at the plant's 6.6 kV board. Return to Figure 17.1 with $S_b = 100$ MVA and zone bases 132, 33 and 6.6 kV.

Utility infeed: $Z = 100/3000 = 0.0333$ pu at X/R = 10: $0.00332 + j0.0332$ pu.

T1: $0.12 \times 100/60 = 0.200$ pu at X/R = 30: $0.00666 + j0.1999$ pu.

Cable: $5 \times (0.13 + j0.11)/10.89 = 0.0597 + j0.0505$ pu.

T2: $0.08 \times 100/20 = 0.400$ pu at X/R = 15: $0.0266 + j0.3991$ pu.

Thevenin impedance at the 6.6 kV board: the sum, $Z_{th} = 0.0963 + j0.6827$ pu, $|Z_{th}| = 0.6894$ pu.

Three-phase fault level (prefault voltage 1.0 pu, motor contributions ignored): $S_k = 100/0.6894 = 145$ MVA, $I_k = 8.748/0.6894 = 12.69$ kA, using $I_b = 8.748$ kA from Table 17.2. At the 33 kV main board (upstream of the cable) the same arithmetic gives $Z_{th} = 0.0100 + j0.2331$ pu, 429 MVA, 7.50 kA.

Adding the 5 MW motor. At rated output, 88 % power factor and 96 % efficiency the motor draws $5/(0.88 \times 0.96) = 5.92$ MVA; its subtransient reactance 0.17 pu on that rating becomes $0.17 \times 100/5.92 = 2.87$ pu. In parallel with Z_{th} (motor X/R = 10), the fault impedance falls to 0.557 pu, so the initial fault current rises to 15.7 kA, of which about 3.0 kA comes from the motor. If the 6.6 kV switchgear is rated 16 kA, the margin is now under 2 %, too small to accept without the detailed IEC 60909 calculation of Chapter 19.

Voltage drop. The existing load is 15 MW and 7 Mvar, $\bar{S} = 0.15 + j0.07$ pu. The approximation $\Delta V \approx RP + XQ = 0.0963 \times 0.15 + 0.6827 \times 0.07 = 0.062$ pu predicts a 6.2 % drop; solving $\bar{V} = 1 - Z_{th}(\bar{S}/\bar{V})^*$ iteratively gives $|\bar{V}| = 0.928$ pu. Most of the drop is across the two transformer reactances, which is why T1 carries an on-load tap changer. The motor will add more, so the decision depends on the tap range and on a load flow (Chapter 18).

The whole calculation used nothing beyond series addition. Done in ohms, it would have required referring each impedance to the 6.6 kV side by the square of two turns ratios, and a single slip in that referral is the classic error in hand fault calculations.

Kron reduction

When a network is described by its bus admittance matrix (Section 17.7), buses that have no injection, such as a tee point or a busbar with no load or generation, can be eliminated algebraically. Partition the nodal

equations into retained buses r and eliminated buses e with $\mathbf{I}_e = \mathbf{0}$:

$$\begin{bmatrix} \mathbf{I}_r \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{rr} & \mathbf{Y}_{re} \\ \mathbf{Y}_{er} & \mathbf{Y}_{ee} \end{bmatrix} \begin{bmatrix} \mathbf{V}_r \\ \mathbf{V}_e \end{bmatrix} \quad \Longrightarrow \quad \mathbf{Y}_{\text{red}} = \mathbf{Y}_{rr} - \mathbf{Y}_{re}\,\mathbf{Y}_{ee}^{-1}\,\mathbf{Y}_{er}.$$

The reduced matrix reproduces exactly the relation between currents and voltages at the retained buses. Eliminating one bus at a time is Gaussian elimination, and Kron reduction is the matrix form of repeated star–delta transformation.

> **Status of the principle: Kron reduction**
> *Derived*. Exact elimination of nodes with zero injection from linear nodal equations; it remains exact only while the eliminated buses carry no load or generation, so a load bus may be eliminated only if its load is first converted to a constant admittance.

External equivalents

A utility studying its own network cannot model the whole interconnection, yet the neighbouring systems affect its flows and fault levels. It replaces them with an **external equivalent**: in the simplest form a Thevenin source at each boundary bus with the fault level supplied by the neighbour; for load flow a **Ward equivalent** (a Kron reduction of the external network, with external injections converted to equivalent boundary injections), or a REI equivalent that preserves aggregated generation. Equivalents are accurate for the operating point at which they were built and degrade as conditions move away from it, which is why interconnected operators exchange full models through common data formats (Section 17.8).

17.7 Bus and branch representation

From node-breaker to bus-branch

Real substations contain busbars, breakers and isolators in arrangements such as breaker-and-a-half or double busbar. A **node–breaker** model represents every switching device; it is used by control-centre topology processors. A **bus–branch** model merges every set of nodes connected by closed switches into one electrical bus and keeps only impedance branches (lines, cables, transformers, series reactors) between buses, and shunt elements (loads, generators, capacitors, reactors) at buses. Every load-flow, fault and stability program solves the bus–branch model; a topology processor rebuilds it whenever a breaker changes state.

Building the admittance matrix

The bus admittance matrix $\mathbf{Y}_{\text{bus}}$ is the complex nodal matrix of *Circuits, Signals and Communications*, Ch. 8 §8.7, assembled by stamping each branch as in Ch. 3 §3.8. In power-system form, with the ground as reference and every quantity in per unit:

- a π branch between buses i and k with series admittance y_{ik} and total charging jB_{ik} adds $y_{ik} + jB_{ik}/2$ to Y_{ii} and Y_{kk}, and $-y_{ik}$ to Y_{ik} and Y_{ki};
- a transformer branch with tap a adds y/a^2, y and $-y/a$ as in Section 17.5;
- a shunt admittance at bus i adds to Y_{ii} only.

The resulting matrix has properties that every power-system program exploits:

1. **Sparse.** Each bus connects to two to four others on average, so a 10,000-bus model has about 99.96 % zero entries. Sparse storage and ordering are what make large studies feasible.
2. **Symmetric** unless the network contains phase-shifting transformers.
3. **Diagonally dominant** in its imaginary part, because transmission branches are mainly inductive.
4. **Nearly singular** if shunts are small: without connections to ground, every row sums to zero. Generator and load admittances, line charging and fault impedances supply the ground connections.

Worked example — Admittance matrix and fault levels of a four-bus network. A 132 kV regional network (Figure 17.5) has a generator at bus 1, a grid infeed at bus 2, a load substation at bus 3 and a passive tee point at bus 4. Branch impedances on 100 MVA are $z_{14} = 0.02 + j0.08$, $z_{24} = 0.01 + j0.06$, $z_{34} = 0.015 + j0.09$ and $z_{23} = 0.02 + j0.12$ pu; line charging is neglected for this example.

Branch admittances: $y_{14} = 2.941 - j11.765$, $y_{24} = 2.703 - j16.216$, $y_{34} = 1.802 - j10.811$, $y_{23} = 1.351 - j8.108$ pu.

Admittance matrix by stamping:

$$\mathbf{Y}_{\text{bus}} = \begin{bmatrix} 2.941 - j11.765 & 0 & 0 & -2.941 + j11.765 \\ 0 & 4.054 - j24.324 & -1.351 + j8.108 & -2.703 + j16.216 \\ 0 & -1.351 + j8.108 & 3.153 - j18.919 & -1.802 + j10.811 \\ -2.941 + j11.765 & -2.703 + j16.216 & -1.802 + j10.811 & 7.446 - j38.792 \end{bmatrix} \text{ pu.}$$

Each row sums to zero, as expected with no shunt elements; six of the sixteen entries are zero.

Kron reduction of the tee point. Eliminating bus 4 gives a 3×3 matrix with off-diagonal entries $-1.104 + j4.925$ (buses 1–2), $-0.736 + j3.283$ (1–3) and $-1.991 + j12.625$ (2–3). The equivalent branch impedances are the negative reciprocals: $0.043 + j0.193$, $0.065 + j0.290$ and $0.0122 + j0.0773$ pu. The last combines the direct 2–3 line with the path through the tee.

Fault levels. Add the source impedances as shunts to ground: the generator's $X''_d = 0.20$ pu at bus 1 and the grid infeed's 0.05 pu at bus 2 (the admittances $-j5$ and $-j20$). Inverting the reduced matrix gives the **bus impedance matrix** $\mathbf{Z}_{\text{bus}}$, whose diagonal entry Z_{kk} is the Thevenin impedance at bus k. The results are $|Z_{11}| = 0.0947$, $|Z_{22}| = 0.0434$ and $|Z_{33}| = 0.1016$ pu, so the three-phase fault levels are 1056 MVA at bus 1, 2304 MVA at bus 2 and 984 MVA at bus 3, or 4.62, 10.08 and 4.30 kA at 132 kV ($I_b = 437.4$ A). Inverting the full 4×4 matrix gives the same Z_{33}, confirming that Kron reduction lost nothing.

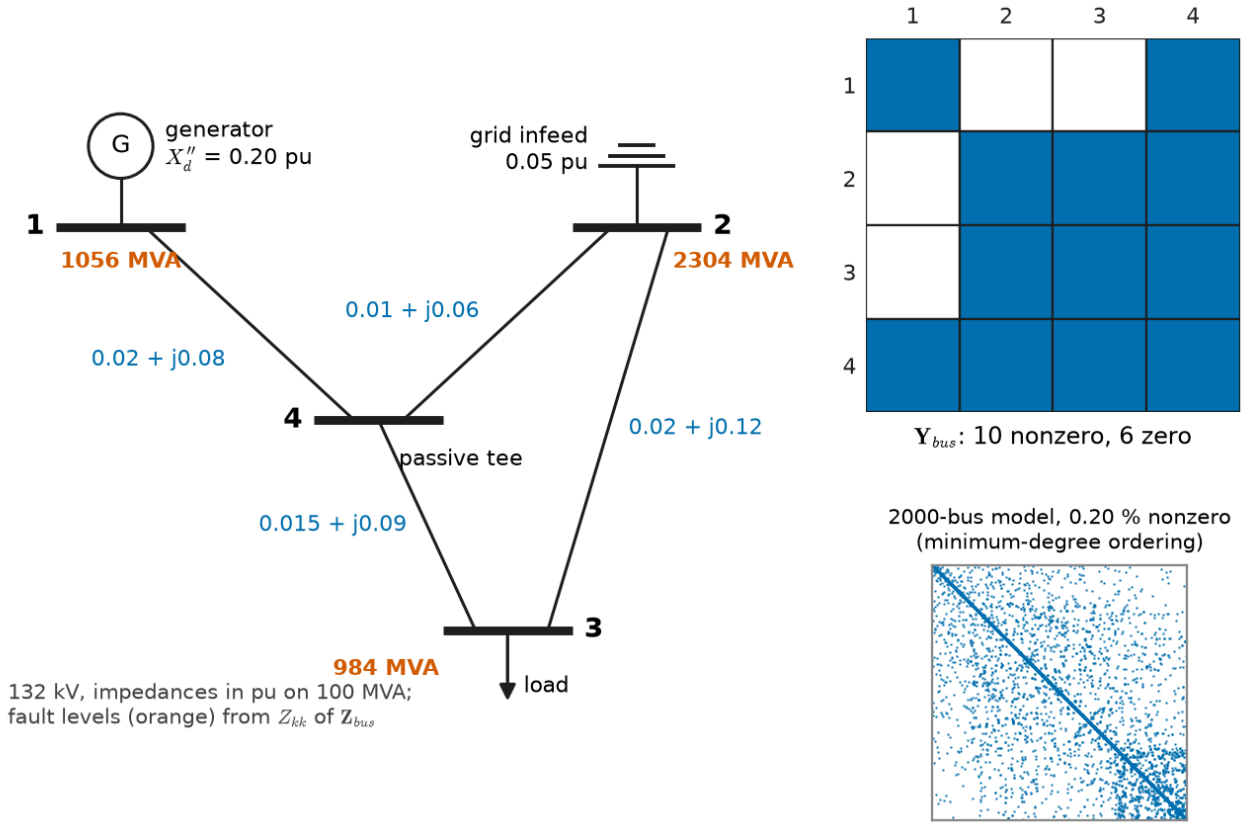


Figure 17.5 — Four-bus 132 kV network of the worked example with its admittance-matrix sparsity pattern.

The bus impedance matrix

$\mathbf{Z}_{\text{bus}} = \mathbf{Y}_{\text{bus}}^{-1}$ is full, not sparse, and is never formed explicitly for a large network; programs compute the columns they need from the sparse factors of $\mathbf{Y}_{\text{bus}}$. Its meaning is direct: Z_{kk} is the Thevenin impedance at bus k, and Z_{ik} is the voltage change at bus i per unit current injected at bus k. Fault calculations (Chapter 19) are built on it, and so are many sensitivity calculations: how much a bus voltage changes when a capacitor bank switches, or how a new generator raises fault levels across a region.

> **Status of the principle: Bus admittance and impedance matrices**
> *Derived*. Nodal analysis of the per-unit, per-phase network; exact for the linear model. Load-flow nonlinearity enters only through the injections (Chapter 18), not through $\mathbf{Y}_{\text{bus}}$.

17.8 System-data preparation

Where the data come from

The study model is only as good as its data, and most errors in power-system studies are data errors, not method errors. Data come from:

- **Nameplates and test reports**: transformer impedance, losses and vector group from factory tests; generator reactances and time constants from type tests; motor ratings.

- **Line and cable constants**: calculated from conductor type, tower geometry or cable construction (Chapter 21) and, where available, checked by field measurement.
- **Utility data**: fault levels and X/R ratios at the connection point, minimum and maximum, for present and future system states.
- **Load data**: metered peak and minimum demand, power factor, and load composition for model selection.
- **Control settings**: tap-changer ranges and set-points, generator voltage set-points and capability curves, inverter control modes required by the grid code.

Formats and exchange

Small studies are entered by hand in a study program. Large models are exchanged in standard formats: the IEEE Common Data Format (from 1973, still used for test systems), vendor formats such as PSS/E RAW, and the Common Information Model of IEC 61970 and IEC 61968, which European transmission operators use to merge national models into a pan-European grid model. The IEEE 14-, 30-, 118- and 300-bus test cases are standard benchmarks for new methods.

A preparation workflow

For the plant of Figure 17.1, as for a 2,000-bus utility model, the steps are the same:

1. **Fix the scope and the study cases.** Which buses, which future years, maximum and minimum load, which generation dispatch, which outages. The utility supplies minimum and maximum fault levels, because the minimum sets protection sensitivity and the maximum sets switchgear duty.
2. **Choose the bases.** One S_b (100 MVA here) and the nominal voltage of each zone; list every transformer whose ratio differs from the zone bases, so that its off-nominal tap is entered deliberately.
3. **Convert every impedance.** Nameplate percent with the change-of-base formula, cable and line ohms with the zone Z_b, infeeds from fault level and X/R. Keep the conversion sheet with the model, so that anyone can trace a per-unit value back to its source document.
4. **Enter loads with a stated model.** Constant power for planning load flows, voltage-dependent models where low voltages are expected, and motor contributions for fault studies.
5. **Check and reconcile.** Run the checks of Table 17.5, then compare with measurement.
6. **Record assumptions.** Tap positions, open points, load power factors and the date of every source document belong in the study report (Chapter 30).

Validation checks

Before a model is used for a decision it is checked. Table 17.5 lists the errors that most often survive into studies and the checks that catch them.

Table 17.5 — Common data errors and how they are caught.

Error	Symptom	Check
Impedance on wrong MVA base (own rating used as system base)	Fault levels wrong by the ratio of the bases	Compare every transformer impedance with Table 17.1 range on own rating
Ohms entered on wrong side of transformer	Impedance wrong by ratio squared (e.g. ×25 for 33/6.6 kV)	Recompute per-unit from ohms with zone Z_b; sanity-check X/R
Percent entered as per unit	Impedance ×100	Values of 5–15 pu for a transformer are impossible
Line charging entered in μS instead of S, or as total instead of half	Large reactive mismatch, voltage rise on light load	Charging Mvar per km against typical (0.05 Mvar/km at 132 kV OHL)
Wrong vector group or omitted phase shift in a loop	Circulating power around loop	Sum of shifts around each loop must be zero
Tap position not matching operation	Voltages off by 1–5 %	Compare with SCADA measurements
Islanded bus or open branch	Load flow fails or singular matrix	Connectivity check, island detection
Load model all constant power at low voltage	Non-convergence near collapse	Use voltage-dependent models below 0.7–0.8 pu

The final validation compares the solved model with measurement: a load flow for a recorded system state should reproduce measured line flows to within a few per cent and bus voltages to within about 0.01 pu. A short-circuit model can be checked against recorded fault currents from protection relays. A model that has never been checked against measurement should be treated as a hypothesis.

> **Status of the principle: Model validation against measurement**
> *Empirical*. No derivation guarantees that a network model represents the real system; agreement with recorded flows, voltages and fault currents is the only evidence, and it holds for the conditions tested.

What remains open

Open question	Where it is answered
How bus voltages and power flows are found when loads are specified as powers, not impedances	Chapter 18
How unbalanced faults are analysed with sequence networks, and how IEC 60909 corrects the simple fault level	Chapter 19
How generator dynamic models are used to judge whether a system stays in synchronism	Chapter 20
How line resistance, inductance and capacitance are computed from conductor geometry; long-line models	Chapter 21
How distribution feeders with high R/X ratios and unbalanced loads are modelled	Chapter 22
How CT and VT ratios and per-unit impedances become relay settings	Chapter 23
How equipment ratings are chosen from fault levels and voltage drops in design	Chapter 30
Transformer equivalent circuit, tests and vector groups	Chapter 4
Synchronous-machine reactances and their physical meaning	Chapter 7
Two-port and transmission-line wave theory	*Circuits, Signals and Communications*, Ch. 12 and Ch. 14

Next: Chapter 18 takes the bus admittance matrix built here and solves the nonlinear power-balance equations that determine voltages and flows when loads and generators are specified as powers.

Chapter 18 — Power-Flow Analysis

Part V · Power-System Analysis

A transmission planner receives a connection request: a 300 MW data-centre campus wants to connect at a 230 kV substation two years from now. Before saying yes, she must answer questions that no single-line diagram can settle by inspection. Will the voltage at the substation stay above 0.95 pu at peak load? Which lines will carry the extra power, and will any exceed its thermal rating? How much reactive power must the nearby generators supply, and will one of them hit its limit? Does the system still meet these limits with any one line out of service? Every one of these questions is answered by a **power-flow** (or **load-flow**) study, the most frequently run calculation in power engineering. Control centres run it every few minutes on the live network; planners run it thousands of times for future years; and every connection agreement for a wind farm, battery plant or factory rests on one. The difficulty is that loads and generators are specified as powers, not impedances or currents, and that turns a linear network into a set of nonlinear equations that must be solved iteratively.

> **What is specified and what is unknown at each bus, and why does specifying power make the network equations nonlinear?**
> **How do the Gauss–Seidel, Newton–Raphson and fast-decoupled methods solve them, and how fast? How are voltage profiles and reactive power read from the solution and managed in practice?**

Concept	Engineering destination
Bus classifications	Modelling generators, loads, the reference machine and voltage-controlled buses
Power-balance equations	The physics every power-flow program solves
Load-flow formulation	Setting up a study: unknowns, mismatches, limits
Gauss–Seidel method	Simple solvers, distribution and teaching tools, initial estimates
Newton–Raphson method	Industry-standard solver in planning and control-centre software
Fast-decoupled load flow	Contingency screening and real-time security analysis
Voltage profiles	Compliance with ANSI C84.1, IEC 60038 and grid-code voltage limits
Reactive-power management	Capacitor banks, tap changers, generator capability, SVCs and STATCOMs
Convergence and interpretation	Trusting, reading and acting on the results

Roadmap. What is known and unknown at each bus (18.1) → the power-balance equations (18.2) → assembling them into a solvable problem, and why a solution may not exist (18.3) → three ways to solve it: Gauss–Seidel (18.4), Newton–Raphson (18.5) and fast-decoupled (18.6) → reading voltage profiles (18.7) → controlling them with reactive power (18.8) → convergence, accuracy and interpretation (18.9).

Boundary.
This chapter uses the per-unit system, equipment models and bus admittance matrix of Chapter 17, the complex power $\bar{S} = \bar{V}\bar{I}^*$ of *Circuits, Signals and Communications*, Ch. 9 §9.4, and its nodal analysis, Ch. 8 §8.7. The numerical methods themselves (fixed-point iteration, Newton's method, sparse LU factorisation) belong to *The Mathematics Universe*; here they are applied, not derived. The system is balanced and at steady state throughout.

18.1 Bus classifications

Four quantities, two specified

At each bus i four real quantities describe the steady state: the voltage magnitude V_i, its angle δ_i, and the net injected active and reactive powers P_i and Q_i (generation minus load). The network equations supply two real relations per bus, so two of the four must be specified. Which two depends on what is connected:

- **PQ bus (load bus).** P_i and Q_i are specified, V_i and δ_i are unknown. Most buses in a transmission model are PQ buses: substations serving load, and buses with no load or generation ($P_i = Q_i = 0$). A generator operating at a fixed power factor, or one that has reached a reactive-power limit, is also modelled as PQ.
- **PV bus (generator or voltage-controlled bus).** P_i and V_i are specified, Q_i and δ_i are unknown. The generator's governor and dispatch set P; its automatic voltage regulator holds V by adjusting excitation, and therefore Q, within the limits $Q_{\min} \le Q_i \le Q_{\max}$. Static var compensators, STATCOMs and inverter-based plants in voltage-control mode are also PV buses.
- **Slack bus (swing or reference bus).** V_i and δ_i are specified (usually $\delta = 0$), P_i and Q_i are unknown. It is needed because the total losses are not known until the solution is found: the slack generator supplies whatever active power balances the system. It also provides the angle reference, since only angle differences are physically meaningful.

Table 18.1 — Bus types in a power-flow study.

Bus type	Specified	Unknown	Typical equipment
Slack	V, δ	P, Q	Largest generator or the external-grid equivalent
PV	P, V	Q, δ	Generators, synchronous condensers, SVC, STATCOM, inverter plants in V control
PQ	P, Q	V, δ	Loads, tee points, generators at Q limit or at fixed power factor

Practical variants

Real programs add refinements. A **distributed slack** shares the mismatch among many generators in proportion to their governor droops, which mimics how the system actually responds to an unknown loss. A **remote-voltage-controlled** bus holds a voltage at a bus other than the generator terminal, usually the high-voltage side of the step-up transformer. Buses can change type during the solution: a PV bus whose required Q exceeds its limit is converted to a PQ bus with Q fixed at the limit (Section 18.8).

The choice of slack matters more than it appears. Because the slack absorbs every error in the specified injections, including the unknown losses, it should be a large unit (or the external-grid equivalent) electrically close to the centre of the system. Choosing a small, remote generator as slack can push its computed output far outside its capability and distort the flows in its neighbourhood, even though the solution is mathematically correct.

18.2 Power-balance equations

Complex form

With the bus admittance matrix $\mathbf{Y}_{\text{bus}}$ of Chapter 17, the injected current at bus i is $\bar{I}_i = \sum_k Y_{ik}\bar{V}_k$, and the complex power injected is

$$\bar{S}_i = P_i + jQ_i = \bar{V}_i\,\bar{I}_i^{*} = \bar{V}_i \sum_{k=1}^{n} Y_{ik}^{*}\,\bar{V}_k^{*}.$$

This is the whole physics of the power-flow problem: KCL at every bus, written with power instead of current.

Polar form

Writing $\bar{V}_i = V_i\angle\delta_i$, $Y_{ik} = G_{ik} + jB_{ik}$ and $\delta_{ik} = \delta_i - \delta_k$ separates the real and imaginary parts:

$$P_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik}\right),$$

$$Q_i = V_i \sum_{k=1}^{n} V_k \left(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik}\right).$$

> **Status of the principle: Power-balance (load-flow) equations**
> *Derived.* KCL and the definition of complex power applied to the linear per-unit network; exact for the balanced, fundamental-frequency model, they become nonlinear only because injections are specified as powers.

What the equations say physically

For a lossless line of reactance X between two buses, the equations reduce to

$$P_{12} = \frac{V_1 V_2}{X}\sin\delta_{12}, \qquad Q_{12} = \frac{V_1^2 - V_1 V_2\cos\delta_{12}}{X}.$$

Active power flows from the leading angle to the lagging one and depends mainly on the angle difference; reactive power flows from the higher voltage magnitude to the lower and depends mainly on the magnitude difference. This **P–δ and Q–V coupling** holds approximately for any network in which $X \gg R$, which covers transmission (X/R of 5–20) but not low-voltage distribution (X/R below 1). It explains why frequency and active power are controlled through prime movers and angles, and voltage through excitation, tap changers and reactive compensation; and it is the basis of the fast-decoupled method of Section 18.6.

18.3 Load-flow formulation

Counting equations and unknowns

In an n-bus system with one slack bus, n_{PV} PV buses and n_{PQ} PQ buses:

- unknowns: $n-1$ angles (all except the slack) and n_{PQ} voltage magnitudes;
- equations: one P equation at every non-slack bus ($n-1$) and one Q equation at every PQ bus (n_{PQ}).

Equations and unknowns both number $n-1+n_{PQ}$. The unknown Q at PV buses and P, Q at the slack are computed afterwards from the solved voltages. The problem is written as a **mismatch** to be driven to zero:

$$\Delta P_i = P_i^{\text{sp}} - P_i(\mathbf{V}, \boldsymbol{\delta}) = 0, \qquad \Delta Q_i = Q_i^{\text{sp}} - Q_i(\mathbf{V}, \boldsymbol{\delta}) = 0.$$

The specified values come from the dispatch and the load forecast; the computed values come from the equations of Section 18.2.

The example system

Figure 18.1 shows a three-bus 230 kV system used for every worked example in this chapter. On a 100 MVA base:

- **Bus 1**, slack: $\bar{V}_1 = 1.05\angle 0^\circ$.
- **Bus 2**, PQ: load 300 MW and 140 Mvar, so $P_2 = -3.0$, $Q_2 = -1.4$ pu.
- **Bus 3**, PV: generator 200 MW at $V_3 = 1.03$ pu, reactive limits -30 to $+50$ Mvar (used in Section 18.8).
- **Lines** (π models): 1–2: $z = 0.02 + j0.10$, $B = 0.04$; 1–3: $z = 0.015 + j0.08$, $B = 0.03$; 2–3: $z = 0.02 + j0.10$, $B = 0.04$ pu.

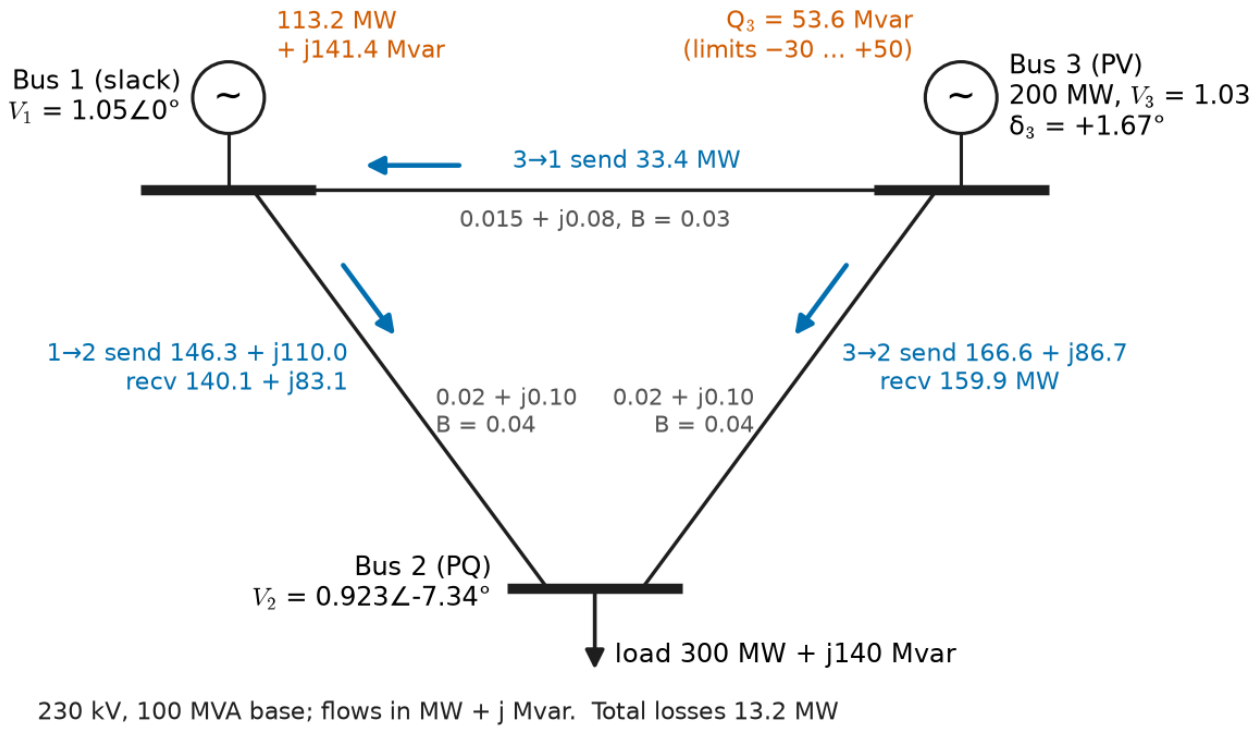


Figure 18.1 — Three-bus 230 kV test system used in the worked examples, with the solved power flow.

The admittance matrix, assembled as in Chapter 17, is

$$\mathbf{Y}_{\text{bus}} = \begin{bmatrix} 4.187 - j21.656 & -1.923 + j9.615 & -2.264 + j12.075 \\ -1.923 + j9.615 & 3.846 - j19.191 & -1.923 + j9.615 \\ -2.264 + j12.075 & -1.923 + j9.615 & 4.187 - j21.656 \end{bmatrix} \text{ pu.}$$

The unknowns are δ_2, δ_3 and V_2; the equations are $\Delta P_2 = 0$, $\Delta P_3 = 0$ and $\Delta Q_2 = 0$.

Nonlinearity, existence and the nose curve

The equations contain products of voltages and trigonometric functions of angles, so they have no closed-form solution and may have one, several or no solutions. The simplest case shows why. A load $P + jQ$ fed from a source E through a lossless reactance X satisfies

$$V^4 + \left(2QX - E^2\right) V^2 + X^2 \left(P^2 + Q^2\right) = 0,$$

a quadratic in V^2. Below a critical loading there are two real solutions: a high-voltage one, at which the system operates, and a low-voltage one that is mathematically valid but practically unstable. At the critical loading they merge, and beyond it there is no solution at all. Plotting V against P traces the **nose curve** (P–V curve) of Figure 18.2. With $E = 1$ and $X = 0.2$ pu, a unity-power-factor load reaches its maximum at $P_{\max} = E^2/(2X) = 2.5$ pu with $V = 0.707$ pu; at 0.9 lagging the maximum falls to 1.57 pu at 0.59 pu voltage; at 0.95 leading (capacitive compensation) it rises to 3.45 pu at 0.85 pu.

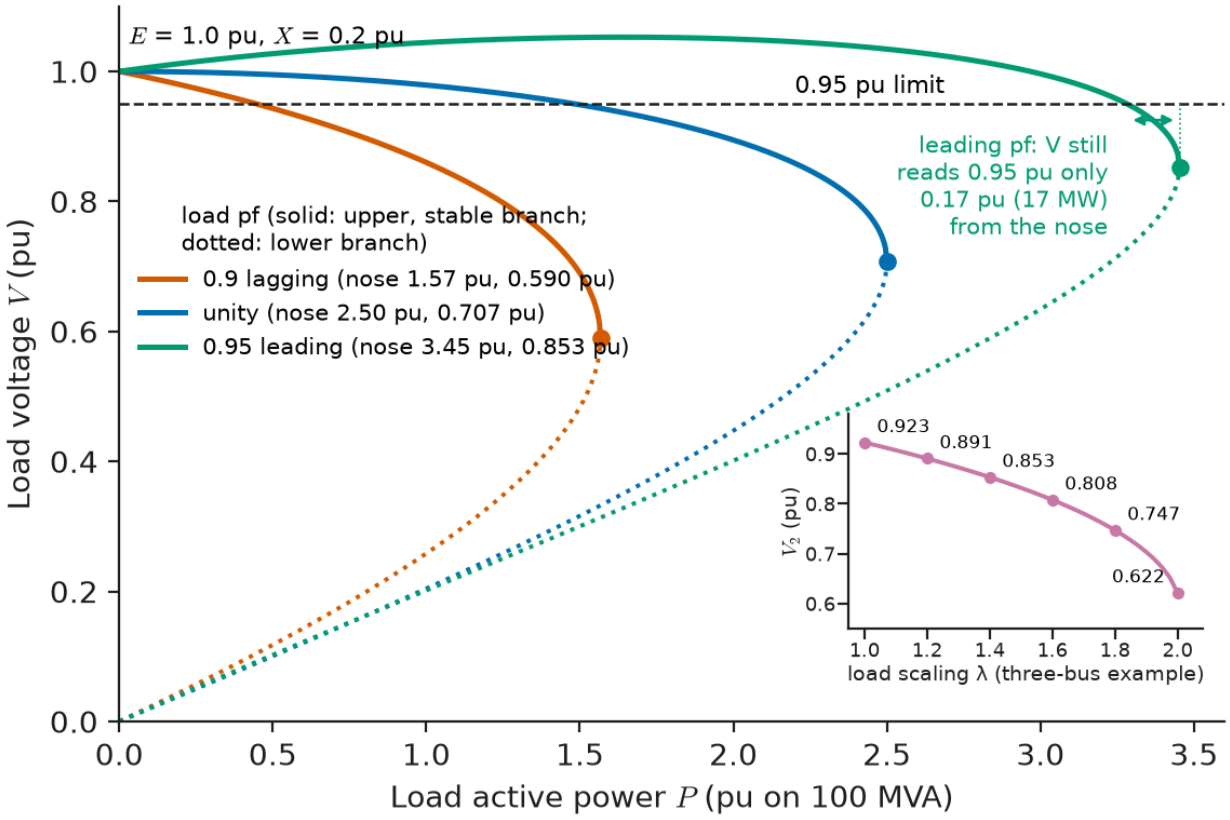


Figure 18.2 — Nose (P-V) curves for a load fed through a reactance of 0.2 pu from a 1.0 pu source at three power factors.

Two lessons follow. First, a power-flow program that fails to converge may be reporting a physical fact, that the specified loading has no steady state, rather than a numerical problem. Second, the voltage at the operating point is a poor indicator of the distance to collapse, especially with heavy capacitive compensation, which raises voltages while moving the nose closer in relative terms. Voltage stability is taken further in Chapter 20.

18.4 Gauss–Seidel method

The iteration

Rearranging the complex power equation at a PQ bus for $\bar{V}_i$ gives a fixed-point form:

$$\bar{V}_i^{(m+1)} = \frac{1}{Y_{ii}}\left[\frac{P_i - jQ_i}{\bar{V}_i^{(m)*}} - \sum_{k<i} Y_{ik}\bar{V}_k^{(m+1)} - \sum_{k>i} Y_{ik}\bar{V}_k^{(m)}\right].$$

Buses are updated in turn, and each update uses the newest values available, which is what distinguishes Gauss–Seidel from Jacobi iteration. At a PV bus Q_i is not known, so it is first computed from the latest voltages, $Q_i = -\operatorname{Im}\{\bar{V}_i^* \sum_k Y_{ik}\bar{V}_k\}$; the voltage update is made with that Q_i; and the magnitude of the result is reset to the specified V_i, keeping only its angle. Iteration stops when the largest power mismatch, or the largest voltage change, falls below a tolerance.

The method needs very little memory and is simple to program, which is why it appeared first (in the 1950s) and why it survives in teaching tools and some distribution software. Its convergence is linear, however, and slows sharply as the network grows: each update spreads information only to neighbouring buses. Large transmission models need hundreds of iterations, and an **acceleration factor** $\alpha \approx 1.4$–1.6, applied as $\bar{V}_i \leftarrow \bar{V}_i^{(m)} + \alpha(\bar{V}_i^{(m+1)} - \bar{V}_i^{(m)})$, is used to speed it up. Systems with series capacitors or a very weak slack connection may not converge at all.

> **Status of the principle: Gauss–Seidel load flow**
> *Derived*. A fixed-point iteration on the exact power-balance equations; its converged solution is exact, but convergence is only linear, depends on bus ordering and diagonal dominance of $\mathbf{Y}_{\text{bus}}$, and is not guaranteed.

Worked example — Gauss–Seidel iterations for the three-bus system. Start from a flat estimate $\bar{V}_2^{(0)} = 1.0\angle 0^\circ$, $\bar{V}_3^{(0)} = 1.03\angle 0^\circ$.

Bus 2, first update. The terms are $(P_2 - jQ_2)/\bar{V}_2^{(0)*} = -3.0 + j1.4$, $Y_{21}\bar{V}_1 = -2.019 + j10.096$ and $Y_{23}\bar{V}_3 = -1.981 + j9.904$. The bracket is $(-3.0 + j1.4) - (-2.019 + j10.096) - (-1.981 + j9.904) = 1.000 - j18.600$, and dividing by $Y_{22} = 3.846 - j19.191$ gives

$$\bar{V}_2^{(1)} = 0.9418 - j0.1367 = 0.9517\angle -8.26^\circ \text{ pu}.$$

Bus 3, first update. With the new $\bar{V}_2$, the reactive power needed at bus 3 is $Q_3 = 0.317$ pu. The voltage update gives $1.0346\angle 1.24^\circ$; resetting the magnitude, $\bar{V}_3^{(1)} = 1.03\angle 1.24^\circ$.

Convergence. Continuing, the largest power mismatch falls as follows:

Iteration	V_2 (pu)	δ_2 (°)	δ_3 (°)	Q_3 (pu)	Largest mismatch (pu)
1	0.9517	−8.255	1.241	0.317	0.602
2	0.9244	−7.195	1.700	0.540	0.0430
3	0.9233	−7.326	1.675	0.530	0.00822
4	0.9229	−7.335	1.670	0.535	0.00217
5	0.9229	−7.341	1.667	0.536	0.000561
6	0.9228	−7.343	1.666	0.536	0.000159
7	0.9228	−7.343	1.666	0.536	0.0000438

The mismatch shrinks by a roughly constant factor of about four per iteration, the signature of linear convergence; seven iterations reach a tolerance of 10^{-4} pu (0.01 MW on a 100 MVA base). On a three-bus system this is quick; on a 2,000-bus system the factor approaches one and the iteration count runs into hundreds.

18.5 Newton–Raphson method

Linearise and correct

The Newton–Raphson method expands the mismatch equations to first order about the current estimate and solves the resulting linear system for a correction:

$$\begin{bmatrix} \Delta \mathbf{P} \\ \Delta \mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{P\delta} & \mathbf{J}_{PV} \\ \mathbf{J}_{Q\delta} & \mathbf{J}_{QV} \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{\delta} \\ \Delta \mathbf{V} \end{bmatrix}, \qquad \boldsymbol{\delta} \leftarrow \boldsymbol{\delta} + \Delta\boldsymbol{\delta}, \quad \mathbf{V} \leftarrow \mathbf{V} + \Delta\mathbf{V}.$$

The **Jacobian J** contains the partial derivatives of the computed P and Q with respect to angles and magnitudes, evaluated at the current estimate; its rows correspond to the mismatch equations of Section 18.3 and its columns to the unknowns. Typical entries, for $i \neq k$, are

$$\frac{\partial P_i}{\partial \delta_k} = V_i V_k \left(G_{ik} \sin\delta_{ik} - B_{ik} \cos\delta_{ik} \right), \qquad \frac{\partial Q_i}{\partial V_k} = V_i \left(G_{ik} \sin\delta_{ik} - B_{ik} \cos\delta_{ik} \right),$$

with similar expressions on the diagonal. The Jacobian has the same sparsity pattern as $\mathbf{Y}_{\text{bus}}$, so it is factorised with sparse LU at every iteration at a cost roughly proportional to the number of buses.

Near the solution the error is squared at each iteration (**quadratic convergence**): if the mismatch is 10^{-2}, the next is about 10^{-4}, then 10^{-8}. From a flat start a well-conditioned system of any size converges in 3–6 iterations. This size-independence, achieved once sparse matrix techniques were introduced in the late 1960s, made Newton–Raphson the industry standard.

> **Status of the principle: Newton–Raphson load flow**
> *Derived.* Newton's method applied to the exact power-balance equations; convergence is quadratic close to a solution, but the Jacobian becomes singular at the nose of the P–V curve, and from a poor starting point the iteration can diverge or converge to the low-voltage solution.

Worked example — Newton–Raphson for the three-bus system. Unknowns δ_2, δ_3, V_2; flat start $\delta_2 = \delta_3 = 0$, $V_2 = 1.0$.

Initial mismatches: $\Delta P_2 = -2.846$, $\Delta P_3 = 1.987$, $\Delta Q_2 = -0.591$ pu.

Initial Jacobian (rows $\Delta P_2, \Delta P_3, \Delta Q_2$; columns δ_2, δ_3, V_2):

$$\mathbf{J}^{(0)} = \begin{bmatrix} 20.000 & -9.904 & 3.692 \\ -9.904 & 22.964 & -1.981 \\ -4.000 & 1.981 & 18.382 \end{bmatrix}.$$

The large diagonal entries in the P–δ and Q–V blocks and the small off-diagonal blocks are the P–δ and Q–V coupling of Section 18.2 made numerical.

First correction: solving gives $\delta_2 = -6.619^\circ$, $\delta_3 = 1.804^\circ$, $V_2 = 0.9393$ pu.

Iterations:

Iteration	δ_2 (°)	δ_3 (°)	V_2 (pu)	Largest mismatch after update (pu)
0 (flat start)	0	0	1.0000	2.85
1	−6.619	1.804	0.9393	0.213
2	−7.326	1.669	0.9232	0.00492
3	−7.343	1.666	0.9228	3.1×10^{-6}
4	−7.343	1.666	0.9228	1.2×10^{-12}

The mismatch exponent roughly doubles each step (10^{-1}, 10^{-3}, 10^{-6}, 10^{-12}), and three iterations reach the accuracy that took Gauss–Seidel seven; the gap widens enormously for large systems.

Solution. $\bar{V}_2 = 0.9228\angle -7.343^\circ$ and $\delta_3 = +1.666^\circ$. From the converged voltages, the slack supplies $P_1 = 113.2$ MW and $Q_1 = 141.4$ Mvar, and the bus-3 generator supplies $Q_3 = 53.6$ Mvar. Generation is $113.2 + 200 = 313.2$ MW against 300 MW of load: the **system losses are 13.2 MW**, which the slack picked up exactly as intended.

Line flows, computed from $\bar{S}_{ik} = \bar{V}_i[(\bar{V}_i - \bar{V}_k)y_{ik} + \bar{V}_i\, jB_{ik}/2]^*$:

Line	Sending end	Receiving end	Loss
1 → 2	146.3 MW, 110.0 Mvar	140.1 MW, 83.1 Mvar	6.17 MW
3 → 1	33.4 MW, −33.0 Mvar	33.1 MW, −31.4 Mvar (into bus 1)	0.30 MW
3 → 2	166.6 MW, 86.7 Mvar	159.9 MW, 56.9 Mvar	6.72 MW

Bus 3 is at a slightly leading angle and exports power both to bus 2 and back towards the slack. Bus 2's voltage of 0.923 pu is below the usual 0.95 pu planning limit; Sections 18.7 and 18.8 deal with that.

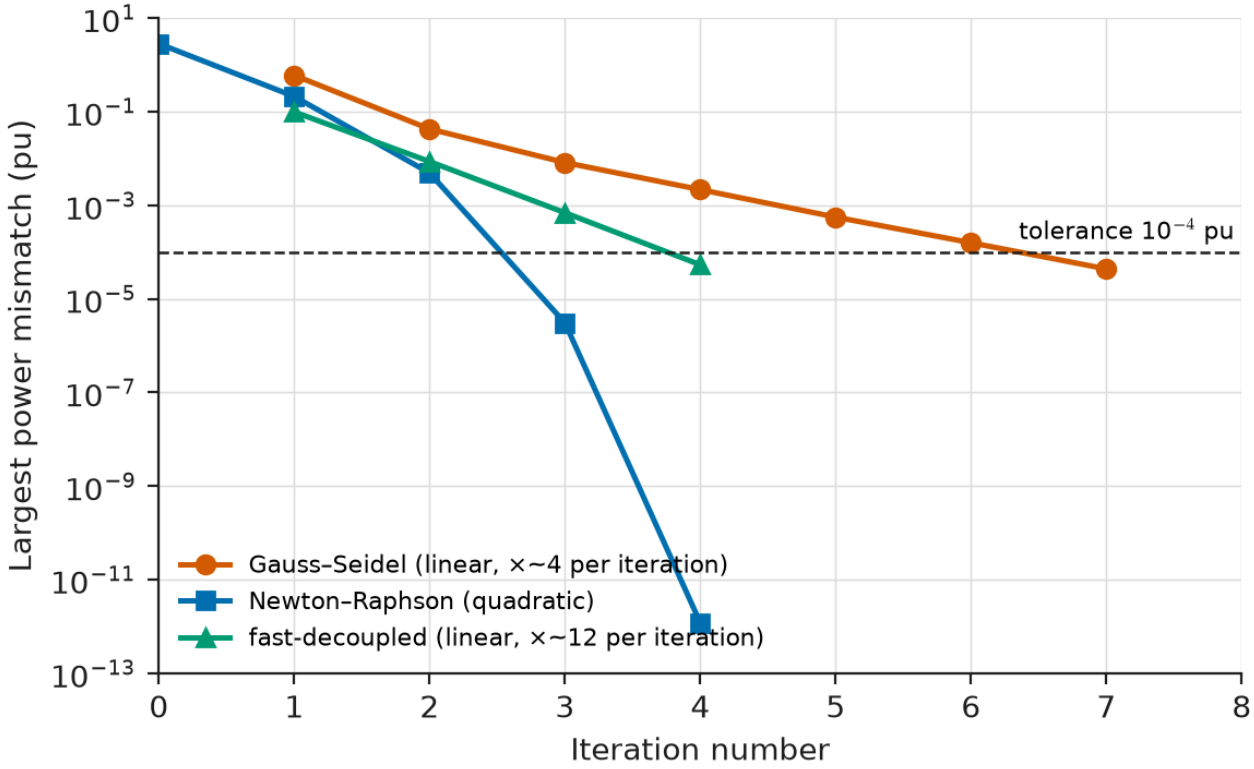


Figure 18.3 — Convergence of Gauss-Seidel, Newton-Raphson and fast-decoupled load flow on the three-bus example.

Flowchart of a production solver

Figure 18.4 shows the loop that every commercial power-flow program runs: build $\mathbf{Y}_{\text{bus}}$; initialise voltages (flat start, or a previous solution); compute mismatches; check convergence; build and factorise the Jacobian; solve for corrections; update; then, after convergence, check generator reactive limits and tap-changer and shunt control actions, change bus types or settings where needed, and re-solve. These "outer loop" controls are where most real-world convergence difficulties arise.

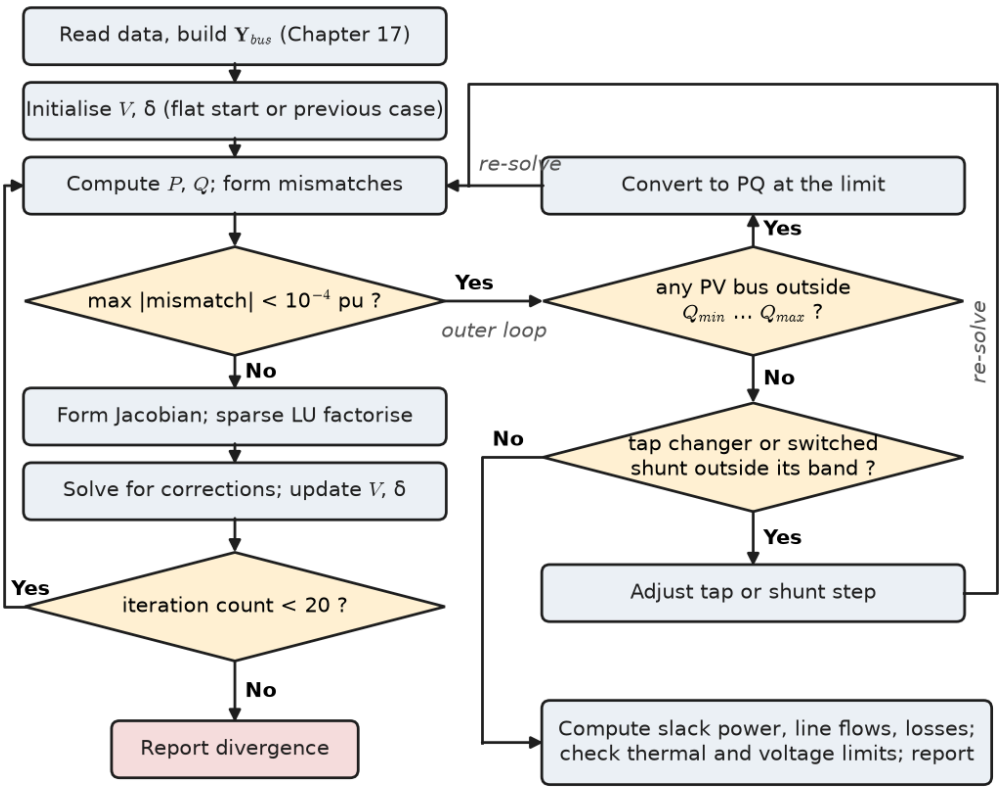


Figure 18.4 — Newton-Raphson load-flow algorithm with reactive-limit and control outer loop.

18.6 Fast-decoupled load flow

Two approximations

The Newton–Raphson Jacobian is recalculated and refactorised at every iteration. In transmission networks two physical facts allow a much cheaper scheme:

1. **Weak coupling.** Because $X \gg R$, $\partial P/\partial V$ and $\partial Q/\partial \delta$ are small compared with $\partial P/\partial \delta$ and $\partial Q/\partial V$ (compare the blocks of $\mathbf{J}^{(0)}$ above), so the off-diagonal blocks are dropped and the P–δ and Q–V problems are solved alternately.
2. **Nearly constant blocks.** Angle differences across lines are small ($\cos\delta_{ik} \approx 1$, $G_{ik}\sin\delta_{ik} \ll B_{ik}$) and $V \approx 1$, so the remaining blocks are nearly constant.

The result, introduced by Stott and Alsaç in 1974, is the **fast-decoupled load flow** (FDLF):

$$\frac{\Delta \mathbf{P}}{\mathbf{V}} = \mathbf{B}'\,\Delta\boldsymbol{\delta}, \qquad \frac{\Delta \mathbf{Q}}{\mathbf{V}} = \mathbf{B}''\,\Delta\mathbf{V}.$$

$\mathbf{B}'$ and $\mathbf{B}''$ are real, constant matrices built once and factorised once. In the common **XB** version, $\mathbf{B}'$ is built from branch reactances alone ($1/x_{ik}$, ignoring resistance, shunts and taps), and $\mathbf{B}''$ is the negative imaginary part of $\mathbf{Y}_{\text{bus}}$ for the PQ buses. Each iteration costs a fraction of a Newton iteration; convergence is linear, but with a small, nearly constant ratio, so 4–8 iterations usually reach 10^{-4} pu.

> **Status of the principle: Fast-decoupled load flow**
> *Approximate.* The approximations affect only the iteration matrices, not the mismatch equations, so a converged result is as accurate as Newton–Raphson; but convergence relies on X/R well above 1 and modest angle differences, and degrades or fails in distribution feeders, cable networks and heavily stressed systems.

Worked example — Fast-decoupled solution of the three-bus system. From the reactances $x_{12} = 0.10$, $x_{13} = 0.08$ and $x_{23} = 0.10$, for the unknown angles at buses 2 and 3,

$$\mathbf{B}' = \begin{bmatrix} 20.0 & -10.0 \\ -10.0 & 22.5 \end{bmatrix}, \qquad \mathbf{B}'' = \begin{bmatrix} 19.191 \end{bmatrix}.$$

Alternating one P–δ and one Q–V half-iteration, the largest mismatch after successive iterations is 0.0993, 0.00865, 0.000697 and 0.0000537 pu: a steady reduction by a factor of about 12 per iteration. After four iterations $V_2 = 0.9228$, $\delta_2 = -7.343^\circ$ and $\delta_3 = +1.666^\circ$, identical to the Newton result. The two constant matrices were factorised once; the Newton solution had to factorise a new 3×3 Jacobian at every iteration. In a 10,000-bus model that difference is what allows control centres to solve thousands of contingency cases every few minutes.

Where decoupling fails

At low and medium voltage, R/X ratios approach or exceed 1: a 400 V cable has R/X of 2–5. The ignored coupling then dominates and FDLF converges slowly or not at all. Distribution programs therefore use full Newton, or methods that exploit radial topology (backward–forward sweep), and treat each phase separately because distribution loads are unbalanced (Chapter 22). The **DC power flow**, the extreme simplification ($V = 1$, $R = 0$, $\sin\delta \approx \delta$, $\mathbf{P} = \mathbf{B}'\boldsymbol{\delta}$), is linear and solves in one step; electricity markets and transmission-expansion studies use it for active-power flows, accepting errors of a few per cent and no information about voltages or reactive power.

18.7 Voltage profiles

Limits

The **voltage profile** is the set of bus voltage magnitudes from the solution. Each must lie within limits set by equipment ratings, by supply standards and by grid codes. Table 18.2 gives representative values.

Table 18.2 — Representative steady-state voltage limits.

Level	Normal limits	Source
North American service voltage, 120 V base	114–126 V (±5 %, Range A); 110–127 V Range B	ANSI C84.1
European low voltage, 230 V nominal	±10 % for 95 % of 10-min means over a week	IEC 60038, EN 50160
Medium-voltage distribution	±5 to ±10 %, after regulators	Utility standards, EN 50160
Transmission, planning (intact system)	0.95–1.05 pu typical	Utility and regional planning criteria
Transmission, after a contingency	0.90–1.10 pu typical, short-term	Planning criteria
400 kV system (Continental Europe)	0.90–1.05 pu unlimited time; highest equipment voltage 420 kV	EU Network Code RfG, IEC 60038

Low voltage reduces motor torque (proportional to V^2), increases current and losses, and can stall motors and trip drives; high voltage stresses insulation, saturates transformers and shortens lamp and electronics life. Transmission systems are often operated deliberately near the upper limit, to minimise losses and maximise margin to voltage collapse.

Reading the profile

A quick estimate of the voltage drop along a branch carrying $P + jQ$ at the receiving end is

$$\Delta V \approx \frac{RP + XQ}{V} \quad \text{(pu)}.$$

On transmission lines $X \gg R$, so reactive flow dominates the drop; on low-voltage cables resistance dominates and active power matters as much. In the three-bus example, bus 2 receives about 140 MW and 83 Mvar from bus 1 over $0.02 + j0.10$ pu, giving $(0.02 \times 1.40 + 0.10 \times 0.83)/0.92 \approx 0.12$ pu, close to the actual 0.127 pu between bus 1 (1.05) and bus 2 (0.923). Most of the drop comes from moving reactive power, not active power, to the load.

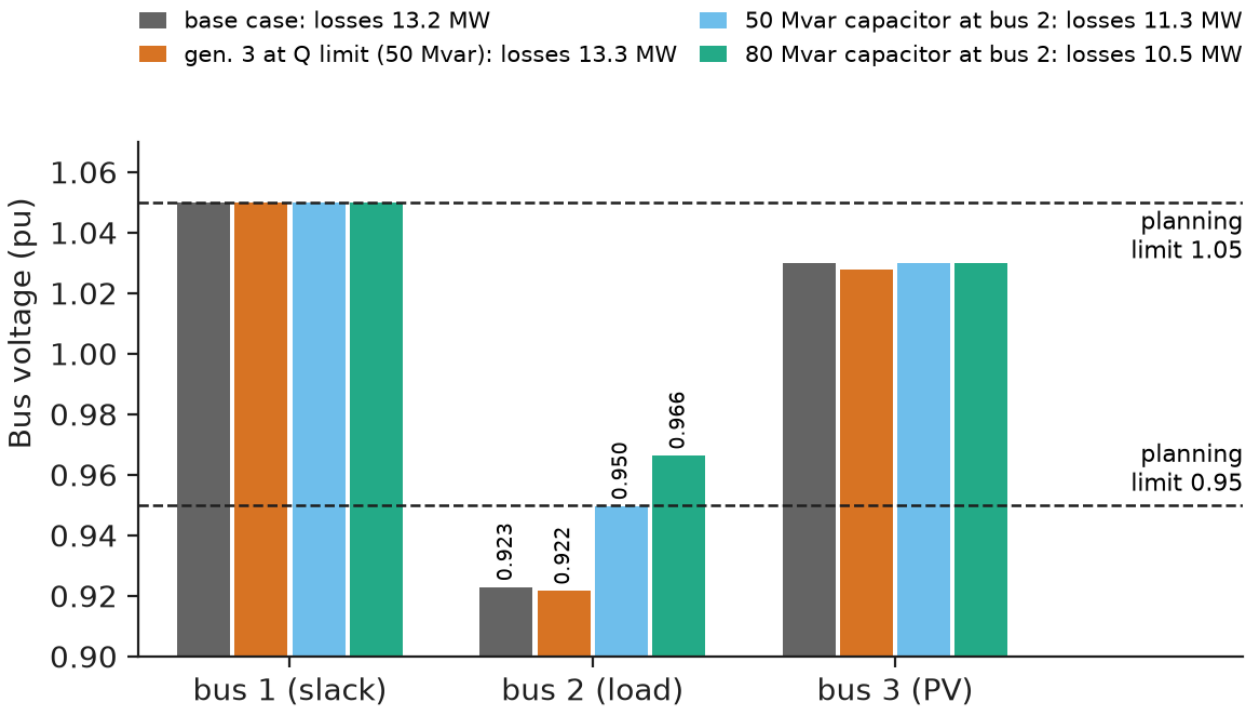


Figure 18.5 — Bus voltage profiles of the three-bus example for four operating cases, against planning limits.

Profiles also expose the opposite problem. At light load, the charging of long lines and cables (Chapter 21) raises voltages, sometimes above 1.05–1.10 pu at remote ends. Distribution feeders with high rooftop PV show the same effect at midday, when reverse power flow lifts voltages at the feeder end (Chapters 22 and 27).

18.8 Reactive-power management

Sources and sinks of reactive power

Reactive power must be balanced at every bus, and unlike active power it cannot be moved far: transmission branches consume I^2X of reactive power and their voltage drops grow with reactive flow. In the base case of the example, the network consumes 55 Mvar net (series I^2X minus line charging) while losing only 13.2 MW of active power. Reactive power is therefore supplied locally, by:

- **Synchronous generators and condensers**, within their capability curves (Figure 18.6);
- **Shunt capacitors**, switched in steps, supplying $Q = BV^2$, which falls just when it is most needed;
- **Shunt reactors**, absorbing reactive power on long lines and cables at light load;
- **On-load tap changers**, which do not create reactive power but redistribute it and hold voltages at selected buses;
- **Static var compensators** (thyristor-controlled reactors with fixed or switched capacitors) and **STATCOMs** (voltage-source converters), which vary output continuously in tens of milliseconds; a STATCOM's output is limited by current, so it falls only in proportion to V, not V^2;
- **Inverter-based resources** (PV, wind, batteries), which grid codes increasingly require to provide reactive power, for example at 0.95 leading to 0.95 lagging at rated output (Chapter 27).

Generator capability

A generator's reactive limits are not fixed numbers. The capability curve of Figure 18.6 is bounded by the armature (stator) current limit, a circle of radius equal to rated MVA; the field current limit, which caps over-excited (lagging) output; the prime-mover limit on active power; and, on the under-excited (leading) side, the stator-end heating and steady-state stability limits. A 600 MVA, 0.85 power-factor turbo-generator rated 510 MW can produce about 316 Mvar lagging at full load, but only about 150–200 Mvar leading. The load flow models these limits as $Q_{\min}$ and $Q_{\max}$, sometimes as functions of P.

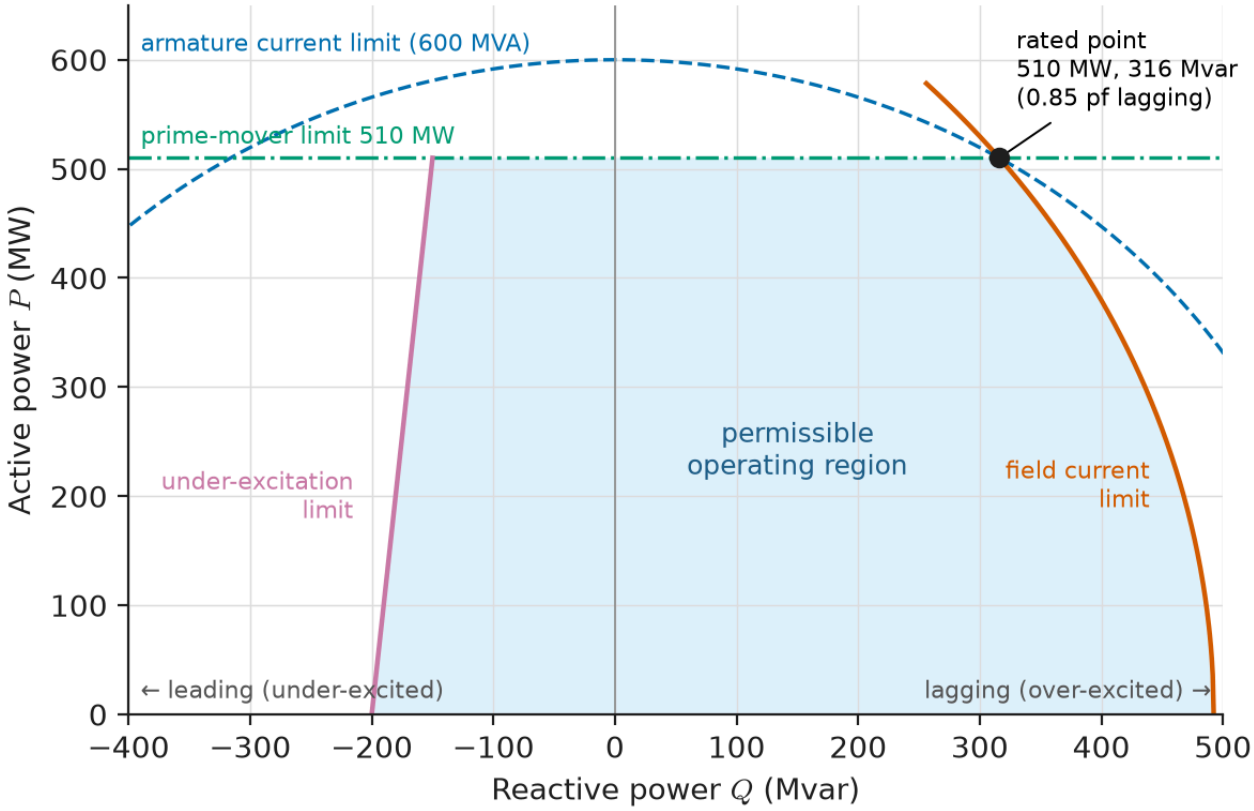


Figure 18.6 — Capability curve of a 600 MVA, 0.85 power factor turbo-generator.

Enforcing reactive limits in the load flow

Worked example — Generator 3 reaches its reactive limit. In the base solution the bus-3 generator must supply 53.6 Mvar to hold 1.03 pu, but its limit is 50 Mvar. The program converts bus 3 to a PQ bus with $Q_3 = 0.50$ pu and re-solves. Newton–Raphson converges in four iterations to $V_3 = 1.0279$ pu (bus 3 can no longer hold its set-point), $V_2 = 0.9216$ pu, $\delta_2 = -7.348^\circ$, with the slack supplying 145.4 Mvar (up from 141.4) and losses 13.26 MW. The change is small here because the limit is only just exceeded, but in a stressed system the loss of voltage control at one generator shifts reactive demand to its neighbours, which may hit their own limits in turn: a cascade of limit hits is the classic precursor of voltage collapse (Chapter 20).

Worked example — Sizing a capacitor bank for bus 2. Bus 2 must be raised to at least 0.95 pu. A shunt capacitor of susceptance B pu adds jB to Y_{22} and supplies BV_2^2.

50 Mvar bank ($B = 0.50$ pu): re-solving gives $V_2 = 0.9496$ pu, just below 0.95, with the bank delivering only $0.50 \times 0.9496^2 = 45.1$ Mvar. Generator 3's reactive output falls to 26.6 Mvar, slack reactive output to 114.0 Mvar, and losses to 11.34 MW.

80 Mvar bank ($B = 0.80$ pu): $V_2 = 0.9663$ pu, delivered 74.7 Mvar; generator 3 at 9.7 Mvar, slack at 96.8 Mvar; losses 10.48 MW.

Interpretation. An 80 Mvar bank, typically installed as two or three switched steps so that voltage can be held at both peak and light load, brings bus 2 comfortably inside limits, frees about 44 Mvar of generator 3's capability for contingencies and cuts losses by $13.18 - 10.48 = 2.7$ MW. Valued at \$60/MWh and a 50 % average load factor, that loss reduction alone is worth about $2.7 \times 8760 \times 0.5 \times 60 \approx$ \$0.71 M per year. The sensitivity from these cases, about 0.044 pu of voltage for 0.75 pu of delivered reactive power, or $\partial V_2/\partial Q_2 \approx 0.058$ pu per 100 Mvar, is what an operator uses to choose how many capacitor steps to switch.

> **Status of the principle: Local reactive-power compensation**
> *Derived*. Follows from the Q–V coupling of the power-balance equations: supplying reactive power at the bus that needs it reduces reactive flow, I^2X losses and voltage drop along the path; the size of the effect depends on the network's Thevenin reactance at that bus and on the voltage dependence of the compensating device.

Tap changers in the load flow

An on-load tap changer (OLTC) holds the voltage of a controlled bus, usually the transformer's low-voltage side, within a deadband of about ±1 % around its set-point by stepping the ratio a of Chapter 17 in steps of about 1.25 %. The load flow represents it in one of two ways. In the **outer-loop** method the transformer is solved at a fixed tap; after convergence the controlled voltage is compared with the band, the tap is moved by the number of steps the sensitivity suggests, $\mathbf{Y}_{\text{bus}}$ is updated and the case is re-solved. In the **integrated** method the tap ratio becomes an extra unknown in the Newton equations, with the controlled voltage magnitude as the extra specified quantity, and is rounded to the nearest physical step at the end. Either way, a tap changer does not create reactive power: raising the ratio to lift the load-side voltage draws more reactive power through the transformer from the upstream network. If the upstream network is already short of reactive power, the tap action lowers the high-voltage side further. This is why tap changers are blocked in emergency conditions, and why reactive sources must be planned together with them.

In the plant of Chapter 17, the 6.6 kV board fell to 0.928 pu at full load with T1 on its nominal tap. Moving T1's tap to boost the secondary by six steps (7.5 %) restores the board to about 1.0 pu ($0.928 \times 1.075 \approx 1.00$, neglecting the small extra drop); the extra reactive power is drawn from the 132 kV grid, whose 3000 MVA fault level makes the upstream voltage change negligible.

Coordinating the controls

In practice the controls act together and on different time scales: generator AVRs and STATCOMs in under a second, switched capacitors and reactors in seconds to minutes, tap changers in tens of seconds per step, and operator dispatch over minutes to hours. Load-flow programs reproduce the steady result by adjusting taps and switched shunts in an outer loop (Figure 18.4); **optimal power flow** goes further and chooses the set-points that minimise losses or cost subject to all limits. Poor coordination, for example tap changers restoring load-side voltage while transmission voltages fall, has contributed to several historical voltage collapses.

18.9 Convergence and interpretation

Tolerances and accuracy

A mismatch tolerance of 10^{-4} pu on a 100 MVA base means 0.01 MW and 0.01 Mvar at every bus, far finer than the accuracy of the input data. Load forecasts are uncertain by several per cent, line impedances by a few per cent, and the voltage dependence of loads is often unknown. The solution is precise; its accuracy is set by the data (Chapter 17).

Why a load flow fails to converge

Non-convergence is information. The usual causes, roughly in order of frequency:

1. **Data errors**: an impedance on the wrong base, a missing branch that islands a load, a transformer tap far from nominal, a load entered in kW as MW (Table 17.5).
2. **No physical solution**: the case is beyond the nose of the P–V curve. Reducing the load until the case converges and tracing the voltage as load is increased (a **continuation power flow**) locates the limit; the three-bus example, with load at bus 2 scaled by a factor λ, still converges at $\lambda = 2.0$ with $V_2 = 0.62$ pu, but needs more iterations and generator 3 supplies 403 Mvar, far beyond any real limit.
3. **Poor starting point**: a flat start far from the solution, for example with large phase-shifting transformers or long series-compensated lines. Starting from a previous solution, or a DC power flow for the angles, usually cures it.
4. **Control interactions**: PV–PQ switching or tap adjustments that oscillate between outer-loop iterations.

Reading the results

A converged solution is checked and interpreted, not just accepted:

- **Balance check.** Total generation equals load plus losses. In transmission systems losses are 1.5–3 % of load; much higher values point to a data error or an unrealistic dispatch.
- **Slack output.** The slack should supply a plausible amount. A slack generator producing negative or several times its rated power means the dispatch does not balance the load.
- **Voltages.** Compare with Table 18.2; unusually high or low values are usually explained by the reactive balance.
- **Branch loading.** Each line and transformer flow is compared with its thermal rating (normal and emergency). In the example, line 3–2 carries $|\bar{S}| = \sqrt{166.6^2 + 86.7^2} = 188$ MVA at its sending end, which must be compared with the conductor rating (Chapter 21).
- **Generator reactive output.** Units at or near their limits have no reserve for contingencies.

Contingency analysis

A system planned to the **N−1 criterion** must remain within thermal and voltage limits after the loss of any single element. The planner of the opening paragraph therefore runs not one load flow but one for each credible outage, for several load levels and generation dispatches, and for future years: often tens of thousands of cases. Fast-decoupled solvers and sensitivity methods screen the cases, and full Newton solutions confirm the critical ones. The result is a list of violations and the reinforcements (new lines, transformers, capacitor banks, or special protection schemes) that remove them, which is the input to the planning decisions of Chapter 13.

> **Status of the principle: Steady-state power-flow solution**
> *Approximate.* A load flow gives one balanced, steady-state operating point for specified injections; it says nothing about whether the system can reach that point after a disturbance (Chapter 20), about unbalanced operation (Chapter 22) or about harmonics (Chapter 25).

What remains open

Open question	Where it is answered
How base values, equipment models and the admittance matrix are prepared	Chapter 17
How the network responds to a short circuit, and how fault currents are calculated	Chapter 19
Whether generators stay in synchronism after a disturbance, and how voltage collapse develops dynamically	Chapter 20
How line parameters, thermal ratings and long-line effects enter the power flow	Chapter 21
How radial, unbalanced distribution feeders are analysed, and how distributed generation changes their voltage profile	Chapter 22
How SCADA measurements are turned into a real-time power-flow model (state estimation)	Chapter 26
How inverter-based resources provide reactive power and voltage control under grid codes	Chapter 27
How network planning uses power-flow results to decide on reinforcements	Chapter 13
Newton's method, sparse matrices and iterative solvers as mathematics	*The Mathematics Universe*
Complex power and nodal analysis	*Circuits, Signals and Communications*, Ch. 8 and Ch. 9

Next: Chapter 19 turns from the normal operating state to the abnormal one, and uses the same per-unit network to calculate the currents that flow when it is short-circuited.

Chapter 19 — Fault Analysis

Part V · Power-System Analysis

A gas-processing plant wants to add a 12.5 MVA gas-turbine generator to its 11 kV switchboard to cut its electricity bill and ride through grid outages. The switchboard was bought fifteen years ago with circuit breakers rated 25 kA. The protection engineer runs the numbers and finds that a short circuit on the busbar, which today would draw about 21 kA from the grid and the running motors, would draw about 28 kA once the generator is added. Nothing about normal operation changes: the load currents are a few hundred amperes. But on the one day a cable termination fails, the breaker asked to clear the fault would be asked to do something it was never tested to do, and a breaker that fails to interrupt becomes the next fault, often with an arc that destroys the switchroom. The same arithmetic appears when a utility connects a wind farm to a 132 kV substation, when a data centre adds a second 2.5 MVA transformer in parallel, and when a ship's designers put four generators on one main switchboard. Fault analysis is the calculation behind every one of those decisions.

> **How large is the current when a network is short-circuited, and how does it change in the first few cycles?**
> **How are unbalanced faults (one phase to earth, two phases together) calculated with balanced-network tools? What ratings must breakers, cables and switchgear have to survive the worst fault, and how is a fault found after it has been cleared?**

Concept	Engineering destination
Short-circuit fundamentals	Fault levels at every bus, subtransient and transient generator response, X/R ratio
Symmetrical faults	Three-phase fault duty of switchgear, the bus impedance matrix
Symmetrical components	Unbalanced-fault calculation, negative-sequence and earth-fault relays
Sequence networks	Equipment data sheets, transformer vector groups, neutral-earthing design
Unsymmetrical faults	Single-line-to-ground, line-to-line and double-line-to-ground currents and voltages
Fault-current calculation	IEC 60909 and IEEE C37 short-circuit studies, peak and breaking currents
Equipment fault ratings	Breaker interrupting and making ratings, cable and busbar withstand, fault-current limiters
Fault-location concepts	Impedance-based and travelling-wave locators, cable fault finding, restoration time

Roadmap. What a short circuit does to a network and to a generator (19.1) → the balanced three-phase fault and the bus impedance matrix (19.2) → symmetrical components as an engineering tool (19.3) and the sequence networks of real equipment (19.4) → connecting those networks for unbalanced faults (19.5) → the standard calculation procedures (19.6) → what the numbers must be compared with (19.7) → finding the fault afterwards (19.8).

19.1 Short-circuit fundamentals

What a fault is

A **fault** is any unintended connection between conductors, or between a conductor and earth, that bypasses the load. On overhead lines most faults are flashovers of insulator strings caused by lightning, pollution, birds or vegetation; in cables they are insulation failures, joint failures and excavator damage; in machines and transformers they are winding insulation failures. The fault path may be a bolted metallic contact with almost zero impedance, an arc with a voltage drop of a few hundred volts to a few kilovolts, or a high-impedance path through a tree or dry ground.

Faults are classified by which conductors are involved:

Fault type	Share of overhead-line faults (typical)	Comment
Single-line-to-ground (SLG)	70–80 %	Insulator flashover to the tower; the most common fault
Line-to-line (LL)	8–10 %	Conductor clashing in wind, galloping
Double-line-to-ground (LLG)	5–10 %	Often an SLG that spreads
Three-phase (LLL, LLLG)	2–5 %	Rare, but usually the most severe; also caused by earthing clamps left on after maintenance

About 80–90 % of overhead-line faults are **transient**: once the arc is extinguished by tripping the line, the insulation recovers and the line can be reclosed. Cable and equipment faults are almost always **permanent**. This difference shapes protection practice (Chapter 23).

Why the current is large and what it does

In normal operation the load impedance dominates the circuit. During a fault only the source, transformer and line impedances remain, and these are designed to be small so that voltage drop under load is small. A transformer with 10 % impedance therefore delivers about ten times its rated current into a fault at its terminals; a network bus fed by several such paths delivers more. The consequences are thermal (heating proportional to $I^2 t$), mechanical (forces between conductors proportional to the square of the instantaneous current) and electrical (voltage collapse over a wide area while the fault persists, and arc energy at the fault itself).

The magnitude used to describe the strength of a network at a bus is the **fault level** or **short-circuit capacity**,

$$S_{\mathrm{sc}} = \sqrt{3}\, V_{\mathrm{n}}\, I_{\mathrm{sc}}$$

where V_{n} is the nominal line voltage and I_{sc} the symmetrical three-phase fault current. In per unit (Chapter 17), $S_{\mathrm{sc}} = 1/Z_{\mathrm{th}}$ for a pre-fault voltage of 1 pu, so a 132 kV bus with a fault level of 4000 MVA has a Thévenin impedance of 0.025 pu on a 100 MVA base. The same quantity measures how "stiff" a bus is: voltage sag when a motor starts, flicker from an arc furnace and the stability of an inverter connection all improve with fault level (Chapters 25 and 27).

Location	Typical fault level	Typical symmetrical fault current
400 V main switchboard of a building or data centre	25–70 MVA	35–100 kA
11 kV primary distribution bus	250–500 MVA	13–25 kA
33 kV bus of a mining or industrial substation	750–1500 MVA	13–25 kA
132 kV transmission substation	3–9 GVA	15–40 kA
400 kV transmission substation	20–40 GVA	30–60 kA
Ship main switchboard, 6.6 kV with four generators	200–400 MVA	18–35 kA

The DC offset

When a fault occurs, the circuit seen from the source is predominantly inductive, with reactance-to-resistance ratio X/R of 3–10 in distribution and 15–50 near large generators and transformers. The current cannot jump, so in addition to the steady sinusoid a decaying **DC offset** appears, whose size depends on the instant in the voltage cycle at which the fault strikes.

> **Boundary.**
> The RL transient behind the offset, including the worst-case first peak of 1.73 times the symmetrical peak for $X/R = 10$, is worked in *Circuits, Signals and Communications*, Ch. 6 §6.4, with the Laplace-domain version in Ch. 11. This chapter takes the result and applies it to breaker duties in Section 19.6.

The offset decays with time constant $\tau = X/(\omega R)$: 10–30 ms in distribution, 50–150 ms close to generators. While it lasts, the current has a higher peak (setting the mechanical forces and the breaker's **making** duty) and may not cross zero for several cycles, which matters because AC circuit breakers interrupt only at a current zero.

The decaying AC component near generators

A second effect is specific to rotating machines. A synchronous generator does not behave as a constant voltage behind a constant reactance during a fault. The flux linkages of its rotor circuits (field winding and damper bars) cannot change instantly, so at the first instant the machine presents its small **subtransient reactance** X''_d. As the damper currents decay (time constant T''_d, a few cycles) the reactance rises to the **transient reactance** X'_d, and as the field current decays toward its steady value (time constant T'_d, about a second) it rises to the **synchronous reactance** X_d. The RMS AC component of a terminal fault current from an unloaded generator with internal voltage E is

$$I_{\mathrm{ac}}(t) = E\left[\left(\frac{1}{X''_d} - \frac{1}{X'_d}\right)e^{-t/T''_d} + \left(\frac{1}{X'_d} - \frac{1}{X_d}\right)e^{-t/T'_d} + \frac{1}{X_d}\right]$$

> **Status of the principle: Generator short-circuit decrement**
> *Approximate.* Derived from the two-axis machine model with constant field voltage (Chapter 7); it neglects saturation, the action of the voltage regulator (which raises the field and holds the current up) and the difference between the direct- and quadrature-axis subtransient reactances.

For the 12.5 MVA generator of the opening, with $X''_d = 0.14$, $X'_d = 0.22$ and $X_d = 1.8$ pu, the AC component of a terminal fault starts at $1/0.14 = 7.1$ times rated current, falls to $1/0.22 = 4.5$ times after a few cycles, and

would settle at $1/1.8 = 0.56$ times rated current if the excitation stayed constant. A steady fault current smaller than full-load current is why a generator's own overcurrent protection uses voltage-restrained or voltage-controlled elements, and why the excitation systems of island generators (ships, oil platforms, mines) carry a **short-circuit support** feature that holds the field up to deliver about three times rated current for a few seconds, enough to operate downstream protection. Induction motors also feed a fault: their rotor flux supplies a current about equal to their locked-rotor current, which decays within two to five cycles because the motor has no field source.

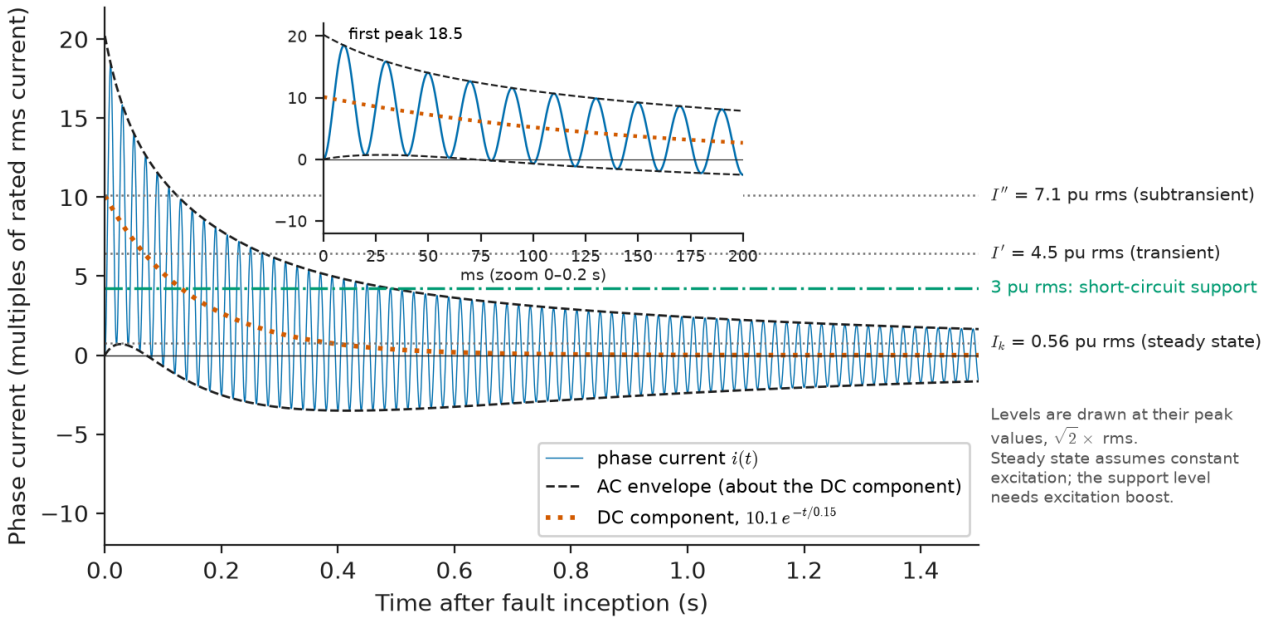


Figure 19.1 — Phase current of a 12.5 MVA generator for a three-phase terminal fault at the instant of maximum DC offset, showing the subtransient, transient and steady-state envelopes.

Figure 19.1 shows the three effects together: the AC envelope decays from subtransient to transient to steady values, and the DC offset displaces the waveform during the first few hundred milliseconds.

The Thévenin method and its assumptions

Practical fault calculation uses one simplification that makes it tractable: **superposition of the pre-fault state and a fault-only network**. A fault at bus k is equivalent to inserting, at that bus, a voltage source equal and opposite to the pre-fault voltage $V_k^{(0)}$. The currents caused by that source alone, flowing in the network with all other sources short-circuited and machines replaced by their subtransient or transient impedances, are the changes due to the fault; adding the pre-fault load currents gives the totals. For a bolted fault,

$$I_f = \frac{V_k^{(0)}}{Z_{\text{th}}}$$

> **Status of the principle: Thévenin fault calculation**
> *Approximate.* Exact for a linear network by superposition; the approximations are in the model, namely machines as constant voltages behind fixed reactances, loads ignored or represented as constant impedances, and a pre-fault voltage that is assumed rather than taken from a load flow. Standards cover the uncertainty with a voltage factor c (Section 19.6).

Pre-fault load currents are usually small compared with fault currents and are neglected, which is the standard practice of IEC 60909. What cannot be neglected are the sources that feed the fault: utility infeed, generators, motors and, increasingly, inverters.

19.2 Symmetrical faults

The balanced fault and the per-phase network

A three-phase fault, with or without earth, keeps the network balanced. The per-phase equivalent of Chapter 17 therefore applies unchanged: one positive-sequence network with each source represented by its Thévenin equivalent and each machine by its subtransient reactance, solved for the current into the short at the faulted bus. This is the calculation that sets the **fault duty** of most switchgear, because three-phase faults are usually the most severe, and it is the starting point for all others.

Worked example — adding a generator to a 25 kA switchboard. The plant of the opening has an 11 kV switchboard fed from the 132 kV grid (fault level 4000 MVA) through a 40 MVA, 132/11 kV transformer with 12 % impedance. Running motors total 15 MVA, with subtransient reactance 0.17 pu on their own rating (a locked-rotor current of about six times rated). The new generator is 12.5 MVA with $X''_d = 0.14$ pu. Work on a 100 MVA base at 11 kV, where the base current is $100 \times 10^6/(\sqrt{3} \times 11 \times 10^3) = 5249$ A, and use reactances only.

Convert each impedance to the common base:

$$X_{\text{grid}} = \frac{100}{4000} = 0.025, \quad X_T = 0.12 \times \frac{100}{40} = 0.300, \quad X_G = 0.14 \times \frac{100}{12.5} = 1.120, \quad X_M = 0.17 \times \frac{100}{15} = 1.13$$

With a pre-fault voltage of 1.0 pu, each path delivers its own contribution into a bolted busbar fault:

Source	Path reactance (pu)	Contribution (pu)	Contribution (kA)
Grid through transformer	0.325	3.077	16.15
Generator	1.120	0.893	4.69
Motors	1.133	0.882	4.63
Total	0.206 (parallel)	4.852	25.47

Before the generator, the fault current was $16.15 + 4.63 = 20.78$ kA. With it, the total is 25.47 kA, a fault level of 485 MVA, already above the 25 kA rating before any safety margin. Applying the IEC voltage factor $c = 1.1$ for maximum fault current at medium voltage gives 22.86 kA without the generator and 28.01 kA with it. The switchboard is adequate today and overdutied tomorrow. Section 19.6 completes this example with the peak and breaking currents; Section 19.7 discusses the remedies.

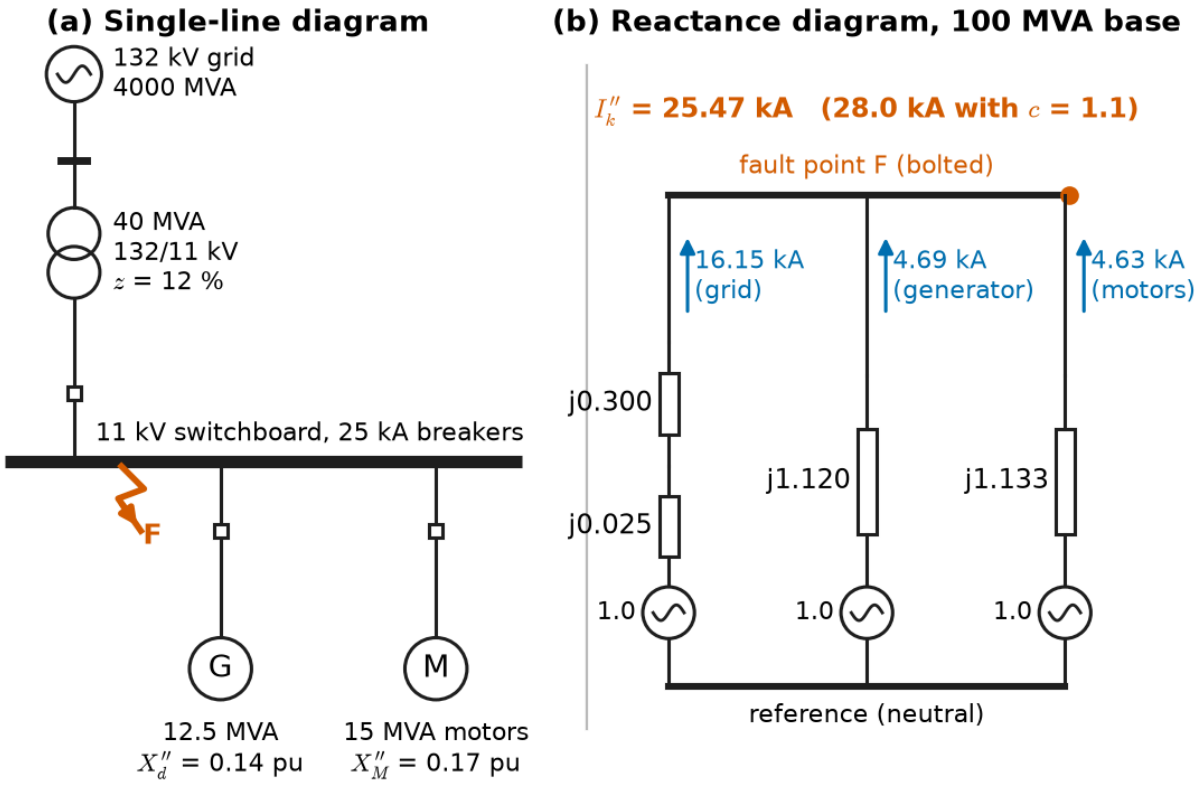


Figure 19.2 — Single-line diagram and positive-sequence reactance diagram of the 11 kV plant switchboard with grid, generator and motor contributions to a busbar fault.

Figure 19.2 shows the single-line diagram and its reactance diagram. The three branches are in parallel at the fault, which is why every new source raises the fault level of every bus electrically close to it.

The bus impedance matrix

In a meshed network with hundreds of buses, faults are calculated at every bus, and the method of choice is the **bus impedance matrix** $\mathbf{Z}_{\text{bus}} = \mathbf{Y}_{\text{bus}}^{-1}$, built from the same branch data as the power-flow admittance matrix (Chapter 18) but with machine subtransient reactances added as shunt branches to the reference. Its elements have direct physical meaning:

- the diagonal element Z_{kk} is the Thévenin impedance at bus k, so the bolted three-phase fault current there is $I_f = V_k^{(0)} / Z_{kk}$;
- the off-diagonal element Z_{ik} gives the voltage change at bus i caused by injecting current at bus k, so during a fault at k the voltage at bus i is

$$V_i = V_i^{(0)} - \frac{Z_{ik}}{Z_{kk}} V_k^{(0)}$$

and the current in any branch between buses i and j follows from the voltage difference across its impedance.

> **Status of the principle: Fault calculation with $\mathbf{Z}_{\text{bus}}$**
> *Derived.* A direct consequence of nodal analysis and superposition in a linear network; one column of $\mathbf{Z}_{\text{bus}}$ gives the voltages everywhere for a fault at the corresponding bus.

In practice $\mathbf{Z}_{\text{bus}}$ is never inverted explicitly for large networks. Programs factorise the sparse $\mathbf{Y}_{\text{bus}}$ once and solve for the columns needed, or build $\mathbf{Z}_{\text{bus}}$ branch by branch with the building algorithm. The voltage profile during the fault, from the formula above, is used to judge which motors will stall, which contactors will drop out (most fall out below 60–70 % voltage) and which inverters must ride through.

Faults with impedance

A fault through an impedance Z_f (an arc, or a tree branch) simply adds Z_f to the Thévenin impedance: $I_f = V_k^{(0)} / (Z_{kk} + Z_f)$. On transmission networks arc resistance is a few ohms and changes the current little; at low voltage it matters a great deal. An arc at 400 V has a voltage drop of roughly 100–200 V, a large fraction of the 230 V phase voltage, so an **arcing fault** may draw only 40–60 % of the bolted current, too little to trip a breaker set for the bolted value quickly. That is why arc-flash studies (Chapter 29) compute both the bolted and the reduced arcing current and check the protection clearing time at each.

19.3 Symmetrical components

The tool and where it comes from

Most faults are unbalanced, and an unbalanced fault on a balanced network breaks the symmetry that the per-phase method relies on. The way back is **symmetrical components**.

> **Boundary.**
> The decomposition of three unbalanced phasors into positive-, negative- and zero-sequence sets, with the operator $a = 1\angle 120^\circ$, is stated in *Circuits, Signals and Communications*, Ch. 10 §10.6, with a numerical example on an unbalanced residential feeder. That chapter also states the key property: in a network whose elements are themselves symmetrical, each sequence set flows in its own uncoupled network. This section adds what a power engineer needs to use it.

In matrix form, with $\mathbf{A}$ the transformation matrix,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}, \qquad \mathbf{V}_{abc} = \mathbf{A}\,\mathbf{V}_{012}$$

Applying $\mathbf{A}^{-1}$ to the impedance matrix of a three-phase element, $\mathbf{Z}_{012} = \mathbf{A}^{-1}\mathbf{Z}_{abc}\mathbf{A}$, diagonalises it whenever $\mathbf{Z}_{abc}$ has equal self-impedances Z_s and equal mutual impedances Z_m, which a transposed line or a symmetrically built machine has. For a static element such as a line,

$$Z_1 = Z_2 = Z_s - Z_m, \qquad Z_0 = Z_s + 2Z_m$$

The positive- and negative-sequence impedances of static equipment are equal because reversing the phase order does not change how a stationary conductor links flux. The zero-sequence impedance is larger because three in-phase currents add, return through earth or a neutral, and enclose a much larger loop.

> **Status of the principle: Decoupling of sequence networks**
> *Derived*. Exact when every element of the network is phase-symmetrical; it is approximate for untransposed lines (mutual impedances unequal by 5–15 %) and fails for single-phase loads, open conductors and saturated equipment, which must be modelled in phase coordinates.

Power, measurement and the phase shift through transformers

Three practical facts complete the toolkit. First, with the transformation as written, complex power is $S = 3(V_0 I_0^* + V_1 I_1^* + V_2 I_2^*)$, so power calculations in sequence quantities need the factor 3. Second, sequence quantities are directly measurable and are used as protection signals: the residual current $3I_0 = I_a + I_b + I_c$ is what a core-balance CT or the summation of three phase CTs measures, and numerical relays compute I_2 and V_2 to detect unbalance and broken conductors.

Third, a transformer whose windings are connected differently on its two sides shifts sequence quantities by a phase angle. In a Dyn11 or YNd1 transformer, positive-sequence voltages and currents are shifted by 30° one way and negative-sequence quantities by 30° the other way; zero-sequence quantities are not transferred at all. When sequence currents are referred through such a transformer, the shift must be applied before recombining to phases, or the phase currents on the far side come out wrong. Worked example 19.5 shows the consequence: an earth fault on the secondary side of a delta–star transformer appears on the primary side as a current in only two lines.

19.4 Sequence networks

Building the three networks

Each piece of equipment has three sequence impedances, and the network for each sequence is assembled from them with the same topology as the power system, with three differences:

- the **positive-sequence network** contains the sources (generator internal voltages); the other two are passive;
- the **negative-sequence network** has the same topology as the positive, with negative-sequence impedances and no sources;
- the **zero-sequence network** has a topology that depends on how each piece of equipment is earthed and how transformer windings are connected, because zero-sequence current can only flow where there is a return path.

Figure 19.3 shows the zero-sequence equivalents of the common transformer connections.

Sequence impedances of real equipment

Equipment	Z_1	Z_2	Z_0
Turbo-generator (round rotor)	X''_d = 0.10–0.25 pu	≈ X''_d, 0.10–0.25 pu	0.03–0.10 pu, plus $3Z_n$ of the neutral earthing
Salient-pole hydro generator with dampers	X''_d = 0.18–0.30 pu	0.20–0.35 pu	0.05–0.15 pu, plus $3Z_n$
Induction motor	Locked-rotor reactance 0.15–0.20 pu	≈ Z_1	Infinite (star or delta, unearthed)
Two-winding power transformer	Leakage impedance 0.06–0.20 pu, rising with rating	= Z_1	Depends on connection; ≈ 0.8–1.0 Z_1 through YNd; open circuit from a delta or unearthed star side
Overhead line, 132–400 kV	0.25–0.40 Ω/km	= Z_1	2.5–3.5 × Z_1 in reactance, with larger resistance (earth return)
Underground XLPE cable	0.08–0.20 Ω/km	= Z_1	1–3 × Z_1, strongly dependent on sheath bonding and earth path
Inverter-based resource (PV, battery, full-converter wind)	Current source limited to 1.0–1.5 pu	Controlled; grid codes now require injection (IEEE 2800, ENTSO-E RfG)	Usually blocked by a delta or unearthed transformer winding

The values are typical; data sheets and test reports govern actual studies.

Generators and their earthing

A synchronous machine's negative-sequence reactance is close to its subtransient reactance: a negative-sequence current produces a field rotating backward at twice synchronous speed relative to the rotor, which the damper windings oppose much as they oppose a sudden change. That same double-frequency rotor current heats the rotor surface, which is why generators have a negative-sequence rating (typically continuous I_2 of 5–10 % and a short-time limit $I_2^2 t$ of 5–30 s in per unit squared seconds, per IEC 60034-1) and a negative-sequence relay to enforce it.

The zero-sequence reactance of a generator winding is small, often smaller than X''_d, because three in-phase currents in distributed windings produce almost no air-gap field. If a generator neutral were solidly earthed, a terminal earth fault could therefore exceed the three-phase fault current and damage the winding. Generators are earthed through an impedance Z_n, which appears in the zero-sequence network as $3Z_n$, because all three sequence-zero currents flow through it: $I_n = 3I_0$, so the neutral voltage is $3I_0 Z_n$.

Transformers and the zero-sequence path

The zero-sequence network of a transformer follows two rules. Zero-sequence current can flow in a star winding only if its neutral is earthed. It can be transferred from one side to the other only if both windings can carry it; a delta winding cannot pass zero-sequence current to its line terminals, but it can circulate it inside the delta, which balances ampere-turns and makes the transformer look like its leakage impedance to earth from the earthed-star side.

Connection	Zero-sequence equivalent seen from side 1	Zero-sequence equivalent seen from side 2
YNyn, both neutrals earthed	Series leakage impedance through to side 2	Series leakage impedance through to side 1
YNd (star earthed on side 1)	Leakage impedance to reference (delta circulates current)	Open circuit
Yd or Yyn with unearthed star	Open circuit	Open circuit (Yd); through only if side 1 is earthed (Yyn)
Dd	Open circuit	Open circuit
Zigzag earthing transformer	Low zero-sequence impedance to reference	—

The YNd connection is the workhorse of power systems because of this table: the step-up transformer of a power station (generator on the delta) isolates the generator from transmission earth faults, and the Dyn distribution transformer gives a solidly earthed LV neutral while presenting an open zero-sequence circuit to the MV system. Three-limb core-type transformers without a delta add a further subtlety: zero-sequence flux has no return path in the core and closes through the tank, giving a zero-sequence impedance far lower than the magnetising impedance (typically 0.3–1 pu), which a fault study must not treat as infinite.

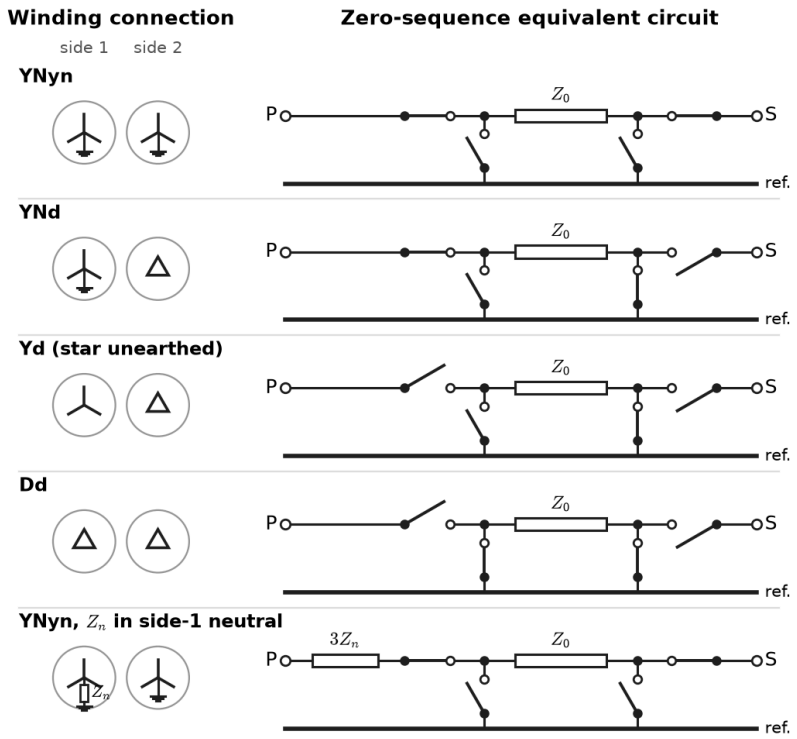


Figure 19.3 — Zero-sequence equivalent circuits of two-winding transformers for the common winding connections.

Lines and cables

The zero-sequence impedance of an overhead line is dominated by the earth return. Carson's equations replace the earth by an equivalent return conductor at a depth of several hundred metres, which depends on soil resistivity, and give zero-sequence reactances about three times the positive-sequence value. Shield wires bonded to every tower carry part of the return current and reduce Z_0. For a 132 kV line with $z_1 = 0.12 + j0.40$ Ω/km, a typical zero-sequence value is $z_0 = 0.35 + j1.25$ Ω/km. Because Z_0 also depends on soil resistivity, which varies with season, earth-fault calculations on long lines carry an uncertainty of 10–20 % that distance-relay settings must allow for (Chapter 23). Line parameters themselves are derived in Chapter 21.

Neutral earthing practice

Because the zero-sequence network contains the earthing impedances, the choice of earthing sets the earth-fault current, the overvoltage on healthy phases and the way the fault must be detected:

Earthing method	Earth-fault current	Healthy-phase voltage during fault	Typical use
Solid	Comparable to, or above, three-phase fault current	≤ 1.4 pu (effectively earthed, IEC 60071)	Transmission ≥ 110 kV, LV TN systems, North American MV four-wire
Low-resistance	100–1000 A (400 A and 1000 A common)	Up to ≈ 1.7 pu	Industrial and mining MV systems
High-resistance	< 10 A, set slightly above system capacitive current	Up to ≈ 1.73 pu	Generator neutrals, continuous-process LV and MV plants (alarm on first fault)
Resonant (Petersen coil)	Residual of a few amperes; arcs self-extinguish	≈ 1.73 pu	Rural MV networks in Germany, Scandinavia, China
Unearthed	System capacitive current only	1.73 pu, with intermittent-arc overvoltages much higher	Older industrial and marine systems

High-resistance and resonant earthing let a plant run on with one earth fault, at the price of insulation stressed to line voltage; solid earthing gives fast, selective tripping at the price of large fault currents.

19.5 Unsymmetrical faults

Fault boundary conditions

Each fault type is a set of boundary conditions on the phase quantities at the fault point. Transformed into sequence quantities, these conditions tell us how to interconnect the three sequence networks at that point. Let Z_1, Z_2 and Z_0 be the Thévenin impedances of the three networks seen from the fault, V_f the pre-fault phase voltage and Z_f the fault impedance. Phase a is chosen as the reference (the faulted phase for SLG, the healthy phase for LL and LLG), which makes the connections symmetrical.

Single-line-to-ground, phase a to earth through Z_f. The boundary conditions $I_b = I_c = 0$ and $V_a = Z_f I_a$ give $I_0 = I_1 = I_2$: the three networks are connected **in series** with $3Z_f$:

$$I_1 = I_2 = I_0 = \frac{V_f}{Z_1 + Z_2 + Z_0 + 3Z_f}, \qquad I_a = 3I_0$$

Line-to-line, phases b and c through Z_f. With $I_a = 0$, $I_b = -I_c$ and $V_b - V_c = Z_f I_b$, the zero-sequence network is not involved and the positive and negative networks are connected **in parallel** (in opposition) through Z_f:

$$I_1 = -I_2 = \frac{V_f}{Z_1 + Z_2 + Z_f}, \qquad I_b = -I_c = -j\sqrt{3}\, I_1$$

Double-line-to-ground, phases b and c to earth. For a bolted fault the three networks are connected **in parallel**:

$$I_1 = \frac{V_f}{Z_1 + \dfrac{Z_2 Z_0}{Z_2 + Z_0}}, \qquad I_2 = -I_1 \frac{Z_0}{Z_2 + Z_0}, \qquad I_0 = -I_1 \frac{Z_2}{Z_2 + Z_0}$$

with the earth current $3I_0$. A fault impedance to earth appears as $3Z_f$ in series with Z_0.

> **Status of the principle: Sequence-network connections for unbalanced faults**
> *Derived*. Each connection follows algebraically from the fault's phase boundary conditions and the symmetrical-component transformation; the results are exact for the linear model of Section 19.1 and inherit its approximations.

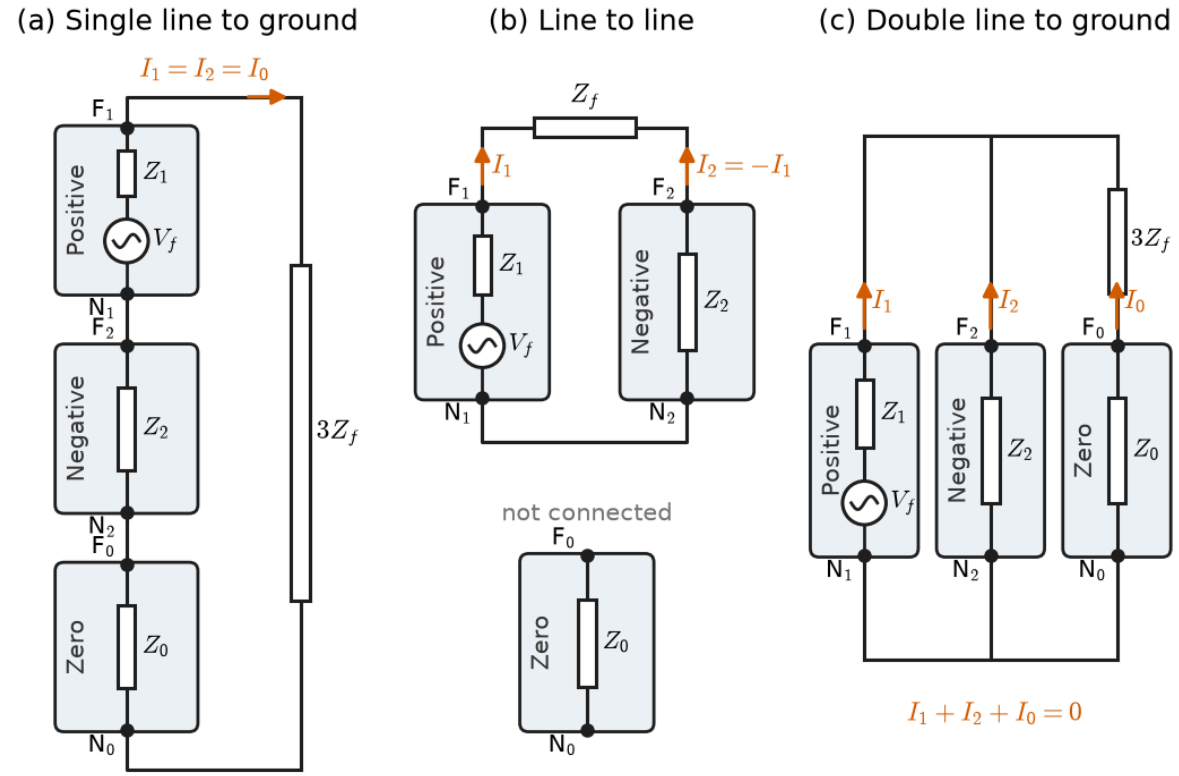


Figure 19.4 — Interconnection of the positive-, negative- and zero-sequence networks at the fault point for single-line-to-ground, line-to-line and double-line-to-ground faults.

Figure 19.4 shows the three connections. Once the sequence currents are known, sequence voltages at the fault follow from $V_1 = V_f - Z_1 I_1$, $V_2 = -Z_2 I_2$ and $V_0 = -Z_0 I_0$, and phase quantities from the transformation matrix. Elsewhere in the network, the sequence currents in each branch come from the columns of the three sequence $\mathbf{Z}_{\text{bus}}$ matrices, exactly as in Section 19.2.

Which fault is the worst?

For a fault far from generators, where $Z_1 = Z_2$, the SLG current compared with the three-phase current is

$$\frac{I_{\text{SLG}}}{I_{\text{3ph}}} = \frac{3Z_1}{2Z_1 + Z_0}$$

so the SLG fault exceeds the three-phase fault whenever $Z_0 < Z_1$. That happens at the LV terminals of a Dyn transformer, at generator terminals with a solid neutral, and at transmission substations with many earthed autotransformers. The LL fault current is $\sqrt{3}/2 = 0.87$ times the three-phase current when $Z_1 = Z_2$, so it never governs breaker ratings far from machines, but it is the minimum-current case that protection must still detect.

Worked example — faults on the low-voltage side of a data-centre transformer. A data-centre electrical room is fed from an 11 kV ring through a 2.5 MVA, 11/0.415 kV Dyn11 transformer with 6 % impedance and a solidly earthed LV neutral. The 11 kV fault level at the ring main unit is 250 MVA. On the transformer base (2.5 MVA, 415 V; base current 3478 A) the source impedance is $2.5/250 = 0.01$ pu and the transformer 0.06 pu. Take all impedances to have the same angle so that magnitudes add, and use $Z_0 = 0.95\,Z_T = 0.057$ pu for the zero-sequence impedance seen from the LV side (the 11 kV source does not appear in it, because the delta blocks zero-sequence current). Then $Z_1 = Z_2 = 0.07$ pu.

- Three-phase: $I = 1/0.07 = 14.29$ pu $= 49.7$ kA.
- SLG: $I = 3/(0.07 + 0.07 + 0.057) = 15.23$ pu $= 53.0$ kA, larger than the three-phase fault because $Z_0 < Z_1$.
- LL: $I = \sqrt{3}/0.14 = 12.37$ pu $= 43.0$ kA.
- LLG: $I_1 = 1/(0.07 + 0.07 \parallel 0.057) = 9.86$ pu; the faulted-phase currents are 14.8 pu (51.5 kA), and the earth current $3I_0$ is 16.3 pu (56.7 kA).

The LV switchboard must therefore be rated for about 53 kA of symmetrical current, not the 50 kA a three-phase calculation suggests, and the earth-return path (neutral bar, main earthing conductor and bonding) must carry up to 57 kA for the clearing time. The view from the 11 kV side is very different. A three-phase LV fault draws $14.29 \times 131.2 = 1875$ A in all three HV lines (131.2 A is the HV base current). An SLG fault on the LV side, transferred through the delta, appears as current in only two HV lines, each $15.23/\sqrt{3} = 8.79$ pu, or 1154 A, and nothing in the third. The HV fuses or overcurrent relays see 62 % of the three-phase current, clear the LV earth fault more slowly, and may not clear an arcing LV earth fault at all. This is why LV switchboards carry their own earth-fault protection (and, in North America, the ground-fault protection of equipment required by NFPA 70 for 480 V services of 1000 A and above).

19.6 Fault-current calculation

The standard procedures

Two calculation standards dominate practice. **IEC 60909** (used in most of the world) computes the fault current from the Thévenin impedance at the fault with an equivalent voltage source $cV_n/\sqrt{3}$ at the fault point,

where the **voltage factor** c covers the unknown pre-fault voltage, tap positions and load: $c_{\max} = 1.10$ for maximum currents at medium and high voltage (1.05 or 1.10 at low voltage depending on tolerance), $c_{\min} = 1.00$ for minimum currents (0.95 at LV). Correction factors are applied to generator and power-station-unit impedances. The **IEEE C37.010/C37.13 and IEEE 551** procedures, used in North America, compute a first-cycle (momentary) duty with subtransient reactances and an interrupting duty with motor reactances multiplied by factors of 1.5–3 to represent their decay, then apply multiplying factors that depend on X/R and on whether the fault is "local" (fed predominantly by nearby generators) or "remote".

Both methods compute, from one network model, a family of currents that answer different questions:

Quantity (IEC 60909 name)	What it is	What it is compared with
Initial symmetrical current I''_k	RMS AC component at fault inception, subtransient reactances	Breaker breaking rating (for far-from-generator faults), relay settings
Peak current i_p	Highest instantaneous current including DC offset	Breaker making rating, busbar and support mechanical strength
Symmetrical breaking current I_b	AC component at the instant of contact separation	Breaker breaking rating (near generators and motors)
DC component i_{DC} at contact separation	Decayed offset at that instant	Breaker's rated DC time constant
Thermal equivalent current I_{th}	RMS current giving the same I^2t as the real decaying current over the fault duration	Short-time withstand of cables, CTs, switchgear
Steady-state current I_k	Current after all transients, with excitation effects	Back-up protection of generator-fed systems

Peak current and the factor κ

The peak current is written as a multiple of the symmetrical peak:

$$i_p = \kappa\sqrt{2}\, I''_k, \qquad \kappa \approx 1.02 + 0.98\, e^{-3R/X}$$

> **Status of the principle: IEC 60909 peak factor**
> *Empirical.* A curve fitted to the exact first peak of the RL transient of *Circuits, Signals and Communications* Ch. 6 over the practical range of R/X; in meshed networks with branches of different X/R the standard adds rules (method (b) multiplies κ by 1.15, method (c) uses an equivalent frequency) because no single X/R describes the network.

κ runs from 1.02 for a purely resistive circuit to 2.0 for a purely inductive one (Figure 19.5). At $X/R = 10$ it is 1.75; at $X/R = 20$ it is 1.86.

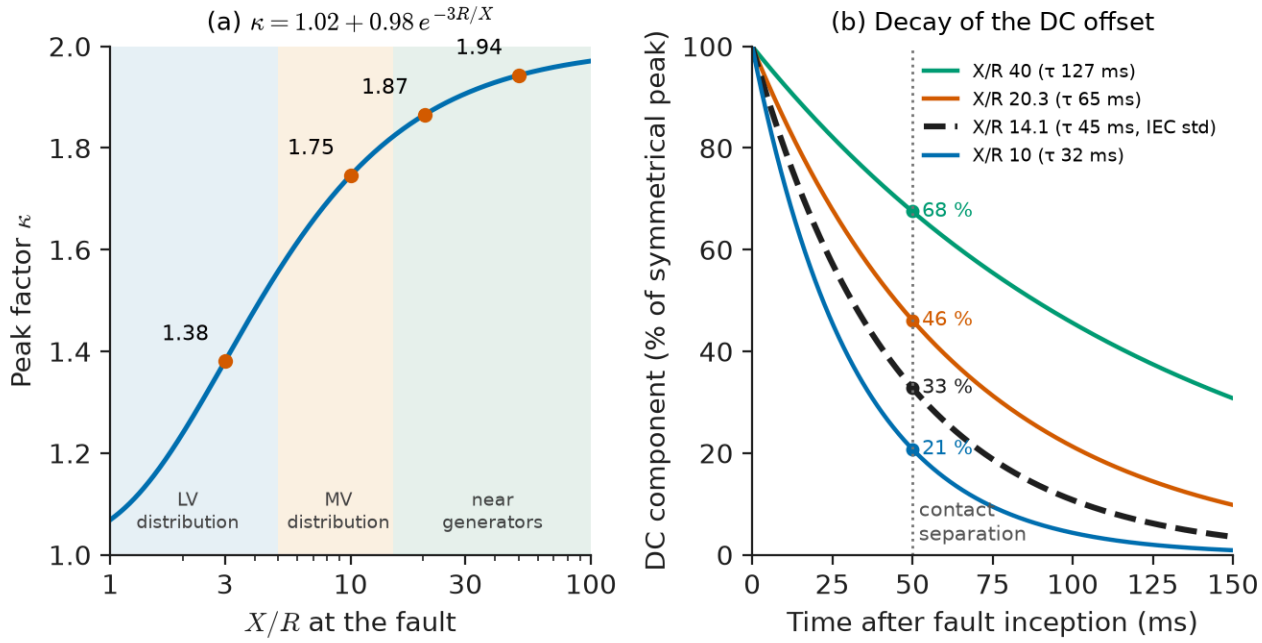


Figure 19.5 — IEC 60909 peak factor kappa versus X/R, and the decay of the DC component for different X/R ratios compared with the 45 ms standard breaker time constant.

Worked example — peak and breaking duty at the plant switchboard. Return to the 11 kV board of Section 19.2 and give each branch a realistic X/R: grid 15, transformer 25, generator 40, motors 10. Combining the complex branch impedances gives a Thévenin impedance $Z_{\mathrm{th}} = 0.0102 + j0.2062$ pu, so $X/R = 20.3$ at the bus. With $c = 1.1$:

$$I''_k = \frac{1.1}{|Z_{\mathrm{th}}|} \times 5249 = 27.96\ \mathrm{kA}, \qquad \kappa = 1.02 + 0.98\,e^{-3\times 0.0494} = 1.865, \qquad i_p = 1.865\sqrt{2} \times 27.96 = 73.8\ \mathrm{kA}$$

IEC method (b) would multiply κ by 1.15, giving 2.14, which the standard caps at 2.0 for this voltage level, and $i_p = 79.1$ kA. A 25 kA breaker has a rated making current of $2.5 \times 25 = 62.5$ kA (50 Hz), so it fails on making duty as well as on breaking duty.

The breaking current, at a minimum time delay of 50 ms (relay plus breaker opening time), decays in the generator and motor branches. With the IEC decay factor $\mu = 0.71 + 0.51\,e^{-0.30 I''_{kG}/I_{rG}}$ for the generator, where $I''_{kG} = 5.15$ kA and the rated current $I_{rG} = 656$ A give a ratio of 7.85 and $\mu = 0.76$, and with the motor contribution roughly halved by 50 ms, the breaking current is about $17.75 + 0.76 \times 5.15 + 0.5 \times 5.07 = 24.2$ kA. The DC component at 50 ms, with the bus time constant $\tau = X/(\omega R) = 64.5$ ms, is $e^{-50/64.5} = 46$ % of the symmetrical peak, well above the 33 % that a breaker tested at the IEC 62271-100 standard time constant of 45 ms is proven to interrupt. The replacement switchgear must therefore be specified for 31.5 kA breaking current, 80 kA making current and a DC time constant of 75 ms (one of the standard's alternative values for special cases), or the generator contribution must be limited (Section 19.7).

Contributions that are easy to get wrong

- **Motors.** Large induction and synchronous motors contribute for the first few cycles; IEC 60909 includes them when their total exceeds 5 % of the fault current without them. In process plants and on ships they can supply a third of the peak current.
- **Inverter-based resources.** PV inverters, battery converters and full-converter wind turbines limit their current electronically to 1.0–1.5 times rated, with no DC offset of their own. A 100 MW solar plant therefore adds roughly 100–150 MVA to a fault level, compared with 500–700 MVA from a

synchronous generator of the same rating. Grid codes specify how fast they must inject reactive current during a fault (for example within 30–60 ms) and, under IEEE 2800-2022 and the ENTSO-E requirements for generators, negative-sequence current. A network dominated by inverters has a lower and more controller-dependent fault level, which degrades overcurrent protection (Chapter 27).

- **Doubly fed induction generators.** Wind turbines of this type contribute like an induction machine until the rotor-side converter's crowbar or chopper acts, typically 5–8 times rated current for the first cycle.
- **Parallel transformers.** Two transformers in parallel halve the source impedance and roughly double the LV fault level; paralleling them is often the decision that overdutied a switchboard.

19.7 Equipment fault ratings

Circuit breakers and switchgear

High-voltage AC breakers are specified under IEC 62271-100 (or IEEE C37.04/C37.06) by a set of ratings that map one-to-one onto the computed fault quantities:

Rated voltage	Typical rated short-circuit breaking currents	Rated making current (50 Hz)
12 kV	25, 31.5, 40 kA	2.5 × breaking current
36 kV	25, 31.5 kA	2.5 ×
145 kV	31.5, 40 kA	2.5 ×
245 kV	40, 50 kA	2.5 ×
420 kV	50, 63 kA	2.5 ×

The making factor is 2.5 at 50 Hz and 2.6 at 60 Hz because it corresponds to $\kappa\sqrt{2}$ at the standard time constant of 45 ms. The **short-time withstand current** equals the breaking current for 1 s (IEC preferred) or 3 s, and covers switchgear, disconnectors and earthing switches. Generator circuit breakers, which interrupt currents with DC time constants of 130 ms or more and may see delayed current zeros, are tested under the dedicated IEC/IEEE 62271-37-013 standard. Metal-enclosed switchgear additionally carries an **internal arc classification** (IEC 62271-200, IAC AFLR 25 kA 1 s, for instance) stating that an internal arc of that current and duration will not eject hot gases at people standing at the front, lateral and rear sides.

Low-voltage devices are rated by their **breaking capacity**: for moulded-case and air circuit breakers under IEC 60947-2 the ultimate (I_{cu}) and service (I_{cs}) breaking capacities, typically 25–150 kA, and for LV switchboards under IEC 61439 the rated short-time withstand I_{cw} and peak I_{pk}. Current-limiting fuses (IEC 60269) interrupt within the first half-cycle and have breaking capacities of 80–120 kA; they are often used as **back-up** protection for breakers whose own rating is lower than the prospective fault current, a combination that must be verified by the manufacturer's tested cascade tables.

Cables, busbars and transformers

Conductors must survive the heat of the fault until it is cleared. Because the fault is short, heat has no time to leave the conductor, and the **adiabatic** criterion of IEC 60949 applies:

$$I^2 t \le k^2 S^2$$

> **Status of the principle: Adiabatic short-circuit rating**
> *Approximate.* It assumes all heat is stored in the conductor, which is conservative for durations up to about 5 s; k (in A·s½/mm²) is computed from the conductor material and the initial and final temperatures permitted by the insulation.

Typical values of k are 143 for copper with XLPE or EPR insulation (90 °C to 250 °C), 115 for copper with PVC (70 °C to 160 °C), 94 for aluminium with XLPE and 76 for aluminium with PVC. For the 11 kV plant, with a clearing time of 0.5 s and the 28 kA of Section 19.2, a copper XLPE cable needs a cross-section of at least

$$S = \frac{I\sqrt{t}}{k} = \frac{28\,000 \times \sqrt{0.5}}{143} = 138 \text{ mm}^2$$

A 95 mm² cable withstands only $143 \times 95/\sqrt{0.5} = 19.2$ kA for 0.5 s and would be damaged even though its continuous rating is ample for a 3 MW motor feeder; 150 mm² withstands 30.3 kA. On long feeders, this fault-withstand requirement, not the load current, often sets the size of the first cable leaving a substation. The metallic screen of an MV cable has its own earth-fault rating (for example 3–10 kA for 1 s), which must be checked against the earth-fault current set by the neutral earthing.

Busbars and their supports must withstand the electromagnetic force between conductors at the **peak** current i_p (IEC 60865); the force grows with i_p^2, so the 73.8 kA of the plant example produces about 39 % more force than the 62.5 kA the original board was built for. Power transformers must withstand external faults for 2 s (IEC 60076-5); repeated through-faults loosen windings, so operators track the cumulative through-fault duty of critical units. Current transformers must not saturate during the DC offset before the protection has operated, which is why protection CTs are dimensioned with a transient factor that grows with the primary time constant (IEC 61869-2, classes TPX, TPY, TPZ).

Living with rising fault levels

Fault levels rise as networks are meshed, transformers paralleled and generators added; they fall as synchronous plants retire and inverters take their place. When they rise above equipment ratings, the options are:

Measure	How it works	Trade-off
Replace switchgear	Higher breaking, making and withstand ratings	Cost, outage time
Split the busbar (open bus-section)	Halves the number of parallel sources	Lower reliability, uneven loading
Higher-impedance transformers	Raises Z_1 of the main infeed	More voltage regulation, more losses
Series (current-limiting) reactor	Adds impedance to a feeder or generator circuit	Continuous voltage drop and reactive losses
Pyrotechnic current limiter (I_s-limiter)	Detects the rising current and inserts a fuse in under 1 ms, before the first peak	Single-shot device, replaced after operation
Superconducting fault-current limiter	Superconductor quenches to a high resistance during the fault	Cryogenic system, still at demonstration-to-early-commercial stage
Neutral impedance	Limits earth-fault current only	Higher overvoltages on healthy phases

For the plant example, a series reactor large enough to hold the bus fault current at 25 kA with $c = 1.1$ would need about 1.58 pu on 100 MVA, or 1.91 Ω, in the generator circuit. At the generator's rated current of 656 A it would drop 1.25 kV, about 20 % of the phase voltage, and absorb reactive power continuously: an unacceptable penalty. An I_s-limiter in the bus-section or generator circuit, or new 31.5 kA switchgear, is the practical answer. Superconducting limiters rely on the transition of a superconductor to the normal state, whose physics is treated in *Engineering Physics*, Ch. 7.

19.8 Fault-location concepts

Why location matters

Protection trips the faulted section; someone must then find the fault. On a 120 km line crossing mountains, a 10 % location error means patrolling 12 km of route; on a buried cable it means digging in the wrong street. Two families of methods exist: impedance-based methods using the 50/60 Hz phasors recorded by relays, and travelling-wave methods timing the high-frequency transients launched by the fault.

Impedance-based location

A distance relay already computes the impedance from its location to the fault (Chapter 23). If the line impedance per kilometre is known, the apparent impedance gives the distance. For an earth fault, the measurement uses the faulted-phase voltage divided by the phase current compensated for the zero-sequence path, $I_a + k_0\,3I_0$ with $k_0 = (Z_0 - Z_1)/(3Z_1)$, so that the result is in positive-sequence ohms. Two errors limit accuracy:

- **Fault resistance and remote infeed.** The voltage drop across the fault resistance is caused by the total fault current, which includes current from the remote end. Seen from one end, the fault resistance is multiplied by the ratio of total fault current to local current, and acquires a phase shift if the two ends' currents are not in phase. The resulting reactance error is the main weakness of one-ended methods.

- **Line parameters.** Errors in Z_0 (soil resistivity, Section 19.4), untransposed sections, mutual coupling with a parallel circuit and non-homogeneous lines (part cable, part overhead) all bias the result.

Worked example — one-ended and two-ended location on a 132 kV line. A 60 km, 132 kV line has $z_1 = 0.12 + j0.40\ \Omega$/km. An earth fault occurs at 25 km from substation A through a tower-footing and arc resistance of 5 Ω. The remote end feeds the fault strongly, so the total current through the fault resistance is 1.5 times the compensated local current and lags it by 10°. The apparent impedance seen from A is

$$Z_{\text{app}} = 25(0.12 + j0.40) + 5 \times 1.5\angle -10^\circ = (3.00 + j10.00) + (7.39 - j1.30) = 10.39 + j8.70\ \Omega$$

A **reactance method**, which divides the measured reactance by the line reactance per kilometre to ignore the (supposedly resistive) fault term, gives $8.70/0.40 = 21.7$ km: an error of 3.3 km, or 5.4 % of the line, entirely due to the phase shift of the remote infeed. Two-ended methods remove it. With synchronised phasors from both ends (from relays time-stamped by GPS, Chapter 26), the fault resistance and remote infeed become known quantities, and accuracy of 1–2 % of line length is routine. A **travelling-wave** locator does better still. The fault launches voltage and current waves in both directions at close to the speed of light; with a propagation speed of 2.95×10^5 km/s, they arrive at A after 84.7 μs and at B, 35 km away, after 118.6 μs. From the difference in arrival times $\Delta t = t_A - t_B = -33.9$ μs,

$$d_A = \frac{\ell + v\,\Delta t}{2} = \frac{60 + (2.95 \times 10^5)(-33.9 \times 10^{-6})}{2} = 25.0\ \text{km}$$

A timing error of 1 μs moves the result by only $v \times 1\ \mu\text{s}/2 = 0.15$ km, which is why travelling-wave locators routinely achieve ±300 m on lines hundreds of kilometres long, about one tower span.

> **Boundary.**
> The propagation of surges as waves on lines, their reflection at discontinuities, and time-domain reflectometry are developed in *Circuits, Signals and Communications*, Ch. 14 §14.5. Travelling-wave locators apply that theory to the transients a fault launches by itself.

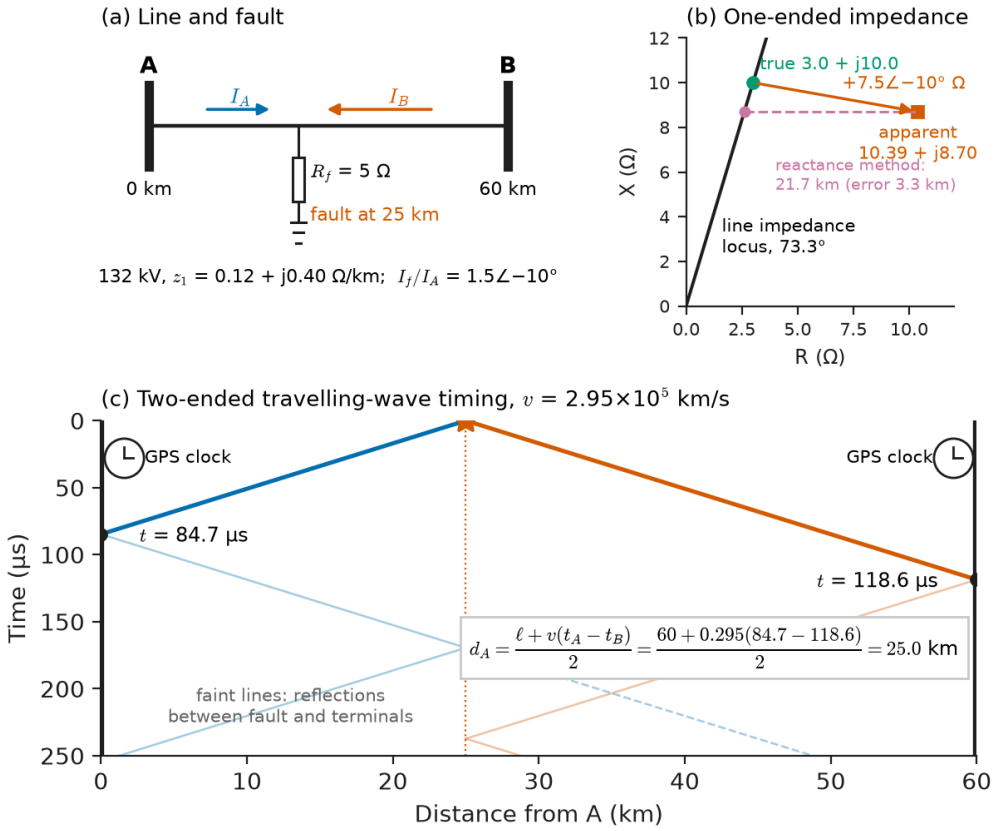


Figure 19.6 — Two fault-location methods on a 60 km, 132 kV line: one-ended impedance measurement with remote infeed, and two-ended travelling-wave timing.

Figure 19.6 compares the two approaches. Single-ended travelling-wave methods, timing the reflection from the fault, exist but are less robust.

Distribution networks and cables

Radial distribution feeders branch, so a distance from the substation identifies several candidate locations. Utilities combine the computed distance with **fault passage indicators** (devices on poles or in ring main units that flag whether fault current passed them), with smart-meter last-gasp messages, and with automatic sectionalising that isolates the faulted section within a minute and restores the rest (Chapters 22 and 26).

Underground cables need a further step because the computed distance is along a route that is not visible. The sequence is **pre-location** (time-domain reflectometry; for high-resistance faults the arc-reflection method, in which a surge strikes an arc at the fault so that the reflectometer sees a short circuit), **route tracing**, and **pinpointing** by "thumping": a surge generator discharges a capacitor of a few kilojoules into the cable every few seconds, the fault flashes over, and an acoustic receiver on the surface locates the bang to within a metre. Repeated high-energy surges stress aged XLPE insulation, so thumping is kept short.

Method	Network	Typical accuracy	Needs
One-ended impedance (reactance, Takagi)	Transmission, sub-transmission	2–10 % of line length	Relay records at one end, line data
Two-ended impedance	Transmission	1–2 %	Synchronised or aligned records from both ends
Travelling-wave, two-ended	Transmission, HVDC lines	±150–300 m	High-speed recorders (MHz sampling), GPS timing
Fault passage indicators and smart meters	Distribution	Section between devices	Indicator placement, communications
TDR, arc reflection, thumping	Cables	Pre-location to a few %, pinpoint to < 1 m	Specialist test van, cable access

What remains open

Open question	Where it is answered
How are relays set to detect each fault type and trip only the faulted section?	Chapter 23 (Power-System Protection)
How do line parameters, including zero-sequence impedance, follow from tower geometry?	Chapter 21 (Transmission Lines)
Will the generators stay in step while the fault is on and after it is cleared?	Chapter 20 (Power-System Stability)
How does arc current translate into incident energy on a worker, and which PPE is required?	Chapter 29 (Electrical Safety)
How are earthing systems designed so that earth-fault currents do not create dangerous touch and step voltages?	Chapters 24 and 30
How do voltage sags caused by remote faults affect sensitive loads?	Chapter 25 (Power Quality)
How do inverter-based resources behave during faults, and what do grid codes require?	Chapter 27 (Grid Integration of Renewable Energy)
How are the per-unit models and $\mathbf{Y}_{\text{bus}}$ used here built?	Chapters 17 and 18
What is the RL transient behind the DC offset?	*Circuits, Signals and Communications*, Ch. 6

Next: Chapter 20 asks what happens to the generators while the fault of this chapter is on the system, and whether the network returns to a steady state after it is cleared.

Chapter 20 — Power-System Stability

Part V · Power-System Analysis

On the afternoon of 9 August 2019, lightning struck a 400 kV line north of London. The protection cleared the fault in less than 100 ms, exactly as designed, and the line reclosed successfully. Within the next second, an offshore wind farm reduced its output by 737 MW because its controls reacted badly to the voltage dip, a steam turbine at a combined-cycle plant tripped, and about 500 MW of small generators embedded in distribution networks disconnected because their loss-of-mains relays misread the disturbance. The frequency fell faster than the reserves could respond. Over the next minute two gas turbines at the same plant tripped, the total loss approached 1.9 GW, and the frequency reached 48.8 Hz, the first stage of automatic under-frequency load shedding. About 1 GW of demand, roughly 1.1 million customers, was disconnected; trains stopped, some of them for hours, because their on-board protection reacted to the frequency. Every individual device did what it was set to do. The event was a stability problem: the question of whether a power system, disturbed from its operating point, returns to an acceptable steady state, and how far it travels on the way.

> **Will the generators stay in synchronism after a fault, and how fast must the fault be cleared?**
> **How much more load can a network carry before its voltage collapses, and how far can the frequency fall after losing a large generator? How are stability margins measured, improved and checked by simulation?**

Concept	Engineering destination
Rotor-angle stability	Synchronous generators staying in step after disturbances; transfer limits on corridors
Small-signal stability	Power-system stabilisers, damping of inter-area oscillations, controller tuning
Transient stability	Critical clearing times, protection speed requirements, generator tripping schemes
Voltage stability	Load-area import limits, reactive reserves, under-voltage load shedding
Frequency stability	Inertia and reserve requirements, under-frequency load shedding, battery fast frequency response
Disturbance response	Sequence of events after a fault or trip; ride-through requirements
Stability margins	Planning criteria (N-1), operating limits, grid-code requirements
Stability improvement	Fast protection, FACTS, PSS, synchronous condensers, grid-forming inverters
Simulation and interpretation	RMS and EMT studies, model validation, reading swing curves and frequency traces

Roadmap. A classification and the rotor-angle problem (20.1) → its linear form, small-signal stability (20.2) → its nonlinear form, transient stability and the equal-area criterion (20.3) → voltage stability (20.4) → frequency stability (20.5) → how a real system responds to a disturbance over milliseconds to minutes (20.6) → how margins are defined (20.7) and improved (20.8) → how studies are done and read (20.9).

20.1 Rotor-angle stability

A classification

Power-system stability is the ability of the system, for a given initial operating condition, to regain a state of operating equilibrium after a physical disturbance, with most system variables bounded so that practically the whole system remains intact. The IEEE/CIGRE classification (2004, extended in 2021 for inverter-dominated systems) divides the problem by the variable that goes wrong, the size of the disturbance and the time scale:

Category	Variable that drifts	Disturbance size	Time scale	Main analysis tool
Rotor-angle, small-signal	Generator angles, oscillating	Small (load changes)	1–30 s	Eigenvalues of the linearised model
Rotor-angle, transient	Generator angles, first swing or later	Large (faults)	0.5–10 s	Time-domain simulation, equal-area criterion
Voltage, short-term	Bus voltages	Large	0.5–10 s	Simulation with dynamic load (motor) models
Voltage, long-term	Bus voltages	Small or large	Minutes	Continuation power flow, PV and QV curves
Frequency	System frequency	Large (loss of generation or load)	1 s–minutes	Aggregated frequency models, simulation
Resonance and converter-driven (2021 additions)	Torsional modes, converter control interactions	Small or large	ms–seconds	EMT simulation, impedance-based analysis

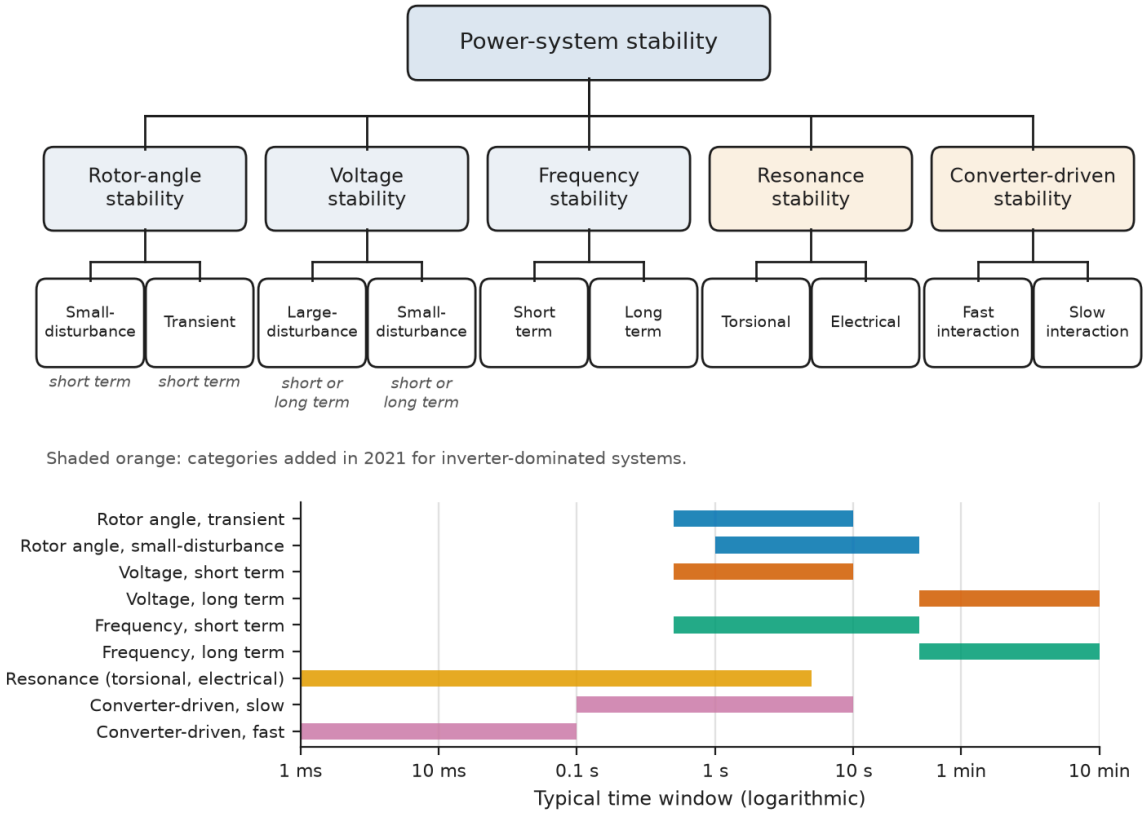


Figure 20.1 — Classification of power-system stability by variable, disturbance size and time scale (IEEE/CIGRE 2004 with 2021 extensions).

The categories are linked: a voltage collapse makes generators lose synchronism, and an angle separation splits the system into islands with frequency problems. Figure 20.1 shows the tree; this chapter follows its branches.

The swing equation and the inertia constant

A synchronous generator stays in step when its rotor turns at synchronous speed, on average, with every other generator. If the mechanical power P_m from the turbine and the electrical power P_e to the network are unequal, the difference accelerates or decelerates the rotor, and the rotor angle δ (the angle of the internal EMF relative to a synchronously rotating reference) changes. In per unit,

$$\frac{2H}{\omega_s}\frac{d^2\delta}{dt^2} = P_m - P_e - D\frac{d\delta}{dt}$$

> **Boundary.**
> The swing equation is Newton's second law for rotation, $J\dot{\omega} = T_m - T_e$ (*Engineering Physics*, Ch. 9 §9.7 and Ch. 19 §19.2), written in electrical angle and per-unit power. Its linearisation into a second-order model with a synchronising coefficient, natural frequency and damping ratio is worked in *Circuits, Signals and Communications*, Ch. 33 §33.2. Here we use both results and supply the engineering data and consequences.

The **inertia constant** H is the kinetic energy stored in the rotating mass at synchronous speed divided by the machine rating, $H = \tfrac{1}{2}J\omega_{sm}^2 / S_{\text{rated}}$, in seconds. It answers a physical question: for how many seconds could the stored energy alone supply rated power? Because it is normalised to rating, it varies surprisingly little across machine sizes:

Machine	Typical H on own rating
Two-pole steam or nuclear turbo-generator (3000/3600 r/min), including turbine	2.5–6 s
Four-pole nuclear turbo-generator (1500/1800 r/min)	4–10 s
Gas turbine and combined-cycle units	4–8 s
Hydro generator (slow-speed salient pole)	2–4 s
Synchronous condenser (with flywheel)	1–2 s (up to 8 s with added flywheel)
Large induction motor load	0.5–2 s
Wind turbine (rotor and generator, behind a converter)	2–6 s physically, but decoupled from the grid unless the controls emulate inertia
PV inverter, battery converter	0 physically; grid-forming control can provide synthetic inertia

The power-angle curve and synchronising power

For a generator with internal voltage E' behind a transient reactance, connected through a total reactance X to an infinite bus at voltage V, the electrical power is $P_e = (E'V/X)\sin\delta = P_{\max}\sin\delta$ (Chapter 7). Steady operation needs $P_e = P_m$, which has a solution only if $P_m < P_{\max}$, and the solution with $\delta < 90^\circ$ is stable: if the rotor advances, P_e increases, the rotor decelerates and returns. The slope

$$P_s = \frac{dP_e}{d\delta} = P_{\max}\cos\delta_0$$

is the **synchronising power coefficient**. A network that is electrically weak (large X) has a small $P_{\max}$, operates at a larger angle for the same power, and has a small P_s: it is held in synchronism by a weak spring. A tripped line, which raises X, weakens the spring further.

Rotor-angle instability shows up in two ways. Lack of synchronising torque gives a non-oscillatory drift of the angle (aperiodic instability, now rare because automatic voltage regulators hold E' up). Lack of **damping torque** gives oscillations of growing amplitude. Both are small-signal phenomena when the disturbance is small and transient phenomena when it is large.

20.2 Small-signal stability

Modes of oscillation

Linearised about an operating point, the power system becomes a linear state-space model $\Delta\dot{\mathbf{x}} = \mathbf{A}\,\Delta\mathbf{x}$, and its stability is decided by the eigenvalues of $\mathbf{A}$: every eigenvalue must have a negative real part (*Circuits, Signals and Communications*, Ch. 34 §34.4). A real system has thousands of states: rotor angles and speeds, flux linkages, exciter and governor states, converter controls. A complex pair of eigenvalues $\sigma \pm j\omega_d$ is an **oscillation mode** with frequency $f = \omega_d/2\pi$ and damping ratio

$$\zeta = \frac{-\sigma}{\sqrt{\sigma^2 + \omega_d^2}}$$

The mode's **eigenvector** shows which generators swing and against whom, and its **participation factors** show which states, and hence which controllers, influence it. The modes that matter in practice are:

Mode	Frequency	What swings	Typical cause of poor damping
Local (plant) mode	0.7–2.0 Hz	One plant against the rest of the system	Fast high-gain AVR on a plant with a weak connection
Inter-area mode	0.1–0.8 Hz	Groups of generators in one region against another	Heavy power transfer across weak tie-lines
Control modes	Varies	Exciters, governors, HVDC and FACTS controls, inverter PLLs	Poorly tuned controllers
Torsional modes	10–50 Hz	Turbine-generator shaft sections against each other	Series-capacitor resonance (subsynchronous resonance), converter controls

The Western North American system has a well-known inter-area mode near 0.25 Hz between the Pacific Northwest and the Southwest; continental Europe has east–west modes near 0.15–0.25 Hz. Their undamped growth in 1996 contributed to the separation of the western US system into islands, and their damping is now monitored in real time with phasor measurement units (Chapter 26).

> **Status of the principle: Small-signal (modal) stability**
> *Derived*. Lyapunov's first method: the stability of the linearised model decides the local stability of the nonlinear system; valid only for disturbances small enough that the linearisation holds, and silent about what happens after a large fault.

Why voltage regulators reduce damping, and how stabilisers restore it

A fast, high-gain automatic voltage regulator (AVR) is good for transient stability, because it pushes E' up quickly after a fault and raises $P_{\max}$. It is bad for damping. When the rotor swings, the terminal voltage swings with it; the AVR responds by changing the field, and through the field winding's time lag the resulting change in electrical torque arrives partly in phase opposition to rotor speed, a negative damping torque. The Heffron–Phillips analysis of a single machine shows that this negative damping grows with AVR gain and with the external reactance, so the problem is worst on remote plants with long lines.

A **power-system stabiliser** (PSS) is an added control loop that measures rotor speed deviation (or electrical power, or a combination), passes it through phase-lead compensation, and adds it to the AVR reference so that the AVR produces a torque component in phase with speed: positive damping. Grid codes in most countries require a PSS on large synchronous generators and specify its tuning tests (IEEE 421.5 provides standard models such as PSS2B and PSS4B).

Worked example — damping of a remote 600 MVA unit. A 600 MVA steam unit with $H = 4.0$ s is connected through its step-up transformer and two parallel lines to a strong grid. On the machine base the reactances are $X'_d = 0.30$, transformer 0.15 and each line 0.50 pu; the internal voltage is $E' = 1.15$ pu, the grid voltage 1.0 pu and the unit delivers $P_m = 0.85$ pu.

With both lines, $X = 0.30 + 0.15 + 0.25 = 0.70$ pu and $P_{\max} = 1.15/0.70 = 1.643$ pu. The operating angle is $\delta_0 = \arcsin(0.85/1.643) = 31.2^\circ$ and $P_s = 1.643\cos 31.2^\circ = 1.406$ pu/rad. With $M = 2H/\omega_s = 8/314.16 = 0.0255$ s²/rad, the undamped natural frequency is

$$\omega_n = \sqrt{\frac{P_s}{M}} = \sqrt{\frac{1.406}{0.0255}} = 7.43 \text{ rad/s} \quad (1.18 \text{ Hz})$$

With one line out, $X = 0.95$ pu, $P_{\max} = 1.211$ pu, $\delta_0 = 44.6^\circ$, $P_s = 0.862$ pu/rad and the mode falls to 5.82 rad/s (0.93 Hz). Represent all damping effects by an equivalent coefficient D in pu power per rad/s. Natural damping from the damper windings and the load, $D = 0.02$, gives $\zeta = D/(2\sqrt{P_s M}) = 0.053$ with both lines and 0.067 with one: acceptable. Now suppose a high-gain static exciter contributes -0.025 to D, a magnitude typical of such exciters on a weak connection. The net $D = -0.005$ gives $\zeta = -0.017$ with one line out: the oscillation grows, doubling in amplitude every $\ln 2/(|\zeta|\omega_n) = 7.1$ s, and the unit will be tripped by its out-of-step protection within a minute. A PSS adding $+0.040$ makes $D = 0.035$ and $\zeta = 0.118$, comfortably above the usual planning target of 0.05. The example illustrates the engineering pattern: damping is a small difference between larger positive and negative contributions, so it must be designed for, not assumed.

20.3 Transient stability

The first swing

A large disturbance, typically a three-phase fault near a generator, collapses the electrical power the generator can deliver while the turbine's mechanical power stays constant (governors are far too slow to act in a fraction of a second). The rotor accelerates. When the fault is cleared, usually by tripping a line, the electrical power jumps back to a value on a post-fault power-angle curve. If the rotor angle at that moment is not too large, the electrical power exceeds the mechanical power, the rotor decelerates, and the angle reaches a maximum and swings back. If it is too large, the rotor passes the point beyond which electrical power again falls short of mechanical power, and the machine **slips a pole**: its angle runs away, it loses synchronism, and its out-of-step protection must trip it.

The equal-area criterion

For a single machine against an infinite bus, multiplying the swing equation by $d\delta/dt$ and integrating shows that the rotor stops accelerating (its speed deviation returns to zero) when the accelerating energy gained since the disturbance has been returned. On the power-angle diagram this is a statement about areas: the area between P_m and the during-fault curve from the initial angle δ_0 to the clearing angle δ_c (accelerating area A_1) must be matched by the area between the post-fault curve and P_m from δ_c to the maximum angle (decelerating area A_2). The largest available decelerating area ends at $\delta_{\max} = 180^\circ - \arcsin(P_m/P_{\max 3})$, where the post-fault curve falls below P_m again. Setting $A_1 = A_2$ at that limit gives the **critical clearing angle**: with pre-fault, during-fault and post-fault amplitudes $P_{\max 1}$, $P_{\max 2}$ and $P_{\max 3}$,

$$\cos\delta_{\text{cr}} = \frac{P_m(\delta_{\max} - \delta_0) + P_{\max 3}\cos\delta_{\max} - P_{\max 2}\cos\delta_0}{P_{\max 3} - P_{\max 2}}$$

When the fault reduces the electrical power to zero ($P_{\max 2} = 0$, a fault at the generator's HV bus), the swing equation integrates directly during the fault, $\delta(t) = \delta_0 + \omega_s P_m t^2/(4H)$, and the **critical clearing time** follows:

$$t_{\text{cr}} = \sqrt{\frac{4H(\delta_{\text{cr}} - \delta_0)}{\omega_s P_m}}$$

Status of the principle: Equal-area criterion
Approximate. Derived exactly from the swing equation for one machine against an infinite bus, under the classical model (constant voltage behind transient reactance, constant mechanical power, no damping); it extends only approximately to multimachine systems (through equivalents such as SIME) and ignores the AVR, governor and the decay of field flux, which matter for swings longer than about one second.

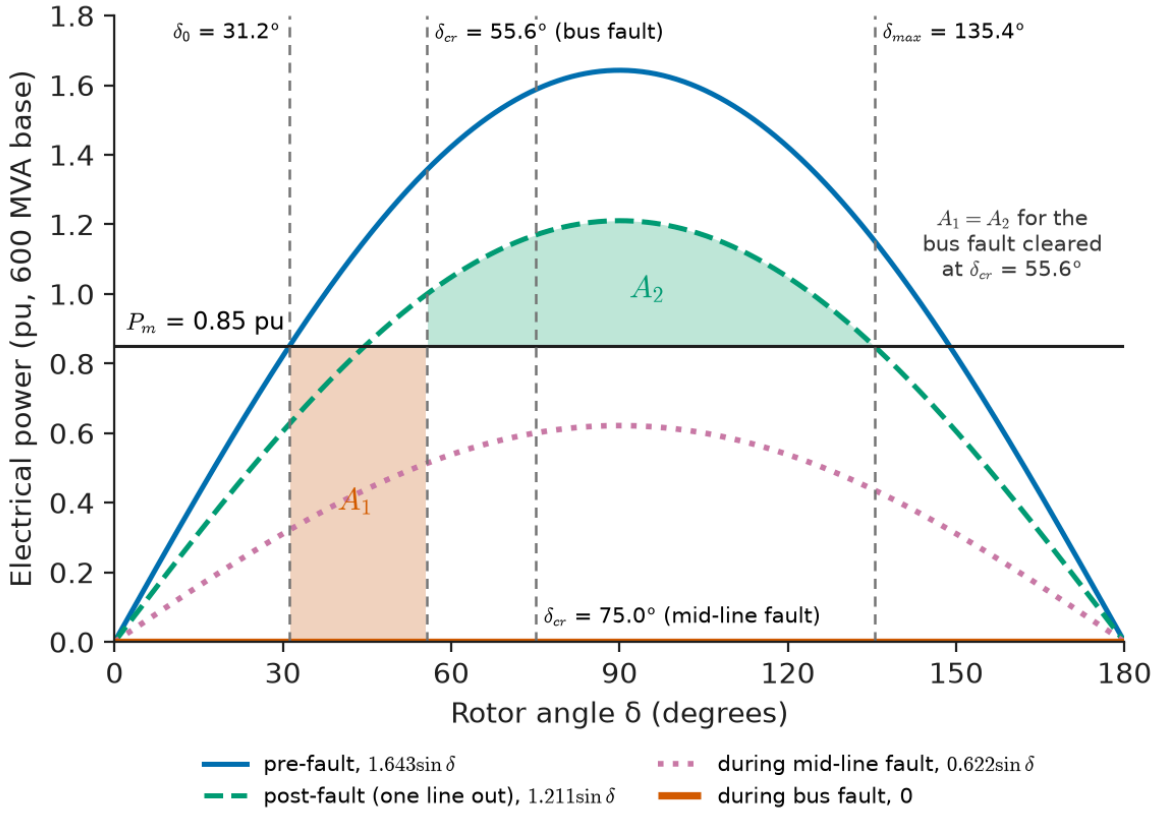


Figure 20.2 — Power-angle curves before, during and after a fault, with the accelerating and decelerating areas of the equal-area criterion at the critical clearing angle.

Figure 20.2 shows the construction for the example that follows.

Worked example — critical clearing time of the 600 MVA unit. Take the unit of Section 20.2 at $P_m = 0.85$ pu ($\delta_0 = 31.2^\circ$, $P_{\max 1} = 1.643$ pu). A three-phase fault occurs on one of the two lines, and is cleared by tripping that line, so the post-fault amplitude is $P_{\max 3} = 1.211$ pu and $\delta_{\max} = 180^\circ - 44.6^\circ = 135.4^\circ$.

Fault at the HV bus end of the line. The generator's HV bus is short-circuited, so $P_{\max 2} = 0$. The formula gives $\delta_{\mathrm{cr}} = 55.6^\circ$ and

$$t_{\mathrm{cr}} = \sqrt{\frac{4 \times 4.0 \times (0.9698 - 0.5438)}{314.16 \times 0.85}} = 0.160 \text{ s}$$

Fault at the middle of the line. The faulted line's two halves (0.25 pu each) now connect the generator side to earth through 0.25 pu; eliminating that node gives a transfer reactance of 1.85 pu and $P_{\max 2} = 1.15/1.85 = 0.622$ pu. The critical clearing angle rises to 75.0°. There is no closed-form time now, so the swing equation is integrated numerically: the critical clearing time is 0.290 s.

A numerical integration of the bus-fault case confirms the analytic result (0.160 s) and shows the swing curves: cleared at 0.10 s, the rotor reaches 40.7° at clearing and a first maximum of 82.9° after 0.45 s; cleared at 0.15 s,

it reaches 52.7° at clearing and swings to 114.6°, stable but with little margin; cleared at 0.20 s, it is at 69.4° at clearing and passes 180° after 0.56 s. A 400 kV substation with modern protection clears close-in faults in 80–100 ms (relay 20–30 ms plus breaker 40–60 ms), which leaves a margin of about 60 ms for the bus fault. If a breaker fails and breaker-failure protection must trip the adjacent breakers after a further 150–200 ms, the total clearing time exceeds 0.16 s, and the unit is lost. Planning studies therefore check both normal and back-up clearing times.

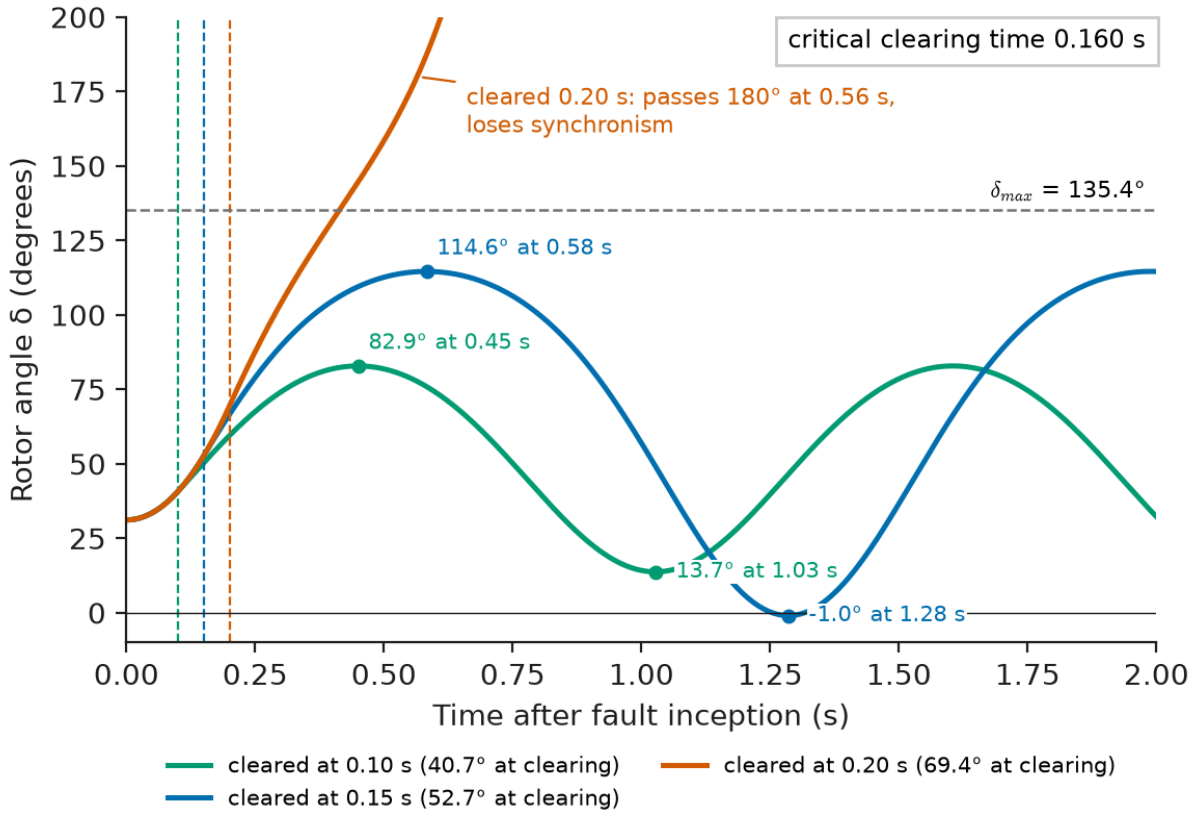


Figure 20.3 — Rotor-angle swing curves of the 600 MVA unit for a three-phase fault at the HV bus cleared at 0.10, 0.15 and 0.20 s (classical model, no damping).

Figure 20.3 shows the three swings. The undamped classical model keeps oscillating forever; a real machine's damping and AVR shrink the swings within a few cycles, and a second or later swing can still be unstable in multimachine systems when inter-area modes are poorly damped. Transient-stability studies therefore run for at least 10–20 s, not just the first swing.

20.4 Voltage stability

The nose of the PV curve

Voltage instability is a different mechanism from loss of synchronism: it can happen with every generator perfectly in step. It comes from the inability of the network and the generators to deliver the reactive power that loads demand at acceptable voltage. Consider a load supplied from a source of voltage E through a reactance X. As the load grows, the current grows and the voltage across X grows, so the load voltage falls. The power delivered, $P = VI\cos\varphi$, first rises with current and then falls, because the voltage falls faster than

the current rises. There is a maximum, at which the load impedance magnitude equals the line reactance (the maximum-power-transfer condition of circuit theory). For a load of constant power factor $\cos\varphi$ (lagging),

$$P_{\max} = \frac{E^2}{2X}\,\frac{\cos\varphi}{1+\sin\varphi}, \qquad V_{\mathrm{cr}} = \frac{E}{\sqrt{2(1+\sin\varphi)}}$$

> **Status of the principle: Maximum loadability of a radial supply**
> *Derived.* Follows from the power-flow equations of a two-bus system with constant source voltage and constant-power-factor load; real networks compute the same "nose point" by continuation power flow, with generator reactive limits and tap changers included.

Plotting load voltage against load power gives the **PV curve** with its characteristic "nose" (Figure 20.4). For every power below the nose there are two voltages; normal operation is on the upper branch. At the nose, the Jacobian of the power-flow equations (Chapter 18) becomes singular, which is why load-flow programs fail to converge near voltage collapse, and why that failure should not be dismissed as a numerical nuisance. The **QV curve**, which plots the reactive power that must be injected at a bus to hold each voltage, carries the same information in a form that operators use to size reactive reserves: the depth of the minimum below zero is the reactive margin.

Worked example — import limit of an industrial area. An industrial area is supplied over a double-circuit 132 kV line and two 132/33 kV transformers. On a 100 MVA base the equivalent reactance from the strong grid to the 33 kV load bus is 0.20 pu with everything in service and 0.30 pu with one circuit out; the source voltage is 1.0 pu. The area's load is 130 MW.

Case	$P_{\max}$	Critical voltage	Load voltage at 130 MW
Both circuits, unity power factor	250 MW	0.707 pu	0.963 pu
Both circuits, 0.95 lagging	181 MW	0.617 pu	0.852 pu
One circuit, unity power factor	167 MW	0.707 pu	0.902 pu
One circuit, 0.95 lagging	121 MW	0.617 pu	No solution: beyond the nose

At 0.95 power factor, the area is already at an unacceptable 0.85 pu with all circuits in service, and the loss of one circuit leaves no operating point at all: the voltage would collapse. With capacitor banks correcting the load to unity power factor at the 33 kV bus, the margin is $1 - 130/250 = 48\ \%$ intact and $1 - 130/167 = 22\ \%$ after the outage, both above the 5 % post-contingency real-power margin that the WECC voltage-stability criterion requires. The calculation also shows the trap of shunt capacitors: they hold voltage up on the upper branch, but their output falls with the square of voltage, so a heavily compensated area operates with near-normal voltage right up to a nose that arrives suddenly. A normal voltage is not evidence of a healthy margin.

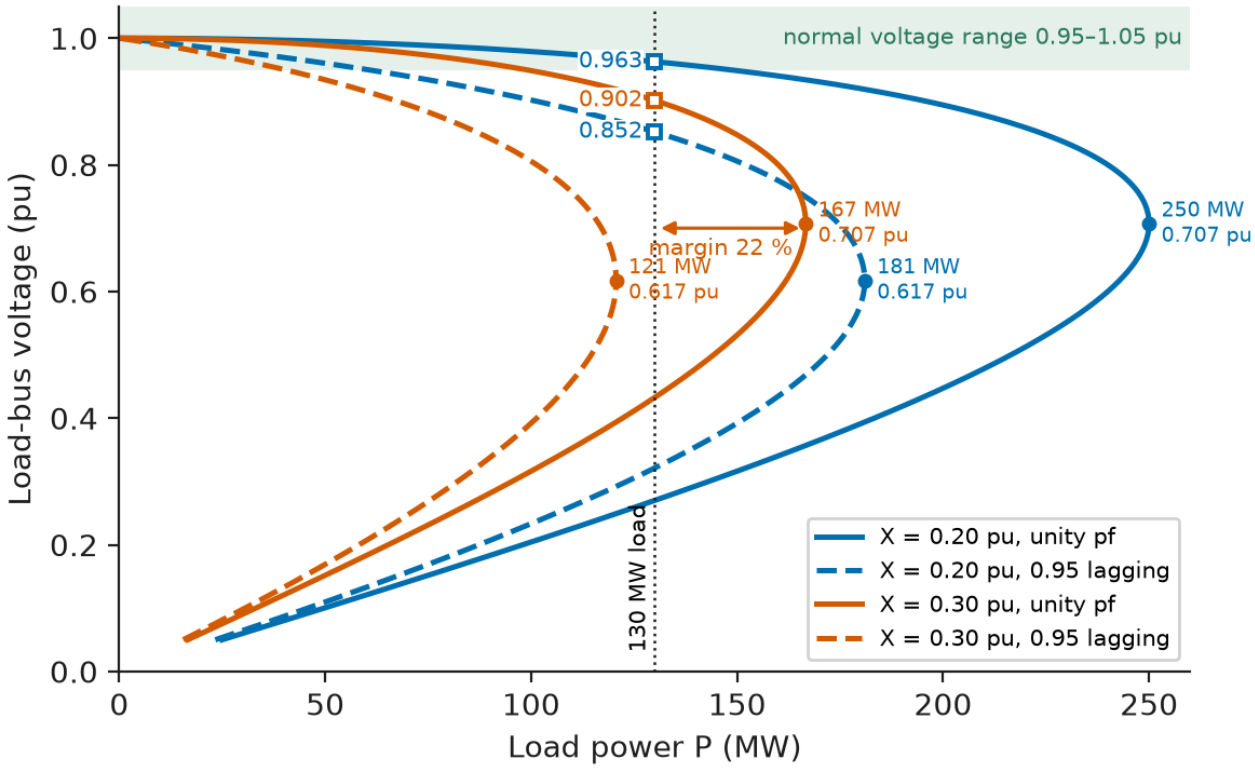


Figure 20.4 — PV curves of a load area supplied through 0.20 pu and 0.30 pu equivalent reactances for unity and 0.95 lagging power factor, showing the nose points and margins at 130 MW.

Dynamics of collapse

Real voltage collapses unfold over seconds to minutes, driven by devices that try to restore load:

- **On-load tap changers** on distribution transformers raise secondary voltage step by step (every 30–60 s), restoring constant-power behaviour of the load and drawing more current from an already strained transmission system.
- **Generator over-excitation limiters** reduce field current after 10–60 s of overload, withdrawing reactive support exactly when it is needed.
- **Induction motors**, especially single-phase air-conditioning compressors with low inertia, stall when voltage stays below about 0.6–0.7 pu for a few cycles; stalled motors draw four to six times their running current at low power factor. The resulting **fault-induced delayed voltage recovery** (FIDVR) has held voltages at 0.7–0.8 pu for 10–30 s after cleared faults in summer-peaking systems in the southern United States.
- **Thermostatically controlled loads** (heaters, refrigeration) run longer at low voltage, so their aggregate power recovers after minutes.

The 1978 France and 1987 Tokyo blackouts were voltage collapses; the August 2003 North American blackout involved voltage decline as lines tripped one after another over an hour before the final cascade. Countermeasures include fast reactive sources (STATCOMs, synchronous condensers), tap-changer blocking, and **under-voltage load shedding**, which disconnects load when voltage stays below a threshold (typically 0.85–0.9 pu for several seconds).

20.5 Frequency stability

Inertia and the first seconds

When a large generator trips, the power balance of the whole interconnected system is upset at once. The missing power is supplied in the first instants from the kinetic energy of every synchronously connected rotor, which slows down together. Aggregating the swing equations of all machines around their centre of inertia gives the initial **rate of change of frequency** (RoCoF):

$$\left.\frac{df}{dt}\right|_{t=0^+} = -\frac{\Delta P\, f_0}{2E_k}, \qquad E_k = \sum_i H_i S_i$$

where E_k is the total stored kinetic energy of the synchronised machines (in MW·s) and ΔP the lost power.

> **Status of the principle: Aggregate frequency model**
> *Approximate.* Assumes a single uniform system frequency; in reality frequency differs between regions by tens of millihertz during the first seconds as inter-area modes swing, which matters for RoCoF relays and for measurements near the disturbance.

The kinetic energy falls as synchronous plants are replaced by inverter-based wind, solar and batteries. Great Britain's synchronous kinetic energy, typically 200–250 GW·s a decade ago, falls below 130 GW·s on windy low-demand days, and the system operator now buys inertia (from synchronous condensers and grid-forming batteries) as a service. Ireland, with an even smaller system, limits the instantaneous share of non-synchronous generation (the SNSP limit, raised in steps from 50 % to 75 %) for the same reason.

Primary response, the nadir and load shedding

Within a few seconds, governors of generators with headroom open their valves in proportion to the frequency deviation (their **droop**, typically 4–5 %: a 4 % droop unit raises output by 100 % of rating for a 4 % fall in frequency). Loads help too: motor loads draw less power at lower frequency, a **load damping** of about 1–2 % of load per hertz. The frequency stops falling at the **nadir**, where the sum of primary response and load relief equals the lost power, and then recovers to a quasi-steady value set by the combined droop. Secondary control (automatic generation control, Chapter 26) restores nominal frequency over minutes.

If the nadir reaches a protection threshold, automatic **under-frequency load shedding** (UFLS) acts. The ENTSO-E network code on emergency and restoration requires shedding to start at 49.0 Hz and to reach 45 % of load by 48.0 Hz in steps of no more than 10 %; Great Britain's low-frequency demand disconnection starts at 48.8 Hz. These schemes are the last line of defence against a collapse of the whole interconnection.

Worked example — frequency after the loss of 1.8 GW. A 35 GW system loses 1.8 GW of generation. Its load damping is 0.7 GW/Hz (2 % per hertz), its primary response is represented as a gain of 3.0 GW/Hz with a first-order lag of 6 s, and its stored kinetic energy depends on the generation mix. The initial RoCoF and the result of integrating the aggregated model are:

Synchronous kinetic energy	Initial RoCoF	Nadir	Time of nadir
200 GW·s (high inertia)	0.225 Hz/s	49.21 Hz	7.0 s
130 GW·s	0.346 Hz/s	49.10 Hz	5.2 s
80 GW·s (low inertia)	0.563 Hz/s	48.96 Hz	3.8 s
80 GW·s plus 0.6 GW of battery fast frequency response	0.563 Hz/s	49.28 Hz	3.3 s
130 GW·s plus 0.6 GW of battery fast frequency response	0.346 Hz/s	49.38 Hz	4.6 s

In every case without fast response the quasi-steady frequency is the same, $50 - 1.8/(0.7 + 3.0) = 49.51$ Hz, because it depends only on droop and load damping. Inertia changes how fast and how far frequency falls before the governors catch up. At 80 GW·s the nadir is only 0.16 Hz above the 48.8 Hz load-shedding threshold, and the initial RoCoF of 0.56 Hz/s exceeds the old 0.125 Hz/s RoCoF settings of embedded generators' loss-of-mains relays, which would trip them and deepen the event exactly as in August 2019 (GB has since raised these settings to 1 Hz/s). Batteries that inject 0.6 GW within about a second, triggered at 49.8 Hz, raise the nadir by about 0.3 Hz: fast response substitutes for inertia, although not for its instantaneous action in the first few hundred milliseconds.

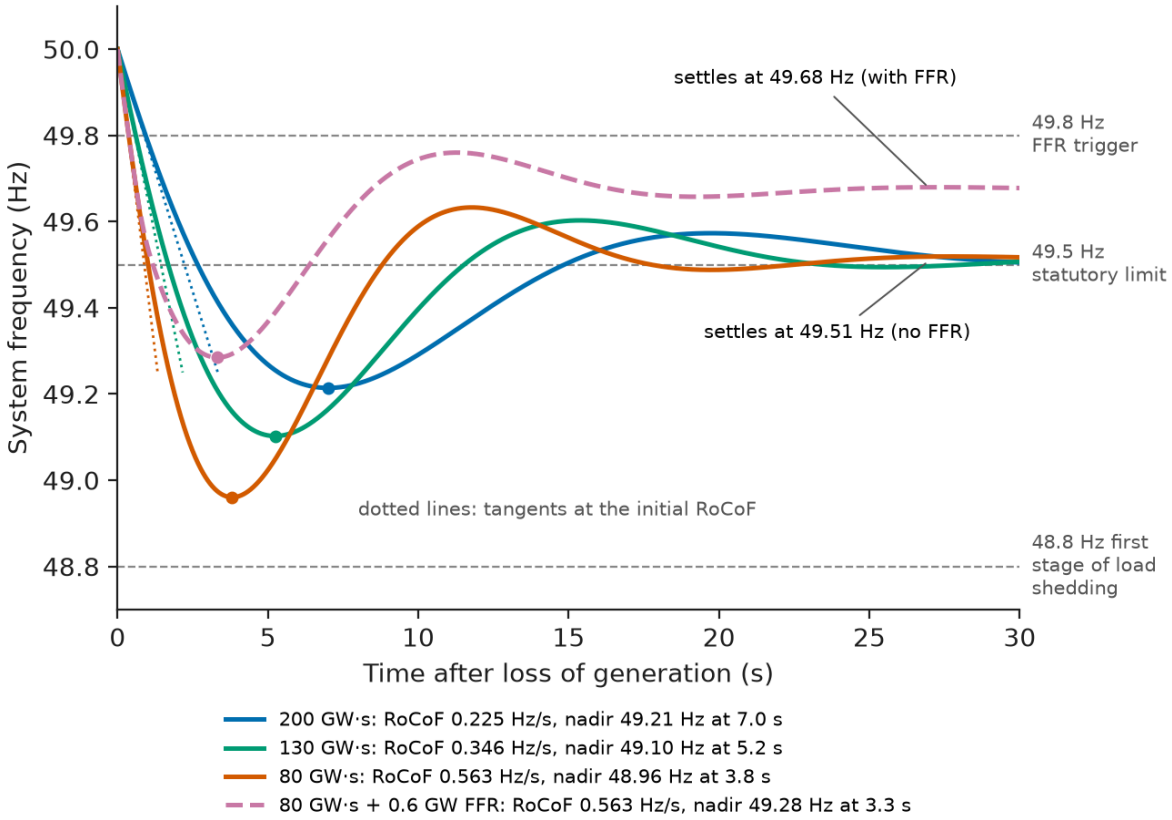


Figure 20.5 — System frequency after the loss of 1.8 GW in a 35 GW system for three levels of kinetic energy and with battery fast frequency response.

Figure 20.5 shows the traces. Over-frequency is the mirror problem: the loss of a large load or an export interconnector raises frequency, and generators must reduce output (limited frequency-sensitive mode, typically from 50.2 Hz) rather than trip, a requirement that was added to grid codes after millions of PV inverters in Germany were found to share a single 50.2 Hz trip setting.

20.6 Disturbance response

A sequence of events across time scales

A disturbance does not excite one stability problem at a time. Following one event through a large interconnection shows the layers, each governed by different physics and different equipment:

Time after the fault	What happens	Governing equipment and physics
0–20 ms	Fault current flows; voltages collapse near the fault; DC offset; travelling waves	Network impedances (Chapter 19), surge behaviour (Chapter 24)
20–150 ms	Protection detects and trips; breakers interrupt; inverters switch to fault ride-through, inject reactive current	Relays, breakers, converter controls
0.1–3 s	First swing of rotor angles; AVRs boost fields; voltage recovers or stalls motors	Generator inertia, exciters, motor dynamics
1–30 s	Electromechanical oscillations damp out (or grow); frequency nadir; governors and fast frequency response	PSS, governors, batteries, HVDC controls
30 s–10 min	Tap changers restore load voltages; over-excitation limiters act; secondary frequency control	OLTCs, OELs, automatic generation control
10 min–hours	Operator actions, thermal limits of lines and transformers, re-dispatch	Control-room procedures, dynamic ratings

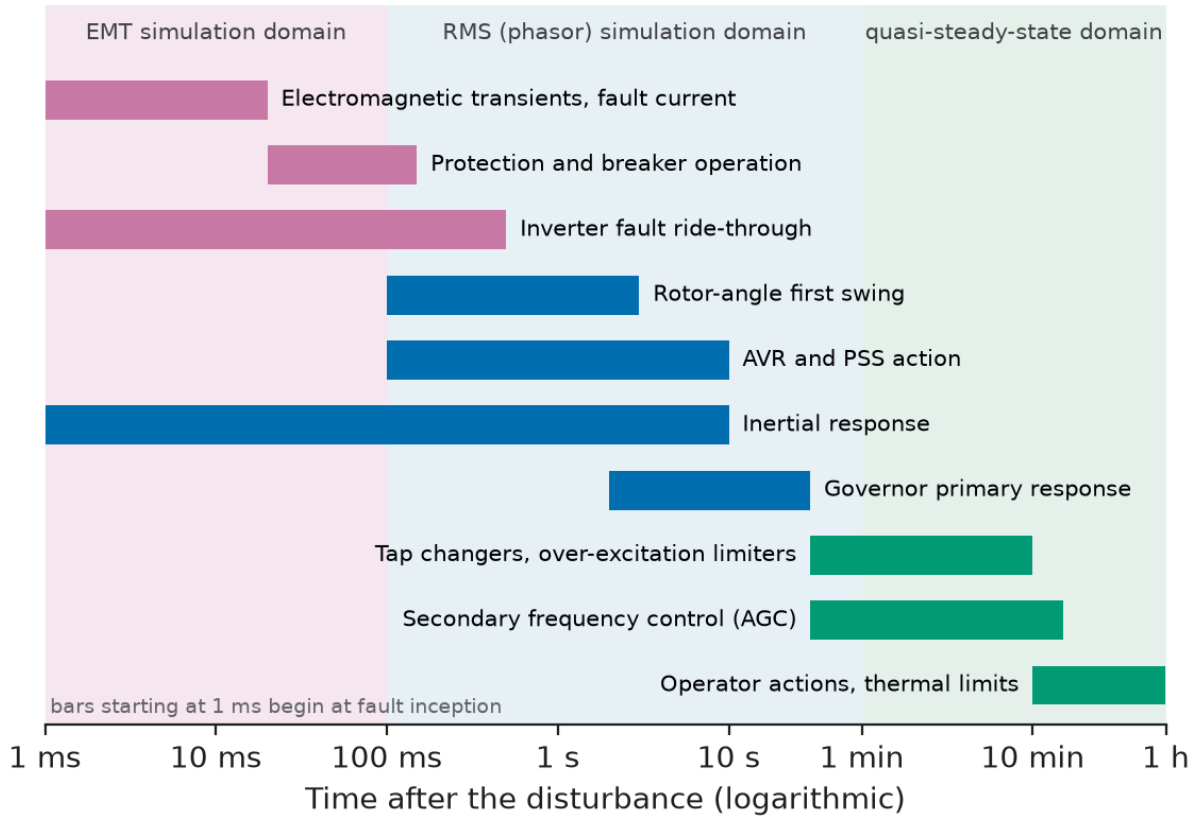


Figure 20.6 — Time scales of power-system response to a disturbance, from fault inception to operator action.

Figure 20.6 sets these on one axis. The layers interact: the reactive current that inverters inject during a fault helps voltage recovery; the active current they withhold while recovering affects frequency; the speed at which they restore active power after the fault (grid codes require 90 % within 0.5–1 s, for example) changes the first swing of nearby synchronous machines.

Inverter-based resources in the disturbance

Most wind, solar and battery plants are **grid-following**: their converters measure the grid voltage angle with a phase-locked loop and inject a controlled current in step with it. They need a stable voltage to follow, ride through faults within prescribed voltage-time envelopes (IEEE 2800-2022, NERC PRC-024, ENTSO-E RfG), and contribute no inherent inertia. In weak networks (short-circuit ratio below about 3) their control loops can interact with each other and with the network, producing the converter-driven oscillations at 2–20 Hz and above that were observed in West Murray (Australia) in 2019 and in Texas wind regions with series compensation. **Grid-forming** converters instead behave as a controlled voltage source behind an impedance, synchronise through a power-frequency relation that mimics the swing equation, and can supply inertia, fault current up to their current limit and system strength. Chapter 27 develops their control; for stability studies they are the main new modelling challenge.

20.7 Stability margins

What "how stable" means

A system that is merely stable is not good enough: loads grow, lines fail and models are imperfect. Planning and operating criteria are therefore expressed as **margins**, measured against defined **contingencies**. The basic contingency criterion, **N-1**, requires the system to remain stable, within voltage and thermal limits, after the loss of any single element (a line, transformer, generator or HVDC pole) with normal protection clearing; many systems add N-1-1 (a second outage after re-dispatch) and selected N-2 events such as a double-circuit line on common towers or a busbar fault. In North America these requirements are formalised in NERC TPL-001; in Europe in the ENTSO-E system operation guideline and national security standards.

Stability type	Margin measure	Typical requirement
Transient	Critical clearing time compared with actual clearing time	Stable for N-1 faults with normal clearing; many utilities also require stability with breaker-failure (back-up) clearing
Transient (transfer)	Maximum stable transfer across a corridor compared with scheduled transfer	Operate at least 5–10 % below the limit found by simulation
Small-signal	Damping ratio of the least-damped mode	≥ 5 % in planning (3 % minimum in some operating guides)
Steady-state angle	Angle across a long corridor	Typically below 30–45 °, giving a 30–40 % margin on the angle limit
Voltage	Real-power distance to the nose of the PV curve	≥ 5 % for N-1 and ≥ 2.5 % for N-2 (WECC criterion); reactive reserves on key buses
Voltage recovery	Post-fault voltage trajectory	Recover above 0.8–0.9 pu within 0.5–2 s (grid-code templates)
Frequency	Nadir and RoCoF after the largest credible infeed loss	Nadir above first load-shedding stage with margin; ENTSO-E Continental Europe: maximum dynamic deviation 800 mHz for 3 GW reference incident

Limits in operation

These margins become operating limits that the control room enforces: a transfer limit on the California–Oregon intertie that varies with the number of lines and series capacitors in service; a minimum number of synchronous units or minimum kinetic energy on the system; a maximum size of single generator or interconnector infeed. **Dynamic security assessment** tools now run hundreds of contingencies against the real-time state every few minutes, so that limits follow the actual system rather than seasonal worst cases. The economic consequence is direct: a stability limit on a corridor forces cheaper generation on one side to be curtailed and more expensive generation on the other to run, and stability improvement (Section 20.8) is often justified by the value of the transfer it releases.

20.8 Stability improvement

Methods and what they act on

Measure	Mechanism	Main stability benefit
Faster protection and breakers	Reduce the accelerating area	Transient
Single-pole tripping and auto-reclosing	Two phases keep transmitting during the most common fault (SLG)	Transient
Series capacitors on long lines	Reduce transfer reactance, raise $P_{\max}$ and synchronising power	Transient, small-signal, voltage (Chapter 21)
Shunt compensation: SVC, STATCOM, synchronous condensers	Hold voltage at key buses, supply reactive power fast	Voltage, transient (by raising mid-line voltage)
Power-system stabilisers	Add damping torque through the excitation	Small-signal
High initial-response excitation systems	Raise E' quickly after the fault	Transient
Fast valving and braking resistors	Reduce turbine power or absorb surplus power during the swing	Transient
Generator tripping and load shedding schemes (special protection schemes)	Rebalance power within 100–200 ms after a critical contingency	Transient, frequency
HVDC and FACTS modulation	Modulate power flow to damp inter-area modes	Small-signal
Synchronous condensers and grid-forming converters	Add inertia, short-circuit strength and voltage-source behaviour	Frequency, converter-driven, voltage
Battery fast frequency response	Inject power within 0.5–1 s of a frequency deviation	Frequency

Two quantitative illustrations

Series compensation. For the 600 MVA unit of Section 20.3, the two lines contribute 0.25 pu of the 0.70 pu transfer reactance. Compensating each line by 50 % with series capacitors reduces the line reactance to 0.125 pu with both lines and 0.25 pu with one, so $P_{\max 1}$ rises from 1.643 to $1.15/0.575 = 2.0$ pu and $P_{\max 3}$ from 1.211 to $1.15/0.70 = 1.643$ pu. The pre-fault angle falls to 25.2°, $\delta_{\max}$ rises to 148.8°, and the equal-area criterion for the bus fault gives $\delta_{\mathrm{cr}} = 74.9^{\circ}$ and a critical clearing time of 0.228 s, against 0.160 s without compensation: enough to survive back-up clearing. The price is subsynchronous resonance: the series capacitor and the line inductance form an electrical resonance below 50 Hz that can lock onto a torsional mode of a turbine shaft. The 1970 and 1971 shaft failures at the Mohave plant in the western United States were caused this way, and series-compensated lines near thermal plants now carry SSR damping controls or thyristor-controlled series capacitors.

Synchronous condensers in a low-inertia system. In the frequency example of Section 20.5, adding five 250 MVA synchronous condensers with flywheels ($H = 6$ s) adds $5 \times 250 \times 6 = 7.5$ GW·s of kinetic energy, together with roughly 0.9 GVA of short-circuit contribution each at the HV side of its step-up transformer ($250/(0.15 + 0.12) = 926$ MVA for a subtransient reactance of 0.15 pu and a 12 % transformer), about 4.6 GVA in total. On an 80 GW·s day that is a 9 % increase in inertia: helpful, but small compared with the effect of 0.6

GW of fast battery response. Condensers are justified chiefly by system strength (fault level and voltage stiffness for inverter-based plants), with inertia as a by-product; Great Britain, Denmark, South Australia and Texas have all installed them for this combination of services.

20.9 Simulation and interpretation

Which simulation

Stability is studied almost entirely by simulation, because real systems are too large and too nonlinear for closed-form criteria. Two families of tools are used:

	RMS (phasor) simulation	Electromagnetic-transient (EMT) simulation
What is represented	Fundamental-frequency phasors; network as algebraic equations; machines and controls as differential equations	Instantaneous three-phase waveforms; network inductances and capacitances as differential equations
Time step	1–10 ms (a quarter to half cycle)	1–50 μs
Typical scope	Whole interconnection, thousands of buses, 10–60 s	A plant and its local network, up to a few hundred buses, a few seconds
Typical tools	PSS/E, PowerFactory (RMS mode), PSLF, TSAT	PSCAD/EMTDC, EMTP, PowerFactory (EMT mode), RTDS for real-time hardware testing
Good for	Rotor-angle, voltage and frequency stability; planning studies; operator security assessment	Converter controls, weak grids, subsynchronous resonance, protection performance

The 2021 addition of converter-driven stability to the classification follows from this split: phenomena that live in converter control loops are invisible in RMS models unless the models are explicitly validated against EMT results or field measurements.

Models and data

Results are only as good as the models. Standard model libraries describe generators (GENROU, GENSAL), exciters (IEEE 421.5 types such as ST1A, AC7B), stabilisers (PSS2B), governors (GGOV1, HYGOV), wind and solar plants (the WECC generic renewable models REGC, REEC, REPC) and loads (ZIP static models or the composite load model CMLD with motor components). Grid codes now require plant owners to supply validated models and to repeat validation after controller changes (NERC MOD-026 and MOD-027, for example). The load model is the largest single uncertainty: a static constant-impedance load model makes voltage recovery look far better than a model with stalling air-conditioner motors, and studies of FIDVR became credible only after composite load models were adopted.

Reading the results

A stability study is judged on a few plots:

- **Rotor angles**, plotted relative to the centre of inertia or to a reference machine (never absolute angles, which drift with frequency). Stable cases show bounded, decaying swings; a machine whose angle

separates from the group has lost synchronism.
- **Bus voltages**, compared with the grid code's voltage-recovery envelope; a slow recovery or a second dip points to motor stalling or reactive-power shortage.
- **Frequency** at several buses, for the nadir, RoCoF and the quasi-steady value; regional differences in the first seconds show the inter-area modes.
- **Active and reactive power** of key plants and interconnectors, for oscillations and limiter actions.
- **Damping estimates** from ringdown analysis (Prony or matrix-pencil fitting of oscillation frequency and damping ratio to a simulated or measured response), compared with the eigenvalue analysis of Section 20.2.

Two habits separate sound interpretation from plotting. First, check that the initial conditions are flat: a well-initialised simulation with no disturbance should stay still for several seconds, and drift indicates a model or data error. Second, look for the mechanism, not just the outcome: a case that is "stable" only because a limiter happened to act, or "unstable" only because a protection model tripped a unit at a questionable setting, tells the planner more about the model than about the system. Comparison with measurements from phasor measurement units after real disturbances is the final test, and the 2019 GB event, the 2016 South Australia blackout and the 2021 Texas winter event each led to model corrections that changed subsequent studies.

What remains open

Open question	Where it is answered
How are the fault currents and clearing times used here calculated and achieved?	Chapters 19 and 23
How do the line reactance, series compensation and SIL used for transfer limits follow from line design?	Chapter 21 (Transmission Lines)
How do grid-forming and grid-following inverters provide frequency and voltage support?	Chapter 27 (Grid Integration of Renewable Energy)
How do energy-management systems, AGC and phasor measurements keep the system secure in real time?	Chapter 26 (Smart Grids)
How do synchronous machines produce the power-angle characteristic and respond to excitation?	Chapter 7 (Synchronous Machines)
How do batteries, flywheels and pumped hydro provide fast reserve?	Chapter 16 (Energy Storage)
How are linear models analysed with eigenvalues, root locus and frequency response?	*Circuits, Signals and Communications*, Ch. 34 and Ch. 35
How do voltage sags and frequency deviations appear to customers?	Chapter 25 (Power Quality)

Next: Chapter 21 opens Part VI with the transmission line itself: how its parameters follow from conductors and towers, and how its length, loading and compensation set the limits that this chapter's stability studies assume.

Part VI — Transmission, Distribution, and Protection

Part V treated the power system as a network of per-unit impedances. Part VI looks at the hardware those impedances stand for and the equipment that keeps it safe. A transmission line is a set of conductors on towers whose geometry fixes its reactance, charging and corona. A distribution feeder is a radial chain of cables, transformers and loads whose end voltage sags every evening and, increasingly, rises at noon when rooftop solar exports. Protection is the system of instrument transformers, relays, breakers and fuses that must find a fault within tens of milliseconds and remove only the faulted part. High-voltage engineering is the discipline that keeps insulation intact against lightning, switching surges and decades of electrical stress. One idea links the four chapters: every component must carry normal power efficiently and survive its abnormal duty (fault current, overvoltage, lightning) without endangering people or the network.

By the end of this part you should understand

- How the resistance, inductance and capacitance of overhead lines and cables follow from conductors, bundles and tower geometry, and which line model is accurate enough for a given length.
- Why long AC lines are limited by voltage regulation, the Ferranti effect and stability rather than by conductor heating, and how shunt reactors, series capacitors, SVCs and STATCOMs relax those limits.
- When high-voltage DC is the better choice, and how line-commutated and voltage-source converters differ in what they demand from the AC system.
- How radial and ring distribution networks are arranged, how feeder voltage drop and losses are calculated, and how tap changers, regulators and distributed generation change the voltage profile.
- How overcurrent, differential, distance and earth-fault protection detect faults, and how relays, fuses and breakers are coordinated so that protection is both selective and sensitive.
- How insulation is coordinated against lightning and switching surges, how surge arresters and earthing work, and how high-voltage equipment is tested and worked on safely.

Chapters

Chapter	Central question	What it delivers
21 — Transmission Lines	How much power can a line carry over a given distance, at what voltage, efficiency and cost, and when is DC better?	Line parameters from geometry, model selection and errors, regulation and loadability, losses, Ferranti effect and reactor sizing, corona design limits, compensation, AC voltage classes, HVDC fundamentals
22 — Distribution Systems	How is power delivered from the substation to millions of customers within voltage limits and at acceptable loss?	Radial and ring architectures, distribution transformers, feeder analysis, voltage drop and regulation, loss calculation, effects of distributed generation, planning methods
23 — Power-System Protection	How is a fault detected and cleared quickly while leaving every healthy part of the network in service?	Protection principles, overcurrent, differential, distance and earth-fault schemes, breakers, fuses, relays and instrument transformers, coordination, selectivity and sensitivity
24 — High-Voltage Engineering	How is insulation designed to survive lightning, switching surges and long-term stress, and how is that verified?	Generation and measurement of high voltages, insulation coordination, breakdown and partial discharge, surges and arresters, grounding, test procedures and safety

How this part connects to the rest of the book

Part VI uses the analytical tools of Part V throughout. The per-unit models and power-flow solutions of Chapters 17 and 18 describe the networks whose lines and feeders are designed here. The fault currents of Chapter 19 are the duties that breakers, cables and earthing systems must survive and the signals that protection relays measure. The stability limits of Chapter 20 are what make long-line compensation and fast protection worth their cost. The transformers of Chapter 4 reappear as distribution and instrument transformers. Looking forward, the voltage and harmonic problems that distribution passes to customers are taken up in Chapter 25 (Power Quality); the automation and distributed resources that are changing distribution and protection in Chapters 26 and 27; and the hazards to people from faults, arcs and touch voltages in Chapters 29 and 30.

Two sibling books supply the foundations this part builds on. *Circuits, Signals and Communications* (Book 3) owns the wave theory of lines, including the telegrapher's equations, surge impedance, reflection and the exact ABCD constants of a long line (Ch. 14), the two-port and nominal-π line models (Ch. 12), and the three-phase circuit theory and symmetrical components that protection relies on (Ch. 10). *Engineering Physics: Top Down* owns the electromagnetic physics: the field definitions of inductance and capacitance (Ch. 11), electromagnetic waves and their propagation, which underlie lightning and switching surges (Ch. 12), and the transport and heat-transfer laws that set conductor resistance and thermal ratings (Ch. 13 and Ch. 19). This part takes those results as given.

Chapter 21 — Transmission Lines

Part VI · Transmission, Distribution, and Protection

A utility must bring 3 GW from a cluster of hydro and wind plants to a city 1000 km away. A planner sketches the obvious answer, a 400 kV double-circuit AC line, and runs into trouble at once. At night, when the line is lightly loaded, its own capacitance would push the receiving-end voltage 10 % or more above the sending end unless large reactors are switched in. At full load, the angle across the line would approach the point at which the two ends fall out of step after a fault. The loss alone, at a few percent of 3 GW for every hour of the year, would cost tens of millions of dollars a year. The alternatives each carry their own numbers: more circuits, a higher voltage, series capacitors halfway along, or a high-voltage DC link with converter stations at each end. Choosing among them requires knowing how a line's resistance, inductance and capacitance follow from its conductors and towers, how those parameters set voltage regulation, efficiency and transfer capability, what limits corona places on the conductor, and where DC wins.

> **How do the conductors and the tower geometry fix a line's resistance, inductance and capacitance?**
> **How much power can a line of given length carry, at what voltage regulation and efficiency, and how is it compensated? When should power be transmitted as high-voltage AC, and when as HVDC?**

Concept	Engineering destination
Line parameters	Conductor and tower selection, data for power-flow, fault and stability studies
Resistance, inductance and capacitance	Losses, reactance, charging current, bundle design
Short, medium and long line models	Choosing the right model for planning and protection studies
Voltage regulation	Receiving-end voltage at light and heavy load, transfer limits
Efficiency	Loss evaluation, conductor economics, loss capitalisation
Ferranti effect	Shunt-reactor sizing, line energisation, cable charging
Corona	Conductor size and bundling, losses, audible noise and radio interference
Compensation	Shunt reactors, series capacitors, SVC and STATCOM
High-voltage AC transmission	Voltage classes, loadability, right-of-way, dynamic ratings
HVDC fundamentals	Long-distance bulk links, submarine cables, asynchronous interconnections

Roadmap. What sets the parameters of a real line (21.1) → resistance, inductance and capacitance from conductors and geometry (21.2) → which line model to use, and how large its error is (21.3) → voltage regulation and transfer capability (21.4) → losses and efficiency (21.5) → the Ferranti effect and its control (21.6) → corona as a limit on conductor design (21.7) → compensation (21.8) → the practice of high-voltage AC transmission (21.9) → HVDC (21.10).

Boundary.
The wave theory of lines is owned by *Circuits, Signals and Communications*, Ch. 14: the distributed $RLGC$ model and telegrapher's equations (§14.1–14.2), characteristic (surge) impedance and surge impedance loading (§14.3), the propagation constant and electrical length (§14.4), and the exact ABCD constants $A = \cosh\gamma\ell$, $B = Z_c \sinh\gamma\ell$, $C = \sinh(\gamma\ell)/Z_c$ of a long line (§14.9). The nominal-π model, with a worked 150 km, 220 kV example of sending-end conditions, efficiency, regulation and the Ferranti effect, is in Ch. 12 §12.10. This chapter uses those results without re-deriving them and supplies what a power engineer adds: parameters from hardware, model choice, loadability, losses, corona, compensation and HVDC.

21.1 Line parameters

The physical line

An overhead transmission line consists of phase conductors, usually stranded aluminium around a steel core, hung from insulator strings on steel lattice towers 30–60 m high, with one or two earthed **shield wires** at the top to intercept lightning. Spans are 300–450 m. At 220 kV and above each phase is usually a **bundle** of two, three, four or more sub-conductors held 400–450 mm apart by spacers. Each of these choices, made for mechanical, thermal and insulation reasons, also fixes the electrical parameters.

The **primary line constants** per unit length are the series resistance R, series inductance L, shunt capacitance C and shunt conductance G. For overhead lines G (leakage over insulators and corona loss) is negligible for voltage calculations and is accounted for separately as a loss (Section 21.7).

Conductors

Conductor (code name)	Construction	Overall diameter	DC resistance at 20 °C	Typical use
ACSR "Zebra"	54 Al / 7 steel, 428 mm² Al	28.6 mm	0.0674 Ω/km	132–220 kV single, 400 kV twin
ACSR "Moose"	54 Al / 7 steel, 528 mm² Al	31.8 mm	0.0560 Ω/km	400 kV twin, 765 kV quad (India)
ACSR "Drake"	26 Al / 7 steel, 403 mm² Al	28.1 mm	0.0718 Ω/km	115–345 kV (North America)
AAAC (all-aluminium alloy)	Alloy strands, no steel	20–40 mm	Slightly higher than ACSR of same area	Coastal lines (no galvanic corrosion), distribution
HTLS (ACCC, ACSS, GZTACSR)	Composite or annealed-aluminium designs	As the conductor replaced	Similar or lower	Reconductoring: 1.5–2 × ampacity at 150–210 °C

Values are typical manufacturer data. The **ampacity** (thermal rating) of a conductor is set by a heat balance between Joule heating and solar gain on one side, and convection and radiation on the other, at a maximum conductor temperature (75–100 °C for ACSR) chosen so that the hot conductor's sag keeps the statutory

clearance to ground. IEEE 738 and CIGRE TB 601 give the calculation; a single Moose conductor carries about 800–900 A at 75 °C in typical summer conditions.

> **Boundary.**
> The resistivity of aluminium and its temperature dependence come from transport theory (*Engineering Physics*, Ch. 13 §13.4); the heat-balance physics behind ampacity is in Ch. 19 §19.6.

Geometry

The second ingredient is geometry: the distances between phases and the size and arrangement of each bundle. A 400 kV single-circuit line with a horizontal (flat) phase arrangement typically has 11–12 m between adjacent phases, the conductors 15–25 m above ground at mid-span, and a twin bundle with 450 mm sub-conductor spacing. Figure 21.1 shows this tower; it is the reference line used in the worked examples of this chapter.

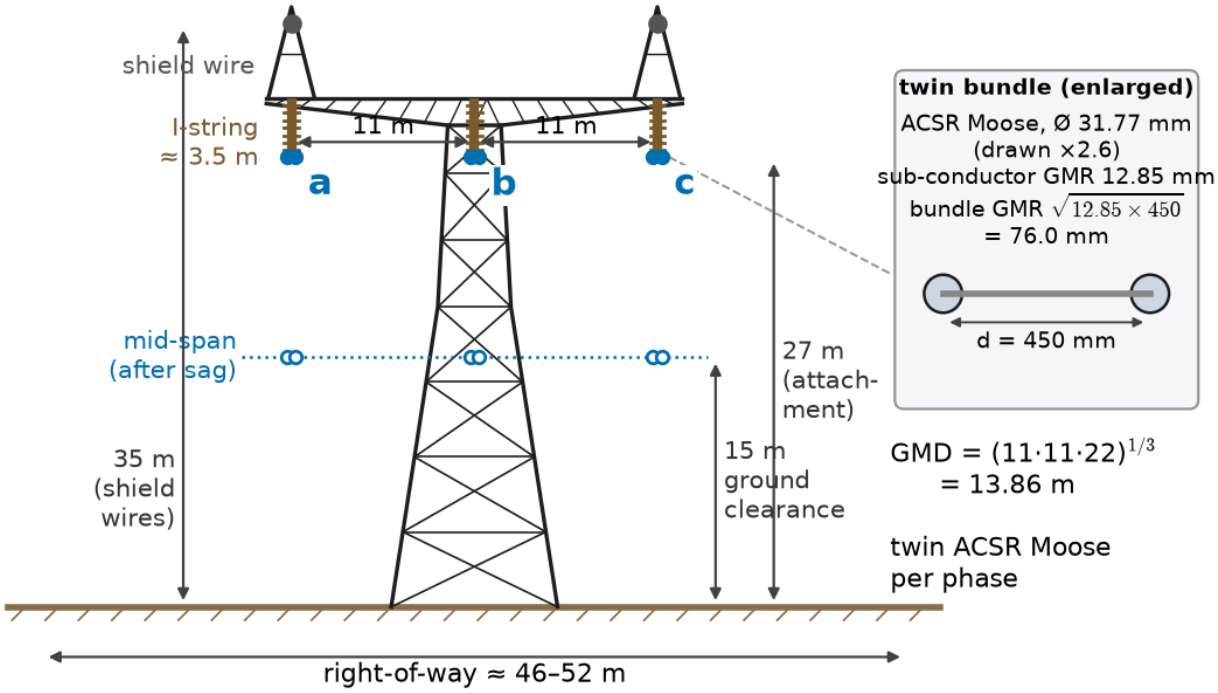


Figure 21.1 — Cross-section of the reference 400 kV single-circuit horizontal tower with twin-Moose bundles and two shield wires.

21.2 Resistance, inductance, and capacitance

Resistance

The DC resistance of a stranded conductor is that of its aluminium cross-section, increased by 1–2 % because the strands spiral and are therefore longer than the conductor. Two corrections raise it further in service. Temperature:

$$R_T = R_{20}\,[1 + \alpha_{20}(T - 20)]\,, \qquad \alpha_{20} = 0.00403\ \mathrm{K}^{-1} \text{ for hard-drawn aluminium}$$

so a conductor at 75 °C has 22 % more resistance than at 20 °C. And alternating current: the skin effect, together with the magnetisation of the steel core in conductors with an odd number of aluminium layers, raises the AC resistance at 50 Hz by 2–10 % for large ACSR. For the Moose conductor at 75 °C, $R = 0.0560 \times (1 + 0.00403 \times 55) = 0.0684\ \Omega/\text{km}$, and with a 2 % AC correction 0.0697 Ω/km; a twin bundle per phase has half of this, **0.0349 Ω/km**.

> **Status of the principle: Linear temperature coefficient of resistance**
> *Empirical.* A linear fit to measured resistivity, accurate within about 1 % between −20 °C and 200 °C for aluminium; it is the reason line losses are 10–20 % higher in summer than in winter at the same current.

Inductance

The inductance per phase of a three-phase line follows from the flux linking each conductor, counted out to the other conductors. The engineering result, for a transposed line, is

$$L = 2 \times 10^{-7} \ln \frac{D_m}{D_s} \quad \text{H/m}$$

where D_m is the **geometric mean distance** (GMD) between phases, $D_m = \sqrt[3]{D_{ab} D_{bc} D_{ca}}$, and D_s is the **geometric mean radius** (GMR) of the phase conductor, an equivalent radius that accounts for flux inside the conductor. For a solid round conductor $D_s = 0.7788 r$; for stranded conductors manufacturers tabulate it (12.85 mm for Moose, about 0.81 times its radius). For a bundle of n sub-conductors of GMR D_s' on a circle with spacing d,

$$D_s^{(2)} = \sqrt{D_s' d}, \qquad D_s^{(3)} = \sqrt[3]{D_s' d^2}, \qquad D_s^{(4)} = 1.09 \sqrt[4]{D_s' d^3}$$

> **Boundary.**
> The flux-linkage definition of inductance, $L = \Lambda / I$, and the magnetostatics behind it are in *Engineering Physics*, Ch. 11 §11.11; the corresponding capacitance definition is in §11.10. This section uses the engineering results.

> **Status of the principle: GMD–GMR inductance formula**
> *Approximate.* Exact for long parallel conductors with uniform current, perfectly transposed and far from the earth; it neglects the earth-return path (which matters only for zero-sequence quantities, Chapter 19) and is within 1–2 % of field calculations for practical lines.

Transposition rotates the phase positions at two points along the line so that each phase occupies each position for a third of the length. It equalises the three phase inductances and capacitances; without it a flat line's unequal mutual couplings produce negative-sequence voltages of 1–2 % at full load, enough to load generators with negative-sequence current. Modern long lines are transposed at substations or at a few towers, or are left untransposed and modelled in phase coordinates.

Capacitance

The capacitance of each phase to neutral uses the same geometric mean distance, but the outer radius of the conductor (or an equivalent bundle radius built with the physical radius r instead of the GMR), because the electric field ends at the conductor surface:

$$C = \frac{2\pi\varepsilon_0}{\ln(D_m/r_{\text{eq}})} \quad \text{F/m}, \qquad r_{\text{eq}}^{(2)} = \sqrt{rd}$$

The earth raises C by 1–3 % for typical heights, a correction usually neglected in planning. The capacitance draws a **charging current** $I_c = \omega C V_{\text{ph}}$ per unit length and generates reactive power $Q_c = \omega C V_{\text{LL}}^2$ (three-phase) per unit length whenever the line is energised, whether or not it carries load.

Worked example — parameters of the reference 400 kV line. For the line of Figure 21.1: phase spacing 11 m, flat, so $D_m = \sqrt[3]{11 \times 11 \times 22} = 13.86$ m. Twin Moose bundles: $D_s' = 12.85$ mm, $r = 15.89$ mm, $d = 0.45$ m.

$$D_s = \sqrt{0.01285 \times 0.45} = 0.0760 \text{ m}, \qquad L = 2 \times 10^{-7} \ln \frac{13.86}{0.0760} = 1.041 \text{ mH/km}, \qquad X = \omega L = 0.327\ \Omega/\text{km}$$

$$r_{\text{eq}} = \sqrt{0.01589 \times 0.45} = 0.0845 \text{ m}, \qquad C = \frac{2\pi \times 8.854 \times 10^{-12}}{\ln(13.86/0.0845)} = 10.91 \text{ nF/km}$$

With $R = 0.0349\ \Omega$/km, the surge impedance is $\sqrt{L/C} = 309\ \Omega$ if resistance is neglected and $|Z_c| = 310\ \Omega$ with it, the surge impedance loading $400^2/310 = 516$ MW, and the wave speed $1/\sqrt{LC} = 2.97 \times 10^5$ km/s, 99 % of the speed of light. The line generates $Q_c = (400 \times 10^3)^2 \times 314.16 \times 10.91 \times 10^{-9} = 54.8$ MVAr per 100 km, drawing 79 A of charging current per 100 km. For comparison, a single Moose conductor per phase at the same spacing gives $L = 1.397$ mH/km ($X = 0.439\ \Omega$/km), $C = 8.22$ nF/km and a surge impedance of 412 Ω: bundling reduces the reactance by 25 %, raises the capacitance by 33 % and raises SIL by a third. That increase, together with the reduction of corona (Section 21.7), is why EHV lines are bundled.

Representative parameters

Line	R (Ω/km per phase)	X (Ω/km)	C (nF/km)	Surge impedance (Ω)	SIL (MW)
132 kV, single ACSR	0.07–0.16	0.38–0.42	8.5–9.5	370–400	44–47
220 kV, single ACSR	0.06–0.08	0.40–0.42	8.8–9.3	375–395	120–130
400 kV, twin bundle	0.03–0.04	0.31–0.33	10.9–11.6	295–310	515–545
765 kV, quad bundle	0.01–0.015	0.27–0.29	13–14	250–260	2250–2350
1000 kV AC (China), eight-conductor bundle	≈ 0.008	≈ 0.26	≈ 14	≈ 240	≈ 4200
400 kV XLPE cable, 2500 mm² Cu	≈ 0.01	0.12–0.20	150–250	30–50	3000–5000 (not reachable thermally)

The surge impedance of overhead lines varies little with voltage, because D_m and the bundle size grow together; SIL therefore rises roughly with the square of voltage. Cables are the opposite case: their thin insulation gives a capacitance 15–25 times that of an overhead line, a surge impedance an order of magnitude lower, and a charging current that dominates their behaviour (Section 21.6).

21.3 Short, medium, and long line models

Three models and why they exist

The exact model of a uniform line is the distributed one, expressed through the hyperbolic ABCD constants of Book 3 Ch. 14. Planners still use simpler models because they slot into network programs as ordinary branches and make hand calculation transparent:

- **Short line** (below about 80 km at 50/60 Hz): series impedance $Z = z\ell$ only; capacitance ignored. $A = D = 1$, $B = Z$, $C = 0$.
- **Medium line** (about 80–250 km): **nominal π**, the total impedance in series with half the total admittance $Y/2$ at each end (Book 3 Ch. 12 §12.10).
- **Long line** (above about 250 km): the **equivalent π**, a π network with corrected elements that reproduces the exact ABCD constants at the line terminals:

$$Z' = Z\,\frac{\sinh\gamma\ell}{\gamma\ell}, \qquad \frac{Y'}{2} = \frac{Y}{2}\,\frac{\tanh(\gamma\ell/2)}{\gamma\ell/2}$$

> **Status of the principle: Short, medium and long line models**
> *Approximate*. The short and nominal-π models are truncations of the series expansions of the exact hyperbolic functions in powers of $\gamma\ell$; the equivalent π is exact at the terminals at one frequency but says nothing about voltages along the line, and none of the three represents travelling-wave behaviour during switching or lightning surges, which needs frequency-dependent distributed models (Chapter 24).

How large the errors are

The length thresholds are not arbitrary: they follow from the electrical length $\beta\ell$, which is 6° per 100 km at 50 Hz (7.2° at 60 Hz). Comparing the models with the exact solution for the reference 400 kV line:

Length	$\beta\ell$	$\lvert A\rvert$ exact	Error of short model in $\lvert A\rvert$ (relative to exact)	Error of nominal π in $\lvert B\rvert$	Error of nominal π in $\lvert C\rvert$
80 km	4.9°	0.9964	0.4 %	+0.1 %	−0.1 %
150 km	9.1°	0.9874	1.3 %	+0.4 %	−0.2 %
250 km	15.2°	0.9652	3.6 %	+1.2 %	−0.6 %
400 km	24.3°	0.9117	9.7 %	+3.1 %	−1.6 %
600 km	36.4°	0.8051	24.2 %	+7.1 %	−3.7 %
800 km	48.6°	0.6629	50.9 %	+13.0 %	−7.2 %

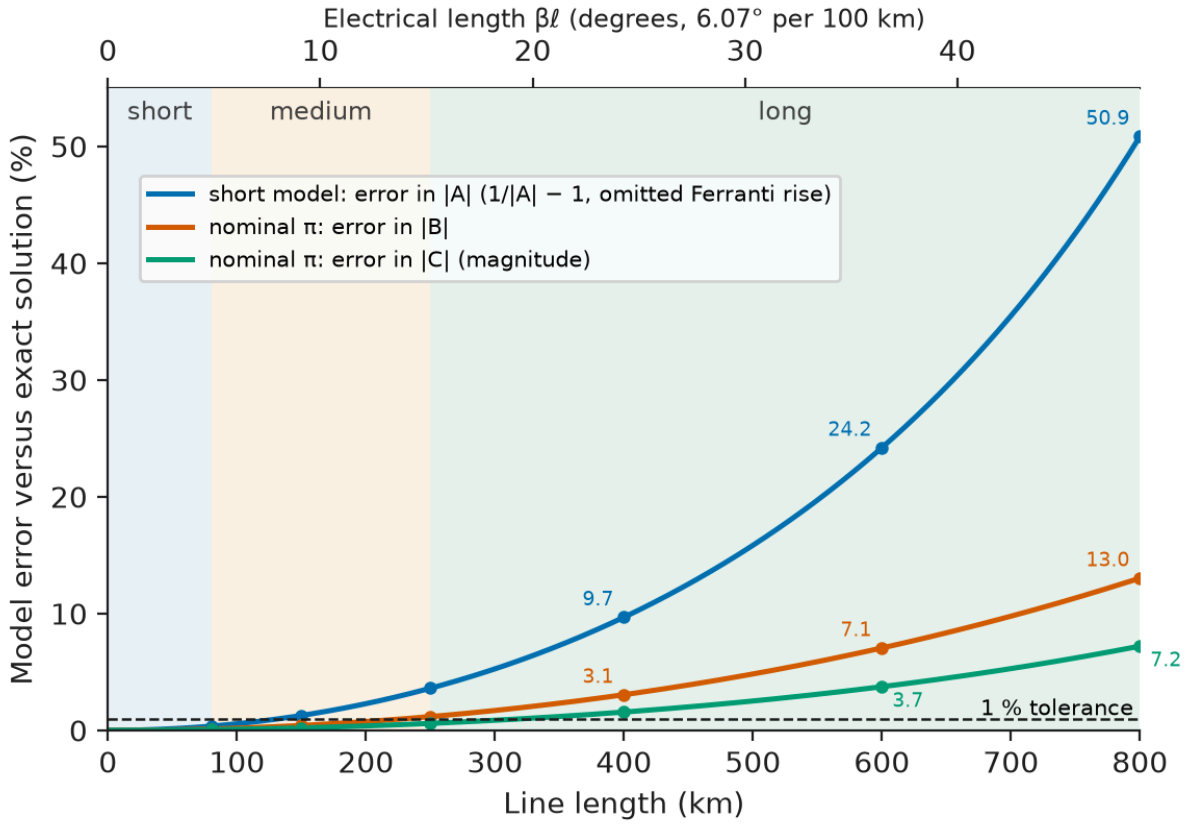


Figure 21.2 — Error of the short-line and nominal-pi models relative to the exact distributed model for the reference 400 kV line at 50 Hz.

Figure 21.2 plots these errors. The short model's error in A, $1/|A| - 1$, is the omitted Ferranti rise, and grows roughly as $(\beta\ell)^2/2$; the nominal π's error in B grows as $(\gamma\ell)^2/6$. A 1 % tolerance gives the conventional thresholds. For the 400 km reference line the equivalent-π correction factors are $\sinh\gamma\ell/\gamma\ell = 0.970$ and $\tanh(\gamma\ell/2)/(\gamma\ell/2) = 1.015$, so the equivalent π has $Z' = 13.12 + j127.0\ \Omega$ in place of $Z = 13.94 + j130.8\ \Omega$, and $Y'/2 = j695.9\ \mu\text{S}$ in place of $j685.5\ \mu\text{S}$. Power-flow and stability programs use the equivalent π for every line regardless of length, since it costs nothing extra once the parameters are known.

Two practical caveats matter more than the model choice. First, the thresholds scale with frequency and with $\sqrt{LC}$: an underground cable, with a much larger capacitance, needs a distributed or multi-section model at lengths of a few tens of kilometres, and harmonic studies at the 13th harmonic need one for a 30 km overhead line. Second, the parameter data are rarely better than 2–5 %: conductor temperature, sag and soil resistivity all move them. A model more exact than its data gains nothing.

21.4 Voltage regulation

Receiving-end voltage under load

Voltage regulation is the rise in receiving-end voltage when full load is removed with the sending-end voltage held constant, expressed as a percentage of the full-load value:

$$\text{VR} = \frac{V_{R,\text{NL}} - V_{R,\text{FL}}}{V_{R,\text{FL}}} \times 100\ \%, \qquad V_{R,\text{NL}} = \frac{V_S}{|A|}$$

For a short line the drop at full load is approximately $\Delta V \approx (RP + XQ)/V$, the familiar distribution-feeder estimate (Chapter 22): reactive power, not active power, causes most of the drop on an inductive line. On long lines this estimate fails, because the line's own charging and the phase rotation along it change the picture.

Worked example — the reference line at 400 km. For the 400 km reference line the exact constants are $A = 0.9116 + j0.0093$ and $B = 13.12 + j127.0\ \Omega$. Holding the sending end at 400 kV and solving $V_S = AV_R + BI_R$ for constant-power loads gives:

Load at receiving end	Receiving-end voltage	Angle across line
No load	1.097 pu	0.5°
0.5 SIL (258 MW), unity power factor	1.052 pu	11.8°
0.5 SIL, 0.95 lagging	0.972 pu	12.3°
SIL (516 MW), unity power factor	0.930 pu	26.7°
SIL, 0.95 lagging	No operating point	—

At half SIL the regulation is $(1.097 - 1.052)/1.052 = 4.3\ \%$; at SIL it is $(1.097 - 0.930)/0.930 = 18\ \%$; and a lagging load of SIL cannot be supplied at all from a sending end held at 1.0 pu. No long line is operated this way. Both ends are held near 1.0 pu by reactive sources (generators, reactors, capacitors, SVCs), and the regulation problem becomes a reactive-power problem: how many MVAr must each end supply or absorb at each loading.

Power transfer with both ends regulated

For a lossless line, with $B = jZ_c \sin\beta\ell$, the active power transferred between ends at voltages V_S and V_R separated by angle δ is

$$P = \frac{V_S V_R}{Z_c \sin\beta\ell}\sin\delta$$

> **Status of the principle: Power-angle relation of a long line**
> *Derived.* Follows from the exact ABCD constants of a lossless line; for a short line $Z_c \sin\beta\ell \to X$ and it reduces to the familiar $P = V_S V_R \sin\delta / X$ of Chapter 7. Losses reduce the receiving-end power by a few percent and shift the maximum slightly.

Neglecting resistance, with both ends at 1.0 pu, the 400 km line transfers SIL at $\delta = 24.3^\circ$ (exactly the electrical length, because a matched line rotates the phase by $\beta\ell$), 1.5 SIL at 38.1°, and 2 SIL at 55.4°. Its theoretical maximum is SIL$/\sin\beta\ell = 2.43$ SIL, or 1255 MW. The reactive power the receiving-end bus must supply to hold 1.0 pu rises with loading: at no load it must *absorb* 111 MVAr (half the line's charging); at SIL, nothing; at 1.5 SIL it must *supply* 156 MVAr. With the line's resistance included the numbers grow: SIL needs 25.1°, and the receiving end must supply about 61 MVAr at SIL and 267 MVAr at 1.5 SIL. Figure 21.3 shows the receiving-end voltage with a fixed sending end and the reactive requirement with both ends fixed.

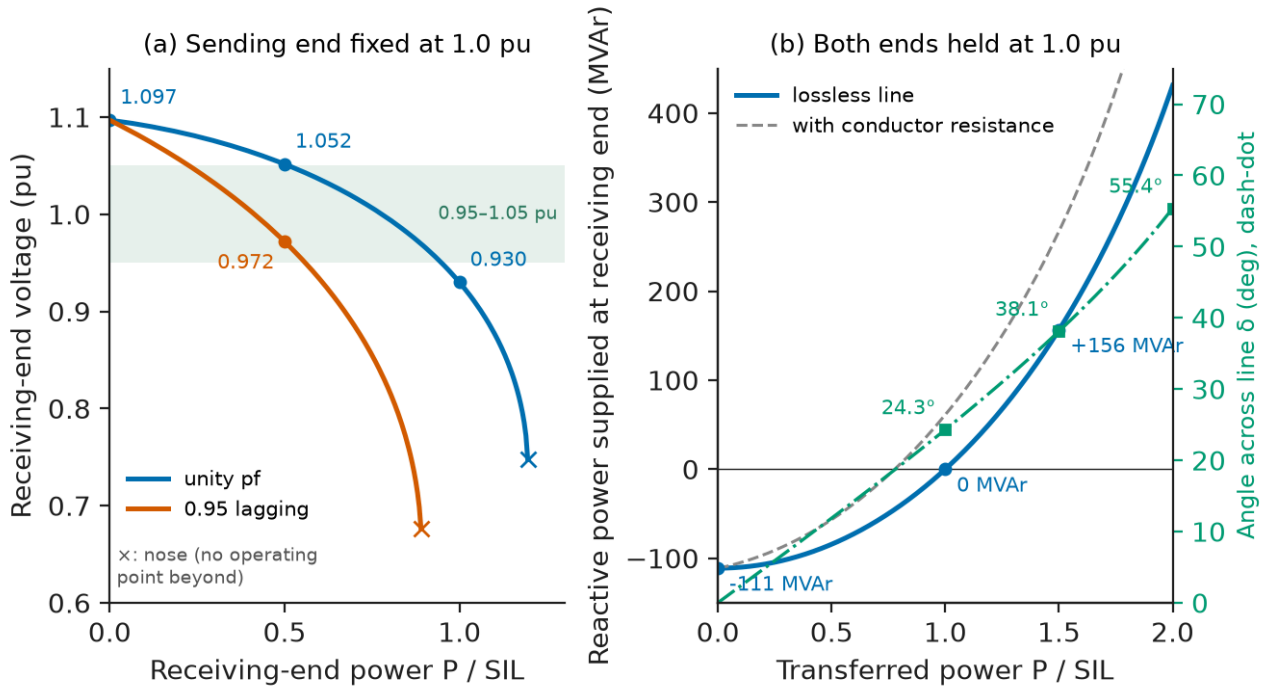


Figure 21.3 — Receiving-end behaviour of the 400 km, 400 kV reference line: voltage with the sending end fixed, and reactive power needed to hold both ends at 1.0 pu.

Loadability

How much power a line may carry in practice depends on which limit binds first, and that depends on length:

- **Short lines** (up to about 80 km) are limited **thermally**, by the conductor's ampacity.
- **Medium lines** (80–300 km) are limited by **voltage drop**, typically 5 % between the ends.
- **Long lines** are limited by **steady-state stability margin**: the angle across the line plus the reactances of the systems at each end must leave a margin (typically 30–35 %) below the maximum transfer, so that the system survives a fault and the loss of a circuit (Chapter 20).

The resulting **St. Clair curve**, first published in 1953 and generalised for EHV lines by Dunlop, Gutman and Marchenko in 1979, gives loadability in multiples of SIL:

Line length	Loadability (multiple of SIL)	Binding limit
80 km	≈ 3.0	Thermal
160 km	≈ 2.0	Voltage drop
320 km	≈ 1.3	Voltage drop and stability
480 km	≈ 1.0	Stability
640 km	≈ 0.8	Stability

> **Status of the principle: St. Clair loadability curve**
> *Empirical.* Fitted to the experience of North American utilities with assumed terminal system strengths, a 5 % voltage-drop criterion and a 30 % stability margin; it is a planning guide, and specific lines are rated by studies of their own systems.

For comparison, a 30° angle across the lossless line alone corresponds to 3.0 SIL at 160 km, 1.5 SIL at 320 km and 1.0 SIL at 480 km; the curve lies below this at short and medium lengths because thermal and voltage-drop limits bind first, and its stability margin includes the reactances of the networks behind each terminal. Figure 21.4 plots the curve. It explains why long AC lines are compensated or replaced by HVDC: an uncompensated 400 kV line 640 km long carries only about 400 MW, a fraction of what its conductors could carry thermally (about 1200 MW for twin Moose).

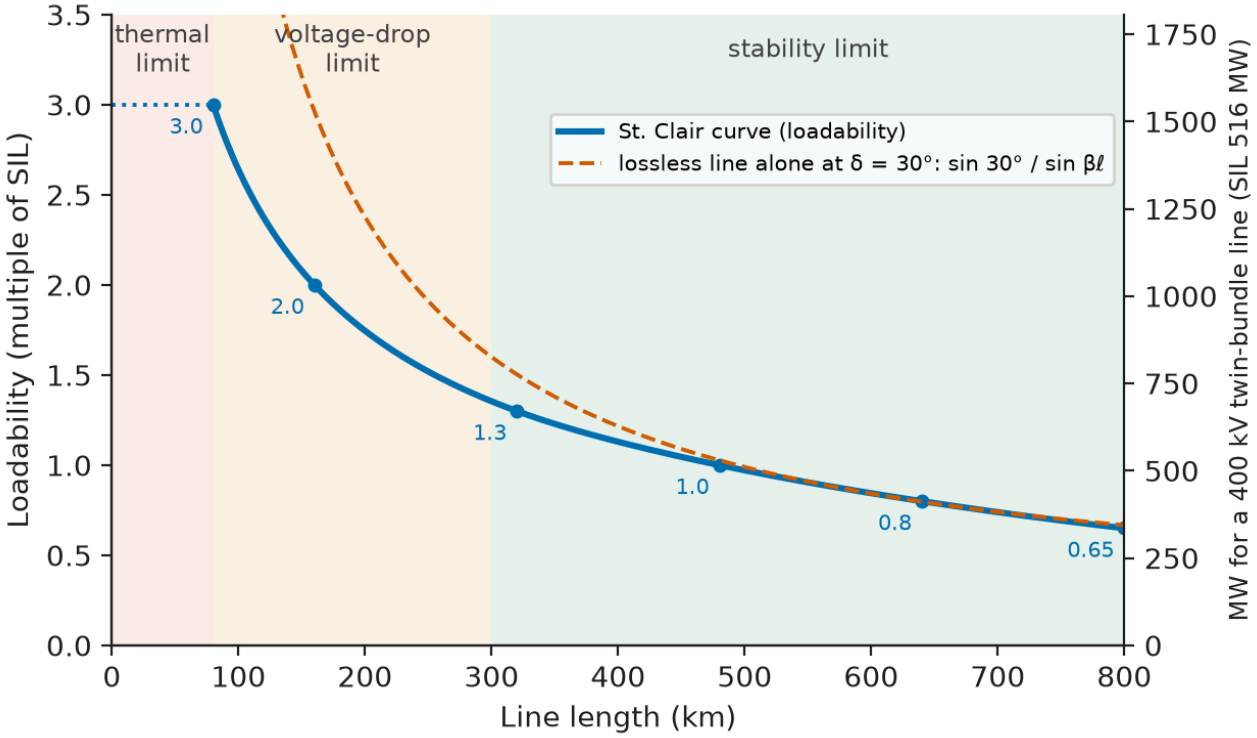


Figure 21.4 — Line loadability in multiples of surge impedance loading versus line length (St. Clair curve), showing the thermal, voltage-drop and stability regions.

21.5 Efficiency

Where the losses are

Transmission **efficiency** is the ratio of receiving-end to sending-end active power. Three loss mechanisms contribute:

- **Joule (I^2R) loss** in the conductors, proportional to the square of current and therefore to the square of load. It dominates.
- **Corona loss**, nearly independent of load, small in fair weather and much larger in rain, snow and fog (Section 21.7).
- **Insulator leakage**, negligible except on polluted insulators in wet weather.

For a fixed power and power factor, I^2R loss falls as $1/V^2$, which is the oldest argument for high transmission voltage (Book 3 Ch. 2 makes it for a single circuit). Reactive power flow also costs losses: current carrying reactive power heats conductors as much as current carrying active power, which is why reactive power is compensated close to where it is consumed.

Worked example — losses and their cost on the 400 km line. With the receiving end held at 400 kV and unity power factor, the exact ABCD solution gives:

Receiving-end power	Sending-end power	Loss	Efficiency	Sending-end voltage needed
0.5 SIL, 258 MW	265.1 MW	6.8 MW	97.4 %	0.957 pu
SIL, 516 MW	539.8 MW	23.3 MW	95.7 %	1.042 pu
1.5 SIL, 775 MW	825.4 MW	50.7 MW	93.9 %	1.158 pu (needs reactive support at the receiving end)

At SIL, the loss is about 1.1 % per 100 km, a figure close to practice for 400 kV lines. Over a year the line is not at SIL all the time. With a peak loading of SIL and a **loss load factor** (ratio of average to peak loss) of 0.45, the I^2R energy loss is $23.3 \times 0.45 \times 8760 = 91.8$ GWh per year; average corona loss of 3 kW/km over 400 km adds 10.5 GWh. At an energy value of USD 60 per MWh the losses cost 6.1 million dollars per year. Doubling the aluminium (a quad bundle of the same conductor) would halve the I^2R loss, saving 45.9 GWh and 2.75 million dollars a year; over a 40-year life at 8 % discount rate (present-worth factor 11.9) that is worth 32.8 million dollars today, to be compared with the extra cost of conductor, heavier towers and foundations.

Economic conductor choice

The comparison in the example is **Kelvin's law** in modern form: the economic conductor is the one that minimises the present worth of capital cost plus capitalised losses. Because energy prices and interest rates change, utilities define a **loss capitalisation** figure (dollars per kW of peak loss) and compare conductor options with it. IEC 60287-3-2 formalises the same idea for cables. Typical results are that economic current densities are well below thermal limits: overhead conductors are often loaded to 0.5–1.0 A/mm² on average against a thermal capability of 1.5–2 A/mm². System-wide, transmission losses are typically 1.5–3 % of energy delivered, and transmission plus distribution losses 5–10 % in well-run systems; Chapter 28 treats loss reduction as an efficiency measure.

21.6 Ferranti effect

The light-load voltage rise

The open-ended long line has a receiving-end voltage $V_S / \cos \beta\ell$ above its sending end (Book 3 Ch. 14 §14.3): the **Ferranti effect**, caused by the line's charging current flowing through its own series inductance. For the 400 km reference line, $1/|A| = 1.097$: a 9.7 % rise, so a line energised at 400 kV from one end sits at 439 kV at the far end, above the 420 kV maximum operating voltage of IEC 60038. The rise grows roughly with the square of length: 0.4 % at 80 km, 3.6 % at 250 km, 24 % at 600 km.

Two practical aggravations matter. First, the sending end is not an infinite bus. The charging current flows through the source impedance and raises the sending-end voltage as well. If the 400 km line is energised from a substation with a fault level of 4000 MVA (source reactance 40 Ω), the sending-end voltage rises to 1.062 times the source EMF and the open end to 1.165 times: the Ferranti rise compounds with the source rise. Line energisation from the weaker end, or from a single generator during black-start restoration, is therefore planned carefully, sometimes with generators absorbing reactive power at reduced voltage. Second, a sudden

load rejection at the receiving end of a long, heavily loaded line produces a temporary overvoltage that combines the Ferranti rise with the generators' internal voltage before their excitation can reduce it, often 1.3–1.5 pu for a few hundred milliseconds, which sets the duty of surge arresters (Chapter 24).

Shunt reactors

The standard remedy is the **shunt reactor**, connected at one or both ends of the line (directly on the line side of the breaker, so that it is energised with the line) or on the tertiary winding of a substation transformer. It absorbs reactive power and cancels part of the line's charging. The degree of shunt compensation, the ratio of reactor MVAr to line charging MVAr, is typically 50–80 % on long EHV lines.

Worked example — reactor for the reference line. The 400 km line generates $4 \times 54.8 = 219$ MVAr. A shunt reactor admittance Y_r at the open receiving end changes the receiving-end voltage to $V_R = V_S / |A + BY_r|$. With the sending end at 1.0 pu:

Reactor at receiving end	Degree of compensation	Open-end voltage
None	0 %	1.097 pu
100 MVAr	46 %	1.009 pu
150 MVAr	68 %	0.970 pu
200 MVAr	91 %	0.934 pu

A 100 MVAr reactor brings the open-end voltage to about 1.01 pu. In practice the compensation is split between the two ends (for example 2 × 63 MVAr or 2 × 80 MVAr, both standard 420 kV ratings) so that either end can energise the line. The reactors stay connected at full load, where they are not wanted, unless they are switched; switched reactors or controllable reactors (thyristor-controlled or saturable types) are used where the loading swings widely. A further benefit of line reactors is in single-pole auto-reclosing: a **neutral reactor** in the star point of the line reactor bank is tuned to cancel the inter-phase capacitive coupling that otherwise keeps a secondary arc burning in the de-energised phase.

Cables

Cables show the Ferranti effect and charging far more severely. A 400 kV XLPE cable with 0.2 µF/km generates $(400 \times 10^3)^2 \times 314.16 \times 0.2 \times 10^{-6} = 10.1$ MVAr per kilometre and draws 14.5 A/km of charging current. Fed from one end, a cable with a thermal rating of 1500 A would be fully occupied by its own charging current at about 103 km, leaving nothing for load. This is the practical limit of AC cable transmission: long AC cables need reactors at both ends and at intermediate points (the Hornsea offshore wind farm connections in the North Sea use an offshore reactive compensation station halfway along their 220 kV export cables), and beyond about 80–100 km at high voltage, HVDC becomes the only option (Section 21.10).

21.7 Corona

What corona is and why it limits design

When the electric field at the surface of a conductor exceeds the breakdown strength of air (about 30 kV/cm peak, 21 kV/cm RMS, at sea level for a smooth surface), the air immediately around it ionises. The result is **corona**: a faint violet glow visible at night, a hissing and crackling audible under the line especially in rain,

radio-frequency noise, ozone, and an energy loss. Corona does not short-circuit the line (the ionised region is confined to a thin layer where the field is high), but it limits conductor design: the field at the surface must be held below the corona onset level in fair weather, and the losses, noise and interference in foul weather must be acceptable.

The surface field depends on the charge per unit length, $q = CV_{\mathrm{ph}}$, and on the conductor radius. For a bundle of n sub-conductors of radius r on a circle of radius R_b, the average and maximum surface gradients are approximately

$$E_{\mathrm{av}} = \frac{q}{2\pi\varepsilon_0 n r}, \qquad E_{\max} = E_{\mathrm{av}}\left[1 + \frac{(n-1)r}{R_b}\right]$$

Peek's formulas

Peek's experiments in the 1910s gave the formulas still used for onset. The **visual corona onset gradient** (RMS) for a conductor of radius r in centimetres is

$$E_v = 21.1\, m\, \delta \left(1 + \frac{0.3}{\sqrt{\delta r}}\right) \ \mathrm{kV/cm}$$

and the **critical disruptive voltage** to neutral is $V_d = 21.1\, m\, \delta\, r \ln(D/r)$ kV, where δ is the relative air density (1.0 at 25 °C and sea level, 0.89 at 1000 m altitude) and m is the **surface irregularity factor** (1.0 for a polished cylinder, 0.82–0.9 for weathered stranded conductors, lower for wet or dirty ones). Peek's fair-weather loss formula is

$$P = \frac{241}{\delta}(f + 25)\sqrt{\frac{r}{D}}\,(V_{\mathrm{ph}} - V_d)^2 \times 10^{-5} \ \mathrm{kW/km\ per\ phase}$$

> **Status of the principle: Peek's corona formulas**
> *Empirical.* Fitted to laboratory measurements on single smooth and stranded conductors; they predict onset well but overestimate loss for bundled EHV conductors and say nothing about foul weather, for which utilities use measured loss and noise data (EPRI, CIGRE) for the specific conductor and climate.

Worked example — why 400 kV lines are bundled. At the maximum operating voltage of 420 kV, the reference twin-Moose line carries $q = CV_{\mathrm{ph}} = 10.91 \times 10^{-12} \times 242.5 \times 10^{3}$ C/m per phase. The average gradient on each sub-conductor is 15.0 kV/cm and the maximum, with $R_b = 0.225$ m, is $15.0 \times (1 + 0.01589/0.225) = 16.0$ kV/cm RMS, below a typical design limit of 17 kV/cm. A single Moose conductor per phase at the same spacing would carry 22.5 kV/cm, above its visual onset gradient of 21.4 kV/cm (Peek, $m = 0.82$, $\delta = 1$). Peek's critical disruptive voltage for the single conductor with $m = 0.85$ is 192.9 kV to neutral (334 kV line-to-line), so at 400 kV it would be in continuous corona; the fair-weather loss from Peek's formula is 26.5 kW/km for the three phases at 400 kV and 45 kW/km at 420 kV, compared with 1–5 kW/km measured on well-designed 400 kV twin bundles. At high altitude ($\delta = 0.89$ at 1000 m) every margin shrinks by about 11 %, which is why lines crossing the Andes or the Tibetan plateau use larger or more numerous sub-conductors.

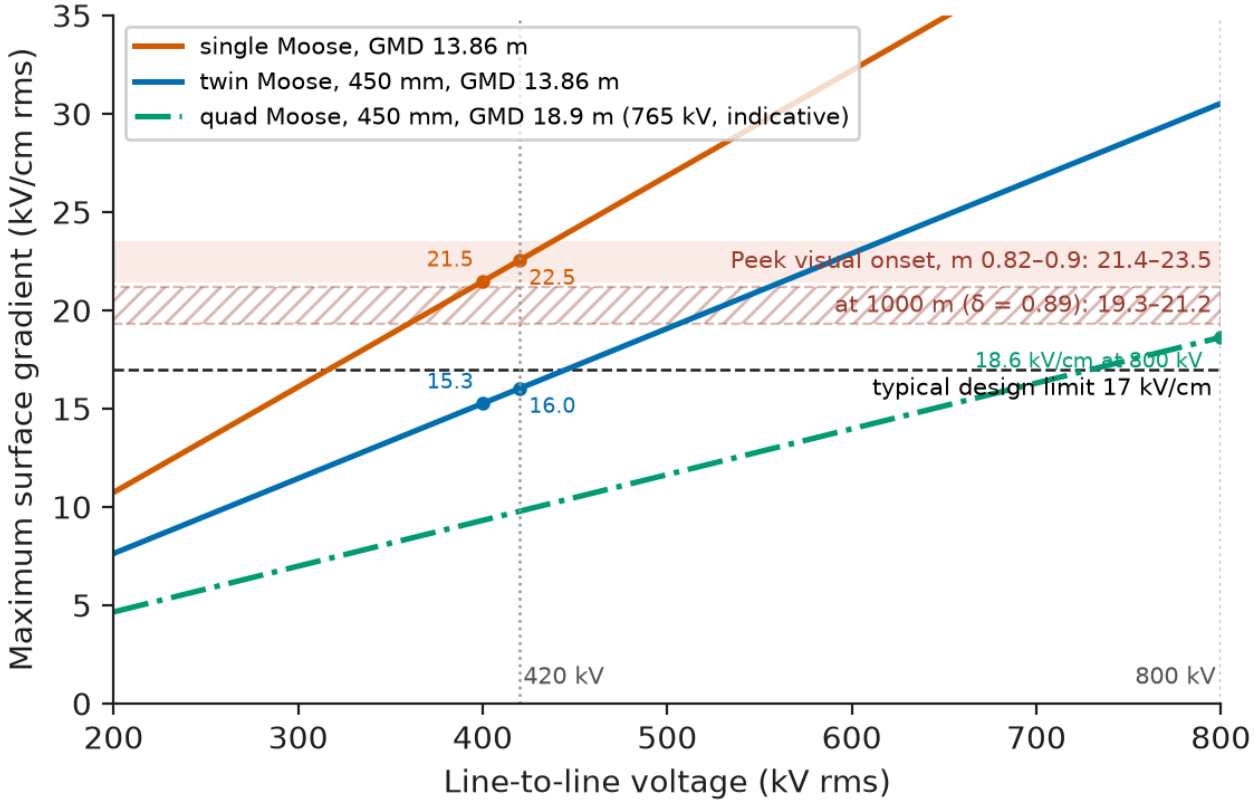


Figure 21.5 — Maximum conductor surface gradient versus line voltage for single, twin and quad Moose bundles, compared with the corona onset gradient.

Figure 21.5 shows the comparison.

Audible noise, radio interference and the environment

The visible and measurable consequences of corona set environmental limits that often bind before losses do. **Audible noise**, a broadband crackle with a 100 Hz (120 Hz) hum component, is typically limited to 40–55 dB(A) at the edge of the right-of-way in wet conditions, depending on national regulations. **Radio interference** in the AM broadcast band is assessed under CISPR 18. Both scale steeply with surface gradient (audible noise rises roughly 3–4 dB per kV/cm), so the gradient limit of 16–18 kV/cm used in EHV design is really a noise limit. Insulator fittings and spacers have their own corona rings to keep their gradients down. The ionised air also produces small amounts of ozone and nitrogen oxides, well below health limits at ground level.

21.8 Compensation

Why compensate

A line's reactive behaviour is set by two numbers, its surge impedance and its electrical length. Compensation changes both. **Shunt reactors** (inductive shunt compensation) reduce the effective shunt capacitance; **shunt capacitors** increase it; **series capacitors** reduce the effective series inductance. If the compensation is spread uniformly along the line (an idealisation that a few concentrated banks approximate), with a degree of shunt compensation k_{sh} and series compensation k_{se},

$$Z_c' = Z_c\sqrt{\frac{1-k_{\mathrm{se}}}{1-k_{\mathrm{sh}}}}, \qquad \beta'\ell = \beta\ell\sqrt{(1-k_{\mathrm{se}})(1-k_{\mathrm{sh}})}, \qquad \mathrm{SIL}' = \mathrm{SIL}\sqrt{\frac{1-k_{\mathrm{sh}}}{1-k_{\mathrm{se}}}}$$

> **Status of the principle: Effective parameters of a compensated line**
> *Approximate.* Exact only for uniformly distributed compensation; with two or three lumped banks the terminal behaviour is close to these values, but the voltage along the line deviates locally, and the series capacitors introduce a subsynchronous resonance that no fundamental-frequency formula captures.

Applied to the 400 km reference line:

Compensation	Effective Z_c	Effective SIL	Electrical length	Maximum transfer (lossless, 1.0 pu ends)
None	310 Ω	516 MW	24.3°	1255 MW
70 % shunt reactors	566 Ω	283 MW	13.3°	1229 MW
50 % series capacitors	219 Ω	730 MW	17.2°	2470 MW
70 % shunt and 50 % series	400 Ω	400 MW	9.4°	2450 MW

The table separates the roles. Shunt reactors solve the light-load voltage problem by lowering the natural load to match the actual load, but do nothing for transfer capability. Series capacitors almost double the maximum transfer, because they shorten the line electrically, and with it the angle across the line for a given power.

Series capacitors

Series capacitors are installed in banks on insulated platforms at line ends or at mid-line switching stations, typically compensating 25–70 % of the line reactance. They are protected against fault currents by metal-oxide varistors in parallel, a triggered spark gap and a bypass breaker. Their large-scale use (in Brazil, Sweden, western North America, India and China) has made it possible to carry power over 800–1000 km with AC. Two complications follow from their resonance with the line inductance at a frequency $f_0\sqrt{k_{\mathrm{se}}}$, below 50 Hz (35 Hz for 50 % compensation): **subsynchronous resonance** with turbine shafts (Chapter 20) and **subsynchronous control interaction** with wind-turbine converters, first seen in Texas in 2009 when a series-compensated line was left radial with a wind farm. The cures are thyristor-controlled series capacitors (TCSC), which make the capacitor's apparent impedance inductive at subsynchronous frequencies, and filters or damping controllers at the affected plants. Series capacitors also complicate distance protection, because a fault behind a capacitor appears closer than it is (Chapter 23).

Static and dynamic shunt compensation

Fixed and mechanically switched shunt capacitors and reactors provide steady compensation cheaply. Fast-acting devices handle contingencies and stability:

Device	Principle	Response	Typical rating
Mechanically switched reactor or capacitor	Breaker-switched fixed bank	Seconds (with switching limits)	20–200 MVAr
Static VAR compensator (SVC)	Thyristor-controlled reactor plus thyristor-switched capacitors	2–3 cycles	±100–400 MVAr
STATCOM	Voltage-source converter (now modular multilevel) acting as a controlled voltage source	1–2 cycles	±50–400 MVAr
Synchronous condenser	Unloaded synchronous machine with fast excitation	Inherent subtransient response, then excitation control	50–350 MVA

An SVC's output falls with the square of voltage (it is a controlled susceptance), so it supports least when voltage is lowest. A STATCOM's current is limited rather than its susceptance, so its reactive output falls only linearly with voltage, a decisive advantage for voltage stability. Synchronous condensers add short-circuit strength and inertia, which converters do not. A mid-point STATCOM or SVC on a long line also raises transfer capability: holding the midpoint voltage effectively splits the line into two lines of half the electrical length, nearly doubling the theoretical maximum transfer. The converter internals of SVCs and STATCOMs belong to *Electronics & Computing*.

21.9 High-voltage AC transmission

Voltage classes

AC transmission voltages are standardised (IEC 60038, with highest voltages for equipment in IEC 60071):

Nominal voltage	Highest equipment voltage	Typical surge impedance loading	Typical thermal rating per circuit	Typical right-of-way width
132 kV	145 kV	≈ 45 MW	150–250 MVA	25–30 m
220–230 kV	245 kV	≈ 125 MW	300–600 MVA	35 m
345 kV (North America)	362 kV	≈ 400 MW	800–1500 MVA	40–45 m
400 kV	420 kV	≈ 520 MW	1000–2200 MVA	46–52 m
500 kV	550 kV	≈ 850–950 MW	1500–3000 MVA	50–60 m
765 kV	800 kV	≈ 2300 MW	3000–5000 MVA	60–70 m
1000 kV (China, UHV AC)	1100 kV	≈ 4200 MW	Up to ≈ 9000 MVA	75–90 m

Raising the voltage raises SIL roughly as V^2 and therefore the power a long line can carry, while the right-of-way grows only modestly; MW per metre of corridor width therefore rises steeply with voltage. Against that,

higher voltages need larger towers, longer insulator strings, larger substations and transformers, and more reactive compensation. The voltage class chosen for a new corridor is the one that minimises the total cost of lines, substations and losses over the planned power and distance, subject to reliability (a single 765 kV circuit carrying 2.5 GW is a single contingency of 2.5 GW, which the frequency criteria of Chapter 20 must allow).

Insulation and clearances

Line insulation is designed for three stresses: continuous power-frequency voltage in polluted conditions, switching overvoltages (dominant above about 300 kV) and lightning (dominant below). Insulator strings are sized by **specific creepage distance**, the length of surface path per kV of line-to-line voltage: about 16 mm/kV for light pollution, 25 mm/kV for heavy and 31 mm/kV for very heavy (IEC TS 60815), which is why coastal and desert lines have longer strings or composite (silicone rubber) insulators. Air clearances and tower dimensions follow from switching-surge withstand, which is reduced by closing resistors or controlled switching of breakers. Insulation coordination is the subject of Chapter 24.

Ratings in operation

The thermal rating of a line is not a fixed number. Conductor temperature depends on ambient temperature, wind speed and direction, and solar radiation; a static seasonal rating assumes conservative values (for example 0.6 m/s wind, 35 °C, full sun). **Dynamic line rating** (DLR) measures or estimates the actual weather and conductor sag in real time and typically finds 10–30 % more capacity for most of the year, more on windy days, which coincides usefully with high wind-farm output. Reconductoring with HTLS conductors raises the thermal rating by 50–100 % on existing towers. Both are among the cheapest ways to increase transmission capacity, but they help only on lines limited thermally, which by the loadability curve of Section 21.4 means short ones.

Underground and gas-insulated alternatives

Underground XLPE cables at 220–400 kV are used in cities, river crossings and sensitive landscapes, at a cost typically 5–15 times that of overhead lines per kilometre, and with the charging limits of Section 21.6. **Gas-insulated lines** (GIL), conductors in SF_6/N_2-filled tubes, have lower capacitance than cables and very high ratings (2–4 GW) for short tunnel or substation connections. For long submarine and underground connections, the answer is DC.

21.10 HVDC fundamentals

Why DC

A DC line has no inductive reactance and no capacitive charging in steady state. Three consequences follow, and they are the reasons HVDC exists:

1. **No distance limit from reactive power or stability.** The power transferred is set by controls, not by an angle across the line. A DC line's capability is its thermal rating at any length.
2. **Cables of any length.** Without charging current, submarine and underground DC cables of hundreds of kilometres are routine (NordLink, 623 km between Norway and Germany; the 720 km North Sea Link between Norway and Britain).
3. **Asynchronous connection.** A DC link connects systems at different frequencies (50 and 60 Hz in Japan; back-to-back links between the US Eastern and Western interconnections) or unsynchronised

systems of the same frequency, and its power can be controlled precisely and quickly, including to damp oscillations in the AC systems around it.

Conductor utilisation and losses

For the same insulation (set by the peak voltage to earth, V_p) and the same conductor current I, a three-conductor AC circuit carries $3(V_p/\sqrt{2})I$ while a two-conductor DC bipole carries $2V_pI$. The DC line carries 94 % of the AC power with two-thirds of the conductors, and its loss as a fraction of power transmitted is $(2/0.943)/3 = 71$ % of the AC line's.

> **Status of the principle: Conductor utilisation of DC versus AC**
> *Approximate*. Assumes that insulation is dimensioned by peak voltage to earth and conductors by RMS current; in practice DC insulators need 30–50 % more creepage because DC fields attract and hold pollution, and AC lines are rarely loaded to their thermal limit at long distances, so the real comparison is made on total cost.

The DC line has other advantages: no skin effect, narrower towers and right-of-way for the same power, and a continued ability to operate one pole with earth or metallic return when the other is out. Its cost is in the terminals. Each converter station costs as much as hundreds of kilometres of line, loses 0.6–1 % of the power converted, and needs filters (for LCC) and transformers designed for DC stresses. The total cost falls more slowly with distance than AC's, giving a **break-even distance** beyond which DC is cheaper: roughly 500–800 km for overhead lines and 50–100 km for submarine or underground cables.

Worked example — a 1000 km, 3 GW bulk link. A bipole carries 3000 MW over 1000 km on overhead lines with four large ACSR sub-conductors per pole, giving 0.009 Ω/km per pole. At ±500 kV, the pole current is $3000/(2 \times 0.5) = 3000$ A and the line loss is $2 \times 3000^2 \times (0.009 \times 1000) = 162$ MW, 5.4 % of the power. Two LCC converter stations at about 0.7 % each add 42 MW, for a total of about 6.8 %. At ±800 kV on the same conductors, the current falls to 1875 A and the line loss to 63.3 MW (2.1 %): raising the DC voltage pays for itself on long links exactly as it does for AC, which is why China built its long links at ±800 kV and ±1100 kV. The AC alternative at 1000 km would need at least two 765 kV circuits with about 50 % series compensation, intermediate switching stations every 300–400 km and large shunt reactors, on a corridor twice as wide; the decisive differences are corridor width, controllability and the size of the single contingency, not the loss alone.

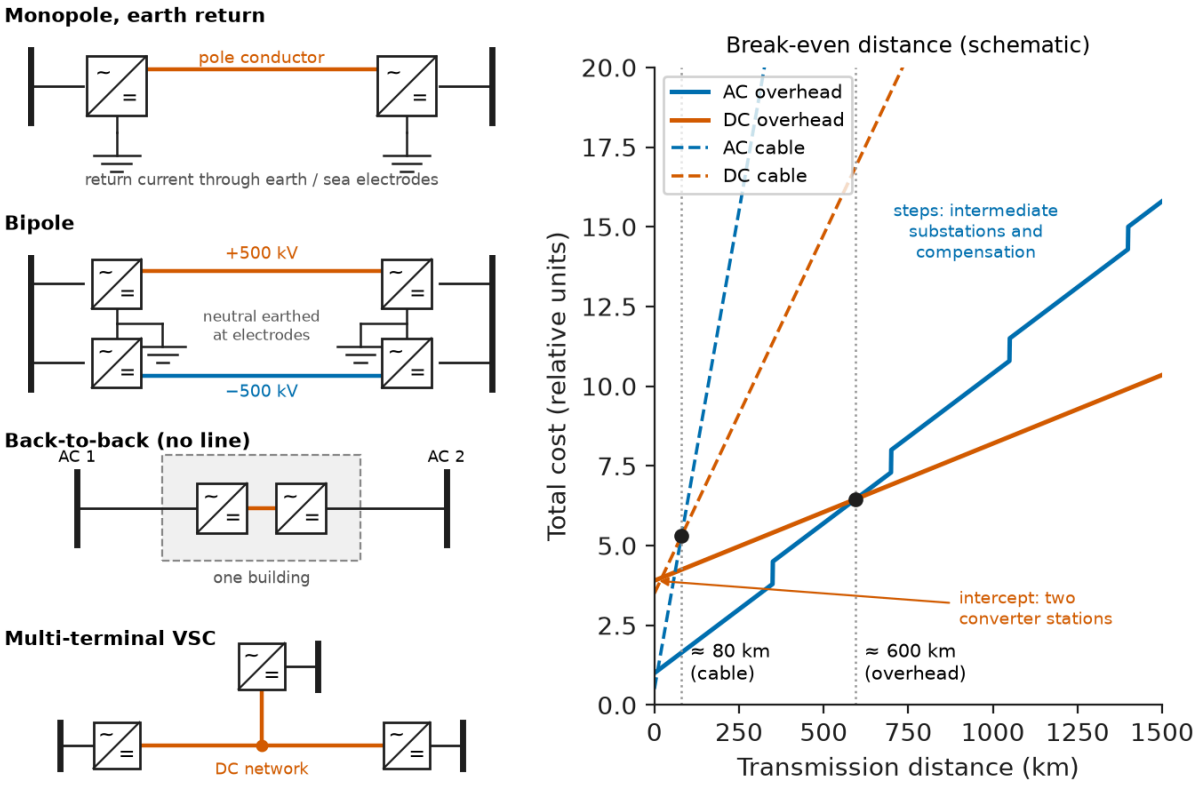


Figure 21.6 — HVDC configurations and the break-even distance between AC and DC transmission.

Figure 21.6 sketches the configurations and the break-even comparison.

Two converter technologies

	Line-commutated converter (LCC)	Voltage-source converter (VSC)
Switching device	Thyristor valves, 12-pulse bridges	IGBT modules in a modular multilevel converter (MMC)
Commutation	Needs an AC voltage to turn off valves	Self-commutated
Reactive power	Consumes 50–60 % of the active power; supplied by filters and capacitor banks	Controls reactive power independently in each station
AC system strength needed	Short-circuit ratio above about 2.5–3	Works into weak systems, can black-start a dead network
Power reversal	By reversing DC voltage	By reversing DC current; suits multi-terminal grids
Converter losses per station	≈ 0.6–0.8 %	≈ 0.8–1.0 % (MMC)
Largest ratings in service	±1100 kV, 12 GW (Changji–Guquan, 3293 km)	±525 kV, 1.4–2 GW per link
Typical use	Bulk overhead links, very long distance	Offshore wind, submarine and underground cables, interconnectors, multi-terminal

The **short-circuit ratio** (SCR), the AC system's fault level at the converter bus divided by the converter rating, is a direct use of Chapter 19's fault levels. An LCC link needs a stiff AC system because its valves commutate with the AC voltage; a fault that depresses that voltage causes **commutation failure** and a momentary collapse of DC power, which a weak system feels as a large disturbance. VSC links avoid this and are the technology of choice for connecting offshore wind farms (the German North Sea clusters at ±320 kV and the 2 GW, ±525 kV standard platforms now being built) and for the multi-terminal DC grids under development. Their remaining obstacle is DC fault interruption: there is no current zero, so a multi-terminal DC grid needs DC circuit breakers (hybrid mechanical–semiconductor designs that interrupt in 2–5 ms) or converters that can block fault current (Chapter 23).

Project	Configuration	Rating	Length	Technology	In service
Itaipu (Brazil)	Two bipoles	±600 kV, 6.3 GW	≈ 800 km overhead	LCC	1984–1987
Xiangjiaba–Shanghai (China)	Bipole	±800 kV, 6.4 GW	1907 km overhead	LCC	2010
Changji–Guquan (China)	Bipole	±1100 kV, 12 GW	3293 km overhead	LCC	2019
NordLink (Norway–Germany)	Bipole	±525 kV, 1.4 GW	623 km, mostly submarine	VSC	2021
North Sea Link (Norway–Britain)	Bipole	±515 kV, 1.4 GW	720 km submarine	VSC	2021

The converter circuits themselves, from thyristor valve design to MMC submodules, belong to *Electronics & Computing*; their effect on harmonics is treated in Chapter 25 and their role in grid integration in Chapter 27.

What remains open

Open question	Where it is answered
How are telegrapher's equations, surge impedance and the exact ABCD constants derived?	*Circuits, Signals and Communications*, Ch. 14
How are lines represented in network studies in per unit?	Chapters 17 and 18
How do line zero-sequence parameters set earth-fault currents?	Chapter 19 (Fault Analysis)
How do transfer limits from line reactance and compensation affect stability?	Chapter 20 (Power-System Stability)
How is power delivered from the transmission substation to customers, and how is feeder voltage controlled?	Chapter 22 (Distribution Systems)
How are lines and series-compensated lines protected?	Chapter 23 (Power-System Protection)
How are line insulation, lightning shielding and switching overvoltages coordinated?	Chapter 24 (High-Voltage Engineering)
How do HVDC converters and SVCs affect harmonics?	Chapter 25 (Power Quality)
How do converter controls of HVDC and wind farms interact with the AC grid?	Chapter 27 (Grid Integration of Renewable Energy)
How do the converter valves and MMC submodules work inside?	*Electronics & Computing*

Next: Chapter 22 follows the power from the transmission substation into the distribution network, where radial feeders, voltage drop and distributed generation set the problems.

Chapter 22 — Distribution Systems

Part VI · Transmission, Distribution, and Protection

Every weekday at about seven in the evening the customers at the far end of an 11 kV rural feeder phone the utility: lights dim, a borehole pump trips on undervoltage, a heat pump refuses to restart. The substation voltmeter reads a healthy 11.2 kV. Eight kilometres away, the same feeder is delivering 10.2 kV. Twelve hours later the problem reverses on the low-voltage side of a suburban street: at noon, with fifty rooftop photovoltaic systems exporting into a lightly loaded cable, the end-of-street voltage climbs past 253 V and the inverters disconnect one by one on overvoltage, throwing away the energy their owners paid for. Both complaints come from the part of the power system that most engineers never see on a single-line diagram of the grid: the distribution network, where most of the conductor length, most of the transformers, most of the interruptions and the larger share of the losses actually are. This chapter treats distribution as an engineering system in its own right: how it is laid out, how its feeders and transformers are modelled, why its voltage drops and where its energy goes, how its voltage is held inside the band the standards require, and what happens when generation appears at the wrong end of it.

> **How are distribution networks laid out, and why does a utility choose radial, ring or meshed operation for a given load?**
> **How much voltage and energy is lost along a feeder, how is the voltage held inside its statutory band, and what changes when distributed generation reverses the power flow?**

Concept	Engineering destination
Distribution architectures	Utility networks, industrial plants, data centres, mines, ships
Radial and ring systems	Rural feeders, urban ring-main units, hospital and data-centre dual supplies
Distribution transformers	Pole-mounted and ground-mounted substations, loss-evaluated procurement
Feeder analysis	Ladder (backward–forward sweep) load flow, unbalance studies, connection assessments
Voltage drop	Conductor sizing, cable selection in buildings, statutory voltage compliance
Distribution losses	Loss factor, loss capitalisation, regulatory loss targets
Voltage regulation	On-load tap changers, line-drop compensation, step regulators, capacitor banks
Distributed generation	Rooftop PV hosting capacity, reverse power flow, volt–var control
Distribution planning	Load forecasts, diversity, firm capacity, EV and heat-pump growth

Roadmap. Where distribution sits and what voltages it uses (22.1) → how feeders are connected and operated (22.2) → the transformers that step down to utilisation voltage (22.3) → modelling a feeder (22.4) → voltage drop along it (22.5) → where the energy goes (22.6) → holding the voltage in its band (22.7) → generation inside the network (22.8) → planning the network for the next twenty years (22.9).

22.1 Distribution architectures

Where distribution begins and ends

Transmission (Chapter 21) moves bulk power at 132–765 kV over meshed networks operated by a system operator. Distribution begins at the **bulk supply point** or **grid substation**, where power is stepped down to subtransmission or primary distribution voltage, and ends at the customer's meter. A typical European chain is 132/33 kV at the grid substation, 33/11 kV at **primary substations**, 11 kV primary feeders, and 11/0.4 kV **secondary substations** feeding 400/230 V low-voltage (LV) mains along the streets (Figure 22.1). North American practice merges the middle levels: substations step 69–138 kV down to 12.47–34.5 kV primary feeders, and small single-phase pole-mounted transformers (10–100 kVA) supply a few customers each at 120/240 V split phase.

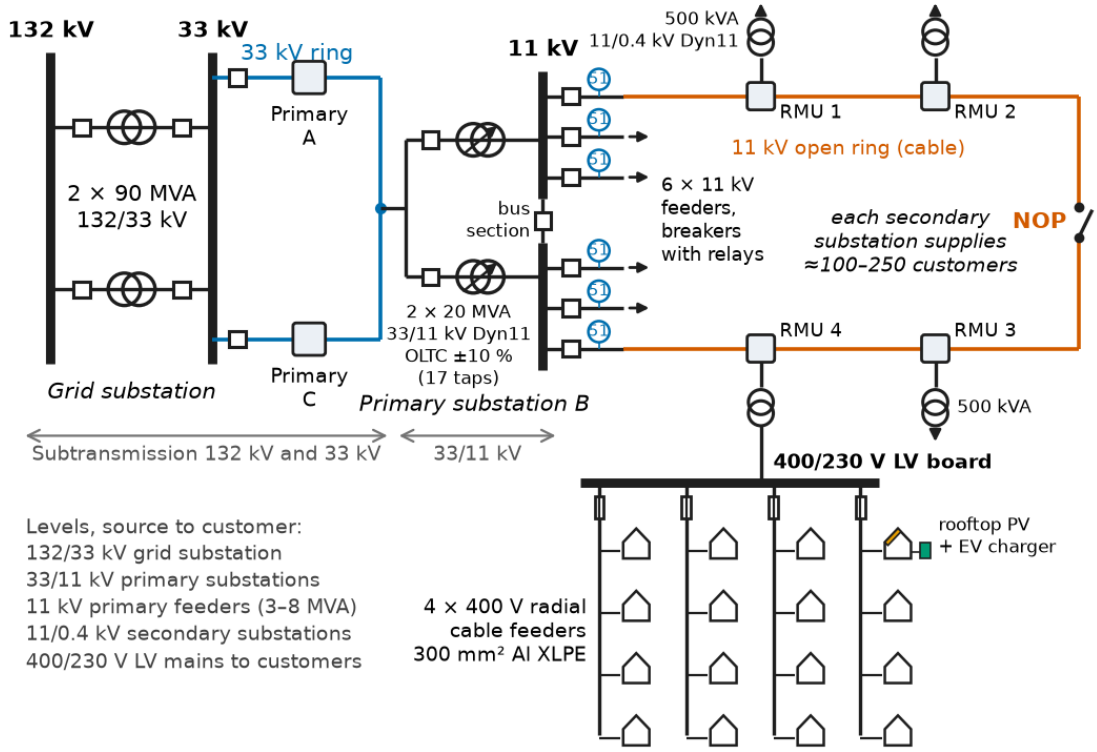


Figure 22.1 — A typical European-style distribution chain from the grid substation to the customer, with representative ratings at each level.

Representative voltage levels are summarised in Table 22.1. Nominal values follow IEC 60038 in most of the world and ANSI C84.1 in North America.

Table 22.1 — Representative distribution voltage levels (nominal, line-to-line unless stated).

Level	IEC practice	North American practice	Typical function
Subtransmission	33, 66, 110, 132 kV	34.5, 46, 69, 115, 138 kV	Grid substation to primary substations; large industry
Primary (MV)	6.6, 10, 11, 20, 22 kV	4.16, 12.47, 13.2, 13.8, 24.9, 34.5 kV	Feeders to secondary transformers; mines; large buildings
Secondary (LV)	400/230 V three-phase four-wire	120/240 V split phase; 208Y/120, 480Y/277 V	Homes, commercial, light industry
Industrial LV	690 V, 1000 V	600 V (Canada), 480 V	Large drives, mining machinery, wind turbines
Typical primary-substation rating	2 × 10 to 2 × 40 MVA	1 × 20 to 3 × 50 MVA	
Typical feeder load	3–8 MVA at 11 kV	5–15 MVA at 12.47–13.8 kV	

Overhead and underground construction

Rural networks are overhead: bare aluminium conductor steel-reinforced (ACSR) or all-aluminium alloy conductors on wood or concrete poles, cheap to build (a few tens of thousands of dollars per km at 11 kV) and quick to repair, but exposed to wind, trees, lightning and vehicles. Urban networks are underground: XLPE-insulated aluminium or copper cables in ducts or direct-buried, typically five to ten times the cost per km, nearly immune to weather, but with fault-location and repair times measured in hours or days. Cables have far more capacitance than overhead lines (about 0.3–0.5 μF/km for an 11 kV XLPE cable against 0.01 μF/km for an overhead line), so a cable network generates a noticeable amount of reactive power at light load and its earth-fault currents are larger (Chapter 23).

Neutral and earthing arrangement

Distribution systems differ sharply in how the neutral is treated. European MV networks are three-wire, with the transformer neutral earthed through a resistor or a Petersen (arc-suppression) coil, or left isolated; every load is three-phase or phase-to-phase. North American MV networks are four-wire multi-grounded: the neutral is carried along the feeder and earthed at every pole transformer, so single-phase laterals can branch off to supply customers. The choice determines how earth faults are detected (Chapter 23 §23.5) and how much current the neutral carries. At LV the IEC 60364 system classes (TN-S, TN-C-S, TT, IT) set how exposed metalwork is connected to earth; they are a safety topic and return in Chapters 29 and 30.

Other distribution systems

The same architectures appear outside utilities. An **industrial plant** takes a 33 or 66 kV supply into its own main substation and distributes at 6.6 or 11 kV to motor control centres and unit substations. A **data centre** of 30 MW typically takes two independent utility feeds, runs dual "A" and "B" MV rings or radials, and supplies each server rack from two separate UPS-backed paths, so that any single component can fail without dropping load. An **underground mine** distributes at 11 kV or 3.3 kV through trailing cables to mobile substations, with resistance-earthed (often IT-type) systems that limit earth-fault current to a few amperes for shock and ignition safety. A **ship** or offshore platform distributes at 6.6 or 11 kV from its own generators with integrated electric propulsion, and an **aircraft** uses 115/200 V at 400 Hz (or variable frequency, 360–800 Hz) to save

transformer and generator mass. All of them face the same trade-offs between cost, losses, voltage drop and the consequences of a single failure.

> **Boundary.**
> Three-phase phasor relations, star and delta connections and per-phase equivalents are taken from *Circuits, Signals and Communications*, Ch. 10; active, reactive and complex power from Ch. 9. The per-unit system and network models used below come from Chapter 17 of this book.

22.2 Radial and ring systems

Radial feeders

In a **radial** system every load has exactly one path to the source. A primary feeder leaves the substation through a circuit breaker, runs as a **main** (trunk) along a road, and splits into **laterals** protected by fuses or reclosers. Radial networks are cheap, simple to protect (current flows one way, so time-graded overcurrent relays and fuses discriminate naturally, Chapter 23) and simple to analyse. Their weakness is that a fault anywhere on the main interrupts every customer downstream of the operating device until it is repaired or bypassed. Rural overhead networks accept this, using **auto-reclosers** to restore supply after the 80–90 % of overhead faults that are transient (lightning flashover, a branch touching the line), and **sectionalisers** to isolate the faulted section of permanent faults.

Open rings and ring-main units

Urban cable networks are built as rings but operated radially. Each 11 kV feeder runs from one substation (or one busbar section) through a chain of **ring-main units** (RMUs) and returns to another, with one switch in the chain left open: the **normally open point** (NOP). Under normal conditions the two halves are two radial feeders. When a cable section faults, the feeder breaker trips, the faulted section is isolated by opening the RMU switches on either side, the breaker is reclosed, and the NOP is closed to restore the customers beyond the fault from the other end. With manual switching this takes 30–90 minutes; with remotely controlled RMUs and fault-passage indicators it takes one to three minutes; with automatic **fault location, isolation and service restoration** (FLISR, Chapter 26) it can take under a minute. The ring costs more cable than a radial (each half must be able to carry the whole ring's load) but customer minutes lost drop by an order of magnitude.

Closed rings and meshed networks

Running the ring closed means every load always has two paths and a single cable fault interrupts nobody, but it needs protection that can tell which of two infeeds is feeding a fault: directional overcurrent, distance, or unit (differential) protection with communication (Chapter 23). Closed rings are standard at 33–132 kV subtransmission and in industrial plants with critical processes. At LV, the dense downtown cores of some North American cities use **secondary networks**: a meshed 208Y/120 V or 480Y/277 V grid fed from several primary feeders through **network protectors**, reverse-power relays that open a transformer's LV breaker if power flows back toward a faulted primary feeder. Such networks give reliability approaching that of transmission, at a high price.

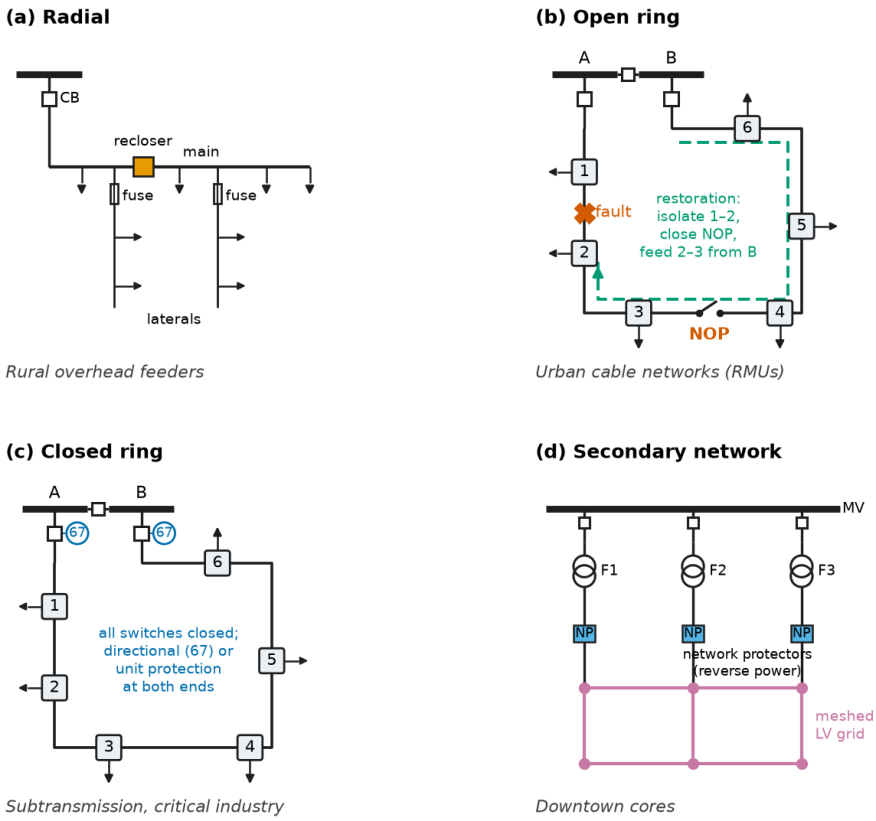


Figure 22.2 — Four ways to connect distribution loads: radial, open ring with a normally open point, closed ring, and a meshed secondary network with network protectors.

Figure 22.2 compares the four arrangements. The choice is an economic one: the cost of the extra cable, switchgear and protection against the value placed on interruptions avoided.

Measuring reliability

Utilities and regulators measure the result with the indices of IEEE 1366:

$$\mathrm{SAIFI} = \frac{\sum N_i}{N_T}, \qquad \mathrm{SAIDI} = \frac{\sum r_i N_i}{N_T}, \qquad \mathrm{CAIDI} = \frac{\mathrm{SAIDI}}{\mathrm{SAIFI}}$$

where N_i is the number of customers interrupted by event i, r_i its duration and N_T the number of customers served. SAIFI counts interruptions per customer per year, SAIDI minutes lost per customer per year, CAIDI the average restoration time. Interruptions shorter than 5 minutes (IEEE 1366) or 3 minutes (European practice) are counted separately as momentary events.

Table 22.2 — Typical reliability of distribution arrangements (excluding major-event days; values vary widely between utilities).

Arrangement	SAIFI (per year)	SAIDI (min per year)	Typical setting
Rural overhead radial with reclosers	1.5–4	150–600	Farmland, mining towns
Suburban mixed overhead/cable, open ring, manual switching	0.5–1.5	60–150	Suburbs
Urban cable, open ring, remote-controlled RMUs	0.2–0.6	15–60	City centres in Europe and East Asia
Secondary network (meshed LV)	below 0.1	below 10	Downtown cores, hospitals
Dual-fed data centre with UPS (at the IT load)	effectively 0	effectively 0	Tier III–IV facilities

> **Status of the principle: Reliability indices**
> *Empirical.* SAIFI and SAIDI are defined bookkeeping quantities; their values come from outage records, and the figures above are typical observations, not predictions. Planning studies predict them from component failure rates and repair times, which are themselves empirical.

22.3 Distribution transformers

Construction and ratings

A distribution transformer is the device studied in Chapter 4 built in very large numbers: a utility with five million customers may own half a million of them. Three-phase ground-mounted units are typically 200–2500 kVA, 11 or 22 kV to 400 V, connected **Dyn11** (delta primary, earthed-star secondary, 30° phase shift) so that the LV side has a neutral for single-phase loads, triplen harmonic currents circulate in the delta rather than in the MV network, and an earth fault on the LV side is not seen as zero-sequence current upstream. Pole-mounted units are smaller: 25–315 kVA three-phase in Europe, 10–167 kVA single-phase in North America. Most are oil-immersed with natural cooling (ONAN); dry-type cast-resin units are used inside buildings, ships and wind-turbine towers where oil is a fire hazard. Tappings are **off-circuit**: typically ±2 × 2.5 % on the HV winding, set once at installation to suit where the transformer sits on its feeder.

Losses, efficiency and loss evaluation

The no-load (core) loss P_0 flows 8760 hours a year whatever the load; the load (copper) loss P_k at rated current scales with the square of the per-unit load k. Efficiency at load k and power factor $\cos\varphi$ is

$$\eta = \frac{k S_r \cos\varphi}{k S_r \cos\varphi + P_0 + k^2 P_k},$$

with its maximum at $k_{\max} = \sqrt{P_0/P_k}$ (Chapter 4). Because distribution transformers are lightly loaded on average (mean loads of 20–40 % of rating are common), the no-load loss dominates their annual energy loss, which is why amorphous-metal and high-grade grain-oriented steel cores pay back. Minimum efficiencies are now regulated: in the European Union by the Ecodesign regulation (EU) 548/2014 with its Tier 2 limits, in the United States by the Department of Energy rules (10 CFR 431), and in many other countries by equivalent standards. Utilities buy transformers on **capitalised loss** rather than first price, adding $A \cdot P_0 + B \cdot P_k$ to the tender price, with A and B (in USD per kW) set by the present value of the energy cost.

Table 22.3 — Typical three-phase oil-immersed distribution transformers, 11 kV/400 V, 50 Hz (ranges span ordinary to low-loss designs).

Rating	Short-circuit impedance	No-load loss P_0	Load loss P_k (75 °C)	Efficiency at 50 % load, PF 1
100 kVA	4 %	0.15–0.32 kW	1.3–2.1 kW	98.4–99.0 %
400 kVA	4 %	0.40–0.93 kW	3.3–4.6 kW	99.0–99.4 %
630 kVA	4–6 %	0.55–1.3 kW	5.4–6.5 kW	99.1–99.4 %
1000 kVA	6 %	0.70–1.7 kW	9–13 kW	99.0–99.4 %
2500 kVA	6 %	1.5–3.2 kW	22–27 kW	99.2–99.4 %

Worked example — All-day efficiency and the value of a low-loss core. A 1000 kVA, 11/0.4 kV transformer supplies a commercial district with a daily cycle of 6 hours at 0.9 per unit, 10 hours at 0.5 per unit and 8 hours at 0.2 per unit, all at power factor 0.95. Compare a low-loss unit ($P_0 = 0.70$ kW, $P_k = 9.0$ kW) with a 1990s unit ($P_0 = 1.70$ kW, $P_k = 10.5$ kW).

Energy delivered per day: $1000 \times 0.95 \times (6 \times 0.9 + 10 \times 0.5 + 8 \times 0.2) = 11\,400$ kWh.

Low-loss unit: no-load energy $24 \times 0.70 = 16.8$ kWh; load-loss energy $9.0 \times (6 \times 0.81 + 10 \times 0.25 + 8 \times 0.04) = 9.0 \times 7.68 = 69.1$ kWh; total 85.9 kWh. All-day efficiency $11\,400/(11\,400 + 85.9) = 99.25\,\%$.

Old unit: no-load energy $24 \times 1.70 = 40.8$ kWh; load-loss energy $10.5 \times 7.68 = 80.6$ kWh; total 121.4 kWh; all-day efficiency 98.95 %.

The difference is 35.5 kWh per day, or about 13 000 kWh per year, worth roughly USD 1000–1300 a year at USD 0.08–0.10 per kWh, for a transformer that will be in service for 30–40 years. Two-thirds of the saving comes from the core. The maximum-efficiency load of the low-loss unit is $\sqrt{0.70/9.0} = 0.28$ per unit, close to its overnight load. Figure 22.3 shows the two efficiency curves.

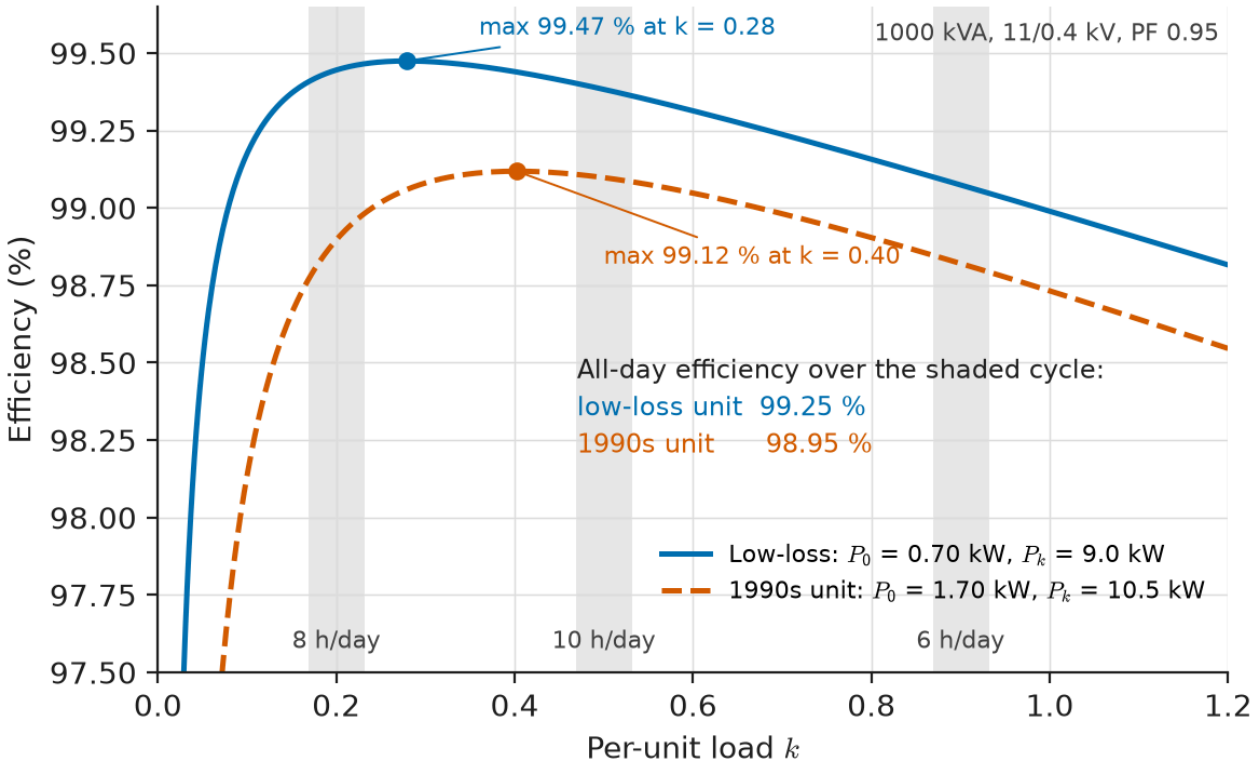


Figure 22.3 — Efficiency versus load of two 1000 kVA distribution transformers: a low-loss design and an older design.

Loading beyond nameplate

A distribution transformer's rating is thermal: IEC 60076-7 and IEEE C57.91 give the hot-spot temperature as a function of load and ambient temperature, and the ageing rate of the paper insulation doubles for every 6 K of hot-spot rise above 98 °C (Chapter 9). A unit that runs at 0.5 per unit most of the day can carry 1.3–1.5 per unit for two hours in a winter evening without exceeding its normal life consumption. This cyclic rating is what makes the after-diversity planning of Section 22.9 work, and it is also what EV charging threatens: a long, flat overnight peak removes the cool-down periods that the cyclic rating relies on.

> **Status of the principle: Insulation ageing doubles per 6 K**
> *Empirical.* The Arrhenius-based ageing law of IEC 60076-7 for non-upgraded kraft paper (thermally upgraded paper ages more slowly and is referenced to 110 °C) is fitted to accelerated-ageing tests; it holds between about 80 °C and 140 °C hot-spot and says nothing about mechanical or moisture-driven failure.

22.4 Feeder analysis

The per-phase feeder model

A distribution feeder is short compared with a wavelength (an 11 kV feeder of 10 km is 0.003 of the 50 Hz wavelength), so the short-line model of Chapter 21 (§21.3) applies: a series impedance $z = r + jx$ per km and, for overhead lines, negligible shunt admittance. Cables add shunt capacitance that matters for reactive balance at light load but rarely for voltage drop. Typical parameters are given in Table 22.4.

Table 22.4 — Representative feeder parameters at 50 Hz (positive sequence, conductor at about 50 °C for overhead, 70 °C for cable).

Conductor	r (Ω/km)	x (Ω/km)	x/r	Thermal rating
11 kV ACSR 50 mm² ("Rabbit")	0.60	0.37	0.6	190 A
11 kV ACSR 100 mm² ("Dog")	0.27–0.30	0.35	1.2–1.3	300 A
33 kV ACSR 150 mm² ("Wolf")	0.19–0.21	0.38	1.9	400 A
11 kV XLPE 3 × 185 mm² Al cable	0.21	0.10	0.5	330 A
11 kV XLPE 3 × 300 mm² Al cable	0.13	0.09	0.7	430 A
0.4 kV XLPE 4 × 95 mm² Al cable	0.32–0.41	0.075	0.2	210 A
0.4 kV XLPE 4 × 300 mm² Al cable	0.10–0.13	0.07	0.6	400 A

Two facts in the table drive almost everything that follows. First, distribution conductors have x/r of order one or less, against 5–15 for transmission lines, so the active power flow affects voltage as strongly as the reactive flow; the decoupling between P and δ and between Q and V that underlies fast-decoupled load flow (Chapter 18) does not hold. Second, resistance depends on temperature, rising about 0.4 % per kelvin in aluminium and copper (*Engineering Physics*, Ch. 14 §14.1), so a cable at 90 °C has 28 % more resistance than at 20 °C, and voltage-drop and loss figures must state the conductor temperature.

Loads on a feeder

A feeder serves hundreds of transformers whose individual loads are rarely metered in real time. Analysis therefore starts with **load allocation**: the measured feeder-head demand is shared among the transformers in proportion to their kVA rating or their customers' metered monthly energy. Each load is then modelled as constant power, constant current or constant impedance (the **ZIP** model, Chapter 18), the mix chosen from measurements: residential loads behave roughly as a blend with active power varying about as V^{1} to $V^{1.5}$, motors as constant power, and resistive heating as constant impedance.

The ladder (backward–forward sweep) method

Newton–Raphson load flow works for distribution networks but is wasteful: radial feeders have no loops, and their high r/x ratios make the Jacobian poorly conditioned. Distribution engineers use the **ladder** or **backward–forward sweep** method instead:

1. Assume all node voltages equal to the substation voltage.
2. **Backward sweep:** compute each load current from its voltage; walking from the feeder ends toward the substation, add currents with KCL to obtain every branch current.
3. **Forward sweep:** walking from the substation outward, subtract each branch's voltage drop with KVL to obtain new node voltages.
4. Repeat until the voltage at every node changes by less than a tolerance (typically 10^{-4} per unit).

The method converges in a handful of iterations for normally loaded radial feeders and handles weakly meshed networks by breaking loops at chosen points and injecting compensating currents. Real feeders are **unbalanced**: single-phase laterals, single-phase loads and untransposed conductors make the three phases carry different currents. Distribution analysis software therefore carries three-phase (or four-wire) phase-

frame models with full 3×3 or 4×4 impedance matrices for each line section (Carson's equations with Kron reduction of the neutral), rather than the positive-sequence per-phase model of transmission studies.

> **Status of the principle: Backward–forward sweep load flow**
> *Derived.* A fixed-point iteration of KCL and KVL on a radial graph; it converges for radial and weakly meshed networks at normal loading, and slows or fails near voltage collapse or with many loops, where Newton-type methods are used instead.

22.5 Voltage drop

The approximate drop formula

For a line section of impedance $R + jX$ carrying a balanced three-phase load $P + jQ$ at the receiving end of line voltage V, the phasor drop per phase is $\mathbf{I}(R + jX)$. Projecting it onto the receiving-end voltage gives the familiar approximation for the magnitude drop,

$$\Delta V \approx I\,(R\cos\varphi + X\sin\varphi) \quad \text{(per phase)}, \qquad \frac{\Delta V}{V} \approx \frac{RP + XQ}{V^2} \quad \text{(per unit, three-phase)}$$

The second form is the one engineers use most: R and X in ohms per phase, P and Q in W and var for all three phases, V line-to-line. It neglects the quadrature component of the drop, which adds a correction of order $\big((XP - RQ)/V^2\big)^2/2$, under 0.5 % of V for drops up to about 10 %.

> **Status of the principle: Approximate voltage-drop formula**
> *Approximate.* Valid for short lines with negligible shunt capacitance, balanced load and drops below about 10 %; it underestimates the true drop slightly, and fails for long cables at light load (where capacitive rise dominates) and for unbalanced feeders, where each phase must be computed separately and neutral voltage drop added.

Lumped and distributed loads

Real feeders tap off load continuously. If a total current I is drawn uniformly along a feeder of length ℓ and impedance z per unit length, the current at distance s from the source is $I(1 - s/\ell)$, and the drop to distance s is

$$\Delta V(s) = I\,z\left(s - \frac{s^2}{2\ell}\right), \qquad \Delta V(\ell) = \frac{I\,z\,\ell}{2}.$$

A uniformly distributed load causes the same end-of-feeder voltage drop as the whole load lumped at the **midpoint**. The power loss, which goes with the square of the current, is $\int_0^\ell I^2(1 - s/\ell)^2 r\,ds = I^2 r\ell/3$: the same as the whole load lumped at **one third** of the length. These two rules of thumb are used every day to size feeders before any software is opened.

Worked example — Voltage drop and losses on a rural 11 kV feeder. An 11 kV overhead feeder of 8 km uses ACSR "Dog" conductor ($r = 0.273\ \Omega$/km, $x = 0.35\ \Omega$/km). It supplies 3 MW at power factor 0.9 lagging. Find the voltage drop and losses (a) with all the load at the far end, (b) with the load uniformly distributed.

Line current: $I = 3 \times 10^6 / (\sqrt{3} \times 11\,000 \times 0.9) = 175.0$ A. Feeder impedance: $R = 2.184\ \Omega$, $X = 2.80\ \Omega$. With $Q = P \tan\varphi = 1.453$ Mvar,

$$\frac{\Delta V}{V} \approx \frac{2.184 \times 3.0 \times 10^6 + 2.80 \times 1.453 \times 10^6}{(11\,000)^2} = \frac{6.552 + 4.068}{121} = 8.78\,\%.$$

(a)

(b) Uniformly distributed: drop $8.78/2 = 4.39\,\%$; losses $200.6/3 = 66.9$ kW, 2.2 % of the load.

Most rural feeders lie between these limits, with load concentrated in villages. A design limit of about 6 % on the 11 kV feeder (leaving room for the transformer and LV drops within the ±10 % band at the customer) is met only if the load is spread out; a new 1 MW cold store at the feeder end would push the drop beyond it. Figure 22.4 shows both profiles.

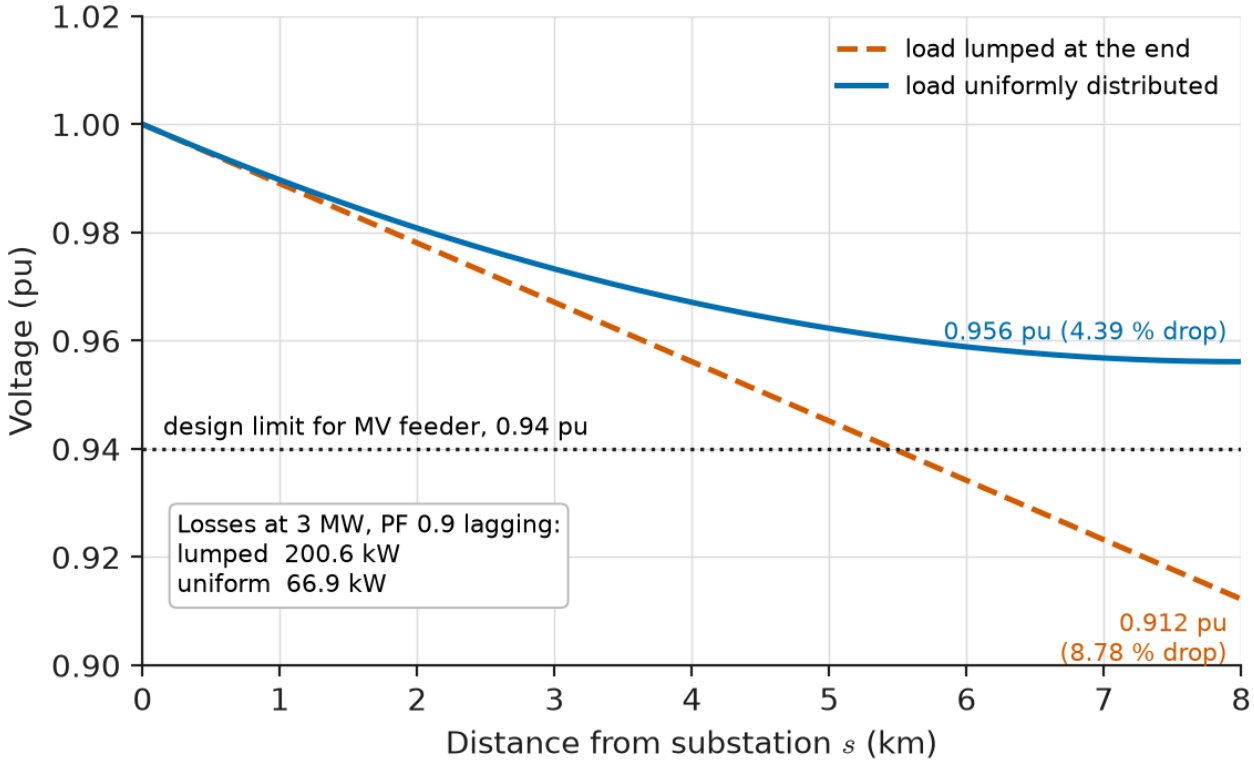


Figure 22.4 — Voltage profile along an 8 km, 11 kV feeder carrying 3 MW at power factor 0.9, with the load lumped at the end and uniformly distributed.

Voltage drop in the low-voltage network and in buildings

The customer sees the sum of MV drop, transformer drop and LV drop. A typical European voltage budget at peak load allows about 2–5 % on the MV feeder, 2–3 % across the secondary transformer (whose 4 % impedance is mostly reactive), and 4–6 % on the LV mains and service cable, with the off-circuit tap and the substation tap changer lifting the no-load voltage to make room. Inside the customer's installation, IEC 60364-5-52 Annex G recommends a further maximum of 3 % for lighting and 5 % for other uses (6 % and 8 % when supplied from a private LV substation); the US National Electrical Code (NFPA 70) recommends 3 % for a branch circuit and 5 % for feeder plus branch. The customer-terminal limits are set by EN 50160 (±10 % of 230 V for 95 % of 10-minute averages over a week) or by ANSI C84.1 (Range A: ±5 % of nominal, 114–126 V on a 120 V base). Table 22.5 collects these limits; the design procedure for cables inside a plant is in Chapter 30.

Table 22.5 — Voltage limits and drop allowances used in distribution design.

Quantity	Typical value	Source
LV supply voltage band (Europe)	230 V ±10 % (207–253 V), 95 % of 10-min means	EN 50160, IEC 60038
MV supply voltage band (Europe)	U_c ±10 %, 95 % of 10-min means	EN 50160
Service voltage, Range A (North America)	±5 % (114–126 V on 120 V)	ANSI C84.1
Service voltage, Range B (North America)	−8.3 % to +5.8 % (110–127 V)	ANSI C84.1
Drop inside installation, lighting	3 % (6 % private substation)	IEC 60364-5-52 Annex G
Drop inside installation, other loads	5 % (8 % private substation)	IEC 60364-5-52 Annex G
Branch circuit / feeder plus branch	3 % / 5 % (recommendation)	NFPA 70 informational notes
Motor starting dip at motor terminals	10–15 % typical design limit	Practice; Chapter 6

22.6 Distribution losses

Where the energy goes

Distribution networks lose far more energy than transmission networks because they carry the same power at lower voltage over far more conductor. In a well-run European or East Asian system, total network losses are 4–8 % of energy delivered, two-thirds of it in distribution; in systems with old networks, long rural feeders or widespread theft, the figure reaches 15–25 %.

Table 22.6 — Typical breakdown of technical losses in a utility distribution network (share of total network losses; varies with network design).

Component	Share	Character
LV mains and service cables	25–40 %	Load-dependent (I^2R), high on long rural LV
Distribution transformer no-load (core) losses	15–30 %	Constant, 8760 h/yr
Distribution transformer load losses	10–15 %	Load-dependent
MV feeders	15–25 %	Load-dependent
Primary substation transformers and subtransmission	10–15 %	Mostly load-dependent
Meters, capacitor banks, dielectric losses in cables, corona	1–3 %	Constant

Losses are split into **technical losses** (physics: I^2R, core loss, dielectric loss) and **non-technical losses** (theft, unmetered supplies, billing and metering errors). Smart meters (Chapter 26) have made the second

class measurable by comparing the energy entering each secondary substation with the sum of its customers' meters.

Load factor and loss factor

A feeder's peak loss is easy to compute from its peak current; its annual energy loss is not, because loss goes as I^2 and the current varies through the day and year. Two ratios are defined from the load curve:

$$\mathrm{LF} = \frac{\text{average load}}{\text{peak load}}, \qquad \mathrm{LsF} = \frac{\text{average loss}}{\text{peak loss}} = \frac{\overline{I^2}}{I_{\text{peak}}^2}.$$

Because $\overline{I^2} \geq \bar{I}^2$, the loss factor lies between LF^2 (a load that is flat at its average value except for a brief peak) and LF (a load that is either at peak or zero). When only the load factor is known, the empirical Buller–Woodrow relation is used:

$$\mathrm{LsF} \approx 0.3\,\mathrm{LF} + 0.7\,\mathrm{LF}^2$$

with coefficients ranging from 0.15/0.85 to 0.3/0.7 in different utilities' fits. With half-hourly smart-meter or SCADA data the loss factor is simply computed from the measured curve.

> **Status of the principle: Loss-factor formula**
> *Empirical.* The coefficients are fitted to typical utility load curves; the relation is a bracketed interpolation between the exact limits LF^2 and LF and should be replaced by direct integration of I^2 when interval data exist.

Worked example — Annual energy loss of the rural feeder. The 8 km feeder of Section 22.5 with uniformly distributed load has a peak loss of 66.9 kW at its 3 MW peak. Its annual load factor is 0.55. Estimate the annual energy loss and its cost at USD 0.08 per kWh.

$\mathrm{LsF} = 0.3 \times 0.55 + 0.7 \times 0.55^2 = 0.165 + 0.212 = 0.377$. Annual energy loss $= 66.9 \times 0.377 \times 8760 = 220.9$ MWh, costing about USD 17 700 a year. Energy delivered $= 3 \times 0.55 \times 8760 = 14\,454$ MWh, so the feeder loses 1.53 % of its energy, although its peak loss is 2.2 % of peak load. Reconductoring with a conductor of half the resistance would save about 110 MWh a year; worth about USD 8 800 a year over a 25-year life at 7 % discount rate (annuity factor 11.65) that justifies about USD 103 000 of investment, roughly the cost of restringing 2–3 km of line. Losses alone rarely pay for reconductoring a whole feeder; losses plus a voltage-drop problem often do.

Reducing losses

Technical losses are reduced by raising voltage (converting 6.6 kV networks to 11 kV quarters the I^2R loss for the same power), shortening LV circuits by adding more, smaller secondary substations, balancing load across phases (an unbalanced LV feeder carries neutral current and higher peak-phase current, both raising loss), correcting power factor locally with capacitors, replacing old transformers with low-loss ones, and reconfiguring meshed networks so that normally open points sit where losses are minimised. The economic assessment, with the life-cycle cost of a kilowatt of loss, is part of Chapter 28.

22.7 Voltage regulation

The regulation problem

The voltage at a customer must stay inside its band at peak load, when drops are largest, and at minimum load, when drops are smallest and, with distributed generation, may become rises. The total drop from substation busbar to the farthest customer at peak might be 12 %, against a band of 20 % (±10 %); the job of voltage regulation is to place that 12 % inside the band at all times. The tools, from source to load, are the on-load tap changer at the primary substation, line-drop compensation, feeder voltage regulators, shunt capacitors, off-circuit taps on the secondary transformers, and increasingly the reactive power of inverters (Section 22.8).

On-load tap changers and line-drop compensation

Primary substation transformers carry an **on-load tap changer** (OLTC) on the HV winding, typically giving ±10 % in 17 or 19 steps of about 1.25 %. An automatic voltage control (AVC) relay compares the busbar voltage with a set point; if it stays outside a deadband of ±1–1.5 % for a time delay (30–120 s, longer at upstream levels so that tap changers at different levels do not "hunt" against each other), it commands one tap step. The deadband must be wider than one tap step or the controller would oscillate.

Holding the busbar voltage constant is not what customers need: at peak load the busbar should be higher to offset the feeder drop. **Line-drop compensation** (LDC) adds to the measured voltage a simulated drop $\mathbf{I}(R_{set} + jX_{set})$ computed from the transformer current, so that the controller regulates the voltage at a notional load centre:

$$V_{\text{regulated}} = \left| \mathbf{V}_{\text{bus}} - \mathbf{I}\,(R_{set} + jX_{set}) \right|$$

LDC works well when all feeders from the busbar have similar load shapes. It fails when one feeder carries reverse power from a solar farm while the others import: the controller then sees a reduced net current, lowers the busbar voltage, and drops the voltage at the ends of the load feeders. This interaction is one of the first symptoms of high DG penetration.

Step voltage regulators

On long rural feeders, especially in North America, single-phase **step voltage regulators** (autotransformers with a 32-step tap changer giving ±10 % in steps of 5/8 %) are installed part-way along the feeder, each with its own controller and LDC. A regulator at the point where the voltage has fallen to 0.95 per unit can lift the downstream section back to 1.03 per unit at peak. Two or three regulators in cascade allow feeders of 30–60 km, with time delays graded (for example 30, 45, 60 s) from the substation outward.

Shunt capacitors

A shunt capacitor bank of Q_C reduces the reactive flow upstream of it, raising the voltage at its point of connection by approximately

$$\frac{\Delta V}{V} \approx \frac{Q_C\, X}{V^2},$$

where X is the reactance from the source to the bank. A 600 kvar bank 6 km along the feeder of Section 22.5 ($X = 2.10\ \Omega$) raises the voltage there by $600 \times 10^3 \times 2.10 / 11\,000^2 = 1.04\,\%$, and reduces losses by cutting the reactive current upstream. Feeder banks of 300–1200 kvar are either fixed (sized for the minimum load) or switched by time, voltage, temperature or var controllers. On feeders with x/r below one, capacitors are much

less effective for voltage support than tap changers, but still useful for losses and power factor; the sizing and power-factor penalty economics are in Chapter 25.

Conservation voltage reduction

Many loads consume less power at lower voltage. **Conservation voltage reduction** (CVR) runs feeders deliberately near the bottom of the band. The CVR factor, the percentage change in energy per percentage change in voltage, is typically 0.5–1.0 for residential feeders, so lowering the average voltage by 3 % saves 1.5–3 % of energy. It requires good knowledge of the lowest voltage on the feeder, which smart meters now provide (Chapter 26), and is less effective where loads are dominated by constant-power electronics and thermostatically controlled heaters.

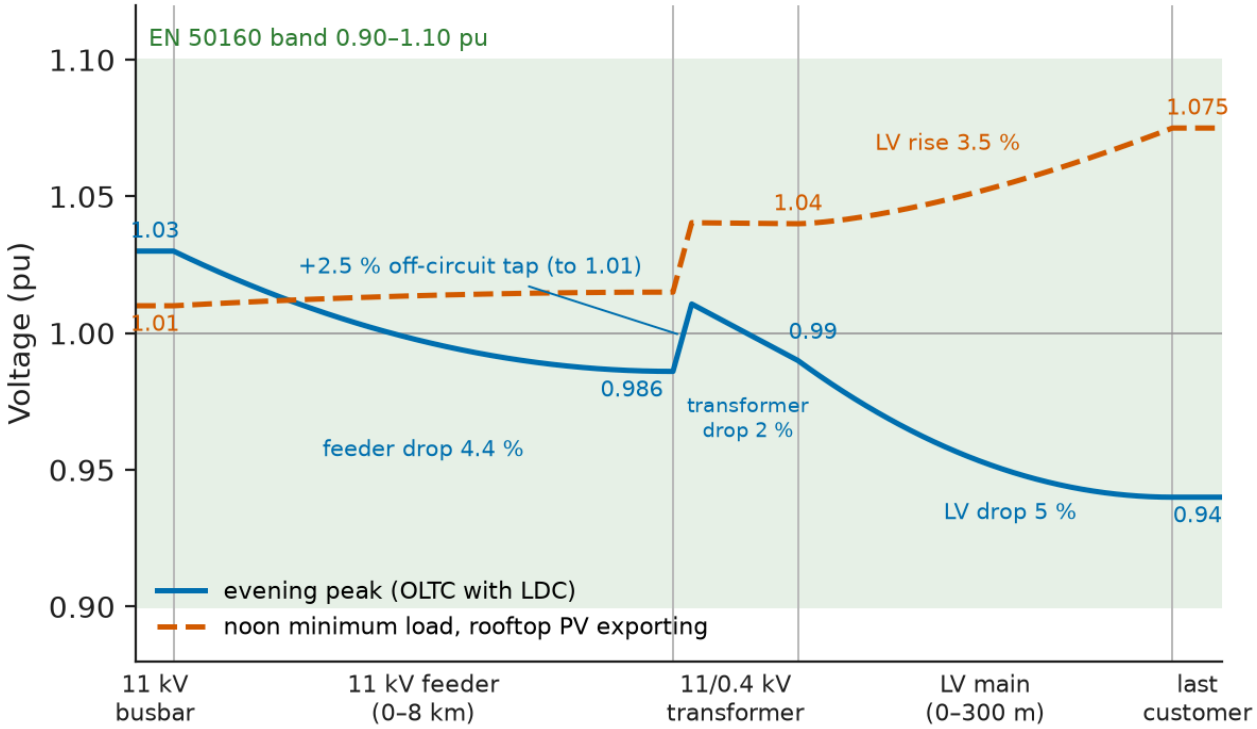


Figure 22.5 — Voltage from primary substation to the last customer at evening peak and at noon minimum load with rooftop PV, showing how tap settings place the profile inside the plus-or-minus 10 percent band.

Figure 22.5 shows the regulation problem end to end: the evening profile must stay above 0.90 per unit and the noon profile below 1.10, with the same off-circuit tap position for both.

22.8 Distributed generation

Reverse power flow and voltage rise

Distribution networks were designed for power flowing one way. **Distributed generation** (DG), from rooftop PV (3–10 kW per house) and commercial PV (50–500 kW) to 1–20 MW solar and wind farms, combined heat and power and small hydro on MV feeders, and battery storage at every scale, sends power the other way for part of the day. The voltage-drop formula of Section 22.5 then changes sign. For a generator exporting P_G and absorbing Q_G at the end of a line,

$$\frac{\Delta V}{V} \approx \frac{R\,P_G - X\,Q_G}{V^2}$$

is a voltage **rise** toward the generator. On LV cables with R/X of 2–5, it is the active power that raises the voltage, and absorbing reactive power is a weak remedy.

Worked example — Voltage rise on a suburban LV feeder at noon. A 250 m, 400 V feeder of 4 × 95 mm² Al XLPE cable ($r = 0.320\ \Omega$/km, $x = 0.075\ \Omega$/km) serves 20 houses. At noon on a spring day the local load is 5 kW and the rooftop PV produces 75 kW, so 70 kW is exported, concentrated by geography toward the far end. The LV busbar sits at 1.04 per unit (239 V) because the off-circuit tap was raised to support the evening peak. Find the end-of-feeder voltage with inverters at unity power factor and at power factor 0.90 absorbing.

Cable impedance: $R = 0.080\ \Omega$, $X = 0.01875\ \Omega$, $R/X = 4.3$.

Unity power factor: $\Delta V = (0.080 \times 70\,000)/400 = 14.0$ V line-to-line, a 3.50 % rise. End voltage $1.04 + 0.035 = 1.075$ per unit, 247 V.

At power factor 0.90 absorbing: $Q_G = 70 \times \tan(\cos^{-1} 0.9) = 33.9$ kvar, $\Delta V = (0.080 \times 70\,000 - 0.01875 \times 33\,900)/400 = 12.4$ V, a 3.10 % rise. Absorbing 34 kvar buys only 0.4 % of voltage on the cable. The same reactive flow through the 400 kVA, 4 % secondary transformer ($R = 1.15\,\%$, $X = 3.83\,\%$) lowers the busbar by a further $3.83 \times 33.9/400 - 1.15 \times 70/400 = 0.12\,\%$ net, versus a 0.20 % rise at unity power factor, so the total gain is about 0.7 %.

At 247 V the feeder is inside the EN 50160 limit, but the inverters' own overvoltage protection (EN 50549-1 sets a 10-minute mean limit of 1.10 per unit by default) leaves little margin, and on a hotter, lighter-load day with the MV busbar at 1.05 per unit the end voltage reaches 1.085–1.10. Fixes, in increasing cost: lower the off-circuit tap by one step in summer (at the price of the evening margin), enable volt–watt curtailment in the inverters, add a battery, fit an on-load tap-changing secondary transformer, or reinforce the cable.

Hosting capacity

The **hosting capacity** of a feeder is the amount of DG it can accept before some limit is violated: overvoltage (usually the first limit on long LV and rural MV feeders), thermal rating of cables and transformers under reverse flow (first on short, stiff urban feeders), protection mis-operation, or power-quality limits on harmonics and flicker. Typical deterministic studies give LV hosting capacities of 30–80 % of the secondary transformer rating for PV at unity power factor; volt–var and volt–watt control can roughly double the voltage-limited figure. Because the result depends on where the DG is, probabilistic (Monte Carlo) hosting-capacity studies are now standard for large-scale connection planning.

Inverter voltage control

Modern grid codes require inverters to help. IEEE 1547-2018 (Category B) and EN 50549-1 specify **volt–var** curves (absorb reactive power above about 1.03 per unit, inject below about 0.97, with a deadband between), **volt–watt** curves (reduce active output linearly from about 1.06 to 1.10 per unit), and fixed power factor or $Q(P)$ modes. These are local, autonomous and fast; the full treatment of inverter control and grid-code compliance belongs to Chapter 27.

Protection and fault-level effects

DG also disturbs protection designed for one-way flow:

- **Protection blinding:** a generator between the substation and a fault supplies part of the fault current, so the feeder relay sees less current and may not trip, or trips late.
- **Sympathetic tripping:** a generator on a healthy feeder feeds a fault on an adjacent feeder through the common busbar; a non-directional relay on the healthy feeder may trip.
- **Fuse-recloser miscoordination:** extra fault current can make a lateral fuse melt before the recloser's fast shot, defeating fuse saving.
- **Unintentional islanding:** a section of feeder disconnected from the substation may remain energised by DG, endangering line crews and causing out-of-phase reclosing. Anti-islanding detection is required by IEEE 1547 and EN 50549 (Chapter 27).

Synchronous machines contribute 4–7 times rated current to a fault; inverter-based DG contributes only 1.1–1.5 times rated current, so the blinding problem is mainly a synchronous-DG problem, while islanding and voltage rise are universal. These effects are taken up in Chapter 23.

> **Status of the principle: Voltage rise from distributed generation**
> *Approximate*. The same short-line approximation as the voltage-drop formula with the sign of P and Q reversed; it assumes a stiff upstream source and balanced injection, and must be replaced by a three-phase load flow when single-phase PV is unevenly spread across phases, which is common and raises the worst phase voltage further.

22.9 Distribution planning

What a plan must answer

Distribution planning decides, five to twenty years ahead, which substations, transformers, feeders and switchgear to build or reinforce so that demand is met within voltage, thermal, fault-level and reliability limits at minimum life-cycle cost. Its inputs are a load forecast, the existing network model, planning standards (security of supply, voltage limits, ratings), unit costs and the value of losses and of lost load. Its outputs are reinforcement projects with dates.

Diversity and after-diversity maximum demand

No two customers peak at the same moment. The **diversity factor** is the ratio of the sum of individual maximum demands to the maximum demand of the group; its inverse is the **coincidence factor**. For planning, utilities use the **after-diversity maximum demand** (ADMD) per customer, which falls with the number of customers and settles toward a constant for large groups. Typical values for gas-heated European homes are 1.5–2.5 kW per home for groups of 100 or more; for all-electric homes with heat pumps 3–6 kW; for North American homes with air conditioning 4–7 kW.

> **Status of the principle: After-diversity maximum demand**
> *Empirical*. ADMD curves are fitted to measured coincident demand of customer groups; they shift when the load mix changes (heat pumps, EVs, PV) and must be re-measured rather than extrapolated.

Worked example — Sizing a secondary substation for a new estate with EV charging. A new estate of 200 gas-heated homes has an ADMD of 2.0 kW per home at the evening peak. Eighty homes will have 7 kW

chargers; measured charging diversity at the evening peak is about 0.25 for groups of this size. Choose the transformer rating, assuming power factor 0.95 at peak.

Domestic demand: $200 \times 2.0 = 400$ kW. EV demand: $80 \times 7 \times 0.25 = 140$ kW. Total 540 kW, or $540/0.95 = 568$ kVA. A 630 kVA unit at 90 % peak loading leaves cyclic-rating headroom (Section 22.3); a 500 kVA unit, sufficient before EVs, would run at 114 % for several hours each evening, which is acceptable only if the overnight period is cool and short. If the estate's homes switch to heat pumps in the next decade (ADMD rising to perhaps 4 kW), the same group reaches about 940 kW, or 990 kVA, and the plan should reserve space and duct routes for a second transformer now. Smart charging that shifts EV demand past midnight (Chapter 26) can defer that reinforcement by years.

Security standards

Planning standards state how much of the load must be supplied after outages. A common rule for primary substations is **N−1**: with any one transformer or incoming circuit out, the remaining equipment, using its emergency cyclic rating, must carry the peak demand. A primary substation with two 20 MVA transformers whose emergency rating is 1.3 per unit therefore has a **firm capacity** of 26 MVA, not 40. Smaller groups of load (a few MW) may be allowed N−0 with restoration by switching in a stated time, which is why open rings with remote control are planned for urban cable networks. In the UK this is codified in Engineering Recommendation P2; other utilities have equivalent internal standards.

Feeder and conductor selection

A planner chooses, for each feeder, a voltage, a route, a conductor size and the switch positions, against several limits at once: thermal rating (normal and after a ring is closed), voltage drop at peak and rise at minimum load with DG, fault-level withstand of cables and switchgear (Chapter 19), and the economic conductor size, at which the capitalised cost of losses over the life of the asset balances the extra cost of a larger conductor. In practice utilities standardise on a few sizes (for example 95, 185 and 300 mm² Al cable at 11 kV) and lay the largest economic size at first construction, because digging a street twice costs far more than the extra aluminium.

The changing load

Planning methods built on slowly growing, predictable residential and commercial load are being stretched by electrification. EV charging, heat pumps and rooftop PV change both the size and the timing of the peak; data centres request 50–500 MW connections with two-year lead times; mines and ports electrify their haul trucks and cranes; railways add 25 kV traction supplies. Planners respond with scenario-based forecasts, probabilistic load flow, flexibility contracts (paying customers to reduce load at peak instead of building), and active network management that curtails DG at a few critical hours rather than reinforcing for them. These tools belong to the smart grid of Chapter 26, and the design procedures for an individual installation to Chapter 30.

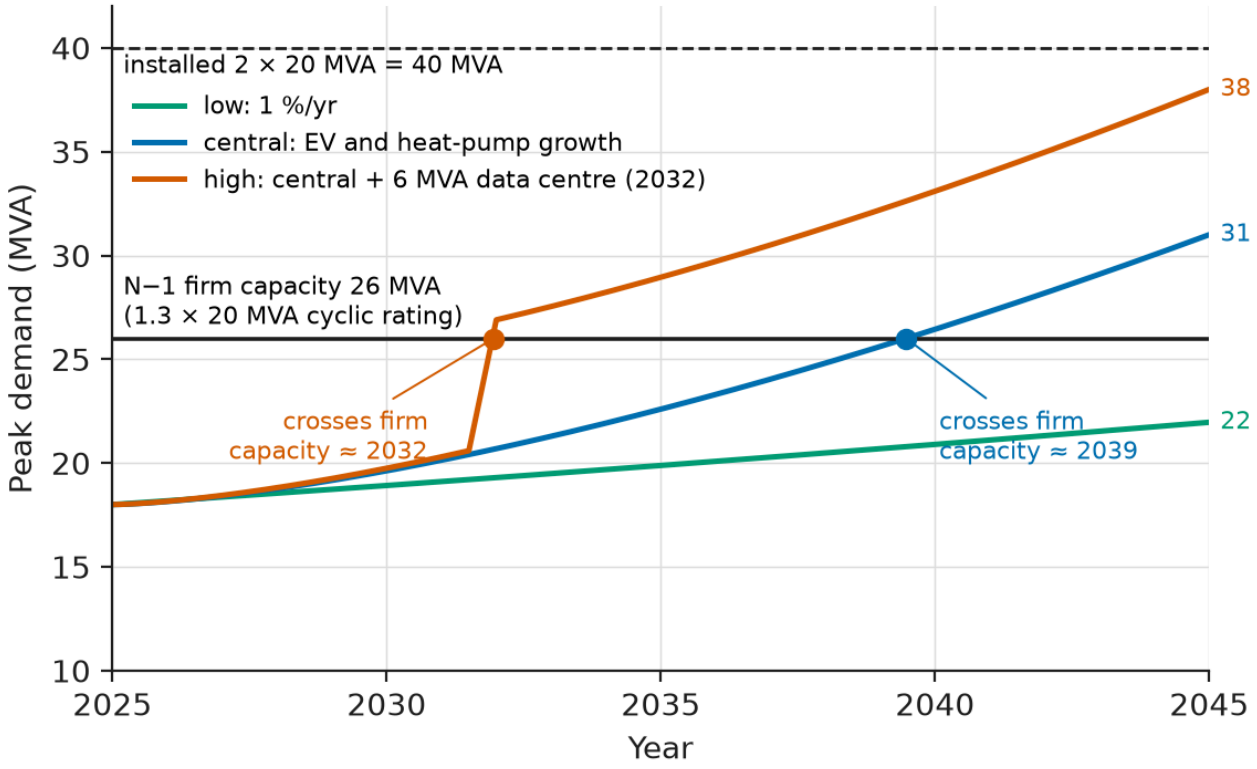


Figure 22.6 — Peak demand forecast for a 2 x 20 MVA primary substation under three electrification scenarios, against its N-1 firm capacity.

Figure 22.6 shows how the choice of scenario moves the reinforcement date of a single primary substation by a decade or more: the firm capacity is exceeded in 2032 in the high scenario, about 2039 in the central one, and not before 2045 in the low one.

What remains open

Open question	Where it is answered
How are the overcurrent relays, fuses and reclosers of a radial feeder set and graded?	Chapter 23
How are distribution networks earthed, and how are earth faults detected in resistance- and coil-earthed systems?	Chapter 23 §23.5; Chapter 24 §24.6
How do harmonics, flicker, sags and unbalance arise and propagate on distribution feeders?	Chapter 25
How do smart meters, FLISR and active network management change distribution operation?	Chapter 26
How must inverter-based DG behave under grid codes (volt–var, ride-through, anti-islanding)?	Chapter 27
How is the life-cycle cost of losses used to justify efficiency investments?	Chapter 28
How are cables inside a plant or building sized for voltage drop, current and fault withstand?	Chapter 30
How are fault levels on distribution busbars computed?	Chapter 19 (back)
Where do the per-unit models of feeders and transformers come from?	Chapter 17 (back); Chapter 4 (back)

Next: Chapter 23 turns from carrying power to interrupting it, asking how a protection system finds a fault among thousands of kilometres of feeder and clears it in a fraction of a second without disconnecting anything healthy.

Chapter 23 — Power-System Protection

Part VI · Transmission, Distribution, and Protection

At 14:07 on a summer afternoon a tree limb falls across one phase of an 11 kV feeder, 5 km from the substation. The fault current, about 4 kA, is twenty times the feeder's normal load. Within 0.3 s a relay at the substation has measured it, recognised it as a fault on its own feeder and not on one of the five healthy feeders beside it, and opened a circuit breaker; two seconds later it recloses, the limb has burnt clear, and most customers notice only a flicker. Had the relay been set a little too sensitive it would have tripped every winter morning on the inrush of cold loads; a little too slow, and the 33/11 kV transformer upstream would have tripped first, blacking out 12 000 customers instead of 1500; slower still, and the arc at the fault would have burnt down the conductor and set the field alight. On a 400 kV line the same story plays out in under 80 ms, because a slower clearance would let the generators nearby swing out of synchronism (Chapter 20). Protection is the part of the power system that does nothing for years and then must act correctly within milliseconds. This chapter explains how it decides what to trip, how fast, and how the devices that detect and interrupt faults are rated.

> **How does a protection system detect a fault, decide whether it is inside its own zone, and remove it without tripping anything healthy?**
> **How are relays, fuses, circuit breakers and instrument transformers chosen and set so that protection is fast, sensitive and selective at the same time?**

Concept	Engineering destination
Protection principles	Zones of protection, main and backup, clearing-time budgets
Overcurrent protection	Radial distribution feeders, motor and transformer backup, LV switchboards
Differential protection	Transformers, generators, busbars, cables and short lines
Distance protection	Transmission and subtransmission lines, interconnectors
Earth-fault protection	Resistance- and coil-earthed MV networks, mines, generator stator earth faults
Circuit breakers	Switchgear ratings at 0.4–800 kV, generator breakers, DC breakers
Fuses and relays	LV distribution, MV laterals, numerical IEDs, IEC 61850 substations
Instrument transformers	CT saturation, accuracy-limit factor, CVT transients
Protection coordination	Grading of relays, fuses and reclosers; time-current studies
Selectivity and sensitivity	Pickup settings, high-impedance faults, arc-flash reduction

Roadmap. What protection must do and how it is organised in zones (23.1) → the oldest and commonest principle, overcurrent (23.2) → comparing currents in and out, differential (23.3) → measuring impedance to the fault, distance (23.4) → faults to earth (23.5) → the circuit breaker that interrupts (23.6) → fuses and the relays that make the decisions (23.7) → the instrument transformers that feed them (23.8) → making devices in series work together (23.9) → the trade-off between catching every fault and tripping only the right breaker (23.10).

23.1 Protection principles

What protection must do

A 25 kA arcing fault in 11 kV switchgear dissipates tens of megawatts in the arc. Left uncleared it melts conductors, ruptures switchgear, burns people and, on a transmission system, destabilises the generators. Protection must **detect** the fault from measurements, **locate** it to a zone, and **disconnect** the smallest part of the system that removes it. The requirements are usually listed as five:

- **Reliability**, divided into *dependability* (it trips for every fault in its zone) and *security* (it never trips for anything else). The two pull against each other: duplicated protections increase dependability and decrease security.
- **Selectivity** (discrimination): only the devices nearest the fault operate.
- **Speed**: short enough to limit damage and keep the system stable.
- **Sensitivity**: it detects the smallest fault current that can occur in its zone, including high-resistance faults.
- **Economy**: the protection costs a small fraction of what it protects. A 400 kV line or a 500 MVA generator justifies duplicated schemes; a 100 kVA pole transformer gets a fuse.

The protection chain

Every protection scheme is a chain: **instrument transformers** (CTs and VTs) scale the primary current and voltage down to 1 or 5 A and 100–120 V; a **relay** (now usually a numerical intelligent electronic device, IED) samples them, applies the protection algorithm and closes a trip contact; a **DC auxiliary supply** (a 110 or 125 V substation battery, independent of the AC system it protects) energises the **trip coil**; and the **circuit breaker** separates its contacts and interrupts the current at a current zero. Any weak link defeats the chain, which is why trip-circuit supervision, battery monitoring and breaker-failure protection exist.

Zones of protection

The system is divided into overlapping **zones**: each generator, transformer, busbar, line, cable and large motor sits inside at least one zone, bounded by the CTs that feed its protection and the breakers that its protection trips (Figure 23.1). Adjacent zones overlap around each breaker, by placing the CTs for each zone on the far side of the breaker, so that no fault falls into a gap between zones. A fault in an overlap trips more breakers than strictly necessary, which is the accepted price.

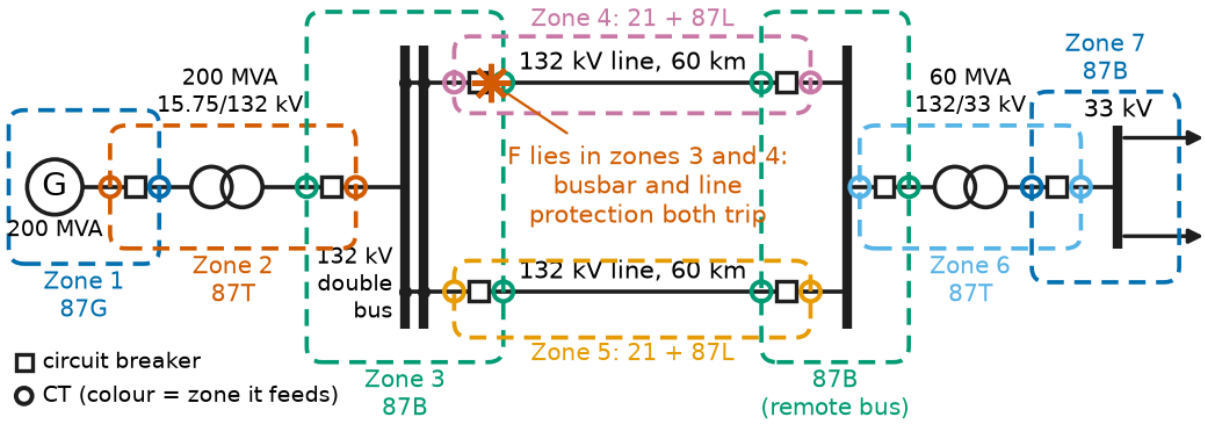


Figure 23.1 — Overlapping zones of protection around a generator, step-up transformer, 132 kV busbars, two lines and a 33 kV busbar.

Main and backup protection

Main (primary) protection is set to clear faults in its zone as fast as possible. **Backup** protection covers the failure of the main protection or its breaker. *Local backup* duplicates the main protection on separate CTs, relays and trip coils (at transmission level, "main 1" and "main 2" from different manufacturers or principles); *breaker-failure* protection (ANSI 50BF) trips all breakers around a stuck breaker if current is still flowing 150–250 ms after a trip command; *remote backup* is provided by slower protection one zone upstream, such as zone 3 of a distance relay or a time-graded overcurrent relay at the substation incomer.

Table 23.1 — Typical fault-clearing times (relay operating time plus breaker interrupting time).

System	Main protection	Backup	What sets the limit
400–765 kV transmission	50–80 ms (2.5–4 cycles)	250–500 ms (breaker failure, zone 2)	Transient stability (Chapter 20), grid code
132–220 kV	70–120 ms	300–600 ms	Stability, equipment damage
11–33 kV distribution	0.05–1 s (graded)	0.4–1.5 s	Selectivity, cable thermal withstand
LV switchboards and final circuits	5–100 ms (fuses, MCBs)	0.1–1 s	Shock protection (IEC 60364-4-41), arc flash
Generator and transformer internal faults	20–40 ms relay, plus breaker	Overcurrent, 0.5–1 s	Winding and core damage

> **Boundary.**
> Fault currents, sequence networks and symmetrical-component analysis are developed in Chapter 19 of this book, building on the first look at symmetrical components in *Circuits, Signals and Communications*, Ch. 10. Here the fault currents are taken as given; this chapter is about what is done with them.

23.2 Overcurrent protection

Definite-time and inverse-time relays

The simplest fault indicator is current magnitude. An **overcurrent relay** (ANSI 51 for time-delayed, 50 for instantaneous) picks up when the current exceeds a **setting** I_s and trips after a delay. In **definite-time** relays the delay is fixed. In **inverse definite minimum time** (IDMT) relays the delay falls as the current rises, because large currents mean close-in faults that must be cleared fast, while small overloads can be tolerated longer. IEC 60255-151 standardises the curves as

$$t = \mathrm{TMS} \times \frac{k}{M^{\alpha} - 1}, \qquad M = \frac{I}{I_s}$$

with the time multiplier setting (TMS) shifting the whole curve vertically. IEEE C37.112 defines similar curves with an additive constant.

Table 23.2 — IEC 60255-151 inverse-time characteristics and operating time at $M = 10$.

Curve	k	α	t at $M = 10$, TMS = 1	Typical use
Standard inverse (SI)	0.14	0.02	2.97 s	General distribution feeders where fault level varies with source
Very inverse (VI)	13.5	1	1.50 s	Feeders where fault current falls strongly with distance
Extremely inverse (EI)	80	2	0.81 s	Grading with fuses; feeders with cold-load inrush
Long-time inverse (LTI)	120	1	13.3 s	Earth-fault backup, thermal-type protection

> **Status of the principle: Inverse-time overcurrent curves**
> *Empirical.* The curve shapes were chosen historically to mimic induction-disc relays and to grade conveniently with fuses; they are standardised conventions, not laws, and are valid from about $M = 1.1$ to $M = 20$, beyond which numerical relays clamp the curve to a definite minimum time.

Worked example — Operating time of an IDMT relay. An 11 kV feeder relay is fed from a 400/1 A CT, with $I_s = 400$ A (100 % setting), standard-inverse curve and TMS = 0.1. A fault draws 4000 A. Then $M = 10$ and

$$t = 0.1 \times \frac{0.14}{10^{0.02} - 1} = 0.1 \times \frac{0.14}{0.04713} = 0.297 \text{ s}.$$

With a very-inverse curve at the same TMS, $t = 0.1 \times 13.5/9 = 0.150$ s; extremely inverse, $0.1 \times 80/99 = 0.081$ s. Figure 23.2 shows the three curves at TMS = 0.1. The steeper curves trip faster for close-in faults but much slower for small overcurrents ($M = 2$: SI 1.00 s, VI 1.35 s, EI 2.67 s), which is why EI curves tolerate the cold-load pickup of a feeder re-energised after a long outage.

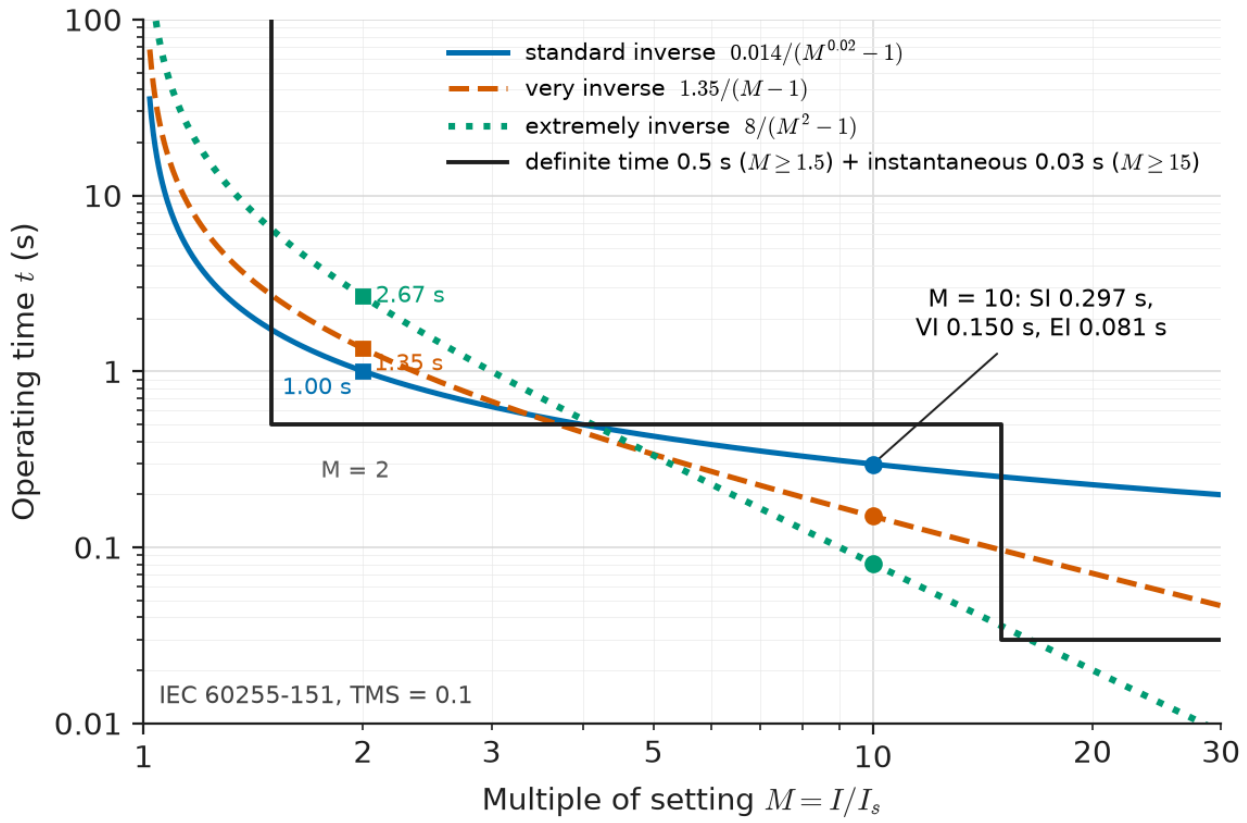


Figure 23.2 — IEC standard-inverse, very-inverse and extremely-inverse relay curves at TMS 0.1, with a definite-time characteristic for comparison.

Instantaneous elements

A high-set **instantaneous** element (50) trips with no intentional delay (20–40 ms relay time) when the current exceeds a level that can only occur for faults close to the relay. On a feeder supplying a distribution transformer, it is set above the maximum fault current on the transformer's LV side (times a factor of 1.2–1.3 for DC-offset overreach), so that it never trips for faults that the transformer's own fuse or LV breaker should clear. Where the source impedance is small compared with the feeder impedance, fault current falls steeply along the feeder and the instantaneous element covers a useful fraction of it; where the source is weak, it covers almost nothing and is set off.

Directional overcurrent

In a closed ring, a parallel feeder pair or a feeder with DG, fault current can flow either way through a relay. **Directional overcurrent** (67) adds a polarising quantity, usually a voltage phasor shifted by a **characteristic angle** (typically 30° or 45° for phase faults), and allows tripping only if the current lies in the forward half-plane. For a three-phase close-in fault the voltage collapses to near zero; numerical relays use pre-fault voltage memory for a few cycles to keep a valid direction.

23.3 Differential protection

The Kirchhoff principle

For any healthy piece of equipment with no internal source or sink of current, the currents entering it sum to zero at every instant: Kirchhoff's current law. A **differential relay** (87) compares the currents at all terminals of its zone, via CTs, and trips when their phasor sum, the **differential** or **operating** current, exceeds a threshold. A through fault (outside the zone) produces large but equal currents in and out, and no differential current; an internal fault produces a differential current equal to the fault current. The zone is precisely bounded by the CT positions, and no time grading with other zones is needed, so differential protection is both fast (one to two cycles) and absolutely selective. It is **unit protection**: it protects one unit and gives no backup to anything else.

> **Status of the principle: Differential protection**
> *Fundamental.* Rests on charge conservation (KCL) for the protected zone; its practical accuracy is limited not by the principle but by CT errors, magnetising current, charging current and communication delays.

Biased (percentage) differential

Real CTs do not match: ratio errors of a few per cent at load grow to tens of per cent when one CT saturates during a heavy through fault. A **biased** differential relay therefore makes its threshold rise with the through current. It computes

$$I_{\mathrm{diff}} = \left| \sum_{k} \mathbf{I}_k \right|, \qquad I_{\mathrm{bias}} = \frac{1}{2} \sum_{k} |\mathbf{I}_k|$$

and trips when I_{diff} exceeds a minimum pickup (typically 0.2–0.3 per unit) plus a slope times I_{bias}. A dual-slope characteristic (Figure 23.3) uses a gentle first slope (20–30 %) for sensitivity to internal faults at load currents and a steep second slope (60–80 %) above about 2 per unit bias to ride through CT saturation on heavy external faults. A saturating CT's output both shrinks and shifts in phase, and the phase shift alone is enough to create a large spurious differential: two 8 pu terminal currents of equal magnitude but 22° apart give $I_{\mathrm{bias}} = 8$ pu and $I_{\mathrm{diff}} = 2 \times 8 \sin 11^\circ = 3.0$ pu.

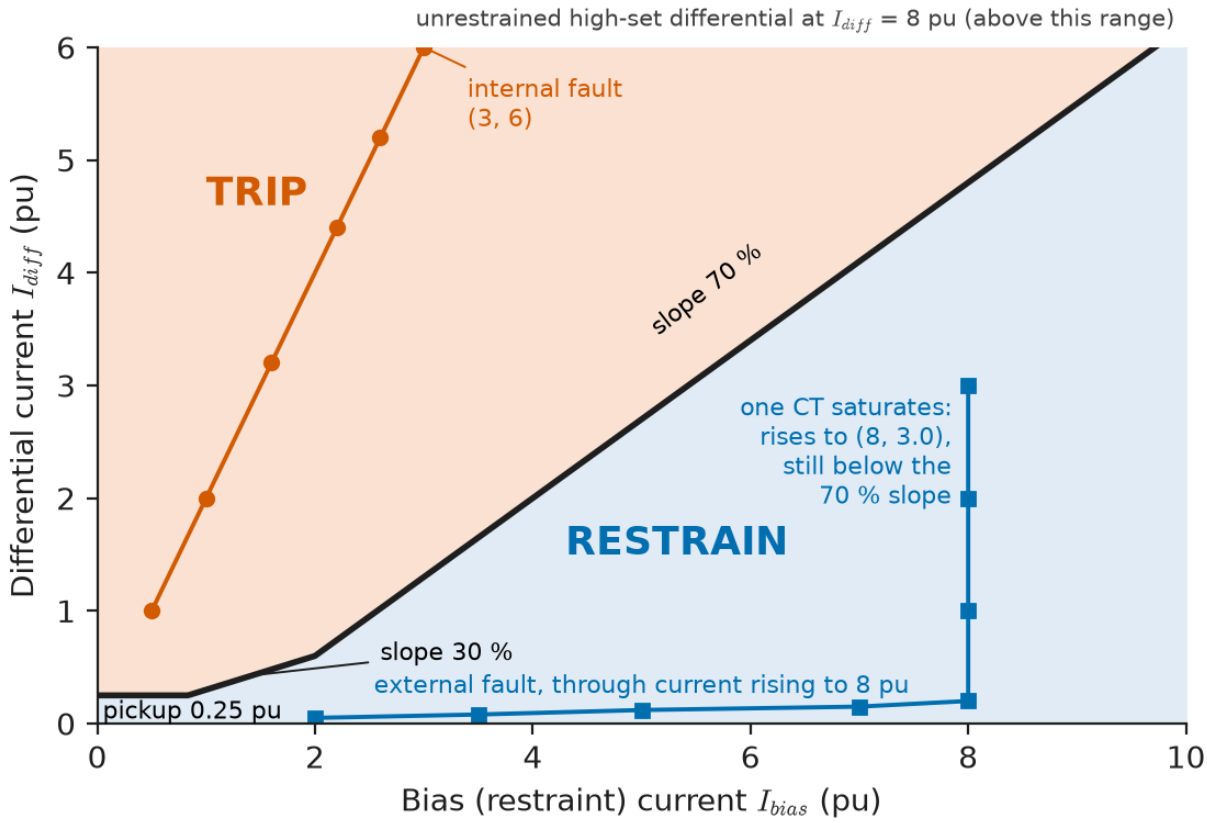


Figure 23.3 — Dual-slope biased differential characteristic, with the trajectories of an internal fault and an external fault with CT saturation.

Transformer differential

A transformer differential relay must cope with three things that a line or busbar relay does not:

- **Ratio and vector-group mismatch.** Currents on the HV and LV sides differ by the turns ratio and, for a Dyn11 transformer, by 30°. Numerical relays correct both in software; older schemes connected the CTs in delta on the star side. Zero-sequence currents must also be removed on an earthed-star winding, otherwise an external earth fault would produce a false differential.
- **Tap-changer range.** An OLTC of ±10 % shifts the ratio, adding a standing differential that the first slope must cover.
- **Magnetising inrush.** Energising a transformer can draw 6–10 times rated current, mostly on one side and containing a large second harmonic (typically 15–60 % of the fundamental) because the core saturates asymmetrically (Chapter 4). The relay **blocks** when the second harmonic in the differential current exceeds about 15–20 % of the fundamental. Overexcitation, from high voltage or low frequency, produces a strong fifth harmonic and is blocked or alarmed in the same way.

A separate **restricted earth fault** (REF, 64REF) element compares the neutral current with the residual of the phase currents on one winding; it detects earth faults close to the neutral end of a star winding, where the fault current is too small for the overall differential.

Busbar, line and generator differential

High-impedance busbar differential connects all the CTs of a busbar in parallel across a relay with a high stabilising resistor. On an external fault with one CT fully saturated, the saturated CT behaves as its winding

resistance alone, and the healthy CTs drive current through it; the voltage across the relay is at most $V_s = I_f(R_{CT} + 2R_L)$, where I_f is the secondary fault current. Setting the relay voltage above V_s guarantees stability; an internal fault drives the CTs to their knee point, far above it.

Worked example — Setting a high-impedance busbar scheme. A 33 kV busbar has six circuits, each with a 2000/1 A class PX CT of secondary resistance $R_{CT} = 6\ \Omega$, and a maximum lead-loop resistance $2R_L = 3\ \Omega$. The maximum through-fault current is 40 kA. Set the scheme and estimate its primary sensitivity, given a relay operating current of 0.1 A and CT magnetising current of 0.02 A each at the setting voltage.

Secondary fault current: $40\,000/2000 = 20$ A. Stability voltage: $V_s = 20 \times (6 + 3) = 180$ V; choose a setting of 200 V. The CTs must have a knee-point voltage of at least $2V_s$, about 400 V, so that an internal fault reliably drives the relay. Primary operating current: $2000 \times (0.1 + 6 \times 0.02) = 440$ A, about 1.1 % of the maximum fault current, sensitive enough for any phase fault and for earth faults on a solidly or low-resistance earthed system. A non-linear resistor (metrosil) across the relay clamps the very high voltages (several kV peak) that an internal fault would otherwise develop across the high-impedance circuit.

Line differential (87L) compares the currents at both ends of a cable or line through a communication channel, now almost always dedicated fibre or a multiplexed digital link, with time alignment by GPS or by echo timing of channel delays. It is the main protection for short lines, cables (where distance protection struggles with the low and variable impedance), and for lines with tapped loads. **Generator differential** (87G) compares the neutral-end and terminal-end currents of each stator phase with settings as low as 5–10 % of rated current.

23.4 Distance protection

Measuring impedance to the fault

On a meshed transmission network, fault currents vary with the generation pattern and flow in both directions, so overcurrent grading fails. A **distance relay** (21) measures the voltage and current at its terminal and computes the apparent impedance $\mathbf{Z} = \mathbf{V}/\mathbf{I}$. For a bolted fault at distance d along a line of impedance z_1 per km, $\mathbf{Z} = d\,z_1$, independent of the source strength. The relay trips when the measured impedance falls inside its **reach**. Because the line impedance is a fixed property of the line, the reach is fixed in kilometres even when the fault level changes.

For phase-to-phase and three-phase faults the relay uses phase-to-phase voltages and the difference of phase currents. For earth faults the measured loop includes the return path, so the current is compensated with the **residual compensation factor**

$$k_0 = \frac{Z_0 - Z_1}{3Z_1}, \qquad \mathbf{Z} = \frac{\mathbf{V}_a}{\mathbf{I}_a + k_0\,3\mathbf{I}_0},$$

where Z_0 and Z_1 are the zero- and positive-sequence line impedances (Chapter 19). For a typical overhead line with $Z_0 \approx 3Z_1$, $k_0 \approx 0.67$.

> **Status of the principle: Distance measurement**
> *Approximate*. The apparent impedance equals the impedance to the fault only for a homogeneous line, a solid (zero-resistance) fault and no infeed between relay and fault; fault resistance, intermediate infeed, mutual coupling with parallel lines, series compensation and CVT transients all distort it, and settings leave margins for these errors.

Characteristics and zones

The trip region is drawn on the R–X plane. The classic **mho** characteristic is a circle through the origin, directional by construction; **quadrilateral** characteristics allow independent resistive reach, needed to cover arc and tower-footing resistance on short lines and earth faults. Distance relays use stepped zones (Figure 23.4):

- **Zone 1**: 80–85 % of the line, instantaneous. Not 100 %, because errors in CTs, VTs, line data and CVT transients could make the relay overreach into the next line.
- **Zone 2**: 120 % of the line, or 100 % plus 50 % of the shortest adjacent line, delayed 0.3–0.4 s. Covers the last 15–20 % of the line and gives backup for the remote busbar.
- **Zone 3**: remote backup for adjacent lines, often 100 % plus 120 % of the longest adjacent line, delayed 0.8–1.5 s. Its large reach is dangerous under heavy load and power swings, and zone-3 trips on load featured in the 2003 North American blackout; modern practice limits it with **load-encroachment** blinders.

Teleprotection

A fault in the last 15–20 % of a line is cleared instantly by the far-end relay's zone 1 but only in zone-2 time by the near end. **Teleprotection** schemes send a signal over fibre, power-line carrier or microwave so that both ends trip at high speed for any fault on the line: in **permissive underreach transfer trip** (PUTT) a zone-1 operation at one end lets the other end trip on zone 2 without delay; in **permissive overreach** (POTT) each end trips only if both see the fault forward; in **directional comparison blocking** (DCB) a reverse-looking element sends a blocking signal. Transmission lines at 220 kV and above normally use teleprotected distance plus line differential as main 1 and main 2.

Power swings and fault resistance

After a disturbance the generator angles oscillate (Chapter 20), and the apparent impedance at a line terminal moves across the R–X plane. A swing that passes through zone 1 looks like a fault. **Power-swing blocking** (68) detects the slow movement of the impedance (a fault jumps into the characteristic in milliseconds, a swing takes tens or hundreds of milliseconds to cross between two concentric characteristics) and blocks the distance zones; **out-of-step tripping** (78) deliberately separates the system at chosen points when a swing is unstable. Fault resistance, by contrast, pushes the apparent impedance to the right: an arc is typically a few ohms or less (an empirical formula, Warrington's, gives about 0.6 Ω for a 5 kA arc across 3 m), but an earth fault through a tree or high tower-footing resistance can reach tens of ohms, which only quadrilateral characteristics with generous resistive reach detect.

Worked example — Distance relay settings for a 132 kV line. A 100 km, 132 kV line has positive-sequence impedance $z_1 = 0.10 + j0.40$ Ω/km. The relay is fed from a 600/1 A CT and a 132 kV/110 V VT. The maximum load is 150 MVA at 0.95 per unit voltage. Find the zone reaches in secondary ohms and check zone 3 against load.

Line impedance: $\mathbf{Z}_L = 10 + j40 = 41.2\angle 76.0^\circ\ \Omega$ primary. The relay sees secondary impedance $Z_{\text{sec}} = Z_{\text{pri}} \times \text{CTR}/\text{VTR} = Z_{\text{pri}} \times 600/1200 = 0.5\, Z_{\text{pri}}$, so the line is 20.6 Ω secondary.

Zone 1 (80 %): 33.0 Ω primary, 16.5 Ω secondary, instantaneous. Zone 2 (120 %): 49.5 Ω primary, 24.7 Ω secondary, 0.35 s. Zone 3 (200 %, say): 82.5 Ω primary, 41.2 Ω secondary, 1.0 s.

Minimum load impedance: $Z_{\text{load}} = V^2/S = (0.95 \times 132)^2/150 = 104.8\ \Omega$ primary (52.4 Ω secondary), at a power-factor angle of up to about 30°. A mho circle with diameter 82.5 Ω along 76° reaches $82.5\cos(76^\circ - 30^\circ) = 57.3\ \Omega$ in the 30° direction, well inside 104.8 Ω, so the margin against load encroachment is a factor of 1.8. Raising the line loading to 250 MVA would cut Z_{load} to 62.9 Ω and the margin to 1.1, at which point a load blinder becomes essential.

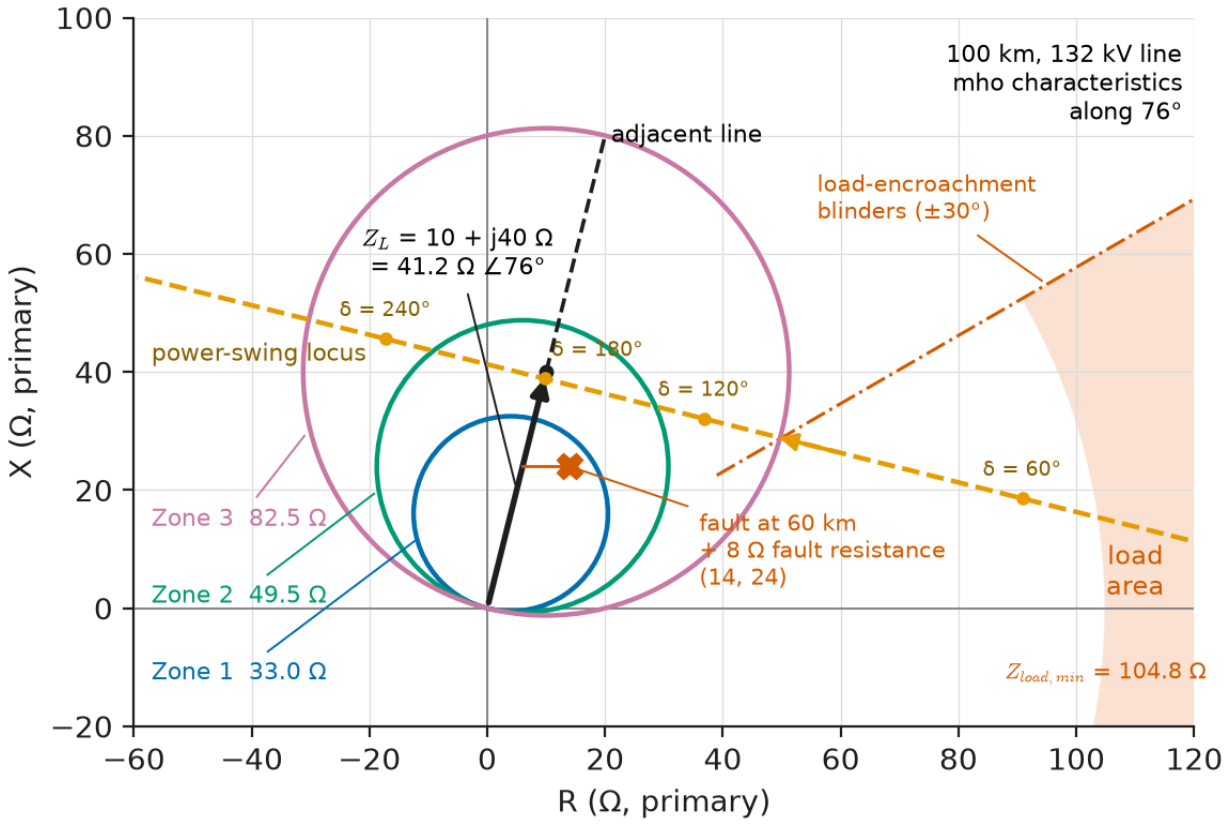


Figure 23.4 — Three-zone mho distance characteristic on the R-X plane for a 100 km, 132 kV line, with the load area and a power-swing trajectory.

23.5 Earth-fault protection

Why earth faults need their own protection

Between 70 % and 90 % of faults on overhead lines involve earth. How much current an earth fault draws depends on how the system neutral is earthed, which ranges from solid connection (fault current comparable with phase faults) to no connection at all (fault current only through the network's capacitance to earth). Earth-fault current is often far below load current, so phase overcurrent relays cannot see it. Earth-fault relays (50N/51N) respond instead to the **residual current** $3\mathbf{I}_0 = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c$, which is zero for balanced load and equal to the earth-fault current during a fault. It is measured either by summing three phase CTs (the residual

connection, sensitive down to about 5–10 % of CT rating because of CT mismatch) or by a single **core-balance CT** around all three phase conductors (sensitive down to a few amperes).

Neutral earthing methods

Table 23.3 — System earthing methods and their earth-fault behaviour.

Method	Typical earth-fault current	Where used	Protection approach
Solidly earthed	1–1.5 × three-phase fault current	Transmission ≥ 110 kV; LV TN systems; North American MV multi-grounded	Distance and overcurrent earth elements; fuses at LV
Low-resistance earthed	300–2000 A (limited by resistor)	Industrial and utility MV networks (6.6–33 kV)	Residual or core-balance overcurrent, graded
High-resistance earthed	1–10 A (just above capacitive current)	Generators, industrial plants needing continuity, mines	Alarm, or trip on neutral voltage or sensitive EF
Resonant (Petersen coil)	Residual current of a few A after compensation	Central and Northern European MV, rural overhead	Transient or admittance-based directional EF, wattmetric EF
Isolated (unearthed)	Capacitive current, 1–100 A depending on network size	Small industrial networks, ships, IT systems at LV	Insulation monitoring, neutral displacement alarm

A **neutral earthing resistor** (NER) is sized to limit the fault current to a chosen value, typically around the full-load current of the transformer: for an 11 kV system limited to 1000 A, $R = 6351/1000 = 6.35\ \Omega$. The resistor limits damage and touch voltage and keeps the fault current large enough for selective relaying. With an earth-fault setting of 40 A primary on a 400/1 A CT, the relay detects faults with total resistance up to $6351/40 = 159\ \Omega$, that is a fault resistance of up to about 152 Ω.

Resonant earthing

An isolated network has a capacitive earth-fault current $I_C = 3\omega C_0 V_{\text{ph}}$, where C_0 is each phase's capacitance to earth. An 11 kV cable network with $C_0 = 10$ µF per phase gives $I_C = 3 \times 314.2 \times 10 \times 10^{-6} \times 6351 = 59.9$ A. Arcs of this size do not extinguish themselves and can escalate through repeated restrikes to overvoltages of 3–4 per unit. A **Petersen coil** (arc-suppression coil) in the neutral, tuned to $L = 1/(3\omega^2 C_0) = 0.338$ H, draws an equal inductive current that cancels the capacitive current at the fault, leaving only a few amperes of resistive and harmonic residual, at which most arcing faults on overhead lines self-extinguish. The system can run with a sustained earth fault for hours while it is located. Detecting *which* feeder is faulted is then hard, because the fault current is tiny; relays use the transient at fault inception, the active component of residual current (wattmetric), or a short-time injection of a parallel resistor.

Restricted, sensitive and LV earth-fault protection

Sensitive earth fault (SEF) elements with settings of a few amperes detect high-impedance faults such as a conductor lying on dry ground (5–50 A), a major cause of wildfires and electrocutions; they must be time-delayed to ride through load unbalance. **Restricted earth fault** protection of transformer windings was introduced in Section 23.3. For generators, 100 % stator earth-fault protection combines a fundamental-frequency neutral overvoltage element (covering the outer 90–95 % of the winding) with a third-harmonic or

injection element for the part near the neutral. At LV, protection against indirect contact is provided by fuses and circuit breakers in TN systems and by **residual-current devices** (RCDs, 30 mA for personal protection, IEC 60364-4-41) in TT systems and final circuits; these belong to Chapter 29.

23.6 Circuit breakers

Interrupting at current zero

A circuit breaker carries rated current continuously, closes onto a fault, and interrupts fault current. When its contacts part under load an arc forms, a column of plasma at 10 000–20 000 K whose conductance keeps the current flowing. In AC the current passes through zero every half-cycle, and at each zero the arc input power vanishes. Whether the gap recovers depends on a race: the **dielectric recovery** of the gap, as the arc channel cools and is replaced by fresh insulating medium, against the **transient recovery voltage** (TRV) that the network impresses across the opening contacts. If the dielectric strength wins, the current is interrupted; if not, the arc reignites and the race repeats at the next zero. Every AC breaker therefore interrupts at a current zero, and the art of breaker design is to cool and deionise the gap within microseconds of that zero.

> **Status of the principle: Arc models of interruption**
> *Empirical.* Cassie's and Mayr's arc equations, and the black-box arc models used to design and test breakers, are fitted to measured arc voltages and conductances; they describe the thermal and dielectric recovery well near current zero but carry no first-principles prediction of a new interrupting medium.

Transient recovery voltage

The TRV is the network's response to the sudden removal of current, a damped oscillation between the source inductance and the stray capacitances at the breaker (Figure 23.5). Its peak and its rate of rise (RRRV) set the duty. For a three-phase ungrounded terminal fault the first pole to clear sees the highest voltage, raised by the **first-pole-to-clear factor** (1.5 for non-effectively earthed systems, 1.3 for effectively earthed systems at 72.5 kV and above), and the oscillation overshoots by an **amplitude factor** of about 1.4. For a 145 kV breaker the standard TRV peak is

$$u_c = k_{pp}\, k_{af}\, U_r \sqrt{\tfrac{2}{3}} = 1.3 \times 1.4 \times 145 \times 0.816 = 215 \text{ kV},$$

reached in a few hundred microseconds. A **short-line fault**, a few kilometres down the line, is harder still: the line side of the breaker produces a sawtooth travelling-wave TRV (Chapter 24) with an initial rate of rise of several kV/μs.

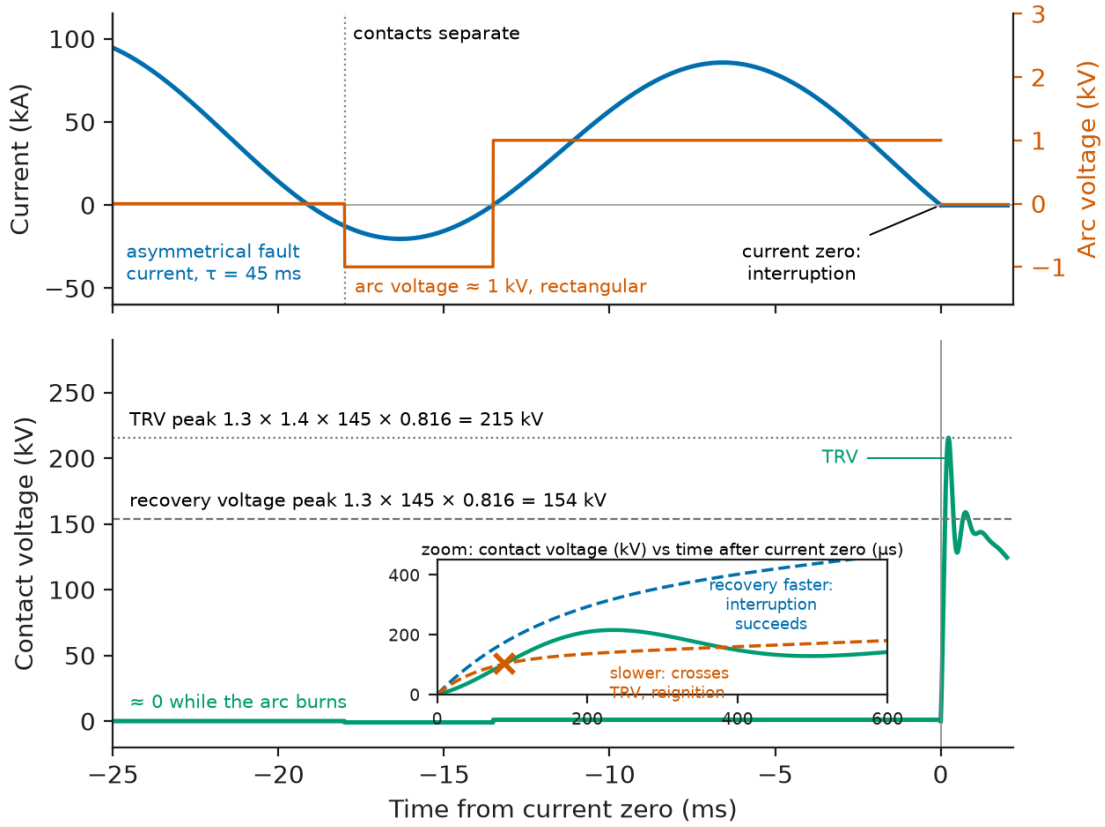


Figure 23.5 — Fault current interruption at a current zero and the transient recovery voltage across the first pole to clear of a 145 kV breaker.

Interrupting media

Table 23.4 — Circuit-breaker technologies and typical ratings.

Type	Voltage range	Rated short-circuit breaking current	Notes
Air circuit breaker (ACB)	LV, up to 1 kV	40–150 kA	Main LV switchboards; arc chutes
Moulded-case (MCCB) and miniature (MCB)	LV	6–150 kA	Current-limiting designs; IEC 60947-2, IEC 60898
Vacuum	3.6–52 kV (now up to 145 kV)	16–63 kA	Dominant at MV; long contact life, low maintenance
SF_6 puffer and self-blast	72.5–800 kV (also MV)	31.5–80 kA	Dominant at HV; SF_6 has a global warming potential about 23 500 times CO_2
SF_6-free (fluoronitrile mixtures, "clean air" with vacuum)	12–420 kV, growing	31.5–63 kA	Replacing SF_6 under F-gas rules
Generator circuit breaker	15–33 kV	63–300 kA	Very high X/R; IEC/IEEE 62271-37-013
DC breaker (hybrid, mechanical with resonant circuit)	1.5–3 kV traction; 320–525 kV HVDC	10–25 kA	No natural current zero; must create one

Oil and air-blast breakers, once standard, survive only in old installations.

Ratings

IEC 62271-100 and IEEE C37.04 define a breaker's duty. The main ratings are the rated voltage, rated normal current, rated short-circuit breaking current (the RMS AC component it can interrupt, for example 31.5 or 40 kA), the DC component it can tolerate at contact separation, the rated short-circuit making current (peak), the rated short-time withstand (1 or 3 s), the TRV, the operating sequence (for example O–0.3 s–CO–3 min–CO for auto-reclosing), and the break time (typically 3 cycles at HV, 2 cycles at EHV).

Worked example — Making current and DC component. A 145 kV breaker is rated 40 kA at 50 Hz, with the IEC standard DC time constant $\tau = 45$ ms ($X/R = \omega\tau = 14.1$). Find the peak current it must close onto and the DC component at contact separation 40 ms after fault inception.

The worst-case peak occurs about half a cycle (10 ms) after a fault at voltage zero:

$$i_p \approx \sqrt{2}\, I_k \left(1 + e^{-0.010/0.045}\right) = \sqrt{2} \times 40 \times 1.80 = 101.9\ \text{kA},$$

so the peak factor is 2.55; IEC 62271-100 rounds this to a rated making current of $2.5 \times 40 = 100$ kA at 50 Hz (2.6 at 60 Hz). At contact separation the DC component, as a fraction of the AC peak, is $e^{-0.040/0.045} = 41\,\%$. Near large generators, where X/R may reach 50–100, the DC component can exceed the AC peak, and the current may not pass through zero for several cycles: the "delayed current zero" problem that generator breakers are specifically designed for.

23.7 Fuses and relays

Fuses

A **fuse** is the simplest and cheapest protective device: a silver or copper element in a sand-filled body that melts, arcs and quenches. High-rupturing-capacity (HRC) fuses to IEC 60269 interrupt 50–120 kA at LV and are **current limiting**: they melt so quickly on a large fault that the current is cut off before it reaches its first prospective peak. A 100 A gG fuse facing a 50 kA prospective fault typically limits the peak let-through current to about 10 kA, against an unlimited peak of over 100 kA, and limits the let-through energy I^2t to tens of kA²s. Downstream equipment need withstand only these let-through values, and arc-flash energy falls dramatically.

Fuse classes are named by letters: **gG** (general purpose, full-range breaking), **aM** (motor circuits, short-circuit only, tolerates starting current), **gR/aR** (semiconductor protection, very fast). Two gG fuses in series discriminate if the upstream rating is at least 1.6 times the downstream rating (IEC 60269 selectivity ratio). At MV, **expulsion fuses** on overhead laterals and pole transformers (11–33 kV) interrupt by gas generated from the fuse tube, and current-limiting MV fuses protect ground-mounted transformers in RMUs. Fuses are single-shot, cannot be set, and can leave a three-phase motor single-phasing unless backed by a relay.

From electromechanical to numerical relays

Single-function **electromechanical** relays (induction discs, attracted armatures) survive in older substations, and **static** electronic relays followed in the 1970s–1990s. Today's **numerical** relays, or IEDs, sample CT and VT signals at 1–4 kHz, compute phasors by discrete Fourier filtering (*Circuits, Signals and Communications*, Ch. 22), and run many protection functions, fault recording, metering and self-supervision in one box. Functions are identified by ANSI/IEEE C37.2 device numbers.

Table 23.5 — Common ANSI device numbers in protection.

Number	Function	Number	Function
21	Distance	59	Overvoltage
25	Synchronism check	64	Earth fault (REF, stator earth fault)
27	Undervoltage	67 / 67N	Directional phase / earth overcurrent
32	Reverse or directional power	68 / 78	Power-swing blocking / out-of-step trip
46	Negative-sequence current	79	Auto-recloser
49	Thermal overload	81	Under/over frequency (load shedding)
50 / 51	Instantaneous / time overcurrent	86	Lockout relay
50N / 51N	Earth (residual) overcurrent	87	Differential (87T, 87G, 87B, 87L)
50BF	Breaker failure	24	Overexcitation (volts per hertz)

Modern substations built to **IEC 61850** replace much of the copper wiring with Ethernet: process-bus merging units digitise CT and VT signals and publish **sampled values**, and relays exchange trips and interlocks as **GOOSE** messages with transfer times under 4 ms. This brings flexibility and supervision, and also cybersecurity requirements (Chapter 26).

Reclosers and sectionalisers

An **auto-recloser** (79) is a breaker with built-in relay that trips and recloses typically up to three times (for example fast trip, 0.5 s dead time, fast trip, 15 s, delayed trip, 15 s, delayed trip, lockout). Because most overhead faults are transient, 80–90 % are cleared by the first reclose. **Sectionalisers** downstream count the recloser's interruptions and open during a dead time after a set count, isolating a permanently faulted section without needing fault-interrupting capability themselves.

23.8 Instrument transformers

The protection duty

Chapter 4 developed the current and voltage transformer as devices, and *Circuits, Signals and Communications* Ch. 9 §9.9 showed their ratio and phase errors at normal load. Protection asks a different question: does the CT reproduce a fault current of 20–50 times rating, possibly with a large DC offset, long enough for the relay to decide? Metering CTs are designed to saturate early to protect meters; protection CTs are designed not to.

Accuracy classes

IEC 61869-2 specifies protection CTs by a composite-error class and an **accuracy-limit factor** (ALF): a **5P20, 15 VA** CT has composite error within 5 % up to 20 times rated current when connected to its rated burden of 15 VA. Class **PX** CTs are specified instead by turns ratio, knee-point voltage E_k, magnetising current at E_k and secondary resistance, the data needed for high-impedance differential schemes. Classes **TPX, TPY, TPZ** add transient performance with DC offset. IEEE C57.13 uses C-classes: a **C800** CT delivers 800 V at its terminals at 20 times rated current (100 A for a 5 A CT) without exceeding 10 % ratio error.

Table 23.6 — Typical protection CT specifications.

Application	Typical specification	Secondary
Distribution feeder overcurrent	5P20, 10–15 VA (IEC); C200 (IEEE)	1 A or 5 A
Transformer and line differential	5P20 or PX; C400–C800	1 A preferred
High-impedance busbar differential	Class PX, E_k 300–1500 V, equal ratios	1 A
Transmission distance	TPY or 5P20 with transient check; C800	1 A
Generator differential	PX or TPY with matched ratios	1 A or 5 A
Revenue metering (for comparison)	Class 0.2S or 0.5, security factor FS ≤ 5	1 A or 5 A

Burden, effective ALF and saturation

The CT's ability to drive current depends on the voltage it must develop, which is the secondary current times the total loop resistance (CT winding plus leads plus relay). If the connected burden is smaller than the rated burden, the CT can reproduce a correspondingly higher current. The **effective ALF** is

$$\mathrm{ALF}' = \mathrm{ALF} \times \frac{R_{\mathrm{rated}} + R_{CT}}{R_{\mathrm{actual}} + R_{CT}}.$$

Worked example — Will the CT saturate? A 400/1 A, 5P20, 10 VA CT ($R_{\text{rated}} = 10\ \Omega$ at 1 A) with $R_{CT} = 3\ \Omega$ feeds a numerical relay (0.1 VA, so 0.1 Ω) over a 50 m run of 2.5 mm² copper (7.41 Ω/km, loop 0.741 Ω). The maximum fault current is 20 kA with $X/R = 20$.

Actual burden: $0.1 + 0.741 = 0.841\ \Omega$. Effective ALF: $20 \times (10 + 3)/(0.841 + 3) = 67.7$, so the CT reproduces symmetrical currents up to about $67.7 \times 400 = 27$ kA without exceeding 5 % error, comfortably above 20 kA. With a fully offset fault, however, the flux needed to avoid saturation is larger by a transient factor of about $1 + X/R = 21$: the CT would have to reproduce $50 \times 21 = 1050$ times rated current, far beyond its capability. It will saturate, typically within one to two cycles. This is acceptable because numerical relays complete their measurement within the first cycle and include saturation detection; where it is not acceptable (high-impedance busbar schemes, generator protection, transmission distance with long DC time constants) the CTs are sized with transient classes or the scheme is designed to be stable with a saturated CT, as in Section 23.3. Remanent flux left in the core after a fault can add up to 80 % of saturation flux and makes the next saturation earlier, which is why TPY and TPZ cores have air gaps.

> **Status of the principle: Effective accuracy-limit factor**
> *Approximate*. Treats burden and CT winding as resistive and the core as linear up to a sharp saturation point; it ignores DC offset, remanence and the curved knee of the magnetising characteristic, which transient-performance calculations (IEC 61869-2 TPx classes, IEEE C37.110) add.

Voltage transformers and CVTs

At 132 kV and above, voltage is usually measured by **capacitor voltage transformers** (CVTs): a capacitive divider down to about 10–20 kV followed by a small tuned magnetic transformer. CVTs are cheap and double as coupling capacitors for power-line carrier, but after a sudden voltage collapse their output contains a decaying transient lasting one to two cycles that can make a distance relay overreach for faults just beyond its zone-1 reach. This is one of the reasons zone 1 is set to 80 % rather than 95 %. Non-conventional optical CTs and Rogowski coils with digital outputs, which do not saturate, are increasingly used in IEC 61850 process-bus substations.

23.9 Protection coordination

Grading in time

In a radial network, relays and fuses in series must operate in sequence: the device nearest the fault first, each upstream device only if the one below fails. With inverse-time relays this is arranged by choosing settings so that at every fault current the upstream relay is slower by a **grading margin**. The margin must cover the downstream breaker's interrupting time (40–100 ms), the upstream relay's overshoot after the current disappears (10–50 ms for electromechanical, near zero for numerical), timing errors of both relays (a few per cent of operating time), CT errors and a safety margin. Typical values are 0.4 s with electromechanical relays and 0.25–0.3 s with numerical relays and vacuum breakers.

> **Status of the principle: Grading margin**
> *Empirical*. The 0.25–0.4 s margins are accumulated from manufacturer tolerances and field experience rather than derived; they are checked at the maximum fault current at the downstream relay, where the curves are closest.

Worked example — Grading two IDMT relays on a radial feeder. A 33/11 kV substation incomer relay A (CT 600/1 A, $I_s = 600$ A) backs up feeder relay B (CT 200/1 A, $I_s = 200$ A, TMS 0.10). Both use standard-inverse curves. The maximum fault current at B's location is 5 kA and at the 11 kV busbar 8 kA. Set A with a 0.3 s margin.

Relay B at 5 kA: $M_B = 25$, $t_B = 0.10 \times 0.14/(25^{0.02} - 1) = 0.211$ s. Relay A must operate at 5 kA in at least $0.211 + 0.3 = 0.511$ s. With $M_A = 5000/600 = 8.33$, the time at TMS 1 is $0.14/(8.33^{0.02} - 1) = 3.23$ s, so TMS $_A \geq 0.511/3.23 = 0.158$; choose 0.16.

Check: at 5 kA $t_A = 0.517$ s (margin 0.306 s); at a 2 kA fault far down the feeder $t_B = 0.297$ s and $t_A = 0.919$ s (margin larger, because the curves diverge at low multiples); for a busbar fault at 8 kA, only A sees the current and trips in 0.421 s. The same pattern continues upward: the 33 kV relay must be slower than A at the 11 kV busbar fault level, and so on, so the clearing time accumulates toward the source exactly where the fault level is highest. Figure 23.6 shows the time-current curves. An instantaneous element on B set at, say, 1.3 times the maximum fault current at the downstream transformer's LV side, and an instantaneous on A blocked by a signal from B (logic selectivity, Section 23.10), remove most of this accumulation.

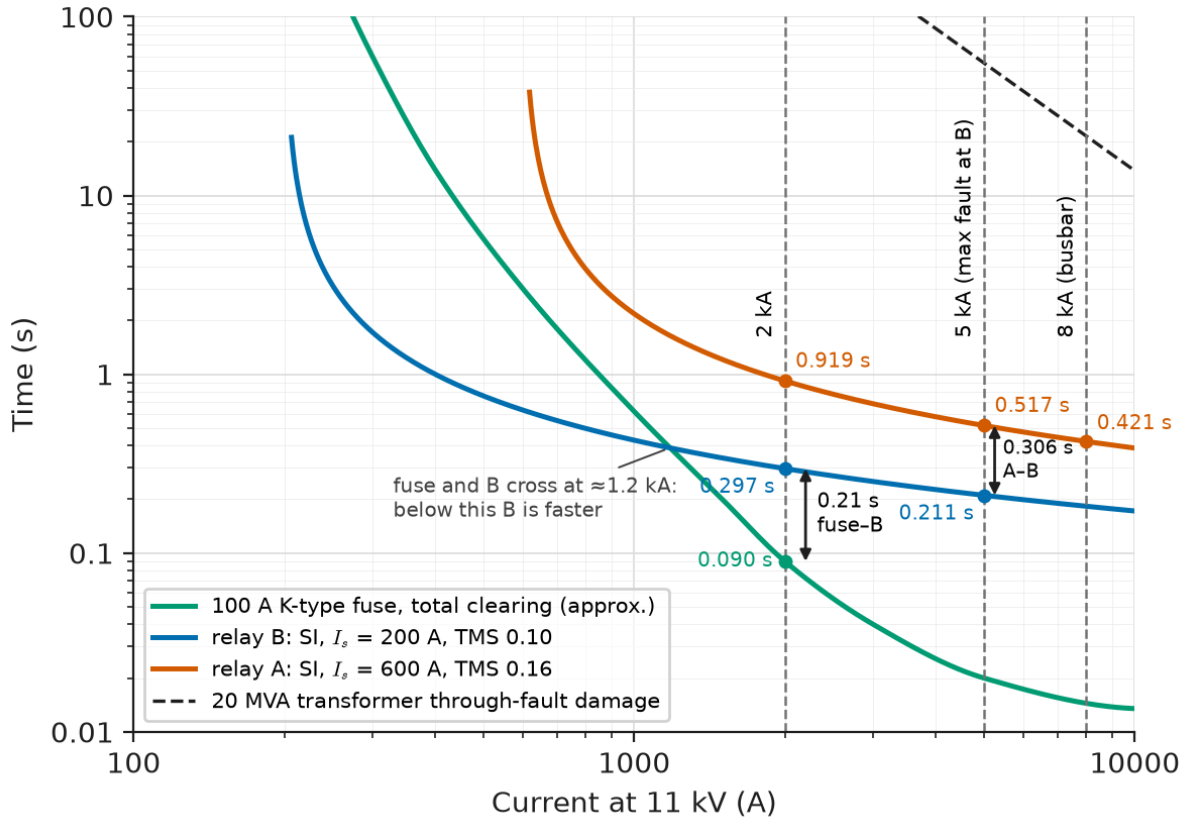


Figure 23.6 — Time-current coordination of a feeder relay B, an incomer relay A and a 100 A lateral fuse on an 11 kV radial feeder.

Fuses, reclosers and relays together

Grading a relay with a downstream fuse needs care because the fuse curve is much steeper than an SI relay curve: the fuse is very fast at high currents and slow at low ones, so the margin must hold over the whole range of fault currents the fuse sees, not only at its maximum. Figure 23.6 shows the difficulty. A 100 A K-type lateral fuse under relay B clears a 2 kA fault in about 0.09 s, 0.21 s ahead of B, but the two curves cross at about 1.2 kA, and a lateral fault below that would trip the whole feeder before the fuse blows. The remedies are an EI

curve on the relay, which follows the fuse's shape, a higher relay pickup or time multiplier, or a smaller fuse. On overhead feeders a recloser and the lateral fuses are coordinated by one of two philosophies. In **fuse saving**, the recloser's first fast trip clears the fault before the fuse melts, giving transient faults a chance to clear without blowing the fuse; if the fault persists, the recloser's delayed curve lets the fuse blow. In **fuse blowing**, preferred where momentary interruptions of the whole feeder are unwelcome (for example, feeders with many data centres or industrial process loads), the fuse clears first. Distributed generation raises the fault current through fuses and can destroy fuse-saving coordination (Chapter 22).

Coordination studies

A coordination study combines fault levels (Chapter 19), equipment damage curves (transformer through-fault curves to IEEE C57.109, cable withstand $I^2t = k^2S^2$ from IEC 60364-4-43, motor starting curves) and every device's curve, and is revised whenever the fault level changes, for instance when inverter-based generation displaces synchronous machines.

23.10 Selectivity and sensitivity

The setting window

A relay's pickup must lie in a window bounded below by the largest current it must not trip on and above by the smallest current it must trip on:

$$k_1\, I_{\text{load,max}} \;\le\; I_s \;\le\; \frac{I_{\text{f,min}}}{k_2}$$

with $k_1 \approx 1.2$–1.5 (to allow for overload, cold-load pickup and the relay's drop-off ratio) and $k_2 \approx 1.5$–2 (so that the minimum fault, usually a phase-to-phase fault at the end of the backup zone under minimum generation, operates the relay reliably). If the window is empty, overcurrent protection cannot both carry the load and see the fault, and another principle is needed: voltage-controlled or voltage-restrained overcurrent, negative-sequence overcurrent (unbalanced faults produce negative sequence, balanced load does not), distance or differential.

> **Status of the principle: Setting window**
> *Empirical.* The factors k_1 and k_2 are conventions from utility practice and standards guidance; the existence of a window is a property of the network, and its absence is the signal that the protection principle must change.

Sensitivity is the ratio of minimum fault current to pickup, sometimes required to be at least 1.5 for main protection and 1.3 for backup. On long rural feeders with weak sources, and on inverter-dominated networks where fault current is only 1.1–1.5 times rated, sensitivity is the binding constraint. Earth faults through high resistance, such as a broken conductor on asphalt or sand, may draw less current than the load unbalance and are found only by sensitive earth-fault elements with long delays, or by specialised high-impedance fault detection algorithms using harmonic and arc signatures.

Methods of achieving selectivity

Method	Principle	Where it is used	Limitation
Time	Upstream devices slower by a grading margin	Radial feeders, LV switchboards	Clearing time accumulates toward the source
Current	Upstream instantaneous set above downstream fault level	Feeders feeding transformers, MCB/MCCB cascades	Needs a large impedance (transformer, long cable) between devices
Logic (zone-selective interlocking)	Downstream relay that sees the fault blocks the upstream fast element by a signal	MV switchboards, LV main boards, data centres	Needs wiring or GOOSE messages and supervision
Directional	Trip only for faults in the forward direction	Rings, parallel feeders, DG feeders	Needs VTs; close-in three-phase faults need voltage memory
Unit (differential)	Compare currents at all boundaries of a zone	Transformers, generators, busbars, cables	Protects only its zone; needs backup
Distance (impedance)	Trip when impedance to fault is inside reach	Transmission and subtransmission	Fault resistance, infeed, swings

Speed, selectivity and arc flash

Time grading buys selectivity with time, and time costs energy at the fault. The incident energy of an arc flash at a switchboard is roughly proportional to the clearing time (IEEE 1584, Chapter 29): an incomer relay graded at 0.5 s may expose a worker at the main switchboard to 20–40 cal/cm², while an instantaneous trip at 50 ms would cut it tenfold. Engineers reconcile selectivity with safety by zone-selective interlocking, bus differential protection on switchboards, arc-flash relays that combine a light sensor with a current check and trip in 2–5 ms, and maintenance switches that temporarily enable fast settings while people work on live equipment. The protection of a whole installation, from source to final circuit, is drawn together in the design procedure of Chapter 30.

What remains open

Open question	Where it is answered
How are the fault currents used for settings computed, for balanced and unbalanced faults?	Chapter 19 (back)
How fast must transmission faults be cleared to keep generators in synchronism?	Chapter 20 (back)
What insulation, surge arresters and earthing does the protected system need against overvoltages?	Chapter 24
How do substation earth grids limit touch and step voltage during an earth fault?	Chapter 24 §24.6
How do inverter-based resources change fault current, grid-code ride-through and anti-islanding?	Chapter 27
How are RCDs, shock protection and arc-flash incident energy treated for personnel safety?	Chapter 29
How is protection selected and documented when designing an installation?	Chapter 30
How are CTs and VTs built and analysed as transformers?	Chapter 4 (back)
How are communication-assisted protection and IEC 61850 substations secured against cyber attack?	Chapter 26

Next: Chapter 24 turns from currents to voltages, asking how insulation is designed and tested to survive the lightning and switching surges that every protected system must also withstand.

Chapter 24 — High-Voltage Engineering

Part VI · Transmission, Distribution, and Protection

A 132 kV substation in a tropical region loses its main 90 MVA transformer in the first thunderstorm of the wet season. The post-mortem finds a flashover between turns near the line end of the HV winding. The transformer had passed its factory impulse test at 650 kV, and the surge arrester at the line entrance had operated correctly, limiting the incoming lightning surge to under 300 kV at the arrester. But the arrester stood 35 m away from the transformer bushing, on the far side of a disconnector, and over that distance the steep surge front reflected from the transformer's high impedance and built up a voltage well above the arrester's residual voltage. A few metres of busbar decided whether a USD 3 million transformer survived. Meanwhile, at a cable factory, a 132 kV XLPE cable fails its routine partial-discharge test because a single 0.1 mm void near the conductor screen discharges at a third of the test voltage; left in service, it would have grown an electrical tree and punctured within a few years. High-voltage engineering is the discipline that makes such decisions predictable: what electric stress each insulation can bear, what overvoltages the system will impose on it, how the two are matched with the help of arresters and earthing, and how the result is proved by test without endangering the people doing the testing.

> **Why does insulation fail, and how are the overvoltages from lightning and switching matched to the strength of every piece of equipment?**
> **How are high voltages generated, measured and applied in tests, and how is high-voltage work made safe?**

Concept	Engineering destination
Dielectric breakdown	Clearances in air, SF_6 pressure in GIS, cable and bushing stress design
Partial discharge	Factory acceptance of cables, transformers and GIS; online condition monitoring
Lightning and switching surges	Shield wires, tower footing resistance, line arresters, breaker closing resistors
Insulation coordination	Selection of LIWL, SIWL and clearances for every voltage class
Surge arresters	Transformer protection, arrester ratings and placement
Grounding	Substation earth grids, touch and step voltages, tower footings
High-voltage generation and measurement	Test transformers, resonant sets, voltage multipliers, impulse generators, dividers
High-voltage testing	Type, routine, commissioning and diagnostic tests
Safety procedures	Test-area design, earthing and discharge, the five safety rules, approach distances

Roadmap. How insulation fails (24.1) → the local discharges that precede failure in solid and liquid insulation (24.2) → the overvoltages that stress it (24.3) → matching stress to strength (24.4) → the arresters that cut the stress (24.5) → the earthing that carries surge and fault currents safely into the ground (24.6) → generating and

measuring test voltages (24.7) → the tests themselves (24.8) → the procedures that keep people alive while doing all of this (24.9).

24.1 Dielectric breakdown

Stress and strength

An insulation system fails when the local electric field somewhere exceeds what the material can bear for the duration and waveshape applied. High-voltage design is therefore two calculations: the **stress**, the field distribution set by geometry and permittivities, and the **strength**, an empirical property of the material that depends on gap length, pressure, temperature, waveshape, time under stress, area and volume stressed, and contamination. The field calculation itself is electrostatics, and the polarisation of a dielectric is described by its permittivity.

> **Boundary.**
> Electrostatics, the constitutive relation $\mathbf{D} = \varepsilon \mathbf{E}$ and the polarisation of matter are taken from *Engineering Physics*, Ch. 11 §11.9–11.10. The microscopic structure of insulating materials belongs to the Materials Core. Neither describes breakdown: discharge physics and the engineering data below are the subject of this section.

The ratio of mean to maximum field, the **field utilisation factor** $\eta = E_{\text{mean}}/E_{\max}$, measures how well a geometry uses its insulation: $\eta = 1$ for parallel plates, about 0.5–0.7 for well-designed GIS conductors, and below 0.1 for a sharp point against a plane. For a coaxial geometry (cable, GIS busbar) with inner radius r and outer radius R,

$$E(x) = \frac{U}{x\,\ln(R/r)}, \qquad E_{\max} = \frac{U}{r\,\ln(R/r)} \quad \text{at } x = r,$$

and for a fixed outer radius $E_{\max}$ is smallest when $R/r = e$. A 132 kV XLPE cable with $U_0 = 76$ kV phase-to-earth, conductor-screen radius 15 mm and insulation-screen radius 32 mm has $E_{\max} = 76/(15 \ln 2.133) = 6.7$ kV/mm at the conductor screen and 3.1 kV/mm at the insulation screen: typical design stresses for HV extruded cables.

Breakdown in gases

In a gas, free electrons accelerated by the field ionise neutral molecules, each new electron starting its own avalanche. Breakdown in a uniform field occurs when each avalanche regenerates at least one new starting electron through secondary processes (Townsend's criterion), and its voltage depends only on the product of pressure p and gap d: **Paschen's law** (Figure 24.1). For air the curve has a minimum of about 330 V near $pd \approx 0.75$ Pa·m; no gap in air, however small, breaks down below this. On the right-hand branch, where all power equipment operates, breakdown voltage rises nearly linearly with pd. For air at standard conditions (20 °C, 101.3 kPa) and uniform field an empirical fit is

$$U_b \approx 24.4\,\delta d + 6.53\sqrt{\delta d}\ \text{kV} \qquad (d \text{ in cm}),$$

where δ is the relative air density. It gives 30.9 kV for a 1 cm gap and 265 kV for a 10 cm gap: the familiar "3 kV/mm" of air, slowly falling with gap length.

Status of the principle: Paschen's law and the air breakdown formula
Empirical. Paschen's similarity law follows from Townsend's avalanche model with empirical ionisation coefficients; the formula above is a fit to uniform-field data for 0.1–10 cm gaps and does not apply to non-uniform fields, very small gaps (field emission), or long gaps where leader discharges dominate.

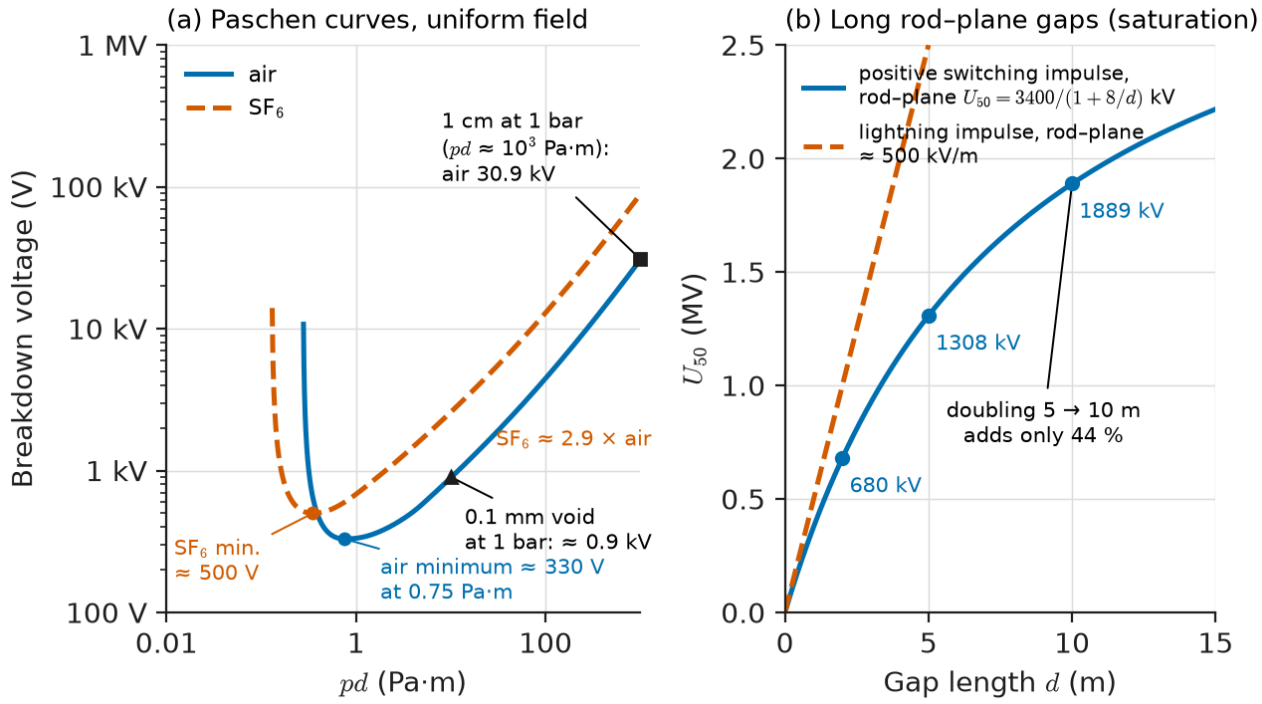


Figure 24.1 — Paschen curves for air and SF6, and breakdown voltage of long rod-plane air gaps under switching impulse.

SF6 is electronegative: its molecules capture free electrons and quench avalanches, giving a critical field of about 89 kV/cm per bar, roughly three times that of air. At 4–6 bar it lets gas-insulated switchgear (GIS) occupy 10–20 % of the volume of an air-insulated substation; its global warming potential of about 23 500 is driving replacement by fluoronitrile mixtures, pressurised dry air and vacuum interrupters.

Long air gaps and the saturation problem

Gaps longer than about 1 m in non-uniform fields break down by a **streamer–leader** mechanism, a hot conducting channel growing in steps at a much lower average field. Positive switching impulses, with fronts of a few hundred microseconds, give the leader most time to develop and are the most dangerous. The empirical Gallet formula for rod-plane gaps,

$$U_{50} \approx \frac{3400\, k_g}{1 + 8/d} \ \text{kV} \qquad (d \ \text{in m}),$$

with gap factor $k_g = 1$ for rod-plane and 1.2–1.5 for conductor-to-tower geometries, gives 1.31 MV at 5 m and only 1.89 MV at 10 m. Doubling the gap adds only 45 % to the strength. This **saturation** is why switching surges, not lightning, dimension the towers of 500 kV and higher lines, and why limiting switching overvoltages (Section 24.3) is worth so much money at EHV and UHV.

Atmospheric correction. Air strength falls with density. External insulation for installations above 1000 m is corrected by the altitude factor of IEC 60071-2,

$$K_a = e^{m\,H/8150},$$

with H in metres and $m = 1$ for lightning impulse and power frequency (less for switching impulse on long gaps). At 2000 m (mining sites in the Andes, substations in Ethiopia or Tibet) $K_a = 1.28$: external insulation must be designed for 28 % more withstand than at sea level.

Liquids and solids

The strength of mineral oil and esters is limited by moisture, fibres and bubbles, which align in the field and bridge the gap: a standard IEC 60156 cell (2.5 mm) gives 30 kV for poor oil and over 70 kV for dried, filtered oil, while transformer ducts are designed for 5–10 kV/mm.

Solid insulation rarely reaches its **intrinsic** strength of hundreds of kV/mm. It fails thermally, when dielectric loss ($\omega C U^2 \tan\delta$) heats it faster than it cools and $\tan\delta$ rises with temperature, or, far more often, by long-term **ageing**: partial discharges erode voids (Section 24.2), electrical trees grow from defects, and in wet cables **water trees** grow over years. The empirical inverse-power law

$$t_{\text{life}} \propto E^{-n}, \qquad n \approx 9\text{–}15 \text{ for XLPE}$$

means that a 10 % increase in stress shortens life by a factor of 1.1^n, 2.4 to 4.2.

> **Status of the principle: Inverse-power life law**
> *Empirical.* Fitted to accelerated-ageing tests at elevated stress and extrapolated to service stress; the exponent depends on material, temperature and test method, and the extrapolation is the main source of uncertainty in cable and machine-insulation life prediction.

Table 24.1 — Representative insulating materials (room temperature, 50 Hz, short-term strength in uniform field unless noted).

Material	Relative permittivity	$\tan\delta$	Short-term strength	Typical design stress
Air, 1 bar	1.0	~0	3 kV/mm (1 cm gap)	0.3–0.5 kV/mm in non-uniform fields
SF_6, 5 bar abs.	1.0	~0	40 kV/mm (approx.)	10–20 kV/mm
Mineral oil (treated)	2.2	0.001	20–30 kV/mm (2.5 mm cell)	5–10 kV/mm in ducts
Oil-impregnated paper	3.3–3.6	0.002–0.003	40–60 kV/mm	10–15 kV/mm (cables)
XLPE	2.3	0.0002–0.0005	50–100 kV/mm (thin samples)	3–7 kV/mm MV/HV, up to 15 kV/mm EHV
Epoxy resin (filled)	3.5–5	0.005–0.02	20–30 kV/mm	2–5 kV/mm
Electrical porcelain	5.5–6.5	0.02	15–25 kV/mm (puncture)	Set by surface flashover
Mica-epoxy (machine windings)	4–5	0.005–0.02	30–50 kV/mm	2–3 kV/mm
Vacuum (contacts, 1–10 mm)	1.0	0	10–40 kV/mm	Set by conditioning

Surface flashover and pollution

Outdoor insulators rarely puncture; they flash over along a surface layer of salt or dust wetted by fog, at normal operating voltage. The design variable is **creepage distance**: IEC TS 60815 recommends unified specific creepage distances of roughly 22 to 53 mm per kV of phase-to-earth voltage from "very light" to "very heavy" pollution, so a coastal 132 kV string needs over 3.5 m. Hydrophobic silicone-rubber insulators and washing are the usual remedies.

24.2 Partial discharge

What a partial discharge is

A **partial discharge** (PD) is a breakdown that bridges only part of the insulation between conductors: a spark in a gas-filled void inside a cable, in a delamination of a machine winding, along the surface of a bushing, or a corona glow from a sharp point in air. Each discharge transfers a small charge, from below 1 pC to thousands of pC, in nanoseconds. A single discharge does no harm; millions of them, at every half-cycle for years, erode polymers chemically and by bombardment, start electrical trees, and end in complete breakdown. PD is therefore both a cause of insulation failure and its earliest measurable symptom.

The void model

A void in solid insulation sees an enhanced field. For a flat void normal to the field, continuity of D gives $E_{\text{void}} = \varepsilon_r E_{\text{ins}}$, and the gas in the void has a lower strength than the solid. The void is modelled by the **abc** (three-capacitance) circuit of Figure 24.2: C_a for the void, C_b for the solid in series with it, and C_c for the rest of the insulation in parallel. When the void voltage reaches its breakdown value it collapses, and the charge redistribution is seen at the terminals as an **apparent charge** $q \approx C_b\,\Delta U_a$, much smaller than the true charge in the void. Apparent charge is what IEC 60270 instruments measure, calibrated by injecting a known charge at the terminals.

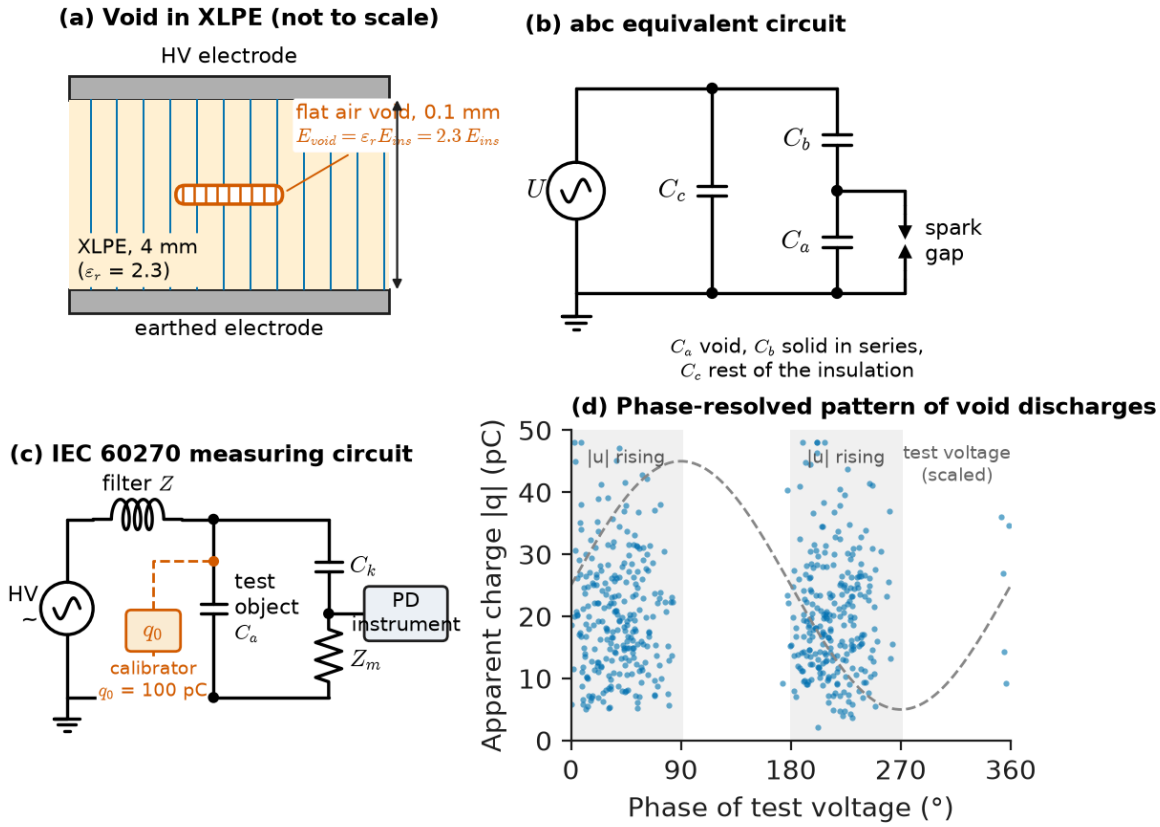


Figure 24.2 — Partial discharge in a void: cross-section, the abc equivalent circuit, and the IEC 60270 measuring circuit with a phase-resolved pattern.

Worked example — Inception field of a void in XLPE. A flat air void 0.1 mm thick sits inside XLPE ($\varepsilon_r = 2.3$) at atmospheric pressure. At what insulation stress does it start to discharge?

The void has $pd = 101.3\ \text{kPa} \times 10^{-4}\ \text{m} \approx 10$ Pa·m (7.6 torr·cm), on the right-hand branch of the Paschen curve, where the Paschen expression with standard air coefficients gives a breakdown voltage of about 0.87 kV, or 8.7 kV/mm across the void. With $E_{\text{void}} = 2.3\,E_{\text{ins}}$, discharges start when the field in the surrounding XLPE reaches $8.7/2.3 = 3.8$ kV/mm. The 132 kV cable of Section 24.1 runs at 6.7 kV/mm at its conductor screen: such a void there would discharge continuously, which is why HV insulation is extruded and degassed under clean-room conditions and every length is PD-tested. A 10 µm void ($pd \approx 1$ Pa·m, near the Paschen minimum of about 330 V) needs about 33 kV/mm in the gas, or 14 kV/mm in the XLPE, before it discharges: the breakdown field of a gas rises steeply as the gap shrinks, which is how clean cable survives its design stress.

> **Status of the principle: abc model of a void**
> *Approximate.* A lumped-capacitance picture that relates true and apparent charge and explains the phase-resolved pattern; it ignores surface charge on the void walls, statistical time lags and changes in the gas over time, so apparent charge is a comparative quality measure, not a direct measure of damage.

Measuring and judging PD

IEC 60270 defines the conventional measuring circuit: a coupling capacitor in parallel with the test object, a measuring impedance, a band-pass detector (typically 100–400 kHz), and calibration with a known charge before every test. Acceptance limits are set per product: MV extruded cables must show no discharge above 10 pC at $1.73U_0$ (IEC 60502-2); large power transformers are typically limited to 100–250 pC at about 1.5–1.6

times rated phase voltage during a long-duration induced test (IEC 60076-3); GIS and cable accessories to a few pC.

In service, PD is detected without an outage by **UHF** antennas in GIS and transformer tanks, **acoustic** sensors for locating sources, **transient earth voltage** probes on MV switchgear and **high-frequency CTs** on cable earth bonds. The **phase-resolved pattern** (Figure 24.2) identifies the defect: void discharges cluster on the rising quarter of each half-cycle, corona from a sharp point appears first near the negative peak. Online PD monitoring is now part of condition-based maintenance of generators, GIS and critical cables (Chapter 9).

24.3 Lightning and switching surges

Classes of overvoltage

IEC 60071-1 classifies the voltage stresses on insulation by shape and duration:

Table 24.2 — Classes of overvoltage and their standard test shapes (IEC 60071-1, IEC 60060-1).

Class	Typical origin	Duration and shape	Standard test shape
Continuous (power frequency)	Normal operation	Continuous, up to $U_m/\sqrt{3}$	Power-frequency, long duration
Temporary (TOV)	Earth faults, load rejection, Ferranti effect, ferroresonance	0.03 s to hours, 1.0–1.7 pu (higher with resonance)	Short-duration power-frequency, 1 min
Slow-front	Switching of lines, capacitors, faults and their clearing	Front 20–5000 µs, tail to 20 ms; 1.5–3.5 pu	Switching impulse 250/2500 µs
Fast-front	Lightning	Front 0.1–20 µs, tail to 300 µs; up to several MV	Lightning impulse 1.2/50 µs
Very-fast-front	Disconnector operation in GIS, vacuum-breaker restrikes	Front 3–100 ns, oscillations 0.3–100 MHz	No general standard shape

Lightning

A cloud-to-ground lightning flash typically consists of a downward stepped leader, an upward connecting leader from the struck object, and a **return stroke** that drains the charge, often followed by subsequent strokes down the same channel. Measured stroke currents are statistically distributed; the first negative return stroke has a median peak of about 31 kA, a median front of a few microseconds, and 5 % of strokes exceed about 96 kA. An empirical fit commonly used in line design (IEEE Std 1243 and 1410) is

$$P(I_p > i) = \frac{1}{1 + (i/31\ \text{kA})^{2.6}}.$$

The rate of flashes to ground per km² per year, the **ground flash density** N_g, is measured by lightning location networks: below 0.1 in polar and some desert regions, 1–4 over much of Europe and North America, and 10–50 in parts of central Africa, Southeast Asia, Florida and northern South America. The number of flashes intercepted by a line of height h (m) is estimated by Eriksson's model as $N_L = N_g(28h^{0.6} + b)/10$ per 100 km per year, b being the separation of the outer conductors.

> **Status of the principle: Lightning statistics**
> *Empirical.* The stroke-current distribution and the attractive-radius models are fits to measurements on instrumented towers and lightning location data; they vary with latitude, terrain and stroke polarity and carry factor-of-two uncertainties in tails that dominate design.

Surges as travelling waves

A lightning surge on a line is a travelling wave: it propagates at close to the speed of light (300 m/μs on overhead lines, 150–200 m/μs in cables), with voltage and current related by the line's **surge impedance** Z_0, about 300–450 Ω for overhead conductors and 20–60 Ω for cables. At every junction it reflects and refracts with the coefficients $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ and $\tau = 2Z_2/(Z_1 + Z_2)$. An open end or a transformer (whose surge impedance is thousands of ohms) nearly doubles the voltage; a surge passing from a 400 Ω line into a 40 Ω cable is transmitted at only $2 \times 40/440 = 0.18$ of its amplitude, which is why a short cable entry can shield a substation.

> **Boundary.**
> Wave propagation, surge impedance, reflection coefficients and lattice diagrams are developed in *Circuits, Signals and Communications*, Ch. 14 §14.3 and §14.5, from the field picture in *Engineering Physics*, Ch. 12 §12.7.3. Steady-state line models are in Chapter 21. Here they are used, not rederived.

A stroke of current I that terminates on a phase conductor divides into two waves of $I/2$ travelling in each direction, each producing a voltage $U = Z_0 I/2$. **Shield wires** (overhead earth wires) above the phase conductors intercept strokes and carry them to earth through the towers. A stroke to a tower or shield wire raises the tower-top potential by roughly the stroke current times the tower footing resistance (plus an inductive term for tall towers); if the difference between tower and phase conductor exceeds the insulator strength, the insulator flashes over from the tower to the phase, a **back-flashover**. Footing resistance is therefore the main design variable for shielded lines.

Worked example — Lightning performance of a 132 kV line. A 132 kV line on 30 m towers ($b = 10$ m) crosses a region with $N_g = 4$ flashes/km²/yr. Its insulator strings have a critical flashover voltage (CFO) of 700 kV and its conductors $Z_0 = 400$ Ω.

Unshielded. A stroke to a phase conductor produces $U = 400I/2 = 200I$; flashover occurs for $I > 2 \times 700/400 = 3.5$ kA. With $P(I_p > 3.5\,\text{kA}) = 1/(1 + (3.5/31)^{2.6}) = 0.997$, practically every stroke to a phase causes an outage. The line intercepts $N_L = 4 \times (28 \times 30^{0.6} + 10)/10 = 90$ flashes per 100 km per year, so an unshielded line would trip about 90 times per 100 km per year: unacceptable.

Shielded. With a shield wire the main risk is back-flashover. Crudely, the insulator sees $(1 - c) I R_f$, where $c \approx 0.25$ is the coupling factor between shield wire and phase. With footing resistance $R_f = 20$ Ω flashover needs $I > 700/(0.75 \times 20) = 46.7$ kA, exceeded by 25.7 % of strokes; with $R_f = 10$ Ω, $I > 93.3$ kA, exceeded by 5.4 %. Taking about 60 % of the intercepted strokes as terminating near enough to a tower, the back-flashover rate is about $0.6 \times 90 \times 0.257 = 13.9$ per 100 km per year at 20 Ω and 2.9 at 10 Ω. This model ignores the tower surge impedance and reflections from adjacent towers, which reduce the voltage, so it overestimates; detailed simulation gives lower numbers but the same lesson. Halving footing resistance with counterpoise wires or deep rods cuts outages roughly fivefold, and where soil resistivity makes 10 Ω impossible (rocky mountain routes), line surge arresters across the insulators of every tower or every few towers are fitted instead. Typical utility targets are 0.5–2 outages per 100 km per year.

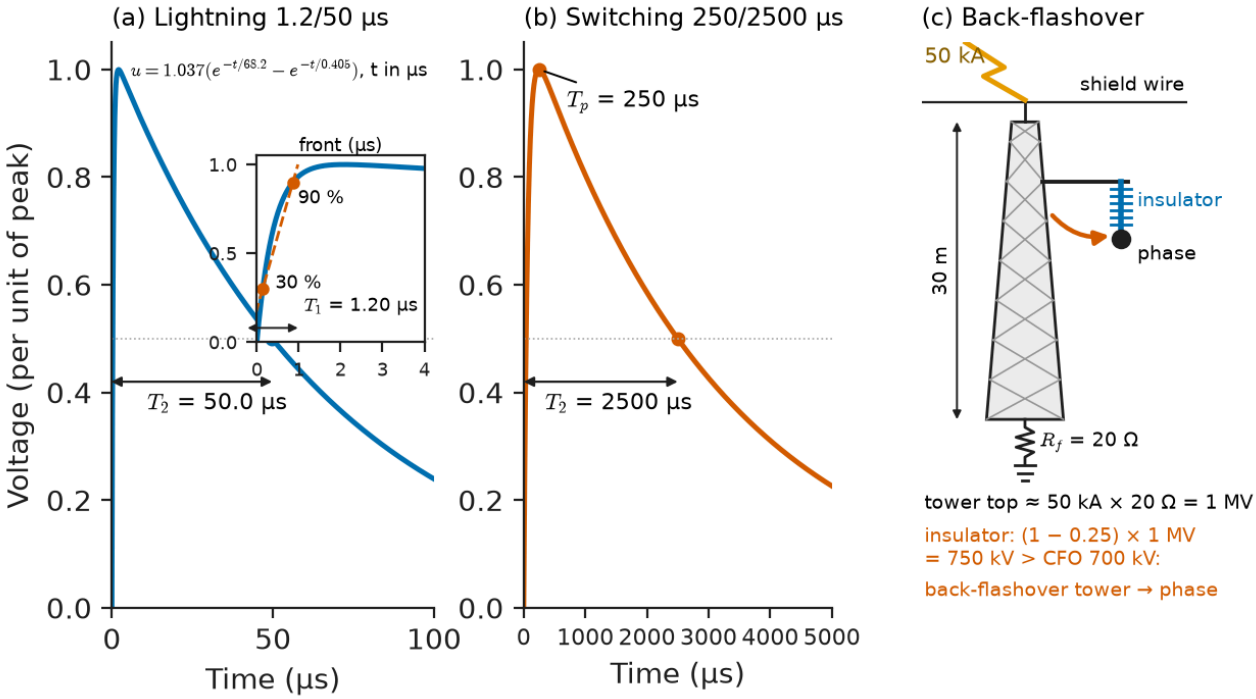


Figure 24.3 — Standard lightning and switching impulse waveshapes, and the tower-top voltage of a back-flashover.

Figure 24.3 shows the standard impulse shapes used to represent these stresses in tests.

Switching surges and temporary overvoltages

Energising a line launches a voltage step that doubles at the open far end, giving about 2 per unit; **high-speed reclosing** onto trapped charge can give 3–3.5 per unit. At 300 kV and above, where switching surges set tower clearances (Section 24.1), they are cut to 1.5–2 per unit by **pre-insertion resistors** (a few hundred ohms for 8–12 ms before the main contacts close), **controlled (point-on-wave) switching** and line-end arresters. Capacitor-bank energisation, restrikes, current chopping by vacuum breakers (hence RC snubbers on motors) and disconnector transients in GIS are other sources.

Temporary overvoltages are long enough to stress arresters thermally rather than insulation dielectrically. A single-line-to-earth fault raises the healthy-phase voltages by the **earth-fault factor**: at most 1.4 on effectively earthed systems ($X_0/X_1 \leq 3$, $R_0/X_1 \leq 1$) and up to 1.73 or more on isolated and resonant-earthed systems (Chapter 23). Load rejection at a long line end and ferroresonance between a transformer's magnetising inductance and cable capacitance (a classic risk when a single fuse of an unloaded cable-fed transformer blows) can cause larger TOVs.

24.4 Insulation coordination

The principle

Insulation coordination is the selection of the dielectric strength of equipment in relation to the voltages that can appear on the system, taking into account the protective devices and the environment, so that the probability of insulation failure is economically and operationally acceptable (IEC 60071-1). It is a matching problem: the stress distribution (overvoltages, after arresters have limited them) on one side, the strength distribution (withstand of insulation) on the other, and the overlap between them gives the risk of failure.

Standard insulation levels

Equipment is tested and specified by standard withstand voltages, chosen from tables by the highest system voltage U_m. In Range I ($U_m \leq 245$ kV) these are the short-duration power-frequency withstand and the **lightning impulse withstand level** (LIWL, also called the basic insulation level, BIL); in Range II ($U_m > 245$ kV) they are the **switching impulse withstand level** (SIWL) and the LIWL. Table 24.3 lists common choices.

Table 24.3 — Typical standard insulation levels (IEC 60071-1; one of several standardised combinations per voltage).

U_m (kV RMS)	Typical system	Power-frequency withstand, 1 min (kV RMS)	SIWL (kV peak)	LIWL (kV peak)
12	10–11 kV	28	—	75
24	20–22 kV	50	—	125
36	33 kV	70	—	170
72.5	66 kV	140	—	325
145	132 kV	275	—	650
245	220 kV	460	—	1050
420	400 kV	—	1050	1425
550	500 kV	—	1175	1550
800	765 kV	—	1425	2100

The procedure

IEC 60071-2 proceeds in four steps:

1. Determine the **representative overvoltages** U_{rp} of each class at each location, from system studies and arrester protective levels.
2. Apply a **coordination factor** K_c to obtain the coordination withstand voltage $U_{cw} = K_c U_{rp}$. In the deterministic method K_c covers modelling uncertainty (1.0–1.1 for lightning with arresters, larger to account for the distance effect); in the statistical method K_c follows from a chosen risk of failure, typically 10^{-3} to 10^{-4} per switching operation.
3. Multiply by a **safety factor** K_s (1.15 for internal insulation, which cannot be tested to destruction and ages; 1.05 for external, self-restoring insulation) and, for external insulation, by the altitude factor K_a, to obtain the **required withstand voltage** U_{rw}.
4. Select the next standard level at or above U_{rw}.

> **Status of the principle: Insulation coordination procedure**
> *Approximate.* A structured margin calculation; the statistical method assumes Gaussian-like distributions of overvoltage and withstand whose tails are poorly known, and the deterministic factors are conventions. It gives an acceptable risk, not zero risk.

Self-restoring insulation (air) recovers after a flashover, and its strength is described statistically by a 50 % flashover voltage U_{50} and a standard deviation (about 3 % for lightning and 6 % for switching impulses), with withstand defined at 10 % flashover probability. Non-self-restoring insulation (oil-paper, XLPE, windings) is permanently damaged by breakdown, so its withstand is proved only by tests without breakdown.

The ratio between the withstand level and the protective level of the arrester is the **protective ratio**. IEEE C62.22 recommends margins of at least 20 % for lightning (LIWL at least 1.2 times the voltage at the protected equipment) and 15 % for switching; practice for important transformers often uses 1.4 or more, which leaves room for the separation effect of Section 24.5 and for ageing. Figure 24.4 sets the levels side by side for a 132 kV system.

24.5 Surge arresters

Metal-oxide arresters

A **surge arrester** is a non-linear resistor from phase to earth that conducts negligible current at operating voltage and diverts surge current with a limited **residual voltage**. Modern arresters are **gapless metal-oxide (ZnO) varistors** whose current varies roughly as U^{α} with α of 30–50: about 1 mA (mostly capacitive) at continuous voltage, and at 10 kA a peak voltage of about 2.3–2.6 times the rated voltage, rising only 15–30 % for a further tenfold current.

Ratings and selection

IEC 60099-4 and IEEE C62.11 specify:

- **Continuous operating voltage** U_c (IEEE: MCOV): the RMS voltage the arrester can carry indefinitely. It must exceed the highest phase-to-earth operating voltage, $U_m/\sqrt{3}$, with a margin of a few per cent for harmonics.
- **Rated voltage** U_r: the RMS voltage it can withstand for 10 s after a surge; it sets TOV capability. Typically $U_c \approx 0.8 U_r$.
- **Nominal discharge current** (5, 10 or 20 kA, 8/20 µs) and **residual voltages** (lightning protective level U_{pl} at nominal current; switching protective level U_{ps} at 0.5–2 kA, 30/60 µs).
- **Energy or charge rating**: thermal energy rating W_{th} in kJ/kV of U_r, or charge transfer rating, determining which line-discharge and capacitor-switching duties it survives.
- **TOV capability**: a curve of allowed power-frequency overvoltage versus time (for example 1.15–1.2 U_r for 1 s).

A higher U_r survives TOVs better but protects worse; a lower U_r risks thermal runaway during a long earth fault.

The separation effect

An arrester limits voltage at its own terminals. Equipment some distance away sees more, because a surge with steepness S (kV/µs) travels to the equipment and reflects back from its high impedance before the arrester's limiting action reaches it. For a line of length L between arrester and an open-circuit-like transformer, the peak at the transformer is approximately

$$U_t \approx U_{pl} + \frac{2\,S\,L}{v},$$

where $v \approx 300$ m/μs. The arrester's own leads to earth add $L\,\mathrm{d}i/\mathrm{d}t$ in the same way (about 1 μH/m, so a 10 kA/μs surge adds 10 kV per metre of lead), which is why arrester earth leads are short and straight.

> **Status of the principle: Separation-distance formula**
> *Approximate.* Assumes a linearly rising surge, lossless lines, a purely capacitive or open-circuit termination and an ideal arrester; it is conservative for short distances and loses accuracy when the travel time is comparable with the surge front, when transformer capacitance resonates with lead inductance, or when corona attenuates the incoming surge. Detailed studies use electromagnetic transient (EMT) simulation.

Worked example — Protecting a 132 kV transformer. A 132 kV system ($U_m = 145$ kV) is effectively earthed with an earth-fault factor of 1.4 and fault clearance within 1 s. The transformer has LIWL 650 kV. Select an arrester and find the maximum separation distance for an incoming surge of steepness 1000 kV/μs.

Rating. $U_c \geq 1.05 \times 145/\sqrt{3} = 87.9$ kV. TOV during earth faults: $1.4 \times 145/\sqrt{3} = 117$ kV for up to 1 s. A station-class arrester with $U_r = 120$ kV has $U_c \approx 96$ kV and a 1 s TOV capability of about $1.15 \times 120 = 138$ kV, both adequate. Its lightning protective level at 10 kA is about $2.42U_r = 290$ kV.

Margin. The required withstand for internal insulation is the LIWL divided by $K_s = 1.15$: $650/1.15 = 565$ kV. With the arrester 15 m from the transformer, $U_t \approx 290 + 2 \times 1000 \times 15/300 = 390$ kV, a protective ratio of $565/390 = 1.45$, or $650/390 = 1.67$ against the LIWL. The largest separation that still meets 565 kV is $L_{\max} = (565 - 290) \times 300/(2 \times 1000) = 41$ m. In the opening story the arrester stood 35 m away: on paper inside the limit, but with a lead length of 8 m and a steeper surge from a close stroke (2000 kV/μs halves $L_{\max}$ to 21 m), the margin vanished. The lesson is the one every substation designer learns: mount transformer arresters on the tank or within a few metres of the bushings.

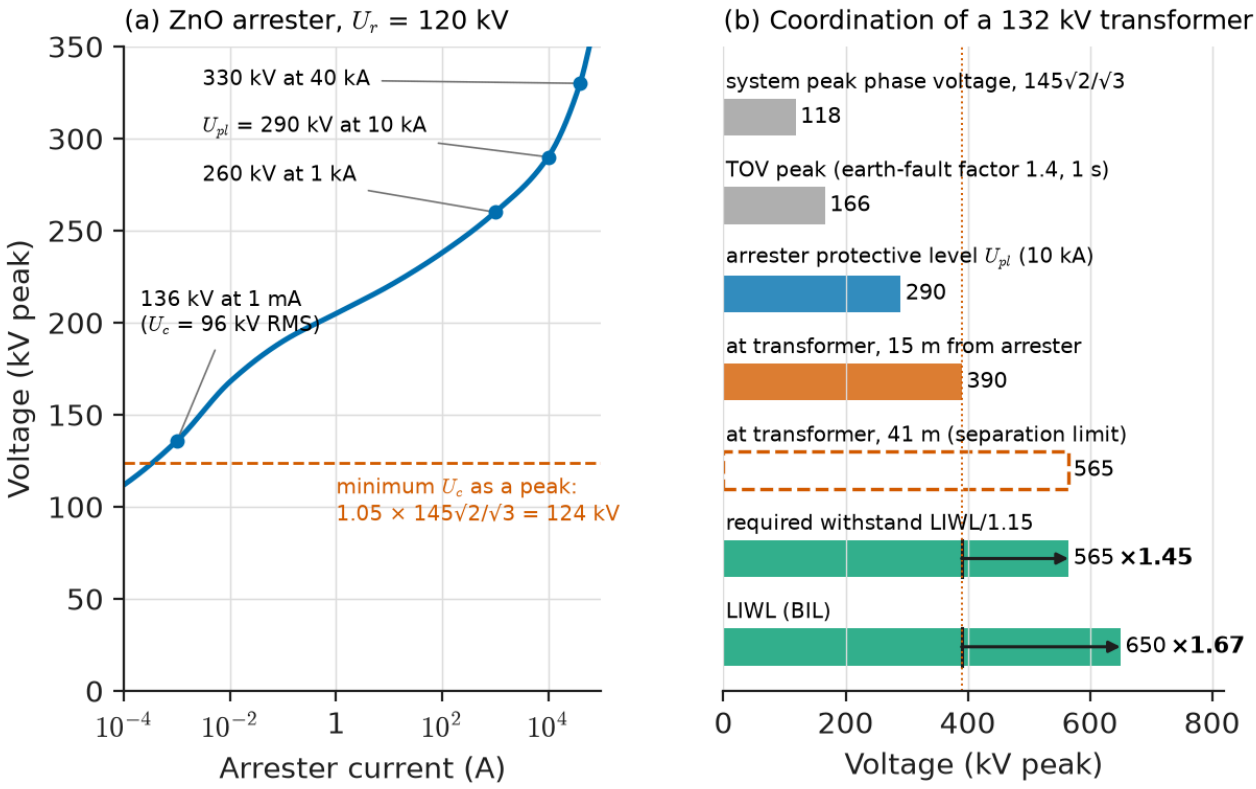


Figure 24.4 — Metal-oxide arrester V-I characteristic and the insulation coordination of a 132 kV transformer.

Where arresters are used

Besides transformers, arresters protect substation line entrances, cable terminations, rotating machines (together with surge capacitors that slow the front), PV and wind plants on both AC and DC sides (surge protective devices to IEC 61643), railway catenaries and data-centre LV intakes. **Line arresters** across insulators reduce back-flashover where footing resistance is high (Section 24.3).

24.6 Grounding

Why earthing matters

A grounding (earthing) system carries fault and surge currents into the earth so that protection operates (Chapter 23), arresters work, and the voltages on metalwork and on the ground surface during a fault are too low to injure people. The last requirement usually binds in substations; it is a problem of current spreading in a resistive medium, the soil.

Soil resistivity and electrode resistance

Soil resistivity spans five orders of magnitude.

Table 24.4 — Typical soil resistivities (vary strongly with moisture, temperature and salt content).

Soil	Resistivity (Ω·m)
Seawater	0.2–0.3
Wet organic soil, marsh	10–50
Clay, moist loam	20–100
Sand and gravel, moist	100–1000
Dry sand, gravel	1000–10 000
Granite, weathered rock	1000–10 000 and above
Crushed rock surface layer (dry)	3000–10 000

Resistivity is measured with the **Wenner** four-electrode method: four equally spaced probes at spacing a, current through the outer pair and voltage across the inner pair, giving the apparent resistivity $\rho = 2\pi a R$. Repeating at several spacings reveals layered soil, which is common and important: a dry top layer over wet clay, or soil over rock.

A single vertical rod of length L and diameter d in uniform soil has resistance (Dwight's formula)

$$R = \frac{\rho}{2\pi L}\left[\ln\frac{8L}{d} - 1\right].$$

A 3 m rod of 16 mm diameter in 100 Ω·m soil has $R = 33.5\ \Omega$. Rods in parallel help only if spaced by more than about their length, because their potential fields overlap. Substations therefore use a buried **grid** of bare

copper (or copper-clad steel) conductors at 0.5–1 m depth, meshed at 3–10 m spacing, supplemented by rods. Its resistance is estimated by Sverak's formula (IEEE 80):

$$R_g = \rho \left[\frac{1}{L_T} + \frac{1}{\sqrt{20A}} \left(1 + \frac{1}{1 + h\sqrt{20/A}} \right) \right],$$

with L_T the total buried conductor length, A the grid area and h the depth. A 60 m × 60 m grid with 2400 m of conductor at 0.5 m depth in 100 Ω·m soil has $R_g = 0.77\ \Omega$. The grid resistance is dominated by its area, roughly $\rho/(4\sqrt{A/\pi})$, and adding conductor beyond a point mainly improves the voltage distribution inside the grid rather than its resistance.

> **Status of the principle: Electrode resistance formulas**
> *Approximate.* Derived from current spreading in a uniform, isotropic half-space with idealised electrode shapes; real soils are layered and seasonal, so formulas give first estimates and two-layer or multilayer computer models fitted to Wenner data are used for final design.

Ground potential rise, touch and step voltage

When an earth-fault current I_G flows into the grid, the whole grid rises to the **ground potential rise** $\mathrm{GPR} = I_G R_g$ relative to remote earth, while the soil surface potential falls away from the grid conductors. A person touching earthed metal while standing on the surface between conductors bridges a **touch voltage**; a person walking bridges a **step voltage** between their feet; a person touching a fence or a telecom cable connected to remote earth may be exposed to a **transferred potential** close to the full GPR (Figure 24.5).

IEEE 80 sets the tolerable voltages from the empirical body-current limit $I_B = 0.116/\sqrt{t_s}$ A for a 50 kg person (Dalziel), a body resistance of 1000 Ω, and the foot resistance contributed by the surface layer:

$$E_{\text{touch}} = (1000 + 1.5\, C_s\, \rho_s)\frac{0.116}{\sqrt{t_s}}, \qquad E_{\text{step}} = (1000 + 6\, C_s\, \rho_s)\frac{0.116}{\sqrt{t_s}},$$

where ρ_s is the surface-layer resistivity, C_s a derating factor for a thin layer and t_s the fault duration. The European approach (EN 50522, IEC 60479-1) uses a curve of permissible touch voltage versus duration that leads to similar values.

> **Status of the principle: Tolerable touch and step voltage**
> *Empirical.* Built on Dalziel's fibrillation threshold fitted to animal experiments and on body-resistance statistics; IEC 60479-1 gives a different, current-path-dependent model, and the two methods can differ by a factor of two at some durations.

Worked example — Is the substation safe to touch? The grid above carries $I_G = 5$ kA (the part of a 132 kV earth-fault current that returns through earth rather than through shield wires and cable sheaths) for $t_s = 0.5$ s. Soil resistivity is 100 Ω·m; the surface is covered by a 0.1 m layer of crushed rock with $\rho_s = 3000$ Ω·m.

$\mathrm{GPR} = 5000 \times 0.774 = 3.87$ kV. Surface-layer factor: $C_s = 1 - 0.09(1 - 100/3000)/(2 \times 0.1 + 0.09) = 0.700$.

Tolerable touch voltage: $(1000 + 1.5 \times 0.700 \times 3000) \times 0.116/\sqrt{0.5} = 681$ V. Tolerable step voltage: $(1000 + 6 \times 0.700 \times 3000) \times 0.116/\sqrt{0.5} = 2231$ V. Without the crushed-rock layer ($\rho_s = 100$ Ω·m, $C_s = 1$), tolerable touch voltage falls to 189 V.

The GPR is nearly six times the tolerable touch voltage, so safety depends on how evenly the grid holds the surface potential. The worst touch voltage is the **mesh voltage**, at the centre of a corner mesh. For this grid (20 × 20 conductors of 10 mm diameter at 3.16 m spacing) IEEE 80's mesh-voltage formula gives about 440 V and a numerical uniform-soil model about 460 V (Figure 24.5), 11–12 % of GPR: below the 681 V limit with the crushed rock, but far above the 189 V limit without it. The crushed-rock layer, which raises the tolerable touch voltage 3.6-fold, is what makes this grid safe. Mesh voltages of 10–25 % of GPR are typical: a coarser 6 m mesh using only 1320 m of conductor would reach about 17–19 % (680–730 V), at or above the limit, so every design is checked and, where needed, refined with closer conductor spacing at the corners and periphery and rods around the perimeter. The perimeter fence is bonded to the grid with a conductor buried 1 m outside it, and communication circuits leaving the site are isolated against the transferred GPR. The full IEEE 80 design procedure is part of Chapter 30.

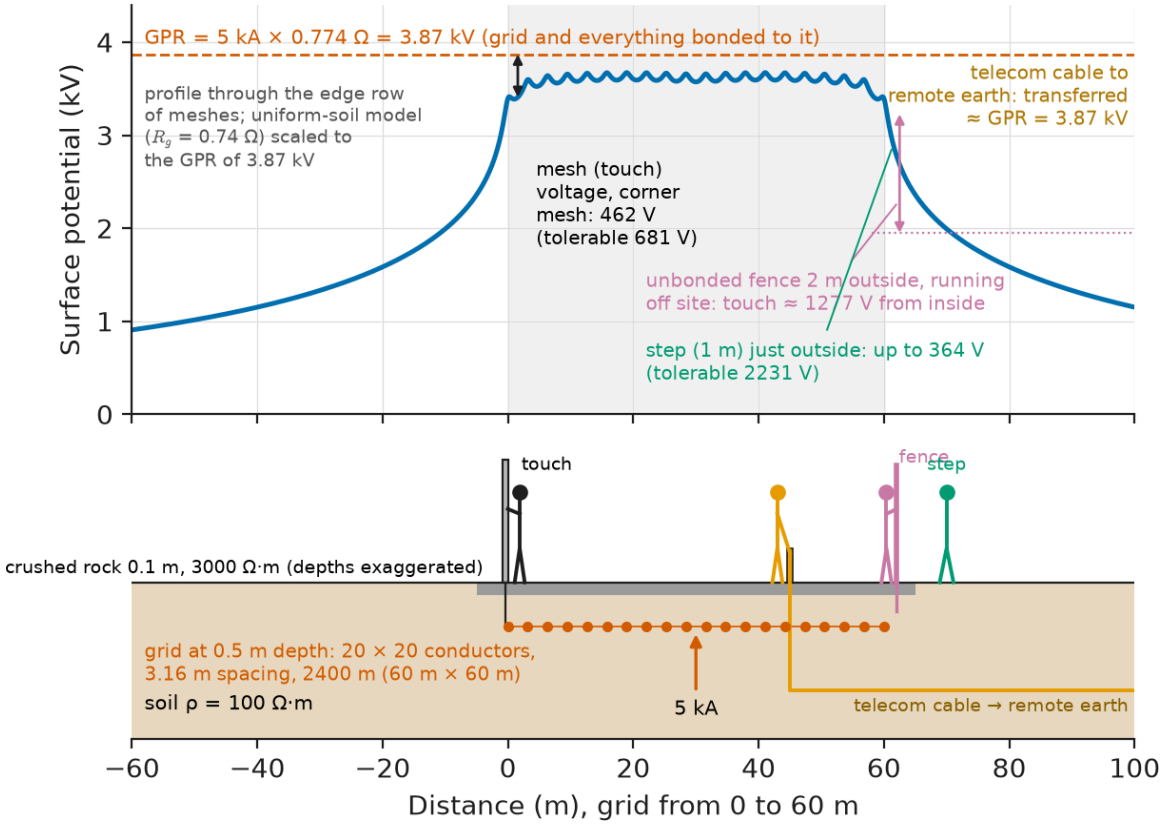


Figure 24.5 — Ground potential rise of a substation grid during an earth fault, with touch, step, mesh and transferred voltages.

Earthing of lines, LV systems and special installations

Transmission towers are earthed with rods or counterpoise wires to reach the 10–20 Ω footing resistance needed for lightning performance (Section 24.3). LV earthing follows the TN/TT/IT arrangements of Chapters 29 and 30. Wind farms interconnect all turbine foundations, PV plants bond structures over square kilometres, and building lightning protection (IEC 62305) shares the same electrode.

24.7 High-voltage generation and measurement

Alternating voltages

Power-frequency test voltages up to about 600–800 kV are generated by single-unit **test transformers** with a low-loss, low-PD design and an excitation winding fed from a regulating transformer. Higher voltages use **cascaded** transformers, each stage's tertiary winding exciting the next, to 2–3 MV. Testing large capacitive objects (cables, GIS, generator stators) with a test transformer would require a source rated at the full reactive power of the object. **Series resonant** test sets avoid this: a variable reactor in series with the capacitive test object is tuned to resonance, either by varying the inductance at fixed frequency or, more commonly for cables, by using a fixed reactor and varying the frequency between 20 and 300 Hz. The source then supplies only the losses, about $1/Q$ of the reactive power, with quality factors Q of 40–200.

Worked example — Resonant test of a 132 kV cable. A 5 km, 132 kV XLPE cable has capacitance 0.2 µF/km per phase, 1.0 µF in total. It is to be tested at $2U_0 = 152$ kV with a 20 H reactor and a variable-frequency source. Find the test frequency, the reactive power, the current and the supply power for $Q = 80$.

$f = 1/(2\pi\sqrt{LC}) = 1/(2\pi\sqrt{20 \times 1.0 \times 10^{-6}}) = 35.6$ Hz, inside the 20–300 Hz band allowed by IEC 60840 for after-installation tests. Current $I = 2\pi f C U = 2\pi \times 35.6 \times 10^{-6} \times 152\,000 = 34.0$ A. Reactive power $2\pi f C U^2 = 5.17$ Mvar. Supply power $5.17/80 \approx 65$ kW, which a trailer-mounted set can provide; a 50 Hz test transformer would need to be rated for 7.3 Mvar at 152 kV.

Direct voltages

High direct voltages for testing HVDC cables, insulation-resistance measurements and research are generated by rectifier circuits, usually the **Cockcroft–Walton** voltage multiplier: a ladder of n stages of capacitors and diodes fed from a transformer of peak voltage U_m, giving an ideal no-load output of $2nU_m$. Under a load current I at supply frequency f with stage capacitance C, the output ripple and the voltage drop are

$$2\delta U = \frac{I}{fC}\,\frac{n(n+1)}{2}, \qquad \Delta U = \frac{I}{fC}\left(\frac{2n^3}{3} + \frac{n^2}{2} - \frac{n}{6}\right).$$

> **Status of the principle: Cockcroft–Walton ripple and regulation**
> *Derived.* Follows from charge balance on the stage capacitors with ideal diodes and equal capacitances, and assumes the ripple is small compared with the stage voltage; it explains why stage count, not voltage per stage, limits practical multipliers.

Worked example — A 1 MV DC test set. A five-stage multiplier with $C = 0.1$ µF per stage is fed at 50 Hz from a transformer of 100 kV peak, and loaded with 2 mA. The ideal output is $2 \times 5 \times 100 = 1000$ kV. With $I/(fC) = 0.002/(50 \times 10^{-7}) = 400$ V, the ripple is $400 \times 15 = 6.0$ kV peak-to-peak and the drop is $400 \times (83.3 + 12.5 - 0.83) = 38$ kV, so the output is 962 kV with 0.6 % ripple. Feeding the multiplier at 500 Hz from a frequency converter cuts both tenfold, to 0.6 kV and 3.8 kV, which is why modern DC test sets use medium-frequency supplies. Because ΔU grows as n^3, doubling the number of stages at the same current increases the drop roughly eightfold.

Impulse voltages

Lightning and switching impulses are produced by the **Marx generator** (Figure 24.6): n capacitors charged in parallel to a DC voltage U_0 through high resistors, then switched into series by triggered spark gaps to give

nearly nU_0. The waveshape is set by a **front resistor** R_f in series with the load capacitance C_b (test object plus divider) and a **tail resistor** R_t discharging the generator capacitance C_s. Approximately,

$$T_1 \approx 3 R_f \frac{C_s C_b}{C_s + C_b}, \qquad T_2 \approx 0.7\, R_t\,(C_s + C_b),$$

so for $C_s = 20$ nF (erected) and $C_b = 2$ nF, a 1.2/50 µs impulse needs $R_f \approx 220\ \Omega$ and $R_t \approx 3.2$ kΩ, and the voltage efficiency is about $C_s/(C_s + C_b) \approx 0.9$. Large laboratories have Marx generators of 3–6 MV and 200–500 kJ; switching impulses use larger front and tail resistors, or the test transformer itself driven by a capacitor discharge.

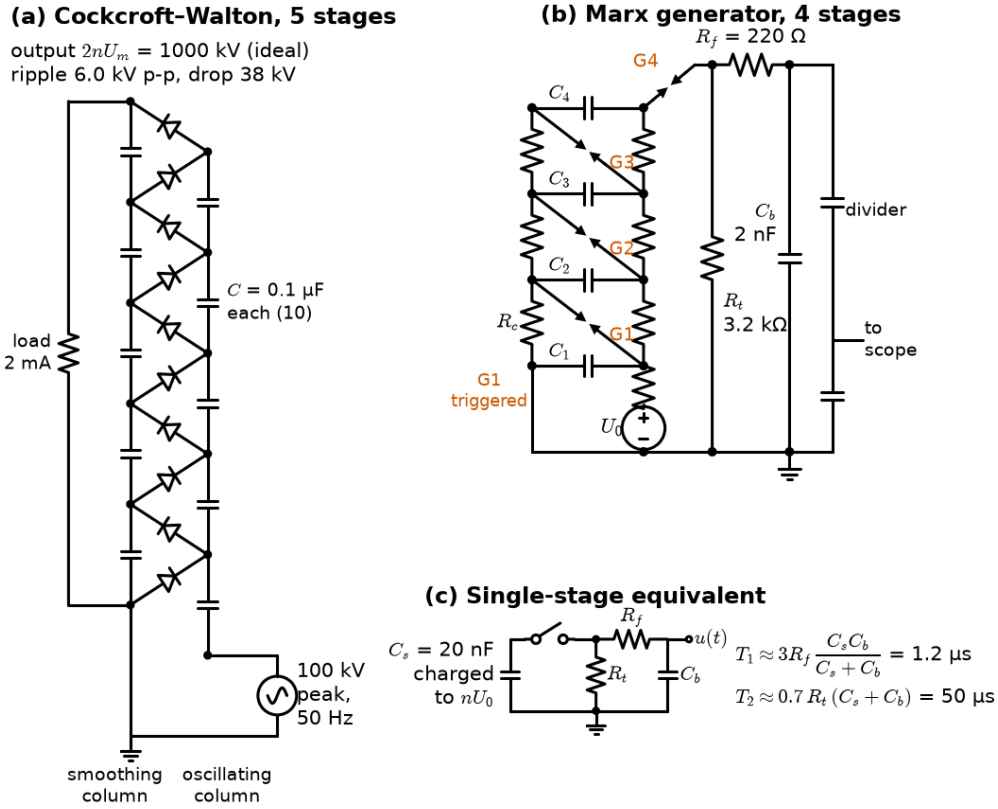


Figure 24.6 — A five-stage Cockcroft-Walton multiplier and a Marx impulse generator with its single-stage equivalent circuit.

Measuring high voltages

IEC 60060-2 requires approved measuring systems with an expanded uncertainty of 3 % or better for peak values. AC is measured with **capacitive dividers** (compressed-gas capacitors as the HV arm) and DC with **resistive dividers**. Impulses need **damped capacitive** or **mixed RC dividers** with response times of tens of nanoseconds, calibrated together with their leads and digital recorder (IEC 61083-1). **Sphere gaps** (IEC 60052), which break down at a known peak voltage for a given diameter and spacing, serve as an independent check to about 3 %.

24.8 High-voltage testing

Kinds of tests

High-voltage tests are grouped by purpose. **Type tests** on one unit prove a design (full impulse withstand, temperature rise, short-circuit withstand). **Routine tests** on every unit find manufacturing defects (power-frequency withstand, PD, capacitance and tan δ). **Commissioning tests** prove that transport and installation did no damage. **Diagnostic tests** in service track ageing ($\tan\delta$ and PD trends, polarisation index, dissolved gas analysis of transformer oil). Techniques are standardised in IEC 60060-1 and IEEE Std 4; levels come from IEC 60071-1 and the product standards.

Table 24.5 — Representative high-voltage tests.

Test	Typical object	Waveshape and duration	Acceptance
Power-frequency withstand	All equipment	50/60 Hz, 1 min at the short-duration withstand level	No breakdown or flashover
Lightning impulse withstand	Transformers, switchgear, bushings	1.2/50 µs, 3 or 15 shots per polarity; chopped waves for transformers	No breakdown; waveform comparison with reduced-level reference shots
Switching impulse withstand	Equipment with $U_m >$ 245 kV	250/2500 µs, 15 shots	≤ 2 flashovers in 15 on self-restoring insulation
Partial discharge	Cables, transformers, GIS, instrument transformers	AC at 1.1–1.8 times phase voltage	Product limits, e.g. ≤ 10 pC at $1.73U_0$ for MV cables
Capacitance and tan δ	Bushings, CTs, cables, machines	AC at 2–12 kV or rated phase voltage	Absolute value and trend; e.g. tan δ rising with voltage indicates PD or moisture
Insulation resistance and polarisation index	Machines, transformers, cables	DC 0.5–10 kV, 1 and 10 min	PI ≥ 2 for class F/H machine windings (IEEE 43)
AC after-installation test	HV and EHV cables	20–300 Hz, about 1.4–2 U_0 for up to 1 h	No breakdown, optionally with PD
VLF test	MV cables	0.1 Hz cosine-rectangular or sine, about 3 U_0 (IEEE 400.2), 15–60 min	No breakdown; tan δ criteria

Testing transformers and cables

A power transformer's **induced voltage test** runs at 100–200 Hz, so that the core is not overfluxed at about twice rated voltage, with PD measured throughout. In its **lightning impulse test**, winding faults are found by comparing the neutral-current waveforms of reduced-level (50–75 %) and full-level shots: a change of shape means an internal turn or section breakdown.

Extruded (XLPE) cables should not be tested with DC, which leaves space charge that can cause failure when AC is reapplied. On-site tests use resonant AC (Section 24.7) or VLF at 0.1 Hz, which needs 500 times less reactive power than 50 Hz at the cost of a less representative stress.

Test voltages on external insulation are corrected to standard atmosphere (20 °C, 101.3 kPa, 11 g/m³) by IEC 60060-1 factors; wet and pollution tests qualify outdoor insulators.

24.9 Safety procedures

Why high-voltage work is different

High voltage kills without contact: air breaks down, and approaching a 132 kV conductor within about a metre can draw an arc to the body. Capacitive objects stay charged after disconnection: a 5 km cable charged to 150 kV DC stores about $\frac{1}{2}CU^2 = 11$ kJ, enough to kill, and can recover a dangerous voltage after being briefly earthed, through dielectric absorption. Induced voltages from adjacent energised lines appear on de-energised conductors. Every procedure is built around these three facts.

The five safety rules

European practice (EN 50110-1) codifies the sequence for making equipment safe before work, which, under different names, is universal:

1. **Disconnect completely** from all sources, including backfeeds from DG, generators, VTs and parallel circuits.
2. **Secure against reconnection**: lock out and tag the isolating devices (lockout/tagout, Chapter 29).
3. **Verify the absence of voltage** with an approved detector, tested on a known live source before and after use.
4. **Earth and short-circuit** all phases at the point of work with rated portable earthing equipment, connecting the earth end first and the line end last with an insulated stick.
5. **Protect against adjacent live parts** with barriers, covers and marked boundaries.

Removal follows the reverse order, under the control of the person in charge, with a formal permit-to-work system recording who is working where and on what.

Approach distances

Standards define zones around exposed live parts that only authorised and equipped people may enter. NFPA 70E (and OSHA 1910.269 in the US, EN 50110 in Europe) gives approach boundaries that rise with voltage.

Table 24.6 — Approximate shock approach boundaries for exposed fixed AC conductors (NFPA 70E; check the current edition for exact values).

Nominal voltage, phase-to-phase	Limited approach boundary	Restricted approach boundary
751 V – 15 kV	1.5 m	0.66 m
15.1 – 36 kV	1.8 m	0.78 m
46.1 – 72.5 kV	2.5 m	1.0 m
138 – 145 kV	3.0 m	1.2 m
230 – 242 kV	4.0 m	1.7 m
345 – 362 kV	4.7 m	2.8 m
500 – 550 kV	5.8 m	3.6 m
765 – 800 kV	7.2 m	4.9 m

Arc-flash boundaries, often larger than shock boundaries at medium voltage, are determined by incident-energy analysis (Chapter 29).

High-voltage test areas

Test areas add interlocked fences or barriers that remove supply when a gate opens, "energised" and "safe" warning lights, earthing switches and sticks at the test object, emergency stops outside the area, clearances scaled with test voltage, and a single person in charge of energising. After a DC or VLF test a cable is discharged through a resistor, then solidly earthed and kept earthed for several times the test duration so that absorbed charge drains. On site, the far end of a cable, often kilometres away and out of sight, must be confirmed isolated and guarded.

Most high-voltage accidents are failures of procedure, not of insulation: a wrong circuit isolated, an earth removed early, a far end left unguarded. Authorisation by role, written switching programmes, two-person verification and the rule that no conductor is dead until it is earthed are the defences; Chapter 29 develops them with the shock and arc-flash hazards behind them.

What remains open

Open question	Where it is answered
How do transient overvoltages, notches and switching transients appear as power-quality disturbances at customers' terminals?	Chapter 25
How do inverter-based plants contribute to TOV and how do grid codes limit it?	Chapter 27
What current through the body is dangerous, and how are shock and arc-flash hazards assessed and controlled?	Chapter 29
How are substation earthing grids, cable insulation levels and surge protection specified in a full design?	Chapter 30
How are relays and earthing methods chosen for earth-fault detection?	Chapter 23 §23.5 (back)
How do insulation ageing and partial discharges limit machine life and rating?	Chapter 9 (back)
How are corona and the Ferranti effect handled on transmission lines?	Chapter 21 §21.6–21.7 (back)
How do travelling waves reflect and refract at junctions?	*Circuits, Signals and Communications*, Ch. 14

Next: Chapter 25 opens Part VII by following voltage disturbances, from slow variations through harmonics and flicker to sags and transients, to the customers who feel them, and asks how power quality is measured and improved.

Part VII — Power Quality and Grid Modernization

Parts IV to VI built the power system of the twentieth century: large synchronous generators, transmission, one-way distribution to passive customers, and protection. This part deals with what has changed. Customers' loads are now dominated by power electronics that distort current and are sensitive to brief disturbances; generation is moving to inverter-connected wind, solar and storage, much of it on distribution feeders and behind customers' meters; the network is wrapped in sensors, communication links and software that let it observe and control itself; and energy use itself has become something to be measured, managed and reduced. The common thread is that delivering energy is no longer enough. The quality of the voltage waveform, the behaviour of inverters in the first second after a fault, the flexibility of demand and the efficiency of use are now engineering quantities with limits, standards, measurements and costs of their own.

By the end of this part you should understand

- How supply voltage departs from the ideal balanced sinusoid (magnitude and frequency variation, harmonics, flicker, transients, sags and swells, unbalance) and what limits IEC, IEEE and EN standards set on each.
- How harmonic voltages, sag depths, flicker and unbalance are estimated from fault levels and load data, measured to IEC 61000-4-30, and mitigated, including the harmonic resonance of capacitor banks.
- How smart meters, grid communication networks, substation and distribution automation, state estimation and synchrophasors give an operator visibility and control down to the customer.
- How distributed resources, microgrids and demand response add flexibility, how energy management systems from the control centre to the building close the loop, and what cybersecurity and resilience demand of them.
- How grid-connected inverters synchronise to the grid, control active and reactive power, and meet grid codes for ride-through, frequency and voltage support and anti-islanding.
- Why falling inertia and system strength limit the instantaneous share of inverter-based generation, and how storage, forecasting, grid-forming control and synchronous condensers push that limit back.
- How energy is audited and managed in plants and buildings, how motor efficiency, power-factor correction and loss reduction are evaluated, and how life-cycle cost decides between options.

Chapters

Chapter	Central question	What it delivers
Chapter 25 — Power Quality	In what ways does the delivered voltage depart from the ideal sinusoid, and how are those departures predicted, limited, measured and mitigated?	Voltage and frequency variation, harmonics and IEEE 519, flicker, switching transients, sags, unbalance, detuned power-factor correction, measurement and mitigation
Chapter 26 — Smart Grids	What does it take, in sensors, communications and software, for the network to observe and control itself, and what new risks come with it?	AMI, grid communications, DER and VPPs, microgrids, demand response, FLISR and volt/var control, PMUs, EMS and AGC, cybersecurity, resilience
Chapter 27 — Grid Integration of Renewable Energy	How must inverter-based plants behave so that the grid stays stable as they replace synchronous machines?	Inverter control and PLLs, grid strength and grid-forming control, grid codes and ride-through, anti-islanding, frequency and voltage support, storage, forecasting, penetration limits
Chapter 28 — Energy Efficiency and Management	Where is electrical energy wasted at the point of use, and how are savings found, verified and justified?	Energy audits and load profiles, demand management, industrial and motor efficiency, power-factor correction economics, monitoring, loss reduction, life-cycle energy cost and improvement strategies

How this part connects to the rest of the book

Part VII stands on almost everything before it. Harmonic resonance and sag calculations use the fault levels and network models of Chapters 17 and 19; frequency support and penetration limits rest on the stability theory of Chapter 20; voltage rise from distributed generation and feeder automation extend Chapter 22; ride-through and inverter fault current return to the protection of Chapter 23; switching transients connect to the surge and insulation coordination of Chapter 24. The machines of Part II are what harmonics heat and unbalance derates, the drives of Part III are the main harmonic sources, and the plants and batteries of Chapters 15 and 16 are the resources whose grid behaviour Chapter 27 specifies. Part VIII takes the same installations into safety and design practice.

Among the sibling books, the circuit and signal theory comes from *Circuits, Signals and Communications*: AC power, power factor and harmonics (Ch. 9), three-phase circuits and symmetrical components (Ch. 10), resonance and switching transients (Ch. 7), the DFT used by power-quality instruments (Ch. 22), the communication theory beneath grid networks (Part VI) and the feedback theory beneath PLLs, droop and AGC (Part VII). Power-electronic converters are outlined in *Engineering Physics* (Ch. 18 §18.8) and their circuits belong to *Electronics & Computing*.

Chapter 25 — Power Quality

Part VII · Power Quality and Grid Modernization

A semiconductor fab in an industrial park loses a day of production twice a year, and each time the utility's records show that the supply was never interrupted. What the fab's monitor recorded was a dip to 62 % of nominal voltage lasting 110 ms, caused by a lightning flashover on a 33 kV line feeding a different town and cleared by protection in under six cycles; the fab's vacuum pumps and process support systems dropped out. In the same park a plastics plant has installed new variable-speed drives on its extruders, and since then a 3 Mvar capacitor bank that had worked for twenty years blows fuses every few weeks. Across the road a steelworks' arc furnace makes the lights in the neighbouring housing estate flicker visibly in the evening. In each case the energy was delivered; what failed was its *quality*, the departure of the waveform from a balanced sinusoid at nominal magnitude and frequency, and each costs real money.

> **What are the standard ways in which supply voltage departs from the ideal sinusoid, and what limits do standards set on each?**
> **How are disturbances such as harmonics, sags, flicker and unbalance predicted from network data, and how are they measured and mitigated at the source, in the network or at the sensitive load?**

Concept	Engineering destination
Voltage variation	Statutory supply limits, tap-changer and regulator settings, motor and lighting performance
Frequency variation	Generator and motor protection, island and marine systems, aircraft 400 Hz supplies
Harmonics	IEEE 519 compliance of drive installations, transformer derating, capacitor-bank failures
Flicker	Arc furnaces, welders, wind farms, compensation with SVC and STATCOM
Transients	Capacitor switching, breaker duty, drive overvoltage trips, surge protection
Voltage sags and swells	Process ride-through, SEMI F47 compliance, contactor drop-out, DVR and UPS selection
Unbalance	Traction supplies, motor derating, negative-sequence protection
Power-factor correction	Tariff penalties, detuned capacitor banks, static compensators
Measurement and mitigation	IEC 61000-4-30 monitoring, filters, active front ends, custom power devices

Roadmap. Slow deviations of voltage (25.1) and frequency (25.2) → waveform distortion by harmonics (25.3) → flicker (25.4) → transients (25.5) → sags and swells (25.6) → unbalance (25.7) → power-factor correction in a harmonic-rich plant (25.8) → measurement and mitigation (25.9).

Boundary.
The circuit theory of this chapter is taken from *Circuits, Signals and Communications*: the Fourier series of a periodic waveform (Ch. 18), THD, active power as a sum over harmonics, and the split of power factor into displacement and distortion factors (Ch. 9 §9.8), capacitor sizing for power-factor correction (Ch. 9 §9.6), triplen harmonics as zero sequence and the first look at symmetrical components and unbalance (Ch. 10 §10.6, §10.9), the inrush of a single capacitor bank and the resonance $f_r = f_1\sqrt{S_{sc}/Q_C}$ (Ch. 7 §7.7–7.8), and DFT measurement with harmonic grouping (Ch. 22). They are used here, not re-derived. Rectifier and inverter circuits are outlined in *Engineering Physics*, Ch. 18 §18.8; their internals belong to *Electronics & Computing*.

25.1 Voltage variation

What is being regulated

Power quality is overwhelmingly a property of **voltage**. The customer controls the current it draws; a distorted or fluctuating current becomes a problem when it flows through the supply impedance and distorts the voltage seen by everyone else. The point where the network operator's responsibility meets the customer's is the **point of common coupling** (PCC), and almost every limit in this chapter is stated there.

The slowest departure is a change of the RMS magnitude itself, lasting minutes to hours: the evening sag at the end of a long rural feeder, the noon rise on a feeder full of rooftop PV, the drift as an on-load tap changer hunts between steps. IEEE 1159 calls an RMS deviation lasting more than one minute an **undervoltage** (0.8–0.9 pu) or **overvoltage** (1.1–1.2 pu). Supply standards define the acceptable band statistically, because the voltage at any customer varies continuously with load (Table 25.1).

Table 25.1 — Steady-state voltage limits in common supply standards.

Standard	System	Limit	Assessment
EN 50160 (Europe)	LV and MV public supply	$U_n \pm 10\,\%$	95 % of 10-min means in each week; all 10-min means within $+10\,\%/-15\,\%$ at LV
ANSI C84.1 (North America)	120 V service	Range A 114–126 V ($\pm 5\,\%$); Range B 110–127 V	Range A normal; Range B infrequent, must be corrected
Typical transmission grid code	400 kV, 220 kV	$\pm 5\,\%$ to $\pm 10\,\%$	Continuous operating range
IEC 60092 (ships)	Shipboard AC	$+6\,\%/-10\,\%$ steady state	Main switchboard

The bands are wider than any one piece of equipment would like. They exist because the voltage drop $\Delta V \approx (RP + XQ)/V$ along a feeder (Chapter 22) must be shared between the MV network, the distribution transformer, the LV feeder and the service cable, and each takes a few per cent.

Consequences for equipment

The bands reflect how equipment responds. An induction motor's torque scales roughly as V^2: at 90 % voltage it develops 81 % of its torque at a given slip, so a loaded motor runs at higher slip and current and its windings run hotter (Chapter 6); at 110 % the core saturates and magnetising current rises steeply. A 3 kW heater gives 2.43 kW at 90 % voltage. LED drivers and electronic power supplies regulate their output and are insensitive inside the band, but they draw constant power, so their current rises as voltage falls and deepens the drop on a weak feeder. Capacitor kvar rises as V^2 and sustained overvoltage shortens capacitor life (IEC 60831 permits 1.10 pu for 8 h in 24 h).

Control

Sustained voltage is controlled by on-load tap changers, line regulators, switched feeder capacitors and, increasingly, inverter reactive power (Chapters 18, 22 and 27). Power-quality engineering adds the statistics: a voltage inside $\pm 10\,\%$ on average but outside it for two hours every evening is a compliance problem even if nobody complains.

25.2 Frequency variation

Why frequency is a system-wide quantity

Voltage varies from bus to bus; frequency, in an interconnected AC system in steady state, is the same everywhere. It is set by the balance between mechanical power into the synchronous generators and electrical power out, through the inertia of every rotating mass on the system (Chapter 20). A frequency deviation is therefore never a local power-quality problem in a large grid. It is a symptom of a generation–load imbalance that the whole system's governors, reserves and, ultimately, under-frequency load shedding are working to correct.

Table 25.2 — Frequency limits in representative systems.

System	Normal range	Wider range
Continental Europe (EN 50160, synchronous)	50 Hz ±1 % (49.5–50.5 Hz) for 99.5 % of a year	47–52 Hz (+4 %/−6 %) for 100 % of the time
Island system (EN 50160, not synchronously connected)	50 Hz ±2 % for 95 % of a week	42.5–57.5 Hz (±15 %) for 100 % of the time
Ships (IEC 60092-101)	±5 % steady state	±10 % transient, recovery in 5 s
Aircraft (MIL-STD-704F, 400 Hz fixed frequency)	393–407 Hz steady state	Variable-frequency systems 360–800 Hz

In a large interconnection frequency rarely leaves ±0.1 Hz, well inside Table 25.2. Frequency variation is a genuine quality problem in small systems: island grids (a Pacific island, a mining camp, an oil platform), microgrids in islanded mode (Chapter 26), ships with a few diesel generators where starting a 1 MW bow thruster swings the frequency by a hertz, and grids with low inertia, where the rate of change of frequency after a large loss of infeed is fast (Chapter 27).

Who notices

Motors change speed in proportion to frequency, and pumps and fans change power roughly as its cube. Transformers and motors are flux-limited by V/f (Chapters 4 and 6), so low frequency at rated voltage over-fluxes them; generator protection includes a volts-per-hertz element alarming at 1.05–1.10 pu. Steam-turbine blades have resonances near running speed, and operation outside about 47.5–51.5 Hz is time-limited by the turbine maker, which often sets the frequency protection of conventional plant. Inverters and loss-of-mains relays trip on frequency and on rate of change of frequency (ROCOF), and their mass tripping during an excursion is a stability problem in its own right (Chapter 27). Frequency is measured and reported as a power-quality parameter (IEC 61000-4-30 Class A: ± 10 mHz over 10 s), but it is controlled by the balancing machinery of Chapters 20, 26 and 27, not by anything a single customer can install.

25.3 Harmonics

Sources and their spectra

Circuits, Signals and Communications (Ch. 9 §9.8) analysed one distorted current; the engineer needs the spectra of real sources, because the spectrum decides which limits are violated and which filters work. The workhorse source is the **six-pulse bridge rectifier** at the front of most drives, UPS units, chargers and DC traction substations. With a balanced supply and smooth DC current its line current contains only the **characteristic harmonics**

$$h = kp \pm 1, \qquad k = 1, 2, 3, \ldots$$

where p is the pulse number. For $p = 6$ these are the 5th, 7th, 11th, 13th, 17th, 19th, … Ideally $I_h \approx I_1/h$; commutation overlap and finite DC inductance modify this (Table 25.3). Twelve-pulse converters, two bridges fed through star and delta windings 30° apart, cancel the 5th, 7th, 17th and 19th, leaving $h = 11, 13, 23, 25, \ldots$

Table 25.3 — Typical harmonic current spectra, % of fundamental.

Source	3rd	5th	7th	11th	13th	THD_I
Ideal six-pulse ($1/h$)	0	20	14.3	9.1	7.7	30 (to 50th)
Six-pulse drive, DC choke or 3 % AC reactor	0–2	30–40	8–15	5–8	3–6	33–45
Twelve-pulse drive	0–1	2–5	1–3	6–9	4–6	10–15
Active-front-end drive (PWM rectifier)	<1	1–3	1–2	<1	<1	3–5
Single-phase SMPS without PFC	70–85	45–60	25–40	5–10	5–8	80–150
Arc furnace, melting phase	4–8	3–6	2–4	1–2	1	5–15, plus interharmonics

Other sources produce non-characteristic or non-integer components: arc furnaces and cycloconverters produce **interharmonics** between the harmonic lines; transformer energisation produces a decaying inrush rich in the 2nd harmonic (used to block differential relays, Chapter 23); and PV and wind inverters inject little at low order but emit at their switching frequencies, in the **supraharmonic** range of 2–150 kHz.

Harmonic voltage at the PCC

The harmonic current from a non-linear load is, to a first approximation, independent of the supply: the load is modelled as a **harmonic current source** I_h at each order. The harmonic voltage it produces at the PCC is that current times the network's harmonic impedance. For a network dominated by inductive source impedance,

$$V_h \approx h\, X_s\, I_h, \qquad X_s = \frac{V^2}{S_{sc}}$$

so harmonic voltage distortion grows with harmonic order, with the load's harmonic current, and inversely with fault level. A drive that is harmless on a 500 MVA bus can be a nuisance on a 50 MVA one.

> **Status of the principle: Current-source model of harmonic loads with inductive network impedance**
> *Approximate*. Valid when the load's harmonic current depends weakly on supply distortion and the network is a lumped inductance; it fails near network resonances, at high orders where cable capacitance matters, and for rectifier–capacitor loads, whose harmonics partly cancel when many are aggregated.

Limits

Equipment limits cap what a product may emit (IEC 61000-3-2 up to 16 A per phase, IEC 61000-3-12 for 16–75 A). **System** limits cap what an installation may inject at the PCC and the voltage distortion the network must keep; the most widely used is IEEE 519-2022 (Tables 25.4 and 25.5), with IEC/TR 61000-3-6 following similar logic and EN 50160 limiting public-supply THD_V to 8 %.

Table 25.4 — IEEE 519-2022 voltage distortion limits at the PCC.

Bus voltage at PCC	Individual harmonic (%)	THD_V (%)
$V \le 1$ kV	5.0	8.0
1 kV $< V \le$ 69 kV	3.0	5.0
69 kV $< V \le$ 161 kV	1.5	2.5
$V >$ 161 kV	1.0	1.5

Table 25.5 — IEEE 519-2022 current distortion limits, 120 V to 69 kV, in % of maximum demand current I_L (odd harmonics; even harmonics limited to 25 % of these).

I_{sc}/I_L	$3 \le h < 11$	$11 \le h < 17$	$17 \le h < 23$	$23 \le h < 35$	$35 \le h \le 50$	TDD
< 20	4.0	2.0	1.5	0.6	0.3	5.0
20–50	7.0	3.5	2.5	1.0	0.5	8.0
50–100	10.0	4.5	4.0	1.5	0.7	12.0
100–1000	12.0	5.5	5.0	2.0	1.0	15.0
> 1000	15.0	7.0	6.0	2.5	1.4	20.0

The current limits are normalised to the **maximum demand current** I_L (the average of monthly peak demands), and the summary quantity is the **total demand distortion** $\text{TDD} = \sqrt{\sum I_h^2}/I_L$, so a lightly loaded plant does not fail merely because its fundamental is small. The limits grow with I_{sc}/I_L because a customer small relative to the network takes a small share of its distortion budget.

> **Status of the principle: IEEE 519 harmonic limits**
> *Empirical.* Engineering consensus, set so that many compliant customers together keep network distortion below the levels at which equipment heating and malfunction become common; assessed statistically (95th and 99th percentiles), not as instantaneous caps.

Worked example — An 11 kV drive installation against IEEE 519. A plastics plant takes supply at 11 kV from a bus with a three-phase fault level of 250 MVA. Its maximum demand is 8 MVA. A new extruder line adds six-pulse drives (with DC chokes) totalling 3 MW at a displacement power factor of 0.95, drawing the spectrum $I_5 = 30\,\%$, $I_7 = 12\,\%$, $I_{11} = 7\,\%$, $I_{13} = 5\,\%$, $I_{17} = 3\,\%$, $I_{19} = 2.5\,\%$ of the drives' fundamental. Assess compliance, first without, then with, the plant's existing 3 Mvar capacitor bank on the 11 kV bus.

Currents. The drives' fundamental apparent power is $3/0.95 = 3.158$ MVA, so $I_1 = 3.158 \times 10^6/(\sqrt{3} \times 11\,000) = 165.7$ A. The maximum demand current is $I_L = 8 \times 10^6/(\sqrt{3} \times 11\,000) = 419.9$ A and the fault current $I_{sc} = 250 \times 10^6/(\sqrt{3} \times 11\,000) = 13.12$ kA, so $I_{sc}/I_L = 31.2$: the 20–50 row of Table 25.5 applies.

h	I_h (A)	% of I_L	Limit (%)
5	49.7	11.8	7.0
7	19.9	4.7	7.0
11	11.6	2.8	3.5
13	8.3	2.0	3.5
17	5.0	1.2	2.5
19	4.1	1.0	2.5

$$\text{TDD} = \frac{\sqrt{49.7^2 + 19.9^2 + 11.6^2 + 8.3^2 + 5.0^2 + 4.1^2}}{419.9} = 13.3\,\% > 8.0\,\%$$

The installation fails on the 5th harmonic and on TDD.

Voltage without the capacitor. $X_s = 11^2/250 = 0.484\ \Omega$ per phase; the phase voltage is 6351 V. The 5th-harmonic voltage is $5 \times 0.484 \times 49.7 = 120$ V, or 1.9 %; the 7th is 1.1 %, the 11th 1.0 %. Summing all six in quadrature gives $\text{THD}_V = 2.7\,\%$, inside the 5 % limit of Table 25.4. The network operator's concern is the current; the voltage is still acceptable.

With the capacitor bank. The bank's reactance is $X_C = 11^2/3 = 40.3\ \Omega$ at 50 Hz, X_C/h at order h. Seen from the drive, it is in parallel with the source, and the parallel combination of jhX_s and $-jX_C/h$ has magnitude $hX_s(X_C/h)/|hX_s - X_C/h|$. The resonance is at $h_r = \sqrt{X_C/X_s} = \sqrt{250/3} = 9.13$ (the 456 Hz of *Circuits, Signals and Communications*, Ch. 7 §7.8). At the 7th the bus impedance rises from 3.39 Ω to 8.22 Ω (×2.43), at the 11th from 5.32 Ω to 11.78 Ω (×2.21), at the 5th from 2.42 Ω to 3.46 Ω (×1.43). The 5th, 7th and 11th voltages become 2.7 %, 2.6 % and 2.2 %, and THD_V rises to 4.4 %, close to the limit. The capacitor also takes most of the 7th and 11th harmonic currents itself, since it is the lower-impedance branch above resonance: its

RMS current and peak voltage rise, its internal fuses see the extra heating, and this is the plant's blown-fuse problem. The remedies, a detuned bank or a tuned filter, are developed in Sections 25.8 and 25.9.

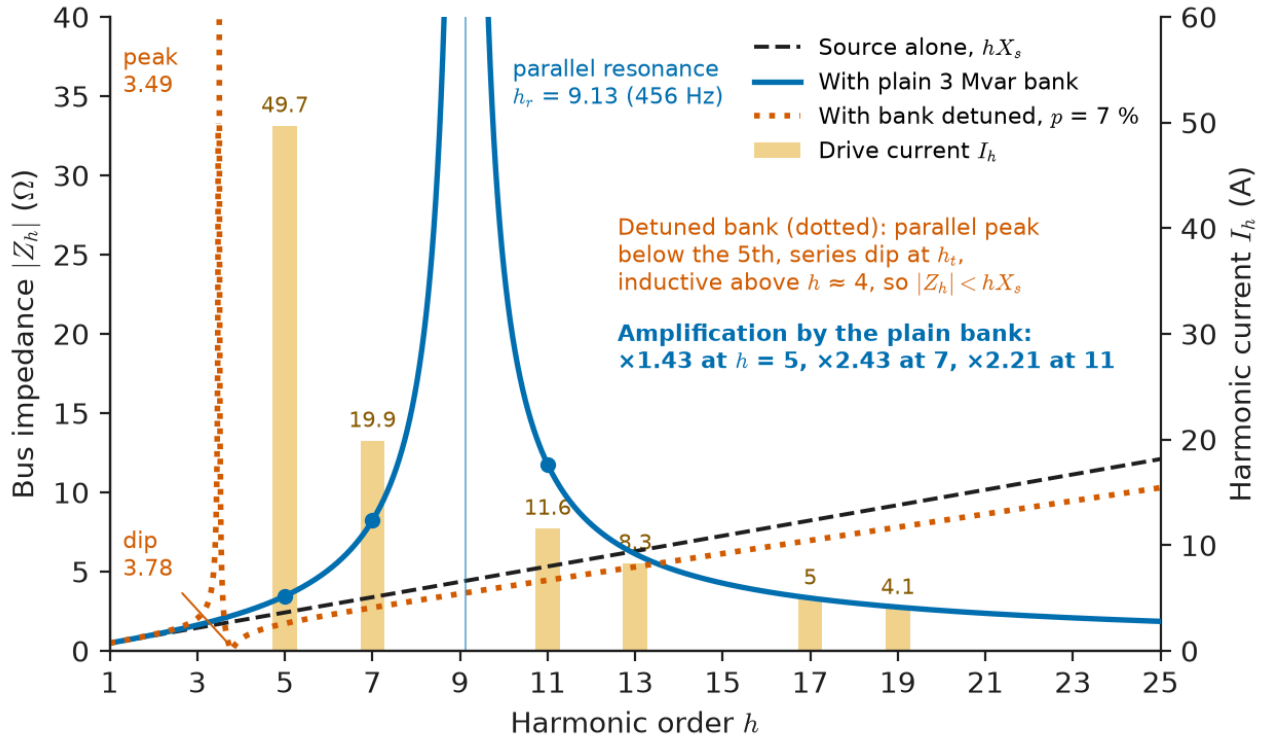


Figure 25.1 — Harmonic impedance of an 11 kV, 250 MVA bus with and without a 3 Mvar capacitor bank, with the drive spectrum overlaid.

Effects on equipment

Harmonics damage equipment by heating, dielectric stress and malfunction. Induction motors suffer extra rotor loss (the 5th, 11th, … are negative sequence and brake the rotor) and torque pulsation at $6f_1$; capacitors, whose admittance rises with h, attract harmonic current and are limited by IEC 60831 to 1.3 pu RMS current; standby generators feeding drive loads are derated for rotor heating; peak-sensing relays and zero-crossing controls misoperate. The most expensive and most calculable effect is transformer heating.

Transformer derating. IEEE C57.110 quantifies transformer heating with the **harmonic loss factor**

$$F_{HL} = \frac{\sum_h (I_h/I_1)^2\, h^2}{\sum_h (I_h/I_1)^2}$$

which multiplies the winding eddy-current loss. If $P_{EC\text{-}R}$ is the rated eddy loss as a fraction of rated I^2R loss, the maximum permissible RMS load current in per unit of rating, keeping total winding loss at its rated value, is

$$I_{max} = \sqrt{\frac{1 + P_{EC\text{-}R}}{1 + F_{HL}\, P_{EC\text{-}R}}}$$

For the office load of *Circuits, Signals and Communications*, Ch. 9 §9.8 ($I_3/I_1 = 0.8$, $I_5/I_1 = 0.6$, $I_7/I_1 = 0.4$, $I_9/I_1 = 0.2$), $\sum (I_h/I_1)^2 = 2.20$ and $\sum h^2 (I_h/I_1)^2 = 26.84$, so $F_{HL} = 12.2$. A dry-type transformer with $P_{EC\text{-}R} = 8\,\%$ may then carry only $\sqrt{1.08/1.976} = 0.74$ pu of its rated current, and one with 15 % only 0.64 pu. This is the origin of **K-rated** transformers (UL 1561 defines the K-factor, numerically equal to F_{HL} when the

currents are normalised to the RMS value): a K-13 transformer is built with subdivided, transposed conductors and an oversized neutral so that it can carry such a load at nameplate.

> **Status of the principle: Harmonic loss factor and transformer derating**
> *Approximate*. Assumes winding eddy loss scales exactly as h^2 (true when the conductor is thin compared with the skin depth at each harmonic) and that other stray losses scale as $h^{0.8}$, as IEEE C57.110 prescribes; for large conductors at high orders the h^2 law overestimates the loss, so the derating is conservative.

25.4 Flicker

From voltage fluctuation to visible flicker

A load whose current fluctuates at a few hertz modulates the supply voltage through the source impedance. The relative voltage change produced by a change ΔS in a load's apparent power is approximately

$$d = \frac{\Delta V}{V} \approx \frac{\Delta S}{S_{sc}} \cos(\psi_k - \varphi)$$

where ψ_k is the network impedance angle and φ the angle of the load change; for the common worst case of a mainly reactive change on an inductive network the cosine is near 1 and $d \approx \Delta S / S_{sc}$. Fluctuations of half a per cent are harmless to motors and electronics; they matter because of the human eye, which is most sensitive to luminance fluctuation near 8.8 Hz, where a sinusoidal modulation of only about 0.25 % of a 230 V lamp's voltage irritates half of observers. What is regulated is therefore a **perceptibility index**.

IEC 61000-4-15 specifies the **flickermeter**: a model of a 60 W, 230 V incandescent lamp (120 V version for 60 Hz systems) followed by a weighting filter that reproduces the eye's frequency response, a squaring and smoothing stage that models the brain's memory, and a statistical classifier. Its output is the **short-term flicker severity** P_{st}, computed every 10 minutes, and the **long-term severity**

$$P_{lt} = \sqrt[3]{\frac{1}{12}\sum_{i=1}^{12} P_{st,i}^3}$$

over 2 hours. $P_{st} = 1$ is the threshold of irritability. The incandescent reference lamp is disappearing, but the index is kept as a reproducible measure of fluctuation; LED lamps range from much less sensitive to, with some cheap drivers, more.

The $P_{st} = 1$ curve of Figure 25.2 shows the amplitude of rectangular voltage changes that gives $P_{st} = 1$ at each repetition rate. It falls from about 2.7 % at one change per minute to a minimum of about 0.3 % near 1000 changes per minute (8.8 Hz) and rises again at higher rates, where the lamp's thermal inertia filters the fluctuation.

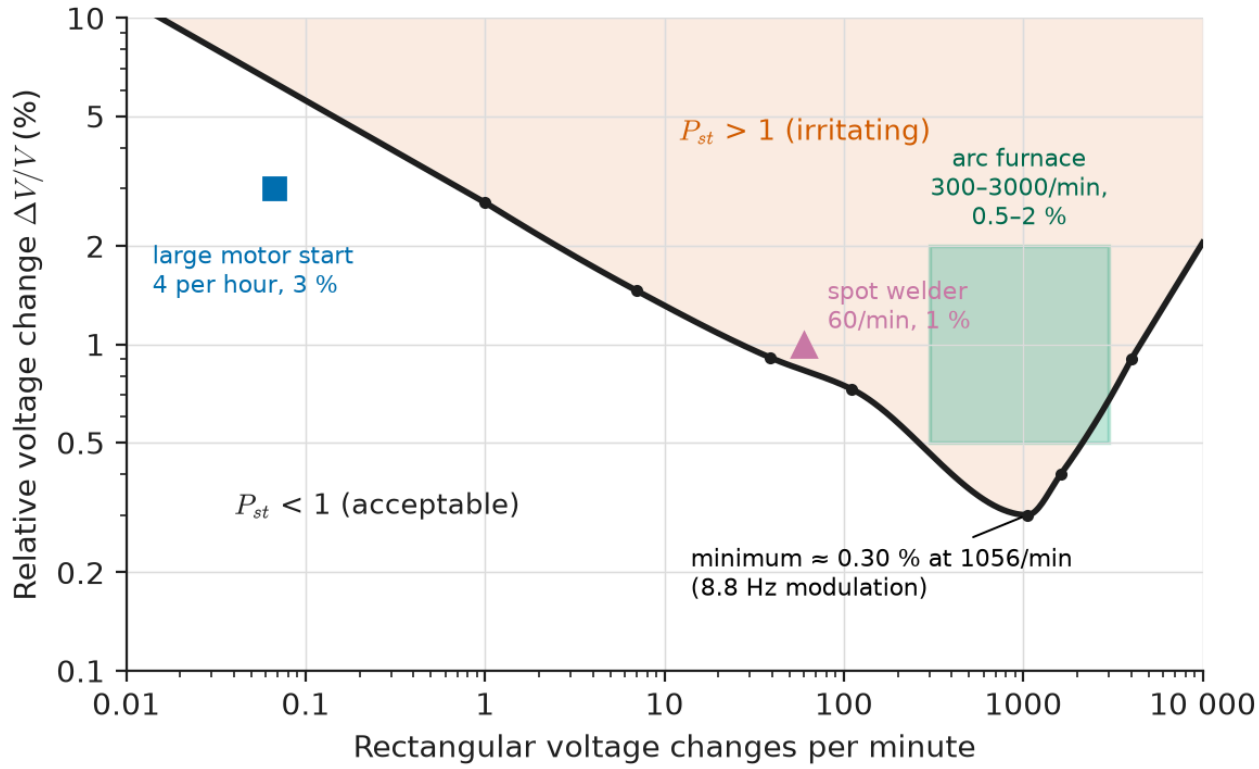


Figure 25.2 — The Pst = 1 curve for rectangular voltage changes on a 230 V lamp, with typical fluctuating loads.

Sources and assessment

The classic flicker source is the **electric arc furnace**, whose arcs strike moving, collapsing scrap so that its reactive power fluctuates randomly between near zero and near short-circuit, with energy concentrated between 1 and 15 Hz. Others are welders, crushers, large motor starts, rolling mills and wind turbines (tower-shadow modulation at three times rotor speed).

For an arc furnace, the empirical planning formula is

$$P_{st,99\,\%} \approx K_{st}\,\frac{S_{scf}}{S_{sc}}$$

where S_{scf} is the furnace's short-circuit power (the apparent power drawn with the electrodes shorted on the scrap, roughly twice the transformer rating) and K_{st} is an emission coefficient, typically 48–85 for AC furnaces (about 60–70 for most), lower for DC furnaces and ladle furnaces. Wind turbines have an analogous coefficient: IEC 61400-21 gives $P_{st} = c(\psi_k, v_a)\,S_n/S_k$ with flicker coefficients c of typically 2–10 for modern variable-speed machines.

> **Status of the principle: Arc-furnace flicker coefficient**
> *Empirical.* The K_{st} values are fitted to measurements on operating furnaces; the linear scaling with S_{scf}/S_{sc} holds because P_{st} is linear in modulation depth, but the coefficient depends on scrap quality, furnace operation and control and has a scatter of $\pm 30\,\%$.

Limits come from IEC/TR 61000-3-7 planning levels (indicative $P_{st} = 0.9$, $P_{lt} = 0.7$ at MV; $P_{st} = 0.8$, $P_{lt} = 0.6$ at HV and EHV), EN 50160 ($P_{lt} \le 1$ for 95 % of a week at LV and MV), and for small equipment IEC 61000-3-3 ($P_{st} \le 1.0$, $P_{lt} \le 0.65$, maximum relative voltage change 4 %).

Worked example — Siting an arc furnace. A steelmaker plans a 60 MVA AC arc furnace with a short-circuit power $S_{scf} = 110$ MVA, to be connected at a 220 kV substation whose minimum fault level is 5000 MVA. Take $K_{st} = 70$. Is the planning level $P_{st} = 0.8$ met, and what can be done?

$$P_{st} \approx 70 \times \frac{110}{5000} = 1.54$$

The emission is nearly twice the planning level. Meeting 0.8 by fault level alone would require $S_{sc} \geq 70 \times 110/0.8 = 9625$ MVA, a much stronger connection point, typically at 400 kV and possibly many kilometres away. The alternative is compensation at the furnace. A thyristor-controlled **SVC** reduces flicker by a factor of typically 1.5–2.5, limited by its half-cycle response; a **STATCOM**, with a response of a few milliseconds, achieves reduction factors of 3–5. With a STATCOM giving a factor of 3, $P_{st} \approx 0.51$. The compensator, rated at roughly the furnace MVA, doubles as the furnace's power-factor and filter installation; the choice is an economic one between network reinforcement and equipment.

25.5 Transients

Categories

A transient is a departure lasting less than about one cycle, from nanoseconds to tens of milliseconds. IEEE 1159 divides transients into **impulsive** (unidirectional, such as the 1.2/50 μs lightning impulse) and **oscillatory** (bipolar ringing), as summarised in Table 25.6; sags (0.1–0.9 pu), swells (1.1–1.8 pu) and interruptions (below 0.1 pu) are the RMS events of Section 25.6. Figure 25.3 locates them on the magnitude–duration map together with the RMS events of Section 25.6.

Table 25.6 — IEEE 1159 categories of transients.

Category	Typical spectral content	Typical duration	Typical magnitude
Impulsive transient, μs	5 ns–0.1 ms rise	50 ns–1 ms	Up to several pu
Oscillatory transient, low frequency	< 5 kHz	0.3–50 ms	0–4 pu
Oscillatory transient, medium frequency	5–500 kHz	20 μs	0–8 pu
Oscillatory transient, high frequency	0.5–5 MHz	5 μs	0–4 pu

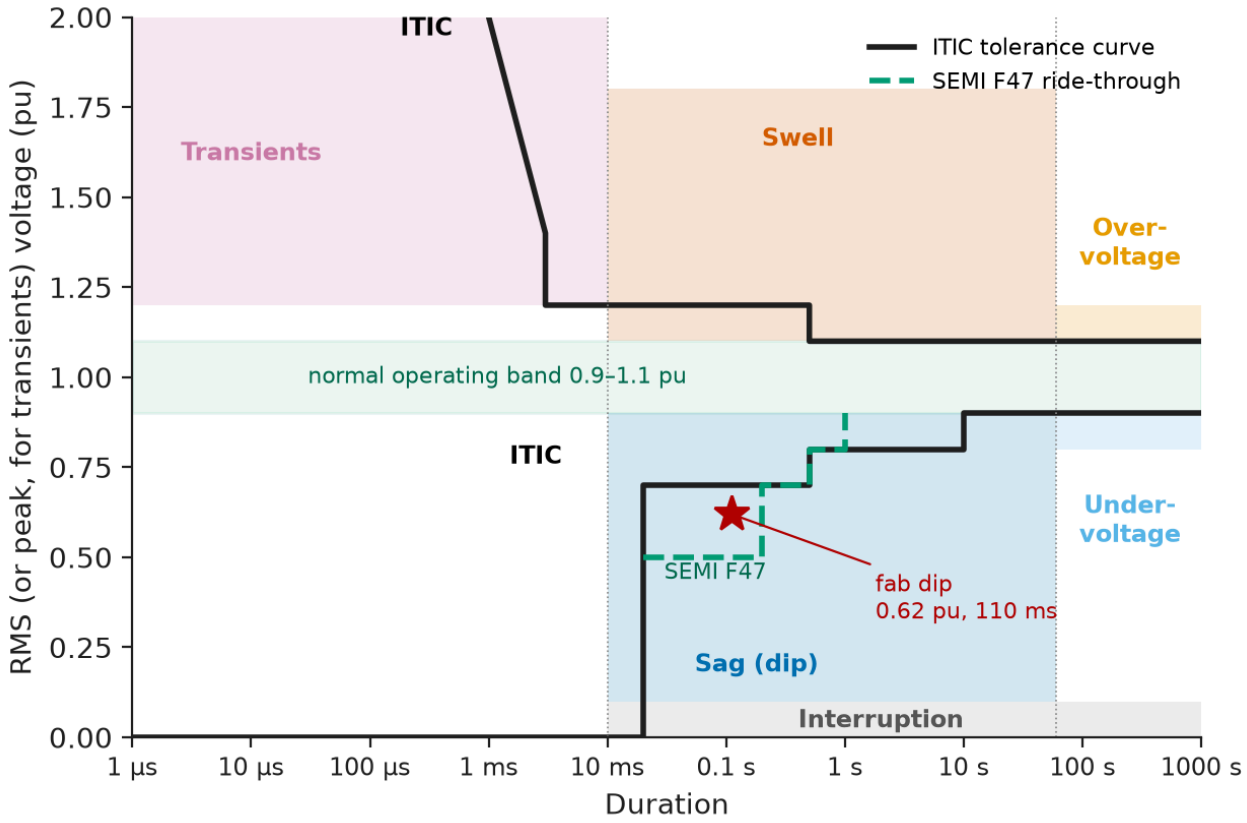


Figure 25.3 — Magnitude-duration map of power-quality disturbances after IEEE 1159, with the ITIC and SEMI F47 tolerance curves.

Sources

Lightning produces impulsive transients by direct strikes and by induction into lines; its travelling-wave behaviour and the surge arresters that limit it are the subject of Chapter 24. On a customer's LV installation, residual lightning surges of a few kilovolts are limited by surge-protective devices coordinated to IEC 61643 and IEC 60364-5-53.

Capacitor switching is the most common oscillatory transient on distribution systems, and the one most often behind unexplained drive trips. Energising an isolated bank gives an inrush oscillating at $f_1\sqrt{S_{sc}/Q_C}$, typically 300–1000 Hz, and a voltage overshoot towards 2 pu (*Circuits, Signals and Communications*, Ch. 7 §7.7). Two variants matter in practice:

- **Back-to-back switching.** When a second bank is energised next to one already charged, the inrush is fed from the first bank through only the small inductance of the bus and cables between them, not through the source. The inrush is far larger and its frequency far higher (worked example below).
- **Voltage magnification.** When an MV bank is energised, its 1–2 pu ringing at several hundred hertz can excite the resonance of a downstream LV capacitor (a power-factor bank or a drive's DC-link filter) with the step-down transformer. If the two resonant frequencies are close, the LV overvoltage can reach 3–4 pu, enough to trip drives on DC-bus overvoltage or fail LV surge devices. Utility switching of feeder banks at 7:00 every morning, followed by a plant's drives tripping at 7:00 every morning, is the classic signature.

Other switching transients include line energisation (Chapter 24), current chopping by vacuum breakers on small inductive currents, and **notching** of the voltage by thyristor-rectifier commutation, whose depth IEEE

519 also limits.

Worked example — Back-to-back capacitor switching at 11 kV. The 11 kV bus of Section 25.3 has a 3 Mvar bank in service. A second, identical 3 Mvar bank is added in the same switchroom and energised while the first is charged. The inductance of the bus and cable between the two banks is 20 μH per phase. Find the inrush peak and frequency, and the series inductance needed to keep it below 6 kA peak.

Each bank has $C = 3\times10^{6}/(2\pi\cdot 50\times 11\,000^{2}) = 78.9\ \mu$F per phase (star equivalent). In the back-to-back circuit the charged bank discharges into the uncharged one through L, and the effective capacitance is the series combination $C_{eq} = C_1C_2/(C_1+C_2) = 39.5\ \mu$F. Worst case is closing at the voltage peak, $V_{pk} = 11\,000\sqrt{2/3} = 8.98$ kV. Treating the circuit as an undamped series LC (as in the Book 3 analysis of the isolated case),

$$\hat{I} = V_{pk}\sqrt{\frac{C_{eq}}{L}} = 8981\sqrt{\frac{39.5\times10^{-6}}{20\times10^{-6}}} = 12.6\ \text{kA}, \qquad f = \frac{1}{2\pi\sqrt{LC_{eq}}} = 5.67\ \text{kHz}$$

Against the isolated bank's 2.03 kA at 456 Hz, that is six times the current at twelve times the frequency. The product $\hat{I}f$ (here 7.1×10^{7} A/s) is what erodes breaker contacts, and IEEE C37.04/C37.06 definite-purpose capacitor-switching breakers are rated for back-to-back inrush of typically 15–20 kA peak at up to 4–6 kHz. To limit the peak to 6 kA, the total inductance must be

$$L = C_{eq}\left(\frac{V_{pk}}{\hat{I}}\right)^{2} = 39.5\times10^{-6}\times\left(\frac{8981}{6000}\right)^{2} = 88\ \mu\text{H}$$

so a 70 μH air-core **inrush-limiting reactor** is added in series with each bank. The inrush frequency falls to 2.7 kHz. Alternatives are pre-insertion resistors and **point-on-wave (synchronous) closing**, which closes each pole near its voltage zero and can reduce the inrush to a small fraction of the worst case.

> **Status of the principle: Back-to-back inrush from lumped LC model**
> *Approximate.* Assumes the source is stiff during the few-kilohertz ringing (so only the inter-bank inductance matters), neglects damping, and treats each phase independently; real inrush is somewhat lower because of resistance and the closing instant, and for long cables between banks the travelling-wave behaviour of Chapter 24 replaces the lumped inductance.

25.6 Voltage sags and swells

The most costly disturbance

A **sag** (IEC: **dip**) is a reduction of RMS voltage to between 0.1 and 0.9 pu for half a cycle to one minute; a **swell** is an increase to 1.1–1.8 pu for the same duration. Sags cause most of industry's power-quality losses: a typical MV customer sees 10–100 per year below 0.9 pu against 1–5 interruptions, and a sag that resets a production line costs nearly as much as an interruption.

Causes and magnitude

Almost all deep sags are caused by **short circuits** somewhere in the network, and the sag lasts until the protection clears the fault. A smaller number come from starting large motors and energising transformers (shallow, longer sags with slow recovery). The depth at a given bus follows from the **voltage divider model**:

with Z_s the source impedance at the PCC and Z_f the impedance from the PCC to the fault (along a feeder that branches off at the PCC), the retained voltage during a three-phase fault is

$$V_{sag} = \frac{Z_f}{Z_s + Z_f} \text{ pu}$$

> **Status of the principle: Voltage-divider model of sag magnitude**
> *Approximate.* Assumes pre-fault voltage of 1 pu, neglects load current and the contribution of motors and generators downstream, and assumes a bolted three-phase fault on a radial branch; for unbalanced faults the same idea is applied to the sequence networks of Chapter 19, and motor back-feed makes real sags somewhat shallower at first and slower to recover.

Worked example — Sags at a fab from faults on neighbouring feeders. The fab of the chapter opening takes supply from a 33 kV bus whose fault level is 750 MVA. Several 33 kV overhead lines leave the same bus to other towns, each with a positive-sequence reactance of 0.4 Ω/km (resistance neglected). Find the sag depth for three-phase faults at various distances along these lines, and the "area of vulnerability" for equipment that drops out below 0.5 pu and 0.7 pu.

$$Z_s = \frac{33^2}{750} = 1.452\ \Omega$$

Fault distance	Z_f (Ω)	V_{sag} (pu)
1 km	0.4	0.22
2 km	0.8	0.36
5 km	2.0	0.58
10 km	4.0	0.73
20 km	8.0	0.85

The critical distance for a retained voltage V is $d = \frac{V}{1-V}\frac{Z_s}{0.4}$: 3.6 km for 0.5 pu, 8.5 km for 0.7 pu and 32.7 km for 0.9 pu, so equipment tripping at 0.7 pu is exposed to the first 8.5 km of every line. With five such lines, a fault rate of 5 per 100 km per year gives about $5 \times 8.5 \times 0.05 = 2.1$ trips per year from three-phase faults alone; single-line-to-ground faults, more frequent but producing shallower sags at the fab behind a delta–star transformer, add more. The duration is the protection's clearing time: about 100 ms for instantaneous protection near the bus, 300–500 ms for time-graded protection further out. The opening's 0.62 pu, 110 ms event is a fault about 6 km out, cleared by instantaneous protection.

Sag type, phase shift and swell

Unbalanced faults produce unbalanced sags, altered by the transformers between fault and load: a single-line-to-ground fault on the MV side of a Dyn transformer appears at LV as a two-phase sag without zero sequence (Chapter 19). Sags also carry a **phase-angle jump** of up to tens of degrees, which upsets thyristor converters and phase-locked loops.

Swells are mostly the healthy-phase voltage rise during a single-line-to-ground fault. On a solidly earthed system the rise is small (up to about 1.2–1.3 pu, set by the earth-fault factor of Chapter 24); on an ungrounded or resonant-earthed system the healthy phases rise towards $\sqrt{3} = 1.73$ pu for the duration of the fault. Load rejection and capacitor energisation produce smaller swells.

Equipment sensitivity

Equipment tolerance is expressed as a voltage–duration curve (Figure 25.3). The **ITIC curve** (successor to the CBEMA curve) describes what 120 V information-technology equipment should tolerate: zero voltage for 20 ms, 70 % for 0.5 s, 80 % for 10 s. **SEMI F47** demands that semiconductor process tools ride through 50 % for 200 ms, 70 % for 0.5 s and 80 % for 1 s. IEC 61000-4-11 (up to 16 A) and IEC 61000-4-34 (above 16 A) set the corresponding immunity test levels.

Table 25.7 gives representative sensitivities, and explains the fab's problem: the dip to 0.62 pu for 110 ms is inside SEMI F47, so the tools themselves should have survived, but the AC contactors, auxiliary relays and older drives in the support systems did not.

Table 25.7 — Typical sag sensitivity of equipment.

Equipment	Drop-out threshold	Time
AC contactor or relay coil	0.5–0.75 pu	10–20 ms (some ride through up to 100 ms)
PLC, power supply without hold-up	0.5–0.8 pu	10–50 ms
Variable-speed drive (default settings)	0.7–0.85 pu (DC-bus undervoltage)	5–20 ms
Variable-speed drive with kinetic buffering	0.5 pu or lower	Several seconds while load inertia lasts
Induction motor (direct-on-line)	Rides through most sags; re-acceleration current on recovery	—

Mitigation of sags

In order of increasing cost per kVA protected: make the process tolerant (DC-fed or held-in contactor coils, drive ride-through and automatic restart, SEMI F47 equipment), which is the cheapest step and the one most often omitted; protect only the control power with a small UPS; protect the critical load with an online UPS, a **dynamic voltage restorer** (DVR, a series converter that injects the missing voltage for a few hundred milliseconds from stored energy) or a **static transfer switch** between two independent feeders; or change the network with faster protection, fewer exposed kilometres and loop supply.

25.7 Unbalance

Measuring it properly

The decomposition of an unbalanced three-phase set into positive-, negative- and zero-sequence components was introduced in *Circuits, Signals and Communications*, Ch. 10 §10.6, together with the quick NEMA definition (maximum deviation from the average line voltage, divided by the average). The quantity that standards actually regulate is the **voltage unbalance factor**

$$\mathrm{VUF} = \frac{|V_2|}{|V_1|} \times 100\,\%$$

It can be computed from the three line-voltage magnitudes alone (line voltages contain no zero sequence), which is how many meters do it:

$$\mathrm{VUF} = \sqrt{\frac{1-\sqrt{3-6\beta}}{1+\sqrt{3-6\beta}}}, \qquad \beta = \frac{V_{ab}^4 + V_{bc}^4 + V_{ca}^4}{\left(V_{ab}^2 + V_{bc}^2 + V_{ca}^2\right)^2}$$

For the line voltages 400, 396 and 388 V of the Book 3 example, β gives VUF $= 1.78\,\%$, against 1.69 % by the NEMA approximation. The difference is small here, but the two definitions diverge for large unbalance, and contracts should state which one applies.

Limits are typically 2 % VUF for 95 % of 10-minute values at LV and MV (EN 50160, with up to 3 % in areas with many single-phase loads), and 1–1.4 % planning levels at HV (IEC/TR 61000-3-13).

Sources

Unbalance comes from unequal single-phase loads, single-phase 25 kV traction supplies connected phase-to-phase on the HV grid, induction furnaces and welders, untransposed lines and open phases. A single-phase load of apparent power S_L connected line-to-line produces a negative-sequence current equal in magnitude to its positive-sequence current, and so a VUF at its connection point of approximately

$$\mathrm{VUF} \approx \frac{S_L}{S_{sc}}$$

> **Status of the principle: Unbalance from a single-phase load, VUF ≈ S_L / S_sc**
> *Approximate*. Assumes equal positive- and negative-sequence source impedances (true for networks dominated by lines and transformers, less so near rotating machines, whose negative-sequence impedance is lower) and a load small compared with the fault level.

Effects on induction motors

The negative-sequence voltage drives a field rotating backwards relative to the rotor, at a slip of nearly 2. The motor's impedance to it is close to its locked-rotor impedance, so the negative-sequence current is

$$\frac{I_2}{I_{rated}} \approx \mathrm{VUF} \times \frac{I_{start}}{I_{rated}}$$

A motor with a starting current of 6 pu therefore draws 6 % negative-sequence current per 1 % of voltage unbalance. At about 100 Hz in the rotor, skin effect makes it heat the bars out of proportion to its size. NEMA MG 1 derating factors are 0.98 at 1 %, 0.95 at 2 %, 0.88 at 3 %, 0.82 at 4 % and 0.75 at 5 % unbalance, and operation above 5 % is not recommended. Generators carry negative-sequence relays set against their $I_2^2 t$ capability (Chapter 23).

Worked example — A traction supply. A high-speed rail operator connects a 30 MVA single-phase 25 kV traction transformer phase-to-phase on a 132 kV bus with a minimum fault level of 2500 MVA. Estimate the VUF and the effect on a large induction motor fed from the same bus through a transformer.

$$\mathrm{VUF} \approx \frac{30}{2500} = 1.2\,\%$$

This is within the 2 % public-supply limit but above a typical 1 % emission allocation for a single customer at HV, and it adds to the background unbalance already present. A motor with 6 pu starting current sees $I_2 \approx 6 \times 1.2 = 7.2\,\%$ negative-sequence current, and by NEMA MG 1 should be derated to about 0.97 of nameplate.

Remedies are to rotate successive substations around the phase pairs, connect at a higher fault level, or install a balancer: the static **Steinmetz circuit** (for a resistive load P between two lines, a capacitor and an inductor of $P/\sqrt{3}$ var each across the other two line pairs) or, for a varying traction load, a STATCOM.

25.8 Power-factor correction

Why it is still a power-quality topic

The arithmetic of power-factor correction, $Q_C = P(\tan\varphi_1 - \tan\varphi_2)$, and the reasons for stopping at 0.95–0.98 are in *Circuits, Signals and Communications*, Ch. 9 §9.6; the tariff and energy-management side is in Chapter 28. This section turns a kvar figure into hardware that survives harmonic resonance and switching in a modern plant.

Where and how

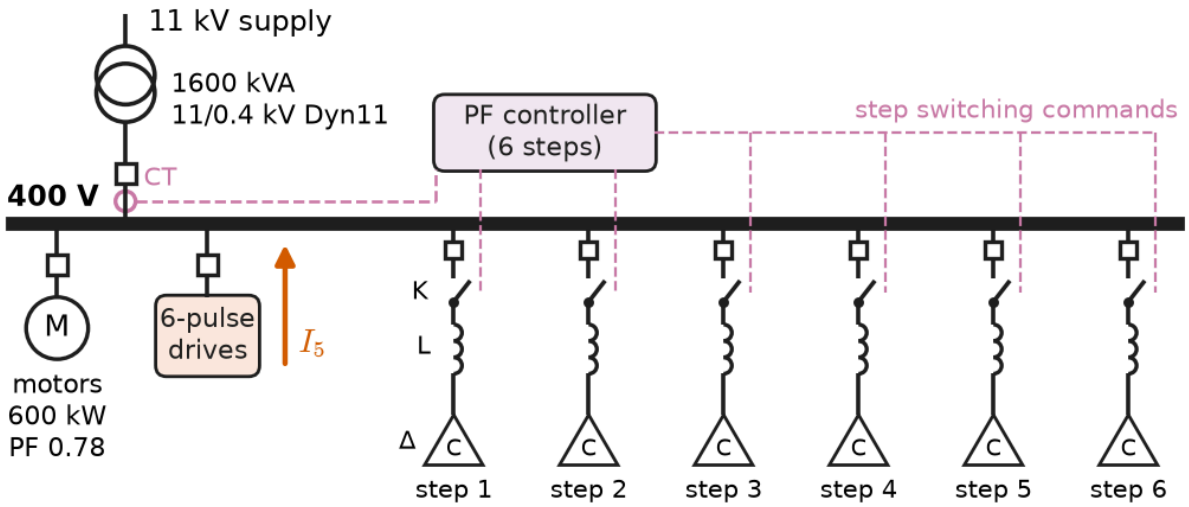
Correction is applied individually at large motors (limited to about 90 % of no-load magnetising kvar to avoid self-excitation), at motor control centres, or centrally by automatic banks whose controller switches steps of 25–50 kvar (LV) or 0.5–3 Mvar (MV) to follow the load. Central correction is cheapest per kvar and is the norm. In a plant with harmonic sources, a plain capacitor bank creates a parallel resonance with the supply transformer and network at $h_r = \sqrt{S_{sc}/Q_C}$ (Section 25.3), and because the bank's steps switch in and out, the resonance sweeps over a range of harmonic orders during the day. Somewhere in that range there is usually a characteristic harmonic. The standard remedy is the **detuned** bank: each step has a series reactor sized so that the step's own series resonance lies below the lowest significant harmonic. The reactor is specified by its **detuning factor** $p = X_L/X_C$ at the fundamental, giving a series tuning order

$$h_t = \frac{1}{\sqrt{p}}$$

Common values are $p = 7\,\%$ ($h_t = 3.78$, 189 Hz at 50 Hz), $p = 5.67\,\%$ ($h_t = 4.2$, 210 Hz) and $p = 14\,\%$ ($h_t = 2.67$, 134 Hz, used where third-harmonic currents are significant). Above h_t the step is inductive, so it cannot form a parallel resonance with the network at any characteristic harmonic, and it draws only a modest share of the harmonic current. Figure 25.1 shows the bus impedance with such a bank.

Two consequences of the series reactor must be designed in. The capacitor's fundamental voltage rises to $V/(1-p)$, and its fundamental kvar for a given capacitance rises by $1/(1-p)$; and harmonic currents still flowing in the step add to the capacitor's RMS current and peak voltage. Capacitors in detuned banks are therefore rated well above the system voltage.

> **Status of the principle: Detuned capacitor bank**
> *Approximate*. The design rests on the current-source harmonic model and a single dominant network inductance; it guarantees the absence of parallel resonance above h_t only if all the plant's capacitance (including drive input filters and other banks) is detuned, and h_t drifts upward as capacitors age and lose capacitance, which is why p is chosen with margin below the 5th.



Each step: contactor K, 7 % reactor L,	X_C = 1.685 Ω, X_L = 0.07 X_C = 0.118 Ω
delta-connected capacitors rated 480 V	306 kvar effective (PF 0.78 → 0.96)
Bank total per delta branch:	Tuning h_t = 1/√0.07 = 3.78 → 189 Hz
L = 0.375 mH, C = 1889 µF	Capacitor voltage 400 / 0.93 = 430 V

Figure 25.4 — A 400 V automatic detuned power-factor-correction bank in a plant with drives.

Worked example — A detuned bank for a 400 V plant. A plant draws 600 kW at a power factor of 0.78 lagging from a 400 V, 50 Hz supply and has a significant drive load. Size a detuned bank with $p = 7\,\%$ to correct to 0.96.

Reactive power. $\tan\varphi_1 = \tan(\arccos 0.78) = 0.802$ and $\tan\varphi_2 = \tan(\arccos 0.96) = 0.292$, so the effective compensation required is

$$Q_{eff} = 600\,(0.802 - 0.292) = 306 \text{ kvar}$$

The apparent power falls from 769 kVA to 625 kVA and the line current from 1110 A to 902 A.

Per-branch reactances (delta, line voltage across each branch). The effective branch reactance is $X_C - X_L = X_C(1-p)$:

$$X_C(1-p) = \frac{3V^2}{Q_{eff}} = \frac{3\times 400^2}{306\,000} = 1.567\ \Omega \quad\Rightarrow\quad X_C = \frac{1.567}{0.93} = 1.685\ \Omega, \quad X_L = 0.07 X_C = 0.118\ \Omega$$

so $C = 1/(2\pi\cdot 50\times 1.685) = 1889\ \mu$F and $L = 0.118/(2\pi\cdot 50) = 0.375$ mH per delta branch (the total bank, split across the six steps).

Capacitor voltage and rating. The fundamental voltage across the capacitor is $400/(1-0.07) = 430$ V. Allowing for the harmonic voltage and normal overvoltage, capacitors rated at 480 V are chosen; at their rated voltage this capacitance is $3\times 480^2/1.685 = 410$ kvar of nameplate, which is why a detuned "306 kvar" bank is built from about 410 kvar of 480 V capacitors.

Behaviour at harmonics. At the 5th harmonic the branch reactance is $5X_L - X_C/5 = 0.590 - 0.337 = +0.253\ \Omega$, inductive; at the 7th $+0.585\ \Omega$. The bank cannot resonate with the supply at any characteristic

harmonic. Correcting the last 4 % to unity power factor would need a further $600 \times 0.292 = 175$ kvar: 57 % more equipment for a negligible current reduction.

Beyond capacitors

Where reactive demand fluctuates quickly (welders, crushers, furnaces, wind farms), switched capacitors are too slow. **SVCs** give continuous control within one to two cycles, **STATCOMs** (Chapter 27) respond in milliseconds and hold rated current down to low voltage, and **synchronous condensers** add inertia and fault current as well (Chapter 7). Inside equipment, active front ends and active PFC correct both displacement and distortion at the source.

25.9 Measurement and mitigation

Instruments and standards

Power-quality data are only comparable if they are measured the same way. IEC 61000-4-30 defines the methods, and its **Class A** instruments agree within stated uncertainty regardless of manufacturer ($\pm 0.1\,\%$ of declared voltage, ± 10 mHz). The basic measurement is over 10 cycles at 50 Hz (12 at 60 Hz), about 200 ms, aggregated to 3 s, 10 minutes and 2 hours; the 10-minute values, usually as a weekly 95th percentile, are what EN 50160 and most contracts assess. The window is locked to the measured mains frequency and harmonics are computed by the grouped DFT of IEC 61000-4-7 (*Circuits, Signals and Communications*, Ch. 22). Sags, swells and interruptions are detected from the RMS value updated every half cycle, and intervals containing them are **flagged** so that a fault's transients do not contaminate the frequency and harmonic statistics.

Transients are outside IEC 61000-4-30 and need separate high-speed capture (MS/s sampling). System-level sag performance is summarised by indices such as **SARFI-X**, the number of sags per year with retained voltage below X % per customer served (SARFI-70, SARFI-90), and by tables of events by residual voltage and duration as in EN 50160 and IEC/TR 61000-2-8.

Monitors are placed at the PCC for compliance, at sensitive loads to correlate trips with events, and at suspected emitters to allocate responsibility.

The mitigation toolkit

Mitigation follows a hierarchy that applies to every phenomenon in this chapter: reduce the **emission** at the source, increase the **immunity** of the sensitive equipment, or change the **network** between them (Table 25.8). The cheapest point of intervention is usually the equipment specification, before anything is bought.

Table 25.8 — Mitigation options by phenomenon.

Phenomenon	At the source	In the network	At the sensitive load
Harmonics	12/18-pulse rectifiers, AC/DC chokes, active front ends, active PFC	Tuned or high-pass passive filters, active harmonic filters, detuned banks, higher fault level, zigzag transformers for triplens	K-rated transformers, oversized neutrals, derating
Flicker	Furnace electrode control, series reactors, soft starters	SVC, STATCOM, connection at higher fault level	— (lighting with low-sensitivity drivers)
Transients	Point-on-wave switching, pre-insertion resistors, inrush reactors	Surge arresters, avoiding tuned LV/MV capacitor pairs	Surge-protective devices, drive line reactors
Sags	Faster protection, fewer faults (covered conductors, cables)	Fault-current limiters, loop supply, static transfer switch	Ride-through settings, coil hold-in, UPS, DVR
Unbalance	Phase rotation of single-phase loads, balanced traction connection	Steinmetz balancer, STATCOM	Motor derating, negative-sequence protection
Low power factor	Correct motor sizing, synchronous motors	Detuned banks, SVC, STATCOM, synchronous condensers	—

Passive filters. A **single-tuned filter** is a series LC branch tuned slightly below a target harmonic, usually 4.7 for the 5th, so that it presents a low impedance at that order and diverts the harmonic current from the network. It is also a capacitor bank at the fundamental and supplies reactive power. Sized at 1.5 Mvar effective on the 11 kV bus of Section 25.3 and tuned to $h_t = 4.7$, it needs $X_C = 84.5\ \Omega$ ($C = 37.7\ \mu$F) and $X_L = 3.82\ \Omega$ ($L = 12.2$ mH) per phase. With the 0.484 Ω source, the filter creates a parallel resonance at

$$h_p = \sqrt{\frac{X_C}{X_s + X_L}} = \sqrt{\frac{84.5}{0.484 + 3.82}} = 4.43$$

which lies safely between harmonics that the six-pulse drives produce. This is the general rule: every shunt filter introduces a parallel resonance below its tuning frequency, and filters are applied from the lowest significant harmonic upward (5th, then 7th, then a high-pass branch for the 11th and above). Tuning a little below the target allows for capacitor ageing and network frequency variation, which would otherwise move the series resonance above the harmonic and the parallel resonance onto it.

Active and custom-power devices. An **active harmonic filter** is a shunt voltage-source converter that measures the load's harmonic current and injects the opposite, up to about the 50th, within a cycle. It cannot create a resonance and adapts to load, at the price of cost, 2–3 % losses and its own switching emission; units of 100–600 A are common at LV in buildings and data centres, while passive filters still dominate at MV. The DVR, the STATCOM and the unified power-quality conditioner (series plus shunt) are its relatives, and all share the converter control of Chapter 27.

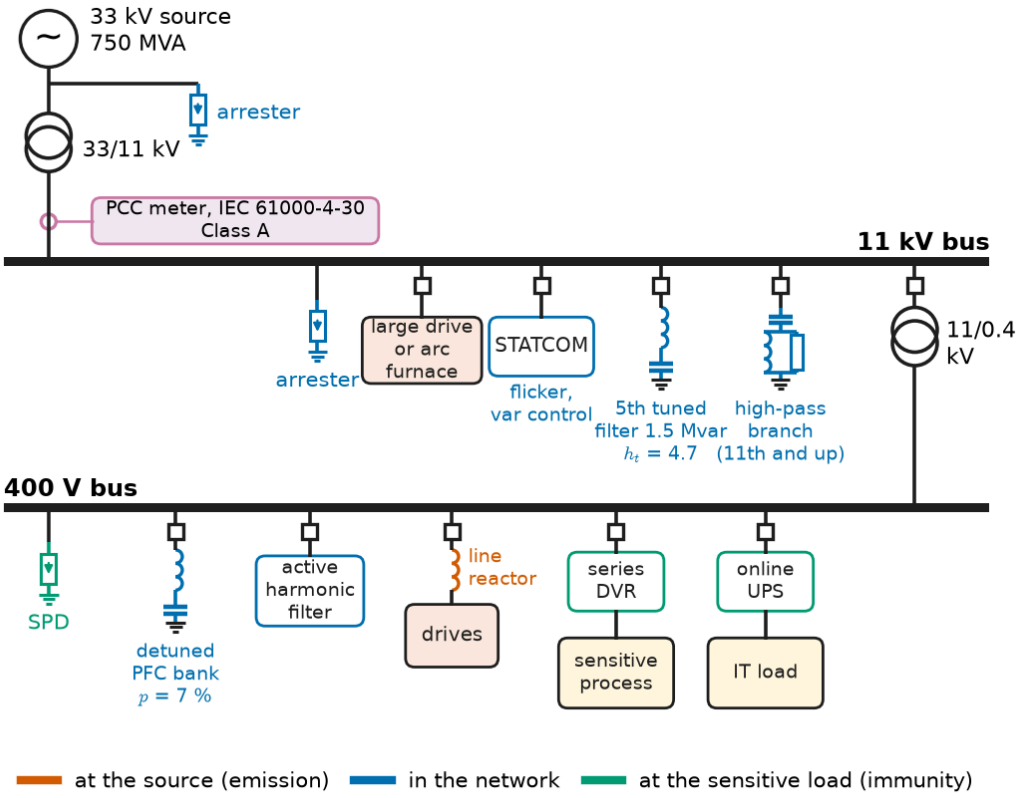


Figure 25.5 — Where power-quality mitigation devices sit in an MV/LV industrial supply.

Worked example — Choosing a remedy for the extruder drives. The plant of Section 25.3 fails IEEE 519 on its 5th harmonic (11.8 % against 7 %) and TDD (13.3 % against 8 %), and its plain capacitor bank is failing. Three options are costed:

1. *Detune the existing bank* with 7 % reactors. This stops the fuse failures but not the injection; the plant still fails.
2. *Replace the bank by a 1.5 Mvar 5th-harmonic tuned filter plus a 1.5 Mvar 7 % detuned bank.* At the 5th the filter branch has a reactance of $5 \times 3.82 - 84.5/5 = 2.23\ \Omega$ against the network's 2.42 Ω (it is tuned at 4.7, not at 5, for the reasons above), so together with the detuned bank it diverts 56 % of the 49.7 A: 21.9 A reaches the PCC, 5.2 % of I_L. The 7th and 11th fall to 3.5 % and 2.2 % of I_L because the detuned branch and the filter are now both inductive there and take a modest share, and TDD falls to 7.0 %. The installation passes, with little margin; a second branch tuned to the 7th would add some. Reactive compensation is kept at 3 Mvar.
3. *Specify active-front-end drives* (THD_I 3–5 %), removing the problem at source at higher drive cost and losses.

For a new installation option 3 is usually right; for a retrofit, option 2. Option 1 alone is the common mistake: it saves the capacitors and passes the problem to the network operator.

What remains open

Open question	Where it is answered
How are tap changers, regulators and feeder capacitors set to hold steady-state voltage?	Chapter 22; reactive-power management in Chapter 18
How is system frequency held, and what happens after a large loss of infeed?	Chapter 20; low-inertia grids in Chapter 27
How are unbalanced fault currents and sag types computed with sequence networks?	Chapter 19
How do lightning and switching surges travel, and how are arresters coordinated?	Chapter 24
How do grid-connected inverters control current, inject reactive power and ride through sags?	Chapter 27
How are power-quality monitors and smart meters networked and their data used?	Chapter 26
What is power-factor correction worth on a tariff, and how is it justified alongside other efficiency measures?	Chapter 28

Next: Chapter 26 turns from the quality of the delivered waveform to the information layer that now surrounds it: smart meters, grid communications, automation and the control of distributed resources.

Chapter 26 — Smart Grids

Part VII · Power Quality and Grid Modernization

At 02:40 on a winter night a tree falls across an 11 kV overhead feeder serving 3000 customers in a suburban district. Twenty years ago the first sign at the control room would have been the telephone: customers calling in, a crew sent to patrol the line with binoculars, sections isolated by hand one switch at a time, power restored to most customers after an hour and a half and to the rest after the repair. Today the substation breaker trips and recloses, the fault persists, and within a minute automated switches on the feeder have reported which section saw fault current, opened around it, and closed a normally open tie to an adjacent feeder; three quarters of the customers never notice more than a flicker. The smart meters in the faulted section send "last gasp" messages as their supply collapses, the outage-management system draws the outage on a map, and the crew drives straight to the span. The copper, the transformers and the physics of the feeder are the same as twenty years ago. What has changed is that the network can now measure itself, communicate and act.

> **What does it take, in sensors, communications and software, for a power network to observe and control itself down to the customer level?**
> **How do distributed resources, microgrids and flexible demand change the way the grid is operated, and what new risks, from cyberattack to loss of observability, come with that change?**

Concept	Engineering destination
Smart-metering systems	Interval billing, outage detection, LV voltage monitoring, theft detection
Grid communication	Protection signalling, SCADA, AMI backhaul, DER control
Distributed energy resources	Rooftop PV, home batteries, EV charging, virtual power plants
Microgrids	Campuses, hospitals, mines, islands, military bases
Demand response	Peak reduction, reserves from load, tariff design
Grid automation	Substation automation, FLISR, volt/var optimisation
Grid observability	State estimation, PMUs, distribution-system visibility
Energy management systems	AGC, economic dispatch, ADMS, DERMS, building EMS
Cybersecurity fundamentals	IEC 62351, IEC 62443, NERC CIP compliance, secure architectures
Resilience	Storm hardening, black start, islanding of critical loads

Roadmap. The sensing layer first: smart meters (26.1) and the communication networks that carry their data and every other signal (26.2). Then the new actors at the edge: distributed resources (26.3), microgrids (26.4) and responsive demand (26.5). Then the control layer: automation (26.6), observability (26.7) and the energy management systems that close the loop (26.8). Two cross-cutting concerns finish the chapter: cybersecurity (26.9) and resilience (26.10).

Boundary.
The communication theory this chapter relies on (channels, modulation, link budgets, fading, fibre, error control) is in *Circuits, Signals and Communications*, Ch. 25, Ch. 27, Ch. 30 and Ch. 31; digital power and energy measurement is in Ch. 9 §9.9; feedback and digital control in Ch. 34–35. This chapter uses those results to design and operate power networks. Meter and controller electronics belong to *Electronics & Computing*.

26.1 Smart-metering systems

From meter reading to metering infrastructure

An electromechanical meter integrated energy on a spinning disc and was read by a person once a month or once a quarter. It answered one question: how many kilowatt-hours since the last visit. A **smart meter** is a sampled-data power analyser (the signal chain of *Circuits, Signals and Communications*, Ch. 9 §9.9) with a clock, memory, a communication module and, usually, a remotely operable disconnect switch. The system around it is **advanced metering infrastructure** (AMI): meters, a **neighbourhood network** to data concentrators, a **backhaul** to the utility, a **head-end system** that manages communication, and a **meter data management system** (MDMS) that validates, estimates and stores the data and passes it to billing, outage management and network planning (Figure 26.1). Automatic meter reading (AMR), its predecessor, was one-way and served billing only; AMI is two-way.

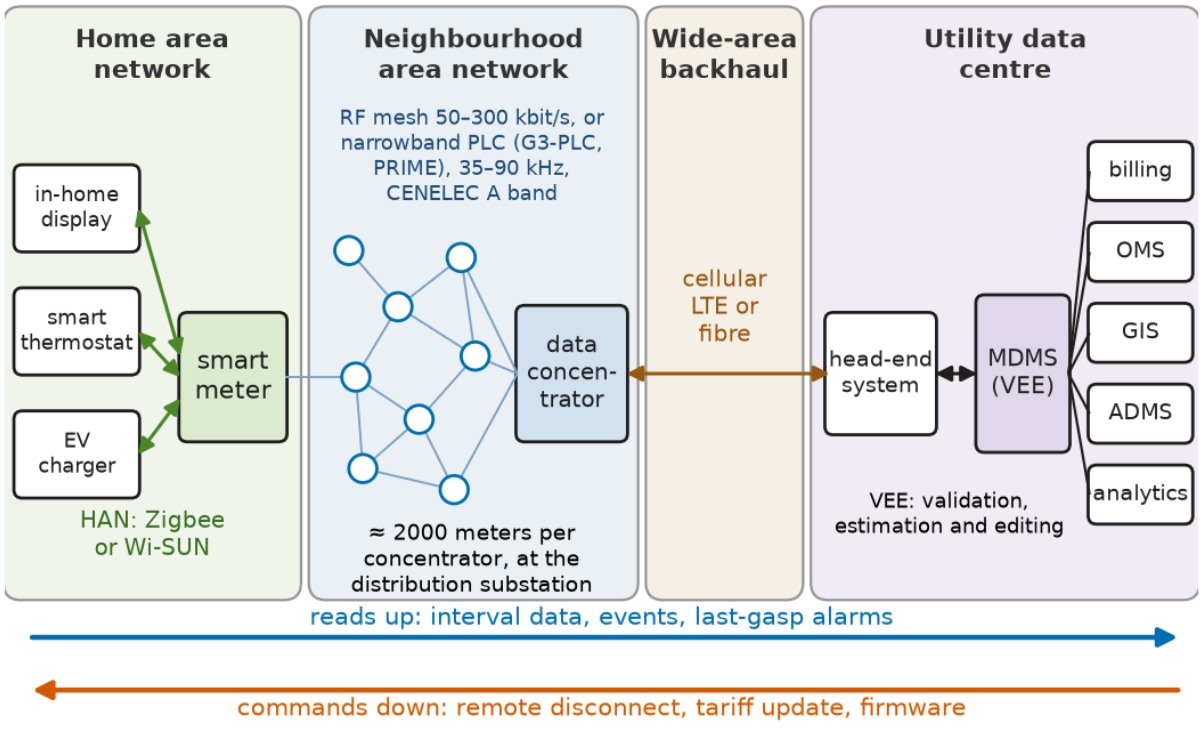


Figure 26.1 — Architecture of an advanced metering infrastructure.

What the meter measures

Modern meters record, at intervals of 15 or 30 minutes (5 minutes in some markets), active import and export energy, reactive energy in four quadrants, RMS voltage per phase (average, minimum and maximum), and event logs: supply interruptions with time stamps, voltage sags and swells, tamper alarms and cover openings. Accuracy is specified by IEC 62053 (class 1 or 0.5S for residential and commercial meters, 0.2S for large customers and grid exchange points) or ANSI C12.20 (accuracy classes 0.5 and 0.2). The data model is standardised in IEC 62056 (DLMS/COSEM) or ANSI C12.19, so that meters and head-ends from different vendors can work together.

What the utility does with the data

- **Billing and tariffs.** Interval data make time-of-use and dynamic tariffs possible (Section 26.5) and end estimated bills.
- **Remote operations.** Connection, disconnection and load limiting without a site visit.
- **Outage management.** A meter's "last gasp" message, sent from the energy stored in its power-supply capacitor as voltage collapses, and its "power restored" message locate outages and confirm restoration, including the single customer whose service fuse remains blown after the feeder is re-energised.
- **Voltage monitoring.** Thousands of meter voltage readings give the first real view of LV voltage profiles, revealing where rooftop PV lifts voltage at noon (Chapter 27) and where evening load pulls it below the statutory band.
- **Network model correction.** Correlating meter voltage time series reveals which customers are on which phase and transformer, correcting the connectivity errors that plague every distribution GIS.
- **Loss and theft detection.** Comparing the energy through a distribution transformer with the sum of its customers' meters exposes non-technical losses, which exceed 10–20 % of energy in some networks.

Worked example — Data volume of an AMI. A utility deploys 1.2 million meters reading every 15 minutes. Each interval record (four channels of energy and voltage, time stamp, status flags and protocol overhead) occupies about 100 bytes. Estimate the daily data volume and the average throughput needed at a data concentrator serving 2000 meters.

Per meter per day: $96 \times 100 = 9600$ bytes. For the fleet:

$$1.2 \times 10^{6} \times 9.6\ \text{kB} = 11.5\ \text{GB/day} \approx 4.2\ \text{TB/year}$$

before events, power-quality logs and redundancy. At the concentrator, $2000 \times 9600 \times 8 = 153.6$ Mbit per day, an average of only 1.8 kbit/s. If the utility wants the previous day's data collected in a four-hour night window, the requirement rises to 10.7 kbit/s of *useful* throughput. That looks modest, but an RF mesh or narrowband PLC network delivers a small fraction of its raw rate after multi-hop relaying, retries and medium-access contention, and the same network must also carry on-demand reads, firmware updates (hundreds of kilobytes per meter) and outage messages that arrive all at once. Sizing an AMI network is therefore dominated by the peaks: a feeder outage generates thousands of last-gasp messages in the same second, most of which are lost, and the system is designed to infer the outage from the few that get through.

26.2 Grid communication

Requirements set by the application

Every function in a smart grid is a control loop with a communication link in it, and the loop's timing sets what the link must do. Table 26.1 groups the main applications by their latency and reliability needs; the spread is six orders of magnitude, from protection at a few milliseconds to billing at a day.

Table 26.1 — Communication requirements of grid applications (typical).

Application	Latency	Data rate per node	Availability	Typical medium
Line differential and teleprotection	2–10 ms	64 kbit/s–1 Mbit/s	99.999 %	Fibre (OPGW), dedicated channels
IEC 61850 GOOSE trip within a substation	≤ 3 ms (transfer class TT6)	Bursty, 10–100 Mbit/s LAN	99.999 % with redundancy (PRP/HSR)	Substation Ethernet
Sampled values from merging units	< 3 ms	5–10 Mbit/s per stream	99.999 %	Process-bus Ethernet
Synchrophasor (PMU) data	20–100 ms	10–100 kbit/s	99.9–99.99 %	Fibre, MPLS networks
SCADA telemetry and control	1–4 s	1–10 kbit/s	99.9–99.99 %	Fibre, microwave, radio, cellular
Distribution automation (FLISR)	0.1–10 s	10–100 kbit/s	99.9 %	Private LTE, RF mesh, fibre
DER and demand-response dispatch	1 s–5 min	1–10 kbit/s	99–99.9 %	Cellular, internet (OpenADR, IEEE 2030.5)
AMI meter reading	Minutes–hours	0.1–10 kbit/s	99–99.5 %	RF mesh, narrowband PLC, cellular

Media

- **Optical fibre** is the backbone of transmission utilities, often run inside the earth wire of overhead lines (optical ground wire, OPGW) or wrapped on phase conductors. It is immune to electromagnetic interference, supports protection-grade latency over hundreds of kilometres, and its link budget is computed as in *Circuits, Signals and Communications*, Ch. 31.
- **Power-line carrier (PLC)** uses the power conductors themselves. On HV lines, analogue PLC at 30–500 kHz has carried teleprotection and voice since the 1930s. In distribution, narrowband PLC in the CENELEC A band (3–95 kHz) at tens of kbit/s (G3-PLC, PRIME) reads meters, especially in Europe; it is cheap because the channel already exists, but it is attenuated by capacitor banks and noisy loads and blocked by transformers.
- **Radio**: licensed point-to-point microwave for backhaul; 900 MHz and 2.4 GHz RF mesh for meters and feeder devices; private LTE (in dedicated utility bands) and public 4G/5G cellular for distribution automation and DER. Fading and outage are handled with the margins of *Circuits, Signals and Communications*, Ch. 30.

- **Satellite** for remote substations, mines and island systems where nothing else reaches.

A utility rarely chooses one: it builds a layered network, with fibre and microwave in the core, MPLS routing across it, and a mix of field-area technologies at the edge.

Protocols

The protocols define what the bits mean. In substations, **IEC 61850** defines an object model of every device (logical nodes such as XCBR for a breaker, MMXU for measurements), the MMS client–server protocol for SCADA, **GOOSE** multicast messages for fast peer-to-peer trips and interlocks, and **sampled values** that replace copper CT and VT wiring with Ethernet from merging units. Between substations and control centres, **DNP3** (North America) and **IEC 60870-5-104** (Europe and much of Asia) dominate SCADA, with **ICCP/TASE.2** between control centres. **IEEE C37.118** carries synchrophasors. At the edge, **OpenADR** and **IEEE 2030.5** carry demand-response and DER signals, **IEC 61850-7-420** and **SunSpec Modbus** describe inverters, and **DLMS/COSEM** carries meter data.

> **Status of the principle: Latency budgeting of grid control loops**
> *Approximate*. A loop is stable and effective if total delay (measurement, communication, computation, actuation) is a small fraction of the time constant of the process controlled, as feedback theory requires; worst-case rather than average delay must be used, and packet-switched networks with retransmission and queuing have delay distributions with long tails that must be bounded by engineering (priority tagging, dedicated bandwidth), not assumed.

26.3 Distributed energy resources

What counts as a DER

A **distributed energy resource** is any generation, storage or controllable load connected to the distribution network or behind a customer's meter: rooftop and commercial PV (typically 3 kW to 5 MW), small wind, combined heat and power units, standby diesel generators, home and commercial batteries, EV chargers (7–22 kW AC at home, 50–350 kW DC on highways), heat pumps, and controllable industrial processes. Table 26.2 gives typical sizes and controllability.

Table 26.2 — Typical distributed energy resources.

Resource	Typical size	Interface	Controllable quantities
Residential PV	3–15 kW	Single- or three-phase inverter	Active-power curtailment, reactive power, volt-var
Commercial and industrial PV	50 kW–5 MW	Three-phase inverters	As above, plus ramp limits, remote setpoints
Home battery	5–15 kWh, 3–10 kW	Hybrid or AC-coupled inverter	Charge/discharge schedule, P and Q
EV charger (AC / DC)	7–22 kW / 50–350 kW	On-board / off-board charger	Charging rate; bidirectional (V2G) on some
Heat pump, water heater	1–10 kW	Motor drive or element	On/off, setpoint shifting
CHP engine	50 kW–5 MW	Synchronous generator	Active and reactive power, island capability
Standby diesel	100 kW–3 MW	Synchronous generator	Peak lopping, emergency supply

Why DERs change the network

Distribution networks were designed for one-way power flow from substation to customer. DERs bring four changes, each treated in its own chapter:

1. **Reverse power flow** and voltage rise on feeders at times of high generation and low load, which limit a feeder's **hosting capacity** (Chapters 22 and 27).
2. **Protection problems**: DER fault contribution can blind feeder relays or cause sympathetic tripping, and inverter-based DER contribute only 1.1–1.5 pu fault current (Chapters 23 and 27).
3. **Invisibility**: behind-the-meter PV masks load, so the substation sees "net load" and the operator does not know how much generation would vanish if a disturbance tripped it (Section 26.7).
4. **New controllability**: a million inverters and chargers that can respond to signals within seconds are a resource that conventional plant cannot match in speed, if they can be coordinated.

Aggregation and the virtual power plant

A single home battery is too small to bid in a wholesale or reserve market, which typically has minimum bids of 0.1–1 MW. An **aggregator** combines thousands into a **virtual power plant** (VPP), forecasts their available flexibility, bids it as one unit and dispatches each device. A fleet of 5000 home batteries of 5 kW and 13.5 kWh offers 25 MW for up to about 2.7 hours, less the reserve each owner keeps for backup, and can reach full output within a second. VPPs of this kind provide frequency reserves in Australia, Germany and the UK, and peak capacity in California.

The utility side of the same coordination is the **DER management system** (DERMS), which knows where each resource is connected and ensures that the aggregator's dispatch does not violate local voltage and thermal limits (Section 26.8). Interconnection rules define what every DER must do automatically: in North America, IEEE 1547-2018 specifies voltage and frequency ride-through, volt-var and frequency-watt functions and communication interfaces, in categories (I, II, III) of increasing capability; in Europe the requirements follow the Requirements for Generators network code (EU 2016/631) and national standards such as VDE-AR-N 4105. The control behind these functions is the subject of Chapter 27.

26.4 Microgrids

Definition and forms

A **microgrid** is a group of interconnected loads and distributed resources within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid and can connect and disconnect from it, operating either grid-connected or islanded (the definition used by IEEE 2030.7 and the US Department of Energy). The defining capability is the controlled transition to and from **island** operation, which distinguishes a microgrid from a site that merely has generation.

Microgrids appear wherever continuity is worth more than its cost or no grid exists:

- **Campuses and hospitals** with CHP, PV, batteries and standby generators, islanding during grid outages.
- **Military bases and data centres**, for mission continuity.
- **Remote mines and islands**, which are permanently islanded diesel systems progressively displaced by PV, wind and batteries; a mine at 50 MW and an island at 5 MW face the same control problems.
- **Ships and aircraft**, which are, electrically, islanded microgrids with tight frequency and voltage limits (IEC 60092, MIL-STD-704).
- **Community microgrids** on utility feeders, created by sectionalising a feeder around a substation battery.

Figure 26.2 shows a typical architecture. Microgrids may be AC (most), DC (data centres, some PV-plus-storage sites, ships with DC distribution) or hybrid.

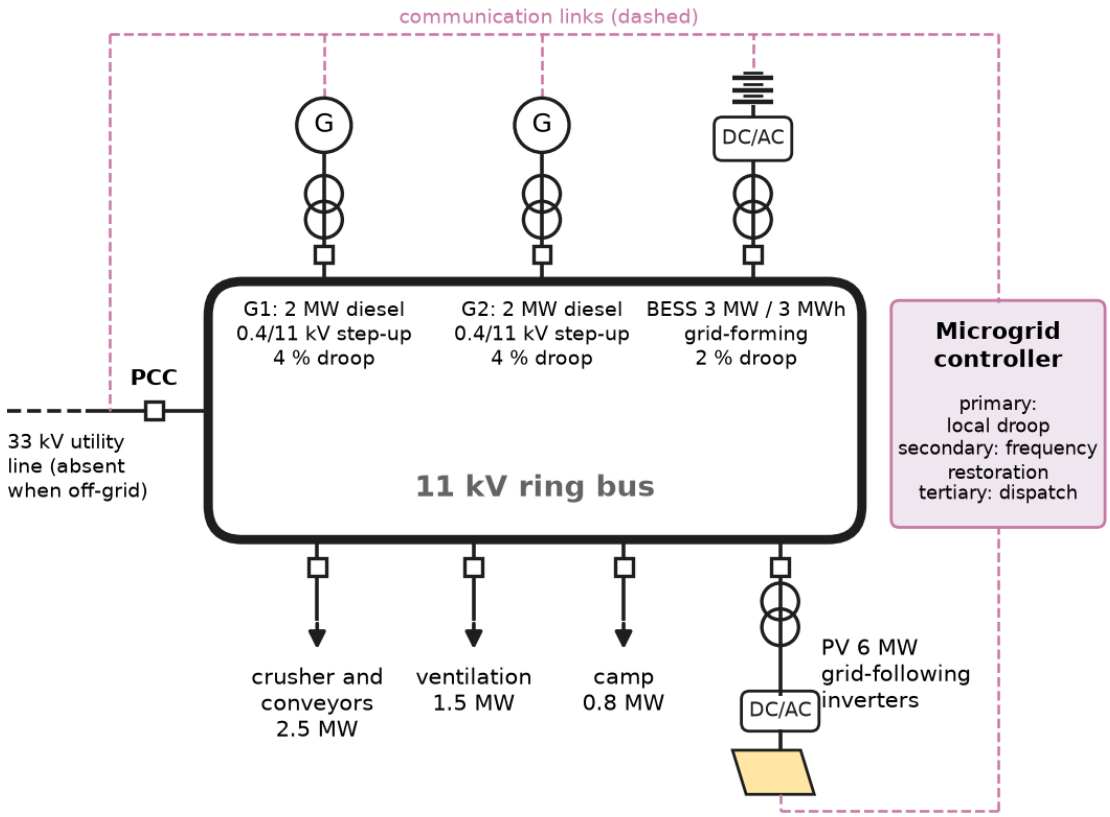


Figure 26.2 — Single-line diagram of an islanded mine microgrid with diesel, PV and battery.

Control hierarchy

Microgrid control is organised in three layers, separated by time scale:

- **Primary control** (milliseconds to seconds) acts locally at each source without communication, sharing load changes and holding voltage and frequency. With synchronous generators this is the governor and AVR droop of Chapter 7; inverters emulate it with **frequency–active-power** and **voltage–reactive-power droop** (Chapter 27).
- **Secondary control** (seconds to minutes) restores frequency and voltage to nominal after a load change by shifting the droop setpoints, through a central microgrid controller or by distributed consensus.
- **Tertiary control** (minutes to hours) dispatches the units economically, manages battery state of charge, schedules generator starts and stops, and handles the import/export with the main grid.

In **islanded** mode at least one source must be **grid-forming**: it must set the voltage and frequency for the others. That role was always played by synchronous generators; in a high-renewable microgrid a grid-forming battery inverter takes it over, allowing all diesels to stop on sunny days.

Worked example — Droop sharing in an islanded mine microgrid. The mine microgrid of Figure 26.2 is islanded at 50 Hz with both 2 MW gensets (4 % droop) and the 3 MW battery (2 % droop) on line. The ventilation fans start a further stage, adding 2.5 MW of load in a few seconds. Find the steady-state frequency after primary control and the share taken by each unit.

A droop R in per unit means that a change of the unit's rated power produces a frequency change of R pu. Each unit's response is therefore $\Delta P_i = -(P_{rated,i}/R_i)\,\Delta f_{pu}$, and the "stiffness" of each is

$$K_{genset} = \frac{2\ \text{MW}}{0.04} = 50\ \text{MW/pu}, \qquad K_{BESS} = \frac{3\ \text{MW}}{0.02} = 150\ \text{MW/pu}$$

for a total of $50 + 50 + 150 = 250$ MW/pu. The frequency falls by

$$\Delta f_{pu} = \frac{2.5}{250} = 0.01\ \text{pu} = 0.5\ \text{Hz}$$

to 49.5 Hz. The gensets each pick up $50 \times 0.01 = 0.5$ MW and the battery $150 \times 0.01 = 1.5$ MW: power is shared in proportion to rating divided by droop. The battery takes the largest share by design, because it responds in milliseconds and its smaller droop makes it stiffer; the diesels take less of the fast transient, which avoids their black-smoke and turbo-lag limitations. Secondary control then raises the setpoints to bring the frequency back to 50 Hz over 10–30 s, and the tertiary layer may start a third genset or let the battery carry the load until its state of charge reaches a floor. Figure 26.3 shows the droop lines. The same arithmetic applies to voltage–reactive-power droop, with the complication that reactive power sharing is also affected by the impedance between sources, so it is less exact.

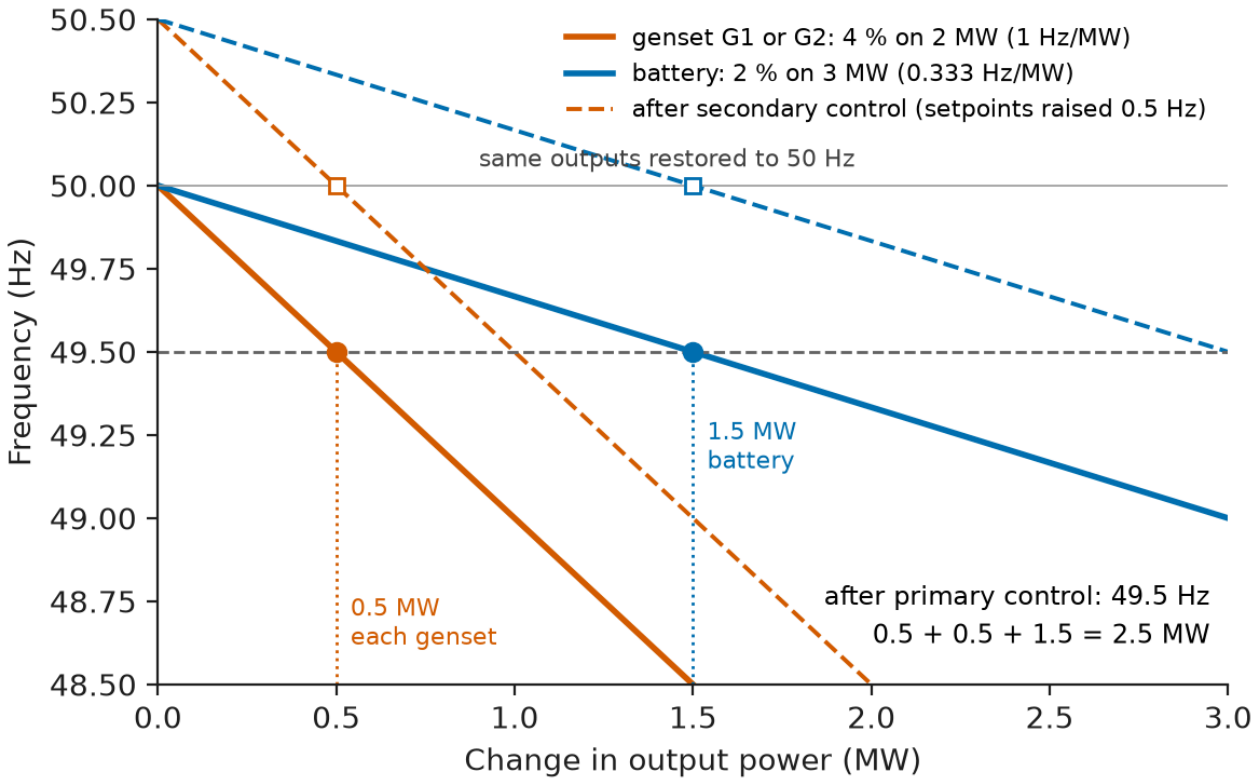


Figure 26.3 — Frequency-power droop characteristics of two gensets and a battery sharing a 2.5 MW load step.

> **Status of the principle: Steady-state droop sharing**
> *Derived*. Follows directly from the linear droop laws and the common steady-state frequency of a synchronous island; it neglects governor and inverter limits, frequency-dependent load and the transient (inertia and damping) behaviour, which govern the first seconds (Chapter 20).

Transitions and protection

Planned islanding (opening the PCC breaker after reducing the exchange to near zero) is easy. Unplanned islanding after a grid fault requires the microgrid to survive a voltage dip, detect the loss of grid, open the PCC within a few cycles and pick up the full island load on its own sources; batteries and grid-forming inverters make this seamless, while diesel-only microgrids usually black out and restart. Reconnection requires synchronisation across the PCC breaker, exactly as for a generator (Chapter 7). Protection is the hardest part of microgrid design: in grid-connected mode the fault current is high, in islanded mode with inverter sources it may be only 1.2–2 times load current, so overcurrent relays set for one mode fail in the other. Adaptive protection with switched setting groups, differential protection and voltage-based schemes are used (Chapter 23).

26.5 Demand response

Load as a resource

Demand response (DR) is the deliberate change of electricity consumption in response to price or to a signal from the system operator. It has always existed for large industrial customers on interruptible contracts; what

is new is that controllable residential and commercial loads, now numbered in millions, can be aggregated and dispatched like a generator. The value is large because peaks are short: in many systems the top 1 % of hours (about 88 hours a year) accounts for 8–15 % of peak capacity, which is built and maintained for those hours alone.

DR programmes divide into two families:

- **Price-based**: time-of-use tariffs (fixed peak and off-peak prices), critical-peak pricing (a very high price on 10–15 announced days a year), and real-time pricing that follows the wholesale price. Customers or their devices respond voluntarily.
- **Incentive-based**: direct load control (the utility cycles water heaters or air conditioners by radio or internet signal), interruptible and curtailable contracts for large customers, and participation of aggregated loads in capacity, energy and reserve markets.

The best DR resources have **thermal or energy storage built in**: water heaters, space heating and cooling, refrigeration and cold stores, EV charging, pumping (water utilities, irrigation), and industrial processes with buffer storage (cement mills, electrolysers, aluminium smelters, which can shed hundreds of megawatts for an hour). Their consumption can be shifted in time without loss of service.

Payback and verification

Deferred energy is not saved; it is moved, and it comes back. When a group of water heaters or air conditioners is released after a control period, their thermostats all call for heat or cooling at once and the aggregate load rises above its normal level, the **payback** or **rebound** peak. Poorly managed, the rebound can create a new peak an hour after the old one.

Worked example — Direct control of water heaters. A utility controls 20,000 residential water heaters of 3 kW. At the evening peak, on average 25 % are on at any instant. The utility switches them all off for 2 hours. Estimate the peak reduction, the deferred energy and the rebound if, on release, 60 % switch on together.

Load removed at the peak: $20\,000 \times 3\ \text{kW} \times 0.25 = 15\ \text{MW}$.

Deferred energy (ignoring the extra standing losses saved while tanks cool): $15\ \text{MW} \times 2\ \text{h} = 30\ \text{MWh}$.

On release, $0.6 \times 20\,000 \times 3\ \text{kW} = 36\ \text{MW}$ is drawn, 21 MW above the normal 15 MW, for about $30/21 = 1.4$ hours until the deferred energy is recovered. A 15 MW peak reduction has been bought at the cost of a 21 MW rebound. In practice the utility releases the heaters in groups staggered over an hour or two, so that the rebound is spread below the original peak, and it can make the control period itself rotate between groups (each heater off for 30 minutes in turn) to deliver a smaller but flat reduction.

Measuring what a customer *would* have consumed without the DR event, the **baseline**, is the central problem of paying for DR. Baselines are computed from recent similar days (for example the average of the highest 5 of the last 10 working days, adjusted for the morning of the event), and they can be gamed. Direct load control with device-level telemetry, and statistical control groups, are the more robust methods. **OpenADR 2.0** and **IEEE 2030.5** are the standard signalling protocols between operators, aggregators and devices.

26.6 Grid automation

Substation automation

A modern substation replaces the hard-wired relay panels, mimic boards and RTUs of the past with intelligent electronic devices (IEDs) on an IEC 61850 network. The architecture has three levels: the **process level** (merging units and breaker interface units in the yard, connected by a process bus carrying sampled values and trip commands over fibre), the **bay level** (protection and control IEDs), and the **station level** (station computer, HMI and gateway to the control centre). Benefits are less copper wiring between yard and control building (a 400 kV substation may have thousands of kilometres of it), self-supervision of every link, and engineering by configuration files (SCL) rather than wiring diagrams. The cost is a new class of failure: a misconfigured network switch can disable protection, so redundancy (PRP, HSR) and rigorous testing are mandatory.

Distribution automation and FLISR

Most customer interruptions originate on the distribution network, so automating it has the largest effect on reliability indices. The key function is **fault location, isolation and service restoration** (FLISR). On a feeder with remotely controlled sectionalising switches or reclosers and normally open ties to neighbouring feeders, the sequence after a permanent fault is:

1. The feeder breaker or recloser trips; reclosing fails.
2. Fault passage indicators at each switch report whether fault current passed through it. The fault lies between the last switch that saw it and the first that did not.
3. The switches either side of the faulted section open.
4. The feeder breaker recloses to restore the upstream sections; the normally open tie to an adjacent feeder closes to restore the downstream sections, after checking that the adjacent feeder has capacity.

This can be centralised (in the ADMS), distributed (switches communicating peer to peer) or local (reclosers using voltage and time logic, without communication). Completion within one to three minutes is typical. Figure 26.4 shows the sequence.

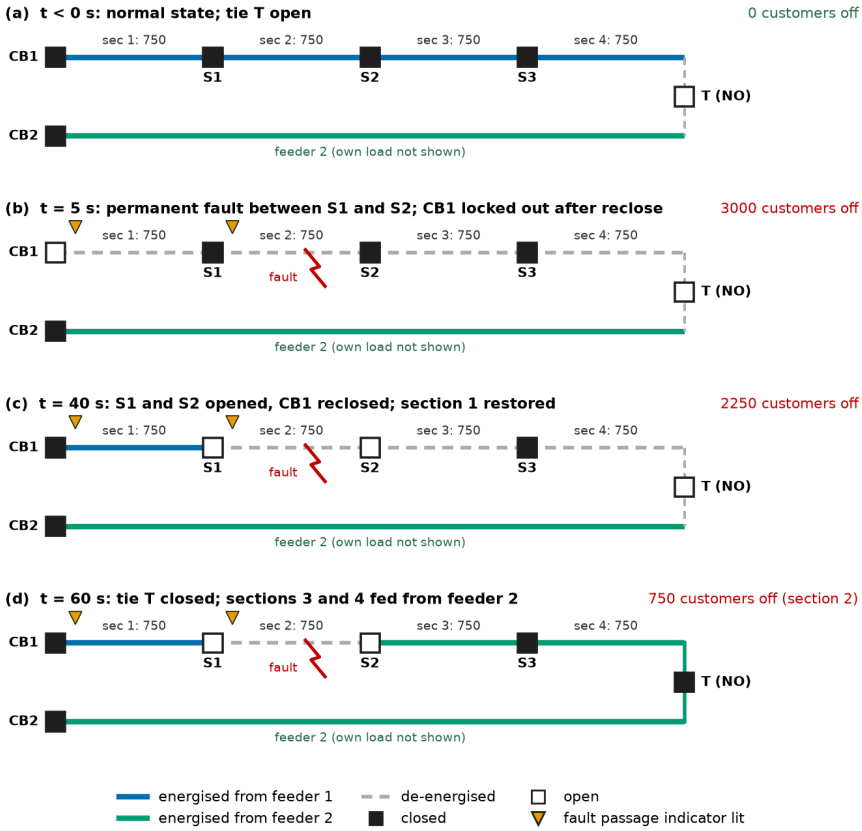


Figure 26.4 — Fault location, isolation and service restoration on two 11 kV feeders with a normally open tie.

Worked example — Reliability benefit of FLISR. An 11 kV feeder serves 3000 customers in four equal sections, and suffers 2 permanent faults per year. Without automation, after a fault the unfaulted three quarters of the customers are restored by manual switching after 1.5 h on average, and the faulted quarter after repair, 4 h. With FLISR the unfaulted sections are restored within one minute. Compare the feeder's contribution to SAIDI and SAIFI.

Without automation, customer-hours of interruption per year:

$$2 \times (2250 \times 1.5 + 750 \times 4) = 2 \times (3375 + 3000) = 12\,750 \text{ customer} \cdot \text{h}$$

SAIDI for these customers $= 12\,750 \times 60/3000 = 255$ minutes per year; SAIFI $= 2$ interruptions per customer per year.

With FLISR, the 2250 unfaulted customers see a one-minute interruption, which under IEEE 1366 is a momentary event (less than 5 minutes) and does not count in SAIDI or SAIFI. Customer-hours $= 2 \times 750 \times 4 = 6000$, SAIDI $= 120$ minutes; SAIFI $= 2 \times 750/3000 = 0.5$. FLISR halves SAIDI and cuts SAIFI by three quarters, without touching the fault rate. The momentary-interruption index MAIFI rises, which matters to customers with sensitive processes (Chapter 25), so FLISR complements, and does not replace, fault reduction.

Volt/var optimisation and conservation voltage reduction

The second major automation function is **volt/var optimisation** (VVO): coordinated control of substation tap changers, line regulators, switched capacitors and inverter reactive power to hold every customer's voltage inside the band while minimising losses or demand. A common objective is **conservation voltage reduction** (CVR): operating in the lower half of the statutory band, because many loads draw less power at lower voltage. The **CVR factor** is the percentage energy reduction per percentage voltage reduction, measured

at 0.5–1.0 for typical residential and commercial feeders (lower where loads are constant-power electronics or thermostatic). A 3 % voltage reduction on a feeder delivering 40 GWh per year with a CVR factor of 0.8 saves $0.8 \times 3 = 2.4\,\%$, or 0.96 GWh per year. CVR depends on knowing the voltage at the end of every feeder, which is exactly what AMI voltage data now provide.

> **Status of the principle: CVR factor**
> *Empirical*. Fitted from field trials on specific feeders; it reflects the voltage dependence of the load mix (resistive and motor loads respond, electronic and thermostatically controlled loads largely do not, or recover energy later) and is falling as constant-power electronic loads replace resistive ones.

26.7 Grid observability

What an operator can see

A power system is **observable** when the available measurements are sufficient to determine the complex voltage at every bus. Transmission systems have been observable since the 1970s: SCADA RTUs report breaker states, flows and voltages every 2–4 seconds, and a **state estimator** fits a consistent network solution to these redundant, noisy measurements by weighted least squares, detects and rejects bad data, and provides the base case for contingency analysis and security assessment (Section 26.8). Redundancy, typically 1.5 to 3 measurements per state variable, is what allows bad data to be identified.

Distribution networks were, until recently, almost unobservable below the substation: a feeder head current and perhaps a few line-sensor readings for hundreds of transformers and thousands of customers. Load was inferred from monthly energy and typical load profiles (**pseudo-measurements**). AMI voltage and interval data, line sensors, and inverter telemetry are changing this, but data latency (meters report hours later) and the cost of real-time communication to millions of points mean that distribution state estimation still mixes a few real-time measurements with many forecasts and pseudo-measurements.

The hardest blind spot is **behind-the-meter generation**. A substation that sees 20 MW of net load may be serving 35 MW of gross load with 15 MW of rooftop PV hidden behind it. If a voltage dip trips that PV (older inverters disconnect on small disturbances), the load seen by the grid jumps by 15 MW at once. Estimating hidden PV from irradiance data, sample metering and inverter fleet data is now a routine operating task.

Synchrophasors

A **phasor measurement unit** (PMU) samples voltages and currents, estimates their phasors (magnitude and angle) and stamps each estimate with GPS-derived time accurate to better than 1 µs, so that phasors from substations hundreds of kilometres apart can be compared directly. Reporting rates are 10–60 per second (25 or 50 at 50 Hz, 30 or 60 at 60 Hz), against one sample per 2–4 s for SCADA. Accuracy is specified by IEEE C37.118.1 through the **total vector error**

$$\mathrm{TVE} = \frac{|\hat{\mathbf{X}} - \mathbf{X}|}{|\mathbf{X}|}$$

with a steady-state limit of 1 %. A 1 % TVE corresponds to a pure angle error of 0.573°, which at 50 Hz is a time error of only $0.573/360 \times 20\ \mathrm{ms} = 31.8\ \mu\mathrm{s}$ (26.5 µs at 60 Hz); this is why GPS or IEEE 1588 precision time is essential.

PMUs made possible what SCADA could not: direct observation of angle differences across a grid (the stress indicator behind the power-angle relationship of Chapter 7), real-time detection of inter-area oscillations at 0.1–1 Hz (Chapter 20), linear state estimation at the reporting rate, post-event analysis with exact timing, and wide-area protection schemes. **Distribution PMUs** (micro-PMUs), with angle resolution of 0.01° or better, extend the idea to feeders, where angle differences are tiny.

> **Status of the principle: Total vector error as a PMU accuracy measure**
> *Derived*. TVE combines magnitude, phase and timing errors into a single vector distance; it is exact as a definition, but a TVE limit says nothing about which component dominates, so frequency and ROCOF accuracy are specified separately, and dynamic performance (during ramps and oscillations) has its own, looser limits.

26.8 Energy management systems

At the transmission control centre

An **energy management system** (EMS) is the software suite with which a transmission operator runs the bulk power system. Its layers are:

- **SCADA**: telemetry, alarms and remote control of breakers and setpoints.
- **Network analysis**: topology processing, state estimation (Section 26.7), power flow (Chapter 18), **contingency analysis** that tests every credible outage ($N-1$) against thermal and voltage limits every few minutes, and online stability assessment (Chapter 20).
- **Generation control**: **automatic generation control** (AGC) and **economic dispatch**, and in market systems the security-constrained unit commitment and dispatch that set prices.

AGC is the secondary frequency control of a whole interconnection. Each balancing area computes its **area control error**

$$\mathrm{ACE} = (NI_A - NI_S) - 10B\,(F_A - F_S)$$

where NI_A and NI_S are the actual and scheduled net interchange (MW, export positive), F_A and F_S the actual and scheduled frequency, and B the area's frequency bias (negative, in MW per 0.1 Hz, as defined in NERC practice). AGC adjusts the setpoints of participating generators every 2–6 s to drive ACE towards zero. The two terms are arranged so that each area corrects only its own imbalance while all areas share the primary frequency response.

> **Status of the principle: Tie-line bias control (area control error)**
> *Approximate*. Assumes each area's frequency response is linear and correctly represented by its bias B; when B differs from the area's true response, areas partly correct each other's imbalances, and the scheme takes no account of the dynamics within the first seconds, which belong to primary control (Chapter 20).

Worked example — Area control error after a generator trip. A balancing area has a frequency bias of $B = -120$ MW/0.1 Hz and a scheduled export of 500 MW. After one of its own units trips, the actual export falls to 460 MW and the interconnection frequency to 59.97 Hz (scheduled 60.00 Hz). Find the ACE.

$$\text{ACE} = (460 - 500) - 10 \times (-120) \times (59.97 - 60.00) = -40 - 36 = -76 \text{ MW}$$

A negative ACE means the area is short of generation: AGC raises its units' output by 76 MW. The 40 MW drop in export is the help the area is receiving from its neighbours through their primary response; the 36 MW term is the area's own share of the frequency deviation. Had the trip occurred in a *neighbouring* area, this area's export would have *risen* by about its own response (36 MW, to 536 MW), and ACE $= 36 - 36 = 0$: correctly, this area's AGC would do nothing, leaving the neighbour to correct its own loss.

In distribution, buildings and microgrids

The distribution counterpart is the **advanced distribution management system** (ADMS), which combines distribution SCADA, the outage management system fed by AMI and customer calls, the network model from GIS, FLISR, VVO, distribution state estimation and, increasingly, a **DERMS** that forecasts, monitors and dispatches distributed resources within network limits. The ADMS is the platform on which most of Sections 26.1–26.7 come together.

At smaller scales the same functions recur. A **microgrid controller** performs dispatch, state-of-charge management and island transitions. A **building energy management system** (BEMS) schedules HVAC, lighting, storage and EV charging against tariffs and comfort constraints, and is how commercial buildings participate in demand response; data centres and industrial plants use the same principles with process constraints (Chapter 28). A **home EMS** coordinates PV, battery, heat pump and EV. Every one of them is an optimiser with a forecast at its input and a control loop at its output, and every one depends on the communication and observability layers beneath it.

26.9 Cybersecurity fundamentals

Why the grid is a target

Connecting protection relays, substations, meters and millions of inverters to communication networks exposes the grid to remote attack. The threat is not hypothetical. In December 2015, attackers who had spent months inside three Ukrainian distribution companies' networks opened breakers remotely through the companies' own SCADA systems, interrupting about 225,000 customers for several hours, and overwrote the firmware of serial-to-Ethernet converters to prevent remote restoration. In December 2016 the Industroyer (CrashOverride) malware, which spoke IEC 60870-5-104, IEC 61850 and OPC natively, opened breakers at a Kyiv transmission substation. Stuxnet (discovered 2010) showed that malware could manipulate industrial controllers while reporting normal values to operators.

Security for operational technology

Information security is usually summarised as **confidentiality, integrity and availability**. In operational technology (OT), the priorities are reversed: **availability** and **integrity** of control come first, confidentiality last. A protection IED must never be delayed by a security check: GOOSE trips must arrive within 3 ms, which rules out many conventional cryptographic handshakes. Field devices last 20–30 years and cannot be patched weekly. Many legacy protocols (Modbus, early DNP3, IEC 60870-5-101/104) have no authentication: any device that can reach them can issue commands.

Defences follow a few principles:

- **Defence in depth and segmentation.** Networks are divided into zones by function and criticality, connected only through controlled conduits (IEC 62443's zones and conduits; the layered "Purdue model" of enterprise, operations and control networks), with demilitarised zones between corporate IT and OT. No direct path from the internet or the corporate network to a protection relay.
- **Authentication and integrity of commands.** IEC 62351 adds TLS to MMS and IEC 60870-5-104, authentication codes to GOOSE and sampled values, and role-based access control; DNP3 Secure Authentication adds challenge–response to critical commands.
- **Least functionality.** Unused ports and services disabled; firmware signed; engineering access through jump hosts with multi-factor authentication.
- **Monitoring.** OT-aware intrusion detection that knows what normal SCADA traffic looks like, logging of every configuration change, and the ability to operate manually if networks are compromised.
- **Supply chain and DER.** Inverter and charger fleets controlled through vendor clouds are a new attack surface: a compromised cloud platform could, in principle, change the settings of a gigawatt of inverters at once, which is a larger and faster disturbance than any single generator trip. Standards (IEEE 1547.3, UL 2941) and regulators increasingly address this.

Regulation makes some of this mandatory: NERC CIP standards in North America for the bulk electric system, the EU NIS2 directive and network codes on cybersecurity, and national equivalents. They define asset categorisation, electronic and physical perimeters, access management, incident response and recovery planning.

26.10 Resilience

Reliability and resilience

Reliability describes performance under routine, credible events: the frequency and duration of interruptions caused by ordinary faults, measured by SAIFI and SAIDI (Chapter 13). **Resilience** describes the ability to withstand, adapt to and recover rapidly from high-impact, low-probability events: hurricanes, ice storms, floods, wildfires, earthquakes, geomagnetic storms, extreme cold or heat, and coordinated physical or cyberattack. These events are usually excluded from reliability statistics as "major event days", yet they cause most of the customer-hours lost in many systems. Examples are Hurricane Maria in Puerto Rico (2017), which left most of the island without power for months; the February 2021 Texas cold snap, when tens of gigawatts of generation were forced out by cold and fuel shortages and rotating outages lasted days; and the 2016 South Australian blackout, when a storm brought down transmission towers and a cascade of wind-farm protection settings tripped 456 MW.

The resilience trapezoid

Resilience is often pictured with the **resilience trapezoid** (Figure 26.5): system performance (for example the fraction of load served) during an event falls (disruption), stays degraded (the post-disruption state) and is restored (recovery). The quantities that matter are how far performance falls, how long it stays low, and how fast it recovers, and the area of lost performance is the measure of total impact. Each phase has its own engineering measures.

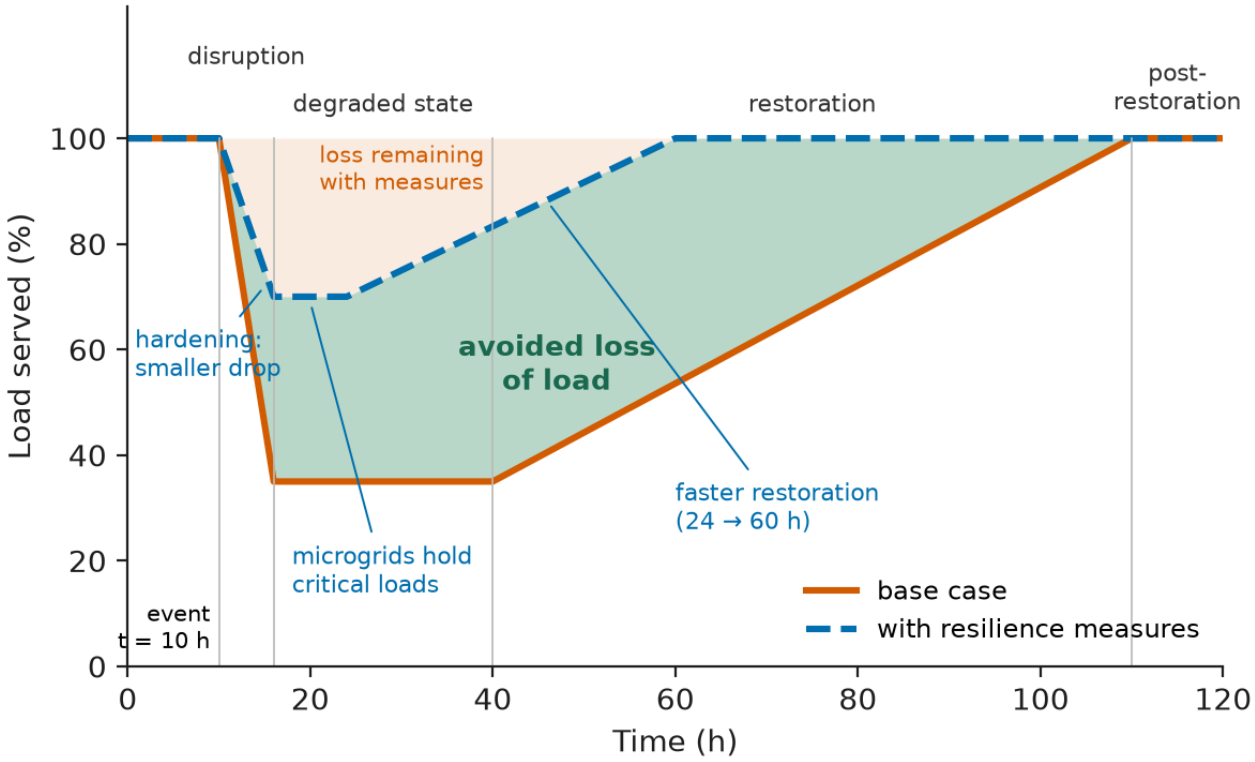


Figure 26.5 — The resilience trapezoid: performance of a network during and after an extreme event, with and without resilience measures.

- **Withstand (reduce the drop).** Hardening: undergrounding in high-risk areas, stronger poles and towers, vegetation management, covered conductors, flood walls around substations, fire-safe designs and pre-emptive de-energisation in wildfire weather (public safety power shut-off).
- **Adapt (sustain critical service).** Microgrids and backup supply for hospitals, water treatment, communications and emergency services; network reconfiguration to supply critical loads; islanding of healthy parts of the grid; demand response to stretch scarce generation; firm fuel supplies and winterised plant.
- **Recover (shorten the outage).** Black-start capability (Chapter 14), restoration plans and drills, mutual aid between utilities, mobile substations and generators, spare transformers, and situational awareness from AMI and drones to direct crews.

Resilience investments are justified differently from reliability investments: by the expected cost of rare, severe events, the value of lost load to critical customers, and scenario analysis, rather than by average indices. A hospital microgrid that does nothing in an average year but keeps an intensive-care unit running through a four-day outage may be worth more than a feeder upgrade that improves SAIDI by ten minutes.

Smart-grid technology cuts both ways here. Automation, microgrids, DER and storage make the grid more adaptable and faster to restore. But the same technology adds dependence on communications, on GPS time, on software and on the electricity supply of the communication network itself (cellular base stations typically have only 2–8 hours of battery), and it adds the cyber exposure of Section 26.9. A resilient smart grid is one that is designed to degrade gracefully to manual and local operation when its information layer fails.

What remains open

Open question	Where it is answered
How do grid-connected inverters synchronise, control P and Q, and meet grid codes?	Chapter 27
How is the voltage rise from distributed PV calculated, and what limits hosting capacity?	Chapter 22, Chapter 27
How do inertia, governor droop and reserves determine frequency after a disturbance?	Chapter 20, Chapter 27
How are batteries in VPPs and microgrids rated, managed and protected?	Chapter 16
How is protection adapted to low fault currents from inverter sources?	Chapter 23
How are reliability indices defined and used in planning?	Chapter 13, Chapter 22
How do building and industrial energy management translate into measured savings?	Chapter 28
How are channels, modulation and link budgets designed for grid communications?	*Circuits, Signals and Communications*, Ch. 25–31

Next: Chapter 27 goes inside the inverters that most distributed and renewable resources connect through, and asks what they must do to keep the grid stable as they replace synchronous machines.

Chapter 27 — Grid Integration of Renewable Energy

Part VII · Power Quality and Grid Modernization

On 9 August 2019 a lightning strike on a transmission line north of London caused a routine fault, cleared in under 100 ms. Within the next second an offshore wind farm's control system reacted badly to the voltage dip and reduced its output by more than 700 MW, a gas-fired plant tripped a steam turbine, and around 500 MW of small distributed generation disconnected because its loss-of-mains protection mistook the system's rate of change of frequency for an island. Frequency fell to 48.8 Hz, automatic demand disconnection shed about a million customers, and trains across the south-east stopped. No equipment was damaged and nothing was overloaded; the system lost more power faster than its inertia and reserves could absorb, and much of what it lost was connected through inverters whose behaviour during a disturbance had not been specified, or not tested, well enough. The incident is a compact summary of this chapter: renewable plants are now large enough that how their inverters behave in the first second after a fault matters as much as how much energy they produce in a year.

> **How does a grid-connected inverter synchronise to the grid and control the active and reactive power it delivers, and what do grid codes require it to do in normal operation and during disturbances?**
> **What limits the share of inverter-based generation a power system can accept, and how do frequency support, voltage support, storage and forecasting push that limit back?**

Concept	Engineering destination
Grid-connected inverters	PV plants, wind turbines, battery systems, HVDC and STATCOM converters
Synchronization	Phase-locked loops, weak-grid stability, grid-forming control
Active and reactive power control	Plant controllers, curtailment, reactive capability at the point of connection
Grid-code fundamentals	IEEE 1547, IEEE 2800, ENTSO-E RfG, compliance testing
Anti-islanding	Rooftop PV, distribution protection, utility worker safety
Frequency support	Fast frequency response, synthetic inertia, frequency-watt functions
Voltage support	Volt-var control, fault-current injection, hosting capacity
Storage coordination	Ramp-rate control, firming, co-located PV and battery plants
Forecasting	Reserve sizing, unit commitment, curtailment planning
Renewable penetration limits	Inertia floors, system strength, SNSP limits, synchronous condensers

Roadmap. The device first: what a grid-connected inverter is as a power-system component (27.1), and how it locks onto the grid (27.2) and controls P and Q (27.3). Then the rules it must follow: grid codes (27.4) and anti-islanding (27.5). Then the services the system needs from it: frequency support (27.6) and voltage support (27.7), and the storage that makes those services firm (27.8). Finally the system view: forecasting (27.9) and the limits to renewable penetration (27.10).

Boundary.
Inverter circuits, PWM and switching devices are outlined in *Engineering Physics*, Ch. 18 §18.8, and belong in detail to *Electronics & Computing*; PWM inverters for drives are treated in Chapter 12. Three-phase phasors and complex power are from *Circuits, Signals and Communications*, Ch. 9 and Ch. 10; the feedback theory behind PLLs and current controllers is in Ch. 34–35. The rotating kinetic energy that inertia refers to is the rigid-body result of *Engineering Physics*, Ch. 9 §9.7; the swing equation is in Chapter 20 of this book.

27.1 Grid-connected inverters

The inverter as a power-system component

A synchronous generator is a voltage source behind a reactance whose frequency is fixed by a spinning rotor and whose internal voltage is set by its field current (Chapter 7). A grid-connected **voltage-source converter** (VSC) is, at the fundamental frequency, the same thing without the rotor: a three-phase voltage **E** synthesised by pulse-width modulation from a DC link, connected to the grid voltage **V** through a filter inductance (or an LCL filter) and usually a step-up transformer, together a reactance X. The power it delivers follows the same power-angle relations as for the synchronous machine:

$$P = \frac{EV}{X}\sin\delta, \qquad Q = \frac{EV\cos\delta - V^2}{X}$$

Two differences matter. First, E and δ are not physical states of a rotor but outputs of a control algorithm, updated every switching period (tens to hundreds of microseconds); the converter has no inertia and no natural tendency to stay synchronised. Second, its semiconductors have almost no thermal capacity, so its current is strictly limited, typically to 1.0–1.2 pu continuously and 1.1–1.5 pu for a few seconds. A synchronous machine can deliver 5–7 pu fault current for a few cycles; an inverter cannot.

Status of the principle: Inverter as a controlled voltage source behind a reactance
Approximate. Valid for the fundamental-frequency, averaged behaviour over many switching periods when the DC link is stiff and the converter is within its current limit; it fails for harmonic and switching-frequency phenomena, during current limiting (when the converter becomes a current source), and on time scales comparable with its control bandwidth.

Architectures in renewable plants

Table 27.1 summarises how the main renewable and storage technologies connect.

Table 27.1 — Grid interfaces of renewable and storage plants.

Plant	Typical unit size	Converter arrangement	Grid side
Residential PV	3–10 kW	String or micro-inverters	230 V single-phase or 400 V
Utility PV	3–5 MW blocks, plants to 1 GW	1500 V DC strings, central or string inverters (630–800 V AC)	MV transformers, HV substation
Wind, Type 3 (DFIG)	2–5 MW	Partial-rated converter (about 30 %) in rotor circuit	690 V stator direct, MV transformer
Wind, Type 4 (full converter)	3–15 MW	Full-rated back-to-back converter, PMSG or induction generator	690 V–3.3 kV, MV transformer
Battery storage	1–5 MW blocks, plants to several GWh	Bidirectional inverter, DC- or AC-coupled	MV transformers, HV substation
HVDC (VSC)	0.5–2 GW	Modular multilevel converter	220–525 kV AC

Utility-scale plants have a **power plant controller** at the point of connection that measures P, Q, V and f there and sends setpoints to the hundreds of inverters below it, so that the plant behaves as one unit; grid codes are written for this point, not for the individual inverter.

Worked example — Converter voltage needed for reactive capability. A 690 V wind-turbine converter is rated 1.0 pu active power and must deliver ± 0.33 pu reactive power at full output over a grid-voltage range of 0.9–1.1 pu. The filter and transformer reactance is $X = 0.15$ pu. Find the internal voltage E the converter must synthesise, and hence the DC-link voltage.

With $\mathbf{V}$ as reference and $S = P + jQ = \mathbf{V}\mathbf{I}^{*}$, the current is $\mathbf{I} = (P - jQ)/V$ and $\mathbf{E} = \mathbf{V} + jX\mathbf{I}$.

At $V = 1.0$, $P = 1$, $Q = +0.33$ (supplying vars): $\mathbf{I} = 1 - j0.33$, $\mathbf{E} = 1 + j0.15(1 - j0.33) = 1.0495 + j0.150$, so $E = 1.060$ pu at $\delta = 8.1^{\circ}$.

At $V = 1.1$, $P = 1$, $Q = -0.33$ (absorbing vars at high grid voltage): $\mathbf{I} = (1 + j0.33)/1.1$, $\mathbf{E} = 1.1 + j0.15\mathbf{I} = 1.055 + j0.136$, so $E = 1.064$ pu.

At $V = 0.9$, $P = 1$, $Q = +0.33$: $E = 0.969$ pu, but the current rises to $|\mathbf{I}| = 1.17$ pu, so the converter must either be rated for 1.17 pu current or reduce P at low voltage.

The largest E is 1.064 pu, a line-to-line RMS voltage of $1.064 \times 690 = 734$ V. With space-vector modulation the peak line-to-line voltage a converter can synthesise in the linear range equals its DC-link voltage, so

$$V_{dc} \geq \sqrt{2} \times 734 = 1038 \text{ V}$$

and with margin for control headroom and harmonic compensation, 690 V converters run DC links of 1050–1200 V. This is one reason the PV industry moved from 1000 V to 1500 V string voltages: higher AC output voltage per inverter, less current and less copper for the same power.

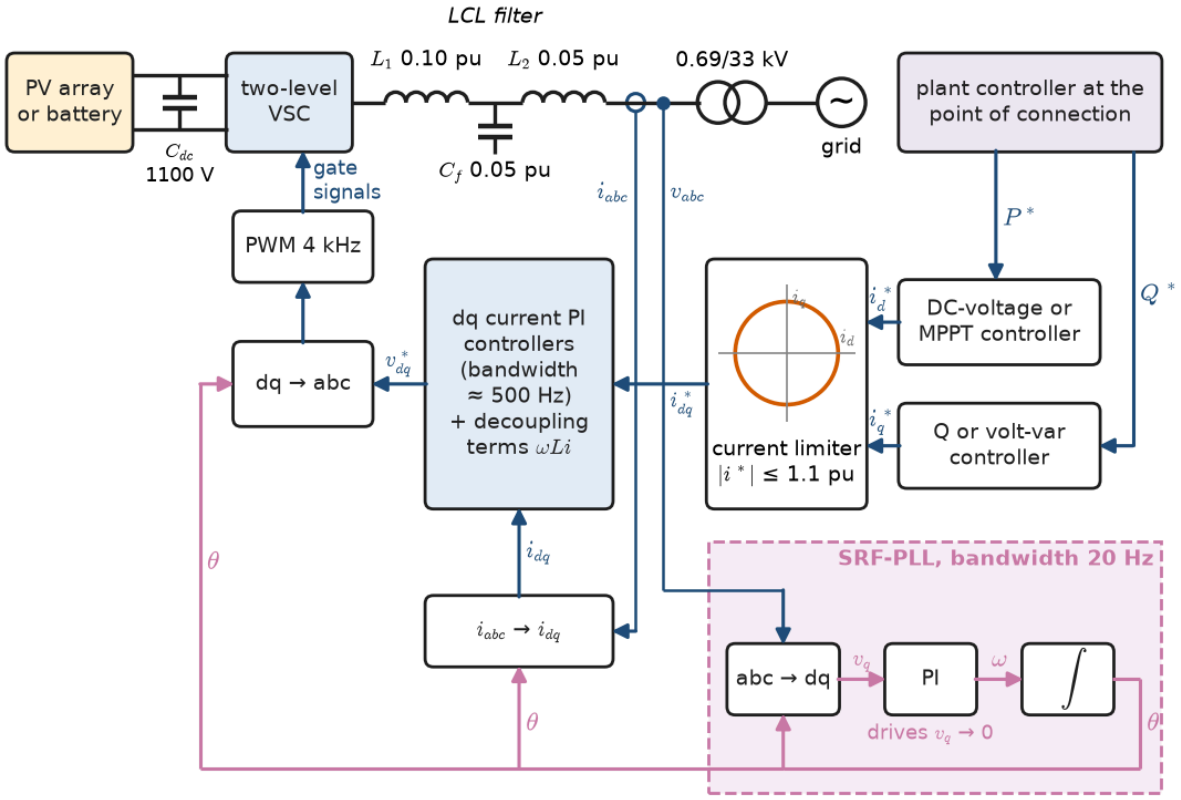


Figure 27.1 — Control structure of a grid-following inverter.

27.2 Synchronization

The phase-locked loop

To deliver current in a defined phase relation to the grid voltage, an inverter must know the grid voltage's phase angle at every instant. The standard method is the **synchronous-reference-frame phase-locked loop** (SRF-PLL). The three measured voltages are transformed into a frame rotating at the PLL's estimated angle θ; if θ equals the true grid angle, the quadrature component v_q is zero. A PI controller adjusts the estimated frequency to drive v_q to zero, and an integrator turns frequency into angle. Locked, the PLL delivers θ, the frequency and the voltage magnitude v_d. The current controller then works in the same rotating frame, where the d-axis current sets active power and the q-axis current reactive power (Section 27.3). An inverter controlled this way is called **grid-following**: it measures the grid voltage and injects a current relative to it, and it needs an existing voltage to follow.

PLL design is a feedback-design problem of the kind treated in *Circuits, Signals and Communications*, Ch. 34–35. A fast PLL (bandwidth of 50–100 Hz) tracks phase jumps quickly, which helps fault ride-through, but passes harmonics and noise into the angle estimate and interacts with the current controller and the grid. A slow PLL (5–20 Hz) is smooth but lags during disturbances. Under unbalanced faults the negative-sequence voltage produces a 100 Hz ripple in v_q; practical PLLs separate the sequences (DSOGI-PLL, decoupled double-frame PLL) so that they lock onto the positive sequence.

Weak grids and the short-circuit ratio

A PLL locks onto the voltage at the inverter's terminals, but that voltage is itself moved by the inverter's own current through the grid impedance. On a stiff grid the effect is negligible. On a **weak grid** (high impedance, low fault level) the inverter is partly locking onto itself, the loop gains change with operating point, and grid-following plants can oscillate (sub-synchronous oscillations at 5–40 Hz have been observed at wind farms in Texas, China and Australia) or lose synchronism after a fault. Grid strength at the connection point is expressed by the **short-circuit ratio**

$$\mathrm{SCR} = \frac{S_{sc}}{P_{rated}}$$

where S_{sc} is the three-phase fault level at the point of connection without the plant's own contribution. Above about 3 the grid is considered strong; between 2 and 3 weak; below 2 very weak, and conventional grid-following control generally needs special tuning or supplementary equipment. Where several plants are electrically close, a weighted SCR that accounts for their interaction is used.

Worked example — Is the connection point strong enough? A developer proposes a 200 MW solar plant at a remote 132 kV substation whose fault level, with the minimum generation dispatch, is 600 MVA. Find the SCR and the voltage sensitivity of the bus to the plant's reactive power.

$$\mathrm{SCR} = \frac{600}{200} = 3.0$$

which is at the boundary of "weak". The bus voltage changes by about $\Delta V/V \approx \Delta Q/S_{sc}$, i.e. 0.17 % per Mvar, or 1 % per 6 Mvar. A plant swinging its reactive output by ±66 Mvar (±0.33 pu) moves the voltage by ±11 %: voltage control must be slow, coordinated and carefully tuned to avoid oscillation with other controllers nearby. If a second plant of the same size connects at the same bus, the effective SCR halves to 1.5, and the network operator will require either a synchronous condenser (which raises the fault level), grid-forming inverters, or detailed electromagnetic-transient studies before connection.

> **Status of the principle: Short-circuit ratio as a measure of grid strength**
> *Empirical.* The thresholds of about 3 and 2 come from experience with HVDC and inverter plants; SCR is a static measure that ignores control design, the X/R ratio and interactions with nearby converters, so it is a screening index, and a low SCR calls for electromagnetic-transient studies rather than deciding the matter.

Grid-forming control

A **grid-forming** inverter reverses the logic: it controls the magnitude and phase of its own internal voltage **E**, like a synchronous machine, and lets the current follow from the network. It synchronises not by a PLL but by a power–frequency relation: if it delivers more power than its setpoint, it slows its internal frequency (droop control), or it emulates the swing equation of a synchronous machine with a chosen virtual inertia (**virtual synchronous machine**). Because it sets a voltage, it can energise an island, it stiffens weak grids, it responds instantly to a change of grid frequency or phase angle with a change of current (providing inertia-like response without measurement delay), and it shares load with other grid-forming sources as in Section 26.4. Its constraint is current: during a fault it must switch to a current-limiting mode without losing synchronism. Grid-forming batteries are now specified by system operators in Australia and Great Britain for system-strength and inertia services, and are mandatory in many microgrids.

27.3 Active and reactive power control

Current control in the rotating frame

In the PLL-aligned frame, the grid voltage is $v_d = V$, $v_q = 0$, and the power delivered is

$$P = \tfrac{3}{2} v_d i_d, \qquad Q = -\tfrac{3}{2} v_d i_q$$

(with amplitude-invariant transformation; the sign of Q depends on the convention adopted). Active and reactive power are therefore controlled independently by the d- and q-axis current references. Fast PI current controllers (bandwidths of a few hundred hertz to about 1 kHz) make the actual currents follow these references. The references in turn come from outer loops:

- **Active power**: for PV, the maximum-power-point tracker sets the DC voltage that extracts maximum power, and a DC-link voltage controller sets i_d to export whatever the array delivers; for wind, the turbine's speed and pitch control decides the power; for batteries, the dispatch setpoint.
- **Reactive power**: a fixed Q or power-factor setpoint, a volt-var characteristic (Section 27.7), or a voltage setpoint from the plant controller.

A **current limiter** caps the magnitude of the reference vector, with a priority rule: in normal operation active current usually has priority; during voltage disturbances many grid codes require reactive current to take priority.

Capability chart

The P–Q capability of an inverter plant is bounded by its apparent-power (current) limit, a circle of radius S_{max}, and by the available primary power. Unlike a synchronous generator, whose capability chart is limited by field heating and stability (Chapter 7), the inverter's chart is nearly a full circle and is the same in every quadrant; a battery can operate in all four. Grid codes specify a required reactive range as a function of active output and voltage. A common requirement is reactive power of $\pm 0.33 P_{max}$ (power factor 0.95 at full output) available down to 20 % or 10 % of active output, and many codes require reactive capability even at zero output, including at night for PV plants ("Q at night"). Meeting ± 0.33 pu at full active output requires an inverter rating of $\sqrt{1 + 0.33^2} = 1.053$ pu, so PV plants are typically built with inverter capacity 5–10 % above their export limit (Figure 27.2).

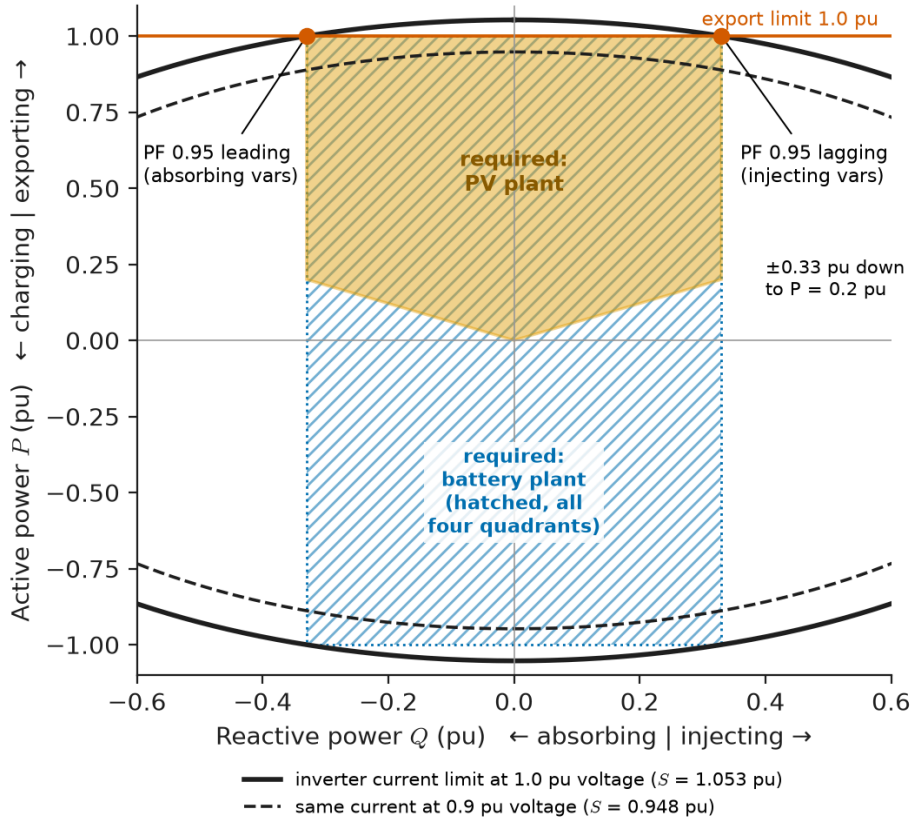


Figure 27.2 — P-Q capability of an inverter-based plant compared with a grid-code requirement.

Plant-level functions

The plant controller adds functions that individual inverters cannot provide: **curtailment** to a setpoint ordered by the system operator or to an export limit; **ramp-rate limits** on active power (typically 10 % of capacity per minute, Section 27.8); **frequency response** (Section 27.6); and **voltage or reactive-power control** at the point of connection with a slope (droop) that coordinates the plant with nearby voltage controllers. Response times are specified: for example, 90 % of a step change in reactive power within 1–2 s at plant level, and active-power frequency response within a few seconds.

27.4 Grid-code fundamentals

Why grid codes exist and who writes them

A **grid code** is the set of technical rules that any generator, storage plant or large load must meet to connect to a network. For synchronous generators many of the rules were implicit in the physics of the machine; inverter behaviour is entirely set by software and must therefore be specified. Codes are written by system operators and regulators, increasingly harmonised by standards:

- **IEEE 1547-2018** (and test standard IEEE 1547.1-2020) for distributed resources connected to distribution systems in North America.
- **IEEE 2800-2022** for inverter-based resources connected to transmission and sub-transmission in North America.

- **ENTSO-E Requirements for Generators** (Commission Regulation EU 2016/631), which classifies plants from Type A (800 W to 1 MW) to Type D (above 75 MW or connected at 110 kV and above in continental Europe), with requirements growing by type; national implementations (VDE-AR-N 4105/4110/4120 in Germany, the GB Grid Code and EREC G99, and others) fill in the parameters.
- **National Electricity Rules** in Australia with their generator performance standards, and national grid codes across Asia, Africa and South America, often adapted from these.

What they require

Table 27.2 summarises the main requirements with representative values. They divide into requirements for **normal operation** (ranges of voltage and frequency for continuous operation, reactive capability, power quality, controllability) and for **disturbances** (ride-through, fast fault current, frequency response, ROCOF withstand).

Table 27.2 — Typical grid-code requirements for large inverter-based plants (representative values).

Requirement	Typical specification	Example source
Continuous frequency range	49.0–51.0 Hz unlimited; 47.5–48.5 Hz for at least 30 min	ENTSO-E RfG (continental Europe)
Continuous voltage range	0.90–1.10 pu (some to 1.118 pu at 400 kV)	RfG, IEEE 2800
ROCOF withstand	2 Hz/s (500 ms window) typical; 5 Hz/s in IEEE 2800	RfG implementations, IEEE 2800
Reactive capability	$\pm 0.33 P_{max}$ (PF 0.95) at full output; voltage-dependent profile	RfG Type D, IEEE 2800, national codes
Low-voltage ride-through	Remain connected for retained voltage 0–0.15 pu for 140–250 ms, recovery to 0.85 pu within about 1.5–3 s	RfG Type D (TSO-chosen parameters)
High-voltage ride-through	1.2 pu for about 1 s; 1.1 pu for longer	IEEE 2800, national codes
Fast fault current	Additional reactive current $\Delta i_q = k\,\Delta v$, $k \approx 2$, rise time below 30–60 ms	VDE-AR-N 4120, IEEE 2800
Active power recovery after fault	90 % of pre-fault power within 0.5–1 s	Various
Over-frequency response (LFSM-O)	Reduce power above 50.2 Hz with 2–12 % droop (default 5 %)	RfG
Primary frequency response	Droop 2–5 %, deadband about ±17–36 mHz	IEEE 2800, NERC, national codes
Power quality	Harmonics, flicker and unbalance emission limits at the PCC	IEC/TR 61000-3-6/-7, IEEE 519

Fault ride-through

The most consequential requirement is **fault ride-through** (FRT). Early wind turbines and PV inverters were required to disconnect at any voltage dip, on the reasoning that small generators should get out of the way of protection. Once wind and solar reached gigawatts, that rule became dangerous: a single transmission fault depresses voltage over hundreds of kilometres, and if every plant in that area disconnected, the system would lose more generation than its largest single unit, which is the event its reserves are sized for. Codes now require plants to stay connected for any voltage–time trajectory above a specified **FRT curve** (Figure 27.3), to inject reactive current during the dip to support the voltage, and to restore active power quickly afterwards. The August 2019 event of the chapter opening was, in part, a failure of plants to behave as their FRT and control requirements intended.

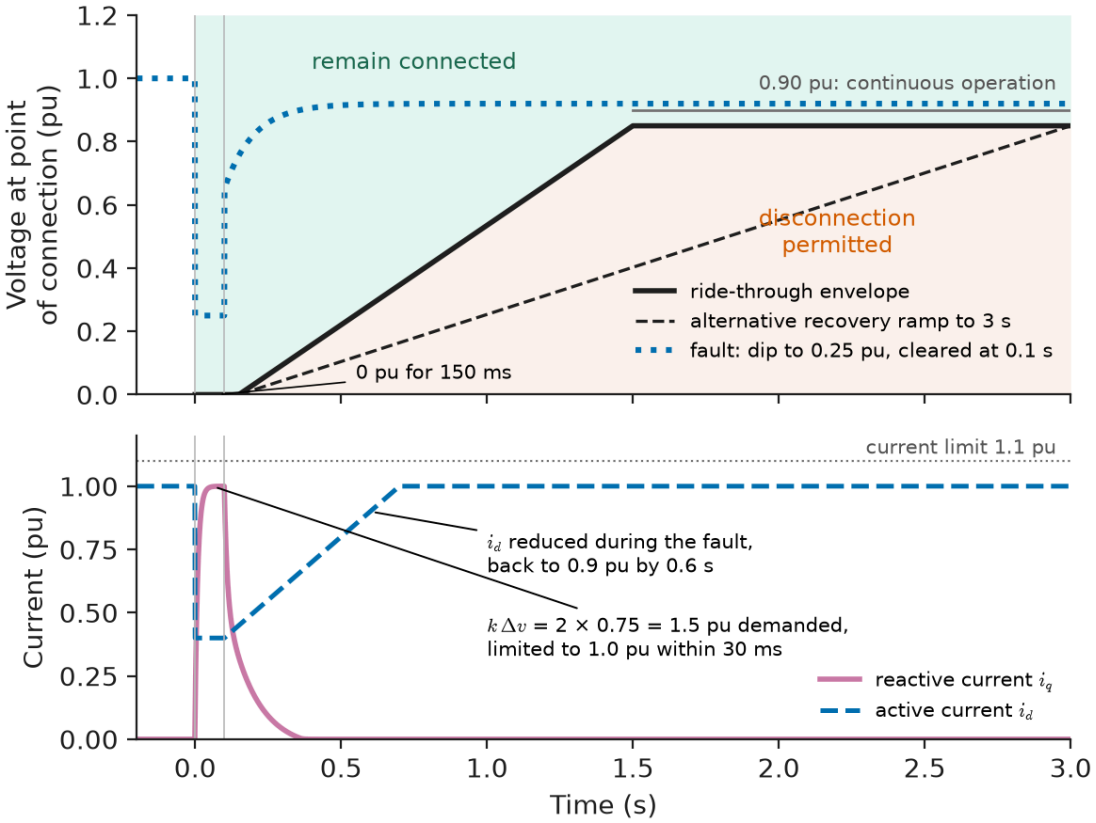


Figure 27.3 — Low-voltage ride-through envelope for a large inverter-based plant, with a fault event and the required reactive current.

Compliance

Compliance is demonstrated by **type testing** of inverters (certified test reports per IEEE 1547.1, IEC 62116 or national schemes), by **plant-level simulation models** (positive-sequence RMS models for power-flow and stability studies, and detailed electromagnetic-transient models for weak-grid and control-interaction studies), validated against measurements, and by **commissioning tests** at the point of connection. Increasingly, operators also require ongoing performance monitoring with high-resolution recorders so that the plant's behaviour in real disturbances can be compared with its models.

> **Status of the principle: Grid-code ride-through and response envelopes**
> *Empirical*. The numbers are engineering choices by system operators, calibrated by studies of their own system's fault clearing times, largest infeed loss and stability margins; they differ between systems and are revised as the generation mix changes, so the code of the actual connection point always governs.

27.5 Anti-islanding

Why unintentional islands are unwanted

An **island** forms when part of a distribution network is disconnected from the main grid (a feeder breaker opens for a fault or for maintenance) while the distributed generation on it continues to supply its loads. An unintentional island is dangerous and harmful: line workers may find a supposedly dead line live; voltage and frequency in the island are uncontrolled and may damage customer equipment; and when the utility's breaker recloses onto an island that has drifted out of phase, the out-of-phase closure can damage generators, motors and the breaker itself. Planned islands with a designated grid-forming source and proper protection are microgrids (Section 26.4); unplanned ones must be detected and shut down. IEEE 1547-2018 requires DER to **cease to energize** within 2 s of island formation; IEC 62116 defines the corresponding inverter test.

Passive methods and the non-detection zone

The simplest detection uses the over/under-voltage and over/under-frequency trips that every inverter has. If the local generation and load are mismatched when the island forms, voltage or frequency moves outside the trip band and the inverter disconnects. But if they are closely matched, nothing changes enough. For a constant-current inverter feeding a parallel RLC load, the island voltage and frequency settle where the load absorbs exactly the inverter's active and reactive power:

$$\frac{V_{island}}{V_n} = \sqrt{\frac{P_{inv}}{P_{load}}}, \qquad \frac{Q_{inv} - Q_{load}}{P} \approx Q_f \left[1 - \left(\frac{f_n}{f_{island}}\right)^2\right]$$

where Q_f is the load's quality factor (the ratio of reactive power stored in its inductance or capacitance to its active power at resonance). The set of mismatches that leaves voltage and frequency inside the trip limits is the **non-detection zone** (NDZ).

Worked example — Non-detection zone of passive protection. A PV inverter supplying a feeder section is set to trip at 0.88 and 1.10 pu voltage and at 59.3 and 60.5 Hz (the classic IEEE 1547-2003 thresholds). The load is modelled as a parallel RLC with $Q_f = 1$, the test value of IEEE 1547.1. Find the NDZ.

Voltage. The island voltage stays within $0.88 \le V \le 1.10$ pu when P_{inv}/P_{load} lies between $0.88^2 = 0.774$ and $1.10^2 = 1.21$. Expressed as the active-power mismatch the grid was supplying before the island, $\Delta P/P_{inv} = P_{load}/P_{inv} - 1$, the NDZ spans

$$\frac{1}{1.21} - 1 = -17.4\,\% \quad \text{to} \quad \frac{1}{0.774} - 1 = +29.1\,\%$$

Frequency. The reactive mismatch that keeps $59.3 \le f \le 60.5$ Hz spans $Q_f[1 - (60/59.3)^2] = -2.4\,\%$ to $Q_f[1 - (60/60.5)^2] = +1.6\,\%$ of P.

So if, at the moment the feeder breaker opens, the local load happens to absorb within about -17 to $+29\,\%$ of the inverter's active power and within about 2 % of its reactive power, passive protection will not detect the

island. On a residential feeder with high PV at midday, that coincidence is not rare.

> **Status of the principle: Non-detection zone from the parallel RLC load model**
> *Approximate*. The RLC model with a single quality factor is the agreed test load, not a description of real feeders, whose motors and electronics behave differently in an island; real NDZs depend on load composition, and multiple inverters with different active methods can interfere with each other.

Active methods and communication

Active methods deliberately perturb the inverter's output so that an island, lacking a stiff grid to hold voltage and frequency, drifts away quickly: active frequency drift and **Sandia frequency shift** (a small, positively fed-back phase shift of the current that pushes the island frequency away from nominal), Sandia voltage shift, and impedance measurement by injected harmonic or pulse. They shrink the NDZ to near zero for a single inverter, at the cost of slight power-quality degradation and the risk of interaction among many inverters. **Communication-based** methods (direct transfer trip from the feeder breaker, or loss of a power-line carrier signal) have no NDZ and are used for large DER, at the cost of a communication channel.

The tension with ride-through

Anti-islanding and ride-through pull in opposite directions: anti-islanding wants DER to disconnect on abnormal voltage or frequency; ride-through wants them to stay connected through the same abnormalities when the grid is still there. Loss-of-mains protection based on rate of change of frequency, set at 0.125 Hz/s on older small generators in Great Britain, tripped hundreds of megawatts in the August 2019 event when the whole grid's frequency fell faster than that. The British settings were subsequently raised to 1 Hz/s, and vector-shift protection was removed. IEEE 1547-2018 likewise moved from "trip fast" to "ride through, then cease to energize if islanded", relying on active methods to tell the two cases apart.

27.6 Frequency support

The declining inertia of the system

When a generator trips, the immediate deficit is supplied by the kinetic energy of every synchronous machine on the system, and frequency starts to fall at a rate set by the system's inertia (the swing equation of Chapter 20). The **rate of change of frequency** immediately after a loss of infeed ΔP is

$$\mathrm{ROCOF} = \frac{\Delta P \, f_0}{2 \, E_k}$$

where $E_k = \sum H_i S_i$ is the total stored kinetic energy of the synchronised machines (GW·s, from inertia constants H of typically 2–9 s on machine rating). Inverter-based generation contributes nothing to E_k unless it is deliberately controlled to. As wind and solar displace synchronous plant at times of high output, E_k falls and ROCOF rises. Governors take several seconds to respond, so the **frequency nadir**, the lowest point reached, depends on how much energy the system loses before primary response arrives (Figure 27.4).

Status of the principle: Initial ROCOF from aggregate inertia
Derived. Follows from the aggregate swing equation at the instant after the disturbance, before any governor or load response; it assumes a single coherent system frequency, whereas in reality machines near the disturbance decelerate faster and ROCOF measured over short windows varies by location.

Worked example — ROCOF in a low-inertia system. A 50 Hz system's largest credible loss is 1.8 GW. Find the initial ROCOF when the synchronous kinetic energy is 150, 100 and 60 GW·s, the time to reach the 48.8 Hz under-frequency load-shedding threshold if nothing responded, and the minimum kinetic energy to keep ROCOF below 1 Hz/s.

$$\text{ROCOF} = \frac{1.8 \times 50}{2E_k}: \qquad 0.30\ \text{Hz/s}\ (150\ \text{GW}\cdot\text{s}),\quad 0.45\ \text{Hz/s}\ (100),\quad 0.75\ \text{Hz/s}\ (60)$$

The 1.2 Hz margin to 48.8 Hz is used up in 4.0 s, 2.7 s and 1.6 s respectively. Conventional primary response delivers its full output in about 10 s; with low inertia it is far too slow, and **fast frequency response** (full output within about 0.5–1 s) is needed. Keeping ROCOF at or below 1 Hz/s requires

$$E_k \geq \frac{1.8 \times 50}{2 \times 1} = 45\ \text{GW}\cdot\text{s}$$

System operators in small, renewable-rich systems (Ireland, Great Britain, South Australia) set exactly such minimum inertia or ROCOF constraints and dispatch synchronous plant or synchronous condensers to meet them when wind and solar are high.

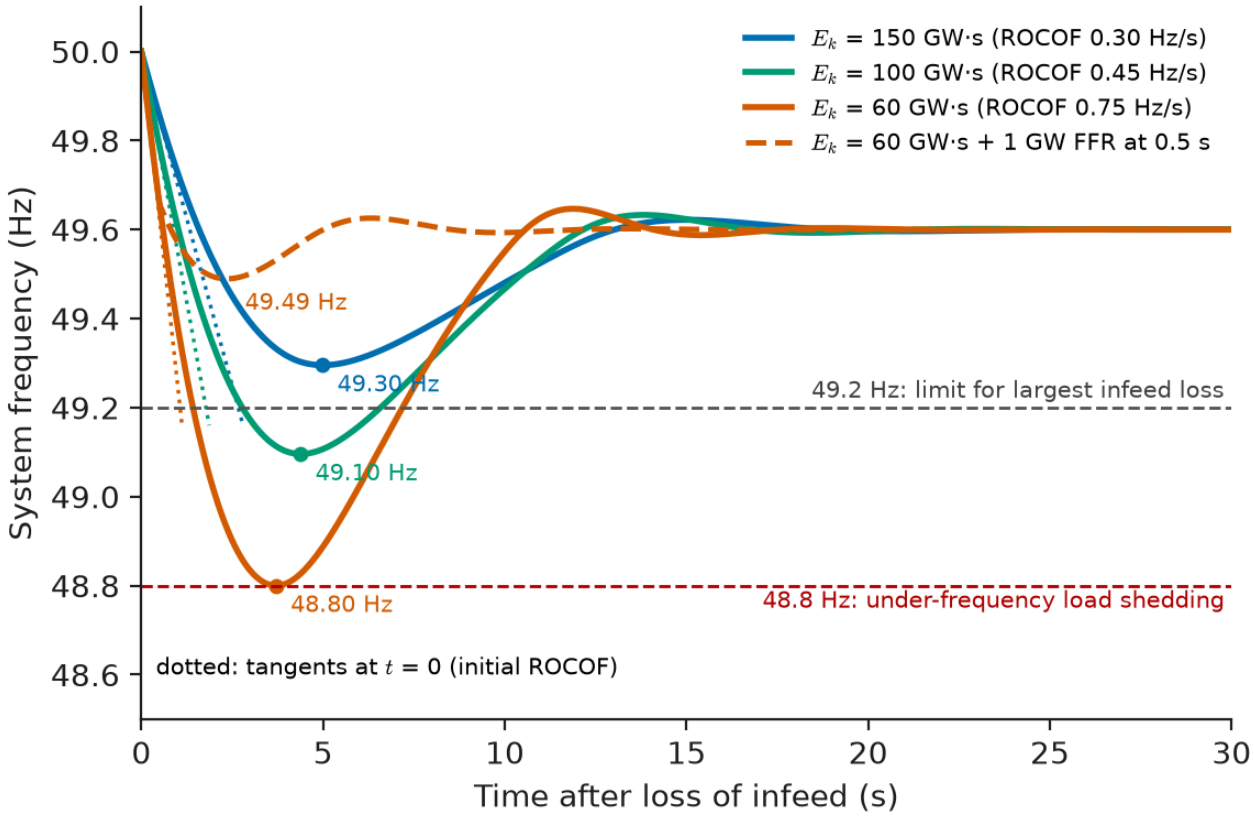


Figure 27.4 — System frequency after a 1.8 GW loss of infeed for three levels of inertia, with and without fast frequency response.

What inverters can provide

Inverter-based resources can support frequency in several ways, all specified in modern grid codes:

- **Over-frequency curtailment** (LFSM-O in ENTSO-E codes; frequency-watt in IEEE 1547). Above a threshold (50.2 Hz in Europe), active power is reduced along a droop of typically 5 %. This is cheap, because it only spills energy, and is mandatory for nearly all new generation. Its origin is instructive: in the early 2010s, German rules had required all small PV to trip at 50.2 Hz, so that a frequency excursion could have disconnected several gigawatts at once; tens of thousands of inverters were retrofitted to the droop characteristic.
- **Under-frequency response**. Raising output requires headroom: a PV or wind plant must be curtailed in advance (operating below its available power, at an energy cost), or it must be a battery. Batteries are the natural provider, with response in under a second and droops of 1–5 %.
- **Fast frequency response** (FFR). A step or proportional increase of active power within 0.15–1 s of a frequency trigger, from batteries, HVDC links or curtailed renewables. FFR arrests the fall before conventional primary response arrives.
- **Synthetic (virtual) inertia**. Power proportional to ROCOF ($\Delta P = -2H_{virtual} S\,\mathrm{d}f/\mathrm{d}t/f_0$), emulated by grid-following controls with a measurement delay, or provided intrinsically, without measurement, by grid-forming inverters. A wind turbine can also release the kinetic energy of its rotor for a few seconds, at the cost of a recovery period with reduced output.

27.7 Voltage support

Voltage rise from distributed generation

Active power injected at the end of a feeder raises the local voltage by approximately

$$\Delta V \approx \frac{RP + XQ}{V}$$

(Chapter 22). On transmission and MV overhead networks, X is comparable to or larger than R, so reactive power is an effective lever on voltage. On LV cable networks, R/X is typically 2–5, and reactive power is a weak lever: voltage there is dominated by active power, which is why rooftop PV hits voltage limits before thermal limits on many LV feeders.

Worked example — Reactive power against PV voltage rise, LV versus MV.

LV feeder. 100 kW of PV is concentrated near the end of a 300 m, 400 V cable with $R = 0.32\ \Omega$/km and $X = 0.075\ \Omega$/km, so $R = 0.096\ \Omega$ and $X = 0.0225\ \Omega$. At unity power factor, $\Delta V = 0.096 \times 100\,000/400 = 24$ V, a rise of 6.0 % of 400 V, which together with the transformer's no-load voltage setting would exceed the +10 % limit. With the inverters absorbing reactive power at power factor 0.9 ($Q = -0.484P = -48.4$ kvar), $\Delta V = (9600 - 1090)/400 = 21.3$ V, or 5.3 %. Absorbing 48 kvar, and the extra current and losses that come with it, buys only 0.7 percentage points.

MV feeder. 3 MW of PV at the end of a 10 km, 11 kV overhead line with $R = 3.0\ \Omega$ and $X = 3.3\ \Omega$. At unity power factor, $\Delta V = 3.0 \times 3 \times 10^6/11\,000 = 818$ V, 7.4 %. At power factor 0.95 absorbing ($Q = -0.986$ Mvar), $\Delta V = (9.0 \times 10^6 - 3.25 \times 10^6)/11\,000 = 522$ V, 4.7 %: a reduction of more than a third.

The contrast explains grid-code practice: reactive voltage support (volt-var) is effective and standard at MV and above, while on LV networks it helps somewhat and is combined with **volt-watt** curtailment, which

reduces active power when voltage approaches the limit, and with network measures (on-load tap-changing distribution transformers, phase balancing, export limits).

Volt-var and related functions

IEEE 1547-2018 defines a family of voltage-support functions for DER, selectable by the utility: constant power factor, constant reactive power, **volt-var** (reactive power as a piecewise-linear function of local voltage), **watt-var** (reactive power as a function of active output) and **volt-watt**. The default volt-var curve for Category B equipment injects up to 44 % of rated apparent power as reactive power at 0.92 pu voltage and below, has a deadband between 0.98 and 1.02 pu, and absorbs up to 44 % at 1.08 pu and above (Figure 27.5). Because each inverter reacts only to its own terminal voltage, volt-var is a decentralised droop control; its slope must be gentle enough that thousands of inverters on the same feeder, each reacting to the others' effect, do not oscillate.

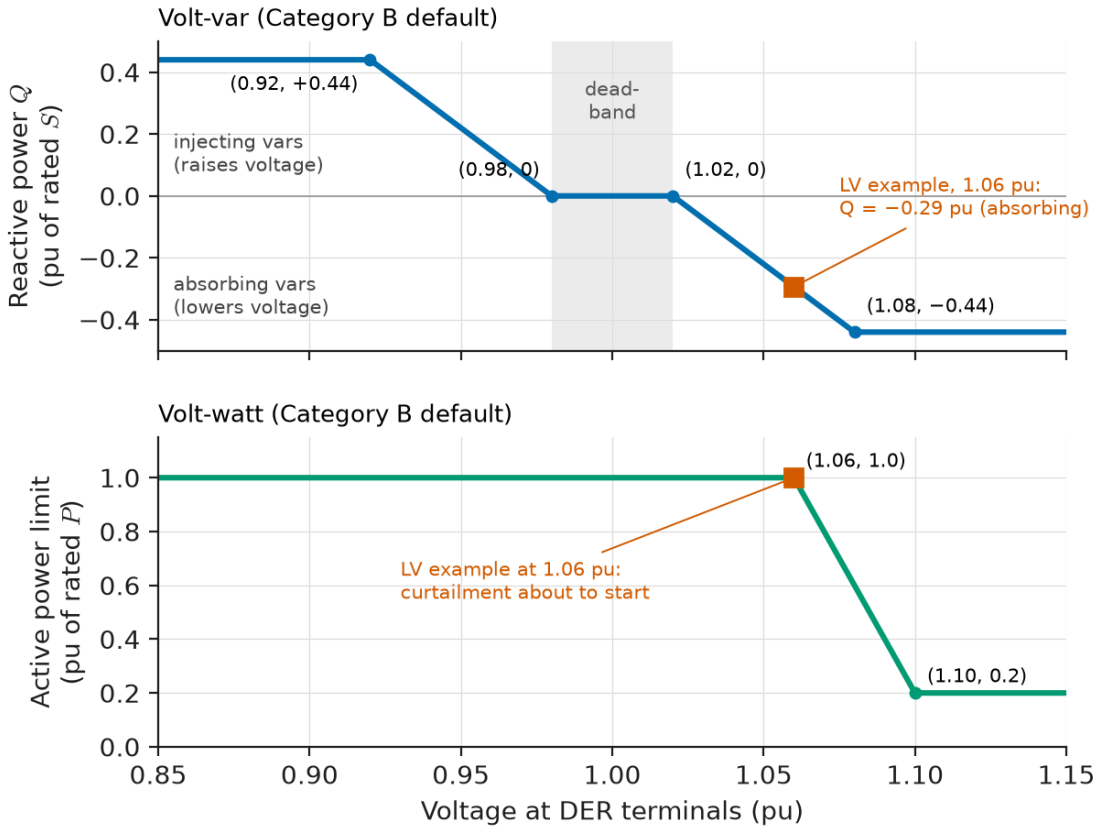


Figure 27.5 — Default IEEE 1547-2018 Category B volt-var characteristic and a volt-watt characteristic.

Dynamic voltage support during faults

During a fault, voltage support means injecting reactive current quickly. Codes (VDE-AR-N 4120 in Germany, IEEE 2800, others) require additional reactive current proportional to the voltage deviation, $\Delta i_q = k\,\Delta v$ with k typically 2 (adjustable 0–10), up to the converter's current limit, with a rise time of 30–60 ms. A 50 % dip with $k = 2$ requires 1.0 pu reactive current, which with a 1.1 pu current limit leaves only $\sqrt{1.1^2 - 1.0^2} = 0.46$ pu for active current during the fault. Because inverter fault current is limited to 1.1–1.5 pu and is controlled rather than physical, protection relays that rely on large fault currents and on the negative-sequence behaviour of synchronous machines may misoperate as inverter penetration grows; IEEE 2800 now also specifies negative-sequence current injection during unbalanced faults, to keep directional and distance protection working (Chapter 23).

27.8 Storage coordination

What storage does for a renewable plant

Battery energy storage co-located with or near wind and solar plants performs several functions that differ in the ratio of power to energy they need:

Function	Duration	Typical sizing (power, energy)
Fast frequency response, synthetic inertia	Seconds to minutes	High power, 0.25–1 h energy
Ramp-rate control	Minutes	High power, 0.1–0.5 h
Firming and forecast-error correction	Tens of minutes to hours	1–2 h
Energy time-shifting (solar to evening peak)	Hours	2–4 h, increasingly 4–8 h
Black start and grid forming	Minutes to hours	Depends on restoration plan

Plants may be **AC-coupled** (battery and PV with separate inverters on a common MV bus) or **DC-coupled** (battery connected through a DC/DC converter to the PV inverter's DC bus). DC coupling captures PV energy that would otherwise be clipped when the array's DC output exceeds the inverter rating (PV plants are commonly built with DC/AC ratios of 1.2–1.4), and shares one inverter; AC coupling is simpler to retrofit and lets the battery operate independently. In both, the **point-of-connection limit** (the export capacity agreed with the network operator) is shared, and the plant controller must coordinate PV and battery to stay within it while meeting frequency, ramp and voltage obligations. **State-of-charge management** is the key coordination problem: a battery committed to frequency response must stay away from full and empty, so its energy-shifting schedule leaves headroom in both directions (Chapter 16).

Worked example — Sizing a battery for ramp-rate control. A 100 MW solar plant must limit ramps to 10 % of capacity per minute (10 MW/min), as several island and Australian grid codes require. A cloud front drops the array output from 100 MW to 30 MW linearly over one minute, and it stays at 30 MW. Find the battery power and energy needed.

The plant output may fall no faster than $100 - 10t$ MW (t in minutes), reaching 30 MW at $t = 7$ min. The array delivers $100 - 70t$ for $t \le 1$, then 30 MW. The battery supplies the difference:

- At $t = 1$ min, $(100 - 10) - 30 = 60$ MW: the maximum power.
- Energy: $\int_0^1 60t\,\mathrm{d}t + \int_1^7 (70 - 10t)\,\mathrm{d}t = 30 + 180 = 210$ MW·min $= 3.5$ MWh.

The battery needs 60 MW but only 3.5 MWh (a 3.5-minute battery at full power), plus margins for state-of-charge range, degradation and repeated events: in practice about 60 MW / 10–15 MWh. Ramp control is a power-intensive service, unlike energy shifting. Two refinements reduce the requirement: an upward ramp limit can be met by curtailment alone, with no battery; and a short-term irradiance forecast from sky cameras lets the plant start ramping down before the cloud arrives, which can halve the battery's energy (Figure 27.6).

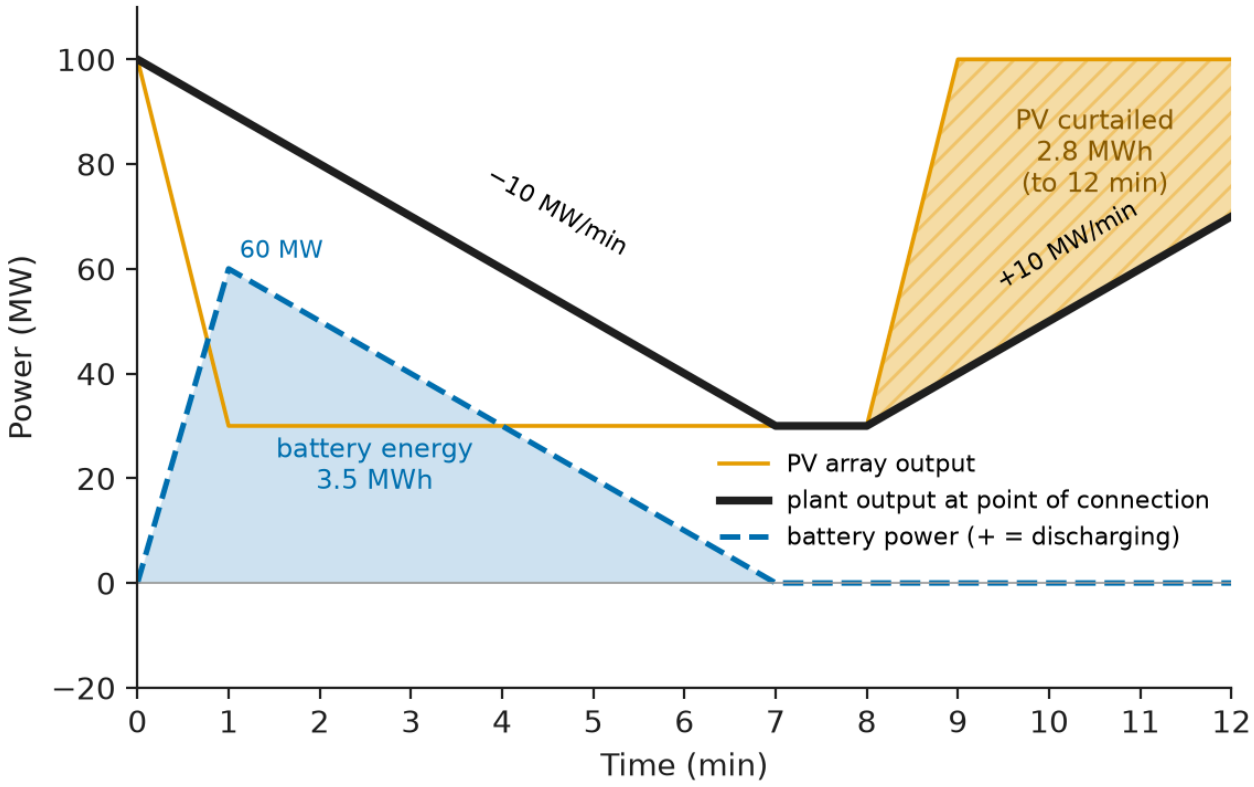


Figure 27.6 — Ramp-rate control of a 100 MW PV plant by a battery during a cloud passage.

27.9 Forecasting

Horizons and uses

Wind and solar output is variable but increasingly predictable, and the value of a forecast depends on its horizon:

Horizon	Uses	Main methods	Typical error (normalised MAE, % of capacity)
Seconds–minutes	Ramp control, battery dispatch, FFR readiness	Persistence, sky cameras, turbine SCADA	Small, but large for single-site solar in broken cloud
15 min–6 h (intraday)	Real-time dispatch, reserve activation, intraday trading	Satellite cloud motion, statistical and machine-learning models on live data	3–8 %
Day-ahead	Unit commitment, market bids, reserve procurement	Numerical weather prediction plus statistical post-processing	Wind 6–12 %, solar 4–8 % (portfolio)
Weeks–seasons	Maintenance, hydro scheduling, fuel planning	Climatology, seasonal ensembles	Large; probabilistic only

Errors fall sharply with **aggregation**: a single wind farm's day-ahead error may be 15 % of its capacity, a national fleet's 4–6 %, because errors at sites hundreds of kilometres apart are only partly correlated. Errors

also depend on the weather regime: frontal passages, convective storms, fog, snow on panels and icing on blades produce the largest misses, and **ramp events** (large changes of wind output over a few hours) are what operators fear most. Modern forecasts are **probabilistic**, giving quantiles or ensembles rather than a single value, so that the operator can size reserves for a chosen risk.

Forecasts and reserves

Reserves cover the combined uncertainty of load, renewable output and generator outages. If the forecast errors are independent and approximately normal, their standard deviations add in quadrature, and reserve is set to cover a chosen number of standard deviations (or a chosen percentile of the combined distribution).

Worked example — Reserve requirement from forecast errors. A system has 10 GW of load (day-ahead error standard deviation 2 %), 2000 MW of wind (8 % of capacity) and 1500 MW of solar (5 % of capacity). Estimate the reserve needed to cover forecast error at the 3σ level, assuming independent errors.

$$\sigma_{load} = 200\ \text{MW}, \quad \sigma_{wind} = 160\ \text{MW}, \quad \sigma_{solar} = 75\ \text{MW}$$

$$\sigma_{total} = \sqrt{200^2 + 160^2 + 75^2} = 267\ \text{MW}, \qquad 3\sigma = 800\ \text{MW}$$

Adding the three 3σ values arithmetically would give 1305 MW; treating errors as independent saves about 500 MW of reserve. The saving is real only if the independence assumption holds: wind and load errors are often correlated in cold, windy weather, and real error distributions have heavier tails than the normal, so operators use empirical distributions of their own past errors. Forecast improvement translates directly into less reserve held, less fuel burned in part-loaded plant, and less curtailment.

> **Status of the principle: Quadrature addition of forecast errors**
> *Approximate*. Valid for independent, approximately normal errors; real errors are skewed, heavy-tailed and partly correlated (by weather systems that affect load, wind and solar together), so the quadrature sum understates extreme requirements and is used as a first estimate before empirical analysis.

27.10 Renewable penetration limits

What limits the share

Annual energy share and instantaneous share are different things. Denmark obtains more than half its annual electricity from wind with strong interconnection to its neighbours; South Australia regularly runs on more than 100 % instantaneous renewable output, exporting the surplus; Ireland operates with up to 75 % of instantaneous demand from non-synchronous sources. The limits that bind are technical and specific to each system:

1. **Inertia and frequency stability.** Minimum synchronous kinetic energy or maximum ROCOF constraints (Section 27.6) require some synchronous machines on line.
2. **System strength.** Grid-following inverters need a minimum fault level to remain stable (Section 27.2); protection needs fault current to detect faults (Chapter 23).
3. **Voltage control.** Synchronous plant provides dynamic reactive power at transmission level; displacing it requires reactive capability from inverters, STATCOMs or condensers.
4. **Thermal and congestion limits.** Renewables are built where the resource is, often far from load and on weak networks, so curtailment for congestion rises long before system-wide limits are reached.

5. **Ramping and the net-load profile.** High solar creates the "duck curve": net load falls at midday and rises steeply at sunset, requiring flexible plant, storage or demand response to follow ramps of several gigawatts per hour (California's evening ramp exceeds 15 GW in three hours).
6. **Minimum generation.** Must-run plant (for inertia, voltage, heat supply, or contractual reasons) occupies part of demand at low-load times, so renewables beyond the remainder are curtailed.

Ireland's operator expresses the binding constraint as the **system non-synchronous penetration**,

$$\text{SNSP} = \frac{P_{wind} + P_{solar} + P_{HVDC,import}}{P_{demand} + P_{HVDC,export}}$$

held below a limit that has been raised in steps from 50 % (2010) to 75 % as studies, new ROCOF standards (1 Hz/s), fast frequency services and synchronous condensers made each step secure. With demand at 5000 MW, 500 MW of HVDC export and 4000 MW of wind, SNSP is $4000/5500 = 72.7\,\%$: just inside the 75 % limit, and any further wind would be curtailed.

> **Status of the principle: Instantaneous penetration limits such as SNSP**
> *Empirical.* The limit values come from system studies and operating experience specific to each grid, its largest infeed, its protection and the capabilities of its inverter fleet; they move as those capabilities change, and are operating constraints rather than physical laws.

Pushing the limits back

Each limit has known remedies, and the history of high-penetration systems is the history of applying them:

- **Synchronous condensers** (Chapter 7) supply inertia, fault current and dynamic reactive power without burning fuel; South Australia installed four large units in 2021, and converting retired thermal generators to condensers is common.
- **Grid-forming inverters**, especially on batteries, provide inertia-like response, system strength and black-start capability, and are the long-run path to systems that can run at 100 % instantaneous inverter-based generation, already demonstrated in island systems such as Kauai, King Island and parts of the Hawaiian grid.
- **Storage and demand flexibility** absorb midday surplus and supply evening ramps.
- **Transmission and interconnection** spread variability over larger areas and relieve congestion; HVDC links with VSC converters add controllability.
- **Better forecasts and faster markets** (5-minute dispatch, intraday trading) reduce reserve needs and let flexible resources respond.
- **Updated grid codes and protection** so that inverters ride through disturbances, support frequency and voltage, and are modelled accurately.

The engineering question has shifted from "how much variable renewable energy can the grid accept?" to "which services does a stable grid need, which devices can provide them, and how are they procured?" The services (inertia or fast frequency response, fault current, voltage control, balancing, black start) are the same as they always were; what is new is that they must be specified, paid for and verified explicitly, because they no longer come for free with the spinning machines that used to supply the energy.

What remains open

Open question	Where it is answered
How are the swing equation, inertia and frequency nadir modelled in stability studies?	Chapter 20
How do PWM inverters and their current control work inside, for drives?	Chapter 12; circuit internals in *Electronics & Computing*
How are PV, wind and their capacity factors and resource assessment treated?	Chapter 15
How are batteries rated, and how are SOC and SOH managed?	Chapter 16
How does protection cope with low, controlled inverter fault currents?	Chapter 23
How is PV voltage rise and hosting capacity assessed on distribution feeders?	Chapter 22
How are VSC-HVDC links and FACTS devices applied in transmission?	Chapter 21
What harmonic and flicker emission limits apply at the point of connection?	Chapter 25
How are DER aggregated, dispatched and secured?	Chapter 26

Next: Chapter 28 turns from the supply side to the demand side: how energy use in plants and buildings is audited, managed and reduced, and how efficiency investments are judged over their life.

Chapter 28 — Energy Efficiency and Management

Part VII · Power Quality and Grid Modernization

A frozen-food plant receives its monthly electricity bill and the finance director asks a simple question: why did it go up by 9 % when production went up by only 2 %? The answer is not on the bill. The bill shows 1080 MWh of energy, a peak demand of 2400 kW and a reactive-energy surcharge, but it does not show that the peak was set by three compressors and a defrost cycle that happened to coincide for twenty minutes on one afternoon, that a 90 kW chilled-water pump runs flat out against a half-closed valve for 6000 hours a year, that a 1990s transformer in the yard wastes 15 MWh a year doing nothing at all, or that a compressed-air ring main leaks a quarter of everything the compressors make. Each of these is an electrical-engineering problem with a number attached: a load factor, an affinity law, a no-load loss, a leak rate. Energy management is the discipline of finding those numbers, ranking them by what they cost, fixing the worthwhile ones and proving afterwards that the saving is real.

> **How do we find out where the energy in a plant, building or network actually goes, and which of the losses are worth removing?**
> **How are savings measured, valued over the life of the equipment and sustained once the project team has gone home?**

Concept	Engineering destination
Energy auditing	ISO 50002 audits of factories, hospitals, data centres, mines and ports
Load profiles	Tariff selection, transformer and generator sizing, storage sizing
Demand management	Peak-demand charges, capacity limits on weak supplies, EV-depot charging
Industrial energy efficiency	Pumps, fans, compressed air, process heat, lighting, refrigeration
Motor efficiency	IE3/IE4 motor replacement, right-sizing, rewind policy
Power-factor correction	Reactive-energy charges, cable and transformer losses, freed capacity
Energy monitoring	Submetering, energy performance indicators, PUE, CUSUM tracking
Loss reduction	Transformer replacement, cable upsizing, voltage optimisation
Life-cycle energy cost	Total-owning-cost tenders for transformers, motors, drives and cables
Efficiency improvement strategies	Energy-management systems (ISO 50001), measurement and verification

Roadmap. How an audit is organised and what it measures (28.1) → reading load profiles and the factors that describe them (28.2) → shaving and shifting the peak (28.3) → the big industrial end uses: pumps, fans, compressed air, heat and light (28.4) → the motor itself (28.5) → power-factor correction as an energy measure (28.6) → monitoring, baselines and CUSUM (28.7) → losses in the plant's own network (28.8) → valuing all of it over a life cycle (28.9) → how the pieces become a strategy that lasts (28.10).

28.1 Energy auditing

Purpose and levels

An energy audit is a systematic account of where the energy entering a site goes, how much of it does useful work, and what it would cost to waste less. It is an engineering survey, not a financial one: its output is a list of measures, each with an estimated saving in kWh and kW, a cost, and a confidence level. ISO 50002 describes the process; ASHRAE (for buildings) and most national energy agencies grade audits into three levels of increasing depth (Table 28.1).

Table 28.1 — Audit levels and what they deliver (typical).

Level	Name	Data used	Accuracy of savings	Typical effort for a 5 GWh/yr site
1	Walk-through	12–36 months of bills, nameplates, a site visit	±30–50 %	1–3 days
2	Energy survey and analysis	Bills plus interval data, spot measurements, end-use breakdown	±15–25 %	1–3 weeks
3	Investment-grade audit	Logged measurements on each major system, process models, vendor quotes	±5–10 %	1–3 months

A walk-through finds the obvious: lights on in empty warehouses, compressors running at weekends, steam traps blowing, capacitor banks switched off since a fuse failed. A level-2 survey quantifies the end uses. A level-3 audit is what a bank or an energy-services company needs before financing a project whose repayments come out of the savings.

The energy balance

The backbone of every audit is a balance: metered input equals the sum of the end uses plus losses. In a typical industrial site the electrical end-use split looks like Table 28.2; the balance is built by measuring the largest consumers directly and estimating the rest from nameplate, run hours and load factor, then checking that the total matches the utility meter within a few per cent. A balance that does not close indicates an unmetered load, a wrong run-hour assumption, or a meter fault, and finding it is often the first saving.

Table 28.2 — Typical share of electricity by end use (indicative ranges).

Sector	Motor-driven systems	Lighting	Process heat and electrolysis	HVAC and refrigeration	IT and electronics
General manufacturing	60–70 %	5–10 %	5–20 %	10–20 %	2–5 %
Food and cold storage	40–50 % (incl. refrigeration compressors)	5–8 %	5–10 %	30–45 %	2–4 %
Office building	15–25 % (fans, pumps)	15–25 %	under 5 %	30–45 %	15–30 %
Data centre	10–15 % (fans, pumps)	1–2 %	—	15–35 % (cooling)	50–75 %
Underground mine	60–80 % (ventilation, hoisting, pumping, crushing)	2–5 %	—	5–15 % (cooling at depth)	under 2 %

Motors dominate. Worldwide, electric motor systems take roughly 45 % of all electricity, which is why Sections 28.4 and 28.5 receive the most space.

Measurement in an audit

An audit lives on measurements that are good enough, not perfect. Portable three-phase power analysers (IEC 61000-4-30 Class S or A) logged for one to two weeks on each major feeder give the load shape and the power factor. Clamp-on current loggers are cheaper but miss the voltage and power factor and so overstate kW on lightly loaded motors. Ultrasonic flow meters, thermal imaging of panels and steam lines, and pressure loggers on compressed-air mains supply the process side. The audit report converts every finding into the same currency: annual kWh, peak kW, cost, and payback.

> **Boundary.**
> The definitions of active, reactive and apparent power, power factor with and without harmonics, kWh and kvarh metering and the structure of demand charges are in *Circuits, Signals and Communications*, Ch. 9 (§9.2–9.9), and three-phase power measurement in Ch. 10 (§10.7–10.8). They are used here without derivation.

28.2 Load profiles

What a profile shows

A load profile is the site's demand plotted against time, usually as 15- or 30-minute averages from the revenue meter. It is the single most informative document in an audit, because it shows not just how much energy is used but *when*, and therefore what the tariff charges for and what the supply equipment must carry. Figure 28.1 shows a weekday profile for the frozen-food plant of the opening and, on the right, the same year of data sorted into a **load duration curve**: demand on the vertical axis, the number of hours per year for which that demand is exceeded on the horizontal.

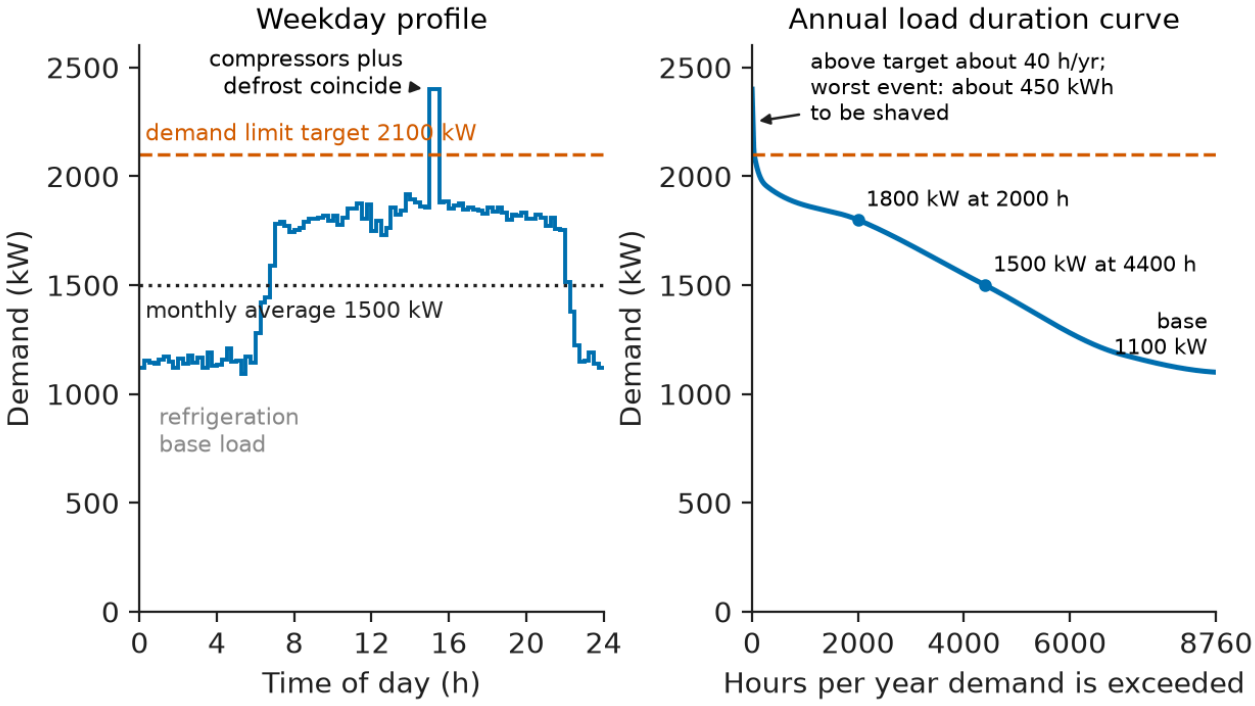


Figure 28.1 — Weekday load profile of a frozen-food plant (left) and its annual load duration curve (right).

The profile is summarised by a few ratios. With P_{max} the maximum demand over the period, $\bar{P}$ the average demand, P_{conn} the sum of nameplate ratings and $\sum P_{max,i}$ the sum of the individual maximum demands of the sub-loads:

$$\text{load factor} = \frac{\bar{P}}{P_{max}}, \qquad \text{demand factor} = \frac{P_{max}}{P_{conn}}, \qquad \text{diversity factor} = \frac{\sum_i P_{max,i}}{P_{max}} \geq 1$$

The load factor measures how well the site uses the capacity it pays for. A continuous-process plant or a data centre runs at 0.75–0.9; a single-shift workshop at 0.25–0.35; an EV fast-charging hub, before storage, often below 0.15. A low load factor means a large demand charge per kWh and supply equipment idle most of the time. The diversity factor says how unlikely it is that every sub-load peaks at once; it is what allows a transformer to be smaller than the sum of the feeders it supplies (Chapter 30).

Table 28.3 — Representative load factors (annual, typical).

Facility	Load factor	Remarks
Hyperscale data centre	0.80–0.90	IT load nearly flat; cooling varies with weather
Aluminium smelter	0.95–0.98	Potline is a constant DC load
Three-shift paper mill	0.75–0.85	Dips during maintenance shutdowns
Frozen-food plant	0.55–0.70	Refrigeration base plus daytime production
Office building	0.30–0.45	Day-only HVAC and lighting
Single-shift engineering workshop	0.25–0.35	Little base load
EV fast-charging hub (no storage)	0.08–0.20	Short, high peaks

> **Status of the principle: Load, demand and diversity factors**
> *Empirical*. The definitions are exact, but the values used for planning are statistics of past behaviour; they hold only while the mix and scheduling of loads stay as they were when the data were collected.

Worked example — Reading a plant's bill through its profile. The frozen-food plant used 1080 MWh in a 30-day month (720 h) with a maximum 15-minute demand of 2400 kW. Its connected load is 4800 kW, and the sum of the maximum demands of its five main feeders, read from submeters, is 3300 kW. The tariff includes a demand charge of 12 USD per kW of monthly maximum demand.

Average demand is $1\,080\,000/720 = 1500$ kW, so the load factor is $1500/2400 = 0.625$. The demand factor is $2400/4800 = 0.50$ and the diversity factor $3300/2400 = 1.375$. The demand charge is $2400 \times 12 = 28\,800$ USD for the month. The profile in Figure 28.1 shows that the plant exceeds 2100 kW for only about 40 hours a year; holding the peak at 2100 kW would save $300 \times 12 = 3600$ USD per month, or 43 200 USD per year, without saving a single kWh. That is the subject of the next section.

Profiles in different sectors

Profiles are signatures. A data centre is flat with a small afternoon bump from chillers. An office rises at 07:00, plateaus, and falls at 19:00, with a weekend base that should be 25–35 % of the weekday peak; if it is 60 %, something is running that should not be. A mine hoist produces a saw-tooth of 5–10 MW pulses every two minutes. An electric bus depot shows a huge evening peak when 100 buses plug in at 150 kW each. A cruise ship in port has a hotel load of 8–12 MW that shore power must supply. In each case the profile tells the engineer which equipment to look at first.

28.3 Demand management

What demand management is, and what it is not

Demand management is the customer-side control of *when* and *how fast* energy is drawn, to cut peak-related charges, stay within a capacity limit, or avoid reinforcing a supply. It differs from demand response (Chapter 26), in which the grid operator or an aggregator asks the customer to change consumption for the grid's benefit and pays for it. The same equipment often does both.

The tools are, in order of cost:

- **Scheduling.** Staggering start-up of production lines, running batch processes such as grinding, pumping to storage or charging forklifts at night, separating defrost cycles in cold stores.
- **Demand limiting.** A controller reads the revenue meter's pulse output, predicts the demand at the end of the current 15-minute interval and, if it will exceed the target, sheds pre-agreed non-critical loads (water heaters, HVAC set-points, EV chargers, snow-melt) for a few minutes.
- **Thermal storage.** Ice banks or chilled-water tanks charged at night supply the afternoon cooling peak; cold stores themselves can be pre-cooled by 2–3 K and allowed to drift.
- **Electrical storage.** A battery discharges during the peak and recharges off-peak (Chapter 16). Batteries also serve as UPS, reactive-power sources and frequency-response assets, and stacking these revenues is what usually makes the economics work.
- **On-site generation.** Diesel or gas gensets run for the few peak hours; increasingly constrained by emissions permits.

Table 28.4 — Demand-management measures (typical characteristics).

Measure	Peak reduction	Energy effect	Capital cost	Response time
Load scheduling	5–15 %	Neutral	Very low	Planned
Automatic demand limiter	5–10 %	Neutral to slight loss	Low	Minutes
Ice or chilled-water storage	10–30 % of cooling peak	5–10 % loss	Medium	Minutes
Battery storage (Li-ion)	Sized to need	8–15 % round-trip loss	High	Under 1 s
Peak-lopping genset	Sized to need	Fuel use, emissions	Medium	10–60 s
Managed EV charging	30–60 % of charging peak	Neutral	Low	Seconds

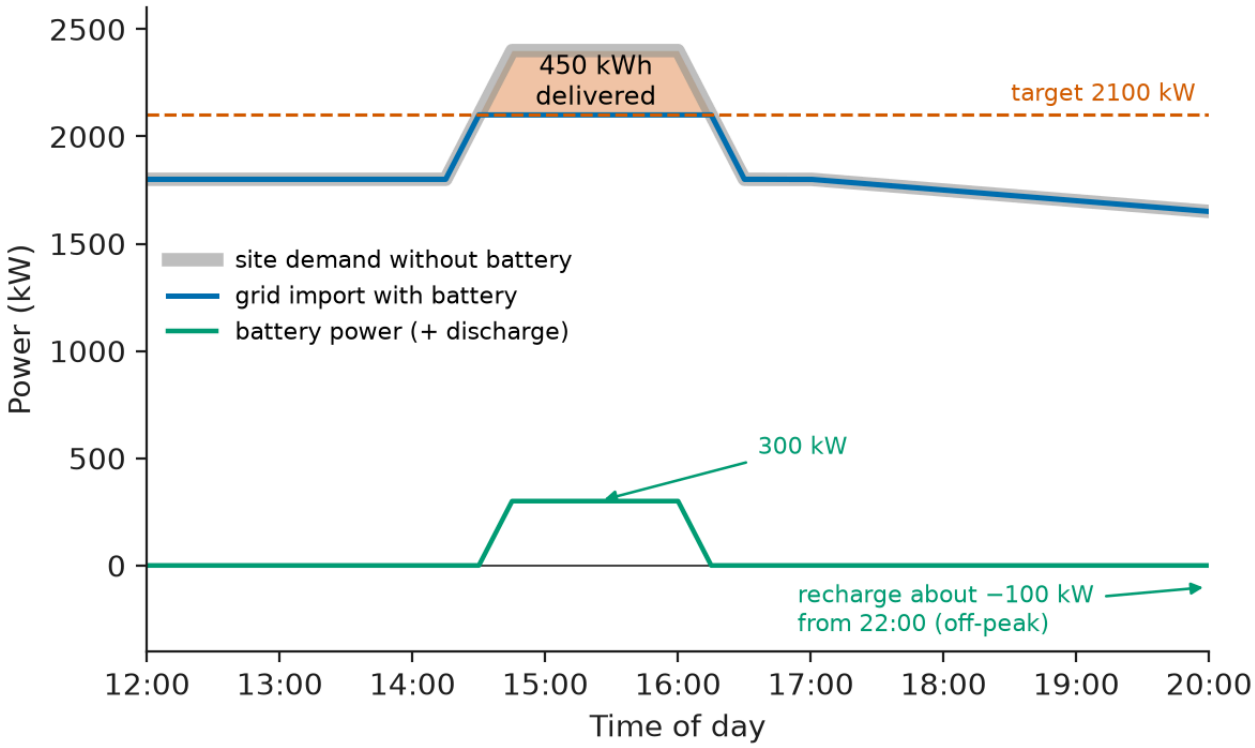


Figure 28.2 — Battery peak shaving on the plant's worst afternoon: site demand without and with the battery, and the battery's power.

Worked example — Sizing a peak-shaving battery. For the plant of Section 28.2, the worst afternoon (Figure 28.2) requires 450 kWh to be delivered above the 2100 kW target, at up to 300 kW. The lithium-iron-phosphate battery may be cycled over 90 % of its nameplate energy, and the inverter efficiency is 95 % one way. Peak-shaving events occur on about 25 days a year (the 40 hours above target in Figure 28.1); most are smaller than the worst afternoon. The battery costs 350 USD per kWh installed; off-peak energy costs 0.08 USD per kWh.

Nameplate energy needed: $450/(0.90 \times 0.95) = 526$ kWh; a standard 300 kW / 600 kWh unit leaves margin for ageing (Chapter 16). With a round-trip efficiency of 0.88, recharging 450 kWh draws $450/0.88 = 511$ kWh,

a loss of at most 61 kWh per event, or at most $61 \times 25 = 1525$ kWh per year costing about 120 USD. Net saving: $43\,200 - 120 \approx 43\,100$ USD per year. Capital: $600 \times 350 = 210\,000$ USD. Simple payback: 4.9 years.

Two cautions temper this. The demand charge is set by the *single* worst interval, so a battery that is empty on the one day when two peaks occur back to back saves nothing that month; the controller needs a load forecast, and the energy rating needs margin. And a battery that only shaves peaks is idle 95 % of the time; adding frequency response or backup service shortens the payback considerably.

> **Status of the principle: Peak shaving with storage**
> *Approximate*. Sizing from a historical worst day assumes future peaks resemble past ones; the saving is a step function of the monthly maximum, so a single forecasting failure can remove a month's benefit.

28.4 Industrial energy efficiency

Systems, not components

The largest savings in industry come from looking at whole systems — motor, drive, transmission, pump or fan, pipework or ductwork, and the process need — rather than at any single component. A premium-efficiency motor driving a pump that pushes water against a throttling valve to supply a process that needs half the flow wastes far more than any motor can save. The order of attack is: reduce the demand (does the process need this flow, this pressure, this temperature?), match the supply to the demand (speed control, right-sized machines), then improve component efficiencies.

Pumps and fans

Centrifugal pumps and fans take roughly 40 % of industrial motor energy. When the flow is controlled by a throttle valve or damper, the machine runs at full speed and the excess head is burned across the valve. When it is controlled by speed, the affinity laws — flow proportional to speed, head to speed squared, power to speed cubed, taken from *Engineering Physics*, Ch. 19 §19.7.3 — give a power that falls steeply with flow. In a system dominated by friction, $P \propto Q^3$; at 80 % flow the shaft power is about half, at 60 % about one fifth. Figure 28.3 compares the two control methods.

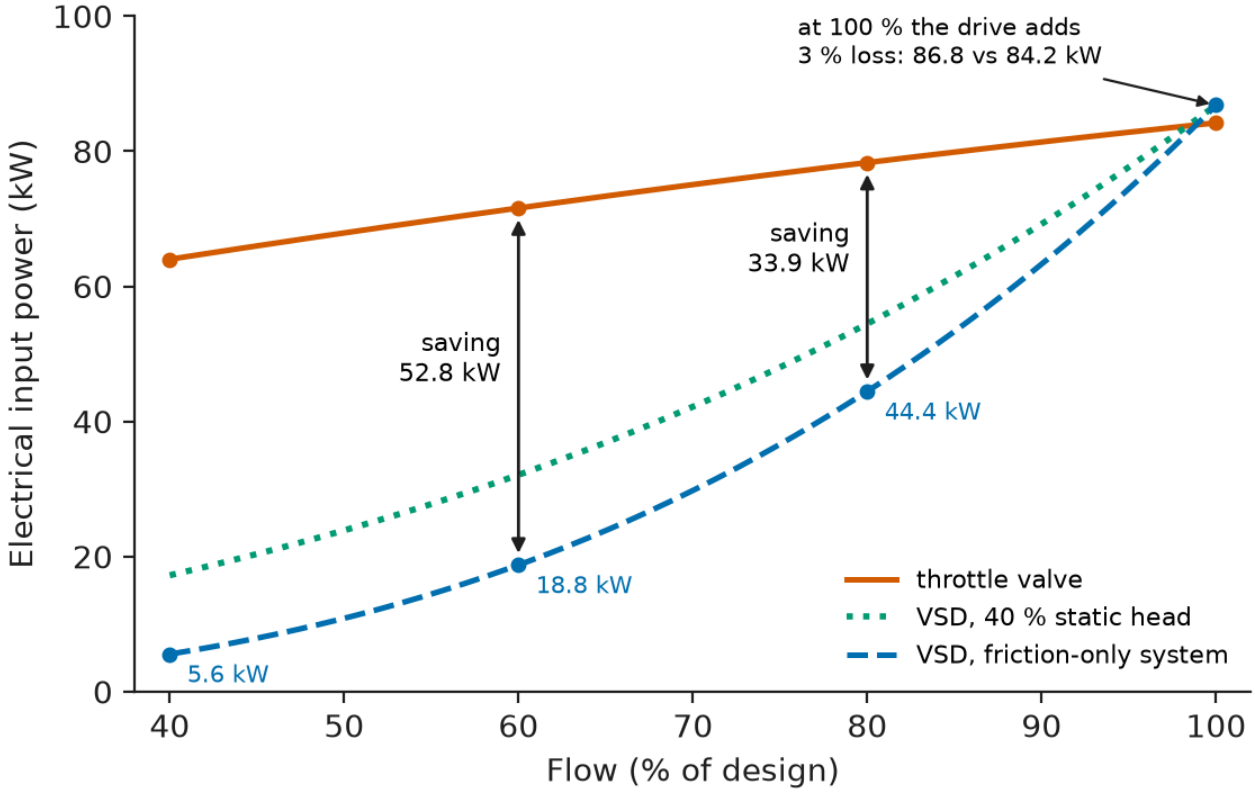


Figure 28.3 — Electrical input power of a 90 kW pump set against flow for throttle control and for variable-speed control.

Worked example — Speed control of a chilled-water pump. A pump needs 80 kW of shaft power at design flow and runs 6000 h/yr: 20 % of the time at 100 % flow, 40 % at 80 % and 40 % at 60 %. At present it is throttled; from the pump curve the shaft power is 100 %, 93 % and 85 % of 80 kW at those flows. The motor efficiency is 95 %. A variable-frequency drive (efficiency 97 %) would replace the valve; the system is a closed loop with negligible static head. Electricity costs 0.10 USD per kWh.

Throttled input powers: $80 \times (1.00,\ 0.93,\ 0.85)/0.95 = 84.2,\ 78.3,\ 71.6$ kW. Variable-speed input powers: $80 \times (1,\ 0.8^3,\ 0.6^3)/(0.95 \times 0.97) = 86.8,\ 44.4,\ 18.8$ kW. Annual energy:

$$E_{thr} = 6000\,(0.2 \times 84.2 + 0.4 \times 78.3 + 0.4 \times 71.6) = 460.8 \text{ MWh}$$

$$E_{VSD} = 6000\,(0.2 \times 86.8 + 0.4 \times 44.4 + 0.4 \times 18.8) = 255.9 \text{ MWh}$$

The saving is 204.9 MWh, 44 % of the pump's energy, worth about 20 500 USD per year. A 90 kW low-voltage drive installed costs roughly 12 000–18 000 USD, so the payback is under a year. Two caveats: at full flow the drive *adds* 3 % loss, so a pump that genuinely runs at design flow all year should not get one; and in systems with large static head (lifting water to a reservoir, boiler feed) the head does not fall with speed, and the saving is much smaller (the dotted curve of Figure 28.3).

> **Status of the principle: Cube-law savings from speed control**
> *Approximate.* The affinity laws hold for a fixed impeller at constant efficiency on a friction-only system curve; static head, efficiency drift away from the best-efficiency point and minimum-speed limits (cooling, bearings, NPSH) reduce the savings.

Compressed air

Compressed air is the most expensive utility in most factories: of the electrical energy into a compressor, only about 10 % reaches the tool as useful work, the rest leaving as heat. A typical system wastes 20–30 % of its output in leaks; a single 3 mm hole at 7 bar costs roughly 3 kW of compressor input, continuously. The standard measures are leak surveys with ultrasonic detectors, reducing the generation pressure (every 1 bar reduction saves about 6–8 % of compressor energy), variable-speed trim compressors, sequencing controllers, and heat recovery from oil-cooled compressors (up to 70–90 % of input power recoverable as hot water at 50–70 °C).

Process heat, refrigeration and lighting

Electric process heat (resistance, induction, arc and infrared) is already near 100 % efficient at the point of conversion; the savings are in insulation, scheduling and avoiding idle holding. Where low-temperature heat is needed, an industrial heat pump with a coefficient of performance of 3–5 replaces resistance heating at a third to a fifth of the electricity — the thermodynamic limit is in *Engineering Physics*, Ch. 19 §19.5.5. Refrigeration plants save through floating head pressure, variable-speed compressors and evaporator fans, and closing doors and strip curtains. Lighting is now almost entirely an LED question: LED high-bay luminaires reach 150–200 lm/W against 80–100 lm/W for metal halide and 85–100 lm/W for T8 fluorescent, and add instant restart, dimming and occupancy control.

Table 28.5 — Typical industrial efficiency measures (indicative).

Measure	Typical saving in the system	Typical simple payback
VSD on variable-flow pump or fan	20–50 %	0.5–3 years
Compressed-air leak repair	10–25 % of air energy	Under 0.5 year
Reduce compressor pressure by 1 bar	6–8 %	Immediate
LED retrofit with controls	50–70 % of lighting	1–4 years
IE2 to IE4 motor at replacement	30–35 % of motor losses	2–5 years
Heat pump instead of resistance heating	65–80 %	2–6 years
Compressor heat recovery	Displaces fuel	1–3 years
Power-factor correction	Removes kvarh charges	0.2–2 years

28.5 Motor efficiency

Efficiency classes

The motor itself converts 85–97 % of its input to shaft power; the rest is stator and rotor copper loss, core loss, friction and windage, and stray load loss (Chapters 6 and 9). Because a motor running 6000 h/yr consumes its own purchase price in electricity in a few weeks, small differences in efficiency are worth a lot. IEC 60034-30-1 defines efficiency classes IE1 (standard) to IE4 (super premium) for line-operated motors; IE5 (ultra premium, about 20 % lower losses than IE4) is defined for converter-fed motors in IEC TS 60034-30-2 and is typically reached with permanent-magnet or synchronous-reluctance rotors (Chapter 8). Efficiencies are measured by

the loss-segregation method of IEC 60034-2-1 (Chapter 9). Table 28.6 gives representative minimum values and Figure 28.4 plots them.

Table 28.6 — Minimum nominal efficiencies, 4-pole 50 Hz motors (approximate, per IEC 60034-30-1).

Rated output	IE1	IE2	IE3	IE4
0.75 kW	72.1 %	79.6 %	82.5 %	85.7 %
1.1 kW	75.0 %	81.4 %	84.1 %	87.2 %
7.5 kW	86.0 %	88.7 %	90.4 %	91.9 %
11 kW	87.6 %	89.8 %	91.4 %	92.6 %
75 kW	93.1 %	94.7 %	95.6 %	96.4 %
200–1000 kW	94.0 %	95.1 %	96.0 %	96.7 %

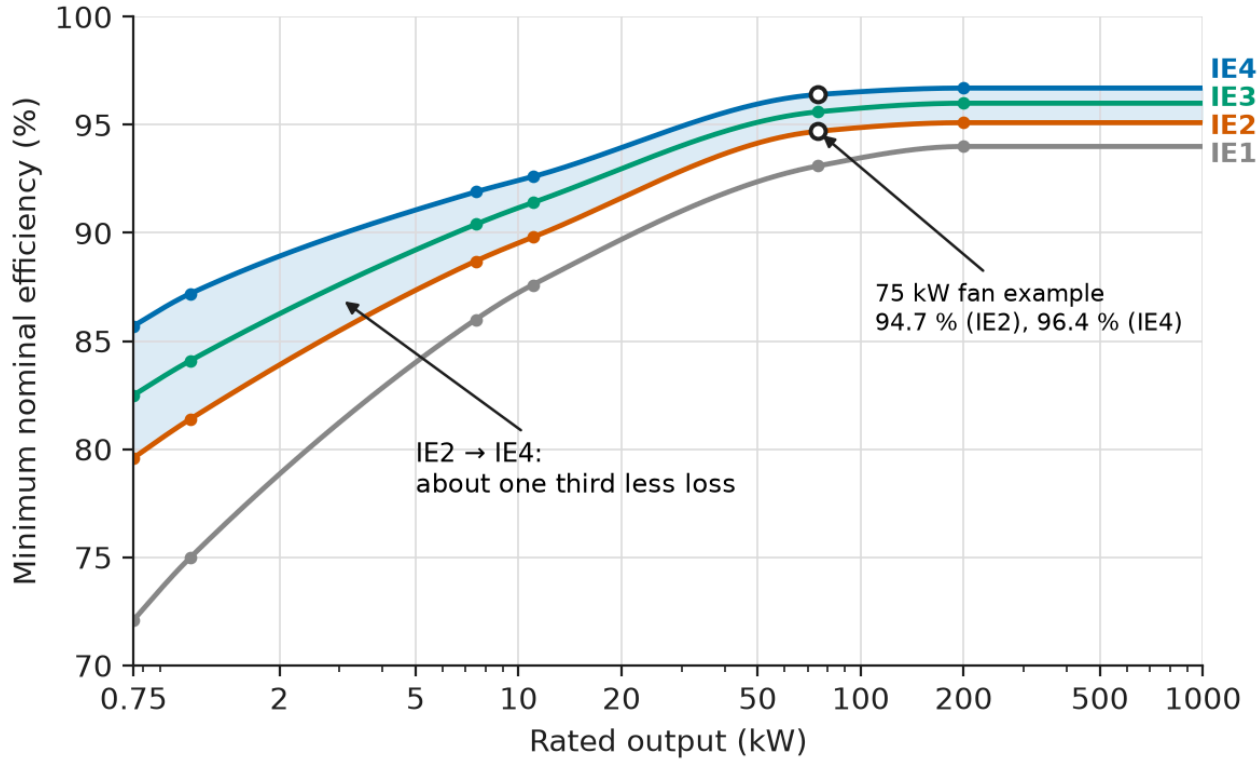


Figure 28.4 — Minimum efficiency against rated output for IE1 to IE4 four-pole 50 Hz motors.

Regulation has moved the market: in the European Union the Ecodesign regulation (EU) 2019/1781 has required IE3 for 0.75–1000 kW motors since 2021 and IE4 for 75–200 kW since 2023; the US DOE rules require levels equivalent to IE3 (NEMA Premium). The gap between classes is larger in small motors, where losses are proportionally higher, but the energy at stake is larger in big ones.

> **Status of the principle: IE efficiency classes**
> *Empirical*. The class limits are fitted to the best practical designs of their time and are defined at rated load, voltage and frequency; part-load efficiency, supply unbalance, harmonics and converter supply are outside the class definition.

Worked example — IE2 or IE4 for a 75 kW fan motor? A 75 kW, 4-pole motor driving a process fan runs 6000 h/yr at 75 % load (56.25 kW shaft). Assume the efficiency at 75 % load equals the nominal value (true within a few tenths of a per cent for these motors). Electricity costs 0.10 USD per kWh; the IE4 motor costs 2500 USD more than the IE2.

Input power: $56.25/0.947 = 59.40$ kW for IE2 and $56.25/0.964 = 58.35$ kW for IE4. The difference, 1.05 kW, over 6000 h is 6285 kWh, worth 628 USD per year: a payback of four years on a machine that lasts fifteen to twenty. In loss terms the IE4 motor dissipates $(1/0.964 - 1)/(1/0.947 - 1) = 0.67$ of the IE2 motor's losses, so it also runs about a third cooler, which extends insulation and bearing life (Chapter 9).

Part load and oversizing

Motor efficiency is roughly flat from 50 % to 100 % load and falls quickly below about 40 %, because the core and friction losses become a large fraction of a small output; the power factor falls even faster as the magnetising current stays while the active current shrinks. Surveys regularly find a third of industrial motors running below 50 % load because they were sized for a worst case, then given a "safety margin", then the next frame size up. Right-sizing at replacement, or adding a VSD, recovers both efficiency and power factor.

Rewind or replace

A failed motor may be rewound for perhaps 50–60 % of the price of a new one. A careless rewind — stripping the old winding by burning at too high a temperature, which damages the interlaminar insulation, or using fewer turns of thinner wire — can add 1–2 percentage points of loss. Good practice (IEC 60034-23 and the EASA/AEMT rewind guide) holds efficiency within 0.1–0.5 point. A common policy: replace rather than rewind below about 30–50 kW, and always replace with IE3 or better when the motor runs more than 4000 h/yr.

28.6 Power-factor correction

Correction as an energy measure

The circuit theory of power-factor correction — the capacitor kvar $Q_C = P(\tan\phi_1 - \tan\phi_2)$, the hazards of over-correction, self-excitation and harmonic resonance, and the effect on the bill — is developed in *Circuits, Signals and Communications*, Ch. 9 §9.6–9.7; the power-quality side, with detuned filters and harmonic interaction, is in Chapter 25 of this book. From the energy manager's side three questions remain: where to put the capacitors, how much energy the correction saves (as opposed to how much money it saves on the tariff), and when correction is the wrong answer.

Where to correct

Central correction at the main LV switchboard with an automatic multi-step bank removes the utility's reactive charges and relieves the transformer, but not the cables inside the plant. **Group correction** at motor

control centres also relieves the main distribution feeders. **Individual correction** at large, long-running motors relieves everything up to the motor terminals and switches with the motor, so it can never over-correct — but must be limited to about 90 % of the motor's no-load kvar to avoid self-excitation. Variable-speed drives with diode front ends already present a displacement power factor near 0.95–0.98 to the supply (their problem is harmonics, Chapter 25), and active-front-end drives can run at unity or even supply reactive power. In plants where more than about 15–20 % of the load is non-linear, every capacitor bank should be detuned (typically 7 % reactor, tuned near 189 Hz on a 50 Hz system) so that it is inductive at the 5th harmonic and above.

> **Status of the principle: Loss reduction by local correction**
> *Approximate.* Losses scale with I^2 only for sinusoidal currents; with harmonics present the RMS current and therefore the loss depend on the distortion, which an undetuned capacitor can make worse.

Worked example — Correcting at the motor control centre. A group of compressors drawing 250 kW at a power factor of 0.80 is supplied from the main switchboard of a 400 V plant through a 150 m run of 240 mm² copper cable per phase. Its resistance is 0.0754 Ω/km at 20 °C; take the operating temperature as 70 °C. The plant already has central correction, so the utility sees no reactive charge. Is it worth adding 125 kvar of correction at the MCC to raise the group's power factor to 0.97? The group runs 6000 h/yr.

Cable resistance per phase: $R = 0.150 \times 0.0754 \times (1 + 0.00393 \times 50) = 0.01353\ \Omega$. Current before and after:

$$I_1 = \frac{250\,000}{\sqrt{3} \times 400 \times 0.80} = 451.1\ \text{A}, \qquad I_2 = \frac{250\,000}{\sqrt{3} \times 400 \times 0.97} = 372.0\ \text{A}$$

Loss: $3I_1^2 R = 8.26$ kW before, $3I_2^2 R = 5.62$ kW after, a saving of 2.64 kW or 15.8 MWh per year (about 1600 USD at 0.10 USD/kWh). The required kvar is $250(\tan\arccos 0.80 - \tan\arccos 0.97) = 124.8$ kvar, and a detuned 125 kvar bank costs perhaps 5000–7000 USD installed, a payback of three to four years from losses alone. It also lowers the voltage drop on the feeder, frees about 18 % of its thermal capacity for new load, and lets the central bank shrink. The tariff saving was already captured by the central bank; the local bank's justification is energy and capacity.

28.7 Energy monitoring

The metering hierarchy

What is not measured cannot be managed, but what is measured badly is managed wrongly. A good monitoring system has a hierarchy: the revenue meter at the point of supply (Class 0.5S or 0.2S, IEC 62053-22, read by the utility); main-feeder and large-load submeters (IEC 61557-12 power metering and monitoring devices, Class 0.5 or 1 for active energy); and process-level meters on individual machines, often built into drives and motor protection relays. The rule of thumb is to submeter every load or group above about 5–10 % of site consumption, so that the sum of submeters accounts for at least 80–90 % of the main meter. Data are collected at 1–15 minute intervals over Modbus, BACnet or IEC 61850 and stored in an energy-management information system.

Energy performance indicators

Raw kWh mean little because they move with production, weather and occupancy. Energy performance indicators (EnPIs, ISO 50006) normalise them: kWh per tonne of product, kWh per m³ of water pumped, kWh

per m² per year for a building, kWh per kilometre for an electric bus fleet, kWh per tonne of ore hoisted. For data centres the standard indicator is **power usage effectiveness** (ISO/IEC 30134-2):

$$\mathrm{PUE} = \frac{E_{facility}}{E_{IT}}$$

A legacy enterprise data centre runs at 1.6–2.0, a modern colocation site at 1.3–1.4, a hyperscale site with free cooling at 1.1–1.2. For a 1 MW IT load, improving the PUE from 1.6 to 1.3 removes 0.3 MW of overhead continuously: 2628 MWh per year.

Baselines and CUSUM

To prove a saving, consumption after a measure must be compared with what it *would have been* without the measure under the same conditions. The usual tool is a regression baseline fitted to 12 or more months of pre-measure data, with the drivers that matter — production, degree-days, occupancy — as variables. For many plants a straight line is adequate:

$$E_{expected} = E_0 + k\,X$$

with E_0 the fixed (production-independent) energy and k the specific energy per unit of driver X. The cumulative sum of the differences between actual and expected consumption, **CUSUM**, makes trends visible that monthly noise would hide: a flat CUSUM means performance is unchanged; a steadily falling one, a sustained saving; a sudden rise, a fault (a stuck damper, a failed trap, a VSD put on bypass). The International Performance Measurement and Verification Protocol (IPMVP) formalises this as "Option C, whole facility" when savings are large compared with the noise of the main meter, and "Option B, retrofit isolation" when the measure is metered on its own.

> **Status of the principle: Regression baseline**
> *Empirical*. The baseline is a statistical fit valid only within the range of the drivers it was fitted to and only while nothing else about the site changes; a new product line or a new building wing requires a baseline adjustment.

Worked example — CUSUM after a refrigeration retrofit. A cold store's baseline, fitted to two years of monthly data, is $E_{expected} = 180\ \mathrm{MWh} + 0.35\ \mathrm{MWh/t} \times X$ where X is tonnes of product frozen. Variable-speed compressors and floating head pressure were commissioned at the end of month 1. Table 28.7 gives the monitoring data.

Table 28.7 — Monitoring a refrigeration retrofit.

Month	Production (t)	Expected (MWh)	Actual (MWh)	Difference (MWh)	CUSUM (MWh)
1	950	512.5	520	+7.5	+7.5
2	1020	537.0	510	−27.0	−19.5
3	980	523.0	480	−43.0	−62.5
4	1100	565.0	520	−45.0	−107.5
5	1000	530.0	470	−60.0	−167.5
6	1050	547.5	480	−67.5	−235.0

Month 2 shows a partial saving while the controls were tuned; from month 3 the CUSUM falls by 43–68 MWh per month, settling near 65 MWh; the average slope over months 3–6 is about 58 MWh per month, 11–12 % of expected consumption. Note that actual consumption in month 4 (520 MWh) equals month 1, yet the measure is clearly working: production rose 16 %. Without the baseline, the saving would be invisible. Figure 28.5 shows the CUSUM line; its slope is the saving rate.

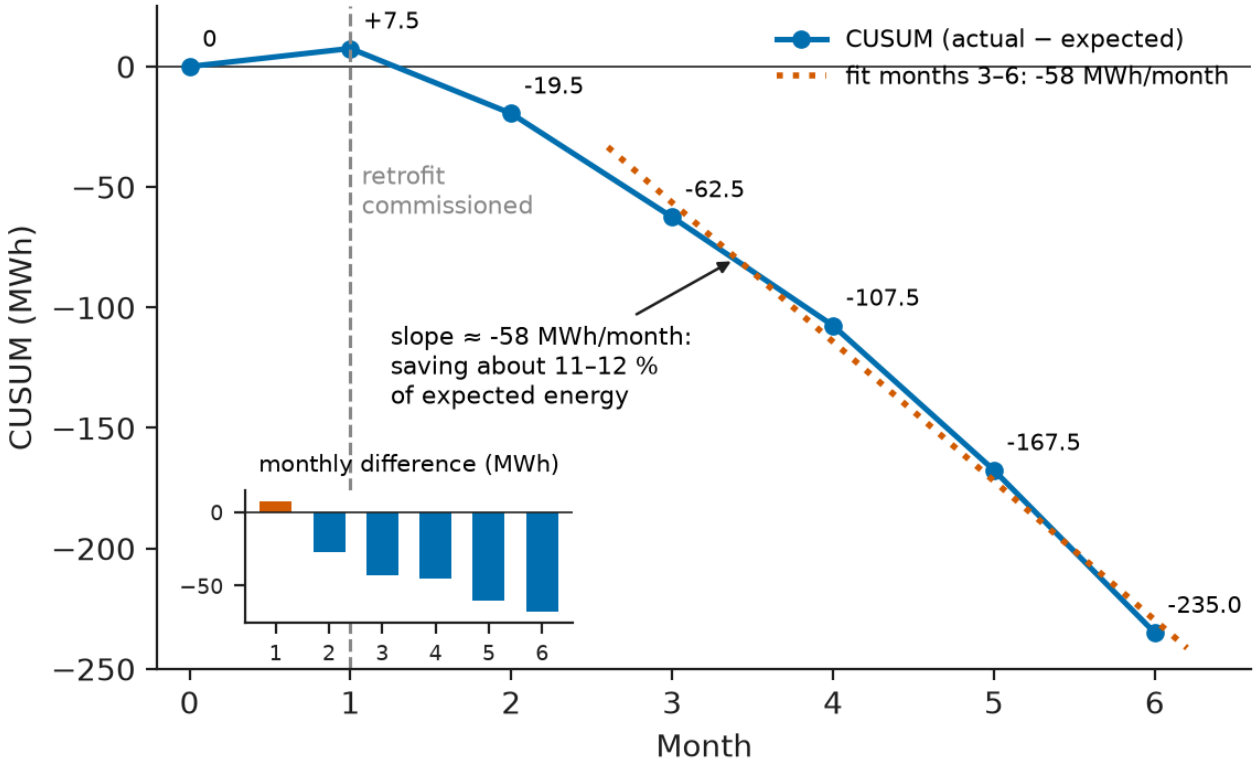


Figure 28.5 — CUSUM of actual minus expected energy for the cold-store retrofit; the slope after commissioning is the saving rate.

28.8 Loss reduction

Where a plant's own network loses energy

Between the revenue meter and the loads, a well-designed industrial network loses 2–4 % of the energy it carries: roughly half in transformers and half in cables, busbars and switchgear. In a poorly designed or overloaded one it can be 6–8 %. For a site using 20 GWh/yr, each percentage point is 200 MWh. The utility network upstream loses more: 6–8 % in a typical developed-country transmission and distribution system and 15–20 % or more where non-technical losses (theft and metering errors) are high (Chapter 22).

Transformer losses

A transformer has two losses (Chapter 4): the **no-load loss** P_0 in the core, present every hour it is energised, independent of load; and the **load loss** P_k at rated current, scaling with the square of the per-unit load x. Efficiency at load x and power factor $\cos\phi$ is

$$\eta(x) = \frac{x\,S_r\cos\phi}{x\,S_r\cos\phi + P_0 + x^2 P_k}$$

and is maximum where the two losses are equal, $x_{opt} = \sqrt{P_0/P_k}$. Modern low-loss distribution transformers have x_{opt} around 0.3–0.45, which suits distribution duty where the average load is well below rating. Annual loss energy is $P_0 \times 8760 + P_k\,\overline{x^2} \times 8760$, with $\overline{x^2}$ the mean square per-unit load (the square of the RMS loading).

> **Status of the principle: Maximum efficiency at equal fixed and variable loss**
> *Derived.* Follows from differentiating $\eta(x)$ with P_0 and P_k constant; holds only while core loss is independent of load (constant voltage) and load loss follows I^2 (no significant harmonic or stray-loss increase).

Table 28.8 — Losses of 1000 kVA oil-immersed distribution transformers (typical).

Design	No-load loss P_0	Load loss P_k	Annual loss at RMS load 0.5
1990s conventional CRGO	1.70 kW	13.0 kW	43.4 MWh
EU Ecodesign Tier 1 (2015)	0.77 kW	10.5 kW	29.7 MWh
EU Ecodesign Tier 2 (2021)	0.69 kW	7.6 kW	22.7 MWh
Amorphous core	0.30–0.45 kW	8–10 kW	19–25 MWh

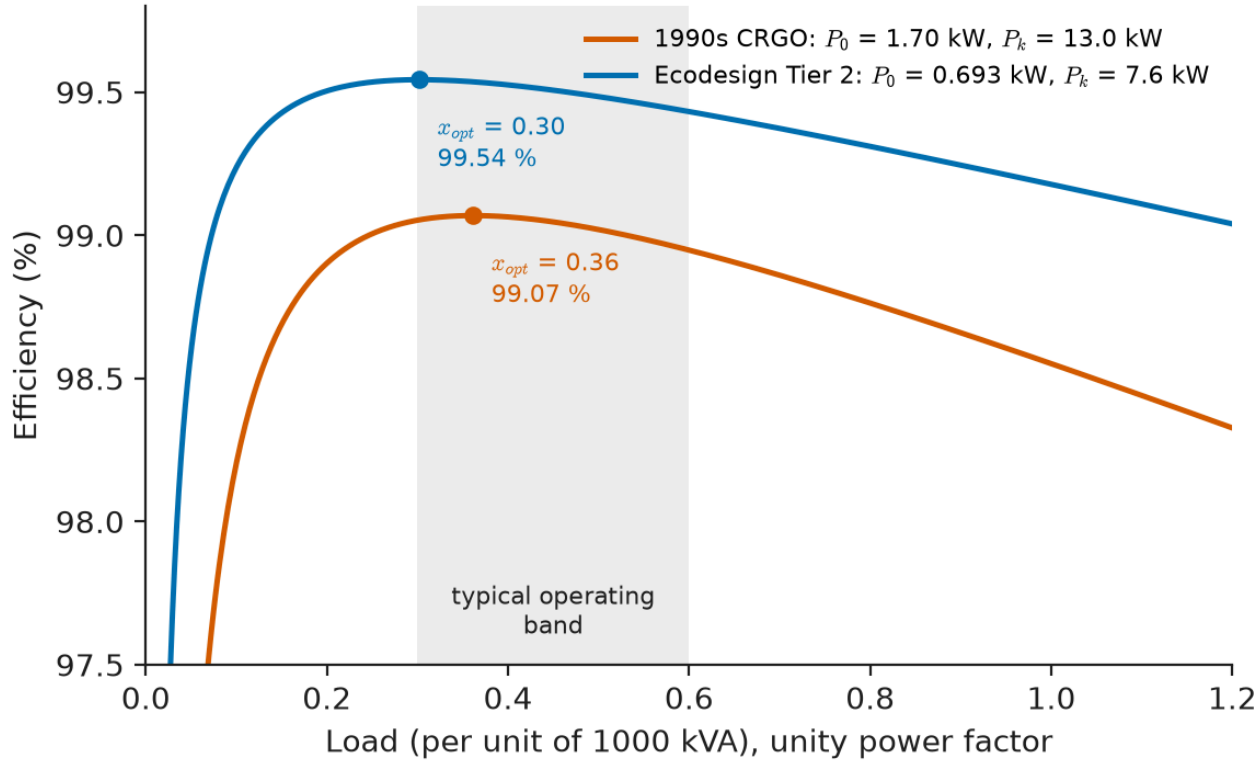


Figure 28.6 — Efficiency against load of a 1990s 1000 kVA transformer and an Ecodesign Tier 2 replacement, at unity power factor.

Figure 28.6 shows why replacing an old transformer can pay even when it still works: across the typical operating band the new unit is about 0.5 percentage point more efficient, which for a 1000 kVA unit is roughly 20 MWh per year (worked out in Section 28.9). Other transformer measures: switching off one of two parallel

transformers at light load (when x for the remaining unit stays below about $\sqrt{2P_0/P_k}$), and de-energising seasonal-load transformers such as irrigation or grain-drying supplies out of season.

Cables, busbars and voltage

Cables are sized for current-carrying capacity and voltage drop (Chapter 30), which gives a cross-section at which the conductor runs hot — up to 70 °C for PVC and 90 °C for XLPE insulation — and loses 2–4 % of the power it carries on a long run. Choosing the next size up halves nothing but reduces loss by roughly the ratio of cross-sections, and because copper resistance rises 0.393 % per kelvin, the cooler cable gains again. IEC 60287-3-2 formalises **economic cable sizing**, which minimises purchase cost plus discounted loss cost; on loads with high utilisation it usually selects one or two sizes above the thermal minimum (worked example in Chapter 30).

Other network losses are in poor joints and terminations (found by thermography, and a fire risk as well as a loss), circulating currents in incorrectly bonded single-core cables, eddy-current heating in steel enclosures around single-core conductors, and harmonic currents, which add $I_h^2 R$ loss in every conductor and extra eddy loss in transformers (Chapter 25).

Voltage optimisation — lowering the supply voltage within the statutory band — saves energy only on loads whose power falls with voltage: resistive heating that is not thermostatically controlled, older lighting, and induction motors that are lightly loaded (where reduced magnetising current lowers core loss). It does nothing for thermostatic heaters, which simply run longer, nor for switch-mode supplies and drives, which draw constant power. Claims of 10–15 % savings on whole sites should be treated with caution; 1–4 % is typical where it works.

28.9 Life-cycle energy cost

Discounting

Energy-consuming equipment costs money to buy and much more to run. Comparing options fairly requires putting future costs on the same footing as present ones. With a discount rate r and a life of n years, an annual cost C has a present value $C \times \mathrm{PWF}$, where the present-worth factor is

$$\mathrm{PWF} = \frac{1 - (1+r)^{-n}}{r}$$

At $r = 6$ % the PWF is 9.71 for 15 years, 12.78 for 25 years and 13.76 for 30 years. The **life-cycle cost** is then

$$\mathrm{LCC} = C_{capital} + C_{install} + \mathrm{PWF}\,(C_{energy} + C_{maintenance}) - \mathrm{PV}(\text{residual value})$$

Energy-price escalation can be folded in by using a reduced effective discount rate. Simple payback (capital divided by annual saving) is easy to explain but ignores everything after the payback date, and so systematically undervalues long-lived efficient equipment.

> **Status of the principle: Present-worth evaluation**
> *Derived*. Exact arithmetic for a constant annual cost and constant discount rate; the result is only as good as the assumed energy price, operating hours and life, which is why sensitivity to each should be reported.

The motor example

For the 75 kW IE4 fan motor of Section 28.5, the annual energy bill is $58.35 \times 6000 \times 0.10 = 35\,000$ USD. Over 15 years at 6 % its present value is $35\,000 \times 9.71 = 340\,000$ USD. Against a purchase price of perhaps 7000 USD and maintenance of 3000 USD present value, energy is 97 % of the life-cycle cost. The purchase price, which dominates procurement discussions, is 2 %.

Loss capitalisation for transformers

Utilities and large industrial buyers evaluate transformer tenders by the **total owning cost**:

$$\mathrm{TOC} = C_{price} + A\,P_0 + B\,P_k$$

where A (in USD per kW of no-load loss) is the present value of a kilowatt lost for every hour of the life, $A = c_e \times 8760 \times \mathrm{PWF}$, and $B = A\,\overline{x^2}$ weights the load loss by the mean-square loading. A manufacturer can then trade more core steel (lower P_0) or copper (lower P_k) against price in a way the buyer has priced in advance.

Worked example — Replace a working 1990s transformer? A 1000 kVA transformer with $P_0 = 1.70$ kW and $P_k = 13.0$ kW feeds a plant whose RMS loading is 0.5 per unit. A Tier 2 replacement ($P_0 = 0.693$ kW, $P_k = 7.6$ kW) costs 17 000 USD installed. Energy costs 0.10 USD/kWh, the evaluation period is 30 years at 6 %.

$\mathrm{PWF} = 13.765$, so $A = 0.10 \times 8760 \times 13.765 = 12\,058$ USD/kW and $B = 0.25\,A = 3014$ USD/kW. Keeping the old unit (no price): $\mathrm{TOC} = 12\,058 \times 1.70 + 3014 \times 13.0 = 59\,690$ USD. The new unit: $17\,000 + 12\,058 \times 0.693 + 3014 \times 7.6 = 48\,270$ USD. Replacing it is worth about 11 400 USD in present value, with annual loss energy falling from 43.4 to 22.7 MWh. The case is real but not overwhelming; it becomes compelling if the old unit also needs a refurbishment, if the plant runs heavier, or if carbon is priced. The same calculation explains why utilities with thousands of transformers specify loss-capitalisation factors in every tender.

Table 28.9 — Share of life-cycle cost due to energy (typical, 6 % discount, long running hours).

Equipment	Typical life	Energy share of LCC
Industrial motor, over 4000 h/yr	15–20 years	95–98 %
Centrifugal pump set	15–20 years	80–90 %
Air compressor	10–15 years	75–85 %
Distribution transformer (losses only)	30–40 years	50–70 %
LED lighting installation	15 years	60–75 %
Power cable on a heavily loaded feeder	30–40 years	30–60 %

28.10 Efficiency improvement strategies

A hierarchy of measures

A sustained programme follows a hierarchy:

1. **Avoid.** Switch off what is not needed: idle conveyors, lighting in unoccupied zones, compressed air at weekends, transformers out of season. Typically 5–10 % of site energy for almost no capital.
2. **Reduce the need.** Lower pressure set-points, wider temperature dead-bands, better insulation, fewer process steps.
3. **Match supply to need.** Variable-speed drives, right-sized motors, sequencing of multiple compressors or chillers, demand-controlled ventilation.
4. **Improve conversion efficiency.** IE4 motors, low-loss transformers, LED lighting, heat pumps, high-efficiency UPS (98–99 % in eco mode against 92–95 % for older double-conversion units).
5. **Recover and reuse.** Compressor and process heat recovery, regenerative drives on hoists, cranes, lifts and test rigs (Chapter 10), data-centre waste heat into district heating.
6. **Supply cleanly.** On-site solar PV, power purchase agreements, storage to shift renewable energy (Chapters 15 and 16).

The order matters: speed control of a fan that need not run at all saves nothing, and recovering heat from a compressor whose air is leaking away recovers waste from waste.

Energy-management systems

Technical measures decay. Set-points drift back, drives are put on bypass after a fault and left there, leaks reappear. ISO 50001 provides a management framework (policy, energy review, baseline, EnPIs, objectives, action plans, monitoring, internal audit, management review) on the same plan–do–check–act cycle as ISO 9001 and ISO 14001. Sites that implement it typically find 10–20 % savings over the first three to five years, much of it from operational changes the monitoring system reveals. Large companies in the EU must audit at least every four years (Energy Efficiency Directive) unless they run a certified ISO 50001 system.

Pitfalls

- **Double counting.** A VSD and an IE4 motor on the same pump do not each save their stand-alone percentages; savings in series multiply, they do not add.
- **Rebound.** Cheaper lighting tends to be left on longer; quote savings from measured, not assumed, hours.
- **Power-quality side effects.** Drives and LED drivers add harmonic current, which can overheat neutrals and resonate with capacitor banks (Chapter 25). An efficiency programme should include a harmonic study.
- **Reliability.** A higher-efficiency motor with lower slip runs slightly faster, so a fan or pump on a fixed-speed drive draws *more* power ($P \propto n^3$); a 0.5 % speed increase costs 1.5 % more power and can eat the efficiency gain unless the impeller is trimmed or the pulley changed.

Worked example — Ranking a programme. The audit of the frozen-food plant produced these measures (savings already corrected for interaction): compressed-air leak repair, 150 MWh/yr for 4000 USD; pump VSD, 205 MWh/yr for 15 000 USD; LED retrofit, 260 MWh/yr for 90 000 USD; battery peak shaving, 43 100 USD/yr net for 210 000 USD; transformer replacement, 20.6 MWh/yr for 17 000 USD. At 0.10 USD/kWh the simple paybacks are 0.27, 0.73, 3.5, 4.9 and 8.2 years. The leak repair and the VSD fund themselves within a year and should be done first; their savings then help to finance the LED retrofit. The battery is a demand-charge project, not an energy project — it uses *more* energy — and should be justified on its stacked revenues. The transformer replacement, poor on simple payback, is positive on life-cycle cost (Section 28.9) and should be scheduled for the next outage or failure. Together the first three measures cut site energy by 615 MWh, about 4.7 % of the plant's 13 GWh annual use, before the monitoring system finds the operational savings.

What remains open

Open question	Where it is answered
How are motor losses separated and measured, and how do temperature and insulation life depend on them?	Chapter 9 (loss calculations, thermal limits, performance testing)
How does a variable-frequency drive control an induction or PM motor, and what are its own losses?	Chapter 12 (VFDs, scalar and vector control); Chapter 10 (drive efficiency)
How are battery energy and power ratings, state of charge and cycle life related, and how are batteries kept safe?	Chapter 16 (energy storage)
How do harmonics from drives and LED drivers interact with capacitor banks, and how are detuned filters designed?	Chapter 25 (harmonics, power-factor correction, mitigation)
How do smart meters, demand response and microgrids extend site energy management to the grid?	Chapter 26 (smart grids)
How are cables, transformers and protective devices actually selected for a new installation, including economic sizing?	Chapter 30 (electrical system design)
What are the losses of a feeder or a utility distribution network, and how are they reduced at network scale?	Chapter 22 (distribution losses)
How does the measurement of power and energy work in revenue meters and analysers?	*Circuits, Signals and Communications*, Ch. 9 §9.9

Next: Chapter 29 opens Part VIII by turning from the energy that equipment wastes to the energy it can release into a person — shock, arc flash and the protective systems and working practices that keep electrical work survivable.

Part VIII — Electrical Safety and Engineering Practice

The first seven parts of this book followed energy from the magnetic field of a machine, through generators, storage, networks and protection, to the quality and management of the supply. Part VIII answers the questions of the engineer who must put a real installation into service and keep it there: will anyone be hurt by it, and will it do what it was designed to do for the next thirty years? Every result of the earlier parts reappears here as a design input or a safety limit. The fault current of Chapter 19 sets the arc-flash energy at a switchboard and the breaking capacity of the breaker in front of it. The protection characteristics of Chapter 23 set how long a person is exposed to a touch voltage and how long an arc burns. The grounding of Chapter 24 decides whether a fault becomes a shock. The losses of Chapter 28 decide which cable size is economic. Safety and design are the two places where the whole book is applied at once.

The part has two chapters because it builds almost entirely on what came before. Chapter 29 treats safety as an engineering quantity: the body current that causes fibrillation, the incident energy that causes a burn, the touch voltage a device must clear in time. Chapter 30 treats design as a chain of calculations that must all close, from a load estimate to transformer, generator, cables, protection and earthing, each checked and recorded so that others can verify, build and operate the installation.

By the end of this part you should understand

- how alternating and direct current act on the human body, what the time–current zones of IEC 60479-1 mean, and why disconnection time matters as much as current;
- how arc-flash incident energy depends on fault current, clearing time and working distance, and how it is estimated, labelled and reduced;
- how TN, TT and IT earthing systems, equipotential bonding and residual-current devices turn an insulation fault into a disconnection rather than a shock;
- how protective devices are coordinated with cables for overload and short circuit, and how isolation, lockout and safe testing make work on de-energised equipment genuinely de-energised;
- how a hazard assessment combines shock and arc-flash boundaries, risk controls and personal protective equipment, following NFPA 70E, EN 50110 and IEC 60364;
- how an electrical installation is designed from load estimation through transformer and generator selection, cable sizing, voltage drop, protection and earthing, to documentation and compliance with the applicable standards.

Chapters

Chapter	Central question	What it delivers
Chapter 29 — Electrical Safety	How does electricity injure people, and which engineering measures and working practices prevent it?	Shock physiology and touch-voltage limits, arc-flash incident energy and boundaries, earthing systems and bonding, protective devices and overcurrent coordination, isolation and lockout, safe testing, installation safety and hazard assessment
Chapter 30 — Electrical System Design	How is a complete, safe and economical installation designed and proven from a list of loads?	Load estimation, transformer and generator selection, equipment ratings, cable sizing and voltage drop, protection selection and selectivity, grounding design, design documentation, standards and compliance

How this part connects to the rest of the book

Part VIII is the book's point of application. Chapter 29 draws on fault analysis (Chapter 19) for prospective currents, on protection (Chapter 23) for device characteristics and clearing times, and on high-voltage engineering (Chapter 24) for earthing grids, step and touch potentials and high-voltage test safety; it supplies the safety limits that Chapter 30 must respect. Chapter 30 draws on nearly everything: transformers (Chapter 4) and synchronous generators (Chapter 7) for equipment selection, motor starting (Chapters 6 and 10) for voltage dips, per-unit methods and fault calculation (Chapters 17 and 19) for ratings, distribution practice (Chapter 22) for architectures and voltage drop, power quality (Chapter 25) for harmonics and capacitor banks, and energy management (Chapter 28) for loss evaluation and life-cycle cost.

From the sibling books, the circuit theory — phasors, AC and three-phase power, neutral currents and the residual-current device as a hardware form of KCL — is taken from *Circuits, Signals and Communications*, Ch. 2, Ch. 9 and Ch. 10. The heat transfer behind conductor ratings and burn injury, and the resistivity behind conductor resistance, are taken from *Engineering Physics*, Ch. 14 §14.7 and Ch. 19 §19.6. Converter internals of drives, inverters and chargers belong to *Electronics & Computing*. What this part adds is the engineering judgement that joins them into an installation that is safe to work on and fit for its purpose.

Chapter 29 — Electrical Safety

Part VIII · Electrical Safety and Engineering Practice

Two incidents from the same industrial estate, a year apart. In the first, a maintenance electrician opens the door of a 400 V main switchboard to tighten a loose busbar connection that thermography had flagged. The board is live because the plant cannot stop. A dropped spanner bridges two phases; the resulting arc burns for half a second, until the upstream breaker's short-time delay expires, and releases about 17 cal/cm² at his face and chest. He was wearing a cotton shirt rated for nothing. In the second, an apprentice replacing a light fitting in a farm workshop isolates the lighting circuit at the board, but the fitting has been wired from a different circuit by a previous owner. He touches the line conductor with one hand while kneeling on a damp concrete floor. The 6 A miniature circuit breaker on that circuit never trips — the current through him is a fraction of an ampere — but a 30 mA residual-current device at the origin does, in about 25 ms, and he walks away. The two cases bracket the subject: electricity injures by current through the body and by the energy of an arc, and the difference between an injury and a near miss is almost always an engineering decision made long before the incident — a protective device, a clearing time, an isolation procedure, a test before touch.

> **How much current, for how long, injures a person, and how much arc energy burns one? Which installation measures and working practices keep those quantities below their limits, and how is the remaining risk assessed?**

Concept	Engineering destination
Shock hazards	Touch-voltage limits, RCD ratings, disconnection times, special locations
Arc-flash hazards	Incident-energy studies, labels, PPE, arc-flash relays, arc-resistant switchgear
Grounding and bonding	TN, TT and IT systems, protective conductors, substation earth grids
Protective devices	MCBs, fuses, RCDs, AFDDs, insulation monitors, surge protective devices
Isolation and lockout	Five safety rules, LOTO procedures, stored-energy discharge
Safe testing	Measurement categories, test-before-touch, insulation and loop tests
Overcurrent protection	Cable–device coordination, adiabatic withstand, automatic disconnection
Electrical installation safety	IEC 60364 and NEC compliance, IP ratings, hazardous areas, EV and PV installations
Hazard assessment	Shock and arc-flash risk assessment, approach boundaries, hierarchy of controls

Roadmap. What current does to the body (29.1) → what an arc does (29.2) → how earthing and bonding turn faults into disconnections (29.3) → the devices that disconnect (29.4) → how those devices are coordinated with cables (29.5) → making equipment dead and keeping it dead (29.6) → proving it is dead and testing safely (29.7) → the installation rules that build all this in (29.8) → how the remaining hazard is assessed and controlled (29.9).

> **Boundary.**
> The residual-current device as a hardware form of Kirchhoff's current law is introduced in *Circuits, Signals and Communications*, Ch. 2, and neutral currents in Ch. 10 §10.9. Prospective fault currents come from Chapter 19, protective-device characteristics and coordination from Chapter 23, and high-voltage earthing grids, surges and high-voltage test safety from Chapter 24. This chapter uses those results to set and check safety limits.

29.1 Shock hazards

Current, not voltage

The physiological effect of an electric shock is set by the current through the body, its path, its frequency and its duration. Voltage matters only because it drives that current through the body's impedance. IEC 60479-1 (with IEC TS 60479-2 for other frequencies and impulses) is the reference used by every installation standard; its values, summarised in Table 29.1, come from measurements on cadavers and volunteers and from animal experiments, and are statistical.

Table 29.1 — Effects of 50/60 Hz current on the body, left hand to feet (approximate, after IEC 60479-1).

Current	Effect
0.5 mA	Threshold of perception (line a)
5–10 mA	Threshold of let-go: above this the hand cannot release a gripped conductor
20–40 mA for seconds	Respiratory paralysis possible, strong muscular contraction, pain
40–50 mA for over 1 s	Ventricular fibrillation becomes possible (curve c1)
500 mA for 10 ms	Fibrillation possible even for very short exposure
Several amperes	Cardiac arrest, deep burns at entry and exit, tissue damage along the path

Fibrillation, the uncoordinated quivering of the heart's ventricles that stops blood circulation, is the main cause of death from low-voltage shock. It is most likely when the shock overlaps the vulnerable period of the cardiac cycle (the T-wave), which is why the threshold falls sharply when the shock lasts longer than one heartbeat (about 0.75–1 s). Figure 29.1 shows the time–current zones.

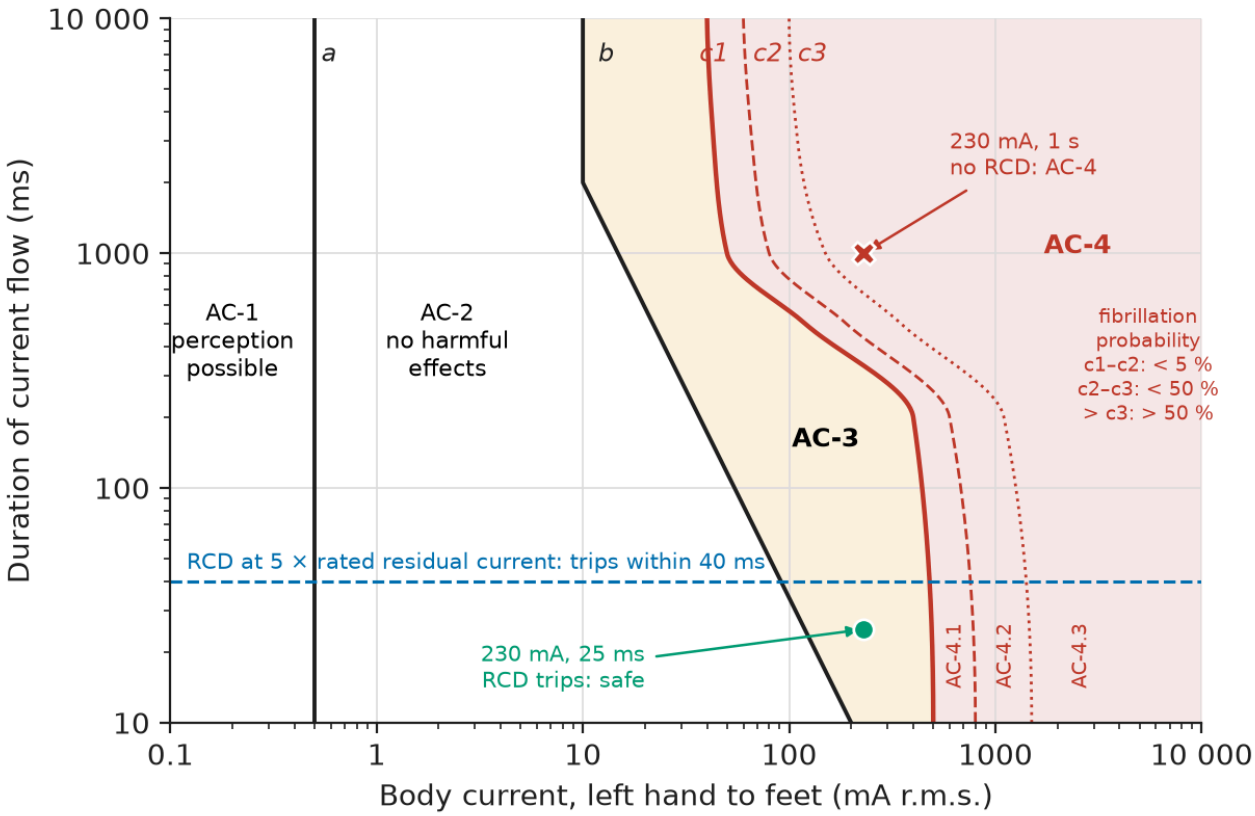


Figure 29.1 — Time-current zones for 50/60 Hz current, left hand to feet, after IEC 60479-1.

> **Status of the principle: IEC 60479-1 time–current zones**
> *Empirical.* The boundaries are statistical fits to animal experiments and limited human data for adults in normal health; they apply to 15–100 Hz sinusoidal current along the stated path and are scaled for other paths by heart-current factors.

Body impedance and current path

The body's impedance is a skin impedance at entry and exit (resistive and capacitive, falling sharply once the skin breaks down above about 50–100 V) in series with an internal impedance of roughly 500 Ω hand to hand. At 230 V, for a large dry contact area, the total hand-to-hand impedance is about 1000 Ω for the 5 % of the population with the lowest values, 1350 Ω for the median, and 2100 Ω for the 95th percentile; at 50 V the median is about 2500 Ω, and in wet conditions the values fall by 25–50 %. Shoes, floor and gloves add resistance in series, which is why dry, insulating floors and footwear matter.

The path matters through the heart-current factor F: the same current is about as dangerous left hand to feet ($F = 1.0$), somewhat less right hand to feet ($F = 0.8$), much less hand to hand ($F = 0.4$), and more chest to left hand ($F = 1.5$). The equivalent left-hand-to-feet current is F times the actual current.

Touch voltage limits

Installation standards turn these data into a **conventional touch-voltage limit** U_L: 50 V AC or 120 V ripple-free DC in normal dry conditions (IEC 60364-4-41, IEC 61140), reduced to 25 V AC in agricultural buildings, medical locations and other places where the body impedance is lower. A touch voltage at or below U_L may persist indefinitely. Above it, the protective device must disconnect within a time that keeps the body

current outside the dangerous zone: 0.4 s for 230 V final circuits in TN systems and 0.2 s in TT systems (IEC 60364-4-41, Table 41.1), and 5 s (TN) or 1 s (TT) for distribution circuits, where people are less likely to be in contact.

DC and other frequencies

For DC the thresholds are higher: perception at about 2 mA, no clear let-go threshold (the muscle reacts at make and break rather than continuously), and fibrillation thresholds two to four times the AC values for long exposures. This is why 120 V DC is treated like 50 V AC. With 400–1500 V DC now common in PV strings, battery storage, EV charging and traction, DC shock is a mainstream hazard, not a niche one. At higher frequencies (400 Hz in aircraft, kilohertz in induction heating) the fibrillation threshold rises but burns become more likely, and at radio frequencies current flows near the skin surface (IEC TS 60479-2).

Worked example — Why the MCB cannot protect a person. A person touches a 230 V line conductor with the left hand while standing barefoot on a wet floor, so that the only significant impedance is the body's, taken as 1000 Ω (5th percentile, conservatively). The circuit is protected by a 16 A type B MCB and, in a second case, also by a 30 mA RCD.

Body current: $230/1000 = 230$ mA. The MCB's instantaneous trip begins at $5 \times 16 = 80$ A; at 230 mA it will never operate, since the person draws less current than a light bulb. From Figure 29.1, 230 mA lies in zone AC-3 for durations below about 0.3 s and in AC-4 beyond, so a sustained contact carries a high risk of fibrillation. The RCD sees a residual current of 230 mA, more than five times its 30 mA rating, and must trip within 40 ms (IEC 61008-1/61009-1); typical devices take 15–30 ms. At 230 mA for 40 ms the point is well left of c1. If the person wore dry shoes on dry concrete adding about 2000 Ω, the current would be $230/3000 = 77$ mA: still above the let-go threshold and dangerous, still above $2 \times 30 = 60$ mA, so the RCD would trip within its 150 ms limit at twice rated current.

The conclusion is general: overcurrent devices protect cables and equipment; only residual-current devices, extra-low voltage, insulation and isolation protect people from direct contact.

29.2 Arc-flash hazards

What an arc releases

When a fault current flows through air instead of metal — a tool dropped across busbars, a racking failure, an insulation breakdown on contaminated surfaces, a rodent — the arc column reaches 10 000–20 000 K, vaporises copper (which expands about 67 000 times in volume), and radiates intense heat and ultraviolet light. The effects are burns from radiation and from molten metal, a pressure wave that can throw a worker and rupture eardrums, shrapnel, and toxic copper oxide. Burns, not shock, are the dominant arc injury. The quantity used to rate the hazard is the **incident energy** E received on a surface at the worker's working distance, in J/cm² or cal/cm² (1 cal/cm² = 4.184 J/cm²). The onset of a just-curable second-degree burn on bare skin is taken as 1.2 cal/cm² (5.0 J/cm²), the Stoll-curve value used by NFPA 70E and IEEE 1584.

Incident energy grows with three quantities the engineer controls:

- the **arcing current** — at low voltage typically 40–70 % of the bolted fault current because of the arc's own voltage, at medium voltage closer to 90 %;
- the **arcing time**, set almost entirely by the protective device that clears the fault;

- the **distance** from the arc, which enters roughly as $1/D^2$ in open air and as $1/D^{1.5}$–$1/D^2$ in a box that focuses the energy outward.

Estimating incident energy

The theoretical upper bound is due to Ralph Lee (1982), who assumed the arc delivers the maximum possible power (half the bolted fault MVA) radiating spherically. In its IEEE 1584 form,

$$E = 2.142 \times 10^{6}\, V\, I_{bf}\, \frac{t}{D^{2}} \quad \mathrm{J/cm^2}$$

with V the system voltage in kV, I_{bf} the bolted three-phase fault current in kA, t the arcing time in s and D the distance in mm. It is conservative at low voltage and is now used mainly for open-air arcs above 15 kV. For 208 V–15 kV equipment the standard method is IEEE 1584-2018, whose empirical equations were fitted to some 1800 laboratory tests. They take as inputs the voltage, bolted fault current, electrode gap, enclosure size and one of five electrode configurations (vertical electrodes in a box, with or without a barrier below; horizontal electrodes in a box; and the two open-air cases), and compute first the arcing current, then the incident energy at the working distance and the **arc-flash boundary**: the distance at which E falls to 1.2 cal/cm². Because a lower arcing current may delay the protective device, the method also repeats the calculation with a reduced arcing current and keeps the worse result. Standard working distances are 455 mm for low-voltage panelboards and MCCs, 610 mm for low-voltage switchgear and 910 mm for medium-voltage switchgear.

> **Status of the principle: Arc-flash incident-energy equations**
> *Empirical.* IEEE 1584-2018 equations are regressions on tests within stated ranges (208 V–15 kV, three-phase, 50/60 Hz, bolted fault 0.5–106 kA at low voltage, specified gaps and enclosures); the Lee equation is a theoretical upper bound. Neither covers single-phase arcs, DC arcs or the blast pressure.

Worked example — Incident energy at a main switchboard. A 1600 kVA, 11/0.4 kV transformer with 6 % impedance, supplied from an 11 kV network of 250 MVA fault level, feeds a main switchboard. The worker stands 455 mm from the busbars in an MCC-style section. The upstream breaker clears faults in the board with a short-time delay giving an arcing time of 0.5 s. Estimate the incident energy by the Lee method and the effect of reducing the clearing time.

Source impedance on the transformer base: $1.6/250 = 0.0064$ pu. Bolted fault current:

$$I_{bf} = \frac{1600}{\sqrt{3} \times 0.4 \times (0.06 + 0.0064)} = 34.8 \ \mathrm{kA}$$

At $t = 0.5$ s:

$$E = 2.142 \times 10^{6} \times 0.4 \times 34.8 \times \frac{0.5}{455^{2}} = 72.0 \ \mathrm{J/cm^2} = 17.2 \ \mathrm{cal/cm^2}$$

and the arc-flash boundary, where $E = 5.0$ J/cm², is $D_B = \sqrt{2.142 \times 10^{6} \times 0.4 \times 34.8 \times 0.5/5.0} = 1726$ mm. With the short-time delay reduced so that $t = 0.2$ s: 28.8 J/cm² (6.9 cal/cm²), boundary 1.09 m. With an arc-flash relay (light plus overcurrent) tripping the breaker in a total of 0.05 s: 7.2 J/cm² (1.7 cal/cm²), boundary 0.55 m. Energy is proportional to clearing time, so a tenfold faster clearance takes the task from the PPE category 3 range to category 1, and this — not heavier clothing — is the engineering answer.

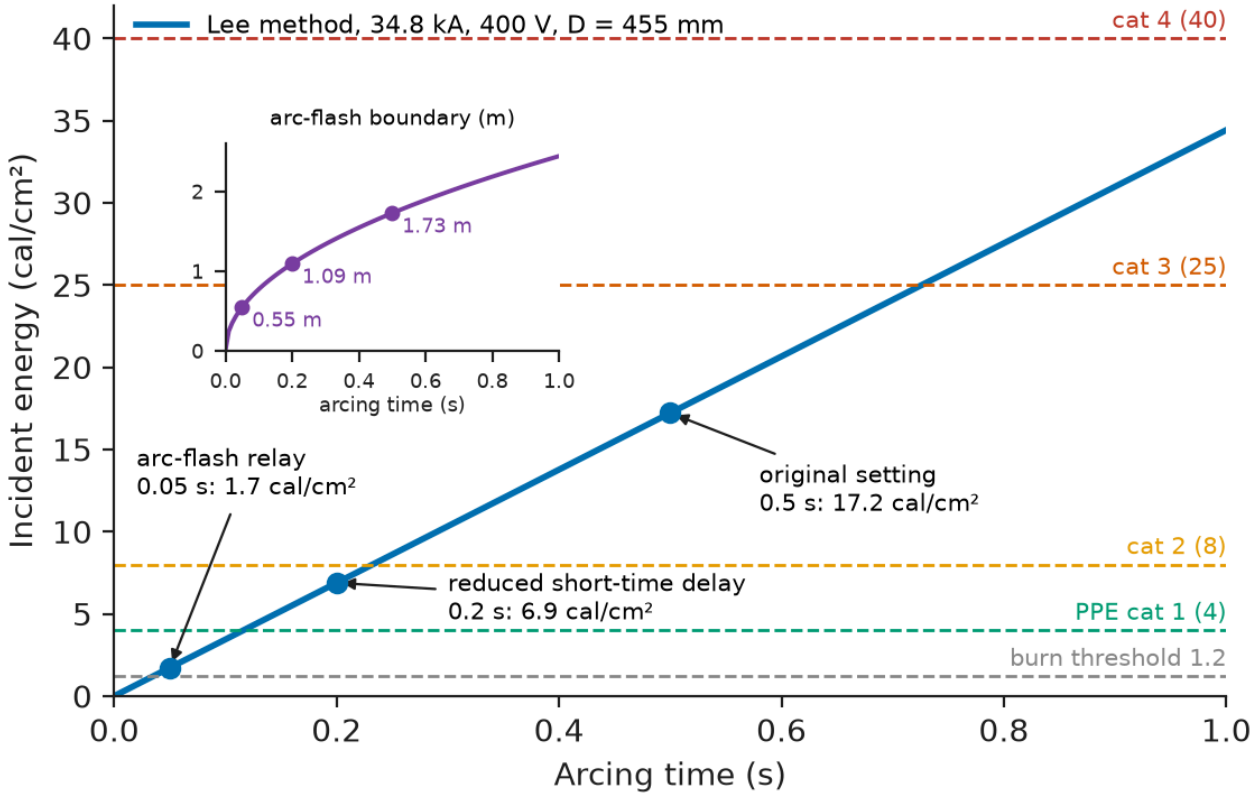


Figure 29.2 — Incident energy at 455 mm for a 34.8 kA, 400 V bolted fault against arcing time, with PPE category thresholds.

Reducing the hazard

In order of effectiveness:

- **Do not work live.** De-energising (Section 29.6) removes the arc-flash hazard except during the test for absence of voltage.
- **Clear faster.** Arc-flash relays with optical sensors (trip in 2–5 ms, total clearing 35–60 ms with the breaker), zone-selective interlocking between breakers, energy-reducing maintenance switches that temporarily set the instantaneous trip low while someone works (required by NEC 240.87 for breakers of 1200 A and above), and current-limiting fuses, which clear high faults within a half-cycle.
- **Reduce the fault current** with transformer impedance or by splitting buses, recognising that this conflicts with voltage regulation and motor starting (Chapter 30) and that a lower current may lengthen the clearing time.
- **Increase the distance.** Remote racking devices, remote switching, and infrared windows that let thermography be done with doors closed.
- **Contain the arc.** Arc-resistant switchgear tested to IEC 62271-200 (internal arc classification) or IEEE C37.20.7 vents the arc products away from the operator when doors are closed. It offers no protection with the doors open.

DC arcs in battery rooms, PV combiner boxes and traction systems are harder to extinguish, because DC has no current zero, and are not covered by IEEE 1584; NFPA 70E gives simplified methods, and the field is still developing.

29.3 Grounding and bonding

Two different jobs

Earthing (grounding) connects parts of the installation to the general mass of the earth through electrodes. **Bonding** connects conductive parts to each other so that they stay at nearly the same potential whatever the earth does. Most electrical shock protection depends on bonding more than on earthing: a person touching a faulty appliance and a water pipe is safe if the two are at the same potential, whatever their potential relative to remote earth. A building's **main equipotential bonding** joins the protective conductors, the main earthing terminal, metallic water and gas pipes, structural steel and lightning-protection down-conductors at the point of entry; **supplementary bonding** does the same locally in bathrooms, swimming pools, agricultural buildings and medical locations.

Earthing systems

IEC 60364-1 classifies low-voltage systems with two letters (Figure 29.3). The first describes the source: T, one point (usually the neutral) directly earthed; I, all live parts isolated from earth or earthed through a high impedance. The second describes the exposed conductive parts of the installation: T, earthed locally and independently of the source earth; N, connected to the source's earthed point by a protective conductor. TN systems are subdivided by whether the neutral and protective functions share a conductor: TN-S (separate throughout), TN-C (combined PEN conductor) and TN-C-S (combined in the supply network, separate inside the installation — the common arrangement known as protective multiple earthing in the UK).

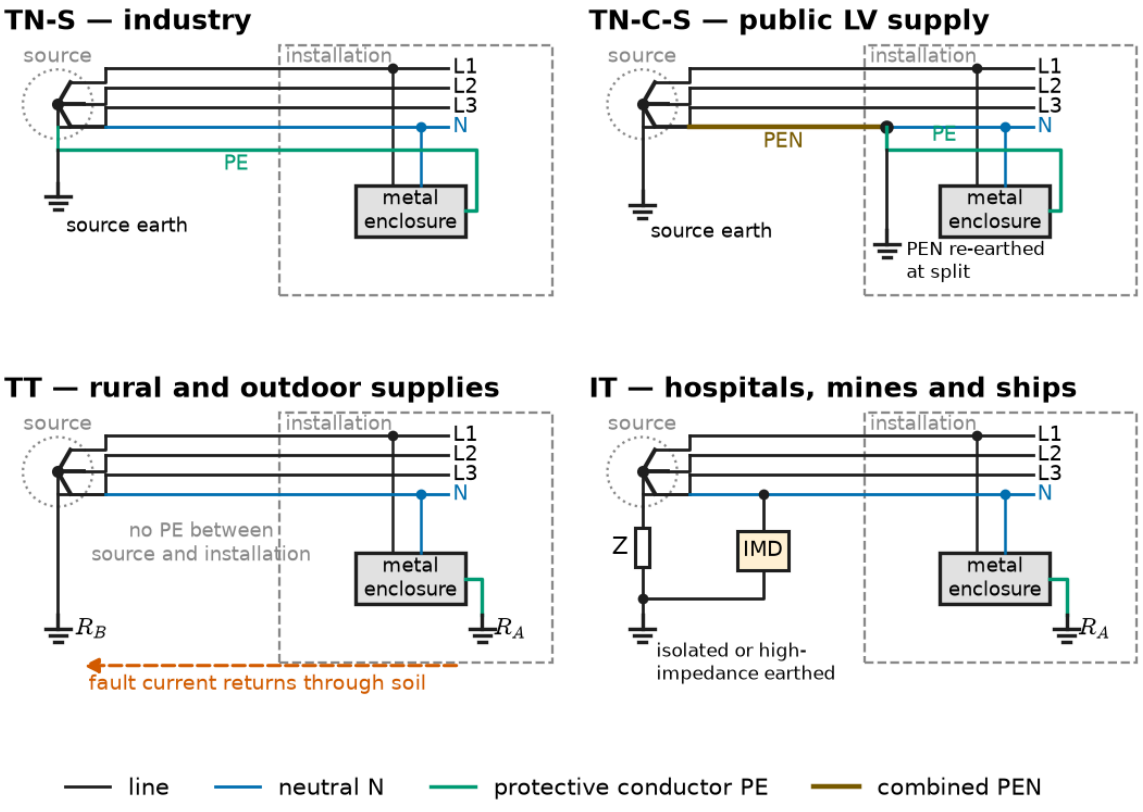


Figure 29.3 — The four low-voltage earthing systems: TN-S, TN-C-S, TT and IT.

Table 29.2 — Low-voltage earthing systems compared.

System	First earth fault	Fault current	Protection	Typical use
TN-S	Line-to-PE short circuit	Hundreds of A to kA	Overcurrent device or RCD	Industrial plants, data centres, buildings with own substation
TN-C-S	As TN-S inside installation	Hundreds of A to kA	Overcurrent device or RCD; risk if PEN breaks	Public LV supply in UK, Europe, Australia
TT	Current returns through soil	A few A to tens of A	RCD essential	Rural supplies, outdoor installations, EV chargers on some networks
IT	Small capacitive current only; operation continues	mA to a few A	Insulation monitor alarms; second fault must trip	Hospitals (operating theatres), mines, ships, continuous processes

The TN-C and TN-C-S systems carry one particular hazard: if the PEN conductor breaks upstream, every exposed conductive part bonded to the PE downstream rises toward line voltage through the load. Multiple re-earthing of the PEN by the utility limits this; for EV chargers outdoors, where a person stands on true earth beside a car bonded to the PE, IEC 60364-7-722 requires extra measures such as PEN-loss detection or a TT arrangement.

Automatic disconnection in TN systems

In a TN system an insulation fault from line to an exposed part is a short circuit through the line conductor and the protective conductor. The fault current must be large enough to operate the protective device within the required time:

$$Z_s \, I_a \leq U_0$$

where Z_s is the earth-fault loop impedance (source, line conductor and protective conductor), I_a the current that operates the device within the required disconnection time, and U_0 the nominal line-to-earth voltage. In practice a factor $C_{min} = 0.95$ is applied to U_0 for the lowest supply voltage, and conductor resistances are taken at their operating temperature.

> **Status of the principle: Automatic disconnection of supply (TN)**
> *Approximate*. The condition treats the fault as a bolted connection with negligible fault resistance and neglects reactance in small conductors; an arcing or high-resistance fault produces less current and may not be cleared in time, which is one reason RCDs are added.

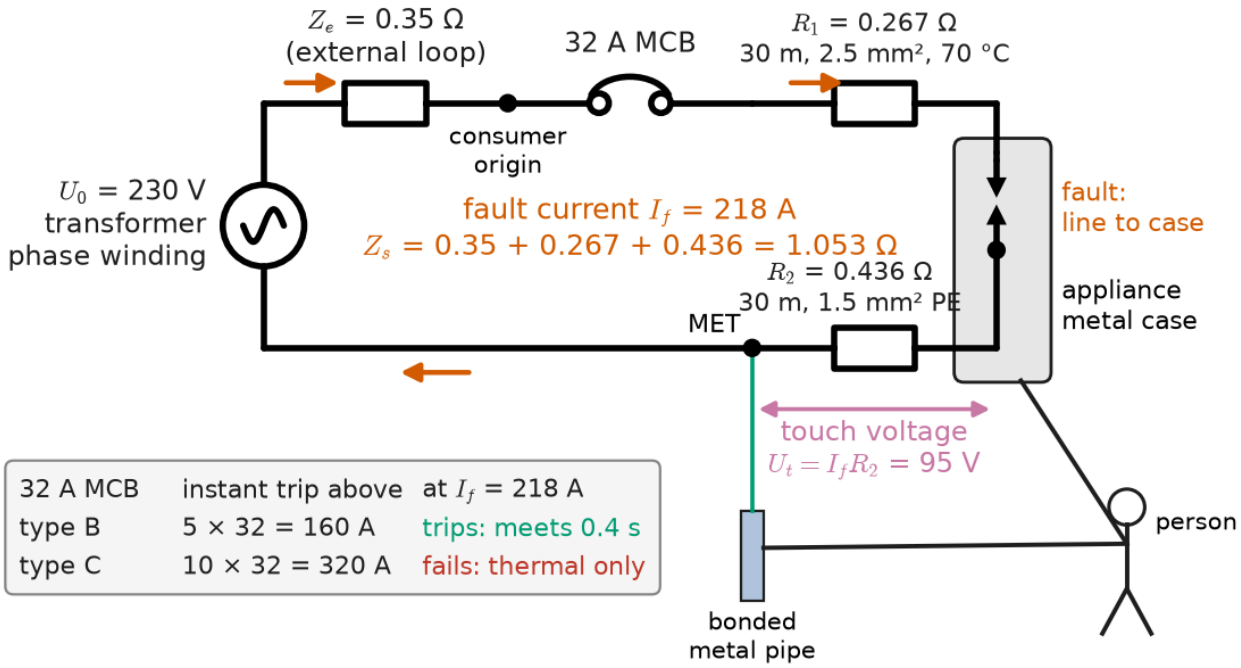


Figure 29.4 — Earth-fault loop in a TN-S final circuit and the resulting touch voltage.

Worked example — Touch voltage and disconnection in a TN-S circuit. A 230 V final circuit in a TN-S installation is 30 m long, with a 2.5 mm² copper line conductor (7.41 mΩ/m at 20 °C) and a 1.5 mm² protective conductor (12.10 mΩ/m). The external loop impedance Z_e is 0.35 Ω. Resistances are multiplied by 1.20 for the 70 °C operating temperature. Check the circuit with a 32 A type B MCB and with a 32 A type C MCB (Figure 29.4).

$R_1 = 7.41 \times 30 \times 1.20 = 267$ mΩ, $R_2 = 12.10 \times 30 \times 1.20 = 436$ mΩ, so

$$Z_s = 0.35 + 0.267 + 0.436 = 1.053\ \Omega, \qquad I_f = \frac{230}{1.053} = 218\ \text{A}$$

The type B MCB operates instantaneously (within 0.1 s) above $5 \times 32 = 160$ A, so the condition is met; the maximum allowed loop impedance is $0.95 \times 230/160 = 1.37\ \Omega$. The type C MCB needs $10 \times 32 = 320$ A, permitting at most 0.68 Ω: the circuit fails, because at 218 A the type C device would rely on its thermal element and could take several seconds. While the fault persists, the case rises to $U_t = I_f R_2 = 218 \times 0.436 = 95$ V above the main earthing terminal. For 0.1 s that is acceptable by Figure 29.1; for several seconds it is not. The remedies are a type B device, a larger protective conductor, a shorter run, or an RCD.

Automatic disconnection in TT systems

In a TT system the fault current returns through the consumer's electrode R_A and the source electrode R_B, and is usually too small to trip an overcurrent device. An RCD is used instead, and the condition is that the touch voltage at the operating current stays within U_L:

$$R_A\, I_{\Delta n} \le 50\ \text{V}$$

Worked example — Why a TT installation needs an RCD. A farm workshop is supplied TT with $R_A = 50\ \Omega$ and $R_B = 10\ \Omega$. An insulation fault occurs in a 230 V machine protected by a 20 A MCB.

Fault current: $230/(50+10) = 3.83$ A — a fifth of the MCB's rating, so it never trips. The machine frame rises to $3.83 \times 50 = 192$ V and stays there. With a 30 mA RCD the condition requires $R_A \le 50/0.03 = 1667\ \Omega$ (easily met), and with a 300 mA time-delayed RCD at the origin, $R_A \le 167\ \Omega$. The RCD trips at a few milliamperes of the 3.83 A fault, within 0.2 s.

Substation earthing: step and touch voltages

At high-voltage substations the earth-fault current is kiloamperes and returns partly through the soil, so the ground surface around the grid rises in potential. A person touching a structure (**touch voltage**) or straddling a gradient with the feet 1 m apart (**step voltage**) is exposed. IEEE 80 sets tolerable limits from Dalziel's fibrillation criterion $I_B = 0.116/\sqrt{t_s}$ A for a 50 kg person (valid for 0.03–3 s), giving for touch and step respectively

$$E_{touch,50} = (1000 + 1.5\,C_s\rho_s)\frac{0.116}{\sqrt{t_s}}, \qquad E_{step,50} = (1000 + 6\,C_s\rho_s)\frac{0.116}{\sqrt{t_s}}$$

where 1000 Ω is the assumed body resistance, ρ_s the surface-layer resistivity and C_s a derating factor for a thin surface layer. The grid design itself, its resistance and the ground-potential rise belong to Chapter 24.

Worked example — Value of a crushed-rock surface. A 132 kV substation clears earth faults in 0.5 s. The native soil is 100 Ω·m. Find the tolerable touch voltage with a 0.1 m layer of 3000 Ω·m crushed rock ($C_s = 1 - 0.09(1-\rho/\rho_s)/(2h_s+0.09) = 0.70$) and without it.

With rock: $(1000 + 1.5 \times 0.70 \times 3000) \times 0.116/\sqrt{0.5} = 681$ V; tolerable step voltage 2231 V. Without rock ($\rho_s = 100$ Ω·m, $C_s = 1$): $(1000 + 150) \times 0.164 = 189$ V. The 10 cm of gravel raises the allowed touch voltage 3.6 times, which is why every outdoor substation has it.

> **Status of the principle: IEEE 80 tolerable touch and step voltages**
> *Empirical.* Based on Dalziel's fibrillation data for a 50 kg (or 70 kg) person, a 1000 Ω body and a 0.03–3 s shock; it is a design criterion with deliberate conservatism, not a prediction for an individual.

29.4 Protective devices

The family of devices

Each protective device answers a different question about the current in a circuit. Table 29.3 summarises them.

Table 29.3 — Low-voltage protective devices (typical characteristics).

Device	Senses	Protects against	Key ratings	Standard
Miniature circuit breaker (MCB)	Overcurrent (thermal and magnetic)	Overload, short circuit, earth fault in TN	In 0.5–125 A; type B 3–5 In, C 5–10 In, D 10–20 In; Icn 6–25 kA	IEC 60898-1
Moulded-case / air circuit breaker	Overcurrent, often earth fault, electronic trip	Feeders, mains, motors	In 16–6300 A; Icu up to 150 kA; Icw for air breakers	IEC 60947-2
Fuse (gG, aM, gR)	Overcurrent (melting)	Short circuit with current limitation; gG also overload	In up to 1250 A; breaking 50–120 kA	IEC 60269
Residual-current device (RCD)	Residual (earth-leakage) current	Shock, earth-fault fire	IΔn 10, 30, 100, 300, 500 mA; types AC, A, F, B	IEC 61008, 61009, 62423
Arc-fault detection device (AFDD)	Arc signature in current waveform	Series and parallel arcing faults, fire	Final circuits up to 63 A	IEC 62606
Insulation monitoring device (IMD)	Insulation resistance to earth in IT systems	First fault (alarm)	Response value typically 50–100 Ω/V	IEC 61557-8
Surge protective device (SPD)	Overvoltage	Lightning and switching surges	Type 1, 2, 3; Up, In, Iimp	IEC 61643-11
Motor protection relay	Current, thermal image, unbalance	Motor overload, stall, phase loss	Trip classes 10, 20, 30	IEC 60947-4-1

RCD types and DC leakage

An RCD of type AC detects only sinusoidal residual current. Type A also detects pulsating DC (from single-phase rectifiers); type F adds mixed frequencies from single-phase drives; type B detects smooth DC as well, which appears with three-phase rectifiers in drives, EV chargers and PV inverters. A smooth DC residual current of only a few milliamperes can saturate the core of a type A device and blind it. EV chargers therefore need either a type B RCD or a type A RCD together with a 6 mA DC residual-current detector (IEC 62955) in the charger, and the RCD selection for PV inverters depends on whether the inverter itself guarantees no DC leakage.

RCDs at 30 mA are required as additional protection for socket-outlets up to 32 A, mobile equipment outdoors, bathrooms and many special locations (IEC 60364-4-41 and Part 7). Time-delayed (type S) RCDs of 100–500 mA upstream provide fire protection and selectivity with the 30 mA devices below. Too many circuits on one RCD cause nuisance tripping from the sum of normal leakage currents (filters in IT equipment leak 0.5–3.5 mA each); good design keeps the standing leakage below about 30 % of $I_{\Delta n}$.

Insulation monitoring in IT systems

An IT system tolerates a first earth fault without tripping, because the only fault-current path is the system's capacitance to earth. The insulation monitoring device injects a measuring signal between the system and earth and alarms when the insulation resistance falls below a set value, so the fault can be found (with insulation-fault location systems) and repaired before a second fault on another phase creates a line-to-line

short circuit through two earthed frames. This is why operating theatres (medical IT, IEC 60364-7-710), ships and mines use IT: continuity of supply through the first fault.

29.5 Overcurrent protection

Two duties

Overcurrent protection has two separate duties, which IEC 60364-4-43 and NEC Article 240 treat separately. **Overload protection** prevents a cable from running persistently above its maximum insulation temperature (70 °C for PVC, 90 °C for XLPE), which halves its life for roughly every 8–10 K of excess (Chapter 9). **Short-circuit protection** interrupts a fault before the conductor reaches its short-time limit (160 °C for PVC, 250 °C for XLPE) and before the electromagnetic forces damage equipment.

Overload coordination

With I_B the design current of the circuit, I_n the rated (or set) current of the device, I_Z the current-carrying capacity of the cable under its installation conditions, and I_2 the current that ensures operation of the device within its conventional time (1 or 2 h):

$$I_B \le I_n \le I_Z, \qquad I_2 \le 1.45\, I_Z$$

For MCBs to IEC 60898-1, $I_2 = 1.45 I_n$, so the second condition is automatically met when $I_n \le I_Z$. For gG fuses $I_2 = 1.6 I_n$, so I_n must not exceed $1.45/1.6 = 0.906$ of I_Z. A 40 A MCB on a cable rated 41 A gives $I_2 = 58$ A against a limit of $1.45 \times 41 = 59.5$ A and is acceptable; a 40 A gG fuse on the same cable would require $I_Z \ge 44.1$ A.

> **Status of the principle: Overload coordination**
> *Empirical.* The factor 1.45 reflects test experience that a cable can carry 45 % overload for the conventional time without serious damage; it presumes infrequent overloads and does not protect against long-term moderate overloads below I_2.

Short-circuit withstand: the adiabatic equation

During a short circuit lasting less than about 5 s, the heat cannot leave the conductor and all goes into raising its temperature. The conductor survives if

$$I^2 t \le k^2 S^2 \qquad \Longleftrightarrow \qquad S \ge \frac{\sqrt{I^2 t}}{k}$$

with S the cross-section in mm², $I^2 t$ the energy let through by the protective device in A²s, and k a constant combining the conductor's resistivity, heat capacity and the initial and final temperatures (IEC 60364-4-43 and -5-54). Representative values: copper with PVC insulation 115, copper with XLPE 143, aluminium with PVC 76, aluminium with XLPE 94, and 143 for a separate PVC-insulated copper protective conductor starting at 30 °C. The underlying adiabatic heating is the heat-capacity limit of a thermal RC network whose time constant is long compared with the fault (*Engineering Physics*, Ch. 14 §14.7.3).

> **Status of the principle: Adiabatic short-circuit withstand**
> *Approximate.* Assumes no heat loss to the insulation or surroundings during the fault and constant material properties; it is conservative for faults over about 0.1 s in small conductors and unsafe for durations over 5 s, where heat loss and different limits apply.

Worked example — Is a 16 mm² sub-main protected? A 16 mm² copper/PVC sub-main ($k = 115$) can see a prospective fault of 5 kA at its origin. Case (a): it is protected by an upstream breaker with a 0.2 s short-time delay. Case (b): a current-limiting MCCB whose let-through I^2t at 5 kA is 0.3 MA²s.

(a) $I^2t = 5000^2 \times 0.2 = 5 \times 10^6$ A²s, so $S \geq \sqrt{5 \times 10^6}/115 = 19.4$ mm²: the 16 mm² cable would be overheated and its insulation damaged. (b) $S \geq \sqrt{0.3 \times 10^6}/115 = 4.8$ mm²: well protected. The same cable is safe or unsafe depending only on the device in front of it. For the final circuit of Section 29.3, $I_f = 218$ A cleared in 0.1 s requires $S \geq \sqrt{218^2 \times 0.1}/115 = 0.60$ mm² for the 1.5 mm² protective conductor, which passes.

Selectivity and back-up

Overcurrent devices in series must be selective: the device nearest the fault should operate and those upstream should not (Chapter 23 for relays; Chapter 30 for low-voltage breakers and fuses). Where a downstream breaker's breaking capacity is below the prospective fault current, an upstream current-limiting device can provide **back-up** (cascading) protection, verified only by tested manufacturer combinations. Back-up and selectivity often conflict — cascading relies on both devices opening — and the designer must decide which matters more for each circuit, a question of safety and continuity together.

29.6 Isolation and lockout

The five safety rules

Most work on electrical equipment should be done dead. EN 50110-1, which governs the operation of electrical installations in Europe, sets out the five safety rules, to be applied in this order:

1. **Disconnect completely** — open every source: main supply, generator, UPS, PV, battery, back-feed through control or auxiliary circuits, interconnections.
2. **Secure against reconnection** — lock the isolating devices open, attach a personal lock and tag for each worker, remove fuses, block motorised operators.
3. **Verify the absence of operating voltage** — on every conductor, with a tested detector of the right category and voltage range, proven on a known live source immediately before and after.
4. **Carry out earthing and short-circuiting** — mandatory at high voltage and on long LV cables or overhead lines, so that induced voltages, capacitive charge or an accidental re-energisation is diverted through a short circuit that trips the protection.
5. **Provide protection against adjacent live parts** — barriers, insulating covers, defined work zones.

The same logic is expressed in North America as establishing an **electrically safe work condition** (NFPA 70E Article 120) and as lockout/tagout under OSHA 29 CFR 1910.147 and 1910.333. Until step 3 is complete, the equipment must be treated as live, with the shock and arc-flash precautions of Section 29.9. Figure 29.5 summarises the sequence.

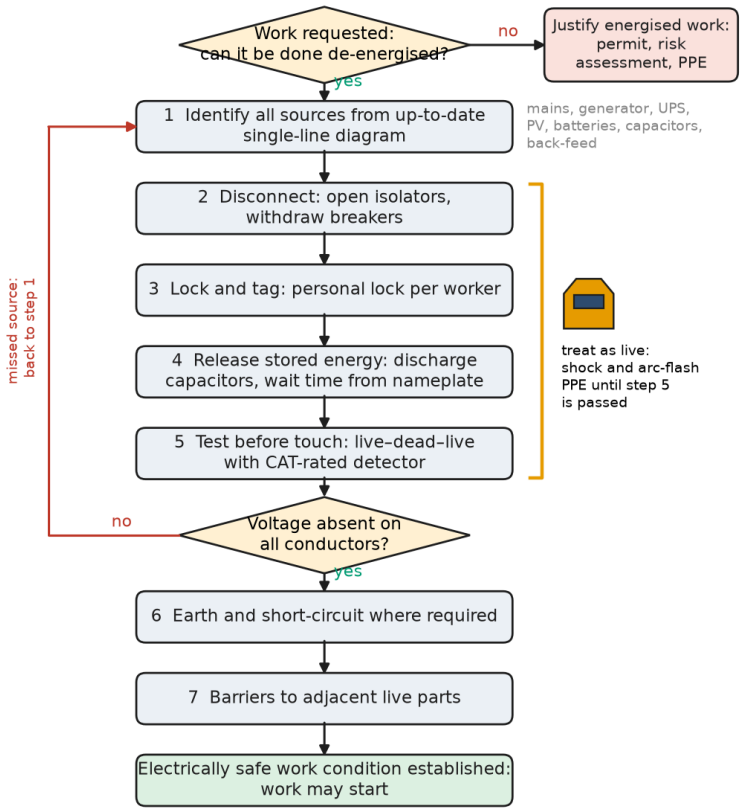


Figure 29.5 — Establishing an electrically safe work condition: the five safety rules with decision points.

Stored and back-fed energy

Opening a switch does not remove every source of energy. The most common surprises are:

- **Capacitors** in drives, UPS DC links, power-factor-correction banks and cable capacitance after insulation testing. Bleeder resistors are required to bring a PFC capacitor below 75 V within 3 minutes (IEC 60831-1), and drive manufacturers specify waiting times of 5–15 minutes.
- **PV arrays**, which generate whenever light falls on them; a string cannot be switched off at the module, which is why module-level rapid shutdown (NEC 690.12) exists.
- **Batteries**, which cannot be isolated internally; their terminals are always live and capable of tens of kiloamperes.
- **Back-feed** from standby generators, neighbouring feeders through open points, or a UPS output feeding back into a "dead" distribution board through a bypass.
- **Induced voltages** on de-energised overhead lines and cables running parallel to live circuits, which can reach hundreds or thousands of volts — the reason for working earths.

Worked example — How long to wait before opening a drive. A 690 V drive has a DC link of 975 V and a total capacitance of 4.7 mF, discharged by bleeder resistors totalling 47 kΩ. How long after isolation does the DC-link voltage fall below 50 V?

Time constant $\tau = RC = 47\,000 \times 0.0047 = 221$ s. The voltage falls as $V(t) = 975\,e^{-t/\tau}$, so $t = \tau \ln(975/50) = 221 \times 2.97 = 656$ s, about 11 minutes. The stored energy at the start, $\tfrac{1}{2}CV^2 = 2.23$ kJ, is enough to cause serious burns and a lethal shock. A drive with a failed bleeder never discharges, which is why rule 3 — test, do not assume — applies to the DC link too.

29.7 Safe testing

Measurement categories

A meter used on a power system must survive not only the system voltage but the transient overvoltages that appear on it. IEC 61010-1 assigns measurement categories by location (Table 29.4): CAT II for circuits fed from socket-outlets, CAT III for the fixed installation (distribution boards, busbars, permanently connected motors), CAT IV for the origin of the installation (service entrance, meters, overhead lines). A CAT IV 600 V meter withstands an 8 kV impulse; a cheap CAT II meter on a main switchboard may fail explosively during a switching surge, starting an arc in the user's hand.

Table 29.4 — Rated impulse withstand of test instruments by measurement category (IEC 61010-1).

Line-to-earth voltage	CAT II	CAT III	CAT IV
150 V	1500 V	2500 V	4000 V
300 V	2500 V	4000 V	6000 V
600 V	4000 V	6000 V	8000 V
1000 V	6000 V	8000 V	12 000 V

Leads and probes must carry the same category, with fused leads, shrouded connectors and probe tips with no more than 4 mm of exposed metal. For the test of absence of voltage, purpose-built two-pole voltage detectors (IEC 61243-3) are preferred over multimeters, which can be left in the wrong range or have a blown fuse. The test itself is **live–dead–live**: prove the detector on a known source, test every conductor combination (line–line, line–neutral, line–earth), prove the detector again.

Verification tests of an installation

Before an installation is energised, and periodically afterwards, IEC 60364-6 (and NEC/NFPA 70B for maintenance) requires inspection and tests in a fixed sequence, dead tests before live:

1. Continuity of protective conductors and bonding (low-resistance ohmmeter, 200 mA or more).
2. Insulation resistance between live conductors and between live conductors and earth, at 250 V DC for SELV/PELV circuits (limit 0.5 MΩ), 500 V for circuits up to 500 V (limit 1 MΩ) and 1000 V above (limit 1 MΩ). Surge protective devices and electronic equipment are disconnected first, and long cables are discharged afterwards.
3. Polarity and phase sequence.
4. Earth-electrode resistance (TT systems).
5. Earth-fault loop impedance, measured live, compared with the maximum Z_s of Section 29.3 corrected to operating temperature.
6. Prospective fault current at each board, compared with the devices' breaking capacities.
7. RCD operation at $I_{\Delta n}$ and $5I_{\Delta n}$, checking trip times against IEC 61008/61009.
8. Functional tests of switchgear, interlocks and emergency stops.

For motors and generators, polarisation index (ratio of 10-minute to 1-minute insulation resistance, at least 2 for class F and H insulation per IEEE 43) and, for high-voltage plant, the tests of Chapter 24. High-voltage testing — AC withstand, VLF and partial-discharge tests on cables — creates its own hazard zone and requires barriers, earthing sticks and a written test procedure.

29.8 Electrical installation safety

Design rules that build safety in

The installation standards — IEC 60364 (and national versions such as BS 7671), NFPA 70 (the National Electrical Code) in North America, AS/NZS 3000 in Australia — are collections of engineering solutions to recurring hazards. Their core structure is common: protection against electric shock (basic protection by insulation, barriers and enclosures; fault protection by automatic disconnection, double insulation, electrical separation or extra-low voltage), protection against thermal effects and fire, protection against overcurrent, protection against overvoltage, and selection and erection rules for equipment. Part 7 of IEC 60364 adds rules for special locations: bathrooms, swimming pools, construction sites, agricultural premises, medical locations, marinas, PV systems (712), EV charging (722), data-processing installations and others.

Enclosures and environments

Enclosures are rated for ingress protection by IEC 60529: the first digit for solids (2 = finger-safe, 5 = dust-protected, 6 = dust-tight) and the second for water (4 = splashing, 5 = jets, 7 = temporary immersion). Switchboards are tested as assemblies under IEC 61439, including temperature rise, short-circuit withstand and internal separation forms (Form 1 to Form 4b) that limit how much of a board is exposed when one compartment is open.

In **hazardous areas** where flammable gas or dust may be present — refineries, chemical plants, mines, grain handling, battery rooms, hydrogen installations — IEC 60079 classifies zones by the likelihood of an explosive atmosphere (Zone 0, 1, 2 for gas; 20, 21, 22 for dust) and requires equipment with a matching protection concept: flameproof enclosures (Ex d), increased safety (Ex e), intrinsic safety (Ex i), pressurisation (Ex p). Underground coal mines add firedamp requirements (Group I equipment). North America uses the Class/Division system of NEC Articles 500–506, with zones as an alternative.

Sector practice

- **Data centres**: high fault levels at 400–480 V with many parallel sources (utility, generators, UPS), making arc-flash the dominant hazard; remote racking and arc-flash relays are common.
- **Renewable plants**: PV DC circuits at 1000–1500 V that cannot be de-energised in daylight, arc-fault detection on DC (UL 1699B), and wind-turbine towers with medium voltage and confined spaces.
- **Traction and EV charging**: DC traction at 750 V–3 kV with return-current and stray-current corrosion issues (EN 50122); DC fast chargers up to 1000 V with type B residual-current protection and insulation monitoring on the vehicle side.
- **Marine**: IT systems with insulation monitoring, IEC 60092 and classification-society rules, and medium voltage (6.6–11 kV) on large ships with shore connection to IEC/IEEE 80005-1.
- **Mining**: trailing cables with earth-continuity monitoring, restricted earth-fault current through neutral-earthing resistors (typically limited to 5–25 A), and flameproof equipment underground.
- **Aircraft**: 115/200 V 400 Hz and 270 V DC systems, with arc-fault circuit breakers on wiring bundles to prevent fires.

Fire

Electrical faults cause a substantial share of building fires, mostly through high-resistance connections, damaged cables and overloaded flexible cords rather than through short circuits, which overcurrent devices

clear. A loose termination dissipating 20–50 W at a contact the size of a fingernail can reach several hundred degrees without any device noticing. AFDDs, thermography during maintenance, correct torque on terminals and RCDs of 300 mA for fire protection address this gap.

29.9 Hazard assessment

Risk assessment before every task

NFPA 70E Article 110 requires an electrical safety program in which every task on or near exposed conductors is preceded by a **shock risk assessment** and an **arc-flash risk assessment**. Each identifies the hazard, estimates the likelihood and severity of injury, decides whether additional protective measures, including PPE, are needed, and records the result. EN 50110-1 and national regulations require the equivalent. The assessment asks first whether the work can be done de-energised; energised work is justified only where de-energising creates a greater hazard (life-support equipment, emergency alarm systems) or is infeasible (diagnostic testing that requires voltage), never merely because an outage is inconvenient.

Approach boundaries

For shock, NFPA 70E defines two boundaries around exposed live parts (Figure 29.6 and Table 29.5). The **limited approach boundary** is the distance inside which an unqualified person may enter only when escorted by a qualified person. The **restricted approach boundary** is the distance inside which there is an increased likelihood of shock through arc-over or inadvertent movement; only qualified persons with shock-protective equipment and a documented plan may cross it. The arc-flash boundary of Section 29.2 is separate and, at high fault levels, often larger.

Table 29.5 — Shock approach boundaries for AC systems (after NFPA 70E Table 130.4(E)(a), rounded metric values).

Nominal voltage (phase to phase)	Limited approach, exposed movable conductor	Limited approach, fixed circuit part	Restricted approach
50–150 V	3.0 m	1.0 m	Avoid contact
151–750 V	3.0 m	1.0 m	0.3 m
751 V–15 kV	3.0 m	1.5 m	0.7 m
15.1–36 kV	3.0 m	1.8 m	0.8 m

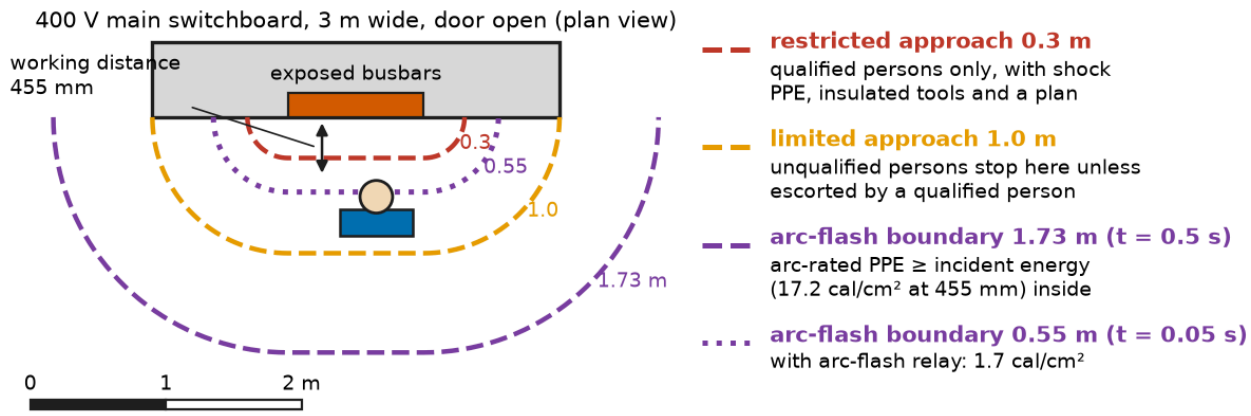


Figure 29.6 — Plan view of the approach boundaries around an open 400 V switchboard with the incident energy of Section 29.2.

PPE and labels

Where an arc-flash hazard remains, PPE is selected either from the incident energy calculated for the equipment (the preferred method: arc-rated clothing with an arc rating at least equal to the incident energy) or from the NFPA 70E tables of tasks and PPE categories, which are valid only within stated limits of fault current and clearing time. The categories require minimum arc ratings of 4, 8, 25 and 40 cal/cm² for categories 1 to 4. Above 40 cal/cm² the blast and pressure effects are such that energised work should not be performed; the engineering solution is a faster protective device or de-energising. Equipment likely to be examined or serviced energised must carry a label stating the nominal voltage, the arc-flash boundary and either the incident energy at the working distance or the PPE category, with the date of the study, which must be reviewed at least every five years and whenever the system changes.

The hierarchy of risk control

PPE is the last line of defence, not the first. NFPA 70E, ISO 45001 and occupational-safety regulations rank controls in order of effectiveness:

1. **Elimination** — de-energise; design the installation so that routine tasks (metering, thermography, setting changes) never need open doors.
2. **Substitution** — lower voltage or lower fault-current systems; 24 V control circuits; current-limiting devices.
3. **Engineering controls** — arc-flash relays, maintenance switches, arc-resistant switchgear, remote racking, finger-safe terminals, infrared windows, insulated tools.
4. **Awareness** — labels, signs, barriers, marking of boundaries.
5. **Administrative controls** — procedures, permits to work, training, qualification, supervision.
6. **PPE** — arc-rated clothing, face shields and hoods, voltage-rated gloves (IEC 60903 / ASTM D120 classes 00 to 4), insulating mats.

Worked example — Assessing a task on the main switchboard. A thermographic survey of the 400 V main switchboard of Section 29.2 must be done under load, with doors open, because the board has no infrared windows. The study gives 17.2 cal/cm² at 455 mm with the present 0.5 s clearing time and a 1.73 m arc-flash boundary; the shock boundaries are 1.0 m (limited) and 0.3 m (restricted).

The task is diagnostic and requires energised equipment, so it may be justified. With the present settings it requires arc-rated clothing of at least 17.2 cal/cm² — category 3 PPE (25 cal/cm²), a hood and heavy gloves — and an exclusion zone of 1.73 m around the open door, through a 3 m wide aisle. Applying the hierarchy changes the answer: installing infrared windows (an engineering control) removes the need to open the door at all; an energy-reducing maintenance switch that drops the clearing time to 0.05 s during the survey reduces the energy to 1.7 cal/cm², within category 1 PPE, and shrinks the boundary to 0.55 m. Both cost less than a single serious burn injury, and the assessment should record that they were considered and why they were or were not adopted. Hazard assessment is not a calculation done once for a label; it is the place where the engineering choices of this chapter are tested against a real task.

> **Status of the principle: Arc-flash risk assessment**
> *Approximate.* The incident energy is computed for a specific configuration, fault current and protective-device setting; any change of transformer, source, cable or setting invalidates it, and the human factors — likelihood of the initiating event — are judged, not calculated.

What remains open

Open question	Where it is answered
How are the prospective fault currents that set touch voltages and arc energy calculated, including unsymmetrical faults?	Chapter 19 (fault analysis)
How are relays, breakers and fuses coordinated so that only the faulted section is disconnected?	Chapter 23 (protection coordination); Chapter 30 (protection selection)
How is a substation earth grid designed to meet IEEE 80 touch and step limits, and how are surges limited?	Chapter 24 (grounding, surge arresters, safety procedures)
How are cables, protective devices and earthing systems chosen for a new installation so that the checks of this chapter pass by design?	Chapter 30 (cable sizing, protection selection, grounding design)
How do drives, PV inverters and EV chargers produce DC leakage, and how are their converters designed?	*Electronics & Computing*; Chapter 12 (drives) and Chapter 27 (grid-connected inverters)
How does the residual-current device follow from Kirchhoff's current law?	*Circuits, Signals and Communications*, Ch. 2
How do battery systems fail thermally and how are battery rooms protected?	Chapter 16 (storage economics and safety)

Next: Chapter 30 brings the whole book together in the design of an installation, from the load estimate through transformer, generator, cables, protection and earthing to the documents that prove it complies.

Chapter 30 — Electrical System Design

Part VIII · Electrical Safety and Engineering Practice

A developer has bought land on the edge of a city to build a cold-storage and packaging facility: four 110 kW refrigeration compressors, a hundred evaporator fans and pumps, packaging lines, offices, ten electric-vehicle chargers for the delivery vans, and a standby generator because a four-hour outage would spoil twenty million dollars of frozen stock. The utility offers an 11 kV connection and asks for the maximum demand. The architect asks for the size of the substation room. The insurer asks how the refrigeration survives a utility failure. The fire officer asks where the main isolators are. The contractor asks for cable sizes, breaker settings and an earthing drawing. Every one of these questions has an answer in the earlier chapters of this book, but none of those chapters answers any of them alone. Electrical system design is the craft of putting the answers together in the right order, checking that they are consistent with each other, and writing them down so that the installation can be built, inspected, operated and, years later, safely changed.

> **How is a complete installation designed from a list of loads: which transformer, which generator, which cables, which protective devices, which earthing?**
> **How does the designer prove — to a client, an inspector and a future engineer — that every one of those choices is safe, economical and compliant?**

Concept	Engineering destination
Load estimation	Utility application, maximum demand, substation and generator sizing
Transformer selection	Rating, impedance, losses, vector group, cooling, redundancy
Generator selection	Standby and prime ratings, motor starting, non-linear loads, autonomy
Equipment ratings	Switchboard current and short-circuit ratings, breaking and making capacity
Cable sizing	Current-carrying capacity, derating, short-circuit withstand, economic size
Voltage-drop calculations	Steady-state and motor-starting voltage at every load
Protection selection	Breaker and fuse ratings, settings, selectivity, arc-flash reduction
Grounding design	Earthing system, protective conductors, electrodes, combined HV/LV earth
Design documentation	Single-line diagrams, schedules, calculation reports, studies, as-builts
Standards and compliance	IEC 60364, IEC 61439, NFPA 70, NFPA 70E, IEEE practice, grid codes, verification

Roadmap. Estimate the load (30.1) → choose the transformer (30.2) and the standby generator (30.3) → set the short-circuit and current ratings of the equipment (30.4) → size the cables (30.5) and check voltage drop (30.6) → select and set the protective devices (30.7) → design the earthing (30.8) → document the design (30.9) → demonstrate compliance (30.10). Each step is worked on the same facility, so the numbers carry from one section to the next.

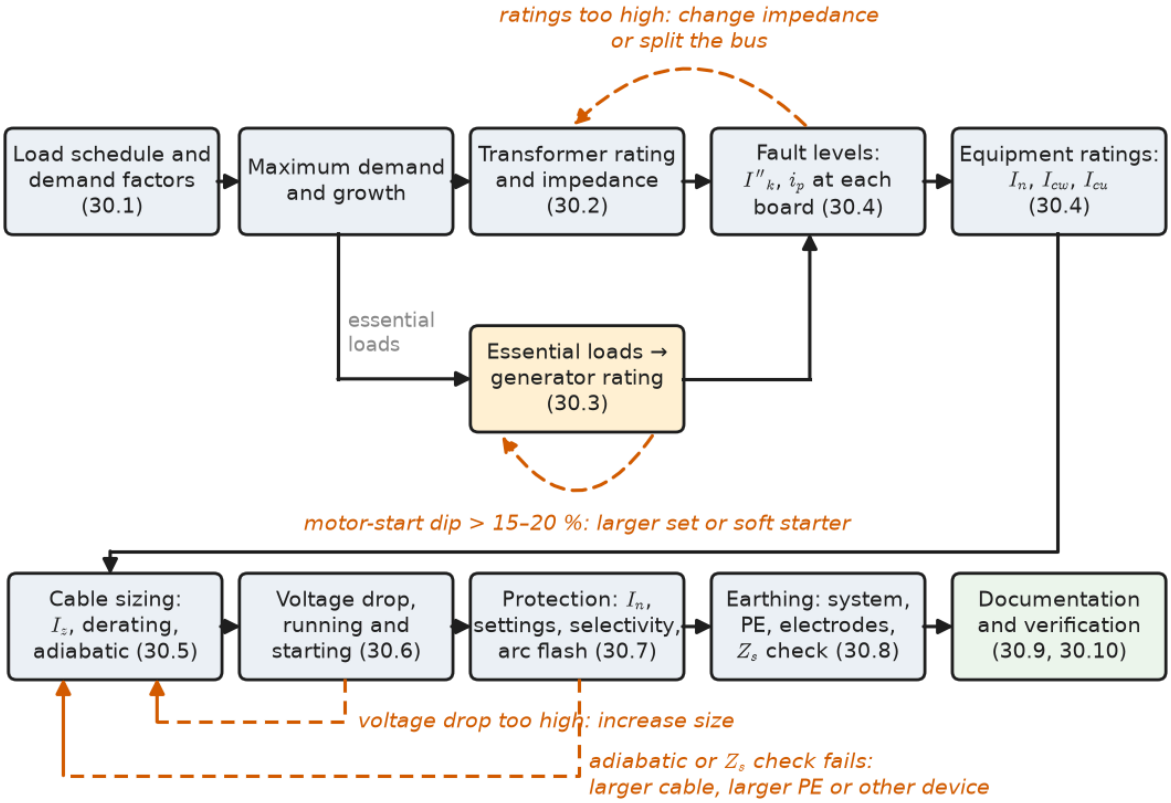


Figure 30.1 — The design sequence of a low-voltage installation, with the iteration loops that close it.

Boundary.
The circuit theory used throughout — complex power, three-phase power and neutral currents — is from *Circuits, Signals and Communications*, Ch. 9 and Ch. 10. Transformer and generator models come from Chapters 4 and 7, per-unit and fault calculations from Chapters 17 and 19, protection principles from Chapter 23, safety limits from Chapter 29. Cable ratings rest on the thermal-resistance networks of *Engineering Physics*, Ch. 14 §14.7 and Ch. 19 §19.6. This chapter assembles them; it does not re-derive them.

30.1 Load estimation

From connected load to maximum demand

Load estimation starts from a **load schedule**: every load or group of loads, its rated power, efficiency, power factor, and an estimate of how it will be used. Connected load — the sum of nameplates — is always much larger than the demand the supply will ever see, because not every load runs at once and few run at full rating. Two factors from Chapter 28 bridge the gap. The **demand factor** of a group is the ratio of its maximum demand to its connected load; the **diversity factor** between groups accounts for their maxima not coinciding. Codes give default values where data are lacking — NEC Article 220 for dwellings, offices and motors; IEC 60364-3 and national guides for European practice — but for industrial plant the designer's knowledge of the process, or measurements from a similar site, are better.

The estimate must be made with active and reactive power separately, not by adding kVA arithmetically, because loads at different power factors do not add in phase. Motor loads are entered at their electrical input (P_{shaft}/η), not their shaft rating. Drives, EV chargers and LED drivers bring harmonic currents whose RMS value exceeds the fundamental (Chapter 25), so their feeders are sized on the true RMS current.

Table 30.1 — Load schedule of the cold-storage facility.

Load group	Connected input (kW)	Demand factor	Demand (kW)	Power factor	Demand (kvar)	Demand (kVA)
Refrigeration compressors, 4 × 110 kW on VSDs	463.2	0.75	347.4	0.95	114.2	365.7
Evaporator fans and pumps, 30 × 4 kW	136.4	0.80	109.1	0.85	67.6	128.3
Packaging lines	250.0	0.70	175.0	0.85	108.5	205.9
LED lighting	60.0	1.00	60.0	0.95	19.7	63.2
Office HVAC	80.0	0.80	64.0	0.90	31.0	71.1
EV chargers, 10 × 22 kW (managed)	220.0	0.50	110.0	0.99	15.7	111.1
Small power and IT	100.0	0.50	50.0	0.95	16.4	52.6
Forklift battery chargers	60.0	0.80	48.0	0.95	15.8	50.5
Sum	**1369.5**		**963.5**	**0.93**	**388.8**	**1039.0**

Worked example — Maximum demand of the facility. From Table 30.1 the sum of group demands is $P = 963.5$ kW and $Q = 388.8$ kvar, so $S = \sqrt{963.5^2 + 388.8^2} = 1039$ kVA at a power factor of 0.927. A diversity factor of 1.10 between the groups — the packaging lines and EV chargers peak at different times of day, and the compressors at full duty mainly on summer afternoons — gives a maximum demand of $1039/1.10 = 945$ kVA. The client expects to add a fifth compressor and more chargers within ten years, so a 25 % growth allowance is added: $945 \times 1.25 = 1181$ kVA. This is the figure for sizing the transformer. The utility application states the present maximum demand, 945 kVA, and the requested capacity, 1250 kVA.

Note how far the result is from the connected load. A designer who simply added nameplates (1370 kW of input, about 1500 kVA) would buy a 1600 kVA transformer, raise the fault level by 25 % (Section 30.4) and the arc-flash energy with it, and run the transformer at 60 % of its rating for a decade.

> **Status of the principle: Demand and diversity factors in design**
> *Empirical.* Factors are drawn from codes or from measurements on comparable sites; they are a judgement about future use and should be revisited with real data (Chapter 28) once the installation is running.

Growth, spare capacity and redundancy

Three separate margins are often confused. **Growth** allows for load added later (20–30 % is common for industry, more for data centres planned in phases). **Spare capacity** in switchboards — empty ways, spare breakers, space for additional sections — costs little at build time and much later. **Redundancy** is a reliability decision: a hospital or a Tier III data centre is supplied by N+1 transformers and generators so that any one can be lost or maintained without dropping load (Chapter 13). Each margin should be stated explicitly in the design basis, so that it is not added three times.

30.2 Transformer selection

Rating

A distribution transformer is selected on its continuous rating under the site's ambient conditions (IEC 60076-1 assumes a 30 °C monthly average and 40 °C maximum), with the load estimate including growth as the target. Standard sizes follow the R10 series: 500, 630, 800, 1000, 1250, 1600, 2000, 2500 kVA. Short-duration overloads are permitted by the thermal loading guide IEC 60076-7 at the cost of accelerated insulation ageing; they are useful for emergencies, not for routine planning. For the facility, 1181 kVA leads to a 1250 kVA unit, loaded to $945/1250 = 76$ % on day one.

Impedance: a compromise

The short-circuit impedance u_k sets two things that pull in opposite directions (Chapter 4): the voltage regulation under load and the voltage dip when a large motor starts (both better with low u_k), and the fault current on the LV switchboard (lower with high u_k). Standard values are about 4 % up to 630 kVA and 5–6 % from 800 to 2500 kVA (IEC 60076-5 minimum values; Ecodesign-compliant units typically 6 % above 630 kVA). Specifying 8 % to keep a 2500 kVA board below 50 kA is a legitimate design choice; so is splitting a large load into two independent 1600 kVA sections with a normally open bus-coupler.

Other choices

- **Losses**: in the EU, Ecodesign Tier 2 (Regulation 548/2014 as amended) sets maximum P_0 and P_k; above that, loss capitalisation (Chapter 28) decides.
- **Type**: liquid-immersed (mineral oil, or less-flammable natural or synthetic esters with fire point above 300 °C, IEC 61099/62770) for outdoor and substation rooms; cast-resin dry type (IEC 60076-11, class F1 fire behaviour) indoors near people, in hospitals, high-rise buildings and data centres.
- **Vector group**: Dyn11 for distribution, giving a solidly earthed LV neutral, blocking triplen harmonics in the delta and isolating zero-sequence between voltage levels (Chapter 19).
- **Tappings**: off-circuit $\pm 2 \times 2.5$ % to centre the LV voltage on a high or low MV feeder; on-load tap changers only for larger or voltage-critical sites.
- **Harmonics**: where a large share of load is non-linear, the transformer is derated or specified with a K-factor or to IEC 60076-1 harmonic loss data (Chapter 25).

Worked example — 1000, 1250 or 1600 kVA? The 11 kV supply has a fault level of 250 MVA. Each candidate transformer has $u_k = 6$ % and $u_r = 1.0$ %; the LV system is 400 V.

Rating	Rated current	Initial symmetrical fault current I_k'' (c = 1.05)	Loading today
1000 kVA	1443 A	23.7 kA	94 %
1250 kVA	1804 A	29.1 kA	76 %
1600 kVA	2309 A	36.5 kA	59 %

The fault current is $I_k'' = c\,S_r/(\sqrt{3}\,U_n\,(u_k + S_r/S_{k,MV}))$; for the 1250 kVA unit, $1.05 \times 1250/(\sqrt{3} \times 0.4 \times (0.06 + 0.005)) = 29.1$ kA. The 1000 kVA unit has no room for growth. The 1600 kVA unit raises the fault level to 36.5 kA, which pushes the main switchboard into a higher (and more expensive) short-circuit rating and raises arc-flash energy in proportion (Chapter 29), while running underloaded. The 1250 kVA unit is chosen. Its regulation at today's load and power factor 0.927 is approximately $0.756 \times (1.0 \times 0.927 + 5.92 \times 0.375) = 2.4$ %, which the +2.5 % tap can offset if the MV supply runs low.

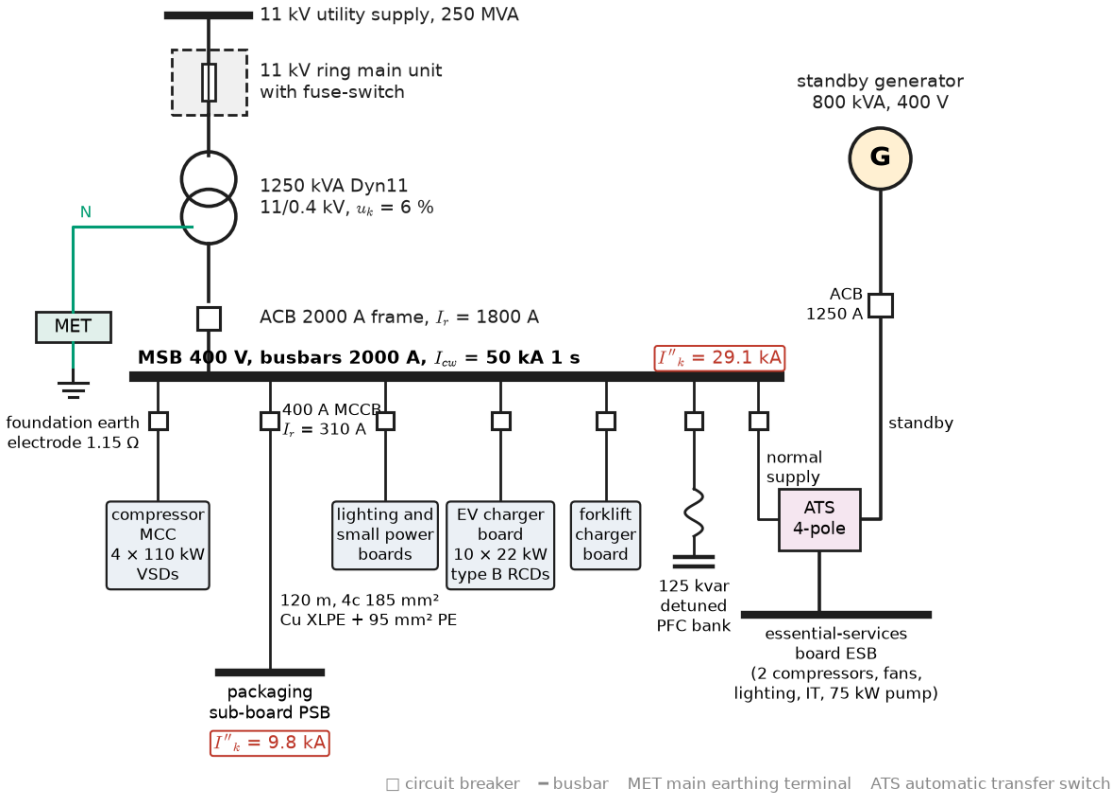


Figure 30.2 — Single-line diagram of the cold-storage facility.

30.3 Generator selection

Ratings and duty

A standby generator is not rated like a transformer. ISO 8528-1 defines several power ratings for the same engine-alternator set, depending on duty:

- **Emergency standby power (ESP)**: the maximum power available for a variable load during a utility outage, typically limited to 200 h per year with an average load not exceeding 70 % of rating; no

overload.

- **Prime power (PRP)**: unlimited hours per year at variable load, average not exceeding 70 % of rating over 24 h.
- **Continuous operating power (COP)**: constant load, unlimited hours.
- Manufacturers also publish **data-centre continuous** ratings, at which the set can carry a constant load for unlimited hours during a long outage.

The ESP rating is typically 10 % above PRP and 20–30 % above COP for the same set. The choice depends on what the outage looks like: a hospital in a city with a reliable grid uses ESP; a mine or island grid with no utility uses PRP or COP; a data centre that must survive a multi-day regional blackout should be sized on continuous ratings. Altitude and temperature derating apply above typically 1000 m and 25–40 °C, by 1–4 % per 300 m and per 5 K depending on the engine.

What the generator must survive

A generator is a source with a high, time-varying internal impedance (subtransient reactance x''_d about 0.10–0.20 pu, transient x'_d 0.15–0.30 pu, synchronous x_d 1.5–3 pu; Chapter 7), behind an engine whose power cannot rise instantly. Three load cases usually size it more than steady kW:

1. **Motor starting.** A direct-on-line motor draws five to seven times its rated current at a power factor of 0.2–0.4. The voltage dip must stay within what contactors, drives and lighting tolerate (typically 15–20 %) and recover within a few seconds. ISO 8528-5 performance classes G1–G4 specify permissible transient voltage and frequency deviations and recovery times.
2. **Load steps.** Turbocharged diesels accept only 30–60 % of rating as a single block step; emergency loads are applied in stages within the time allowed (10 s for NFPA 110 Type 10, 15 s for hospital safety services in IEC 60364-7-710).
3. **Non-linear and regenerative loads.** UPS rectifiers and drives distort the generator voltage because its impedance is much higher than the utility's; low x''_d and permanent-magnet excitation help. Regenerating hoists and lifts push power back into a generator that cannot absorb it and must be given braking resistors.

A simple screening estimate of the dip models the starting motor as a constant impedance $z_m = S_{gen}/S_{start}$ (per unit on the generator base) in series with an effective generator reactance x between x''_d and x'_d:

$$\Delta V \approx \frac{x}{z_m + x} = \frac{x\,S_{start}}{S_{gen} + x\,S_{start}}$$

> **Status of the principle: Screening estimate of generator voltage dip**
> *Approximate.* Ignores the existing load, the AVR response and engine speed droop, and treats the motor as a fixed impedance; it is adequate for selecting a frame size, while a time-domain simulation or the manufacturer's motor-start data confirm the final choice.

Worked example — Sizing the standby generator. The essential loads that must run during an outage are two compressors (231.6 kW at 0.95), lighting (60 kW), small power and IT (50 kW), the evaporator fans at their 0.8 demand factor (109.1 kW at 0.85) and a 75 kW direct-on-line ammonia-transfer pump with efficiency 0.95 and power factor 0.86. Total steady load: $P = 529.6$ kW, $Q = 226.7$ kvar, $S = 576$ kVA. The pump starts at 6.5 times its rated kVA; the effective generator reactance is 0.20 pu.

The pump's rated input is $75/(0.95 \times 0.86) = 91.8$ kVA, and its starting demand $6.5 \times 91.8 = 597$ kVA. For three candidate prime-rated sets:

Generator	Steady loading	z_m	Estimated dip
600 kVA	96 %	1.005	16.6 %
800 kVA	72 %	1.340	13.0 %
1000 kVA	58 %	1.676	10.7 %

The 600 kVA set is too heavily loaded for prime duty and gives a dip above 15 %. The 800 kVA set meets both constraints and leaves headroom for one additional compressor in an extended outage. It is selected, with the pump started last in the load-step sequence. Fitting the pump with a soft starter would allow a smaller set; the trade-off is cost against flexibility. Figure 30.3 plots the dip against generator rating for DOL and soft-start cases. Fuel storage for 48 hours at the average essential load of about 450 kW, at roughly 0.25 L/kWh, is about 5400 L.

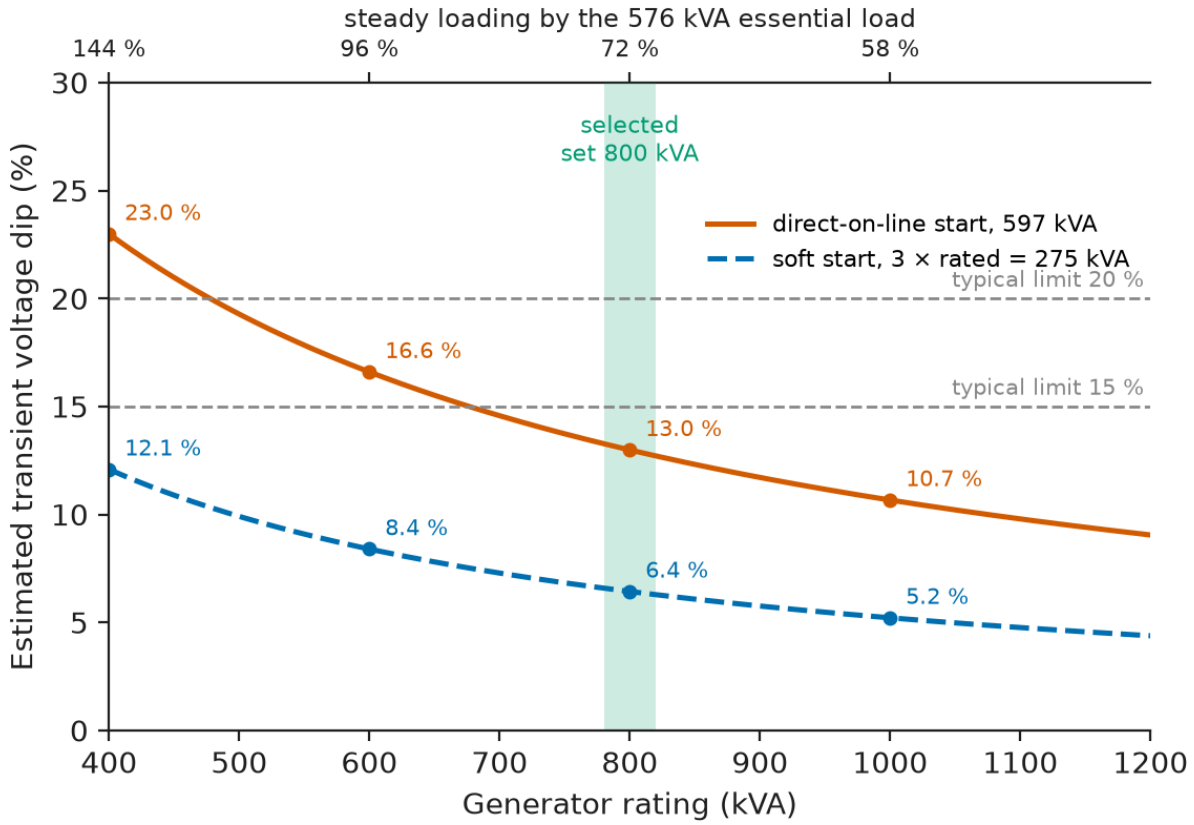


Figure 30.3 — Estimated transient voltage dip when the 75 kW pump starts, against generator rating.

30.4 Equipment ratings

What every item of switchgear must be rated for

Each item of equipment in the installation must withstand, continuously and during faults, what the system can impose on it. The ratings to check (IEC 61439 for assemblies, IEC 60947-2 for breakers, IEC 62271 for medium voltage) are:

- **Rated voltage** U_e and **rated insulation voltage** U_i; **impulse withstand** U_{imp} matched to the overvoltage category at that location (Chapter 24).
- **Rated current** I_n (or the assembly's rated diversity factor), derated for ambient temperature above 35 °C, altitude above 2000 m (or 1000 m for MV), enclosure ventilation and harmonic content.
- **Short-time withstand current** I_{cw} (usually for 1 s) of busbars and air circuit breakers: the RMS current they carry without damage while a downstream device clears.
- **Peak withstand current** I_{pk} of busbars and the **making capacity** I_{cm} of breakers: the first asymmetrical peak, set by the X/R ratio of the source.
- **Ultimate and service breaking capacity** I_{cu} and I_{cs} of breakers: I_{cu} is the maximum the breaker can interrupt once; I_{cs} (25–100 % of I_{cu}) is what it can interrupt and remain in service. Critical boards should have $I_{cs} = I_{cu}$ above the prospective current.
- **Internal arc classification** (IEC 62271-200 for MV, IEC TR 61641 for LV) and **form of separation**, which govern safety during faults and maintenance (Chapter 29).

The peak current follows from IEC 60909: $i_p = \kappa\sqrt{2}\, I''_k$, with $\kappa = 1.02 + 0.98\, e^{-3R/X}$ ranging from 1.0 for a resistive source to 2.0 for a purely reactive one.

> **Status of the principle: IEC 60909 short-circuit currents**
> *Approximate.* An equivalent voltage source $cU_n/\sqrt{3}$ at the fault replaces the pre-fault load flow, and the peak factor κ is an empirical fit for a single R–X branch; the method is conservative for design and is not a substitute for a dynamic study where motors or inverters contribute significantly.

Worked example — Ratings of the main switchboard. For the 1250 kVA transformer ($u_k = 6$ %, $u_r = 1$ %), the transformer impedance on a 400 V base is $Z_T = 0.06 \times 0.4^2/1.25 = 7.68$ mΩ, with $R_T = 1.28$ mΩ and $X_T = 7.57$ mΩ; the 250 MVA source adds $0.4^2/250 = 0.64$ mΩ, almost all reactive. Then $R/X = 1.28/(7.57 + 0.64) = 0.156$, so

$$\kappa = 1.02 + 0.98\, e^{-3\times 0.156} = 1.634, \qquad i_p = 1.634 \times \sqrt{2} \times 29.1 = 67.2 \text{ kA}$$

Motor contribution is small here, because the compressors are on VSDs that contribute little fault current. The main switchboard is specified as 2000 A, $I_{cw} = 50$ kA for 1 s, $I_{pk} = 105$ kA, form 3b, with an ACB of $I_{cs} = I_{cu} = 50$ kA, giving margin for a future second transformer only if the bus-coupler is interlocked so that the two never run in parallel (two such units in parallel would raise the fault level to about 54 kA).

Table 30.2 — Typical LV equipment ratings by transformer size (400 V, u_k as shown, infinite MV source).

Transformer	u_k	Rated current	I''_k approx.	Typical board rating
630 kVA	4 %	909 A	24 kA	1000 A, 25–36 kA
1000 kVA	6 %	1443 A	25 kA	1600 A, 36 kA
1250 kVA	6 %	1804 A	32 kA	2000 A, 50 kA
1600 kVA	6 %	2309 A	40 kA	2500 A, 50 kA
2500 kVA	6 %	3608 A	63 kA	4000 A, 65–80 kA

(The values in Table 30.2 are upper bounds with an infinite source and $c = 1.05$; a real MV source reduces them, as the worked example shows.)

30.5 Cable sizing

Three independent criteria

A cable must satisfy, at once:

1. **Current-carrying capacity**: under its installation conditions, its conductor must not exceed the insulation's maximum continuous temperature (70 °C PVC, 90 °C XLPE/EPR) when carrying the design current I_B, and it must be protected against overload ($I_B \le I_n \le I_Z$, Chapter 29).
2. **Short-circuit withstand**: $S \ge \sqrt{I^2 t}/k$ for the energy let through by its protective device (Chapter 29).
3. **Voltage drop**: within the limits of Section 30.6, running and starting.

A fourth, economic sizing, often selects a larger size still, and a fifth, the earth-fault loop impedance of the protective conductor (Section 30.8), can force a larger PE or a different device.

Current-carrying capacity and derating

The tabulated current-carrying capacities in IEC 60364-5-52 (and in NEC Table 310.16 and its successors) are calculated by IEC 60287, which models the cable and its surroundings as a thermal-resistance network — conductor, insulation, sheath, the air or soil around it — through which the conductor's I^2R loss must flow at a temperature rise the insulation can tolerate. That network is the heat-conduction model of *Engineering Physics*, Ch. 14 §14.7, applied to a cylinder in a medium. The tables are given for reference conditions (30 °C in air, 20 °C in ground, soil thermal resistivity 2.5 K·m/W, one circuit), and **correction factors** adjust them:

- **Ambient temperature**: for XLPE in air, 1.0 at 30 °C, 0.96 at 35 °C, 0.91 at 40 °C, 0.87 at 45 °C.
- **Grouping**: several loaded circuits heat each other; four multicore cables touching on a perforated tray, about 0.77; six, about 0.73; nine bunched in conduit, about 0.50.
- **Soil thermal resistivity and depth** for buried cables, which can reduce capacity by 10–30 % in dry sand.
- **Harmonics**: when triplen harmonics exceed 15 % of the phase current, the neutral carries current and the cable is sized on the neutral (IEC 60364-5-52 Annex E).

The required tabulated capacity is $I_t \ge I_B / (C_a\, C_g\, C_s \cdots)$.

Table 30.3 — Current-carrying capacity of multicore copper XLPE cables, three loaded conductors, on a perforated tray in free air at 30 °C (approximate, after IEC 60364-5-52, method E).

Cross-section	16 mm²	35 mm²	70 mm²	95 mm²	120 mm²	150 mm²	185 mm²	240 mm²
Capacity	100 A	158 A	246 A	298 A	346 A	399 A	456 A	538 A
R at 20 °C (Ω/km)	1.15	0.524	0.268	0.193	0.153	0.124	0.0991	0.0754

Worked example — The packaging sub-board feeder. The packaging sub-board has a design current $I_B = 300$ A at a power factor of 0.85. Its feeder is a 120 m four-core copper XLPE cable on a perforated tray shared with three other loaded circuits, in a plant room at 40 °C. The prospective fault at the main switchboard is 29.1 kA; the feeder's MCCB clears high faults in under 20 ms.

Correction factors: grouping 0.77, ambient 0.91. Required tabulated capacity: $300/(0.77 \times 0.91) = 428$ A. From Table 30.3, 185 mm² (456 A) is the smallest that qualifies; its corrected capacity is $456 \times 0.77 \times 0.91 = 320$ A. A 400 A frame MCCB with its electronic trip set to $I_r = 310$ A satisfies $300 \le 310 \le 320$ A.

Short circuit: with $k = 143$, a 29.1 kA fault at the feeder's origin cleared in 0.02 s requires at most $S \ge \sqrt{29\,100^2 \times 0.02}/143 = 28.8$ mm² (a current-limiting MCCB lets through less); even if the upstream ACB's 0.2 s short-time delay had to clear it, $S \ge 91$ mm². 185 mm² passes. The protective conductor, from IEC 60364-5-54 ($S_{PE} = S/2$ for phase conductors above 35 mm²), is 95 mm². Voltage drop is checked in Section 30.6.

> **Status of the principle: Tabulated current-carrying capacities**
> *Approximate.* The tables come from steady-state thermal-network calculations (IEC 60287) for idealised installations; correction factors are treated as independent and multiplicative, which is conservative in some combinations and not in others, and cyclic loading or intermittent duty can justify higher ratings.

Economic sizing

The thermal minimum is the cheapest cable to buy, not to own. IEC 60287-3-2 minimises purchase cost plus the present value of losses. **Worked example — Upsizing the feeder.** At 300 A the 185 mm² cable runs at about 84 °C, with a resistance of 0.124 Ω/km and losses of $3 \times 300^2 \times 0.124 \times 0.12 = 4.02$ kW. A 240 mm² cable at the same current runs at about 72 °C, $R = 0.0907$ Ω/km, losing 2.94 kW. With a loss-load factor of 0.4 the saving is $1.08 \times 0.4 \times 8760 = 3780$ kWh per year, worth 378 USD at 0.10 USD/kWh, or 4830 USD present value over 25 years at 6 % (PWF 12.78). If the larger cable and its glands cost less than about 4800 USD more for the 120 m run, it is the economic choice; typically it costs 2500–3500 USD more, so it is selected. Economic sizing is most powerful on long, heavily loaded feeders — which is also where voltage drop bites.

30.6 Voltage-drop calculations

The engineering formula

For a three-phase feeder of length L with resistance r and reactance x per unit length carrying current I at power factor $\cos\phi$, the voltage drop (line-to-line) is, to within a fraction of a per cent for drops below about 10 %,

$$\Delta V \approx \sqrt{3}\, I\, L\, (r \cos\phi + x \sin\phi)$$

This is the projection of the series impedance drop on the sending-end voltage phasor; its derivation from the phasor diagram is the short-line model of Chapter 21. Resistance is taken at the conductor's operating temperature; reactance of multicore LV cables is 0.07–0.09 Ω/km, of single-core cables in trefoil about 0.10 Ω/km, and of single cores spaced apart up to 0.2 Ω/km. For single-phase circuits, replace $\sqrt{3}$ by 2.

> **Status of the principle: Approximate voltage-drop formula**
> *Approximate.* Neglects the quadrature component of the drop and the shunt capacitance of the cable; it is accurate for LV and short MV circuits and underestimates the drop only when x/r is large and the angle across the circuit is several degrees.

Limits

IEC 60364-5-52 Annex G recommends maximum drops from the origin of the installation to any load of 3 % for lighting and 5 % for other uses when the supply is from a public LV network, and 6 % and 8 % when the installation has its own transformer (as the facility does), excluding the transformer. NFPA 70 recommends (in informational notes) 3 % for a branch circuit and 5 % for feeder plus branch. Equipment standards set the real limit: motors are designed for ±5 % at rated output (IEC 60034-1 zone A) and must start at the reduced voltage of the starting dip; contactor coils may drop out below 65–80 % of rated voltage; drives trip on undervoltage around 80–85 % of nominal DC-link voltage. During motor starting a drop of 10–15 % at the motor terminals is usually acceptable if the motor's torque at reduced voltage (proportional to V^2, Chapter 6) still exceeds the load torque.

Worked example — Voltage at a packaging-line motor, running and starting. A 22 kW motor (efficiency 0.91, power factor 0.86) on the packaging sub-board is fed by 40 m of 10 mm² copper XLPE cable ($r = 2.33$ Ω/km at 90 °C, $x = 0.08$ Ω/km). The motor starts direct on line at 6.5 times rated current at power factor 0.30.

Feeder (Section 30.5), running: with $r = 0.124$ Ω/km at the estimated 84 °C operating temperature and $x = 0.075$ Ω/km,

$$\Delta V_{feeder} = \sqrt{3} \times 300 \times 0.12 \times (0.124 \times 0.85 + 0.075 \times 0.527) = 9.0 \text{ V } (2.26\ \%)$$

Motor circuit, running: $I = 22\,000/(\sqrt{3} \times 400 \times 0.86 \times 0.91) = 40.6$ A, so $\Delta V = \sqrt{3} \times 40.6 \times 0.04 \times (2.33 \times 0.86 + 0.08 \times 0.51) = 5.8$ V (1.44 %). The total running drop from the main switchboard is 3.7 %, well within the 8 % recommendation.

Starting: the motor circuit carries $6.5 \times 40.6 = 264$ A at power factor 0.30, a drop of $\sqrt{3} \times 264 \times 0.04 \times (2.33 \times 0.30 + 0.08 \times 0.954) = 14.2$ V (3.5 %). The feeder carries the rest of the sub-board's load plus the starting current, phasor-summed to about 491 A at a lower power factor, a drop of 13.8 V (3.4 %). Total at the motor terminals during starting: about 7.0 %, plus the transformer's own dip. Acceptable: the motor sees 93 % of voltage and 86 % of its rated starting torque. Figure 30.4 shows the voltage profile.

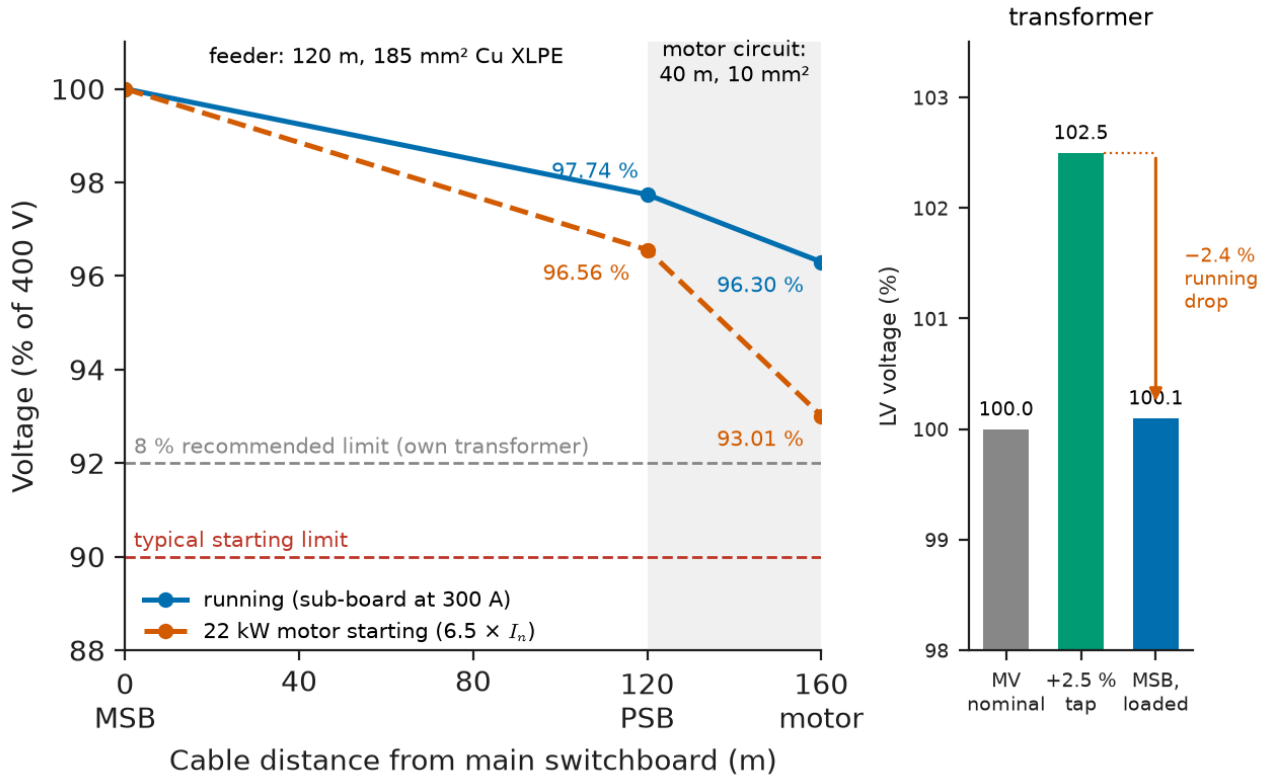


Figure 30.4 — Voltage profile from the transformer terminals to the 22 kW packaging-line motor, running and while the motor starts.

30.7 Protection selection

Requirements

Protective devices are chosen to satisfy, simultaneously:

1. **Protection of the cable** against overload and short circuit (Section 30.5 and Chapter 29).
2. **Protection of people** by automatic disconnection within the times of IEC 60364-4-41 and by RCDs where required.
3. **Breaking capacity** at least equal to the prospective fault current at the point of installation, or a tested back-up combination.
4. **Selectivity** (discrimination): for any fault, only the device immediately upstream should operate.
5. **Arc-flash reduction** where people may work on or near energised equipment (Chapter 29).
6. **Equipment-specific protection**: transformer (Buchholz relay, winding temperature, MV fuses or relay), generator (overcurrent with voltage restraint, reverse power, earth fault, loss of excitation), motors (thermal overload class, stall, phase loss), capacitor banks (unbalance, overcurrent).

Selectivity between low-voltage breakers

Low-voltage selectivity is achieved by three mechanisms, often combined:

- **Current selectivity**: the upstream device's instantaneous threshold is above the maximum fault current at the downstream device's location. Simple, but limited when fault currents are high and close together.

- **Time selectivity**: the upstream breaker has a short-time delay (0.1–0.4 s) and a short-time withstand I_{cw} high enough to wait while the downstream one clears. Typical of air circuit breakers on main boards.
- **Energy selectivity**: a current-limiting downstream MCCB lets through so little energy that the upstream device, though in its instantaneous range, does not trip. Valid only from manufacturer tables.

For fuses, selectivity holds when the upstream rating is at least 1.6 times the downstream one (gG fuses to IEC 60269), and for overload regions of breakers a ratio of setting currents of at least 1.5–2 is usual. **Zone-selective interlocking** passes a restraint signal from the downstream breaker that sees a fault to the upstream one; an upstream breaker that sees the fault but receives no restraint knows the fault is in its own zone and trips without its delay — selectivity and fast clearing together.

Worked example — Settings at the main switchboard. The transformer's rated LV current is 1804 A. The incomer is a 2000 A ACB; the packaging feeder a 400 A MCCB at $I_r = 310$ A (Section 30.5).

Table 30.4 — Settings chosen for the main incomer and the packaging feeder.

Device	Long time I_r	Short time I_{sd}, delay	Instantaneous I_i	Earth fault
Incomer ACB, 2000 A	1800 A (0.9)	4 × Ir = 7.2 kA, 0.2 s	Off (or above 50 kA)	1200 A, 0.4 s
Packaging MCCB, 400 A	310 A	6 × Ir = 1.86 kA, 0.1 s	10 × In = 4 kA	Via short-time

Checks:

- **Overload ratio**: $1800/310 = 5.8$, ample selectivity in the long-time region.
- **Short circuit at the sub-board**: the three-phase fault at the end of the 120 m feeder is about 9.8 kA (from the source, transformer and feeder impedances); the MCCB clears it instantaneously in under 20 ms, while the ACB's short-time element picks up at 7.2 kA but waits 0.2 s. Time selectivity is achieved, and the 0.2 s × 9.8 kA is far inside the ACB's 50 kA 1 s withstand.
- **Earth fault at the sub-board**: loop impedance through the transformer, the 185 mm² phase conductor and the 95 mm² PE is about 51 mΩ, giving $0.95 \times 230/0.051 = 4.3$ kA. This is only marginally above the MCCB's 4 kA instantaneous setting (tolerance ±20 %), so instantaneous operation is not guaranteed; the short-time element at 1.86 kA and 0.1 s clears it well within the 5 s permitted for a distribution circuit. A designer who relied on the instantaneous setting alone would have missed this.
- **Arc flash at the main board**: an arcing fault on the main busbar (about 50 % of 29.1 kA, i.e. 14.6 kA) is above I_{sd} and is cleared in about 0.25 s including breaker opening. The Lee estimate of Chapter 29 gives about 7.2 cal/cm² at 455 mm for 0.25 s, still category 2 PPE. Zone-selective interlocking between the incomer and the outgoing MCCBs would let the ACB trip without delay for busbar faults and cut this to about 1.5 cal/cm²; it is specified.

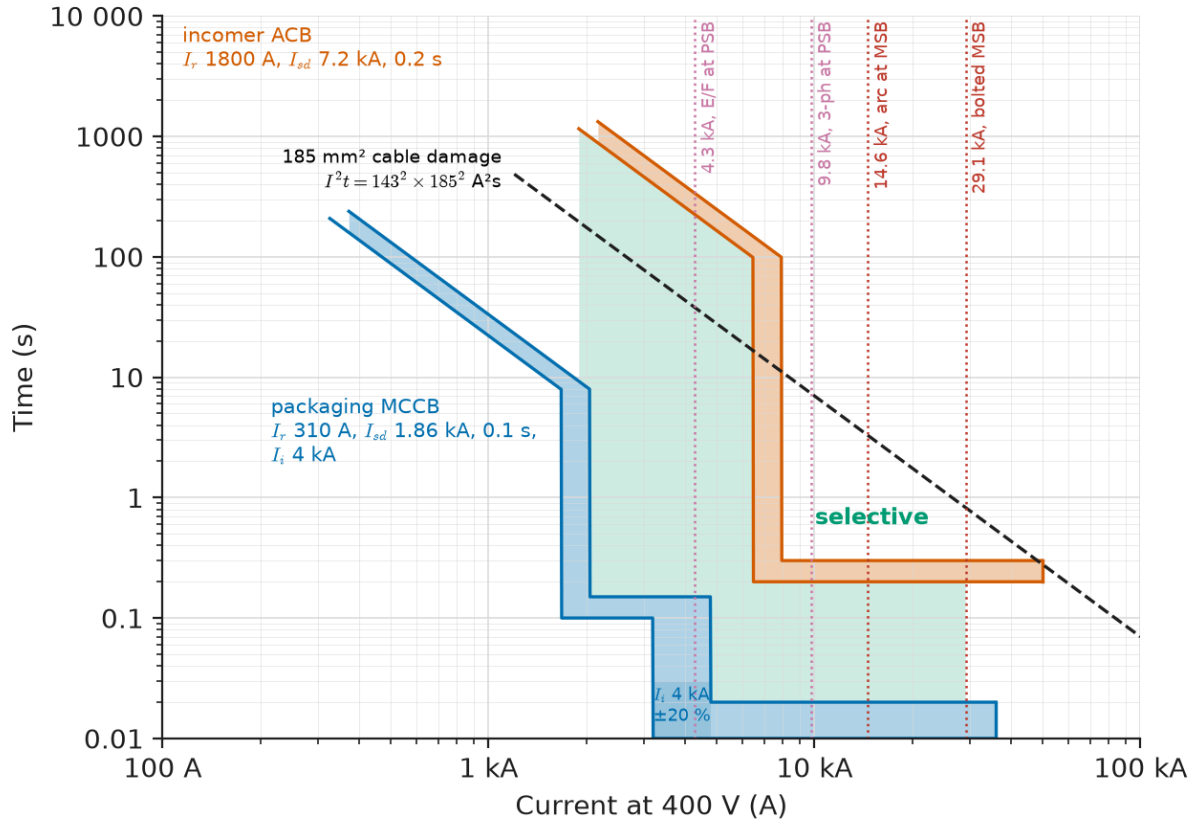


Figure 30.5 — Time-current curves of the main incomer ACB and the packaging feeder MCCB, showing selectivity and the fault currents at each board.

30.8 Grounding design

Choosing the earthing system

The earthing system (Chapter 29) is a design decision with consequences for safety, continuity and equipment:

- **TN-S** for sites with their own transformer: high fault currents allow overcurrent devices to provide fault protection, a separate PE keeps load current off the protective conductors (important for EMC in data centres and control rooms), and neutral currents are measurable.
- **TN-C-S** is the usual arrangement where the supply is a public LV network; inside the installation it is TN-S.
- **TT** where the supply earth cannot be relied upon (rural overhead networks, some EV-charging and outdoor installations), with RCDs everywhere.
- **IT** where a first earth fault must not interrupt supply: operating theatres, process plants with continuous processes, mines, ships, some data-centre UPS outputs.

For the facility, with its own 11/0.4 kV substation, TN-S is chosen. Where a standby generator feeds part of the installation through a transfer switch, the neutral must be switched (4-pole changeover) or earthed at only one point; otherwise neutral current returns through parallel neutral and PE paths, upsetting earth-fault relays and RCDs.

Protective conductors

The cross-section of a protective conductor is either taken from IEC 60364-5-54 Table 54.2 — equal to the phase conductor up to 16 mm², 16 mm² for phase conductors of 16–35 mm², and half the phase conductor above that — or calculated from the adiabatic equation for the actual let-through energy, which often allows smaller sizes. The loop-impedance check of Chapter 29 must then be met with the chosen PE. Bonding conductors to extraneous metalwork are sized as half the largest PE, with a minimum of 6 mm² copper and a maximum of 25 mm² in IEC practice.

Earth electrodes

The earth electrode connects the system's earthing point to the soil. Its resistance depends almost entirely on soil resistivity (from 10–50 Ω·m in wet clay to 1000–10 000 Ω·m in dry rock and sand) and on the electrode's size. For a single vertical rod of length L and diameter d in uniform soil of resistivity ρ, the classic engineering result (Dwight) is

$$R_{rod} = \frac{\rho}{2\pi L}\left(\ln\frac{8L}{d} - 1\right)$$

and for a mesh or foundation electrode of area A with total buried conductor length L_T, Laurent's approximation

$$R_{grid} \approx \frac{\rho}{4r} + \frac{\rho}{L_T}, \qquad r = \sqrt{A/\pi}$$

Rods in parallel interfere with each other because their potential zones overlap. With spacing equal to the rod length, n rods give $R_n \approx F\,R_{rod}/n$ with multiplying factors F of about 1.16 for 2 rods, 1.36 for 4 and 1.68 for 8 (IEEE 142).

> **Status of the principle: Earth-electrode resistance formulas**
> *Approximate*. Assume homogeneous soil, which real sites rarely have; layered soil, seasonal moisture and frost change ρ by factors of two to ten, so designs are based on measured resistivity (Wenner four-pin method) and verified by measurement after construction.

Worked example — Earthing the facility's substation. The site's soil resistivity, measured by the Wenner method, is 100 Ω·m. The 11 kV network is resistance-earthed; for an MV earth fault at the substation, after current returning via cable sheaths is deducted, 180 A flows into the local electrode. For a combined HV/LV earth (the LV neutral and the MV equipment earthed to the same electrode, as is normal in urban substations), the earth potential rise must stay within the national limit derived from IEC 61936-1/EN 50522; take 430 V. What electrode is needed?

Required resistance: $R_E \le 430/180 = 2.39\ \Omega$. A single 3 m rod of 16 mm diameter gives

$$R_{rod} = \frac{100}{2\pi \times 3}\left(\ln\frac{8\times 3}{0.016} - 1\right) = 5.31 \times 6.31 = 33.5\ \Omega$$

Four rods give about 11.4 Ω and eight rods 7.0 Ω (Figure 30.6): rods alone cannot reach 2.4 Ω in this soil without dozens of them. The building's reinforced-concrete foundation, however, covers 60 m × 40 m. Bonded into a foundation earth electrode with 400 m of conductor in a mesh, its resistance is $r = \sqrt{2400/\pi} = 27.6$ m, $R \approx 100/(4\times 27.6) + 100/400 = 0.91 + 0.25 = 1.15\ \Omega$. The substation earth mat is connected to it, and the combined earth meets the limit with margin. This is why foundation earth electrodes are required in new

buildings in several European countries (DIN 18014, for example) and are the first choice for any large building.

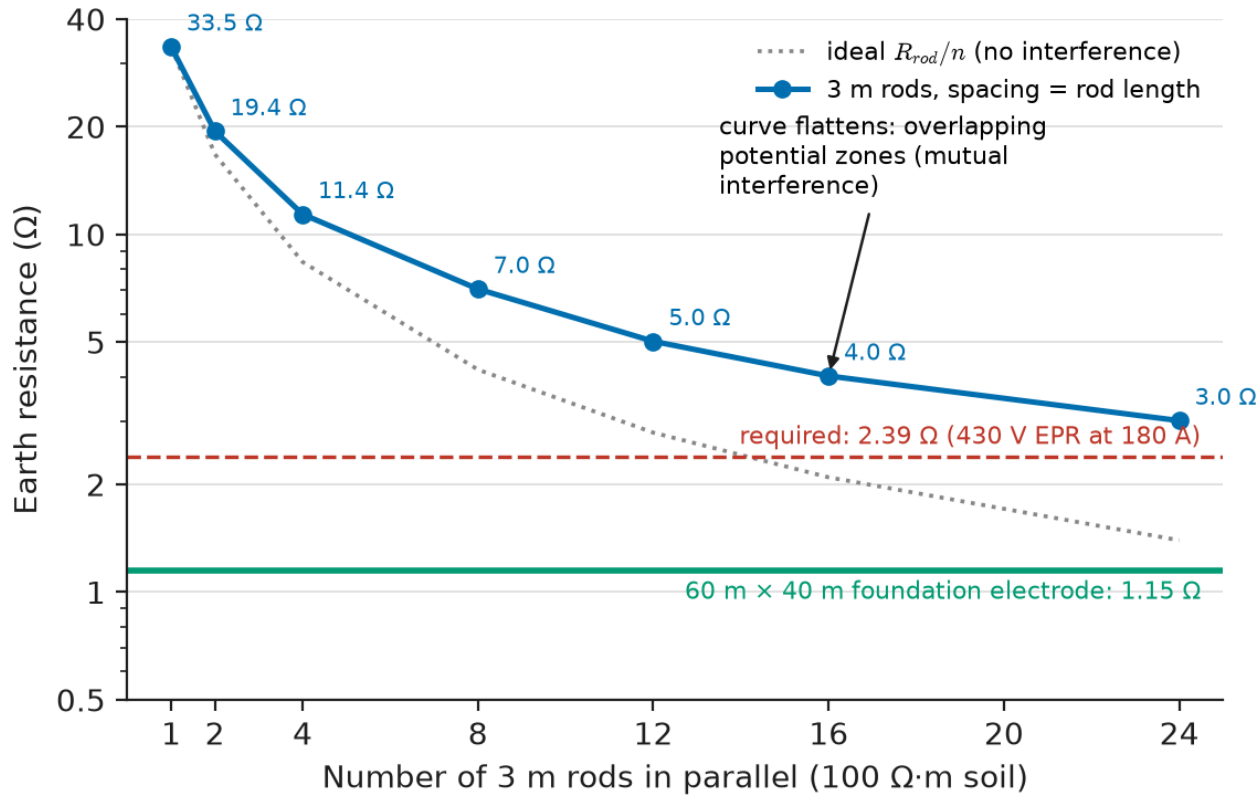


Figure 30.6 — Earth resistance of rod groups and of a foundation electrode in 100 ohm-metre soil, against the 2.39 ohm requirement.

30.9 Design documentation

Why documentation is part of the design

An installation outlives its designers. The documents are how the contractor builds it, how the inspector verifies it, how the operator switches it safely, and how the engineer who extends it in fifteen years knows what the fault level is and which breaker protects which cable. NFPA 70E and EN 50110 both rely on up-to-date single-line diagrams for isolation; arc-flash labels are invalid the moment the system changes without the study being updated. Documentation is a safety device.

Table 30.5 — Typical design deliverables.

Document	Content	Used by
Design basis	Supply data, earthing system, design standards, ambient conditions, margins, redundancy philosophy	Everyone; the reference for every later decision
Single-line diagram	Every source, transformer, board, feeder, device rating and setting, cable size, earthing points	Contractor, operator, inspector, isolation planning
Load schedule	Each load with rating, demand and diversity factors, maximum demand per board	Utility application, transformer and generator sizing
Cable schedule	Each cable: from, to, type, size, length, route, installation method, derating, voltage drop	Contractor, cable pulling, verification
Short-circuit study	I''_k, i_p and minimum fault current at every board (IEC 60909 or IEEE C37/ANSI method)	Equipment ratings, protection settings
Protection coordination study	Time-current curves, settings table, selectivity analysis	Commissioning, future modifications
Arc-flash study	Incident energy and boundaries at each location, labels (IEEE 1584)	Safety programme, PPE selection
Earthing design	Earthing system, electrode design and calculations, bonding schedule	Contractor, inspector
Layout drawings	Substation, switchrooms (with clearances and escape routes), containment routes	Architect, contractor, fire officer
Specifications	Equipment technical requirements and test requirements	Procurement, factory tests
Verification records	Test results to IEC 60364-6, commissioning records, certificates	Owner, insurer, authority
As-built and O&M manuals	Final drawings, settings, manuals, maintenance schedules	Operator for the life of the installation

Conventions

Drawings follow IEC 60617 graphical symbols and IEC 61082 documentation rules; equipment is identified with reference designations to IEC 81346 so that "=MSB-Q12" means the same breaker on every drawing, label and maintenance record. Calculations are done in recognised software (power-system analysis packages implementing IEC 60909, IEC 60364 and IEEE 1584) but must still be checked by hand for a few points, as the worked examples of this chapter do, because an input error propagates silently through every result. Increasingly the design lives in a building information model (BIM), from which schedules and drawings are generated, which makes consistency easier and errors in the model more far-reaching.

30.10 Standards and compliance

The role of standards

Standards encode the accumulated failures of the industry into rules that can be checked. They do not replace engineering judgement — every standard has scope limits and many leave choices to the designer — but they

define the minimum, make designs reviewable by others, and in most jurisdictions are made legally binding through building regulations, electricity safety regulations and insurance conditions. A design that complies with the applicable standards and documents how is defensible; one that departs from them needs a documented engineering justification.

Table 30.6 — Principal standards in low- and medium-voltage installation design.

Subject	IEC / international	North America	Notes
LV installations	IEC 60364 series (BS 7671, NF C 15-100, AS/NZS 3000 as national versions)	NFPA 70 (NEC), CSA C22.1	Shock, overcurrent, selection and erection, verification
Work safety	EN 50110-1	NFPA 70E, OSHA 29 CFR 1910 Subpart S	Safe work practices, arc-flash assessment
Short-circuit calculation	IEC 60909	IEEE C37.010, IEEE 551	Equivalent-source method
Arc-flash calculation	—	IEEE 1584	Used internationally
LV switchboards	IEC 61439	UL 891, UL 1558	Temperature rise, short-circuit, forms
Breakers and fuses	IEC 60947-2, IEC 60898-1, IEC 60269	UL 489, UL 248	Breaking capacity, characteristics
Transformers	IEC 60076 series	IEEE C57 series	Ratings, losses, thermal loading
Generator sets	ISO 8528 series	NFPA 110, IEEE 446	Ratings, performance classes, emergency supply
Cable ratings	IEC 60287, IEC 60364-5-52	NEC Article 310, IEEE 835	Ampacity, derating, economic sizing
Industrial practice	—	IEEE 3000 series (formerly the "Color Books": 141, 142, 242, 399, 446, 493)	Design guidance
MV installations	IEC 61936-1, EN 50522	IEEE C2 (NESC), IEEE 80	Clearances, earthing
Hazardous areas	IEC 60079	NEC Articles 500–506	Zones and equipment protection
Marine	IEC 60092, IEC/IEEE 80005	ABS, DNV, Lloyd's rules	IT systems, shore power
Grid connection	National grid and distribution codes, IEC 61727, EN 50549	IEEE 1547	Generators and inverters on the network (Chapter 27)

Demonstrating compliance

Compliance is demonstrated, not asserted, through four layers:

1. **Design review** against the design basis and standards, ideally by an engineer independent of the designer.
2. **Type-tested and certified equipment**: switchboards verified to IEC 61439 by test or by the standard's design rules, breakers with test certificates for their declared ratings, Ex equipment with IECEx or ATEX certification.
3. **Inspection and testing** of the completed installation to IEC 60364-6 (or the NEC and local authority inspection), in the dead-then-live sequence of Chapter 29, with every result recorded against its limit.
4. **Certification and handover**: an electrical installation certificate signed by the designer, the installer and the inspector, with the schedule of test results, the as-built documents and the arc-flash labels in place. Periodic inspection then repeats the tests at intervals set by the risk (typically one to five years for industrial sites).

Grid-connected generation and storage add a fifth layer: compliance tests and simulation models to the grid code, witnessed by the network operator before the plant may export (Chapter 27).

Worked example — The facility, summarised. The design of this chapter can be stated in one paragraph: a maximum demand of 945 kVA (1181 kVA with growth) supplied by a 1250 kVA Dyn11 6 % transformer giving 29.1 kA and 67.2 kA peak at a 2000 A, 50 kA main switchboard; an 800 kVA prime-rated generator limiting the worst motor-start dip to 13 %; a 185 mm² feeder (240 mm² pays back on losses and is the recommended option; the voltage-drop and fault figures here are for 185 mm²), with 3.7 % running and 7.0 % starting voltage drop at the furthest motor; selectivity by time grading with zone-selective interlocking reducing busbar arc energy to category 1; a TN-S system with a 1.15 Ω foundation earth electrode satisfying a 430 V earth-potential-rise limit. Each figure in that paragraph traces back to a calculation in this chapter, and each calculation back to a model in an earlier chapter. That traceability — from a number on a drawing to the physics that justifies it — is what distinguishes engineering design from assembling equipment.

What remains open

Open question	Where it is answered
How are transformer impedance, losses and regulation derived from the equivalent circuit?	Chapter 4 (transformers)
How does a synchronous generator respond to a sudden load, and what sets x''_d and x'_d?	Chapter 7 (synchronous machines); Chapter 20 (stability)
How are induction-motor starting current and torque related to voltage, and which starting methods reduce the dip?	Chapter 6 (starting methods); Chapter 10 (drive selection)
How are fault currents computed for unsymmetrical faults and networks with many sources?	Chapters 17 and 19 (per-unit modelling, fault analysis)
How do relays protect MV feeders, transformers and generators, and how are they coordinated?	Chapter 23 (power-system protection)
How are substation earth grids designed against step and touch voltages, and how is insulation coordinated?	Chapter 24 (high-voltage engineering)
How do harmonics from drives and chargers affect cable, transformer and generator sizing?	Chapter 25 (power quality)
How are inverter-based generation and storage connected under grid codes?	Chapter 27 (grid integration of renewable energy)
How are losses valued over the life of the installation, and how is the design kept efficient in operation?	Chapter 28 (energy efficiency and management)
How do the converters inside drives, UPS and EV chargers work, and what are their device limits?	*Electronics & Computing*
How do conductor and insulation materials age, and how are their properties set by their structure?	*The Materials Core*

This is where the book ends and practice begins: from the flux in a single core in Chapter 1 to a complete installation that is safe, efficient and documented, every result has been a step in following energy from where it is converted to where it is used — and the engineer who can trace each number on a drawing back to the physics behind it is ready to design the systems that will carry the energy transition.